Mathematical Methods for Physicists

Problems and Worked Solutions — Arfken, Weber & Harris, 7th ed.

Dark red text marks published errata (from the Instructor's Manual, *Errata and Revision Status*, 6 April 2012) and additional corrections or necessary qualifications explained in the corresponding solutions.

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1 Chapter 1. Mathematical Preliminaries

Problems

1.1 Infinite Series

1.1.1

(a) Prove that if $\lim_{n\to\infty} n^p u_n = A < \infty$, $p > 1$, the series $\sum_{n=1}^{\infty} u_n$ converges.

(b) Prove that if $\lim_{n\to\infty} n u_n = A > 0$, the series diverges. (The test fails for $A = 0$.)

These two tests, known as **limit tests**, are often convenient for establishing the convergence of a series. They may be treated as comparison tests, comparing with

$$\sum_n n^{-q}, \quad 1 \le q < p.$$

Solution.

(a) Interpret A as a finite real or complex number. By the definition of limit there is N such that $|n^p u_n - A| < 1$ for $n \ge N$. Hence

$$|u_n| \le \frac{|A|+1}{n^p}, \qquad \sum_{n=N}^{\infty} |u_n| \le (|A|+1) \sum_{n=N}^{\infty} n^{-p} < \infty.$$

The last series converges because $\int_N^{\infty} x^{-p} dx = N^{1-p}/(p-1) < \infty$. Adding finitely many initial terms proves absolute convergence. This also covers $A = 0$.

(b) If $n u_n \to A > 0$, choose N so that $n u_n > A/2$ for $n \ge N$. Then $\sum_{n=N}^{M} u_n \ge (A/2) \sum_{n=N}^{M} n^{-1} \to +\infty$. When $A = 0$, both $u_n = n^{-2}$ (convergent) and $u_n = 1/(n \ln n)$, $n \ge 2$ (divergent), meet the limit condition; the test therefore cannot decide that case.

1.1.2 For eventually positive sequences, if $\lim_{n\to\infty} \dfrac{b_n}{a_n} = K$, a constant with $0 < K < \infty$, show that $\sum_n b_n$ converges or diverges with $\sum a_n$.

Hint. If $\sum a_n$ converges, rescale b_n to $b_n' = \dfrac{b_n}{2K}$. If $\sum_n a_n$ diverges, rescale to $b_n'' = \dfrac{2b_n}{K}$.

Solution.

The limit-comparison statement requires eventually positive terms (as in the positive-series tests of this section). For sufficiently large n, the limit gives

$$\frac{K}{2} < \frac{b_n}{a_n} < \frac{3K}{2}, \qquad \frac{K}{2} a_n < b_n < \frac{3K}{2} a_n.$$

If $\sum a_n$ converges, the upper bound proves convergence of $\sum b_n$. If $\sum a_n$ diverges, the lower bound proves divergence of $\sum b_n$. Finite initial segments do not change either conclusion. The sign hypothesis matters: with $a_n = (-1)^n/\sqrt{n}$ and $b_n = a_n + 1/n$, one has $b_n/a_n = 1 + (-1)^n/\sqrt{n} \to 1$, but $\sum a_n$ converges whereas the partial sums of $\sum b_n$ tend to $+\infty$. Thus the unrestricted statement is false.

1.1.3

(a) Show that the series $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ converges.

(b) By direct addition $\sum_{n=2}^{100,000}[n(\ln n)^2]^{-1} = 2.02288$. Use Eq. (1.9) to make a five-significant-figure estimate of the sum of this series.

Solution.

(a) Set $f(x) = 1/[x(\ln x)^2]$. For $x > 1$,

$$f'(x) = -\frac{\ln x + 2}{x^2(\ln x)^3} < 0, \qquad \int_2^{\infty} f(x)\,dx = \left[-\frac{1}{\ln x}\right]_2^{\infty} = \frac{1}{\ln 2} < \infty.$$

The integral test proves convergence.

(b) For $S_N = \sum_{n=2}^{N} f(n)$, rectangle comparisons give

$$\int_{N+1}^{\infty} f(x)dx < S - S_N < \int_N^{\infty} f(x)dx, \quad S_N + \frac{1}{\ln(N+1)} < S < S_N + \frac{1}{\ln N}.$$

The supplementary script `ch01_check.py` gives

$$S_{100000} = 2.022883942578507\ldots.$$

Consequently

$$2.10974276351492 < S < 2.10974283895916.$$

Both bounds round to $\boxed{2.1097}$ (five significant figures). Even an uncertainty of 5×10^{-6} in the supplied rounded partial sum leaves this five-significant-figure conclusion unchanged.

1.1.4 Gauss' test is often given in the form of a test of the ratio

$$\frac{u_n}{u_{n+1}} = \frac{n^2 + a_1 n + a_0}{n^2 + b_1 n + b_0}.$$

Assume the sequence is eventually positive and nonzero, and the displayed ratio is defined for all sufficiently large n. For what values of the parameters a_1 and b_1 is there convergence? divergence?

ANS. Convergent for $a_1 - b_1 > 1$, divergent for $a_1 - b_1 \le 1$.

Solution.

For an eventually positive nonzero sequence put $d = a_1 - b_1$. Dividing numerator and denominator by n^2 and using $(1+z)^{-1} = 1 - z + O(z^2)$ gives

$$\frac{u_n}{u_{n+1}} = 1 + \frac{d}{n} + O(n^{-2}), \qquad \log\frac{u_{n+1}}{u_n} = -\frac{d}{n} + O(n^{-2}).$$

Summing from a fixed large N to $n-1$,

$$\log u_n = \log u_N - d\sum_{k=N}^{n-1}\frac{1}{k} + \sum_{k=N}^{n-1} O(k^{-2}) = -d\log n + C + o(1).$$

Here the last error sum converges, as does the difference between the harmonic sum and $\log n$. Thus $u_n \sim C_1 n^{-d}$ with $C_1 > 0$. Limit comparison with a p-series proves convergence for $d > 1$ and divergence for $d \le 1$, including the borderline $d = 1$. The parameters a_0, b_0 affect the constant, but not this classification.

1.1.5 Test for convergence

(a) $\displaystyle\sum_{n=2}^{\infty}(\ln n)^{-1}$

(b) $\displaystyle\sum_{n=1}^{\infty}\frac{n!}{10^n}$

(c) $\displaystyle\sum_{n=1}^{\infty}\frac{1}{2n(2n+1)}$

(d) $\displaystyle\sum_{n=1}^{\infty}[n(n+1)]^{-1/2}$

(e) $\displaystyle\sum_{n=0}^{\infty}\frac{1}{2n+1}$

Solution.

(a) For $n \geq 2$, $\ln n < n$, so $1/\ln n > 1/n$. Comparison with the harmonic series proves divergence.

(b) For $u_n = n!/10^n$, $u_{n+1}/u_n = (n+1)/10$. For $n \geq 19$ this is at least 2, so the terms grow instead of tending to zero. The series diverges.

(c) Since $0 < 1/[2n(2n+1)] < 1/(4n^2)$, the series converges. As a check, partial fractions give $1/[2n(2n+1)] = 1/(2n) - 1/(2n+1)$, whose sum is $1 - \ln 2$.

(d) With $u_n = [n(n+1)]^{-1/2}$, $nu_n = (1+1/n)^{-1/2} \to 1$. Limit comparison with $1/n$ proves divergence.

(e) For $n \geq 1$, $1/(2n+1) \geq 1/(3n)$, so the series diverges. Its first term $n = 0$ has no effect on this conclusion.

1.1.6 Test for convergence

(a) $\displaystyle\sum_{n=1}^{\infty}\frac{1}{n(n+1)}$

(b) $\displaystyle\sum_{n=2}^{\infty}\frac{1}{n\ln n}$

(c) $\displaystyle\sum_{n=1}^{\infty}\frac{1}{n\,2^n}$

(d) $\displaystyle\sum_{n=1}^{\infty}\ln\left(1+\frac{1}{n}\right)$

(e) $\displaystyle\sum_{n=1}^{\infty}\frac{1}{n\cdot n^{1/n}}$

Solution.

(a) $1/[n(n+1)] = 1/n - 1/(n+1)$, so the partial sum through N is $1 - 1/(N+1) \to 1$. The series converges.

(b) The positive function $1/(x \ln x)$ decreases for $x > 1$, but $\int_2^M dx/(x \ln x) = \ln\ln M - \ln\ln 2 \to \infty$. The series diverges.

(c) $0 < 1/(n2^n) \le 2^{-n}$ proves convergence. Integrating $\sum_{n=0}^{\infty} x^n = (1-x)^{-1}$ from 0 to 1/2 gives $\sum_{n=1}^{\infty} 1/(n2^n) = -\ln(1-1/2) = \ln 2$.

(d) The partial sum is $\sum_{n=1}^{N} \ln[(n+1)/n] = \ln \prod_{n=1}^{N} (n+1)/n = \ln(N+1)$, which tends to infinity.

(e) $n^{1/n} = \exp[(\ln n)/n] \to 1$, so $n/[nn^{1/n}] \to 1$. Limit comparison with the harmonic series proves divergence.

1.1.7 For what values of p and q will $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n^p (\ln n)^q}$ converge?

ANS. Convergent for $\begin{cases} p > 1, \text{ all } q, \\ p = 1, \ q > 1, \end{cases}$ divergent for $\begin{cases} p < 1, \text{ all } q, \\ p = 1, \ q \le 1. \end{cases}$

Solution.

Let p, q be real. We use the elementary fact that, for every $\epsilon > 0$ and fixed real q, $(\ln n)^{|q|} < n^{\epsilon}$ for all sufficiently large n; taking logarithms reduces it to $|q| \ln\ln n < \epsilon \ln n$. If $p > 1$, take $\epsilon = (p-1)/2$. Eventually

$$\frac{1}{n^p (\ln n)^q} \le \frac{(\ln n)^{|q|}}{n^p} \le n^{-(p+1)/2},$$

and the comparison series converges. If $p < 1$, take $\epsilon = (1-p)/2$. Eventually $n^p (\ln n)^q \le n^{(p+1)/2}$, so the terms dominate a divergent p-series. For $p = 1$ make the substitution $t = \ln x$:

$$\int_2^M \frac{dx}{x(\ln x)^q} = \int_{\ln 2}^{\ln M} t^{-q} dt.$$

This is finite at its upper limit exactly when $q > 1$. The function used in this last integral is eventually decreasing, so the integral test applies. These three cases give all the stated ranges.

1.1.8 Given $\sum_{n=1}^{1,000} n^{-1} = 7.485\,470\ldots$ set upper and lower bounds on the Euler-Mascheroni constant.

ANS. $0.5767 < \gamma < 0.5778$.

Solution.

Write $H_N = \sum_{n=1}^{N} 1/n$. Rectangle comparisons on each interval $[n, n+1]$ show that $H_N - \ln N$ decreases and $H_N - \ln(N+1)$ increases. Moreover $H_N - \ln N > H_N - \ln(N+1) \ge 1 - \ln 2$ and $H_N - \ln(N+1) \le H_N - \ln N \le 1$, so the monotone sequences have finite limits. Their difference is $\ln(1 + 1/N) \to 0$, so both approach the same constant γ. Consequently

$$H_N - \ln(N+1) < \gamma < H_N - \ln N.$$

For $N = 1000$, direct summation gives $H_{1000} = 7.485470860550345\ldots$, and the bounds are

$$0.57671608123512 < \gamma < 0.57771558156821.$$

Rounding the lower bound downward and the upper bound upward gives $0.5767 < \gamma < 0.5778$.

1.1.9 (From **Olbers' paradox**.) Assume a static universe in which the stars are uniformly distributed. Divide all space into shells of constant thickness; the stars in any one shell by themselves subtend a solid angle of ω_0. **Allowing for the blocking out of distant stars by nearer stars**, show that the total net solid angle subtended by all stars, shells extending to infinity, is **exactly** 4π with probability one in the model of independent uniform angular coverage, where $0 < \omega_0 \leq 4\pi$. [Therefore the night sky should be ablaze with light. For more details, see E. Harrison, *Darkness at Night: A Riddle of the Universe*. Cambridge, MA: Harvard University Press (1987).]

Solution.

Use the intended model of uniform, independent angular coverage by successive shells, with $0 < \omega_0 \leq 4\pi$. Let $f = \omega_0/(4\pi)$. For a fixed direction the probability of remaining unobscured after $n-1$ independent shells is $(1-f)^{n-1}$. The probability that shell n supplies its first visible star is therefore $f(1-f)^{n-1}$. Integrating this indicator probability over all directions gives the expected visible solid angle of shell n:

$$\mathbb{E}[\Delta\Omega_n] = 4\pi f(1-f)^{n-1}.$$

For $0 < f < 1$ the finite geometric sum is

$$\mathbb{E}[\Omega_N] = 4\pi f \frac{1-(1-f)^N}{1-(1-f)} = 4\pi[1-(1-f)^N] \longrightarrow 4\pi.$$

This also proves the almost-sure statement about the limiting solid angle. Let $U_N = 4\pi - \Omega_N$ be the uncovered area. It is nonnegative and decreases to U_∞, so for every $\epsilon > 0$,

$$\mathbb{P}(U_\infty > \epsilon) \leq \mathbb{P}(U_N > \epsilon) \leq \frac{\mathbb{E}[U_N]}{\epsilon} = \frac{4\pi(1-f)^N}{\epsilon} \longrightarrow 0.$$

Thus $U_\infty = 0$ with probability one and $\Omega_\infty = 4\pi$. In the deterministic idealization where each shell covers exactly fraction f of every remaining open region, the same geometric formulas hold without the expectation symbols. If $f = 1$, the first shell already covers the sky. Independent angular coverage, or this equivalent idealization, is essential: placing every shell behind the same patch would not establish the conclusion merely from the value of ω_0.

1.1.10 Test for convergence

$$\sum_{n=1}^{\infty} \left[\frac{1\cdot 3\cdot 5\cdots(2n-1)}{2\cdot 4\cdot 6\cdots(2n)}\right]^2 = \frac{1}{4} + \frac{9}{64} + \frac{25}{256} + \cdots.$$

Solution.

Let u_n denote the squared product. Cancelling the common factors gives

$$\frac{u_n}{u_{n+1}} = \left(\frac{2n+2}{2n+1}\right)^2 = \frac{n^2+2n+1}{n^2+n+1/4}.$$

Thus $a_1 - b_1 = 2 - 1 = 1$, the divergent borderline case of Exercise 1.1.4. For an independent check, write

$$u_n = \left[\frac{(2n)!}{2^{2n}(n!)^2}\right]^2.$$

Stirling's formula gives $(2n)!/[2^{2n}(n!)^2] \sim 1/\sqrt{\pi n}$, so $u_n \sim 1/(\pi n)$ and the harmonic comparison again proves divergence. The manual (p. 8) prints $u_{n+1}/u_n = 1+1/n+O(n^{-2})$. For that orientation of the ratio the sign is minus: $u_{n+1}/u_n = [(2n+1)/(2n+2)]^2 = 1-1/n+O(n^{-2})$. The reciprocal used above has the plus sign.

1.1.11 Determine whether each of these series is convergent, and if so, whether it is absolutely convergent:

(a) $\dfrac{\ln 2}{2} - \dfrac{\ln 3}{3} + \dfrac{\ln 4}{4} - \dfrac{\ln 5}{5} + \dfrac{\ln 6}{6} - \cdots$,

(b) $\dfrac{1}{1} + \dfrac{1}{2} - \dfrac{1}{3} - \dfrac{1}{4} + \dfrac{1}{5} + \dfrac{1}{6} - \dfrac{1}{7} - \dfrac{1}{8} + \cdots$,

(c) $1 - \dfrac{1}{2} - \dfrac{1}{3} + \dfrac{1}{4} + \dfrac{1}{5} + \dfrac{1}{6} - \dfrac{1}{7} - \dfrac{1}{8} - \dfrac{1}{9} - \dfrac{1}{10} + \dfrac{1}{11} + \cdots + \dfrac{1}{15} - \dfrac{1}{16} - \cdots - \dfrac{1}{21} + \cdots$.

Solution.

(a) For $a(x) = \ln x/x$, $a'(x) = (1 - \ln x)/x^2 < 0$ when $x > e$, and $a(x) \to 0$. The alternating-series test, ignoring finitely many initial terms, proves convergence. The absolute series diverges since $\int_e^M (\ln x)/x\,dx = [(\ln x)^2 - 1]/2 \to \infty$.

(b) Combine consecutive pairs to obtain

$$\sum_{k=0}^{\infty} (-1)^k b_k, \qquad b_k = \frac{1}{2k+1} + \frac{1}{2k+2}.$$

Both summands decrease, and $b_k \to 0$, so the grouped series converges. An unpaired last term is at most $1/(2k+1) \to 0$; hence the original partial sums have the same limit. Removing signs gives the harmonic series, so convergence is conditional.

(c) Put $T_k = k(k+1)/2$. The kth block has sign $(-1)^{k+1}$, length k, and magnitude

$$B_k = \sum_{j=1}^{k} \frac{1}{T_{k-1} + j}.$$

We must prove its decrease, not merely bound its size. Write $a_j = T_{k-1} + j$, so that

$$B_k - B_{k+1} = \sum_{j=1}^{k} \frac{k}{a_j(a_j + k)} - \frac{1}{T_{k+1}} \geq \frac{k^2/T_k - 1}{T_{k+1}} = \frac{k-1}{(k+1)T_{k+1}} > 0 \quad (k > 1).$$

Here $a_j \leq T_k$ and $a_j + k < T_{k+1}$. Also $B_1 = 1 > B_2 = 5/6$, and $0 < B_k \leq k/(T_{k-1} + 1) \to 0$. The block series thus converges by the alternating-series test. Every incomplete block has magnitude at most $B_k \to 0$, proving convergence of all original partial sums. The absolute series is again harmonic. All three series are therefore conditionally, but not absolutely, convergent.

1.1.12 Catalan's constant $\beta(2)$ is defined by

$$\beta(2) = \sum_{k=0}^{\infty} (-1)^k (2k+1)^{-2} = \frac{1}{1^2} - \frac{1}{3^2} + \frac{1}{5^2} \cdots.$$

Calculate $\beta(2)$ to six-digit accuracy.

Hint. The rate of convergence is enhanced by pairing the terms,

$$(4k-1)^{-2} - (4k+1)^{-2} = \frac{16k}{(16k^2-1)^2}.$$

If you have carried enough digits in your summation, $\sum_{1\le k\le N} 16k/(16k^2-1)^2$, additional significant figures may be obtained by setting upper and lower bounds on the tail of the series, $\sum_{k=N+1}^{\infty}$. These bounds may be set by comparison with integrals, as in the Maclaurin integral test.

ANS. $\beta(2) = 0.9159\,6559\,4177\cdots$.

Solution.

Pair the terms after the first:

$$G = 1 - \sum_{k=1}^{\infty}\left[\frac{1}{(4k-1)^2} - \frac{1}{(4k+1)^2}\right] = 1 - \sum_{k=1}^{\infty} f(k), \quad f(x) = \frac{16x}{(16x^2-1)^2}.$$

For $x \ge 1$, $f'(x) = -16(48x^2+1)/(16x^2-1)^3 < 0$, and

$$\int_N^{\infty} f(x)dx = \frac{1}{2(16N^2-1)}.$$

If $G_N = 1 - \sum_{k=1}^{N} f(k)$, integral comparison gives

$$G_N - \frac{1}{2(16N^2-1)} < G < G_N - \frac{1}{2(16(N+1)^2-1)}.$$

With $N = 100$, the verified interval is $0.91596556308 < G < 0.91596562466$. Thus to six decimal places (and also to six significant figures), $\boxed{G = 0.915966}$. The script checks this interval against direct numerical integration of $\arctan t/t$ as well.

1.1.13 Show how to combine $\zeta(2) = \sum_{n=1}^{\infty} n^{-2}$ with α_1 and α_2 to obtain a series converging as n^{-4}.

Note. $\zeta(2)$ has the known value $\pi^2/6$. See Eq. (12.66).

Solution.

Use $\alpha_1 = \sum[n(n+1)]^{-1} = 1$ and $\alpha_2 = \sum[n(n+1)(n+2)]^{-1} = 1/4$. With undetermined c_1, c_2,

$$\frac{1}{n^2} - \frac{c_1}{n(n+1)} - \frac{c_2}{n(n+1)(n+2)} = \frac{(1-c_1)n^2 + (3-2c_1-c_2)n + 2}{n^2(n+1)(n+2)}.$$

Set $1 - c_1 = 0$ and $3 - 2c_1 - c_2 = 0$, giving $c_1 = c_2 = 1$. Summing the exact termwise identity yields

$$\boxed{\zeta(2) = \frac{5}{4} + \sum_{n=1}^{\infty} \frac{2}{n^2(n+1)(n+2)}}.$$

The new summand is asymptotic to $2n^{-4}$, as requested.

1.1.14 Give a method of computing

$$\lambda(3) = \sum_{n=0}^{\infty} \frac{1}{(2n+1)^3}$$

that converges at least as fast as n^{-8} and obtain a result good to six decimal places.

ANS. $\lambda(3) = 1.051800$.

Solution.

Separate the even terms of $\zeta(3)$: $\lambda(3) = \zeta(3) - 2^{-3}\zeta(3) = 7\zeta(3)/8$. To accelerate the evaluation, define, for $n \geq 4$,

$$A_2(n) = \frac{1}{n(n^2-1)}, \quad A_4(n) = \frac{1}{n(n^2-1)(n^2-4)}, \quad A_6(n) = \frac{1}{n(n^2-1)(n^2-4)(n^2-9)}.$$

Use the common denominator $D = n^3(n^2-1)(n^2-4)(n^2-9)$. With $z = n^2$, the numerator of $n^{-3} - A_2(n) + A_4(n) - 4A_6(n)$ becomes

$$\begin{aligned}
&(z-1)(z-4)(z-9) - z(z-4)(z-9) + z(z-9) - 4z \\
&\quad = -(z-4)(z-9) + z(z-9) - 4z \\
&\quad = (-z^2 + 13z - 36) + (z^2 - 9z) - 4z = -36.
\end{aligned}$$

Rearranging therefore gives the exact identity

$$\frac{1}{n^3} = A_2(n) - A_4(n) + 4A_6(n) - \frac{36}{n^3(n^2-1)(n^2-4)(n^2-9)}.$$

To spell out the shifted sums, $A_2(n)$ is $[(n-1)n(n+1)]^{-1}$, $A_4(n)$ is the reciprocal of the five consecutive factors from $n-2$ to $n+2$, and $A_6(n)$ is the reciprocal of the seven from $n-3$ to $n+3$. Their complete sums therefore start at $n = 2, 3, 4$ and equal $\alpha_2, \alpha_4, \alpha_6$, respectively. The telescoping formula $\alpha_p = 1/(p\,p!)$ (proved in Exercise 1.5.3) implies

$$\sum_{n=4}^{\infty} A_2(n) = \frac{1}{4} - \frac{1}{6} - \frac{1}{24} = \frac{1}{24}, \quad \sum_{n=4}^{\infty} A_4(n) = \frac{1}{96} - \frac{1}{120} = \frac{1}{480}, \quad \sum_{n=4}^{\infty} A_6(n) = \frac{1}{4320}.$$

Adding $n = 1, 2, 3$ therefore gives

$$\boxed{\lambda(3) = \frac{7}{8}\left[\frac{1039}{864} - 36\sum_{n=4}^{\infty} \frac{1}{n^3(n^2-1)(n^2-4)(n^2-9)}\right]}.$$

The summand is $O(n^{-9})$, faster than required. A cutoff at $N = 10$ gives $L_{10} = 1.051799819028417\ldots$. For $n > N$ put $C_N = \prod_{j=1}^{3}[1 - j^2/(N+1)^2]$. Then

$$0 < L_N - \lambda(3) < \frac{63}{2C_N}\int_N^{\infty} x^{-9}dx = \frac{63}{16C_N N^8}.$$

At $N = 10$ this is less than 4.5×10^{-8}. The whole resulting interval rounds to $\boxed{1.051800}$ at six decimal places.

1.1.15 Show that (a) $\sum_{n=2}^{\infty}[\zeta(n) - 1] = 1$, (b) $\sum_{n=2}^{\infty}(-1)^n[\zeta(n) - 1] = \frac{1}{2}$, where $\zeta(n)$ is the Riemann zeta function.

Solution.

Since $\zeta(n) - 1 = \sum_{k=2}^{\infty} k^{-n}$, the nonnegative double series can first be summed over n:

$$\sum_{n=2}^{\infty}[\zeta(n) - 1] = \sum_{k=2}^{\infty}\sum_{n=2}^{\infty} k^{-n} = \sum_{k=2}^{\infty} \frac{k^{-2}}{1 - k^{-1}} = \sum_{k=2}^{\infty}\left(\frac{1}{k-1} - \frac{1}{k}\right) = 1.$$

This also proves absolute convergence of the signed double series in (b), so its order may be exchanged:

$$\sum_{n=2}^{\infty}(-1)^n[\zeta(n) - 1] = \sum_{k=2}^{\infty} \frac{k^{-2}}{1 + k^{-1}} = \sum_{k=2}^{\infty}\left(\frac{1}{k} - \frac{1}{k+1}\right) = \frac{1}{2}.$$

Both final sums telescope; their partial sums are respectively $1 - 1/K$ and $1/2 - 1/(K+1)$.

1.1.16 The convergence improvement of Example 1.1.11 may be carried out more expediently (in this special case) by putting α_2, from Eq. (1.26), into a more symmetric form: Replacing n by $n-1$, we have

$$\alpha_2' = \sum_{n=2}^{\infty} \frac{1}{(n-1)n(n+1)} = \frac{1}{4}.$$

(a) Combine $\zeta(3)$ and α_2' to obtain convergence as n^{-5}.

(b) Let α_4' be α_4 with $n \to n-2$. Combine $\zeta(3)$, α_2', and α_4' to obtain convergence as n^{-7}.

(c) If $\zeta(3)$ is to be calculated to six-decimal place accuracy (error 5×10^{-7}), how many terms are required for $\zeta(3)$ alone? combined as in part (a)? combined as in part (b)?

Note. The error may be estimated using the corresponding integral.

ANS. (a) $\zeta(3) = \dfrac{5}{4} - \displaystyle\sum_{n=2}^{\infty} \frac{1}{n^3(n^2-1)}$.

Solution.

The opening reference is to Example 1.1.11 (the acceleration of $\zeta(3)$), not Exercise 1.1.11. (a) For $n \geq 2$,

$$\frac{1}{n^3} - \frac{1}{n(n^2-1)} = -\frac{1}{n^3(n^2-1)}.$$

Separate $n=1$ and use $\alpha_2' = 1/4$ to obtain $\zeta(3) = 5/4 - \sum_{n=2}^{\infty}[n^3(n^2-1)]^{-1}$, whose summand is $O(n^{-5})$.

(b) Separate the $n=2$ term in this last sum and add/subtract $\alpha_4' = \sum_{n=3}^{\infty}[n(n^2-1)(n^2-4)]^{-1} = 1/96$. The exact algebra is

$$-\frac{1}{n^3(n^2-1)} + \frac{1}{n(n^2-1)(n^2-4)} = \frac{4}{n^3(n^2-1)(n^2-4)}.$$

Thus

$$\boxed{\zeta(3) = \frac{5}{4} - \frac{1}{24} - \frac{1}{96} + \sum_{n=3}^{\infty} \frac{4}{n^3(n^2-1)(n^2-4)} = \frac{115}{96} + \sum_{n=3}^{\infty} \frac{4}{n^3(n^2-1)(n^2-4)}}.$$

The summand is $O(n^{-7})$. The manual's final formula in part (b), p. 10, has n instead of n^3 in the denominator. The factor n^3 is required by the displayed subtraction and by the claimed n^{-7} rate.

(c) Specify the cutoff by the largest included index N, avoiding ambiguity about the separately summed initial terms. Leading integral estimates of the three tails are $1/(2N^2)$, $1/(4N^4)$, and $2/(3N^6)$, respectively. Exact positive tails, checked by the script, give the decisive neighboring cutoffs:

	N	error	$N_{\min}$	error
$\sum n^{-3}$	999	$5.0050025\,10^{-7}$	1000	$4.9950025\,10^{-7}$
(a)	26	$5.0681999\,10^{-7}$	27	$4.370325 4\,10^{-7}$
(b)	10	$5.0671099\,10^{-7}$	11	$2.9266119\,10^{-7}$

These cutoffs correspond to 1000, 26, and 9 summands, respectively, with constants 5/4 or 115/96 evaluated exactly. Positivity makes the error decrease with N, so checking the two neighboring values identifies the minimum. The manual's "999" for the unaccelerated sum is one term too small for an error below 5×10^{-7}.

1.2 Series of Functions

1.2.1 Find the range of uniform convergence of the series

(a) $\eta(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^x}$,

(b) $\zeta(x) = \sum_{n=1}^{\infty} \frac{1}{n^x}$.

ANS. (a) $0 < s \le x < \infty$. (b) $1 < s \le x < \infty$.

Solution.

(a) For fixed $x > 0$, n^{-x} decreases to zero. The alternating-series remainder therefore obeys

$$|\eta(x) - S_N(x)| \le (N+1)^{-x} \le (N+1)^{-s}, \qquad x \ge s > 0.$$

The last expression is independent of x and tends to zero, proving uniform convergence on every $[s, \infty)$ with $s > 0$. On $(0, \infty)$ it is not uniform: a necessary condition is uniform convergence of the terms to zero, whereas $\sup_{x>0} n^{-x} = 1$ for every n. At $x \le 0$ the terms do not even tend to zero.

(b) If $x \ge s > 1$, then $|n^{-x}| \le n^{-s}$, and the convergent p-series supplies the Weierstrass bound. Thus convergence is uniform on every $[s, \infty)$, $s > 1$. It is not uniform on $(1, \infty)$: for every N,

$$\sup_{x>1} \sum_{n=N+1}^{2N} n^{-x} = \sum_{n=N+1}^{2N} \frac{1}{n} \ge \frac{1}{2},$$

contradicting the uniform Cauchy criterion. For $x \le 1$ the positive series diverges. The lower bounds $s > 0$ and $s > 1$ cannot be omitted.

1.2.2 For what range of x is the geometric series $\sum_{n=0}^{\infty} x^n$ uniformly convergent?

ANS. $-1 < -s \le x \le s < 1$.

Solution.

For $|x| < 1$ the finite-sum identity gives the exact remainder

$$\sum_{n=N+1}^{\infty} x^n = \frac{x^{N+1}}{1-x}.$$

If $|x| \le s < 1$, its absolute value is at most $s^{N+1}/(1-s)$, which tends to zero independently of x. Thus the series is uniformly convergent on every $[-s, s]$, $s < 1$, or more generally any set bounded away from both endpoints ± 1. If a set has points with $|x|$ arbitrarily close to 1, then $\sup |x^n| = 1$ and the necessary uniform term test fails. The full interval $(-1, 1)$ is therefore not a range of uniform convergence, though it is the range of pointwise convergence.

1.2.3 For what range of positive values of x is $\sum_{n=0}^{\infty} 1/(1+x^n)$

(a) convergent? (b) uniformly convergent?

Solution.

The term $n = 0$ is $1/2$. If $0 < x < 1$, the terms approach 1; if $x = 1$ they all equal $1/2$. Both cases diverge. If $x > 1$,

$$0 < \frac{1}{1 + x^n} \leq x^{-n},$$

so geometric comparison proves convergence. For $x \geq 1 + \epsilon$, $\epsilon > 0$, the terms are bounded by $(1 + \epsilon)^{-n}$, proving uniform convergence on $[1 + \epsilon, \infty)$. Uniform convergence fails on $(1, \infty)$, since $\sup_{x>1}(1 + x^n)^{-1} = 1/2$ for each n. Thus a range extending arbitrarily close to 1 from above must be excluded.

1.2.4 If the series of the coefficients $\sum a_n$ and $\sum b_n$ are absolutely convergent, show that the Fourier series

$$\sum (a_n \cos nx + b_n \sin nx)$$

is **uniformly** convergent for $-\infty < x < \infty$.

Solution.

For every real x,

$$|a_n \cos nx + b_n \sin nx| \leq |a_n|\,|\cos nx| + |b_n|\,|\sin nx| \leq |a_n| + |b_n|.$$

The hypotheses give $\sum_n (|a_n| + |b_n|) < \infty$. In particular, for any $M > N$,

$$\sup_{x \in \mathbb{R}} \left| \sum_{n=N+1}^{M} (a_n \cos nx + b_n \sin nx) \right| \leq \sum_{n=N+1}^{M} (|a_n| + |b_n|) \longrightarrow 0.$$

This is the uniform Cauchy criterion on the entire real line, and proves both absolute pointwise convergence and uniform convergence.

1.2.5 Assume a nonzero initial term and real l. The Legendre series $\sum_{j \text{ even}} u_j(x)$ satisfies the recurrence relations

$$u_{j+2}(x) = \frac{(j+1)(j+2) - l(l+1)}{(j+2)(j+3)}\, x^2 u_j(x),$$

in which the index j is even and l is some constant (but, in this problem, **not** a nonnegative odd integer, nor a negative even integer if negative l is allowed). Find the range of values of x for which this Legendre series is convergent. Test the endpoints.

ANS. $-1 < x < 1$.

Solution.

The quoted answer concerns a nonzero, nonterminating series. If arbitrary real l is allowed, also exclude $l = -2, -4, -6, \ldots$; these values terminate just as $l = 1, 3, 5, \ldots$ do. Write $j = 2r$. For a nonterminating series the magnitude of the ratio of successive nonzero terms tends to x^2. Hence the ratio test proves absolute convergence for $|x| < 1$ and divergence for $|x| > 1$. At $x = \pm 1$, the ratio is eventually positive, and its reciprocal is

$$\frac{(2r+2)(2r+3)}{(2r+1)(2r+2) - l(l+1)} = \frac{4r^2 + 10r + 6}{4r^2 + 6r + 2 - l(l+1)} = 1 + \frac{1}{r} + O(r^{-2}).$$

The argument of Exercise 1.1.4 shows that the terms have one eventual sign and magnitude asymptotic to C/r, so both endpoints diverge. To check the exception, the numerator vanishes exactly when $l(l+1) = (j+1)(j+2)$, or $l = j + 1$ or $l = -j - 2$. For even $j \geq 0$ these are precisely the excluded parameter values. A terminated series, or the identically zero series, converges for all x.

1.2.6 A series solution of the Chebyshev equation leads to successive terms having the ratio

$$\frac{u_{j+2}(x)}{u_j(x)} = \frac{(k+j)^2 - n^2}{(k+j+1)(k+j+2)}\, x^2,$$

with $k = 0$ and $k = 1$. Test for convergence at $x = \pm 1$.

ANS. Convergent.

Solution.

Within either parity branch put $j = 2r$ and $m = k + 2r$. At $x = \pm 1$ the reciprocal ratio of successive terms is

$$\frac{(m+1)(m+2)}{m^2 - n^2} = \frac{m^2 + 3m + 2}{m^2 - n^2} = 1 + \frac{3}{m} + O(m^{-2}) = 1 + \frac{3}{2r} + O(r^{-2}).$$

For a nonterminating branch the terms have magnitude asymptotic to $Cr^{-3/2}$ by the product/logarithm argument of Exercise 1.1.4. The series therefore converges absolutely at both endpoints. If $m^2 = n^2$ at an index of the branch, the recurrence terminates; that polynomial also converges at both endpoints. This covers $k = 0$ and $k = 1$, including terminating cases.

1.2.7 A series solution for the ultraspherical (Gegenbauer) function $C_n^\alpha(x)$ leads to the recurrence

$$a_{j+2} = a_j \frac{(k+j)(k+j+2\alpha) - n(n+2\alpha)}{(k+j+1)(k+j+2)}.$$

Take real n, α, $k = 0$ or 1, and even $j = 0, 2, \ldots$ in each parity branch. Investigate the convergence of each of these series at $x = \pm 1$ as a function of the parameter α.

ANS. For a nonterminating branch: convergent for $\alpha < 1/2$, divergent for $\alpha \geq 1/2$. A terminated branch converges at both endpoints.

Solution.

For the stated recurrence the nonterminating series converges at $x = \pm 1$ for $\alpha < 1/2$, and diverges for $\alpha \geq 1/2$. The book's threshold 1 is incorrect. Terminated series converge for all α. Here α, n are real and $k = 0$ or 1. Indeed, put $j = 2r$ and $m = k + 2r$. For large r the ratio is positive, and at either endpoint its reciprocal is

$$\frac{(m+1)(m+2)}{m(m+2\alpha) - n(n+2\alpha)} = 1 + \frac{3 - 2\alpha}{m} + O(m^{-2}) = 1 + \frac{3/2 - \alpha}{r} + O(r^{-2}).$$

Thus $|a_{2r}| \sim Cr^{\alpha - 3/2}$ in a nonzero nonterminating branch. The p-series criterion requires $3/2 - \alpha > 1$, or $\alpha < 1/2$. At equality the tail is harmonic, and larger α also diverges. Since the parity is fixed, replacing 1 by -1 only multiplies the whole branch by $(-1)^k$; it cannot produce alternating convergence. Finally,

$$m(m+2\alpha) - n(n+2\alpha) = (m-n)(m+n+2\alpha),$$

so the branch terminates if an allowed $m = k + 2r$ equals n or $-n - 2\alpha$. Such branches are the explicit exception to the endpoint test. As a check, $\alpha = 1/2$ gives Legendre's recurrence and its nonterminating logarithmic endpoint divergence.

1.2.8 Show that

(a) $\sin x = \sum_{n=0}^{\infty} (-1)^n \dfrac{x^{2n+1}}{(2n+1)!}$,

(b) $\cos x = \sum_{n=0}^{\infty} (-1)^n \dfrac{x^{2n}}{(2n)!}$.

Solution.

Successive derivatives of sine repeat as $\sin x, \cos x, -\sin x, -\cos x$, while those of cosine repeat as $\cos x, -\sin x, -\cos x, \sin x$. At $x = 0$ this gives

$$\sin^{(2n)}(0) = 0, \quad \sin^{(2n+1)}(0) = (-1)^n, \qquad \cos^{(2n)}(0) = (-1)^n, \quad \cos^{(2n+1)}(0) = 0.$$

Insert these values into Taylor's formula $f(x) = \sum_{j=0}^{N} f^{(j)}(0)x^j/j! + R_N(x)$. For real x all derivatives have magnitude at most one on the segment from 0 to x, so $|R_N(x)| \le |x|^{N+1}/(N+1)! \to 0$. The two stated series therefore equal sine and cosine for every real x. Their term ratios tend to zero for every complex x as well, giving entire power-series extensions.

1.2.9 Derive a series expansion of $\cot x$ in increasing powers of x by dividing the power series for $\cos x$ by that for $\sin x$.

Note. The resultant series that starts with $1/x$ is known as a Laurent series ($\cot x$ does not have a Taylor expansion about $x = 0$, although $\cot(x) - x^{-1}$ does). Although the two series for $\sin x$ and $\cos x$ were valid for all x, the convergence of the series for $\cot x$ is limited by the zeros of the denominator, $\sin x$.

Solution.

Sine is odd and cosine is even, so write $\cot x = x^{-1} + a_1 x + a_3 x^3 + a_5 x^5 + \cdots$. Multiply by $\sin x = x - x^3/6 + x^5/120 - x^7/5040 + \cdots$ and equate the result to $\cos x = 1 - x^2/2 + x^4/24 - x^6/720 + \cdots$:

$$\begin{aligned}
x^2: &\quad a_1 - \tfrac{1}{6} = -\tfrac{1}{2} && \Rightarrow a_1 = -\tfrac{1}{3},\\
x^4: &\quad a_3 - \tfrac{a_1}{6} + \tfrac{1}{120} = \tfrac{1}{24} && \Rightarrow a_3 = -\tfrac{1}{45},\\
x^6: &\quad a_5 - \tfrac{a_3}{6} + \tfrac{a_1}{120} - \tfrac{1}{5040} = -\tfrac{1}{720} && \Rightarrow a_5 = -\tfrac{2}{945},\\
x^8: &\quad a_7 - \tfrac{a_5}{6} + \tfrac{a_3}{120} - \tfrac{a_1}{5040} + \tfrac{1}{362880} = \tfrac{1}{40320} && \Rightarrow a_7 = -\tfrac{1}{4725}.
\end{aligned}$$

Consequently

$$\cot x = \frac{1}{x} - \frac{x}{3} - \frac{x^3}{45} - \frac{2x^5}{945} - \frac{x^7}{4725} - \cdots.$$

More generally, if $\cot x = \sum_{m \ge 0} c_m x^{2m-1}$ and $c_0 = 1$, coefficient comparison gives the recursion

$$c_m = \frac{(-1)^m}{(2m)!} - \sum_{j=1}^{m} \frac{(-1)^j c_{m-j}}{(2j+1)!}, \qquad m \ge 1.$$

The nearest nonzero zeros of $\sin x$ are $\pm\pi$. Since the quotient has poles there and no other singularity closer to zero, its Laurent series converges for $0 < |x| < \pi$.

1.2.10 Show by series expansion that

$$\frac{1}{2}\ln\frac{\eta_0+1}{\eta_0-1} = \coth^{-1}\eta_0, \qquad |\eta_0|>1.$$

This identity may be used to obtain a second solution for Legendre's equation.

Solution.

Put $t = 1/\eta_0$, so $|t|<1$. Subtracting the two logarithm series gives

$$\frac{1}{2}\ln\frac{\eta_0+1}{\eta_0-1} = \frac{1}{2}[\ln(1+t)-\ln(1-t)] = t+\frac{t^3}{3}+\frac{t^5}{5}+\cdots.$$

To identify the inverse, set this expression equal to y. Then $e^{2y} = (1+t)/(1-t)$ and hence

$$\coth y = \frac{e^{2y}+1}{e^{2y}-1} = \frac{(1+t)+(1-t)}{(1+t)-(1-t)} = \frac{1}{t} = \eta_0.$$

For real $|\eta_0|>1$ the logarithm's argument is positive and y has the same sign as η_0, giving the real inverse $\coth^{-1}\eta_0$. For complex arguments a consistent logarithm branch is required.

1.2.11 Show that $f(x) = x^{1/2}$ (a) has no Maclaurin expansion but (b) has a Taylor expansion about any point $x_0>0$ for the real-valued square root (or any nonzero complex point with a chosen local branch). Find the range of convergence of the Taylor expansion about $x = x_0$.

Solution.

(a) For real $x>0$, $f'(x) = 1/(2\sqrt{x})$ diverges as $x\downarrow 0$. A convergent Maclaurin power series would have a finite first derivative at its center, so none represents $\sqrt{x}$ near zero.

(b) For a real-valued square root, take $x_0>0$. At a complex $x_0\neq 0$, choose a local branch. Set $h = (x-x_0)/x_0$. The binomial expansion gives

$$\sqrt{x} = \sqrt{x_0}(1+h)^{1/2} = \sqrt{x_0}\left[1+\frac{h}{2}-\frac{h^2}{8}+\frac{h^3}{16}-\frac{5h^4}{128}+\cdots\right],$$

with general coefficient $\binom{1/2}{m} = (1/2)(1/2-1)\cdots(1/2-m+1)/m!$. The ratio test gives radius $|h|<1$, or $|x-x_0|<|x_0|$; the branch point at zero prevents a larger radius. For real $x_0>0$, the two endpoints also converge absolutely: the absolute coefficient ratio is $(m-1/2)/(m+1) = 1-3/(2m)+O(m^{-2})$, so the coefficients are $O(m^{-3/2})$. By continuity their sums are 0 at $x=0$ and $\sqrt{2x_0}$ at $x=2x_0$. Thus the real convergence interval is $[0,2x_0]$, with analyticity only in its interior.

1.2.12 Prove l'Hôpital's rule, Eq. (1.58). For the 0/0 version, assume real f,g are continuous at x_0, differentiable in a punctured one-sided neighborhood, $f(x_0) = g(x_0) = 0$, and $g'\neq 0$ there. Assume that $f'/g'\to L$, where L may be an extended real limit. Prove $f/g\to L$.

Solution.

The rule needs hypotheses: assume f,g are continuous at x_0, differentiable in a punctured one-sided neighborhood, $f(x_0) = g(x_0) = 0$, $g'(x)\neq 0$ there, and $\lim_{x\to x_0} f'(x)/g'(x) = L$ exists (possibly infinite). For $x\neq x_0$ sufficiently close to x_0, Cauchy's mean value theorem supplies ξ strictly between x and x_0 such that

$$[f(x)-f(x_0)]g'(\xi) = [g(x)-g(x_0)]f'(\xi).$$

Rolle's theorem and $g' \neq 0$ ensure $g(x) \neq g(x_0) = 0$. Dividing the equality gives

$$\frac{f(x)}{g(x)} = \frac{f'(\xi)}{g'(\xi)}.$$

As $x \to x_0$, $\xi \to x_0$, and the right side tends to L. This proves the one-sided 0/0 rule in Eq. (1.58). If both one-sided limits agree, the two-sided version follows. This proof does not require f, g to have convergent Taylor series.

1.2.13 With $n > 1$, show that

$$\text{(a)}\ \frac{1}{n} - \ln\left(\frac{n}{n-1}\right) < 0, \qquad \text{(b)}\ \frac{1}{n} - \ln\left(\frac{n+1}{n}\right) > 0.$$

Use these inequalities to show that the limit defining the Euler-Mascheroni constant, Eq. (1.13), is finite.

Solution.

Since $1/t > 1/n$ for $n-1 < t < n$ and $1/t < 1/n$ for $n < t < n+1$,

$$\ln\frac{n}{n-1} = \int_{n-1}^{n} \frac{dt}{t} > \frac{1}{n}, \qquad \ln\frac{n+1}{n} = \int_{n}^{n+1} \frac{dt}{t} < \frac{1}{n}.$$

These are the two requested strict inequalities. Let $a_n = H_n - \ln n$ and $b_n = H_n - \ln(n+1)$. Then

$$a_n - a_{n-1} = \frac{1}{n} - \ln\frac{n}{n-1} < 0, \quad b_n - b_{n-1} = \frac{1}{n} - \ln\frac{n+1}{n} > 0.$$

Moreover $a_n > b_n$ and $a_n - b_n = \ln(1 + 1/n) \to 0$. Thus a_n is decreasing and bounded below by $b_1 = 1 - \ln 2$; b_n is increasing and bounded above by $a_1 = 1$. Both have finite limits, and their vanishing difference proves that the limits agree. That common limit is γ.

1.2.14 In numerical analysis it is often convenient to approximate $d^2\psi(x)/dx^2$ by

$$\frac{d^2}{dx^2}\psi(x) \approx \frac{1}{h^2}[\psi(x+h) - 2\psi(x) + \psi(x-h)].$$

Find the error in this approximation. Assume $\psi \in C^6$ near x, and define error as approximation minus exact value.

ANS. Error $= \dfrac{h^2}{12}\psi^{(4)}(x) + O(h^4)$.

Solution.

With the stated C^6 regularity, Taylor's remainder through degree five is $O(h^6)$. Taylor expansion on both sides gives

$$\psi(x \pm h) = \psi(x) \pm h\psi'(x) + \frac{h^2}{2}\psi''(x) \pm \frac{h^3}{6}\psi^{(3)}(x) + \frac{h^4}{24}\psi^{(4)}(x) \pm \frac{h^5}{120}\psi^{(5)}(x) + O(h^6).$$

Adding cancels the odd powers. Subtracting $2\psi(x)$ and dividing by h^2 yields

$$D_h\psi = \frac{\psi(x+h) - 2\psi(x) + \psi(x-h)}{h^2} = \psi''(x) + \frac{h^2}{12}\psi^{(4)}(x) + O(h^4).$$

Thus the printed error means approximation minus exact value, $D_h\psi - \psi''(x) = h^2\psi^{(4)}(x)/12 + O(h^4)$. If error is defined as exact minus approximation, its sign is reversed. For $\psi(x) = x^4$ the difference is exactly $2h^2$, confirming the coefficient $24/12 = 2$ and the stated sign convention. The printed answer is the leading error, not generally the whole error. For $\psi(x) = x^6$ at $x = 0$, the displayed fourth derivative vanishes, but $D_h\psi - \psi''(0) = 2h^4 \neq 0$. This verifies why the remainder term must be retained.

1.2.15 Evaluate $\lim_{x\to 0}\left[\dfrac{\sin(\tan x)-\tan(\sin x)}{x^7}\right]$.

ANS. $-\dfrac{1}{30}$.

Solution.

Put $t=\tan x = x + x^3/3 + 2x^5/15 + 17x^7/315 + O(x^9)$ and $s=\sin x = x - x^3/6 + x^5/120 - x^7/5040 + O(x^9)$. Multiplication gives

$$t^3 = x^3 + x^5 + \tfrac{11}{15}x^7 + O(x^9), \qquad t^5 = x^5 + \tfrac{5}{3}x^7 + O(x^9),$$
$$s^3 = x^3 - \tfrac{1}{2}x^5 + \tfrac{13}{120}x^7 + O(x^9), \quad s^5 = x^5 - \tfrac{5}{6}x^7 + O(x^9).$$

Also $t^7 = s^7 = x^7 + O(x^9)$. Substituting into the outer series,

$$\sin t = t - \frac{t^3}{6} + \frac{t^5}{120} - \frac{t^7}{5040} + O(x^9) = x + \frac{x^3}{6} - \frac{x^5}{40} - \frac{55x^7}{1008} + O(x^9),$$
$$\tan s = s + \frac{s^3}{3} + \frac{2s^5}{15} + \frac{17s^7}{315} + O(x^9) = x + \frac{x^3}{6} - \frac{x^5}{40} - \frac{107x^7}{5040} + O(x^9).$$

Their difference is $[-275/5040 + 107/5040]x^7 + O(x^9) = -x^7/30 + O(x^9)$. Division by x^7 therefore gives $\boxed{-1/30}$ in the limit.

1.2.16 A power series has actual radius of convergence $R>0$ (possibly $R=\infty$) and converges for $-R<x<R$. Show that the differentiated series and the integrated series have the same interval of convergence. (Do not bother about the endpoints $x=\pm R$.)

Solution.

Write $f(x)=\sum_{n=0}^{\infty} a_n x^n$. Fix r,ρ with $0<r<\rho<R$. Convergence at ρ implies $|a_n|\rho^n \le M$ for some finite M. Consequently, on $|x|\le r$,

$$\sum_{n=1}^{\infty} n|a_n|r^{n-1} \le \frac{M}{\rho}\sum_{n=1}^{\infty} n(r/\rho)^{n-1} < \infty,$$

so termwise differentiation is justified on each smaller interval. Similarly the integral series $C+\sum_{n=0}^{\infty} a_n x^{n+1}/(n+1)$ converges for $|x|<R$. Neither resulting radius can exceed R: if the differentiated series converged on a larger open interval, integrating it there would recover the original series on that interval; if the integrated series did so, differentiating it on smaller subintervals would have the same effect. Both would contradict the original radius. Thus both radii equal R. Here R denotes the actual radius of convergence, not merely an arbitrary smaller interval on which the original series converges. Endpoint behavior may change, as the question explicitly permits.

1.3 Binomial Theorem

1.3.1 The classical Langevin theory of paramagnetism leads to an expression for the magnetic polarization,

$$P(x) = c\left(\frac{\cosh x}{\sinh x} - \frac{1}{x}\right).$$

Expand $P(x)$ as a power series for small x (low fields, high temperature).

Solution.

Write $\coth x = x^{-1}+ax+bx^3+dx^5+\cdots$ and multiply by $\sinh x = x+x^3/6+x^5/120+x^7/5040+\cdots$. Equating to $\cosh x = 1 + x^2/2 + x^4/24 + x^6/720 + \cdots$ gives

$$a+\frac{1}{6}=\frac{1}{2}, \qquad b+\frac{a}{6}+\frac{1}{120}=\frac{1}{24}, \qquad d+\frac{b}{6}+\frac{a}{120}+\frac{1}{5040}=\frac{1}{720}.$$

Solving successively yields $a = 1/3$, $b = -1/45$, $d = 2/945$. The apparent pole cancels in P, leaving

$$\boxed{P(x) = c\left(\frac{x}{3}-\frac{x^3}{45}+\frac{2x^5}{945}+O(x^7)\right)}.$$

In particular $P(0) = 0$ by continuous extension, and the leading small-field response is $cx/3$.

1.3.2 Given that

$$\int_0^1 \frac{dx}{1+x^2} = \tan^{-1} x\,\Big|_0^1 = \frac{\pi}{4},$$

expand the integrand into a series and integrate term by term obtaining[1]

$$\frac{\pi}{4} = 1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\frac{1}{9}-\cdots+(-1)^n\frac{1}{2n+1}+\cdots,$$

which is Leibniz's formula for π. Compare the convergence of the integrand series and the integrated series at $x = 1$. Leibniz's formula converges so slowly that it is quite useless for numerical work.

Solution.

For $|x| < 1$, the geometric expansion is $(1 + x^2)^{-1} = \sum_{n=0}^{\infty}(-1)^n x^{2n}$. To justify integrating all the way to 1, use the finite identity

$$\frac{1}{1+x^2} = \sum_{n=0}^{N}(-1)^n x^{2n} + \frac{(-1)^{N+1}x^{2N+2}}{1+x^2}.$$

Integrating from 0 to 1 gives

$$\frac{\pi}{4} = \sum_{n=0}^{N}\frac{(-1)^n}{2n+1} + R_N, \qquad |R_N| \le \int_0^1 x^{2N+2}dx = \frac{1}{2N+3} \to 0.$$

Thus the stated Leibniz series follows. At $x = 1$ the integrand series is $1 - 1 + 1 - 1 + \cdots$, whose terms fail to tend to zero. The integrated series has decreasing term magnitudes $1/(2n + 1) \to 0$ and converges conditionally; its absolute series diverges by harmonic comparison.

1.3.3 Expand the incomplete gamma function $\gamma(n + 1, x) \equiv \int_0^x e^{-t}t^n\,dt$ in a series of powers of x, initially for a nonnegative integer n. What is the range of convergence of the resulting series?

ANS. $\displaystyle\int_0^x e^{-t}t^n\,dt = x^{n+1}\left[\frac{1}{n+1}-\frac{x}{n+2}+\frac{x^2}{2!(n+3)}-\cdots+\frac{(-1)^p x^p}{p!(n+p+1)}+\cdots\right].$

[1] The series expansion of $\tan^{-1} x$ (upper limit 1 replaced by x) was discovered by James Gregory in 1671, 3 years before Leibniz. See Peter Beckmann's entertaining book, *A History of Pi*, 2nd ed., Boulder, CO: Golem Press (1971), and L. Berggren, J. Borwein, and P. Borwein, *Pi: A Source Book*, New York: Springer (1997).

Solution.

For a nonnegative integer n, expand e^{-t} and integrate:

$$\gamma(n+1,x) = \int_0^x \sum_{p=0}^{\infty} \frac{(-1)^p t^{n+p}}{p!}\,dt$$
$$= \sum_{p=0}^{\infty} \frac{(-1)^p x^{n+p+1}}{p!(n+p+1)} = x^{n+1} \sum_{p=0}^{\infty} \frac{(-1)^p x^p}{p!(n+p+1)}.$$

On every bounded integration segment the exponential series is uniformly absolutely convergent, which justifies the interchange. The absolute ratio of successive nonzero terms is

$$|x| \frac{n+p+1}{(p+1)(n+p+2)} \longrightarrow 0.$$

Thus the resulting power series converges for every real or complex x. For nonintegral n with $\Re n > -1$ the same coefficient formula holds with a consistent branch of x^{n+1}; it is then a power series times that prefactor rather than an ordinary Maclaurin series.

1.3.4 Develop a series expansion of $y = \sinh^{-1} x$ (that is, $\sinh y = x$) in powers of x by

(a) inversion of the series for $\sinh y$,

(b) a direct Maclaurin expansion.

Solution.

(a) Seek $y = x + ax^3 + bx^5 + dx^7 + O(x^9)$ in $x = \sinh y = y + y^3/6 + y^5/120 + y^7/5040 + O(y^9)$. Since

$$y^3 = x^3 + 3ax^5 + (3b + 3a^2)x^7 + O(x^9), \quad y^5 = x^5 + 5ax^7 + O(x^9), \quad y^7 = x^7 + O(x^9),$$

the coefficients of x^3, x^5, x^7 give

$$a + \frac{1}{6} = 0, \quad b + \frac{a}{2} + \frac{1}{120} = 0, \quad d + \frac{b + a^2}{2} + \frac{a}{24} + \frac{1}{5040} = 0.$$

Hence $a = -1/6$, $b = 3/40$, $d = -5/112$.

(b) Differentiating $\sinh y = x$ gives $\cosh y\, y' = 1$. As $\cosh y = \sqrt{1 + x^2}$ on the branch through zero,

$$y' = (1 + x^2)^{-1/2} = \sum_{r=0}^{\infty} (-1)^r \frac{\binom{2r}{r}}{4^r} x^{2r}.$$

Integrating and using $y(0) = 0$ produces the full Maclaurin series

$$\boxed{\operatorname{arsinh} x = \sum_{r=0}^{\infty} (-1)^r \frac{\binom{2r}{r}}{4^r(2r+1)} x^{2r+1} = x - \frac{x^3}{6} + \frac{3x^5}{40} - \frac{5x^7}{112} + \cdots}.$$

Its radius is 1 (branch points at $\pm i$); at the real endpoints $x = \pm 1$ the coefficient magnitudes are $O(r^{-3/2})$, so it also converges absolutely there.

1.3.5 Show that for integral $n \geq 0$ and $|x| < 1$, $\dfrac{1}{(1-x)^{n+1}} = \displaystyle\sum_{m=n}^{\infty} \binom{m}{n} x^{m-n}$.

Solution.

For $|x| < 1$, differentiate the geometric series n times:

$$\frac{d^n}{dx^n}\frac{1}{1-x} = \frac{n!}{(1-x)^{n+1}}, \qquad \frac{d^n}{dx^n}\sum_{m=0}^{\infty} x^m = \sum_{m=n}^{\infty} \frac{m!}{(m-n)!} x^{m-n}.$$

Termwise differentiation is valid inside the radius of convergence, as established in Exercise 1.2.16. Divide by $n!$ to obtain

$$\frac{1}{(1-x)^{n+1}} = \sum_{m=n}^{\infty} \binom{m}{n} x^{m-n}.$$

The case $n = 0$ is the geometric series itself. The expansion's ordinary convergence domain is $|x| < 1$.

1.3.6 Show that $(1+x)^{-m/2} = \displaystyle\sum_{n=0}^{\infty} (-1)^n \frac{(m+2n-2)!!}{2^n\, n!\, (m-2)!!}\, x^n$, for $m = 1, 2, 3, \ldots$ and $|x| < 1$, with $(-1)!! = 0!! = 1$.

Solution.

The generalized binomial coefficient is

$$\binom{-m/2}{n} = \frac{(-m/2)(-m/2-1)\cdots(-m/2-n+1)}{n!} = (-1)^n \frac{m(m+2)\cdots(m+2n-2)}{2^n n!}.$$

Canceling the initial factors in the double factorial gives $m(m+2)\cdots(m+2n-2) = (m+2n-2)!!/(m-2)!!$. Substitute into $(1+x)^{-m/2} = \sum_{n\geq 0} \binom{-m/2}{n} x^n$ to obtain the requested formula for $|x| < 1$. The conventions $(-1)!! = 0!! = 1$ cover $m = 1, 2$, and an empty product is 1, so the $n = 0$ term is correctly equal to 1 in every case.

1.3.7 Using binomial expansions, compare the three Doppler shift formulas:

(a) $\nu' = \nu\left(1 \mp \dfrac{v}{c}\right)^{-1}$ moving source;

(b) $\nu' = \nu\left(1 \pm \dfrac{v}{c}\right)$ moving observer;

(c) $\nu' = \nu\left(1 \pm \dfrac{v}{c}\right)\left(1 - \dfrac{v^2}{c^2}\right)^{-1/2}$ relativistic.

Note. The relativistic formula agrees with the classical formulas if terms of order v^2/c^2 can be neglected.

Solution.

Put $\beta = v/c$ and use a single sign $\sigma = +1$ for the upper signs and $\sigma = -1$ for the lower signs. Then

(a) $\nu'/\nu = (1-\sigma\beta)^{-1} = 1 + \sigma\beta + \beta^2 + \sigma\beta^3 + O(\beta^4)$,

(b) $\nu'/\nu = 1 + \sigma\beta$,

(c) $\nu'/\nu = (1+\sigma\beta)(1-\beta^2)^{-1/2}$

$= (1+\sigma\beta)(1 + \frac{1}{2}\beta^2 + O(\beta^4)) = 1 + \sigma\beta + \frac{1}{2}\beta^2 + \frac{1}{2}\sigma\beta^3 + O(\beta^4)$.

All three share $1+\sigma\beta$ at first order. Their second-order coefficients are respectively $1, 0, 1/2$, so distinguishing the formulas requires retaining terms of order v^2/c^2.

1.3.8 In the theory of general relativity there are various ways of relating (defining) a velocity of recession of a galaxy to its red shift, δ. Milne's model (kinematic relativity) gives

(a) $v_1 = c\delta\left(1+\dfrac{\delta}{2}\right)$,

(b) $v_2 = c\delta\left(1+\dfrac{\delta}{2}\right)(1+\delta)^{-2}$,

(c) $1+\delta = \left[\dfrac{1+v_3/c}{1-v_3/c}\right]^{1/2}$.

1. Show that for $\delta \ll 1$ (and $v_3/c \ll 1$), all three formulas reduce to $v = c\delta$.
2. Compare the three velocities through terms of order δ^2.

Note. In special relativity (with δ replaced by z), the ratio of observed wavelength λ to emitted wavelength λ_0 is given by

$$\frac{\lambda}{\lambda_0} = 1+z = \left(\frac{c+v}{c-v}\right)^{1/2}.$$

Solution.

(a) The first formula already reads $v_1/c = \delta + \delta^2/2$. (b) Since $(1+\delta)^{-2} = 1 - 2\delta + 3\delta^2 + O(\delta^3)$,

$$\frac{v_2}{c} = (\delta + \tfrac{1}{2}\delta^2)(1 - 2\delta + 3\delta^2 + \cdots) = \delta - \frac{3}{2}\delta^2 + O(\delta^3).$$

(c) Squaring and solving the third formula for $\beta_3 = v_3/c$ gives

$$(1+\delta)^2(1-\beta_3) = 1+\beta_3, \quad \beta_3 = \frac{2\delta+\delta^2}{2+2\delta+\delta^2} = \frac{\delta+\delta^2/2}{1+\delta+\delta^2/2} = \delta - \frac{1}{2}\delta^2 + O(\delta^3).$$

Thus all three reduce to $v = c\delta$ at first order. Through second order their coefficients of $c\delta^2$ are $+1/2, -3/2, -1/2$; for small positive nonzero δ, $v_1 > v_3 > v_2$.

1.3.9 The relativistic sum w of two velocities u and v in the same direction is given by

$$\frac{w}{c} = \frac{u/c + v/c}{1 + uv/c^2}.$$

If

$$\frac{v}{c} = \frac{u}{c} = 1 - \alpha,$$

where $0 \le \alpha \le 1$, find w/c in powers of α through terms in α^3.

Solution.

Substitution first gives an exact rational function:

$$\frac{w}{c} = \frac{2(1-\alpha)}{1+(1-\alpha)^2} = 1 - \frac{\alpha^2}{2 - 2\alpha + \alpha^2}.$$

Near $\alpha = 0$,

$$\frac{1}{2 - 2\alpha + \alpha^2} = \frac{1}{2}\frac{1}{1 - \alpha + \alpha^2/2} = \frac{1}{2}[1 + \alpha + O(\alpha^2)].$$

Therefore

$$\boxed{w/c = 1 - \frac{\alpha^2}{2} - \frac{\alpha^3}{2} + O(\alpha^4)}.$$

There is no linear term. This is an expansion about $\alpha = 0$, not an assertion that truncation is accurate throughout $0 \le \alpha \le 1$. The unexpanded expression gives $w/c = 1$ at $\alpha = 0$ and 0 at $\alpha = 1$, consistent with the limiting input velocities.

1.3.10 The displacement x of a particle of rest mass m_0, resulting from a constant force $m_0 g$ along the x-axis, is

$$x = \frac{c^2}{g}\left\{\left[1 + \left(g\frac{t}{c}\right)^2\right]^{1/2} - 1\right\},$$

including relativistic effects. Find the displacement x as a power series in time t. Compare with the classical result,

$$x = \frac{1}{2}gt^2.$$

Solution.

Set $q = (gt/c)^2$ and use $(1+q)^{1/2} = 1 + q/2 - q^2/8 + q^3/16 - 5q^4/128 + \cdots$. After subtraction of 1 and multiplication by c^2/g,

$$\boxed{x = \frac{gt^2}{2} - \frac{g^3t^4}{8c^2} + \frac{g^5t^6}{16c^4} - \frac{5g^7t^8}{128c^6} + \cdots}.$$

The first term is the classical displacement. For $|gt/c| \ll 1$ the leading relative correction is $-g^2t^2/(4c^2)$. The binomial series converges for $|gt/c| < 1$ and at the real endpoints; the original expression, unlike the small-time truncation, also describes larger times.

1.3.11 By use of Dirac's relativistic theory, the fine structure formula of atomic spectroscopy is given by

$$E = mc^2\left[1 + \frac{\gamma^2}{(s + n - |k|)^2}\right]^{-1/2},$$

where

$$s = (|k|^2 - \gamma^2)^{1/2}, \quad k = \pm 1, \pm 2, \pm 3, \ldots.$$

Take the principal quantum number $n > 0$ and $|\gamma| < |k|$. Expand in powers of γ^2 through order γ^4 ($\gamma = Ze^2/(4\pi\epsilon_0\hbar c)$, with Z the atomic number). This expansion is useful in comparing the predictions of the Dirac electron theory with those of a relativistic Schrödinger electron theory. Experimental results support the Dirac theory.

Solution.

In the physical definition the parameter is $\gamma = Ze^2/(4\pi\epsilon_0\hbar c) = Z\alpha$, not $\gamma^2 = Ze^2/(4\pi\epsilon_0\hbar c)$. This follows by matching the given denominator with the standard Dirac formula containing $(Z\alpha)^2$; see Kramida's NIST compilation, Sec. 2, Eq. (3). For the algebra set $K = |k| > 0$ and $q = \gamma^2$. Then

$$s = K\sqrt{1 - q/K^2} = K - \frac{q}{2K} - \frac{q^2}{8K^3} + O(q^3), \quad s + n - K = n - \frac{q}{2K} + O(q^2).$$

Consequently

$$\frac{q}{(s+n-K)^2} = \frac{q}{n^2}\left(1 - \frac{q}{2Kn} + O(q^2)\right)^{-2} = \frac{q}{n^2} + \frac{q^2}{Kn^3} + O(q^3).$$

Now apply $(1+u)^{-1/2} = 1 - u/2 + 3u^2/8 + O(u^3)$:

$$\boxed{E = mc^2\left[1 - \frac{\gamma^2}{2n^2} - \frac{\gamma^4}{2n^4}\left(\frac{n}{|k|} - \frac{3}{4}\right) + O(\gamma^6)\right]}.$$

The expansion is for $|\gamma| < |k|$ and small γ at fixed positive n. For $n = |k| = 1$ the exact expression reduces to $mc^2\sqrt{1-\gamma^2}$, whose expansion $mc^2(1 - \gamma^2/2 - \gamma^4/8 + \cdots)$ checks the result.

1.3.12 In a head-on proton-proton collision, the ratio of the kinetic energy in the center of mass system to the incident kinetic energy is

$$R = \left[\sqrt{2mc^2(E_k + 2mc^2)} - 2mc^2\right]/E_k.$$

Find the value of this ratio of kinetic energies for

(a) $E_k \ll mc^2$ (nonrelativistic),

(b) $E_k \gg mc^2$ (extreme-relativistic).

ANS. (a) $\frac{1}{2}$, (b) 0. The latter answer is a sort of law of diminishing returns for high-energy particle accelerators (with stationary targets).

Solution.

Let $q = E_k/(mc^2) > 0$. Rationalizing removes the cancellation at $q = 0$:

$$R = \frac{\sqrt{4+2q} - 2}{q} = \frac{2}{\sqrt{4+2q}+2} = \frac{1}{\sqrt{1+q/2}+1}.$$

(a) For $q \ll 1$, $\sqrt{1+q/2} = 1 + q/4 + O(q^2)$, giving $R = \frac{1}{2}(1 + q/8 + O(q^2))^{-1} = 1/2 - q/16 + O(q^2)$. Thus the nonrelativistic limit is $1/2$.

(b) For $q \gg 1$, the original numerator gives

$$R = \sqrt{\frac{2}{q}}\sqrt{1 + \frac{2}{q}} - \frac{2}{q} = \sqrt{\frac{2}{q}} - \frac{2}{q} + O(q^{-3/2}) \longrightarrow 0.$$

The positive leading term also specifies how the ratio approaches zero.

1.3.13 With binomial expansions

$$\frac{x}{1-x} = \sum_{n=1}^{\infty} x^n, \qquad \frac{x}{x-1} = \frac{1}{1-x^{-1}} = \sum_{n=0}^{\infty} x^{-n}.$$

Adding these two series yields $\sum_{n=-\infty}^{\infty} x^n = 0$.

Hopefully, we can agree that this is nonsense, but what has gone wrong?

Solution.

The first equality uses a geometric series with ratio x, and thus requires $|x| < 1$. The second uses ratio x^{-1} and requires $|x| > 1$. There is no common point where the two infinite series both converge, so adding their sums termwise is unjustified. Indeed, for any nonzero x, the positive-power tail fails to converge when $|x| \geq 1$, while the negative-power tail fails when $|x| \leq 1$. On $|x| = 1$ neither tail's terms tend to zero. At $x = 0$ negative powers are undefined. Thus the proposed bilateral series has no ordinary convergence domain, although the two rational functions algebraically cancel wherever they are defined.

1.3.14

(a) Planck's theory of quantized oscillators leads to an average energy

$$\langle \varepsilon \rangle = \frac{\displaystyle\sum_{n=1}^{\infty} n\varepsilon_0 \exp(-n\varepsilon_0/kT)}{\displaystyle\sum_{n=0}^{\infty} \exp(-n\varepsilon_0/kT)},$$

where ε_0 is a fixed energy. Identify the numerator and denominator as binomial expansions and show that the ratio is

$$\langle \varepsilon \rangle = \frac{\varepsilon_0}{\exp(\varepsilon_0/kT) - 1}.$$

(b) Show that the $\langle \varepsilon \rangle$ of part (a) reduces to kT, the classical result, for $kT \gg \varepsilon_0$.

Solution.

(a) Put $q = e^{-\varepsilon_0/(kT)}$, so $0 < q < 1$ for $\varepsilon_0, kT > 0$. The geometric and differentiated geometric sums are

$$\sum_{n=0}^{\infty} q^n = \frac{1}{1-q}, \qquad \sum_{n=1}^{\infty} n q^n = q\frac{d}{dq}\frac{1}{1-q} = \frac{q}{(1-q)^2}.$$

Their ratio, including the energy factor, is

$$\langle \varepsilon \rangle = \varepsilon_0 \frac{q/(1-q)^2}{1/(1-q)} = \frac{\varepsilon_0 q}{1-q} = \frac{\varepsilon_0}{e^{\varepsilon_0/(kT)} - 1}.$$

(b) With $z = \varepsilon_0/(kT) \ll 1$, $e^z - 1 = z(1 + z/2 + z^2/6 + \cdots)$, hence

$$\langle \varepsilon \rangle = kT\left(1 - \frac{z}{2} + \frac{z^2}{12} + O(z^4)\right) = kT - \frac{\varepsilon_0}{2} + O(\varepsilon_0^2/(kT)).$$

For clarity, the reciprocal series was found by setting $(1 + z/2 + z^2/6 + \cdots)(1 + az + bz^2 + \cdots) = 1$: $a + 1/2 = 0$ and $b + a/2 + 1/6 = 0$, so $a = -1/2$ and $b = 1/12$. Its leading high-temperature value is kT. The manual (p. 14) gives a plus sign in $z/(e^z - 1)$ and a positive constant correction to the energy. The expansion above fixes the first correction as $-\varepsilon_0/2$.

1.3.15 Expand by the binomial theorem and integrate term by term to obtain the Gregory series for $y = \tan^{-1} x$ (note $\tan y = x$):

$$\tan^{-1} x = \int_0^x \frac{dt}{1+t^2} = \int_0^x \left\{1 - t^2 + t^4 - t^6 + \cdots\right\} dt$$
$$= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}, \quad -1 \le x \le 1.$$

Solution.

For $|x| < 1$ the expansion $(1 + t^2)^{-1} = \sum_{n=0}^{\infty} (-1)^n t^{2n}$ is uniformly convergent on the segment from 0 to x. Termwise integration therefore gives

$$\arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}.$$

At $x = 1$, the finite geometric remainder after integration is bounded by $1/(2N + 3)$ as in Exercise 1.3.2, proving convergence to $\pi/4$. Oddness gives convergence to $-\pi/4$ at $x = -1$. At either endpoint convergence is conditional; in the interior it is absolute. For $|x| > 1$, $|x|^{2n+1}/(2n + 1)$ does not tend to zero, so no further real points belong to the convergence interval.

1.3.16 The Klein-Nishina formula for the scattering of photons by electrons contains a term of the form

$$f(\varepsilon) = \frac{(1+\varepsilon)}{\varepsilon^2}\left[\frac{2+2\varepsilon}{1+2\varepsilon} - \frac{\ln(1+2\varepsilon)}{\varepsilon}\right].$$

Here $\varepsilon = h\nu/mc^2$, the ratio of the photon energy to the electron rest mass energy. Find $\lim_{\varepsilon\to 0} f(\varepsilon)$.

ANS. $\dfrac{4}{3}$.

Solution.

The prefactor contains ε^{-2}, so expand the bracket at least through ε^2:

$$\frac{2+2\varepsilon}{1+2\varepsilon} = 2(1+\varepsilon)(1-2\varepsilon+4\varepsilon^2-8\varepsilon^3+\cdots) = 2-2\varepsilon+4\varepsilon^2-8\varepsilon^3+O(\varepsilon^4),$$
$$\frac{\ln(1+2\varepsilon)}{\varepsilon} = 2-2\varepsilon+\frac{8}{3}\varepsilon^2-4\varepsilon^3+O(\varepsilon^4).$$

The constant and linear terms cancel, leaving $\frac{4}{3}\varepsilon^2 - 4\varepsilon^3 + O(\varepsilon^4)$. Multiplication by $(1 + \varepsilon)/\varepsilon^2$ now gives

$$f(\varepsilon) = \frac{4}{3} - \frac{8}{3}\varepsilon + O(\varepsilon^2), \qquad \boxed{\lim_{\varepsilon\to 0} f(\varepsilon) = \frac{4}{3}}.$$

1.3.17 The behavior of a neutron losing energy by colliding elastically with nuclei of mass A is described by a parameter ξ_1,

$$\xi_1 = 1 + \frac{(A-1)^2}{2A}\ln\frac{A-1}{A+1}.$$

An approximation, good for large A, is

$$\xi_2 = \frac{2}{A+\frac{2}{3}}.$$

Expand ξ_1 and ξ_2 in powers of A^{-1}. Show that ξ_2 agrees with ξ_1 through $(A^{-1})^2$. Find the difference in the coefficients of the $(A^{-1})^3$ term.

Solution.

Set $x = 1/A$, with $|x| < 1$. Then

$$\ln\frac{A-1}{A+1} = \ln(1-x) - \ln(1+x) = -2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots\right), \qquad \frac{(A-1)^2}{2A} = \frac{1-2x+x^2}{2x}.$$

Retain through x^4 in the product (the x^{-1} prefactor requires the logarithm through x^5):

$$\begin{aligned}\xi_1 &= 1 - (1-2x+x^2)(1+x^2/3+x^4/5+\cdots)\\ &= 2x - \frac{4}{3}x^2 + \frac{2}{3}x^3 - \frac{8}{15}x^4 + O(x^5),\\ \xi_2 &= \frac{2x}{1+2x/3} = 2x - \frac{4}{3}x^2 + \frac{8}{9}x^3 - \frac{16}{27}x^4 + O(x^5).\end{aligned}$$

Thus they agree through A^{-2}, and $\xi_2 - \xi_1 = (8/9 - 2/3)A^{-3} + O(A^{-4}) = \boxed{(2/9)A^{-3} + O(A^{-4})}$. The manual (p. 14) prints $4x^3/3$ in its intermediate expansion of ξ_1. The coefficient is $2/3$, as follows from the product above; its final coefficient difference $2/9$ is correct.

1.3.18 Show that each of these two integrals equals Catalan's constant:

(a) $\displaystyle\int_0^1 \arctan t\,\frac{dt}{t}$, (b) $\displaystyle -\int_0^1 \ln x\,\frac{dx}{1+x^2}$.

Note. The definition and numerical computation of Catalan's constant was addressed in Exercise 1.1.12.

Solution.

(a) Divide the Gregory series by t and integrate:

$$\int_0^1 \frac{\arctan t}{t}dt = \sum_{n=0}^{\infty}\frac{(-1)^n}{2n+1}\int_0^1 t^{2n}dt = \sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)^2} = G.$$

The apparent singularity at zero is removable since $\arctan t/t \to 1$. Interchanging summation and integration is justified by $\sum_n \int_0^1 t^{2n}/(2n+1)dt = \sum_n (2n+1)^{-2} < \infty$.

(b) Similarly expand $1/(1+x^2)$ and use integration by parts:

$$\int_0^1 x^{2n}\ln x\,dx = \left[\frac{x^{2n+1}\ln x}{2n+1}\right]_0^1 - \frac{1}{2n+1}\int_0^1 x^{2n}dx = -\frac{1}{(2n+1)^2}.$$

Therefore $-\int_0^1 \ln x/(1+x^2)dx = \sum_{n\ge 0}(-1)^n/(2n+1)^2 = G$. The same summable absolute integrals justify this interchange too.

1.4 Mathematical Induction

1.4.1 Show that $\displaystyle\sum_{j=1}^{n} j^4 = \frac{n}{30}(2n+1)(n+1)(3n^2+3n-1)$.

Solution.

Let $F(n) = n(n+1)(2n+1)(3n^2+3n-1)/30$. Multiplying the factors gives

$$F(n) = \frac{1}{5}n^5 + \frac{1}{2}n^4 + \frac{1}{3}n^3 - \frac{1}{30}n.$$

The base case is $F(1) = 1$. For the inductive step use the binomial expansions of $(n+1)^j$:

$$\begin{aligned} F(n+1) - F(n) &= \tfrac{1}{5}(5n^4+10n^3+10n^2+5n+1) \\ &\quad + \tfrac{1}{2}(4n^3+6n^2+4n+1) + \tfrac{1}{3}(3n^2+3n+1) - \tfrac{1}{30} \\ &= n^4+4n^3+6n^2+4n+1 = (n+1)^4. \end{aligned}$$

If $\sum_{j=1}^{n} j^4 = F(n)$, adding $(n+1)^4$ consequently gives $F(n+1)$. The formula follows for every positive integer n.

1.4.2 Prove the Leibniz formula for the repeated differentiation of a product:

$$\left(\frac{d}{dx}\right)^n [f(x)g(x)] = \sum_{j=0}^{n} \binom{n}{j} \left[\left(\frac{d}{dx}\right)^j f(x)\right] \left[\left(\frac{d}{dx}\right)^{n-j} g(x)\right].$$

Solution.

For $n = 0$ the identity is $fg = fg$, and for $n = 1$ it is the product rule. Assume it holds at order n, with the needed derivatives existing. Differentiating each term gives

$$(fg)^{(n+1)} = \sum_{j=0}^{n} \binom{n}{j} f^{(j+1)} g^{(n-j)} + \sum_{j=0}^{n} \binom{n}{j} f^{(j)} g^{(n+1-j)}.$$

In the first sum replace $j+1$ by r. The boundary terms are $f^{(n+1)}g$ and $fg^{(n+1)}$, and for $1 \le r \le n$ the coefficient of $f^{(r)}g^{(n+1-r)}$ is

$$\binom{n}{r-1} + \binom{n}{r} = \frac{n!r + n!(n+1-r)}{r!(n+1-r)!} = \binom{n+1}{r}.$$

The boundary coefficients also equal the corresponding binomial coefficients, both 1. This is exactly the formula at order $n+1$, completing induction.

1.5 Operations on Series Expansions of Functions

1.5.1 Using a partial fraction expansion, show that for $0 < x < 1$,

$$\int_{-x}^{x} \frac{dt}{1-t^2} = \ln\left(\frac{1+x}{1-x}\right).$$

Solution.

Solve $1/(1-t^2) = A/(1-t) + B/(1+t)$ by multiplying by $1-t^2$: $1 = A(1+t) + B(1-t)$. Constant and linear coefficients give $A + B = 1$ and $A - B = 0$, hence $A = B = 1/2$. Therefore

$$\begin{aligned}\int_{-x}^{x} \frac{dt}{1-t^2} &= \frac{1}{2}[-\ln(1-t) + \ln(1+t)]_{-x}^{x} \\ &= \frac{1}{2}\left[\ln\frac{1+x}{1-x} - \ln\frac{1-x}{1+x}\right] = \ln\frac{1+x}{1-x}.\end{aligned}$$

For $0 < x < 1$ the denominators stay nonzero and all logarithms have positive arguments, so no principal-value or branch convention is needed.

1.5.2 Prove the partial fraction expansion

$$\frac{1}{n(n+1)\cdots(n+p)} = \frac{1}{p!}\left[\binom{p}{0}\frac{1}{n} - \binom{p}{1}\frac{1}{n+1} + \binom{p}{2}\frac{1}{n+2} - \cdots + (-1)^p\binom{p}{p}\frac{1}{n+p}\right],$$

where p is a positive integer.

Hint. Use mathematical induction. Two binomial coefficient formulas of use here are

$$\frac{p+1}{p+1-j}\binom{p}{j} = \binom{p+1}{j}, \qquad \sum_{j=1}^{p+1}(-1)^{j-1}\binom{p+1}{j} = 1.$$

Solution.

The first identity in the transcribed hint should be $\frac{p+1}{p+1-j}\binom{p}{j} = \binom{p+1}{j}$ for $0 \le j \le p$, as printed in the textbook (p. 44). For an induction convenient here, let $F_p(n) = 1/[n(n+1)\cdots(n+p)]$. The base case $p = 0$ is $F_0(n) = 1/n$. Directly bringing two fractions to a common denominator gives

$$F_p(n) - F_p(n+1) = \frac{(n+p+1) - n}{n(n+1)\cdots(n+p+1)} = (p+1)F_{p+1}(n).$$

Assume $F_p(n) = \frac{1}{p!}\sum_{j=0}^{p}(-1)^j\binom{p}{j}/(n+j)$. Apply it at both n and $n+1$, subtract, and divide by $p+1$:

$$F_{p+1}(n) = \frac{1}{(p+1)!}\left[\sum_{j=0}^{p}\frac{(-1)^j\binom{p}{j}}{n+j} - \sum_{j=1}^{p+1}\frac{(-1)^{j-1}\binom{p}{j-1}}{n+j}\right].$$

For internal indices the numerator becomes $(-1)^j[\binom{p}{j} + \binom{p}{j-1}] = (-1)^j\binom{p+1}{j}$. The endpoint numerators are 1 and $(-1)^{p+1}$, as required. Thus the formula holds at $p+1$ and for all p by induction. All formulas are rational identities away from the poles $n = 0, -1, \ldots$.

1.5.3 The formula for α_p, Eq. (1.26), is a summation of the form $\sum_{n=1}^{\infty} u_n(p)$, with

$$u_n(p) = \frac{1}{n(n+1)\cdots(n+p)}.$$

Applying a partial fraction decomposition to the first and last factors of the denominator, i.e.,

$$\frac{1}{n(n+p)} = \frac{1}{p}\left[\frac{1}{n} - \frac{1}{n+p}\right],$$

show that $u_n(p) = \dfrac{u_n(p-1) - u_{n+1}(p-1)}{p}$ and that $\sum_{n=1}^{\infty} u_n(p) = \dfrac{1}{p\,p!}$.

Hint. It is useful to note that $u_1(p-1) = 1/p!$.

Solution.

Multiplying the supplied partial fraction decomposition by $1/[(n+1)\cdots(n+p-1)]$ gives

$$u_n(p) = \frac{1}{p}\left[\frac{1}{n(n+1)\cdots(n+p-1)} - \frac{1}{(n+1)\cdots(n+p)}\right] = \frac{u_n(p-1) - u_{n+1}(p-1)}{p}.$$

Now sum only to a finite N, so the cancellation is unambiguous:

$$\sum_{n=1}^{N} u_n(p) = \frac{1}{p}[u_1(p-1) - u_{N+1}(p-1)] = \frac{1}{p\,p!} - \frac{1}{p(N+1)(N+2)\cdots(N+p)}.$$

For every integer $p \ge 1$ the last term tends to zero. Taking the limit gives $\alpha_p = 1/(p\,p!)$, and the same expression supplies the exact finite-sum remainder.

1.5.4 Proof of Euler transformation: By substituting Eq. (1.88) into Eq. (1.87), verify that Eq. (1.86) is recovered.

Hint. It may help to rearrange the resultant double series so that both indices are summed on the range $(0, \infty)$. Then the summation not containing the coefficients c_j can be recognized as a binomial expansion.

Solution.

In the book's notation put $y = x/(1+x)$ and $a_n = \sum_{j=0}^{n}(-1)^j\binom{n}{j}c_{n-j}$. Substitute this in the transformed series and set $k = n - j$:

$$\frac{1}{1+x}\sum_{n=0}^{\infty}(-1)^n a_n y^n = \frac{1}{1+x}\sum_{k=0}^{\infty}\sum_{j=0}^{\infty}(-1)^k c_k\binom{k+j}{j}y^{k+j}$$
$$= \frac{1}{1+x}\sum_{k=0}^{\infty}(-1)^k c_k y^k \sum_{j=0}^{\infty}\binom{k+j}{j}y^j.$$

Exercise 1.3.5 identifies the inner sum as $(1-y)^{-k-1}$. Because $1 - y = 1/(1+x)$, its complete prefactor simplifies to

$$\frac{y^k}{(1+x)(1-y)^{k+1}} = x^k.$$

Thus the result is $\sum_{k\geq 0}(-1)^k c_k x^k$, precisely Eq. (1.86). The exchanges hold wherever the double series converges absolutely; if the original radius is $R > 0$, it suffices that $|y| < 1$ and $|y|/(1-|y|) < R$. This includes a neighborhood of zero. The identity then extends to any common connected analytic domain containing zero where the resulting series converge. It is not a license to rearrange an arbitrary divergent double series.

1.5.5 Carry out the Euler transformation on the series for $\arctan(x)$:

$$\arctan(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \frac{x^9}{9} - \cdots.$$

Check your work by computing $\arctan(1) = \pi/4$ and $\arctan(3^{-1/2}) = \pi/6$.

Solution.

Apply Exercise 1.5.4 to $\arctan x/x$ with variable $u = x^2$ and $c_j = 1/(2j+1) = \int_0^1 t^{2j}dt$. Its difference coefficient is

$$a_n = \int_0^1 \sum_{j=0}^{n}(-1)^j\binom{n}{j}t^{2(n-j)}dt = \int_0^1 (t^2-1)^n dt = (-1)^n I_n, \quad I_n = \int_0^1 (1-t^2)^n dt.$$

Integration by parts gives, for $n \geq 1$, $I_n = 2n\int_0^1 t^2(1-t^2)^{n-1}dt = 2n(I_{n-1} - I_n)$. Since $I_0 = 1$, this yields $I_n = \frac{2n}{2n+1}I_{n-1} = (2n)!!/(2n+1)!!$. The transformed series is therefore

$$\boxed{\arctan x = \frac{x}{1+x^2}\sum_{n=0}^{\infty}\frac{(2n)!!}{(2n+1)!!}\left(\frac{x^2}{1+x^2}\right)^n}.$$

For every finite real x, $0 \leq q = x^2/(1+x^2) < 1$ and $0 < I_n \leq 1$, so this series converges. Summing its integral representation gives $\frac{x}{1+x^2}\int_0^1[1-q(1-t^2)]^{-1}dt = \int_0^1 x/(1+x^2t^2)dt = \arctan x$, confirming its full real domain. Keeping $n = 0, \ldots, 39$ gives

x	q	computed value
1	1/2	0.7853981633973235
$1/\sqrt{3}$	1/4	0.5235987755982989

These agree with $\pi/4$ and $\pi/6$; the absolute remainder is at most $|x|q^{40}$ (respectively 9.10×10^{-13} and 4.78×10^{-25}).

1.6 Some Important Series

1.6.1 Show that $\ln\left(\dfrac{1+x}{1-x}\right) = 2\left(x + \dfrac{x^3}{3} + \dfrac{x^5}{5} + \cdots\right), \quad -1 < x < 1.$

Solution.

For $|x| < 1$,

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}x^n}{n}, \qquad \ln(1-x) = -\sum_{n=1}^{\infty} \frac{x^n}{n}.$$

Subtracting cancels every even power and doubles every odd power:

$$\ln\frac{1+x}{1-x} = \sum_{n=1}^{\infty} \frac{[(-1)^{n-1}+1]x^n}{n} = 2\sum_{r=0}^{\infty} \frac{x^{2r+1}}{2r+1}.$$

The subtraction is justified by absolute convergence in $|x| < 1$. At $x = \pm 1$ the odd harmonic series diverges, agreeing with the logarithmic singularities at the interval boundaries.

1.7 Vectors

1.7.1 The vector $\mathbf{A}$ whose magnitude is 1.732 units makes equal angles with the positive coordinate axes. Find A_x, A_y, and A_z.

Solution.

If the common angle with the positive coordinate axes is θ, then $A_x = A_y = A_z = A\cos\theta$. Taking the squared magnitude gives $A^2 = 3A^2\cos^2\theta$, so $\cos\theta = \pm 1/\sqrt{3}$. Thus

$$(A_x, A_y, A_z) = \pm\frac{1.732}{\sqrt{3}}(1,1,1) \simeq \pm(0.999971, 0.999971, 0.999971).$$

The usual first-octant interpretation selects approximately $(1,1,1)$. Equal angles alone permits the opposite vector as well; a first-octant or acute-angle condition is needed to select the plus sign. The value 1.732 is a rounded magnitude, so components exactly equal to 1 correspond to the unrounded magnitude $\sqrt{3}$.

1.7.2 A triangle is defined by the vertices of three vectors $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ that extend from the origin. In terms of $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ show that the vector sum of the successive sides of the triangle $(\mathbf{AB} + \mathbf{BC} + \mathbf{CA})$ is zero, where the side $\mathbf{AB}$ is from $\mathbf{A}$ to $\mathbf{B}$, etc.

Solution.

A displacement from a point to another is final position minus initial position. Therefore

$$\mathbf{AB} = \mathbf{B} - \mathbf{A}, \quad \mathbf{BC} = \mathbf{C} - \mathbf{B}, \quad \mathbf{CA} = \mathbf{A} - \mathbf{C}.$$

Adding cancels each position vector once with its negative: $\mathbf{AB} + \mathbf{BC} + \mathbf{CA} = (\mathbf{B} - \mathbf{A}) + (\mathbf{C} - \mathbf{B}) + (\mathbf{A} - \mathbf{C}) = \mathbf{0}$. This is the zero net displacement on traversing the closed triangle.

1.7.3 A sphere of radius a is centered at a point $\mathbf{r}_1$.

(a) Write out the algebraic equation for the sphere.

(b) Write out a vector equation for the sphere.

ANS. (a) $(x-x_1)^2+(y-y_1)^2+(z-z_1)^2=a^2$.
(b) $\mathbf{r}=\mathbf{r}_1+\mathbf{a}$, where $\mathbf{a}$ takes on all directions but has a fixed magnitude a.

Solution.

A point $\mathbf{r}=(x,y,z)$ is on the sphere precisely when its distance from $\mathbf{r}_1=(x_1,y_1,z_1)$ equals a. Squaring that condition yields

$$|\mathbf{r}-\mathbf{r}_1|^2=(x-x_1)^2+(y-y_1)^2+(z-z_1)^2=a^2.$$

This is the Cartesian equation in (a). In vector form one may use $(\mathbf{r}-\mathbf{r}_1)\cdot(\mathbf{r}-\mathbf{r}_1)=a^2$, or $\mathbf{r}=\mathbf{r}_1+\mathbf{a}$ with $|\mathbf{a}|=a$ and all orientations of $\mathbf{a}$ allowed. A fixed direction for $\mathbf{a}$ would describe only one point, not the whole sphere.

1.7.4 Hubble's law. Hubble found that distant galaxies are receding with a velocity proportional to their distance from where we are on Earth. For the ith galaxy,

$$\mathbf{v}_i=H_0\mathbf{r}_i$$

with us at the origin. Show that this recession of the galaxies from us does not imply that we are at the center of the universe. Specifically, take the galaxy at $\mathbf{r}_1$ as a new origin and show that Hubble's law is still obeyed.

Solution.

An observer moving with galaxy 1 measures the simultaneous relative position $\mathbf{r}_i'=\mathbf{r}_i-\mathbf{r}_1$ and relative velocity $\mathbf{v}_i'=\mathbf{v}_i-\mathbf{v}_1$. Substitution gives

$$\mathbf{v}_i'=H_0\mathbf{r}_i-H_0\mathbf{r}_1=H_0(\mathbf{r}_i-\mathbf{r}_1)=H_0\mathbf{r}_i'.$$

Thus the same instantaneous linear recession law holds with every comoving galaxy as origin. The equation does not select our position as a distinguished center. This argument uses the relative-position and relative-velocity description assumed by the exercise.

1.7.5 Find the diagonal vectors of a unit cube with one corner at the origin and its three sides lying along Cartesian coordinates axes. Show that there are four diagonals with length $\sqrt{3}$. Representing these as vectors, what are their components? Show that the diagonals of the cube's faces have length $\sqrt{2}$ and determine their components.

Solution.

Take the cube to have vertices (a,b,c) with each coordinate 0 or 1. Opposite vertices are (a,b,c) and $(1-a,1-b,1-c)$; the connecting vector is $(1-2a,1-2b,1-2c)$, with each component ±1. Opposite orientations describe the same geometric diagonal, leaving four representatives:

$$(1,1,1),\quad (1,1,-1),\quad (1,-1,1),\quad (1,-1,-1).$$

Each has squared length $1+1+1=3$. A face diagonal changes exactly two coordinates, giving the six unoriented direction representatives

$$(1,\pm1,0),\qquad (1,0,\pm1),\qquad (0,1,\pm1),$$

all of length $\sqrt{2}$. Each direction occurs on two opposite faces, so there are twelve face-diagonal segments, two on each of six faces. Reversing any listed vector supplies its opposite orientation.

1.7.6 The vector $\mathbf{r}$, starting at the origin, terminates at and specifies the point in space (x, y, z). Find the surface swept out by the tip of $\mathbf{r}$ if

(a) $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{a} = 0$. Characterize $\mathbf{a}$ geometrically.

(b) $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{r} = 0$. Describe the geometric role of $\mathbf{a}$.

The vector $\mathbf{a}$ is constant (in magnitude and direction).

Solution.

(a) Expanding gives $\mathbf{r} \cdot \mathbf{a} = |\mathbf{a}|^2$. For $\mathbf{a} \neq 0$ this is a plane perpendicular to $\mathbf{a}$ and passing through its tip. Its signed unit-normal equation is $\mathbf{r} \cdot \widehat{\mathbf{a}} = |\mathbf{a}|$, so $\mathbf{a}$ is the perpendicular displacement from the origin to the plane. If $\mathbf{a} = 0$, the equation is $0 = 0$ and the set is all space.

(b) Expanding and completing the square gives

$$|\mathbf{r}|^2 - \mathbf{a} \cdot \mathbf{r} = 0 \quad \Longleftrightarrow \quad \left|\mathbf{r} - \frac{\mathbf{a}}{2}\right|^2 = \frac{|\mathbf{a}|^2}{4}.$$

This is the sphere centered at $\mathbf{a}/2$ of radius $|\mathbf{a}|/2$; the segment from 0 to $\mathbf{a}$ is a diameter. For $\mathbf{a} = 0$ it degenerates to the single point at the origin.

1.7.7 A pipe comes diagonally down the south wall of a building, making an angle of 45° with the horizontal. Coming into a corner, the pipe turns and continues diagonally down a west-facing wall, still making an angle of 45° with the horizontal. What is the angle between the south-wall and west-wall sections of the pipe?

ANS. 120°.

Solution.

Choose x east, y north, and z upward. From the southwest corner, a ray back along the south-wall pipe can be represented by $\mathbf{a} = (1, 0, 1)$, while the ray forward down the west-wall pipe is $\mathbf{b} = (0, 1, -1)$. Each has equal horizontal and vertical component magnitudes, as required by the 45° slopes. Their included angle is

$$\cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|} = \frac{-1}{\sqrt{2}\sqrt{2}} = -\frac{1}{2}, \qquad \boxed{\theta = 120^\circ}.$$

The change of direction of travel is the supplementary angle 60°, since the incoming travel direction is $-\mathbf{a}$. The quoted answer is the angle between the two rays of pipe extending from the corner.

1.7.8 Find the shortest distance of an observer at the point $(2, 1, 3)$ from a rocket in free flight with velocity $(1, 2, 3)$ km/s. The rocket was launched at time $t = 0$ from $(1, 1, 1)$. Lengths are in kilometers.

Solution.

At time $t \geq 0$ seconds the rocket is at $\mathbf{r}(t) = (1 + t, 1 + 2t, 1 + 3t)$. Its squared distance from the observer is

$$d^2(t) = (t - 1)^2 + (2t)^2 + (3t - 2)^2 = 14t^2 - 14t + 5 = 14(t - \tfrac{1}{2})^2 + \frac{3}{2}.$$

The minimum is attained at the permitted time $t = 1/2$ s, with

$$\boxed{d_{\min} = \sqrt{3/2} \text{ km} \simeq 1.224745 \text{ km}}.$$

At that time the displacement to the rocket is $(-1/2, 1, -1/2)$. Its dot product with $(1, 2, 3)$ is $-1/2 + 2 - 3/2 = 0$, independently confirming that the shortest connecting segment is perpendicular to the trajectory.

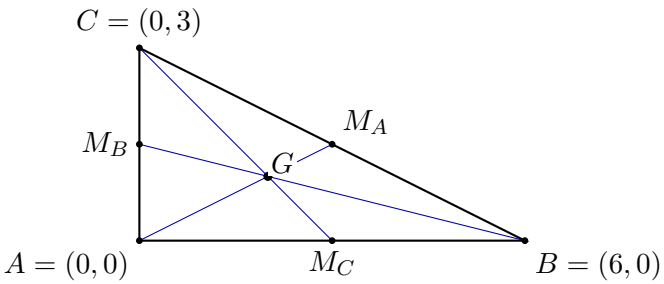
1.7.9 Show that the medians of a triangle intersect in the center which is 2/3 of the median's length from each vertex. Construct a numerical example and plot it.

Solution.

Let the position vectors of the vertices be $\mathbf{a}, \mathbf{b}, \mathbf{c}$. The midpoint opposite $\mathbf{a}$ is $\mathbf{m}_a = (\mathbf{b}+\mathbf{c})/2$. At fraction 2/3 of this median from $\mathbf{a}$ the position is

$$\mathbf{a} + \frac{2}{3}(\mathbf{m}_a - \mathbf{a}) = \frac{\mathbf{a}+\mathbf{b}+\mathbf{c}}{3} = \mathbf{g}.$$

Cyclically interchanging the vertex labels gives exactly the same $\mathbf{g}$ on each of the other two medians. Thus all three intersect there, and each median is divided in the length ratio $2:1$. For a concrete example use $A=(0,0)$, $B=(6,0)$, $C=(0,3)$. The midpoints are $M_A=(3,3/2)$, $M_B=(0,3/2)$, $M_C=(3,0)$, and $G=(2,1)$. Substitution in the three parametric medians gives G at parameter 2/3 in every case.



1.7.10 Prove the law of cosines starting from $A^2 = |\mathbf{B}-\mathbf{C}|^2$.

Solution.

The starting equation is a vector identity: $A^2 = |\mathbf{B}-\mathbf{C}|^2$, where A, B, C denote side lengths. Place the tails of $\mathbf{B}$ and $\mathbf{C}$ at the same triangle vertex, and let θ be the included angle. The opposite side is $\mathbf{A} = \mathbf{B} - \mathbf{C}$ up to orientation. Therefore

$$\begin{aligned} A^2 &= (\mathbf{B}-\mathbf{C})\cdot(\mathbf{B}-\mathbf{C}) \\ &= \mathbf{B}\cdot\mathbf{B} + \mathbf{C}\cdot\mathbf{C} - 2\mathbf{B}\cdot\mathbf{C} = B^2 + C^2 - 2BC\cos\theta. \end{aligned}$$

This is the law of cosines, reducing to Pythagoras when $\theta = \pi/2$.

1.7.11 Given the three vectors,

$$\mathbf{P} = 3\hat{\mathbf{e}}_x + 2\hat{\mathbf{e}}_y - \hat{\mathbf{e}}_z,$$
$$\mathbf{Q} = -6\hat{\mathbf{e}}_x - 4\hat{\mathbf{e}}_y + 2\hat{\mathbf{e}}_z,$$
$$\mathbf{R} = \hat{\mathbf{e}}_x - 2\hat{\mathbf{e}}_y - \hat{\mathbf{e}}_z,$$

find two that are perpendicular and two that are parallel or antiparallel.

Solution.

The component vectors are $\mathbf{P} = (3,2,-1)$, $\mathbf{Q} = (-6,-4,2)$, and $\mathbf{R} = (1,-2,-1)$. First $\mathbf{Q} = -2\mathbf{P}$, so $\mathbf{P}$ and $\mathbf{Q}$ are antiparallel. Next

$$\mathbf{P}\cdot\mathbf{R} = 3(1) + 2(-2) + (-1)(-1) = 3 - 4 + 1 = 0,$$

so $\mathbf{P}$ and $\mathbf{R}$ are perpendicular. Also $\mathbf{Q}\cdot\mathbf{R} = -2(\mathbf{P}\cdot\mathbf{R}) = 0$, so $\mathbf{Q}$ and $\mathbf{R}$ form a second perpendicular pair.

1.8 Complex Numbers and Functions

1.8.1 Find the reciprocal of $x+iy$, working in polar form but expressing the final result in Cartesian form.

Solution.

For $z = x + iy \neq 0$, set $r = \sqrt{x^2+y^2}$ and $z = r(\cos\theta + i\sin\theta) = re^{i\theta}$. The reciprocal has magnitude $1/r$ and angle $-\theta$:

$$\frac{1}{z} = r^{-1}e^{-i\theta} = \frac{1}{r}(\cos\theta - i\sin\theta) = \boxed{\frac{x-iy}{x^2+y^2}}.$$

Multiplication by $x+iy$ gives $(x^2+y^2)/(x^2+y^2) = 1$, checking the result. No reciprocal exists at $x = y = 0$.

1.8.2 Show that complex numbers have square roots and that the square roots are contained in the complex plane. What are the square roots of i?

Solution.

Write a nonzero complex number as $z = re^{i\theta}$ with $r > 0$. The two numbers $w_k = \sqrt{r}\, e^{i(\theta/2+k\pi)}$, $k = 0, 1$, satisfy $w_k^2 = re^{i\theta}e^{2\pi ik} = z$. They are opposites and are distinct. Conversely, if $w = \rho e^{i\phi}$ satisfies $w^2 = z$, then $\rho^2 = r$ and $2\phi = \theta + 2\pi k$, giving precisely these two roots. Zero has the single distinct square root zero. For $i = e^{i\pi/2}$ this gives

$$\boxed{\sqrt{i} = \pm e^{i\pi/4} = \pm\frac{1+i}{\sqrt{2}}}.$$

Squaring either value gives $(1 + 2i - 1)/2 = i$.

1.8.3 For a nonnegative integer n and real θ, show that

(a) $\cos n\theta = \cos^n\theta - \binom{n}{2}\cos^{n-2}\theta\sin^2\theta + \binom{n}{4}\cos^{n-4}\theta\sin^4\theta - \cdots$,

(b) $\sin n\theta = \binom{n}{1}\cos^{n-1}\theta\sin\theta - \binom{n}{3}\cos^{n-3}\theta\sin^3\theta + \cdots$.

Solution.

For a nonnegative integer n, de Moivre's formula and the binomial expansion give

$$\cos n\theta + i\sin n\theta = (\cos\theta + i\sin\theta)^n = \sum_{j=0}^{n}\binom{n}{j}\cos^{n-j}\theta\, i^j \sin^j\theta.$$

For $j = 2r$, $i^{2r} = (-1)^r$, while for $j = 2r+1$, $i^{2r+1} = i(-1)^r$. Equating real and imaginary parts therefore yields

$$\cos n\theta = \sum_{r=0}^{\lfloor n/2\rfloor}(-1)^r\binom{n}{2r}\cos^{n-2r}\theta\sin^{2r}\theta,$$

$$\sin n\theta = \sum_{r=0}^{\lfloor (n-1)/2\rfloor}(-1)^r\binom{n}{2r+1}\cos^{n-2r-1}\theta\sin^{2r+1}\theta.$$

These are the two requested finite expansions. For $n = 0$ the second sum is empty and gives zero.

1.8.4 Prove that

(a) $$\sum_{n=0}^{N-1} \cos nx = \frac{\sin(Nx/2)}{\sin(x/2)} \cos\left(\frac{(N-1)x}{2}\right),$$

(b) $$\sum_{n=0}^{N-1} \sin nx = \frac{\sin(Nx/2)}{\sin(x/2)} \sin\left(\frac{(N-1)x}{2}\right).$$

Here N is a positive integer and x is real; at $x = 2\pi m$, use the continuous limits N and 0, respectively. These series occur in the analysis of the multiple-slit diffraction pattern.

Solution.

For $e^{ix} \neq 1$, sum the finite geometric progression:

$$\sum_{n=0}^{N-1} e^{inx} = \frac{1 - e^{iNx}}{1 - e^{ix}}.$$

Using $1 - e^{it} = -2ie^{it/2} \sin(t/2)$ in numerator and denominator reduces this to

$$\sum_{n=0}^{N-1} e^{inx} = e^{i(N-1)x/2} \frac{\sin(Nx/2)}{\sin(x/2)}.$$

For real x the sine quotient is real, so taking real and imaginary parts gives (a) and (b). At $x = 2\pi m$ the displayed quotient is $0/0$, but the original sums are N and 0; these values are also the continuous limits of the right sides. This supplies the removable exception to the formulas.

1.8.5 Assume that the trigonometric functions and the hyperbolic functions are defined for complex argument by the appropriate power series. Show that

$$i \sin z = \sinh iz, \quad \sin iz = i \sinh z,$$

$$\cos z = \cosh iz, \quad \cos iz = \cosh z.$$

Solution.

The first identity was mistranscribed: the textbook (p. 61) has $i \sin z = \sinh(iz)$, not $i \sin(iz) = \sinh(iz)$. Direct substitution into the entire power series gives

$$\sinh(iz) = \sum_{n=0}^{\infty} \frac{i^{2n+1} z^{2n+1}}{(2n+1)!} = i \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!} = i \sin z,$$

$$\sin(iz) = \sum_{n=0}^{\infty} \frac{(-1)^n i^{2n+1} z^{2n+1}}{(2n+1)!} = i \sum_{n=0}^{\infty} \frac{z^{2n+1}}{(2n+1)!} = i \sinh z,$$

$$\cosh(iz) = \sum_{n=0}^{\infty} \frac{i^{2n} z^{2n}}{(2n)!} = \cos z,$$

$$\cos(iz) = \sum_{n=0}^{\infty} \frac{(-1)^n i^{2n} z^{2n}}{(2n)!} = \cosh z.$$

No restriction on complex z is required because all four power series converge absolutely everywhere.

1.8.6 Using the identities

$$\cos z = \frac{e^{iz} + e^{-iz}}{2}, \quad \sin z = \frac{e^{iz} - e^{-iz}}{2i},$$

established from comparison of power series, show that

(a) $\sin(x + iy) = \sin x \cosh y + i \cos x \sinh y$,
$\cos(x + iy) = \cos x \cosh y - i \sin x \sinh y$,

(b) $|\sin z|^2 = \sin^2 x + \sinh^2 y, \quad |\cos z|^2 = \cos^2 x + \sinh^2 y.$

This demonstrates that we may have $|\sin z|, |\cos z| > 1$ in the complex plane.

Solution.

(a) For real x, y, $e^{i(x+iy)} = e^{-y}(\cos x + i \sin x)$ and $e^{-i(x+iy)} = e^{y}(\cos x - i \sin x)$. Substituting these in the exponential definitions and collecting terms gives

$$\sin(x + iy) = \sin x \frac{e^y + e^{-y}}{2} + i \cos x \frac{e^y - e^{-y}}{2} = \sin x \cosh y + i \cos x \sinh y,$$
$$\cos(x + iy) = \cos x \cosh y - i \sin x \sinh y.$$

(b) Add the squares of each expression's real and imaginary parts:

$$|\sin z|^2 = \sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y = \sin^2 x + (\sin^2 x + \cos^2 x) \sinh^2 y = \sin^2 x + \sinh^2 y,$$
$$|\cos z|^2 = \cos^2 x \cosh^2 y + \sin^2 x \sinh^2 y = \cos^2 x + \sinh^2 y.$$

Here $\cosh^2 y = 1 + \sinh^2 y$ was used. For example $|\cos(iy)| = \cosh y > 1$ for $y \neq 0$.

1.8.7 From the identities in Exercises 1.8.5 and 1.8.6 show that

(a) $\sinh(x + iy) = \sinh x \cos y + i \cosh x \sin y$,
$\cosh(x + iy) = \cosh x \cos y + i \sinh x \sin y$,

(b) $|\sinh z|^2 = \sinh^2 x + \sin^2 y, \quad |\cosh z|^2 = \cosh^2 x - \sin^2 y.$

Solution.

(a) Insert $e^{x+iy} = e^x(\cos y + i \sin y)$ and $e^{-x-iy} = e^{-x}(\cos y - i \sin y)$ in the definitions of sinh and cosh. Subtracting and adding respectively yields

$$\sinh(x + iy) = \sinh x \cos y + i \cosh x \sin y, \quad \cosh(x + iy) = \cosh x \cos y + i \sinh x \sin y.$$

(b) Consequently

$$|\sinh z|^2 = \sinh^2 x \cos^2 y + \cosh^2 x \sin^2 y = \sinh^2 x + \sin^2 y,$$
$$|\cosh z|^2 = \cosh^2 x \cos^2 y + \sinh^2 x \sin^2 y = \cosh^2 x - \sin^2 y.$$

The last sign is minus, correcting an error also present in the textbook (p. 61). At $x = 0, y = \pi/2$, $\cosh(i\pi/2) = 0$; the corrected right side is $1 - 1 = 0$, whereas the printed plus sign would give 2. This independently checks the correction.

1.8.8 Put $z = x + iy$ with real x, y, and exclude the poles of each side. Show that

(a) $\tanh \dfrac{z}{2} = \dfrac{\sinh x + i \sin y}{\cosh x + \cos y}$,

(b) $\coth \dfrac{z}{2} = \dfrac{\sinh x - i \sin y}{\cosh x - \cos y}$.

Solution.

Let $z = x + iy$ with x, y real. From the exponential definitions, $\tanh(z/2) = (e^z - 1)/(e^z + 1)$. Multiply numerator and denominator by the conjugate $e^{x-iy} + 1$ to obtain

$$\tanh \frac{z}{2} = \frac{e^{2x} - 1 + 2ie^x \sin y}{e^{2x} + 1 + 2e^x \cos y} = \frac{\sinh x + i \sin y}{\cosh x + \cos y}.$$

Likewise $\coth(z/2) = (e^z + 1)/(e^z - 1)$. Multiplication by $e^{x-iy} - 1$ gives

$$\coth \frac{z}{2} = \frac{e^{2x} - 1 - 2ie^x \sin y}{e^{2x} + 1 - 2e^x \cos y} = \frac{\sinh x - i \sin y}{\cosh x - \cos y}.$$

The divisions exclude the poles: $z = (2m + 1)\pi i$ for $\tanh(z/2)$ and $z = 2m\pi i$ for $\coth(z/2)$, $m \in \mathbb{Z}$.

1.8.9 By comparing series expansions, show that $\tan^{-1} x = \dfrac{i}{2} \operatorname{Log}\left(\dfrac{1 - ix}{1 + ix}\right)$, for real x, with the principal inverse tangent and principal logarithm.

Solution.

For $|x| < 1$, subtract the logarithm series on their branches through $\log 1 = 0$:

$$\log(1 - ix) - \log(1 + ix) = -2\left(ix + \frac{(ix)^3}{3} + \frac{(ix)^5}{5} + \cdots\right) = -2i\left(x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots\right).$$

Multiplication by $i/2$ gives exactly the Gregory series for $\arctan x$. For all real x, $(1 - ix)/(1 + ix)$ has unit modulus and argument $-2 \arctan x \in (-\pi, \pi)$, so the principal logarithm satisfies

$$\boxed{\arctan x = \frac{i}{2} \operatorname{Log} \frac{1 - ix}{1 + ix}}.$$

Other logarithm branches differ by $2\pi i m$ and change the right side by $-\pi m$, the corresponding multivalued inverse-tangent branches. For complex x, branch choices and the singularities $x = \pm i$ must be retained.

1.8.10 Find the Cartesian form for all values of

(a) $(-8)^{1/3}$,

(b) $i^{1/4}$,

(c) $e^{i\pi/4}$.

Solution.

(a) Write $-8 = 8e^{i(\pi+2\pi k)}$. The cube roots have magnitude 2 and arguments $(\pi + 2\pi k)/3$, $k = 0, 1, 2$:

$$\boxed{1 + i\sqrt{3}, \quad -2, \quad 1 - i\sqrt{3}}.$$

(b) The fourth roots of $i = e^{i(\pi/2+2\pi k)}$ have arguments $\pi/8+k\pi/2$, $k = 0, 1, 2, 3$. Set $A = \sqrt{2+\sqrt{2}}/2$ and $B = \sqrt{2-\sqrt{2}}/2$; half-angle identities give $\cos(\pi/8) = A$, $\sin(\pi/8) = B$. The four Cartesian values are

$$\boxed{A + iB, \quad -B + iA, \quad -A - iB, \quad B - iA}.$$

(c) The exponential is single-valued: $\boxed{e^{i\pi/4} = (1 + i)/\sqrt{2}}$. Adding 2π to its angle reproduces this same value; it does not create additional roots. Cubing the values in (a) or taking fourth powers in (b) returns the specified original number.

1.8.11 Find the polar form for all values of

(a) $(1 + i)^3$,

(b) $(-1)^{1/5}$.

Solution.

(a) Since $1 + i = \sqrt{2}e^{i\pi/4}$,

$$\boxed{(1 + i)^3 = 2\sqrt{2}e^{3\pi i/4}}.$$

This is one value, equal to $-2 + 2i$ in Cartesian form. Its argument may be written $3\pi/4 + 2\pi m$, but all such choices represent the same complex number.

(b) Write $-1 = e^{i(\pi+2\pi k)}$. Its fifth roots are

$$\boxed{e^{i(2k+1)\pi/5}, \qquad k = 0, 1, 2, 3, 4}.$$

They all have unit modulus. Their arguments are $\pi/5, 3\pi/5, \pi, 7\pi/5, 9\pi/5$, and raising any one to the fifth power gives $e^{i(2k+1)\pi} = -1$. Values of k outside this range repeat the roots.

1.9 Derivatives and Extrema

1.9.1 Derive the following formula for the Maclaurin expansion of a function of two variables analytic near $(0, 0)$, for (x, y) within the convergence neighborhood of its Maclaurin series:

$$\begin{aligned} f(x,y) &= f(0,0) + x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} \\ &+ \frac{1}{2!}\left[\binom{2}{0}x^2\frac{\partial^2 f}{\partial x^2} + \binom{2}{1}xy\frac{\partial^2 f}{\partial x\,\partial y} + \binom{2}{2}y^2\frac{\partial^2 f}{\partial y^2}\right] \\ &+ \frac{1}{3!}\left[\binom{3}{0}x^3\frac{\partial^3 f}{\partial x^3} + \binom{3}{1}x^2y\frac{\partial^3 f}{\partial x^2\,\partial y} + \binom{3}{2}xy^2\frac{\partial^3 f}{\partial x\,\partial y^2} + \binom{3}{3}y^3\frac{\partial^3 f}{\partial y^3}\right] + \cdots, \end{aligned}$$

where all the partial derivatives are to be evaluated at the point $(0, 0)$.

Solution.

An infinite Taylor expansion represents the function only where its remainder tends to zero; analyticity near the origin is a sufficient hypothesis. Smoothness alone is not sufficient. Fix x, y and define the one-variable function $F(t) = f(tx, ty)$. The chain rule gives $F'(t) = (x\partial_x + y\partial_y)f(tx, ty)$. Iterating, using commutation of the mixed partial derivatives, yields

$$F^{(n)}(0) = \sum_{j=0}^{n} \binom{n}{j} x^{n-j} y^j \frac{\partial^n f}{\partial x^{n-j} \partial y^j}(0, 0).$$

Taylor-expand F about $t = 0$ and then set $t = 1$:

$$f(x, y) = \sum_{n=0}^{\infty} \sum_{j=0}^{n} \frac{x^{n-j} y^j}{(n-j)!\, j!} \frac{\partial^n f}{\partial x^{n-j} \partial y^j}(0, 0).$$

For $n = 1$ this is $xf_x + yf_y$; for $n = 2$ it is $(x^2 f_{xx} + 2xyf_{xy} + y^2 f_{yy})/2!$; for $n = 3$ it is $(x^3 f_{xxx} + 3x^2 y f_{xxy} + 3xy^2 f_{xyy} + y^3 f_{yyy})/3!$. These are exactly the displayed terms. For a finite order N, ordinary Taylor's theorem also provides a remainder $F^{(N+1)}(\theta)/(N+1)!$ for some $0 < \theta < 1$, under its usual real-variable hypotheses.

1.9.2 The result in Exercise 1.9.1 can be generalized to larger numbers of independent variables. Prove that for an m-variable system analytic near the origin, at points sufficiently close to it, the Maclaurin expansion can be written in the symbolic form

$$f(x_1, \ldots, x_m) = \sum_{n=0}^{\infty} \frac{t^n}{n!} \left[\left(\sum_{i=1}^{m} \alpha_i \frac{\partial}{\partial x_i} \right)^n f(\mathbf{x}) \right]_{\mathbf{x}=0},$$

where in the right-hand side we have made the substitutions $x_j = \alpha_j t$.

Solution.

Define $F(t) = f(\alpha_1 t, \ldots, \alpha_m t)$ with fixed α_i. The chain rule and induction give

$$F^{(n)}(t) = \left(\sum_{i=1}^{m} \alpha_i \partial_{x_i} \right)^n f(\alpha_1 t, \ldots, \alpha_m t).$$

Therefore Taylor's formula in t is

$$F(t) = \sum_{n=0}^{\infty} \frac{t^n}{n!} \left[\left(\sum_{i=1}^{m} \alpha_i \partial_{x_i} \right)^n f(\mathbf{x}) \right]_{\mathbf{x}=0}.$$

The book's symbolic $D^n f(0)$ means that derivatives act on f before evaluation at zero; the brackets in the corrected statement make this order explicit. Differentiating the constant $f(0)$ would instead give zero for $n > 0$. Since $x_i = \alpha_i t$, this is the required symbolic form. Explicitly the multinomial theorem gives

$$f(\mathbf{x}) = \sum_{r_1, \ldots, r_m \geq 0} \frac{x_1^{r_1} \cdots x_m^{r_m}}{r_1! \cdots r_m!} (\partial_{x_1}^{r_1} \cdots \partial_{x_m}^{r_m} f)(0).$$

As in Exercise 1.9.1, equality to the infinite expansion requires convergence to f; assume analyticity on the neighborhood under consideration, or retain a finite Taylor remainder.

1.10 Evaluation of Integrals

1.10.1 Use a recursive method to show that, for all positive integers n, $\Gamma(n) = (n-1)!$.
Evaluate the integrals in Exercises 1.10.2 through 1.10.9.

Solution.

Start from $\Gamma(n) = \int_0^\infty t^{n-1}e^{-t}dt$ for $n > 0$. For an integer $n \geq 1$, integration by parts gives

$$\Gamma(n+1) = \int_0^\infty t^n e^{-t}dt = [-t^n e^{-t}]_0^\infty + n\int_0^\infty t^{n-1}e^{-t}dt = n\Gamma(n).$$

The boundary term vanishes because the exponential dominates any fixed power at infinity and $t^n \to 0$ at zero. The initial value is $\Gamma(1) = \int_0^\infty e^{-t}dt = 1$. Repeated application of the recurrence yields $\Gamma(n) = (n-1)(n-2)\cdots 1\,\Gamma(1) = \boxed{(n-1)!}$, including $n = 1$ with $0! = 1$.

1.10.2 $\displaystyle\int_0^\infty \frac{\sin x}{x}\,dx.$

Hint. Multiply integrand by e^{-ax} and take the limit $a \downarrow 0$.

Solution.

For $a > 0$ define $I(a) = \int_0^\infty e^{-ax}\sin x\,dx/x$. Exponential decay permits differentiation under the integral:

$$I'(a) = -\int_0^\infty e^{-ax}\sin x\,dx = -\operatorname{Im}\frac{1}{a-i} = -\frac{1}{1+a^2}.$$

Also $|I(a)| \leq \int_0^\infty e^{-ax}dx = 1/a$, so $I(a) \to 0$ as $a \to \infty$. Integrating this derivative from a to infinity gives

$$I(a) = \int_a^\infty \frac{dt}{1+t^2} = \frac{\pi}{2} - \arctan a.$$

It remains to justify removing the damping. For $a \geq 0$ the decreasing function $g(x) = e^{-ax}/x$ satisfies $g' \leq 0$. On $[R, T]$,

$$\int_R^T g(x)\sin x\,dx = [-g(x)\cos x]_R^T + \int_R^T g'(x)\cos x\,dx.$$

Its absolute value is at most $g(R)+g(T)+\int_R^T |g'(x)|dx = g(R)+g(T)+g(R)-g(T) = 2g(R) \leq 2/R$. Taking $T \to \infty$ proves the same bound for the tail. This tail bound is uniform in a, and the integral on $[0, R]$ tends to its undamped value by dominated convergence. Thus the ordinary improper integral equals $\lim_{a\downarrow 0} I(a)$:

$$\boxed{\int_0^\infty \frac{\sin x}{x}dx = \frac{\pi}{2}}.$$

The tail argument also proves convergence, rather than merely assigning an Abel-summed value to the expression.

1.10.3 $\displaystyle\int_0^\infty \frac{dx}{\cosh x}.$

Hint. Expand the integrand using e^{-x} for $x > 0$, and control the integrated remainder at $x = 0$.

Solution.

For $x > 0$,

$$\frac{1}{\cosh x} = \frac{2e^{-x}}{1+e^{-2x}} = 2\sum_{n=0}^{N}(-1)^n e^{-(2n+1)x} + \frac{2(-1)^{N+1}e^{-(2N+3)x}}{1+e^{-2x}}.$$

The integral of the last term has absolute value at most $2/(2N+3)$, so integrating and taking the limit is justified even though the geometric series itself does not converge at $x = 0$. Thus the original hint's claim of series convergence at every relevant x needs this endpoint qualification. Therefore

$$\int_0^\infty \frac{dx}{\cosh x} = 2\sum_{n=0}^{\infty}\frac{(-1)^n}{2n+1} = \boxed{\frac{\pi}{2}}.$$

An independent substitution $t = e^{-x}$ gives $2\int_0^1 dt/(1+t^2) = 2[\arctan t]_0^1 = \pi/2$.

1.10.4 $\displaystyle\int_0^\infty \frac{dx}{e^{ax}+1}$, for $a > 0$.

Solution.

Set $u = e^{-ax}$ for $a > 0$. Then $dx = -du/(au)$, $e^{ax} = 1/u$, and the limits $x = 0, \infty$ become $u = 1, 0$. Consequently

$$\int_0^\infty \frac{dx}{e^{ax}+1} = \frac{1}{a}\int_0^1 \frac{du}{1+u} = \boxed{\frac{\ln 2}{a}}.$$

As a series check, for $x > 0$ the integrand is $e^{-ax} - e^{-2ax} + e^{-3ax} - \cdots$; integrating its finite geometric remainder gives an error at most $1/[a(N+1)]$. Hence its integral is $(1/a)\sum_{n\geq 1}(-1)^{n-1}/n = (\ln 2)/a$ as well.

1.10.5 $\displaystyle\int_\pi^\infty \frac{\sin x}{x^2}\,dx$.

Solution.

Differentiate $\sin x/x$ to obtain $\sin x/x^2 = \cos x/x - d(\sin x/x)/dx$. On a finite interval $[\pi, R]$,

$$\int_\pi^R \frac{\sin x}{x^2}dx = \int_\pi^R \frac{\cos x}{x}dx - \left[\frac{\sin x}{x}\right]_\pi^R.$$

The boundary term tends to zero because $\sin\pi = 0$ and $|\sin R/R| \leq 1/R$. With the convention $\mathrm{Ci}(z) = -\int_z^\infty \cos t\,dt/t$ for $z > 0$,

$$\boxed{\int_\pi^\infty \frac{\sin x}{x^2}dx = -\,\mathrm{Ci}(\pi) = -0.0736679120464255\ldots}.$$

The negative sign is consistent with the negative first lobe on $(\pi, 2\pi)$ and the decreasing weights of subsequent lobes. The verification script checks the result by oscillatory quadrature.

1.10.6 $\displaystyle\int_0^\infty \frac{e^{-x}\sin x}{x}\,dx.$

Solution.

Use $I(a)$ derived in Exercise 1.10.2:

$$I(a)=\int_0^\infty \frac{e^{-ax}\sin x}{x}dx=\frac{\pi}{2}-\arctan a \quad (a>0).$$

At $a=1$, this gives $\boxed{I(1)=\pi/4}$. For a direct check, write $\sin x/x=\int_0^1\cos(tx)dt$; absolute integrability of the exponentially damped double integral permits interchanging integrals, giving

$$\int_0^1 dt\int_0^\infty e^{-x}\cos(tx)dx=\int_0^1\frac{dt}{1+t^2}=\frac{\pi}{4}.$$

1.10.7 $\displaystyle\int_0^x \operatorname{erf}(t)\,dt.$

The result can be expressed in terms of special functions in Table 1.2.

Solution.

From the definition of the error function, $\operatorname{erf}'(t)=2e^{-t^2}/\sqrt{\pi}$. Integration by parts gives

$$\begin{aligned}\int_0^x \operatorname{erf}(t)dt&=[t\operatorname{erf}(t)]_0^x-\frac{2}{\sqrt{\pi}}\int_0^x te^{-t^2}dt\\&=\boxed{x\operatorname{erf}(x)+\frac{e^{-x^2}-1}{\sqrt{\pi}}}.\end{aligned}$$

Differentiating the answer cancels $2xe^{-x^2}/\sqrt{\pi}$ against $-2xe^{-x^2}/\sqrt{\pi}$, leaving $\operatorname{erf}(x)$; at $x=0$ the answer is zero. Both checks fix the additive constant.

1.10.8 $\displaystyle\int_1^x E_1(t)\,dt.$

Obtain a result in which the only special function is E_1, for real $x>0$.

Solution.

For $t>0$, $E_1(t)=\int_t^\infty e^{-u}du/u$ and hence $E_1'(t)=-e^{-t}/t$. Integration by parts now gives

$$\begin{aligned}\int_1^x E_1(t)dt&=[tE_1(t)]_1^x-\int_1^x tE_1'(t)dt\\&=xE_1(x)-E_1(1)+\int_1^x e^{-t}dt\\&=\boxed{xE_1(x)-e^{-x}-E_1(1)+e^{-1}},\qquad x>0.\end{aligned}$$

Its derivative is $E_1(x)+x(-e^{-x}/x)+e^{-x}=E_1(x)$, and its value at $x=1$ is zero. Thus it has both the correct derivative and the prescribed lower-limit constant.

1.10.9 $\displaystyle\int_0^\infty \frac{e^{-x}}{x+1}\,dx.$

Solution.

Set $u = x+1$, so $e^{-x} = e\,e^{-u}$ and $dx = du$. The limits are $u = 1$ and $u = \infty$, and therefore

$$\int_0^\infty \frac{e^{-x}}{x+1}dx = e\int_1^\infty \frac{e^{-u}}{u}du = \boxed{eE_1(1) = 0.596347362323194\ldots}.$$

The integral is convergent because its integrand is bounded at zero and bounded above by e^{-x} for $x \geq 0$. Direct positive-integrand quadrature in the supplementary script checks the numerical value.

1.10.10 Show that $\displaystyle\int_0^\infty \left(\frac{\tan^{-1} x}{x}\right)^2 dx = \pi \ln 2.$

Hint. Integrate by parts, to linearize in $\tan^{-1}$. Then replace $\tan^{-1} x$ by $\tan^{-1} ax$ and evaluate for $a = 1$.

Solution.

Integrate by parts with $u = (\arctan x)^2$ and $dv = dx/x^2$:

$$I = \left[-\frac{(\arctan x)^2}{x}\right]_0^\infty + 2\int_0^\infty \frac{\arctan x}{x(1+x^2)}dx.$$

The boundary term vanishes: near zero its magnitude is $O(x)$, and at infinity it is $O(1/x)$. Define, for $a \geq 0$,

$$J(a) = 2\int_0^\infty \frac{\arctan(ax)}{x(1+x^2)}dx, \qquad I = J(1), \qquad J(0) = 0.$$

Differentiation is dominated by $2/(1+x^2)$ and yields

$$J'(a) = 2\int_0^\infty \frac{dx}{(1+x^2)(1+a^2x^2)}.$$

For $a > 0$, $a \neq 1$, partial fractions give

$$\frac{1}{(1+x^2)(1+a^2x^2)} = \frac{1}{1-a^2}\left(\frac{1}{1+x^2} - \frac{a^2}{1+a^2x^2}\right),$$

so $J'(a) = 2(\pi/2 - a\pi/2)/(1-a^2) = \pi/(1+a)$. Continuity of the integral extends this to $a = 0, 1$. Integrating from 0 to a gives $J(a) = \pi\ln(1+a)$, and hence $\boxed{I = \pi \ln 2}$.

1.10.11 By direct integration in Cartesian coordinates, find the area of the ellipse with positive semiaxes a, b, defined by

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Solution.

Take $a, b > 0$. In the first quadrant the ellipse has upper boundary $y = b\sqrt{1 - x^2/a^2}$, $0 \leq x \leq a$. Reflection symmetry gives

$$A = 4b\int_0^a \sqrt{1 - x^2/a^2}\,dx.$$

Use $x = a\sin\theta$, $dx = a\cos\theta\,d\theta$, with $0 \leq \theta \leq \pi/2$:

$$A = 4ab\int_0^{\pi/2} \cos^2\theta\,d\theta = 2ab\left[\theta + \frac{1}{2}\sin 2\theta\right]_0^{\pi/2} = \boxed{\pi ab}.$$

For $a = b = r$ this reduces to the circle area πr^2, checking the scale and numerical coefficient.

1.10.12 A unit circle is divided into two pieces by a straight line whose distance of closest approach to the center is $1/2$ unit. By evaluating a suitable integral, find the area of the smaller piece thereby produced. Then use simple geometric considerations to verify your answer.

Solution.

Rotate coordinates so that the cutting line is $x = 1/2$ in the unit circle $x^2 + y^2 = 1$. The smaller cap has vertical extent $2\sqrt{1-x^2}$, so

$$\begin{aligned} A &= 2\int_{1/2}^{1} \sqrt{1-x^2}\,dx \\ &= \Big[x\sqrt{1-x^2} + \arcsin x\Big]_{1/2}^{1} = \frac{\pi}{2} - \left(\frac{\sqrt{3}}{4} + \frac{\pi}{6}\right) = \boxed{\frac{\pi}{3} - \frac{\sqrt{3}}{4}}. \end{aligned}$$

For the geometric check, the chord's endpoints make angles $\pm\pi/3$ with the positive x axis. The sector between them has angle $2\pi/3$ and area $(1/2)(2\pi/3) = \pi/3$. The triangle cut off from that sector has base $\sqrt{3}$ and altitude $1/2$, hence area $\sqrt{3}/4$. Subtracting gives the same cap area.

1.11 Dirac Delta Function

1.11.1 Let

$$\delta_n(x) = \begin{cases} 0, & x < -\dfrac{1}{2n}, \\ n, & -\dfrac{1}{2n} < x < \dfrac{1}{2n}, \\ 0, & \dfrac{1}{2n} < x. \end{cases}$$

Show that

$$\lim_{n\to\infty} \int_{-\infty}^{\infty} f(x)\delta_n(x)\,dx = f(0),$$

assuming that $f(x)$ is locally integrable and continuous at $x = 0$.

Solution.

The rectangle has width $1/n$ and height n, so its integral is 1. Subtracting $f(0)$ under the integral gives the useful explicit bound

$$\begin{aligned} \left|\int_{\mathbb{R}} f(x)\delta_n(x)dx - f(0)\right| &= \left|n\int_{-1/(2n)}^{1/(2n)} [f(x) - f(0)]dx\right| \\ &\le \sup_{|x|\le 1/(2n)} |f(x) - f(0)|. \end{aligned}$$

Continuity at zero makes the supremum tend to zero: for each $\epsilon > 0$, take $\eta > 0$ with $|f(x)-f(0)| < \epsilon$ when $|x| < \eta$, then take $n > 1/(2\eta)$. This proves the desired limit. The unspecified values at the two endpoints do not affect an integral.

1.11.2 For

$$\delta_n(x) = \frac{n}{\pi}\,\frac{1}{1+n^2x^2},$$

show that

$$\int_{-\infty}^{\infty} \delta_n(x)\,dx = 1.$$

Solution.

For $n > 0$ use $u = nx$, so $dx = du/n$. Then

$$\int_{-\infty}^{\infty} \delta_n(x)dx = \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{du}{1+u^2} = \frac{1}{\pi}[\arctan u]_{-\infty}^{\infty} = \frac{1}{\pi}\left(\frac{\pi}{2}+\frac{\pi}{2}\right) = 1.$$

This proves normalization for each n, not merely in the limit. It also displays why the numerator must contain the factor n.

1.11.3 Fejer's method of summing series is associated with the function

$$\delta_n(t) = \frac{1}{2\pi n}\left[\frac{\sin(nt/2)}{\sin(t/2)}\right]^2.$$

Show that $\delta_n(t)$ is a delta distribution on one period, for continuous f, in the sense that

$$\lim_{n\to\infty}\frac{1}{2\pi n}\int_{-\pi}^{\pi} f(t)\left[\frac{\sin(nt/2)}{\sin(t/2)}\right]^2 dt = f(0).$$

Solution.

The printed whole-line assertion is false. On one period, $\int_{-\pi}^{\pi} f(t)\delta_n(t)dt \to f(0)$ for continuous f. On the whole real line the distributional limit is instead the comb $\sum_{m\in\mathbb{Z}}\delta(t-2\pi m)$. For integer $n \geq 1$, the finite geometric sum gives

$$\delta_n(t) = \frac{1}{2\pi n}\left|\sum_{j=0}^{n-1} e^{ijt}\right|^2 = \frac{1}{2\pi}\sum_{k=-(n-1)}^{n-1}\left(1-\frac{|k|}{n}\right)e^{ikt}.$$

The modulus-square expression proves positivity; integrating the finite Fourier sum over $[-\pi,\pi]$ leaves only its constant term, giving $\int_{-\pi}^{\pi}\delta_n(t)dt = 1$. For any $0 < \eta < \pi$, if $\eta \leq |t| \leq \pi$ then

$$0 \leq \delta_n(t) \leq \frac{1}{2\pi n\sin^2(\eta/2)}, \qquad \int_{\eta\leq|t|\leq\pi}\delta_n(t)dt \leq \frac{1}{n\sin^2(\eta/2)} \to 0.$$

Subtract $f(0)$ and split the integral into $|t| < \eta$ and its complement. The first part is at most $\sup_{|t|<\eta}|f(t)-f(0)|$, which continuity makes arbitrarily small. For bounded continuous f the second is at most $2\|f\|_\infty/[n\sin^2(\eta/2)] \to 0$. This proves the one-period result. For a smooth compactly supported test function on the whole line, partition its support into periods and use 2π-periodicity of δ_n. The same argument gives

$$\lim_{n\to\infty}\int_{\mathbb{R}} f(t)\delta_n(t)dt = \sum_{m\in\mathbb{Z}} f(2\pi m).$$

Only finitely many terms occur for such a test function. Choosing it supported near 2π, with $f(2\pi) = 1$ and $f(0) = 0$, directly contradicts the original claim and verifies the correction.

1.11.4 Prove that

$$\delta[a(x-x_1)] = \frac{1}{|a|}\,\delta(x-x_1), \qquad a \neq 0.$$

Note. The absolute value is required for all real nonzero a, including negative a.

Solution.

For every real $a \neq 0$ the factor is $1/|a|$; evenness is not an optional extra convention for the Dirac distribution. Test the identity against a smooth compactly supported ϕ and make the change of variable $u = a(x - x_1)$. The orientation of the integration limits changes when $a < 0$, producing the absolute Jacobian:

$$\int_{\mathbb{R}} \phi(x)\delta(a(x-x_1))dx = \frac{1}{|a|}\int_{\mathbb{R}} \phi(x_1 + u/a)\delta(u)du = \frac{\phi(x_1)}{|a|}.$$

This equals the action of $\delta(x - x_1)/|a|$, proving the distribution identity. For $a > 0$ it reduces to the printed factor $1/a$. The value $a = 0$ is excluded because $\delta(0)$ is not a distribution obtained by this change of variable.

1.11.5 For distinct real x_1, x_2, show that

$$\delta[(x-x_1)(x-x_2)] = [\delta(x-x_1) + \delta(x-x_2)]/|x_1 - x_2|.$$

Hint. Try using Exercise 1.11.4.

Solution.

Assume distinct real roots $x_1 \neq x_2$. Let $g(x) = (x - x_1)(x - x_2)$. Near either simple root, use the local variable $u = g(x)$, with $dx = du/|g'(x)|$ in a real integral. Since

$$g'(x) = 2x - x_1 - x_2, \quad g'(x_1) = x_1 - x_2, \quad g'(x_2) = x_2 - x_1,$$

a test function ϕ has integral

$$\int_{\mathbb{R}} \phi(x)\delta(g(x))dx = \frac{\phi(x_1)}{|x_1 - x_2|} + \frac{\phi(x_2)}{|x_2 - x_1|}.$$

There is no contribution away from the roots, where g stays away from zero locally. Recognizing the two evaluation functionals proves

$$\delta((x-x_1)(x-x_2)) = \frac{\delta(x-x_1) + \delta(x-x_2)}{|x_1 - x_2|}.$$

For coincident roots the Jacobian vanishes and this formula does not apply; its singular denominator cannot simply be assigned a value.

1.11.6 Using the Gauss error curve delta sequence $\delta_n = \dfrac{n}{\sqrt{\pi}} e^{-n^2x^2}$, show that

$$x\frac{d}{dx}\delta(x) = -\delta(x),$$

treating $\delta(x)$ and its derivative as in Eq. (1.157).

Solution.

The textbook (p. 81) asks for $x\delta'(x) = -\delta(x)$. The transcribed derivative of the product is incorrect: $(x\delta(x))' = 0$, since $x\delta(x) = 0$. For the specified Gaussian sequence,

$$\delta_n'(x) = -\frac{2n^3 x}{\sqrt{\pi}} e^{-n^2x^2}.$$

Against a smooth compactly supported test function ϕ, substitute $y = nx$ to obtain

$$\int_{\mathbb{R}} x\delta_n'(x)\phi(x)dx = -\frac{2}{\sqrt{\pi}} \int_{\mathbb{R}} y^2 e^{-y^2} \phi(y/n)dy.$$

Dominated convergence is justified by the integrable bound $\|\phi\|_\infty y^2 e^{-y^2}$. Integration by parts gives

$$\int_{\mathbb{R}} y^2 e^{-y^2} dy = -\frac{1}{2}[ye^{-y^2}]_{-\infty}^{\infty} + \frac{1}{2}\int_{\mathbb{R}} e^{-y^2} dy = \frac{\sqrt{\pi}}{2}.$$

Hence the limit is $-\phi(0)$, exactly the action of $-\delta$. This proves $x\delta' = -\delta$. The product rule then gives $(x\delta)' = \delta + x\delta' = 0$, an independent consistency check.

1.11.7 Show that

$$\int_{-\infty}^{\infty} \delta'(x) f(x)\, dx = -f'(0).$$

Here we assume that $f'(x)$ is continuous at $x = 0$.

Solution.

A distributional derivative is defined by its action on test functions, $\langle T', f\rangle = -\langle T, f'\rangle$. Taking $T = \delta$ gives

$$\boxed{\int_{\mathbb{R}} \delta'(x) f(x) dx = -f'(0)}.$$

To check this from regular functions, use a smooth delta sequence δ_n and a smooth compactly supported f. Ordinary integration by parts gives

$$\int_{\mathbb{R}} \delta_n'(x) f(x) dx = [\delta_n(x) f(x)]_{-\infty}^{\infty} - \int_{\mathbb{R}} \delta_n(x) f'(x) dx = -\int_{\mathbb{R}} \delta_n(x) f'(x) dx.$$

The boundary term vanishes by compact support. Continuity of f' at zero then makes the last integral tend to $-f'(0)$, reproducing the distributional definition and fixing the minus sign. Only the values near zero matter: multiply any locally C^1 function by a smooth cutoff equal to 1 there. Since δ' is supported at zero, this leaves its action unchanged. The identity therefore extends to the local regularity stated in the question without imposing any behavior on f at infinity.

1.11.8 Prove that

$$\delta(f(x)) = \frac{1}{|f\prime(x_0)|}\delta(x - x_0),$$

where f is real and C^1, and x_0 is its unique simple real zero, so $f(x_0) = 0$ and $f\prime(x_0) \neq 0$.

Hint. Use the local variable $u = f(x)$ and the absolute Jacobian $1/|f\prime(x_0)|$ at each simple zero.

Solution.

For a unique simple real zero x_0, the factor must be $1/|f'(x_0)|$. For multiple simple zeros, add their contributions. Let f be a real C^1 function on the domain under consideration, with $f'(x_i) \neq 0$

at every zero x_i. The inverse function theorem supplies a local inverse $x = x_i(u)$ for $u = f(x)$. Restricting a test function ϕ to this neighborhood and changing variables gives

$$\int \phi(x)\delta(f(x))dx = \int \frac{\phi(x_i(u))}{|f'(x_i(u))|}\delta(u)du = \frac{\phi(x_i)}{|f'(x_i)|}.$$

A partition into neighborhoods of the zeros (only finitely many in a compact test support) therefore yields

$$\boxed{\delta(f(x)) = \sum_{i:f(x_i)=0} \frac{\delta(x - x_i)}{|f'(x_i)|}}.$$

The printed single-root formula is recovered only with the additional positive-derivative assumption. A negative derivative reverses limits, which is why the absolute value is necessary. A repeated zero has $f'(x_i) = 0$ and lies outside this formula's hypotheses.

1.11.9

(a) If we define a sequence $\delta_n(x) = n/(2\cosh^2 nx)$, show that

$$\int_{-\infty}^{\infty} \delta_n(x)\, dx = 1, \quad \text{independent of } n.$$

(b) Continuing this analysis, show that[2]

$$\int_{-\infty}^{x} \delta_n(t)\, dt = \frac{1}{2}[1 + \tanh nx] \equiv u_n(x)$$

and

$$\lim_{n\to\infty} u_n(x) = \begin{cases} 0, & x < 0, \\ 1, & x > 0. \end{cases}$$

This is the Heaviside unit step function (Fig. 1.23).

Solution.

(a) For $n > 0$, substitute $u = nx$:

$$\int_{-\infty}^{\infty} \frac{n}{2\cosh^2(nx)}dx = \frac{1}{2}\int_{-\infty}^{\infty} \operatorname{sech}^2 u\, du = \frac{1}{2}[\tanh u]_{-\infty}^{\infty} = 1.$$

(b) Use a separate dummy variable in the integral. Since $d(\tanh nt)/dt = n\operatorname{sech}^2(nt)$,

$$u_n(x) = \int_{-\infty}^{x} \delta_n(t)dt = \frac{1}{2}[\tanh(nt)]_{t=-\infty}^{t=x} = \frac{1}{2}[1 + \tanh(nx)].$$

If $x > 0$, $nx \to +\infty$ and $u_n(x) \to 1$; if $x < 0$, $nx \to -\infty$ and $u_n(x) \to 0$. At $x = 0$ every $u_n(0) = 1/2$, so this symmetric sequence selects the convention $u(0) = 1/2$. The value at a single point does not affect its distributional derivative, which is δ.

[2]Many other symbols are used for this function. This is the AMS-55 notation (in Additional Readings, see Abramowitz and Stegun): u for unit.

2 Chapter 2. Determinants and Matrices

Problems

2.1 Determinants

2.1.1 Evaluate the following determinants:

(a) $\begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix}$,

(b) $\begin{vmatrix} 1 & 2 & 0 \\ 3 & 1 & 2 \\ 0 & 3 & 1 \end{vmatrix}$,

(c) $\dfrac{1}{\sqrt{2}} \begin{vmatrix} 0 & \sqrt{3} & 0 & 0 \\ \sqrt{3} & 0 & 2 & 0 \\ 0 & 2 & 0 & \sqrt{3} \\ 0 & 0 & \sqrt{3} & 0 \end{vmatrix}$.

Solution.

Expand along a row or column containing as many zeros as possible.

(a) Only the middle entry of row 2 contributes:

$$D_a = 1(-1)^{2+2} \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} = 1(1 \cdot 0 - 1 \cdot 1) = \boxed{-1}.$$

(b) Expansion along row 1 gives

$$D_b = 1 \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} - 2 \begin{vmatrix} 3 & 2 \\ 0 & 1 \end{vmatrix} = 1(1-6) - 2(3-0) = \boxed{-11}.$$

(c) Write $s = \sqrt{3}$. The factor $1/\sqrt{2}$ in the question multiplies the determinant once; it is not a factor on each row. Expand row 1 and then column 1 of the resulting minor:

$$D_c = -\frac{s}{\sqrt{2}} \begin{vmatrix} s & 2 & 0 \\ 0 & 0 & s \\ 0 & s & 0 \end{vmatrix} = -\frac{s^2}{\sqrt{2}} \begin{vmatrix} 0 & s \\ s & 0 \end{vmatrix}$$

$$= -\frac{3}{\sqrt{2}}(0 - 3) = \boxed{\frac{9}{\sqrt{2}}}.$$

2.1.2 Test the set of linear homogeneous equations

$$x + 3y + 3z = 0, \quad x - y + z = 0, \quad 2x + y + 3z = 0$$

to see if it possesses a nontrivial solution. In any case, find a solution to this equation set.

Solution.

The coefficient determinant is

$$D = \begin{vmatrix} 1 & 3 & 3 \\ 1 & -1 & 1 \\ 2 & 1 & 3 \end{vmatrix}$$
$$= 1(-3-1) - 3(3-2) + 3(1+2) = -4 - 3 + 9 = 2.$$

Since $D \neq 0$, only the trivial solution is possible. Directly, the second equation gives $x = y - z$. The first and third then become

$$(y - z) + 3y + 3z = 4y + 2z = 0, \qquad 2(y - z) + y + 3z = 3y + z = 0.$$

Thus $z = -2y$ and $z = -3y$, so $y = 0$, $z = 0$, and $x = 0$: $\boxed{(x, y, z) = (0, 0, 0)}$. Substitution into each original equation gives zero.

2.1.3 Given the pair of equations

$$x + 2y = 3, \quad 2x + 4y = 6,$$

(a) Show that the determinant of the coefficients vanishes.

(b) Show that the numerator determinants, see Eq. (2.18), also vanish.

(c) Find at least two solutions.

Solution.

The three determinants in Cramer's rule are

$$D = \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = 4 - 4 = 0, \quad D_x = \begin{vmatrix} 3 & 2 \\ 6 & 4 \end{vmatrix} = 12 - 12 = 0, \quad D_y = \begin{vmatrix} 1 & 3 \\ 2 & 6 \end{vmatrix} = 6 - 6 = 0.$$

The quotients $D_x/D, D_y/D$ are undefined and cannot determine a unique solution. The second equation is twice the first, so put $y = t$. Then $x = 3 - 2t$, and every real solution is

$$\boxed{(x, y) = (3 - 2t, t), \quad t \in \mathbb{R}}.$$

(Complex t is allowed if complex solutions are sought.) For example $t = 0$ gives $(3, 0)$, and $t = 1$ gives $(1, 1)$. Both satisfy $x + 2y = 3$ and hence $2x + 4y = 6$.

2.1.4 If C_{ij} is the cofactor of element a_{ij}, formed by striking out the ith row and jth column and including a sign $(-1)^{i+j}$, show that

(a) $\sum_i a_{ij} C_{ij} = \sum_i a_{ji} C_{ji} = |\mathsf{A}|$, where $|\mathsf{A}|$ is the determinant with the elements a_{ij},

(b) $\sum_i a_{ij}C_{ik} = \sum_i a_{ji}C_{ki} = 0,\ j \neq k.$

Solution.

For fixed j, Laplace expansion down column j is $\det A = \sum_i a_{ij}(-1)^{i+j} \det A_{\widehat{i,j}} = \sum_i a_{ij}C_{ij}$. Expansion across row j similarly gives $\det A = \sum_i a_{ji}C_{ji}$. This proves (a), without assuming invertibility.

For (b), replace column k of A by its column j, with $j \neq k$, and call the result B. Columns j and k of B are identical, so $\det B = 0$. Expanding down column k, each minor is unchanged from A because that column is deleted; its cofactor is C_{ik}. The entry being multiplied by that cofactor is now a_{ij}. Therefore

$$0 = \det B = \sum_i a_{ij}C_{ik}.$$

Replacing row k by row j and expanding across row k gives, by the same argument, $0 = \sum_i a_{ji}C_{ki}$.

2.1.5 A determinant with all elements of order unity may be surprisingly small. The Hilbert determinant $H_{ij} = (i + j - 1)^{-1}$, $i, j = 1, 2, \ldots, n$ is notorious for its small values.

(a) Calculate the value of the Hilbert determinants of order n for $n = 1$, 2, and 3.

(b) If an appropriate subroutine is available, find the Hilbert determinants of order n for $n = 4$, 5, and 6.

	n	$\mathrm{Det}(H_n)$
	1	1.
	2	8.33333×10^{-2}
ANS.	3	4.62963×10^{-4}
	4	1.65344×10^{-7}
	5	3.74930×10^{-12}
	6	5.36730×10^{-18}.

Solution.

Use exact fractions so that small determinants are not lost by subtracting rounded numbers. The first three determinants are

$$D_1 = 1, \qquad D_2 = 1 \cdot \frac{1}{3} - \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{12},$$

$$\begin{aligned} D_3 &= \left(\frac{1}{15} - \frac{1}{16}\right) - \frac{1}{2}\left(\frac{1}{10} - \frac{1}{12}\right) + \frac{1}{3}\left(\frac{1}{8} - \frac{1}{9}\right) \\ &= \frac{1}{240} - \frac{1}{120} + \frac{1}{216} = \frac{9 - 18 + 10}{2160} = \frac{1}{2160}. \end{aligned}$$

For the larger matrices, Gaussian elimination uses

$$R_i \longleftarrow R_i - \frac{u_{ik}}{u_{kk}} R_k \quad (i = k + 1, \ldots, n;\ k = 1, \ldots, n - 1).$$

Each operation preserves the determinant. For example, the first step on H_6 changes an entry with $i, j \geq 2$ into

$$\frac{1}{i + j - 1} - \frac{1}{i}\frac{1}{j} = \frac{(i - 1)(j - 1)}{ij(i + j - 1)}.$$

For completeness, the multipliers u_{ik}/u_{kk} in the successive steps are listed below. At step k, subtract the indicated multiple of the current row k from each listed row i:

k	i	u_{ik}/u_{kk} in the same order
1	$2,3,4,5,6$	$1/2,\ 1/3,\ 1/4,\ 1/5,\ 1/6$
2	$3,4,5,6$	$1,\ 9/10,\ 4/5,\ 5/7$
3	$4,5,6$	$3/2,\ 12/7,\ 25/14$
4	$5,6$	$2,\ 25/9$
5	6	$5/2$

For example, the second pivot is $1/12$ and the entry below it in row 4 is $3/40$, giving multiplier $(3/40)/(1/12) = 9/10$. These operations produce

$$U_6 = \begin{pmatrix} 1 & 1/2 & 1/3 & 1/4 & 1/5 & 1/6 \\ 0 & 1/12 & 1/12 & 3/40 & 1/15 & 5/84 \\ 0 & 0 & 1/180 & 1/120 & 1/105 & 5/504 \\ 0 & 0 & 0 & 1/2800 & 1/1400 & 1/1008 \\ 0 & 0 & 0 & 0 & 1/44100 & 1/17640 \\ 0 & 0 & 0 & 0 & 0 & 1/698544 \end{pmatrix}.$$

Elimination on a leading $n \times n$ block uses the same leading pivots, so D_n is the product of the first n diagonal entries. In particular

$$D_4 = \frac{D_3}{2800}, \qquad D_5 = \frac{D_4}{44100}, \qquad D_6 = \frac{D_5}{698544}.$$

The resulting exact values and six-significant-figure approximations are

n	D_n	decimal value
1	1	1
2	1/12	$8.33333\,10^{-2}$
3	1/2160	$4.62963\,10^{-4}$
4	1/6048000	$1.65344\,10^{-7}$
5	1/266716800000	$3.74930\,10^{-12}$
6	1/186313420339200000	$5.36730\,10^{-18}$

The supplementary program reproduces the elimination and independently checks its products against exact determinant evaluation.

2.1.6 Prove that the determinant consisting of the coefficients from a set of linearly dependent forms has the value zero.

Solution.

Write the n forms as $L_i(x) = \sum_{j=1}^{n} a_{ij}x_j$. Linear dependence means that constants c_i, not all zero, satisfy $\sum_i c_i L_i(x) = 0$ for every x. Equating coefficients gives $\sum_i c_i a_{ij} = 0$ for each j. Choose r with $c_r \neq 0$. Then the row vectors obey

$$R_r = -\sum_{i \neq r} \frac{c_i}{c_r} R_i.$$

Replace R_r by $R_r + \sum_{i \neq r}(c_i/c_r)R_i$. Adding multiples of other rows leaves the determinant unchanged, but the new row is zero. Expansion across it therefore gives $\boxed{\det A = 0}$. The same proof applies if the forms are recorded as columns.

2.1.7 Solve the following set of linear simultaneous equations. Give the results to five decimal places.

$$\begin{aligned}
1.0x_1 + 0.9x_2 + 0.8x_3 + 0.4x_4 + 0.1x_5 &= 1.0\\
0.9x_1 + 1.0x_2 + 0.8x_3 + 0.5x_4 + 0.2x_5 + 0.1x_6 &= 0.9\\
0.8x_1 + 0.8x_2 + 1.0x_3 + 0.7x_4 + 0.4x_5 + 0.2x_6 &= 0.8\\
0.4x_1 + 0.5x_2 + 0.7x_3 + 1.0x_4 + 0.6x_5 + 0.3x_6 &= 0.7\\
0.1x_1 + 0.2x_2 + 0.4x_3 + 0.6x_4 + 1.0x_5 + 0.5x_6 &= 0.6\\
0.1x_2 + 0.2x_3 + 0.3x_4 + 0.5x_5 + 1.0x_6 &= 0.5.
\end{aligned}$$

Note. These equations may also be solved by matrix inversion, as discussed in Section 2.2.

Solution.

Multiply every equation by 10 and put them in the order $5, 6, 3, 4, 1, 2$:

$$\left[\begin{array}{cccccc|c}
1 & 2 & 4 & 6 & 10 & 5 & 6\\
0 & 1 & 2 & 3 & 5 & 10 & 5\\
8 & 8 & 10 & 7 & 4 & 2 & 8\\
4 & 5 & 7 & 10 & 6 & 3 & 7\\
10 & 9 & 8 & 4 & 1 & 0 & 10\\
9 & 10 & 8 & 5 & 2 & 1 & 9
\end{array}\right].$$

Here R_i refers to this new ordering. Apply $R_3 - 8R_1$, $R_4 - 4R_1$, $R_5 - 10R_1$, and $R_6 - 9R_1$ to rows $3, 4, 5, 6$. The last four rows, omitting the zero first column, become

$$\left[\begin{array}{ccccc|c}
-8 & -22 & -41 & -76 & -38 & -40\\
-3 & -9 & -14 & -34 & -17 & -17\\
-11 & -32 & -56 & -99 & -50 & -50\\
-8 & -28 & -49 & -88 & -44 & -45
\end{array}\right].$$

Next add $8R_2, 3R_2, 11R_2, 8R_2$ to those rows. In columns x_3, x_4, x_5, x_6 this gives

$$\left[\begin{array}{cccc|c}
-6 & -17 & -36 & 42 & 0\\
-3 & -5 & -19 & 13 & -2\\
-10 & -23 & -44 & 60 & 5\\
-12 & -25 & -48 & 36 & -5
\end{array}\right].$$

To eliminate x_3, replace

$$R_4 \leftarrow -2R_4 + R_3, \quad R_5 \leftarrow -3R_5 + 5R_3, \quad R_6 \leftarrow -R_6 + 2R_3.$$

Rows $4, 5, 6$, in columns x_4, x_5, x_6, are now

$$\left[\begin{array}{ccc|c}
-7 & 2 & 16 & 4\\
-16 & -48 & 30 & -15\\
-9 & -24 & 48 & 5
\end{array}\right].$$

Eliminate x_4 by $R_5 \leftarrow -7R_5 + 16R_4$ and $R_6 \leftarrow -7R_6 + 9R_4$. The final two equations are

$$368x_5 + 46x_6 = 169, \qquad 186x_5 - 192x_6 = 1.$$

Finally $R_6 \leftarrow (368R_6 - 186R_5)/14$ gives $-5658x_6 = -2219$. Back substitution gives every component:

$$\begin{aligned}
x_6 &= \frac{2219}{5658},\\
x_5 &= \frac{169 - 46x_6}{368} = \frac{2321}{5658},\\
x_4 &= \frac{2x_5 + 16x_6 - 4}{7} = \frac{417}{943},\\
x_3 &= \frac{-17x_4 - 36x_5 + 42x_6}{6} = -\frac{2741}{2829},\\
x_2 &= 5 - 2x_3 - 3x_4 - 5x_5 - 10x_6 = -\frac{89}{246},\\
x_1 &= 6 - 2x_2 - 4x_3 - 6x_4 - 10x_5 - 5x_6 = \frac{3551}{1886}.
\end{aligned}$$

Thus, to five decimal places,

$$\boxed{x = (1.88282, -0.36179, -0.96889, 0.44221, 0.41022, 0.39219)^T}.$$

For an exact check, let B be ten times the original coefficient matrix. In a common denominator,

$$x = \frac{1}{5658}(10653, -2047, -5482, 2502, 2321, 2219)^T.$$

Direct multiplication gives

$$\begin{aligned}
B(10653, -2047, -5482, 2502, 2321, 2219)^T &= (56580, 50922, 45264, 39606, 33948, 28290)^T\\
&= 5658(10, 9, 8, 7, 6, 5)^T,
\end{aligned}$$

verifying all six equations without rounding.

2.1.8 Show that (in 3-D space)

(a) $\displaystyle\sum_i \delta_{ii} = 3,$

(b) $\displaystyle\sum_{ij} \delta_{ij}\varepsilon_{ijk} = 0,$

(c) $\displaystyle\sum_{pq} \varepsilon_{ipq}\varepsilon_{jpq} = 2\delta_{ij},$

(d) $\displaystyle\sum_{ijk} \varepsilon_{ijk}\varepsilon_{ijk} = 6.$

Note. The symbol δ_{ij} is the Kronecker delta, defined in Eq. (1.164), and ε_{ijk} is the Levi-Civita symbol, Eq. (2.8).

Solution.

(a) Since $\delta_{11} = \delta_{22} = \delta_{33} = 1$, $\sum_i \delta_{ii} = 1 + 1 + 1 = 3$.

(b) The factor δ_{ij} eliminates every term with $i \neq j$, so $\sum_{ij} \delta_{ij}\varepsilon_{ijk} = \sum_i \varepsilon_{iik} = 0$. Every remaining Levi-Civita symbol has a repeated index.

(c) If $i = j$, the only nonzero terms have (p, q) equal to the two orderings of the other two indices. Their squares are both 1, so the sum is 2. If $i \neq j$, a nonzero ε_{ipq} requires $\{i, p, q\} = \{1, 2, 3\}$. For the same p, q, ε_{jpq} then has a repeated index and vanishes. Thus the sum is 2 for $i = j$ and 0 otherwise: $\sum_{pq} \varepsilon_{ipq}\varepsilon_{jpq} = 2\delta_{ij}$.

(d) There are $3! = 6$ permutations of $(1, 2, 3)$. For each, $\varepsilon_{ijk}^2 = 1$, and every other term vanishes. Thus $\sum_{ijk} \varepsilon_{ijk}\varepsilon_{ijk} = 6$.

2.1.9 Show that (in 3-D space)

$$\sum_k \varepsilon_{ijk}\varepsilon_{pqk} = \delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp}.$$

Note. See Exercise 2.1.8 for definitions of δ_{ij} and ε_{ijk}.

Solution.

The textbook identity is correct. The manual's final line has $\delta_{ip}\delta_{jp}$ in place of $\delta_{ip}\delta_{jq}$. For $i = p = 1$, $j = q = 2$, the left side is 1, whereas that misprinted right side is 0. If $i = j$ or $p = q$, a Levi-Civita factor on the left vanishes for every k; the two delta products on the right are identical and cancel. Now take $i \neq j$ and $p \neq q$. A nonzero product requires a k different from both i, j and from both p, q. In three dimensions this requires the unordered pairs $\{i, j\}$ and $\{p, q\}$ to agree. If they differ, both sides are zero. If $(p, q) = (i, j)$, the single allowed k contributes $\varepsilon_{ijk}^2 = 1$, and the right side is $1 - 0$. If $(p, q) = (j, i)$, antisymmetry gives $\varepsilon_{ijk}\varepsilon_{jik} = -1$, and the right side is $0 - 1$. These cases exhaust all indices, proving

$$\boxed{\sum_k \varepsilon_{ijk}\varepsilon_{pqk} = \delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp}}.$$

2.2 Matrices

2.2.1 Show that matrix multiplication is associative, $(\mathsf{AB})\mathsf{C} = \mathsf{A}(\mathsf{BC})$.

Solution.

Let A, B, C have sizes $m \times n$, $n \times p$, and $p \times q$. For every allowed i, j,

$$\begin{aligned}((AB)C)_{ij} &= \sum_{k=1}^{p} (AB)_{ik}c_{kj} = \sum_{k=1}^{p}\sum_{\ell=1}^{n} a_{i\ell}b_{\ell k}c_{kj} \\ &= \sum_{\ell=1}^{n} a_{i\ell}\left(\sum_{k=1}^{p} b_{\ell k}c_{kj}\right) = \sum_{\ell=1}^{n} a_{i\ell}(BC)_{\ell j} = (A(BC))_{ij}.\end{aligned}$$

The sums are finite, so their order can be interchanged. Equality of every entry proves $(AB)C = A(BC)$.

2.2.2 Show that

$$(\mathsf{A} + \mathsf{B})(\mathsf{A} - \mathsf{B}) = \mathsf{A}^2 - \mathsf{B}^2$$

if and only if A and B commute,

$$[\mathsf{A}, \mathsf{B}] = 0.$$

Solution.

Distribute while preserving factor order:

$$(A+B)(A-B) = A^2 - AB + BA - B^2 = A^2 - B^2 - [A,B].$$

If $[A,B]=0$, the required equality follows. Conversely, if that equality holds, subtract A^2-B^2 from both sides to get $-[A,B]=0$. This proves both directions.

2.2.3

(a) Complex numbers, $a+ib$, with a and b real, may be represented by (or are isomorphic with) 2×2 matrices:

$$a+ib \longleftrightarrow \begin{pmatrix} a & b \\ -b & a \end{pmatrix}.$$

Show that this matrix representation is valid for (i) addition and (ii) multiplication.

(b) Find the matrix corresponding to $(a+ib)^{-1}$, assuming $a+ib \neq 0$.

Solution.

Define $F(a+ib) = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$. For real a,b,c,d,

$$F(a+ib)+F(c+id) = \begin{pmatrix} a+c & b+d \\ -(b+d) & a+c \end{pmatrix} = F((a+c)+i(b+d)).$$

Multiplication gives

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix}\begin{pmatrix} c & d \\ -d & c \end{pmatrix} = \begin{pmatrix} ac-bd & ad+bc \\ -(ad+bc) & ac-bd \end{pmatrix},$$

which is $F((a+ib)(c+id))$ because $(a+ib)(c+id) = (ac-bd)+i(ad+bc)$. The representation is one-to-one, since its first row recovers a,b, and $F(1)=I_2$.

For $a+ib\neq 0$,

$$\frac{1}{a+ib} = \frac{a-ib}{a^2+b^2}, \qquad \boxed{F((a+ib)^{-1}) = \frac{1}{a^2+b^2}\begin{pmatrix} a & -b \\ b & a \end{pmatrix}}.$$

Multiplication by $F(a+ib)$ gives $(a^2+b^2)I_2/(a^2+b^2) = I_2$; if $a=b=0$, neither inverse exists.

2.2.4 If A is an $n\times n$ matrix, show that

$$\det(-\mathsf{A}) = (-1)^n \det \mathsf{A}.$$

Solution.

Changing A into $-A$ multiplies each of its n rows by -1. Linearity of the determinant in each row gives n such factors:

$$\det(-A) = \underbrace{(-1)(-1)\cdots(-1)}_{n \text{ rows}} \det A = \boxed{(-1)^n \det A}.$$

No nonsingularity assumption is needed.

2.2.5

(a) The matrix equation $\mathsf{A}^2 = 0$ does not imply $\mathsf{A} = 0$. Show that the most general 2×2 matrix whose square is zero may be written as

$$\begin{pmatrix} ab & b^2 \\ -a^2 & -ab \end{pmatrix},$$

where a and b are real or complex numbers. Allow complex parameters for this complete parametrization, including when A is real. A parametrization using only real parameters needs the additional sign derived below.

(b) If $\mathsf{C} = \mathsf{A} + \mathsf{B}$, in general

$$\det \mathsf{C} \neq \det \mathsf{A} + \det \mathsf{B}.$$

Construct a specific numerical example to illustrate this inequality.

Solution.

In (a), complex parameters are needed to cover every matrix by the displayed formula. Restricting both parameters to real numbers does not cover every real square-zero matrix.

(a) Let $A = \begin{pmatrix} u & v \\ w & t \end{pmatrix}$. Then

$$A^2 = \begin{pmatrix} u^2 + vw & v(u+t) \\ w(u+t) & t^2 + vw \end{pmatrix}.$$

If $v = w = 0$, the equations $u^2 = t^2 = 0$ force $A = 0$, represented by $a = b = 0$. Otherwise one of v, w is nonzero, so the off-diagonal equations give $t = -u$. The remaining condition is $u^2 + vw = 0$. If $v \neq 0$, choose a complex square root b of v and set $a = u/b$. Then $ab = u$, $b^2 = v$, and $-a^2 = -u^2/v = w$. If $v = 0, w \neq 0$, the condition gives $u = t = 0$; choose $b = 0$ and $a^2 = -w$. These cases prove completeness over $\mathbb{C}$. Conversely,

$$\begin{pmatrix} ab & b^2 \\ -a^2 & -ab \end{pmatrix}^2 = \begin{pmatrix} a^2b^2 - a^2b^2 & ab^3 - ab^3 \\ -a^3b + a^3b & -a^2b^2 + a^2b^2 \end{pmatrix} = 0.$$

For real matrices a complete real parametrization is $s \begin{pmatrix} ab & b^2 \\ -a^2 & -ab \end{pmatrix}$, with $a, b \in \mathbb{R}$ and $s = \pm 1$. For $v \neq 0$, take $s = \operatorname{sgn} v$, $b = \sqrt{|v|}$, $a = u/(sb)$; for $v = 0, w \neq 0$, take $s = -\operatorname{sgn} w$, $a = \sqrt{|w|}$, $b = 0$. For example $\begin{pmatrix} 0 & -1 \\ 0 & 0 \end{pmatrix}$ would require $b^2 = -1$ in the unmodified real formula.

(b) Take $A = B = I_2$. Then $\det A + \det B = 2$ but $\det(A + B) = \det(2I_2) = 4$. Thus the determinant is not additive.

2.2.6 Given

$$\mathsf{K} = \begin{pmatrix} 0 & 0 & i \\ -i & 0 & 0 \\ 0 & -1 & 0 \end{pmatrix},$$

show that

$$\mathsf{K}^n = \mathsf{K}\mathsf{K}\mathsf{K}\cdots(n \text{ factors}) = \mathbf{1}$$

(with the proper choice of n, $n \neq 0$).

Solution.

Multiplication gives

$$K^2 = \begin{pmatrix} 0 & -i & 0 \\ 0 & 0 & 1 \\ i & 0 & 0 \end{pmatrix}, \qquad K^3 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = -I_3.$$

For example $(K^2)_{12} = K_{13}K_{32} = i(-1) = -i$; each row has only one nonzero product. Multiplying K^2 by K gives diagonal entries $(-i)(-i) = -1$, $1(-1) = -1$, and $i(i) = -1$. Therefore $\boxed{K^6 = (K^3)^2 = I_3}$. This is the smallest positive exponent: $K, K^2, K^4 = -K, K^5 = -K^2$ are off-diagonal, and $K^3 = -I_3 \neq I_3$. In general $K^n = I_3$ exactly when n is a multiple of 6.

2.2.7 Verify the Jacobi identity,

$$[\mathsf{A}, [\mathsf{B}, \mathsf{C}]] = [\mathsf{B}, [\mathsf{A}, \mathsf{C}]] - [\mathsf{C}, [\mathsf{A}, \mathsf{B}]].$$

Solution.

Expand without reordering factors:

$$\begin{aligned} [A, [B, C]] &= ABC - ACB - BCA + CBA, \\ [B, [A, C]] &= BAC - BCA - ACB + CAB, \\ [C, [A, B]] &= CAB - CBA - ABC + BAC. \end{aligned}$$

Subtracting the third line from the second cancels BAC and CAB, leaving the first line. This proves the identity for arbitrary square matrices of the same size.

2.2.8 Show that the matrices

$$\mathsf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathsf{B} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathsf{C} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

satisfy the commutation relations

$$[\mathsf{A}, \mathsf{B}] = \mathsf{C}, \quad [\mathsf{A}, \mathsf{C}] = 0, \quad \text{and } [\mathsf{B}, \mathsf{C}] = 0.$$

Solution.

Let E_{ij} have one 1 at position (i, j). Entrywise,

$$(E_{ij}E_{kl})_{rs} = \sum_t \delta_{ri}\delta_{tj}\delta_{tk}\delta_{sl} = \delta_{jk}\delta_{ri}\delta_{sl}, \quad E_{ij}E_{kl} = \delta_{jk}E_{il}.$$

Here $A = E_{12}$, $B = E_{23}$, $C = E_{13}$. Therefore

$$\begin{array}{lll} AB = E_{13} = C, & BA = 0, & [A, B] = C, \\ AC = 0, & CA = 0, & [A, C] = 0, \\ BC = 0, & CB = 0, & [B, C] = 0. \end{array}$$

For example $BA = E_{23}E_{12} = 0$ because $3 \neq 1$.

2.2.9 Let

$$\mathsf{i} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix}, \quad \mathsf{j} = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

and

$$\mathsf{k} = \begin{pmatrix} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}.$$

Show that

(a) $\mathsf{i}^2 = \mathsf{j}^2 = \mathsf{k}^2 = -\mathbf{1}$, where $\mathbf{1}$ is the unit matrix.

(b) $\mathsf{ij} = -\mathsf{ji} = \mathsf{k}$,
$\mathsf{jk} = -\mathsf{kj} = \mathsf{i}$,
$\mathsf{ki} = -\mathsf{ik} = \mathsf{j}$.

These three matrices (i, j, and k) plus the unit matrix $\mathbf{1}$ form a basis for **quaternions**. An alternate basis is provided by the four 2×2 matrices, $i\sigma_1$, $i\sigma_2$, $-i\sigma_3$, and $\mathbf{1}$, where the σ_i are the Pauli spin matrices of Example 2.2.1.

Solution.

Track the four standard basis columns. The given matrices act as

	e_1	e_2	e_3	e_4
i	$-e_2$	e_1	$-e_4$	e_3
j	e_4	e_3	$-e_2$	$-e_1$
k	e_3	$-e_4$	$-e_1$	e_2

Applying each row twice sends every e_r to $-e_r$; hence $\mathsf{i}^2 = \mathsf{j}^2 = \mathsf{k}^2 = -I_4$. For example $\mathsf{i}^2 e_1 = \mathsf{i}(-e_2) = -e_1$ and $\mathsf{j}^2 e_1 = \mathsf{j}(e_4) = -e_1$. For the cyclic products the successive images are

	e_1	e_2	e_3	e_4
ij	e_3	$-e_4$	$-e_1$	e_2
jk	$-e_2$	e_1	$-e_4$	e_3
ki	e_4	e_3	$-e_2$	$-e_1$

Thus these products equal $\mathsf{k}, \mathsf{i}, \mathsf{j}$. For the reversed products, for example, $\mathsf{ji} = (\mathsf{ki})\mathsf{i} = \mathsf{k}(-I_4) = -\mathsf{k}$. Likewise $\mathsf{kj} = (\mathsf{ij})\mathsf{j} = -\mathsf{i}$ and $\mathsf{ik} = (\mathsf{jk})\mathsf{k} = -\mathsf{j}$. The complete multiplication table is

	i	j	k
i	$-I_4$	k	$-\mathsf{j}$
j	$-\mathsf{k}$	$-I_4$	i
k	j	$-\mathsf{i}$	$-I_4$

.

For the alternative basis, put $q_1 = i\sigma_1$, $q_2 = i\sigma_2$, $q_3 = -i\sigma_3$. The Pauli products in 2.2.11 give $q_r^2 = -I_2$ and

$$q_1 q_2 = -\sigma_1\sigma_2 = -i\sigma_3 = q_3, \quad q_2 q_3 = \sigma_2\sigma_3 = i\sigma_1 = q_1, \quad q_3 q_1 = \sigma_3\sigma_1 = i\sigma_2 = q_2.$$

The reversed products change sign. If $a_0 I_2 + a_1 q_1 + a_2 q_2 + a_3 q_3 = 0$, its entries are

$$\begin{pmatrix} a_0 - ia_3 & ia_1 + a_2 \\ ia_1 - a_2 & a_0 + ia_3 \end{pmatrix} = 0.$$

Adding and subtracting the diagonal equations gives $a_0 = a_3 = 0$; the off-diagonal equations give $a_1 = a_2 = 0$. Thus these four matrices are linearly independent and realize the same quaternion algebra with real coefficients.

2.2.10 A matrix with elements $a_{ij} = 0$ for $j < i$ may be called upper right triangular. The elements in the lower left (below and to the left of the main diagonal) vanish. Show that the product of two upper right triangular matrices is an upper right triangular matrix.

Solution.

For $i > j$ the (i, j) entry is $(AB)_{ij} = \sum_k a_{ik} b_{kj}$. If $k < i$, then $a_{ik} = 0$ by upper triangularity of A. If $k \geq i$, then $k > j$ and $b_{kj} = 0$ by upper triangularity of B. Thus every summand is zero, and $(AB)_{ij} = 0$ whenever $i > j$. This is precisely upper triangularity of AB.

2.2.11 The three Pauli spin matrices are

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \text{and} \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Show that

(a) $(\sigma_i)^2 = \mathbf{1}_2$,

(b) $\sigma_i \sigma_j = i\sigma_k$, $(i, j, k) = (1, 2, 3)$ or a cyclic permutation thereof,

(c) $\sigma_i \sigma_j + \sigma_j \sigma_i = 2\delta_{ij} \mathbf{1}_2$; $\mathbf{1}_2$ is the 2×2 unit matrix.

Solution.

Direct multiplication gives

$$\sigma_1^2 = I_2, \qquad \sigma_2^2 = \begin{pmatrix} (-i)i & 0 \\ 0 & i(-i) \end{pmatrix} = I_2, \qquad \sigma_3^2 = \begin{pmatrix} 1 & 0 \\ 0 & (-1)^2 \end{pmatrix} = I_2.$$

The six products with distinct indices are

$$\begin{aligned} \sigma_1\sigma_2 &= \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = i\sigma_3, \quad \sigma_2\sigma_1 = -i\sigma_3, \\ \sigma_2\sigma_3 &= \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} = i\sigma_1, \qquad \sigma_3\sigma_2 = -i\sigma_1, \\ \sigma_3\sigma_1 &= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = i\sigma_2, \quad \sigma_1\sigma_3 = -i\sigma_2. \end{aligned}$$

Hence the anticommutator is zero for distinct indices; for equal indices it is $2\sigma_i^2 = 2I_2$. Together,

$$\boxed{\sigma_i \sigma_j + \sigma_j \sigma_i = 2\delta_{ij} I_2}, \qquad \sigma_i \sigma_j = \delta_{ij} I_2 + i \sum_k \varepsilon_{ijk} \sigma_k.$$

2.2.12 One description of spin-1 particles uses the matrices

$$\mathsf{M}_x = \frac{1}{\sqrt{2}}\begin{pmatrix}0&1&0\\1&0&1\\0&1&0\end{pmatrix}, \quad \mathsf{M}_y = \frac{1}{\sqrt{2}}\begin{pmatrix}0&-i&0\\i&0&-i\\0&i&0\end{pmatrix},$$

and

$$\mathsf{M}_z = \begin{pmatrix}1&0&0\\0&0&0\\0&0&-1\end{pmatrix}.$$

Show that

(a) $[\mathsf{M}_x, \mathsf{M}_y] = i\mathsf{M}_z$, and so on (cyclic permutation of indices). Using the Levi-Civita symbol, we may write

$$[\mathsf{M}_i, \mathsf{M}_j] = i\sum_k \varepsilon_{ijk}\mathsf{M}_k.$$

(b) $\mathsf{M}^2 \equiv \mathsf{M}_x^2 + \mathsf{M}_y^2 + \mathsf{M}_z^2 = 2\,\mathbf{1}_3$, where $\mathbf{1}_3$ is the 3×3 unit matrix.

(c) $[\mathsf{M}^2, \mathsf{M}_i] = 0$,
$[\mathsf{M}_z, \mathsf{L}^+] = \mathsf{L}^+$,
$[\mathsf{L}^+, \mathsf{L}^-] = 2\mathsf{M}_z$,
where $\mathsf{L}^+ \equiv \mathsf{M}_x + i\mathsf{M}_y$ and $\mathsf{L}^- \equiv \mathsf{M}_x - i\mathsf{M}_y$.

Solution.

(a) The two products involving x, y are

$$M_xM_y = \frac{i}{2}\begin{pmatrix}1&0&-1\\0&0&0\\1&0&-1\end{pmatrix}, \quad M_yM_x = \frac{i}{2}\begin{pmatrix}-1&0&-1\\0&0&0\\1&0&1\end{pmatrix}.$$

Their difference is $i\,\mathrm{diag}(1,0,-1) = iM_z$. For the other cyclic pairs,

$$M_yM_z = \frac{i}{\sqrt{2}}\begin{pmatrix}0&0&0\\1&0&1\\0&0&0\end{pmatrix}, \quad M_zM_y = \frac{i}{\sqrt{2}}\begin{pmatrix}0&-1&0\\0&0&0\\0&-1&0\end{pmatrix},$$
$$M_zM_x = \frac{1}{\sqrt{2}}\begin{pmatrix}0&1&0\\0&0&0\\0&-1&0\end{pmatrix}, \quad M_xM_z = \frac{1}{\sqrt{2}}\begin{pmatrix}0&0&0\\1&0&-1\\0&0&0\end{pmatrix}.$$

Subtraction gives $[M_y, M_z] = iM_x$ and $[M_z, M_x] = iM_y$. Reversed commutators change sign and $[M_i, M_i] = 0$, yielding the Levi-Civita expression for all indices.

(b) The squares are

$$M_x^2 = \frac{1}{2}\begin{pmatrix}1&0&1\\0&2&0\\1&0&1\end{pmatrix}, \quad M_y^2 = \frac{1}{2}\begin{pmatrix}1&0&-1\\0&2&0\\-1&0&1\end{pmatrix}, \quad M_z^2 = \mathrm{diag}(1,0,1).$$

Adding cancels the off-diagonal entries and gives $\boxed{M^2 = 2I_3}$.

(c) Since $M^2 = 2I_3$, $[M^2, M_i] = 2M_i - 2M_i = 0$. The raising and lowering matrices are

$$L^+ = \sqrt{2}\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad L^- = \sqrt{2}\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

For a diagonal $D = \mathrm{diag}(d_1, d_2, d_3)$, $[D, L]_{ij} = (d_i - d_j)L_{ij}$. On the nonzero entries of L^+ the difference for M_z is 1, so $[M_z, L^+] = L^+$. Finally

$$L^+L^- = \mathrm{diag}(2, 2, 0), \qquad L^-L^+ = \mathrm{diag}(0, 2, 2),$$

whose difference is $\mathrm{diag}(2, 0, -2) = 2M_z$.

2.2.13 Repeat Exercise 2.2.12, using the matrices for a spin of 3/2,

$$\mathsf{M}_x = \frac{1}{2}\begin{pmatrix} 0 & \sqrt{3} & 0 & 0 \\ \sqrt{3} & 0 & 2 & 0 \\ 0 & 2 & 0 & \sqrt{3} \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix}, \quad \mathsf{M}_y = \frac{i}{2}\begin{pmatrix} 0 & -\sqrt{3} & 0 & 0 \\ \sqrt{3} & 0 & -2 & 0 \\ 0 & 2 & 0 & -\sqrt{3} \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix},$$

and

$$\mathsf{M}_z = \frac{1}{2}\begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix}.$$

In repeating part (b), determine the new value of M^2: for these matrices the claim is $M^2 = \frac{15}{4}I_4$. The numerical spin-1 value is not retained.

Solution.

For spin 3/2, the analogue of 2.2.12(b) is $M^2 = \frac{15}{4}I_4$, not $2I_3$ (nor $2I_4$). The commutator identities are unchanged, with 4×4 matrices. Form the sums and differences:

$$L^+ = M_x + iM_y = \begin{pmatrix} 0 & \sqrt{3} & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad L^- = (L^+)^T.$$

The diagonal entries of M_z are $3/2, 1/2, -1/2, -3/2$. Each nonzero entry of L^+ connects adjacent entries whose difference is 1. Thus $[M_z, L]_{ij} = ((M_z)_{ii} - (M_z)_{jj})L_{ij}$ gives

$$[M_z, L^+] = L^+, \qquad [M_z, L^-] = -L^-.$$

Sparse multiplication gives

$$L^+L^- = \mathrm{diag}(3, 4, 3, 0), \qquad L^-L^+ = \mathrm{diag}(0, 3, 4, 3),$$

so $[L^+, L^-] = \mathrm{diag}(3, 1, -1, -3) = 2M_z$. Now $M_x = (L^+ + L^-)/2$ and $M_y = (L^+ - L^-)/(2i)$, giving

$$\begin{aligned} [M_x, M_y] &= \frac{1}{4i}\big(-[L^+, L^-] + [L^-, L^+]\big) = -\frac{4M_z}{4i} = iM_z, \\ [M_z, M_x] &= \tfrac{1}{2}(L^+ - L^-) = iM_y, \\ [M_y, M_z] &= \tfrac{1}{2i}(-L^+ - L^-) = iM_x. \end{aligned}$$

Expand the squares:

$$\begin{aligned} M_x^2 + M_y^2 &= \tfrac{1}{4}\big((L^+)^2 + L^+L^- + L^-L^+ + (L^-)^2\big) \\ &\quad - \tfrac{1}{4}\big((L^+)^2 - L^+L^- - L^-L^+ + (L^-)^2\big) \\ &= \tfrac{1}{2}(L^+L^- + L^-L^+) = \mathrm{diag}(3/2, 7/2, 7/2, 3/2). \end{aligned}$$

Adding $M_z^2 = \mathrm{diag}(9/4, 1/4, 1/4, 9/4)$ gives $\boxed{M^2 = \frac{15}{4} I_4}$. This commutes with every M_i; the other identities in part (c) were established directly above.

2.2.14 If A is a diagonal matrix, with all diagonal elements different, and A and B commute, show that B is diagonal.

Solution.

Write $A = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$. Then $(AB)_{ij} = \lambda_i b_{ij}$ and $(BA)_{ij} = b_{ij}\lambda_j$, so $AB = BA$ implies

$$(\lambda_i - \lambda_j) b_{ij} = 0.$$

For $i \neq j$ the hypothesis gives $\lambda_i - \lambda_j \neq 0$, so $b_{ij} = 0$. Diagonal entries b_{ii} are unrestricted. Therefore B is diagonal.

2.2.15 If A and B are diagonal, show that A and B commute.

Solution.

For $A = \mathrm{diag}(a_1, \ldots, a_n)$ and $B = \mathrm{diag}(b_1, \ldots, b_n)$,

$$(AB)_{ij} = \sum_k a_i \delta_{ik} b_k \delta_{kj} = a_i b_i \delta_{ij}.$$

Similarly $(BA)_{ij} = b_i a_i \delta_{ij}$. Scalar multiplication commutes, so the entries agree and $AB = BA$.

2.2.16 Show that $\mathrm{trace}(\mathsf{ABC}) = \mathrm{trace}(\mathsf{CBA})$ if any two of the three matrices commute.

Solution.

For square matrices of the same size,

$$\mathrm{tr}(XY) = \sum_{i,j} x_{ij} y_{ji} = \sum_{j,i} y_{ji} x_{ij} = \mathrm{tr}(YX).$$

Taking $X = AB, Y = C$, and then $X = A, Y = BC$, gives the cyclic rule $\mathrm{tr}(ABC) = \mathrm{tr}(BCA) = \mathrm{tr}(CAB)$. There are three possible commuting pairs:

$$\begin{aligned} AB = BA: &\quad \mathrm{tr}(ABC) = \mathrm{tr}(BAC) = \mathrm{tr}(CBA), \\ BC = CB: &\quad \mathrm{tr}(ABC) = \mathrm{tr}(ACB) = \mathrm{tr}(CBA), \\ CA = AC: &\quad \mathrm{tr}(ABC) = \mathrm{tr}(BCA) = \mathrm{tr}(BAC) = \mathrm{tr}(CBA). \end{aligned}$$

Every possible hypothesis therefore gives the required equality.

2.2.17 Angular momentum matrices satisfy a commutation relation

$$[\mathsf{M}_j, \mathsf{M}_k] = i\mathsf{M}_l, \quad j,k,l \text{ cyclic}.$$

Show that the trace of each angular momentum matrix vanishes. Use finite matrices, so all products and traces below exist.

Solution.

For finite square matrices, cyclicity gives $\operatorname{tr}[A,B] = \operatorname{tr}(AB) - \operatorname{tr}(BA) = 0$. Apply this to each cyclic angular momentum relation:

$$\begin{aligned} i \operatorname{tr} M_z &= \operatorname{tr}[M_x, M_y] = 0, \\ i \operatorname{tr} M_x &= \operatorname{tr}[M_y, M_z] = 0, \\ i \operatorname{tr} M_y &= \operatorname{tr}[M_z, M_x] = 0. \end{aligned}$$

Division by i proves that every trace vanishes. This uses ordinary finite-dimensional traces, not undefined traces of infinite-dimensional operators.

2.2.18 A and B anticommute: $\mathsf{AB} = -\mathsf{BA}$. Also, $\mathsf{A}^2 = \mathbf{1}$, $\mathsf{B}^2 = \mathbf{1}$. Show that $\operatorname{trace}(\mathsf{A}) = \operatorname{trace}(\mathsf{B}) = 0$.

Note. The Pauli and Dirac matrices are specific examples.

Solution.

From $AB = -BA$ and $B^2 = I$, we obtain $BAB = -AB^2 = -A$. Taking traces and cycling the first B to the end,

$$-\operatorname{tr} A = \operatorname{tr}(BAB) = \operatorname{tr}(ABB) = \operatorname{tr} A.$$

Hence $2 \operatorname{tr} A = 0$. Similarly $ABA = -BA^2 = -B$ and $\operatorname{tr}(ABA) = \operatorname{tr}(BAA) = \operatorname{tr} B$. Thus $\boxed{\operatorname{tr} A = \operatorname{tr} B = 0}$.

2.2.19

(a) If two nonsingular matrices anticommute, show that the trace of each one is zero. (Nonsingular means that the determinant of the matrix is nonzero.)

(b) For the conditions of part (a) to hold, A and B must be $n \times n$ matrices with n **even**. Show that if n is **odd**, a contradiction results.

Solution.

The manual's left multiplication by B^{-1} proves the vanishing of $\operatorname{tr} A$, not of $\operatorname{tr} B$ as printed. Interchanging the roles of A and B supplies the second trace. **(a)** Invertibility of B and $AB = -BA$ give $BAB^{-1} = -A$. Taking the trace,

$$-\operatorname{tr} A = \operatorname{tr}(BAB^{-1}) = \operatorname{tr}(AB^{-1}B) = \operatorname{tr} A,$$

so $\operatorname{tr} A = 0$. Likewise invertibility of A gives $ABA^{-1} = -B$, and $-\operatorname{tr} B = \operatorname{tr}(ABA^{-1}) = \operatorname{tr}(BA^{-1}A) = \operatorname{tr} B$. Hence $\operatorname{tr} B = 0$.

(b) Taking determinants gives

$$\det A \det B = \det(AB) = \det(-BA) = (-1)^n \det B \det A.$$

Both determinants are nonzero; division by their product yields $1 = (-1)^n$. Odd n would give $1 = -1$, a contradiction over the real or complex numbers. Therefore $\boxed{n \text{ is even}}$.

2.2.20 If A^{-1} has elements

$$(\mathsf{A}^{-1})_{ij} = a_{ij}^{(-1)} = \frac{C_{ji}}{|\mathsf{A}|},$$

where C_{ji} is the jith cofactor of $|\mathsf{A}|$, show that

$$\mathsf{A}^{-1}\mathsf{A} = \mathbf{1}.$$

Hence A^{-1} is the inverse of A (if $|\mathsf{A}| \neq 0$).

Solution.

Let $B_{ij} = C_{ji}/\det A$, with $\det A \neq 0$. Then

$$(BA)_{ij} = \sum_k \frac{C_{ki}}{\det A} a_{kj} = \frac{1}{\det A} \sum_k a_{kj} C_{ki}.$$

For $i = j$, the numerator is expansion down column i and equals $\det A$. For $i \neq j$, it is the determinant with column i replaced by column j, so it is zero. Thus $(BA)_{ij} = \delta_{ij}$ and $BA = I$. The corresponding row-cofactor identity gives

$$(AB)_{ij} = \frac{1}{\det A} \sum_k a_{ik} C_{jk} = \delta_{ij}.$$

The formula therefore produces the two-sided inverse $B = A^{-1}$.

2.2.21 Find the matrices M_L such that the product $\mathsf{M}_L\mathsf{A}$ will be A but with:

(a) The ith row multiplied by a constant k ($a_{ij} \to ka_{ij}$, $j = 1, 2, 3, \ldots$);

(b) The ith row replaced by the original ith row minus a multiple of the mth row ($a_{ij} \to a_{ij} - Ka_{mj}$, $j = 1, 2, 3, \ldots$);

(c) The ith and mth rows interchanged ($a_{ij} \to a_{mj}$, $a_{mj} \to a_{ij}$, $j = 1, 2, 3, \ldots$).

Solution.

In (b), the varying index in the parentheses must be j, not i: i labels the one row being changed. For (c), the manual's $M_{mi} - M_{im} = 1$ must read $M_{mi} = M_{im} = 1$ when $i \neq m$. Both inserted entries are positive. Let E_{ab} have a single 1 in position (a, b). Its action is

$$(E_{ab}A)_{rj} = \sum_s \delta_{ra}\delta_{sb}a_{sj} = \delta_{ra}a_{bj}.$$

Thus $E_{ab}A$ has row b of A in row a and all other rows zero. For distinct i, m, the requested matrices are

(a)	$M_L = I + (k-1)E_{ii}$,
(b)	$M_L = I - KE_{im}$,
(c)	$M_L = I - E_{ii} - E_{mm} + E_{im} + E_{mi}$.

In (a), row i becomes $R_i + (k-1)R_i = kR_i$. In (b), it becomes $R_i - KR_m$. In (c), the two subtractions remove rows i, m; the last two terms insert R_m, R_i in their places. Every other row retains only the identity contribution. If $i = m$, (b) reduces to scaling by $1 - K$, and the formula in (c) reduces to I.

2.2.22 Find the matrices M_R such that the product AM_R will be A but with:

(a) The ith column multiplied by a constant k ($a_{ji} \to ka_{ji}$, $j = 1, 2, 3, \ldots$);

(b) The ith column replaced by the original ith column minus a multiple of the mth column ($a_{ji} \to a_{ji} - ka_{jm}$, $j = 1, 2, 3, \ldots$);

(c) The ith and mth columns interchanged ($a_{ji} \to a_{jm}$, $a_{jm} \to a_{ji}$, $j = 1, 2, 3, \ldots$).

Solution.

For right multiplication,

$$(AE_{ab})_{rj} = \sum_s a_{rs}\delta_{sa}\delta_{jb} = a_{ra}\delta_{jb}.$$

Thus AE_{ab} puts column a of A in column b, giving

$$\boxed{\begin{array}{ll} \text{(a)} & M_R = I + (k-1)E_{ii}, \\ \text{(b)} & M_R = I - kE_{mi}, \\ \text{(c)} & M_R = I - E_{ii} - E_{mm} + E_{im} + E_{mi}. \end{array}}$$

Column i in (a) is $C_i + (k-1)C_i = kC_i$; in (b), it is $C_i - kC_m$; in (c), the two columns are removed and reinserted in exchanged positions. The off-diagonal entry in (b) is (m, i), the transpose of the row-operation entry (i, m); the manual's statement that the answers are the same requires this transpose. For rectangular A, M_L has order equal to the number of rows of A, whereas M_R has order equal to the number of columns.

2.2.23 Find the inverse of

$$\mathsf{A} = \begin{pmatrix} 3 & 2 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 4 \end{pmatrix}.$$

Solution.

Expansion along row 1 gives

$$\det A = 3(8-1) - 2(8-1) + (2-2) = 21 - 14 = 7.$$

The signed minors are

$$\begin{array}{lll} C_{11} = 8 - 1 = 7, & C_{12} = -(8-1) = -7, & C_{13} = 2 - 2 = 0, \\ C_{21} = -(8-1) = -7, & C_{22} = 12 - 1 = 11, & C_{23} = -(3-2) = -1, \\ C_{31} = 2 - 2 = 0, & C_{32} = -(3-2) = -1, & C_{33} = 6 - 4 = 2. \end{array}$$

The cofactor matrix is symmetric, so its transpose is unchanged:

$$\boxed{A^{-1} = \frac{1}{7}\begin{pmatrix} 7 & -7 & 0 \\ -7 & 11 & -1 \\ 0 & -1 & 2 \end{pmatrix}}.$$

An independent multiplication checks every entry:

$$A\begin{pmatrix} 7 & -7 & 0 \\ -7 & 11 & -1 \\ 0 & -1 & 2 \end{pmatrix} = \begin{pmatrix} 21-14 & -21+22-1 & -2+2 \\ 14-14 & -14+22-1 & -2+2 \\ 7-7 & -7+11-4 & -1+8 \end{pmatrix} = 7I_3.$$

2.2.24 Matrices are far too useful to remain the exclusive property of physicists. They may appear wherever there are linear relations. For instance, in a study of population movement the initial fraction of a fixed population in each of n areas (or industries or religions, etc.) is represented by an n-component column vector P. The movement of people from one area to another in a given time is described by an $n \times n$ (stochastic) matrix T. Here T_{ij} is the fraction of the population in the jth area that moves to the ith area. (Those not moving are covered by $i = j$.) With P describing the initial population distribution, the final population distribution is given by the matrix equation $\mathsf{TP} = \mathsf{Q}$. From its definition, $\sum_{i=1}^{n} P_i = 1$.

(a) Show that conservation of people requires that

$$\sum_{i=1}^{n} T_{ij} = 1, \quad j = 1, 2, \ldots, n.$$

(b) Prove that

$$\sum_{i=1}^{n} Q_i = 1$$

continues the conservation of people.

Solution.

(a) A source area j containing N_j people contributes $T_{1j}N_j, \ldots, T_{nj}N_j$ to the destination areas. Conservation requires the sum to be N_j; for $N_j > 0$, division gives $\sum_i T_{ij} = 1$. This requirement holds for every source area, even one initially empty, because conservation must hold for every possible initial distribution. Equivalently, start with $P = e_j$; the final total $\sum_i T_{ij}$ must be 1.

(b) Since $Q_i = \sum_j T_{ij}P_j$,

$$\sum_i Q_i = \sum_i \sum_j T_{ij}P_j = \sum_j P_j \left(\sum_i T_{ij} \right) = \sum_j P_j = 1.$$

Nonnegative fractions and $P_j \geq 0$ also imply $Q_i \geq 0$, so Q is again a valid population distribution.

2.2.25 Given a 6×6 matrix A with elements $a_{ij} = 0.5^{|i-j|}$, $i, j = 0, 1, 2, \ldots, 5$, find A^{-1}.

$$ANS.\ \mathsf{A}^{-1} = \frac{1}{3} \begin{pmatrix} 4 & -2 & 0 & 0 & 0 & 0 \\ -2 & 5 & -2 & 0 & 0 & 0 \\ 0 & -2 & 5 & -2 & 0 & 0 \\ 0 & 0 & -2 & 5 & -2 & 0 \\ 0 & 0 & 0 & -2 & 5 & -2 \\ 0 & 0 & 0 & 0 & -2 & 4 \end{pmatrix}.$$

Solution.

First retain a parameter r in $A_{ij} = r^{|i-j|}$ for $i, j = 0, \ldots, 5$. Subtract r times original row $i-1$ from row i for each $i = 1, \ldots, 5$. For $j < i$, the new entry is $r^{i-j} - r\, r^{i-1-j} = 0$; for $j \geq i$, it is $r^{j-i} - r\, r^{j-i+1} = (1 - r^2)r^{j-i}$. This suggests that a second difference can isolate a diagonal entry.

Define tridiagonal T with end diagonal entries 1, interior diagonal entries $1 + r^2$, and adjacent entries $-r$. For an interior column $j = 1, \ldots, 4$,

$$(AT)_{ij} = -rA_{i,j-1} + (1 + r^2)A_{ij} - rA_{i,j+1}.$$

If $i<j$, put $d=j-i\geq 1$ to obtain $-rr^{d-1}+(1+r^2)r^d-rr^{d+1}=0$. If $i>j$, put $d=i-j\geq 1$ to obtain $-rr^{d+1}+(1+r^2)r^d-rr^{d-1}=0$. For $i=j$, the result is $-r^2+(1+r^2)-r^2=1-r^2$. At the first column,

$$(AT)_{i0}=A_{i0}-rA_{i1}=\begin{cases}1-r^2, & i=0,\\ r^i-rr^{i-1}=0, & i>0.\end{cases}$$

Reversing the indices proves the same statement for the last column. Thus $AT=(1-r^2)I_6$. At $r=1/2$,

$$\boxed{A^{-1}=\frac{T}{1-r^2}=\frac{1}{3}\begin{pmatrix}4&-2&0&0&0&0\\-2&5&-2&0&0&0\\0&-2&5&-2&0&0\\0&0&-2&5&-2&0\\0&0&0&-2&5&-2\\0&0&0&0&-2&4\end{pmatrix}}.$$

This product calculation both derives and verifies the inverse.

2.2.26 Show that the product of two orthogonal matrices is orthogonal.

Solution.

Let $A^TA=I$ and $B^TB=I$. Since transposition reverses products,

$$(AB)^T(AB)=B^TA^TAB=B^T(A^TA)B=B^TIB=B^TB=I.$$

This is exactly the orthogonality condition for AB.

2.2.27 If A is orthogonal, show that its determinant $=\pm 1$.

Solution.

Taking determinants of $A^TA=I$ and using $\det(A^T)=\det A$ gives

$$1=\det(A^TA)=\det(A^T)\det A=(\det A)^2.$$

The two roots of $d^2=1$ are $1,-1$, so $\boxed{\det A=\pm 1}$.

2.2.28 Show that the trace of the product of a symmetric and an antisymmetric matrix is zero.

Solution.

Let $S^T=S$ and $K^T=-K$. Transposition preserves the trace:

$$\operatorname{tr}(SK)=\operatorname{tr}((SK)^T)=\operatorname{tr}(K^TS^T)=-\operatorname{tr}(KS)=-\operatorname{tr}(SK).$$

Thus $2\operatorname{tr}(SK)=0$ and $\boxed{\operatorname{tr}(SK)=0}$. Entrywise, each pair with $i<j$ cancels: $s_{ij}k_{ji}+s_{ji}k_{ij}=s_{ij}(-k_{ij})+s_{ij}k_{ij}=0$, while all diagonal terms vanish because $k_{ii}=0$.

2.2.29 A is 2×2 and orthogonal. Find the most general form of

$$\mathsf{A}=\begin{pmatrix}a&b\\c&d\end{pmatrix}.$$

Solution.

For real orthogonal A, its first row is a real unit vector, so write $(a, b) = (\cos\theta, \sin\theta)$. The second row is a unit vector perpendicular to the first. That one-dimensional subspace is spanned by the unit vector $(-\sin\theta, \cos\theta)$. Hence

$$\boxed{A = \begin{pmatrix} \cos\theta & \sin\theta \\ -s\sin\theta & s\cos\theta \end{pmatrix}, \quad s = \pm 1, \quad \theta \in \mathbb{R}}.$$

The two row norms are 1, their inner product is 0, and $\det A = s(\cos^2\theta + \sin^2\theta) = s$. Both signs are required. The manual assumes $\det A = 1$ and omits the $\det A = -1$ reflection family. If orthogonality is extended to complex A with $A^T A = I$, the algebraic form is $A = \begin{pmatrix} a & b \\ -sb & sa \end{pmatrix}$ with $a^2 + b^2 = 1$, $a, b \in \mathbb{C}$, and $s = \pm 1$.

2.2.30 Show that

$$\det(\mathsf{A}^*) = (\det \mathsf{A})^* = \det(\mathsf{A}^\dagger).$$

Solution.

The permutation formula is

$$\det A = \sum_{\pi \in S_n} \operatorname{sgn}(\pi) \prod_{r=1}^{n} a_{r,\pi(r)}.$$

Because each permutation sign is real, conjugating the finite sum gives

$$(\det A)^* = \sum_{\pi} \operatorname{sgn}(\pi) \prod_{r} a^*_{r,\pi(r)} = \det(A^*).$$

Finally $A^\dagger = (A^*)^T$, and determinants are invariant under transposition. Thus $\det(A^\dagger) = \det(A^*) = (\det A)^*$.

2.2.31 Three angular momentum matrices satisfy the basic commutation relation

$$[\mathsf{J}_x, \mathsf{J}_y] = i\mathsf{J}_z$$

(and cyclic permutation of indices). If two of the matrices have real elements, show that the elements of the third must be pure imaginary.

Solution.

Suppose the real matrices are J_x, J_y. Each entry

$$R_{ab} := ([J_x, J_y])_{ab} = \sum_c \big((J_x)_{ac}(J_y)_{cb} - (J_y)_{ac}(J_x)_{cb}\big)$$

is real. The commutation relation gives $(J_z)_{ab} = R_{ab}/i = -iR_{ab}$, which is pure imaginary (possibly zero). For either other real pair, use the corresponding cyclic commutation relation in the same calculation.

2.2.32 Show that $(\mathsf{AB})^\dagger = \mathsf{B}^\dagger\mathsf{A}^\dagger$.

Solution.

For conformable matrices,

$$((AB)^\dagger)_{ij} = (AB)^*_{ji} = \left(\sum_k a_{jk}b_{ki}\right)^* = \sum_k a^*_{jk}b^*_{ki}$$
$$= \sum_k (B^\dagger)_{ik}(A^\dagger)_{kj} = (B^\dagger A^\dagger)_{ij}.$$

The scalar entries commute; no commutation of the matrices was assumed. Equality of all entries proves the reversed product rule.

2.2.33 A matrix $\mathsf{C} = \mathsf{S}^\dagger\mathsf{S}$. Show that the trace is positive definite unless S is the null matrix, in which case trace $(\mathsf{C}) = 0$.

Solution.

If S has entries s_{ki}, the diagonal entries of $C = S^\dagger S$ are

$$C_{ii} = \sum_k (S^\dagger)_{ik}S_{ki} = \sum_k s^*_{ki}s_{ki} = \sum_k |s_{ki}|^2.$$

Therefore

$$\boxed{\mathrm{tr}(S^\dagger S) = \sum_{i,k} |s_{ki}|^2}.$$

Every summand is nonnegative. The sum is zero exactly when every entry is zero; otherwise it is strictly positive. The trace is thus a positive-definite quadratic form in the entries of S. This does not imply that C itself is positive definite: $S = \mathrm{diag}(1, 0)$ gives $C = \mathrm{diag}(1, 0)$, which has a zero eigenvalue.

2.2.34 If A and B are Hermitian matrices, show that $(\mathsf{AB}+\mathsf{BA})$ and $i(\mathsf{AB}-\mathsf{BA})$ are also Hermitian.

Solution.

Using $A^\dagger = A$, $B^\dagger = B$, and reversing the product order,

$$(AB + BA)^\dagger = B^\dagger A^\dagger + A^\dagger B^\dagger = BA + AB = AB + BA.$$

For the second expression also conjugate the scalar i:

$$\big(i(AB - BA)\big)^\dagger = -i(B^\dagger A^\dagger - A^\dagger B^\dagger) = -i(BA - AB) = i(AB - BA).$$

Both expressions equal their adjoints and are therefore Hermitian.

2.2.35 The matrix C is not Hermitian. Show that then $\mathsf{C} + \mathsf{C}^\dagger$ and $i(\mathsf{C} - \mathsf{C}^\dagger)$ are Hermitian. This means that a non-Hermitian matrix may be resolved into two Hermitian parts,

$$\mathsf{C} = \frac{1}{2}(\mathsf{C} + \mathsf{C}^\dagger) + \frac{1}{2i}i(\mathsf{C} - \mathsf{C}^\dagger).$$

This decomposition of a matrix into two Hermitian matrix parts parallels the decomposition of a complex number z into $x + iy$, where $x = (z + z^*)/2$ and $y = (z - z^*)/2i$.

Solution.

The transcription omitted the extra factor i in the second term of the displayed decomposition on textbook p. 119. The corrected expression above equals $H + iK$, with H, K Hermitian; the whole second summand iK need not be Hermitian. For any square C, whether or not it is Hermitian, set

$$H = \frac{C + C^\dagger}{2}, \qquad K = \frac{C - C^\dagger}{2i}.$$

Taking adjoints,

$$H^\dagger = \frac{C^\dagger + C}{2} = H, \qquad K^\dagger = \frac{C^\dagger - C}{-2i} = \frac{C - C^\dagger}{2i} = K.$$

Also $C + C^\dagger = 2H$ and $i(C - C^\dagger) = -2K$, so both matrices requested in the first sentence are Hermitian. Recombining,

$$H + iK = \frac{C + C^\dagger}{2} + \frac{C - C^\dagger}{2} = C.$$

This is the matrix analogue of $z = \operatorname{Re} z + i \operatorname{Im} z$. It is unique: if $C = H_1 + iK_1$ with Hermitian H_1, K_1, then $C^\dagger = H_1 - iK_1$; addition and subtraction recover the displayed formulas for H, K.

2.2.36 A and B are two noncommuting Hermitian matrices:

$$\mathsf{AB} - \mathsf{BA} = i\mathsf{C}.$$

Prove that C is Hermitian.

Solution.

Taking adjoints of $AB - BA = iC$ gives

$$(AB - BA)^\dagger = B^\dagger A^\dagger - A^\dagger B^\dagger = BA - AB = -iC,$$

whereas $(iC)^\dagger = -iC^\dagger$. Therefore $-iC^\dagger = -iC$, and division by $-i$ proves $\boxed{C^\dagger = C}$.

2.2.37 Two matrices A and B are each Hermitian. Find a necessary and sufficient condition for their product AB to be Hermitian.

ANS. $[\mathsf{A}, \mathsf{B}] = 0$.

Solution.

Hermiticity of A, B gives $(AB)^\dagger = B^\dagger A^\dagger = BA$. Thus AB is Hermitian precisely when $AB = (AB)^\dagger = BA$, or $\boxed{[A, B] = 0}$. This proves necessity and sufficiency.

2.2.38 Show that the reciprocal (that is, inverse) of a unitary matrix is unitary.

Solution.

Unitary U satisfies $U^{-1} = U^\dagger$. Put $V = U^{-1}$. Then $V^\dagger = (U^\dagger)^\dagger = U$, so

$$V^\dagger V = UU^{-1} = I, \qquad VV^\dagger = U^{-1}U = I.$$

Hence V is unitary.

2.2.39 Prove that the direct product of two unitary matrices is unitary.

Solution.

"Direct product" here means the tensor product defined in the text. The manual instead displays the proof for the ordinary product UV; the double-index proof below supplies the requested result. Let unitary U, V have orders m, n. Their direct (Kronecker) product has entries $W_{(i,a),(j,b)} = U_{ij}V_{ab}$. Thus

$$\begin{aligned}(W^\dagger W)_{(j,b),(k,c)} &= \sum_{i=1}^{m}\sum_{a=1}^{n} U_{ij}^* V_{ab}^* U_{ik} V_{ac} \\ &= \left(\sum_i U_{ij}^* U_{ik}\right)\left(\sum_a V_{ab}^* V_{ac}\right) = \delta_{jk}\delta_{bc}.\end{aligned}$$

Therefore $W^\dagger W = I_{mn}$. Equivalently,

$$(U \otimes V)^\dagger (U \otimes V) = (U^\dagger U) \otimes (V^\dagger V) = I_m \otimes I_n = I_{mn}.$$

The reverse product is the identity by the same calculation, proving unitarity of the direct product.

2.2.40 If $\boldsymbol{\sigma}$ is the vector with the σ_i as components given in Eq. (2.61), and $\mathbf{p}$ is an ordinary vector, show that

$$(\boldsymbol{\sigma}\cdot\mathbf{p})^2 = \mathbf{p}^2 \mathbf{1}_2,$$

where $\mathbf{1}_2$ is a 2×2 unit matrix.

Solution.

In the manual's expansion, the term multiplying $p_y p_z$ must contain $\sigma_2\sigma_3 + \sigma_3\sigma_2$, not the repeated $\sigma_1\sigma_2 + \sigma_2\sigma_1$. Both vanish, so the printed final result is unaffected. The scalar components p_i commute. Group the cross terms:

$$\begin{aligned}(\boldsymbol{\sigma}\cdot\mathbf{p})^2 = {} & p_1^2\sigma_1^2 + p_2^2\sigma_2^2 + p_3^2\sigma_3^2 \\ & + p_1p_2(\sigma_1\sigma_2 + \sigma_2\sigma_1) + p_1p_3(\sigma_1\sigma_3 + \sigma_3\sigma_1) \\ & + p_2p_3(\sigma_2\sigma_3 + \sigma_3\sigma_2).\end{aligned}$$

Each square is I_2 and each mixed anticommutator is zero:

$$\boxed{(\boldsymbol{\sigma}\cdot\mathbf{p})^2 = (p_1^2 + p_2^2 + p_3^2) I_2}.$$

The same reasoning applies to the mutually commuting free-particle momentum operators $-i\hbar\partial_i$ on functions for which the mixed derivatives commute.

2.2.41 Use the equations for the properties of direct products, Eqs. (2.57) and (2.58), to show that the four matrices $\boldsymbol{\gamma}^\mu$, $\mu = 0, 1, 2, 3$, satisfy the conditions listed in Eqs. (2.74) and (2.75).

Solution.

Explicitly, the referenced definitions are

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = i\sigma_2, \qquad \gamma^0 = \sigma_3 \otimes I_2, \qquad \gamma^i = J \otimes \sigma_i \quad (i = 1, 2, 3).$$

The direct-product rule is $(A \otimes B)(C \otimes D) = AC \otimes BD$. Since $J^2 = -I_2$ and $\sigma_i^2 = I_2$,

$$(\gamma^0)^2 = I_2 \otimes I_2 = I_4, \qquad (\gamma^i)^2 = -I_2 \otimes I_2 = -I_4.$$

Also $\sigma_3 J = \sigma_1$ and $J\sigma_3 = -\sigma_1$, giving

$$\gamma^0\gamma^i + \gamma^i\gamma^0 = (\sigma_3 J + J\sigma_3) \otimes \sigma_i = 0.$$

For spatial indices,

$$\gamma^i\gamma^j + \gamma^j\gamma^i = J^2 \otimes (\sigma_i\sigma_j + \sigma_j\sigma_i) = -2\delta_{ij} I_4.$$

These are the conditions (2.74)–(2.75), compactly

$$\boxed{\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} I_4, \quad g = \mathrm{diag}(1, -1, -1, -1)}.$$

2.2.42 Show that $\boldsymbol{\gamma}^5$, Eq. (2.76), anticommutes with all four $\boldsymbol{\gamma}^\mu$.

Solution.

By definition, $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$. From 2.2.41 and $\sigma_1\sigma_2\sigma_3 = iI_2$,

$$\gamma^1\gamma^2\gamma^3 = J^3 \otimes (\sigma_1\sigma_2\sigma_3) = (-J) \otimes (iI_2) = -iJ \otimes I_2.$$

Hence

$$\gamma^5 = (\sigma_3 J) \otimes I_2 = \sigma_1 \otimes I_2 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}.$$

Since $\sigma_1\sigma_3 = -\sigma_3\sigma_1$ and $\sigma_1 J = -J\sigma_1$,

$$\{\gamma^5, \gamma^0\} = (\sigma_1\sigma_3 + \sigma_3\sigma_1) \otimes I_2 = 0,$$

$$\{\gamma^5, \gamma^i\} = (\sigma_1 J + J\sigma_1) \otimes \sigma_i = 0 \quad (i = 1, 2, 3).$$

Thus γ^5 anticommutes with all four γ^μ.

2.2.43 In this problem, the summations are over $\mu = 0, 1, 2, 3$. Define $g_{\mu\nu} = g^{\mu\nu}$ by the relations

$$g_{00} = 1; \quad g_{kk} = -1, \quad k = 1, 2, 3; \quad g_{\mu\nu} = 0, \quad \mu \neq \nu;$$

and define γ_μ as $\sum_{\nu=0}^{3} g_{\nu\mu}\gamma^\nu$. Using these definitions, show that

(a) $\sum \gamma_\mu \gamma^\alpha \gamma^\mu = -2\gamma^\alpha$,

(b) $\sum \gamma_\mu \gamma^\alpha \gamma^\beta \gamma^\mu = 4g^{\alpha\beta}\mathbf{1}_4$,

(c) $\sum \gamma_\mu \gamma^\alpha \gamma^\beta \gamma^\nu \gamma^\mu = -2\gamma^\nu\gamma^\beta\gamma^\alpha$.

Solution.

The lowering definition uses the dummy index ν while μ is free. The textbook prints γ^μ inside an unindexed sum; the explicit contraction above removes that ambiguity. For this diagonal metric it also equals $g_{\mu\mu}\gamma^\mu$ with no sum on μ. In (b), a scalar denotes that scalar times I_4, made explicit in the statement. The sums in (a)–(c) run over $\mu = 0, 1, 2, 3$. Lowering an index gives $\gamma_0 = \gamma^0$, $\gamma_i = -\gamma^i$. The Clifford relation implies

$$\{\gamma_\mu, \gamma^\alpha\} = 2\delta_\mu^\alpha I_4, \qquad \sum_\mu \gamma_\mu \gamma^\mu = 4I_4.$$

For the second identity, the time term is I_4 and each spatial term is $-(\gamma^i)^2 = I_4$.

(a) Move γ_μ past the first factor:

$$\sum_\mu \gamma_\mu \gamma^\alpha \gamma^\mu = \sum_\mu (2\delta^\alpha_\mu I_4 - \gamma^\alpha \gamma_\mu)\gamma^\mu$$
$$= 2\gamma^\alpha - \gamma^\alpha (4I_4) = \boxed{-2\gamma^\alpha}.$$

(b) Use the same step and then (a):

$$\sum_\mu \gamma_\mu \gamma^\alpha \gamma^\beta \gamma^\mu = 2\gamma^\beta \gamma^\alpha - \gamma^\alpha \sum_\mu \gamma_\mu \gamma^\beta \gamma^\mu$$
$$= 2\gamma^\beta \gamma^\alpha + 2\gamma^\alpha \gamma^\beta = \boxed{4g^{\alpha\beta} I_4}.$$

(c) Use (b) on the contracted part:

$$\sum_\mu \gamma_\mu \gamma^\alpha \gamma^\beta \gamma^\nu \gamma^\mu = 2\gamma^\beta \gamma^\nu \gamma^\alpha - \gamma^\alpha \sum_\mu \gamma_\mu \gamma^\beta \gamma^\nu \gamma^\mu$$
$$= 2\gamma^\beta \gamma^\nu \gamma^\alpha - 4g^{\beta\nu}\gamma^\alpha$$
$$= 2(\gamma^\beta \gamma^\nu - 2g^{\beta\nu} I_4)\gamma^\alpha = \boxed{-2\gamma^\nu \gamma^\beta \gamma^\alpha}.$$

These derivations include coincident free indices; no distinctness assumption was made.

2.2.44 If $\mathsf{M} = \frac{1}{2}(1 + \boldsymbol{\gamma}^5)$, where $\boldsymbol{\gamma}^5$ is given in Eq. (2.76), show that

$$\mathsf{M}^2 = \mathsf{M}.$$

This equation remains valid for a replacement matrix Γ precisely when $\Gamma^2 = \mathbf{1}_4$; it does not hold for every Dirac matrix listed in Eq. (2.76).

Solution.

Since $\gamma^5 = \sigma_1 \otimes I_2$, $(\gamma^5)^2 = I_4$. Therefore

$$M^2 = \tfrac{1}{4}(I_4 + 2\gamma^5 + (\gamma^5)^2) = \tfrac{1}{4}(2I_4 + 2\gamma^5) = \tfrac{1}{2}(I_4 + \gamma^5) = M.$$

The printed note is too broad: the conclusion does not hold for every matrix of Eq. (2.76). For $M_\Gamma = (I_4 + \Gamma)/2$, the necessary and sufficient condition is $\Gamma^2 = I_4$. Indeed,

$$M_\Gamma^2 - M_\Gamma = \frac{\Gamma^2 - I_4}{4}.$$

The squares of the full basis are

Γ	Γ^2
$I_4,\ \gamma^5,\ \gamma^0,\ \gamma^5\gamma^i,\ \sigma^{ij}\ (1 \le i < j \le 3)$	$+I_4$
$\gamma^i,\ \gamma^5\gamma^0,\ \sigma^{0i}\ (i = 1, 2, 3)$	$-I_4$

where $\sigma^{\mu\nu} = i\gamma^\mu\gamma^\nu$ for $\mu < \nu$. For example $(\gamma^5\gamma^\mu)^2 = -(\gamma^\mu)^2$, and $(i\gamma^\mu\gamma^\nu)^2 = (\gamma^\mu)^2(\gamma^\nu)^2$. For $\Gamma^2 = -I_4$, the error is $M_\Gamma^2 - M_\Gamma = -I_4/2$. Multiplying such a Γ by i makes its square $+I_4$ and restores the projector construction.

2.2.45 Prove that the 16 Dirac matrices form a linearly independent set.

Solution.

Let $\sigma_0 = I_2$. From the Pauli products,

$$\mathrm{tr}(\sigma_a^\dagger \sigma_b) = 2\delta_{ab}, \qquad a, b = 0, 1, 2, 3.$$

For $a = b$ the product is I_2; for distinct indices the product is a traceless Pauli matrix or i times one. Also

$$\mathrm{tr}(X \otimes Y) = \sum_{r,s} X_{rr} Y_{ss} = \mathrm{tr}\, X \ \mathrm{tr}\, Y.$$

Therefore the 16 tensor products $T_{ab} = \sigma_a \otimes \sigma_b$ obey

$$\mathrm{tr}(T_{ab}^\dagger T_{cd}) = 4\delta_{ac}\delta_{bd}.$$

The explicit correspondence with Eq. (2.76) is

Dirac matrix	tensor form
I_4	$\sigma_0 \otimes \sigma_0$
γ^5	$\sigma_1 \otimes \sigma_0$
γ^0	$\sigma_3 \otimes \sigma_0$
$\gamma^5\gamma^0$	$-i\sigma_2 \otimes \sigma_0$
γ^i $(i = 1, 2, 3)$	$i\sigma_2 \otimes \sigma_i$
$\gamma^5\gamma^i$ $(i = 1, 2, 3)$	$-\sigma_3 \otimes \sigma_i$
σ^{0i} $(i = 1, 2, 3)$	$i\sigma_1 \otimes \sigma_i$
$\sigma^{12},\ \sigma^{13},\ \sigma^{23}$	$\sigma_0 \otimes \sigma_3,\ -\sigma_0 \otimes \sigma_2,\ \sigma_0 \otimes \sigma_1$

For example $\gamma^5\gamma^i = \sigma_1(i\sigma_2) \otimes \sigma_i = -\sigma_3 \otimes \sigma_i$, and $\sigma^{ij} = -iI_2 \otimes \sigma_i\sigma_j = \varepsilon_{ijk} I_2 \otimes \sigma_k$. The table uses each pair (a, b) once with a phase of magnitude 1, so $\mathrm{tr}(\Gamma_r^\dagger \Gamma_s) = 4\delta_{rs}$. If $\sum_s c_s \Gamma_s = 0$, multiply by $\Gamma_r^\dagger$ and take the trace: $4c_r = 0$ for every r. This proves linear independence over $\mathbb{C}$, and hence over $\mathbb{R}$. The complex 4×4 matrix space has dimension 16, so these matrices also span it.

2.2.46 If we assume that a given 4×4 matrix A (with constant elements) can be written as a linear combination of the 16 Dirac matrices (which we denote here as Γ_i)

$$\mathsf{A} = \sum_{i=1}^{16} c_i \Gamma_i,$$

show that

$$c_i \sim \mathrm{trace}(\mathsf{A}\Gamma_i).$$

Solution.

Write $\Gamma_r^2 = s_r I_4$, where $s_r = \pm 1$ as in 2.2.44. The tensor forms in 2.2.45 give $\Gamma_r^\dagger = s_r \Gamma_r$ and

$$\mathrm{tr}(\Gamma_r \Gamma_j) = 4 s_r \delta_{rj}.$$

Multiplying $A = \sum_j c_j \Gamma_j$ by Γ_r and taking the trace,

$$\mathrm{tr}(A\Gamma_r) = \sum_j c_j \,\mathrm{tr}(\Gamma_j \Gamma_r) = 4 s_r c_r.$$

Thus the full proportionality, with its normalization and sign, is

$$\boxed{c_r = \frac{1}{4s_r}\,\mathrm{tr}(A\Gamma_r) = \frac{s_r}{4}\,\mathrm{tr}(A\Gamma_r) = \frac{1}{4}\,\mathrm{tr}(\Gamma_r^\dagger A)}.$$

The sign matters for basis matrices squaring to $-I_4$.

2.2.47 The matrix $\mathsf{C} = i\boldsymbol{\gamma}^2\boldsymbol{\gamma}^0$ is sometimes called the charge conjugation matrix. Show that $\mathsf{C}\boldsymbol{\gamma}^\mu\mathsf{C}^{-1} = -(\boldsymbol{\gamma}^\mu)^T$.

Solution.

Use the specified Dirac representation $\gamma^0 = \sigma_3 \otimes I_2$, $\gamma^i = i\sigma_2 \otimes \sigma_i$. Then

$$C = i\gamma^2\gamma^0 = -\sigma_2\sigma_3 \otimes \sigma_2 = -i\sigma_1 \otimes \sigma_2, \qquad C^2 = -I_4, \quad C^{-1} = -C.$$

The Pauli transposes are $\sigma_1^T = \sigma_1$, $\sigma_2^T = -\sigma_2$, $\sigma_3^T = \sigma_3$, giving

$$(\gamma^0)^T = \gamma^0, \quad (\gamma^2)^T = \gamma^2, \quad (\gamma^1)^T = -\gamma^1, \quad (\gamma^3)^T = -\gamma^3.$$

The product $\gamma^2\gamma^0$ anticommutes with γ^0 and γ^2; for example $C\gamma^0 = i\gamma^2$ but $\gamma^0 C = -i\gamma^2$. It commutes with γ^1, γ^3, since moving either past its two factors gives two minus signs. Therefore

$$C\gamma^\mu C^{-1} = \begin{cases} -\gamma^\mu, & \mu = 0, 2, \\ +\gamma^\mu, & \mu = 1, 3. \end{cases}$$

Comparison with the transposes proves $\boxed{C\gamma^\mu C^{-1} = -(\gamma^\mu)^T}$ for every μ.

2.2.48

(a) Show that, by substitution of the definitions of the $\boldsymbol{\gamma}^\mu$ matrices from Eqs. (2.70) and (2.72), that the Dirac equation, Eq. (2.73), takes the following form when written as 2×2 blocks (with ψ_L and ψ_S column vectors of dimension 2). Here L and S stand, respectively, for "large" and "small" because of their relative size in the nonrelativistic limit):

$$\begin{pmatrix} mc^2 - E & c(\sigma_1 p_1 + \sigma_2 p_2 + \sigma_3 p_3) \\ +c(\sigma_1 p_1 + \sigma_2 p_2 + \sigma_3 p_3) & -mc^2 - E \end{pmatrix} \begin{pmatrix} \psi_L \\ \psi_S \end{pmatrix} = 0.$$

(b) Take $m > 0$ and the positive-energy branch. To reach the nonrelativistic limit, make the substitution $E = mc^2 + \varepsilon$ and approximate $-2mc^2 - \varepsilon$ by $-2mc^2$. Then write the matrix equation as two simultaneous two-component equations and show that they can be rearranged to yield

$$\frac{1}{2m}\left(p_1^2 + p_2^2 + p_3^2\right)\psi_L = \varepsilon\psi_L,$$

which is just the Schrödinger equation for a free particle.

(c) Explain why is it reasonable to call ψ_L and ψ_S "large" and "small."

Solution.

The lower-left block in the printed question (textbook p. 121) has an incorrect minus sign. Equation (2.73) requires the positive sign shown in the corrected question above. The printed sign would give negative free-particle kinetic energy upon elimination of the small component: its second row would imply $\psi_S \simeq -P\psi_L/(2mc)$, and its first would then give $\varepsilon\psi_L = -\mathbf{p}^2\psi_L/(2m)$, where $P = \boldsymbol{\sigma} \cdot \mathbf{p}$.

The manual's block Hamiltonian also omits c from its momentum terms. Equation (2.73) has $c\alpha_i p_i$; all factors of c are retained below.

(a) Put $P = \boldsymbol{\sigma} \cdot \mathbf{p}$. The definitions in Eqs. (2.70)–(2.73) give

$$\gamma^0 = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix}, \quad \alpha_i = \gamma^0\gamma^i = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix}.$$

Thus $(mc^2\gamma^0 + c\sum_i \alpha_i p_i)\Psi = E\Psi$ is

$$\boxed{\begin{pmatrix} (mc^2 - E)I_2 & cP \\ cP & (-mc^2 - E)I_2 \end{pmatrix} \begin{pmatrix} \psi_L \\ \psi_S \end{pmatrix} = 0}.$$

Both off-diagonal blocks are the same.

(b) On the positive-energy branch set $E = mc^2 + \varepsilon$ with $|\varepsilon| \ll mc^2$. The two equations become

$$-\varepsilon\psi_L + cP\psi_S = 0, \qquad cP\psi_L - (2mc^2 + \varepsilon)\psi_S = 0.$$

The second gives $\psi_S = cP\psi_L/(2mc^2 + \varepsilon)$. Substitution into the first yields

$$\varepsilon\psi_L = \frac{c^2P^2}{2mc^2 + \varepsilon}\psi_L = \frac{c^2\mathbf{p}^2}{2mc^2 + \varepsilon}\psi_L.$$

At leading nonrelativistic order, $(2mc^2 + \varepsilon)^{-1} \simeq (2mc^2)^{-1}$, giving

$$\boxed{\frac{\mathbf{p}^2}{2m}\psi_L = \varepsilon\psi_L}.$$

With $p_i = -i\hbar\partial_i$, this is $-\hbar^2\nabla^2\psi_L/(2m) = \varepsilon\psi_L$. As an exact check before approximation, for a nonzero momentum eigenstate multiply by $2mc^2 + \varepsilon$ to obtain $\varepsilon(2mc^2 + \varepsilon) = c^2\mathbf{p}^2$, or $E^2 = m^2c^4 + c^2\mathbf{p}^2$.

(c) Since $P^\dagger P = \mathbf{p}^2 I_2$, the exact eliminated expression gives, for such a state,

$$\frac{\|\psi_S\|}{\|\psi_L\|} = \frac{c|\mathbf{p}|}{E + mc^2} \simeq \frac{|\mathbf{p}|}{2mc} \ll 1 \quad \text{when } |\mathbf{p}| \ll mc.$$

This explains the names on the positive-energy, nonrelativistic branch. At rest the small component vanishes; the ratio need not be small in the relativistic regime.

2.2.49 Show that it is consistent with the requirements that they must satisfy to take the Dirac gamma matrices to be (in 2×2 block form)

$$\boldsymbol{\gamma}^0 = \begin{pmatrix} 0 & \mathbf{1}_2 \\ \mathbf{1}_2 & 0 \end{pmatrix}, \quad \boldsymbol{\gamma}^i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix}, \ (i = 1, 2, 3).$$

This choice for the gamma matrices is called the Weyl representation.

Solution.

Write the proposed Weyl matrices as

$$\gamma^0 = \sigma_1 \otimes I_2, \qquad \gamma^i = J \otimes \sigma_i, \quad J = i\sigma_2.$$

Then $(\gamma^0)^2 = I_4$ and $(\gamma^i)^2 = J^2 \otimes I_2 = -I_4$. Since $\sigma_1 J = -\sigma_3$ and $J\sigma_1 = \sigma_3$,

$$\{\gamma^0, \gamma^i\} = (\sigma_1 J + J\sigma_1) \otimes \sigma_i = 0.$$

For spatial indices, $\{\gamma^i, \gamma^j\} = J^2 \otimes \{\sigma_i, \sigma_j\} = -2\delta_{ij} I_4$. All the required Clifford relations hold. Also $(\gamma^0)^\dagger = \gamma^0$ and $(\gamma^i)^\dagger = -\gamma^i$, so $\alpha_i^\dagger = -\gamma^i\gamma^0 = \gamma^0\gamma^i = \alpha_i$. The free Dirac Hamiltonian has the required Hermiticity as well.

2.2.50 Show that the Dirac equation separates into independent 2×2 blocks in the Weyl representation (see Exercise 2.2.49) in the limit that the mass m approaches zero. This observation is important in the ultra relativistic regime where the rest mass is inconsequential, or for particles of negligible mass (e.g., neutrinos).

Solution.

The manual's momentum blocks again lack their factors of c. With the units of Eq. (2.73), the decoupled equations contain $\pm c\boldsymbol{\sigma} \cdot \mathbf{p}$. In the Weyl representation,

$$\alpha_i = \gamma^0\gamma^i = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}\begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix} = \begin{pmatrix} -\sigma_i & 0 \\ 0 & \sigma_i \end{pmatrix}.$$

With $\Psi = (\psi_1, \psi_2)^T$, Eq. (2.73) is

$$\begin{pmatrix} -c\boldsymbol{\sigma} \cdot \mathbf{p} & mc^2 I_2 \\ mc^2 I_2 & c\boldsymbol{\sigma} \cdot \mathbf{p} \end{pmatrix}\begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = E\begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}.$$

Equivalently,

$$-c\boldsymbol{\sigma} \cdot \mathbf{p}\,\psi_1 + mc^2\psi_2 = E\psi_1, \qquad mc^2\psi_1 + c\boldsymbol{\sigma} \cdot \mathbf{p}\,\psi_2 = E\psi_2.$$

At $m = 0$ the coupling terms disappear:

$$\boxed{-c\boldsymbol{\sigma} \cdot \mathbf{p}\,\psi_1 = E\psi_1, \qquad c\boldsymbol{\sigma} \cdot \mathbf{p}\,\psi_2 = E\psi_2}.$$

Each two-component equation can be solved independently. Squaring either and using $(\boldsymbol{\sigma}\cdot\mathbf{p})^2 = \mathbf{p}^2 I_2$ gives the massless relation $E^2 = c^2\mathbf{p}^2$.

2.2.51

(a) Given $\mathbf{r}' = \mathsf{U}\mathbf{r}$, with U a unitary matrix and $\mathbf{r}$ a (column) vector with complex elements, show that the magnitude of $\mathbf{r}$ is invariant under this operation.

(b) The square matrix U transforms any column vector $\mathbf{r}$ with complex elements into $\mathbf{r}'$, leaving the magnitude invariant: $\mathbf{r}^\dagger\mathbf{r} = \mathbf{r}'^\dagger\mathbf{r}'$. Show that U is unitary.

Solution.

(a) For a complex vector, $\|\mathbf{r}\|^2 = \mathbf{r}^\dagger\mathbf{r}$. If $\mathbf{r}' = U\mathbf{r}$ with $U^\dagger U = I$, then

$$\|\mathbf{r}'\|^2 = (U\mathbf{r})^\dagger U\mathbf{r} = \mathbf{r}^\dagger U^\dagger U\mathbf{r} = \mathbf{r}^\dagger\mathbf{r}.$$

Taking nonnegative square roots proves invariance of the magnitude.

(b) The manual's $\mathbf{r}'^\dagger\mathbf{r}$ must be $\mathbf{r}'^\dagger\mathbf{r}'$. Here U is square and acts on vectors of the same dimension. For rectangular U, norm preservation implies only $U^\dagger U = I$, not two-sided unitarity. Set $H = U^\dagger U - I$. The hypothesis is $\mathbf{r}^\dagger H\mathbf{r} = 0$ for every complex $\mathbf{r}$; also $H^\dagger = H$. Choose $\mathbf{r} = e_i$ to get $H_{ii} = 0$. For $i \neq j$, choose $\mathbf{r} = e_i + e_j$:

$$0 = H_{ii} + H_{ij} + H_{ji} + H_{jj} = H_{ij} + H_{ji} = 2\,\mathrm{Re}\,H_{ij}.$$

Next choose $\mathbf{r} = e_i + ie_j$:

$$0 = H_{ii} + iH_{ij} - iH_{ji} + H_{jj} = i(H_{ij} - H_{ji}) = -2\,\mathrm{Im}\,H_{ij}.$$

Every entry vanishes, so $H = 0$ and $U^\dagger U = I$. Since U is square, taking determinants yields $|\det U|^2 = 1$. It is invertible, and multiplication by U^{-1} gives $U^\dagger = U^{-1}$ and also $UU^\dagger = I$. Thus U is unitary.

3 Chapter 3. Vector Analysis

Problems

3.2 Vectors in 3-D Space

3.2.1 If $\mathbf{P} = \hat{\mathbf{e}}_x P_x + \hat{\mathbf{e}}_y P_y$ and $\mathbf{Q} = \hat{\mathbf{e}}_x Q_x + \hat{\mathbf{e}}_y Q_y$ are any two nonparallel (also nonantiparallel) vectors in the xy-plane, show that $\mathbf{P} \times \mathbf{Q}$ is in the z-direction.

Solution.

Distribute the cross product, using $\hat{\mathbf{e}}_x \times \hat{\mathbf{e}}_y = \hat{\mathbf{e}}_z$, $\hat{\mathbf{e}}_y \times \hat{\mathbf{e}}_x = -\hat{\mathbf{e}}_z$, and $\hat{\mathbf{e}}_i \times \hat{\mathbf{e}}_i = 0$:

$$\mathbf{P} \times \mathbf{Q} = P_x Q_y \hat{\mathbf{e}}_z - P_y Q_x \hat{\mathbf{e}}_z = \boxed{(P_x Q_y - P_y Q_x)\hat{\mathbf{e}}_z}.$$

The coefficient is nonzero for nonparallel nonzero vectors. Its sign determines whether the direction is $+z$ or $-z$.

3.2.2 Prove that $(\mathbf{A} \times \mathbf{B}) \cdot (\mathbf{A} \times \mathbf{B}) = (AB)^2 - (\mathbf{A} \cdot \mathbf{B})^2$.

Solution.

For nonzero $\mathbf{A}, \mathbf{B}$, let θ be their angle. Then $|\mathbf{A} \times \mathbf{B}| = AB \sin\theta$ and $\mathbf{A} \cdot \mathbf{B} = AB \cos\theta$, so

$$(\mathbf{A} \times \mathbf{B})^2 = A^2 B^2 \sin^2\theta = A^2 B^2 (1 - \cos^2\theta) = \boxed{A^2 B^2 - (\mathbf{A} \cdot \mathbf{B})^2}.$$

If either vector is zero, both sides are zero directly. An algebraic check expands the left side as $(A_y B_z - A_z B_y)^2 + (A_z B_x - A_x B_z)^2 + (A_x B_y - A_y B_x)^2$. The squared terms are $\sum_{i \neq j} A_i^2 B_j^2$ and the mixed terms are $-2 \sum_{i<j} A_i A_j B_i B_j$, exactly the expansion of the right side.

3.2.3 Using the vectors

$$\begin{aligned} \mathbf{P} &= \hat{\mathbf{e}}_x \cos\theta + \hat{\mathbf{e}}_y \sin\theta, \\ \mathbf{Q} &= \hat{\mathbf{e}}_x \cos\varphi - \hat{\mathbf{e}}_y \sin\varphi, \\ \mathbf{R} &= \hat{\mathbf{e}}_x \cos\varphi + \hat{\mathbf{e}}_y \sin\varphi, \end{aligned}$$

prove the familiar trigonometric identities

$$\sin(\theta + \varphi) = \sin\theta \cos\varphi + \cos\theta \sin\varphi, \quad \cos(\theta + \varphi) = \cos\theta \cos\varphi - \sin\theta \sin\varphi.$$

Solution.

All three vectors have unit length. Their directed angles from $+x$ are $\theta, -\varphi, \varphi$, respectively; in particular the directed angle from $\mathbf{Q}$ to $\mathbf{P}$ is $\theta + \varphi$. Geometrically the dot product and signed z-component of the cross product are

$$\mathbf{Q} \cdot \mathbf{P} = \cos(\theta + \varphi), \qquad (\mathbf{Q} \times \mathbf{P})_z = \sin(\theta + \varphi).$$

Evaluation from their Cartesian components gives

$$\mathbf{Q} \cdot \mathbf{P} = \cos\varphi \cos\theta - \sin\varphi \sin\theta,$$

$$(\mathbf{Q} \times \mathbf{P})_z = Q_x P_y - Q_y P_x = \cos\varphi \sin\theta + \sin\varphi \cos\theta.$$

Equating the geometric and component expressions proves both identities, including their signs. Directed angles extend the argument to all real θ, φ.

3.2.4

(a) Find a vector $\mathbf{A}$ that is perpendicular to

$$\mathbf{U} = 2\hat{\mathbf{e}}_x + \hat{\mathbf{e}}_y - \hat{\mathbf{e}}_z, \quad \mathbf{V} = \hat{\mathbf{e}}_x - \hat{\mathbf{e}}_y + \hat{\mathbf{e}}_z.$$

(b) What is $\mathbf{A}$ if, in addition to this requirement, we demand that it have unit magnitude?

Solution.

(a) Write $\mathbf{A} = (a, b, c)$. The two perpendicularity conditions are $2a + b - c = 0$ and $a - b + c = 0$. Adding gives $3a = 0$, so $a = 0$ and then $b = c$. Thus every common perpendicular is

$$\mathbf{A} = t(0, 1, 1).$$

A nonzero choice is $(0, 1, 1)$; the cross product $\mathbf{U} \times \mathbf{V} = (0, -3, -3)$ gives the same direction. **(b)** Its squared norm is $2t^2$. Unit magnitude requires $t = \pm 1/\sqrt{2}$, giving

$$\boxed{\mathbf{A} = \pm(\hat{\mathbf{e}}_y + \hat{\mathbf{e}}_z)/\sqrt{2}}.$$

Both orientations meet the requirements.

3.2.5 If four vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, and $\mathbf{d}$ all lie in the same plane, show that

$$(\mathbf{a} \times \mathbf{b}) \times (\mathbf{c} \times \mathbf{d}) = 0.$$

Hint. Consider the directions of the cross-product vectors.

Solution.

Let $\hat{\mathbf{n}}$ be a unit normal to the common plane. Both $\mathbf{a} \times \mathbf{b}$ and $\mathbf{c} \times \mathbf{d}$ are normal to it, so they have the forms $\lambda\hat{\mathbf{n}}$ and $\mu\hat{\mathbf{n}}$ (including zero if a pair is dependent). Hence their cross product is $\lambda\mu(\hat{\mathbf{n}} \times \hat{\mathbf{n}}) = 0$.

3.2.6 Derive the law of sines (see Fig. 3.4):

$$\frac{\sin\alpha}{|A|} = \frac{\sin\beta}{|B|} = \frac{\sin\gamma}{|C|}.$$

Solution.

Take a nondegenerate triangle with side lengths A, B, C opposite α, β, γ. Its area Δ is half the magnitude of the cross product of either pair of adjacent side vectors:

$$2\Delta = BC\sin\alpha = CA\sin\beta = AB\sin\gamma.$$

Divide each expression by the nonzero product ABC:

$$\boxed{\frac{\sin\alpha}{A} = \frac{\sin\beta}{B} = \frac{\sin\gamma}{C} = \frac{2\Delta}{ABC}}.$$

This also follows by dropping an altitude: its length is either adjacent side times the sine of the opposite angle.

3.2.7 The magnetic induction $\mathbf{B}$ is defined by the Lorentz force equation,

$$\mathbf{F} = q(\mathbf{v} \times \mathbf{B}).$$

Carrying out three experiments, we find that if

$$\mathbf{v} = \hat{\mathbf{e}}_x, \quad \frac{\mathbf{F}}{q} = 2\hat{\mathbf{e}}_z - 4\hat{\mathbf{e}}_y,$$
$$\mathbf{v} = \hat{\mathbf{e}}_y, \quad \frac{\mathbf{F}}{q} = 4\hat{\mathbf{e}}_x - \hat{\mathbf{e}}_z,$$
$$\mathbf{v} = \hat{\mathbf{e}}_z, \quad \frac{\mathbf{F}}{q} = \hat{\mathbf{e}}_y - 2\hat{\mathbf{e}}_x.$$

From the results of these three separate experiments calculate the magnetic induction $\mathbf{B}$.

Solution.

Put $\mathbf{B} = (B_x, B_y, B_z)$. The three unit-velocity cross products are

$$\hat{\mathbf{e}}_x \times \mathbf{B} = (0, -B_z, B_y), \quad \hat{\mathbf{e}}_y \times \mathbf{B} = (B_z, 0, -B_x), \quad \hat{\mathbf{e}}_z \times \mathbf{B} = (-B_y, B_x, 0).$$

The first measurement gives $B_z = 4, B_y = 2$; the second gives $B_z = 4, B_x = 1$; the third independently gives $B_y = 2, B_x = 1$. All are consistent, so $\boxed{\mathbf{B} = \hat{\mathbf{e}}_x + 2\hat{\mathbf{e}}_y + 4\hat{\mathbf{e}}_z}$.

3.2.8 You are given the three vectors $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$,

$$\mathbf{A} = \hat{\mathbf{e}}_x + \hat{\mathbf{e}}_y, \quad \mathbf{B} = \hat{\mathbf{e}}_y + \hat{\mathbf{e}}_z, \quad \mathbf{C} = \hat{\mathbf{e}}_x - \hat{\mathbf{e}}_z.$$

(a) Compute the scalar triple product, $\mathbf{A} \cdot \mathbf{B} \times \mathbf{C}$. Noting that $\mathbf{A} = \mathbf{B} + \mathbf{C}$, give a geometric interpretation of your result for the scalar triple product.

(b) Compute $\mathbf{A} \times (\mathbf{B} \times \mathbf{C})$.

Solution.

First compute

$$\mathbf{B} \times \mathbf{C} = \begin{vmatrix} \hat{\mathbf{e}}_x & \hat{\mathbf{e}}_y & \hat{\mathbf{e}}_z \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{vmatrix} = (-1, 1, -1).$$

(a) $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = (1, 1, 0) \cdot (-1, 1, -1) = 0$. Since $\mathbf{A} = \mathbf{B} + \mathbf{C}$, the three vectors lie in one plane; the parallelepiped they span has zero volume. **(b)** Direct multiplication gives

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (1, 1, 0) \times (-1, 1, -1) = \boxed{(-1, 1, 2)}.$$

The triple-product identity also gives $\mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) = \mathbf{B} - \mathbf{C} = (-1, 1, 2)$.

3.2.9 Prove Jacobi's identity for vector products:

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) + \mathbf{c} \times (\mathbf{a} \times \mathbf{b}) = 0.$$

Solution.

Expand the three vector triple products:

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b}),$$
$$\mathbf{b} \times (\mathbf{c} \times \mathbf{a}) = \mathbf{c}(\mathbf{b} \cdot \mathbf{a}) - \mathbf{a}(\mathbf{b} \cdot \mathbf{c}),$$
$$\mathbf{c} \times (\mathbf{a} \times \mathbf{b}) = \mathbf{a}(\mathbf{c} \cdot \mathbf{b}) - \mathbf{b}(\mathbf{c} \cdot \mathbf{a}).$$

Dot products commute, so the coefficient of each of $\mathbf{a}, \mathbf{b}, \mathbf{c}$ in the sum is zero. The sum vanishes.

3.2.10 A vector $\mathbf{A}$ is decomposed into a radial vector $\mathbf{A}_r$ and a tangential vector $\mathbf{A}_t$. If $\hat{\mathbf{r}}$ is a unit vector in the radial direction, show that

(a) $\mathbf{A}_r = \hat{\mathbf{r}}(\mathbf{A}\cdot\hat{\mathbf{r}})$ and

(b) $\mathbf{A}_t = -\hat{\mathbf{r}}\times(\hat{\mathbf{r}}\times\mathbf{A})$.

Solution.

The radial component must be $\mathbf{A}_r = a\hat{\mathbf{r}}$. The remainder is tangential if $0 = \hat{\mathbf{r}}\cdot(\mathbf{A}-a\hat{\mathbf{r}}) = \hat{\mathbf{r}}\cdot\mathbf{A}-a$. Thus

$$\boxed{\mathbf{A}_r = \hat{\mathbf{r}}(\mathbf{A}\cdot\hat{\mathbf{r}})}.$$

The triple-product identity and $\hat{\mathbf{r}}\cdot\hat{\mathbf{r}} = 1$ give

$$-\hat{\mathbf{r}}\times(\hat{\mathbf{r}}\times\mathbf{A}) = -\hat{\mathbf{r}}(\hat{\mathbf{r}}\cdot\mathbf{A}) + \mathbf{A} = \mathbf{A} - \mathbf{A}_r = \boxed{\mathbf{A}_t}.$$

It is perpendicular to $\hat{\mathbf{r}}$ and the two parts sum to $\mathbf{A}$, verifying the decomposition.

3.2.11 Prove that a necessary and sufficient condition for the three (nonvanishing) vectors $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ to be coplanar is the vanishing of the scalar triple product

$$\mathbf{A}\cdot\mathbf{B}\times\mathbf{C} = 0.$$

Solution.

If $\mathbf{B},\mathbf{C}$ are independent, their cross product is a nonzero normal to their plane. Consequently $\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) = 0$ says exactly that $\mathbf{A}$ lies in that plane. This proves necessity and sufficiency in this case. If $\mathbf{B},\mathbf{C}$ are dependent, their cross product is zero, and all three vectors lie in a plane containing $\mathbf{A}$ and their common line. Conversely any three vectors in a plane have one normal perpendicular to all of them, so the triple product vanishes. The argument also covers collinear triples.

3.2.12 Three vectors $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ are given by

$$\mathbf{A} = 3\hat{\mathbf{e}}_x - 2\hat{\mathbf{e}}_y + 2\hat{\mathbf{z}}, \quad \mathbf{B} = 6\hat{\mathbf{e}}_x + 4\hat{\mathbf{e}}_y - 2\hat{\mathbf{z}}, \quad \mathbf{C} = -3\hat{\mathbf{e}}_x - 2\hat{\mathbf{e}}_y - 4\hat{\mathbf{z}}.$$

Compute the values of $\mathbf{A}\cdot\mathbf{B}\times\mathbf{C}$ and $\mathbf{A}\times(\mathbf{B}\times\mathbf{C})$, $\mathbf{C}\times(\mathbf{A}\times\mathbf{B})$ and $\mathbf{B}\times(\mathbf{C}\times\mathbf{A})$.

Solution.

The needed cross products and dot products are

$$\begin{aligned}
\mathbf{B}\times\mathbf{C} &= (-16+(-4),\ 6+24,\ -12+12) = (-20,30,0),\\
\mathbf{A}\cdot\mathbf{B} &= 18-8-4 = 6,\\
\mathbf{A}\cdot\mathbf{C} &= -9+4-8 = -13, \qquad \mathbf{B}\cdot\mathbf{C} = -18-8+8 = -18.
\end{aligned}$$

Therefore the scalar triple product is

$$\boxed{\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) = 3(-20)+(-2)(30)+2(0) = -120}.$$

Using $\mathbf{X}\times(\mathbf{Y}\times\mathbf{Z}) = \mathbf{Y}(\mathbf{X}\cdot\mathbf{Z}) - \mathbf{Z}(\mathbf{X}\cdot\mathbf{Y})$,

$$\begin{aligned}
\mathbf{A}\times(\mathbf{B}\times\mathbf{C}) &= -13\mathbf{B} - 6\mathbf{C} = \boxed{(-60,-40,50)},\\
\mathbf{C}\times(\mathbf{A}\times\mathbf{B}) &= -18\mathbf{A} + 13\mathbf{B} = \boxed{(24,88,-62)},\\
\mathbf{B}\times(\mathbf{C}\times\mathbf{A}) &= 6\mathbf{C} + 18\mathbf{A} = \boxed{(36,-48,12)}.
\end{aligned}$$

Their sum is $(0,0,0)$, as required by Jacobi's identity.

3.2.13 Show that

$$(\mathbf{A}\times\mathbf{B})\cdot(\mathbf{C}\times\mathbf{D}) = (\mathbf{A}\cdot\mathbf{C})(\mathbf{B}\cdot\mathbf{D}) - (\mathbf{A}\cdot\mathbf{D})(\mathbf{B}\cdot\mathbf{C}).$$

Solution.

Write components using ε_{ijk}. Summing every repeated index from 1 to 3 and using the contraction of two Levi-Civita symbols,

$$\begin{aligned}(\mathbf{A}\times\mathbf{B})\cdot(\mathbf{C}\times\mathbf{D}) &= \varepsilon_{ijk}\varepsilon_{i\ell m}A_jB_kC_\ell D_m \\ &= (\delta_{j\ell}\delta_{km} - \delta_{jm}\delta_{k\ell})A_jB_kC_\ell D_m \\ &= \sum_{j,k} A_jC_jB_kD_k - \sum_{j,k} A_jD_jB_kC_k \\ &= (\mathbf{A}\cdot\mathbf{C})(\mathbf{B}\cdot\mathbf{D}) - (\mathbf{A}\cdot\mathbf{D})(\mathbf{B}\cdot\mathbf{C}).\end{aligned}$$

No independence or nonzero-vector assumption is required.

3.2.14 Show that

$$(\mathbf{A}\times\mathbf{B})\times(\mathbf{C}\times\mathbf{D}) = (\mathbf{A}\cdot\mathbf{B}\times\mathbf{D})\mathbf{C} - (\mathbf{A}\cdot\mathbf{B}\times\mathbf{C})\mathbf{D}.$$

Solution.

Apply the triple-product identity with $\mathbf{X} = \mathbf{A}\times\mathbf{B}$:

$$\mathbf{X}\times(\mathbf{C}\times\mathbf{D}) = \mathbf{C}(\mathbf{X}\cdot\mathbf{D}) - \mathbf{D}(\mathbf{X}\cdot\mathbf{C}).$$

Cyclicity of the scalar triple product gives $(\mathbf{A}\times\mathbf{B})\cdot\mathbf{D} = \mathbf{A}\cdot(\mathbf{B}\times\mathbf{D})$ and the analogous formula with $\mathbf{C}$. Substitution gives exactly the stated result, with the first scalar multiplying $\mathbf{C}$ and the second multiplying $-\mathbf{D}$.

3.2.15 An electric charge q_1 moving with velocity $\mathbf{v}_1$ produces a magnetic induction $\mathbf{B}$ given by

$$\mathbf{B} = \frac{\mu_0}{4\pi} q_1 \frac{\mathbf{v}_1\times\hat{\mathbf{r}}}{r^2} \quad \text{(mks units)},$$

where $\hat{\mathbf{r}}$ is a unit vector that points from q_1 to the point at which $\mathbf{B}$ is measured (Biot and Savart law).

(a) Show that the magnetic force exerted by q_1 on a second charge q_2, velocity $\mathbf{v}_2$, is given by the vector triple product

$$\mathbf{F}_2 = \frac{\mu_0}{4\pi}\frac{q_1q_2}{r^2}\,\mathbf{v}_2\times(\mathbf{v}_1\times\hat{\mathbf{r}}).$$

(b) Write out the corresponding magnetic force $\mathbf{F}_1$ that q_2 exerts on q_1. Define your unit radial vector. How do $\mathbf{F}_1$ and $\mathbf{F}_2$ compare?

(c) Calculate $\mathbf{F}_1$ and $\mathbf{F}_2$ for the case of q_1 and q_2 moving along parallel trajectories side by side.

ANS. (b) $\mathbf{F}_1 = -\dfrac{\mu_0}{4\pi}\dfrac{q_1q_2}{r^2}\,\mathbf{v}_1\times(\mathbf{v}_2\times\hat{\mathbf{r}})$. In general, there is no simple relation between $\mathbf{F}_1$ and $\mathbf{F}_2$. Specifically, Newton's third law, $\mathbf{F}_1 = -\mathbf{F}_2$, does not hold.

(c) $\mathbf{F}_1 = \dfrac{\mu_0}{4\pi}\dfrac{q_1q_2}{r^2}(\mathbf{v}_1\cdot\mathbf{v}_2)\hat{\mathbf{r}} = -\mathbf{F}_2$. Mutual attraction if $q_1q_2(\mathbf{v}_1\cdot\mathbf{v}_2) > 0$; repulsion if it is negative.

Solution.

Set $K = \mu_0 q_1 q_2/(4\pi r^2)$ and let $\hat{\mathbf{r}}$ point from charge 1 to charge 2. The magnetic Lorentz force is

$$\mathbf{F}_2 = q_2 \mathbf{v}_2 \times \mathbf{B}_1 = K \mathbf{v}_2 \times (\mathbf{v}_1 \times \hat{\mathbf{r}}) = K\{\mathbf{v}_1(\mathbf{v}_2 \cdot \hat{\mathbf{r}}) - \hat{\mathbf{r}}(\mathbf{v}_1 \cdot \mathbf{v}_2)\}.$$

For the field of charge 2 evaluated at charge 1, the radial unit vector is $-\hat{\mathbf{r}}$, so

$$\mathbf{F}_1 = -K \mathbf{v}_1 \times (\mathbf{v}_2 \times \hat{\mathbf{r}}) = K\{\hat{\mathbf{r}}(\mathbf{v}_1 \cdot \mathbf{v}_2) - \mathbf{v}_2(\mathbf{v}_1 \cdot \hat{\mathbf{r}})\}.$$

Thus $\mathbf{F}_1 + \mathbf{F}_2 = K[\mathbf{v}_1(\mathbf{v}_2 \cdot \hat{\mathbf{r}}) - \mathbf{v}_2(\mathbf{v}_1 \cdot \hat{\mathbf{r}})]$, generally nonzero. This concerns the magnetic forces on the particles; it is not a claim that particle-plus-field momentum fails to be conserved.

For side-by-side trajectories, both velocities are perpendicular to $\hat{\mathbf{r}}$. Hence

$$\boxed{\mathbf{F}_1 = K(\mathbf{v}_1 \cdot \mathbf{v}_2)\hat{\mathbf{r}} = -\mathbf{F}_2}.$$

The printed v^2 assumes equal velocities. Attraction requires $q_1 q_2(\mathbf{v}_1 \cdot \mathbf{v}_2) > 0$; opposite sign gives repulsion. For equal velocities and like charges this reduces to the printed attractive result.

3.3 Coordinate Transformations

3.3.1 A rotation $\varphi_1 + \varphi_2$ about the z-axis is carried out as two successive rotations φ_1 and φ_2, each about the z-axis. Use the matrix representation of the rotations to derive the trigonometric identities

$$\cos(\varphi_1 + \varphi_2) = \cos\varphi_1 \cos\varphi_2 - \sin\varphi_1 \sin\varphi_2, \quad \sin(\varphi_1 + \varphi_2) = \sin\varphi_1 \cos\varphi_2 + \cos\varphi_1 \sin\varphi_2.$$

Solution.

For a passive rotation of axes about z,

$$Z(\varphi) = \begin{pmatrix} \cos\varphi & \sin\varphi & 0 \\ -\sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The two successive rotations give $Z(\varphi_2)Z(\varphi_1)$. Its upper left block is

$$\begin{pmatrix} c_2c_1 - s_2s_1 & c_2s_1 + s_2c_1 \\ -s_2c_1 - c_2s_1 & -s_2s_1 + c_2c_1 \end{pmatrix}, \qquad c_i = \cos\varphi_i, \ \ s_i = \sin\varphi_i.$$

The overall rotation is $Z(\varphi_1 + \varphi_2)$. Equating the $(1,1)$ and $(1,2)$ entries gives the stated cosine and sine addition formulas.

3.3.2 A corner reflector is formed by three mutually perpendicular reflecting surfaces. Show that a ray of light incident upon the corner reflector (striking all three surfaces) is reflected back along a line parallel to the line of incidence.

Hint. Consider the effect of a reflection on the components of a vector describing the direction of the light ray.

Solution.

Choose x, y, z perpendicular to the three reflecting planes. Reflection reverses the normal component of a ray's direction and preserves its two tangential components. The three reflection matrices are therefore

$$R_x = \mathrm{diag}(-1, 1, 1), \quad R_y = \mathrm{diag}(1, -1, 1), \quad R_z = \mathrm{diag}(1, 1, -1).$$

They commute and $R_x R_y R_z = -I_3$. After one reflection from each surface, the propagation direction is $-\mathbf{v}$. The outgoing line is thus parallel to, and traversed oppositely from, the incident line; it can be displaced from that line.

3.3.3 Let $\mathbf{x}$ and $\mathbf{y}$ be column vectors. Under an orthogonal transformation S, they become $\mathbf{x}' = \mathsf{S}\mathbf{x}$ and $\mathbf{y}' = \mathsf{S}\mathbf{y}$. Show that $(\mathbf{x}')^T\mathbf{y}' = \mathbf{x}^T\mathbf{y}$, a result equivalent to the invariance of the dot product under a rotational transformation.

Solution.

Orthogonality means $S^T S = I$. Consequently

$$(\mathbf{x}')^T\mathbf{y}' = (S\mathbf{x})^T(S\mathbf{y}) = \mathbf{x}^T S^T S\mathbf{y} = \boxed{\mathbf{x}^T\mathbf{y}}.$$

This argument works for both proper rotations and orthogonal transformations with determinant -1.

3.3.4 Given the orthogonal transformation matrix S and vectors $\mathbf{a}$ and $\mathbf{b}$,

$$\mathsf{S} = \begin{pmatrix} 0.80 & 0.60 & 0.00 \\ -0.48 & 0.64 & 0.60 \\ 0.36 & -0.48 & 0.80 \end{pmatrix}, \quad \mathbf{a} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix},$$

(a) Calculate $\det(\mathsf{S})$.

(b) Verify that $\mathbf{a} \cdot \mathbf{b}$ is invariant under application of S to $\mathbf{a}$ and $\mathbf{b}$.

(c) Determine what happens to $\mathbf{a} \times \mathbf{b}$ under application of S to $\mathbf{a}$ and $\mathbf{b}$. Is this what is expected?

Solution.

(a) Expansion along the first row gives

$$\det S = 0.80(0.64 \cdot 0.80 + 0.60 \cdot 0.48) - 0.60(-0.48 \cdot 0.80 - 0.60 \cdot 0.36) = 0.80(0.80) - 0.60(-0.60) = 1.$$

(b) Matrix multiplication gives

$$\mathbf{a}' = (0.80, 0.12, 1.16), \quad \mathbf{b}' = (1.20, 0.68, -1.76).$$

Then $\mathbf{a} \cdot \mathbf{b} = -1$, and $\mathbf{a}' \cdot \mathbf{b}' = 0.96 + 0.0816 - 2.0416 = -1$. **(c)** The original cross product is $(-2, 1, 2)$, so

$$S(\mathbf{a} \times \mathbf{b}) = (-1.00, 2.80, 0.40).$$

The cross product after transformation is

$$\begin{aligned} \mathbf{a}' \times \mathbf{b}' &= (0.12(-1.76) - 1.16(0.68),\ 1.16(1.20) - 0.80(-1.76),\ 0.80(0.68) - 0.12(1.20)) \\ &= (-1.00, 2.80, 0.40). \end{aligned}$$

This is expected: for orthogonal S, $(S\mathbf{a}) \times (S\mathbf{b}) = (\det S)S(\mathbf{a} \times \mathbf{b})$, and here $\det S = 1$.

3.3.5 Using $\mathbf{a}$ and $\mathbf{b}$ as defined in Exercise 3.3.4, but with

$$\mathsf{S} = \begin{pmatrix} 0.60 & 0.00 & 0.80 \\ -0.64 & -0.60 & 0.48 \\ -0.48 & 0.80 & 0.36 \end{pmatrix} \quad \text{and} \quad \mathbf{c} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix},$$

(a) Calculate $\det(\mathsf{S})$.

Apply S to $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$, and determine what happens to

(b) $\mathbf{a} \times \mathbf{b}$,

(c) $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$,

(d) $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$.

(e) Classify the expressions in (b) through (d) as scalar, vector, pseudovector, or pseudoscalar.

Solution.

Expansion gives

$$\det S = 0.60(-0.60 \cdot 0.36 - 0.48 \cdot 0.80) + 0.80(-0.64 \cdot 0.80 - (-0.60)(-0.48)) = -0.36 - 0.64 = -1.$$

Multiplication of the three columns gives

$$\mathbf{a}' = (1.40, -0.16, -0.12), \quad \mathbf{b}' = (-0.80, -1.68, 1.24), \quad \mathbf{c}' = (3.60, -0.44, 0.92).$$

(b) $\mathbf{a} \times \mathbf{b} = (-2, 1, 2)$, whereas

$$\mathbf{a}' \times \mathbf{b}' = (-0.40, -1.64, -2.48) = -S(-2, 1, 2).$$

For example its first component is $(-0.16)(1.24) - (-0.12)(-1.68) = -0.40$. **(c)** The original triple product is $-4 + 1 + 6 = 3$; the new one is $-1.44 + 0.7216 - 2.2816 = -3$. **(d)** Since $\mathbf{a} \cdot \mathbf{c} = 5$ and $\mathbf{a} \cdot \mathbf{b} = -1$,

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = 5\mathbf{b} + \mathbf{c} = (2, 11, -2).$$

Dot products are unchanged, so the transformed product is $5\mathbf{b}' + \mathbf{c}' = (-0.40, -8.84, 7.12) = S(2, 11, -2)$. **(e)** Thus (b) is a pseudovector, (c) a pseudoscalar, and (d) an ordinary vector. A pseudovector acquires the extra factor $\det S$, and a pseudoscalar changes sign under this improper rotation.

The solutions manual, p. 36, prints $S_{32} = -0.80$ in its matrices; the textbook has $+0.80$, as used here. Its listed transformed vectors also require the positive sign. For example, $b'_z = 0.80(2) + 0.36(-1) = 1.24$, whereas the negative sign gives -1.96.

3.4 Rotations in $\mathbb{R}^3$

3.4.1 Another set of Euler rotations in common use is

(1) a rotation about the x_3-axis through an angle φ, counterclockwise,

(2) a rotation about the x_1'-axis through an angle θ, counterclockwise,

(3) a rotation about the x_3''-axis through an angle ψ, counterclockwise.

If

$$\begin{aligned} \alpha &= \varphi - \pi/2 & \varphi &= \alpha + \pi/2 \\ \beta &= \theta \qquad \text{or} & \theta &= \beta \\ \gamma &= \psi + \pi/2 & \psi &= \gamma - \pi/2, \end{aligned}$$

show that the final systems are identical.

Solution.

Use passive rotation matrices

$$Z(t) = \begin{pmatrix} c_t & s_t & 0 \\ -s_t & c_t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \; Y(t) = \begin{pmatrix} c_t & 0 & -s_t \\ 0 & 1 & 0 \\ s_t & 0 & c_t \end{pmatrix}, \; X(t) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_t & s_t \\ 0 & -s_t & c_t \end{pmatrix},$$

where $c_t = \cos t, s_t = \sin t$. The textbook convention is $S(\alpha, \beta, \gamma) = Z(\gamma)Y(\beta)Z(\alpha)$. The alternative is $Z(\psi)X(\theta)Z(\varphi)$. Direct multiplication of the two quarter-turn matrices gives

$$Z(\pi/2)Y(\theta)Z(-\pi/2) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_\theta & s_\theta \\ 0 & -s_\theta & c_\theta \end{pmatrix} = X(\theta).$$

Consequently, under the stated substitutions,

$$\begin{aligned} S &= Z(\psi + \pi/2)Y(\theta)Z(\varphi - \pi/2) \\ &= Z(\psi)\left[Z(\pi/2)Y(\theta)Z(-\pi/2)\right]Z(\varphi) = Z(\psi)X(\theta)Z(\varphi). \end{aligned}$$

The matrices, and therefore the final frames, coincide. For use below, the explicit product is

$$S = \begin{pmatrix} c_\gamma c_\beta c_\alpha - s_\gamma s_\alpha & c_\gamma c_\beta s_\alpha + s_\gamma c_\alpha & -c_\gamma s_\beta \\ -s_\gamma c_\beta c_\alpha - c_\gamma s_\alpha & -s_\gamma c_\beta s_\alpha + c_\gamma c_\alpha & s_\gamma s_\beta \\ s_\beta c_\alpha & s_\beta s_\alpha & c_\beta \end{pmatrix}. \tag{1}$$

3.4.2 Suppose the Earth is moved (rotated) so that the north pole goes to 30° north, 20° west (original latitude and longitude system) and the 10° west meridian points due south (also in the original system).

(a) What are the Euler angles describing this rotation?

(b) Find the corresponding direction cosines.

ANS. (b) $\mathsf{S} = \begin{pmatrix} 0.9551 & -0.2552 & -0.1504 \\ 0.0052 & 0.5221 & -0.8529 \\ 0.2962 & 0.8138 & 0.5000 \end{pmatrix}$. To reproduce the printed matrix, take $+y$ along the equator at Greenwich and $+x$ at 90° west; the longitude convention otherwise leaves the Cartesian axes unspecified.

Solution.

A longitude reference for the Cartesian axes is needed. The printed matrix is consistent with the $+y$-axis through Greenwich and the $+x$-axis through 90° west; use that convention here. East longitude λ then has azimuth $\alpha = 90° + \lambda$. The new pole has $\lambda = -20°$ and colatitude $\beta = 90° - 30° = 60°$, so $\alpha = 70°$.

Before the last rotation, let $\mathbf{e}_\theta$ point due south at the new pole and $\mathbf{e}_\varphi$ point east. The final equatorial axes are

$$\mathbf{e}_1' = \cos\gamma\,\mathbf{e}_\theta + \sin\gamma\,\mathbf{e}_\varphi, \qquad \mathbf{e}_2' = -\sin\gamma\,\mathbf{e}_\theta + \cos\gamma\,\mathbf{e}_\varphi.$$

The body's 10° west meridian has azimuth 80° measured from its x-axis. Its tangent away from the pole is therefore

$$\cos 80°\,\mathbf{e}_1' + \sin 80°\,\mathbf{e}_2' = \cos(\gamma + 80°)\mathbf{e}_\theta + \sin(\gamma + 80°)\mathbf{e}_\varphi.$$

Requiring this tangent to point south gives $\gamma = -80°$ (equivalently 280°). Thus

$$\boxed{(\alpha, \beta, \gamma) = (70°, 60°, -80°)}.$$

Substitution into (1) gives

$$\boxed{S = \begin{pmatrix} 0.95511217 & -0.25523613 & -0.15038373 \\ 0.00523613 & 0.52209946 & -0.85286853 \\ 0.29619813 & 0.81379768 & 0.50000000 \end{pmatrix}},$$

before rounding to the four places printed. The third row is the pole direction and its z-component is $\sin 30° = 1/2$, providing a direct geometric check. If instead $+x$ is Greenwich and $+y$ is 90° east, the same physical orientation has Euler angles $(-20°, 60°, 10°)$; its numerical matrix changes because the reference axes change.

3.4.3 Verify that the Euler angle rotation matrix, Eq. (3.37), is invariant under the transformation

$$\alpha \to \alpha + \pi, \quad \beta \to -\beta, \quad \gamma \to \gamma - \pi.$$

Solution.

In (1), the substitutions give

$$(c_\alpha, s_\alpha) \mapsto (-c_\alpha, -s_\alpha), \quad (c_\beta, s_\beta) \mapsto (c_\beta, -s_\beta), \quad (c_\gamma, s_\gamma) \mapsto (-c_\gamma, -s_\gamma).$$

Each term in the upper left 2×2 block has either one α factor and one γ factor, or those two factors times c_β; it acquires two minus signs. The $(1,3), (2,3)$ entries contain one γ factor and s_β, again two signs. The $(3,1), (3,2)$ entries contain one α factor and s_β, also two signs. Finally $(3,3) = c_\beta$ is unchanged. Every matrix entry is therefore invariant.

3.4.4 Show that the Euler angle rotation matrix $\mathsf{S}(\alpha,\beta,\gamma)$ satisfies the following relations:

(a) $\mathsf{S}^{-1}(\alpha,\beta,\gamma) = \tilde{\mathsf{S}}(\alpha,\beta,\gamma)$,

(b) $\mathsf{S}^{-1}(\alpha,\beta,\gamma) = \mathsf{S}(-\gamma,-\beta,-\alpha)$.

Solution.

(a) For each elementary matrix, $Z(t)^T = Z(-t) = Z(t)^{-1}$ and $Y(t)^T = Y(-t) = Y(t)^{-1}$; multiplying the 2×2 blocks verifies these using $\cos^2 t + \sin^2 t = 1$. Therefore

$$S^T S = Z(\alpha)^T Y(\beta)^T Z(\gamma)^T Z(\gamma) Y(\beta) Z(\alpha) = I,$$

after canceling the three adjacent inverse pairs. Hence $S^{-1} = S^T$. **(b)** Inverting reverses the order:

$$S^{-1} = Z(-\alpha)Y(-\beta)Z(-\gamma) = \boxed{S(-\gamma,-\beta,-\alpha)}.$$

3.4.5 The coordinate system (x,y,z) is rotated through an angle Φ counterclockwise about an axis defined by the unit vector $\hat{\mathbf{n}}$ into system (x',y',z'). In terms of the new coordinates the radius vector becomes

$$\mathbf{r}' = \mathbf{r}\cos\Phi + \mathbf{r}\times\hat{\mathbf{n}}\sin\Phi + \hat{\mathbf{n}}(\hat{\mathbf{n}}\cdot\mathbf{r})(1-\cos\Phi).$$

(a) Derive this expression from geometric considerations.

(b) Show that it reduces as expected for $\hat{\mathbf{n}} = \hat{\mathbf{e}}_z$. The answer, in matrix form, appears in Eq. (3.35).

(c) Verify that $r'^2 = r^2$.

Solution.

Split $\mathbf{r} = \mathbf{r}_\parallel + \mathbf{r}_\perp$, where $\mathbf{r}_\parallel = \hat{\mathbf{n}}(\hat{\mathbf{n}}\cdot\mathbf{r})$. A passive rotation preserves $\mathbf{r}_\parallel$ and rotates the perpendicular components through $-\Phi$:

$$\mathbf{r}' = \mathbf{r}_\parallel + \mathbf{r}_\perp\cos\Phi + (\mathbf{r}_\perp\times\hat{\mathbf{n}})\sin\Phi.$$

Here $\mathbf{r}_\perp\times\hat{\mathbf{n}} = \mathbf{r}\times\hat{\mathbf{n}}$. Substitute $\mathbf{r}_\perp = \mathbf{r} - \mathbf{r}_\parallel$ to get the required formula.

For $\hat{\mathbf{n}} = \hat{\mathbf{e}}_z$, $\mathbf{r}\times\hat{\mathbf{e}}_z = (y,-x,0)$, so

$$\mathbf{r}' = (x\cos\Phi + y\sin\Phi,\ -x\sin\Phi + y\cos\Phi,\ z),$$

which is $Z(\Phi)\mathbf{r}$. Finally the three terms in the perpendicular decomposition are mutually orthogonal, and $|\mathbf{r}_\perp\times\hat{\mathbf{n}}| = |\mathbf{r}_\perp|$. Thus $|\mathbf{r}'|^2 = |\mathbf{r}_\parallel|^2 + |\mathbf{r}_\perp|^2(\cos^2\Phi + \sin^2\Phi) = |\mathbf{r}|^2$.

3.5 Differential Vector Operators

3.5.1 If $S(x,y,z) = \left(x^2+y^2+z^2\right)^{-3/2}$, find

(a) ∇S at the point $(1,2,3)$,

(b) the magnitude of the gradient of S, $|\nabla S|$ at $(1,2,3)$, and

(c) the direction cosines of ∇S at $(1,2,3)$.

Solution.

For $r = (x^2+y^2+z^2)^{1/2} > 0$, differentiate by the chain rule:

$$\partial_x S = -\tfrac{3}{2}(x^2+y^2+z^2)^{-5/2}(2x) = -3xr^{-5},$$

and similarly for y, z. Hence $\nabla S = -3(x,y,z)/r^5$. At $(1,2,3)$, $r^2 = 14$, giving

$$\boxed{\nabla S = -\frac{3}{14^{5/2}}(1,2,3)}, \qquad \boxed{|\nabla S| = \frac{3\sqrt{14}}{14^{5/2}} = \frac{3}{196}}.$$

Divide the vector by its magnitude to obtain the direction cosines $\boxed{(-1,-2,-3)/\sqrt{14}}$, whose squares sum to 1.

3.5.2

(a) Find a unit vector perpendicular to the surface

$$x^2+y^2+z^2 = 3$$

at the point $(1,1,1)$.

(b) Derive the equation of the plane tangent to the surface at $(1,1,1)$.

ANS. (a) $\left(\hat{\mathbf{e}}_x+\hat{\mathbf{e}}_y+\hat{\mathbf{e}}_z\right)/\sqrt{3}$, (b) $x+y+z=3$.

Solution.

Let $f = x^2+y^2+z^2-3$. A displacement tangent to $f=0$ satisfies $df = \nabla f \cdot d\mathbf{r} = 0$, so ∇f is a normal. At $(1,1,1)$, $\nabla f = (2,2,2)$ with length $2\sqrt{3}$. An outward unit normal is therefore $(1,1,1)/\sqrt{3}$; its negative is the inward unit normal. The tangent plane through $\mathbf{r}_0 = (1,1,1)$ satisfies

$$(2,2,2)\cdot(\mathbf{r}-\mathbf{r}_0) = 0,$$

or $2(x-1)+2(y-1)+2(z-1)=0$. Thus $\boxed{x+y+z=3}$.

3.5.3 Given a vector $\mathbf{r}_{12} = \hat{\mathbf{e}}_x(x_1-x_2)+\hat{\mathbf{e}}_y(y_1-y_2)+\hat{\mathbf{e}}_z(z_1-z_2)$, show that $\nabla_1 r_{12}$ (gradient with respect to x_1, y_1, and z_1 of the magnitude r_{12}) is a unit vector in the direction of $\mathbf{r}_{12}$. Assume $r_{12} > 0$.

Solution.

For separated points $r_{12} > 0$,

$$r_{12} = [(x_1-x_2)^2+(y_1-y_2)^2+(z_1-z_2)^2]^{1/2}.$$

Keeping point 2 fixed gives $\partial_{x_1} r_{12} = (x_1-x_2)/r_{12}$ and the analogous two derivatives. Therefore

$$\boxed{\nabla_1 r_{12} = \frac{(x_1-x_2, y_1-y_2, z_1-z_2)}{r_{12}} = \hat{\mathbf{r}}_{12}}.$$

Its norm is $r_{12}/r_{12} = 1$. At coincidence $r_{12} = 0$, the distance is not differentiable and no radial unit direction is defined.

3.5.4 If a vector function $\mathbf{F}$ depends on both space coordinates (x, y, z) and time t, show that

$$d\mathbf{F} = (d\mathbf{r} \cdot \nabla)\mathbf{F} + \frac{\partial \mathbf{F}}{\partial t}\, dt.$$

Solution.

In a fixed Cartesian basis, apply the total differential to each component:

$$dF_i = (\partial_x F_i)dx + (\partial_y F_i)dy + (\partial_z F_i)dz + (\partial_t F_i)dt.$$

Because $d\mathbf{r} \cdot \nabla = dx\,\partial_x + dy\,\partial_y + dz\,\partial_z$, the first three terms are $[(d\mathbf{r} \cdot \nabla)\mathbf{F}]_i$. Collecting the three components proves the stated identity. Along a path $\mathbf{r}(t)$ it gives $d\mathbf{F}/dt = \partial_t \mathbf{F} + (\mathbf{v} \cdot \nabla)\mathbf{F}$.

3.5.5 Show that $\nabla(uv) = v\nabla u + u\nabla v$, where u and v are differentiable scalar functions of x, y, and z.

Solution.

The scalar product rule holds for each coordinate: $\partial_i(uv) = v\,\partial_i u + u\,\partial_i v$. Multiply by the constant Cartesian unit vector $\hat{\mathbf{e}}_i$ and sum:

$$\nabla(uv) = \sum_i \hat{\mathbf{e}}_i \partial_i (uv) = v \sum_i \hat{\mathbf{e}}_i \partial_i u + u \sum_i \hat{\mathbf{e}}_i \partial_i v = \boxed{v\nabla u + u\nabla v}.$$

3.5.6 For a particle moving in a circular orbit $\mathbf{r} = \hat{\mathbf{e}}_x r\cos\omega t + \hat{\mathbf{e}}_y r \sin\omega t$:

(a) Evaluate $\mathbf{r} \times \dot{\mathbf{r}}$, with $\dot{\mathbf{r}} = d\mathbf{r}/dt = \mathbf{v}$.

(b) Show that $\ddot{\mathbf{r}} + \omega^2 \mathbf{r} = 0$ with $\ddot{\mathbf{r}} = d\mathbf{v}/dt$.

Hint. The radius r and the angular velocity ω are constant.

ANS. (a) $\hat{\mathbf{e}}_z \omega r^2$.

Solution.

Differentiating, with r, ω constant,

$$\dot{\mathbf{r}} = \omega r(-\sin\omega t, \cos\omega t, 0).$$

Thus

$$\mathbf{r} \times \dot{\mathbf{r}} = \hat{\mathbf{e}}_z \omega r^2 (\cos^2\omega t + \sin^2\omega t) = \boxed{\hat{\mathbf{e}}_z \omega r^2}.$$

One further derivative gives $\ddot{\mathbf{r}} = -\omega^2 r(\cos\omega t, \sin\omega t, 0) = -\omega^2 \mathbf{r}$, proving $\boxed{\ddot{\mathbf{r}} + \omega^2 \mathbf{r} = 0}$.

3.5.7 Vector $\mathbf{A}$ satisfies the vector transformation law, Eq. (3.26). Show directly that its time derivative $d\mathbf{A}/dt$ also satisfies Eq. (3.26) and is therefore a vector. Use a fixed, time-independent orthogonal transformation.

Solution.

The transformation is a fixed, time-independent change of Cartesian axes. A frame rotating with time requires an additional term. For $A'_i = \sum_j S_{ij} A_j$ with constant S_{ij},

$$\frac{dA'_i}{dt} = \sum_j S_{ij} \frac{dA_j}{dt}.$$

This is precisely the same vector transformation law. If $S = S(t)$ instead, the product rule gives $\dot{\mathbf{A}}' = S\dot{\mathbf{A}} + \dot{S}\mathbf{A}$; it is not legitimate to omit the second term.

3.5.8 Show, by differentiating components, that

(a) $\dfrac{d}{dt}(\mathbf{A}\cdot\mathbf{B}) = \dfrac{d\mathbf{A}}{dt}\cdot\mathbf{B} + \mathbf{A}\cdot\dfrac{d\mathbf{B}}{dt}$,

(b) $\dfrac{d}{dt}(\mathbf{A}\times\mathbf{B}) = \dfrac{d\mathbf{A}}{dt}\times\mathbf{B} + \mathbf{A}\times\dfrac{d\mathbf{B}}{dt}$,

just like the derivative of the product of two algebraic functions.

Solution.

For the dot product, differentiate its component sum:

$$\frac{d}{dt}(\mathbf{A}\cdot\mathbf{B}) = \sum_i \frac{d}{dt}(A_iB_i) = \sum_i (\dot{A}_iB_i + A_i\dot{B}_i) = \dot{\mathbf{A}}\cdot\mathbf{B} + \mathbf{A}\cdot\dot{\mathbf{B}}.$$

For each cross-product component,

$$\frac{d}{dt}(\mathbf{A}\times\mathbf{B})_i = \sum_{jk} \varepsilon_{ijk}(\dot{A}_jB_k + A_j\dot{B}_k) = (\dot{\mathbf{A}}\times\mathbf{B} + \mathbf{A}\times\dot{\mathbf{B}})_i.$$

The order of the cross-product factors is preserved, as required by its antisymmetry.

3.5.9 Prove $\nabla\cdot(\mathbf{a}\times\mathbf{b}) = \mathbf{b}\cdot(\nabla\times\mathbf{a}) - \mathbf{a}\cdot(\nabla\times\mathbf{b})$.

Hint. Treat as a scalar triple product.

Solution.

Apply the product rule with repeated indices summed:

$$\begin{aligned}\nabla\cdot(\mathbf{a}\times\mathbf{b}) &= \partial_i(\varepsilon_{ijk}a_jb_k)\\ &= \varepsilon_{ijk}(\partial_ia_j)b_k + \varepsilon_{ijk}a_j(\partial_ib_k)\\ &= b_k\varepsilon_{kij}\partial_ia_j - a_j\varepsilon_{jik}\partial_ib_k\\ &= \boxed{\mathbf{b}\cdot(\nabla\times\mathbf{a}) - \mathbf{a}\cdot(\nabla\times\mathbf{b})}.\end{aligned}$$

The first sign follows from the cyclic permutation $(i,j,k)\to(k,i,j)$; the second from swapping i,j.

3.5.10 Classically, orbital angular momentum is given by $\mathbf{L} = \mathbf{r}\times\mathbf{p}$, where $\mathbf{p}$ is the linear momentum. To go from classical mechanics to quantum mechanics, $\mathbf{p}$ is replaced (in units with $\hbar = 1$) by the operator $-i\nabla$. Show that the quantum mechanical angular momentum operator has Cartesian components

$$L_x = -i\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right), \quad L_y = -i\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z}\right), \quad L_z = -i\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right).$$

Solution.

Apply the operator to an arbitrary differentiable scalar f:

$$\mathbf{L}f = -i\,\mathbf{r}\times\nabla f = -i\begin{vmatrix}\hat{\mathbf{e}}_x & \hat{\mathbf{e}}_y & \hat{\mathbf{e}}_z\\ x & y & z\\ \partial_x f & \partial_y f & \partial_z f\end{vmatrix}.$$

Expanding gives $L_xf = -i(yf_z - zf_y)$, $L_yf = -i(zf_x - xf_z)$, and $L_zf = -i(xf_y - yf_x)$. Because f is arbitrary, the stated three operator formulas follow. The coordinate factors multiply on the left of the derivatives.

3.5.11 Using the angular momentum operators previously given, show that they satisfy commutation relations of the form

$$[L_x, L_y] \equiv L_x L_y - L_y L_x = iL_z$$

and hence

$$\mathbf{L} \times \mathbf{L} = i\mathbf{L}.$$

These commutation relations will be taken later as the defining relations of an angular momentum operator.

Solution.

Let f have commuting mixed second derivatives. The product rule gives

$$\begin{aligned} L_x L_y f &= -\{y f_x + yz f_{xz} - xy f_{zz} - z^2 f_{xy} + xz f_{yz}\}, \\ L_y L_x f &= -\{zy f_{xz} - z^2 f_{xy} - xy f_{zz} + x f_y + xz f_{yz}\}. \end{aligned}$$

All second-derivative terms cancel on subtraction, leaving

$$[L_x, L_y] f = x f_y - y f_x = iL_z f.$$

Cyclic relabeling of x, y, z in this explicit computation gives $[L_y, L_z] = iL_x$ and $[L_z, L_x] = iL_y$. Thus

$$\mathbf{L} \times \mathbf{L} = \big([L_y, L_z], [L_z, L_x], [L_x, L_y]\big) = \boxed{i\mathbf{L}}.$$

Unlike ordinary numbers, the components here do not commute, so a vector operator need not have zero cross product with itself.

In the manual's last line on p. 39, the coefficient of ∂_x is misprinted as $-u$; it is $-y$, as the preceding subtraction shows.

3.5.12 With the aid of the results of Exercise 3.5.11, show that if two vectors $\mathbf{a}$ and $\mathbf{b}$ commute with each other and with $\mathbf{L}$, that is, $[\mathbf{a}, \mathbf{b}] = [\mathbf{a}, \mathbf{L}] = [\mathbf{b}, \mathbf{L}] = 0$, show that

$$[\mathbf{a} \cdot \mathbf{L}, \mathbf{b} \cdot \mathbf{L}] = i(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{L}.$$

Solution.

Interpret the commuting hypotheses componentwise. They permit a_i, b_j to be moved through each other and through every L_k:

$$\begin{aligned} [\mathbf{a} \cdot \mathbf{L}, \mathbf{b} \cdot \mathbf{L}] &= \sum_{ij} (a_i L_i b_j L_j - b_j L_j a_i L_i) \\ &= \sum_{ij} a_i b_j (L_i L_j - L_j L_i) \\ &= i \sum_{ijk} a_i b_j \varepsilon_{ijk} L_k = \boxed{i(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{L}}. \end{aligned}$$

The last equality uses $\varepsilon_{ijk} = \varepsilon_{kij}$.

3.5.13 Prove that the stream lines of $\mathbf{b}$ in Example 3.5.7 with nonzero distance from the z-axis are counterclockwise circles; points on the z-axis are stationary.

Solution.

Example 3.5.7 defines $\mathbf{b} = (-y, x, 0)$. A streamline parametrized in the flow direction obeys $dx/ds = -y$, $dy/ds = x$, $dz/ds = 0$. Therefore

$$\frac{d}{ds}(x^2 + y^2) = 2x(-y) + 2yx = 0, \qquad z = z_0.$$

Its nonconstant curves are circles $x^2 + y^2 = R^2$ in planes $z = z_0$. Explicitly $x = R\cos(s + s_0)$, $y = R\sin(s + s_0)$; increasing s is counterclockwise as viewed from $+z$. At $(R, 0)$ the velocity is $(0, R)$, confirming the sense. On the z-axis the field vanishes, so these are stationary points rather than nonzero-radius circular streamlines.

3.6 Differential Vector Operators: Further Properties

3.6.1 Show that $\mathbf{u} \times \mathbf{v}$ is solenoidal if $\mathbf{u}$ and $\mathbf{v}$ are each irrotational.

Solution.

Irrotationality means $\nabla \times \mathbf{u} = \nabla \times \mathbf{v} = 0$. By the product rule proved in 3.5.9,

$$\nabla \cdot (\mathbf{u} \times \mathbf{v}) = \mathbf{v} \cdot (\nabla \times \mathbf{u}) - \mathbf{u} \cdot (\nabla \times \mathbf{v}) = 0 - 0 = 0.$$

Thus $\mathbf{u} \times \mathbf{v}$ is solenoidal.

3.6.2 If $\mathbf{A}$ is irrotational, show that $\mathbf{A} \times \mathbf{r}$ is solenoidal.

Solution.

In Cartesian coordinates $\mathbf{r} = (x, y, z)$, and each component of $\nabla \times \mathbf{r}$ is a difference of zero cross derivatives. Therefore

$$\nabla \cdot (\mathbf{A} \times \mathbf{r}) = \mathbf{r} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{r}) = 0.$$

The first term vanishes by the hypothesis, the second by the calculation.

3.6.3 A rigid body is rotating with constant angular velocity $\boldsymbol{\omega}$. Show that the linear velocity $\mathbf{v}$ is solenoidal.

Solution.

The velocity is $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$, or

$$\mathbf{v} = (\omega_y z - \omega_z y,\ \omega_z x - \omega_x z,\ \omega_x y - \omega_y x).$$

Each component is independent of its matching coordinate when $\boldsymbol{\omega}$ is spatially constant. Hence $\nabla \cdot \mathbf{v} = \partial_x v_x + \partial_y v_y + \partial_z v_z = 0$. A spatially uniform translational velocity can be added without changing this result.

3.6.4 If a vector function $\mathbf{V}(x, y, z)$ is not irrotational, show that if there exists a nowhere-zero scalar function $g(x, y, z)$ such that $g\mathbf{V}$ is irrotational, then

$$\mathbf{V} \cdot \nabla \times \mathbf{V} = 0.$$

Solution.

The integrating factor must be nonzero on the region where the conclusion is asserted; allowing $g \equiv 0$ makes the statement false. The product rule gives

$$0 = \nabla \times (g\mathbf{V}) = \nabla g \times \mathbf{V} + g\, \nabla \times \mathbf{V}.$$

Dot with $\mathbf{V}$. The scalar triple product with two factors $\mathbf{V}$ is zero, so $g\, \mathbf{V} \cdot (\nabla \times \mathbf{V}) = 0$. Where $g \neq 0$, division proves $\boxed{\mathbf{V} \cdot (\nabla \times \mathbf{V}) = 0}$. To check why the qualification is necessary, take $\mathbf{V} = (-y, x, 1)$. Then $\nabla \times \mathbf{V} = (0, 0, 2)$ and $\mathbf{V} \cdot \nabla \times \mathbf{V} = 2$, yet $g\mathbf{V} = 0$ is irrotational when $g = 0$.

3.6.5 Verify the vector identity

$$\nabla \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} - \mathbf{B}(\nabla \cdot \mathbf{A}) + \mathbf{A}(\nabla \cdot \mathbf{B}).$$

Solution.

For component i, contract the Levi-Civita symbols:

$$\begin{aligned}
[\nabla \times (\mathbf{A} \times \mathbf{B})]_i &= \varepsilon_{ijk} \partial_j (\varepsilon_{k\ell m} A_\ell B_m) \\
&= (\delta_{i\ell}\delta_{jm} - \delta_{im}\delta_{j\ell}) \partial_j (A_\ell B_m) \\
&= \partial_j (A_i B_j) - \partial_j (A_j B_i) \\
&= B_j \partial_j A_i + A_i \partial_j B_j - B_i \partial_j A_j - A_j \partial_j B_i.
\end{aligned}$$

The four terms are respectively the i-components of $(\mathbf{B} \cdot \nabla)\mathbf{A}$, $\mathbf{A}(\nabla \cdot \mathbf{B})$, $-\mathbf{B}(\nabla \cdot \mathbf{A})$, and $-(\mathbf{A} \cdot \nabla)\mathbf{B}$, proving the identity.

3.6.6 As an alternative to the vector identity of Example 3.6.4 show that

$$\nabla(\mathbf{A} \cdot \mathbf{B}) = (\mathbf{A} \times \nabla) \times \mathbf{B} + (\mathbf{B} \times \nabla) \times \mathbf{A} + \mathbf{A}(\nabla \cdot \mathbf{B}) + \mathbf{B}(\nabla \cdot \mathbf{A}).$$

Solution.

The derivatives in $(\mathbf{A} \times \nabla) \times \mathbf{B}$ act on $\mathbf{B}$ to their right, not on the coefficient $\mathbf{A}$. Thus

$$\begin{aligned}
[(\mathbf{A} \times \nabla) \times \mathbf{B}]_i &= \varepsilon_{ijk} \varepsilon_{j\ell m} A_\ell \partial_m B_k \\
&= (\delta_{im}\delta_{k\ell} - \delta_{i\ell}\delta_{km}) A_\ell \partial_m B_k \\
&= A_k \partial_i B_k - A_i \partial_k B_k.
\end{aligned}$$

Interchanging A, B gives $B_k \partial_i A_k - B_i \partial_k A_k$. Adding the i-components of $\mathbf{A}(\nabla \cdot \mathbf{B}) + \mathbf{B}(\nabla \cdot \mathbf{A})$ cancels the two divergence terms. What remains is $A_k \partial_i B_k + B_k \partial_i A_k = \partial_i (A_k B_k)$, the i-component of $\nabla(\mathbf{A} \cdot \mathbf{B})$.

3.6.7 Verify the identity

$$\mathbf{A} \times (\nabla \times \mathbf{A}) = \frac{1}{2}\nabla(A^2) - (\mathbf{A} \cdot \nabla)\mathbf{A}.$$

Solution.

For each component,

$$\begin{aligned}
[\mathbf{A} \times (\nabla \times \mathbf{A})]_i &= \varepsilon_{ijk}\varepsilon_{k\ell m} A_j \partial_\ell A_m \\
&= A_j \partial_i A_j - A_j \partial_j A_i \\
&= \tfrac{1}{2}\partial_i \left(\sum_j A_j^2 \right) - (\mathbf{A} \cdot \nabla) A_i.
\end{aligned}$$

Here $\partial_i(A_j^2) = 2A_j \partial_i A_j$. Collecting components gives the stated identity.

3.6.8 If $\mathbf{A}$ and $\mathbf{B}$ are constant vectors, show that

$$\nabla(\mathbf{A} \cdot \mathbf{B} \times \mathbf{r}) = \mathbf{A} \times \mathbf{B}.$$

Solution.

Cyclicity changes the scalar triple product to $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{r}) = (\mathbf{A} \times \mathbf{B}) \cdot \mathbf{r}$. Write the constant vector $\mathbf{A} \times \mathbf{B} = \mathbf{C}$. Then $\nabla(C_x x + C_y y + C_z z) = (C_x, C_y, C_z) = \mathbf{C}$, which proves the result. Constancy matters because otherwise derivatives of $\mathbf{C}$ would also appear.

3.6.9 Verify Eq. (3.70),

$$\nabla \times (\nabla \times \mathbf{V}) = \nabla(\nabla \cdot \mathbf{V}) - \nabla \cdot \nabla \mathbf{V},$$

by direct expansion in Cartesian coordinates.

Solution.

Assume the components have commuting mixed second derivatives. Directly, the x-component of the double curl is

$$\begin{aligned}
[\nabla \times (\nabla \times \mathbf{V})]_x &= \partial_y(\partial_x V_y - \partial_y V_x) - \partial_z(\partial_z V_x - \partial_x V_z) \\
&= \partial_x \partial_y V_y + \partial_x \partial_z V_z - \partial_y^2 V_x - \partial_z^2 V_x \\
&= \partial_x(\partial_x V_x + \partial_y V_y + \partial_z V_z) - (\partial_x^2 + \partial_y^2 + \partial_z^2) V_x.
\end{aligned}$$

Cyclic replacement gives $[\nabla \times (\nabla \times \mathbf{V})]_y = \partial_y(\nabla \cdot \mathbf{V}) - \nabla^2 V_y$ and $[\nabla \times (\nabla \times \mathbf{V})]_z = \partial_z(\nabla \cdot \mathbf{V}) - \nabla^2 V_z$. Together these are $\nabla(\nabla \cdot \mathbf{V}) - \nabla^2 \mathbf{V}$, where the Cartesian vector Laplacian acts componentwise.

3.6.10 Prove that $\nabla \times (\varphi \nabla \varphi) = 0$.

Solution.

The product rule gives $\varphi \nabla \varphi = \frac{1}{2}\nabla(\varphi^2)$. For a C^2 scalar f, the i-component of $\nabla \times \nabla f$ is $\varepsilon_{ijk}\partial_j \partial_k f = 0$, since the derivatives are symmetric and ε_{ijk} antisymmetric. Hence $\nabla \times (\varphi \nabla \varphi) = \frac{1}{2}\nabla \times \nabla(\varphi^2) = 0$.

3.6.11 You are given that the curl of $\mathbf{F}$ equals the curl of $\mathbf{G}$. Show that $\mathbf{F}$ and $\mathbf{G}$ may differ by (a) a constant and (b) a gradient of a scalar function.

Solution.

Put $\mathbf{H} = \mathbf{F} - \mathbf{G}$. The hypothesis is $\nabla \times \mathbf{H} = 0$. A constant vector $\mathbf{c}$ has zero derivatives, so $\mathbf{F} = \mathbf{G} + \mathbf{c}$ has the same curl as $\mathbf{G}$. Likewise $\nabla \times \nabla f = 0$, so $\mathbf{F} = \mathbf{G} + \nabla f$ has the same curl. The constant case is included by $f = \mathbf{c} \cdot \mathbf{r}$.

If one further asserts that every such difference is a globally single-valued gradient, a simply connected domain is sufficient and topology must be considered. On that domain set $f(\mathbf{r}) = \int_{\mathbf{r}_0}^{\mathbf{r}} \mathbf{H} \cdot d\boldsymbol{\ell}$; Stokes' theorem makes this path-independent and differentiation gives $\nabla f = \mathbf{H}$. On the punctured plane, $\mathbf{H} = (-y, x, 0)/(x^2 + y^2)$ has zero curl but circulation 2π, so it is not a global single-valued gradient there.

3.6.12 The Navier-Stokes equation of hydrodynamics contains a nonlinear term of the form $(\mathbf{v} \cdot \nabla)\mathbf{v}$. Show that the curl of this term may be written as $-\nabla \times [\mathbf{v} \times (\nabla \times \mathbf{v})]$.

Solution.

Exercise 3.6.7 with $\mathbf{A} = \mathbf{v}$ gives

$$(\mathbf{v} \cdot \nabla)\mathbf{v} = \tfrac{1}{2}\nabla(\mathbf{v} \cdot \mathbf{v}) - \mathbf{v} \times (\nabla \times \mathbf{v}).$$

Take the curl. The gradient term has zero curl, so

$$\boxed{\nabla \times [(\mathbf{v} \cdot \nabla)\mathbf{v}] = -\nabla \times [\mathbf{v} \times (\nabla \times \mathbf{v})]}.$$

No incompressibility assumption is needed.

The manual, p. 41, omits the factor $1/2$ in front of $\nabla(\mathbf{v} \cdot \mathbf{v})$. For $\mathbf{v} = (x, 0, 0)$, the correct difference is $(x, 0, 0) - (x, 0, 0) = 0$; without $1/2$ it is $(x, 0, 0)$. Taking a curl hides this error because either gradient has zero curl.

3.6.13 Prove that $(\nabla u) \times (\nabla v)$ is solenoidal, where u and v are C^2 scalar functions.

Solution.

For a classical pointwise calculation assume u, v are C^2, so the required mixed derivatives exist and commute. Using 3.5.9,

$$\nabla \cdot (\nabla u \times \nabla v) = \nabla v \cdot (\nabla \times \nabla u) - \nabla u \cdot (\nabla \times \nabla v) = 0.$$

Each curl of a gradient vanishes by equality of mixed partial derivatives, so the cross product is solenoidal.

3.6.14 The function φ is a scalar satisfying Laplace's equation, $\nabla^2 \varphi = 0$. Show that $\nabla\varphi$ is both solenoidal and irrotational.

Solution.

For $\mathbf{A} = \nabla\varphi$, $\nabla \cdot \mathbf{A} = \nabla^2\varphi = 0$ by the given equation. Also

$$(\nabla \times \mathbf{A})_i = \varepsilon_{ijk}\partial_j\partial_k\varphi = 0$$

by commuting mixed derivatives. Thus the gradient is both solenoidal and irrotational.

3.6.15 Show that any solution of the equation

$$\nabla \times (\nabla \times \mathbf{A}) - k^2 \mathbf{A} = 0$$

automatically satisfies the vector Helmholtz equation

$$\nabla^2 \mathbf{A} + k^2 \mathbf{A} = 0$$

and the solenoidal condition

$$\nabla \cdot \mathbf{A} = 0.$$

Hint. Let $\nabla\cdot$ operate on the first equation. Assume that k is constant and $k \neq 0$.

Solution.

The conclusion requires a constant $k \neq 0$. Taking divergence of the first equation gives $0 - k^2 \nabla \cdot \mathbf{A} = 0$, since the divergence of a curl vanishes. Division by k^2 gives $\nabla \cdot \mathbf{A} = 0$. Substitution into the double-curl identity then yields

$$0 = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} - k^2 \mathbf{A} = -\nabla^2 \mathbf{A} - k^2 \mathbf{A}.$$

This is the required vector Helmholtz equation. For $k = 0$, take $\mathbf{A} = \nabla(x^3) = (3x^2, 0, 0)$. Its curl and double curl vanish, but $\nabla \cdot \mathbf{A} = 6x$ and $\nabla^2 \mathbf{A} = (6, 0, 0)$; both proposed conclusions fail in general.

3.6.16 The theory of heat conduction leads to an equation

$$\nabla^2 \Psi = k \left| \nabla \Phi \right|^2,$$

where Φ is a potential satisfying Laplace's equation: $\nabla^2 \Phi = 0$. Show that a solution of this equation is $\Psi = k\Phi^2/2$. Take Φ real-valued and k constant.

Solution.

For the real potential Φ and constant k, differentiate $\Psi = k\Phi^2/2$:

$$\partial_i \Psi = k\Phi \, \partial_i \Phi, \qquad \partial_i^2 \Psi = k(\partial_i \Phi)^2 + k\Phi \, \partial_i^2 \Phi.$$

Summing over the three coordinates gives

$$\nabla^2 \Psi = k|\nabla \Phi|^2 + k\Phi \nabla^2 \Phi = \boxed{k|\nabla \Phi|^2}.$$

Thus it is a solution. One can add any harmonic function to Ψ because its Laplacian contributes zero.

3.6.17 Given the three matrices

$$\mathsf{M}_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad \mathsf{M}_y = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix},$$

and

$$\mathsf{M}_z = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

show that the matrix-vector equation

$$\left(\mathsf{M}\cdot\nabla+\mathbf{1}_3\frac{1}{c}\frac{\partial}{\partial t}\right)\boldsymbol{\psi}=0$$

reproduces the two curl equations of Maxwell's vacuum system. The full system additionally requires $\nabla\cdot\mathbf{E}=\nabla\cdot\mathbf{B}=0$ initially. Here $\boldsymbol{\psi}$ is a column vector with components $\psi_j=B_j-iE_j/c$, $j=x,y,z$. Note that $\epsilon_0\mu_0=1/c^2$ and that $\mathbf{1}_3$ is the 3×3 unit matrix.

Solution.

The matrix differential operator is

$$\mathsf{M}\cdot\nabla=\begin{pmatrix}0 & -i\partial_z & i\partial_y\\ i\partial_z & 0 & -i\partial_x\\ -i\partial_y & i\partial_x & 0\end{pmatrix}.$$

Its action on a column $\boldsymbol{\psi}$ is $i\nabla\times\boldsymbol{\psi}$. The given equation is therefore $i\nabla\times\boldsymbol{\psi}+c^{-1}\partial_t\boldsymbol{\psi}=0$. Insert $\boldsymbol{\psi}=\mathbf{B}-i\mathbf{E}/c$ and separate real and imaginary parts:

$$\frac{1}{c}(\nabla\times\mathbf{E}+\partial_t\mathbf{B})+i\left(\nabla\times\mathbf{B}-\frac{1}{c^2}\partial_t\mathbf{E}\right)=0.$$

For real fields this gives

$$\boxed{\nabla\times\mathbf{E}=-\partial_t\mathbf{B},\qquad\nabla\times\mathbf{B}=\frac{1}{c^2}\partial_t\mathbf{E}}.$$

The matrix equation supplies the two vacuum curl equations. The full vacuum Maxwell system also needs $\nabla\cdot\mathbf{E}=\nabla\cdot\mathbf{B}=0$ as initial constraints; they are not consequences of the matrix equation alone. Taking divergence shows $\partial_t(\nabla\cdot\mathbf{B})=\partial_t(\nabla\cdot\mathbf{E})=0$, so initial zero divergences are preserved. As a counterexample without those initial conditions, the static fields $\mathbf{B}=\mathbf{r}$, $\mathbf{E}=0$ solve the matrix equation but have $\nabla\cdot\mathbf{B}=3$.

The manual, p. 42, labels its full operator matrix $\mathsf{M}\cdot\nabla$ even though it includes $c^{-1}\partial_t I$. After multiplying that equation by c, the imaginary part is $c[\nabla\times\mathbf{B}-c^{-2}\partial_t\mathbf{E}]$, so its displayed imaginary bracket is also missing an overall factor c. These notation and normalization errors do not change the two zero curl-equation residuals.

3.6.18 Using the Pauli matrices σ_i of Eq. (2.28), show that

$$(\boldsymbol{\sigma}\cdot\mathbf{a})(\boldsymbol{\sigma}\cdot\mathbf{b})=(\mathbf{a}\cdot\mathbf{b})\mathbf{1}_2+i\boldsymbol{\sigma}\cdot(\mathbf{a}\times\mathbf{b}).$$

Here

$$\boldsymbol{\sigma}\equiv\hat{\mathbf{e}}_x\sigma_1+\hat{\mathbf{e}}_y\sigma_2+\hat{\mathbf{e}}_z\sigma_3,$$

$\mathbf{a}$ and $\mathbf{b}$ are ordinary vectors, and $\mathbf{1}_2$ is the 2×2 unit matrix.

Solution.

Expand and use the Pauli multiplication law:

$$\begin{aligned}(\boldsymbol{\sigma}\cdot\mathbf{a})(\boldsymbol{\sigma}\cdot\mathbf{b}) &= \sum_{ij} a_i b_j \sigma_i \sigma_j \\ &= \sum_{ij} a_i b_j \left(\delta_{ij} I_2 + i\sum_k \varepsilon_{ijk}\sigma_k\right) \\ &= \left(\sum_i a_i b_i\right) I_2 + i\sum_k \sigma_k \left(\sum_{ij} \varepsilon_{kij} a_i b_j\right) \\ &= \boxed{(\mathbf{a}\cdot\mathbf{b})I_2 + i\boldsymbol{\sigma}\cdot(\mathbf{a}\times\mathbf{b})}.\end{aligned}$$

The ordinary vector components commute with the Pauli matrices.

3.7 Vector Integration

3.7.1 The origin and the three vectors $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ (all of which start at the origin) define a tetrahedron. Taking the outward direction as positive, calculate the total vector area of the four tetrahedral surfaces.

Solution.

Suppose first $\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) > 0$. The outward vector areas of faces OAB, OBC, OCA are respectively $-\mathbf{A}\times\mathbf{B}/2$, $-\mathbf{B}\times\mathbf{C}/2$, $-\mathbf{C}\times\mathbf{A}/2$; their negatives would point toward the opposite vertex. For face ABC the outward vector area is

$$\frac{1}{2}(\mathbf{B}-\mathbf{A})\times(\mathbf{C}-\mathbf{A}) = \frac{1}{2}(\mathbf{B}\times\mathbf{C}+\mathbf{C}\times\mathbf{A}+\mathbf{A}\times\mathbf{B}).$$

Adding all four cancels every term: $\boxed{\sum_{\text{faces}} \boldsymbol{\sigma} = 0}$. For negative scalar triple product, every outward orientation reverses relative to these expressions, and the sum remains zero. This is the sum of directed areas; the sum of their positive magnitudes does not vanish.

The manual, p. 42, sums the positive scalar face areas instead of the requested vector areas. For the tetrahedron with vertices $0, \hat{\mathbf{e}}_x, \hat{\mathbf{e}}_y, \hat{\mathbf{e}}_z$, its scalar total is $(3+\sqrt{3})/2$, while the outward vector areas still sum to zero.

3.7.2 Find the work $-\int_C \mathbf{F}\cdot d\mathbf{r}$ done moving on a unit circle in the xy-plane, doing work against a force field given by

$$\mathbf{F} = \frac{-\hat{\mathbf{e}}_x y}{x^2+y^2} + \frac{\hat{\mathbf{e}}_y x}{x^2+y^2}:$$

(a) Counterclockwise from 0 to π,

(b) Clockwise from 0 to $-\pi$.

Note that the work done depends on the path.

Solution.

These paths are semicircles, not closed loops. Also, work against the specified field is $-\int_C \mathbf{F}\cdot d\mathbf{r}$; $\int_C \mathbf{F}\cdot d\mathbf{r}$ is the work done by the field. Parametrize the unit circle by $\mathbf{r}(\varphi) = (\cos\varphi, \sin\varphi, 0)$. Then $\mathbf{F} = (-\sin\varphi, \cos\varphi, 0)$ and $d\mathbf{r} = (-\sin\varphi, \cos\varphi, 0)d\varphi$, so $\mathbf{F}\cdot d\mathbf{r} = d\varphi$. Thus the work by the field is

$$W_F^{(a)} = \int_0^{\pi} d\varphi = \pi, \qquad W_F^{(b)} = \int_0^{-\pi} d\varphi = -\pi.$$

The requested work against it is consequently

$$\boxed{W^{(a)}_{\text{against}} = -\pi, \qquad W^{(b)}_{\text{against}} = +\pi}.$$

Both paths join $(1,0)$ to $(-1,0)$ but give different work.

3.7.3 Calculate the work you do in going from point $(1,1)$ to point $(3,3)$. The force you exert is given by

$$\mathbf{F} = \hat{\mathbf{e}}_x(x-y) + \hat{\mathbf{e}}_y(x+y).$$

Specify clearly the path you choose. Note that this force field is nonconservative.

Solution.

Choose the straight path $x = y = t$, $1 \le t \le 3$. On this path, $\mathbf{F} = (0, 2t)$ and $d\mathbf{r} = (1,1)dt$, giving

$$\boxed{W = \int_1^3 2t\,dt = [t^2]_1^3 = 8}.$$

To exhibit path dependence, instead go horizontally from $(1,1)$ to $(3,1)$, then vertically to $(3,3)$:

$$W = \int_1^3 (x-1)\,dx + \int_1^3 (3+y)\,dy = 2 + 10 = 12.$$

The curl is $\hat{\mathbf{e}}_z[\partial_x(x+y) - \partial_y(x-y)] = 2\hat{\mathbf{e}}_z$, consistent with nonconservative behavior.

3.7.4 Evaluate $\oint \mathbf{r}\cdot d\mathbf{r}$ for a closed path of your choosing.

Solution.

Since $d(\mathbf{r}\cdot\mathbf{r}) = 2\mathbf{r}\cdot d\mathbf{r}$,

$$\oint_C \mathbf{r}\cdot d\mathbf{r} = \frac{1}{2}\oint_C d(r^2) = \frac{1}{2}(r^2_{\text{final}} - r^2_{\text{initial}}) = \boxed{0}.$$

This holds for every piecewise smooth closed path. For a concrete choice, a circle $\mathbf{r} = (R\cos t, R\sin t, 0)$ has $\mathbf{r}\cdot d\mathbf{r} = R^2(-\cos t\sin t + \sin t\cos t)dt = 0$ pointwise.

3.7.5 Evaluate

$$\frac{1}{3}\int_s \mathbf{r}\cdot d\boldsymbol{\sigma}$$

over the unit cube defined by the point $(0,0,0)$ and the unit intercepts on the positive x-, y-, and z-axes. Note that $\mathbf{r}\cdot d\boldsymbol{\sigma}$ is zero for three of the surfaces and that each of the three remaining surfaces contributes the same amount to the integral.

Solution.

Use outward normals. On $x=0, y=0, z=0$, respectively, $\mathbf{r}\cdot\hat{\mathbf{n}} = -x, -y, -z = 0$. On each of $x=1, y=1, z=1$, it is 1, and the face area is 1. Thus

$$\boxed{\frac{1}{3}\oint_S \mathbf{r}\cdot d\boldsymbol{\sigma} = \frac{1}{3}(0+0+0+1+1+1) = 1}.$$

As a volume check, $\nabla\cdot\mathbf{r} = 3$ and the cube has volume 1.

3.8 Integral Theorems

3.8.1 Using Gauss' theorem, prove that

$$\oint_S d\boldsymbol{\sigma} = 0$$

if $S = \partial V$ is a closed surface.

Solution.

For each constant Cartesian unit vector $\hat{\mathbf{e}}_i$, Gauss' theorem gives

$$\hat{\mathbf{e}}_i \cdot \oint_S d\boldsymbol{\sigma} = \oint_S \hat{\mathbf{e}}_i \cdot d\boldsymbol{\sigma} = \int_V \nabla\cdot\hat{\mathbf{e}}_i \, d\tau = 0.$$

All three components vanish, so $\boxed{\oint_S d\boldsymbol{\sigma} = 0}$. The surface is oriented outward.

3.8.2 Show that

$$\frac{1}{3}\oint_S \mathbf{r}\cdot d\boldsymbol{\sigma} = V,$$

where V is the volume enclosed by the closed surface $S = \partial V$.

Note. This is a generalization of Exercise 3.7.5.

Solution.

Apply Gauss' theorem to $\mathbf{r} = (x, y, z)$:

$$\frac{1}{3}\oint_S \mathbf{r}\cdot d\boldsymbol{\sigma} = \frac{1}{3}\int_V (\partial_x x + \partial_y y + \partial_z z)\, d\tau = \frac{1}{3}\int_V 3\, d\tau = \boxed{V}.$$

The result is independent of the position of the origin because adding a constant to $\mathbf{r}$ adds zero flux by 3.8.1.

3.8.3 If $\mathbf{B} = \nabla\times\mathbf{A}$, show that

$$\oint_S \mathbf{B}\cdot d\boldsymbol{\sigma} = 0$$

for any closed surface S. Assume S bounds a region on a neighborhood of which $\mathbf{A}$ is a single-valued C^2 field.

Solution.

Assume $\mathbf{A}$ is a single-valued C^2 field on a neighborhood of the enclosed region; singular potentials require separate treatment. The divergence of its curl is $\varepsilon_{ijk}\partial_i\partial_j A_k = 0$, since mixed derivatives are symmetric in i, j. Therefore Gauss' theorem gives

$$\boxed{\oint_S \mathbf{B}\cdot d\boldsymbol{\sigma} = \int_V \nabla\cdot(\nabla\times\mathbf{A})\, d\tau = 0}.$$

3.8.4 From Eq. (3.72), with $\mathbf{V}$ the electric field $\mathbf{E}$ and f the electrostatic potential φ, show that, for integration over all space,

$$\int \rho\varphi \, d\tau = \epsilon_0 \int E^2 \, d\tau.$$

This corresponds to a 3-D integration by parts.

Hint. $\mathbf{E} = -\nabla\varphi$, $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$. You may assume that φ vanishes at large r at least as fast as r^{-1}. Assume both volume integrals converge and $\lim_{R\to\infty} \oint_{r=R} \varphi\mathbf{E} \cdot d\boldsymbol{\sigma} = 0$. For a regular localized charge distribution, $\varphi = O(R^{-1})$ and $\mathbf{E} = O(R^{-2})$ suffice for this boundary condition.

Solution.

The product rule and the electrostatic equations give

$$\nabla \cdot (\varphi\mathbf{E}) = \nabla\varphi \cdot \mathbf{E} + \varphi\nabla \cdot \mathbf{E} = -E^2 + \frac{\rho\varphi}{\epsilon_0}.$$

Integrate over a ball of radius R and apply Gauss' theorem:

$$\int_{r<R} \rho\varphi \, d\tau = \epsilon_0 \int_{r<R} E^2 \, d\tau + \epsilon_0 \oint_{r=R} \varphi\mathbf{E} \cdot d\boldsymbol{\sigma}.$$

The all-space identity requires convergence and a vanishing outer surface term. For a localized regular charge distribution, $\varphi = O(R^{-1})$ and $\mathbf{E} = O(R^{-2})$ ensure this; ideal point charges have divergent self-energy integrals. Under these conditions the surface term is $O(R^{-1})$ because its area is $4\pi R^2$. Taking $R \to \infty$ gives $\boxed{\int \rho\varphi \, d\tau = \epsilon_0 \int E^2 \, d\tau}$.

3.8.5 A particular steady-state electric current distribution is localized in space. Choosing a bounding surface far enough out so that the current density $\mathbf{J}$ is zero everywhere on the surface, show that

$$\int \mathbf{J} \, d\tau = 0.$$

Hint. Take one component of $\mathbf{J}$ at a time. With $\nabla \cdot \mathbf{J} = 0$, show that $J_i = \nabla \cdot (x_i\mathbf{J})$ and apply Gauss' theorem.

Solution.

Steady current obeys $\nabla \cdot \mathbf{J} = 0$. For each component,

$$\nabla \cdot (x_i\mathbf{J}) = \sum_j [(\partial_j x_i)J_j + x_i\partial_j J_j] = J_i + x_i\nabla \cdot \mathbf{J} = J_i.$$

Hence $\int_V J_i \, d\tau = \oint_{\partial V} x_i\mathbf{J} \cdot d\boldsymbol{\sigma} = 0$ because $\mathbf{J} = 0$ on the chosen boundary. All three components are zero, giving $\boxed{\int_V \mathbf{J} \, d\tau = 0}$.

3.8.6 Given a vector $\mathbf{t} = -\hat{\mathbf{e}}_x y + \hat{\mathbf{e}}_y x$, show, with the help of Stokes' theorem, that the integral of $\mathbf{t}$ around a simple, piecewise smooth, counterclockwise oriented closed curve in the xy-plane satisfies

$$\frac{1}{2} \oint \mathbf{t} \cdot d\boldsymbol{\lambda} = \frac{1}{2} \oint (x \, dy - y \, dx) = A,$$

where A is the area enclosed by the curve.

Solution.

Take a simple, piecewise smooth loop oriented counterclockwise as viewed from $+z$. Reversing orientation changes the sign; a self-intersecting loop gives signed area with multiplicity. For $\mathbf{t} = (-y, x, 0)$,

$$\nabla \times \mathbf{t} = (0, 0, \partial_x x - \partial_y(-y)) = 2\hat{\mathbf{e}}_z.$$

Stokes' theorem with upward surface normal gives

$$\frac{1}{2}\oint_C \mathbf{t} \cdot d\boldsymbol{\lambda} = \frac{1}{2}\int_S 2\hat{\mathbf{e}}_z \cdot \hat{\mathbf{e}}_z \, dA = \boxed{A}.$$

Since $\mathbf{t} \cdot d\boldsymbol{\lambda} = x\,dy - y\,dx$, this also proves the second displayed expression.

3.8.7 The calculation of the magnetic moment of a current loop leads to the line integral

$$\oint \mathbf{r} \times d\mathbf{r}.$$

(a) Integrate around the perimeter of a current loop (in the xy-plane) and show that the scalar magnitude of this line integral is twice the area of the enclosed surface.

(b) The perimeter of an ellipse is described by $\mathbf{r} = \hat{\mathbf{e}}_x a\cos\theta + \hat{\mathbf{e}}_y b\sin\theta$. From part (a) show that the area of the ellipse is πab.

In (a), take a simple, piecewise smooth loop; in (b), take $a, b > 0$.

Solution.

(a) In the xy-plane, $\mathbf{r} \times d\mathbf{r} = (x\,dy - y\,dx)\hat{\mathbf{e}}_z$. For a simple positively oriented loop, 3.8.6 gives

$$\oint_C \mathbf{r} \times d\mathbf{r} = 2A\hat{\mathbf{e}}_z,$$

so the magnitude is $2A$. Reverse traversal reverses the vector but not its magnitude. **(b)** For the ellipse with $a, b > 0$, $dx = -a\sin\theta\,d\theta$ and $dy = b\cos\theta\,d\theta$. Then $x\,dy - y\,dx = ab(\cos^2\theta + \sin^2\theta)d\theta = ab\,d\theta$. Integrating $0 \le \theta \le 2\pi$ gives $2\pi ab = 2A$, hence $\boxed{A = \pi ab}$.

3.8.8 Evaluate $\oint \mathbf{r} \times d\mathbf{r}$ by using the alternate form of Stokes' theorem given by Eq. (3.95):

$$\int_S (d\boldsymbol{\sigma} \times \nabla) \times \mathbf{P} = \oint d\boldsymbol{\lambda} \times \mathbf{P}.$$

Take the loop to be entirely in the xy-plane.

Solution.

Take $\mathbf{P} = \mathbf{r}$ in the alternate Stokes formula. Treat the normal area element as the coefficient of the differential operator. Component contraction gives

$$[(d\boldsymbol{\sigma} \times \nabla) \times \mathbf{r}]_i = d\sigma_j\,\partial_i r_j - d\sigma_i\,\partial_j r_j = d\sigma_i - 3d\sigma_i = -2d\sigma_i.$$

Thus $\oint d\mathbf{r} \times \mathbf{r} = -2\int_S d\boldsymbol{\sigma}$. Reversing the order of the cross product gives, for upward normal,

$$\boxed{\oint \mathbf{r} \times d\mathbf{r} = 2A\hat{\mathbf{e}}_z}.$$

The sign is reversed for clockwise traversal.

3.8.9 Prove that

$$\oint u\nabla v \cdot d\boldsymbol{\lambda} = -\oint v\nabla u \cdot d\boldsymbol{\lambda}.$$

Assume u, v are single-valued C^1 functions along the closed, piecewise smooth path.

Solution.

By the product rule, $u\nabla v + v\nabla u = \nabla(uv)$. Its closed line integral is

$$\oint_C (u\nabla v + v\nabla u) \cdot d\boldsymbol{\lambda} = \oint_C d(uv) = 0$$

for single-valued differentiable u, v. Move the second integral to the right to obtain the stated integration-by-parts identity.

3.8.10 Prove that

$$\oint u\nabla v \cdot d\boldsymbol{\lambda} = \int_S (\nabla u) \times (\nabla v) \cdot d\boldsymbol{\sigma}.$$

Here the line integral is over the positively oriented boundary ∂S of an oriented, piecewise smooth surface, and u, v are C^2 on a neighborhood of S.

Solution.

Apply Stokes' theorem to $u\nabla v$:

$$\oint_{\partial S} u\nabla v \cdot d\boldsymbol{\lambda} = \int_S \nabla \times (u\nabla v) \cdot d\boldsymbol{\sigma}.$$

The product rule gives $\nabla \times (u\nabla v) = \nabla u \times \nabla v + u\,\nabla \times \nabla v = \nabla u \times \nabla v$. Substitution proves the result with the compatible boundary orientation. The fields must be regular on the spanning surface.

The first integral in the manual's displayed derivation on p. 45 must be $2\oint_{\partial S} u\nabla v \cdot d\boldsymbol{\lambda}$, not $2\int_S u\nabla v \cdot d\boldsymbol{\sigma}$. Stokes' theorem relates that line integral to the surface integral of its curl.

3.8.11 Prove that

$$\oint_{\partial V} d\boldsymbol{\sigma} \times \mathbf{P} = \int_V \nabla \times \mathbf{P}\, d\tau.$$

Solution.

Take a constant test vector $\mathbf{a}$. Scalar triple-product cyclicity and Gauss' theorem give

$$\begin{aligned}
\mathbf{a} \cdot \oint_{\partial V} d\boldsymbol{\sigma} \times \mathbf{P} &= \oint_{\partial V} (\mathbf{P} \times \mathbf{a}) \cdot d\boldsymbol{\sigma} \\
&= \int_V \nabla \cdot (\mathbf{P} \times \mathbf{a})\, d\tau \\
&= \int_V \mathbf{a} \cdot (\nabla \times \mathbf{P})\, d\tau,
\end{aligned}$$

because $\nabla \times \mathbf{a} = 0$. Taking $\mathbf{a}$ to be each Cartesian unit vector proves the vector identity.

3.8.12 Prove that

$$\int_S d\boldsymbol{\sigma} \times \nabla\varphi = \oint_{\partial S} \varphi \, d\mathbf{r}.$$

Solution.

For any constant $\mathbf{a}$, Stokes' theorem applied to $\varphi\mathbf{a}$ gives

$$\begin{aligned}\mathbf{a} \cdot \oint_{\partial S} \varphi \, d\mathbf{r} &= \int_S [\nabla \times (\varphi\mathbf{a})] \cdot d\boldsymbol{\sigma} \\ &= \int_S (\nabla\varphi \times \mathbf{a}) \cdot d\boldsymbol{\sigma} \\ &= \mathbf{a} \cdot \int_S d\boldsymbol{\sigma} \times \nabla\varphi.\end{aligned}$$

The last equality is a cyclic scalar triple product. Equality for all $\mathbf{a}$ proves the claimed vector equation.

3.8.13 Prove that

$$\int_S (d\boldsymbol{\sigma} \times \nabla) \times \mathbf{P} = \oint_{\partial S} d\mathbf{r} \times \mathbf{P}.$$

Solution.

Let $\mathbf{a}$ be constant. Then

$$\mathbf{a} \cdot \oint_{\partial S} d\mathbf{r} \times \mathbf{P} = \oint_{\partial S} (\mathbf{P} \times \mathbf{a}) \cdot d\mathbf{r} = \int_S [\nabla \times (\mathbf{P} \times \mathbf{a})] \cdot d\boldsymbol{\sigma}.$$

The curl-product rule reduces the integrand to $[(\mathbf{a} \cdot \nabla)\mathbf{P} - \mathbf{a}(\nabla \cdot \mathbf{P})] \cdot d\boldsymbol{\sigma} = a_i[d\sigma_j \partial_i P_j - d\sigma_i \partial_j P_j]$. On the other hand, direct Levi-Civita contraction gives

$$[(d\boldsymbol{\sigma} \times \nabla) \times \mathbf{P}]_i = d\sigma_j \partial_i P_j - d\sigma_i \partial_j P_j.$$

Both sides have the same dot product with every constant $\mathbf{a}$, so the vector integrals agree.

3.9 Potential Theory

3.9.1 If a force $\mathbf{F}$ is given by

$$\mathbf{F} = \left(x^2 + y^2 + z^2\right)^n \left(\hat{\mathbf{e}}_x x + \hat{\mathbf{e}}_y y + \hat{\mathbf{e}}_z z\right),$$

find

(a) $\nabla \cdot \mathbf{F}$.

(b) $\nabla \times \mathbf{F}$.

(c) A scalar potential $\varphi(x, y, z)$ so that $\mathbf{F} = -\nabla\varphi$.

(d) For what value of the exponent n does the scalar potential diverge at both the origin and infinity?

ANS. (a) $(2n+3)r^{2n}$ (b) 0 (c) $-r^{2n+2}/(2n+2),\ n\neq -1$ (d) $n=-1,\ \varphi=-\ln r$. Take $n\in\mathbb{R}$ and initially work at $r>0$; any extension across the origin requires a separate regularity check.

Solution.

Write $\mathbf{F}=r^{2n}\mathbf{r}$, with $r>0$. **(a)** Since $\partial_i r = x_i/r$,

$$\nabla\cdot\mathbf{F}=\sum_i[2nr^{2n-2}x_i^2+r^{2n}]=2nr^{2n}+3r^{2n}=\boxed{(2n+3)r^{2n}}.$$

(b) The product curl is $\nabla(r^{2n})\times\mathbf{r}+r^{2n}\nabla\times\mathbf{r}=2nr^{2n-2}\mathbf{r}\times\mathbf{r}=0$. **(c)** A radial potential must obey $-\varphi'(r)=r^{2n+1}$. Integration gives

$$\boxed{\varphi(r)=\begin{cases}-r^{2n+2}/(2n+2)+C, & n\neq -1,\\ -\ln(r/r_0)+C, & n=-1,\end{cases}}$$

where $r_0>0$ fixes the additive reference. **(d)** A positive power of r is finite at zero and divergent at infinity; a negative power has the reverse behavior. Only the logarithmic case $n=-1$ diverges in magnitude at both ends. These differential formulas hold away from singularities. For instance, at $n=-3/2$ the divergence is zero for $r>0$, but its distributional extension is $4\pi\delta^{(3)}(\mathbf{r})$, as the flux through every sphere is 4π.

3.9.2 A sphere of radius a is uniformly charged (throughout its volume). Construct the electrostatic potential $\varphi(r)$ for $0\le r<\infty$.

Solution.

Let the total charge be Q, so the uniform density is $\rho_0=3Q/(4\pi a^3)$. Choose $\varphi(\infty)=0$. Spherical symmetry and Gauss' law give

$$4\pi r^2E_r(r)=\frac{Q_{\text{enclosed}}(r)}{\epsilon_0},\qquad Q_{\text{enclosed}}(r)=\begin{cases}Qr^3/a^3, & r\le a,\\ Q, & r\ge a.\end{cases}$$

Thus $E_r=Qr/(4\pi\epsilon_0 a^3)$ inside and $E_r=Q/(4\pi\epsilon_0 r^2)$ outside. Integrating inward from infinity,

$$\varphi(r)=\int_r^\infty E_r(s)\,ds=\frac{Q}{4\pi\epsilon_0 r}\quad(r\ge a).$$

Inside,

$$\begin{aligned}\varphi(r)&=\varphi(a)+\int_r^a\frac{Qs}{4\pi\epsilon_0 a^3}\,ds\\ &=\frac{Q}{4\pi\epsilon_0 a}+\frac{Q(a^2-r^2)}{8\pi\epsilon_0 a^3}.\end{aligned}$$

Consequently

$$\boxed{\varphi(r)=\begin{cases}\dfrac{Q}{8\pi\epsilon_0 a}(3-r^2/a^2), & 0\le r\le a,\\ \dfrac{Q}{4\pi\epsilon_0 r}, & r\ge a.\end{cases}}$$

Both φ and $-\varphi'=E_r$ agree at a, and $\varphi(0)$ is finite.

3.9.3 The origin of the Cartesian coordinates is at the Earth's center. The moon is on the z-axis, a fixed distance R away (center-to-center distance). The tidal force exerted by the moon on a particle at the Earth's surface (point x, y, z) is given by

$$F_x = -GMm\frac{x}{R^3}, \quad F_y = -GMm\frac{y}{R^3}, \quad F_z = +2GMm\frac{z}{R^3}.$$

Find the potential that yields this tidal force.

ANS. $-\dfrac{GMm}{R^3}\left(z^2 - \dfrac{1}{2}x^2 - \dfrac{1}{2}y^2\right)$.

Solution.

Set $K = GMm/R^3$. From $\mathbf{F} = -\nabla\varphi$, $\partial_x\varphi = Kx$, so $\varphi = Kx^2/2 + f(y,z)$. The y equation gives $\partial_y f = Ky$, hence $f = Ky^2/2 + g(z)$. The z equation gives $g' = -2Kz$, so $g = -Kz^2 + C$. Thus

$$\boxed{\varphi = \frac{GMm}{R^3}\left(\frac{x^2+y^2}{2} - z^2\right) + C}.$$

Its negative gradient is $(-Kx, -Ky, 2Kz)$, exactly the given force.

The manual, p. 46, gives the opposite overall sign for the potential; applying $-\nabla$ to its expression reverses every required force component. Its intermediate expression also includes an extra factor r^2 before $(3z^2 - r^2)$. The consistent chain is $V = -\dfrac{GmM}{2R^3}(3z^2 - r^2) = -\dfrac{GmMr^2}{R^3}P_2(\cos\theta)$, as in the textbook answer.

3.9.4 A long, straight wire carrying a current I produces a magnetic induction $\mathbf{B}$ with components

$$\mathbf{B} = \frac{\mu_0 I}{2\pi}\left(\frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2}, 0\right).$$

Find a magnetic vector potential $\mathbf{A}$.

ANS. $\mathbf{A} = -\hat{\mathbf{z}}(\mu_0 I/4\pi)\ln(x^2+y^2)$. (This solution is not unique.) Work off the ideal wire, where $x^2 + y^2 > 0$.

Solution.

Choose $\mathbf{A} = A_z(x,y)\hat{\mathbf{e}}_z$. Then $\nabla \times \mathbf{A} = (\partial_y A_z, -\partial_x A_z, 0)$. With $K = \mu_0 I/(2\pi)$, the required y-component implies $\partial_x A_z = -Kx/(x^2+y^2)$. Integration in x gives $A_z = -(K/2)\ln(x^2+y^2) + f(y)$. The required x-component sets $f'(y) = 0$. Equivalently, with $\rho = \sqrt{x^2+y^2} > 0$ and reference length ρ_0,

$$\boxed{\mathbf{A} = K\ln(\rho_0/\rho)\hat{\mathbf{e}}_z}.$$

Indeed $\partial_y A_z = -Ky/\rho^2$ and $-\partial_x A_z = Kx/\rho^2$. Adding any regular gradient leaves the curl unchanged. The ideal-wire axis is excluded.

3.9.5 If

$$\mathbf{B} = \frac{\hat{\mathbf{r}}}{r^2} = \left(\frac{x}{r^3}, \frac{y}{r^3}, \frac{z}{r^3}\right),$$

find a local vector potential, for example off the z-axis, $\mathbf{A}$ such that $\nabla \times \mathbf{A} = \mathbf{B}$.

ANS. One possible solution is $\mathbf{A} = \hat{\mathbf{e}}_x \dfrac{yz}{r(x^2+y^2)} - \hat{\mathbf{e}}_y \dfrac{xz}{r(x^2+y^2)}$.

Solution.

The displayed potential is valid off the z-axis. No globally smooth, single-valued potential exists on all of $\mathbb{R}^3\backslash\{0\}$ for this field. Write $\rho = \sqrt{x^2+y^2}$ and $r = \sqrt{\rho^2+z^2}$. Then $\mathbf{B} = (\rho/r^3)\hat{\mathbf{e}}_\rho + (z/r^3)\hat{\mathbf{e}}_z$.

Try $\mathbf{A} = A_\varphi(\rho, z)\hat{\mathbf{e}}_\varphi$. Its curl has components $B_\rho = -\partial_z A_\varphi$ and $B_z = \rho^{-1}\partial_\rho(\rho A_\varphi)$. Choosing $A_\varphi = -z/(\rho r)$ gives

$$-\partial_z A_\varphi = \frac{1}{\rho}\left(\frac{1}{r} - \frac{z^2}{r^3}\right) = \frac{\rho}{r^3}, \qquad \frac{1}{\rho}\partial_\rho(\rho A_\varphi) = \frac{1}{\rho}\frac{z\rho}{r^3} = \frac{z}{r^3}.$$

Since $\hat{\mathbf{e}}_\varphi = (-y/\rho, x/\rho, 0)$,

$$\boxed{\mathbf{A} = \left(\frac{yz}{r\rho^2}, -\frac{xz}{r\rho^2}, 0\right)}.$$

For the global obstruction, the flux through a sphere is $\int r^{-2} dS = 4\pi$. If a regular single-valued potential existed on that sphere, applying Stokes' theorem to its upper and lower hemispheres would express the flux as two opposite equatorial line integrals, whose sum is zero. The contradiction proves the stated global limitation.

3.9.6 Show that the pair of equations

$$\mathbf{A} = \frac{1}{2}(\mathbf{B} \times \mathbf{r}), \quad \mathbf{B} = \nabla \times \mathbf{A},$$

is satisfied by any constant magnetic induction $\mathbf{B}$.

Solution.

Use the curl-product rule with constant $\mathbf{B}$:

$$\nabla \times (\mathbf{B} \times \mathbf{r}) = (\mathbf{r} \cdot \nabla)\mathbf{B} - (\mathbf{B} \cdot \nabla)\mathbf{r} - \mathbf{r}(\nabla \cdot \mathbf{B}) + \mathbf{B}(\nabla \cdot \mathbf{r}).$$

The four terms are $0, -\mathbf{B}, 0, 3\mathbf{B}$, because $(\mathbf{B} \cdot \nabla)(x, y, z) = \mathbf{B}$. Thus the curl is $2\mathbf{B}$, and $\boxed{\nabla \times [\tfrac{1}{2}(\mathbf{B} \times \mathbf{r})] = \mathbf{B}}$.

3.9.7 Vector $\mathbf{B}$ is formed by the product of two gradients

$$\mathbf{B} = (\nabla u) \times (\nabla v),$$

where u and v are C^2 scalar functions.

(a) Show that $\mathbf{B}$ is solenoidal.

(b) Show that

$$\mathbf{A} = \frac{1}{2}(u\nabla v - v\nabla u)$$

is a vector potential for $\mathbf{B}$, in that

$$\mathbf{B} = \nabla \times \mathbf{A}.$$

Solution.

With commuting mixed derivatives,

$$\nabla \cdot \mathbf{B} = \nabla v \cdot (\nabla \times \nabla u) - \nabla u \cdot (\nabla \times \nabla v) = 0.$$

For the proposed potential,

$$\begin{aligned} 2\nabla \times \mathbf{A} &= \nabla \times (u\nabla v) - \nabla \times (v\nabla u) \\ &= (\nabla u \times \nabla v) - (\nabla v \times \nabla u) \\ &= 2\nabla u \times \nabla v = 2\mathbf{B}. \end{aligned}$$

The curl-of-gradient terms vanish separately. Thus $\mathbf{A}$ is a vector potential and $\mathbf{B}$ is solenoidal.

3.9.8 The magnetic induction $\mathbf{B}$ is related to the magnetic vector potential $\mathbf{A}$ by $\mathbf{B} = \nabla \times \mathbf{A}$. By Stokes' theorem

$$\int \mathbf{B} \cdot d\boldsymbol{\sigma} = \oint \mathbf{A} \cdot d\mathbf{r}.$$

Show that each side of this equation is invariant under the gauge transformation, $\mathbf{A} \to \mathbf{A} + \nabla\varphi$.

Note. Take the function φ to be single-valued.

Solution.

Let $\mathbf{A}' = \mathbf{A} + \nabla\varphi$, with regular single-valued φ on the relevant surface. Then $\mathbf{B}' = \nabla \times \mathbf{A}' = \mathbf{B} + \nabla \times \nabla\varphi = \mathbf{B}$, so the flux does not change. On the closed boundary,

$$\oint \mathbf{A}' \cdot d\mathbf{r} - \oint \mathbf{A} \cdot d\mathbf{r} = \oint \nabla\varphi \cdot d\mathbf{r} = \varphi(\text{end}) - \varphi(\text{start}) = 0.$$

Single-valuedness makes the final equality valid.

3.9.9 Show that the value of the electrostatic potential φ at any point P is equal to the average of the potential over any spherical surface centered on P, provided that there are no electric charges on or within the sphere.

Hint. Use Green's theorem, Eq. (3.85), with $u = r^{-1}$, where r is the distance from P, and $v = \varphi$. Equation (3.120) will also be useful.

Solution.

Place P at the origin and let the sphere have radius a. The potential is harmonic throughout the ball. To avoid applying Green's identity through the singularity of $u = 1/r$, use the shell $\epsilon < r < a$. Both $\nabla^2 u$ and $\nabla^2\varphi$ vanish there:

$$0 = \oint_{\partial\text{shell}} \left(\frac{1}{r}\partial_n\varphi - \varphi\partial_n\frac{1}{r} \right) dS.$$

On the outer sphere, $\partial_n = \partial_r$, so its contribution is $a^{-1}\oint_{r=a} \partial_r\varphi\, dS + a^{-2}\oint_{r=a} \varphi\, dS$. The first integral is zero by Gauss' theorem and harmonicity in the ball. On the inner sphere the outward shell normal is $-\hat{\mathbf{e}}_r$, giving

$$-\epsilon^{-1}\oint_{r=\epsilon} \partial_r\varphi\, dS - \epsilon^{-2}\oint_{r=\epsilon} \varphi\, dS.$$

The first term tends to zero (bounded gradient times area gives $O(\epsilon)$); the second tends to $-4\pi\varphi(P)$ by continuity. Letting $\epsilon \to 0$ gives $a^{-2}\oint_{r=a} \varphi\, dS = 4\pi\varphi(P)$, or

$$\boxed{\varphi(P) = \frac{1}{4\pi a^2}\oint_{r=a} \varphi\, dS}.$$

3.9.10 Using Maxwell's equations, show that for a system (steady current) the magnetic vector potential $\mathbf{A}$ satisfies a vector Poisson equation,

$$\nabla^2\mathbf{A} = -\mu\mathbf{J},$$

provided we require $\nabla \cdot \mathbf{A} = 0$. Use magnetostatics in a medium with spatially constant scalar permeability μ.

Solution.

Use a magnetostatic field in a medium with spatially constant scalar permeability μ. Ampere's law is $\nabla \times \mathbf{B} = \mu \mathbf{J}$. Insert $\mathbf{B} = \nabla \times \mathbf{A}$ and expand the double curl:

$$\mu \mathbf{J} = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}.$$

The stated gauge condition $\nabla \cdot \mathbf{A} = 0$ removes the first term, leaving $\boxed{\nabla^2 \mathbf{A} = -\mu \mathbf{J}}$. Taking the divergence gives $0 = -\mu \nabla \cdot \mathbf{J}$, consistent with steady-current conservation.

3.9.11 Derive, assuming the Lorentz gauge, Eq. (3.109):

$$\frac{1}{c^2} \frac{\partial^2 \mathbf{A}}{\partial t^2} - \nabla^2 \mathbf{A} = \mu_0 \mathbf{J}.$$

Hint. Eq. (3.70) will be helpful.

Solution.

Use $\mathbf{B} = \nabla \times \mathbf{A}$, $\mathbf{E} = -\nabla \varphi - \partial_t \mathbf{A}$, and $\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + c^{-2} \partial_t \mathbf{E}$. Substitution yields

$$\nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} = \mu_0 \mathbf{J} - \frac{1}{c^2} \nabla \partial_t \varphi - \frac{1}{c^2} \partial_t^2 \mathbf{A}.$$

The Lorenz gauge is $\nabla \cdot \mathbf{A} + c^{-2} \partial_t \varphi = 0$. It makes the two gradient terms identical on the two sides, so they cancel. Rearrangement gives

$$\boxed{\frac{1}{c^2} \partial_t^2 \mathbf{A} - \nabla^2 \mathbf{A} = \mu_0 \mathbf{J}}.$$

3.9.12 Prove that an arbitrary solenoidal vector $\mathbf{B}$ can be described as $\mathbf{B} = \nabla \times \mathbf{A}$, with

$$\mathbf{A} = -\hat{\mathbf{e}}_x \int_{y_0}^{y} B_z(x, y, z)\, dy - \hat{\mathbf{e}}_z \left[\int_{x_0}^{x} B_y(x, y_0, z)\, dx - \int_{y_0}^{y} B_x(x, y, z)\, dy \right].$$

Take $\mathbf{B}$ to be C^1 on a coordinate box containing the indicated integration paths; the formula is a local construction on more general regions.

Solution.

The construction is valid on a coordinate box, or on a region containing all the indicated integration paths with a regular solenoidal field. Solenoidality alone does not ensure a global potential on an arbitrary punctured domain; see 3.9.5. Use distinct dummy variables:

$$A_x = -\int_{y_0}^{y} B_z(x, \eta, z)\, d\eta, \quad A_y = 0, \quad A_z = -\int_{x_0}^{x} B_y(\xi, y_0, z)\, d\xi + \int_{y_0}^{y} B_x(x, \eta, z)\, d\eta.$$

The fundamental theorem of calculus immediately gives $(\nabla \times \mathbf{A})_x = \partial_y A_z = B_x$ and $(\nabla \times \mathbf{A})_z = -\partial_y A_x = B_z$. For the remaining component, differentiate both integrals:

$$\begin{aligned}
(\nabla \times \mathbf{A})_y &= \partial_z A_x - \partial_x A_z \\
&= -\int_{y_0}^{y} \partial_z B_z\, d\eta + B_y(x, y_0, z) - \int_{y_0}^{y} \partial_x B_x\, d\eta \\
&= B_y(x, y_0, z) + \int_{y_0}^{y} \partial_\eta B_y(x, \eta, z)\, d\eta \\
&= B_y(x, y_0, z) + B_y(x, y, z) - B_y(x, y_0, z) = B_y(x, y, z).
\end{aligned}$$

The third line uses $\partial_x B_x + \partial_\eta B_y + \partial_z B_z = 0$. All three curl components are correct.

3.10 Curvilinear Coordinates

The coordinate differential formulas below are classical formulas on regular charts: $\rho > 0$ in cylindrical coordinates and $r > 0$, $0 < \theta < \pi$ in spherical coordinates. At a coordinate axis or the origin, an extension must be checked from the underlying Cartesian field.

3.10.1 The u-, v-, z-coordinate system frequently used in electrostatics and in hydrodynamics is defined by

$$xy = u, \quad x^2 - y^2 = v, \quad z = z.$$

This u-, v-, z-system is orthogonal.

(a) In words, describe briefly the nature of each of the three families of coordinate surfaces.

(b) Sketch the system in the xy-plane showing the intersections of surfaces of constant u and surfaces of constant v with the xy-plane.

(c) Indicate the directions of the unit vectors $\hat{\mathbf{e}}_u$ and $\hat{\mathbf{e}}_v$ in all four quadrants.

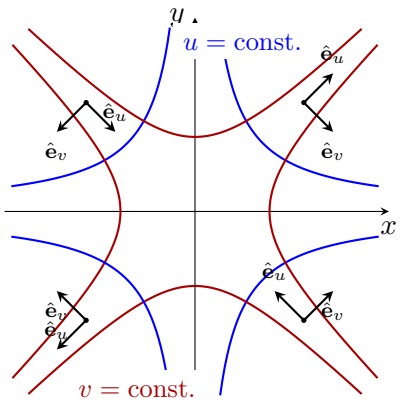
(d) Finally, is this u-, v-, z-system right-handed ($\hat{\mathbf{e}}_u \times \hat{\mathbf{e}}_v = +\hat{\mathbf{e}}_z$) or left-handed ($\hat{\mathbf{e}}_u \times \hat{\mathbf{e}}_v = -\hat{\mathbf{e}}_z$)?

Use a regular local branch with $x^2 + y^2 > 0$.

Solution.

(a) The surfaces $xy = u$ are rectangular hyperbolic cylinders parallel to z: for $u > 0$ their sections occupy quadrants I and III, and for $u < 0$ quadrants II and IV. Their asymptotes are the coordinate axes. The surfaces $x^2 - y^2 = v$ are another family of rectangular hyperbolic cylinders, with asymptotes $y = \pm x$; they open along x for $v > 0$ and along y for $v < 0$. Constant z gives horizontal planes. The zero levels are the corresponding intersecting planes.

(b) Here blue curves have $u = \pm 1$ and red curves have $v = \pm 1.2$. Arrows show increasing coordinates:



(c) For $\rho = \sqrt{x^2 + y^2} > 0$, $\nabla u = (y, x, 0)$ and $\nabla v = (2x, -2y, 0)$. They are orthogonal, since their dot product is $2xy - 2xy = 0$. The increasing-coordinate unit vectors are

$$\boxed{\hat{\mathbf{e}}_u = \frac{(y, x, 0)}{\rho}, \qquad \hat{\mathbf{e}}_v = \frac{(x, -y, 0)}{\rho}}.$$

Their (x, y) component signs in quadrants I, II, III, IV are respectively $(+,+), (+,-), (-,-), (-,+)$ for $\hat{\mathbf{e}}_u$ and $(+,-), (-,-), (-,+), (+,+)$ for $\hat{\mathbf{e}}_v$. **(d)** $\hat{\mathbf{e}}_u \times \hat{\mathbf{e}}_v = -(x^2 + y^2)\hat{\mathbf{e}}_z/\rho^2 = -\hat{\mathbf{e}}_z$: the ordered

system (u, v, z) is left-handed. The chart is singular on the z-axis and is only local unless a branch is chosen, because (x, y) and $(-x, -y)$ have the same (u, v).

One can also check handedness by the Jacobian. Differentiating the definitions gives

$$\begin{pmatrix} du \\ dv \end{pmatrix} = \begin{pmatrix} y & x \\ 2x & -2y \end{pmatrix} \begin{pmatrix} dx \\ dy \end{pmatrix}, \qquad \det \frac{\partial(u, v)}{\partial(x, y)} = -2(x^2 + y^2).$$

Inverting this 2×2 matrix explicitly gives

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{pmatrix} y/\rho^2 & x/(2\rho^2) \\ x/\rho^2 & -y/(2\rho^2) \end{pmatrix}, \qquad \det \frac{\partial(x, y)}{\partial(u, v)} = -\frac{1}{2\rho^2}.$$

The manual, p. 49, omits the factor $1/2$ in this inverse determinant. Its earlier differential must likewise read $dv = 2x\,dx - 2y\,dy$, with a minus sign.

3.10.2 The elliptic cylindrical coordinate system consists of three families of surfaces:

$$(1)\ \frac{x^2}{a^2 \cosh^2 u} + \frac{y^2}{a^2 \sinh^2 u} = 1; \quad (2)\ \frac{x^2}{a^2 \cos^2 v} - \frac{y^2}{a^2 \sin^2 v} = 1; \quad (3)\ z = z.$$

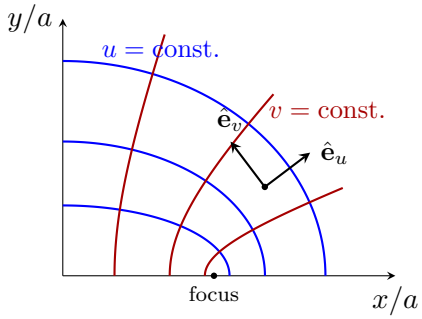
Sketch the coordinate surfaces $u =$ constant and $v =$ constant as they intersect the first quadrant of the xy-plane. Show the unit vectors $\hat{\mathbf{e}}_u$ and $\hat{\mathbf{e}}_v$. The range of u is $0 \le u < \infty$. The range of v is $0 \le v \le 2\pi$. Take $a > 0$. Interpret the zero-denominator limiting surfaces by the parametrization below; the unit basis is undefined at the foci, where $\sinh^2 u + \sin^2 v = 0$.

Solution.

Use the parametrization

$$x = a \cosh u \cos v, \qquad y = a \sinh u \sin v, \qquad z = z, \quad a > 0.$$

Dividing its squared coordinates by $a^2 \cosh^2 u, a^2 \sinh^2 u$ and adding gives 1; dividing by $a^2 \cos^2 v, a^2 \sin^2 v$ and subtracting also gives 1. Thus it generates both stated families. Constant $u > 0$ gives ellipses with foci $(\pm a, 0)$, and constant v gives confocal hyperbolas. In the first quadrant use $u > 0$, $0 < v < \pi/2$.



Differentiate the position and normalize:

$$\boxed{\begin{aligned} \hat{\mathbf{e}}_u &= \frac{(\sinh u \cos v,\ \cosh u \sin v,\ 0)}{\sqrt{\sinh^2 u + \sin^2 v}}, \\ \hat{\mathbf{e}}_v &= \frac{(-\cosh u \sin v,\ \sinh u \cos v,\ 0)}{\sqrt{\sinh^2 u + \sin^2 v}}. \end{aligned}}$$

The common scale factor is $h_u = h_v = a\sqrt{\sinh^2 u + \sin^2 v}$; the dot product of the two numerators is zero. The first vector points outward and northeast, the second along an ellipse toward increasing v (northwest) in this quadrant. At limiting coordinate surfaces where a displayed denominator of the implicit equations vanishes, use the parametrization and its limit.

3.10.3 Develop arguments to show that dot and cross products (not involving ∇) in orthogonal curvilinear coordinates in $\mathbb{R}^3$ proceed, as in Cartesian coordinates, **with no involvement of scale factors**. Use physical components along orthonormal unit vectors; the usual cross-product signs require a right-handed ordering.

Solution.

At a fixed regular point, expand in the local orthonormal basis: $\mathbf{A} = \sum_i A_i \hat{\mathbf{e}}_i$ and $\mathbf{B} = \sum_j B_j \hat{\mathbf{e}}_j$. Then

$$\mathbf{A} \cdot \mathbf{B} = \sum_{ij} A_i B_j \delta_{ij} = \sum_i A_i B_i.$$

For a right-handed basis, $\hat{\mathbf{e}}_i \times \hat{\mathbf{e}}_j = \sum_k \varepsilon_{ijk} \hat{\mathbf{e}}_k$, so

$$\mathbf{A} \times \mathbf{B} = \sum_{ijk} \varepsilon_{ijk} A_i B_j \hat{\mathbf{e}}_k.$$

These are exactly the Cartesian component rules. No differentiation of the varying basis is involved. The usual cross-product sign assumes a right-handed unit basis; a left-handed ordering, such as (u, v, z) in 3.10.1, contributes an overall minus sign. Also these formulas use physical components along unit vectors. Components along the unnormalized coordinate tangents $\partial \mathbf{r}/\partial q_i = h_i \hat{\mathbf{e}}_i$ would acquire the corresponding scale factors.

3.10.4 With $\hat{\mathbf{e}}_1$ a unit vector in the direction of increasing q_1, show that

(a) $\nabla \cdot \hat{\mathbf{e}}_1 = \dfrac{1}{h_1 h_2 h_3} \dfrac{\partial (h_2 h_3)}{\partial q_1}$

(b) $\nabla \times \hat{\mathbf{e}}_1 = \dfrac{1}{h_1} \left[\hat{\mathbf{e}}_2 \dfrac{1}{h_3} \dfrac{\partial h_1}{\partial q_3} - \hat{\mathbf{e}}_3 \dfrac{1}{h_2} \dfrac{\partial h_1}{\partial q_2} \right]$.

Note that even though $\hat{\mathbf{e}}_1$ is a unit vector, its divergence and curl **do not necessarily vanish**. Use a regular, right-handed orthogonal coordinate chart with positive scale factors h_i.

Solution.

For physical components V_i in an orthogonal system, the flux through opposite q_i faces gives

$$\nabla \cdot \mathbf{V} = \frac{1}{h_1 h_2 h_3} \sum_i \partial_{q_i} (h_j h_k V_i), \qquad \{i, j, k\} = \{1, 2, 3\}.$$

For $\mathbf{V} = \hat{\mathbf{e}}_1$, $V_1 = 1, V_2 = V_3 = 0$, so $\boxed{\nabla \cdot \hat{\mathbf{e}}_1 = (h_1 h_2 h_3)^{-1} \partial_{q_1} (h_2 h_3)}$. For the curl, use $\hat{\mathbf{e}}_1 = h_1 \nabla q_1$:

$$\nabla \times \hat{\mathbf{e}}_1 = \nabla h_1 \times \nabla q_1 = \left(\sum_j \frac{\hat{\mathbf{e}}_j}{h_j} \partial_{q_j} h_1 \right) \times \frac{\hat{\mathbf{e}}_1}{h_1}.$$

The $j = 1$ term vanishes. In a right-handed basis the other two terms give

$$\boxed{\nabla \times \hat{\mathbf{e}}_1 = \frac{1}{h_1} \left[\frac{\hat{\mathbf{e}}_2}{h_3} \partial_{q_3} h_1 - \frac{\hat{\mathbf{e}}_3}{h_2} \partial_{q_2} h_1 \right]}.$$

The basis direction can vary even though its magnitude is fixed.

3.10.5 Show that a set of orthogonal unit vectors $\hat{\mathbf{e}}_i$ may be defined by

$$\hat{\mathbf{e}}_i = \frac{1}{h_i}\frac{\partial \mathbf{r}}{\partial q_i}.$$

In particular, show that $\hat{\mathbf{e}}_i \cdot \hat{\mathbf{e}}_i = 1$ leads to an expression for h_i in agreement with Eq. (3.131).

The above equation for $\hat{\mathbf{e}}_i$ may be taken as a starting point for deriving

$$\frac{\partial \hat{\mathbf{e}}_i}{\partial q_j} = \hat{\mathbf{e}}_j \frac{1}{h_i}\frac{\partial h_j}{\partial q_i}, \quad i \neq j$$

and

$$\frac{\partial \hat{\mathbf{e}}_i}{\partial q_i} = -\sum_{j \neq i} \hat{\mathbf{e}}_j \frac{1}{h_j}\frac{\partial h_i}{\partial q_j}.$$

Assume the coordinates form a regular orthogonal chart and $h_i > 0$; arbitrary curvilinear coordinates need not be orthogonal.

Solution.

On a regular orthogonal chart, the tangent to increasing q_i is $\mathbf{r}_i = \partial_{q_i}\mathbf{r}$. Thus

$$h_i = |\mathbf{r}_i| = \sqrt{\sum_{a=x,y,z} (\partial_{q_i} a)^2}, \qquad \hat{\mathbf{e}}_i = \mathbf{r}_i / h_i.$$

The normalization follows because $\hat{\mathbf{e}}_i^2 = |\mathbf{r}_i|^2/h_i^2 = 1$; orthogonality of the coordinate tangents gives $\hat{\mathbf{e}}_i \cdot \hat{\mathbf{e}}_j = 0$ for $i \neq j$.

To derive the basis derivatives, differentiate $\mathbf{r}_i = h_i \hat{\mathbf{e}}_i$ and commute mixed partials:

$$(\partial_j h_i)\hat{\mathbf{e}}_i + h_i \partial_j \hat{\mathbf{e}}_i = (\partial_i h_j)\hat{\mathbf{e}}_j + h_j \partial_i \hat{\mathbf{e}}_j. \tag{*}$$

Normalization gives $\hat{\mathbf{e}}_i \cdot \partial_j \hat{\mathbf{e}}_i = 0$. For three distinct indices i, j, k, (*) and differentiated orthogonality successively give

$$\begin{aligned} h_i \hat{\mathbf{e}}_k \cdot \partial_j \hat{\mathbf{e}}_i = h_j \hat{\mathbf{e}}_k \cdot \partial_i \hat{\mathbf{e}}_j &= -h_j \hat{\mathbf{e}}_j \cdot \partial_i \hat{\mathbf{e}}_k \\ &= -h_k \hat{\mathbf{e}}_j \cdot \partial_k \hat{\mathbf{e}}_i = h_k \hat{\mathbf{e}}_i \cdot \partial_k \hat{\mathbf{e}}_j \\ &= h_i \hat{\mathbf{e}}_i \cdot \partial_j \hat{\mathbf{e}}_k = -h_i \hat{\mathbf{e}}_k \cdot \partial_j \hat{\mathbf{e}}_i. \end{aligned}$$

That component is zero. Dotting (*) with $\hat{\mathbf{e}}_j$ for $i \neq j$ gives the only remaining component: $h_i \hat{\mathbf{e}}_j \cdot \partial_j \hat{\mathbf{e}}_i = \partial_i h_j$. Therefore

$$\boxed{\partial_j \hat{\mathbf{e}}_i = \hat{\mathbf{e}}_j \frac{\partial_i h_j}{h_i} \quad (i \neq j)}.$$

Finally differentiate $\hat{\mathbf{e}}_i \cdot \hat{\mathbf{e}}_j = 0$ with respect to q_i and use this formula: $\hat{\mathbf{e}}_j \cdot \partial_i \hat{\mathbf{e}}_i = -\hat{\mathbf{e}}_i \cdot \partial_i \hat{\mathbf{e}}_j = -(\partial_j h_i)/h_j$. Expanding in the perpendicular basis gives $\boxed{\partial_i \hat{\mathbf{e}}_i = -\sum_{j \neq i} \hat{\mathbf{e}}_j (\partial_j h_i)/h_j}$.

3.10.6 Resolve the circular cylindrical unit vectors into their Cartesian components (see Fig. 3.23).

ANS.

$$\begin{aligned} \hat{\mathbf{e}}_\rho &= \hat{\mathbf{e}}_x \cos\varphi + \hat{\mathbf{e}}_y \sin\varphi, \\ \hat{\mathbf{e}}_\varphi &= -\hat{\mathbf{e}}_x \sin\varphi + \hat{\mathbf{e}}_y \cos\varphi, \\ \hat{\mathbf{e}}_z &= \hat{\mathbf{e}}_z. \end{aligned}$$

Solution.

Start with $\mathbf{r} = (\rho\cos\varphi, \rho\sin\varphi, z)$. Its coordinate derivatives are

$$\partial_\rho \mathbf{r} = (\cos\varphi, \sin\varphi, 0), \quad \partial_\varphi \mathbf{r} = \rho(-\sin\varphi, \cos\varphi, 0), \quad \partial_z \mathbf{r} = (0,0,1).$$

Their lengths are $1, \rho, 1$ for $\rho > 0$. Dividing by these lengths gives

$$\boxed{\hat{\mathbf{e}}_\rho = (\cos\varphi, \sin\varphi, 0), \quad \hat{\mathbf{e}}_\varphi = (-\sin\varphi, \cos\varphi, 0), \quad \hat{\mathbf{e}}_z = (0,0,1)}.$$

Their dot products are δ_{ij} and $\hat{\mathbf{e}}_\rho \times \hat{\mathbf{e}}_\varphi = \hat{\mathbf{e}}_z$.

3.10.7 Resolve the Cartesian unit vectors into their circular cylindrical components (see Fig. 3.23).

$$\begin{aligned} &\hat{\mathbf{e}}_x = \hat{\mathbf{e}}_\rho \cos\varphi - \hat{\mathbf{e}}_\varphi \sin\varphi, \\ \textit{ANS.}\quad &\hat{\mathbf{e}}_y = \hat{\mathbf{e}}_\rho \sin\varphi + \hat{\mathbf{e}}_\varphi \cos\varphi, \\ &\hat{\mathbf{e}}_z = \hat{\mathbf{e}}_z. \end{aligned}$$

Solution.

Take the combinations of the vectors in 3.10.6:

$$\begin{aligned} \cos\varphi\,\hat{\mathbf{e}}_\rho - \sin\varphi\,\hat{\mathbf{e}}_\varphi &= (\cos^2\varphi + \sin^2\varphi, 0, 0) = \hat{\mathbf{e}}_x, \\ \sin\varphi\,\hat{\mathbf{e}}_\rho + \cos\varphi\,\hat{\mathbf{e}}_\varphi &= (0, \sin^2\varphi + \cos^2\varphi, 0) = \hat{\mathbf{e}}_y. \end{aligned}$$

Together with unchanged $\hat{\mathbf{e}}_z$, these are the requested inverse formulas. The inverse matrix is the transpose of the orthogonal matrix in 3.10.6.

3.10.8 From the results of Exercise 3.10.6, show that

$$\frac{\partial \hat{\mathbf{e}}_\rho}{\partial\varphi} = \hat{\mathbf{e}}_\varphi, \quad \frac{\partial \hat{\mathbf{e}}_\varphi}{\partial\varphi} = -\hat{\mathbf{e}}_\rho$$

and that all other first derivatives of the circular cylindrical unit vectors with respect to the circular cylindrical coordinates vanish.

Solution.

Differentiating the Cartesian expressions gives

$$\partial_\varphi \hat{\mathbf{e}}_\rho = (-\sin\varphi, \cos\varphi, 0) = \hat{\mathbf{e}}_\varphi,$$

$$\partial_\varphi \hat{\mathbf{e}}_\varphi = (-\cos\varphi, -\sin\varphi, 0) = -\hat{\mathbf{e}}_\rho.$$

Neither depends on ρ or z, so those derivatives vanish. The vector $\hat{\mathbf{e}}_z = (0,0,1)$ is constant, so all three of its coordinate derivatives vanish.

3.10.9 Compare $\nabla \cdot \mathbf{V}$ as given for cylindrical coordinates in Eq. (3.148) with the result of its computation by applying to $\mathbf{V}$ the operator

$$\nabla = \hat{\mathbf{e}}_\rho \frac{\partial}{\partial\rho} + \hat{\mathbf{e}}_\varphi \frac{1}{\rho}\frac{\partial}{\partial\varphi} + \hat{\mathbf{e}}_z \frac{\partial}{\partial z}$$

Note that ∇ acts both on the unit vectors and on the components of $\mathbf{V}$.

Solution.

Write $\mathbf{V} = V_\rho \hat{\mathbf{e}}_\rho + V_\varphi \hat{\mathbf{e}}_\varphi + V_z \hat{\mathbf{e}}_z$. The basis has no ρ or z dependence, giving $\hat{\mathbf{e}}_\rho \cdot \partial_\rho \mathbf{V} = \partial_\rho V_\rho$ and $\hat{\mathbf{e}}_z \cdot \partial_z \mathbf{V} = \partial_z V_z$. For the azimuthal derivative, differentiate coefficients and basis:

$$\partial_\varphi \mathbf{V} = (\partial_\varphi V_\rho - V_\varphi)\hat{\mathbf{e}}_\rho + (\partial_\varphi V_\varphi + V_\rho)\hat{\mathbf{e}}_\varphi + (\partial_\varphi V_z)\hat{\mathbf{e}}_z.$$

Taking the specified dot product yields

$$\boxed{\nabla \cdot \mathbf{V} = \partial_\rho V_\rho + \frac{V_\rho}{\rho} + \frac{1}{\rho}\partial_\varphi V_\varphi + \partial_z V_z = \frac{1}{\rho}\partial_\rho(\rho V_\rho) + \frac{1}{\rho}\partial_\varphi V_\varphi + \partial_z V_z}.$$

The V_ρ/ρ term is exactly the contribution of the varying basis.

The manual, p. 51, calls the exercise's operator incorrect. The textbook explicitly says that it acts on the unit vectors as well as the components, so the operator as stated is correct: the basis derivative supplies exactly the V_ρ/ρ term computed above.

3.10.10

(a) Show that $\mathbf{r} = \hat{\mathbf{e}}_\rho \rho + \hat{\mathbf{e}}_z z$.

(b) Working entirely in circular cylindrical coordinates, show that

$$\nabla \cdot \mathbf{r} = 3 \quad \text{and} \quad \nabla \times \mathbf{r} = 0.$$

Solution.

Since $x = \rho\cos\varphi, y = \rho\sin\varphi$, $x\hat{\mathbf{e}}_x + y\hat{\mathbf{e}}_y = \rho\hat{\mathbf{e}}_\rho$, so $\mathbf{r} = \rho\hat{\mathbf{e}}_\rho + z\hat{\mathbf{e}}_z$. Its physical cylindrical components are $(\rho, 0, z)$. Therefore

$$\nabla \cdot \mathbf{r} = \frac{1}{\rho}\partial_\rho(\rho^2) + \frac{1}{\rho}\partial_\varphi 0 + \partial_z z = 2 + 0 + 1 = 3.$$

For the curl,

$$(\nabla \times \mathbf{r})_\rho = \rho^{-1}\partial_\varphi z - \partial_z 0 = 0, \quad (\nabla \times \mathbf{r})_\varphi = \partial_z \rho - \partial_\rho z = 0,$$

and $(\nabla \times \mathbf{r})_z = \rho^{-1}[\partial_\rho(\rho\, 0) - \partial_\varphi \rho] = 0$. Thus $\boxed{\nabla \times \mathbf{r} = 0}$.

3.10.11

(a) Show that the parity operation (reflection through the origin) on a point (ρ, φ, z) relative to fixed x-, y-, z-axes consists of the transformation

$$\rho \to \rho, \quad \varphi \to \varphi \pm \pi, \quad z \to -z.$$

(b) Show that $\hat{\mathbf{e}}_\rho$ and $\hat{\mathbf{e}}_\varphi$ have odd parity (reversal of direction) and that $\hat{\mathbf{e}}_z$ has even parity.

Note. The Cartesian unit vectors $\hat{\mathbf{e}}_x$, $\hat{\mathbf{e}}_y$, and $\hat{\mathbf{e}}_z$ remain constant.

Solution.

Under $(x, y, z) \mapsto (-x, -y, -z)$, $\rho = \sqrt{x^2 + y^2}$ is unchanged. Both $\cos\varphi = x/\rho$ and $\sin\varphi = y/\rho$ change sign, so $\varphi \mapsto \varphi + \pi$ modulo 2π and $z \mapsto -z$. Substituting this angle into the unit-vector functions gives

$$\hat{\mathbf{e}}_\rho(\varphi + \pi) = -\hat{\mathbf{e}}_\rho(\varphi), \quad \hat{\mathbf{e}}_\varphi(\varphi + \pi) = -\hat{\mathbf{e}}_\varphi(\varphi), \quad \hat{\mathbf{e}}_z(\varphi + \pi) = \hat{\mathbf{e}}_z.$$

This is evaluation of the curvilinear basis at the reflected point relative to fixed Cartesian unit vectors, as specified in the note.

3.10.12 A rigid body is rotating about a fixed axis with a constant angular velocity $\boldsymbol{\omega}$. Take $\boldsymbol{\omega}$ to lie along the z-axis. Express the position vector $\mathbf{r}$ in circular cylindrical coordinates and using circular cylindrical coordinates,

(a) calculate $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$,

(b) calculate $\nabla \times \mathbf{v}$.

ANS. (a) $\mathbf{v} = \hat{\mathbf{e}}_\varphi \omega \rho$ (b) $\nabla \times \mathbf{v} = 2\boldsymbol{\omega}$.

Solution.

With $\boldsymbol{\omega} = \omega \hat{\mathbf{e}}_z$ and $\mathbf{r} = \rho \hat{\mathbf{e}}_\rho + z\hat{\mathbf{e}}_z$,

$$\mathbf{v} = \omega \hat{\mathbf{e}}_z \times (\rho \hat{\mathbf{e}}_\rho + z\hat{\mathbf{e}}_z) = \boxed{\omega \rho \hat{\mathbf{e}}_\varphi}.$$

Its components are $(0, \omega\rho, 0)$ and are independent of z, φ. Hence both transverse curl components vanish, while

$$(\nabla \times \mathbf{v})_z = \frac{1}{\rho} \partial_\rho (\rho\, \omega \rho) = 2\omega.$$

Thus $\boxed{\nabla \times \mathbf{v} = 2\boldsymbol{\omega}}$.

3.10.13 Find the circular cylindrical components of the velocity and acceleration of a moving particle,

$$\begin{aligned} v_\rho &= \dot{\rho}, & a_\rho &= \ddot{\rho} - \rho\dot{\varphi}^2, \\ v_\varphi &= \rho\dot{\varphi}, & a_\varphi &= \rho\ddot{\varphi} + 2\dot{\rho}\dot{\varphi}, \\ v_z &= \dot{z}, & a_z &= \ddot{z}. \end{aligned}$$

Hint. $\mathbf{r}(t) = \hat{\mathbf{e}}_\rho(t)\rho(t) + \hat{\mathbf{e}}_z z(t)$
$= [\hat{\mathbf{e}}_x \cos\varphi(t) + \hat{\mathbf{e}}_y \sin\varphi(t)]\rho(t) + \hat{\mathbf{e}}_z z(t)$.
Note. $\dot{\rho} = d\rho/dt$, $\ddot{\rho} = d^2\rho/dt^2$, and so on.

Solution.

Along the trajectory, the basis derivatives are $\dot{\hat{\mathbf{e}}}_\rho = \dot{\varphi}\hat{\mathbf{e}}_\varphi$, $\dot{\hat{\mathbf{e}}}_\varphi = -\dot{\varphi}\hat{\mathbf{e}}_\rho$, and $\dot{\hat{\mathbf{e}}}_z = 0$. Differentiate $\mathbf{r} = \rho\hat{\mathbf{e}}_\rho + z\hat{\mathbf{e}}_z$:

$$\mathbf{v} = \dot{\rho}\hat{\mathbf{e}}_\rho + \rho\dot{\varphi}\hat{\mathbf{e}}_\varphi + \dot{z}\hat{\mathbf{e}}_z.$$

Differentiate each term once more:

$$\begin{aligned} \mathbf{a} = {} & \ddot{\rho}\hat{\mathbf{e}}_\rho + \dot{\rho}\dot{\varphi}\hat{\mathbf{e}}_\varphi + (\dot{\rho}\dot{\varphi} + \rho\ddot{\varphi})\hat{\mathbf{e}}_\varphi \\ & - \rho\dot{\varphi}^2\hat{\mathbf{e}}_\rho + \ddot{z}\hat{\mathbf{e}}_z. \end{aligned}$$

Collecting coefficients gives

$$\boxed{\begin{array}{lll} v_\rho = \dot{\rho}, & v_\varphi = \rho\dot{\varphi}, & v_z = \dot{z}, \\ a_\rho = \ddot{\rho} - \rho\dot{\varphi}^2, & a_\varphi = \rho\ddot{\varphi} + 2\dot{\rho}\dot{\varphi}, & a_z = \ddot{z}. \end{array}}$$

3.10.14 In right circular cylindrical coordinates, a particular vector function is given by

$$\mathbf{V}(\rho,\varphi)=\hat{\mathbf{e}}_\rho V_\rho(\rho,\varphi)+\hat{\mathbf{e}}_\varphi V_\varphi(\rho,\varphi).$$

Show that $\nabla\times\mathbf{V}$ has only a z-component. Note that this result will hold for any vector confined to a surface $q_3=$ constant as long as the products h_1V_1 and h_2V_2 are each independent of q_3.

Solution.

Use $V_z=0$ and the absence of z dependence:

$$(\nabla\times\mathbf{V})_\rho=\rho^{-1}\partial_\varphi V_z-\partial_z V_\varphi=0,\qquad(\nabla\times\mathbf{V})_\varphi=\partial_z V_\rho-\partial_\rho V_z=0.$$

The remaining component need not vanish:

$$\boxed{\nabla\times\mathbf{V}=\frac{\hat{\mathbf{e}}_z}{\rho}[\partial_\rho(\rho V_\varphi)-\partial_\varphi V_\rho]}.$$

More generally, when $V_3=0$ the first two orthogonal-coordinate curl components are

$$-(h_2h_3)^{-1}\partial_3(h_2V_2),\qquad(h_3h_1)^{-1}\partial_3(h_1V_1).$$

Both vanish under the conditions stated in the problem, leaving only component 3.

For this $V_z=0$ field, the transverse curl components are $-\partial_z V_\varphi$ and $\partial_z V_\rho$. The manual, p. 52, gives the first with the wrong sign and the second with an extra factor $1/\rho$. Both are zero here because the components are independent of z, so the conclusion survives those errors.

3.10.15 A conducting wire along the z-axis carries a current I. The resulting magnetic vector potential is given by

$$\mathbf{A}=\hat{\mathbf{e}}_z\frac{\mu I}{2\pi}\ln\frac{1}{\rho}.$$

Show that the magnetic induction $\mathbf{B}$ is given by

$$\mathbf{B}=\hat{\mathbf{e}}_\varphi\frac{\mu I}{2\pi\rho}.$$

Solution.

Let $K=\mu I/(2\pi)$, so $A_z=-K\ln\rho$, $A_\rho=A_\varphi=0$. The ρ and z curl components vanish. The remaining one is

$$B_\varphi=\partial_z A_\rho-\partial_\rho A_z=0-(-K/\rho)=K/\rho.$$

Thus $\boxed{\mathbf{B}=\mu I\,\hat{\mathbf{e}}_\varphi/(2\pi\rho)}$, valid for $\rho>0$ outside the ideal wire. Replacing $\ln(1/\rho)$ by $\ln(\rho_0/\rho)$ makes the logarithm dimensionless and adds only a constant to A_z, leaving $\mathbf{B}$ unchanged.

3.10.16 A force is described by

$$\mathbf{F}=-\hat{\mathbf{e}}_x\frac{y}{x^2+y^2}+\hat{\mathbf{e}}_y\frac{x}{x^2+y^2}.$$

(a) Express $\mathbf{F}$ in circular cylindrical coordinates.

Operating entirely in circular cylindrical coordinates for (b) and (c),

(b) Calculate the curl of $\mathbf{F}$ and

(c) Calculate the work done by $\mathbf{F}$ in encircling the unit circle once counterclockwise.

(d) How do you reconcile the results of (b) and (c)?

Solution.

(a) Substitute $x = \rho\cos\varphi, y = \rho\sin\varphi$:

$$\mathbf{F} = \frac{1}{\rho}(-\sin\varphi, \cos\varphi, 0) = \boxed{\hat{\mathbf{e}}_\varphi/\rho}.$$

(b) For $\rho > 0$, the transverse curl components vanish, and $(\nabla \times \mathbf{F})_z = \rho^{-1}\partial_\rho(\rho/\rho) = 0$. **(c)** On the unit circle, $d\mathbf{r} = \hat{\mathbf{e}}_\varphi d\varphi$, so $\boxed{W = \int_0^{2\pi} d\varphi = 2\pi}$. **(d)** The field is undefined on the axis. A disk spanning the circle crosses that singularity, so Stokes' theorem cannot use the ordinary zero curl over that disk. On an annulus excluding the axis, the outer counterclockwise circulation 2π and the inner clockwise circulation -2π cancel, in full agreement with zero curl. Locally $\mathbf{F} = \nabla\varphi$, but φ is not a global single-valued potential around the axis.

3.10.17 A calculation of the magnetohydrodynamic pinch effect involves the evaluation of $(\mathbf{B}\cdot\nabla)\mathbf{B}$. If the magnetic induction $\mathbf{B}$ is taken to be $\mathbf{B} = \hat{\mathbf{e}}_\varphi B_\varphi(\rho)$, show that

$$(\mathbf{B}\cdot\nabla)\mathbf{B} = -\hat{\mathbf{e}}_\rho B_\varphi^2/\rho.$$

Solution.

For $\mathbf{B} = B_\varphi(\rho)\hat{\mathbf{e}}_\varphi$, the directional derivative operator is $\mathbf{B}\cdot\nabla = (B_\varphi/\rho)\partial_\varphi$. It must differentiate the basis as well as the scalar component:

$$(\mathbf{B}\cdot\nabla)\mathbf{B} = \frac{B_\varphi}{\rho}\left[(\partial_\varphi B_\varphi)\hat{\mathbf{e}}_\varphi + B_\varphi\partial_\varphi\hat{\mathbf{e}}_\varphi\right] = \boxed{-\frac{B_\varphi^2}{\rho}\hat{\mathbf{e}}_\rho}.$$

Here $\partial_\varphi B_\varphi = 0$ and $\partial_\varphi\hat{\mathbf{e}}_\varphi = -\hat{\mathbf{e}}_\rho$.

3.10.18 Express the spherical polar unit vectors in terms of Cartesian unit vectors.

ANS.
$$\begin{aligned}
\hat{\mathbf{e}}_r &= \hat{\mathbf{e}}_x\sin\theta\cos\varphi + \hat{\mathbf{e}}_y\sin\theta\sin\varphi + \hat{\mathbf{e}}_z\cos\theta,\\
\hat{\mathbf{e}}_\theta &= \hat{\mathbf{e}}_x\cos\theta\cos\varphi + \hat{\mathbf{e}}_y\cos\theta\sin\varphi - \hat{\mathbf{e}}_z\sin\theta,\\
\hat{\mathbf{e}}_\varphi &= -\hat{\mathbf{e}}_x\sin\varphi + \hat{\mathbf{e}}_y\cos\varphi.
\end{aligned}$$

Solution.

The position map is $\mathbf{r} = r(\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$. Its r, θ, φ derivatives have lengths $1, r, r\sin\theta$ on a regular patch $r > 0$, $0 < \theta < \pi$. Normalize each derivative to obtain

$$\boxed{\begin{aligned}
\hat{\mathbf{e}}_r &= (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta),\\
\hat{\mathbf{e}}_\theta &= (\cos\theta\cos\varphi, \cos\theta\sin\varphi, -\sin\theta),\\
\hat{\mathbf{e}}_\varphi &= (-\sin\varphi, \cos\varphi, 0).
\end{aligned}}$$

Their norms are one, their pairwise dot products vanish, and $\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_\theta = \hat{\mathbf{e}}_\varphi$. At the polar axis, the angular basis requires a choice of azimuthal limit and is not a unique coordinate basis.

3.10.19 Resolve the Cartesian unit vectors into their spherical polar components:

$$\begin{aligned}\hat{\mathbf{e}}_x &= \hat{\mathbf{e}}_r \sin\theta\cos\varphi + \hat{\mathbf{e}}_\theta \cos\theta\cos\varphi - \hat{\mathbf{e}}_\varphi \sin\varphi,\\ \hat{\mathbf{e}}_y &= \hat{\mathbf{e}}_r \sin\theta\sin\varphi + \hat{\mathbf{e}}_\theta \cos\theta\sin\varphi + \hat{\mathbf{e}}_\varphi \cos\varphi,\\ \hat{\mathbf{e}}_z &= \hat{\mathbf{e}}_r \cos\theta - \hat{\mathbf{e}}_\theta \sin\theta.\end{aligned}$$

Solution.

Any vector is the sum of its projections on an orthonormal basis. For example, $\hat{\mathbf{e}}_x = \sum_{j=r,\theta,\varphi}(\hat{\mathbf{e}}_x \cdot \hat{\mathbf{e}}_j)\hat{\mathbf{e}}_j$. The projection coefficients are precisely the x entries in 3.10.18. Repeating for y, z gives

$$\boxed{\begin{aligned}\hat{\mathbf{e}}_x &= \sin\theta\cos\varphi\,\hat{\mathbf{e}}_r + \cos\theta\cos\varphi\,\hat{\mathbf{e}}_\theta - \sin\varphi\,\hat{\mathbf{e}}_\varphi,\\ \hat{\mathbf{e}}_y &= \sin\theta\sin\varphi\,\hat{\mathbf{e}}_r + \cos\theta\sin\varphi\,\hat{\mathbf{e}}_\theta + \cos\varphi\,\hat{\mathbf{e}}_\varphi,\\ \hat{\mathbf{e}}_z &= \cos\theta\,\hat{\mathbf{e}}_r - \sin\theta\,\hat{\mathbf{e}}_\theta.\end{aligned}}$$

Thus the inverse change-of-basis matrix is the transpose.

3.10.20

(a) Explain why it is not possible to relate a column vector $\mathbf{r}$ (with components x, y, z) to another column vector $\mathbf{r}'$ (with components r, θ, φ), via a matrix equation of the form $\mathbf{r}' = \mathsf{B}\mathbf{r}$ with one constant matrix B valid at all positions.

(b) One can write a matrix equation relating the Cartesian components of a vector to its components in spherical polar coordinates. Find the transformation matrix and determine whether it is orthogonal.

Solution.

(a) The impossible mapping is by one constant matrix valid for all positions. The radius $r = \sqrt{x^2+y^2+z^2}$ is not a linear function: $r(-\mathbf{x}) = r(\mathbf{x})$, whereas a linear first component would satisfy $(B(-\mathbf{x}))_1 = -(B\mathbf{x})_1$. For $\mathbf{x} \neq 0$ these cannot both equal its positive radius. The angular coordinates are nonlinear as well.

(b) Physical components of a vector at a fixed point obey

$$\boxed{\begin{pmatrix} V_r \\ V_\theta \\ V_\varphi \end{pmatrix} = T \begin{pmatrix} V_x \\ V_y \\ V_z \end{pmatrix}, \quad T = \begin{pmatrix} \sin\theta\cos\varphi & \sin\theta\sin\varphi & \cos\theta \\ \cos\theta\cos\varphi & \cos\theta\sin\varphi & -\sin\theta \\ -\sin\varphi & \cos\varphi & 0 \end{pmatrix}.}$$

The rows are the three unit vectors from 3.10.18, so their inner products give $TT^T = I$ and $\det T = 1$. Thus $T^{-1} = T^T$, but T depends on the observation point. In particular $T(x,y,z)^T = (r,0,0)^T$, the components of the position vector, rather than the coordinate labels $(r,\theta,\varphi)^T$.

3.10.21 Find the transformation matrix that converts the components of a vector in spherical polar coordinates into its components in circular cylindrical coordinates. Then find the matrix of the inverse transformation.

Solution.

The common azimuthal vector is unchanged, while

$$\hat{\mathbf{e}}_\rho = \sin\theta\,\hat{\mathbf{e}}_r + \cos\theta\,\hat{\mathbf{e}}_\theta, \qquad \hat{\mathbf{e}}_z = \cos\theta\,\hat{\mathbf{e}}_r - \sin\theta\,\hat{\mathbf{e}}_\theta.$$

Project $\mathbf{V}$ onto these directions. In the specified component orders,

$$\boxed{\begin{pmatrix} V_\rho \\ V_\varphi \\ V_z \end{pmatrix} = C \begin{pmatrix} V_r \\ V_\theta \\ V_\varphi \end{pmatrix}, \quad C = \begin{pmatrix} \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \\ \cos\theta & -\sin\theta & 0 \end{pmatrix}}.$$

The rows are orthonormal, so

$$\boxed{C^{-1} = C^T = \begin{pmatrix} \sin\theta & 0 & \cos\theta \\ \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \end{pmatrix}}.$$

Multiplication of these matrices gives I using $\sin^2\theta + \cos^2\theta = 1$.

3.10.22

(a) From the results of Exercise 3.10.18, calculate the partial derivatives of $\hat{\mathbf{e}}_r$, $\hat{\mathbf{e}}_\theta$, and $\hat{\mathbf{e}}_\varphi$ with respect to r, θ, and φ.

(b) With ∇ given by

$$\hat{\mathbf{e}}_r \frac{\partial}{\partial r} + \hat{\mathbf{e}}_\theta \frac{1}{r}\frac{\partial}{\partial \theta} + \hat{\mathbf{e}}_\varphi \frac{1}{r\sin\theta}\frac{\partial}{\partial \varphi}$$

(greatest space rate of change), use the results of part (a) to calculate $\nabla \cdot \nabla\psi$. This is an alternate derivation of the Laplacian.

Note. The derivatives of the left-hand ∇ operate on the unit vectors of the right-hand ∇ before the dot product is evaluated.

Solution.

(a) Differentiating the Cartesian expressions in 3.10.18 gives

	∂_r	∂_θ	∂_φ
$\hat{\mathbf{e}}_r$	0	$\hat{\mathbf{e}}_\theta$	$\sin\theta\,\hat{\mathbf{e}}_\varphi$
$\hat{\mathbf{e}}_\theta$	0	$-\hat{\mathbf{e}}_r$	$\cos\theta\,\hat{\mathbf{e}}_\varphi$
$\hat{\mathbf{e}}_\varphi$	0	0	$-\sin\theta\,\hat{\mathbf{e}}_r - \cos\theta\,\hat{\mathbf{e}}_\theta$

For the last entry, the Cartesian derivative is $(-\cos\varphi, -\sin\varphi, 0)$; its projections on r, θ, φ are $-\sin\theta, -\cos\theta, 0$, respectively.

(b) First apply the given ∇ to any vector $\mathbf{W} = W_r\hat{\mathbf{e}}_r + W_\theta\hat{\mathbf{e}}_\theta + W_\varphi\hat{\mathbf{e}}_\varphi$. The derivative table gives

$$\hat{\mathbf{e}}_r \cdot \partial_r\mathbf{W} = \partial_r W_r, \quad \hat{\mathbf{e}}_\theta \cdot \partial_\theta\mathbf{W} = W_r + \partial_\theta W_\theta,$$
$$\hat{\mathbf{e}}_\varphi \cdot \partial_\varphi\mathbf{W} = \sin\theta W_r + \cos\theta W_\theta + \partial_\varphi W_\varphi.$$

Including the scale factors,

$$\nabla \cdot \mathbf{W} = \partial_r W_r + \frac{2W_r}{r} + \frac{\partial_\theta W_\theta + \cot\theta W_\theta}{r} + \frac{\partial_\varphi W_\varphi}{r\sin\theta}.$$

Now put $\mathbf{W} = \nabla\psi$, whose physical components are $\psi_r, \psi_\theta/r, \psi_\varphi/(r\sin\theta)$. Substitution yields

$$\nabla^2\psi = \psi_{rr} + \frac{2}{r}\psi_r + \frac{1}{r^2}(\psi_{\theta\theta} + \cot\theta\,\psi_\theta) + \frac{\psi_{\varphi\varphi}}{r^2\sin^2\theta}$$
$$= \boxed{\frac{1}{r^2}\partial_r(r^2\partial_r\psi) + \frac{1}{r^2\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta\psi) + \frac{1}{r^2\sin^2\theta}\partial_\varphi^2\psi}.$$

The manual, p. 54, prints $\tan\theta/r^2$ instead of $\cot\theta/r^2$ as the coefficient of ∂_θ. Its following divergence-form identity must have $1/(r^2\sin\theta)$, not $1/(r^2\sin^2\theta)$, before $\partial_\theta(\sin\theta\,\partial_\theta)$. For a quick independent check, $\psi = r\cos\theta = z$ has zero Cartesian Laplacian. The correct formula gives $2\cos\theta/r - \cos\theta/r - \cos\theta/r = 0$; the printed $\tan\theta$ version instead gives $\cos(2\theta)/(r\cos\theta)$.

3.10.23 A rigid body is rotating about a fixed axis with a constant angular velocity $\boldsymbol{\omega}$. Take $\boldsymbol{\omega}$ to be along the z-axis. Using spherical polar coordinates,

(a) calculate $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$.

(b) calculate $\nabla \times \mathbf{v}$.

ANS. (a) $\mathbf{v} = \hat{\mathbf{e}}_\varphi \omega r \sin\theta$. (b) $\nabla \times \mathbf{v} = 2\boldsymbol{\omega}$.

Solution.

In spherical components, $\boldsymbol{\omega} = \omega(\cos\theta\,\hat{\mathbf{e}}_r - \sin\theta\,\hat{\mathbf{e}}_\theta)$ and $\mathbf{r} = r\hat{\mathbf{e}}_r$. The cross product is therefore $\boxed{\mathbf{v} = \omega r \sin\theta\,\hat{\mathbf{e}}_\varphi}$. The spherical curl gives

$$(\nabla \times \mathbf{v})_r = \frac{1}{r\sin\theta}\partial_\theta(\omega r \sin^2\theta) = 2\omega\cos\theta,$$

$$(\nabla \times \mathbf{v})_\theta = -\frac{1}{r}\partial_r(\omega r^2 \sin\theta) = -2\omega\sin\theta, \qquad (\nabla \times \mathbf{v})_\varphi = 0.$$

Thus the result is $2\omega(\cos\theta\,\hat{\mathbf{e}}_r - \sin\theta\,\hat{\mathbf{e}}_\theta) = \boxed{2\boldsymbol{\omega}}$.

3.10.24 A certain vector $\mathbf{V}$ has no radial component. Its curl has no tangential components. What does this imply about the radial dependence of the tangential components of $\mathbf{V}$?

Solution.

Since $V_r = 0$, the tangential curl components reduce to

$$(\nabla \times \mathbf{V})_\theta = -\frac{1}{r}\partial_r(rV_\varphi), \qquad (\nabla \times \mathbf{V})_\varphi = \frac{1}{r}\partial_r(rV_\theta).$$

Setting both to zero means rV_θ and rV_φ are independent of r. Hence, on $r > 0$,

$$\boxed{V_\theta = \frac{f(\theta,\varphi)}{r}, \qquad V_\varphi = \frac{g(\theta,\varphi)}{r}}.$$

If the region includes the origin and boundedness there is also required, then these two functions must vanish; that extra regularity was not part of the stated assumptions.

3.10.25 Modern physics lays great stress on the property of parity (whether a quantity remains invariant or changes sign under an inversion of the coordinate system). In Cartesian coordinates this means $x \to -x$, $y \to -y$, and $z \to -z$.

(a) Show that the inversion (reflection through the origin) of a point (r, θ, φ) relative to fixed x-, y-, z-axes consists of the transformation

$$r \to r, \quad \theta \to \pi - \theta, \quad \varphi \to \varphi \pm \pi.$$

(b) Show that $\hat{\mathbf{e}}_r$ and $\hat{\mathbf{e}}_\varphi$ have odd parity (reversal of direction) and that $\hat{\mathbf{e}}_\theta$ has even parity.

Solution.

Inversion leaves $r = \sqrt{x^2+y^2+z^2}$ unchanged. Since $\cos\theta = z/r$ changes sign, $\theta \mapsto \pi - \theta$. Both x, y change sign, so $\varphi \mapsto \varphi + \pi$ modulo 2π. Under these substitutions, $\sin\theta$ is unchanged, $\cos\theta$ changes sign, and $\sin\varphi, \cos\varphi$ both change sign. The component formulas in 3.10.18 then give

$$\hat{\mathbf{e}}_r(\pi-\theta, \varphi+\pi) = -\hat{\mathbf{e}}_r(\theta,\varphi),$$

$$\hat{\mathbf{e}}_\theta(\pi-\theta, \varphi+\pi) = \hat{\mathbf{e}}_\theta(\theta,\varphi), \qquad \hat{\mathbf{e}}_\varphi(\varphi+\pi) = -\hat{\mathbf{e}}_\varphi(\varphi).$$

Thus the stated basis-field parities follow relative to fixed Cartesian axes.

3.10.26 With $\mathbf{A}$ any vector,

$$\mathbf{A}\cdot\nabla\mathbf{r} = \mathbf{A}.$$

(a) Verify this result in Cartesian coordinates.

(b) Verify this result using spherical polar coordinates. Equation (3.156) provides ∇.

Solution.

(a) In Cartesian coordinates, $[(\mathbf{A}\cdot\nabla)\mathbf{r}]_j = \sum_i A_i \partial_i x_j = \sum_i A_i \delta_{ij} = A_j$. **(b)** In spherical coordinates $\mathbf{r} = r\hat{\mathbf{e}}_r$, so

$$\partial_r \mathbf{r} = \hat{\mathbf{e}}_r, \quad \partial_\theta \mathbf{r} = r\hat{\mathbf{e}}_\theta, \quad \partial_\varphi \mathbf{r} = r\sin\theta\,\hat{\mathbf{e}}_\varphi.$$

Therefore

$$(\mathbf{A}\cdot\nabla)\mathbf{r} = A_r\partial_r\mathbf{r} + \frac{A_\theta}{r}\partial_\theta\mathbf{r} + \frac{A_\varphi}{r\sin\theta}\partial_\varphi\mathbf{r} = \boxed{\mathbf{A}}.$$

The derivatives act on $\mathbf{r}$, not on the coefficients of $\mathbf{A}$.

3.10.27 Find the spherical coordinate components of the velocity and acceleration of a moving particle:

$$\begin{aligned} v_r &= \dot r, & a_r &= \ddot r - r\dot\theta^2 - r\sin^2\theta\,\dot\varphi^2,\\ v_\theta &= r\dot\theta, & a_\theta &= r\ddot\theta + 2\dot r\dot\theta - r\sin\theta\cos\theta\,\dot\varphi^2,\\ v_\varphi &= r\sin\theta\,\dot\varphi, & a_\varphi &= r\sin\theta\,\ddot\varphi + 2\dot r\sin\theta\,\dot\varphi + 2r\cos\theta\,\dot\theta\dot\varphi. \end{aligned}$$

Hint. $\mathbf{r}(t) = \hat{\mathbf{e}}_r(t) r(t)$
$= [\hat{\mathbf{e}}_x \sin\theta(t)\cos\varphi(t) + \hat{\mathbf{e}}_y \sin\theta(t)\sin\varphi(t) + \hat{\mathbf{e}}_z\cos\theta(t)]r(t).$

Note. The dot in $\dot r$, $\dot\theta$, $\dot\varphi$ means time derivative: $\dot r = dr/dt$, $\dot\theta = d\theta/dt$, $\dot\varphi = d\varphi/dt$.

Solution.

From the basis derivatives in 3.10.22,

$$\begin{aligned} \dot{\hat{\mathbf{e}}}_r &= \dot\theta\hat{\mathbf{e}}_\theta + \sin\theta\dot\varphi\hat{\mathbf{e}}_\varphi,\\ \dot{\hat{\mathbf{e}}}_\theta &= -\dot\theta\hat{\mathbf{e}}_r + \cos\theta\dot\varphi\hat{\mathbf{e}}_\varphi,\\ \dot{\hat{\mathbf{e}}}_\varphi &= -\sin\theta\dot\varphi\hat{\mathbf{e}}_r - \cos\theta\dot\varphi\hat{\mathbf{e}}_\theta. \end{aligned}$$

Differentiate $\mathbf{r} = r\hat{\mathbf{e}}_r$: $\mathbf{v} = \dot{r}\hat{\mathbf{e}}_r + r\dot{\theta}\hat{\mathbf{e}}_\theta + r\sin\theta\dot{\varphi}\hat{\mathbf{e}}_\varphi$. For clarity, the three differentiated terms contributing to acceleration are

$$\begin{aligned}
\frac{d}{dt}(\dot{r}\hat{\mathbf{e}}_r) &= \ddot{r}\hat{\mathbf{e}}_r + \dot{r}\dot{\theta}\hat{\mathbf{e}}_\theta + \dot{r}\sin\theta\dot{\varphi}\hat{\mathbf{e}}_\varphi,\\
\frac{d}{dt}(r\dot{\theta}\hat{\mathbf{e}}_\theta) &= (\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\mathbf{e}}_\theta - r\dot{\theta}^2\hat{\mathbf{e}}_r + r\cos\theta\dot{\theta}\dot{\varphi}\hat{\mathbf{e}}_\varphi,\\
\frac{d}{dt}(r\sin\theta\dot{\varphi}\hat{\mathbf{e}}_\varphi) &= (\dot{r}\sin\theta\dot{\varphi} + r\cos\theta\dot{\theta}\dot{\varphi} + r\sin\theta\ddot{\varphi})\hat{\mathbf{e}}_\varphi\\
&\quad - r\sin^2\theta\dot{\varphi}^2\hat{\mathbf{e}}_r - r\sin\theta\cos\theta\dot{\varphi}^2\hat{\mathbf{e}}_\theta.
\end{aligned}$$

Collecting each basis vector yields

$$\boxed{\begin{aligned}
a_r &= \ddot{r} - r\dot{\theta}^2 - r\sin^2\theta\dot{\varphi}^2,\\
a_\theta &= r\ddot{\theta} + 2\dot{r}\dot{\theta} - r\sin\theta\cos\theta\dot{\varphi}^2,\\
a_\varphi &= r\sin\theta\ddot{\varphi} + 2\dot{r}\sin\theta\dot{\varphi} + 2r\cos\theta\dot{\theta}\dot{\varphi}.
\end{aligned}}$$

The velocity components are the three coefficients of $\mathbf{v}$ above.

3.10.28 Express $\partial/\partial x$, $\partial/\partial y$, $\partial/\partial z$ in spherical polar coordinates.

ANS.
$$\begin{aligned}
\frac{\partial}{\partial x} &= \sin\theta\cos\varphi\frac{\partial}{\partial r} + \cos\theta\cos\varphi\frac{1}{r}\frac{\partial}{\partial\theta} - \frac{\sin\varphi}{r\sin\theta}\frac{\partial}{\partial\varphi},\\
\frac{\partial}{\partial y} &= \sin\theta\sin\varphi\frac{\partial}{\partial r} + \cos\theta\sin\varphi\frac{1}{r}\frac{\partial}{\partial\theta} + \frac{\cos\varphi}{r\sin\theta}\frac{\partial}{\partial\varphi},\\
\frac{\partial}{\partial z} &= \cos\theta\frac{\partial}{\partial r} - \sin\theta\frac{1}{r}\frac{\partial}{\partial\theta}.
\end{aligned}$$

Hint. Equate ∇_{xyz} and $\nabla_{r\theta\varphi}$.

Solution.

The chain rule is $\partial_x = r_x\partial_r + \theta_x\partial_\theta + \varphi_x\partial_\varphi$, and similarly for y, z. From $r^2 = x^2 + y^2 + z^2$, $r_x = x/r = \sin\theta\cos\varphi$. Differentiating $\cos\theta = z/r$ with respect to x gives $-\sin\theta\,\theta_x = -zx/r^3$, hence $\theta_x = \cos\theta\cos\varphi/r$. Finally differentiating $\varphi = \operatorname{atan2}(y, x)$ on a regular local branch gives $\varphi_x = -y/(x^2 + y^2)$. The full Jacobian obtained this way is

	∂_x	∂_y	∂_z
r	$\sin\theta\cos\varphi$	$\sin\theta\sin\varphi$	$\cos\theta$
θ	$\cos\theta\cos\varphi/r$	$\cos\theta\sin\varphi/r$	$-\sin\theta/r$
φ	$-\sin\varphi/(r\sin\theta)$	$\cos\varphi/(r\sin\theta)$	0

Substitution into the chain rule gives all three displayed operator formulas in the question. They hold on $r > 0, \sin\theta \neq 0$, where these coordinates are nonsingular.

3.10.29 Using results from Exercise 3.10.28, show that

$$-i\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right) = -i\frac{\partial}{\partial\varphi}.$$

This is the quantum mechanical operator corresponding to the z-component of orbital angular momentum.

Solution.

Insert the derivatives from 3.10.28 and $x = r\sin\theta\cos\varphi, y = r\sin\theta\sin\varphi$. In $x\partial_y - y\partial_x$, the coefficient of ∂_r is $r\sin^2\theta(\cos\varphi\sin\varphi - \sin\varphi\cos\varphi) = 0$. The coefficient of ∂_θ is $\sin\theta\cos\theta(\cos\varphi\sin\varphi - \sin\varphi\cos\varphi) = 0$. The coefficient of ∂_φ is $\cos^2\varphi + \sin^2\varphi = 1$. Thus $x\partial_y - y\partial_x = \partial_\varphi$ and $\boxed{L_z = -i\partial_\varphi}$.

3.10.30 With the quantum mechanical orbital angular momentum operator defined as $\mathbf{L} = -i(\mathbf{r} \times \nabla)$, show that

(a) $L_x + iL_y = e^{i\varphi}\left(\dfrac{\partial}{\partial\theta} + i\cot\theta\dfrac{\partial}{\partial\varphi}\right)$,

(b) $L_x - iL_y = -e^{-i\varphi}\left(\dfrac{\partial}{\partial\theta} - i\cot\theta\dfrac{\partial}{\partial\varphi}\right)$.

Solution.

Using $\mathbf{r} = r\hat{\mathbf{e}}_r$ in the spherical gradient,

$$\mathbf{L} = i\left(\hat{\mathbf{e}}_\theta \csc\theta\,\partial_\varphi - \hat{\mathbf{e}}_\varphi\partial_\theta\right).$$

Projection on x, y gives $L_x = i(\sin\varphi\,\partial_\theta + \cot\theta\cos\varphi\,\partial_\varphi)$ and $L_y = i(-\cos\varphi\,\partial_\theta + \cot\theta\sin\varphi\,\partial_\varphi)$. Therefore

$$\begin{aligned} L_x + iL_y &= (\cos\varphi + i\sin\varphi)\partial_\theta + (i\cos\varphi - \sin\varphi)\cot\theta\,\partial_\varphi \\ &= \boxed{e^{i\varphi}(\partial_\theta + i\cot\theta\,\partial_\varphi)}, \\ L_x - iL_y &= (-\cos\varphi + i\sin\varphi)\partial_\theta + (i\cos\varphi + \sin\varphi)\cot\theta\,\partial_\varphi \\ &= \boxed{-e^{-i\varphi}(\partial_\theta - i\cot\theta\,\partial_\varphi)}. \end{aligned}$$

3.10.31 Verify that $\mathbf{L} \times \mathbf{L} = i\mathbf{L}$ in spherical polar coordinates. $\mathbf{L} = -i(\mathbf{r} \times \nabla)$, the quantum mechanical orbital angular momentum operator.

Written in component form, this relation is

$$L_yL_z - L_zL_y = iL_x, \quad L_zL_x - L_xL_z = iL_y, \quad L_xL_y - L_yL_x = iL_z.$$

Using the commutator notation, $[A, B] = AB - BA$, and the definition of the Levi-Civita symbol ϵ_{ijk}, the above can also be written

$$[L_i, L_j] = i\epsilon_{ijk}L_k,$$

where i, j, k are x, y, z in any order.

Hint. Use spherical polar coordinates for $\mathbf{L}$ but Cartesian components for the cross product.

Solution.

The second printed component relation must be $[L_z, L_x] = iL_y$, not $-L_y$; the same error appears on textbook p. 201. Use the Cartesian components expressed in spherical coordinates: $L_x = iX$, $L_y = iY$, $L_z = -i\partial_\varphi$, where

$$X = \sin\varphi\,\partial_\theta + \cot\theta\cos\varphi\,\partial_\varphi, \quad Y = -\cos\varphi\,\partial_\theta + \cot\theta\sin\varphi\,\partial_\varphi.$$

For first-order operators, second-derivative terms in a commutator cancel. Put $t = \cot\theta$ and $s = \csc^2\theta$, so $\partial_\theta t = -s$. The coefficient of ∂_θ in $[X, Y]$ is $t\cos\varphi\sin\varphi - t\sin\varphi\cos\varphi = 0$. The coefficient of ∂_φ is

$$[-s\sin^2\varphi + t^2\cos^2\varphi] - [s\cos^2\varphi - t^2\sin^2\varphi] = t^2 - s = -1.$$

Thus $[L_x, L_y] = -[X, Y] = \partial_\varphi = iL_z$. For the other two commutators, simply differentiate the coefficients:

$$\begin{aligned} [L_z, L_x] &= [\partial_\varphi, X] = \cos\varphi\,\partial_\theta - \cot\theta\sin\varphi\,\partial_\varphi = iL_y, \\ [L_y, L_z] &= [Y, \partial_\varphi] = -\sin\varphi\,\partial_\theta - \cot\theta\cos\varphi\,\partial_\varphi = iL_x. \end{aligned}$$

The cross product of the Cartesian operator components therefore equals

$$\big([L_y, L_z], [L_z, L_x], [L_x, L_y]\big) = \boxed{i\mathbf{L}}.$$

3.10.32

(a) Using Eq. (3.156) show that

$$\mathbf{L} = -i(\mathbf{r} \times \nabla) = i\left(\hat{\mathbf{e}}_\theta \frac{1}{\sin\theta}\frac{\partial}{\partial\varphi} - \hat{\mathbf{e}}_\varphi \frac{\partial}{\partial\theta}\right).$$

(b) Resolving $\hat{\mathbf{e}}_\theta$ and $\hat{\mathbf{e}}_\varphi$ into Cartesian components, determine L_x, L_y, and L_z in terms of θ, φ, and their derivatives.

(c) From $L^2 = L_x^2 + L_y^2 + L_z^2$ show that

$$L^2 = -\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) - \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2} = -r^2\nabla^2 + \frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right).$$

Solution.

(a) In $\mathbf{r} \times \nabla$ the radial derivative term vanishes. Since $\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_\theta = \hat{\mathbf{e}}_\varphi$ and $\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_\varphi = -\hat{\mathbf{e}}_\theta$,

$$\boxed{\mathbf{L} = i(\hat{\mathbf{e}}_\theta \csc\theta\, \partial_\varphi - \hat{\mathbf{e}}_\varphi \partial_\theta)}.$$

(b) Insert the Cartesian basis components:

$$\boxed{\begin{aligned} L_x &= i(\sin\varphi\, \partial_\theta + \cot\theta\cos\varphi\, \partial_\varphi), \\ L_y &= i(-\cos\varphi\, \partial_\theta + \cot\theta\sin\varphi\, \partial_\varphi), \\ L_z &= -i\partial_\varphi. \end{aligned}}$$

(c) Apply the squared operators to an arbitrary smooth f, including derivatives of their coefficients. Write $s = \sin\varphi$, $c = \cos\varphi$, $t = \cot\theta$, $q = \csc^2\theta$. Direct product differentiation gives

$$\begin{aligned} L_x^2 f &= -s^2 f_{\theta\theta} - 2tscf_{\theta\varphi} - t^2c^2 f_{\varphi\varphi} - tc^2 f_\theta + sc(q + t^2)f_\varphi, \\ L_y^2 f &= -c^2 f_{\theta\theta} + 2tscf_{\theta\varphi} - t^2s^2 f_{\varphi\varphi} - ts^2 f_\theta - sc(q + t^2)f_\varphi, \\ L_z^2 f &= -f_{\varphi\varphi}. \end{aligned}$$

Mixed terms and the terms proportional to f_φ cancel. Using $s^2 + c^2 = 1$ and $1 + t^2 = q$,

$$L^2 f = -f_{\theta\theta} - \cot\theta f_\theta - \csc^2\theta f_{\varphi\varphi} = -\frac{1}{\sin\theta}\partial_\theta(\sin\theta f_\theta) - \frac{1}{\sin^2\theta}f_{\varphi\varphi}.$$

Comparison with the Laplacian from 3.10.22 gives

$$\boxed{L^2 = -r^2\nabla^2 + \partial_r(r^2\partial_r)}.$$

In the manual's part (a), p. 56, the coefficient of $\hat{\mathbf{e}}_\theta \csc\theta\, \partial_\varphi$ is printed as -1. It must be $+i$, as follows directly from $-i\hat{\mathbf{e}}_r \times \hat{\mathbf{e}}_\varphi = +i\hat{\mathbf{e}}_\theta$.

3.10.33 With $\mathbf{L} = -i\mathbf{r} \times \nabla$, verify the operator identities

(a) $\nabla = \hat{\mathbf{e}}_r \dfrac{\partial}{\partial r} - i\dfrac{\mathbf{r} \times \mathbf{L}}{r^2}$,

(b) $\mathbf{r}\nabla^2 - \nabla\left(1 + r\dfrac{\partial}{\partial r}\right) = i\nabla \times \mathbf{L}$.

Solution.

In (b), the first r is the position vector $\mathbf{r}$, as printed in bold on textbook p. 202; a scalar $r\nabla^2$ cannot equal the other vector operators. All identities below act on an arbitrary smooth scalar f.
(a) The ordinary triple-product identity applies after ∇ has acted:

$$\mathbf{r} \times \mathbf{L}f = -i\,\mathbf{r} \times (\mathbf{r} \times \nabla f) = -i[\mathbf{r}(\mathbf{r} \cdot \nabla f) - r^2 \nabla f].$$

Since $\mathbf{r} \cdot \nabla f = r\partial_r f$, $-i(\mathbf{r} \times \mathbf{L}f)/r^2 = \nabla f - \hat{\mathbf{e}}_r \partial_r f$. Rearranging proves (a).
(b) Set $D = \mathbf{r} \cdot \nabla = r\partial_r$. Using the curl-product rule on the ordinary vector field ∇f,

$$\begin{aligned} i\nabla \times (\mathbf{L}f) &= \nabla \times (\mathbf{r} \times \nabla f) \\ &= (\nabla f \cdot \nabla)\mathbf{r} - (\mathbf{r} \cdot \nabla)\nabla f - 3\nabla f + \mathbf{r}\,\nabla^2 f \\ &= \mathbf{r}\,\nabla^2 f - 2\nabla f - D(\nabla f). \end{aligned}$$

Componentwise $\partial_i(Df) = \partial_i(x_j \partial_j f) = \partial_i f + x_j \partial_j \partial_i f$, so $\nabla(Df) = \nabla f + D(\nabla f)$. Substitution yields $i\nabla \times (\mathbf{L}f) = \mathbf{r}\nabla^2 f - \nabla(f + Df)$, which is precisely the corrected operator identity.

3.10.34 Show that the following three forms (spherical coordinates) of $\nabla^2\psi(r)$ are equivalent:

$$\text{(a)}\ \frac{1}{r^2}\frac{d}{dr}\left[r^2\frac{d\psi(r)}{dr}\right]; \quad \text{(b)}\ \frac{1}{r}\frac{d^2}{dr^2}[r\psi(r)]; \quad \text{(c)}\ \frac{d^2\psi(r)}{dr^2} + \frac{2}{r}\frac{d\psi(r)}{dr}.$$

The second form is particularly convenient in establishing a correspondence between spherical polar and Cartesian descriptions of a problem.

Solution.

For $r > 0$, expand the first expression:

$$\frac{1}{r^2}\frac{d}{dr}(r^2\psi') = \frac{2r\psi' + r^2\psi''}{r^2} = \psi'' + \frac{2}{r}\psi'.$$

For the second, first differentiate $r\psi$ to get $\psi + r\psi'$, then differentiate again to get $2\psi' + r\psi''$. Division by r again gives $\psi'' + 2\psi'/r$. Thus all three expressions agree. At $r = 0$, values are obtained by regular limits when they exist; the displayed divisions by r do not themselves define a value at the origin.

3.10.35 A certain force field is given in spherical polar coordinates by

$$\mathbf{F} = \hat{\mathbf{e}}_r \frac{2P\cos\theta}{r^3} + \hat{\mathbf{e}}_\theta \frac{P}{r^3}\sin\theta, \quad r \geq P/2.$$

(a) Examine $\nabla \times \mathbf{F}$ to see if a potential exists.

(b) Calculate $\oint \mathbf{F} \cdot d\mathbf{r}$ for a unit circle in the plane $\theta = \pi/2$. What does this indicate about the force being conservative or nonconservative?

(c) If you believe that $\mathbf{F}$ may be described by $\mathbf{F} = -\nabla\psi$, find ψ. Otherwise simply state that no acceptable potential exists.

Take P constant and $r > 0$. In (b), the unit circle belongs to the stated region only if $P/2 \le 1$ in the chosen units.

Solution.

For constant P the only possibly nonzero curl component is

$$(\nabla \times \mathbf{F})_\varphi = \frac{1}{r}[\partial_r(rF_\theta) - \partial_\theta F_r]$$
$$= \frac{1}{r}\left[-\frac{2P\sin\theta}{r^3} - \left(-\frac{2P\sin\theta}{r^3}\right)\right] = 0.$$

The other components vanish because $F_\varphi = 0$ and no component depends on φ. On the equatorial unit circle, $d\mathbf{r} = \hat{\mathbf{e}}_\varphi d\varphi$, so the integrand is zero and $\boxed{\oint \mathbf{F}\cdot d\mathbf{r} = 0}$. This circle lies in the stated region only if $1 \ge P/2$ in the chosen units. The field formulas are evaluated at $r > 0$. A zero integral on one loop alone would not prove conservativeness; here an explicit potential settles it. Integrate $-\partial_r\psi = 2P\cos\theta/r^3$: $\psi = P\cos\theta/r^2 + f(\theta,\varphi)$. The equation $-r^{-1}\partial_\theta\psi = P\sin\theta/r^3$ gives $\partial_\theta f = 0$, and $F_\varphi = 0$ gives $\partial_\varphi f = 0$. Hence

$$\boxed{\psi = P\cos\theta/r^2 + C}.$$

It is single-valued and regular throughout the stated region away from the origin, so the force is conservative there.

3.10.36

(a) Show that $\mathbf{A} = -\hat{\mathbf{e}}_\varphi \cot\theta/r$ is a solution of $\nabla \times \mathbf{A} = \hat{\mathbf{e}}_r/r^2$.

(b) Show that this spherical polar coordinate solution agrees with the solution given for Exercise 3.9.5:

$$\mathbf{A} = \hat{\mathbf{e}}_x \frac{yz}{r(x^2+y^2)} - \hat{\mathbf{e}}_y \frac{xz}{r(x^2+y^2)}.$$

Note that the solution diverges for $\theta = 0, \pi$ corresponding to $x, y = 0$.

(c) Finally, show that $\mathbf{A} = -\hat{\mathbf{e}}_\theta \varphi \sin\theta/r$ is a solution. Note that although this solution does not diverge ($r \ne 0$), it is no longer single-valued for all possible azimuth angles.

Use a regular local spherical chart with a chosen azimuth branch; the potentials are not globally regular single-valued potentials on all of $r > 0$.

Solution.

(a) For $A_\varphi = -\cot\theta/r$, $A_r = A_\theta = 0$,

$$(\nabla \times \mathbf{A})_r = \frac{1}{r\sin\theta}\partial_\theta(\sin\theta A_\varphi) = \frac{1}{r\sin\theta}\partial_\theta(-\cos\theta/r) = \frac{1}{r^2}.$$

The other nontrivial candidate is $(\nabla \times \mathbf{A})_\theta = -r^{-1}\partial_r(rA_\varphi) = 0$, and the φ component is zero. **(b)** Use $\cot\theta = z/\rho$, $\rho^2 = x^2 + y^2$, and $\hat{\mathbf{e}}_\varphi = (-y/\rho, x/\rho, 0)$:

$$\mathbf{A} = -\frac{z}{r\rho}(-y/\rho, x/\rho, 0) = \left(\frac{yz}{r(x^2+y^2)}, -\frac{xz}{r(x^2+y^2)}, 0\right).$$

(c) For $A_\theta = -\varphi \sin\theta/r$, $A_r = A_\varphi = 0$,

$$(\nabla \times \mathbf{A})_r = -\frac{1}{r\sin\theta}\partial_\varphi A_\theta = \frac{1}{r^2}, \qquad (\nabla \times \mathbf{A})_\varphi = \frac{1}{r}\partial_r(rA_\theta) = 0.$$

The θ component is also zero. Both are local potentials: (a) excludes the polar axis, and (c) requires an azimuthal branch. Neither is a globally regular single-valued potential for the monopole field. Indeed increasing φ by 2π changes the latter potential by $-2\pi \sin\theta\,\hat{\mathbf{e}}_\theta/r = \nabla(2\pi\cos\theta)$. The curl is unchanged within overlapping regular patches, while the potential itself is not periodic in φ.

3.10.37 An electric dipole of moment $\mathbf{p}$ is located at the origin. The dipole creates an electric potential at $\mathbf{r}$ given by

$$\psi(\mathbf{r}) = \frac{\mathbf{p}\cdot\mathbf{r}}{4\pi\epsilon_0 r^3}.$$

Find the electric field, $\mathbf{E} = -\nabla\psi$ at $\mathbf{r}$.

Solution.

The dipole moment is constant and $r > 0$. Apply the scalar product rule:

$$\nabla\psi = \frac{1}{4\pi\epsilon_0}\left[r^{-3}\nabla(\mathbf{p}\cdot\mathbf{r}) + (\mathbf{p}\cdot\mathbf{r})\nabla(r^{-3})\right].$$

Here $\nabla(\mathbf{p}\cdot\mathbf{r}) = \mathbf{p}$ and $\nabla r^{-3} = -3r^{-4}\hat{\mathbf{e}}_r = -3\mathbf{r}/r^5$. Thus

$$\boxed{\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0 r^3}\left[3(\mathbf{p}\cdot\hat{\mathbf{e}}_r)\hat{\mathbf{e}}_r - \mathbf{p}\right]}.$$

For $\mathbf{p} = p\hat{\mathbf{e}}_z = p(\cos\theta\,\hat{\mathbf{e}}_r - \sin\theta\,\hat{\mathbf{e}}_\theta)$, this becomes $\mathbf{E} = p(2\cos\theta\,\hat{\mathbf{e}}_r + \sin\theta\,\hat{\mathbf{e}}_\theta)/(4\pi\epsilon_0 r^3)$, providing the component check against 3.10.35.

4 Chapter 4. Tensors and Differential Forms

Problems

4.1 Tensor Analysis

4.1.1 Show that if all the components of any tensor of any rank vanish in one particular coordinate system, they vanish in all coordinate systems.

Note. This point takes on special importance in the four-dimensional (4-D) curved space of general relativity. If a quantity, expressed as a tensor, exists in one coordinate system, it exists in all coordinate systems and is not just a consequence of a choice of a coordinate system (as are centrifugal and Coriolis forces in Newtonian mechanics).

Solution.

Let T have p contravariant indices and q covariant indices, so its rank is $p+q$. At the same point, two nonsingular coordinate systems are related by the tensor transformation law

$$T'^{i_1\cdots i_p}{}_{j_1\cdots j_q} = \frac{\partial x'^{i_1}}{\partial x^{a_1}}\cdots\frac{\partial x'^{i_p}}{\partial x^{a_p}}\frac{\partial x^{b_1}}{\partial x'^{j_1}}\cdots\frac{\partial x^{b_q}}{\partial x'^{j_q}}T^{a_1\cdots a_p}{}_{b_1\cdots b_q}.$$

Every repeated index is summed. If every unprimed component on the right is zero, each term in every sum is zero. Thus every primed component is zero, regardless of the values of the finite Jacobian factors. For rank zero the transformation law is simply $T' = T$, with the same conclusion. This proves the assertion pointwise; if the tensor vanishes throughout a region, apply the argument at each point of that region.

4.1.2 The components of tensor A are equal to the corresponding components of tensor B in one particular coordinate system denoted, by the superscript 0; that is,

$$A^0_{ij} = B^0_{ij}.$$

Show that tensor A is equal to tensor B, $A_{ij} = B_{ij}$, in all coordinate systems.

Solution.

The difference $C_{ij} = A_{ij} - B_{ij}$ is a tensor because the transformation law is linear. In the specified system, $C^0_{ab} = A^0_{ab} - B^0_{ab} = 0$. In any other system,

$$C_{ij} = \frac{\partial x^a_0}{\partial x^i}\frac{\partial x^b_0}{\partial x^j}C^0_{ab} = 0.$$

Consequently $A_{ij} - B_{ij} = 0$ in every system. Here the superscript 0 labels the original coordinate system; it is not a tensor index.

4.1.3 The last three components of a 4-D vector vanish in each of two reference frames. If the second reference frame is not merely a rotation of the first about the x_0 axis, meaning that at least one of the coefficients $\partial(x')^i/\partial x^0$ $(i = 1, 2, 3)$ is nonzero, show that the zeroth component vanishes in all reference frames. Translated into relativistic mechanics, this means that if momentum is conserved in two Lorentz frames, then energy is conserved in all Lorentz frames.

Solution.

Write the vector in the first frame as $V^\mu = (V^0, 0, 0, 0)$. Its spatial components in the second frame are

$$V'^i = \frac{\partial x'^i}{\partial x^\mu} V^\mu = \frac{\partial x'^i}{\partial x^0} V^0, \qquad i = 1, 2, 3.$$

By hypothesis all three left sides vanish, and at least one coefficient $\partial x'^i / \partial x^0$ is nonzero. The corresponding equation therefore gives $V^0 = 0$. All four components vanish in the first frame, so Exercise 4.1.1 gives $V'^\mu = 0$ in every frame.

For the conservation statement, apply this argument to the difference between total final and initial four-momenta,

$$\Delta P^\mu = (\Delta E/c, \Delta p_x, \Delta p_y, \Delta p_z).$$

Conservation of three-momentum in the two frames supplies the two sets of three zero components. The nontrivial mixing of time and space then forces $\Delta E = 0$, and hence conservation of all four components in every Lorentz frame.

4.1.4 From an analysis of the behavior of a general second-rank tensor under 90° and 180° rotations about the coordinate axes, show that an isotropic second-rank tensor in 3-D space must be a multiple of δ^i_j.

Solution.

In orthonormal Cartesian coordinates, write the components as a 3×3 matrix T. A rotation Q gives $T' = QTQ^{\mathsf{T}}$; isotropy means $QTQ^{\mathsf{T}} = T$ for every proper rotation.

First use the 180° rotations

$$R_1 = \mathrm{diag}(1, -1, -1), \qquad R_2 = \mathrm{diag}(-1, 1, -1).$$

If $R = \mathrm{diag}(s_1, s_2, s_3)$, then $(RTR^{\mathsf{T}})_{ij} = s_i s_j T_{ij}$. Thus R_1 changes the signs of $T_{12}, T_{13}, T_{21}, T_{31}$, and invariance forces each to equal its negative, hence to vanish. Similarly R_2 forces $T_{23} = T_{32} = 0$. We have proved, without assuming symmetry in advance, that $T = \mathrm{diag}(a, b, c)$.

Now use 90° rotations about the third and first axes:

$$Q_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad Q_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Direct multiplication gives

$$Q_3 T Q_3^{\mathsf{T}} = \mathrm{diag}(b, a, c), \qquad Q_1 T Q_1^{\mathsf{T}} = \mathrm{diag}(a, c, b).$$

Equality with T requires $a = b$ and $b = c$. Therefore

$$\boxed{T^i{}_j = \lambda \delta^i_j.}$$

Conversely $Q(\lambda I)Q^{\mathsf{T}} = \lambda Q Q^{\mathsf{T}} = \lambda I$, so every tensor of this form is isotropic.

4.1.5 The 4-D fourth-rank Riemann-Christoffel curvature tensor of general relativity, R_{iklm}, satisfies the symmetry relations

$$R_{iklm} = -R_{ikml} = -R_{kilm}.$$

With the indices running from 0 to 3, show that the number of independent components is reduced from 256 to 36 and that the condition

$$R_{iklm} = R_{lmik}$$

further reduces the number of independent components to 21. Finally, if the components satisfy an identity $R_{iklm} + R_{ilmk} + R_{imkl} = 0$, show that the number of independent components is reduced to 20.

Note. The final three-term identity furnishes new information only if all four indices are different.

Solution.

There are initially $4^4 = 256$ components. Antisymmetry in (i, k) makes every component with $i = k$ zero and determines a component with $i > k$ as the negative of one with $i < k$. The six possible ordered representatives of a pair are

$$01, \quad 02, \quad 03, \quad 12, \quad 13, \quad 23.$$

The same argument applies independently to (l, m). Thus the tensor is specified by a 6×6 array $C_{(ik),(lm)} = R_{iklm}$, containing $6^2 = 36$ potentially independent entries.

Pair exchange, $R_{iklm} = R_{lmik}$, says that C is symmetric. Its six diagonal entries and $6 \cdot 5/2 = 15$ entries above the diagonal determine all the others. The number is now $6 + 15 = 21$.

Consider the cyclic identity. If two of k, l, m coincide, cyclically relabel them so that $k = l$. Its left side then has the form

$$R_{ikkm} + R_{ikmk} + R_{imkk} = R_{ikkm} - R_{ikkm} + 0 = 0.$$

If i coincides with one of k, l, m, cyclically arrange $i = k$. Then

$$R_{iilm} + R_{ilmi} + R_{imil} = 0 - R_{ilim} + R_{ilim} = 0,$$

where the last equality uses pair exchange as well as antisymmetry. Thus a new restriction is possible only when all four indices differ. For the four available indices it is

$$R_{0123} + R_{0231} + R_{0312} = 0, \quad \text{or} \quad \boxed{C_{01,23} - C_{02,13} + C_{03,12} = 0.}$$

All other orderings give this same equation, possibly with the opposite sign. To see why, set $B_{iklm} = R_{iklm} + R_{ilmk} + R_{imkl}$. Interchanging k, l changes all three terms' signs, because

$$B_{ilkm} = R_{ilkm} + R_{ikml} + R_{imlk} = -R_{ilmk} - R_{iklm} - R_{imkl} = -B_{iklm}.$$

Interchanging i, k gives

$$B_{kilm} = -R_{iklm} - R_{imkl} - R_{ilmk} = -B_{iklm},$$

using pair exchange in the last two terms. Together with the cyclic symmetry in k, l, m, these interchanges make B completely antisymmetric. Its only independent entry in four dimensions is B_{0123}. The displayed relation is nontrivial, so it removes exactly one of the 21 parameters. The final number is

$$\boxed{21 - 1 = 20.}$$

4.1.6 T_{iklm} is antisymmetric with respect to all pairs of indices. How many independent components has it (in 3-D space)?

Solution.

Four indices chosen from the three values $1, 2, 3$ must include a repeated value. Interchanging the two slots containing that repeated value leaves a component unchanged. Complete antisymmetry says that the same interchange changes its sign. Therefore each component obeys $T_{iklm} = -T_{iklm}$ and is zero. The number of independent components is

$$\boxed{0.}$$

Equivalently, the space of completely antisymmetric rank-four tensors in three dimensions has dimension $\binom{3}{4} = 0$.

4.1.7 If $T_{\ldots i}$ is a tensor of rank n, show that $\partial T_{\ldots i}/\partial x^j$ is a tensor of rank $n+1$ (Cartesian coordinates).

Note. In non-Cartesian coordinate systems the coefficients a_{ij} are, in general, functions of the coordinates, and the derivatives of the components of a tensor of rank n do not form a tensor except in the special case $n = 0$. In this case the derivative does yield a covariant vector (tensor of rank 1).

Solution.

A change between orthonormal Cartesian systems has the form $x'^i = Q_{ia}x^a + b^i$, where Q is constant and $Q^{\mathsf{T}}Q = I$. Because $x^a = Q_{ia}(x'^i - b^i)$, the chain rule gives

$$\partial'_j \equiv \frac{\partial}{\partial x'^j} = Q_{jb}\frac{\partial}{\partial x^b} = Q_{jb}\partial_b.$$

For a rank-n tensor, its Cartesian component law is

$$T'_{i_1\cdots i_n} = Q_{i_1a_1}\cdots Q_{i_na_n}T_{a_1\cdots a_n}.$$

Differentiate this equation. The Q's are constant, so the product rule produces no derivatives of them:

$$\partial'_j T'_{i_1\cdots i_n} = Q_{i_1a_1}\cdots Q_{i_na_n}Q_{jb}\partial_b T_{a_1\cdots a_n}.$$

There is one transformation factor for each of the original n indices and one more for j. This is precisely a tensor law of rank $n+1$. For general coordinates the Jacobian factors depend on position, and differentiating them gives extra terms; a scalar has no such factors, so its ordinary first derivative remains a covector in general coordinates.

4.1.8 If $T_{ijk\ldots}$ is a tensor of rank n, show that $\sum_j \partial T_{ijk\ldots}/\partial x^j$ is a tensor of rank $n-1$ (Cartesian coordinates).

Solution.

Let $D_{ik\cdots} = \partial_j T_{ijk\cdots}$, with the repeated j summed. Use the same constant orthogonal matrix as in Exercise 4.1.7:

$$\begin{aligned}
D'_{ik\cdots} &= \partial'_j T'_{ijk\cdots} \\
&= Q_{jd}\partial_d\big(Q_{ia}Q_{jb}Q_{kc}\cdots T_{abc\cdots}\big) \\
&= Q_{ia}Q_{kc}\cdots\left(\sum_j Q_{jd}Q_{jb}\right)\partial_d T_{abc\cdots} \\
&= Q_{ia}Q_{kc}\cdots\delta_{db}\partial_d T_{abc\cdots} \\
&= Q_{ia}Q_{kc}\cdots\partial_b T_{abc\cdots} = Q_{ia}Q_{kc}\cdots D_{ac\cdots}.
\end{aligned}$$

Differentiation introduces one index and contraction removes two, leaving $n+1-2=n-1$ free indices. The displayed law verifies that the result is a rank-$(n-1)$ tensor. In the rank-one case the same calculation gives the scalar divergence $\partial_j T_j$.

4.1.9 The operator

$$\nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}$$

may be written as

$$\sum_{i=1}^{4} \frac{\partial^2}{\partial x_i^2},$$

using $x_4 = ict$. This is the 4-D Laplacian, sometimes called the d'Alembertian and denoted by $\Box^2$. Show that it is a scalar operator, that is, invariant under Lorentz transformations, represented here by complex orthogonal transformations in the space of vectors (x_1, x_2, x_3, x_4).

Solution.

Since $x_4 = ict$, differentiation at fixed spatial coordinates gives

$$\frac{\partial}{\partial x_4} = \frac{1}{ic}\frac{\partial}{\partial t}, \qquad \frac{\partial^2}{\partial x_4^2} = -\frac{1}{c^2}\frac{\partial^2}{\partial t^2}.$$

Thus $\sum_{a=1}^{4} \partial_a^2$ is the stated operator. In these complex coordinates a Lorentz transformation is represented by a constant complex orthogonal matrix Q: it preserves $x_1^2 + x_2^2 + x_3^2 + x_4^2 = \mathbf{x}^2 - c^2t^2$, so $Q^{\mathsf{T}}Q = I$. These are complex rotations, not arbitrary real Euclidean rotations mixing the physical time and space coordinates. The chain rule gives $\partial'_a = Q_{ab}\partial_b$. Hence, acting on a scalar function,

$$\begin{aligned}\sum_a \partial_a'^2 &= \sum_{a,b,d} Q_{ab}Q_{ad}\partial_b\partial_d \\ &= \sum_{b,d} (Q^{\mathsf{T}}Q)_{bd}\partial_b\partial_d = \sum_{b,d}\delta_{bd}\partial_b\partial_d = \sum_b \partial_b^2.\end{aligned}$$

This proves the operator's Lorentz invariance. No derivative of Q appears because inertial Lorentz transformations are linear.

4.1.10 The double summation $K_{ij}A_iB_j$ is invariant for any two vectors A_i and B_j. Prove that K_{ij} is a second-rank tensor.

Note. In the form $ds^2(\text{invariant}) = g_{ij}\,dx^i\,dx^j$, this result shows that the symmetric matrix g_{ij} is a tensor.

Solution.

Use orthonormal Cartesian coordinates, so vectors transform as $A'_p = Q_{pi}A_i$ and $B'_q = Q_{qj}B_j$. Invariance of the given scalar means

$$K_{ij}A_iB_j = K'_{pq}A'_pB'_q = K'_{pq}Q_{pi}Q_{qj}A_iB_j.$$

Choose A and B successively to be each pair of coordinate unit vectors. Because every such choice is permitted, the coefficient of each A_iB_j must agree:

$$K_{ij} = Q_{pi}Q_{qj}K'_{pq}.$$

Multiply by $Q_{ri}Q_{sj}$ and sum i, j. Orthogonality gives

$$Q_{ri}Q_{sj}K_{ij} = (Q_{ri}Q_{pi})(Q_{sj}Q_{qj})K'_{pq} = \delta_{rp}\delta_{sq}K'_{pq} = K'_{rs}.$$

Thus $K'_{rs} = Q_{ri}Q_{sj}K_{ij}$, the second-rank tensor law.

The line-element application uses the symmetry of the metric. A quadratic expression alone cannot determine an antisymmetric part of g_{ij}, because that part contracts to zero with $dx^i dx^j$. For a symmetric metric, polarization recovers the bilinear expression from the invariant quadratic form $q(U) = g_{ij}U^iU^j$:

$$g_{ij}U^iV^j = \frac{1}{2}\{q(U+V) - q(U) - q(V)\}.$$

To see the polarization step explicitly, expand

$$q(U+V) = q(U) + q(V) + g_{ij}U^iV^j + g_{ij}V^iU^j = q(U) + q(V) + 2g_{ij}U^iV^j,$$

where symmetry and relabeling i, j equate the cross terms. The left side of the polarization formula is consequently invariant for independent U, V. For a general nonsingular coordinate change, write $U^i = J^i{}_a U'^a$, $V^j = J^j{}_b V'^b$, where $J^i{}_a = \partial x^i/\partial x'^a$. Then

$$g'_{ab}U'^aV'^b = g_{ij}J^i{}_a J^j{}_b U'^a V'^b.$$

Taking U', V' to be each pair of coordinate unit vectors yields $g'_{ab} = J^i{}_a J^j{}_b g_{ij}$, the covariant metric transformation law.

4.1.11 The equation $K_{ij}A_{jk} = B_{ik}$ holds for all orientations of the coordinate system. If A and B are arbitrary second-rank tensors, show that K is a second-rank tensor also.

Solution.

Write the contraction as the matrix equation $KA = B$. Under a Cartesian rotation Q, the two known tensors transform as $A' = QAQ^{\mathsf{T}}$ and $B' = QBQ^{\mathsf{T}}$. Consequently

$$B' = QKAQ^{\mathsf{T}} = (QKQ^{\mathsf{T}})(QAQ^{\mathsf{T}}) = (QKQ^{\mathsf{T}})A'.$$

But the hypothesis states that the equation has the same form in the new coordinates: $K'A' = B'$. Subtracting gives

$$(K' - QKQ^{\mathsf{T}})A' = 0.$$

This must hold for every second-rank A'. In particular, set $A' = I$; then $K' - QKQ^{\mathsf{T}} = 0$. Hence

$$\boxed{K' = QKQ^{\mathsf{T}},}$$

which is the required tensor law. The word "arbitrary" is essential: if the equation were supplied only for one singular A, it would not in general determine K; for example $A = B = 0$ imposes no condition.

4.2 Pseudotensors, Dual Tensors

4.2.1 An antisymmetric square array is given by

$$\begin{pmatrix} 0 & C_3 & -C_2 \\ -C_3 & 0 & C_1 \\ C_2 & -C_1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & C_{12} & C_{13} \\ -C_{12} & 0 & C_{23} \\ -C_{13} & -C_{23} & 0 \end{pmatrix},$$

where (C_1, C_2, C_3) form a pseudovector. Assuming that the relation

$$C_i = \frac{1}{2!}\epsilon_{ijk}C_{jk}$$

holds in all orthonormal Cartesian coordinate systems, prove that C_{jk} is a tensor. (This is another form of the quotient theorem.)

Solution.

The numerical Levi-Civita symbol in this question uses orthonormal Cartesian axes. General curvilinear coordinates require the metric-dependent volume tensor, rather than the same numerical symbol. First invert the stated relation. The contraction identity

$$\epsilon_{\ell mi}\epsilon_{ijk} = \delta_{\ell j}\delta_{mk} - \delta_{\ell k}\delta_{mj}$$

gives

$$\epsilon_{\ell mi}C_i = \frac{1}{2}(\delta_{\ell j}\delta_{mk} - \delta_{\ell k}\delta_{mj})C_{jk} = \frac{1}{2}(C_{\ell m} - C_{m\ell}) = C_{\ell m}.$$

Here antisymmetry was essential in the final step. Thus $C_{jk} = \epsilon_{jki}C_i$, in every coordinate system.

For an orthogonal change of Cartesian axes Q, put $d = \det Q = \pm 1$. A pseudovector obeys $C'_i = dQ_{ia}C_a$, while the numerical Levi-Civita symbols obey the determinant identity

$$\epsilon_{jki} = dQ_{jb}Q_{kc}Q_{i\ell}\epsilon_{bc\ell}.$$

Substituting both identities into the inverted relation gives

$$\begin{aligned} C'_{jk} &= \epsilon_{jki}C'_i \\ &= d^2 Q_{jb}Q_{kc}Q_{i\ell}Q_{ia}\epsilon_{bc\ell}C_a \\ &= Q_{jb}Q_{kc}\delta_{\ell a}\epsilon_{bc\ell}C_a \\ &= Q_{jb}Q_{kc}C_{bc}. \end{aligned}$$

We used $d^2 = 1$ and $\sum_i Q_{i\ell}Q_{ia} = \delta_{\ell a}$. This is the ordinary second-rank tensor law, including reflections.

4.2.2 Show that a linear one-to-one correspondence between the full space of antisymmetric second-rank tensors and the space of vectors exists only in three dimensions. This is the sense of uniqueness of the vector product intended here.

Solution.

In n dimensions, an antisymmetric tensor has zero diagonal entries, and each entry below the diagonal is minus the corresponding entry above it. Its independent components are therefore indexed by unordered pairs of distinct indices, giving

$$\dim \Lambda^2 \mathbb{R}^n = \binom{n}{2} = \frac{n(n-1)}{2}.$$

A vector has n independent components. A linear one-to-one correspondence between the full component spaces requires equal dimensions:

$$\frac{n(n-1)}{2} = n \quad \Longrightarrow \quad n(n-3) = 0.$$

For a positive space dimension this forces $n = 3$. In three dimensions the correspondence actually exists: $C_i = \frac{1}{2}\epsilon_{ijk}C_{jk}$ and $C_{jk} = \epsilon_{jki}C_i$ are inverse maps, as verified in Exercise 4.2.1. The associated vector is a pseudovector when orientation reversals are included.

The uniqueness claim concerns this one-to-one correspondence with all antisymmetric second-rank tensors. The dimension count does not exclude other, differently defined products bearing the name cross product.

4.2.3 Write $\nabla \cdot \nabla \times \mathbf{A}$ and $\nabla \times \nabla\varphi$ in tensor (index) notation in $\mathbb{R}^3$ so that it becomes obvious that each expression vanishes. Assume that the fields have continuous second derivatives.

ANS.
$$\nabla \cdot \nabla \times \mathbf{A} = \epsilon_{ijk}\frac{\partial}{\partial x^i}\frac{\partial}{\partial x^j}A^k$$
$$(\nabla \times \nabla\varphi)_i = \epsilon_{ijk}\frac{\partial}{\partial x^j}\frac{\partial}{\partial x^k}\varphi.$$

Solution.

Assume the fields have continuous second derivatives so that mixed partial derivatives commute. With repeated indices summed,

$$\nabla \cdot (\nabla \times \mathbf{A}) = \partial_i(\epsilon_{ijk}\partial_j A^k) = \epsilon_{ijk}\partial_i\partial_j A^k.$$

Interchanging the dummy indices i, j changes ϵ_{ijk} to its negative but leaves $\partial_i\partial_j A^k$ unchanged. More explicitly,

$$\epsilon_{ijk}\partial_i\partial_j A^k = \frac{1}{2}\epsilon_{ijk}(\partial_i\partial_j - \partial_j\partial_i)A^k = 0.$$

Likewise,

$$(\nabla \times \nabla\varphi)_i = \epsilon_{ijk}\partial_j\partial_k\varphi = \frac{1}{2}\epsilon_{ijk}(\partial_j\partial_k - \partial_k\partial_j)\varphi = 0.$$

Both identities are contractions of an antisymmetric pair of indices with a symmetric pair.

4.2.4 Verify that each of the following fourth-rank tensors is isotropic, that is, that it has the same form independent of any rotation of the coordinate systems.

(a) $A^{ik}_{jl} = \delta^i_j\delta^k_l$,

(b) $B^{ij}_{kl} = \delta^i_k\delta^j_l + \delta^i_l\delta^j_k$,

(c) $C^{ij}_{kl} = \delta^i_k\delta^j_l - \delta^i_l\delta^j_k$.

Solution.

The mixed identity tensor is invariant. Indeed, under a rotation,

$$\delta'^i_{\ j} = Q_{ia}Q_{jb}\delta^a_b = \sum_a Q_{ia}Q_{ja} = (QQ^{\mathsf{T}})_{ij} = \delta^i_j.$$

For (a), apply the four-index transformation law and group the factors associated with each Kronecker delta:

$$\begin{aligned} A'^{ik}_{jl} &= Q_{ia}Q_{kc}Q_{jb}Q_{ld}\delta^a_b\delta^c_d \\ &= \left(\sum_a Q_{ia}Q_{ja}\right)\left(\sum_c Q_{kc}Q_{lc}\right) = \delta^i_j\delta^k_l = A^{ik}_{jl}. \end{aligned}$$

The same calculation with the indices arranged as in (b) gives

$$(\delta^i_k\delta^j_l)' = \delta^i_k\delta^j_l, \qquad (\delta^i_l\delta^j_k)' = \delta^i_l\delta^j_k.$$

Since transformation is linear, adding these two equations proves $B'^{ij}_{kl} = B^{ij}_{kl}$, and subtracting them proves $C'^{ij}_{kl} = C^{ij}_{kl}$. Thus all three tensors are isotropic. The argument in fact also allows orthogonal reflections.

4.2.5 Show that the two-index Levi-Civita symbol ϵ_{ij} is a second-rank pseudotensor (in two-dimensional [2-D] space). Does this contradict the uniqueness of δ^i_j (Exercise 4.1.4)?

Solution.

Use the orientation convention $\epsilon_{12} = 1$, so

$$E = (\epsilon_{ij}) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

For $Q = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, direct multiplication gives

$$QEQ^{\mathsf{T}} = \begin{pmatrix} -b & a \\ -d & c \end{pmatrix}\begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} 0 & ad-bc \\ bc-ad & 0 \end{pmatrix} = (\det Q)E.$$

In particular, for orthogonal Q the pseudotensor law is

$$E' = (\det Q)QEQ^{\mathsf{T}} = (\det Q)^2 E = E.$$

Thus the same numerical symbol in every oriented coordinate convention is a second-rank pseudotensor. A proper rotation has $\det Q = 1$; a reflection has $\det Q = -1$ and contributes the extra sign appropriate to a pseudotensor.

There is no contradiction with Exercise 4.1.4, which concerns *three* dimensions. In two dimensions E is a nonzero antisymmetric matrix invariant under all proper rotations; the rotations about three distinct axes used in that exercise are unavailable. Moreover, E is not invariant under the ordinary tensor transformation law for reflections, whereas the identity is.

4.2.6 Represent ϵ_{ij} by a 2×2 matrix, and using the 2×2 rotation matrix of Eq. (3.23), show that ϵ_{ij} is invariant under proper orthogonal similarity transformations.

Solution.

Here "orthogonal similarity transformations" must be restricted to proper rotations ($\det Q = 1$). For general orthogonal Q the correct relation is $QEQ^{-1} = (\det Q)E$.

With $c = \cos\phi$, $s = \sin\phi$, the rotation matrix in the question is

$$R = \begin{pmatrix} c & s \\ -s & c \end{pmatrix}, \qquad R^{-1} = R^{\mathsf{T}} = \begin{pmatrix} c & -s \\ s & c \end{pmatrix}.$$

Since $E = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, multiply explicitly:

$$RER^{-1} = \begin{pmatrix} -s & c \\ -c & -s \end{pmatrix} \begin{pmatrix} c & -s \\ s & c \end{pmatrix} = \begin{pmatrix} 0 & s^2+c^2 \\ -(c^2+s^2) & 0 \end{pmatrix} = E.$$

This proves the desired rotational invariance. To check the necessary qualification, take the reflection $Q = \text{diag}(1, -1)$. Then

$$QEQ^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -E.$$

Under the *pseudotensor* law its additional factor $\det Q = -1$ restores E, exactly as in Exercise 4.2.5.

4.2.7 Given $A_k = \frac{1}{2}\epsilon_{ijk}B^{ij}$ with $B^{ij} = -B^{ji}$, antisymmetric, show that

$$B^{mn} = \epsilon^{mnk}A_k.$$

Solution.

Multiply the given equation by ϵ^{mnk} and sum k:

$$\begin{aligned} \epsilon^{mnk}A_k &= \frac{1}{2}\epsilon^{mnk}\epsilon_{ijk}B^{ij} \\ &= \frac{1}{2}(\delta_i^m\delta_j^n - \delta_j^m\delta_i^n)B^{ij} \\ &= \frac{1}{2}(B^{mn} - B^{nm}) \\ &= \frac{1}{2}(B^{mn} + B^{mn}) = B^{mn}. \end{aligned}$$

This is the stated inverse relation. It also checks the factor 1/2: without it the two antisymmetric terms would produce $2B^{mn}$.

4.3 Tensors in General Coordinates

4.3.1 For the special case of 3-D space ($\boldsymbol{\varepsilon}_1, \boldsymbol{\varepsilon}_2, \boldsymbol{\varepsilon}_3$ defining a right-handed coordinate system, not necessarily orthogonal), show that

$$\boldsymbol{\varepsilon}^i = \frac{\boldsymbol{\varepsilon}_j \times \boldsymbol{\varepsilon}_k}{\boldsymbol{\varepsilon}_j \times \boldsymbol{\varepsilon}_k \cdot \boldsymbol{\varepsilon}_i}, \quad i, j, k = 1, 2, 3 \text{ and cyclic permutations.}$$

Note. These contravariant basis vectors $\boldsymbol{\varepsilon}^i$ define the reciprocal lattice space of Example 3.2.1.

Solution.

The reciprocal basis is defined by $\boldsymbol{\varepsilon}^i \cdot \boldsymbol{\varepsilon}_\ell = \delta_\ell^i$. Let

$$D = (\boldsymbol{\varepsilon}_1 \times \boldsymbol{\varepsilon}_2) \cdot \boldsymbol{\varepsilon}_3.$$

Linear independence implies $D \neq 0$, and the specified right-handedness makes $D > 0$. Cyclic permutations preserve this scalar triple product. For cyclic (i, j, k) define

$$\mathbf{b}^i = \frac{\boldsymbol{\varepsilon}_j \times \boldsymbol{\varepsilon}_k}{D}.$$

The cross product is perpendicular to both its factors, so

$$\mathbf{b}^i \cdot \boldsymbol{\varepsilon}_j = 0, \qquad \mathbf{b}^i \cdot \boldsymbol{\varepsilon}_k = 0.$$

Its remaining scalar product is

$$\mathbf{b}^i \cdot \boldsymbol{\varepsilon}_i = \frac{(\boldsymbol{\varepsilon}_j \times \boldsymbol{\varepsilon}_k) \cdot \boldsymbol{\varepsilon}_i}{D} = 1.$$

These are exactly the three reciprocal-basis conditions. They determine a vector uniquely: the difference of any two solutions would be orthogonal to three linearly independent vectors and hence zero. Therefore $\mathbf{b}^i = \boldsymbol{\varepsilon}^i$, proving the formula.

4.3.2 If the covariant vectors $\boldsymbol{\varepsilon}_i$ are orthogonal, show that

(a) g_{ij} is diagonal,

(b) $g^{ii} = 1/g_{ii}$ (no summation),

(c) $|\boldsymbol{\varepsilon}^i| = 1/|\boldsymbol{\varepsilon}_i|$.

Solution.

Put $h_i = |\boldsymbol{\varepsilon}_i| > 0$. Orthogonality means

$$g_{ij} = \boldsymbol{\varepsilon}_i \cdot \boldsymbol{\varepsilon}_j = \begin{cases} h_i^2, & i = j, \\ 0, & i \neq j. \end{cases}$$

Thus (a) the covariant metric is diagonal. Its inverse must satisfy $\sum_j g^{ij} g_{jk} = \delta^i_k$. For a diagonal matrix this reads $g^{ik} h_k^2 = \delta^i_k$, so

$$g^{ik} = 0 \quad (i \neq k), \qquad \boxed{g^{ii} = h_i^{-2} = 1/g_{ii}} \quad \text{(no sum)}.$$

This proves (b). To obtain (c), observe that $\boldsymbol{\varepsilon}_i / h_i^2$ has scalar product 1 with $\boldsymbol{\varepsilon}_i$ and 0 with all the other basis vectors. It is consequently the reciprocal vector:

$$\boldsymbol{\varepsilon}^i = \frac{\boldsymbol{\varepsilon}_i}{h_i^2}, \qquad \boxed{|\boldsymbol{\varepsilon}^i| = \frac{h_i}{h_i^2} = \frac{1}{|\boldsymbol{\varepsilon}_i|}}.$$

4.3.3 Prove that $(\boldsymbol{\varepsilon}^i \cdot \boldsymbol{\varepsilon}^j)(\boldsymbol{\varepsilon}_j \cdot \boldsymbol{\varepsilon}_k) = \delta^i_k$.

Solution.

The dual-basis expansion of any vector $\mathbf{v}$ is

$$\mathbf{v} = \sum_j (\boldsymbol{\varepsilon}_j \cdot \mathbf{v}) \boldsymbol{\varepsilon}^j.$$

To verify it, take the scalar product with each $\boldsymbol{\varepsilon}_k$; the right side gives $\sum_j (\boldsymbol{\varepsilon}_j \cdot \mathbf{v}) \delta^j_k = \boldsymbol{\varepsilon}_k \cdot \mathbf{v}$. A vector is determined by these three scalar products. Apply the expansion to $\mathbf{v} = \boldsymbol{\varepsilon}_k$ and then dot with $\boldsymbol{\varepsilon}^i$:

$$\boldsymbol{\varepsilon}^i \cdot \boldsymbol{\varepsilon}_k = \sum_j (\boldsymbol{\varepsilon}_j \cdot \boldsymbol{\varepsilon}_k)(\boldsymbol{\varepsilon}^i \cdot \boldsymbol{\varepsilon}^j) = \delta^i_k.$$

This is the desired identity, also written $g^{ij} g_{jk} = \delta^i_k$.

4.3.4 Show that $\Gamma^m_{jk} = \Gamma^m_{kj}$.

Solution.

For a smooth coordinate map $\mathbf{r}(q^1, q^2, q^3)$,

$$\boldsymbol{\varepsilon}_j = \frac{\partial \mathbf{r}}{\partial q^j}, \qquad \frac{\partial \boldsymbol{\varepsilon}_j}{\partial q^k} = \Gamma^m_{jk}\boldsymbol{\varepsilon}_m.$$

Commuting the mixed second derivatives of $\mathbf{r}$ gives

$$\Gamma^m_{jk}\boldsymbol{\varepsilon}_m = \frac{\partial^2 \mathbf{r}}{\partial q^k \partial q^j} = \frac{\partial^2 \mathbf{r}}{\partial q^j \partial q^k} = \Gamma^m_{kj}\boldsymbol{\varepsilon}_m.$$

Linear independence of the basis forces equality of each coefficient: $\boxed{\Gamma^m_{jk} = \Gamma^m_{kj}}$. Equivalently, dot the equality with $\boldsymbol{\varepsilon}^m$. This is the symmetry of the coordinate Christoffel connection used here; it is not a claim that every possible connection has zero torsion.

4.3.5 Derive the covariant and contravariant metric tensors for circular cylindrical coordinates.

Solution.

Choose $q^1 = \rho$, $q^2 = \phi$, $q^3 = z$, with

$$\mathbf{r} = (\rho\cos\phi, \rho\sin\phi, z).$$

Differentiating gives the coordinate basis

$$\boldsymbol{\varepsilon}_\rho = (\cos\phi, \sin\phi, 0), \quad \boldsymbol{\varepsilon}_\phi = (-\rho\sin\phi, \rho\cos\phi, 0), \quad \boldsymbol{\varepsilon}_z = (0, 0, 1).$$

Its squared lengths are $1, \rho^2, 1$. The only potentially nonzero off-diagonal scalar product is $\boldsymbol{\varepsilon}_\rho \cdot \boldsymbol{\varepsilon}_\phi = -\rho\cos\phi\sin\phi + \rho\sin\phi\cos\phi = 0$; the other off-diagonal scalar products also vanish. Therefore

$$\boxed{(g_{ij}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \rho^2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad (g^{ij}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \rho^{-2} & 0 \\ 0 & 0 & 1 \end{pmatrix}.}$$

As checks, their product is the identity and $ds^2 = g_{ij}\, dq^i dq^j = d\rho^2 + \rho^2 d\phi^2 + dz^2$. These coordinate formulas apply for $\rho > 0$; at the axis the angular coordinate is singular.

4.3.6 Show that the covariant derivative of a covariant vector is given by

$$V_{i;j} \equiv \frac{\partial V_i}{\partial q^j} - V_k \Gamma^k_{ij}.$$

Hint. Differentiate

$$\boldsymbol{\varepsilon}^i \cdot \boldsymbol{\varepsilon}_j = \delta^i_j.$$

Solution.

Differentiate the reciprocal-basis identity with respect to q^j:

$$0 = \partial_j(\boldsymbol{\varepsilon}^i \cdot \boldsymbol{\varepsilon}_k) = (\partial_j \boldsymbol{\varepsilon}^i) \cdot \boldsymbol{\varepsilon}_k + \boldsymbol{\varepsilon}^i \cdot (\Gamma^m_{kj}\boldsymbol{\varepsilon}_m).$$

The last term is Γ^i_{kj}, so $(\partial_j \boldsymbol{\varepsilon}^i)\cdot\boldsymbol{\varepsilon}_k = -\Gamma^i_{kj}$. Expanding this derivative in the reciprocal basis therefore gives

$$\partial_j \boldsymbol{\varepsilon}^i = -\Gamma^i_{kj}\boldsymbol{\varepsilon}^k.$$

Now express $\mathbf{V}$ using covariant components, $\mathbf{V} = V_i\boldsymbol{\varepsilon}^i$. Differentiation gives

$$\begin{aligned}\partial_j \mathbf{V} &= (\partial_j V_i)\boldsymbol{\varepsilon}^i + V_i \partial_j \boldsymbol{\varepsilon}^i \\ &= (\partial_j V_i)\boldsymbol{\varepsilon}^i - V_i\Gamma^i_{kj}\boldsymbol{\varepsilon}^k \\ &= (\partial_j V_i - V_k\Gamma^k_{ij})\boldsymbol{\varepsilon}^i.\end{aligned}$$

In the last line the dummy indices in the second term were relabeled. The coefficients of the reciprocal basis are the covariant components of the derivative, hence

$$\boxed{V_{i;j} = \partial_j V_i - V_k\Gamma^k_{ij}.}$$

4.3.7 Verify that $V_{i;j} = g_{ik}V^k_{\ ;j}$ by showing that

$$\frac{\partial V_i}{\partial q^j} - V_k\Gamma^k_{ij} = g_{ik}\left[\frac{\partial V^k}{\partial q^j} + V^m\Gamma^k_{mj}\right].$$

Solution.

Start from $V_i = g_{ik}V^k$. We first need the derivative of the metric. By differentiating its basis-vector definition,

$$\begin{aligned}\partial_j g_{ik} &= \partial_j(\boldsymbol{\varepsilon}_i\cdot\boldsymbol{\varepsilon}_k) \\ &= (\Gamma^m_{ij}\boldsymbol{\varepsilon}_m)\cdot\boldsymbol{\varepsilon}_k + \boldsymbol{\varepsilon}_i\cdot(\Gamma^m_{kj}\boldsymbol{\varepsilon}_m) \\ &= \Gamma^m_{ij}g_{mk} + \Gamma^m_{kj}g_{im}.\end{aligned}$$

Use this identity in the product rule:

$$\begin{aligned}V_{i;j} &= \partial_j(g_{ik}V^k) - V_\ell\Gamma^\ell_{ij} \\ &= g_{ik}\partial_j V^k + (\partial_j g_{ik})V^k - g_{\ell k}V^k\Gamma^\ell_{ij} \\ &= g_{ik}\partial_j V^k + \Gamma^m_{ij}g_{mk}V^k + \Gamma^m_{kj}g_{im}V^k - g_{mk}V^k\Gamma^m_{ij} \\ &= g_{ik}\partial_j V^k + g_{im}\Gamma^m_{kj}V^k \\ &= g_{ik}\big(\partial_j V^k + \Gamma^k_{mj}V^m\big) = \boxed{g_{ik}V^k_{\ ;j}}.\end{aligned}$$

The two terms containing $\Gamma^m_{ij}g_{mk}V^k$ cancel exactly; only dummy-index relabeling is used in the last line.

4.3.8 From the circular cylindrical metric tensor g_{ij}, calculate the Γ^k_{ij} for circular cylindrical coordinates.

Note. There are only three nonvanishing Γ.

Solution.

For $q^i = (\rho, \phi, z)$, Exercise 4.3.5 gives $g_{ij} = \mathrm{diag}(1, \rho^2, 1)$ and $g^{ij} = \mathrm{diag}(1, \rho^{-2}, 1)$. Use

$$\Gamma^k_{ij} = \frac{1}{2}g^{k\ell}(\partial_i g_{\ell j} + \partial_j g_{\ell i} - \partial_\ell g_{ij}).$$

All metric derivatives vanish except $\partial_\rho g_{\phi\phi} = 2\rho$. For an upper ρ index the inverse metric selects $\ell = \rho$; only the last derivative can survive, and only when $i = j = \phi$:

$$\Gamma^\rho_{\phi\phi} = \frac{1}{2} g^{\rho\rho}(0 + 0 - \partial_\rho g_{\phi\phi}) = -\rho.$$

For an upper ϕ index, the first or second derivative survives:

$$\Gamma^\phi_{\rho\phi} = \frac{1}{2} g^{\phi\phi}(\partial_\rho g_{\phi\phi} + 0 - 0) = \frac{1}{2\rho^2}(2\rho) = \frac{1}{\rho}, \qquad \Gamma^\phi_{\phi\rho} = \frac{1}{\rho}.$$

With an upper z index, every term differentiates a constant metric entry or differentiates with respect to z, so every term is zero. There are no other nonzero possibilities. Thus the complete list is

$$\boxed{\Gamma^\rho_{\phi\phi} = -\rho, \qquad \Gamma^\phi_{\rho\phi} = \Gamma^\phi_{\phi\rho} = \rho^{-1}.}$$

4.3.9 Using the Γ^k_{ij} from Exercise 4.3.8, write out the covariant derivatives $V^i_{;j}$ of a vector $\mathbf{V}$ in circular cylindrical coordinates.

Solution.

Use the contravariant derivative formula

$$V^i{}_{;j} = \partial_j V^i + \Gamma^i_{mj} V^m.$$

For $i = \rho$, the only nonzero connection term occurs at $m = j = \phi$ and equals $-\rho V^\phi$. For $i = \phi$, the terms are V^ϕ/ρ when $j = \rho$ and V^ρ/ρ when $j = \phi$. There are no connection terms for $i = z$. Thus all nine derivatives are

$$\boxed{\left(V^i{}_{;j}\right) = \begin{pmatrix} \partial_\rho V^\rho & \partial_\phi V^\rho - \rho V^\phi & \partial_z V^\rho \\ \partial_\rho V^\phi + V^\phi/\rho & \partial_\phi V^\phi + V^\rho/\rho & \partial_z V^\phi \\ \partial_\rho V^z & \partial_\phi V^z & \partial_z V^z \end{pmatrix}.}$$

Rows are labeled by $i = (\rho, \phi, z)$ and columns by $j = (\rho, \phi, z)$. These are components in the coordinate basis, whose angular vector has length ρ. If $\mathbf{V} = V_{\hat\rho}\hat{\mathbf{e}}_\rho + V_{\hat\phi}\hat{\mathbf{e}}_\phi + V_{\hat z}\hat{\mathbf{e}}_z$, then

$$V^\rho = V_{\hat\rho}, \qquad V^\phi = V_{\hat\phi}/\rho, \qquad V^z = V_{\hat z}.$$

For example, the first row's angular entry becomes $\partial_\phi V_{\hat\rho} - V_{\hat\phi}$, agreeing with direct differentiation of the unit vectors $\partial_\phi \hat{\mathbf{e}}_\rho = \hat{\mathbf{e}}_\phi$ and $\partial_\phi \hat{\mathbf{e}}_\phi = -\hat{\mathbf{e}}_\rho$.

4.3.10 Show that for the metric tensor $g_{ij;k} = g^{ij}_{;k} = 0$.

Solution.

The covariant derivative acts on each covariant index with a minus connection term:

$$g_{ij;k} = \partial_k g_{ij} - \Gamma^m_{ik} g_{mj} - \Gamma^m_{jk} g_{im}.$$

Differentiate $g_{ij} = \boldsymbol{\varepsilon}_i \cdot \boldsymbol{\varepsilon}_j$ as in Exercise 4.3.7 to obtain

$$\partial_k g_{ij} = \Gamma^m_{ik} g_{mj} + \Gamma^m_{jk} g_{im}.$$

Substitution cancels both terms, so $\boxed{g_{ij;k} = 0}$.

For the inverse metric, differentiate $g^{ia}g_{ab} = \delta^i_b$:

$$(\partial_k g^{ia})g_{ab} + g^{ia}\partial_k g_{ab} = 0.$$

Multiply by g^{bj} and sum b. This isolates

$$\begin{aligned}\partial_k g^{ij} &= -g^{ia}(\partial_k g_{ab})g^{bj} \\ &= -g^{ia}(\Gamma^m_{ak}g_{mb} + \Gamma^m_{bk}g_{am})g^{bj} \\ &= -g^{ia}\Gamma^j_{ak} - \Gamma^i_{bk}g^{bj}.\end{aligned}$$

The contravariant derivative has a plus connection term for each upper index. Consequently

$$g^{ij}{}_{;k} = \partial_k g^{ij} + \Gamma^i_{mk}g^{mj} + \Gamma^j_{mk}g^{im} = 0.$$

Both cancellations rely on the metric-compatible Christoffel connection used in this section.

4.3.11 Starting with the divergence in tensor notation, Eq. (4.70), develop the divergence of a vector in spherical polar coordinates, Eq. (3.157).

Solution.

The tensor divergence is

$$V^i{}_{;i} = \partial_i V^i + \Gamma^i_{mi}V^m = \frac{1}{\sqrt{g}}\partial_i(\sqrt{g}\,V^i), \qquad g = \det(g_{ij}).$$

For completeness, the contraction in the middle expression satisfies

$$\Gamma^i_{mi} = \frac{1}{2}g^{i\ell}(\partial_m g_{\ell i} + \partial_i g_{\ell m} - \partial_\ell g_{mi}) = \frac{1}{2}g^{i\ell}\partial_m g_{i\ell} = \partial_m \log\sqrt{g}.$$

The last two derivatives in parentheses cancel after interchanging the dummy indices i, ℓ; the determinant derivative $\partial_m g = g\, g^{i\ell}\partial_m g_{\ell i}$ proves the last equality. The product rule then gives the first displayed formula.

In spherical polar coordinates $q^i = (r, \theta, \phi)$,

$$g_{ij} = \mathrm{diag}(1, r^2, r^2\sin^2\theta), \qquad \sqrt{g} = r^2\sin\theta$$

on the usual regular chart $r > 0$, $0 < \theta < \pi$. Let V_r, V_θ, V_ϕ denote components along the *unit* vectors. The coordinate basis has lengths $1, r, r\sin\theta$, so

$$V^r = V_r, \qquad V^\theta = \frac{V_\theta}{r}, \qquad V^\phi = \frac{V_\phi}{r\sin\theta}.$$

Insert these quantities before differentiating:

$$\begin{aligned}\nabla\cdot\mathbf{V} &= \frac{1}{r^2\sin\theta}\left[\partial_r(r^2\sin\theta V_r) + \partial_\theta(r\sin\theta V_\theta) + \partial_\phi(rV_\phi)\right] \\ &= \boxed{\frac{1}{r^2}\frac{\partial(r^2V_r)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial(\sin\theta V_\theta)}{\partial\theta} + \frac{1}{r\sin\theta}\frac{\partial V_\phi}{\partial\phi}}.\end{aligned}$$

For example, expansion of the products gives $\partial_r V_r + 2V_r/r + (\partial_\theta V_\theta)/r + (\cot\theta)V_\theta/r + (\partial_\phi V_\phi)/(r\sin\theta)$, confirming the individual radial and angular terms.

4.3.12 The covariant vector A_i is the gradient of a twice continuously differentiable scalar. Show that the difference of covariant derivatives $A_{i;j} - A_{j;i}$ vanishes.

Solution.

Write $A_i = \partial_i \Phi$ for the scalar potential. Then

$$A_{i;j} = \partial_j \partial_i \Phi - \Gamma^k_{ij} \partial_k \Phi, \qquad A_{j;i} = \partial_i \partial_j \Phi - \Gamma^k_{ji} \partial_k \Phi.$$

Subtract:

$$A_{i;j} - A_{j;i} = (\partial_j \partial_i - \partial_i \partial_j)\Phi - (\Gamma^k_{ij} - \Gamma^k_{ji})\partial_k \Phi = 0.$$

The first term vanishes for a twice continuously differentiable scalar; the second vanishes by Exercise 4.3.4. Thus the covariant Hessian of a scalar is symmetric for this connection.

4.4 Jacobians

4.4.1 Assuming the functions u and v to be continuously differentiable, and interpreting part (a) locally where at least one of $\nabla u, \nabla v$ is nonzero,

(a) Show that a necessary and sufficient condition that $u(x, y, z)$ and $v(x, y, z)$ are related by a regular C^1 relation $f(u, v) = 0$, with $(f_u, f_v) \neq (0, 0)$, is that $(\nabla u) \times (\nabla v) = 0$;

(b) If $u = u(x, y)$ and $v = v(x, y)$, show that the condition $(\nabla u) \times (\nabla v) = 0$ leads to the 2-D Jacobian

$$J = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y} \\ \dfrac{\partial v}{\partial x} & \dfrac{\partial v}{\partial y} \end{vmatrix} = 0.$$

Solution.

The usual equivalence is local and needs regularity. A meaningful relation is a C^1 function f with $(f_u, f_v) \neq (0, 0)$ on the image near the point in question; otherwise $f \equiv 0$ would make the claim vacuous. For the converse, work where at least one of $\nabla u, \nabla v$ is nonzero. Mere differentiability, without these qualifications, does not establish the printed global "necessary and sufficient" statement.

(a) First suppose $f(u(x, y, z), v(x, y, z)) = 0$ is a regular relation. Differentiating it with respect to the three coordinates gives

$$f_u \nabla u + f_v \nabla v = 0.$$

If $f_v \neq 0$, then $\nabla v = -(f_u/f_v)\nabla u$, and their cross product is zero. If $f_v = 0$, regularity forces $f_u \neq 0$, whence $\nabla u = 0$ and the cross product again vanishes. This proves necessity.

Conversely suppose $\nabla u \times \nabla v = 0$ in a neighborhood and $\nabla u \neq 0$ at the point under consideration. Relabel x, y, z if necessary so that $u_x \neq 0$ there. Continuity lets us shrink the neighborhood so that $u_x \neq 0$ throughout. The coordinate change

$$(s, \eta, \zeta) = (u(x, y, z), y, z)$$

has Jacobian $u_x \neq 0$, so the inverse function theorem makes it locally invertible. The vanishing cross product says that $\nabla v = \lambda \nabla u$, with $\lambda = v_x/u_x$; in differential notation this is $dv = \lambda\, du$. Expressing it in the new coordinates,

$$d\widetilde{v} = \frac{\partial \widetilde{v}}{\partial s}\, ds + \frac{\partial \widetilde{v}}{\partial \eta}\, d\eta + \frac{\partial \widetilde{v}}{\partial \zeta}\, d\zeta = \lambda\, ds.$$

Independence of $ds, d\eta, d\zeta$ gives $\widetilde{v}_\eta = \widetilde{v}_\zeta = 0$. On a sufficiently small rectangular coordinate neighborhood, $\widetilde{v}$ consequently depends only on s: $\widetilde{v} = F(s)$. Therefore

$$v = F(u), \qquad f(u, v) = v - F(u) = 0,$$

and this relation is regular because $f_v = 1$. If instead only ∇v is known to be nonzero, interchange u and v in the argument.

The restriction at critical points is substantive. For $u = x^2$, $v = x^3$, the cross product vanishes everywhere, but no regular C^1 relation describes the image near $(u, v) = (0, 0)$. If $f_v \neq 0$, the implicit function theorem would give $v = F(u)$, contradicted by x and $-x$ producing the same u and opposite v. If $f_u \neq 0$, it would give $u = G(v) = |v|^{2/3}$, whose difference quotient at zero is unbounded. The singular relation $v^2 - u^3 = 0$ does exist, illustrating why a regularity statement is needed.

(b) If the functions are independent of z, write their gradients as $(u_x, u_y, 0)$ and $(v_x, v_y, 0)$. Expanding the cross product gives

$$\nabla u \times \nabla v = \big(0, 0, u_x v_y - u_y v_x\big) = \hat{\mathbf{e}}_z \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}.$$

Thus its vanishing is exactly $\boxed{\partial(u, v)/\partial(x, y) = 0}$. This algebraic identity itself does not require the regular-point hypothesis in part (a).

4.4.2 A 2-D orthogonal system is described by the coordinates q_1 and q_2. Show that the Jacobian J satisfies the equation

$$J \equiv \frac{\partial(x, y)}{\partial(q_1, q_2)} \equiv \frac{\partial x}{\partial q_1}\frac{\partial y}{\partial q_2} - \frac{\partial x}{\partial q_2}\frac{\partial y}{\partial q_1} = \pm h_1 h_2.$$

Equivalently, $|J| = h_1 h_2$; the positive sign requires positive orientation. *Hint.* It's easier to work with the square of each side of this equation.

Solution.

The orientation-independent statement is $|J| = h_1 h_2$. The printed positive sign assumes an orientation-preserving coordinate chart.

At a given point set

$$a = x_{q_1}, \quad b = y_{q_1}, \quad c = x_{q_2}, \quad d = y_{q_2}.$$

The coordinate basis vectors are (a, b) and (c, d), so

$$h_1^2 = a^2 + b^2, \qquad h_2^2 = c^2 + d^2, \qquad ac + bd = 0.$$

The last equality is precisely orthogonality. Now expand the squares:

$$\begin{aligned} h_1^2 h_2^2 - (ac + bd)^2 &= (a^2 + b^2)(c^2 + d^2) - (a^2 c^2 + 2abcd + b^2 d^2) \\ &= a^2 d^2 + b^2 c^2 - 2abcd \\ &= (ad - bc)^2 = J^2. \end{aligned}$$

Because $ac + bd = 0$, we obtain $J^2 = h_1^2 h_2^2$. Scale factors are positive on a regular chart, hence $|J| = h_1 h_2$. Positive orientation selects $J = +h_1 h_2$; negative orientation selects $J = -h_1 h_2$. For an explicit check, $x = q_1$, $y = -q_2$ is orthogonal with $h_1 = h_2 = 1$ but $J = -1$, so orthogonality alone cannot fix the sign.

4.4.3 For the transformation $u = x + y$, $v = x/y$, with $x \geq 0$ and $y > 0$, find the Jacobian $\dfrac{\partial(x, y)}{\partial(u, v)}$

(a) By direct computation,

(b) By first computing J^{-1}.

Solution.

The ratio $v = x/y$ requires $y > 0$ on the stated nonnegative quadrant; the boundary $y = 0$ is excluded. Then $u > 0$, $v \geq 0$, and the inverse transformation follows directly from $x = vy$:

$$u = x + y = (v + 1)y, \qquad \boxed{y = \frac{u}{v+1}, \qquad x = \frac{uv}{v+1}.}$$

(a) Differentiate the inverse formulas:

$$x_u = \frac{v}{v+1}, \qquad x_v = \frac{u}{(v+1)^2}, \qquad y_u = \frac{1}{v+1}, \qquad y_v = -\frac{u}{(v+1)^2}.$$

Their determinant is

$$\begin{aligned} J &= x_u y_v - x_v y_u \\ &= -\frac{uv}{(v+1)^3} - \frac{u}{(v+1)^3} = \boxed{-\frac{u}{(v+1)^2}}. \end{aligned}$$

(b) For the forward transformation,

$$u_x = 1, \quad u_y = 1, \quad v_x = \frac{1}{y}, \quad v_y = -\frac{x}{y^2}.$$

Thus

$$J^{-1} = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} 1 & 1 \\ 1/y & -x/y^2 \end{vmatrix} = -\frac{x+y}{y^2} = -\frac{u}{y^2}.$$

It is nonzero on the admissible domain, so inversion is valid:

$$J = -\frac{y^2}{u} = -\frac{1}{u}\left(\frac{u}{v+1}\right)^2 = -\frac{u}{(v+1)^2},$$

in agreement with (a). The negative sign records orientation reversal; an ordinary unoriented area integral uses $|J| = u/(v+1)^2$.

4.5 Differential Forms

4.5.1 Using the rules for the application of the Hodge star operator, verify the results given in Eq. (4.82) for its application to all linearly independent differential forms in Minkowski space. In the rule on p. 236, count the negative metric factors in the input form being starred, rather than in its complement.

Solution.

Use the oriented coframe

$$e^0 = dt, \quad e^1 = dx_1, \quad e^2 = dx_2, \quad e^3 = dx_3, \qquad \eta = \mathrm{diag}(1, -1, -1, -1),$$

and write $e^{a_1\cdots a_p} = e^{a_1} \wedge \cdots \wedge e^{a_p}$, with $\mathrm{vol} = e^{0123}$. For an increasing index set I, let J be its increasing complement, N the number of inversions in the concatenation (I, J), and m the number of spatial indices in I. Then

$$e^I \wedge e^J = (-1)^N \mathrm{vol}, \qquad \langle e^I, e^I \rangle = (-1)^m.$$

The defining equation $e^I \wedge *e^I = \langle e^I, e^I \rangle \mathrm{vol}$ therefore gives

$$\boxed{*e^I = (-1)^{N+m} e^J.}$$

The same definition makes the wedge product zero for unequal basis forms of the same degree, so it determines the Hodge star completely.

The rule in the first paragraph of textbook p. 236 says to count negative metric entries in the complementary form ω'. That is a typographical error: count them in the form ω being starred. The worked example and Eq. (4.82) use this corrected rule. For instance, $*1 = \mathrm{vol}$ has $m = 0$, while $*\mathrm{vol} = -1$ has $m = 3$.

Here is the explicit calculation for every basis form. The complement column is written before its final sign is applied.

input	complement	N	m	Hodge star
1	e^{0123}	0	0	e^{0123}
e^0	e^{123}	0	0	e^{123}
e^1	e^{023}	1	1	e^{023}
e^2	e^{013}	2	1	$-e^{013}$
e^3	e^{012}	3	1	e^{012}
e^{01}	e^{23}	0	1	$-e^{23}$
e^{02}	e^{13}	1	1	e^{13}
e^{03}	e^{12}	2	1	$-e^{12}$
e^{12}	e^{03}	2	2	e^{03}
e^{13}	e^{02}	3	2	$-e^{02}$
e^{23}	e^{01}	4	2	e^{01}
e^{012}	e^3	0	2	e^3
e^{013}	e^2	1	2	$-e^2$
e^{023}	e^1	2	2	e^1
e^{123}	e^0	3	3	e^0
e^{0123}	1	0	3	-1

The table includes $1 + 4 + 6 + 4 + 1 = 16$ independent forms. Take (i, j, k) to be $(1, 2, 3)$, $(2, 3, 1)$, or $(3, 1, 2)$. Rewriting the table in cyclic order absorbs the displayed minus signs associated with reordering spatial indices. It gives exactly

$$*1 = dt \wedge dx_1 \wedge dx_2 \wedge dx_3, \qquad *dt = dx_1 \wedge dx_2 \wedge dx_3,$$
$$*dx_i = dt \wedge dx_j \wedge dx_k,$$
$$*(dx_j \wedge dx_k) = dt \wedge dx_i, \qquad *(dt \wedge dx_i) = -dx_j \wedge dx_k,$$
$$*(dx_1 \wedge dx_2 \wedge dx_3) = dt, \qquad *(dt \wedge dx_i \wedge dx_j) = dx_k,$$
$$*(dt \wedge dx_1 \wedge dx_2 \wedge dx_3) = -1.$$

As an independent sign check, reversing the order of a p-form and its complement contributes $(-1)^{p(4-p)}$, and the metric factors from both sets contribute $(-1)^3$. Hence

$$** \omega_p = (-1)^{p(4-p)+3} \omega_p,$$

which is negative for even p and positive for odd p, in agreement with every pair of rows in the table.

4.5.2 If the force field is constant and moving a particle from the origin to $(3,0,0)$ requires a units of work, from $(-1,-1,0)$ to $(-1,1,0)$ takes b units of work, and from $(0,0,4)$ to $(0,0,5)$ c units of work, find the 1-form of the work.

Solution.

A constant force $\mathbf{F} = (F_x, F_y, F_z)$ has work 1-form $\omega = F_x\,dx + F_y\,dy + F_z\,dz$. Its work between two points is $\mathbf{F}\cdot\Delta\mathbf{r}$, because each force component can be taken outside the line integral. The three given displacements are

$$(3,0,0), \qquad (0,2,0), \qquad (0,0,1).$$

They yield $3F_x = a$, $2F_y = b$, and $F_z = c$. Therefore

$$\boxed{\omega = \frac{a}{3}\,dx + \frac{b}{2}\,dy + c\,dz.}$$

For a direct check, integrating along the three paths multiplies these coefficients by $3, 2, 1$, respectively, returning a, b, c. The form is also exact: $\omega = d(ax/3 + by/2 + cz)$, so the result does not depend on the shapes of the paths.

4.6 Differentiating Forms

4.6.1 Given the two 1-forms $\omega_1 = x\,dy + y\,dx$ and $\omega_2 = x\,dy - y\,dx$, calculate

(a) $d\omega_1$,

(b) $d\omega_2$.

(c) For each of your answers to (a) or (b) that is nonzero, apply the operator d a second time and verify that $d(d\omega_i) = 0$.

Solution.

Use $d(f\alpha) = df \wedge \alpha + f\,d\alpha$ and $d(dx) = d(dy) = 0$.

(a) For the first form,

$$\begin{aligned} d\omega_1 &= d(x\,dy) + d(y\,dx) \\ &= dx \wedge dy + x\,d(dy) + dy \wedge dx + y\,d(dx) \\ &= dx \wedge dy - dx \wedge dy = \boxed{0}. \end{aligned}$$

Indeed $\omega_1 = d(xy)$.

(b) Changing the sign between the terms gives

$$\begin{aligned} d\omega_2 &= d(x\,dy) - d(y\,dx) \\ &= dx \wedge dy - dy \wedge dx = \boxed{2\,dx \wedge dy}. \end{aligned}$$

(c) The nonzero result is a constant-coefficient 2-form. Applying d again, and using the minus sign appropriate to a first-degree factor,

$$d(d\omega_2) = 2d(dx \wedge dy) = 2\{d(dx) \wedge dy - dx \wedge d(dy)\} = 0.$$

The already-zero result for ω_1 also has zero exterior derivative.

4.6.2 Apply the operator d twice to $\omega_3 = xy\,dz + xz\,dy - yz\,dx$. Verify that the second application of d yields a zero result.

Solution.

First differentiate each coefficient and keep every wedge term:

$$\begin{aligned} d\omega_3 &= d(xy) \wedge dz + d(xz) \wedge dy - d(yz) \wedge dx \\ &= (y\,dx + x\,dy) \wedge dz + (z\,dx + x\,dz) \wedge dy - (z\,dy + y\,dz) \wedge dx \\ &= y\,dx \wedge dz + x\,dy \wedge dz + z\,dx \wedge dy + x\,dz \wedge dy \\ &\quad - z\,dy \wedge dx - y\,dz \wedge dx. \end{aligned}$$

The two terms with coefficient x cancel. Reordering the last two terms reverses their signs, so the terms with coefficients y and z each add rather than cancel:

$$\boxed{d\omega_3 = 2y\,dx \wedge dz + 2z\,dx \wedge dy.}$$

Apply d again. Derivatives of the basis differentials vanish, leaving

$$\begin{aligned} d(d\omega_3) &= 2\,dy \wedge dx \wedge dz + 2\,dz \wedge dx \wedge dy \\ &= -2\,dx \wedge dy \wedge dz + 2\,dx \wedge dy \wedge dz = \boxed{0}. \end{aligned}$$

The first permutation needs one interchange; the second needs two, which explains their opposite signs.

4.6.3 For ω_2 and ω_3 the 1-forms with these names in Exercises 4.6.1 and 4.6.2, evaluate $d(\omega_2 \wedge \omega_3)$:

(a) By forming the exterior product and then differentiating, and

(b) Using the formula for differentiating a product of two forms.

Verify that both approaches give the same result.

Solution.

(a) Expand the product, keeping the order of differentials:

$$\begin{aligned} \omega_2 \wedge \omega_3 &= (x\,dy - y\,dx) \wedge (xy\,dz + xz\,dy - yz\,dx) \\ &= x^2y\,dy \wedge dz + x^2z\,dy \wedge dy - xyz\,dy \wedge dx \\ &\quad - xy^2\,dx \wedge dz - xyz\,dx \wedge dy + y^2z\,dx \wedge dx \\ &= x^2y\,dy \wedge dz - xy^2\,dx \wedge dz. \end{aligned}$$

The repeated-differential terms vanish, and the two xyz terms cancel because $dy \wedge dx = -dx \wedge dy$. Differentiating the surviving coefficients gives

$$\begin{aligned} d(\omega_2 \wedge \omega_3) &= (2xy\,dx + x^2\,dy) \wedge dy \wedge dz \\ &\quad - (y^2\,dx + 2xy\,dy) \wedge dx \wedge dz \\ &= 2xy\,dx \wedge dy \wedge dz - 2xy\,dy \wedge dx \wedge dz \\ &= \boxed{4xy\,dx \wedge dy \wedge dz}. \end{aligned}$$

(b) Since ω_2 is a 1-form, the product rule has a minus sign:

$$d(\omega_2 \wedge \omega_3) = d\omega_2 \wedge \omega_3 - \omega_2 \wedge d\omega_3.$$

Use the preceding exercises to compute the terms separately:

$$\begin{aligned} d\omega_2 \wedge \omega_3 &= 2\,dx \wedge dy \wedge (xy\,dz + xz\,dy - yz\,dx) \\ &= 2xy\,dx \wedge dy \wedge dz, \\ \omega_2 \wedge d\omega_3 &= (x\,dy - y\,dx) \wedge (2y\,dx \wedge dz + 2z\,dx \wedge dy) \\ &= 2xy\,dy \wedge dx \wedge dz = -2xy\,dx \wedge dy \wedge dz. \end{aligned}$$

All terms not displayed in the simplified lines contain a repeated dx or dy and are zero. Subtracting the second result from the first gives $(2xy - (-2xy))dx \wedge dy \wedge dz = 4xy\,dx \wedge dy \wedge dz$, exactly as in (a).

4.7 Integrating Forms

4.7.1 Use differential-forms relations to transform the integral $A(x, y, z)\,dx \wedge dy \wedge dz$ to the equivalent expression in $du \wedge dv \wedge dw$, where u, v, w is an invertible linear transformation of x, y, z, and thereby find the determinant that can be identified as the Jacobian of the transformation.

Solution.

To serve as a change of three independent coordinates, the linear transformation must be invertible. Write its inverse as

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix} \begin{pmatrix} u \\ v \\ w \end{pmatrix} \equiv L \begin{pmatrix} u \\ v \\ w \end{pmatrix}.$$

The coefficients are constant, so

$$dx = a_1 du + a_2 dv + a_3 dw, \quad dy = b_1 du + b_2 dv + b_3 dw, \quad dz = c_1 du + c_2 dv + c_3 dw.$$

Expand their exterior product. Any term containing two copies of a differential vanishes; precisely six terms remain:

$$\begin{aligned} dx \wedge dy \wedge dz &= a_1 b_2 c_3\, du \wedge dv \wedge dw + a_1 b_3 c_2\, du \wedge dw \wedge dv \\ &\quad + a_2 b_1 c_3\, dv \wedge du \wedge dw + a_2 b_3 c_1\, dv \wedge dw \wedge du \\ &\quad + a_3 b_1 c_2\, dw \wedge du \wedge dv + a_3 b_2 c_1\, dw \wedge dv \wedge du \\ &= (a_1 b_2 c_3 - a_1 b_3 c_2 - a_2 b_1 c_3 + a_2 b_3 c_1 + a_3 b_1 c_2 - a_3 b_2 c_1)\, du \wedge dv \wedge dw. \end{aligned}$$

The coefficient is the determinant of L, namely

$$J = \det L = \begin{vmatrix} x_u & x_v & x_w \\ y_u & y_v & y_w \\ z_u & z_v & z_w \end{vmatrix} = \frac{\partial(x, y, z)}{\partial(u, v, w)}.$$

Therefore the transformed 3-form is

$$\boxed{A(x(u, v, w), y(u, v, w), z(u, v, w))\, J\, du \wedge dv \wedge dw.}$$

Its integral is taken over the corresponding parameter region with the induced orientation. Differential forms retain the sign of J; a scalar integral written using positive, unoriented volume elements instead uses $|J|$. If the forward transformation has matrix M, then $L = M^{-1}$ and $J = 1/\det M$.

4.7.2 Write Oersted's law,

$$\int_{\partial S} \mathbf{H} \cdot d\mathbf{r} = \int_S \nabla \times \mathbf{H} \cdot d\mathbf{a} \sim I,$$

in differential form notation.

Solution.

Associate the 1-form

$$h = H_x\, dx + H_y\, dy + H_z\, dz$$

with the magnetic field. Its exterior derivative is obtained by expanding $dH_x \wedge dx + dH_y \wedge dy + dH_z \wedge dz$ and grouping terms with the same ordered differential pair:

$$\begin{aligned} dh = {} & (\partial_y H_z - \partial_z H_y)\, dy \wedge dz \\ & + (\partial_z H_x - \partial_x H_z)\, dz \wedge dx \\ & + (\partial_x H_y - \partial_y H_x)\, dx \wedge dy. \end{aligned}$$

These three coefficients are the components of $\nabla \times \mathbf{H}$. On an oriented surface, $dy \wedge dz$, $dz \wedge dx$, and $dx \wedge dy$ pull back to the three components of its vector area form. Stokes' theorem thus gives the requested notation:

$$\boxed{\int_{\partial S} h = \int_S dh \ \sim \ I.}$$

The boundary orientation is the one induced by S. More explicitly, if the current density is $\mathbf{J}$, its flux 2-form is

$$j = J_x\, dy \wedge dz + J_y\, dz \wedge dx + J_z\, dx \wedge dy, \qquad I = \int_S j.$$

With the proportionality constant κ appropriate to the chosen units, the magnetostatic field equation is $dh = \kappa j$, and the integral relation reads $\int_{\partial S} h = \kappa \int_S j = \kappa I$. This preserves the unspecified proportionality convention in the question.

4.7.3 A 1-form $A\, dx + B\, dy$ is defined as closed if $\dfrac{\partial A}{\partial y} = \dfrac{\partial B}{\partial x}$. It is called exact if there is a function f such that $\dfrac{\partial f}{\partial x} = A$ and $\dfrac{\partial f}{\partial y} = B$. Determine which of the following 1-forms are closed, or exact, and find the corresponding functions f for those that are exact. Use their natural real domains for the first four forms, and state explicitly when a primitive is only local. For the fifth, take a C^1 complex-valued coefficient on an open domain in $\mathbb{C}$ and give the conditions for closedness and global exactness:

$$y\, dx + x\, dy, \qquad \frac{y\, dx + x\, dy}{x^2 + y^2}, \qquad [\ln(xy) + 1]\, dx + \frac{x}{y}\, dy,$$

$$-\frac{y\, dx}{x^2 + y^2} + \frac{x\, dy}{x^2 + y^2}, \qquad f(z)\, dz \text{ with } z = x + iy.$$

Solution.

For a C^1 1-form $\omega = A\, dx + B\, dy$,

$$d\omega = dA \wedge dx + dB \wedge dy = (B_x - A_y)\, dx \wedge dy.$$

Thus closedness means $B_x = A_y$. A globally exact form $\omega = d\Phi$ is closed and has zero integral around every closed curve. We use Φ for a potential to distinguish it from the last question's f.

Exactness depends on the domain. In particular the angular form below is only locally exact on the punctured plane. For the last form, closedness requires holomorphic f, and global exactness requires a single-valued primitive on its domain.

First form. For $\omega = y\,dx + x\,dy$, $A_y = B_x = 1$, so the form is closed. Integrate $\Phi_x = y$ with respect to x: $\Phi = xy + h(y)$. The other equation gives $\Phi_y = x + h'(y) = x$, hence $h' = 0$. Therefore

$$\boxed{\omega = d(xy), \qquad \Phi = xy + C,}$$

and the form is globally exact on $\mathbb{R}^2$.

Second form. Its domain excludes the origin. Set $A = y/(x^2+y^2)$ and $B = x/(x^2+y^2)$. The quotient rule gives

$$A_y = \frac{x^2+y^2-2y^2}{(x^2+y^2)^2} = \frac{x^2-y^2}{(x^2+y^2)^2}, \qquad B_x = \frac{x^2+y^2-2x^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}.$$

These derivatives differ except on the lines $y = \pm x$. Those lines contain no nonempty open subset of the plane. Consequently

$$\boxed{\frac{y\,dx + x\,dy}{x^2+y^2} \text{ is neither closed nor exact}}$$

on any nonempty open part of its natural domain.

Third form. For real logarithms the domain is $xy > 0$. The two coefficients obey

$$\partial_y[\ln(xy)+1] = \frac{x}{xy} = \frac{1}{y}, \qquad \partial_x(x/y) = \frac{1}{y},$$

so the form is closed. Integration in x gives $\Phi = x\ln(xy) + h(y)$, because $\partial_x[x\ln(xy)] = \ln(xy)+1$. Then $\Phi_y = x/y + h'(y) = x/y$, so h is constant. Thus

$$\boxed{[\ln(xy)+1]\,dx + \frac{x}{y}\,dy = d[x\ln(xy)].}$$

It is exact on each component of $xy > 0$, and also on their union with the displayed potential; arbitrary additive constants may be chosen independently on the two components.

Fourth form. Here $A = -y/(x^2+y^2)$ and $B = x/(x^2+y^2)$, so

$$A_y = \frac{-(x^2+y^2)+2y^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2} = B_x.$$

The form is closed everywhere on $\mathbb{R}^2 \setminus \{0\}$. On a region with a continuous angle coordinate θ, use $x = r\cos\theta$, $y = r\sin\theta$:

$$\begin{aligned} -y\,dx + x\,dy = &-r\sin\theta(\cos\theta\,dr - r\sin\theta\,d\theta) \\ &+ r\cos\theta(\sin\theta\,dr + r\cos\theta\,d\theta) = r^2\,d\theta. \end{aligned}$$

Thus $\omega = d\theta$ locally. For example, on $x > 0$ a potential is $\Phi = \arctan(y/x) + C$, since

$$\Phi_x = \frac{-y/x^2}{1+y^2/x^2} = -\frac{y}{x^2+y^2}, \qquad \Phi_y = \frac{1/x}{1+y^2/x^2} = \frac{x}{x^2+y^2}.$$

For a complete counterclockwise circle $x = R\cos t$, $y = R\sin t$, $0 \le t \le 2\pi$,

$$\oint \omega = \int_0^{2\pi} \frac{R^2\sin^2 t + R^2\cos^2 t}{R^2}\,dt = 2\pi.$$

A single-valued potential would make this integral zero. Therefore

$$\boxed{\omega \text{ is closed but not globally exact on the punctured plane.}}$$

The Instructor's Manual, p. 65, prints $d\tan(y/x)$; the local primitive is $\arctan(y/x)$, and its assertion of exactness needs the domain qualification just proved.

Fifth form. Write the complex-valued coefficient as $f(x, y)$ and use $dz = dx + i\,dy$. Thus $A = f$ and $B = if$, giving

$$d(f\,dz) = (if_x - f_y)\,dx \wedge dy = 2i\frac{\partial f}{\partial \bar z}\,dx \wedge dy, \qquad \frac{\partial f}{\partial \bar z} = \frac{1}{2}(f_x + if_y).$$

For C^1 coefficients the form is closed exactly when $f_y = if_x$, the Cauchy–Riemann condition for f to be holomorphic. Under that condition it is locally exact. One can construct a primitive on a small rectangle, without presupposing a complex antiderivative:

$$\Phi(x, y) = \int_{x_0}^{x} f(s, y_0)\,ds + i\int_{y_0}^{y} f(x, t)\,dt.$$

Indeed,

$$\Phi_y = if(x, y), \qquad \Phi_x = f(x, y_0) + i\int_{y_0}^{y} f_x(x, t)\,dt = f(x, y_0) + \int_{y_0}^{y} f_y(x, t)\,dt = f(x, y).$$

Hence $d\Phi = f\,dz$. On a simply connected complex domain, every closed curve is contractible, and Stokes' theorem for the closed form gives zero integral around it. Two paths with the same endpoints therefore have equal integrals. The local primitives thus give a global primitive

$$\boxed{F(z) = \int_{z_0}^{z} f(\zeta)\,d\zeta + C, \qquad dF = f(z)\,dz,}$$

where the integral is path independent. More generally, this formula defines a global primitive precisely when every closed-curve period of $f(z)\,dz$ vanishes. The domain condition cannot be omitted: $f(z) = 1/z$ is holomorphic on $\mathbb{C} \setminus \{0\}$, but on $z = Re^{it}$,

$$\oint \frac{dz}{z} = \int_0^{2\pi} i\,dt = 2\pi i \neq 0,$$

so it is not globally exact there. If f is not holomorphic, even closedness can fail: $f = \bar z$ gives $d(f\,dz) = 2i\,dx \wedge dy$. Accordingly, the manual's unqualified claim that $f(z)\,dz$ is exact must be read with both the analyticity and domain conditions above.

5 Chapter 5. Vector Spaces

Problems

5.1 Vectors in Function Spaces

5.1.1 A function $f(x)$ is expanded in a series of orthonormal functions, with convergence in the Hilbert-space norm,

$$f(x) = \sum_{n=0}^{\infty} a_n \varphi_n(x).$$

Show that the series expansion is unique for a given set of $\varphi_n(x)$. The functions $\varphi_n(x)$ are being taken here as the basis vectors in an infinite-dimensional Hilbert space.

Solution.

Write $S_N = \sum_{n=0}^{N} a_n \varphi_n$ and interpret the expansion as $S_N \to f$ in the Hilbert-space norm. For a fixed k and $N \geq k$,

$$\langle \varphi_k | S_N \rangle = \sum_{n=0}^{N} a_n \langle \varphi_k | \varphi_n \rangle = \sum_{n=0}^{N} a_n \delta_{kn} = a_k.$$

The scalar product is continuous: Schwarz's inequality gives

$$|\langle \varphi_k | f - S_N \rangle| \leq \|\varphi_k\| \, \|f - S_N\| = \|f - S_N\| \longrightarrow 0.$$

Taking the limit therefore determines each coefficient uniquely:

$$\boxed{a_k = \langle \varphi_k | f \rangle.}$$

Any other convergent expansion in the same orthonormal set has the same scalar products and hence the same coefficients. Equality of functions in this Hilbert space means equality almost everywhere; changes at isolated points do not produce a different Hilbert-space vector.

5.1.2 A function $f(x)$ is represented by a finite set of basis functions $\varphi_i(x)$,

$$f(x) = \sum_{i=1}^{N} c_i \varphi_i(x).$$

Show that the components c_i are unique, that no different set c_i' exists.

Note. Your basis functions are automatically linearly independent. They are not necessarily orthogonal.

Solution.

Suppose also $f = \sum_{i=1}^{N} c_i' \varphi_i$. Subtract the two finite sums:

$$0 = \sum_{i=1}^{N} (c_i - c_i') \varphi_i.$$

Linear independence means precisely that a zero linear combination of the φ_i has every coefficient zero. Consequently $c_i - c_i' = 0$ for $i = 1, \ldots, N$, proving uniqueness without any orthogonality assumption.

As a scalar-product check, let $G_{ij} = \langle \varphi_i | \varphi_j \rangle$. For any nonzero column d,

$$d^\dagger G d = \left\| \sum_i d_i \varphi_i \right\|^2 > 0.$$

Thus G is invertible, and the equations $\sum_j G_{ij} c_j = \langle \varphi_i | f \rangle$ also have exactly one solution.

5.1.3 A square-integrable function $f(x)$ is approximated by a power series $\sum_{i=0}^{n-1} c_i x^i$ over the interval $[0, 1]$. Show that minimizing the mean square error leads to a set of linear equations

$$\mathsf{A}\mathbf{c} = \mathbf{b},$$

where

$$A_{ij} = \int_0^1 x^{i+j}\, dx = \frac{1}{i+j+1}, \quad i, j = 0, 1, 2, \ldots, n-1$$

and

$$b_i = \int_0^1 x^i f(x)\, dx, \quad i = 0, 1, 2, \ldots, n-1.$$

Note. The A_{ij} are the elements of the Hilbert matrix of order n. The determinant of this Hilbert matrix is a rapidly decreasing function of n. For $n = 5$, $\det \mathsf{A} = 3.7 \times 10^{-12}$ and the set of equations $\mathsf{A}\mathbf{c} = \mathbf{b}$ is becoming ill-conditioned and unstable. The determinant alone does not establish ill-conditioning; here $\kappa_2(\mathsf{A}) \approx 4.7661 \times 10^5$ provides the relevant test.

Solution.

For real f and real coefficients, put $p(x) = \sum_{j=0}^{n-1} c_j x^j$. Expanding the error before differentiating gives

$$\begin{aligned} E(c) &= \int_0^1 (f - p)^2\, dx \\ &= \int_0^1 f^2\, dx - 2\sum_j c_j \int_0^1 x^j f\, dx + \sum_{j,k} c_j c_k \int_0^1 x^{j+k}\, dx. \end{aligned}$$

Since the last matrix is symmetric,

$$\frac{\partial E}{\partial c_i} = -2b_i + 2\sum_{j=0}^{n-1} A_{ij} c_j = 0, \qquad A_{ij} = \left[\frac{x^{i+j+1}}{i+j+1} \right]_0^1 = \frac{1}{i+j+1}.$$

Hence the normal equations are $\boxed{Ac = b}$. They determine the global minimum, not merely a stationary point: for any nonzero vector v,

$$v^T A v = \int_0^1 \left(\sum_{j=0}^{n-1} v_j x^j \right)^2 dx > 0.$$

A nonzero polynomial cannot vanish on an interval, so the Hessian $2A$ is positive definite. If c_* solves the normal equations, direct completion of the square gives

$$E(c) = E(c_*) + (c - c_*)^T A (c - c_*),$$

which proves uniqueness. For complex f, minimize $\int |f-p|^2 dx$; variation of real and imaginary parts gives the same equations, with c, b complex and v^T replaced by $v^\dagger$.

For $n = 5$, exact rational elimination gives

$$\det A = \frac{1}{266716800000} = 3.7492951325\ldots \times 10^{-12}.$$

A small determinant alone is not a condition-number test (rescaling any well-conditioned matrix can make its determinant small). For this particular matrix, the supplementary singular-value calculation gives $\kappa_2(A) \simeq 4.7661 \times 10^5$, confirming sensitivity of the equations.

5.1.4 Let F and φ_n be real, square-integrable functions, with the φ_n orthonormal for a nonnegative weight w. In place of the expansion of $F(x)$ given by

$$F(x) = \sum_{n=0}^{\infty} a_n \varphi_n(x),$$

with

$$a_n = \int_a^b F(x)\varphi_n(x)w(x)\,dx,$$

take the finite series approximation

$$F(x) \approx \sum_{n=0}^{m} c_n \varphi_n(x).$$

Show that the mean square error

$$\int_a^b \left[F(x) - \sum_{n=0}^{m} c_n \varphi_n(x)\right]^2 w(x)\,dx$$

is minimized by taking $c_n = a_n$.

Note. The values of the coefficients are independent of the number of terms in the finite series. This independence is a consequence of orthogonality and would not hold for a least-squares fit using powers of x.

Solution.

Use the real orthonormal functions and nonnegative weight implicit in the stated coefficient formula: $\int_a^b \varphi_j \varphi_k w\,dx = \delta_{jk}$. Expanding the finite error,

$$\begin{aligned} E(c) &= \int_a^b F^2 w\,dx - 2\sum_{n=0}^{m} c_n \int_a^b F\varphi_n w\,dx + \sum_{n,k=0}^{m} c_n c_k \delta_{nk} \\ &= \|F\|^2 - 2\sum_{n=0}^{m} c_n a_n + \sum_{n=0}^{m} c_n^2 \\ &= \|F\|^2 - \sum_{n=0}^{m} a_n^2 + \sum_{n=0}^{m} (c_n - a_n)^2. \end{aligned}$$

Only the last sum depends on the trial coefficients. It is nonnegative and vanishes exactly when

$$\boxed{c_n = a_n \quad (0 \le n \le m)}, \qquad E_{\min} = \|F\|^2 - \sum_{n=0}^{m} a_n^2.$$

Adding another orthonormal function only adds its own square to this calculation; it does not alter the earlier coefficients. If the full expansion converges in norm, $E_{\min} = \sum_{n>m} a_n^2$. For complex functions the corresponding formula uses $a_n = \int \varphi_n^* F w\, dx$ and $E = \int |F - \sum c_n \varphi_n|^2 w\, dx$; the last sum is then $\sum |c_n - a_n|^2$, with the same conclusion.

5.1.5 From Example 5.1.6,

$$f(x) = \begin{cases} \dfrac{h}{2}, & 0 < x < \pi \\ -\dfrac{h}{2}, & -\pi < x < 0 \end{cases} = \frac{2h}{\pi} \sum_{n=0}^{\infty} \frac{\sin(2n+1)x}{2n+1}.$$

(a) Show that

$$\int_{-\pi}^{\pi} [f(x)]^2\, dx = \frac{\pi}{2} h^2 = \frac{4h^2}{\pi} \sum_{n=0}^{\infty} (2n+1)^{-2}.$$

For a finite upper limit this would be Bessel's inequality. For the upper limit ∞, this is Parseval's identity.

(b) Verify that

$$\frac{\pi}{2} h^2 = \frac{4h^2}{\pi} \sum_{n=0}^{\infty} (2n+1)^{-2}$$

by evaluating the series.

Hint. The series can be expressed in terms of the Riemann zeta function $\zeta(2) = \pi^2/6$.

Solution.

(a) Except at the irrelevant endpoints and jump, $f^2 = h^2/4$. Thus

$$\int_{-\pi}^{\pi} f^2\, dx = 2\pi \frac{h^2}{4} = \frac{\pi h^2}{2}.$$

For positive integers k, ℓ, the product-to-sum identity gives

$$\int_{-\pi}^{\pi} \sin kx \sin \ell x\, dx = \frac{1}{2} \int_{-\pi}^{\pi} [\cos(k-\ell)x - \cos(k+\ell)x] dx = \pi \delta_{k\ell}.$$

The norm squared of the partial sum through $n = N$ is therefore

$$\begin{aligned} \int_{-\pi}^{\pi} f_N^2\, dx &= \frac{4h^2}{\pi^2} \sum_{n,m=0}^{N} \frac{\pi \delta_{nm}}{(2n+1)(2m+1)} \\ &= \frac{4h^2}{\pi} \sum_{n=0}^{N} \frac{1}{(2n+1)^2}. \end{aligned}$$

The residual is orthogonal to this partial sum, so $\|f\|^2 = \|f_N\|^2 + \|f - f_N\|^2 \geq \|f_N\|^2$. The Fourier expansion converges in L^2; letting $N \to \infty$ makes the residual norm zero and proves the stated Parseval equality.

(b) Split the positive integers into odd and even ones:

$$\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \sum_{k=1}^{\infty} \frac{1}{k^2} - \sum_{k=1}^{\infty} \frac{1}{(2k)^2} = \left(1 - \frac{1}{4}\right) \zeta(2) = \frac{\pi^2}{8}.$$

Substitution yields $(4h^2/\pi)(\pi^2/8) = \pi h^2/2$, an independent check of the equality.

5.1.6 Derive the Schwarz inequality for real square-integrable functions from the identity

$$\left[\int_a^b f(x)g(x)\,dx\right]^2 = \int_a^b f^2(x)\,dx \int_a^b g^2(x)\,dx - \frac{1}{2}\int_a^b dx \int_a^b dy\,[f(x)g(y) - f(y)g(x)]^2.$$

Solution.

The displayed identity is for real square-integrable functions. To check it, expand its double integral. Fubini's theorem applies since $f, g \in L^2$ and $fg \in L^1$. The latter fact follows from $2|fg| \le f^2 + g^2$, so it does not assume the Schwarz inequality we are proving:

$$\begin{aligned} D &= \int_a^b \int_a^b \{f(x)^2 g(y)^2 + f(y)^2 g(x)^2 - 2f(x)g(x)f(y)g(y)\}\,dx\,dy \\ &= 2\left(\int_a^b f^2 dx\right)\left(\int_a^b g^2 dx\right) - 2\left(\int_a^b fg\,dx\right)^2. \end{aligned}$$

Since the original integrand in D is a square, $D \ge 0$. Rearranging and taking nonnegative square roots proves

$$\boxed{\left|\int_a^b fg\,dx\right| \le \left(\int_a^b f^2 dx\right)^{1/2}\left(\int_a^b g^2 dx\right)^{1/2}.}$$

If $g \neq 0$, set $\lambda = (\int fg)/(\int g^2)$. Then

$$\int_a^b (f - \lambda g)^2 dx = \int f^2 dx - \frac{(\int fg\,dx)^2}{\int g^2 dx}.$$

Thus equality holds exactly when $f = \lambda g$ almost everywhere; the zero-function cases also give equality. For complex functions the Schwarz inequality is $|\int f^* g|^2 \le \int |f|^2 \int |g|^2$; ordinary squares in the displayed real identity must be replaced by appropriate modulus squares.

5.1.7 Starting from

$$I = \left\langle f - \sum_i a_i\varphi_i \,\middle|\, f - \sum_j a_j\varphi_j \right\rangle \ge 0,$$

derive Bessel's inequality, $\langle f|f\rangle \ge \sum_n |a_n|^2$. Here the φ_n are orthonormal and $a_n = \langle\varphi_n|f\rangle$; begin with finite partial sums.

Solution.

Take a finite set of orthonormal functions first, and use the projection coefficients $a_j = \langle\varphi_j|f\rangle$. The scalar product is conjugate-linear in its first argument, so

$$\begin{aligned} I_N &= \left\langle f - \sum_{i=0}^N a_i\varphi_i \,\middle|\, f - \sum_{j=0}^N a_j\varphi_j \right\rangle \\ &= \langle f|f\rangle - \sum_j a_j\langle f|\varphi_j\rangle - \sum_i a_i^*\langle\varphi_i|f\rangle + \sum_{i,j} a_i^* a_j \delta_{ij} \\ &= \|f\|^2 - \sum_j |a_j|^2 - \sum_i |a_i|^2 + \sum_i |a_i|^2 = \|f\|^2 - \sum_{i=0}^N |a_i|^2 \ge 0. \end{aligned}$$

The partial sums of the nonnegative series $\sum |a_i|^2$ increase and are bounded by $\|f\|^2$. Taking their limit gives

$$\boxed{\sum_{i=0}^{\infty} |a_i|^2 \le \langle f|f\rangle.}$$

No completeness assumption on the orthonormal set is needed for this inequality. To check the equality case, let $S_N = \sum_{i=0}^{N} a_i\varphi_i$. For $M > N$, orthonormality gives $\|S_M - S_N\|^2 = \sum_{i=N+1}^{M} |a_i|^2 \to 0$, so completeness of the Hilbert space supplies a limit S in the closed span. Taking limits in the residual identity gives

$$\|f - S\|^2 = \|f\|^2 - \sum_{i=0}^{\infty} |a_i|^2.$$

Thus equality implies $f = S$ in that closed span. Conversely, if f is in the closed span, the residual $f - S$ belongs to it and is orthogonal to every φ_i, hence to the entire closed span. Its squared norm is therefore zero, proving equality.

5.1.8 Expand the function $\sin \pi x$ in a series of functions φ_i that are orthogonal (but not normalized) on the range $0 \le x \le 1$ when the scalar product has definition

$$\langle f|g\rangle = \int_0^1 f^*(x)g(x)\,dx.$$

Keep the first four terms of the expansion. The first four φ_i are:

$$\varphi_0 = 1, \quad \varphi_1 = 2x - 1, \quad \varphi_2 = 6x^2 - 6x + 1, \quad \varphi_3 = 20x^3 - 30x^2 + 12x - 1.$$

Note. The integrals that are needed are the subject of Example 1.10.5.

Solution.

For an orthogonal but unnormalized family, $c_n = \langle\varphi_n|f\rangle/\langle\varphi_n|\varphi_n\rangle$. Direct polynomial integration gives the four norms

$$\int_0^1 \varphi_n^2 dx = 1,\ \frac{1}{3},\ \frac{1}{5},\ \frac{1}{7} \qquad (n = 0, 1, 2, 3).$$

The nonconstant norm calculations are, explicitly,

$$\begin{aligned}
\|\varphi_1\|^2 &= \frac{4}{3} - \frac{4}{2} + 1 = \frac{1}{3},\\
\|\varphi_2\|^2 &= \frac{36}{5} - \frac{72}{4} + \frac{48}{3} - \frac{12}{2} + 1 = \frac{1}{5},\\
\|\varphi_3\|^2 &= \frac{400}{7} - \frac{1200}{6} + \frac{1380}{5} - \frac{760}{4} + \frac{204}{3} - \frac{24}{2} + 1 = \frac{1}{7}.
\end{aligned}$$

To compute all numerators, put $I_j = \int_0^1 x^j \sin \pi x\,dx$. The first two integrals are

$$I_0 = \frac{2}{\pi}, \qquad I_1 = \left[-\frac{x\cos\pi x}{\pi} + \frac{\sin \pi x}{\pi^2}\right]_0^1 = \frac{1}{\pi}.$$

Two integrations by parts, for $j \ge 2$, give

$$\begin{aligned}
I_j &= \frac{1}{\pi} + \frac{j}{\pi}\int_0^1 x^{j-1}\cos\pi x\,dx\\
&= \frac{1}{\pi} - \frac{j(j-1)}{\pi^2} I_{j-2}.
\end{aligned}$$

Hence $I_2 = 1/\pi - 4/\pi^3$ and $I_3 = 1/\pi - 6/\pi^3$. Substituting,

$$\begin{aligned}
\langle\varphi_0|f\rangle &= I_0 = 2/\pi,\\
\langle\varphi_1|f\rangle &= 2I_1 - I_0 = 0,\\
\langle\varphi_2|f\rangle &= 6I_2 - 6I_1 + I_0 = 2/\pi - 24/\pi^3,\\
\langle\varphi_3|f\rangle &= 20I_3 - 30I_2 + 12I_1 - I_0 = 0.
\end{aligned}$$

The requested four-term projection is therefore

$$\boxed{\sin\pi x \;\approx\; \frac{2}{\pi}\varphi_0 + 0\varphi_1 + \frac{10(\pi^2-12)}{\pi^3}\varphi_2 + 0\varphi_3.}$$

The two zero coefficients can also be checked by reflection: $\sin\pi(1-x) = \sin\pi x$, whereas $\varphi_1(1-x) = -\varphi_1(x)$ and $\varphi_3(1-x) = -\varphi_3(x)$. Orthogonality verifies that the residual has zero scalar product with each of the four retained functions.

5.1.9 Expand the function e^{-x} in Laguerre polynomials $L_n(x)$, which are orthonormal on the range $0 \le x < \infty$ with scalar product

$$\langle f|g\rangle = \int_0^\infty f^*(x)g(x)e^{-x}\,dx.$$

Keep the first four terms of the expansion. The first four $L_n(x)$ are

$$L_0 = 1, \quad L_1 = 1 - x, \quad L_2 = \frac{2 - 4x + x^2}{2}, \quad L_3 = \frac{6 - 18x + 9x^2 - x^3}{6}.$$

Solution.

The weight contributes an additional factor e^{-x}, so $a_n = \int_0^\infty L_n(x)e^{-2x}dx$. Set $J_k = \int_0^\infty x^k e^{-2x}dx$. Integration by parts gives $J_0 = 1/2$ and $J_k = (k/2)J_{k-1}$, hence $J_k = k!/2^{k+1}$. The four coefficients are

$$\begin{aligned}
a_0 &= J_0 = \frac{1}{2},\\
a_1 &= J_0 - J_1 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4},\\
a_2 &= J_0 - 2J_1 + \frac{1}{2}J_2 = \frac{1}{2} - \frac{1}{2} + \frac{1}{8} = \frac{1}{8},\\
a_3 &= J_0 - 3J_1 + \frac{3}{2}J_2 - \frac{1}{6}J_3 = \frac{1}{2} - \frac{3}{4} + \frac{3}{8} - \frac{1}{16} = \frac{1}{16}.
\end{aligned}$$

Consequently

$$\boxed{e^{-x} \approx \frac{1}{2}L_0 + \frac{1}{4}L_1 + \frac{1}{8}L_2 + \frac{1}{16}L_3 = \frac{15}{16} - \frac{11x}{16} + \frac{5x^2}{32} - \frac{x^3}{96}.}$$

As a check of both the weight and the normalization, the error norm is

$$\begin{aligned}
\|e^{-x} - f_3\|^2 &= \int_0^\infty e^{-3x}dx - \sum_{n=0}^{3}|a_n|^2\\
&= \frac{1}{3} - \left(\frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \frac{1}{256}\right) = \frac{1}{768}.
\end{aligned}$$

Direct integration of the squared residual with weight e^{-x} gives the same value.

5.1.10 The explicit form of a function f is not known, but the coefficients a_n of its expansion in the orthonormal set φ_n are available. Assuming that the φ_n and the members of another orthonormal set, χ_n, are available, use Dirac notation to obtain a formula for the coefficients for the expansion of f in the χ_n set.

Solution.

Write $|f\rangle = \sum_n a_n|\varphi_n\rangle$. Taking the scalar product with the mth new basis vector gives

$$\boxed{b_m = \langle\chi_m|f\rangle = \sum_n \langle\chi_m|\varphi_n\rangle a_n.}$$

Thus only overlaps between known basis functions are required. In matrix notation $b = Ua$, where $U_{mn} = \langle\chi_m|\varphi_n\rangle$. For an infinite expansion the interchange follows by taking scalar products with its norm-convergent partial sums, as in 5.1.1.

If both sets are complete in the same space, inserting their resolutions of the identity gives

$$(U^\dagger U)_{jk} = \sum_m \langle\varphi_j|\chi_m\rangle\langle\chi_m|\varphi_k\rangle = \langle\varphi_j|\varphi_k\rangle = \delta_{jk},$$

and similarly $UU^\dagger = I$. If the χ_m do not span f, the same formula instead gives the coefficients of its orthogonal projection; it cannot reconstruct a component outside their closed span.

5.1.11 Using conventional vector notation, evaluate $\sum_j |\hat{\mathbf{e}}_j\rangle\langle\hat{\mathbf{e}}_j|\mathbf{a}\rangle$, where $\mathbf{a}$ is an arbitrary vector in the space spanned by the orthonormal $\hat{\mathbf{e}}_j$.

Solution.

For the usual orthonormal unit vectors, $\langle\hat{\mathbf{e}}_j|\mathbf{a}\rangle = \hat{\mathbf{e}}_j \cdot \mathbf{a} = a_j$. Therefore

$$\boxed{\sum_j |\hat{\mathbf{e}}_j\rangle\langle\hat{\mathbf{e}}_j|\mathbf{a}\rangle = \sum_j \hat{\mathbf{e}}_j(\hat{\mathbf{e}}_j \cdot \mathbf{a}) = \sum_j a_j\hat{\mathbf{e}}_j = \mathbf{a}.}$$

In ordinary dyadic notation this is $(\sum_j \hat{\mathbf{e}}_j\hat{\mathbf{e}}_j^{\mathsf{T}})\mathbf{a} = I\mathbf{a}$. Orthonormality is needed here; unit length and spanning alone are insufficient. For example, the unit vectors $(1,0)$ and $(1,1)/\sqrt{2}$ span $\mathbb{R}^2$, but the displayed sum applied to $(1,0)$ is $(3/2, 1/2)$, not $(1,0)$.

5.1.12 Letting $\mathbf{a} = a_1\hat{\mathbf{e}}_1 + a_2\hat{\mathbf{e}}_2$ and $\mathbf{b} = b_1\hat{\mathbf{e}}_1 + b_2\hat{\mathbf{e}}_2$ be vectors in $\mathbb{R}^2$, for what values of k, if any, is

$$\langle\mathbf{a}|\mathbf{b}\rangle = a_1b_1 - a_1b_2 - a_2b_1 + ka_2b_2$$

a valid definition of a scalar product?

Solution.

The proposed form is bilinear and symmetric for every real k, because its matrix is

$$G = \begin{pmatrix} 1 & -1 \\ -1 & k \end{pmatrix} = G^T.$$

It remains to test positive definiteness. Completing the square,

$$\langle\mathbf{a}|\mathbf{a}\rangle = a_1^2 - 2a_1a_2 + ka_2^2 = (a_1 - a_2)^2 + (k-1)a_2^2.$$

If $k > 1$, both terms are nonnegative and their sum can vanish only if $a_2 = 0$ and then $a_1 = 0$. If $k = 1$, the nonzero vector $(1, 1)$ has zero norm. If $k < 1$, this same vector has squared norm $k - 1 < 0$. Thus all scalar-product axioms hold exactly for

$$\boxed{k > 1.}$$

The determinant check $\det G = k - 1 > 0$, together with $G_{11} = 1 > 0$, agrees with the direct square-completion argument.

5.2 Gram-Schmidt Orthogonalization

For the Gram-Schmidt constructions in Exercises 5.2.1 through 5.2.6, use a scalar product of the form given in Eq. (5.7) with the specified interval and weight.

5.2.1 Following the Gram-Schmidt procedure, construct a set of polynomials $P_n^*(x)$ orthogonal (unit weighting factor) over the range $[0, 1]$ from the set $[1, x, x^2, \ldots]$. Scale so that $P_n^*(1) = 1$.

ANS.
$$\begin{aligned} P_0^*(x) &= 1, \\ P_1^*(x) &= 2x - 1, \\ P_2^*(x) &= 6x^2 - 6x + 1, \\ P_3^*(x) &= 20x^3 - 30x^2 + 12x - 1. \end{aligned}$$

These are the first four shifted Legendre polynomials.

Note. The "*" is the standard notation for "shifted": $[0, 1]$ instead of $[-1, 1]$. It does not mean complex conjugate.

Solution.

The first answer on textbook p. 273 is printed $P_n^*(x) = 1$; its subscript must be 0, since the subsequent higher-degree polynomials are not constant. It is convenient first to construct monic orthogonal polynomials q_n:

$$q_n = x^n - \sum_{j=0}^{n-1} \frac{\int_0^1 q_j x^n \, dx}{\int_0^1 q_j^2 \, dx} \, q_j, \qquad P_n^* = \frac{q_n}{q_n(1)}.$$

All the required integrals reduce to $\int_0^1 x^k dx = 1/(k+1)$. Start with $q_0 = 1$, $\|q_0\|^2 = 1$. Then

$$q_1 = x - \frac{1}{2}, \qquad \|q_1\|^2 = \frac{1}{3} - \frac{1}{2} + \frac{1}{4} = \frac{1}{12}.$$

For the next polynomial,

$$\langle q_0 | x^2 \rangle = \frac{1}{3}, \qquad \langle q_1 | x^2 \rangle = \frac{1}{4} - \frac{1}{6} = \frac{1}{12}.$$

Thus

$$q_2 = x^2 - \frac{1}{3} - q_1 = x^2 - x + \frac{1}{6},$$
$$\|q_2\|^2 = \frac{1}{5} - \frac{1}{2} + \frac{4}{9} - \frac{1}{6} + \frac{1}{36} = \frac{1}{180}.$$

For x^3 the three projection numerators are

$$\frac{1}{4}, \qquad \frac{1}{5} - \frac{1}{8} = \frac{3}{40}, \qquad \frac{1}{6} - \frac{1}{5} + \frac{1}{24} = \frac{1}{120}.$$

Dividing by $1, 1/12, 1/180$, respectively, gives

$$\begin{aligned} q_3 &= x^3 - \frac{1}{4} - \frac{9}{10}\left(x - \frac{1}{2}\right) - \frac{3}{2}\left(x^2 - x + \frac{1}{6}\right) \\ &= x^3 - \frac{3}{2}x^2 + \frac{3}{5}x - \frac{1}{20}. \end{aligned}$$

The endpoint values are $q_0(1) = 1$, $q_1(1) = 1/2$, $q_2(1) = 1/6$, and $q_3(1) = 1/20$. Hence

$$\boxed{P_0^* = 1, \quad P_1^* = 2x - 1, \quad P_2^* = 6x^2 - 6x + 1, \quad P_3^* = 20x^3 - 30x^2 + 12x - 1.}$$

The construction extends to the complete set. A formula with the same orthogonality and endpoint normalization is

$$\boxed{P_n^*(x) = \frac{1}{n!}\frac{d^n}{dx^n}[x^n(x-1)^n] = \sum_{k=0}^{n}(-1)^{n-k}\binom{n}{k}\binom{n+k}{k}x^k.}$$

The finite sum follows by first expanding and then differentiating:

$$x^n(x-1)^n = \sum_{k=0}^{n}(-1)^{n-k}\binom{n}{k}x^{n+k}, \qquad \frac{1}{n!}\frac{d^n x^{n+k}}{dx^n} = \frac{(n+k)!}{n!k!}x^k = \binom{n+k}{k}x^k.$$

To verify orthogonality, multiply the derivative formula by x^j, $j < n$, and integrate by parts n times. All boundary terms vanish because $x^n(x-1)^n$ and its first $n-1$ derivatives vanish at both endpoints; the remaining integral vanishes because $d^n x^j/dx^n = 0$. At $x = 1$ only the term differentiating $(x-1)^n$ all n times survives, and it equals $n!$, so $P_n^*(1) = 1$. Uniqueness of Gram–Schmidt with this normalization identifies these as its output.

A useful check is the norm formula. The leading coefficient is $\binom{2n}{n}$ and lower powers are orthogonal to P_n^*, so

$$\begin{aligned} \int_0^1 (P_n^*)^2 dx &= \binom{2n}{n}\int_0^1 x^n P_n^*(x)dx \\ &= \binom{2n}{n}\int_0^1 x^n(1-x)^n dx = \frac{(2n)!}{(n!)^2}\frac{(n!)^2}{(2n+1)!} = \frac{1}{2n+1}. \end{aligned}$$

The middle equality again uses n integrations by parts. To evaluate the last integral without a table, set $J_{a,b} = \int_0^1 x^a(1-x)^b dx$. For integers $a \geq 0$, $b \geq 1$, integration by parts gives

$$J_{a,b} = \left[\frac{x^{a+1}(1-x)^b}{a+1}\right]_0^1 + \frac{b}{a+1}J_{a+1,b-1} = \frac{b}{a+1}J_{a+1,b-1}.$$

Repeating b times and using $J_{a+b,0} = 1/(a+b+1)$ yields $J_{a,b} = a!b!/(a+b+1)!$; take $a = b = n$.

5.2.2 Apply the Gram-Schmidt procedure to form the first three Laguerre polynomials:

$$u_n(x) = x^n, \quad n = 0, 1, 2, \ldots, \quad 0 \leq x < \infty, \quad w(x) = e^{-x}.$$

The conventional normalization is

$$\int_0^\infty L_m(x)L_n(x)e^{-x}\,dx = \delta_{mn}.$$

ANS. $L_0 = 1, \quad L_1 = (1-x), \quad L_2 = \dfrac{2-4x+x^2}{2}.$

Solution.

For $k \geq 1$, integration by parts gives

$$M_k = [-x^k e^{-x}]_0^\infty + kM_{k-1} = kM_{k-1}, \qquad M_0 = 1.$$

The boundary term is zero, so repeated application gives the moments

$$M_k = \int_0^\infty x^k e^{-x} dx = k!, \qquad M_0 = 1,\ M_1 = 1,\ M_2 = 2,\ M_3 = 6,\ M_4 = 24.$$

Begin with $q_0 = 1$ and subtract its projection from x:

$$q_1 = x - \frac{M_1}{M_0} = x - 1, \qquad \|q_1\|^2 = M_2 - 2M_1 + M_0 = 1.$$

For x^2, the projections onto q_0 and q_1 have coefficients

$$\frac{\langle q_0 | x^2 \rangle}{\|q_0\|^2} = 2, \qquad \frac{\langle q_1 | x^2 \rangle}{\|q_1\|^2} = M_3 - M_2 = 4.$$

Therefore

$$q_2 = x^2 - 2 - 4(x-1) = x^2 - 4x + 2,$$

$$\|q_2\|^2 = M_4 - 8M_3 + 20M_2 - 16M_1 + 4M_0 = 24 - 48 + 40 - 16 + 4 = 4.$$

Choose the conventional signs $L_n(0) > 0$. Dividing by the norms gives

$$\boxed{L_0 = 1, \qquad L_1 = -q_1 = 1 - x, \qquad L_2 = q_2/2 = 1 - 2x + \frac{x^2}{2}.}$$

Gram–Schmidt makes the cross integrals zero, and the displayed norm calculations make all three diagonal integrals one. Normalization alone leaves each polynomial's overall sign free.

5.2.3 You are given

(a) a set of functions $u_n(x) = x^n$, $n = 0, 1, 2, \ldots$,

(b) an interval $(0, \infty)$,

(c) a weighting function $w(x) = xe^{-x}$. Use the Gram-Schmidt procedure to construct the first three orthonormal functions from the set $u_n(x)$ for this interval and this weighting function.

ANS. $\varphi_0(x) = 1, \quad \varphi_1(x) = (x-2)/\sqrt{2}, \quad \varphi_2(x) = \dfrac{x^2 - 6x + 6}{2\sqrt{3}}.$

Solution.

With the extra factor x in the weight, the moments are

$$M_k = \int_0^\infty x^k x e^{-x} dx = (k+1)!, \quad (M_0, M_1, M_2, M_3, M_4) = (1, 2, 6, 24, 120).$$

The first function is $q_0 = \varphi_0 = 1$. For the second,

$$q_1 = x - \frac{M_1}{M_0} = x - 2, \qquad \|q_1\|^2 = 6 - 4(2) + 4(1) = 2, \quad \varphi_1 = \frac{x-2}{\sqrt{2}}.$$

For the third polynomial,

$$\langle q_0|x^2\rangle = M_2 = 6, \qquad \frac{\langle q_1|x^2\rangle}{\|q_1\|^2} = \frac{M_3 - 2M_2}{2} = \frac{24-12}{2} = 6.$$

Consequently

$$q_2 = x^2 - 6 - 6(x-2) = x^2 - 6x + 6.$$

Its square is $x^4 - 12x^3 + 48x^2 - 72x + 36$, so

$$\|q_2\|^2 = 120 - 12(24) + 48(6) - 72(2) + 36 = 12.$$

Thus

$$\boxed{\varphi_0 = 1, \qquad \varphi_1 = \frac{x-2}{\sqrt{2}}, \qquad \varphi_2 = \frac{x^2-6x+6}{2\sqrt{3}}.}$$

The printed slash notation for φ_2 means division by the entire quantity $2\sqrt{3}$; writing a fraction removes the ambiguity. As a direct orthogonality check, $\langle 1|q_2\rangle = 6 - 12 + 6 = 0$ and $\langle q_1|q_2\rangle = 24 - 8(6) + 18(2) - 12 = 0$.

5.2.4 Using the Gram-Schmidt orthogonalization procedure, construct the lowest three Hermite polynomials:

$$u_n(x) = x^n, \quad n = 0, 1, 2, \ldots, \quad -\infty < x < \infty, \quad w(x) = e^{-x^2}.$$

For this set of polynomials the usual normalization is

$$\int_{-\infty}^{\infty} H_m(x) H_n(x) w(x)\, dx = \delta_{mn} 2^m m!\, \pi^{1/2}.$$

ANS. $H_0 = 1, \quad H_1 = 2x, \quad H_2 = 4x^2 - 2.$

Solution.

Odd moments vanish by parity. For $j \geq 1$, use $xe^{-x^2}dx = -\frac{1}{2}d(e^{-x^2})$ in integration by parts:

$$M_{2j} = -\frac{1}{2}[x^{2j-1}e^{-x^2}]_{-\infty}^{\infty} + \frac{2j-1}{2}\int_{-\infty}^{\infty} x^{2j-2}e^{-x^2}dx.$$

The boundary term vanishes. The even-moment recurrence is therefore

$$M_{2j} = \int_{-\infty}^{\infty} x^{2j}e^{-x^2}dx = \frac{2j-1}{2}M_{2j-2}, \qquad M_0 = \sqrt{\pi}.$$

Thus $M_2 = \sqrt{\pi}/2$ and $M_4 = 3\sqrt{\pi}/4$. Start with $q_0 = 1$. The projection of x onto 1 vanishes, so $q_1 = x$. For x^2 the projection onto x vanishes, and

$$q_2 = x^2 - \frac{M_2}{M_0} = x^2 - \frac{1}{2}.$$

Their squared norms are

$$\|q_0\|^2 = \sqrt{\pi}, \qquad \|q_1\|^2 = \frac{\sqrt{\pi}}{2}, \qquad \|q_2\|^2 = M_4 - M_2 + \frac{1}{4}M_0 = \frac{\sqrt{\pi}}{2}.$$

The required norms for degrees $0, 1, 2$ are respectively $\sqrt{\pi}, 2\sqrt{\pi}, 8\sqrt{\pi}$. The needed positive scale factors are therefore 1, $\sqrt{2/(1/2)} = 2$, and $\sqrt{8/(1/2)} = 4$. Hence

$$\boxed{H_0 = 1, \qquad H_1 = 2x, \qquad H_2 = 4x^2 - 2.}$$

Parity verifies the odd cross integrals, while $\langle H_0|H_2\rangle = 4M_2 - 2M_0 = 0$ checks the remaining pair.

5.2.5 Use the Gram-Schmidt orthogonalization scheme to construct the first three Chebyshev polynomials (type I):

$$u_n(x) = x^n, \quad n = 0, 1, 2, \ldots, \quad -1 \le x \le 1, \quad w(x) = (1 - x^2)^{-1/2}.$$

Take the normalization

$$\int_{-1}^{1} T_m(x) T_n(x) w(x)\, dx = \begin{cases} \pi, & m = n = 0, \\ \dfrac{\pi}{2}, & m = n \ge 1, \\ 0, & m \ne n. \end{cases}$$

Hint. The needed integrals are given in Exercise 13.3.2.

ANS. $T_0 = 1, \quad T_1 = x, \quad T_2 = 2x^2 - 1, \quad (T_3 = 4x^3 - 3x).$

Solution.

The source gives only the two diagonal cases, multiplied by δ_{mn}; the completed piecewise formula also explicitly defines the value for $m \ne n$. Substitute $x = \cos t$. Since $dx = -\sin t\, dt$ and $\sqrt{1 - x^2} = \sin t$ on $0 \le t \le \pi$,

$$M_{2j} = \int_{-1}^{1} \frac{x^{2j}}{\sqrt{1 - x^2}} dx = \int_0^{\pi} \cos^{2j} t\, dt, \quad M_{2j} = \frac{2j - 1}{2j} M_{2j-2}.$$

To see every term of the reduction, integrate $\cos^{2j-1} t\, d(\sin t)$ by parts:

$$M_{2j} = [\cos^{2j-1} t \sin t]_0^{\pi} + (2j - 1) \int_0^{\pi} \sin^2 t \cos^{2j-2} t\, dt = (2j - 1)(M_{2j-2} - M_{2j}).$$

Moving the last term to the left gives the stated recurrence. It yields $M_0 = \pi$, $M_2 = \pi/2$, $M_4 = 3\pi/8$, and $M_6 = 5\pi/16$; all odd moments vanish. Gram–Schmidt therefore gives

$$q_0 = 1, \qquad q_1 = x, \qquad q_2 = x^2 - \frac{M_2}{M_0} = x^2 - \frac{1}{2}.$$

Their squared norms are

$$\|q_0\|^2 = \pi, \quad \|q_1\|^2 = \frac{\pi}{2}, \quad \|q_2\|^2 = M_4 - M_2 + \frac{1}{4} M_0 = \frac{\pi}{8}.$$

Thus q_0 and q_1 already have the requested normalization, and q_2 needs multiplication by 2:

$$\boxed{T_0 = 1, \qquad T_1 = x, \qquad T_2 = 2x^2 - 1.}$$

To check the additional answer T_3, only the projection of x^3 onto x survives parity:

$$q_3 = x^3 - \frac{M_4}{M_2} x = x^3 - \frac{3}{4} x,$$

$$\|q_3\|^2 = M_6 - \frac{3}{2} M_4 + \frac{9}{16} M_2 = \frac{5\pi}{16} - \frac{9\pi}{16} + \frac{9\pi}{32} = \frac{\pi}{32}.$$

Multiplying q_3 by 4 gives $T_3 = 4x^3 - 3x$ and norm squared $\pi/2$.

5.2.6 Use the Gram-Schmidt orthogonalization scheme to construct the first three Chebyshev polynomials (type II):

$$u_n(x) = x^n, \quad n = 0, 1, 2, \ldots, \quad -1 \le x \le 1, \quad w(x) = (1 - x^2)^{+1/2}.$$

Take the normalization to be

$$\int_{-1}^{1} U_m(x) U_n(x) w(x)\, dx = \delta_{mn} \frac{\pi}{2}.$$

Hint.

$$\int_{-1}^{1} (1 - x^2)^{1/2} x^{2n}\, dx = \frac{\pi}{2} \times \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{4 \cdot 6 \cdot 8 \cdots (2n+2)}, \quad n = 1, 2, 3, \ldots$$
$$= \frac{\pi}{2}, \quad n = 0.$$

ANS. $U_0 = 1, \quad U_1 = 2x, \quad U_2 = 4x^2 - 1.$

Solution.

The hint gives the even moments

$$M_0 = \frac{\pi}{2}, \qquad M_2 = \frac{\pi}{8}, \qquad M_4 = \frac{\pi}{16},$$

and symmetry makes the odd moments zero. The monic Gram–Schmidt polynomials are

$$q_0 = 1, \qquad q_1 = x, \qquad q_2 = x^2 - \frac{\langle 1 | x^2 \rangle}{\langle 1 | 1 \rangle} = x^2 - \frac{1}{4}.$$

There is no projection onto x in the last line because $\langle x | x^2 \rangle = M_3 = 0$. The squared norms are

$$\|q_0\|^2 = \frac{\pi}{2}, \qquad \|q_1\|^2 = \frac{\pi}{8},$$

$$\|q_2\|^2 = M_4 - \frac{1}{2} M_2 + \frac{1}{16} M_0 = \frac{\pi}{16} - \frac{\pi}{16} + \frac{\pi}{32} = \frac{\pi}{32}.$$

Scale these by 1, 2, and 4 so that every squared norm is $\pi/2$:

$$\boxed{U_0 = 1, \qquad U_1 = 2x, \qquad U_2 = 4x^2 - 1.}$$

The remaining even cross integral is $\langle U_0 | U_2 \rangle = 4M_2 - M_0 = 0$; the other cross integrals are odd.

5.2.7 As a modification of Exercise 5.2.4, apply the Gram-Schmidt orthogonalization procedure to the set $u_n(x) = x^n$, $n = 0, 1, 2, \ldots$, $0 \le x < \infty$. Take $w(x)$ to be $\exp(-x^2)$. Find the first two nonvanishing polynomials. Normalize so that the coefficient of the highest power of x is unity. In Exercise 5.2.4, the interval $(-\infty, \infty)$ led to the Hermite polynomials. The functions found here are certainly not the Hermite polynomials.

ANS. $\varphi_0 = 1, \quad \varphi_1 = x - \pi^{-1/2}.$

Solution.

Both references to Exercise 5.2.5 in the original question should be to Exercise 5.2.4, which uses the Gaussian weight and gives Hermite polynomials. The reference error is present in the textbook (p. 275).

On the half-line the moments required for the first two polynomials are

$$M_0 = \int_0^\infty e^{-x^2}dx = \frac{\sqrt{\pi}}{2}, \qquad M_1 = \int_0^\infty xe^{-x^2}dx = \left[-\frac{1}{2}e^{-x^2}\right]_0^\infty = \frac{1}{2}.$$

The monic constant is $\varphi_0 = 1$, and subtraction of its projection from x gives

$$\boxed{\varphi_1 = x - \frac{\langle 1|x\rangle}{\langle 1|1\rangle} = x - \frac{1}{\sqrt{\pi}}.}$$

Indeed, $\langle\varphi_0|\varphi_1\rangle = 1/2 - (1/\sqrt{\pi})(\sqrt{\pi}/2) = 0$. For an independent norm check, $M_2 = \sqrt{\pi}/4$, so

$$\|\varphi_1\|^2 = M_2 - \frac{2}{\sqrt{\pi}}M_1 + \frac{1}{\pi}M_0 = \frac{\pi - 2}{4\sqrt{\pi}} > 0.$$

On the full line the first moment is zero; on the half-line it is $1/2$. This explicitly explains the constant term absent from the odd Hermite polynomial.

5.2.8 Form a set of three orthonormal vectors by the Gram-Schmidt process using these input vectors in the order given:

$$\mathbf{c}_1 = \begin{pmatrix}1\\1\\1\end{pmatrix}, \quad \mathbf{c}_2 = \begin{pmatrix}1\\1\\2\end{pmatrix}, \quad \mathbf{c}_3 = \begin{pmatrix}1\\0\\2\end{pmatrix}.$$

Solution.

Normalize the first vector:

$$\mathbf{e}_1 = \frac{\mathbf{c}_1}{\|\mathbf{c}_1\|} = \frac{1}{\sqrt{3}}(1,1,1)^T.$$

The projection coefficient of $\mathbf{c}_2$ onto it is $\mathbf{e}_1 \cdot \mathbf{c}_2 = 4/\sqrt{3}$. Subtract and normalize:

$$\mathbf{q}_2 = (1,1,2)^T - \frac{4}{3}(1,1,1)^T = \frac{1}{3}(-1,-1,2)^T, \qquad \|\mathbf{q}_2\|^2 = \frac{2}{3},$$

$$\mathbf{e}_2 = \frac{1}{\sqrt{6}}(-1,-1,2)^T.$$

For the third vector,

$$\mathbf{e}_1 \cdot \mathbf{c}_3 = \frac{3}{\sqrt{3}} = \sqrt{3}, \qquad \mathbf{e}_2 \cdot \mathbf{c}_3 = \frac{-1+4}{\sqrt{6}} = \frac{3}{\sqrt{6}}.$$

Thus

$$\begin{aligned}\mathbf{q}_3 &= (1,0,2)^T - (1,1,1)^T - \frac{1}{2}(-1,-1,2)^T \\ &= (1/2,-1/2,0)^T, \qquad \|\mathbf{q}_3\|^2 = \frac{1}{2},\end{aligned}$$

so the requested set is

$$\boxed{\mathbf{e}_1 = \frac{(1,1,1)^T}{\sqrt{3}}, \quad \mathbf{e}_2 = \frac{(-1,-1,2)^T}{\sqrt{6}}, \quad \mathbf{e}_3 = \frac{(1,-1,0)^T}{\sqrt{2}}.}$$

The three cross-dot-product numerators are $-1-1+2 = 0$, $1-1 = 0$, and $-1+1 = 0$; their squared lengths are respectively $3/3$, $6/6$, and $2/2$, independently verifying orthonormality.

5.3 Operators

5.3.1 Show (without introducing matrix representations) that the adjoint of the adjoint of a bounded, everywhere-defined operator restores the original operator, i.e., that $(\mathcal{A}^\dagger)^\dagger = \mathcal{A}$.

Solution.

For the identity as an equality of operators, assume the operators are bounded and everywhere defined, as they automatically are in finite dimension. For a densely defined unbounded operator, the double adjoint is its closure when the operator is closable; equality then additionally requires that it be closed.

Let f, g be arbitrary vectors. The adjoint definition and conjugate symmetry of the scalar product give, step by step,

$$\begin{aligned}\langle f|Ag\rangle &= \langle A^\dagger f|g\rangle = \langle g|A^\dagger f\rangle^* \\ &= \langle (A^\dagger)^\dagger g|f\rangle^* = \langle f|(A^\dagger)^\dagger g\rangle.\end{aligned}$$

Hence $\langle f|[A-(A^\dagger)^\dagger]g\rangle = 0$ for every f. For fixed g, choose $f = [A-(A^\dagger)^\dagger]g$. Its squared norm is zero, so it is the zero vector. Since g was arbitrary,

$$\boxed{(A^\dagger)^\dagger = A.}$$

5.3.2 $\mathcal{U}$ and $\mathcal{V}$ are two arbitrary bounded, everywhere-defined operators. Without introducing matrix representations of these operators, show that

$$(\mathcal{U}\mathcal{V})^\dagger = \mathcal{V}^\dagger\mathcal{U}^\dagger.$$

Note the resemblance to adjoint matrices.

Solution.

The unrestricted equality is valid for bounded, everywhere-defined operators, including all finite matrices. For general unbounded operators their domains matter; the formal algebra alone does not prove equality of the adjoint domains. For arbitrary f, g in the bounded setting, apply the adjoint definition twice, in the order the operators act:

$$\langle f|UVg\rangle = \langle U^\dagger f|Vg\rangle = \langle V^\dagger U^\dagger f|g\rangle.$$

The operator multiplying f in the first slot is thus the adjoint of UV, proving

$$\boxed{(UV)^\dagger = V^\dagger U^\dagger.}$$

The order reverses because U is moved across the scalar product before V. Uniqueness follows from nondegeneracy of the scalar product, as in the previous exercise.

5.3.3 Consider a Hilbert space spanned by the three functions $\varphi_1 = x_1$, $\varphi_2 = x_2$, $\varphi_3 = x_3$, and a scalar product defined by $\langle x_\nu|x_\mu\rangle = \delta_{\nu\mu}$.

(a) Form the 3×3 matrix of each of the following operators:

$$\mathcal{A}_1 = \sum_{i=1}^{3} x_i\left(\frac{\partial}{\partial x_i}\right), \quad \mathcal{A}_2 = x_1\left(\frac{\partial}{\partial x_2}\right) - x_2\left(\frac{\partial}{\partial x_1}\right).$$

(b) Form the column vector representing $\psi = x_1 - 2x_2 + 3x_3$.

(c) Form the matrix equation corresponding to $\chi = (\mathcal{A}_1 - \mathcal{A}_2)\psi$ and verify that the matrix equation reproduces the result obtained by direct application of $\mathcal{A}_1 - \mathcal{A}_2$ to ψ.

Solution.

(a) The jth matrix column is the expansion of $A\varphi_j$ in the ordered basis (x_1, x_2, x_3). Since $\partial x_j/\partial x_i = \delta_{ij}$,

$$A_1 x_j = \sum_i x_i \delta_{ij} = x_j.$$

For the second operator,

$$A_2 x_1 = -x_2, \qquad A_2 x_2 = x_1, \qquad A_2 x_3 = 0.$$

Reading off the columns gives

$$\boxed{A_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.}$$

(b) The coefficient column of ψ is $c = (1, -2, 3)^T$.

(c) Subtraction and multiplication give

$$(A_1 - A_2)c = \begin{pmatrix} 1 & -1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1+2 \\ 1-2 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ 3 \end{pmatrix}.$$

Independently, differentiating the function itself yields

$$A_1\psi = x_1 - 2x_2 + 3x_3, \qquad A_2\psi = x_1(-2) - x_2(1) = -2x_1 - x_2.$$

Thus $\chi = 3x_1 - x_2 + 3x_3$, exactly the function represented by the matrix result.

5.3.4

(a) Obtain the matrix representation of $\mathcal{A} = x(d/dx)$ in a basis of Legendre polynomials, keeping terms through P_3. Use the orthonormal forms of these polynomials as given in Example 5.2.1 and the scalar product defined there.

(b) Expand x^3 in the orthonormal Legendre polynomial basis.

(c) Verify that $\mathcal{A}x^3$ is given correctly by its matrix representation.

Solution.

The source says only "5.2.1." The intended reference is *Example* 5.2.1 (textbook pp. 270–271), which uses $[-1, 1]$. Exercise 5.2.1 instead uses shifted polynomials on $[0, 1]$. Write the orthonormal basis without confusing it with the conventionally normalized P_n:

$$p_0 = \frac{1}{\sqrt{2}}, \quad p_1 = \sqrt{\frac{3}{2}}x, \quad p_2 = \sqrt{\frac{5}{2}}\frac{3x^2 - 1}{2}, \quad p_3 = \sqrt{\frac{7}{2}}\frac{5x^3 - 3x}{2}.$$

(a) Differentiate each polynomial and then re-expand:

$$\begin{aligned} Ap_0 &= 0, \qquad Ap_1 = \sqrt{\frac{3}{2}}x = p_1, \\ Ap_2 &= \sqrt{\frac{5}{2}}3x^2 = 2p_2 + \sqrt{\frac{5}{2}} = 2p_2 + \sqrt{5}\,p_0, \\ Ap_3 &= \sqrt{\frac{7}{2}}\frac{15x^3 - 3x}{2} \\ &= 3\sqrt{\frac{7}{2}}\frac{5x^3 - 3x}{2} + 3\sqrt{\frac{7}{2}}x = 3p_3 + \sqrt{21}\,p_1. \end{aligned}$$

All outputs remain in this four-dimensional span. With row and column order $0, 1, 2, 3$, the matrix is

$$\boxed{A = \begin{pmatrix} 0 & 0 & \sqrt{5} & 0 \\ 0 & 1 & 0 & \sqrt{21} \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.}$$

(b) The coefficients of the even polynomials vanish by parity. The two odd coefficients are

$$\begin{aligned} \langle p_1 | x^3 \rangle &= \sqrt{\frac{3}{2}}\int_{-1}^{1} x^4 dx = \sqrt{\frac{3}{2}}\frac{2}{5} = \frac{\sqrt{6}}{5}, \\ \langle p_3 | x^3 \rangle &= \frac{1}{2}\sqrt{\frac{7}{2}}\left(5\frac{2}{7} - 3\frac{2}{5}\right) = \frac{2\sqrt{14}}{35}. \end{aligned}$$

Therefore

$$\boxed{x^3 = \frac{\sqrt{6}}{5}p_1 + \frac{2\sqrt{14}}{35}p_3}, \qquad c = \begin{pmatrix} 0 \\ \sqrt{6}/5 \\ 0 \\ 2\sqrt{14}/35 \end{pmatrix}.$$

Substitution of the polynomials checks the expansion directly: $(3/5)x + (1/5)(5x^3 - 3x) = x^3$.

(c) Matrix multiplication gives

$$Ac = \begin{pmatrix} 0 \\ \sqrt{6}/5 + \sqrt{21}(2\sqrt{14}/35) \\ 0 \\ 6\sqrt{14}/35 \end{pmatrix} = \begin{pmatrix} 0 \\ 3\sqrt{6}/5 \\ 0 \\ 6\sqrt{14}/35 \end{pmatrix} = 3c.$$

This represents $3x^3$, agreeing with the direct derivative $x(d/dx)x^3 = 3x^3$.

5.4 Self-Adjoint Operators

5.4.1 Work with bounded, everywhere-defined operators.

(a) $\mathcal{A}$ is a non-Hermitian operator. Show that the operators $\mathcal{A}+\mathcal{A}^\dagger$ and $i(\mathcal{A}-\mathcal{A}^\dagger)$ are Hermitian.

(b) Using the preceding result, show that every non-Hermitian operator may be written as a linear combination of two Hermitian operators.

Solution.

Use the adjoint identities $(\lambda A)^\dagger = \lambda^* A^\dagger$, $(A+B)^\dagger = A^\dagger + B^\dagger$, and $(A^\dagger)^\dagger = A$ for bounded, everywhere-defined operators. General unbounded operators require additional domain assumptions, as noted in 5.3.1–5.3.2.

(a) If $B = A + A^\dagger$, then

$$B^\dagger = A^\dagger + (A^\dagger)^\dagger = A^\dagger + A = B.$$

If $C = i(A - A^\dagger)$, conjugating the scalar i is essential:

$$C^\dagger = -i\{A^\dagger - (A^\dagger)^\dagger\} = -i(A^\dagger - A) = i(A - A^\dagger) = C.$$

Thus both operators are Hermitian.

(b) Solving these two definitions for A gives

$$\boxed{A = \frac{1}{2}B - \frac{i}{2}C.}$$

Equivalently, define the Hermitian real and imaginary parts

$$H = \frac{A + A^\dagger}{2}, \qquad K = \frac{A - A^\dagger}{2i}; \qquad \boxed{A = H + iK}.$$

Substitution verifies the reconstruction because the $A^\dagger$ terms cancel and the two $A/2$ terms add. This decomposition works also if A happens already to be Hermitian, in which case $K = 0$.

5.4.2 Prove that the product of two bounded, everywhere-defined Hermitian operators is Hermitian if and only if the two operators commute.

Solution.

For Hermitian A and B, the product-adjoint identity gives

$$(AB)^\dagger = B^\dagger A^\dagger = BA.$$

If AB is Hermitian, then $AB = (AB)^\dagger = BA$, so the operators commute. Conversely, if $AB = BA$, then $(AB)^\dagger = BA = AB$, so the product is Hermitian. Thus

$$\boxed{AB \text{ is Hermitian} \iff [A, B] = 0.}$$

This statement has its usual finite-dimensional or bounded-operator meaning. For unbounded operators one must also establish the domains of the product and its adjoint; formal commutation on a subdomain alone is not a proof of self-adjointness.

5.4.3 $\mathcal{A}$ and $\mathcal{B}$ are noncommuting Hermitian, bounded, everywhere-defined quantum mechanical operators, and $\mathcal{C}$ is given by the formula

$$\mathcal{A}\mathcal{B} - \mathcal{B}\mathcal{A} = i\mathcal{C}.$$

Show that $\mathcal{C}$ is Hermitian. For an unbounded realization, equality of all adjoint domains used in the argument must also be established; a formal boundary-condition assertion alone is insufficient.

Solution.

The conclusion requires A and B themselves to be Hermitian, as when they represent quantum observables. Being merely "quantum mechanical operators" or merely noncommuting does not imply this. With that assumption and the indicated adjoint/domain conditions,

$$\begin{aligned}(AB - BA)^\dagger &= B^\dagger A^\dagger - A^\dagger B^\dagger \\ &= BA - AB = -(AB - BA) = -iC.\end{aligned}$$

But the adjoint of the right side of $AB - BA = iC$ is $-iC^\dagger$. Thus $-iC^\dagger = -iC$, and

$$\boxed{C^\dagger = C.}$$

Equivalently $C = -i[A, B]$ is Hermitian because $[A, B]$ is anti-Hermitian. Noncommutation only ensures $C \neq 0$; it is not needed for Hermiticity. To check why the added hypothesis matters, take

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Then $[A, B] = -A$, and $C = iA$ is not Hermitian. This counterexample rules out the unqualified statement.

5.4.4 The operator $\mathcal{L}$ is Hermitian. Show that $\langle L^2 \rangle \geq 0$, meaning that for all $\psi \in D(L^2)$, $\langle \psi | L^2 | \psi \rangle \geq 0$.

Solution.

For an unbounded L, the expression in the question requires $\psi \in D(L^2) = \{\psi \in D(L) : L\psi \in D(L)\}$, not merely $\psi \in D(L)$. Move the left factor of L across the scalar product and use $L^\dagger = L$:

$$\boxed{\langle \psi | L^2 | \psi \rangle = \langle L^\dagger \psi | L\psi \rangle = \langle L\psi | L\psi \rangle = \|L\psi\|^2 \geq 0.}$$

Equality occurs exactly when $L\psi = 0$. For a normalized state this is the stated expectation value. For an unnormalized nonzero state, dividing by $\langle \psi | \psi \rangle > 0$ preserves the inequality.

5.4.5 Consider a Hilbert space whose members are functions defined on the surface of the unit sphere, with a scalar product of the form

$$\langle f | g \rangle = \int d\Omega \, f^* g,$$

where $d\Omega$ is the element of solid angle. Note that the total solid angle of the sphere is 4π. We work here with the three functions $\varphi_1 = Cx/r$, $\varphi_2 = Cy/r$, $\varphi_3 = Cz/r$, with C assigned a value that makes the φ_i normalized.

(a) Find C, and show that the φ_i are also mutually orthogonal.

(b) Form the 3×3 matrices of the angular momentum operators

$$L_x = -i\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right), \quad L_y = -i\left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z}\right), \quad L_z = -i\left(x\frac{\partial}{\partial y} - y\frac{\partial}{\partial x}\right).$$

(c) Verify that the matrix representations of the components of $\mathbf{L}$ satisfy the angular momentum commutator $[L_x, L_y] = iL_z$.

Solution.

(a) Put $n_x = x/r = \sin\theta\cos\phi$, $n_y = \sin\theta\sin\phi$, $n_z = \cos\theta$, and $d\Omega = \sin\theta\, d\theta\, d\phi$. Then

$$\begin{aligned}\int n_x^2 d\Omega &= \left(\int_0^{2\pi} \cos^2\phi\, d\phi\right)\left(\int_0^{\pi} \sin^3\theta\, d\theta\right) \\ &= \pi\int_{-1}^{1}(1-u^2)du = \frac{4\pi}{3}, \\ \int n_y^2 d\Omega = \frac{4\pi}{3}, &\qquad \int n_z^2 d\Omega = 2\pi\int_{-1}^{1} u^2 du = \frac{4\pi}{3}.\end{aligned}$$

The mixed xy integral contains $\int_0^{2\pi}\sin\phi\cos\phi\, d\phi = 0$; the xz and yz integrals contain respectively the zero integrals of $\cos\phi$ and $\sin\phi$. Choosing the common normalization real and positive gives

$$\boxed{C = \sqrt{\frac{3}{4\pi}}}, \qquad \langle\varphi_i|\varphi_j\rangle = \delta_{ij}.$$

(b) Rotational derivatives annihilate the radial factor. For example,

$$(y\partial_z - z\partial_y)\frac{1}{r} = y\frac{-z}{r^3} - z\frac{-y}{r^3} = 0.$$

The product rule therefore leaves only differentiation of x, y, z. The nine actions are

	φ_1	φ_2	φ_3
L_x	0	$i\varphi_3$	$-i\varphi_2$
L_y	$-i\varphi_3$	0	$i\varphi_1$
L_z	$i\varphi_2$	$-i\varphi_1$	0

and each matrix column is the corresponding action:

$$\boxed{L_x = \begin{pmatrix}0&0&0\\0&0&-i\\0&i&0\end{pmatrix}, \quad L_y = \begin{pmatrix}0&0&i\\0&0&0\\-i&0&0\end{pmatrix}, \quad L_z = \begin{pmatrix}0&-i&0\\i&0&0\\0&0&0\end{pmatrix}.}$$

Conjugate transposition checks that each is Hermitian.

(c) Multiplication gives

$$L_xL_y = \begin{pmatrix}0&0&0\\-1&0&0\\0&0&0\end{pmatrix}, \qquad L_yL_x = \begin{pmatrix}0&-1&0\\0&0&0\\0&0&0\end{pmatrix}.$$

Therefore

$$[L_x, L_y] = \begin{pmatrix}0&1&0\\-1&0&0\\0&0&0\end{pmatrix} = i\begin{pmatrix}0&-i&0\\i&0&0\\0&0&0\end{pmatrix} = \boxed{iL_z}.$$

5.5 Unitary Operators

5.5.1 Show that the corrected matrix for Example 5.5.1,

$$\mathsf{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 & i & 0 & 0 & 0 \\ 1 & i & 0 & 0 & 0 \\ 0 & 0 & -i & 1 & 0 \\ 0 & 0 & i & 1 & 0 \\ 0 & 0 & 0 & 0 & \sqrt{2} \end{pmatrix},$$

correctly transforms the vector $f(\theta,\varphi) = 3\chi_1 + 2i\chi_2 - \chi_3 + \chi_5$ to the $\{\chi'_i\}$ basis by

(a) (1) Making a column vector $\mathbf{c}$ that represents $f(\theta,\varphi)$ in the $\{\chi_i\}$ basis, (2) forming $\mathbf{c}' = \mathsf{U}\mathbf{c}$, and (3) comparing the expansion $\sum_i c'_i\chi'_i(\theta,\varphi)$ with $f(\theta,\varphi)$;

(b) Verifying that U is unitary.

Solution.

The matrix displayed in Example 5.5.1 (p. 290) has four incorrect imaginary signs for the basis functions defined on p. 289. In fact its worked integral gives $u_{22} = +i/\sqrt{2}$, while the displayed matrix has $-i/\sqrt{2}$. The matrix must be corrected before part (a) is true. We derive the correction directly from those functions, without relying on the displayed matrix. Euler's identities give

$$\chi'_1 = -\frac{\chi_1 + i\chi_2}{\sqrt{2}}, \qquad \chi'_2 = \frac{\chi_1 - i\chi_2}{\sqrt{2}},$$
$$\chi'_3 = \frac{\chi_4 + i\chi_3}{\sqrt{2}}, \qquad \chi'_4 = \frac{\chi_4 - i\chi_3}{\sqrt{2}}, \qquad \chi'_5 = \chi_5.$$

For example, $e^{2i\phi} = \cos 2\phi + i\sin 2\phi$ and $\sin 2\phi = 2\sin\phi\cos\phi$ yield the third identity with exactly the normalizing constants in the example. Since $U_{ij} = \langle \chi'_i | \chi_j \rangle$, the coefficients in these identities must be conjugated to obtain the rows of U. Thus

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 & i & 0 & 0 & 0 \\ 1 & i & 0 & 0 & 0 \\ 0 & 0 & -i & 1 & 0 \\ 0 & 0 & i & 1 & 0 \\ 0 & 0 & 0 & 0 & \sqrt{2} \end{pmatrix}.$$

(a) The original and transformed coefficient columns are

$$c = \begin{pmatrix} 3 \\ 2i \\ -1 \\ 0 \\ 1 \end{pmatrix}, \qquad c' = Uc = \frac{1}{\sqrt{2}} \begin{pmatrix} -3 + i(2i) \\ 3 + i(2i) \\ (-i)(-1) \\ i(-1) \\ \sqrt{2} \end{pmatrix} = \boxed{\begin{pmatrix} -5/\sqrt{2} \\ 1/\sqrt{2} \\ i/\sqrt{2} \\ -i/\sqrt{2} \\ 1 \end{pmatrix}}.$$

Substituting the basis relations back into the expansion checks it:

$$\begin{aligned} \sum_j c'_j\chi'_j &= \frac{5}{2}(\chi_1 + i\chi_2) + \frac{1}{2}(\chi_1 - i\chi_2) \\ &\quad + \frac{i}{2}(\chi_4 + i\chi_3) - \frac{i}{2}(\chi_4 - i\chi_3) + \chi_5 \\ &= 3\chi_1 + 2i\chi_2 - \chi_3 + \chi_5 = f. \end{aligned}$$

For comparison, using the matrix as printed gives $\widetilde{c}' = (-1, 5, -i, i, \sqrt{2})^T/\sqrt{2}$ and reconstructs $3\chi_1 - 2i\chi_2 + \chi_3 + \chi_5$, which is not the requested f. This explicitly verifies the need for the correction.

(b) The corrected U has two 2×2 blocks and a final 1. Their products with their adjoints are

$$\frac{1}{2}\begin{pmatrix} -1 & i \\ 1 & i \end{pmatrix}\begin{pmatrix} -1 & 1 \\ -i & -i \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 2 & -1+1 \\ -1+1 & 2 \end{pmatrix} = I_2,$$

$$\frac{1}{2}\begin{pmatrix} -i & 1 \\ i & 1 \end{pmatrix}\begin{pmatrix} i & -i \\ 1 & 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 2 & -1+1 \\ -1+1 & 2 \end{pmatrix} = I_2.$$

The blocks have disjoint support, so $UU^\dagger = U^\dagger U = I_5$. The printed matrix is also unitary; that test by itself cannot detect a matrix that represents the wrong basis transformation.

5.5.2

(a) Given (in $\mathbb{R}^3$) the basis $\varphi_1 = x$, $\varphi_2 = y$, $\varphi_3 = z$, consider the basis transformation $x = z'$, $y = y'$, $z = -x'$ (old coordinate functions written in new coordinates). Find the 3×3 matrix U for this transformation.

(b) This transformation corresponds to a rotation of the coordinate axes. Identify the rotation and reconcile your transformation matrix with an appropriate matrix $\mathsf{S}(\alpha, \beta, \gamma)$ of the form given in Eq. (3.37).

(c) Form the column vector $\mathbf{c}$ representing (in the original basis) $f = 2x - 3y + z$, find the result of applying U to $\mathbf{c}$, and show that this is consistent with the basis transformation of part (a).

Note. You do not need to be able to form scalar products to handle this exercise; a knowledge of the linear relationship between the original and transformed functions is sufficient.

Solution.

The arrows require a convention: here, consistent with the coefficient transformation in Section 5.5, they express the old coordinate functions in new coordinates, $x = z'$, $y = y'$, $z = -x'$. Equivalently the new functions are $x' = -z$, $y' = y$, $z' = x$.

(a) Each column of U contains the coefficients of an old basis function in the new basis (x', y', z'). The identities

$$x = 0x' + 0y' + 1z', \qquad y = 0x' + 1y' + 0z', \qquad z = -1x' + 0y' + 0z'$$

therefore give

$$\boxed{U = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.}$$

(b) The new coordinate axes are $\hat{e}'_x = -\hat{e}_z$, $\hat{e}'_y = \hat{e}_y$, $\hat{e}'_z = \hat{e}_x$. They result from a right-handed rotation of the axes by $\pi/2$ about $+y$. For the Euler convention of Eq. (3.37),

$$S(0, \beta, 0) = \begin{pmatrix} \cos\beta & 0 & -\sin\beta \\ 0 & 1 & 0 \\ \sin\beta & 0 & \cos\beta \end{pmatrix}.$$

Putting $\beta = \pi/2$ gives $S(0, \pi/2, 0) = U$, as required.

(c) The original coefficient column and its transform are

$$c = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}, \qquad c' = Uc = \begin{pmatrix} -1 \\ -3 \\ 2 \end{pmatrix}.$$

The function itself satisfies

$$f = 2x - 3y + z = 2z' - 3y' - x' = -x' - 3y' + 2z',$$

which verifies the coefficient column without scalar products.

If instead the arrows are used to *define* the new ordered basis as $(z, y, -x)$ in the old coordinates, the coefficient transformation for the same fixed f is U^{-1} and gives $(1, -3, -2)^T$. Likewise, active substitution $x \mapsto z$, $z \mapsto -x$ in f produces $-x - 3y + 2z$ in the old labels. Keeping the primes explicit distinguishes these interpretations and their opposite rotation senses.

5.5.3 Construct the matrix representing the inverse of the transformation in Exercise 5.5.2, and show that this matrix and the transformation matrix of that exercise are matrix inverses of each other.

Solution.

Using the convention established in 5.5.2, solve the transformation $x = z'$, $y = y'$, $z = -x'$ for the primed variables: $x' = -z$, $y' = y$, $z' = x$. To express a primed expansion in the old basis, read the coefficients of these three functions as columns:

$$\boxed{V = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix} = U^T = U^\dagger.}$$

Both orders of multiplication give the identity:

$$UV = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

$$VU = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} = I_3.$$

As a vector check, $V(-1, -3, 2)^T = (2, -3, 1)^T$, recovering the original coefficients from 5.5.2.

5.5.4 Consider two successive changes of coefficient columns, first from a basis $\{\varphi_i\}$ to $\{\varphi'_i\}$ by U, then to $\{\chi_i\}$ by V. The bases need not be orthonormal. Take real θ with $\sin\theta\cos\theta(\cos^2\theta - \sin^2\theta) \neq 0$ so both maps are invertible. Their matrix representations are

$$\mathsf{U} = \begin{pmatrix} i\sin\theta & \cos\theta & 0 \\ -\cos\theta & i\sin\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathsf{V} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & i\sin\theta \\ 0 & \cos\theta & -i\sin\theta \end{pmatrix}.$$

Given the function $f(x) = 3\varphi_1(x) - \varphi_2(x) - 2\varphi_3(x)$,

(a) By applying U, form the vector representing $f(x)$ in the $\{\varphi_i'\}$ basis and then by applying V form the vector representing $f(x)$ in the $\{\chi_i\}$ basis. Use this result to write $f(x)$ as a linear combination of the χ_i.

(b) Form the matrix products UV and VU and then apply each to the vector representing $f(x)$ in the $\{\varphi_i\}$ basis. Verify that the results of these applications differ and that only one of them gives the result corresponding to part (a).

Solution.

Put $s = \sin\theta$, $c = \cos\theta$, and $a = (3, -1, -2)^T$. First retain the matrices exactly as printed and evaluate the requested products.

(a) Row-by-column multiplication gives

$$Ua = \begin{pmatrix} 3is - c \\ -3c - is \\ -2 \end{pmatrix}.$$

Applying V to this column gives

$$b = V(Ua) = \begin{pmatrix} 3is - c \\ c(-3c - is) + is(-2) \\ c(-3c - is) - is(-2) \end{pmatrix} = \boxed{\begin{pmatrix} -c + 3is \\ -3c^2 - is(c+2) \\ -3c^2 + is(2-c) \end{pmatrix}}.$$

Where these matrices are invertible and interpreted as successive coefficient changes, the formal expansion is

$$f = (-c + 3is)\chi_1 + [-3c^2 - is(c+2)]\chi_2 + [-3c^2 + is(2-c)]\chi_3.$$

These signs also correct the manual's multiplication: the second entry of the initial vector is -1, so the first two entries of Ua are $3is - c$ and $-3c - is$.

(b) Form both matrix products before acting on a:

$$UV = \begin{pmatrix} is & c^2 & ics \\ -c & ics & -s^2 \\ 0 & c & -is \end{pmatrix}, \qquad VU = \begin{pmatrix} is & c & 0 \\ -c^2 & ics & is \\ -c^2 & ics & -is \end{pmatrix}.$$

Consequently

$$(UV)a = \begin{pmatrix} -c^2 + is(3 - 2c) \\ -3c - ics + 2s^2 \\ -c + 2is \end{pmatrix}, \qquad (VU)a = \begin{pmatrix} -c + 3is \\ -3c^2 - is(c+2) \\ -3c^2 + is(2-c) \end{pmatrix} = b.$$

Only VU corresponds to applying U first and V second. In fact, the two resulting columns differ for every real θ: equality of their first entries would require $c(c-1) + 2isc = 0$, so $c = 0$ or $c = 1$. For $c = 0$, their second entries are 2 and $-2is$ and differ; for $c = 1$, $s = 0$ and their third entries are -1 and -3 and differ.

The original premise that both printed matrices are unitary is false; this is why the corrected question permits general bases and excludes singular angles. An explicit check gives

$$U^\dagger U = \begin{pmatrix} 1 & -2isc & 0 \\ 2isc & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad V^\dagger V = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2c^2 & 0 \\ 0 & 0 & 2s^2 \end{pmatrix}.$$

Thus U is unitary only when $sc = 0$, whereas V is unitary only when $c^2 = s^2 = 1/2$. These conditions never hold simultaneously. Moreover,

$$\det U = c^2 - s^2, \qquad \det V = -2isc.$$

At their zeros even an invertible change of basis is impossible. Away from those zeros the calculations above describe general changes of coefficients, but not successive changes between orthonormal bases.

One consistent repair, not uniquely determined by the printed problem, is to use the following unitary families instead:

$$\widehat{U} = \begin{pmatrix} is & c & 0 \\ c & is & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \widehat{V} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c & is \\ 0 & is & c \end{pmatrix}.$$

Their nontrivial blocks are $isI + c\sigma_1$ and $cI + is\sigma_1$; multiplication by their adjoints cancels the cross terms and leaves $(s^2 + c^2)I = I$. With these repaired data, the full calculation becomes

$$\widehat{U}a = \begin{pmatrix} -c + 3is \\ 3c - is \\ -2 \end{pmatrix}, \qquad \widehat{V}\widehat{U}a = \begin{pmatrix} -c + 3is \\ 3c^2 - is(c+2) \\ s^2 - 2c + 3ics \end{pmatrix},$$

$$\widehat{U}\widehat{V} = \begin{pmatrix} is & c^2 & ics \\ c & ics & -s^2 \\ 0 & is & c \end{pmatrix}, \qquad \widehat{V}\widehat{U} = \begin{pmatrix} is & c & 0 \\ c^2 & ics & is \\ ics & -s^2 & c \end{pmatrix},$$

$$\widehat{U}\widehat{V}a = \begin{pmatrix} -c^2 + is(3 - 2c) \\ 3c - ics + 2s^2 \\ -2c - is \end{pmatrix}.$$

The coefficients of the repaired expansion in its final basis are the entries of $\widehat{V}\widehat{U}a$. These repaired matrices fail to commute in general; for example $c = 0$, $s = 1$ makes the second entries of the two results $-2i$ and 2. At special angles they may commute (e.g. $\theta = 0$), so an assertion of different products for every angle would require its own qualification.

5.5.5 Three functions which are orthogonal with unit weight on the range $-1 \le x \le 1$ are $P_0 = 1$, $P_1 = x$, and $P_2 = \frac{3}{2}x^2 - \frac{1}{2}$. Another set of functions that are orthogonal and span the same space are $F_0 = x^2$, $F_1 = x$, $F_2 = 5x^2 - 3$. Although much of this exercise can be done by inspection, write down and evaluate all the integrals that lead to the results when they are obtained in terms of scalar products.

(a) Normalize each of the P_i and F_i.

(b) Find the unitary matrix U that transforms from the normalized P_i basis to the normalized F_i basis.

(c) Find the unitary matrix V that transforms from the normalized F_i basis to the normalized P_i basis.

(d) Show that U and V are unitary, and that $\mathsf{V} = \mathsf{U}^{-1}$.

(e) Expand $f(x) = 5x^2 - 3x + 1$ in terms of the normalized versions of both bases, and verify that the transformation matrix U converts the P-basis expansion of $f(x)$ into its F-basis expansion.

Solution.

Use $M_k = \int_{-1}^{1} x^k dx$, so $M_k = 0$ for odd k and $M_k = 2/(k+1)$ for even k.

(a) The normalization integrals are

$$\|P_0\|^2 = M_0 = 2, \qquad \|P_1\|^2 = M_2 = \frac{2}{3},$$
$$\|P_2\|^2 = \frac{1}{4}(9M_4 - 6M_2 + M_0) = \frac{1}{4}\left(\frac{18}{5} - 4 + 2\right) = \frac{2}{5},$$
$$\|F_0\|^2 = M_4 = \frac{2}{5}, \qquad \|F_1\|^2 = M_2 = \frac{2}{3},$$
$$\|F_2\|^2 = 25M_4 - 30M_2 + 9M_0 = 10 - 20 + 18 = 8.$$

The normalized functions are therefore

$$p_0 = \frac{1}{\sqrt{2}}, \quad p_1 = \sqrt{\frac{3}{2}}x, \quad p_2 = \sqrt{\frac{5}{2}}\frac{3x^2-1}{2},$$

$$g_0 = \sqrt{\frac{5}{2}}x^2, \quad g_1 = \sqrt{\frac{3}{2}}x, \quad g_2 = \frac{5x^2-3}{2\sqrt{2}}.$$

For completeness, the original off-diagonal integrals are

$$\langle P_0|P_1\rangle = M_1 = 0, \quad \langle P_0|P_2\rangle = (3M_2 - M_0)/2 = 0, \quad \langle P_1|P_2\rangle = (3M_3 - M_1)/2 = 0,$$

$$\langle F_0|F_1\rangle = M_3 = 0, \quad \langle F_0|F_2\rangle = 5M_4 - 3M_2 = 0, \quad \langle F_1|F_2\rangle = 5M_3 - 3M_1 = 0.$$

(b) Evaluate all nine overlaps $U_{ij} = \int_{-1}^{1} g_i p_j dx$:

$$U_{00} = \frac{\sqrt{5}}{2}M_2 = \frac{\sqrt{5}}{3}, \qquad U_{01} = \frac{\sqrt{15}}{2}M_3 = 0,$$
$$U_{02} = \frac{5}{4}(3M_4 - M_2) = \frac{2}{3}, \qquad U_{10} = \frac{\sqrt{3}}{2}M_1 = 0,$$
$$U_{11} = \frac{3}{2}M_2 = 1, \qquad U_{12} = \frac{\sqrt{15}}{4}(3M_3 - M_1) = 0,$$
$$U_{20} = \frac{1}{4}(5M_2 - 3M_0) = -\frac{2}{3}, \qquad U_{21} = \frac{\sqrt{3}}{4}(5M_3 - 3M_1) = 0,$$
$$U_{22} = \frac{\sqrt{5}}{8}(15M_4 - 14M_2 + 3M_0) = \frac{\sqrt{5}}{3}.$$

Thus

$$\boxed{U = \begin{pmatrix} \sqrt{5}/3 & 0 & 2/3 \\ 0 & 1 & 0 \\ -2/3 & 0 & \sqrt{5}/3 \end{pmatrix}.}$$

(c) Reversing the real integrands gives $V_{ij} = \int p_i g_j dx = U_{ji}$; these are the same nine evaluated integrals with their indices interchanged. Therefore

$$\boxed{V = U^T = \begin{pmatrix} \sqrt{5}/3 & 0 & -2/3 \\ 0 & 1 & 0 \\ 2/3 & 0 & \sqrt{5}/3 \end{pmatrix}.}$$

(d) Let $a = \sqrt{5}/3$, $b = 2/3$, so $a^2 + b^2 = 5/9 + 4/9 = 1$. The $0, 2$ block of UV is

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix} \begin{pmatrix} a & -b \\ b & a \end{pmatrix} = \begin{pmatrix} a^2 + b^2 & -ab + ab \\ -ab + ab & a^2 + b^2 \end{pmatrix} = I_2.$$

The middle entry is 1 and other entries vanish, so $UV = VU = I_3$. Since $V = U^\dagger$, both matrices are unitary and $V = U^{-1}$.

(e) Evaluate the six coefficient integrals, retaining their moment expressions to check every term:

$$\begin{aligned}
a_0 &= \int p_0 f \, dx = \frac{5M_2 - 3M_1 + M_0}{\sqrt{2}} = \frac{8\sqrt{2}}{3}, \\
a_1 &= \int p_1 f \, dx = \sqrt{\frac{3}{2}}(5M_3 - 3M_2 + M_1) = -\sqrt{6}, \\
a_2 &= \int p_2 f \, dx = \frac{1}{2}\sqrt{\frac{5}{2}}(15M_4 - 9M_3 - 2M_2 + 3M_1 - M_0) = \frac{2\sqrt{10}}{3}, \\
b_0 &= \int g_0 f \, dx = \sqrt{\frac{5}{2}}(5M_4 - 3M_3 + M_2) = \frac{4\sqrt{10}}{3}, \\
b_1 &= \int g_1 f \, dx = -\sqrt{6}, \\
b_2 &= \int g_2 f \, dx = \frac{25M_4 - 15M_3 - 10M_2 + 9M_1 - 3M_0}{2\sqrt{2}} = -\frac{2\sqrt{2}}{3}.
\end{aligned}$$

(All integrals in this part have limits $-1, 1$.) Thus

$$\boxed{f = \frac{8\sqrt{2}}{3}p_0 - \sqrt{6}\,p_1 + \frac{2\sqrt{10}}{3}p_2 = \frac{4\sqrt{10}}{3}g_0 - \sqrt{6}\,g_1 - \frac{2\sqrt{2}}{3}g_2.}$$

Direct substitution yields, respectively, $8/3 - 3x + (5/3)(3x^2 - 1)$ and $(20/3)x^2 - 3x - (5x^2 - 3)/3$, both equal to f. Finally,

$$Ua = \begin{pmatrix} (\sqrt{5}/3)(8\sqrt{2}/3) + (2/3)(2\sqrt{10}/3) \\ -\sqrt{6} \\ (-2/3)(8\sqrt{2}/3) + (\sqrt{5}/3)(2\sqrt{10}/3) \end{pmatrix} = \begin{pmatrix} 4\sqrt{10}/3 \\ -\sqrt{6} \\ -2\sqrt{2}/3 \end{pmatrix} = b,$$

checking the coefficient transformation independently of reconstruction.

5.6 Transformations of Operators

5.6.1

(a) Using the two spin functions $\varphi_1 = \alpha$ and $\varphi_2 = \beta$ as an orthonormal basis (so $\langle\alpha|\alpha\rangle = \langle\beta|\beta\rangle = 1$, $\langle\alpha|\beta\rangle = 0$), and the relations

$$S_x\alpha = \frac{1}{2}\beta, \quad S_x\beta = \frac{1}{2}\alpha, \quad S_y\alpha = \frac{i}{2}\beta, \quad S_y\beta = -\frac{i}{2}\alpha, \quad S_z\alpha = \frac{1}{2}\alpha, \quad S_z\beta = -\frac{1}{2}\beta,$$

construct the 2×2 matrices of S_x, S_y, and S_z.

(b) Taking now the basis $\varphi_1' = C(\alpha + \beta)$, $\varphi_2' = C(\alpha - \beta)$:

(i) Verify that φ_1' and φ_2' are orthogonal,

(ii) Assign C a value that makes φ_1' and φ_2' normalized,

(iii) Find the unitary matrix for the transformation $\{\varphi_i\} \to \{\varphi_i'\}$.

(c) Find the matrices of S_x, S_y, and S_z in the $\{\varphi_i'\}$ basis.

Solution.

(a) In the order (α, β), an operator's first column gives its action on α and its second gives its action on β. The supplied six relations therefore give

$$\boxed{S_x = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad S_y = \frac{1}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad S_z = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.}$$

For example, $S_y\alpha = (i/2)\beta$ supplies the column $(0, i/2)^T$; this fixes the imaginary signs without an appeal to a memorized matrix.

(b) Expand the scalar products using orthonormality:

$$\langle\alpha + \beta|\alpha - \beta\rangle = 1 - 0 + 0 - 1 = 0,$$
$$\|\alpha \pm \beta\|^2 = 1 \pm 0 \pm 0 + 1 = 2.$$

Hence $C = 1/\sqrt{2}$ is a real positive normalizing choice. The overlap matrix $U_{ij} = \langle\varphi_i'|\varphi_j\rangle$ is

$$\boxed{U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}}, \qquad U^\dagger = U, \quad U^2 = \frac{1}{2}\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = I.$$

(c) Put $H = \sqrt{2}U$. The transformed operator is $S_j' = US_jU^\dagger$. The three unscaled products are

$$H\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}H = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix},$$
$$H\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}H = \begin{pmatrix} i & -i \\ -i & -i \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 2i \\ -2i & 0 \end{pmatrix},$$
$$H\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}H = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}.$$

Each must be multiplied by $1/4$ (two factors $1/\sqrt{2}$ and the operator's $1/2$), giving

$$\boxed{S_x' = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad S_y' = \frac{1}{2}\begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}, \quad S_z' = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.}$$

A direct action check is $S_x(\alpha \pm \beta) = \pm(\alpha \pm \beta)/2$, which confirms the diagonal S_x' and the order of its eigenvalues.

5.6.2 For the basis $\varphi_1 = Cxe^{-r^2}$, $\varphi_2 = Cye^{-r^2}$, $\varphi_3 = Cze^{-r^2}$, where $r^2 = x^2 + y^2 + z^2$, with the scalar product defined as an unweighted integral over $\mathbb{R}^3$ and with C chosen to make the φ_i normalized:

(a) Find the 3×3 matrix of $L_x = -i\left(y\dfrac{\partial}{\partial z} - z\dfrac{\partial}{\partial y}\right)$;

(b) Using the transformation matrix $\mathsf{U} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & -i/\sqrt{2} \\ 0 & 1/\sqrt{2} & i/\sqrt{2} \end{pmatrix}$, find the transformed matrix of L_x;

(c) Find the new basis functions φ_i' defined by the transformation U, and write explicitly (in terms of x, y, and z) the functional forms of $L_x\varphi_i'$, $i = 1, 2, 3$.

Hint. Use $\int e^{-r^2} d^3r = \pi^{3/2}$, $\int x^2 e^{-r^2} d^3r = \frac{1}{2}\pi^{3/2}$; the integrals are over $\mathbb{R}^3$.

Solution.

First determine the common normalization, choosing C real and positive. Scaling the Gaussian integral in the hint gives

$$\int_{\mathbb{R}^3} e^{-ar^2} d^3r = \pi^{3/2} a^{-3/2}, \qquad \int_{\mathbb{R}^3} x^2 e^{-ar^2} d^3r = \frac{\pi^{3/2}}{2a^{5/2}}.$$

For the second formula substitute $(X, Y, Z) = \sqrt{a}(x, y, z)$ in the hint: $x^2 d^3r = X^2 d^3R / a^{5/2}$. The squared wavefunctions involve $a = 2$, not $a = 1$, so

$$1 = C^2 \frac{\pi^{3/2}}{2^{7/2}}, \qquad \boxed{C = \frac{2^{7/4}}{\pi^{3/4}}}.$$

The mixed integrals vanish by odd parity, so the basis is orthonormal.

(a) Write $E = e^{-r^2}$. Its rotational derivative is zero:

$$(y\partial_z - z\partial_y)E = y(-2zE) - z(-2yE) = 0.$$

The product rule then gives

$$L_x(xE) = 0, \qquad L_x(yE) = -i(0 - z)E = izE, \qquad L_x(zE) = -i(y - 0)E = -iyE.$$

The common factor C does not change the coefficients, so

$$\boxed{L_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}}.$$

(b) The nontrivial block of U has rows $(1, -i)/\sqrt{2}$ and $(1, i)/\sqrt{2}$, whose norms are one and whose mutual Hermitian scalar product is $(1 + (-i)(-i))/2 = 0$. Thus $U^{-1} = U^\dagger$. Explicitly,

$$UL_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1/\sqrt{2} & -i/\sqrt{2} \\ 0 & -1/\sqrt{2} & -i/\sqrt{2} \end{pmatrix}, \qquad U^\dagger = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & i/\sqrt{2} & -i/\sqrt{2} \end{pmatrix}.$$

Multiplication gives

$$\boxed{L_x' = U L_x U^\dagger = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.}$$

(c) Because $U_{ij} = \langle \varphi_i' | \varphi_j \rangle$, the new ket functions use the *conjugates* of the rows of U:

$$\boxed{\varphi_1' = CxE, \quad \varphi_2' = \frac{C}{\sqrt{2}}(y + iz)E, \quad \varphi_3' = \frac{C}{\sqrt{2}}(y - iz)E.}$$

Applying the differential operator, using the actions in (a), gives

$$\begin{aligned} L_x \varphi_1' &= 0, \\ L_x \varphi_2' &= \frac{C}{\sqrt{2}}[iz + i(-iy)]E = \frac{C}{\sqrt{2}}(y + iz)E = \varphi_2', \\ L_x \varphi_3' &= \frac{C}{\sqrt{2}}[iz - i(-iy)]E = -\frac{C}{\sqrt{2}}(y - iz)E = -\varphi_3'. \end{aligned}$$

These are exactly the three columns of the diagonal matrix in (b).

5.6.3 The Gram-Schmidt process for converting an arbitrary basis χ_ν into an orthonormal set φ_ν is described in Section 5.2 in a way that introduces coefficients of the form $-\langle \varphi_\mu | \chi_\nu \rangle$. For bases consisting of three functions, convert the formulation so that φ_ν is expressed entirely in terms of the χ_μ, thereby obtaining an expression for the upper-triangular matrix T appearing in Eq. (5.81).

Solution.

Use the complex scalar-product convention of the chapter and define

$$S_{ij} = \langle \chi_i | \chi_j \rangle, \quad S_{ji} = S_{ij}^*, \qquad D_1 = S_{11}, \quad D_2 = S_{11}S_{22} - S_{12}S_{21}, \quad D_3 = \det S.$$

Since the χ_i form a basis, their Gram matrix is positive definite, so $D_1, D_2, D_3 > 0$ and all square roots below are well defined and positive. Gram–Schmidt begins with

$$\varphi_1 = \frac{\chi_1}{\sqrt{D_1}}, \qquad \psi_2 = \chi_2 - \frac{S_{12}}{D_1}\chi_1.$$

Expanding the norm, including complex conjugation, gives

$$\|\psi_2\|^2 = S_{22} - \frac{S_{12}S_{21}}{D_1} - \frac{S_{21}S_{12}}{D_1} + \frac{S_{21}S_{12}S_{11}}{D_1^2} = \frac{D_2}{D_1}.$$

Consequently

$$\varphi_2 = \frac{D_1\chi_2 - S_{12}\chi_1}{\sqrt{D_1 D_2}}.$$

For the third step define $K = D_1 S_{23} - S_{21}S_{13}$. The two projection coefficients, with the conjugation on the first argument accounted for, are

$$\langle \varphi_1 | \chi_3 \rangle = \frac{S_{13}}{\sqrt{D_1}}, \qquad \langle \varphi_2 | \chi_3 \rangle = \frac{D_1 S_{23} - S_{21}S_{13}}{\sqrt{D_1 D_2}} = \frac{K}{\sqrt{D_1 D_2}}.$$

Substitute the already-expanded φ_1, φ_2:

$$\begin{aligned}\psi_3 &= \chi_3 - \frac{S_{13}}{D_1}\chi_1 - \frac{K}{D_1 D_2}(D_1\chi_2 - S_{12}\chi_1) \\ &= \chi_3 - \frac{K}{D_2}\chi_2 + \frac{-D_2 S_{13} + S_{12}K}{D_1 D_2}\chi_1.\end{aligned}$$

The last numerator simplifies without any reality assumption:

$$\begin{aligned}-D_2 S_{13} + S_{12}K &= -(D_1 S_{22} - S_{12}S_{21})S_{13} + S_{12}(D_1 S_{23} - S_{21}S_{13}) \\ &= D_1(S_{12}S_{23} - S_{22}S_{13}).\end{aligned}$$

Thus the unnormalized third vector is

$$\psi_3 = \chi_3 + \frac{S_{21}S_{13} - D_1 S_{23}}{D_2}\chi_2 + \frac{S_{12}S_{23} - S_{22}S_{13}}{D_2}\chi_1.$$

Its norm can be evaluated by subtracting the squared orthogonal projection lengths:

$$\|\psi_3\|^2 = S_{33} - \frac{S_{31}S_{13}}{D_1} - \frac{K^* K}{D_1 D_2} = \frac{D_3}{D_2}.$$

To verify the determinant simplification explicitly, multiply by $D_1 D_2$:

$$\begin{aligned}D_1 D_2\|\psi_3\|^2 &= D_1(D_1 S_{22} - S_{12}S_{21})S_{33} - (D_1 S_{22} - S_{12}S_{21})S_{31}S_{13} \\ &\quad - (D_1 S_{32} - S_{12}S_{31})(D_1 S_{23} - S_{21}S_{13}) \\ &= D_1\{D_1 S_{22}S_{33} - S_{12}S_{21}S_{33} - S_{22}S_{31}S_{13} \\ &\qquad\qquad - D_1 S_{32}S_{23} + S_{32}S_{21}S_{13} + S_{12}S_{31}S_{23}\} \\ &= D_1 \det S.\end{aligned}$$

Dividing ψ_3 by $\sqrt{D_3/D_2}$ now expresses all three φ_j in the original χ_i. Since the coefficients of $\varphi_j = \sum_i T_{ij}\chi_i$ form column j, the answer is

$$\boxed{T = \begin{pmatrix} \dfrac{1}{\sqrt{D_1}} & -\dfrac{S_{12}}{\sqrt{D_1 D_2}} & \dfrac{S_{12}S_{23} - S_{22}S_{13}}{\sqrt{D_2 D_3}} \\ 0 & \sqrt{\dfrac{D_1}{D_2}} & \dfrac{S_{21}S_{13} - D_1 S_{23}}{\sqrt{D_2 D_3}} \\ 0 & 0 & \sqrt{\dfrac{D_2}{D_3}} \end{pmatrix}}.$$

Every column has unit S-norm and is S-orthogonal to the others, so $T^\dagger S T = I$. Finally, if $A_{ij} = \langle\chi_i|\mathcal{A}\chi_j\rangle$, expanding both arguments gives

$$\langle\varphi_\mu|\mathcal{A}\varphi_\nu\rangle = \sum_{i,j} T^*_{i\mu} A_{ij} T_{j\nu} = (T^\dagger A T)_{\mu\nu},$$

which also verifies the orientation of T in Eq. (5.81).

5.7 Invariants

5.7.1 Using the formal properties of unitary transformations, show that the operator commutator $[x, p] = iI$, on a common domain of the products, is invariant under unitary conjugation when the domain is carried along by the same unitary map.

Solution.

Let $x' = UxU^\dagger$, $p' = UpU^\dagger$, with $U^\dagger U = UU^\dagger = I$. Multiply before simplifying:

$$\begin{aligned}[x', p'] &= (UxU^\dagger)(UpU^\dagger) - (UpU^\dagger)(UxU^\dagger) \\ &= Ux(U^\dagger U)pU^\dagger - Up(U^\dagger U)xU^\dagger \\ &= U(xp - px)U^\dagger = U(iI)U^\dagger = iUU^\dagger = \boxed{iI}.\end{aligned}$$

The canonical relation $[x, p] = iI$ cannot hold for finite square matrices: $\operatorname{tr}(xp - px) = 0$, whereas $\operatorname{tr}(iI_N) = iN \neq 0$. It is an operator relation in an infinite-dimensional space. For unbounded position and momentum the calculation holds on a common domain of the products, carried to the new domain by U; it is not a claim that those products are defined on every Hilbert-space vector. For example, the usual relation holds for $x\psi = x\psi(x)$ and $p\psi = -i\psi'(x)$ on smooth rapidly decreasing functions, since

$$x(-i\psi') - [-i(x\psi)'] = -ix\psi' + i(\psi + x\psi') = i\psi.$$

This explicitly supplies a domain and realization for the formal relation.

5.7.2 The Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

have commutator $[\sigma_1, \sigma_2] = 2i\sigma_3$. Show that this relationship continues to be valid if these matrices are transformed by

$$\mathsf{U} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}.$$

Solution.

Let $c = \cos\theta$, $s = \sin\theta$. The columns of U are real orthonormal vectors, so $U^\dagger = U^T = U^{-1}$. Direct multiplication gives

$$\begin{aligned}\sigma_1' = U\sigma_1 U^T &= \begin{pmatrix} s & c \\ c & -s \end{pmatrix}\begin{pmatrix} c & -s \\ s & c \end{pmatrix} = \begin{pmatrix} 2sc & c^2 - s^2 \\ c^2 - s^2 & -2sc \end{pmatrix}, \\ \sigma_2' = U\sigma_2 U^T &= \begin{pmatrix} is & -ic \\ ic & is \end{pmatrix}\begin{pmatrix} c & -s \\ s & c \end{pmatrix} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \\ \sigma_3' = U\sigma_3 U^T &= \begin{pmatrix} c & -s \\ -s & -c \end{pmatrix}\begin{pmatrix} c & -s \\ s & c \end{pmatrix} = \begin{pmatrix} c^2 - s^2 & -2sc \\ -2sc & -(c^2 - s^2) \end{pmatrix}.\end{aligned}$$

Write $C = \cos 2\theta = c^2 - s^2$ and $S = \sin 2\theta = 2sc$. Then

$$\sigma_1'\sigma_2' = \begin{pmatrix} S & C \\ C & -S \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} iC & -iS \\ -iS & -iC \end{pmatrix} = i\sigma_3',$$

$$\sigma_2'\sigma_1' = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} S & C \\ C & -S \end{pmatrix} = \begin{pmatrix} -iC & iS \\ iS & iC \end{pmatrix} = -i\sigma_3'.$$

Subtracting proves

$$\boxed{[\sigma_1', \sigma_2'] = 2i\sigma_3'.}$$

This also follows from $[UAU^\dagger, UBU^\dagger] = U[A,B]U^\dagger$, but the displayed products check the signs and factors directly.

5.7.3

(a) The operator L_x is defined as

$$L_x = -i\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right).$$

Verify that the basis $\varphi_1 = Cxe^{-r^2}$, $\varphi_2 = Cye^{-r^2}$, $\varphi_3 = Cze^{-r^2}$, where $r^2 = x^2+y^2+z^2$, forms a closed set under the operation of L_x, meaning that when L_x is applied to any member of this basis the result is a function within the basis space, and construct the 3×3 matrix of L_x in this basis from the result of the application of L_x to each basis function.

(b) Verify that $L_x(x+iy)e^{-r^2} = -ze^{-r^2}$, and note that this result, using the $\{\varphi_i\}$ basis, can be written $L_x(\varphi_1 + i\varphi_2) = -\varphi_3$.

(c) Express the equation of part (b) in matrix form, and write the matrix equation that results when each of the quantities is transformed using the transformation matrix

$$\mathsf{U} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & -i/\sqrt{2} \\ 0 & 1/\sqrt{2} & i/\sqrt{2} \end{pmatrix}.$$

(d) Regarding the transformation U as producing a new basis $\{\varphi_i'\}$, find the explicit form (in x,y,z) of the φ_i'.

(e) Using the operator form of L_x and the explicit forms of the φ_i', verify the validity of the transformed equation found in part (c).

Hint. The results of Exercise 5.6.2 may be useful.

Solution.

Put $E = e^{-r^2}$ and use the same positive normalization $C = 2^{7/4}/\pi^{3/4}$ as in 5.6.2.

(a) Since $(y\partial_z - z\partial_y)E = 0$, the product rule gives

$$L_x\varphi_1 = 0, \qquad L_x\varphi_2 = i\varphi_3, \qquad L_x\varphi_3 = -i\varphi_2.$$

Every result belongs to the span, so the set is closed under L_x and

$$L = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}.$$

(b) Applying the same actions without the factor C gives

$$L_x[(x+iy)E] = L_x(xE) + iL_x(yE) = 0 + i(izE) = -zE.$$

Multiplying by C produces $L_x(\varphi_1 + i\varphi_2) = -\varphi_3$.

(c) The input and output coefficient columns are $a = (1, i, 0)^T$ and $b = (0, 0, -1)^T$, so the original equation is

$$La = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ i^2 \end{pmatrix} = b.$$

Transform both vectors and the operator, not just the vectors:

$$a' = Ua = \begin{pmatrix} 1 \\ i/\sqrt{2} \\ i/\sqrt{2} \end{pmatrix}, \quad b' = Ub = \begin{pmatrix} 0 \\ i/\sqrt{2} \\ -i/\sqrt{2} \end{pmatrix}, \quad L' = ULU^\dagger = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

The transformed equation is therefore

$$\boxed{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ i/\sqrt{2} \\ i/\sqrt{2} \end{pmatrix} = \begin{pmatrix} 0 \\ i/\sqrt{2} \\ -i/\sqrt{2} \end{pmatrix}.}$$

(d) Conjugating the rows of U to form the new ket functions yields

$$\varphi_1' = CxE, \qquad \varphi_2' = \frac{C}{\sqrt{2}}(y + iz)E, \qquad \varphi_3' = \frac{C}{\sqrt{2}}(y - iz)E.$$

(e) Direct differentiation, including the cancellation of the radial terms as in (a), gives

$$L_x\varphi_1' = 0, \qquad L_x\varphi_2' = \frac{C}{\sqrt{2}}[iz + i(-iy)]E = \varphi_2',$$

$$L_x\varphi_3' = \frac{C}{\sqrt{2}}[iz - i(-iy)]E = -\varphi_3'.$$

The transformed matrix equation thus means exactly

$$L_x\left(\varphi_1' + \frac{i}{\sqrt{2}}\varphi_2' + \frac{i}{\sqrt{2}}\varphi_3'\right) = \frac{i}{\sqrt{2}}\varphi_2' - \frac{i}{\sqrt{2}}\varphi_3'.$$

Substitute their explicit forms. The input is

$$CxE + \frac{iC}{2}[(y + iz) + (y - iz)]E = C(x + iy)E,$$

and the output is

$$\frac{iC}{2}[(y + iz) - (y - iz)]E = -CzE.$$

This is precisely the functional equation from (b), verifying the invariance using both matrices and differential operators.

6 Chapter 6. Eigenvalue Problems

Problems

6.2 Matrix Eigenvalue Problems

Find the eigenvalues and corresponding normalized eigenvectors of the matrices in Exercises 6.2.1 through 6.2.14. Orthogonalize any degenerate eigenvectors.

In this section write $v = (x, y, z)^{\mathsf{T}}$ and $p(\lambda) = \det(A - \lambda I)$. A normalized vector satisfies $v^{\dagger} v = 1$; an overall sign or complex phase does not change its eigendirection.

6.2.1 $\mathsf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$.

ANS. $\lambda = 0, 1, 2$.

Solution.

Expanding the determinant along the second row gives

$$p(\lambda) = (1-\lambda)\big[(1-\lambda)^2 - 1\big] = (1-\lambda)\lambda(\lambda - 2).$$

Thus the eigenvalues are $0, 1, 2$. The component equations are

$$(1-\lambda)x + z = 0, \qquad (1-\lambda)y = 0, \qquad x + (1-\lambda)z = 0.$$

For $\lambda = 0$, they give $y = 0$, $z = -x$. For $\lambda = 1$, the first and third give $z = x = 0$, leaving y free. For $\lambda = 2$, they give $y = 0$, $z = x$. The resulting raw vectors have squared lengths $2, 1, 2$, so the normalized eigenpairs are

λ	u_λ
0	$(1, 0, -1)^{\mathsf{T}}/\sqrt{2}$
1	$(0, 1, 0)^{\mathsf{T}}$
2	$(1, 0, 1)^{\mathsf{T}}/\sqrt{2}$

The first and third vectors have scalar product $(1-1)/2 = 0$, and both are perpendicular to the middle vector. Thus this is an orthonormal basis, with no degeneracy to resolve.

6.2.2 $\mathsf{A} = \begin{pmatrix} 1 & \sqrt{2} & 0 \\ \sqrt{2} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

ANS. $\lambda = -1, 0, 2$.

Solution.

The third coordinate decouples. Expanding on that coordinate,

$$p(\lambda) = -\lambda\big[(1-\lambda)(-\lambda) - 2\big] = -\lambda(\lambda^2 - \lambda - 2) = -\lambda(\lambda - 2)(\lambda + 1).$$

The first two component equations are

$$(1-\lambda)x + \sqrt{2}\,y = 0, \qquad \sqrt{2}\,x - \lambda y = 0,$$

while the last is $-\lambda z = 0$. For $\lambda = -1$, $z = 0$ and $y = -\sqrt{2}\,x$. For $\lambda = 0$, the second equation forces $x = 0$, then the first forces $y = 0$; z is free. For $\lambda = 2$, $z = 0$ and $x = \sqrt{2}\,y$. The nontrivial two-coordinate vectors each have squared length 3:

λ	u_λ
-1	$(1, -\sqrt{2}, 0)^{\mathsf{T}}/\sqrt{3}$
0	$(0, 0, 1)^{\mathsf{T}}$
2	$(\sqrt{2}, 1, 0)^{\mathsf{T}}/\sqrt{3}$

The two vectors in the xy plane have scalar product $(\sqrt{2} - \sqrt{2})/3 = 0$; the third coordinate vector is perpendicular to both.

6.2.3 $\mathsf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$.

ANS. $\lambda = -1, 1, 2$.

Solution.

Expansion along the first row gives

$$\begin{aligned} p(\lambda) &= (1-\lambda)\big[-\lambda(1-\lambda) - 1\big] - (1-\lambda) \\ &= (1-\lambda)(\lambda^2 - \lambda - 2) = (1-\lambda)(\lambda - 2)(\lambda + 1). \end{aligned}$$

The equations for an eigenvector are

$$(1-\lambda)x + y = 0, \quad x - \lambda y + z = 0, \quad y + (1-\lambda)z = 0.$$

For $\lambda = -1$, the first and third yield $y = -2x = -2z$; the middle then holds because $x + y + z = x - 2x + x = 0$. For $\lambda = 1$, $y = 0$ and the middle equation gives $z = -x$. For $\lambda = 2$, the first and third give $y = x = z$, which also satisfies the middle equation. Their squared lengths before normalization are $6, 2, 3$, respectively. Hence

λ	u_λ
-1	$(1, -2, 1)^{\mathsf{T}}/\sqrt{6}$
1	$(1, 0, -1)^{\mathsf{T}}/\sqrt{2}$
2	$(1, 1, 1)^{\mathsf{T}}/\sqrt{3}$

The raw pairwise scalar products are $1 - 1$, $1 - 2 + 1$, and $1 - 1$; all are zero.

6.2.4 $\mathsf{A} = \begin{pmatrix} 1 & \sqrt{8} & 0 \\ \sqrt{8} & 1 & \sqrt{8} \\ 0 & \sqrt{8} & 1 \end{pmatrix}$.

ANS. $\lambda = -3, 1, 5$.

Solution.

Set $q = 1 - \lambda$ and note that $\sqrt{8} = 2\sqrt{2}$. The tridiagonal determinant expands to

$$p(\lambda) = q(q^2 - 8) - 8q = q(q^2 - 16) = q(q-4)(q+4).$$

Thus $q = 4, 0, -4$, giving $\lambda = -3, 1, 5$. The component equations are

$$qx + 2\sqrt{2}\,y = 0, \quad 2\sqrt{2}\,x + qy + 2\sqrt{2}\,z = 0, \quad 2\sqrt{2}\,y + qz = 0.$$

When $q = 4$, they give $y = -\sqrt{2}\,x$ and $z = x$; the middle equation then reads $2\sqrt{2}x - 4\sqrt{2}x + 2\sqrt{2}x = 0$. When $q = 0$, they give $y = 0$ and $z = -x$. When $q = -4$, they give $y = \sqrt{2}\,x$ and $z = x$, and the middle equation again vanishes. Normalization gives

λ	u_λ
-3	$(1, -\sqrt{2}, 1)^{\mathsf{T}}/2$
1	$(1, 0, -1)^{\mathsf{T}}/\sqrt{2}$
5	$(1, \sqrt{2}, 1)^{\mathsf{T}}/2$

The raw squared norms are $4, 2, 4$; the first and last scalar product is $1 - 2 + 1 = 0$, and both are perpendicular to the middle vector.

6.2.5 $\mathsf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$.

ANS. $\lambda = 0, 1, 2$.

Solution.

The first coordinate is separate from the lower 2×2 block:

$$p(\lambda) = (1 - \lambda)\big[(1 - \lambda)^2 - 1\big] = (1 - \lambda)\lambda(\lambda - 2).$$

The equations are

$$(1 - \lambda)x = 0, \qquad (1 - \lambda)y + z = 0, \qquad y + (1 - \lambda)z = 0.$$

At $\lambda = 0$, $x = 0$ and $z = -y$. At $\lambda = 1$, the last two equations give $y = z = 0$, while x is free. At $\lambda = 2$, $x = 0$ and $z = y$. Thus

λ	u_λ
0	$(0, 1, -1)^{\mathsf{T}}/\sqrt{2}$
1	$(1, 0, 0)^{\mathsf{T}}$
2	$(0, 1, 1)^{\mathsf{T}}/\sqrt{2}$

The vectors' squared norms are one; the only nontrivial scalar product is $(1 - 1)/2 = 0$.

6.2.6 $\mathsf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & \sqrt{2} \\ 0 & \sqrt{2} & 0 \end{pmatrix}$.

ANS. $\lambda = -1, 1, 2$.

Solution.

The block determinant is

$$\begin{aligned} p(\lambda) &= (1 - \lambda)\big[(1 - \lambda)(-\lambda) - 2\big] \\ &= (1 - \lambda)(\lambda^2 - \lambda - 2) = (1 - \lambda)(\lambda - 2)(\lambda + 1). \end{aligned}$$

The eigenvector equations are

$$(1 - \lambda)x = 0, \qquad (1 - \lambda)y + \sqrt{2}\,z = 0, \qquad \sqrt{2}\,y - \lambda z = 0.$$

For $\lambda = -1$, $x = 0$ and $z = -\sqrt{2}\,y$. For $\lambda = 1$, the last two equations force $z = y = 0$, leaving x free. For $\lambda = 2$, $x = 0$ and $y = \sqrt{2}\,z$. Therefore

λ	u_λ
-1	$(0, 1, -\sqrt{2})^{\mathsf{T}}/\sqrt{3}$
1	$(1, 0, 0)^{\mathsf{T}}$
2	$(0, \sqrt{2}, 1)^{\mathsf{T}}/\sqrt{3}$

Each two-coordinate raw vector has squared norm 3, and their scalar product is $\sqrt{2} - \sqrt{2} = 0$. The Instructor's Manual, p. 81, prints $(0, \sqrt{2}, 0)^{\mathsf{T}}/\sqrt{3}$ for the eigenvalue 2. Its last component must be $1/\sqrt{3}$, as above: the printed vector has squared norm $2/3$ and does not satisfy $Av = 2v$. Indeed, for the printed vector v_{p},

$$Av_{\mathrm{p}} - 2v_{\mathrm{p}} = \frac{1}{\sqrt{3}}(0, -\sqrt{2}, 2)^{\mathsf{T}} \neq 0.$$

6.2.7 $\mathsf{A} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$.

ANS. $\lambda = -\sqrt{2}, 0, \sqrt{2}$.

Solution.

The determinant is

$$p(\lambda) = (-\lambda)(\lambda^2 - 1) - (-\lambda) = -\lambda(\lambda^2 - 2),$$

so $\lambda = 0, \pm\sqrt{2}$. The three equations are

$$y = \lambda x, \qquad x + z = \lambda y, \qquad y = \lambda z.$$

For $\lambda = 0$, $y = 0$ and $z = -x$. For $\lambda \neq 0$, the first and third imply $z = x$; substituting into the middle gives $2x = \lambda^2 x$. A nonzero vector then has $\lambda = \pm\sqrt{2}$ and $y = \pm\sqrt{2}\,x$. Thus

λ	u_λ
$-\sqrt{2}$	$(1, -\sqrt{2}, 1)^{\mathsf{T}}/2$
0	$(1, 0, -1)^{\mathsf{T}}/\sqrt{2}$
$\sqrt{2}$	$(1, \sqrt{2}, 1)^{\mathsf{T}}/2$

The nonzero-eigenvalue raw vectors have squared length $1 + 2 + 1 = 4$; the zero-eigenvalue vector has squared length 2. Pairwise orthogonality follows from $1 - 2 + 1 = 0$ and $1 - 1 = 0$.

6.2.8 $\mathsf{A} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}$.

ANS. $\lambda = 0, 2, 2$.

Solution.

Expansion on the first coordinate gives

$$p(\lambda) = (2 - \lambda)\big[(1 - \lambda)^2 - 1\big] = -\lambda(\lambda - 2)^2.$$

For $\lambda = 0$, the equations give $x = 0$ and $y + z = 0$, hence $u_0 = (0, 1, -1)^{\mathsf{T}}/\sqrt{2}$. For the repeated eigenvalue $\lambda = 2$,

$$(A - 2I)v = 0 \quad \Longleftrightarrow \quad -y + z = 0,$$

with x free. The entire eigenspace is

$$v = x(1, 0, 0)^{\mathsf{T}} + y(0, 1, 1)^{\mathsf{T}}.$$

Its two displayed spanning vectors are already orthogonal, with lengths 1 and $\sqrt{2}$. Normalizing them gives the required orthonormal choice within the degenerate eigenspace:

λ	orthonormal eigenvectors
0	$(0, 1, -1)^{\mathsf{T}}/\sqrt{2}$
2	$(1, 0, 0)^{\mathsf{T}}, \quad (0, 1, 1)^{\mathsf{T}}/\sqrt{2}$

Both eigenvectors for 2 have zero scalar product with u_0.

6.2.9 $\mathsf{A} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$.

ANS. $\lambda = -1, -1, 2$.

Solution.

The determinant expands as

$$p(\lambda) = -\lambda(\lambda^2 - 1) + (\lambda + 1) + (\lambda + 1) = -\lambda^3 + 3\lambda + 2 = (2 - \lambda)(\lambda + 1)^2.$$

To obtain the eigenspaces without repeated elimination, set $s = x + y + z$. Then

$$Av = (s - x, s - y, s - z)^{\mathsf{T}}.$$

If $x = y = z$, this equals $2v$, giving $u_2 = (1, 1, 1)^{\mathsf{T}}/\sqrt{3}$. For $\lambda = -1$, the equation $Av = -v$ is exactly $s = 0$, a plane of dimension two. Choose the independent vectors

$$a = (1, -1, 0)^{\mathsf{T}}, \qquad b = (1, 0, -1)^{\mathsf{T}}.$$

They are not orthogonal: $a^{\mathsf{T}}b = 1$ and $a^{\mathsf{T}}a = 2$. Gram–Schmidt gives

$$b_\perp = b - \frac{a^{\mathsf{T}}b}{a^{\mathsf{T}}a}a = b - \frac{1}{2}a = (\tfrac{1}{2}, \tfrac{1}{2}, -1)^{\mathsf{T}}, \qquad \|b_\perp\|^2 = \frac{3}{2}.$$

Consequently a normalized orthogonal choice is

λ	orthonormal eigenvectors
-1	$(1, -1, 0)^{\mathsf{T}}/\sqrt{2}, \quad (1, 1, -2)^{\mathsf{T}}/\sqrt{6}$
2	$(1, 1, 1)^{\mathsf{T}}/\sqrt{3}$

The coordinate sum of each vector in the first row is zero, so each is also perpendicular to u_2.

6.2.10 $\mathsf{A} = \begin{pmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix}$.

ANS. $\lambda = -1, 2, 2$.

Solution.

Let J be the matrix whose nine entries are all 1. Here $A = 2I - J$, and $Jv = (x+y+z)(1,1,1)^{\mathsf{T}}$. On the line spanned by $w = (1,1,1)^{\mathsf{T}}$, $Jw = 3w$, so $Aw = (2-3)w = -w$. On the plane $x+y+z = 0$, $Jv = 0$, so $Av = 2v$. The line and plane have dimensions one and two and together span the whole space, proving that these are all the eigenvalues. In particular,

$$p(\lambda) = (-1-\lambda)(2-\lambda)^2.$$

To orthogonalize the degenerate plane, take $a = (1,-1,0)^{\mathsf{T}}$ and $b = (1,0,-1)^{\mathsf{T}}$. Since $a \cdot b = 1$ and $a \cdot a = 2$,

$$b_\perp = b - \frac{1}{2}a = \frac{1}{2}(1,1,-2)^{\mathsf{T}}, \qquad \|a\| = \sqrt{2}, \quad \|b_\perp\| = \sqrt{3/2}.$$

Hence

λ	orthonormal eigenvectors
-1	$(1,1,1)^{\mathsf{T}}/\sqrt{3}$
2	$(1,-1,0)^{\mathsf{T}}/\sqrt{2}$, $(1,1,-2)^{\mathsf{T}}/\sqrt{6}$

The coordinate-sum condition also makes both eigenvectors for 2 perpendicular to the eigenvector for -1.

6.2.11 $\mathsf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$.

ANS. $\lambda = 0, 0, 3$.

Solution.

For this all-ones matrix,

$$Av = (x+y+z)(1,1,1)^{\mathsf{T}}.$$

The vector $w = (1,1,1)^{\mathsf{T}}$ satisfies $Aw = 3w$. Every vector in the plane $x+y+z = 0$ is annihilated, so 0 is an eigenvalue with a two-dimensional eigenspace. These dimensions add to three; thus

$$p(\lambda) = (3-\lambda)\lambda^2.$$

For an explicit orthogonalization of the zero eigenspace, choose $a = (1,-1,0)^{\mathsf{T}}$ and $b = (1,0,-1)^{\mathsf{T}}$. Subtract the projection of b on a:

$$b_\perp = b - \frac{1}{2}a = \tfrac{1}{2}(1,1,-2)^{\mathsf{T}}.$$

Their lengths are $\sqrt{2}$ and $\sqrt{3/2}$, respectively. Therefore

λ	orthonormal eigenvectors
0	$(1,-1,0)^{\mathsf{T}}/\sqrt{2}$, $(1,1,-2)^{\mathsf{T}}/\sqrt{6}$
3	$(1,1,1)^{\mathsf{T}}/\sqrt{3}$

The last vector is perpendicular to the zero eigenspace because its scalar product with v is $(x+y+z)/\sqrt{3}$.

6.2.12 $\mathsf{A} = \begin{pmatrix} 5 & 0 & 2 \\ 0 & 1 & 0 \\ 2 & 0 & 2 \end{pmatrix}$.

ANS. $\lambda = 1, 1, 6$.

Solution.

The middle coordinate decouples, so

$$\begin{aligned} p(\lambda) &= (1-\lambda)\big[(5-\lambda)(2-\lambda) - 4\big] \\ &= (1-\lambda)(\lambda^2 - 7\lambda + 6) = (1-\lambda)^2(6-\lambda). \end{aligned}$$

At $\lambda = 1$, the first and third equations are $4x + 2z = 0$ and $2x + z = 0$, which are equivalent to $z = -2x$; y is free. Thus

$$E_1 = \operatorname{span}\{(1, 0, -2)^{\mathsf{T}}, (0, 1, 0)^{\mathsf{T}}\}.$$

The displayed vectors are already orthogonal, with squared norms 5 and 1. At $\lambda = 6$, the equations give $y = 0$, $x = 2z$, producing a raw vector of squared norm 5. The normalized answer is

λ	orthonormal eigenvectors
1	$(1, 0, -2)^{\mathsf{T}}/\sqrt{5}$, $(0, 1, 0)^{\mathsf{T}}$
6	$(2, 0, 1)^{\mathsf{T}}/\sqrt{5}$

The remaining scalar product is $(2 - 2)/5 = 0$, completing the orthonormality check.

6.2.13 $\mathsf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

ANS. $\lambda = 0, 0, 2$.

Solution.

The third row gives the determinant factor $-\lambda$:

$$p(\lambda) = -\lambda\big[(1-\lambda)^2 - 1\big] = (2-\lambda)\lambda^2.$$

For $\lambda = 0$, the first two equations both reduce to $x + y = 0$, and z is free. Two orthogonal spanning vectors are $(1, -1, 0)^{\mathsf{T}}$ and $(0, 0, 1)^{\mathsf{T}}$, with lengths $\sqrt{2}$ and 1. For $\lambda = 2$, the third equation forces $z = 0$, and the first gives $y = x$. Thus

λ	orthonormal eigenvectors
0	$(1, -1, 0)^{\mathsf{T}}/\sqrt{2}$, $(0, 0, 1)^{\mathsf{T}}$
2	$(1, 1, 0)^{\mathsf{T}}/\sqrt{2}$

The two vectors in the degenerate zero eigenspace are orthonormal as chosen, and each is perpendicular to the eigenvector for 2.

6.2.14 $\mathsf{A} = \begin{pmatrix} 5 & 0 & \sqrt{3} \\ 0 & 3 & 0 \\ \sqrt{3} & 0 & 3 \end{pmatrix}$.

ANS. $\lambda = 2, 3, 6$.

Solution.

Expansion on the middle coordinate gives

$$\begin{aligned} p(\lambda) &= (3-\lambda)\big[(5-\lambda)(3-\lambda)-3\big] \\ &= (3-\lambda)(\lambda^2-8\lambda+12) = (3-\lambda)(\lambda-2)(\lambda-6). \end{aligned}$$

The nontrivial component equations are

$$(5-\lambda)x+\sqrt{3}\,z=0, \qquad (3-\lambda)y=0, \qquad \sqrt{3}\,x+(3-\lambda)z=0.$$

At $\lambda=2$, $y=0$ and $z=-\sqrt{3}\,x$. At $\lambda=3$, the last equation forces $x=0$, then the first forces $z=0$, leaving y free. At $\lambda=6$, $y=0$ and $x=\sqrt{3}\,z$. The first and third raw vectors have squared norm 4, so

λ	u_λ
2	$(1,0,-\sqrt{3})^{\mathsf{T}}/2$
3	$(0,1,0)^{\mathsf{T}}$
6	$(\sqrt{3},0,1)^{\mathsf{T}}/2$

Their only nontrivial scalar product is $(\sqrt{3}-\sqrt{3})/4=0$.

6.2.15 Describe the geometric properties of the surface

$$x^2+2xy+2y^2+2yz+z^2=1.$$

How is it oriented in 3-D space? Classify the quadric surface and its cross sections perpendicular to its axis.

Solution.

The quadratic form is

$$q=\begin{pmatrix}x & y & z\end{pmatrix}\underbrace{\begin{pmatrix}1&1&0\\1&2&1\\0&1&1\end{pmatrix}}_{B}\begin{pmatrix}x\\y\\z\end{pmatrix}.$$

Notice first that $q=(x+y)^2+(y+z)^2$. It is nonnegative, but it vanishes on the entire line $(x,y,z)=t(1,-1,1)$, not only at the origin. Thus the form is positive semidefinite rather than positive definite. The characteristic determinant is

$$\begin{aligned} \det(B-\lambda I) &= (1-\lambda)\big[(2-\lambda)(1-\lambda)-1\big]-(1-\lambda) \\ &= (1-\lambda)(\lambda^2-3\lambda) = \lambda(1-\lambda)(\lambda-3). \end{aligned}$$

For $\lambda=0$, the equations are $x+y=0$, $y+z=0$, giving $(1,-1,1)$. For $\lambda=1$, they give $y=0$, $x+z=0$, giving $(1,0,-1)$. For $\lambda=3$, they give $y=2x$, $z=x$, giving $(1,2,1)$. Normalize these mutually perpendicular vectors:

$$e_1=\frac{(1,0,-1)^{\mathsf{T}}}{\sqrt{2}}, \qquad e_3=\frac{(1,2,1)^{\mathsf{T}}}{\sqrt{6}}, \qquad e_0=\frac{(1,-1,1)^{\mathsf{T}}}{\sqrt{3}}.$$

Their subscripts label the eigenvalues. Define orthogonal coordinates

$$\xi=e_1^{\mathsf{T}}r=\frac{x-z}{\sqrt{2}}, \quad \eta=e_3^{\mathsf{T}}r=\frac{x+2y+z}{\sqrt{6}}, \quad \zeta=e_0^{\mathsf{T}}r=\frac{x-y+z}{\sqrt{3}}.$$

The inverse relation is $r = \xi e_1 + \eta e_3 + \zeta e_0$. Because $Be_1 = e_1$, $Be_3 = 3e_3$, $Be_0 = 0$, all mixed products vanish and

$$q = \xi^2 + 3\eta^2, \qquad \boxed{\xi^2 + 3\eta^2 = 1, \quad \zeta \text{ arbitrary.}}$$

This is an **elliptic cylinder**. Its axis is the line through the origin in direction $(1, -1, 1)$; its transverse semiaxes have lengths 1 along e_1 and $1/\sqrt{3}$ along e_3. A parametrization displaying both the elliptical cross section and the unrestricted axis is

$$r(t, s) = e_1 \cos t + \frac{e_3}{\sqrt{3}} \sin t + s e_0, \qquad 0 \le t < 2\pi, \quad s \in \mathbb{R}.$$

A surface in three-dimensional space is a quadric here, not itself a planar conic section. Its sections perpendicular to the axis are ellipses. The Instructor's Manual, p. 82, incorrectly calls the form positive definite and describes an ellipsoid or ellipse; the zero eigenvalue and unrestricted ζ prove that the full surface is an elliptic cylinder.

6.4 Hermitian Matrix Diagonalization

6.4.1 Show that the eigenvalues of a matrix are unaltered if the matrix is transformed by a similarity transformation—a transformation that need not be unitary, but of the form given in Eq. (5.78).

This property is not limited to symmetric or Hermitian matrices. It holds for any matrix satisfying an eigenvalue equation of the type $\mathsf{A}\mathbf{x} = \lambda\mathbf{x}$. If our matrix can be brought into diagonal form by a similarity transformation, then two immediate consequences are that:

1. The trace (sum of eigenvalues) is invariant under a similarity transformation.
2. The determinant (product of eigenvalues) is invariant under a similarity transformation.

Note. The invariance of the trace and determinant are often demonstrated by using the Cayley-Hamilton theorem, which states that a matrix satisfies its own characteristic (secular) equation.

Solution.

Let S be invertible and write the similar matrix as $B = S^{-1}AS$. If $Ax = \lambda x$ with $x \neq 0$, set $y = S^{-1}x$. Invertibility ensures $y \neq 0$, and

$$By = S^{-1}AS(S^{-1}x) = S^{-1}Ax = \lambda S^{-1}x = \lambda y.$$

Conversely, an eigenvector y of B maps to $x = Sy$ of A, with the same eigenvalue. Algebraic multiplicities are also preserved, since

$$\begin{aligned}\det(B - \lambda I) &= \det\bigl[S^{-1}(A - \lambda I)S\bigr] \\ &= \det(S^{-1}) \det(A - \lambda I) \det S = \det(A - \lambda I).\end{aligned}$$

Thus the entire characteristic polynomial is unchanged.

The determinant follows directly by putting $\lambda = 0$: $\det B = (\det S)^{-1} \det A \det S = \det A$. For the trace, an index calculation avoids any diagonalizability assumption:

$$\operatorname{tr} B = \sum_{i,j,k} (S^{-1})_{ij} A_{jk} S_{ki} = \sum_{j,k} A_{jk} \underbrace{\sum_i S_{ki} (S^{-1})_{ij}}_{\delta_{kj}} = \sum_j A_{jj} = \operatorname{tr} A.$$

These invariances therefore hold even for defective matrices that cannot be diagonalized. When a diagonal form exists, its diagonal entries are the eigenvalues, immediately giving the sum and product interpretations stated in the question.

6.4.2 As a converse of the theorem that Hermitian matrices have real eigenvalues and that eigenvectors corresponding to distinct eigenvalues are orthogonal, show that if

(a) the eigenvalues of a matrix are real and

(b) a complete set of n eigenvectors for the $n \times n$ matrix satisfies $\mathbf{x}_i^\dagger \mathbf{x}_j = \delta_{ij}$,

then the matrix is Hermitian.

Solution.

The orthonormal eigenvectors must form a complete basis: there must be n of them for an $n \times n$ matrix. Let U have those eigenvectors as its columns and set $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. Their orthonormality and the eigenvalue equations give

$$U^\dagger U = U U^\dagger = I, \qquad AU = UD.$$

Multiply the second equation by $U^\dagger$ on the right: $A = UDU^\dagger$. Because every λ_i is real, $D^\dagger = D$. Taking the adjoint now gives

$$A^\dagger = (UDU^\dagger)^\dagger = UD^\dagger U^\dagger = UDU^\dagger = A.$$

Hence A is Hermitian.

Completeness is essential. The matrix $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has the real eigenvalue 1 and a normalized eigenvector $(1,0)^\mathsf{T}$, but its eigenspace has only one dimension and the matrix is not Hermitian. A partial orthonormal set does not provide $UU^\dagger = I$.

6.4.3 Show that a real matrix that is not symmetric cannot be diagonalized by an orthogonal transformation. For unitary diagonalization, add the hypothesis that all eigenvalues are real.

Hint. Assume that the nonsymmetric real matrix can be diagonalized and develop a contradiction.

Solution.

The claim is correct for real orthogonal diagonalization. For unitary diagonalization it is false without the additional hypothesis that the eigenvalues are real.

First suppose a real orthogonal Q diagonalizes the real matrix A: $Q^\mathsf{T} A Q = D$. The matrix D is real and diagonal. Recover A and take its transpose:

$$A = QDQ^\mathsf{T}, \qquad A^\mathsf{T} = (QDQ^\mathsf{T})^\mathsf{T} = QD^\mathsf{T}Q^\mathsf{T} = QDQ^\mathsf{T} = A.$$

This contradicts the assumed nonsymmetry.

For a unitary transformation, however, the diagonal entries need not be real. A counterexample is

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix}.$$

Here $A^\mathsf{T} = -A \neq A$, but the columns $u_+ = (1,-i)^\mathsf{T}/\sqrt{2}$ and $u_- = (1,i)^\mathsf{T}/\sqrt{2}$ satisfy

$$Au_+ = iu_+, \qquad Au_- = -iu_-, \qquad U^\dagger U = I.$$

Consequently $U^\dagger A U = \operatorname{diag}(i,-i)$ is a valid unitary diagonalization of a real nonsymmetric matrix.

If one additionally assumes real eigenvalues, a unitary diagonalization would instead give $A = UDU^\dagger$ with $D^\dagger = D$, implying $A^\dagger = A$; because A is real, this is $A^\mathsf{T} = A$. That proves the corrected unitary version. The distinction between transpose and adjoint prevents the orthogonal proof from being applied to an arbitrary complex unitary matrix.

6.4.4 The matrices representing the angular momentum components L_x, L_y, and L_z are all Hermitian. Show that the eigenvalues of L^2, where $L^2 = L_x^2 + L_y^2 + L_z^2$, are real and nonnegative.

Solution.

For each component $L_\alpha^\dagger = L_\alpha$, so

$$(L_\alpha^2)^\dagger = L_\alpha^\dagger L_\alpha^\dagger = L_\alpha^2.$$

Therefore $L^2 = \sum_{\alpha=x,y,z} L_\alpha^2$ is Hermitian. More explicitly, let $L^2 v = \ell v$ with $v \neq 0$. Multiply by $v^\dagger$ and use the Hermiticity of each component:

$$\begin{aligned} \ell\, v^\dagger v = v^\dagger L^2 v &= \sum_\alpha v^\dagger L_\alpha^2 v \\ &= \sum_\alpha (L_\alpha v)^\dagger (L_\alpha v) = \sum_\alpha \|L_\alpha v\|^2. \end{aligned}$$

Each norm square is real and nonnegative, while $v^\dagger v > 0$. Thus

$$\boxed{\ell = \frac{\|L_x v\|^2 + \|L_y v\|^2 + \|L_z v\|^2}{\|v\|^2} \in [0, \infty).}$$

Equality is possible exactly when all three $L_\alpha v$ vanish.

6.4.5 A has eigenvalues λ_i and corresponding eigenvectors $|x_i\rangle$. Show that A^{-1} has the same eigenvectors but with eigenvalues λ_i^{-1}.

Solution.

The notation A^{-1} presupposes that A is invertible. An eigenvalue of an invertible matrix cannot be zero: $Ax_i = 0$ would imply $x_i = A^{-1}0 = 0$, contradicting the definition of eigenvector. Thus $\lambda_i \neq 0$. Multiply $Ax_i = \lambda_i x_i$ by A^{-1}:

$$x_i = \lambda_i A^{-1} x_i, \qquad \boxed{A^{-1} x_i = \lambda_i^{-1} x_i.}$$

Conversely, multiplying an eigenvalue equation for A^{-1} by A recovers one for A, so the eigendirections are the same in both directions. No complete eigenbasis is required for this argument.

6.4.6 A square matrix with zero determinant is labeled singular.

(a) If A is singular, show that there is at least one nonzero column vector $\mathbf{v}$ such that

$$\mathsf{A}|\mathbf{v}\rangle = 0.$$

(b) If there is a nonzero vector $|\mathbf{v}\rangle$ such that

$$\mathsf{A}|\mathbf{v}\rangle = 0,$$

show that A is a singular matrix. This means that if a matrix (or operator) has zero as an eigenvalue, the matrix (or operator) has no inverse and its determinant is zero in the finite-dimensional matrix case; a general infinite-dimensional operator need not have a determinant.

Solution.

Write the columns of A as $a_1, \ldots, a_n$.

(a) A zero determinant means that these columns are linearly dependent. Hence there exist coefficients $v_1, \ldots, v_n$, not all zero, such that

$$v_1 a_1 + \cdots + v_n a_n = 0.$$

For the nonzero column vector $v = (v_1, \ldots, v_n)^{\mathsf{T}}$, the left side is exactly Av. Thus $Av = 0$.

(b) Conversely, if $Av = 0$ with $v \neq 0$, expanding the matrix-vector product gives precisely such a nontrivial linear dependence among the columns. Their determinant is therefore zero, and A is singular. An inverse is impossible: if A^{-1} existed, $Av = 0$ would give $v = A^{-1}Av = 0$, a contradiction.

The equation $Av = 0v$ is the eigenvalue equation for eigenvalue zero. We have proved the equivalence between a zero eigenvalue, a nontrivial kernel, and zero determinant for a finite square matrix, including matrices that are not diagonalizable. For a general infinite-dimensional operator a determinant need not be defined. A nonzero vector in its kernel still precludes a two-sided inverse; the determinant statement is a finite-dimensional one.

6.4.7 Two Hermitian matrices A and B have the same eigenvalues including multiplicities. Show that A and B are related by a unitary transformation.

Solution.

"The same eigenvalues" must include their algebraic multiplicities. The spectral theorem gives orthonormal eigenbases $a_1, \ldots, a_n$ for A and $b_1, \ldots, b_n$ for B, ordered so that

$$Aa_i = \lambda_i a_i, \qquad Bb_i = \lambda_i b_i.$$

Define

$$W = \sum_i |b_i\rangle\langle a_i|.$$

Its adjoint is $W^\dagger = \sum_i |a_i\rangle\langle b_i|$. Multiplying and using orthonormality gives

$$W^\dagger W = \sum_{i,j} |a_i\rangle \underbrace{\langle b_i | b_j\rangle}_{\delta_{ij}} \langle a_j| = \sum_i |a_i\rangle\langle a_i| = I,$$

and similarly $WW^\dagger = I$. Thus W is unitary and $Wa_i = b_i$. On each basis vector b_i,

$$WAW^\dagger b_i = WAa_i = \lambda_i W a_i = \lambda_i b_i = Bb_i.$$

Because the b_i span the space, $\boxed{B = WAW^\dagger}$. Within a degenerate eigenspace any orthonormal basis may be used. If multiplicities were ignored, the statement would fail, for example for $\mathrm{diag}(1,1,2)$ and $\mathrm{diag}(1,2,2)$, whose traces differ despite having the same two distinct eigenvalues.

6.4.8 Find the eigenvalues and an orthonormal set of eigenvectors for each of the matrices of Exercise 2.2.12.

Solution.

The matrices in Exercise 2.2.12 (textbook p. 116) are

$$M_x = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad M_y = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \qquad M_z = \mathrm{diag}(1, 0, -1).$$

For $M_x v = \lambda v$, the equations are

$$y/\sqrt{2} = \lambda x, \qquad (x+z)/\sqrt{2} = \lambda y, \qquad y/\sqrt{2} = \lambda z.$$

If $\lambda = 0$, then $y = 0$ and $z = -x$. If $\lambda \neq 0$, the first and third equations give $z = x$, and substitution into the middle gives $\lambda^2 = 1$. The nonzero-eigenvalue vectors have the form $(1, \sqrt{2}\lambda, 1)^{\mathsf{T}}$, whose squared norm is 4. Therefore

	$\lambda = 1$	$\lambda = 0$	$\lambda = -1$
M_x	$\dfrac{(1,\sqrt{2},1)^{\mathsf{T}}}{2}$	$\dfrac{(1,0,-1)^{\mathsf{T}}}{\sqrt{2}}$	$\dfrac{(1,-\sqrt{2},1)^{\mathsf{T}}}{2}$

Their pairwise inner products vanish, as follows from $1 - 2 + 1 = 0$ and $1 - 1 = 0$.

For M_y, use the diagonal unitary matrix $D = \mathrm{diag}(1, i, -1)$. Its phases give

$$DM_xD^\dagger = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix} = M_y.$$

If $M_x u = \lambda u$, then $M_y(Du) = DM_x u = \lambda(Du)$. Since $D^\dagger D = I$, multiplication by D preserves all norms and inner products. Thus an orthonormal answer for M_y is

	$\lambda = 1$	$\lambda = 0$	$\lambda = -1$
M_y	$\dfrac{(1,i\sqrt{2},-1)^{\mathsf{T}}}{2}$	$\dfrac{(1,0,1)^{\mathsf{T}}}{\sqrt{2}}$	$\dfrac{(1,-i\sqrt{2},-1)^{\mathsf{T}}}{2}$

Finally, M_z is already diagonal:

$$\boxed{\lambda = 1:\ (1,0,0)^{\mathsf{T}}, \qquad \lambda = 0:\ (0,1,0)^{\mathsf{T}}, \qquad \lambda = -1:\ (0,0,1)^{\mathsf{T}}.}$$

Every matrix has the spectrum $\{1, 0, -1\}$ and the displayed orthonormal eigenbasis.

6.4.9 The unit process in the iterative matrix diagonalization procedure known as the Jacobi method is a unitary transformation that operates on rows/columns i and j of a real symmetric matrix A to make $a_{ij} = a_{ji} = 0$. If this transformation (from basis functions φ_i and φ_j to φ_i' and φ_j') is written

$$\varphi_i' = \varphi_i \cos\theta - \varphi_j \sin\theta, \quad \varphi_j' = \varphi_i \sin\theta + \varphi_j \cos\theta,$$

(a) Show that a_{ij} is transformed to zero if $\tan 2\theta = \dfrac{2a_{ij}}{a_{jj} - a_{ii}}$ when $a_{jj} \neq a_{ii}$; also give a valid angle when $a_{jj} = a_{ii}$ or $a_{ij} = 0$,

(b) Show that $a_{\mu\nu}$ remains unchanged if neither μ nor ν is i or j,

(c) Find a_{ii}' and a_{jj}' and show that the trace of A is not changed by the transformation,

(d) Find $a_{i\mu}'$ and $a_{j\mu}'$ (where μ is neither i nor j) and show that the sum of the squares of the off-diagonal elements of A is reduced by the amount $2a_{ij}^2$.

Solution.

Put $c = \cos\theta$ and $s = \sin\theta$. Only the two basis vectors i, j are altered; the new matrix elements are $a'_{\mu\nu} = \langle\varphi'_\mu|A|\varphi'_\nu\rangle$. All coefficients are real, and $a_{ji} = a_{ij}$.

(a) Expanding the off-diagonal entry gives

$$\begin{aligned} a'_{ij} &= \langle c\varphi_i - s\varphi_j|A|s\varphi_i + c\varphi_j\rangle \\ &= cs\,a_{ii} + c^2 a_{ij} - s^2 a_{ji} - sc\,a_{jj} \\ &= \frac{1}{2}(a_{ii} - a_{jj})\sin 2\theta + a_{ij}\cos 2\theta. \end{aligned}$$

Setting this to zero gives

$$\boxed{\tan 2\theta = \frac{2a_{ij}}{a_{jj} - a_{ii}}}$$

when the denominator is nonzero, with a branch of θ satisfying the preceding sine-cosine equation. The tangent formula needs its limiting cases: if $a_{ii} = a_{jj}$ and $a_{ij} \neq 0$, choose $\theta = \pi/4$; if $a_{ij} = 0$, choose $\theta = 0$ and no elimination is needed. These choices avoid dividing by zero.

(b) If neither μ nor ν is i or j, neither basis vector changes. Thus $a'_{\mu\nu} = \langle\varphi_\mu|A|\varphi_\nu\rangle = a_{\mu\nu}$.

(c) Expand the two diagonal entries:

$$\begin{aligned} a'_{ii} &= \langle c\varphi_i - s\varphi_j|A|c\varphi_i - s\varphi_j\rangle = c^2 a_{ii} + s^2 a_{jj} - 2cs\,a_{ij}, \\ a'_{jj} &= \langle s\varphi_i + c\varphi_j|A|s\varphi_i + c\varphi_j\rangle = s^2 a_{ii} + c^2 a_{jj} + 2cs\,a_{ij}. \end{aligned}$$

Their sum is

$$a'_{ii} + a'_{jj} = (c^2 + s^2)(a_{ii} + a_{jj}) = a_{ii} + a_{jj}.$$

Every other diagonal element is unchanged by (b), so the total trace is unchanged.

(d) For $\mu \neq i, j$, one of the two basis vectors remains fixed:

$$\boxed{a'_{i\mu} = c\,a_{i\mu} - s\,a_{j\mu}, \qquad a'_{j\mu} = s\,a_{i\mu} + c\,a_{j\mu}.}$$

Squaring and adding exhibits the cancellation of cross terms:

$$\begin{aligned} (a'_{i\mu})^2 + (a'_{j\mu})^2 &= c^2 a_{i\mu}^2 + s^2 a_{j\mu}^2 - 2cs\,a_{i\mu}a_{j\mu} \\ &\quad + s^2 a_{i\mu}^2 + c^2 a_{j\mu}^2 + 2cs\,a_{i\mu}a_{j\mu} \\ &= a_{i\mu}^2 + a_{j\mu}^2. \end{aligned}$$

Symmetry gives the same result for the transposed entries $a'_{\mu i}, a'_{\mu j}$. Entries with neither index i, j are unchanged. Thus, in the sum $S = \sum_{\mu\neq\nu} a_{\mu\nu}^2$, the only net change is the removal of a_{ij} and a_{ji} themselves:

$$\boxed{S' = S - a_{ij}^2 - a_{ji}^2 = S - 2a_{ij}^2.}$$

Both members of each symmetric off-diagonal pair are included in S, which accounts for the factor 2.

6.5 Normal Matrices

6.5.1 Find the eigenvalues and corresponding eigenvectors for

$$\begin{pmatrix} 2 & 4 \\ 1 & 2 \end{pmatrix}.$$

Note that the eigenvectors are not orthogonal.

ANS. $\lambda_1 = 0,\ \mathbf{c}_1 = (2,-1); \quad \lambda_2 = 4,\ \mathbf{c}_2 = (2,1).$

Solution.

The characteristic determinant is

$$\det(A - \lambda I) = (2-\lambda)^2 - 4 = \lambda^2 - 4\lambda = \lambda(\lambda - 4).$$

For $\lambda = 0$, the first row gives $2x + 4y = 0$, or $x = -2y$; $(2,-1)^{\mathsf{T}}$ is a nonzero solution. For $\lambda = 4$, the first row gives $-2x + 4y = 0$, or $x = 2y$; choose $(2,1)^{\mathsf{T}}$. In each case the second row is the same equation up to a constant factor. Normalizing the vectors, whose squared norms are 5, gives

$$\boxed{\lambda_0 = 0, \quad u_0 = \frac{(2,-1)^{\mathsf{T}}}{\sqrt{5}}; \qquad \lambda_4 = 4, \quad u_4 = \frac{(2,1)^{\mathsf{T}}}{\sqrt{5}}.}$$

Their inner product is $(4-1)/5 = 3/5 \neq 0$, so they are not orthogonal. This causes no contradiction: the matrix is not Hermitian, and in fact

$$AA^{\mathsf{T}} = \begin{pmatrix} 20 & 10 \\ 10 & 5 \end{pmatrix}, \qquad A^{\mathsf{T}}A = \begin{pmatrix} 5 & 10 \\ 10 & 20 \end{pmatrix}$$

show that it is not normal either.

6.5.2 If A is a 2×2 matrix, show that its eigenvalues λ satisfy the secular equation

$$\lambda^2 - \lambda\,\mathrm{trace}(\mathsf{A}) + \det(\mathsf{A}) = 0.$$

Solution.

Write $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. An eigenvalue makes $A - \lambda I$ singular, so

$$\begin{aligned} 0 = \det(A - \lambda I) &= (a-\lambda)(d-\lambda) - bc \\ &= ad - (a+d)\lambda + \lambda^2 - bc \\ &= \lambda^2 - \lambda\,\mathrm{tr}\,A + \det A. \end{aligned}$$

This proves the formula for any real or complex 2×2 matrix; no symmetry or diagonalizability is needed. It also shows that the two roots, counted with multiplicity, have sum $a + d$ and product $ad - bc$.

6.5.3 Assuming a unitary matrix U to satisfy an eigenvalue equation $\mathsf{Uc} = \lambda\mathbf{c}$, show that the eigenvalues of the unitary matrix have unit magnitude. This same result holds for real orthogonal matrices.

Solution.

Let $Uc = \lambda c$ with $c \neq 0$. Unitarity means $U^\dagger U = I$, so the squared norm is preserved:

$$c^\dagger c = (Uc)^\dagger (Uc) = (\lambda c)^\dagger (\lambda c) = \lambda^* \lambda \, c^\dagger c = |\lambda|^2 c^\dagger c.$$

Divide by $c^\dagger c > 0$ to obtain

$$\boxed{|\lambda|^2 = 1, \qquad |\lambda| = 1.}$$

For a real orthogonal matrix Q, $Q^\dagger = Q^{\mathsf{T}}$ and $Q^{\mathsf{T}} Q = I$, so the identical argument applies, allowing complex eigenvectors and eigenvalues even though the entries of Q are real.

6.5.4 Since an orthogonal matrix describing a rotation in real 3-D space is a special case of a unitary matrix, such an orthogonal matrix can be diagonalized by a unitary transformation.

(a) Show that the sum of the three eigenvalues is $1 + 2\cos\varphi$, where φ is the net angle of rotation about a single fixed axis.

(b) Given that one eigenvalue is 1, show that the other two eigenvalues must be $e^{i\varphi}$ and $e^{-i\varphi}$.

Our orthogonal rotation matrix (real elements) has generally complex eigenvalues.

Solution.

(a) Choose an orthonormal basis whose third axis is the fixed rotation axis. In this basis the matrix is, up to the sign convention for the angle,

$$R_0 = \begin{pmatrix} \cos\varphi & -\sin\varphi & 0 \\ \sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

A change of orthonormal basis is a similarity, so the trace is the same in the original coordinates. Thus

$$\boxed{\lambda_1 + \lambda_2 + \lambda_3 = \operatorname{tr} R_0 = 1 + 2\cos\varphi.}$$

(b) Take $\lambda_1 = 1$, corresponding to the rotation axis. A proper rotation has determinant 1, so $\lambda_2 \lambda_3 = 1$; part (a) gives $\lambda_2 + \lambda_3 = 2\cos\varphi$. Therefore the remaining two roots satisfy

$$\lambda^2 - 2\cos\varphi\,\lambda + 1 = 0.$$

Equivalently the planar block has characteristic determinant $(\cos\varphi - \lambda)^2 + \sin^2\varphi$. Its roots are

$$\boxed{\lambda_2 = \cos\varphi + i\sin\varphi = e^{i\varphi}, \qquad \lambda_3 = \cos\varphi - i\sin\varphi = e^{-i\varphi}.}$$

The unordered pair is unchanged if the passive-rotation convention reverses φ. These two eigenvalues are generally nonreal, but they are real when φ is an integer multiple of π: both are 1 for a full turn and both are -1 for a half-turn.

6.5.5 A is an nth-order Hermitian matrix with orthonormal eigenvectors $|x_i\rangle$ and real eigenvalues $\lambda_1 \le \lambda_2 \le \lambda_3 \le \cdots \le \lambda_n$. Show that for a unit magnitude vector $|y\rangle$,

$$\lambda_1 \le \langle y|\mathsf{A}|y\rangle \le \lambda_n.$$

Solution.

Expand the unit vector in the complete orthonormal eigenbasis:

$$|y\rangle = \sum_i c_i|x_i\rangle, \qquad c_i = \langle x_i|y\rangle, \qquad 1 = \langle y|y\rangle = \sum_i |c_i|^2.$$

Then

$$\begin{aligned}\langle y|A|y\rangle &= \sum_{i,j} c_i^* c_j \langle x_i|A|x_j\rangle \\ &= \sum_{i,j} c_i^* c_j \lambda_j \delta_{ij} = \sum_i \lambda_i |c_i|^2.\end{aligned}$$

Each weight $|c_i|^2$ is nonnegative. Since $\lambda_1 \le \lambda_i \le \lambda_n$, multiplication by these weights and summation give

$$\lambda_1 \sum_i |c_i|^2 \le \sum_i \lambda_i |c_i|^2 \le \lambda_n \sum_i |c_i|^2.$$

The two outer sums equal one, proving the stated bounds. Equality at an endpoint occurs precisely when y has components only in the eigenspace of that endpoint eigenvalue.

6.5.6 A particular matrix is both Hermitian and unitary. Show that its eigenvalues are all ± 1.

Note. The Pauli and Dirac matrices are specific examples.

Solution.

For a Hermitian matrix each eigenvalue is real. For a unitary matrix Exercise 6.5.3 gives $|\lambda| = 1$. Combining these facts yields $\lambda = 1$ or $\lambda = -1$. One may also use both matrix properties directly: $A^\dagger = A$ and $A^\dagger A = I$ imply $A^2 = I$. Thus for $Av = \lambda v$,

$$v = A^2 v = \lambda^2 v.$$

Since $v \neq 0$, $\lambda^2 = 1$, again giving $\boxed{\lambda \in \{-1, 1\}}$. Either sign may occur with any allowed multiplicity; a particular matrix need not contain both signs.

6.5.7 For his relativistic electron theory Dirac required a set of four anticommuting matrices. Assume that these matrices are to be Hermitian and unitary. If these are $n \times n$ matrices, show that n must be even. With 2×2 matrices inadequate (why?), this demonstrates that the smallest possible matrices forming a set of four anticommuting, Hermitian, unitary matrices are 4×4.

Solution.

Write the four matrices as Γ_a, with $\Gamma_a \Gamma_b = -\Gamma_b \Gamma_a$ when $a \neq b$. Unitarity makes every determinant nonzero. Taking determinants of an anticommutation relation gives

$$\det \Gamma_a \det \Gamma_b = \det(-\Gamma_b \Gamma_a) = (-1)^n \det \Gamma_b \det \Gamma_a.$$

Cancel the nonzero determinants. Then $(-1)^n = 1$, so n must be even.

We next exclude $n = 2$, not merely odd n. Hermiticity and unitarity give $\Gamma_a^2 = I$ and eigenvalues ± 1. The first matrix cannot be I or $-I$, because anticommuting with either would force every other matrix to be zero. Thus a unitary change of basis puts it in the form

$$\Gamma_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Write another Hermitian matrix as $B = \begin{pmatrix} a & z \\ z^* & d \end{pmatrix}$, with a, d real. Its anticommutator with Γ_1 is

$$\Gamma_1 B + B\Gamma_1 = \begin{pmatrix} 2a & 0 \\ 0 & -2d \end{pmatrix}.$$

Therefore every remaining matrix must have the form

$$B(z) = \begin{pmatrix} 0 & z \\ z^* & 0 \end{pmatrix}, \qquad B(z)^2 = |z|^2 I,$$

and unitarity forces $|z| = 1$. For two such matrices,

$$B(z)B(w) + B(w)B(z) = 2\Re(zw^*)I.$$

Writing $z = a + ib$ and $w = c + id$ gives $\Re(zw^*) = ac + bd$. Thus the three remaining matrices would require three nonzero, mutually orthogonal real vectors (a, b) in a two-dimensional space. After choosing two independent orthogonal vectors, a vector orthogonal to both is zero; this contradicts $|z| = 1$. Hence four such 2×2 matrices cannot exist.

To prove that 4×4 matrices do exist, define the Pauli matrices

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Each is Hermitian, has square I_2, and anticommutes with either of the other two, as direct multiplication verifies. Use the four block matrices

$$\Gamma_1 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \quad \Gamma_2 = \begin{pmatrix} 0 & -iI_2 \\ iI_2 & 0 \end{pmatrix}, \quad \Gamma_3 = \begin{pmatrix} \sigma_x & 0 \\ 0 & -\sigma_x \end{pmatrix}, \quad \Gamma_4 = \begin{pmatrix} \sigma_y & 0 \\ 0 & -\sigma_y \end{pmatrix}.$$

They are Hermitian and their squares are I_4. Equivalently, $\Gamma_1 = \sigma_x \otimes I_2$, $\Gamma_2 = \sigma_y \otimes I_2$, $\Gamma_3 = \sigma_z \otimes \sigma_x$, and $\Gamma_4 = \sigma_z \otimes \sigma_y$. The first pair anticommutes by the first Pauli factors; either of the first two anticommutes with either of the last two because $\{\sigma_x, \sigma_z\} = \{\sigma_y, \sigma_z\} = 0$; the last pair anticommutes because $\sigma_z^2 \otimes \{\sigma_x, \sigma_y\} = 0$. This checks all six pairs. Therefore the smallest possible size is

$$\boxed{n = 4.}$$

6.5.8 A is a normal matrix with eigenvalues λ_n and orthonormal eigenvectors $|x_n\rangle$. Show that A may be written as

$$\mathsf{A} = \sum_n \lambda_n |x_n\rangle\langle x_n|.$$

Hint. Show that both this eigenvector form of A and the original A give the same result acting on an arbitrary vector $|y\rangle$.

Solution.

Let $B = \sum_n \lambda_n |x_n\rangle\langle x_n|$ be the proposed expression. The normal matrix has the complete orthonormal eigenbasis specified in the question. Expand an arbitrary vector as

$$|y\rangle = \sum_j c_j |x_j\rangle, \qquad c_j = \langle x_j | y\rangle.$$

The original matrix acts as

$$A|y\rangle = \sum_j c_j A|x_j\rangle = \sum_j c_j \lambda_j |x_j\rangle.$$

The proposed matrix acts as

$$\begin{aligned} B|y\rangle &= \sum_{n,j} \lambda_n c_j |x_n\rangle \underbrace{\langle x_n | x_j\rangle}_{\delta_{nj}} \\ &= \sum_j c_j \lambda_j |x_j\rangle = A|y\rangle. \end{aligned}$$

Since this holds for every y, $A = B$, proving

$$\boxed{A = \sum_n \lambda_n |x_n\rangle\langle x_n|.}$$

Repeated eigenvalues cause no difficulty; any orthonormal basis in each degenerate eigenspace gives the same operator.

6.5.9 A has eigenvalues 1 and -1 and corresponding eigenvectors $\begin{pmatrix}1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\end{pmatrix}$. Construct A.

ANS. $\mathsf{A} = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}$.

Solution.

Let $e_1 = (1,0)^{\mathsf{T}}$ and $e_2 = (0,1)^{\mathsf{T}}$. The product Ae_j is the jth column of A. Thus $Ae_1 = e_1$ fixes the first column as $(1,0)^{\mathsf{T}}$, while $Ae_2 = -e_2$ fixes the second as $(0,-1)^{\mathsf{T}}$. Therefore

$$\boxed{A = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}.}$$

The spectral formula gives the same result explicitly:

$$A = |e_1\rangle\langle e_1| - |e_2\rangle\langle e_2| = \begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix} - \begin{pmatrix}0 & 0\\ 0 & 1\end{pmatrix}.$$

6.5.10 A non-Hermitian matrix A has eigenvalues λ_i and corresponding eigenvectors $|u_i\rangle$. The adjoint matrix $\mathsf{A}^\dagger$ has the conjugate eigenvalues λ_i^*, with corresponding eigenvectors $|v_i\rangle$. Show that the eigenvectors form a biorthogonal set in the sense that

$$\langle v_i | u_j\rangle = 0 \quad \text{for } \lambda_i \neq \lambda_j.$$

Solution.

In general the adjoint has the conjugate eigenvalues, not the same eigenvalues. With the consistent pairing $A^\dagger v_i = \lambda_i^* v_i$, the condition for the displayed orthogonality is $\lambda_i \neq \lambda_j$. The conjugation of the spectrum follows from

$$\det(A^\dagger - \mu I) = \overline{\det(A - \overline{\mu} I)}\,:$$

its zeros are the complex conjugates of the zeros of the original characteristic polynomial, including multiplicities.

Using the corrected pairing, take the adjoint of the second eigenvalue equation:

$$Au_j = \lambda_j u_j, \qquad A^\dagger v_i = \lambda_i^* v_i \quad \Longrightarrow \quad \langle v_i | A = \lambda_i \langle v_i |.$$

Evaluate the same scalar in these two ways:

$$\langle v_i | A | u_j \rangle = \lambda_j \langle v_i | u_j \rangle = \lambda_i \langle v_i | u_j \rangle.$$

Subtracting gives

$$(\lambda_j - \lambda_i)\langle v_i | u_j \rangle = 0, \qquad \boxed{\langle v_i | u_j \rangle = 0 \quad \text{if } \lambda_i \neq \lambda_j.}$$

If one instead denotes the eigenvalue of $A^\dagger$ by μ_i, without pairing it in advance, the same calculation yields $(\lambda_j - \mu_i^*)\langle v_i | u_j \rangle = 0$. This is the consistent version of the condition with a star that appears in the original wording. No orthogonality is forced when the two paired eigenvalues coincide, and the argument does not assume that a non-Hermitian matrix has a complete eigenbasis.

6.5.11 You are given a pair of equations:

$$\mathsf{A}|f_n\rangle = \lambda_n |g_n\rangle$$

$$\tilde{\mathsf{A}}|g_n\rangle = \lambda_n |f_n\rangle \quad \text{with } \mathsf{A} \text{ real.}$$

(a) Prove that $|f_n\rangle$ is an eigenvector of $(\tilde{\mathsf{A}}\mathsf{A})$ with eigenvalue λ_n^2.

(b) Prove that $|g_n\rangle$ is an eigenvector of $(\mathsf{A}\tilde{\mathsf{A}})$ with eigenvalue λ_n^2.

(c) State how you know that

(1) The $|f_n\rangle$ can be chosen to form an orthogonal set.

(2) The $|g_n\rangle$ can be chosen to form an orthogonal set.

(3) λ_n^2 is real.

Solution.

Here $\widetilde{A} = A^{\mathsf{T}} = A^\dagger$ because A is real. Take the indicated f_n, g_n to be nonzero.

(a) Apply A^{T} to the first equation:

$$A^{\mathsf{T}} A f_n = \lambda_n A^{\mathsf{T}} g_n = \lambda_n(\lambda_n f_n) = \boxed{\lambda_n^2 f_n}.$$

(b) Apply A to the second equation:

$$A A^{\mathsf{T}} g_n = \lambda_n A f_n = \lambda_n(\lambda_n g_n) = \boxed{\lambda_n^2 g_n}.$$

(c) Both products are real symmetric:

$$(A^{\mathsf{T}}A)^{\mathsf{T}} = A^{\mathsf{T}}A, \qquad (AA^{\mathsf{T}})^{\mathsf{T}} = AA^{\mathsf{T}}.$$

They are also positive semidefinite. In particular,

$$\lambda_n^2 = \frac{f_n^\dagger A^{\mathsf{T}} A f_n}{f_n^\dagger f_n} = \frac{\|Af_n\|^2}{\|f_n\|^2} \geq 0,$$

so the eigenvalue λ_n^2 is real and nonnegative. Since its square is nonnegative, λ_n itself can be taken real; its nonnegative choice is the singular value.

For distinct squared eigenvalues, symmetry proves orthogonality:

$$\lambda_j^2 f_i^\dagger f_j = f_i^\dagger (A^{\mathsf{T}}A) f_j = ((A^{\mathsf{T}}A) f_i)^\dagger f_j = \lambda_i^2 f_i^\dagger f_j.$$

Thus $f_i^\dagger f_j = 0$ when $\lambda_i^2 \neq \lambda_j^2$. The identical argument with AA^{T} proves the assertion for g_i, g_j.

Eigenvectors in a repeated eigenspace are not automatically orthogonal; the correct assertion is that they can be chosen orthonormal. For example, if $A = I_2$, the unit vectors $f_1 = g_1 = (1,0)^{\mathsf{T}}$ and $f_2 = g_2 = (1,1)^{\mathsf{T}}/\sqrt{2}$ satisfy both paired equations with $\lambda_1 = \lambda_2 = 1$, but their inner product is $1/\sqrt{2}$.

To make a consistent orthonormal choice, first orthonormalize the f_n in each eigenspace of $A^{\mathsf{T}}A$. For a nonzero singular value $\sigma_n = |\lambda_n|$, set $g_n = Af_n/\sigma_n$. Then

$$g_i^\dagger g_j = \frac{f_i^\dagger A^{\mathsf{T}} A f_j}{\sigma_i \sigma_j} = \frac{\sigma_j^2}{\sigma_i \sigma_j}\delta_{ij} = \delta_{ij}, \qquad A^{\mathsf{T}} g_n = \sigma_n f_n.$$

This also handles repeated positive singular values. For zero singular values, choose orthonormal vectors separately in $\ker A$ and $\ker A^{\mathsf{T}}$; for a square matrix these kernels have the same dimension and can be paired. This gives the complete orthonormal families used in the following exercises.

6.5.12 Prove that A of the preceding exercise may be written as

$$\mathsf{A} = \sum_n \lambda_n |g_n\rangle\langle f_n|,$$

with complete orthonormal families $|g_n\rangle$ and $|f_n\rangle$ chosen as in Exercise 6.5.11.

Hint. Expand an arbitrary vector as a linear combination of $|f_n\rangle$.

Solution.

The right vectors f_n must form a complete orthonormal basis, as chosen in Exercise 6.5.11; unit norms alone are insufficient. Expand an arbitrary vector using that basis:

$$|y\rangle = \sum_n c_n |f_n\rangle, \qquad c_n = \langle f_n | y\rangle.$$

The paired equation $A|f_n\rangle = \lambda_n |g_n\rangle$ gives

$$A|y\rangle = \sum_n c_n \lambda_n |g_n\rangle.$$

For the proposed expression, orthonormality gives

$$\left(\sum_n \lambda_n |g_n\rangle\langle f_n|\right)|y\rangle = \sum_{n,j} \lambda_n c_j |g_n\rangle \underbrace{\langle f_n|f_j\rangle}_{\delta_{nj}} = \sum_j c_j \lambda_j |g_j\rangle = A|y\rangle.$$

Agreement on every vector proves

$$\boxed{A = \sum_n \lambda_n |g_n\rangle\langle f_n|.}$$

Terms with $\lambda_n = 0$ vanish, but including the corresponding right vectors makes the expansion of an arbitrary y complete. For nonzero λ_n, a unit right vector has a unit left vector when the paired equations hold:

$$\|g_n\|^2 = \frac{f_n^\dagger A^\mathsf{T} A f_n}{\lambda_n^2} = \|f_n\|^2 = 1.$$

The nonnegative choices of λ_n give the usual singular-value decomposition; a negative real factor may instead be absorbed by reversing the sign of its g_n.

The unit-normalized but nonorthogonal example in Exercise 6.5.11 shows why the qualification matters: it would give

$$|f_1\rangle\langle f_1| + |f_2\rangle\langle f_2| = \begin{pmatrix} 3/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix} \neq I_2,$$

even though both paired factors are 1 and the actual matrix is I_2.

6.5.13 Given

$$\mathsf{A} = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & 2 \\ 1 & -4 \end{pmatrix},$$

(a) Construct the transpose $\tilde{\mathsf{A}}$ and the symmetric forms $\tilde{\mathsf{A}}\mathsf{A}$ and $\mathsf{A}\tilde{\mathsf{A}}$.

(b) From $\mathsf{A}\tilde{\mathsf{A}}|g_n\rangle = \lambda_n^2|g_n\rangle$, find λ_n and $|g_n\rangle$. Normalize the $|g_n\rangle$.

(c) From $\tilde{\mathsf{A}}\mathsf{A}|f_n\rangle = \lambda_n^2|f_n\rangle$, find λ_n [same as (b)] and $|f_n\rangle$. Normalize the $|f_n\rangle$.

(d) Verify that $\mathsf{A}|f_n\rangle = \lambda_n|g_n\rangle$ and $\tilde{\mathsf{A}}|g_n\rangle = \lambda_n|f_n\rangle$.

(e) Verify that $\mathsf{A} = \sum_n \lambda_n |g_n\rangle\langle f_n|$.

Solution.

The matrix's first column is $(2,1)^\mathsf{T}/\sqrt{5}$ in the textbook, p. 326; the earlier transcription reversed these two entries. Also, the right side in part (c) must contain f_n, not g_n. The corrections are marked in the question.

(a) For the corrected matrix,

$$A = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & 2 \\ 1 & -4 \end{pmatrix}, \qquad A^\mathsf{T} = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & 1 \\ 2 & -4 \end{pmatrix}.$$

Multiply in both orders, retaining the individual products:

$$A^{\mathsf{T}}A = \frac{1}{5}\begin{pmatrix} 2\cdot 2+1\cdot 1 & 2\cdot 2+1\cdot(-4) \\ 2\cdot 2+(-4)\cdot 1 & 2\cdot 2+(-4)(-4)\end{pmatrix} = \begin{pmatrix}1 & 0\\ 0 & 4\end{pmatrix},$$
$$AA^{\mathsf{T}} = \frac{1}{5}\begin{pmatrix} 2\cdot 2+2\cdot 2 & 2\cdot 1+2\cdot(-4) \\ 1\cdot 2+(-4)\cdot 2 & 1\cdot 1+(-4)(-4)\end{pmatrix} = \frac{1}{5}\begin{pmatrix}8 & -6\\ -6 & 17\end{pmatrix}.$$

(b) Write $\mu = \lambda^2$. The characteristic determinant for AA^{T} is

$$\det(AA^{\mathsf{T}} - \mu I) = \frac{(8-5\mu)(17-5\mu)-36}{25} = \mu^2 - 5\mu + 4 = (\mu-1)(\mu-4).$$

Thus the nonnegative singular values are $\lambda_1 = 1$, $\lambda_2 = 2$. For $\mu = 1$, the first row of the eigenvector equation gives $3g_x - 6g_y = 0$, so $g_x = 2g_y$. For $\mu = 4$, it gives $-12g_x - 6g_y = 0$, so $g_y = -2g_x$. The two raw vectors have squared length 5 and scalar product $2 - 2 = 0$, giving

$$g_1 = \frac{(2,1)^{\mathsf{T}}}{\sqrt{5}}, \qquad g_2 = \frac{(1,-2)^{\mathsf{T}}}{\sqrt{5}}.$$

(c) The matrix $A^{\mathsf{T}}A = \mathrm{diag}(1,4)$ is already diagonal, so the matching orthonormal right vectors are

$$f_1 = (1,0)^{\mathsf{T}}, \qquad f_2 = (0,1)^{\mathsf{T}}.$$

(d) The relative signs of left and right vectors are confirmed directly:

$$Af_1 = \frac{1}{\sqrt{5}}\begin{pmatrix}2\\1\end{pmatrix} = g_1, \qquad Af_2 = \frac{1}{\sqrt{5}}\begin{pmatrix}2\\-4\end{pmatrix} = 2g_2,$$
$$A^{\mathsf{T}}g_1 = \frac{1}{5}\begin{pmatrix}2\cdot 2+1\cdot 1\\ 2\cdot 2-4\cdot 1\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix} = f_1, \qquad A^{\mathsf{T}}g_2 = \frac{1}{5}\begin{pmatrix}2\cdot 1+1\cdot(-2)\\ 2\cdot 1-4\cdot(-2)\end{pmatrix} = \begin{pmatrix}0\\2\end{pmatrix} = 2f_2.$$

(e) Finally, form the two outer products:

$$\lambda_1 g_1 f_1^{\mathsf{T}} + \lambda_2 g_2 f_2^{\mathsf{T}} = \frac{1}{\sqrt{5}}\begin{pmatrix}2 & 0\\ 1 & 0\end{pmatrix} + \frac{1}{\sqrt{5}}\begin{pmatrix}0 & 2\\ 0 & -4\end{pmatrix} = \boxed{\frac{1}{\sqrt{5}}\begin{pmatrix}2 & 2\\ 1 & -4\end{pmatrix} = A.}$$

6.5.14 Given the factors in the paired equations $\lambda_1 = 1$, $\lambda_2 = -1$ and the corresponding right and left vectors

$$|f_1\rangle = \begin{pmatrix}1\\0\end{pmatrix}, \quad |g_1\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, \quad |f_2\rangle = \begin{pmatrix}0\\1\end{pmatrix}, \quad \text{and} \quad |g_2\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix},$$

(a) construct A;

(b) verify that $\mathsf{A}|f_n\rangle = \lambda_n|g_n\rangle$;

(c) verify that $\tilde{\mathsf{A}}|g_n\rangle = \lambda_n|f_n\rangle$.

ANS. $\mathsf{A} = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & -1\\ 1 & 1\end{pmatrix}$.

Solution.

The numbers $1, -1$ are factors in the paired equations $Af_n = \lambda_n g_n$, not ordinary eigenvalues of A. The vectors f_n and g_n are right and left singular vectors with a sign convention. With this

wording correction the exercise is well-defined, even though the Instructor's Manual, p. 86, says to disregard it.

(a) Both given families are orthonormal: the f_n are coordinate vectors, while $g_1^\dagger g_2 = (1-1)/2 = 0$ and each g_n has norm one. Use Exercise 6.5.12:

$$\begin{aligned} A &= g_1 f_1^{\mathsf{T}} - g_2 f_2^{\mathsf{T}} \\ &= \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix} - \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 \\ 0 & -1 \end{pmatrix} = \boxed{\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}}. \end{aligned}$$

(b) Multiplication by the coordinate vectors selects the columns:

$$Af_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = g_1, \qquad Af_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ 1 \end{pmatrix} = -g_2.$$

(c) The transpose is $A^{\mathsf{T}} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$, so

$$A^{\mathsf{T}} g_1 = \frac{1}{2}\begin{pmatrix} 1+1 \\ -1+1 \end{pmatrix} = f_1, \qquad A^{\mathsf{T}} g_2 = \frac{1}{2}\begin{pmatrix} 1-1 \\ -1-1 \end{pmatrix} = -f_2.$$

All paired equations hold with the supplied signs. To check explicitly why the original terminology fails,

$$\det(A - \mu I) = \mu^2 - \sqrt{2}\mu + 1, \qquad \mu = \frac{1 \pm i}{\sqrt{2}}.$$

These are the actual eigenvalues of A, and neither equals 1 or -1. Its ordinary singular values are $1, 1$, since $A^{\mathsf{T}} A = I$; replacing g_2 by $-g_2$ converts the second paired factor to $+1$.

6.5.15 Two matrices U and H are related by

$$\mathsf{U} = e^{ia\mathsf{H}},$$

with a real. For the converses in (b) and (d), the stated property is required for every a in an interval about 0, with H fixed.

(a) If H is Hermitian, show that U is unitary.

(b) If U is unitary, show that H is Hermitian. (H is independent of a.)

(c) If $\operatorname{trace} \mathsf{H} = 0$, show that $\det \mathsf{U} = +1$.

(d) If $\det \mathsf{U} = +1$, show that $\operatorname{trace} \mathsf{H} = 0$.

Hint. When H is Hermitian, it can be diagonalized unitarily; then U is also diagonal in that basis. The corresponding eigenvalues are given by $u_j = \exp(iah_j)$.

Solution.

Parts (b) and (d) require the indicated property of $U(a) = e^{iaH}$ for every real a in an interval about 0, with the same fixed matrix H. Either converse is false for only one fixed value of a.

(a) The matrix exponential is defined by its convergent power series. Taking its adjoint term by term, and using real a and $H^\dagger = H$,

$$U(a)^\dagger = \sum_{k=0}^{\infty} \frac{(-ia)^k (H^\dagger)^k}{k!} = e^{-iaH} = U(a)^{-1}.$$

The last equality follows because iaH and $-iaH$ commute, so the product of their exponentials is $e^0 = I$. Thus $U(a)$ is unitary.

(b) Suppose now that $U(a)^\dagger U(a) = I$ throughout an interval about zero. The power series gives

$$U(0) = I, \qquad U'(0) = iH, \qquad \left.\frac{dU^\dagger}{da}\right|_{a=0} = -iH^\dagger.$$

Differentiate the unitary identity and evaluate at zero:

$$0 = \left.\frac{d}{da}(U^\dagger U)\right|_0 = (-iH^\dagger)I + I(iH) = i(H - H^\dagger).$$

Therefore $\boxed{H = H^\dagger}$.

For parts (c) and (d) we first establish a determinant identity valid for every finite square matrix H, whether or not it is diagonalizable. Let $F(a) = \det U(a)$. The exponential is invertible and $U' = iHU$. If C_{ij} denotes the cofactor of entry U_{ij}, differentiation of the determinant and the inverse-cofactor formula give

$$F' = \sum_{i,j} C_{ij} U'_{ij}, \qquad C_{ij} = F(U^{-1})_{ji}.$$

Consequently

$$F' = F \sum_{i,j} (U^{-1})_{ji} U'_{ij} = F \operatorname{tr}(U^{-1}U'),$$

which is Jacobi's formula with all indices accounted for. Hence

$$F' = i \operatorname{tr}(U^{-1}HU)F = i(\operatorname{tr} H)F, \qquad F(0) = 1,$$

where trace invariance was proved in Exercise 6.4.1. Solving this scalar constant-coefficient equation yields

$$\boxed{\det U(a) = e^{ia \operatorname{tr} H}.}$$

(c) If $\operatorname{tr} H = 0$, this formula immediately gives $\det U(a) = 1$.

(d) If $\det U(a) = 1$ for every a near zero, differentiate at zero: $0 = F'(0) = i \operatorname{tr} H$. Therefore $\boxed{\operatorname{tr} H = 0}$ under the corrected hypothesis. For a single nonzero a_0, however,

$$e^{ia_0 \operatorname{tr} H} = 1 \quad \Longleftrightarrow \quad \operatorname{tr} H = \frac{2\pi k}{a_0}, \qquad k \in \mathbb{Z},$$

which need not vanish. At $a_0 = 0$ there is no restriction at all.

Here is a nonzero-parameter counterexample to the unqualified part (b), as well as to (d). Take

$$P = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}, \qquad P^2 = P, \qquad H = 2\pi P, \qquad a_0 = 1.$$

This H is not Hermitian and has trace 2π. Nevertheless the idempotence allows its exponential to be summed exactly:

$$e^{iH} = I + \sum_{k=1}^{\infty} \frac{(2\pi i)^k}{k!} P = I + (e^{2\pi i} - 1)P = I.$$

Thus $U(1)$ is unitary with determinant one, confirming why information about the whole parameter family is required for both converses.

6.5.16 An $n \times n$ matrix A has n eigenvalues A_i. If $\mathsf{B} = e^{\mathsf{A}}$, show that B has every eigenvector of A as an eigenvector, with the corresponding eigenvalues B_i given by $B_i = \exp(A_i)$.

Solution.

Let $Av = \alpha v$ with $v \neq 0$. Starting from $A^0 v = v$, induction gives

$$A^{k+1}v = A(A^k v) = A(\alpha^k v) = \alpha^k Av = \alpha^{k+1}v,$$

so $A^k v = \alpha^k v$ for every nonnegative integer k. Apply the convergent exponential series:

$$e^A v = \sum_{k=0}^{\infty} \frac{A^k v}{k!} = \sum_{k=0}^{\infty} \frac{\alpha^k}{k!} v = \boxed{e^{\alpha} v}.$$

Thus every eigenvector of A remains an eigenvector of $B = e^A$, with its eigenvalue exponentiated. In the question's notation, $\boxed{B_i = e^{A_i}}$. The proof works for each eigenvector even if A is defective and lacks a complete eigenbasis.

"The same eigenvectors" should mean this forward implication; e^A can have additional eigendirections when different eigenvalues of A have the same exponential. For example,

$$A = \operatorname{diag}(0, 2\pi i), \qquad e^A = I_2.$$

The vector $(1, 1)^{\mathsf{T}}$ is an eigenvector of e^A but not of A. This also explains why exponentiating eigenvalues need not preserve the multiplicities of distinct eigenspaces.

6.5.17 A matrix P is a projection operator satisfying the condition

$$\mathsf{P}^2 = \mathsf{P}.$$

Show that the corresponding eigenvalues $(\rho^2)_\lambda$ and ρ_λ satisfy the relation

$$(\rho^2)_\lambda = (\rho_\lambda)^2 = \rho_\lambda.$$

This means that the eigenvalues of P belong to $\{0, 1\}$.

Solution.

Let $Pv = \rho_\lambda v$ with $v \neq 0$. Applying P once more,

$$P^2 v = P(\rho_\lambda v) = \rho_\lambda Pv = (\rho_\lambda)^2 v.$$

Thus the eigenvalue of P^2 on this same vector is $(\rho^2)_\lambda = (\rho_\lambda)^2$. But $P^2 = P$, so it is also ρ_λ:

$$(\rho_\lambda)^2 = \rho_\lambda, \qquad \rho_\lambda(\rho_\lambda - 1) = 0, \qquad \boxed{\rho_\lambda \in \{0, 1\}.}$$

The interpretation is direct. If v lies in the range of P, say $v = Pw$, then $Pv = P^2 w = Pw = v$. If it lies in the kernel, $Pv = 0$. Every vector decomposes as $w = Pw + (I - P)w$, with the first part in the range and the second in the kernel because $P(I - P) = 0$. The eigenvalues belong to $\{0, 1\}$; both need not occur. For example, $P = 0$ has only eigenvalue 0, and $P = I$ has only eigenvalue 1.

6.5.18 In the matrix eigenvector-eigenvalue equation

$$\mathsf{A}|x_i\rangle = \lambda_i|x_i\rangle,$$

A is an $n \times n$ Hermitian matrix. For simplicity assume that its n real eigenvalues are distinct, λ_1 being the largest. If $|x\rangle$ is an approximation to $|x_1\rangle$,

$$|x\rangle = |x_1\rangle + \sum_{i=2}^{n} \delta_i |x_i\rangle,$$

show that

$$\frac{\langle x|\mathsf{A}|x\rangle}{\langle x|x\rangle} \leq \lambda_1$$

and that the error in λ_1 is of the order $|\delta_i|^2$. Take $|\delta_i| \ll 1$.

Hint. The n vectors $|x_i\rangle$ form a complete orthogonal set spanning the n-dimensional (complex) space.

Solution.

First allow the mutually orthogonal eigenvectors to have arbitrary nonzero norms $s_i = \langle x_i|x_i\rangle > 0$. Inserting the given expansion and using $\langle x_i|x_j\rangle = 0$ for $i \neq j$ gives

$$\langle x|x\rangle = s_1 + \sum_{i=2}^{n} |\delta_i|^2 s_i,$$

$$\langle x|A|x\rangle = \lambda_1 s_1 + \sum_{i=2}^{n} \lambda_i |\delta_i|^2 s_i.$$

There are no linear terms in δ_i because every such term contains an inner product of two distinct eigenvectors. Subtract the Rayleigh quotient $R(x)$ from λ_1, combining the terms over a common denominator:

$$\lambda_1 - R(x) = \frac{\lambda_1\big(s_1 + \sum_{i=2}^{n} |\delta_i|^2 s_i\big) - \big(\lambda_1 s_1 + \sum_{i=2}^{n} \lambda_i |\delta_i|^2 s_i\big)}{s_1 + \sum_{i=2}^{n} |\delta_i|^2 s_i}$$

$$= \boxed{\frac{\sum_{i=2}^{n} (\lambda_1 - \lambda_i)|\delta_i|^2 s_i}{s_1 + \sum_{i=2}^{n} |\delta_i|^2 s_i}} \geq 0.$$

Every gap $\lambda_1 - \lambda_i$ is positive and the denominator is positive. Thus $R(x) \leq \lambda_1$, with equality exactly when every $\delta_i = 0$ under the distinct-eigenvalue assumption.

For an orthonormal choice, all $s_i = 1$. Put $S = \sum_{i=2}^{n} |\delta_i|^2$ and $N = \sum_{i=2}^{n} (\lambda_1 - \lambda_i)|\delta_i|^2$. The exact deficit is $N/(1+S)$. Since S and N are both of second order,

$$\lambda_1 - R(x) = N(1 - S + O(S^2)) = \sum_{i=2}^{n} (\lambda_1 - \lambda_i)|\delta_i|^2 + O(\|\delta\|^4).$$

This proves the claimed quadratic error and explicitly identifies its leading coefficient. The preceding weighted formula proves the same order for any fixed choice of eigenvector norms.

6.5.19 Two equal masses are connected to each other and to walls by springs as shown in Fig. 6.5. The masses are constrained to stay on a horizontal line.

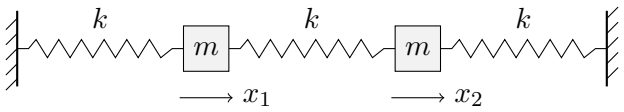


Fig. 6.5 (schematic): equal masses and three equal spring constants.

(a) Set up the Newtonian acceleration equation for each mass.

(b) Solve the secular equation for the eigenvalues.

(c) Determine the eigenvectors and thus the normal modes of motion.

Solution.

The source figure has three springs with the same constant k and two masses of the same mass m. Take $m, k > 0$ and measure both displacements to the right from equilibrium.

(a) The left outer spring exerts $-kx_1$ on mass 1. The middle spring's extension is $x_2 - x_1$, so its forces are $+k(x_2 - x_1)$ on mass 1 and $-k(x_2 - x_1)$ on mass 2. The right outer spring exerts $-kx_2$ on mass 2. Newton's equations are therefore

$$\boxed{m\ddot{x}_1 = -2kx_1 + kx_2, \qquad m\ddot{x}_2 = kx_1 - 2kx_2.}$$

In matrix notation,

$$m\ddot{\boldsymbol{x}} + k\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}\boldsymbol{x} = 0.$$

(b) Seek a normal mode $\boldsymbol{x}(t) = \boldsymbol{X}e^{i\omega t}$ and use $\ddot{\boldsymbol{x}} = -\omega^2\boldsymbol{x}$. Then

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}\boldsymbol{X} = \lambda\boldsymbol{X}, \qquad \lambda = \frac{m\omega^2}{k}.$$

The secular equation is

$$0 = \begin{vmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{vmatrix} = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3).$$

Thus

$$\boxed{\omega_+ = \sqrt{k/m}, \qquad \omega_- = \sqrt{3k/m}.}$$

Part (b) asks for eigenvectors in the source; the secular equation determines the eigenvalues. The eigenvectors are then found in part (c), as follows.

(c) For $\lambda = 1$, the first row is $X_1 - X_2 = 0$; choose the unit vector $u_+ = (1,1)^{\mathsf{T}}/\sqrt{2}$. The masses move in phase with equal displacement, so the middle spring has no changing extension. For $\lambda = 3$, the first row is $-X_1 - X_2 = 0$; choose $u_- = (1,-1)^{\mathsf{T}}/\sqrt{2}$. The masses move with equal amplitudes and opposite signs, stretching and compressing the middle spring. These vectors have norm one and inner product zero.

The corresponding normal coordinates are

$$Q_+ = \frac{x_1 + x_2}{\sqrt{2}}, \qquad Q_- = \frac{x_1 - x_2}{\sqrt{2}}.$$

Adding and subtracting the Newton equations gives

$$m\ddot{Q}_+ + kQ_+ = 0, \qquad m\ddot{Q}_- + 3kQ_- = 0.$$

Therefore the general real motion can be written

$$\boxed{\boldsymbol{x}(t) = C_+ u_+ \cos(\omega_+ t + \delta_+) + C_- u_- \cos(\omega_- t + \delta_-),}$$

where the amplitudes and phases are fixed by the initial positions and velocities.

6.5.20 Given a normal matrix A with eigenvalues λ_j, show that $\mathsf{A}^\dagger$ has eigenvalues λ_j^*, its real part $(\mathsf{A}+\mathsf{A}^\dagger)/2$ has eigenvalues $\Re(\lambda_j)$, and its imaginary part $(\mathsf{A}-\mathsf{A}^\dagger)/2i$ has eigenvalues $\Im(\lambda_j)$.

Solution.

First show that a right eigenvector of a normal matrix is also an eigenvector of its adjoint with the conjugate eigenvalue. If $Ax_j = \lambda_j x_j$, put $B = A - \lambda_j I$. It is normal because

$$BB^\dagger - B^\dagger B = AA^\dagger - A^\dagger A = 0.$$

Using $Bx_j = 0$, compute

$$\|B^\dagger x_j\|^2 = x_j^\dagger B B^\dagger x_j = x_j^\dagger B^\dagger B x_j = \|Bx_j\|^2 = 0.$$

A vector with zero norm is zero, so $B^\dagger x_j = 0$, or $\boxed{A^\dagger x_j = \lambda_j^* x_j}$. Now add and subtract the two eigenvalue equations:

$$\frac{A + A^\dagger}{2} x_j = \frac{\lambda_j + \lambda_j^*}{2} x_j = \boxed{\Re(\lambda_j) x_j},$$
$$\frac{A - A^\dagger}{2i} x_j = \frac{\lambda_j - \lambda_j^*}{2i} x_j = \boxed{\Im(\lambda_j) x_j}.$$

Thus the adjoint and both Hermitian parts share the original orthonormal eigenbasis; the indicated eigenvalues follow on each basis vector, including degenerate eigenspaces.

6.5.21 Consider a rotation given by Euler angles $\alpha = \pi/4$, $\beta = \pi/2$, $\gamma = 5\pi/4$.

(a) Using the formula of Eq. (3.37), construct the matrix U representing this rotation.

(b) Find the eigenvalues and eigenvectors of U, and from them describe this rotation by specifying a single rotation axis and an angle of rotation about that axis.

Note. This technique provides a representation of rotations alternative to the Euler angles.

Solution.

(a) Equation (3.37), verified against textbook p. 141, uses $U = S_3(\gamma) S_2(\beta) S_1(\alpha)$, where the rotations act on coordinates. Set $c = 1/\sqrt{2}$. The specified angles give

$$\cos\alpha = \sin\alpha = c, \quad \cos\beta = 0, \quad \sin\beta = 1, \quad \cos\gamma = \sin\gamma = -c.$$

The three matrices and their first product are

$$S_1 = \begin{pmatrix} c & c & 0 \\ -c & c & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad S_2 = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad S_3 = \begin{pmatrix} -c & -c & 0 \\ c & -c & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

$$S_2 S_1 = \begin{pmatrix} 0 & 0 & -1 \\ -c & c & 0 \\ c & c & 0 \end{pmatrix}.$$

Multiplying by S_3 on the left gives

$$\boxed{U = \begin{pmatrix} 1/2 & -1/2 & 1/\sqrt{2} \\ 1/2 & -1/2 & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{pmatrix}.}$$

Its columns have norm one and pairwise scalar products zero; its determinant is 1, as also follows from the three proper-rotation factors.

(b) To display the characteristic calculation without fractions, expand the determinant of $2(U - \lambda I)$:

$$\begin{aligned} 8\det(U - \lambda I) &= \begin{vmatrix} 1-2\lambda & -1 & \sqrt{2} \\ 1 & -1-2\lambda & -\sqrt{2} \\ \sqrt{2} & \sqrt{2} & -2\lambda \end{vmatrix} \\ &= (1-2\lambda)(4\lambda^2 + 2\lambda + 2) + (2 - 2\lambda) + 4(1+\lambda) \\ &= 8(1 - \lambda^3). \end{aligned}$$

Thus

$$\lambda^3 = 1, \qquad \boxed{\lambda_0 = 1, \quad \lambda_\pm = \frac{-1 \pm i\sqrt{3}}{2} = e^{\pm 2\pi i/3}.}$$

For an eigenvector $(x, y, z)^{\mathsf{T}}$, adding the first two rows of the eigenvalue equation and retaining the third gives

$$(1-\lambda)x - (1+\lambda)y = 0, \qquad x + y = \sqrt{2}\lambda z.$$

At $\lambda = 1$ the first equation gives $y = 0$ and the second gives $x = \sqrt{2}z$. The remaining row also holds. Normalizing yields the fixed-axis vector

$$\boxed{n = \frac{(\sqrt{2}, 0, 1)^{\mathsf{T}}}{\sqrt{3}}.}$$

For $\lambda = \lambda_\pm$, substitute $y = \sqrt{2}\lambda z - x$ into the first equation:

$$2x - \sqrt{2}\lambda(1+\lambda)z = 0.$$

Since $\lambda^2 + \lambda + 1 = 0$, this gives $x = -z/\sqrt{2}$ and then

$$y = \frac{1 + 2\lambda}{\sqrt{2}} z = \pm i \sqrt{\frac{3}{2}}\, z.$$

For $z = 1$ the squared Hermitian norm is $1/2 + 3/2 + 1 = 3$, so

$$\boxed{u_\pm = \begin{pmatrix} -1/\sqrt{6} \\ \pm i/\sqrt{2} \\ 1/\sqrt{3} \end{pmatrix}.}$$

Substitution into the unused row gives $Uu_\pm = \lambda_\pm u_\pm$. Also $n^\dagger u_\pm = -1/3 + 1/3 = 0$ and $u_+^\dagger u_- = 1/6 - 1/2 + 1/3 = 0$, so these three vectors are orthonormal.

The rotation axis is the line through the origin parallel to $(\sqrt{2}, 0, 1)$, and its unsigned angle is $2\pi/3 = 120°$. The sign can be fixed, rather than inferred only from a conjugate pair of eigenvalues. Define $[n]_\times v = n \times v$. Direct subtraction gives

$$U - U^\mathsf{T} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -\sqrt{2} \\ 0 & \sqrt{2} & 0 \end{pmatrix} = \sqrt{3}\,[n]_\times .$$

For an active right-handed rotation, $R - R^\mathsf{T} = 2\sin\varphi[n]_\times$, while $\operatorname{tr} U = 0 = 1 + 2\cos\varphi$. Thus active application of U to a column vector is a rotation by $+120°$ about n. Equation (3.37) is a passive coordinate transformation. The corresponding rotation of the coordinate axes is therefore $-120°$ about n, equivalently $+120°$ about $-n$. The directed angle must be paired with the chosen direction of the axis.

7 Chapter 7. Ordinary Differential Equations

Problems

7.2 First-Order Equations

7.2.1 From Kirchhoff's law the current I in an RC (resistance-capacitance) circuit (Fig. 7.1) obeys the equation

$$R\frac{dI}{dt} + \frac{1}{C}I = 0.$$

(a) Find $I(t)$.

(b) For a capacitance of 10,000 μF charged to 100 V and discharging through a resistance of 1 MΩ, find the current I for $t = 0$ and for $t = 100$ seconds.

Note. The initial voltage is I_0R or Q/C, where $Q = \int_0^\infty I(t)\,dt$.

Solution.

(a) For $R, C > 0$, separation gives

$$\frac{dI}{I} = -\frac{dt}{RC}, \qquad \ln\left|\frac{I(t)}{I_0}\right| = -\frac{t}{RC}, \qquad \boxed{I(t) = I_0 e^{-t/(RC)}}.$$

The identically zero solution is included by $I_0 = 0$. Differentiation checks $RI' + I/C = -I/C + I/C = 0$.

(b) Convert the given units before finding the time constant:

$$C = 10^4 \times 10^{-6}\ \text{F} = 10^{-2}\ \text{F}, \quad R = 10^6\ \Omega, \quad RC = 10^4\ \text{s}.$$

Thus

$$\boxed{I(0) = \frac{100}{10^6}\ \text{A} = 100\ \mu\text{A}}, \qquad \boxed{I(100\ \text{s}) = 100e^{-0.01}\ \mu\text{A} = 99.00498337\ldots\ \mu\text{A}}.$$

As a charge check, $\int_0^\infty I(t)dt = I_0RC = CV_0 = 1$ C.

7.2.2 The Laplace transform of Bessel's equation ($n = 0$) leads to

$$(s^2 + 1)f'(s) + sf(s) = 0.$$

Solve for $f(s)$.

Solution.

For real s, $1 + s^2$ is positive. Separate and integrate:

$$\frac{f'}{f} = -\frac{s}{1+s^2}, \qquad \ln|f| = -\frac{1}{2}\ln(1+s^2) + K.$$

Absorbing the sign into an arbitrary real constant gives

$$\boxed{f(s) = \frac{A}{\sqrt{1+s^2}}}.$$

This includes $f = 0$. Indeed, $f' = -As(1+s^2)^{-3/2}$, so $(1+s^2)f' + sf = 0$. The ODE alone does not determine A. If the intended transform is specifically that of J_0, its normalization $J_0(0) = 1$ gives $sf(s) \to 1$ as $s \to +\infty$, hence $A = 1$. For complex s, the same formula holds with a complex constant on a simply connected domain avoiding $s = \pm i$, after choosing a square-root branch. On the Laplace half-plane $\operatorname{Re} s > 0$, choose the analytic branch positive for positive real s; it has $\sqrt{1+s^2} \sim s$ along the positive real axis.

7.2.3 The decay of a population by catastrophic two-body collisions is described by

$$\frac{dN}{dt} = -kN^2.$$

This is a first-order, nonlinear differential equation. Derive the solution

$$N(t) = N_0 \left(1 + \frac{t}{\tau_0}\right)^{-1},$$

where $\tau_0 = (kN_0)^{-1}$, with $k, N_0 > 0$. The backward extrapolation has a pole as $t \downarrow -\tau_0$; $N(-\tau_0)$ itself is undefined.

Solution.

For $N_0 > 0$ and $k > 0$, integrate from the initial state:

$$\int_{N_0}^{N(t)} \frac{dN}{N^2} = -k \int_0^t ds, \qquad -\frac{1}{N(t)} + \frac{1}{N_0} = -kt.$$

Solving for N gives

$$\boxed{N(t) = \frac{N_0}{1 + kN_0 t} = N_0 \left(1 + \frac{t}{\tau_0}\right)^{-1}}, \qquad \tau_0 = \frac{1}{kN_0}.$$

Direct differentiation yields $N' = -kN_0^2/(1+kN_0t)^2 = -kN^2$. The initial-value solution with positive population has mathematical interval $t > -\tau_0$ and a pole at its left endpoint; physical decay uses $t \geq 0$. The pole is a backward extrapolation of the model, not an attained infinite population at a finite time within its solution interval. If $N_0 = 0$, the separate solution is $N \equiv 0$; if $k = 0$, $N \equiv N_0$.

7.2.4 The rate of a particular chemical reaction $\mathrm{A} + \mathrm{B} \to \mathrm{C}$ is proportional to the concentrations of the reactants A and B:

$$\frac{dC(t)}{dt} = \alpha[A(0) - C(t)][B(0) - C(t)].$$

(a) Find $C(t)$ for $A(0) \neq B(0)$.

(b) Find $C(t)$ for $A(0) = B(0)$.

The initial condition is that $C(0) = 0$.

Solution.

Let $A_0 = A(0)$ and $B_0 = B(0)$, and assume nonnegative initial concentrations and $\alpha \geq 0$.
(a) If $A_0 \neq B_0$ and both are positive, partial fractions give

$$\frac{1}{(A_0 - C)(B_0 - C)} = \frac{1}{B_0 - A_0}\left(\frac{1}{A_0 - C} - \frac{1}{B_0 - C}\right).$$

Integrating with $C(0) = 0$ therefore yields

$$\frac{1}{B_0 - A_0} \ln\left[\frac{B_0 - C}{A_0 - C}\frac{A_0}{B_0}\right] = \alpha t.$$

Set $E = e^{\alpha(B_0 - A_0)t}$. Exponentiation and collection of the terms in C give

$$A_0(B_0 - C) = B_0 E(A_0 - C), \qquad C(B_0 E - A_0) = A_0 B_0(E - 1).$$

Thus

$$\boxed{C(t) = \frac{A_0 B_0(1 - e^{-\alpha(B_0 - A_0)t})}{B_0 - A_0 e^{-\alpha(B_0 - A_0)t}}}.$$

It starts at zero and approaches $\min(A_0, B_0)$ for $\alpha > 0$. If either concentration is zero, $C(t) = 0$ directly; the formula also reduces to that value when $A_0 \neq B_0$.

(b) With $A_0 = B_0 = a > 0$, separate instead as

$$\int_0^{C(t)} \frac{du}{(a - u)^2} = \alpha t, \qquad \frac{1}{a - C} - \frac{1}{a} = \alpha t.$$

Hence

$$\boxed{C(t) = \frac{\alpha a^2 t}{1 + \alpha a t}}.$$

The formula includes $a = 0$ and $\alpha = 0$. In both cases substitution of the displayed expressions verifies $C' = \alpha(A_0 - C)(B_0 - C)$.

7.2.5 A boat, coasting through the water, experiences a resisting force proportional to v^n, v being the boat's instantaneous velocity. Newton's second law leads to

$$m\frac{dv}{dt} = -kv^n.$$

With $v(t = 0) = v_0$, $x(t = 0) = 0$, integrate to find v as a function of time and v as a function of distance. Take $m, k, v_0 > 0$, $n \in \mathbb{R}$, and use the real positive-speed branch until it reaches zero, if it does.

Solution.

Take $m, k, v_0 > 0$ and interpret v as speed while the boat moves forward. Put $\gamma = k/m$. For $n \neq 1$,

$$v^{-n}dv = -\gamma dt, \qquad \frac{v^{1-n} - v_0^{1-n}}{1 - n} = -\gamma t,$$

so

$$\boxed{v(t) = \left[v_0^{1-n} + (n - 1)\gamma t\right]^{1/(1-n)}} \quad (n \neq 1).$$

The real positive branch is used while the bracket is positive. For $n = 1$, the integration is logarithmic:

$$\ln(v/v_0) = -\gamma t, \qquad \boxed{v(t) = v_0 e^{-\gamma t}}.$$

To eliminate time use $dv/dt = (dv/dx)(dx/dt) = v\,dv/dx$ while $v > 0$. Then $v^{1-n}dv = -\gamma dx$, and for $n \neq 2$,

$$\frac{v^{2-n} - v_0^{2-n}}{2 - n} = -\gamma x, \qquad \boxed{v(x) = \left[v_0^{2-n} + (n - 2)\gamma x\right]^{1/(2-n)}}.$$

For the omitted case $n = 2$, $dv/v = -\gamma dx$, giving

$$\boxed{v(x) = v_0 e^{-\gamma x}}.$$

These special cases are essential: $n = 1$ gives $v(x) = v_0 - \gamma x$, and $n = 2$ gives $v(t) = v_0/(1 + \gamma v_0 t)$. Differentiating the two integrated identities checks both equations without choosing special n.

For $n < 1$ the positive-speed branch reaches zero at $t_* = v_0^{1-n}/[(1-n)\gamma]$; for $n < 2$ its limiting travel distance is $x_* = v_0^{2-n}/[(2-n)\gamma]$. For $n \geq 1$ stopping takes infinite time, and for $n \geq 2$ the traveled distance is unbounded. The positive-speed drag model does not authorize continuing a fractional or negative power through $v = 0$. At or after stopping an appropriate force law must be specified; the formulas above state their actual domain.

7.2.6 In the first-order differential equation $dy/dx = f(x, y)$, the function $f(x, y)$ is a function of the ratio y/x:

$$\frac{dy}{dx} = g(y/x).$$

Show that the substitution of $u = y/x$ leads to a separable equation in u and x.

Solution.

Work on an interval with $x \neq 0$. Since $y = xu$,

$$\frac{dy}{dx} = u + x\frac{du}{dx} = g(u), \qquad x\frac{du}{dx} = g(u) - u.$$

Thus, whenever $g(u) - u \neq 0$,

$$\boxed{\frac{du}{g(u) - u} = \frac{dx}{x}}, \qquad \int^u \frac{ds}{g(s) - s} = \ln|x| + C.$$

Division by $g(u) - u$ omits its zeros. Each constant u_* satisfying $g(u_*) = u_*$ supplies the additional straight-line solution $y = u_* x$, which must be checked before declaring the solution set complete.

7.2.7 The differential equation

$$P(x, y)\, dx + Q(x, y)\, dy = 0$$

is exact. Assume P, Q have continuous first partial derivatives and that the rectangular integration path lies inside the domain. Show that its solution is of the form

$$\varphi(x, y) = \int_{x_0}^{x} P(\xi, y)\, d\xi + \int_{y_0}^{y} Q(x_0, \eta)\, d\eta = \text{constant}.$$

Solution.

Use distinct integration variables, so the proposed potential is

$$\varphi(x, y) = \int_{x_0}^{x} P(\xi, y) d\xi + \int_{y_0}^{y} Q(x_0, \eta) d\eta.$$

This is the line integral along the path first from (x_0, y_0) to (x_0, y), then to (x, y). Exactness means $P\,dx + Q\,dy = d\Phi$ for a potential Φ, so the integral equals $\Phi(x, y) - \Phi(x_0, y_0)$. It follows immediately that $d\varphi = P\,dx + Q\,dy$ and the solution curves satisfy

$$\boxed{\varphi(x, y) = C}.$$

More explicitly, if $P_y = Q_x$, differentiation gives

$$\varphi_x = P(x,y), \quad \varphi_y = \int_{x_0}^{x} P_y(\xi,y)d\xi + Q(x_0,y) = Q(x,y) - Q(x_0,y) + Q(x_0,y) = Q(x,y).$$

Along any solution curve $(x(t), y(t))$, the chain rule gives $d\varphi/dt = P\dot{x} + Q\dot{y} = 0$. Conversely this same calculation shows that a differentiable level curve solves the differential form. The displayed rectangular path is understood to lie inside the domain; otherwise one uses an admissible path for the same exact differential.

7.2.8 The differential equation

$$P(x,y)\,dx + Q(x,y)\,dy = 0$$

is exact. Assume P, Q have continuous first partial derivatives and that the rectangular integration path lies inside the domain. If

$$\varphi(x,y) = \int_{x_0}^{x} P(\xi,y)\,d\xi + \int_{y_0}^{y} Q(x_0,\eta)\,d\eta,$$

show that

$$\frac{\partial\varphi}{\partial x} = P(x,y), \quad \frac{\partial\varphi}{\partial y} = Q(x,y).$$

Hence, $\varphi(x,y) =$ constant is a solution of the original differential equation.

Solution.

Write the formula with dummy variables:

$$\varphi(x,y) = \int_{x_0}^{x} P(\xi,y)d\xi + \int_{y_0}^{y} Q(x_0,\eta)d\eta.$$

The fundamental theorem of calculus, with y held fixed, gives $\boxed{\varphi_x = P(x,y)}$. Differentiating with respect to y requires both terms:

$$\begin{aligned}\varphi_y &= \int_{x_0}^{x} P_y(\xi,y)d\xi + Q(x_0,y)\\ &= \int_{x_0}^{x} Q_x(\xi,y)d\xi + Q(x_0,y)\\ &= Q(x,y) - Q(x_0,y) + Q(x_0,y) = \boxed{Q(x,y)}.\end{aligned}$$

The second line uses exactness and continuous first partial derivatives. Therefore $d\varphi = \varphi_x dx + \varphi_y dy = Pdx + Qdy$. A level curve $\varphi = C$ has $d\varphi = 0$ and satisfies the original equation. As in 7.2.7, the chosen paths must lie within the domain.

7.2.9 Prove that Eq. (7.12) is exact in the sense of Eq. (7.9), provided that $\alpha(x)$ satisfies Eq. (7.14).

Solution.

The referenced equations are $\alpha y' + \alpha p y = \alpha q$ and $\alpha' = \alpha p$. In differential form the first is

$$M(x,y)dx + N(x,y)dy = 0, \qquad M = \alpha(x)[p(x)y - q(x)], \quad N = \alpha(x).$$

The exactness test is

$$M_y = \alpha p, \qquad N_x = \alpha',$$

so $M_y = N_x$ follows precisely from Eq. (7.14). A direct potential is

$$\boxed{\Phi(x,y) = \alpha(x)y - \int^x \alpha(s)q(s)ds},$$

because $\Phi_y = \alpha = N$ and $\Phi_x = \alpha' y - \alpha q = \alpha(py - q) = M$. For equivalence with the original ODE choose a nonzero integrating factor, $\alpha = e^{\int p\,dx}$; the identically zero factor would give only $0 = 0$ and discard the equation.

7.2.10 A certain differential equation has the form

$$f(x)\,dx + g(x)h(y)\,dy = 0,$$

with none of the functions $f(x)$, $g(x)$, $h(y)$ identically zero. Show that a necessary and sufficient condition for this equation to be exact is that $g(x) =$ constant. Work on a product of connected intervals, with C^1 coefficients.

Solution.

On a product of connected intervals, with the stated functions continuously differentiable where needed, exactness requires

$$\frac{\partial f(x)}{\partial y} = 0 = \frac{\partial [g(x)h(y)]}{\partial x} = g'(x)h(y).$$

Since h is not identically zero, choose one y_* with $h(y_*) \neq 0$. The equality at (x, y_*) forces $g'(x) = 0$ for every x, so g is a constant. Conversely, if $g = g_0$, then

$$f(x)dx + g_0 h(y)dy = d\left[\int^x f(s)ds + g_0 \int^y h(t)dt\right]$$

is exact. Hence $\boxed{g(x) = g_0}$ is necessary and sufficient. The connected product-domain assumption specifies the usual meaning of this exercise. Without it, h could vanish throughout an entire component, imposing no condition on g there, or different x-components could carry different constants.

7.2.11 Show that

$$y(x) = \exp\left[-\int^x p(t)\,dt\right]\left\{\int^x \left[\exp\left(\int^s p(t)\,dt\right)\right] q(s)\,ds + C\right\}$$

is a solution of

$$\frac{dy}{dx} + p(x)y(x) = q(x)$$

by differentiating the expression for $y(x)$ and substituting into the differential equation.

Solution.

Choose a fixed base point and abbreviate

$$P_0(x) = \int_{x_0}^x p(t)dt, \qquad J(x) = \int_{x_0}^x e^{P_0(s)} q(s)ds + C.$$

Then the proposed solution is $y = e^{-P_0}J$, with $P_0' = p$ and $J' = e^{P_0}q$. The product and chain rules give

$$y' = -pe^{-P_0}J + e^{-P_0}e^{P_0}q = -py + q.$$

Therefore $\boxed{y' + py = q}$. Changing the fixed lower limits merely changes the arbitrary constant or rescales the integrating factor and does not change the resulting family of solutions.

7.2.12 The motion of a body falling in a resisting medium may be described by

$$m\frac{dv}{dt} = mg - bv$$

when the retarding force is proportional to the velocity, v. Find the velocity. Evaluate the constant of integration by demanding that $v(0) = 0$.

Solution.

For $b > 0$, divide by m and multiply by $e^{bt/m}$:

$$v' + \frac{b}{m}v = g, \qquad \frac{d}{dt}(e^{bt/m}v) = ge^{bt/m}.$$

Integrating from 0 to t, using $v(0) = 0$, gives

$$e^{bt/m}v(t) = \frac{mg}{b}(e^{bt/m} - 1), \qquad \boxed{v(t) = \frac{mg}{b}(1 - e^{-bt/m})}.$$

The check is $mv' = mge^{-bt/m} = mg - bv$. The terminal velocity is mg/b; as $b \to 0$, the expansion $1 - e^{-bt/m} = bt/m + O(b^2)$ gives the drag-free solution $v = gt$.

7.2.13 Radioactive nuclei decay according to the law

$$\frac{dN}{dt} = -\lambda N,$$

N being the concentration of a given nuclide and λ, the particular decay constant. In a radioactive series of two different nuclides, with concentrations $N_1(t)$ and $N_2(t)$, we have

$$\frac{dN_1}{dt} = -\lambda_1 N_1, \quad \frac{dN_2}{dt} = \lambda_1 N_1 - \lambda_2 N_2.$$

Find $N_2(t)$ for the conditions $N_1(0) = N_0$ and $N_2(0) = 0$.

Solution.

The first equation integrates to $N_1 = N_0 e^{-\lambda_1 t}$. The second becomes

$$N_2' + \lambda_2 N_2 = \lambda_1 N_0 e^{-\lambda_1 t}, \qquad \frac{d}{dt}(e^{\lambda_2 t}N_2) = \lambda_1 N_0 e^{(\lambda_2 - \lambda_1)t}.$$

Integrate from 0 to t, with $N_2(0) = 0$. If $\lambda_1 \neq \lambda_2$,

$$\boxed{N_2(t) = \frac{\lambda_1 N_0}{\lambda_2 - \lambda_1}(e^{-\lambda_1 t} - e^{-\lambda_2 t})}.$$

The coincident-rate case must be handled separately rather than dividing by $\lambda_2 - \lambda_1$. If both rates equal λ,

$$e^{\lambda t}N_2 = \lambda N_0 \int_0^t ds = \lambda N_0 t, \qquad \boxed{N_2 = \lambda N_0 t e^{-\lambda t}}.$$

This is also the limit of the preceding expression. Both forms satisfy the initial condition and the inhomogeneous equation by differentiation.

7.2.14 The rate of evaporation from a particular spherical drop of liquid (constant density) is proportional to its surface area. Assuming this to be the sole mechanism of mass loss, find the radius of the drop as a function of time.

Solution.

Let ρ be the fixed mass density and let $j > 0$ be the mass lost per unit area per unit time. Then

$$M = \frac{4\pi}{3}\rho r^3, \qquad \frac{dM}{dt} = 4\pi\rho r^2 r' = -j(4\pi r^2).$$

While $r > 0$, cancellation gives $r' = -j/\rho$. With $r(0) = r_0$,

$$\boxed{r(t) = r_0 - \frac{j}{\rho}t, \qquad 0 \le t \le t_* := \frac{\rho r_0}{j}.}$$

At t_* the drop is exhausted. The model's physical continuation has zero radius and mass, not a negative radius. If the proportionality constant is defined for volume loss instead of mass loss, put $\kappa = j/\rho$ and write $r = r_0 - \kappa t$.

7.2.15 In the linear homogeneous differential equation

$$\frac{dv}{dt} = -av$$

the variables are separable. When the variables are separated, the equation is exact. Solve this differential equation subject to $v(0) = v_0$ by the following three methods:

(a) Separating variables and integrating.

(b) Treating the separated variable equation as exact.

(c) Using the result for a linear homogeneous differential equation.

ANS. $v(t) = v_0 e^{-at}$.

Solution.

(a) For $v_0 \neq 0$, separation gives

$$\int_{v_0}^{v(t)} \frac{du}{u} = -a\int_0^t ds, \quad \ln|v(t)/v_0| = -at, \quad v(t) = v_0 e^{-at}.$$

The sign stays that of v_0; $v \equiv 0$ is separately included when $v_0 = 0$.

(b) The separated form $a\,dt + dv/v = 0$ is exact on either half-plane $v > 0$ or $v < 0$, since both cross partial derivatives vanish. Its potential is $\Phi(t, v) = at + \ln|v|$. The level condition $\Phi(t, v) = \Phi(0, v_0) = \ln|v_0|$ reproduces the same exponential. The zero solution was excluded only by division by v and must be restored.

(c) In the linear first-order formula take $p = a$, $q = 0$:

$$v = e^{-\int a\,dt}(C + 0) = Ce^{-at}.$$

The initial condition gives $C = v_0$. Thus all three methods give $\boxed{v = v_0 e^{-at}}$, and $v' = -av$ checks it directly.

7.2.16

(a) Solve Example 7.2.1, assuming that the parachute opens when the parachutist's velocity has reached $v_i = 60$ mi/h (regard this time as $t = 0$). Find $v(t)$.

(b) For a skydiver in free fall use the friction coefficient $b = 0.25$ kg/m and mass $m = 70$ kg. What is the limiting velocity in this case?

Solution.

In Example 7.2.1 the downward speed obeys $mv' = mg - bv^2$. Put $v_T = \sqrt{mg/b}$ and $T = \sqrt{m/(bg)}$, so $v_T/T = g$.

(a) Partial-fraction integration with a nonzero initial speed gives

$$\int_{v_i}^{v} \frac{du}{v_T^2 - u^2} = \frac{b}{m}t, \qquad \frac{1}{2v_T}\ln\left|\frac{v_T + v}{v_T - v}\frac{v_T - v_i}{v_T + v_i}\right| = \frac{b}{m}t.$$

For $v_i \neq v_T$, put $q = (v_i - v_T)/(v_i + v_T)$. Continuity preserves the sign of $v - v_T$ on this solution, so exponentiating the logarithm gives

$$\frac{v - v_T}{v + v_T} = qe^{-2t/T}, \qquad v(1 - qe^{-2t/T}) = v_T(1 + qe^{-2t/T}).$$

Divide by $1 - qe^{-2t/T}$ and multiply numerator and denominator by $v_i + v_T$. This gives

$$\boxed{v(t) = v_T\frac{(v_i + v_T) + (v_i - v_T)e^{-2t/T}}{(v_i + v_T) - (v_i - v_T)e^{-2t/T}}} = v_T\frac{v_i + v_T\tanh(t/T)}{v_T + v_i\tanh(t/T)}.$$

For $v_i = v_T$, separation divided by zero; direct substitution instead gives the equilibrium $v(t) = v_T$, which the boxed formula also includes. For every $v_i \geq 0$ and $t \geq 0$, $-1 \leq q < 1$, so its denominator is positive. It gives $v(0) = v_i$ and approaches v_T. For the example's $m = 70$ kg, $b = 700$ kg/m, $g = 9.8$ m/s^2,

$$v_i = 60(0.44704) = 26.8224 \text{ m/s},$$
$$v_T = \sqrt{0.98} = 0.9899494937\ldots \text{ m/s},$$
$$T = 1/\sqrt{98} = 0.1010152545\ldots \text{ s}.$$

Here $v_i > v_T$. The manual's real expression involving $\operatorname{arctanh}(v_i/v_T)$ is therefore inapplicable. The equivalent real hyperbolic form for this case is

$$v(t) = v_T\coth\left(t/T + \operatorname{arccoth}(v_i/v_T)\right), \quad \operatorname{arccoth} q = \tfrac{1}{2}\ln\frac{q+1}{q-1} \quad (q > 1).$$

Differentiating either real form verifies $v' = g(1 - v^2/v_T^2)$; the speed initially decreases, as it must when the parachute opens above terminal speed.

(b) With the free-fall drag coefficient $b = 0.25$ kg/m,

$$\boxed{v_T = \sqrt{\frac{70(9.8)}{0.25}} = 52.38320341\ldots \text{ m/s} = 188.5795\ldots \text{ km/h}}.$$

7.2.17 Solve the ODE

$$(xy^2 - y)\,dx + x\,dy = 0.$$

Solution.

On $x \neq 0$, divide by x:

$$y' - \frac{y}{x} = -y^2.$$

The solution $y = 0$ is immediate. For $y \neq 0$, put $u = 1/y$; then $u' = -y'/y^2$ and multiplication by $-y^{-2}$ gives

$$u' + \frac{u}{x} = 1, \qquad (xu)' = x.$$

Integrating yields $xu = x^2/2 + C$, hence

$$\boxed{y(x) = \frac{2x}{x^2 + K}}, \qquad \boxed{y \equiv 0},$$

where $K = 2C$. Substitution into the original, undivided equation gives

$$xy' = \frac{2x(K - x^2)}{(x^2 + K)^2} = y - xy^2.$$

The rational solution is valid wherever its denominator is nonzero, including its smooth extension through $x = 0$ when $K \neq 0$. The equation itself forces $y(0) = 0$ if a solution is defined there. For $K = 0$, $y = 2/x$ has a pole at zero and is used on either side separately.

7.2.18 Solve the ODE

$$(x^2 - y^2 e^{y/x})\,dx + (x^2 + xy)e^{y/x}\,dy = 0.$$

Hint. Note that the quantity y/x in the exponents is of combined degree zero and does not affect the determination of homogeneity.

Solution.

The original equation is defined on $x \neq 0$. Set $u = y/x$, so $y = xu$ and $dy = u\,dx + x\,du$. Substitution gives

$$\begin{aligned} 0 &= x^2(1 - u^2 e^u)dx + x^2(1 + u)e^u(u\,dx + x\,du) \\ &= x^2(1 + ue^u)dx + x^3(1 + u)e^u du. \end{aligned}$$

After division by x^2 this is already exact:

$$(1 + ue^u)dx + x(1 + u)e^u du = d[x(1 + ue^u)] = 0.$$

Therefore

$$\boxed{x + ye^{y/x} = C}.$$

A direct check avoids division by $1 + ue^u$ or $1 + u$:

$$d(x + ye^{y/x}) = (1 - u^2 e^u)dx + (1 + u)e^u dy,$$

which is exactly the original differential form divided by x^2. For real u, $ue^u \geq -1/e$, so $1 + ue^u > 0$; no real constant-ratio solution was lost. At $u = -1$ the level curve can have a vertical tangent; the implicit differential-form solution remains valid there.

7.3 ODEs with Constant Coefficients

Find the general solutions to the following ODEs. Write the solutions in forms that are entirely real (i.e., that contain no complex quantities).

7.3.1 $y''' - 2y'' - y' + 2y = 0.$

Solution.

Substitute $y = e^{rx}$ and cancel the nonzero exponential. The characteristic polynomial factors by grouping:

$$r^3 - 2r^2 - r + 2 = r^2(r-2) - (r-2) = (r-2)(r-1)(r+1).$$

Its three distinct real roots give

$$\boxed{y = C_1 e^{2x} + C_2 e^{x} + C_3 e^{-x}}.$$

Each exponential solves the ODE because its exponent is a root. Their Wronskian is $-6e^{2x} \neq 0$, so these three solutions are independent and span the three-dimensional solution space.

7.3.2 $y''' - 2y'' + y' - 2y = 0.$

Solution.

For $y = e^{rx}$ the characteristic equation is

$$r^3 - 2r^2 + r - 2 = r^2(r-2) + (r-2) = (r-2)(r^2+1) = 0.$$

The roots are $2, i, -i$. Euler's identity converts the last pair into real functions, giving

$$\boxed{y = C_1 e^{2x} + C_2 \cos x + C_3 \sin x}.$$

Direct differentiation verifies all three terms. Their Wronskian at $x = 0$ is

$$\det \begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 1 \\ 4 & -1 & 0 \end{pmatrix} = 5 \neq 0,$$

so the three arbitrary real constants provide the general real solution.

7.3.3 $y''' - 3y' + 2y = 0.$

Solution.

The exponential trial gives

$$r^3 - 3r + 2 = (r-1)(r^2 + r - 2) = (r-1)^2(r+2).$$

The repeated root $r = 1$ needs a second solution in addition to e^x. To see its form without assuming a multiplicity rule, set $y = e^x u$. Then $(D-1)(e^x u) = e^x u'$, so $(D-1)^2(D+2)y = e^x(u^{(3)} + 3u'') = 0$. Taking $u = 1$ and $u = x$ supplies e^x and xe^x; the other root gives e^{-2x}. Hence

$$\boxed{y = (C_1 + C_2 x)e^{x} + C_3 e^{-2x}}.$$

The Wronskian at 0 is $\det \left(\begin{smallmatrix} 1 & 0 & 1 \\ 1 & 1 & -2 \\ 1 & 2 & 4 \end{smallmatrix}\right) = 9$, which checks independence and completeness.

7.3.4 $y'' + 2y' + 2y = 0$.

Solution.

The characteristic equation is $r^2 + 2r + 2 = (r+1)^2 + 1 = 0$, with roots $-1 \pm i$. Equivalently, substituting $y = e^{-x}u$ gives

$$y' = e^{-x}(u' - u), \quad y'' = e^{-x}(u'' - 2u' + u), \quad y'' + 2y' + 2y = e^{-x}(u'' + u).$$

Thus $u = C_1 \cos x + C_2 \sin x$, and

$$\boxed{y = e^{-x}(C_1 \cos x + C_2 \sin x)}.$$

The Wronskian of the two displayed real functions is e^{-2x}, which is nonzero for every real x.

7.4 Second-Order Linear ODEs

7.4.1 Show that Legendre's equation has regular singularities at $x = -1$ and 1, and at ∞ when $l(l+1) \neq 0$. Classify ∞ also when $l = 0$ or $l = -1$.

Solution.

Write $\lambda = l(l+1)$ and divide Legendre's equation by $1 - x^2$:

$$y'' + P(x)y' + Q(x)y = 0, \qquad P = -\frac{2x}{1-x^2}, \quad Q = \frac{\lambda}{1-x^2}.$$

At $x = 1$, $(x-1)P = 2x/(x+1)$ and $(x-1)^2 Q = -\lambda(x-1)/(x+1)$ are analytic. At $x = -1$, $(x+1)P = 2x/(x-1)$ and $(x+1)^2 Q = -\lambda(x+1)/(x-1)$ are analytic. The simple poles of P are nonremovable, so both are regular singular points for every λ.

At infinity put $z = 1/x$ and $Y(z) = y(1/z)$. The chain rule gives

$$y' = -z^2 Y', \qquad y'' = z^4 Y'' + 2z^3 Y'.$$

Substitution into the original equation and division by $z^2(z^2-1)$ gives

$$Y'' + \frac{2z}{z^2-1}Y' + \frac{\lambda}{z^2(z^2-1)}Y = 0.$$

Thus $z\widetilde{P} = 2z^2/(z^2-1)$ and $z^2\widetilde{Q} = \lambda/(z^2-1)$ are analytic at zero. For $\lambda \neq 0$ the double pole of $\widetilde{Q}$ makes infinity regular singular. The original unqualified question has an exception: if $l = 0$ or $l = -1$, then $\lambda = 0$ and both transformed coefficients are analytic at $z = 0$; infinity is then an ordinary point.

7.4.2 Show that Laguerre's equation, like the Bessel equation, has a regular singularity at $x = 0$ and an irregular singularity at $x = \infty$.

Solution.

For Laguerre's equation $xy'' + (1-x)y' + ny = 0$,

$$P = \frac{1}{x} - 1, \qquad Q = \frac{n}{x}.$$

At $x = 0$, $xP = 1 - x$ and $x^2Q = nx$ are analytic, whereas P has a simple pole; zero is regular singular. To analyze infinity, put $z = 1/x$, so $y' = -z^2Y'$ and $y'' = z^4Y'' + 2z^3Y'$. Dividing by z^4 transforms a normalized ODE to

$$Y'' + \left[\frac{2}{z} - \frac{P(1/z)}{z^2}\right] Y' + \frac{Q(1/z)}{z^4} Y = 0.$$

Here the transformed coefficients are

$$\widetilde{P} = \frac{1}{z} + \frac{1}{z^2}, \qquad \widetilde{Q} = \frac{n}{z^3}.$$

Since $z\widetilde{P} = 1 + 1/z$ is singular, infinity is irregular even when $n = 0$.

For comparison, Bessel's equation has $P = 1/x$, $Q = 1 - \nu^2/x^2$. At zero, $xP = 1$ and $x^2Q = x^2 - \nu^2$ are analytic. At infinity,

$$\widetilde{P} = \frac{1}{z}, \qquad \widetilde{Q} = \frac{1}{z^4} - \frac{\nu^2}{z^2},$$

so $z^2\widetilde{Q} = 1/z^2 - \nu^2$ is singular. These establish the same classification for Bessel's equation.

7.4.3 Show that Chebyshev's equation, like the Legendre equation when $l(l + 1) \neq 0$, has regular singularities at $x = -1$, 1, and ∞.

Solution.

Chebyshev's equation is $(1 - x^2)y'' - xy' + n^2y = 0$, so

$$P = -\frac{x}{1 - x^2}, \qquad Q = \frac{n^2}{1 - x^2}.$$

At 1, $(x-1)P = x/(x+1)$ and $(x-1)^2Q = -n^2(x-1)/(x+1)$ are analytic; at -1 the corresponding expressions are $x/(x - 1)$ and $-n^2(x + 1)/(x - 1)$. The poles of P ensure both points are regular singular.

Using $z = 1/x$ and the derivative rule proved in 7.4.2 gives

$$\widetilde{P} = \frac{2}{z} - \frac{1}{z(1 - z^2)} = \frac{1 - 2z^2}{z(1 - z^2)}, \qquad \widetilde{Q} = \frac{n^2}{z^2(z^2 - 1)}.$$

Then $z\widetilde{P} = (1-2z^2)/(1-z^2)$ and $z^2\widetilde{Q} = n^2/(z^2-1)$ are analytic at zero. Since $\widetilde{P}$ always has a simple pole, infinity is regular singular even for $n = 0$. The manual's first displayed normalized Chebyshev equation has $+x/(1-x^2)$ multiplying y'. Division of the original equation gives $-x/(1-x^2)$ instead; its transformed P coefficient must therefore also be the one calculated above. The singularity classification is unchanged.

7.4.4 Show that Hermite's equation has no singularity other than an irregular singularity at $x = \infty$.

Solution.

In $y'' - 2xy' + 2\alpha y = 0$, the normalized coefficients $P = -2x$ and $Q = 2\alpha$ are polynomials. They have no singularities at any finite complex x. For $z = 1/x$, the transformed equation is

$$Y'' + \left(\frac{2}{z} + \frac{2}{z^3}\right) Y' + \frac{2\alpha}{z^4} Y = 0.$$

Multiplying the coefficient of Y' by z gives $2 + 2/z^2$, which is not analytic or finite at zero. Thus infinity is an irregular singular point for every α, including $\alpha = 0$.

7.4.5 Show that the substitution

$$x \to \frac{1-x}{2}, \quad a = -l, \quad b = l+1, \quad c = 1$$

converts the hypergeometric equation into Legendre's equation. Use $t(t-1)u'' + [(1+a+b)t - c]u' + ab\,u = 0$; Table 7.1 has the wrong sign on c.

Solution.

Distinguish the hypergeometric variable t from the new variable x:

$$t(t-1)u_{tt} + [(1+a+b)t - c]u_t + ab\,u = 0, \qquad t = \frac{1-x}{2}, \quad y(x) = u(t).$$

The coefficient is $(1+a+b)t - c$, as in Exercise 7.5.5. Table 7.1 of the textbook prints $+c$ in this sign convention; that sign does not give Legendre's equation under the stated substitution. Since $dx/dt = -2$, $u_t = -2y'$ and $u_{tt} = 4y''$. Also $t(t-1) = (x^2-1)/4$. With $a = -l$, $b = l+1$, $c = 1$,

$$1 + a + b = 2, \qquad 2t - 1 = -x, \qquad ab = -l(l+1).$$

Substitution therefore gives

$$(x^2-1)y'' + (-x)(-2y') - l(l+1)y = 0.$$

Multiplying by -1 yields exactly

$$\boxed{(1-x^2)y'' - 2xy' + l(l+1)y = 0}.$$

7.5 Series Solutions—Frobenius' Method

7.5.1 *Uniqueness theorem.* The function $y(x)$ satisfies a second-order, linear, homogeneous differential equation. At $x = x_0$, $y(x) = y_0$ and $dy/dx = y_0'$. Show that $y(x)$ is unique, in that no other solution of this differential equation passes through the points (x_0, y_0) with a slope of y_0'. Assume a connected interval on which the normalized coefficients are continuous. For the Taylor-series hint, assume they are analytic and x_0 is an ordinary point.

Hint. Assume a second solution satisfying these conditions and compare the Taylor series expansions.

Solution.

Uniqueness requires an ordinary initial point, with continuous normalized coefficients near it. The hint using Taylor series additionally requires analytic coefficients. The unqualified claim, and the manual's extension to regular singular points, are false. For example, $xy'' - y' = 0$ has the family $y = Cx^2$; all members have $y(0) = y'(0) = 0$, although they differ elsewhere.

For the Taylor proof, write $y'' + P(x)y' + Q(x)y = 0$, with P, Q analytic at x_0. If y and $\widetilde{y}$ share the initial data, their difference $u = y - \widetilde{y}$ has $u(x_0) = u'(x_0) = 0$. The equation gives $u''(x_0) = -P(x_0)u'(x_0) - Q(x_0)u(x_0) = 0$. Differentiating k times and applying the product rule gives

$$u^{(k+2)} = -\sum_{j=0}^{k} \binom{k}{j} \{P^{(j)}u^{(k-j+1)} + Q^{(j)}u^{(k-j)}\}.$$

If all derivatives through order $k+1$ vanish at x_0, the right side vanishes there. Induction therefore makes every Taylor coefficient of u zero. Analyticity implies $u = 0$ in a neighborhood, and continuation through the connected ordinary interval proves uniqueness throughout it.

For merely continuous P, Q, one can replace the Taylor argument by an integral estimate. With zero initial data,

$$u(x) = \int_{x_0}^{x} u'(t)dt, \qquad u'(x) = -\int_{x_0}^{x} [P(t)u'(t) + Q(t)u(t)]dt.$$

On a compact interval to the right, choose $K \geq 1 + |P(t)| + |Q(t)|$ and put $F(x) = \int_{x_0}^{x} (|u| + |u'|)dt$. Then $F' \leq KF$ and $F(x_0) = 0$. Thus $(e^{-K(x-x_0)}F)' \leq 0$, while $F \geq 0$, forcing $F = 0$ and $u = u' = 0$. Reverse the integration direction for points to the left. This establishes the continuous-coefficient version without assuming that a smooth function equals its Taylor series.

7.5.2 A series solution of Eq. (7.23) is attempted, expanding about the point $x = x_0$. If x_0 is an ordinary point, show that the indicial equation has roots $s = 0, 1$.

Solution.

Set $t = x - x_0$ and write the normalized equation as $y'' + P(t)y' + Q(t)y = 0$, where P, Q are analytic at $t = 0$. For a Frobenius trial

$$y = \sum_{j=0}^{\infty} a_j t^{s+j}, \quad a_0 \neq 0, \quad y' = \sum_{j} (s+j)a_j t^{s+j-1}, \quad y'' = \sum_{j} (s+j)(s+j-1)a_j t^{s+j-2},$$

the lowest possible power is t^{s-2}. Only y'' contributes to it: Py' starts at t^{s-1} and Qy at t^s. Hence

$$a_0 s(s-1) = 0, \qquad \boxed{s = 0 \text{ or } s = 1}.$$

These represent a solution that can be nonzero at x_0 and one with a simple zero there. The coefficient of t^{s-1} is $s(s+1)a_1 + sP(0)a_0 = 0$; for $s = 1$ this gives $a_1 = -P(0)a_0/2$, so ordinary-point exponents alone do not generally force a_1 to vanish.

7.5.3 In the development of a series solution of the simple harmonic oscillator (SHO) equation, the second series coefficient a_1 was neglected except to set it equal to zero. From the coefficient of the next-to-the-lowest power of x, x^{s-1}, develop a second-indicial type equation.

(a) (SHO equation with $s = 0$). Show that a_1, may be assigned any finite value (including zero).

(b) (SHO equation with $s = 1$). Show that a_1 must be set equal to zero.

Solution.

Use $y'' + \omega^2 y = 0$ and $y = \sum_{j\geq 0} a_j x^{s+j}$ with $a_0 \neq 0$. Substitution gives

$$\sum_{j\geq 0} (s+j)(s+j-1)a_j x^{s+j-2} + \omega^2 \sum_{j\geq 0} a_j x^{s+j} = 0.$$

The coefficients of x^{s-2} and x^{s-1} are respectively

$$s(s-1)a_0 = 0, \qquad \boxed{s(s+1)a_1 = 0}.$$

(a) For $s = 0$ the second equation is $0 = 0$, leaving a_1 arbitrary. The subsequent equations give

$$a_{j+2} = -\frac{\omega^2 a_j}{(j+2)(j+1)},$$

so the even coefficients depend on a_0 and the odd coefficients on a_1. In particular, $y = a_0 \cos \omega x + (a_1/\omega) \sin \omega x$ for $\omega \neq 0$.

(b) For $s = 1$ the second equation is $2a_1 = 0$, hence $a_1 = 0$. The recurrence becomes $a_{j+2} = -\omega^2 a_j/[(j+3)(j+2)]$, and the series is $a_0 \sin(\omega x)/\omega$. Thus setting $a_1 = 0$ in the $s = 0$ series is harmless only when the separate $s = 1$ solution is also retained. For $\omega = 0$ the two solutions reduce to $1, x$.

7.5.4 Analyze the series solutions of the following differential equations to see when a_1 may be set equal to zero without irrevocably losing anything and when a_1 must be set equal to zero.

(a) Legendre, (b) Chebyshev, (c) Bessel, (d) Hermite.

ANS. (a) Legendre, (b) Chebyshev, and (d) Hermite: For $s = 0$, a_1 may be set equal to zero; for $s = 1$, a_1 must be set equal to zero.

(c) Bessel: a_1 must be set equal to zero (except for $s = \pm n = -\frac{1}{2}$).

Solution.

Insert $y = \sum_{j \geq 0} a_j x^{s+j}$, $a_0 \neq 0$. For Legendre, Chebyshev, and Hermite the coefficient multiplying y' starts with x, so their lowest two powers come only from y'':

$$s(s-1)a_0 = 0, \qquad s(s+1)a_1 = 0.$$

Their explicit recurrences are, respectively,

$$\begin{aligned}
\text{Legendre:} \quad & a_{j+2} = \frac{(s+j)(s+j+1) - l(l+1)}{(s+j+2)(s+j+1)} a_j, \\
\text{Chebyshev:} \quad & a_{j+2} = \frac{(s+j)^2 - n^2}{(s+j+2)(s+j+1)} a_j, \\
\text{Hermite:} \quad & a_{j+2} = \frac{2(s+j-\alpha)}{(s+j+2)(s+j+1)} a_j.
\end{aligned}$$

Thus (a), (b), and (d) have arbitrary a_1 for $s = 0$, but require $a_1 = 0$ for $s = 1$. Each recurrence preserves parity. Choosing $a_1 = 0$ in the $s = 0$ branch gives the even solution; the omitted odd branch is then supplied separately by $s = 1$. It would be lost if only the even branch were retained.

(c) Bessel's equation is $x^2 y'' + xy' + (x^2 - n^2)y = 0$. Its lowest two coefficients and subsequent recurrence are

$$(s^2 - n^2)a_0 = 0, \quad [(s+1)^2 - n^2]a_1 = 0, \quad [(s+j)^2 - n^2]a_j + a_{j-2} = 0 \quad (j \geq 2).$$

At either indicial root $s^2 = n^2$, the second coefficient reduces to $(2s+1)a_1 = 0$. Hence a_1 is forced to zero except when

$$\boxed{s = -\tfrac{1}{2}, \qquad n^2 = \tfrac{1}{4}}.$$

In that exceptional lower-root branch, a_1 is free and generates the other, $s = +1/2$, solution. This conclusion concerns a_1 only; for integer orders a later vanishing denominator in the lower-root recurrence can require a logarithmic second solution.

7.5.5 Obtain a series solution of the hypergeometric equation

$$x(x-1)y'' + [(1+a+b)x - c]y' + aby = 0.$$

Test your solution for convergence.

Solution.

Multiply the equation by -1 to use $x(1-x)y'' + [c-(a+b+1)x]y' - aby = 0$. Insert $y = \sum_{j\geq 0} A_j x^{s+j}$, $A_0 \neq 0$. The contributions from $xy''+cy'$ at power x^{s+j-1} are $(s+j)(s+j+c-1)A_j$; those from $-x^2y'' - (a+b+1)xy' - aby$ at power x^{s+j} are $-(s+j+a)(s+j+b)A_j$. Thus

$$s(s+c-1) = 0, \qquad \boxed{A_{j+1} = \frac{(s+j+a)(s+j+b)}{(s+j+1)(s+j+c)} A_j}.$$

With $(q)_j = q(q+1)\cdots(q+j-1)$ and $(q)_0 = 1$, the two generic Frobenius solutions are

$$y_1 = \sum_{j=0}^{\infty} \frac{(a)_j (b)_j}{(c)_j j!} x^j = 1 + \frac{ab}{c}x + \frac{a(a+1)b(b+1)}{2c(c+1)}x^2 + \cdots,$$

$$y_2 = x^{1-c} \sum_{j=0}^{\infty} \frac{(a-c+1)_j (b-c+1)_j}{(2-c)_j j!} x^j.$$

They are independent for noninteger c: their leading powers are 1 and x^{1-c}, and their Wronskian starts as $(1-c)x^{-c} \neq 0$. A branch of the fractional power is fixed when complex x is used.

The displayed coefficient formulas require nonzero denominators. At integer c, one cannot automatically assert two independent solutions of these exact forms. A valid first series always comes from the root $s = 0$ if $c \geq 1$ and $s = 1-c$ if $c \leq 0$: its recurrence denominators do not vanish. Lower-root resonance may require a logarithm. To construct its companion, normalize the equation and split the first-derivative coefficient:

$$P(x) = \frac{c-(a+b+1)x}{x(1-x)} = \frac{c}{x} + \frac{c-a-b-1}{1-x}.$$

Consequently $e^{-\int P dx} = x^{-c}(1-x)^{c-a-b-1}$, and

$$y_2 = y_1 \int^x \frac{t^{-c}(1-t)^{c-a-b-1}}{y_1(t)^2}\, dt.$$

Expand the analytic factors and integrate each power near zero; a t^{-1} term produces $\ln t$. On a branch avoiding $0, 1$ and zeros of y_1, its Wronskian is $x^{-c}(1-x)^{c-a-b-1} \neq 0$. This supplies the missing independent solution, including resonances. If a numerator terminates the recurrence before an obstructing denominator, the terminating polynomial is interpreted directly through the original recurrence.

For a nonterminating series, its successive terms satisfy

$$\left|\frac{A_{j+1}x^{j+1}}{A_j x^j}\right| = |x| \left|\frac{(j+s+a)(j+s+b)}{(j+s+1)(j+s+c)}\right| \longrightarrow |x|.$$

It converges absolutely for $|x| < 1$ and diverges for $|x| > 1$. For boundary convergence put $\sigma = c-a-b$. The gamma-function expression for the coefficients gives, in a nonterminating generic case,

$$A_j = K j^{-\sigma-1}[1 + O(j^{-1})], \qquad K \neq 0.$$

This follows by writing $(q)_j = \Gamma(j+q)/\Gamma(q)$ and using $\Gamma(j+u)/\Gamma(j+v) \sim j^{u-v}$. Therefore at $x = 1$ convergence holds precisely when $\operatorname{Re}\sigma > 0$. Indeed, absolute convergence is then comparison with $j^{-1-\operatorname{Re}\sigma}$. If $-1 < \operatorname{Re}\sigma \le 0$, summing the leading powers gives a term proportional to $N^{-\sigma}$, or $\ln N$ when $\sigma = 0$, with no limit; the $O(j^{-\sigma-2})$ correction is summable. For $\operatorname{Re}\sigma \le -1$ even the individual terms fail to tend to zero. At $x = -1$, put $p = \sigma + 1$. When $\operatorname{Re} p > 0$, $j^{-p} \to 0$ and $(j+1)^{-p} - j^{-p} = O(j^{-\operatorname{Re}p-1})$ is absolutely summable. Summation by parts with the bounded partial sums of $(-1)^j$ therefore proves convergence. The coefficient correction is also absolutely summable. Thus the condition there is $\operatorname{Re}\sigma > -1$, with absolute convergence only for $\operatorname{Re}\sigma > 0$. The second series has the same σ, since $(2-c) - (a-c+1) - (b-c+1) = c-a-b$. For any other point on $|x| = 1$ with $x \ne 1$, the partial sums of x^j are bounded by $2/|1-x|$, so the same summation-by-parts argument as at -1 gives convergence precisely when $\operatorname{Re}\sigma > -1$. A terminating power-series factor is a polynomial and converges for every finite x; the second solution still retains its possibly singular or multivalued prefactor x^{1-c}.

7.5.6 Obtain two series solutions of the confluent hypergeometric equation

$$xy'' + (c-x)y' - ay = 0.$$

Test your solutions for convergence. For integer c, allow a generalized Frobenius solution containing a logarithm; two independent pure power-series forms need not exist.

Solution.

Put $y = \sum_{j\ge0} A_j x^{s+j}$. Then $xy'' + cy'$ contributes $(s+j)(s+j+c-1)A_j x^{s+j-1}$, whereas $-xy' - ay$ contributes $-(s+j+a)A_j x^{s+j}$. Matching powers,

$$s(s+c-1) = 0, \qquad \boxed{A_{j+1} = \frac{s+j+a}{(s+j+1)(s+j+c)} A_j}.$$

For noninteger c the independent solutions, each with leading coefficient one, are

$$y_1 = \sum_{j=0}^{\infty} \frac{(a)_j}{(c)_j j!} x^j = 1 + \frac{a}{c}x + \frac{a(a+1)}{2c(c+1)}x^2 + \cdots,$$

$$y_2 = x^{1-c} \sum_{j=0}^{\infty} \frac{(a-c+1)_j}{(2-c)_j j!} x^j.$$

The first is conventionally $M(a,c,x)$; the second is $x^{1-c}M(a-c+1, 2-c, x)$. Their Wronskian has leading term $(1-c)x^{-c}$, verifying independence. For either coefficient sequence,

$$\left|\frac{A_{j+1}x^{j+1}}{A_j x^j}\right| = |x| \left|\frac{s+j+a}{(s+j+1)(s+j+c)}\right| \sim \frac{|x|}{j} \longrightarrow 0.$$

Thus both power-series factors converge for every finite x. The prefactor x^{1-c} may still be singular or multivalued at zero; convergence of its power-series factor does not remove that singularity. A terminating power-series factor is a polynomial; the second solution still includes its displayed power prefactor.

For integer c, two independent pure Frobenius series need not exist; a logarithmic solution may be necessary. The ordinary formulas above must not be used through a zero denominator.

A constructive series prescription for all integer c is as follows. Choose $s = \max(0, 1-c)$ and obtain $y_1 = x^s A(x)$ with $A(0) = 1$ by the displayed recurrence; its denominators are nonzero. Write

$$A(x)^2 = \sum_{j\ge 0} B_j x^j, \quad B_j = \sum_{r=0}^{j} A_r A_{j-r}, \quad \frac{e^x}{A(x)^2} = \sum_{j\ge 0} h_j x^j.$$

Since $B_0 = 1$, coefficient comparison gives $h_0 = 1$ and $h_k = 1/k! - \sum_{j=1}^{k} B_j h_{k-j}$. For $P = c/x - 1$, reduction of order then yields the explicit generalized series

$$y_2 = x^s A(x) \left[\sum_{\substack{k\ge 0 \\ k\ne c+2s-1}} \frac{h_k x^{k-c-2s+1}}{k-c-2s+1} + h_{c+2s-1} \ln x \right].$$

Here $c + 2s - 1$ is a nonnegative integer, and the logarithm is omitted if its coefficient vanishes. This locally convergent expansion follows by integrating $e^x x^{-c}/y_1^2$ term by term near zero where $A(x) \ne 0$; its Wronskian with y_1 is $e^x x^{-c} \ne 0$. For example, $a = c = 1$ gives $y_1 = e^x$ and $y_2 = e^x[\ln x + \sum_{k\ge 1}(-x)^k/(kk!)]$, displaying the necessary logarithm explicitly. Real solutions on the negative half-line use $\ln|x|$ and an appropriate real power convention.

7.5.7 A quantum mechanical analysis of the Stark effect (parabolic coordinates) leads to the differential equation

$$\frac{d}{d\xi}\left(\xi \frac{du}{d\xi}\right) + \left(\frac{1}{2}E\xi + \alpha - \frac{m^2}{4\xi} - \frac{1}{4}F\xi^2\right) u = 0.$$

Here α is a constant, E is the total energy, and F is a constant such that Fz is the potential energy added to the system by the introduction of an electric field.

Using the larger root of the indicial equation, develop a power-series solution about $\xi = 0$. Evaluate the first three coefficients in terms of a_0.

ANS. Indicial equation $s^2 - \dfrac{m^2}{4} = 0$,

$$M = |m|,\ u(\xi) = a_0 \xi^{M/2}\left[1 - \frac{\alpha}{M+1}\xi + \left(\frac{\alpha^2}{2(M+1)(M+2)} - \frac{E}{4(M+2)}\right)\xi^2 + \cdots\right].$$

Note that the perturbation F does not appear until a_3 is included.

Solution.

The larger indicial root is $|m|/2$ for a real azimuthal quantum number. The printed answer implicitly assumes $m \ge 0$; use $M = |m|$ for either sign. Expand the derivative and multiply by ξ:

$$\xi^2 u'' + \xi u' - \frac{m^2}{4}u + \alpha\xi u + \frac{E}{2}\xi^2 u - \frac{F}{4}\xi^3 u = 0.$$

For $u = \sum_{j\ge 0} a_j \xi^{s+j}$, set negative-index coefficients to zero. Matching ξ^{s+j} gives

$$\left[(s+j)^2 - \frac{m^2}{4}\right] a_j + \alpha a_{j-1} + \frac{E}{2}a_{j-2} - \frac{F}{4}a_{j-3} = 0.$$

At $j = 0$ this yields $s^2 - m^2/4 = 0$; choose $s = M/2$. For $j \ge 1$ the nonzero denominator is $j(j+M)$, so

$$a_j = \frac{-\alpha a_{j-1} - (E/2)a_{j-2} + (F/4)a_{j-3}}{j(j+M)}.$$

In particular,

$$a_1 = -\frac{\alpha}{M+1}a_0, \quad a_2 = \frac{-\alpha a_1 - (E/2)a_0}{2(M+2)} = a_0\left[\frac{\alpha^2}{2(M+1)(M+2)} - \frac{E}{4(M+2)}\right].$$

Therefore the requested result is

$$\boxed{u = a_0\xi^{M/2}\left[1 - \frac{\alpha}{M+1}\xi + \left(\frac{\alpha^2}{2(M+1)(M+2)} - \frac{E}{4(M+2)}\right)\xi^2 + \cdots\right]}.$$

The next coefficient is

$$a_3 = \frac{1}{3(M+3)}\left[-\alpha a_2 + \frac{E\alpha}{2(M+1)}a_0 + \frac{F}{4}a_0\right],$$

so F first enters at a_3, as claimed.

7.5.8 For the special case of no azimuthal dependence, the quantum mechanical analysis of the hydrogen molecular ion leads to the equation

$$\frac{d}{d\eta}\left[(1-\eta^2)\frac{du}{d\eta}\right] + \alpha u + \beta\eta^2 u = 0.$$

Develop a power-series solution for $u(\eta)$. Evaluate the first three allowed coefficients in terms of a_0; special parameter values may make some of them zero.

ANS. Indicial equation $s(s-1) = 0$,

$$u_{k=1} = a_0\eta\left[1 + \frac{2-\alpha}{6}\eta^2 + \left(\frac{(2-\alpha)(12-\alpha)}{120} - \frac{\beta}{20}\right)\eta^4 + \cdots\right].$$

Solution.

Expanding the differential operator gives $(1-\eta^2)u'' - 2\eta u' + \alpha u + \beta\eta^2 u = 0$. For $u = \sum_{j\geq 0} a_j\eta^{s+j}$, the two lowest coefficients are

$$s(s-1)a_0 = 0, \qquad s(s+1)a_1 = 0.$$

Thus $s = 0, 1$; take $a_1 = 0$ in each branch to separate the even and odd solutions. Equating coefficients of η^{s+j} then gives

$$\boxed{a_{j+2} = \frac{[(s+j)(s+j+1) - \alpha]a_j - \beta a_{j-2}}{(s+j+2)(s+j+1)}}, \qquad a_{-2} = 0.$$

For $s = 1$,

$$a_2 = \frac{2-\alpha}{6}a_0, \qquad a_4 = \frac{(12-\alpha)a_2 - \beta a_0}{20} = a_0\left[\frac{(2-\alpha)(12-\alpha)}{120} - \frac{\beta}{20}\right].$$

This reproduces the displayed odd solution and fulfills the request for a series solution. For completeness, the even branch has

$$a_2 = -\frac{\alpha}{2}a_0, \qquad a_4 = \frac{(6-\alpha)a_2 - \beta a_0}{12} = a_0\left[\frac{\alpha(\alpha-6)}{24} - \frac{\beta}{12}\right].$$

Thus a normalized pair is

$$u_{\text{even}} = 1 - \frac{\alpha}{2}\eta^2 + \left[\frac{\alpha(\alpha-6)}{24} - \frac{\beta}{12}\right]\eta^4 + \cdots,$$
$$u_{\text{odd}} = \eta\left[1 + \frac{2-\alpha}{6}\eta^2 + \left(\frac{(2-\alpha)(12-\alpha)}{120} - \frac{\beta}{20}\right)\eta^4 + \cdots\right].$$

The initial values $(u, u') = (1, 0)$ and $(0, 1)$ at zero establish independence. Their linear combination is the general local solution; both series converge at least for $|\eta| < 1$, the disk up to the nearest singular coefficients. At special parameter values some displayed coefficients can vanish, so these are the first three allowed coefficients, not necessarily three nonzero numbers.

7.5.9 To a good approximation, the interaction of two nucleons may be described by a mesonic potential

$$V = \frac{Ae^{-ax}}{x},$$

attractive for A negative. Show that the resultant Schrödinger wave equation

$$\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + (E - V)\psi = 0$$

has the following series solution through x^3, allowing a displayed coefficient to vanish at special parameter values:

$$\psi = a_0\left\{x + \frac{1}{2}A'x^2 + \frac{1}{6}\left[\frac{1}{2}A'^2 - E' - aA'\right]x^3 + \cdots\right\},$$

where the prime indicates multiplication by $2m/\hbar^2$.

Solution.

Put $A' = 2mA/\hbar^2$, $E' = 2mE/\hbar^2$. The equation becomes

$$\psi'' + \left(E' - \frac{A'e^{-ax}}{x}\right)\psi = 0.$$

The indicial term from $\psi \sim x^s$ is $s(s-1)$, so the regular vanishing branch has $s = 1$. Write $\psi = \sum_{j\geq 0} a_j x^{j+1}$ and use $e^{-ax} = \sum_{k\geq 0}(-a)^k x^k/k!$. At power x^j, the recurrence is

$$(j+2)(j+1)a_{j+1} + E'a_{j-1} - A'\sum_{k=0}^{j}\frac{(-a)^k}{k!}a_{j-k} = 0, \qquad a_{-1} = 0.$$

For $j = 0$ this gives $2a_1 - A'a_0 = 0$, and for $j = 1$ it gives $6a_2 + E'a_0 - A'(a_1 - aa_0) = 0$. Consequently

$$a_1 = \frac{A'}{2}a_0, \qquad a_2 = \frac{a_0}{6}\left(\frac{A'^2}{2} - E' - aA'\right),$$

and

$$\boxed{\psi = a_0\left[x + \frac{A'}{2}x^2 + \frac{1}{6}\left(\frac{A'^2}{2} - E' - aA'\right)x^3 + \cdots\right]}.$$

The same coefficients follow by substituting $e^{-ax}/x = x^{-1} - a + (a^2/2)x + \cdots$ and equating the constant and linear terms of the differential equation. The other branch may have a logarithm; the displayed series is the regular branch requested here.

7.5.10 If the parameter b^2 in Eq. (7.53) is equal to 2, Eq. (7.53) becomes

$$y'' + \frac{1}{x^2}y' - \frac{2}{x^2}y = 0.$$

From the indicial equation and the recurrence relation, derive a solution $y = 1 + 2x + 2x^2$. Verify that this is indeed a solution by substituting back into the differential equation.

Solution.

Multiplying by x^2 gives $x^2y'' + y' - 2y = 0$ for $x \neq 0$. For $y = \sum_{j\geq 0} a_j x^{s+j}$, the lowest power x^{s-1} comes only from y' and has coefficient sa_0. With $a_0 \neq 0$, $s = 0$. Matching the subsequent coefficients of x^j gives

$$[j(j-1) - 2]a_j + (j+1)a_{j+1} = 0, \qquad \boxed{a_{j+1} = \frac{2 - j(j-1)}{j+1}a_j}.$$

Thus $a_1 = 2a_0$, $a_2 = a_1 = 2a_0$, $a_3 = 0$, and every later coefficient is zero. Taking $a_0 = 1$ yields $\boxed{y = 1 + 2x + 2x^2}$. The direct check is

$$y' = 2 + 4x, \quad y'' = 4, \quad x^2y'' + y' - 2y = 4x^2 + (2 + 4x) - 2(1 + 2x + 2x^2) = 0.$$

The original normalized equation is undefined at zero even though this particular solution extends there analytically. Zero is irregular singular, so finding this terminating series does not imply that every solution admits an ordinary Frobenius series there.

7.5.11 The modified Bessel function $I_0(x)$ satisfies the differential equation

$$x^2\frac{d^2}{dx^2}I_0(x) + x\frac{d}{dx}I_0(x) - x^2I_0(x) = 0.$$

Given that the leading term in an asymptotic expansion as $x \to +\infty$ is known to be

$$I_0(x) \sim \frac{e^x}{\sqrt{2\pi x}},$$

assume a series of the form

$$I_0(x) \sim \frac{e^x}{\sqrt{2\pi x}}\left\{1 + b_1x^{-1} + b_2x^{-2} + \cdots\right\}.$$

Determine the coefficients b_1 and b_2.

ANS. $b_1 = \dfrac{1}{8}, \quad b_2 = \dfrac{9}{128}.$

Solution.

Write $I_0 = Ef$, where $E = e^x/\sqrt{2\pi x}$. The derivatives of the prefactor satisfy

$$\frac{E'}{E} = 1 - \frac{1}{2x}, \qquad \frac{E''}{E} = 1 - \frac{1}{x} + \frac{3}{4x^2}.$$

Since $I_0' = E'f + Ef'$ and $I_0'' = E''f + 2E'f' + Ef''$, substitution into $x^2I_0'' + xI_0' - x^2I_0 = 0$ cancels the terms in x^2f and xf, leaving

$$\boxed{x^2f'' + 2x^2f' + \frac{1}{4}f = 0}.$$

Set $f \sim \sum_{j\ge 0} b_j x^{-j}$, $b_0 = 1$. At power x^{-j} the equation is

$$[j(j+1) + \tfrac{1}{4}]b_j - 2(j+1)b_{j+1} = 0, \quad b_{j+1} = \frac{(j+1/2)^2}{2(j+1)} b_j.$$

Therefore

$$\boxed{b_1 = \frac{1/4}{2} = \frac{1}{8}}, \qquad \boxed{b_2 = \frac{9/4}{4}\frac{1}{8} = \frac{9}{128}}.$$

These are asymptotic coefficients for large positive x, not a convergent series at fixed x: the ratio of successive inverse-power terms eventually behaves as $j/(2x)$ and grows beyond one. Truncation must be understood in the asymptotic sense.

7.5.12 The even power-series solution of Legendre's equation is given by Exercise 8.3.1. Take $a_0 = 1$ and n not an even integer, say $n = 0.5$. Calculate the partial sums of the series through x^{200}, x^{400}, x^{600}, ..., x^{2000} for $x = 0.95(0.01)1.00$. Also, write out the individual term corresponding to each of these powers.

Note. This calculation does not constitute proof of convergence at $x = 0.99$ or divergence at $x = 1.00$, but perhaps you can see the difference in the behavior of the sequence of partial sums for these two values of x.

Solution.

For $n = 1/2$, Legendre's equation is $(1 - x^2)y'' - 2xy' + \frac{3}{4}y = 0$. Insert the even series $y = \sum_{j\ge 0} a_{2j} x^{2j}$ with $a_0 = 1$. The coefficient of x^{2j} gives

$$(2j+2)(2j+1)a_{2j+2} + [\tfrac{3}{4} - 2j(2j+1)]a_{2j} = 0,$$

so

$$\boxed{a_{2j+2} = \frac{2j(2j+1) - 3/4}{(2j+2)(2j+1)} a_{2j}}.$$

For example $a_2 = -3/8$ and $a_4 = -21/128$. For each specified x, update the term and sum without repeatedly forming large powers:

$$T_0 = 1, \quad S_0 = 1, \quad T_{j+1} = T_j x^2 \frac{2j(2j+1) - 3/4}{(2j+2)(2j+1)}, \quad S_{j+1} = S_j + T_{j+1}.$$

The first table gives $S_J = \sum_{j=0}^{J} a_{2j} x^{2j}$ through the power $N = 2J$. The second gives the individual term $T_J = a_N x^N$ at that same power. Calculations used 70 decimal digits before rounding; an independent 90-digit Pochhammer-coefficient sum checked every entry.

Partial sums S_J

N	$x = 0.95$	0.96	0.97	0.98	0.99	1.00
200	0.2154301	0.1476299	0.0605814	−0.06073655	−0.2554509	−0.6500133
400	0.2154292	0.1476211	0.06048901	−0.06176434	−0.2684287	−0.8540902
600	0.2154292	0.147621	0.0604889	−0.06177416	−0.2694032	−0.9735833
800	0.2154292	0.147621	0.0604889	−0.06177427	−0.2694942	−1.058397
1000	0.2154292	0.147621	0.0604889	−0.06177428	−0.2695036	−1.124197
1200	0.2154292	0.147621	0.0604889	−0.06177428	−0.2695046	−1.177967
1400	0.2154292	0.147621	0.0604889	−0.06177428	−0.2695047	−1.223433
1600	0.2154292	0.147621	0.0604889	−0.06177428	−0.2695047	−1.262819
1800	0.2154292	0.147621	0.0604889	−0.06177428	−0.2695047	−1.297563
2000	0.2154292	0.147621	0.0604889	−0.06177428	−0.2695047	−1.328643

Last terms T_J

N	$x = 0.95$	0.96	0.97	0.98	0.99	1.00
200	-1.036298×10^{-7}	-8.414151×10^{-7}	-6.685141×10^{-6}	-5.199707×10^{-5}	-0.000396098	-0.002956403
400	$-1.814542 \times 10^{-12}$	-1.19624×10^{-10}	-7.551248×10^{-9}	-4.568307×10^{-7}	-2.650958×10^{-5}	-0.001476811
600	$-4.238976 \times 10^{-17}$	$-2.269016 \times 10^{-14}$	-1.13799×10^{-11}	-5.354799×10^{-9}	-2.367089×10^{-6}	-0.0009842326
800	$-1.114231 \times 10^{-21}$	$-4.842588 \times 10^{-18}$	-1.92965×10^{-14}	$-7.062389 \times 10^{-11}$	-2.378191×10^{-7}	-0.000738059
1000	$-3.124249 \times 10^{-26}$	$-1.102487 \times 10^{-21}$	$-3.490395 \times 10^{-17}$	$-9.936101 \times 10^{-13}$	-2.548795×10^{-8}	-0.0005903918
1200	$-9.125533 \times 10^{-31}$	$-2.614638 \times 10^{-25}$	$-6.576775 \times 10^{-20}$	$-1.456206 \times 10^{-14}$	-2.845545×10^{-9}	-0.0004919624
1400	$-2.741657 \times 10^{-35}$	$-6.378109 \times 10^{-29}$	$-1.274658 \times 10^{-22}$	$-2.195188 \times 10^{-16}$	$-3.267669 \times 10^{-10}$	-0.0004216632
1600	$-8.408677 \times 10^{-40}$	$-1.588298 \times 10^{-32}$	$-2.521936 \times 10^{-25}$	$-3.378161 \times 10^{-18}$	$-3.830633 \times 10^{-11}$	-0.0003689429
1800	$-2.619901 \times 10^{-44}$	$-4.018045 \times 10^{-36}$	$-5.068939 \times 10^{-28}$	$-5.281188 \times 10^{-20}$	$-4.561899 \times 10^{-12}$	-0.0003279407
2000	$-8.264934 \times 10^{-49}$	$-1.029188 \times 10^{-39}$	$-1.031567 \times 10^{-30}$	$-8.359498 \times 10^{-22}$	$-5.500701 \times 10^{-13}$	-0.0002951405

The convergence distinction can also be proved rather than inferred from these tables. The exact coefficients are

$$a_{2j} = \frac{(-1/4)_j (3/4)_j}{(1/2)_j j!}.$$

Their ratio gives absolute convergence for $|x| < 1$. Gamma-function ratios give

$$a_{2j} \sim \frac{\Gamma(1/2)}{\Gamma(-1/4)\Gamma(3/4)} \frac{1}{j} = -\frac{\sqrt{\pi}}{4\Gamma(3/4)^2} \frac{1}{j}.$$

All coefficients after a_0 are negative, and the nonzero negative constant multiplying $1/j$ implies divergence to $-\infty$ at $x = 1$. For every listed $x < 1$, the additional factor x^{2j} instead makes the terms exponentially small. The finite numerical sequences alone would not establish either conclusion.

7.5.13

(a) The odd power-series solution of Hermite's equation is given by Exercise 8.3.3. Take $a_0 = 1$. Evaluate this series for $\alpha = 0$, $x = 1, 2, 3$. Cut off your calculation after the last term calculated has dropped below the maximum term by a factor of 10^6 or more. Set an upper bound to the error made in ignoring the remaining terms in the infinite series.

(b) As a check on the calculation of part (a), show that the Hermite series $y_{\text{odd}}(\alpha = 0)$ corresponds to $\int_0^x e^{t^2}\, dt$.

(c) Calculate this integral for $x = 1, 2, 3$.

Solution.

(a) For $\alpha = 0$, Hermite's equation is $y'' - 2xy' = 0$. The normalized odd solution has $y(0) = 0$, $y'(0) = 1$. Substituting $y = \sum_{k\geq 0} c_k x^{2k+1}$ gives

$$(2k+3)(2k+2)c_{k+1} = 2(2k+1)c_k, \qquad c_0 = 1.$$

Induction verifies $c_k = 1/[k!(2k+1)]$, since its consecutive ratio is $(2k+1)/[(k+1)(2k+3)]$. Thus

$$\boxed{y(x) = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{k!(2k+1)}}.$$

For a fixed positive x, define $t_k = x^{2k+1}/[k!(2k+1)]$. The iteration uses

$$\frac{t_{k+1}}{t_k} = \frac{x^2(2k+1)}{(k+1)(2k+3)}, \qquad S_K = \sum_{k=0}^{K} t_k.$$

Stop at the first K for which $t_K \le 10^{-6} \max_{j\le K} t_j$. For every subsequent ratio, $t_{j+1}/t_j < x^2/(j+1) \le q := x^2/(K+2)$ when $j \ge K+1$. The cutoffs below have $q < 1$, so the omitted tail has the rigorous bound

$$\boxed{0 < y - S_K \le B_K := \frac{t_{K+1}}{1 - x^2/(K+2)}}.$$

This bounds the entire remaining sum, rather than assuming that the last calculated term is itself an error bound. The tabulated bounds are rounded upward; the sums and last terms are rounded to nearest.

x	K	S_K	t_K	B_K
1	9	1.4626517316055	1.450385×10^{-7}	1.443479×10^{-8}
2	17	16.452627042558	2.760027×10^{-6}	7.349021×10^{-7}
3	28	1444.5450849204	9.034328×10^{-5}	3.869592×10^{-5}

(b) The original integral reused x as its integration variable and upper limit. The dummy variable t makes its intended meaning explicit. Since the exponential series converges uniformly on every finite integration interval, termwise integration is valid:

$$\int_0^x e^{t^2} dt = \sum_{k=0}^{\infty} \frac{1}{k!} \int_0^x t^{2k} dt = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{k!(2k+1)} = y(x).$$

Alternatively $y' = e^{x^2}$ directly gives $y'' = 2xy'$ and the correct initial data.

(c) Independent quadrature, also checked against $(\sqrt{\pi}/2)\,\mathrm{erfi}(x)$, gives

$$\int_0^1 e^{t^2} dt = 1.462651745907181609\ldots,$$
$$\int_0^2 e^{t^2} dt = 16.452627765507230 22\ldots,$$
$$\int_0^3 e^{t^2} dt = 1444.5451228927141547\ldots.$$

The differences from the finite sums in (a) are positive and smaller than the displayed bounds, as an independent numerical check.

7.6 Other Solutions

7.6.1 You know that the three unit vectors $\hat{\mathbf{e}}_x$, $\hat{\mathbf{e}}_y$, and $\hat{\mathbf{e}}_z$ are mutually perpendicular (orthogonal). Show that $\hat{\mathbf{e}}_x$, $\hat{\mathbf{e}}_y$, and $\hat{\mathbf{e}}_z$ are linearly independent. Specifically, show that no relation of the form of Eq. (7.54) exists for $\hat{\mathbf{e}}_x$, $\hat{\mathbf{e}}_y$, and $\hat{\mathbf{e}}_z$.

Solution.

Suppose $a\hat{e}_x + b\hat{e}_y + c\hat{e}_z = 0$. Take the dot product successively with the three unit vectors. Their orthonormality gives

$$a(1) + b(0) + c(0) = 0, \qquad a(0) + b(1) + c(0) = 0, \qquad a(0) + b(0) + c(1) = 0.$$

Thus $a = b = c = 0$. No nontrivial constant linear relation exists, which is exactly linear independence. Equivalently their component matrix is the identity and has determinant one.

7.6.2 The criterion for the linear independence of three vectors $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ is that the equation

$$a\mathbf{A} + b\mathbf{B} + c\mathbf{C} = 0,$$

analogous to Eq. (7.54), has no solution other than the trivial $a = b = c = 0$. Using components $\mathbf{A} = (A_1, A_2, A_3)$, and so on, set up the determinant criterion for the existence or nonexistence of a nontrivial solution for the coefficients a, b, and c. Show that your criterion is equivalent to the scalar triple product $\mathbf{A} \cdot \mathbf{B} \times \mathbf{C} \neq 0$.

Solution.

The vector equation is the homogeneous component system

$$\begin{pmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = 0.$$

It has only the zero solution precisely when its square coefficient matrix is invertible, or equivalently when its determinant is nonzero. Expansion down the first column gives

$$\begin{aligned} \det[A\ B\ C] &= A_1(B_2C_3 - B_3C_2) - A_2(B_1C_3 - B_3C_1) \\ &\quad + A_3(B_1C_2 - B_2C_1) = \mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}). \end{aligned}$$

Therefore

$$\boxed{\mathbf{A}, \mathbf{B}, \mathbf{C} \text{ independent} \iff \mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) \neq 0}.$$

If the determinant is zero, the matrix has a nonzero null vector and there is a nontrivial relation; this proves the converse as well.

7.6.3 Using the Wronskian determinant, show that the set of functions

$$\left\{1, \frac{x^n}{n!}, (n = 1, 2, \ldots, N)\right\}$$

is linearly independent.

Solution.

The source exercise (textbook p. 370) includes the constant function 1; it was omitted in the transcription. The corrected set is $\{1, x, x^2/2!, \ldots, x^N/N!\}$. For columns indexed by $n = 0, \ldots, N$ and rows by derivative order $j = 0, \ldots, N$,

$$\frac{d^j}{dx^j}\frac{x^n}{n!} = \begin{cases} x^{n-j}/(n-j)!, & j \leq n, \\ 0, & j > n. \end{cases}$$

The Wronskian matrix is upper triangular with diagonal entries one:

$$W = \det \begin{pmatrix} 1 & x & x^2/2! & \cdots & x^N/N! \\ 0 & 1 & x & \cdots & x^{N-1}/(N-1)! \\ 0 & 0 & 1 & \cdots & x^{N-2}/(N-2)! \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix} = \boxed{1}.$$

If a constant linear combination of the functions were zero, its first N derivatives would be zero too. Invertibility of this matrix would then force every coefficient to vanish.

The transcribed set without 1 is also independent, but its Wronskian is different. The determinant of the falling-factorial coefficient matrix is the Vandermonde product of the distinct exponents $1, \dots, N$:

$$W\{x, x^2/2!, \dots, x^N/N!\} = \frac{\prod_{1 \le i < j \le N} (j - i)}{\prod_{n=1}^{N} n!} x^{\sum_{n=1}^{N} n - \sum_{j=0}^{N-1} j} = \boxed{\frac{x^N}{N!}}.$$

Here the product in the numerator is $\prod_{j=1}^{N-1} j!$. This Wronskian is nonzero at every $x \neq 0$, enough to prove independence; its isolated zero at zero does not imply dependence.

7.6.4 If the Wronskian of two functions y_1 and y_2 is identically zero, show by direct integration that

$$y_1 = c y_2,$$

that is, that y_1 and y_2 are linearly dependent. Assume the functions have continuous derivatives and that at least one of the functions does not vanish in the connected interval under consideration. Interchange the labels first, if necessary, so that y_2 is nowhere zero; without this convention only the dependence conclusion is unconditional.

Solution.

Suppose first that y_2 is nowhere zero on the connected interval. The quotient rule gives

$$\left(\frac{y_1}{y_2}\right)' = \frac{y_1' y_2 - y_1 y_2'}{y_2^2} = -\frac{W}{y_2^2} = 0.$$

Integrating from any fixed x_* yields $y_1(x)/y_2(x) = y_1(x_*)/y_2(x_*) = c$, hence $\boxed{y_1 = c y_2}$. This argument also allows y_1 to have zeros.

If the nowhere-zero function is instead y_1, the directly valid statement is $y_2 = c y_1$. The literal requested orientation $y_1 = c y_2$ fails when $y_2 \equiv 0$ and $y_1 \neq 0$. In that case the same quotient calculation with roles reversed proves linear dependence. If its constant is nonzero one can invert it and recover the displayed orientation; if it is zero, dependence still holds because one function is identically zero. Thus the dependence conclusion requires no division by a function that may vanish.

7.6.5 The Wronskian of two functions is found to be zero at $x_0 - \epsilon \le x \le x_0 + \epsilon$ for arbitrarily small $\epsilon > 0$. Show that this Wronskian vanishes for all x and that the functions are linearly dependent. Assume both functions are analytic on one connected interval containing these points, or that they solve the same normalized second-order ODE with continuous coefficients there.

Solution.

The original statement is false for general differentiable, or even infinitely differentiable, functions. It needs analyticity on a connected interval, or a common regular second-order ODE. A smooth counterexample on $\mathbb{R}$ is

$$y_1(x) = 1, \qquad y_2(x) = \begin{cases} 0, & x \le 1, \\ e^{-1/(x-1)^2}, & x > 1. \end{cases}$$

To check smoothness at 1, set $u = 1/(x-1)$ on the right. Since $d/dx = -u^2 d/du$, induction shows that every derivative has the form $p(u)e^{-u^2}$ for a polynomial p. Each tends to zero as $u \to +\infty$,

matching every derivative of the zero branch. Thus y_2 is smooth. Its Wronskian with 1 is

$$W = y_2' = \begin{cases} 0, & x \le 1, \\ \dfrac{2e^{-1/(x-1)^2}}{(x-1)^3}, & x > 1. \end{cases}$$

It vanishes in every sufficiently small interval around $x_0 = 0$, but not for $x > 1$. The functions are independent: a relation $a + by_2 = 0$ first forces $a = 0$ on $x \le 1$, then $b = 0$ on $x > 1$.

Under the repaired analytic hypothesis, W is analytic. Its vanishing on a nonempty interval makes all its Taylor coefficients there zero; the analytic identity theorem propagates $W = 0$ through the connected domain. If, say, y_2 is not identically zero, choose a small interval where it is nonzero. Exercise 7.6.4 gives $y_1 = cy_2$ there, and the analytic identity theorem applied to $y_1 - cy_2$ extends that relation throughout the domain. An identically zero member gives dependence immediately.

Alternatively, for two solutions of the same normalized ODE with continuous P, Q, Abel's identity gives $W(x) = W(x_*)e^{-\int_{x_*}^{x} P(t)dt}$. A single zero then makes W identically zero, and initial-value uniqueness gives linear dependence. The ODE or analytic assumption is the missing reason local information can be extended globally.

7.6.6 The three functions $\sin x$, e^x, and e^{-x} are linearly independent. No one function can be written as a linear combination of the other two. Show that the Wronskian of $\sin x$, e^x, and e^{-x} vanishes but only at isolated points.

ANS. $W = 4\sin x$, $\quad W = 0$ for $x = \pm n\pi$, $\quad n = 0, 1, 2, \ldots$.

Solution.

Form the determinant with functions as columns:

$$W = \det \begin{pmatrix} \sin x & e^x & e^{-x} \\ \cos x & e^x & -e^{-x} \\ -\sin x & e^x & e^{-x} \end{pmatrix}.$$

Factor e^x and e^{-x} from the last two columns; their product is one. Expanding the remaining determinant along its first row gives

$$W = 2\sin x - (\cos x - \sin x) + (\cos x + \sin x) = \boxed{4\sin x}.$$

It vanishes exactly at the isolated points $x = k\pi$, $k \in \mathbb{Z}$. Since it is nonzero, for example, at $x = \pi/2$, the functions are linearly independent. The special theorem that an ODE Wronskian is either everywhere zero or nowhere zero concerns solutions of one common regular ODE of the matching order; arbitrary independent functions can have isolated Wronskian zeros, as this example demonstrates.

7.6.7 Consider two functions $\varphi_1 = x$ and $\varphi_2 = |x|$. Since $\varphi_1' = 1$ and $\varphi_2' = x/|x|$, $W(\varphi_1, \varphi_2) = 0$ for any interval, including $[-1, +1]$. Does the vanishing of the Wronskian over $[-1, +1]$ prove that φ_1 and φ_2 are linearly dependent? Clearly, they are not. What is wrong?

Solution.

For $x > 0$, $|x| = x$ and $W = x(1) - 1(x) = 0$. For $x < 0$, $|x| = -x$ and $W = x(-1) - 1(-x) = 0$. But $|x|$ has left derivative -1 and right derivative 1 at zero, so

$$\boxed{W(x, |x|) \text{ is undefined at } x = 0, \text{ not zero there.}}$$

This is the flaw the question asks us to find; the printed argument is an intentional trap. The continuous-derivative hypothesis needed for the usual theorem fails. Furthermore, neither function is nowhere zero on $[-1, 1]$, so the quotient argument cannot connect the constants on the two sides.

They are independent on that full interval: if $ax + b|x| = 0$, positive x gives $a + b = 0$ and negative x gives $a - b = 0$. Therefore $a = b = 0$. On either open half-interval separately they are dependent, with proportionality constants $+1$ and -1, respectively.

7.6.8 Explain that linear independence does not mean the absence of any dependence. Illustrate your argument with $\cosh x$ and e^x.

Solution.

Linear dependence requires a relation with constant coefficients, $a\cosh x + be^x = 0$, valid throughout an interval. Since $\cosh x = (e^x + e^{-x})/2$, it would imply

$$\left(\frac{a}{2} + b\right) e^{2x} + \frac{a}{2} = 0.$$

Differentiation gives $2(a/2 + b)e^{2x} = 0$, hence $a/2 + b = 0$; the original relation then gives $a/2 = 0$ and $b = 0$. Thus they are linearly independent. Their Wronskian checks this directly:

$$W(\cosh x, e^x) = e^x(\cosh x - \sinh x) = 1.$$

They nevertheless satisfy the nonlinear relation

$$\boxed{2\cosh x - e^x - \frac{1}{e^x} = 0}.$$

Writing $Y_1 = \cosh x$, $Y_2 = e^x$, this is $2Y_1 - Y_2 - Y_2^{-1} = 0$. Linear independence excludes constant linear relations only, not all functional or nonlinear relations.

7.6.9 Legendre's differential equation

$$(1 - x^2)y'' - 2xy' + n(n + 1)y = 0$$

has a regular solution $P_n(x)$ and an irregular solution $Q_n(x)$. Show that the Wronskian of P_n and Q_n is given by

$$P_n(x)Q_n'(x) - P_n'(x)Q_n(x) = \frac{A_n}{1 - x^2},$$

with A_n independent of x.

Solution.

Let $u = P_n$, $v = Q_n$ and $W = uv' - u'v$ on an interval not crossing $x = \pm 1$. Differentiate so the cross terms cancel:

$$W' = u'v' + uv'' - u''v - u'v' = uv'' - u''v.$$

The normalized Legendre equation gives $y'' = 2xy'/(1 - x^2) - n(n + 1)y/(1 - x^2)$. Substitution yields

$$W' = \frac{2x}{1 - x^2}(uv' - u'v) = \frac{2x}{1 - x^2}W.$$

Equivalently, $[(1 - x^2)W]' = -2xW + (1 - x^2)W' = 0$. Integrating directly gives

$$\boxed{W(P_n, Q_n) = \frac{A_n}{1 - x^2}},$$

with A_n constant on the interval. Its value depends on the normalization of the chosen solutions; the differential equation alone determines the x dependence.

7.6.10 Show, by means of the Wronskian, that a linear, second-order, homogeneous ODE of the form

$$y''(x) + P(x)y'(x) + Q(x)y(x) = 0$$

cannot have three independent solutions. Assume P, Q are continuous on a connected interval.

Hint. Assume a third solution and show that the Wronskian vanishes for all x.

Solution.

Work on a connected interval where P, Q are continuous, so the normalized second-order initial-value problem is unique. Suppose y_1, y_2, y_3 solve the equation. In their 3×3 Wronskian,

$$W_3 = \det \begin{pmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{pmatrix},$$

the third row equals $-Q(x)$ times the first row minus $P(x)$ times the second. Hence $W_3(x) = 0$ for every x.

To complete the argument without an invalid converse for arbitrary smooth functions, choose an ordinary point x_0. The two homogeneous equations

$$\sum_{j=1}^{3} c_j y_j(x_0) = 0, \qquad \sum_{j=1}^{3} c_j y_j'(x_0) = 0$$

have a nonzero constant solution (c_1, c_2, c_3) because there are three unknowns and only two equations. The function $u = \sum c_j y_j$ solves the same ODE and has $u(x_0) = u'(x_0) = 0$. Uniqueness forces $u \equiv 0$. This supplies a nontrivial constant linear relation, proving that three independent solutions cannot exist.

7.6.11 Show the following when the linear second-order differential equation $py'' + qy' + ry = 0$ is expressed in self-adjoint form: Use the coefficients after conversion, so $q = p'$, and work on an interval with $p \neq 0$; part (b) initially requires $y_1 \neq 0$.

(a) The Wronskian is equal to a constant divided by p:

$$W(x) = \frac{C}{p(x)}.$$

(b) A second solution $y_2(x)$ is obtained from a first solution $y_1(x)$ as

$$y_2(x) = C y_1(x) \int^x \frac{dt}{p(t)[y_1(t)]^2}.$$

Solution.

In self-adjoint form the equation is

$$(py')' + ry = 0, \qquad \text{so } py'' + p'y' + ry = 0.$$

Here p denotes the coefficient after any integrating factor has been applied; the self-adjoint condition is $q = p'$. Work where $p \neq 0$.

(a) For solutions y_1, y_2, subtract y_2 times the first equation from y_1 times the second. The terms involving r cancel:

$$p(y_1y_2'' - y_1''y_2) + p'(y_1y_2' - y_1'y_2) = 0.$$

Thus $pW' + p'W = (pW)' = 0$, and

$$\boxed{W = C/p}.$$

(b) On a subinterval where $y_1 \neq 0$,

$$\left(\frac{y_2}{y_1}\right)' = \frac{y_1y_2' - y_1'y_2}{y_1^2} = \frac{C}{py_1^2}.$$

Integration gives

$$\boxed{y_2 = y_1\left[D + C\int^x \frac{dt}{p(t)y_1(t)^2}\right]}.$$

The D term only repeats the known solution; set $D = 0$ and choose $C \neq 0$ for an independent second solution. The formula is initially local around points with $y_1 \neq 0$ and may then be continued by the ODE.

7.6.12 Transform our linear, second-order ODE

$$y'' + P(x)y' + Q(x)y = 0$$

with P continuously differentiable and Q continuous on the interval, by the substitution

$$y = z\exp\left[-\frac{1}{2}\int^x P(t)\,dt\right]$$

and show that the resulting differential equation for z is

$$z'' + q(x)z = 0,$$

where

$$q(x) = Q(x) - \frac{1}{2}P'(x) - \frac{1}{4}P^2(x).$$

Note. This substitution can be derived by the technique of Exercise 7.6.25.

Solution.

Set $E(x) = \exp[-\frac{1}{2}\int^x P(t)dt]$, so $y = Ez$ and

$$E' = -\frac{P}{2}E, \qquad E'' = \left(-\frac{P'}{2} + \frac{P^2}{4}\right)E.$$

The product rule gives

$$y' = E\left(z' - \frac{P}{2}z\right), \qquad y'' = E\left[z'' - Pz' + \left(-\frac{P'}{2} + \frac{P^2}{4}\right)z\right].$$

Substitute all three terms of the ODE:

$$\begin{aligned} y'' + Py' + Qy &= E\left[z'' - Pz' + \left(-\frac{P'}{2} + \frac{P^2}{4}\right)z + Pz' - \frac{P^2}{2}z + Qz\right] \\ &= E\left[z'' + \left(Q - \frac{P'}{2} - \frac{P^2}{4}\right)z\right]. \end{aligned}$$

Since E never vanishes, division by it proves

$$\boxed{z'' + qz = 0, \qquad q = Q - \tfrac{1}{2}P' - \tfrac{1}{4}P^2}.$$

This calculation requires P differentiable where the substitution is used, in addition to the usual existence assumptions on the coefficients.

7.6.13 Use the result of Exercise 7.6.12 to show that the replacement of $\varphi(r)$ by $r\varphi(r)$ may be expected to eliminate the first derivative from the Laplacian in spherical polar coordinates. See also Exercise 3.10.34.

Solution.

The radial part of the spherical Laplacian is

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d\varphi}{dr}\right) = \varphi'' + \frac{2}{r}\varphi', \qquad r > 0.$$

Thus $P(r) = 2/r$ and the factor in 7.6.12 is $\exp[-\frac{1}{2}\int(2/r)dr] = 1/r$, up to a constant. The new dependent variable is therefore $u = r\varphi$, or $\varphi = u/r$. Direct differentiation confirms the cancellation:

$$\varphi' = \frac{u'}{r} - \frac{u}{r^2}, \qquad \varphi'' = \frac{u''}{r} - \frac{2u'}{r^2} + \frac{2u}{r^3},$$

$$\boxed{\varphi'' + \frac{2}{r}\varphi' = \frac{u''}{r}}.$$

The potential correction from the substitution also vanishes: $-P'/2 - P^2/4 = 1/r^2 - 1/r^2 = 0$. If angular dependence is present, it is unaffected: $\nabla^2(u/r) = u_{rr}/r + \Delta_\Omega u/r^3$. The eliminated derivative is the first *radial* derivative; the origin requires its separate regularity condition.

7.6.14 By direct differentiation and substitution show that

$$y_2(x) = y_1(x)\int^x \frac{\exp\left[-\int^s P(t)\,dt\right]}{[y_1(s)]^2}\,ds$$

satisfies, like $y_1(x)$, the ODE

$$y_2''(x) + P(x)y_2'(x) + Q(x)y_2(x) = 0.$$

Note. The Leibniz formula for the derivative of an integral is

$$\frac{d}{d\alpha}\int_{g(\alpha)}^{h(\alpha)} f(x,\alpha)\,dx = \int_{g(\alpha)}^{h(\alpha)} \frac{\partial f(x,\alpha)}{\partial\alpha}\,dx + f[h(\alpha),\alpha]\frac{dh(\alpha)}{d\alpha} - f[g(\alpha),\alpha]\frac{dg(\alpha)}{d\alpha}.$$

Apply the integral formula on an interval with continuous P, Q and a nowhere-zero first solution y_1; choose all integration limits within that interval.

Solution.

On an interval where $y_1 \neq 0$, abbreviate

$$E = e^{-\int^x P(t)dt}, \qquad h = E/y_1^2, \qquad J = \int^x h(s)ds,$$

so $E' = -PE$, $J' = h$, and $y_2 = y_1 J$. Differentiate explicitly:

$$h' = -Ph - 2\frac{y_1'}{y_1}h, \quad y_2' = y_1' J + y_1 h, \quad y_2'' = y_1'' J + 2y_1' h + y_1 h'.$$

Substitution groups the terms as

$$\begin{aligned} y_2'' + Py_2' + Qy_2 &= J(y_1'' + Py_1' + Qy_1) + (2y_1' + Py_1)h + y_1 h' \\ &= 0 + (2y_1' + Py_1)h + y_1\left(-P - 2\frac{y_1'}{y_1}\right)h = 0. \end{aligned}$$

Thus the displayed expression is a solution. Its Wronskian with y_1 is $y_1(y_1' J + y_1 h) - y_1'(y_1 J) = y_1^2 h = E \neq 0$, independently confirming that it is a second, independent solution.

7.6.15 In the equation

$$y_2(x) = y_1(x)\int^x \frac{\exp\left[-\int^s P(t)\,dt\right]}{[y_1(s)]^2}\,ds,$$

$y_1(x)$ satisfies

$$y_1'' + P(x)y_1' + Q(x)y_1 = 0.$$

The function $y_2(x)$ is a linearly independent second solution of the same equation. Show that the inclusion of lower limits on the two integrals leads to nothing new, that is, that it generates only an overall constant factor and a constant multiple of the known solution $y_1(x)$. Apply the integral formula on an interval with continuous P, Q and a nowhere-zero first solution y_1; choose all integration limits within that interval.

Solution.

Distinguish the inner and outer lower limits by defining

$$E_a(s) = \exp\left[-\int_a^s P(t)dt\right], \qquad Y_{a,c}(x) = y_1(x)\int_c^x \frac{E_a(s)}{y_1(s)^2}ds.$$

For another inner lower limit b,

$$\int_a^s P = \int_b^s P - \int_b^a P, \qquad E_a(s) = KE_b(s), \quad K = \exp\left(\int_b^a P(t)dt\right) \neq 0.$$

Changing the outer lower limit from c to d then gives

$$\begin{aligned} Y_{a,c} &= Ky_1\left[\int_d^x \frac{E_b(s)}{y_1(s)^2}ds - \int_d^c \frac{E_b(s)}{y_1(s)^2}ds\right] \\ &= \boxed{KY_{b,d} + Dy_1}, \qquad D = -K\int_d^c \frac{E_b(s)}{y_1(s)^2}ds. \end{aligned}$$

Both K and D are constants. Hence lower limits only rescale the second solution and add a multiple of the first. The limits must lie in an interval where the integrals and the nonzero-denominator formula are well defined.

7.6.16 Given that one solution of

$$R'' + \frac{1}{r}R' - \frac{m^2}{r^2}R = 0$$

is $R = r^m$, show that Eq. (7.67) predicts a second solution, $R = r^{-m}$ for $m \neq 0$, and $R = \ln r$ for $m = 0$, on $r > 0$.

Solution.

On $r > 0$, $P = 1/r$ and $e^{-\int P dr} = 1/r$, up to a constant. With $R_1 = r^m$, reduction of order gives

$$R_2 = r^m \int^r \frac{ds/s}{s^{2m}} = r^m \int^r s^{-2m-1} ds.$$

For $m \neq 0$,

$$R_2 = -\frac{1}{2m} r^{-m},$$

so rescaling yields the requested independent solution $\boxed{R_2 = r^{-m}}$. Direct substitution gives $m(m+1)r^{-m-2} - mr^{-m-2} - m^2 r^{-m-2} = 0$. When $m = 0$, $r^m = r^{-m} = 1$ does not provide two independent solutions. The integral becomes $\int ds/s$, and the second solution is $\ln r$. Its derivatives $1/r, -1/r^2$ satisfy the $m = 0$ equation.

7.6.17 Using

$$y_1(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

as a solution of the linear oscillator equation, follow the analysis that proceeds through Eq. (7.81) and show that in that equation $c_n = 0$, so that in this case the second solution does not contain a logarithmic term.

Solution.

The given first solution is $y_1 = \sin x = xA(x^2)$ with $A(0) = 1$; the oscillator equation has $P = 0$. Its reduction-of-order integrand is

$$\frac{1}{y_1^2} = \frac{1}{x^2}\frac{1}{A(x^2)^2} = \frac{1}{x^2}\sum_{j\geq 0} b_j x^{2j} = \sum_{j\geq 0} b_j x^{2j-2}.$$

The reciprocal series exists near zero because $A(0) \neq 0$. A logarithm can arise on termwise integration only from a power x^{-1}. But $2j - 2 = -1$ would require $j = 1/2$, not an integer. Thus the logarithm coefficient denoted c_n in the referenced analysis is zero. Indeed,

$$y_2 = \sin x \int^x \frac{ds}{\sin^2 s} = \sin x(-\cot x) = -\cos x,$$

up to a multiple of $\sin x$. The Taylor series $-1 + x^2/2! - x^4/4! + \cdots$ contains no logarithm and verifies the coefficient argument directly.

7.6.18 Show that when n is not an integer in Bessel's ODE, Eq. (7.40), the second solution of Bessel's equation, obtained from Eq. (7.67), does not contain a logarithmic term.

Solution.

For a real noninteger order set $\nu = |n|$; for a complex noninteger order the same proof works with $\nu = n$. Bessel's equation depends on n^2, and the Frobenius solution with exponent ν has the form

$$y_1 = x^\nu A(x^2), \qquad A(0) \neq 0,$$

with only even increments in its Frobenius series. Since $P = 1/x$,

$$\frac{e^{-\int P dx}}{y_1^2} = x^{-2\nu-1}\frac{1}{A(x^2)^2} = \sum_{k\geq 0} b_k x^{2k-2\nu-1}.$$

A logarithm could arise only if $2k - 2\nu - 1 = -1$, or $k = \nu$. There is no such nonnegative integer k when ν is noninteger. Termwise integration therefore gives

$$y_2 = x^\nu A(x^2)\sum_{k\geq 0}\frac{b_k}{2k-2\nu}x^{2k-2\nu} = x^{-\nu} \times \text{a power series in } x^2,$$

with no logarithm. The power-series integrations are local, and a consistent branch of x^ν is used for complex x. The leading coefficient is nonzero, so this is independent of $x^\nu A(x^2)$ and proportional, up to addition of that first solution, to $J_{-\nu}$. This also explains why half-integer orders, though their indicial roots differ by an integer, still produce no logarithm: parity prevents the resonant x^{-1} term.

7.6.19

(a) One solution of Hermite's differential equation

$$y'' - 2xy' + 2\alpha y = 0$$

for $\alpha = 0$ is $y_1(x) = 1$. Find a second solution, $y_2(x)$, using Eq. (7.67). Show that your second solution is equivalent to y_{odd} (Exercise 8.3.3).

(b) Find a second solution for $\alpha = 1$, where $y_1(x) = x$, using Eq. (7.67). Show that your second solution is equivalent to y_{even} (Exercise 8.3.3).

Solution.

For Hermite's equation $P = -2x$, so $e^{-\int P dx} = e^{x^2}$.

(a) When $\alpha = 0$ and $y_1 = 1$,

$$\boxed{y_2 = \int_0^x e^{t^2}\,dt = \sum_{k=0}^{\infty}\frac{x^{2k+1}}{k!(2k+1)}}.$$

The exponential series converges uniformly on finite intervals, justifying its termwise integral. This is the normalized odd solution: $y_2(0) = 0$, $y_2'(0) = 1$, and $y_2'' - 2xy_2' = 0$ directly.

(b) With $\alpha = 1$ and $y_1 = x$, reduction of order gives $x\int^x e^{t^2}/t^2\,dt$. Differentiate e^{t^2}/t:

$$\frac{d}{dt}\left(\frac{e^{t^2}}{t}\right) = 2e^{t^2} - \frac{e^{t^2}}{t^2}, \quad \int^x \frac{e^{t^2}}{t^2}dt = -\frac{e^{x^2}}{x} + 2\int_0^x e^{t^2}\,dt.$$

Multiplying by x gives a solution with value -1 at zero. Reverse its sign to normalize the even solution to one:

$$\boxed{y_{\text{even}} = e^{x^2} - 2x \int_0^x e^{t^2} dt}.$$

Its series is

$$\begin{aligned} y_{\text{even}} &= \sum_{k\geq 0} \frac{x^{2k}}{k!} - 2\sum_{k\geq 0} \frac{x^{2k+2}}{k!(2k+1)} \\ &= 1 + \sum_{k\geq 1} \left[\frac{1}{k!} - \frac{2}{(k-1)!(2k-1)}\right] x^{2k} \\ &= 1 - \sum_{k\geq 1} \frac{x^{2k}}{k!(2k-1)} = 1 - x^2 - \frac{x^4}{6} - \frac{x^6}{30} - \cdots . \end{aligned}$$

This is the even Hermite series at $\alpha = 1$. Setting $J = \int_0^x e^{t^2} dt$ gives $y' = -2J$, $y'' = -2e^{x^2}$, and $y'' - 2xy' + 2y = -2e^{x^2} + 4xJ + 2(e^{x^2} - 2xJ) = 0$. The integral construction is initially used on $x \neq 0$; the displayed even function extends smoothly through zero.

7.6.20 One solution of Laguerre's differential equation

$$xy'' + (1-x)y' + ny = 0$$

for $n = 0$ is $y_1(x) = 1$. Using Eq. (7.67), develop a second, linearly independent solution. Exhibit the logarithmic term explicitly.

Solution.

Normalize the $n = 0$ equation to $y'' + (1/x - 1)y' = 0$ on $x \neq 0$. Since $y_1 = 1$,

$$e^{-\int (1/x - 1)dx} = \frac{e^x}{x}$$

up to a nonzero constant on either real half-line. Hence

$$y_2 = \int^x \frac{e^s}{s} ds.$$

Expand $e^s/s = s^{-1} + \sum_{k\geq 1} s^{k-1}/k!$ and integrate:

$$\boxed{y_2 = \ln|x| + \sum_{k=1}^{\infty} \frac{x^k}{k\,k!}}.$$

An additive constant repeats y_1 and is omitted. The series factor converges for all finite x, by its factorial denominator. Its derivative is e^x/x, so $W(1, y_2) = e^x/x \neq 0$, proving independence. The logarithmic singularity at zero is explicit.

7.6.21 For Laguerre's equation with $n = 0$,

$$y_2(x) = \int^x \frac{e^s}{s}\, ds.$$

(a) Write $y_2(x)$ as a logarithm plus a power series.

(b) Verify that the integral form of $y_2(x)$, previously given, is a solution of Laguerre's equation ($n = 0$) by direct differentiation of the integral and substitution into the differential equation.

(c) Verify that the series form of $y_2(x)$, part (a), is a solution by differentiating the series and substituting back into Laguerre's equation.

Solution.

(a) Separate the integrand's pole before integrating:

$$\frac{e^s}{s} = \frac{1}{s} + \sum_{k=1}^{\infty} \frac{s^{k-1}}{k!}, \qquad \boxed{y_2 = \ln|x| + \sum_{k=1}^{\infty} \frac{x^k}{k\,k!} + C}, \quad x \neq 0.$$

The factorial series and its derivatives converge locally uniformly, so these integrations and differentiations are justified away from zero.

(b) The fundamental theorem gives $y_2' = e^x/x$, and the quotient rule gives $y_2'' = e^x/x - e^x/x^2$. Consequently

$$xy_2'' + (1-x)y_2' = e^x - \frac{e^x}{x} + \frac{e^x}{x} - e^x = 0.$$

(c) Differentiate the series explicitly:

$$y_2' = \frac{1}{x} + \sum_{k\geq 1} \frac{x^{k-1}}{k!}, \qquad y_2'' = -\frac{1}{x^2} + \sum_{k\geq 1} \frac{(k-1)x^{k-2}}{k!}.$$

Then

$$\begin{aligned} xy_2'' + (1-x)y_2' &= -1 + \sum_{k\geq 1} \frac{kx^{k-1}}{k!} - \sum_{k\geq 1} \frac{x^k}{k!} \\ &= -1 + e^x - (e^x - 1) = 0. \end{aligned}$$

This verifies the series independently of recognizing it as the integral.

7.6.22 One solution of the Chebyshev equation

$$(1-x^2)y'' - xy' + n^2 y = 0$$

for $n = 0$ is $y_1 = 1$.

(a) Using Eq. (7.67), develop a second, linearly independent solution.

(b) Find a second solution by direct integration of the Chebyshev equation.

Hint. Let $v = y'$ and integrate. Compare your result with the second solution given in Section 18.4.

ANS. (a) $y_2 = \sin^{-1} x$ on $-1 < x < 1$. (b) The second solution, $V_n(x)$, is not defined for $n = 0$ under the nonzero-n convention of Section 18.4.

Solution.

Work on $-1 < x < 1$. (a) The normalized coefficient is $P = -x/(1 - x^2)$, so

$$\int P dx = \tfrac{1}{2}\ln(1 - x^2), \qquad e^{-\int P dx} = (1 - x^2)^{-1/2}.$$

Since $y_1 = 1$, reduction of order yields

$$\boxed{y_2 = \int_0^x \frac{ds}{\sqrt{1 - s^2}} = \arcsin x}.$$

Its Wronskian with 1 is $1/\sqrt{1 - x^2} \neq 0$.

(b) Put $v = y'$ in the $n = 0$ equation:

$$(1 - x^2)v' - xv = 0, \quad \frac{dv}{v} = \frac{x\,dx}{1 - x^2}, \quad \ln|v| = -\tfrac{1}{2}\ln(1 - x^2) + C.$$

Therefore $v = A/\sqrt{1 - x^2}$ and $\boxed{y = A\arcsin x + B}$, reproducing the same pair.

Section 18.4 explicitly introduces $V_n(x) = \sin(n \arccos x)$ for $n \neq 0$, so the printed answer follows its convention. The sine expression has a continuous extension to $n = 0$, but this gives the zero function, which cannot be an independent second solution. The expression $V_n = \sqrt{1 - x^2}\,U_{n-1}$ also falls outside the usual nonnegative polynomial indexing when $n = 0$. A nonzero limiting second solution is $\lim_{n\to 0} V_n(x)/n = \arccos x = \pi/2 - \arcsin x$, differing only by a sign and a multiple of the first solution.

7.6.23 One solution of the Chebyshev equation

$$(1 - x^2)y'' - xy' + n^2 y = 0$$

for $n = 1$ is $y_1(x) = x$. Set up the Wronskian double integral solution and derive a second solution, $y_2(x)$.

ANS. $y_2 = -(1 - x^2)^{1/2}$.

Solution.

On $-1 < x < 1$, use $P = -x/(1 - x^2)$ and $y_1 = x$. Reduction of order gives, initially on either side of zero,

$$y_2 = x\int^x \frac{\exp[-\int^s (-t/(1 - t^2))dt]}{s^2} ds = x\int^x \frac{ds}{s^2\sqrt{1 - s^2}}.$$

The antiderivative follows directly from the product rule:

$$\frac{d}{ds}\left(-\frac{\sqrt{1 - s^2}}{s}\right) = \frac{1}{\sqrt{1 - s^2}} + \frac{\sqrt{1 - s^2}}{s^2} = \frac{1}{s^2\sqrt{1 - s^2}}.$$

Thus

$$\boxed{y_2 = -\sqrt{1 - x^2}},$$

up to a multiple of x. Its derivatives are $y_2' = x/\sqrt{1 - x^2}$ and $y_2'' = (1 - x^2)^{-3/2}$. Substitution gives $(1 - x^2)y_2'' - xy_2' + y_2 = (1 - x^2)/\sqrt{1 - x^2} - \sqrt{1 - x^2} = 0$. Finally $W(x, y_2) = 1/\sqrt{1 - x^2} \neq 0$. Although the reduction integrand was singular at zero, the final solution and this Wronskian extend there regularly.

7.6.24 The radial Schrödinger wave equation for a spherically symmetric potential can be written in the form

$$\left[-\frac{\hbar^2}{2m}\frac{d^2}{dr^2}+l(l+1)\frac{\hbar^2}{2mr^2}+V(r)\right]y(r)=Ey(r).$$

Take $l=0,1,2,\ldots$ and $r>0$. The potential energy $V(r)$ may be expanded about the origin as

$$V(r)=\frac{b_{-1}}{r}+b_0+b_1r+\cdots.$$

(a) Show that there is one (regular) solution $y_1(r)$ starting with r^{l+1}.

(b) From Eq. (7.69) show that the irregular solution $y_2(r)$ has leading behavior r^{-l}, diverging for $l\geq 1$ but approaching a nonzero constant for $l=0$.

Solution.

Let $\kappa=2m/\hbar^2$ and take the physical $l=0,1,2,\ldots$. The equation is

$$y''-\frac{l(l+1)}{r^2}y+\kappa[E-V(r)]y=0.$$

(a) The potential has at most a simple pole, so the leading r^{s-2} coefficient of a Frobenius trial is

$$[s(s-1)-l(l+1)]a_0=(s-l-1)(s+l)a_0=0.$$

The roots are $l+1$ and $-l$. For $s=l+1$, the coefficient of a_j in the recurrence is

$$(s+j)(s+j-1)-l(l+1)=j(2l+j+1)>0\quad(j\geq 1).$$

In detail, with negative-index coefficients zero,

$$j(2l+j+1)a_j=\kappa\left[\sum_{q=-1}^{j-2}b_qa_{j-2-q}-Ea_{j-2}\right].$$

Thus an arbitrary nonzero a_0 determines every following coefficient; for example $a_1=\kappa b_{-1}a_0/[2(l+1)]$. The regular solution is $\boxed{y_1=a_0r^{l+1}[1+O(r)]}$.

(b) There is no first derivative, so reduction of order is $y_2=y_1\int^r ds/y_1(s)^2$. Its leading terms are

$$\frac{1}{y_1(s)^2}=a_0^{-2}s^{-2l-2}[1+O(s)],\quad \int^r\frac{ds}{y_1(s)^2}\sim-\frac{a_0^{-2}}{2l+1}r^{-2l-1}.$$

Multiplying gives

$$\boxed{y_2(r)\sim-\frac{1}{a_0(2l+1)}r^{-l}}.$$

For $l\geq 1$ this diverges, as stated. For $l=0$, however, y_2 tends to a nonzero constant; it does not diverge. The physical radial function $R=y/r$ still behaves as $1/r$ and is irregular. For $l=0$ the next terms may contain $r\ln r$, which still tend to zero and do not change the constant leading behavior.

7.6.25 Show that if a second solution, y_2, is assumed to be related to the first solution, y_1, according to $y_2(x) = y_1(x)f(x)$, substitution back into the original equation

$$y_2'' + P(x)y_2' + Q(x)y_2 = 0$$

leads to

$$f(x) = \int^x \frac{\exp\left[-\int^s P(t)\,dt\right]}{[y_1(s)]^2}\,ds,$$

in agreement with Eq. (7.67). Apply the integral formula on an interval with continuous P, Q and a nowhere-zero first solution y_1; choose all integration limits within that interval.

Solution.

Set $y_2 = y_1 f$. Two differentiations give

$$y_2' = y_1' f + y_1 f', \qquad y_2'' = y_1'' f + 2y_1' f' + y_1 f''.$$

Substitution and use of the equation for y_1 leave

$$0 = f(y_1'' + Py_1' + Qy_1) + (2y_1' + Py_1)f' + y_1 f'' = y_1 f'' + (2y_1' + Py_1)f'.$$

On an interval with $y_1 \neq 0$, let $g = f'$; then

$$g' + \left(2\frac{y_1'}{y_1} + P\right) g = 0, \qquad \frac{g'}{g} = -2\frac{y_1'}{y_1} - P.$$

Integrating and exponentiating gives $g = Ce^{-\int P dx}/y_1^2$, and a second integration gives

$$\boxed{f = D + C\int^x \frac{e^{-\int^s P(t)dt}}{y_1(s)^2}ds}.$$

Here $C = 0$ recovers the constant-f branch excluded by division by g. For an independent second solution choose $C \neq 0$, and omit the Dy_1 contribution. Taking $C = 1, D = 0$ gives exactly the formula in the question; its Wronskian is $e^{-\int P dx} \neq 0$.

7.6.26 Use $x > 0$ and $a_0 \neq 0$; the primed labels in parts (b),(c) name limiting solutions, not derivatives.

(a) Show that

$$y'' + \frac{1-\alpha^2}{4x^2}y = 0$$

has two solutions:

$$y_1(x) = a_0 x^{(1+\alpha)/2}, \quad y_2(x) = a_0 x^{(1-\alpha)/2}.$$

(b) For $\alpha = 0$ the two linearly independent solutions of part (a) reduce to the single solution $y_1' = a_0 x^{1/2}$. Using Eq. (7.68) derive a second solution,

$$y_2'(x) = a_0 x^{1/2} \ln x.$$

Verify that y_2' is indeed a solution.

(c) Show that the second solution from part (b) may be obtained as a limiting case from the two solutions of part (a):

$$y_2'(x) = \lim_{\alpha\to 0}\left(\frac{y_1 - y_2}{\alpha}\right).$$

Solution.

Take $x > 0$ and $a_0 \neq 0$. (a) For a trial $y = a_0 x^s$,

$$y'' + \frac{1-\alpha^2}{4x^2}y = a_0 x^{s-2}\left[s(s-1) + \frac{1-\alpha^2}{4}\right].$$

The quadratic factors as $[s-(1+\alpha)/2][s-(1-\alpha)/2]$, so both stated powers solve the equation. Their Wronskian is

$$W = a_0^2\left(\frac{1-\alpha}{2} - \frac{1+\alpha}{2}\right)x^{(1+\alpha)/2+(1-\alpha)/2-1} = -\alpha a_0^2.$$

They are independent precisely when $\alpha \neq 0$.

(b) At $\alpha = 0$ the common solution is $u_1 = a_0\sqrt{x}$; here u_1, u_2 denote the primed labels in the question, not derivatives. Since $P = 0$,

$$u_1\int^x \frac{ds}{u_1(s)^2} = a_0\sqrt{x}\int^x \frac{ds}{a_0^2 s} = \frac{1}{a_0}\sqrt{x}\ln x.$$

An arbitrary nonzero rescaling gives $\boxed{u_2 = a_0\sqrt{x}\ln x}$. Explicitly,

$$u_2' = a_0 x^{-1/2}(1 + \tfrac{1}{2}\ln x), \qquad u_2'' = -\frac{a_0}{4}x^{-3/2}\ln x,$$

so $u_2'' + u_2/(4x^2) = 0$. Also $W(u_1, u_2) = a_0^2 \neq 0$.

(c) Write the powers as exponentials:

$$\frac{y_1 - y_2}{\alpha} = a_0\sqrt{x}\frac{e^{(\alpha/2)\ln x} - e^{-(\alpha/2)\ln x}}{\alpha}.$$

Since $e^{hL} = 1 + hL + \frac{1}{2}h^2L^2 + O(h^3)$, the equal quadratic terms cancel and the numerator is $\alpha\ln x + O(\alpha^3)$, and hence

$$\boxed{\lim_{\alpha\to 0}\frac{y_1 - y_2}{\alpha} = a_0\sqrt{x}\ln x = u_2}.$$

The manual drops a_0 and gives an extra factor $1/2$ in this limit. The explicit expansion above retains a_0 and shows that no extra $1/2$ is present.

7.7 Inhomogeneous Linear ODEs

7.7.1 If our linear, second-order ODE is inhomogeneous, that is, of the form of Eq. (7.94), the most general solution is

$$y(x) = C_1 y_1(x) + C_2 y_2(x) + y_p(x),$$

where y_1 and y_2 are independent solutions of the homogeneous equation, on a connected interval where the normalized coefficients and source are continuous. Show that

$$y_p(x) = y_2(x)\int^x \frac{y_1(s)F(s)\,ds}{W\{y_1(s), y_2(s)\}} - y_1(x)\int^x \frac{y_2(s)F(s)\,ds}{W\{y_1(s), y_2(s)\}},$$

with $W\{y_1(x), y_2(x)\}$ the Wronskian of $y_1(s)$ and $y_2(s)$.

Solution.

For fixed independent homogeneous solutions, the general solution is $y = C_1y_1+C_2y_2+y_p$; the two arbitrary coefficients must be explicit. Write the normalized equation as $L[y] = y'' + Py' + Qy = F$ and seek $y_p = u_1y_1 + u_2y_2$. Impose the auxiliary condition $u_1'y_1 + u_2'y_2 = 0$. Then

$$y_p' = u_1y_1' + u_2y_2', \qquad y_p'' = u_1y_1'' + u_2y_2'' + u_1'y_1' + u_2'y_2'.$$

Because $L[y_1] = L[y_2] = 0$, the remaining equation is $u_1'y_1' + u_2'y_2' = F$. Thus

$$\begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} \begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = \begin{pmatrix} 0 \\ F \end{pmatrix}.$$

Its determinant is $W = y_1y_2' - y_1'y_2$, nonzero on the regular interval. Cramer's rule gives

$$u_1' = -\frac{y_2F}{W}, \qquad u_2' = \frac{y_1F}{W}.$$

Integrate and substitute:

$$\boxed{y_p = y_2(x)\int^x \frac{y_1(s)F(s)}{W(s)}ds - y_1(x)\int^x \frac{y_2(s)F(s)}{W(s)}ds}.$$

Constants in the integrals contribute only multiples of y_1, y_2. The calculation of $L[y_p]$ above gives $u_1'y_1' + u_2'y_2' = F$, checking the formula directly. Finally any other solution minus this particular solution satisfies the homogeneous equation, proving that $C_1y_1 + C_2y_2 + y_p$ is the general solution.

Find the general solutions to the following inhomogeneous ODEs:

7.7.2 $y'' + y = 1$.

Solution.

The homogeneous characteristic equation is $r^2 + 1 = 0$, giving $y_h = C_1\cos x + C_2\sin x$. For a constant particular solution $y_p = A$, the equation requires $0 + A = 1$, so $A = 1$. Therefore

$$\boxed{y = C_1\cos x + C_2\sin x + 1}.$$

For a variation-of-parameters check, take $y_1 = \cos x$, $y_2 = \sin x$, $W = 1$, $F = 1$. Then $u_1' = -\sin x$, $u_2' = \cos x$, so one choice is $u_1 = \cos x$, $u_2 = \sin x$. Their combination is $y_p = \cos^2 x + \sin^2 x = 1$, agreeing with the constant trial. The manual labels both derivatives u_1' in the next sentence of its calculation; the second must be $u_2' = \cos x$, as the two equations above show.

7.7.3 $y'' + 4y = e^x$.

Solution.

The homogeneous roots satisfy $r^2+4 = 0$, so the real homogeneous solution is $C_1\cos 2x+C_2\sin 2x$. Try $y_p = Ae^x$; then

$$y_p'' + 4y_p = (A + 4A)e^x = 5Ae^x.$$

Matching the right side sets $A = 1/5$. Thus

$$\boxed{y = C_1\cos 2x + C_2\sin 2x + \frac{1}{5}e^x}.$$

The homogeneous Wronskian is 2, so the two constants are independent. Differentiating the displayed solution twice verifies that its trigonometric contributions cancel and its exponential contribution is exactly e^x. In the manual's variation-of-parameters calculation, $u_2' = e^x\cos(2x)/2$, not $e^x\sin(2x)/2$: Cramer's rule gives $y_1F/W = \cos(2x)e^x/2$. Its printed antiderivative and final answer agree with this corrected derivative.

7.7.4 $y'' - 3y' + 2y = \sin x$.

Solution.

The homogeneous characteristic polynomial is $r^2-3r+2 = (r-1)(r-2)$, giving $y_h = C_1e^x+C_2e^{2x}$. For $y_p = A\sin x + B\cos x$,

$$y_p' = A\cos x - B\sin x, \qquad y_p'' = -A\sin x - B\cos x.$$

Hence

$$y_p'' - 3y_p' + 2y_p = (A+3B)\sin x + (B-3A)\cos x.$$

Matching coefficients gives $A+3B=1$, $B-3A=0$; hence $B=3A$, $10A=1$, $A=1/10$, $B=3/10$. The general solution is

$$\boxed{y = C_1e^x + C_2e^{2x} + \frac{\sin x + 3\cos x}{10}}.$$

The coefficient equations above are also a direct residual check of the particular solution; the homogeneous Wronskian is $e^{3x} \neq 0$.

7.7.5 $xy'' - (1+x)y' + y = x^2$.

Solution.

First find homogeneous solutions. For a linear trial $y = ax+b$, $xy'' - (1+x)y' + y = b - a$, so $y_1 = x+1$ is a solution. The exponential $y_2 = e^x$ also works, since $xe^x - (1+x)e^x + e^x = 0$. Their Wronskian is

$$W = (x+1)e^x - e^x = xe^x,$$

nonzero on either interval $x > 0$ or $x < 0$. After division by x, the source term for variation of parameters is $F = x$, not x^2. Therefore

$$u_1' = -\frac{e^x x}{xe^x} = -1, \qquad u_2' = \frac{(x+1)x}{xe^x} = (x+1)e^{-x}.$$

One choice of antiderivatives is $u_1 = -x$, $u_2 = -(x+2)e^{-x}$, giving

$$y_p = -x(x+1) - (x+2) = -x^2 - 2x - 2.$$

Since $-2(x+1)$ is homogeneous, it can be absorbed into its arbitrary coefficient; take simply $y_p = -x^2$. Thus

$$\boxed{y = C_1(x+1) + C_2e^x - x^2}.$$

The direct check for the particular solution is $x(-2) - (1+x)(-2x) - x^2 = x^2$. The formula extends smoothly through zero, although the normalized ODE and the Wronskian argument are applied on intervals away from zero. It also describes every C^2 solution across zero: write the formula with constants $C_1^{\pm}, C_2^{\pm}$ on the two sides. Continuity of y'' gives $C_2^- = C_2^+$ because $y''(0) = C_2 - 2$; continuity of y then gives $C_1^- = C_1^+$. Thus no additional C^2 branch is lost at zero.

7.8 Nonlinear Differential Equations

7.8.1 Consider the Riccati equation $y' = y^2 - y - 2$. A particular solution to this equation is $y = 2$. Find a more general solution.

Solution.

Use the given solution by setting $y = 2 + u$. Then

$$u' = (2+u)^2 - (2+u) - 2 = u^2 + 3u.$$

The branch $u = 0$ gives the original solution $y = 2$. For $u \neq 0$, put $v = 1/u$; then

$$v' = -\frac{u'}{u^2} = -1 - 3v, \qquad (e^{3x}v)' = -e^{3x}.$$

Integrating gives $v = Ce^{-3x} - 1/3$. Renaming $3C$ as K,

$$\boxed{y = 2 + \frac{3}{Ke^{-3x} - 1}}, \qquad \boxed{y \equiv 2}.$$

The constant solution $y = -1$ is already included at $K = 0$. Every nonconstant expression is used on a maximal interval where its denominator is nonzero. To check it, write $D = Ke^{-3x} - 1$: $u' = 9Ke^{-3x}/D^2$, while $u^2 + 3u = 9(1+D)/D^2 = 9Ke^{-3x}/D^2$. The integrating-constant formula for v must have the particular term $-1/3$; the manual's intermediate $+1/3$ does not satisfy $v' + 3v = -1$.

7.8.2 A particular solution to $y' = y^2/x^3 - y/x + 2x$ is $y = x^2$. Find a more general solution.

Solution.

Work on $x \neq 0$ and set $y = x^2 + u$. Expanding every term gives

$$2x + u' = \frac{x^4 + 2x^2u + u^2}{x^3} - \frac{x^2 + u}{x} + 2x = 2x + \frac{u}{x} + \frac{u^2}{x^3}.$$

Thus $u' = u/x + u^2/x^3$. The branch $u = 0$ is $y = x^2$. For $u \neq 0$, let $v = 1/u$:

$$v' = -\frac{v}{x} - \frac{1}{x^3}, \qquad v' + \frac{v}{x} = -\frac{1}{x^3}, \qquad (xv)' = -\frac{1}{x^2}.$$

Integrating gives $xv = 1/x + C$ and $v = (1 + Cx)/x^2$. Therefore

$$\boxed{y = x^2 + \frac{x^2}{1 + Cx}}, \qquad \boxed{y = x^2}.$$

The first family is valid where $x \neq 0$ and $1 + Cx \neq 0$; the second is the lost $u = 0$ branch. For a check, $u = x^2/(1+Cx)$ gives $u' = (2x + Cx^2)/(1+Cx)^2 = u/x + u^2/x^3$. The manual's intermediate linear equation is missing the factor $1/x$ multiplying v; its displayed final solution agrees with the corrected equation derived here.

7.8.3 Solve the Bernoulli equation $y' + xy = xy^3$.

Solution.

The solution $y = 0$ must be recorded before dividing by a power of y. For a nonzero solution set $u = y^{-2}$. Then

$$u' = -2y^{-3}y' = -2y^{-3}(xy^3 - xy) = -2x + 2xu.$$

Thus

$$u' - 2xu = -2x, \qquad (e^{-x^2}u)' = -2xe^{-x^2} = (e^{-x^2})'.$$

After integration $e^{-x^2}u = e^{-x^2} + C$, so $u = 1 + Ce^{x^2}$. The real solutions are

$$\boxed{y = 0 \quad \text{or} \quad y = \pm\frac{1}{\sqrt{1 + Ce^{x^2}}}}.$$

Both signs and the zero solution are needed; taking only the positive square root loses real solutions. Choose a connected interval where $1 + Ce^{x^2} > 0$ and a fixed sign. In particular $C = 0$ gives $y = \pm 1$. Differentiation gives $y' = -xCe^{x^2}y/(1 + Ce^{x^2})$, whereas $xy^3 - xy = xy[(1 + Ce^{x^2})^{-1} - 1]$ is the same expression.

7.8.4 ODEs of the form $y = xy' + f(y')$ are known as Clairaut equations. The first step in solving an equation of this type is to differentiate it, assuming a twice differentiable solution, yielding

$$y' = y' + xy'' + f'(y')y'', \quad \text{or } y''\left[x + f'(y')\right] = 0.$$

Solutions may therefore be obtained both from $y'' = 0$ and from $f'(y') = -x$. The so-called general solution comes from $y'' = 0$.

For $f(y') = (y')^2$,

(a) Obtain the general solution (note that it contains a single constant).

(b) Obtain the so-called singular solution from $f'(y') = -x$. By substituting back into the original ODE show that this singular solution contains no adjustable constants.

Note. The singular solution is the envelope of the general solutions.

Solution.

For $f(p) = p^2$, differentiating the equation gives $y''(x + 2y') = 0$.

(a) On the branch $y'' = 0$, write $y = ax + b$ and $y' = a$. Substitute into the original, rather than only its derivative:

$$ax + b = xa + a^2 \quad \Longrightarrow \quad b = a^2.$$

Hence the one-parameter family is

$$\boxed{y = ax + a^2}, \qquad a \text{ arbitrary}.$$

(b) On the other branch $x + 2y' = 0$, so $y' = -x/2$ and $y = -x^2/4 + C$. The original equation requires

$$-\frac{x^2}{4} + C = x\left(-\frac{x}{2}\right) + \left(-\frac{x}{2}\right)^2 = -\frac{x^2}{4},$$

so $C = 0$ and

$$\boxed{y = -x^2/4}.$$

To verify the envelope statement, eliminate the parameter a from $y = ax + a^2$ and its parameter derivative $0 = x + 2a$. This gives $a = -x/2$ and $y = -x^2/4$. Each line touches the parabola at $x = -2a$: both values there are $-a^2$ and both slopes are a.

If only C^1 solutions of the original first-order equation are required, the two types can also be joined at a contact point. For example, $y = 0$ for $x \le 0$ and $y = -x^2/4$ for $x \ge 0$ has continuous value and derivative at zero, and each branch satisfies $y = xy' + (y')^2$. The second derivatives there are 0 and $-1/2$, so this joined solution lies outside the differentiation argument. The requested line family and singular curve are the standard smooth branches; they do not exhaust such C^1 joins.

8 Chapter 8. Sturm-Liouville Theory

Problems

8.2 Hermitian Operators

8.2.1 Show that Laguerre's ODE, Table 7.1, may be put into self-adjoint form by multiplying by e^{-x} and that $w(x) = e^{-x}$ is the weighting function.

Solution.

Write the equation as $xy'' + (1-x)y' + ny = 0$. Multiplication by $w = e^{-x}$ gives

$$xe^{-x}y'' + (1-x)e^{-x}y' + ne^{-x}y = 0.$$

The product rule gives $(xe^{-x})' = (1-x)e^{-x}$, so the first two terms combine without changing the eigenvalue:

$$\boxed{(xe^{-x}y')' + ne^{-x}y = 0}, \qquad p(x) = xe^{-x}, \quad w(x) = e^{-x}.$$

Thus $Ay = -w^{-1}(py')' = -xy'' - (1-x)y'$ has eigenvalue n. For functions in a common domain, integrate once in each scalar product:

$$\langle u, Av\rangle_w = -\int_a^b u^*(pv')'\,dx = -[u^*pv']_a^b + \int_a^b p\,u'^*v'\,dx,$$
$$\langle Au, v\rangle_w = -\int_a^b (pu'^*)'v\,dx = -[pu'^*v]_a^b + \int_a^b p\,u'^*v'\,dx.$$

Subtracting cancels the identical integrals and gives

$$\langle u, Av\rangle_w - \langle Au, v\rangle_w = -\big[p(u^*v' - u'^*v)\big]_a^b, \qquad \langle u, v\rangle_w = \int_a^b u^*v\,e^{-x}\,dx.$$

This is the required weighted self-adjoint differential form; vanishing of the displayed boundary form makes the operator Hermitian on that domain. For polynomial solutions on $[0,\infty)$, both endpoint terms vanish, as checked in Exercise 8.2.4.

8.2.2 Show that the Hermite ODE, Table 7.1, may be put into self-adjoint form by multiplying by e^{-x^2} and that this gives $w(x) = e^{-x^2}$ as the appropriate weighting function.

Solution.

Hermite's equation is $y'' - 2xy' + 2ny = 0$. Since

$$\frac{d}{dx}e^{-x^2} = -2xe^{-x^2},$$

multiplication by e^{-x^2} gives

$$e^{-x^2}y'' - 2xe^{-x^2}y' + 2ne^{-x^2}y = (e^{-x^2}y')' + 2ne^{-x^2}y = 0.$$

Therefore

$$\boxed{p = w = e^{-x^2}, \qquad A = -w^{-1}(p\,d/dx)' = -d^2/dx^2 + 2x\,d/dx},$$

where $Ay = 2ny$. In the scalar product $\langle u, v\rangle_w = \int u^*ve^{-x^2}\,dx$, two integrations by parts leave only $-[e^{-x^2}(u^*v' - u'^*v)]_a^b$. It vanishes on $(-\infty,\infty)$ for polynomial solutions and their linear combinations, establishing the claimed weight and Hermiticity.

8.2.3 Show that the Chebyshev ODE, Table 7.1, may be put into self-adjoint form by multiplying by $(1-x^2)^{-1/2}$ and that this gives $w(x) = (1-x^2)^{-1/2}$ as the appropriate weighting function.

Solution.

On $-1 < x < 1$, multiply $(1-x^2)y'' - xy' + n^2 y = 0$ by $(1-x^2)^{-1/2}$. Because

$$\frac{d}{dx}\sqrt{1-x^2} = -\frac{x}{\sqrt{1-x^2}},$$

the resulting equation is

$$\boxed{\left(\sqrt{1-x^2}\,y'\right)' + \frac{n^2}{\sqrt{1-x^2}}y = 0}.$$

Thus $p = \sqrt{1-x^2}$, $w = (1-x^2)^{-1/2}$ and $A = -w^{-1}(p\,d/dx)'$ has eigenvalue n^2. The scalar product is

$$\boxed{\langle u, v\rangle_w = \int_{-1}^{1} \frac{u^*(x)v(x)}{\sqrt{1-x^2}}\,dx}.$$

The singularities of this weight are integrable: near either endpoint they have the form a constant times $t^{-1/2}$, whose integral is $2\sqrt{t}$. Polynomial solutions therefore have finite norm, and p times any product of their values and first derivatives vanishes at either endpoint.

8.2.4 The Legendre, Chebyshev, Hermite, and Laguerre equations, given in Table 7.1, have solutions that are polynomials. Show that ranges of integration that guarantee that the Hermitian operator boundary conditions will be satisfied are

(a) Legendre $[-1, 1]$, (b) Chebyshev $[-1, 1]$,

(c) Hermite $(-\infty, \infty)$, (d) Laguerre $[0, \infty)$.

Solution.

For $(pu')' + \lambda w u = 0$, the boundary condition needed in the orthogonality proof is

$$\big[p(u^*v' - u'^*v)\big]_a^b = 0.$$

If u, v are polynomials, then $u^*v' - u'^*v$ is another polynomial (on the real axis). We can therefore check each endpoint explicitly.

(a) Legendre: $p = 1 - x^2$ and $w = 1$. The polynomial factor is bounded at $x = \pm 1$, while $1 - x^2 \to 0$. Each boundary term is zero, and the norm integral over $[-1, 1]$ is finite.

(b) Chebyshev: $p = \sqrt{1-x^2} \to 0$ at ± 1. Bounded polynomial values and derivatives again make each boundary term zero. The norm exists because $w = (1-x^2)^{-1/2}$ is integrable.

(c) Hermite: $p = w = e^{-x^2}$. Every polynomial is bounded in magnitude by $C(1 + |x|^N)$ for suitable C, N, and $|x|^N e^{-x^2} \to 0$ as $x \to \pm\infty$. Thus both boundary terms vanish and the weighted norms converge.

(d) Laguerre: $p = xe^{-x}$ and $w = e^{-x}$. At 0, the factor x multiplies a bounded polynomial; at ∞, $x^{N+1}e^{-x} \to 0$. Both boundary terms vanish, and the weighted norms on $[0, \infty)$ converge.

These conclusions use the polynomial hypothesis. The same intervals alone do not make arbitrary singular solutions satisfy the boundary conditions, as Exercises 8.2.6 and 8.2.7 demonstrate.

8.2.5 The functions $u_1(x)$ and $u_2(x)$ are eigenfunctions of the same Hermitian operator but for distinct eigenvalues λ_1 and λ_2. Prove that $u_1(x)$ and $u_2(x)$ are linearly independent.

Solution.

Suppose $c_1u_1 + c_2u_2 = 0$, where eigenfunctions are nonzero by definition. Apply A and then subtract λ_2 times the original relation:

$$c_1\lambda_1u_1 + c_2\lambda_2u_2 = 0, \qquad c_1(\lambda_1 - \lambda_2)u_1 = 0.$$

Since $\lambda_1 \neq \lambda_2$ and $u_1 \neq 0$, we obtain $c_1 = 0$. The original relation then implies $c_2 = 0$. Hence the functions are linearly independent. Hermiticity is sufficient but is not needed for this particular conclusion; only linearity and distinct eigenvalues were used.

8.2.6 Given that

$$P_1(x) = x \quad \text{and} \quad Q_0(x) = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$$

are solutions of Legendre's differential equation (Table 7.1) corresponding to different eigenvalues:

(a) Evaluate their orthogonality integral

$$\int_{-1}^{1} \frac{x}{2}\ln\left(\frac{1+x}{1-x}\right)\,dx.$$

(b) Explain why these two functions are not orthogonal, that is, why the proof of orthogonality does not apply.

Solution.

(a) Put $Q = Q_0 = \frac{1}{2}\log((1+x)/(1-x))$. Its derivative is

$$Q' = \frac{1}{2}\left(\frac{1}{1+x} + \frac{1}{1-x}\right) = \frac{1}{1-x^2}.$$

Integrate by parts on a closed subinterval of $(-1, 1)$, choosing the antiderivative $(x^2-1)/2$ of x:

$$\int xQ\,dx = \frac{x^2-1}{2}Q - \frac{1}{2}\int (x^2-1)\frac{dx}{1-x^2} = \frac{x^2-1}{2}Q + \frac{x}{2}.$$

The endpoint products vanish because $t\log t \to 0$ as $t \downarrow 0$. Taking both improper limits therefore gives

$$\boxed{\int_{-1}^{1} P_1Q_0\,dx = 0 + \frac{1}{2}(1-(-1)) = 1}.$$

(b) For $A = -[(1-x^2)d/dx]'$, the eigenvalues of $P_1 = x$ and Q are 2 and 0. Their boundary form is not zero:

$$(1-x^2)(P_1Q' - P_1'Q) = x - (1-x^2)Q \longrightarrow \begin{cases} 1, & x \to 1, \\ -1, & x \to -1. \end{cases}$$

Its endpoint difference is 2. Indeed, subtracting their differential equations gives

$$\big[(1-x^2)(P_1Q' - P_1'Q)\big]_{-1}^{1} = (2-0)\int_{-1}^{1} P_1Q\,dx = 2.$$

Although Q is square integrable, it is singular at the endpoints and does not belong to the same boundary-condition domain as the Legendre polynomials. The Hermitian orthogonality theorem thus does not apply to this pair.

8.2.7 $T_0(x) = 1$ and $V_1(x) = (1 - x^2)^{1/2}$ are solutions of the Chebyshev differential equation corresponding to different eigenvalues. Explain, in terms of the boundary conditions, why these two functions are not orthogonal on the range $(-1, 1)$ with the weighting function found in Exercise 8.2.3.

Solution.

With $u = T_0 = 1$, $v = V_1 = \sqrt{1 - x^2}$, $p = \sqrt{1 - x^2}$ and $w = 1/p$, the weighted inner product is

$$\boxed{\langle u, v\rangle_w = \int_{-1}^{1} 1\, dx = 2 \neq 0}.$$

Here $u' = 0$, $v' = -x/p$, so

$$p(uv' - u'v) = -x, \qquad \big[p(uv' - u'v)\big]_{-1}^{1} = -2.$$

The two eigenvalues are 0 and 1: explicitly, $(pu')' = 0$ and $(pv')' = -1 = -wv$. Consequently the boundary identity gives $-2 = (0 - 1)\langle u, v\rangle_w$, consistently with the integral. The divergence of v' cancels the zero of p; the boundary form does not vanish even though v itself vanishes at both endpoints. They are therefore not eigenfunctions in a common Hermitian domain containing the Chebyshev polynomials.

8.2.8 A set of functions $u_n(x)$ satisfies the Sturm-Liouville equation

$$\frac{d}{dx}\left[p(x)\frac{d}{dx}u_n(x)\right] + \lambda_n w(x) u_n(x) = 0.$$

The functions $u_m(x)$ and $u_n(x)$ satisfy boundary conditions that lead to orthogonality. The corresponding eigenvalues λ_m and λ_n are distinct. Prove that for appropriate boundary conditions, $u_m'(x)$ and $u_n'(x)$ are orthogonal with $p(x)$ as a weighting function.

Solution.

Multiply the equation for u_n by u_m^* and integrate:

$$0 = \int_a^b u_m^*(pu_n')'\, dx + \lambda_n \int_a^b w u_m^* u_n\, dx.$$

One integration by parts gives the exact identity

$$\boxed{\int_a^b p\, u_m'^* u_n'\, dx = \big[pu_m^* u_n'\big]_a^b + \lambda_n \int_a^b w u_m^* u_n\, dx}.$$

For distinct eigenvalues the last integral is zero by the assumed orthogonality. Thus the additional sufficient boundary condition is

$$\big[pu_m^* u_n'\big]_a^b = 0,$$

which yields $\int_a^b p\, u_m'^* u_n'\, dx = 0$. This holds, for example, under common Dirichlet conditions, common Neumann conditions, or a common choice of Dirichlet or Neumann condition at each endpoint. It also holds for periodic values of u and pu', when the endpoint products cancel, or when $p \to 0$ and the products $u_m^* u_n'$ remain bounded.

The vanishing of the antisymmetrized boundary form alone need not suffice. Under real Robin conditions $u'(a) = h_a u(a)$ and $u'(b) = h_b u(b)$ that form vanishes, but the boundary term here is

$$p(b) h_b u_m^*(b) u_n(b) - p(a) h_a u_m^*(a) u_n(a),$$

which requires a separate check. For normalized eigenfunctions and the stronger vanishing condition, the same calculation with $m = n$ also gives $\int_a^b p|u_n'|^2\, dx = \lambda_n$.

8.2.9 Linear operator $\mathcal{A}$ has n distinct eigenvalues and n corresponding eigenfunctions: $\mathcal{A}\psi_i = \lambda_i\psi_i$. Show that the n eigenfunctions are linearly independent. Do not assume $\mathcal{A}$ to be Hermitian.

Hint. Assume linear dependence, i.e., that $\psi_n = \sum_{i=1}^{n-1} a_i\psi_i$. Use this relation and the operator-eigenfunction equation first in one order and then in the reverse order. Show that a contradiction results.

Solution.

We prove the statement by induction, so that comparison of coefficients will not presuppose the independence we are trying to establish. One nonzero eigenfunction is independent. Suppose any $n-1$ eigenfunctions with distinct eigenvalues are independent, and consider

$$\sum_{i=1}^{n} c_i\psi_i = 0.$$

Apply $\mathcal{A} - \lambda_n I$ to this equality. Linearity and the eigenfunction equations give

$$\sum_{i=1}^{n-1} c_i(\lambda_i - \lambda_n)\psi_i = 0.$$

By the induction hypothesis every coefficient in this shorter sum is zero. Since $\lambda_i - \lambda_n \neq 0$, $c_i = 0$ for $i < n$. The original equality becomes $c_n\psi_n = 0$, hence $c_n = 0$. This completes the induction without Hermiticity and without division by any individual eigenvalue (which may be zero).

8.2.10 The ultraspherical polynomials $C_n^{(\alpha)}(x)$ are solutions of the differential equation

$$\left\{(1-x^2)\frac{d^2}{dx^2} - (2\alpha+1)x\frac{d}{dx} + n(n+2\alpha)\right\} C_n^{(\alpha)}(x) = 0.$$

(a) Transform this differential equation into self-adjoint form.

(b) Find an interval of integration and weighting factor that make $C_n^{(\alpha)}(x)$ of the same α but different n orthogonal.

Note. Assume that your solutions are polynomials. Take real $\alpha > -1/2$. At $\alpha = 0$, use nonzero limiting normalizations for degrees $n \geq 1$.

Solution.

For a positive integrable weight supporting all degrees, assume real $\alpha > -\frac{1}{2}$. Eigenfunctions must be nonzero; at $\alpha = 0$ use the nonzero Chebyshev limits rather than the identically zero standard $C_n^{(0)}$ for $n \geq 1$.

(a) If w is the multiplier and $p = (1-x^2)w$, the coefficient of y' must be $p' = -(2\alpha+1)xw$. Expanding this condition gives

$$(1-x^2)w' - 2xw = -(2\alpha+1)xw, \qquad \frac{w'}{w} = \frac{(1-2\alpha)x}{1-x^2}.$$

Since $d\log(1-x^2)/dx = -2x/(1-x^2)$, integration yields

$$w = (1-x^2)^{\alpha-1/2}, \qquad p = (1-x^2)^{\alpha+1/2},$$

where the irrelevant positive constant has been set to one. Hence

$$\boxed{\left[(1-x^2)^{\alpha+1/2}y'\right]' + n(n+2\alpha)(1-x^2)^{\alpha-1/2}y = 0}.$$

(b) On $(-1,1)$ this weight is positive. At either endpoint it behaves as a positive constant times $t^{\alpha-1/2}$, whose integral is finite exactly when $\alpha > -1/2$. The same inequality makes $p \to 0$, so the boundary form vanishes for polynomial solutions. For $m \neq n$ in $\{0,1,2,\ldots\}$,

$$\lambda_n - \lambda_m = (n-m)(n+m+2\alpha) \neq 0,$$

because $n+m \geq 1$ and $2\alpha > -1$. Subtraction of the two self-adjoint equations therefore gives

$$\boxed{\int_{-1}^{1} (1-x^2)^{\alpha-1/2} C_m^{(\alpha)}(x) C_n^{(\alpha)}(x)\, dx = 0 \quad (m \neq n)}.$$

To explain the qualification at zero, the generating function $(1-2xt+t^2)^{-\alpha} = \sum_{n\geq 0} C_n^{(\alpha)}(x)t^n$ is 1 when $\alpha = 0$. For $n \geq 1$ its nonzero limiting normalization is $\lim_{\alpha\to 0} C_n^{(\alpha)}(x)/\alpha = 2T_n(x)/n$. To see the coefficient explicitly, differentiate the generating function with respect to α at zero. Put $x = \cos\theta$ and factor $1-2xt+t^2 = (1-e^{i\theta}t)(1-e^{-i\theta}t)$. For $|t| < 1$,

$$\begin{aligned} -\log(1-2xt+t^2) &= \sum_{n\geq 1} \frac{e^{in\theta}+e^{-in\theta}}{n} t^n \\ &= 2\sum_{n\geq 1} \frac{\cos(n\theta)}{n} t^n = 2\sum_{n\geq 1} \frac{T_n(x)}{n} t^n. \end{aligned}$$

The resulting Chebyshev family has precisely the weight in Exercise 8.2.3.

8.3 ODE Eigenvalue Problems

8.3.1 Solve the Legendre equation

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0$$

by direct series substitution.

(a) Verify that the indicial equation is

$$s(s-1) = 0.$$

(b) Using $s = 0$ and setting the coefficient $a_1 = 0$, obtain a series of even powers of x:

$$y_{\text{even}} = a_0 \left[1 - \frac{n(n+1)}{2!}x^2 + \frac{(n-2)n(n+1)(n+3)}{4!}x^4 + \cdots\right],$$

where

$$a_{j+2} = \frac{j(j+1)-n(n+1)}{(j+1)(j+2)} a_j.$$

(c) Using $s = 1$ and noting that the coefficient a_1 must be zero, develop a series of odd powers of x:

$$y_{\text{odd}} = a_0 \left[x - \frac{(n-1)(n+2)}{3!} x^3 + \frac{(n-3)(n-1)(n+2)(n+4)}{5!} x^5 + \cdots \right],$$

where

$$a_{j+2} = \frac{(j+1)(j+2) - n(n+1)}{(j+2)(j+3)} a_j.$$

(d) Show that both solutions, y_{even} and y_{odd}, diverge for $x = \pm 1$ if the series continue to infinity. (Compare with Exercise 1.2.5.)

(e) Finally, show that by an appropriate choice of n, one series at a time may be converted into a polynomial, thereby avoiding the divergence catastrophe. In quantum mechanics this restriction of n to integral values corresponds to quantization of angular momentum.

Solution.

(a) Substitute $y = \sum_{j \geq 0} a_j x^{j+s}$, with $a_0 \neq 0$. Before combining powers, the equation reads

$$\sum_{j \geq 0} (j+s)(j+s-1) a_j x^{j+s-2} + \sum_{j \geq 0} \{ n(n+1) - (j+s)(j+s+1) \} a_j x^{j+s} = 0.$$

The coefficient of x^{s-2} is $s(s-1)a_0$, and that of x^{s-1} is $s(s+1)a_1$. Thus $s = 0$ or 1; a_1 is free for $s = 0$ and is zero for $s = 1$. Matching the remaining powers gives

$$\boxed{a_{j+2} = \frac{(j+s-n)(j+s+n+1)}{(j+s+1)(j+s+2)} a_j}.$$

The point $x = 0$ is ordinary; the two values of s simply select the even and odd Taylor solutions.

(b) For $s = 0$, set $a_1 = 0$. Then

$$a_2 = -\frac{n(n+1)}{2} a_0, \qquad a_4 = \frac{(2-n)(n+3)}{3 \cdot 4} a_2 = \frac{(n-2)n(n+1)(n+3)}{4!} a_0.$$

In general, with an empty product equal to one,

$$y_e = a_0 \sum_{k=0}^{\infty} \frac{\prod_{r=0}^{k-1} (2r-n)(2r+n+1)}{(2k)!} x^{2k}.$$

This gives the stated even series and its recurrence.

(c) For $s = 1$, $a_1 = 0$ is forced by $2a_1 = 0$. Now

$$a_2 = -\frac{(n-1)(n+2)}{3!} a_0, \qquad a_4 = \frac{(3-n)(n+4)}{4 \cdot 5} a_2 = \frac{(n-3)(n-1)(n+2)(n+4)}{5!} a_0.$$

Thus the independent odd solution is

$$y_o = a_0 \sum_{k=0}^{\infty} \frac{\prod_{r=0}^{k-1} (2r+1-n)(2r+n+2)}{(2k+1)!} x^{2k+1}.$$

The two occurrences of a_0 are independent arbitrary constants.

(d) Let c_k be the coefficient of x^{2k+s} in either nonterminating branch, with $s = 0$ or 1. The exact recurrence implies

$$\frac{c_{k+1}}{c_k} = \frac{(2k+s)(2k+s+1) - n(n+1)}{(2k+s+1)(2k+s+2)} = 1 - \frac{2(2k+s+1) + n(n+1)}{(2k+s+1)(2k+s+2)}.$$

The subtracted numerator is $4k + O(1)$ and the denominator is $4k^2 + O(k)$, so the ratio is $1 - 1/k + O(k^{-2})$. Consequently

$$\frac{(k+1)c_{k+1}}{kc_k} = 1 + O(k^{-2}).$$

Starting beyond all possible zeros, let $b_k = kc_k$. The recurrence for b_k is $b_{k+1}/b_k = 1 + O(k^{-2})$. The logarithms of these ratios have an absolutely convergent sum, so their product has a finite nonzero limit C. Its tail has logarithm $O(\sum_{j\geq k} j^{-2}) = O(k^{-1})$; hence $b_k = C[1 + O(k^{-1})]$ and

$$c_k = \frac{C}{k} + O(k^{-2}), \qquad C \neq 0.$$

At $x = 1$, the C/k terms have divergent partial sums, while the $O(k^{-2})$ remainder is absolutely summable. At $x = -1$, every term acquires the same factor $(-1)^s$, so it diverges there too. The ratio test gives convergence for $|x| < 1$. The manual's observation that $c_{k+1}/c_k \to 1$ is not by itself a divergence proof; the $1/k$ asymptotic above supplies the necessary endpoint argument.

(e) A branch terminates when an exponent $r = 2k + s \geq 0$ satisfies $(r - n)(r + n + 1) = 0$. With the conventional representative $n \geq 0$, this means $n = r$: even nonnegative n truncates the even branch, and odd positive n truncates the odd branch. The other branch remains nonterminating and singular at the endpoints. If negative parameters are allowed, $n = -r - 1$ produces the same eigenvalue $n(n+1) = r(r+1)$ and the same polynomial; restricting to $n \geq 0$ removes this duplicate labeling. Normalizing the surviving polynomial by $P_n(1) = 1$ gives the Legendre polynomial, for example $1, x, (3x^2 - 1)/2, (5x^3 - 3x)/2$ for $n = 0, 1, 2, 3$.

8.3.2 Show that with the weight factor $\exp(-x^2)$ and the interval $-\infty < x < \infty$ for the scalar product, the Hermite ODE eigenvalue problem is Hermitian. Use a common domain with vanishing boundary form at both infinities, such as the span of the polynomial eigenfunctions.

Solution.

A common domain satisfying the endpoint condition is required in addition to the weight and interval. The polynomial eigenfunctions satisfy it. Write the eigenvalue problem as $Ay = 2\alpha y$, where

$$A = -\frac{d^2}{dx^2} + 2x\frac{d}{dx} = -e^{x^2}\frac{d}{dx}\left(e^{-x^2}\frac{d}{dx}\right).$$

For finite endpoints a, b, integration by parts yields

$$\begin{aligned}
\langle u, Av\rangle_w &= -\int_a^b u^*(e^{-x^2}v')'\,dx \\
&= -[u^* e^{-x^2} v']_a^b + \int_a^b u'^* e^{-x^2} v'\,dx, \\
\langle Au, v\rangle_w &= -[u'^* e^{-x^2} v]_a^b + \int_a^b u'^* e^{-x^2} v'\,dx.
\end{aligned}$$

Their difference is

$$-\left[e^{-x^2}(u^*v' - u'^*v)\right]_a^b.$$

For polynomials u, v, the Gaussian defeats every power of $|x|$, so both endpoint limits are zero as $a \to -\infty$, $b \to \infty$. All displayed integrals converge. The identity $\langle u, Av\rangle_w = \langle Au, v\rangle_w$ follows on the invariant domain of finite linear combinations of these polynomials. More generally it holds on any common operator domain where the integrals exist and the same boundary form vanishes.

Weighted square integrability alone, as asserted in the manual, is insufficient to establish this boundary condition. For example $u = 1$ and $v = \sin(e^{x^2})$ are both in $L^2(\mathbb{R}, e^{-x^2}dx)$, since they are bounded, but

$$e^{-x^2}(uv' - u'v) = 2x\cos(e^{x^2})$$

has no zero limit: at $x_k = \sqrt{\log(2\pi k)}$ it equals $2x_k$. Such a function is excluded from the domain used in the Hermitian problem; square integrability does not replace an operator-domain condition.

8.3.3

(a) Develop series solutions for Hermite's differential equation

$$y'' - 2xy' + 2\alpha y = 0.$$

ANS. $s(s-1) = 0$, indicial equation.

For $s = 0$,

$$a_{j+2} = 2a_j \frac{j - \alpha}{(j+1)(j+2)} \quad (j \text{ even}),$$

$$y_{\text{even}} = a_0\left[1 + \frac{2(-\alpha)x^2}{2!} + \frac{2^2(-\alpha)(2-\alpha)x^4}{4!} + \cdots\right].$$

For $s = 1$,

$$a_{j+2} = 2a_j \frac{j + 1 - \alpha}{(j+2)(j+3)} \quad (j \text{ even}),$$

$$y_{\text{odd}} = a_1\left[x + \frac{2(1-\alpha)x^3}{3!} + \frac{2^2(1-\alpha)(3-\alpha)x^5}{5!} + \cdots\right].$$

(b) Show that both series solutions are convergent for all x, the ratio of successive coefficients behaving, for a large index, like the corresponding ratio in the expansion of $\exp(x^2)$.

(c) Show that by appropriate choice of α, the series solutions may be cut off and converted to finite polynomials. (These polynomials, properly normalized, become the Hermite polynomials in Section 18.1.)

Solution.

(a) Substitution of $y = \sum_{j\geq 0} a_j x^{j+s}$ gives

$$\sum_{j\geq 0}(j+s)(j+s-1)a_j x^{j+s-2} + \sum_{j\geq 0} 2(\alpha - j - s)a_j x^{j+s} = 0.$$

Thus $s(s-1) = 0$, $s(s+1)a_1 = 0$, and

$$\boxed{a_{j+2} = \frac{2(j+s-\alpha)}{(j+s+1)(j+s+2)}a_j}.$$

For the even solution take $s = 0$, $a_1 = 0$, $a_0 = A$. Successive substitutions give

$$a_2 = -\alpha A, \qquad a_4 = \frac{2(2-\alpha)}{3\cdot 4}(-\alpha A),$$

and hence

$$y_e = A\sum_{k\geq 0} \frac{2^k \prod_{r=0}^{k-1}(2r-\alpha)}{(2k)!}x^{2k} = A\left[1 - \alpha x^2 + \frac{\alpha(\alpha-2)}{6}x^4 + \cdots\right].$$

For the odd solution take $s = 1$. Its Frobenius coefficient a_1 is forced to zero; call its free a_0 coefficient B, which is the coefficient of x (denoted a_1 in the printed answer). Then

$$a_2 = \frac{2(1-\alpha)}{2\cdot 3}B, \qquad a_4 = \frac{2(3-\alpha)}{4\cdot 5}a_2,$$

so

$$y_o = B\sum_{k\geq 0} \frac{2^k \prod_{r=0}^{k-1}(2r+1-\alpha)}{(2k+1)!}x^{2k+1}.$$

Its first terms are exactly $B[x + 2(1-\alpha)x^3/3! + 2^2(1-\alpha)(3-\alpha)x^5/5! + \cdots]$.

(b) For a fixed x, the ratio of successive nonzero terms of either parity has magnitude

$$|x|^2\left|\frac{2(j+s-\alpha)}{(j+s+1)(j+s+2)}\right| \sim \frac{2|x|^2}{j} \longrightarrow 0.$$

Both series therefore converge absolutely for every finite real or complex x. With $j = 2k$, the ratio is asymptotic to x^2/k, which is the large-k behavior of the ratio $x^2/(k+1)$ for $e^{x^2} = \sum_{k\geq 0} x^{2k}/k!$.

(c) Termination requires $j+s = \alpha$ at a nonnegative integer exponent. Thus $\alpha = N \in \{0, 1, 2, \ldots\}$ gives one polynomial: the even branch for even N and the odd branch for odd N. The other branch does not terminate, and its coefficient must be set to zero if a polynomial is required. Choosing leading coefficient 2^N yields H_N, beginning with

$$H_0 = 1, \quad H_1 = 2x, \quad H_2 = 4x^2 - 2, \quad H_3 = 8x^3 - 12x.$$

8.3.4 Laguerre's ODE is

$$xL_n''(x) + (1-x)L_n'(x) + nL_n(x) = 0.$$

Develop a series solution and select the parameter n to make your series a polynomial.

Solution.

Use $y = \sum_{j\geq 0} a_j x^{j+s}$. The $xy'' + y'$ terms combine as

$$xy'' + y' = \sum_{j\geq 0}(j+s)^2 a_j x^{j+s-1},$$

whereas $-xy' + ny = \sum_{j\geq 0}(n-j-s)a_j x^{j+s}$. The lowest coefficient is $s^2 a_0$, so the repeated indicial root is $s = 0$. Matching x^{j+s} gives

$$(j+s+1)^2 a_{j+1} + (n-j-s)a_j = 0, \qquad \boxed{a_{j+1} = \frac{j-n}{(j+1)^2}a_j \quad (s=0)}.$$

The recurrence advances by one index; the manual's a_{j+2} on the left is a typographical error. Iteration yields

$$a_1 = -na_0, \quad a_2 = \frac{n(n-1)}{(2!)^2}a_0, \quad a_3 = -\frac{n(n-1)(n-2)}{(3!)^2}a_0,$$

and, in general,

$$y = a_0 \sum_{j=0}^{\infty} \frac{(-1)^j n(n-1)\cdots(n-j+1)}{(j!)^2} x^j.$$

For a nonterminating solution the term ratio is asymptotic to x/j, so this regular solution is entire. It becomes a polynomial precisely when $n = N$ is a nonnegative integer; then $a_{N+1} = 0$ and all later coefficients vanish. With $a_0 = 1$,

$$\boxed{L_N(x) = \sum_{j=0}^{N} (-1)^j \binom{N}{j} \frac{x^j}{j!}}.$$

For example $L_0 = 1$, $L_1 = 1 - x$, and $L_2 = 1 - 2x + x^2/2$. There is only one analytic solution near zero up to scale, since a_0 determines every coefficient. Reduction of order supplies a second local solution

$$y_2 = y_1 \int \frac{e^x}{x\, y_1(x)^2}\, dx,$$

where $y_1(0) = 1$. Its integrand is $1/x + O(1)$, so it contains $\log x$ and cannot be a regular polynomial at zero.

8.3.5 Solve the Chebyshev equation

$$(1 - x^2)T_n'' - xT_n' + n^2 T_n = 0,$$

by series substitution. What restrictions are imposed on n if you demand that the series solution converge for $x = \pm 1$?

ANS. The infinite series does converge for $x = \pm 1$ and no restriction on n exists (compare with Exercise 1.2.6).

Solution.

Substitute $y = \sum_{j\ge 0} a_j x^{j+s}$. The coefficients of x^{s-2} and x^{s-1} give $s(s-1) = 0$ and $s(s+1)a_1 = 0$. The other coefficients give

$$(j + s + 1)(j + s + 2)a_{j+2} + [n^2 - (j + s)^2]a_j = 0,$$

so

$$\boxed{a_{j+2} = \frac{(j+s)^2 - n^2}{(j+s+1)(j+s+2)} a_j}.$$

The independent even and odd solutions, normalized to $y_e(0) = 1$ and $y_o'(0) = 1$, are therefore

$$y_e = \sum_{k\ge 0} \frac{\prod_{r=0}^{k-1}[(2r)^2 - n^2]}{(2k)!} x^{2k} = 1 - \frac{n^2 x^2}{2!} + \frac{n^2(n^2-4)x^4}{4!} + \cdots,$$

$$y_o = \sum_{k\ge 0} \frac{\prod_{r=0}^{k-1}[(2r+1)^2 - n^2]}{(2k+1)!} x^{2k+1} = x + \frac{(1-n^2)x^3}{3!} + \frac{(1-n^2)(9-n^2)x^5}{5!} + \cdots.$$

For a nonterminating branch put $c_k = a_{2k}$ in its own Frobenius notation. Its ratio is

$$\frac{c_{k+1}}{c_k} = \frac{(2k+s)^2 - n^2}{(2k+s+1)(2k+s+2)} = 1 - \frac{3(2k+s)+2+n^2}{(2k+s+1)(2k+s+2)}.$$

The fraction subtracted from one is $(6k + O(1))/(4k^2 + O(k)) = 3/(2k) + O(k^{-2})$. Multiplying c_k by $k^{3/2}$ makes the new successive ratio $1 + O(k^{-2})$, whose product converges to a nonzero constant. Thus $c_k \sim Ck^{-3/2}$. Comparison with $\sum k^{-3/2}$ proves absolute convergence at both $x = 1$ and $x = -1$. Terminating branches also converge there. Hence

$$\boxed{\text{endpoint convergence imposes no restriction on } n.}$$

For an independent check, set $x = \sin t$ with $|t| < \pi/2$. Writing $Y(t) = y(\sin t)$, the chain rule gives

$$Y' = y' \cos t, \qquad Y'' = y'' \cos^2 t - y' \sin t = (1 - x^2)y'' - xy'.$$

Thus $Y'' + n^2 Y = 0$. The initial conditions at zero select

$$y_e = \cos(n \arcsin x), \qquad y_o = \begin{cases} \sin(n \arcsin x)/n, & n \neq 0, \\ \arcsin x, & n = 0. \end{cases}$$

Their endpoint values are finite. Polynomial solutions instead require $n^2 = N^2$, N a nonnegative integer, and selection of the branch of parity N. Endpoint convergence and polynomial termination are therefore different requirements.

8.3.6 Solve

$$(1 - x^2)U_n''(x) - 3xU_n'(x) + n(n+2)U_n(x) = 0,$$

choosing the root of the indicial equation to obtain a series of odd powers of x. Since the nonterminating series will diverge for $x = 1$, choose n to convert it into a polynomial.

Solution.

Substitution of $y = \sum_{j \geq 0} a_j x^{j+s}$ gives $s(s-1) = 0$, $s(s+1)a_1 = 0$, and

$$a_{j+2} = \frac{(j+s)(j+s+2) - n(n+2)}{(j+s+1)(j+s+2)} a_j.$$

For the odd branch choose $s = 1$, which forces $a_1 = 0$. Factor the numerator to obtain

$$\boxed{a_{j+2} = \frac{(j+1-n)(j+n+3)}{(j+2)(j+3)} a_j \quad (j = 0, 2, 4, \ldots)}.$$

In particular,

$$a_2 = -\frac{(n-1)(n+3)}{3!} a_0, \qquad a_4 = \frac{(n-3)(n-1)(n+3)(n+5)}{5!} a_0,$$

so

$$y = a_0 \left[x - \frac{(n-1)(n+3)}{3!} x^3 + \frac{(n-3)(n-1)(n+3)(n+5)}{5!} x^5 + \cdots \right].$$

For the nonterminating odd series, $c_k = a_{2k}$ satisfies

$$\frac{c_{k+1}}{c_k} = \frac{(2k+1)(2k+3) - n(n+2)}{(2k+2)(2k+3)} = 1 - \frac{2k+3+n(n+2)}{(2k+2)(2k+3)}.$$

The subtracted fraction is $(2k+O(1))/(4k^2+O(k)) = 1/(2k)+O(k^{-2})$. As in Exercise 8.3.1, a convergent product shows $c_k \sim Ck^{-1/2}$, $C \neq 0$; the terms have a fixed eventual sign for real n, so the series diverges at $x=1$ (and at -1). Termination occurs at an odd exponent $N = 2k+1$ when

$$(N-n)(N+n+2) = 0.$$

With the usual convention $n \geq 0$, choose $n = N = 1, 3, 5, \ldots$; then the degree is N. Allowing negative labels also gives $n = -N-2 = -3, -5, \ldots$, which produce the same eigenvalue and polynomial. The standard second-kind Chebyshev normalization is $a_0 = (-1)^{(N-1)/2}(N+1)$. The first three polynomials are

$$U_1 = 2x, \qquad U_3 = 8x^3 - 4x, \qquad U_5 = 32x^5 - 32x^3 + 6x.$$

This normalization can be checked from $U_N(\cos\theta) = \sin((N+1)\theta)/\sin\theta$. Since $d/dx = -(\sin\theta)^{-1} d/d\theta$, differentiation at $x = 0$ ($\theta = \pi/2$) gives

$$a_0 = U_N'(0) = -(N+1)\cos\frac{(N+1)\pi}{2} = (-1)^{(N-1)/2}(N+1),$$

where the last equality uses odd N.

8.4 Variation Method

8.4.1 A function that is normalized on the interval $0 \leq x < \infty$ with an unweighted scalar product is

$$\psi = 2\alpha^{3/2} x e^{-\alpha x}, \qquad \alpha > 0 \text{ real}.$$

(a) Verify the normalization.

(b) Verify that for this ψ, $\langle x^{-1} \rangle = \alpha$.

(c) Verify that for this ψ, $\langle d^2/dx^2 \rangle = -\alpha^2$.

(d) Use the variation method to find the value of α that minimizes

$$\left\langle \psi \left| -\frac{1}{2}\frac{d^2}{dx^2} - \frac{1}{x} \right| \psi \right\rangle,$$

and find the minimum value of this expectation value.

Solution.

Take $\alpha > 0$ real; this is necessary for the stated normalized trial function. For $\beta > 0$, define $I_k(\beta) = \int_0^\infty x^k e^{-\beta x}\, dx$. Direct integration gives $I_0 = 1/\beta$, and integration by parts gives

$$I_k = \left[-\frac{x^k e^{-\beta x}}{\beta} \right]_0^\infty + \frac{k}{\beta} I_{k-1} = \frac{k}{\beta} I_{k-1} \quad (k \geq 1).$$

Therefore $I_k = k!/\beta^{k+1}$.

(a) The normalization integral is

$$\int_0^\infty |\psi|^2\, dx = 4\alpha^3 I_2(2\alpha) = 4\alpha^3 \frac{2!}{(2\alpha)^3} = 1.$$

(b) Multiplication by x^{-1} removes one power of x:

$$\boxed{\langle x^{-1}\rangle = 4\alpha^3 I_1(2\alpha) = 4\alpha^3 \frac{1!}{(2\alpha)^2} = \alpha}.$$

The apparent singularity at 0 is harmless because the integrand is proportional to $xe^{-2\alpha x}$.

(c) Differentiate with the exponential retained:

$$\psi' = 2\alpha^{3/2}(1-\alpha x)e^{-\alpha x}, \qquad \psi'' = 2\alpha^{3/2}(\alpha^2 x - 2\alpha)e^{-\alpha x}.$$

The factor $e^{-\alpha x}$ is missing from the manual's displayed second derivative, although its following integral is correct. Consequently

$$\begin{aligned}\langle d^2/dx^2\rangle &= 4\alpha^3 \int_0^\infty (\alpha^2 x^2 - 2\alpha x)e^{-2\alpha x}\,dx \\ &= 4\alpha^3\left[\alpha^2\frac{2!}{(2\alpha)^3} - 2\alpha\frac{1!}{(2\alpha)^2}\right] = \boxed{-\alpha^2}.\end{aligned}$$

The sign also follows by integration by parts: $\int \psi\psi'' = [\psi\psi']_0^\infty - \int(\psi')^2 = -\int(\psi')^2$, since $\psi(0)=0$ and the exponential removes the boundary term at infinity.

(d) Combine the two expectation values:

$$E(\alpha) = -\tfrac{1}{2}(-\alpha^2) - \alpha = \frac{\alpha^2}{2} - \alpha = \frac{(\alpha-1)^2}{2} - \frac{1}{2}.$$

Thus the unique minimum for $\alpha > 0$ is

$$\boxed{\alpha_{\text{opt}} = 1, \qquad E_{\min} = -\tfrac{1}{2}}.$$

Equivalently $E' = \alpha - 1 = 0$ and $E'' = 1 > 0$. As a direct check, at $\alpha = 1$ the expression for ψ'' gives $(-\frac{1}{2}d^2/dx^2 - 1/x)\psi = -\frac{1}{2}\psi$; the optimized trial function is an exact eigenfunction in this case.

9 Chapter 9. Partial Differential Equations

Problems

9.2 First-Order Equations

Find the general solutions of the PDEs in Exercises 9.2.1 to 9.2.4.

9.2.1 $\dfrac{\partial\psi}{\partial x} + 2\dfrac{\partial\psi}{\partial y} + (2x - y)\psi = 0.$

Solution.

Let $s = x$ and $u = y - 2x$, so $y = u + 2s$. The chain rule gives

$$\partial_x = \partial_s - 2\partial_u, \qquad \partial_y = \partial_u, \qquad 2x - y = -u.$$

Consequently the PDE reduces, at fixed u, to $\psi_s - u\psi = 0$. Multiplying by e^{-us} gives $\partial_s(e^{-us}\psi) = 0$, so

$$\boxed{\psi(x, y) = e^{x(y-2x)}F(y - 2x)},$$

where F is an arbitrary differentiable function; this includes the zero solution and does not require taking $\log\psi$. Indeed $D = \partial_x + 2\partial_y$ annihilates u and satisfies $D(xu) = u$, hence $D\psi = u\psi = -(2x - y)\psi$. The coordinate map has determinant one, so it loses no local solutions.

9.2.2 $\dfrac{\partial\psi}{\partial x} - 2\dfrac{\partial\psi}{\partial y} + x + y = 0.$

Solution.

Put $s = x$, $u = y + 2x$. Then $\partial_x - 2\partial_y = \partial_s$ and $x + y = u - s$, reducing the PDE to

$$\psi_s = s - u.$$

Integration with u held fixed gives

$$\psi = \tfrac{1}{2}s^2 - us + F(u) = -\tfrac{3}{2}x^2 - xy + F(y + 2x).$$

Since $\tfrac{1}{2}(x + y)^2 - (-\tfrac{3}{2}x^2 - xy) = \tfrac{1}{2}(y + 2x)^2$, the difference is an arbitrary function of u and can be absorbed into F. An equivalent convenient form is

$$\boxed{\psi(x, y) = \tfrac{1}{2}(x + y)^2 + F(y + 2x)}.$$

Direct differentiation gives $\psi_x = (x + y) + 2F'$ and $\psi_y = (x + y) + F'$, so $\psi_x - 2\psi_y + x + y = 0$ exactly.

9.2.3 $\dfrac{\partial\psi}{\partial x} + \dfrac{\partial\psi}{\partial y} = \dfrac{\partial\psi}{\partial z}.$

Solution.

Move the z derivative to the left and put $s = x$, $u = y - x$, $v = z + x$. The inverse transformation is $x = s$, $y = u + s$, $z = v - s$, and

$$\partial_x = \partial_s - \partial_u + \partial_v, \quad \partial_y = \partial_u, \quad \partial_z = \partial_v.$$

Thus $(\partial_x + \partial_y - \partial_z)\psi = \psi_s = 0$, giving

$$\boxed{\psi(x, y, z) = F(y - x, z + x)}.$$

Here F is any differentiable function of two arguments. Its derivatives are $\psi_x = -F_u + F_v$, $\psi_y = F_u$, $\psi_z = F_v$, which directly verify the original equation. The characteristic lines have tangent $(1, 1, -1)$ and keep both arguments of F constant.

9.2.4 $\dfrac{\partial \psi}{\partial x} + \dfrac{\partial \psi}{\partial y} + \dfrac{\partial \psi}{\partial z} = x - y.$

Solution.

Choose $s = z$, $u = x - z$, $v = y - z$, whose inverse is $x = u + s$, $y = v + s$, $z = s$. The sum of the three original derivatives is ∂_s, and $x - y = u - v$. Therefore

$$\psi_s = u - v, \qquad \psi = s(u - v) + F(u, v),$$

or

$$\boxed{\psi(x, y, z) = z(x - y) + F(x - z, y - z)}.$$

The derivative sum annihilates $x - z$, $y - z$ and $x - y$, while it sends z to one. Applied to the boxed expression it gives $x - y$, as required. Every characteristic is a straight line in the direction $(1, 1, 1)$.

9.2.5

(a) Show that the PDE

$$y\frac{\partial \psi}{\partial x} + x\frac{\partial \psi}{\partial y} = 0$$

can be transformed into a readily soluble form by writing it in the new variables $u = xy$, $v = x^2 - y^2$, and find its general local solution away from the origin; distinguish connected characteristic branches when describing global solutions.

(b) Discuss this result in terms of characteristics.

Solution.

(a) For $u = xy$, $v = x^2 - y^2$, the chain rule gives

$$\psi_x = y\psi_u + 2x\psi_v, \qquad \psi_y = x\psi_u - 2y\psi_v.$$

Thus

$$y\psi_x + x\psi_y = (x^2 + y^2)\psi_u.$$

Away from $(0, 0)$ the coefficient is nonzero, so $\psi_u = 0$ and

$$\boxed{\psi = F(v) = F(x^2 - y^2) \quad \text{locally}}.$$

The Jacobian is $\partial(u, v)/\partial(x, y) = -2(x^2 + y^2)$, confirming local invertibility at every such point.

(b) A characteristic parametrized by s satisfies $x' = y$, $y' = x$, $\psi' = 0$. Adding and subtracting the first two relations yields

$$x + y = Ae^s, \qquad x - y = Be^{-s}, \qquad x^2 - y^2 = AB.$$

The nonzero level sets are hyperbolas, and the zero level consists of the lines $y = \pm x$, with each ray a separate characteristic; the origin is a stationary point. The solution is constant along every connected characteristic.

Away from the origin, a single function $F(x^2 - y^2)$ is the general local form. Globally, distinct connected branches of the same level set can carry different functions, subject to smooth matching across the separating lines and at the origin. More explicitly, on the whole plane a C^1 solution can use four functions F_R, F_L for $v > 0$ (right and left sectors) and F_U, F_D for $v < 0$ (upper and lower sectors). These functions must extend C^1 to $v = 0$ with a common value C and a common first derivative. Set $\psi = C$ on both lines $y = \pm x$. The common value follows because each characteristic ray approaches the origin; the common derivative follows by crossing each ray at a nonzero point, where $\nabla v \neq 0$. Conversely these matching conditions make the piecewise formula C^1: off the lines, $\nabla\psi = F'(v)(2x, -2y)$; across a nonzero ray both limits agree, and at the origin the gradient tends to zero. For a smooth solution all one-sided derivatives of the four F's must likewise agree at zero. For example, write $v = x^2 - y^2$ and set

$$\psi = \begin{cases} e^{-1/v^2}, & v > 0,\ x > 0, \\ -e^{-1/v^2}, & v > 0,\ x < 0, \\ 0, & v \leq 0. \end{cases}$$

Each derivative of e^{-1/v^2} is that exponential times a polynomial in $1/v$: differentiating preserves this form because $(e^{-1/v^2})' = 2v^{-3}e^{-1/v^2}$. Every resulting term tends to zero: with $w = 1/v$, $w^m e^{-w^2} \to 0$ for every fixed m. Thus all four branch functions have zero derivatives of every order at zero, so this is smooth and satisfies the PDE everywhere. Its values at (x, y) and $(-x, -y)$ with $v > 0$ differ, proving that it cannot be represented by one global $F(v)$. At the origin the PDE itself is $0 = 0$; continuity and differentiability govern the matching there.

9.2.6 Find the general local solution away from the origin, and account for characteristic branches globally, to the PDE

$$x\frac{\partial\psi}{\partial x} - y\frac{\partial\psi}{\partial y} = 0.$$

Hint. The solution to Exercise 9.2.5 may provide a suggestion as to how to proceed.

Solution.

Use the same coordinates $u = xy$, $v = x^2 - y^2$ as in Exercise 9.2.5. Now

$$x\psi_x - y\psi_y = x(y\psi_u + 2x\psi_v) - y(x\psi_u - 2y\psi_v) = 2(x^2 + y^2)\psi_v.$$

Thus, off the origin,

$$\boxed{\psi(x, y) = F(xy) \quad \text{locally}}.$$

The characteristic equations integrate to

$$x' = x, \quad y' = -y, \quad \psi' = 0, \qquad x = Ae^s, \quad y = Be^{-s}, \quad xy = AB.$$

Consequently the characteristics are connected branches of $xy =$ constant, together with the four open coordinate-axis rays and the stationary origin.

As in Exercise 9.2.5, different branches can have different functions of the invariant; smoothness must be imposed where they meet. A single $F(xy)$ need not describe every global solution. On the whole plane assign functions $F_{\mathrm{I}}, F_{\mathrm{III}}$ on $u > 0$ to the first and third quadrants, and $F_{\mathrm{II}}, F_{\mathrm{IV}}$ on

$u < 0$ to the other quadrants. For a C^1 solution they extend to $u = 0$ with a common value C and first derivative; set $\psi = C$ on the axes. Necessity follows by approaching the origin along each ray and by crossing each nonzero ray transversely. Sufficiency follows from $\nabla\psi = F'(u)(y, x)$, whose limiting values match on each ray and tend to zero at the origin. Smooth solutions require matching derivatives of every order, exactly as in Exercise 9.2.5. For a concrete example let $u = xy$ and take $\psi = \operatorname{sgn}(x)e^{-1/u^2}$ when $u > 0$, and $\psi = 0$ when $u \le 0$. It is smooth because the exponential is flat at $u = 0$. Both u and the sign of x are constant along each of its nonzero characteristics, so it solves the PDE, but its values in the first and third quadrants differ at the same positive u.

9.3 Second-Order Equations

9.3.1 Let a, b, c be real constants with $c > 0$, so the displayed coordinates are real and defined. Show that by making a change of variables to $\xi = c^{1/2}x - c^{-1/2}by$, $\eta = c^{-1/2}y$, the operator L of Eq. (9.18) can be brought to the form

$$L = (ac - b^2)\frac{\partial^2}{\partial\xi^2} + \frac{\partial^2}{\partial\eta^2}.$$

Solution.

The coefficient $ac - b^2$ multiplies only ∂_ξ^2, as in the textbook's Eq. (9.21). The displayed change of coordinates requires $c \neq 0$; take $c > 0$ for real square roots. The coefficients are constants. Expanding Eq. (9.18) gives $L = a\partial_x^2 + 2b\partial_x\partial_y + c\partial_y^2$. For the given ξ, η, the chain rule gives

$$\partial_x = \sqrt{c}\,\partial_\xi, \qquad \partial_y = -\frac{b}{\sqrt{c}}\partial_\xi + \frac{1}{\sqrt{c}}\partial_\eta.$$

As these coefficients are constant, a second differentiation gives

$$\begin{aligned}
\partial_x^2 &= c\partial_\xi^2, \\
\partial_x\partial_y &= -b\partial_\xi^2 + \partial_\xi\partial_\eta, \\
\partial_y^2 &= \frac{b^2}{c}\partial_\xi^2 - \frac{2b}{c}\partial_\xi\partial_\eta + \frac{1}{c}\partial_\eta^2.
\end{aligned}$$

Substituting and collecting coefficients yields

$$L = (ac - 2b^2 + b^2)\partial_\xi^2 + (2b - 2b)\partial_\xi\partial_\eta + \partial_\eta^2 = \boxed{(ac - b^2)\partial_\xi^2 + \partial_\eta^2}.$$

If $\Delta = ac - b^2 > 0$, a further rescaling $\widetilde{\eta} = \sqrt{\Delta}\,\eta$ does give $L = \Delta(\partial_\xi^2 + \partial_{\widetilde{\eta}}^2)$. That is a different coordinate change from the one stated.

9.4 Separation of Variables

9.4.1 By letting the operator $\nabla^2 + k^2$ act on the general form $a_1\psi_1(x,y,z) + a_2\psi_2(x,y,z)$, show that it is linear, i.e., that $(\nabla^2 + k^2)(a_1\psi_1 + a_2\psi_2) = a_1(\nabla^2 + k^2)\psi_1 + a_2(\nabla^2 + k^2)\psi_2$.

Solution.

For constants a_1, a_2, each second derivative obeys

$$\partial_j^2(a_1\psi_1 + a_2\psi_2) = a_1\partial_j^2\psi_1 + a_2\partial_j^2\psi_2 \quad (j = x, y, z).$$

Adding these three identities and distributing the multiplication by k^2 gives

$$\begin{aligned}(\nabla^2 + k^2)(a_1\psi_1 + a_2\psi_2) &= a_1\nabla^2\psi_1 + a_2\nabla^2\psi_2 + a_1k^2\psi_1 + a_2k^2\psi_2 \\ &= a_1(\nabla^2 + k^2)\psi_1 + a_2(\nabla^2 + k^2)\psi_2.\end{aligned}$$

This proves linearity for arbitrary twice-differentiable functions. In particular, a constant linear combination of solutions of the homogeneous equation is another solution. The statement remains true if k^2 is a prescribed position-dependent coefficient, since multiplication by a fixed function is also linear.

9.4.2 Show that the Helmholtz equation,

$$\nabla^2\psi + k^2\psi = 0,$$

is still separable in circular cylindrical coordinates if k^2 is generalized to $k^2 + f(\rho) + (1/\rho^2)g(\varphi) + h(z)$.

Solution.

Set $\psi = R(\rho)\Phi(\varphi)Z(z)$. Substitution of

$$\nabla^2 = \frac{1}{\rho}\partial_\rho(\rho\partial_\rho) + \frac{1}{\rho^2}\partial_\varphi^2 + \partial_z^2$$

and division by the product, where it is nonzero, gives

$$\frac{(\rho R')'}{\rho R} + k^2 + f(\rho) + \frac{1}{\rho^2}\left(\frac{\Phi''}{\Phi} + g(\varphi)\right) + \frac{Z''}{Z} + h(z) = 0.$$

Only the last two terms depend on z, so set $Z''/Z + h = -\kappa^2$. The remaining equation, multiplied by ρ^2, is

$$\frac{\rho(\rho R')'}{R} + \rho^2[k^2 - \kappa^2 + f(\rho)] + \frac{\Phi''}{\Phi} + g(\varphi) = 0.$$

Set the angular part equal to $-\mu$. The three ordinary differential equations are therefore

$$\boxed{\begin{aligned} Z'' + [h(z) + \kappa^2]Z &= 0, \\ \Phi'' + [g(\varphi) + \mu]\Phi &= 0, \\ \rho^2R'' + \rho R' + \{\rho^2[k^2 - \kappa^2 + f(\rho)] - \mu\}R &= 0. \end{aligned}}$$

Their products satisfy the original PDE by direct substitution, including at zeros of individual factors by continuity. The notation κ^2 does not presume positivity; boundary conditions determine the allowed constants. When $g = 0$ and the angular function is single-valued and periodic, $\mu = m^2$ with integer m.

9.4.3 Separate variables in the Helmholtz equation in spherical polar coordinates, splitting off the radial dependence first. Show that your separated equations have the same form as Eqs. (9.74), (9.77), and (9.78).

Solution.

Separate the radius first by writing $\psi = R(r)Y(\theta,\varphi)$. Define the angular differential expression

$$\Delta_\Omega Y = \frac{1}{\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta Y) + \frac{1}{\sin^2\theta}\partial_\varphi^2 Y.$$

The Helmholtz equation becomes

$$\frac{Y}{r^2}(r^2R')' + \frac{R}{r^2}\Delta_\Omega Y + k^2RY = 0.$$

Multiplication by $r^2/(RY)$ isolates the independent variables:

$$\frac{(r^2R')'}{R} + k^2r^2 = -\frac{\Delta_\Omega Y}{Y} = \lambda.$$

Hence

$$(r^2R')' + (k^2r^2 - \lambda)R = 0, \qquad \Delta_\Omega Y + \lambda Y = 0.$$

Now set $Y = \Theta(\theta)\Phi(\varphi)$ and multiply its equation by $\sin^2\theta/(\Theta\Phi)$. The result is

$$\frac{\sin\theta}{\Theta}(\sin\theta\,\Theta')' + \lambda\sin^2\theta = -\frac{\Phi''}{\Phi} = m^2.$$

Thus all three separated equations are

$$\boxed{\begin{aligned} \Phi'' + m^2\Phi &= 0, \\ \frac{1}{\sin\theta}(\sin\theta\,\Theta')' + \left(\lambda - \frac{m^2}{\sin^2\theta}\right)\Theta &= 0, \\ \frac{1}{r^2}(r^2R')' + \left(k^2 - \frac{\lambda}{r^2}\right)R &= 0. \end{aligned}}$$

These are Eqs. (9.74), (9.77), and (9.78), respectively. Periodicity of a nonzero Φ on the full azimuth makes m integral; regularity on the sphere selects $\lambda = l(l+1)$ with $l \geq |m|$, as in the associated Legendre problem. The separation itself does not require these boundary restrictions.

9.4.4 Verify that

$$\nabla^2\psi(r,\theta,\varphi) + \left[k^2 + f(r) + \frac{1}{r^2}g(\theta) + \frac{1}{r^2\sin^2\theta}h(\varphi)\right]\psi(r,\theta,\varphi) = 0$$

is separable (in spherical polar coordinates). The functions f, g, and h are functions only of the variables indicated; k^2 is a constant.

Solution.

The Laplacian acts once on ψ; the square brackets contain only the multiplying coefficient. The original transcription placed the first term inside the brackets and would have made the PDE nonlinear. Substitute $\psi = R(r)\Theta(\theta)\Phi(\varphi)$ in the corrected PDE and multiply by r^2/ψ. We obtain

$$\frac{(r^2R')'}{R} + r^2[k^2 + f(r)] + \frac{(\sin\theta\,\Theta')'}{\Theta\sin\theta} + g(\theta) + \frac{1}{\sin^2\theta}\left(\frac{\Phi''}{\Phi} + h(\varphi)\right) = 0.$$

The radial part depends only on r; set it equal to λ. Multiply the remaining angular equation by $\sin^2\theta$ to find

$$\frac{\sin\theta}{\Theta}(\sin\theta\,\Theta')' + [g(\theta)+\lambda]\sin^2\theta = -\frac{\Phi''}{\Phi} - h(\varphi) = \mu.$$

The desired ODEs are

$$\boxed{\begin{aligned}(r^2R')' + \{r^2[k^2+f(r)] - \lambda\}R &= 0,\\ \frac{1}{\sin\theta}(\sin\theta\,\Theta')' + \left[g(\theta)+\lambda - \frac{\mu}{\sin^2\theta}\right]\Theta &= 0,\\ \Phi'' + [h(\varphi)+\mu]\Phi &= 0.\end{aligned}}$$

Every coefficient in each equation depends on its own independent variable alone. Adding the three equations after multiplication by the corresponding factors reproduces the PDE. For general g, h, the separation constants need not be the unperturbed values $l(l+1)$ and m^2.

9.4.5 An atomic (quantum mechanical) particle is confined inside a rectangular box of sides a, b, and c. The particle is described by a wave function ψ that satisfies the Schrödinger wave equation

$$-\frac{\hbar^2}{2m}\nabla^2\psi = E\psi.$$

The wave function is required to vanish at each surface of the box (but not to be identically zero). This condition imposes constraints on the separation constants and therefore on the energy E. What is the smallest value of E for which such a solution can be obtained?

ANS. $E = \dfrac{\pi^2\hbar^2}{2m}\left(\dfrac{1}{a^2}+\dfrac{1}{b^2}+\dfrac{1}{c^2}\right)$.

Solution.

Place the box at $0<x<a$, $0<y<b$, $0<z<c$, with $a,b,c,m>0$. For a product $\psi = XYZ$, division of the Schrödinger equation by XYZ gives

$$\frac{X''}{X} + \frac{Y''}{Y} + \frac{Z''}{Z} = -\frac{2mE}{\hbar^2}.$$

Each ratio must be a constant, written $-k_x^2, -k_y^2, -k_z^2$. For $X'' = -k_x^2X$ with $X(0) = X(a) = 0$, integration by parts gives

$$k_x^2\int_0^a |X|^2dx = \int_0^a |X'|^2dx > 0$$

for a nonzero X. Thus $X = A\sin k_x x + B\cos k_x x$; $X(0)=0$ makes $B=0$, and $X(a)=0$ requires $k_x = n_x\pi/a$, $n_x = 1,2,\ldots$. The other two directions give $k_y = n_y\pi/b$, $k_z = n_z\pi/c$. Consequently

$$E_{n_xn_yn_z} = \frac{\pi^2\hbar^2}{2m}\left(\frac{n_x^2}{a^2}+\frac{n_y^2}{b^2}+\frac{n_z^2}{c^2}\right).$$

All three contributions are minimized at $n_x = n_y = n_z = 1$, yielding

$$\boxed{E_{\min} = \frac{\pi^2\hbar^2}{2m}\left(\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}\right)}.$$

A normalized state attaining it is

$$\psi_{111} = \sqrt{\frac{8}{abc}}\sin\frac{\pi x}{a}\sin\frac{\pi y}{b}\sin\frac{\pi z}{c},$$

since each sine-squared integral is half the corresponding side length. To see that a nonseparable state cannot have lower energy, put $s(x) = \sin(\pi x/a)$ and write $X = sV$. This allows complex V. Since $s'' = -(\pi/a)^2 s$,

$$|X'|^2 = (s')^2|V|^2 + s^2|V'|^2 + ss'\big(V'^*V + V^*V'\big),$$
$$\frac{d}{dx}(ss'|V|^2) = \left[(s')^2 - \frac{\pi^2}{a^2}s^2\right]|V|^2 + ss'\big(V'^*V + V^*V'\big).$$

Subtracting the second identity from the first and integrating gives

$$\int_0^a \left(|X'|^2 - \frac{\pi^2}{a^2}|X|^2\right) dx = \int_0^a \sin^2(\pi x/a)|V'|^2 dx \geq 0.$$

The boundary term vanishes for regular Dirichlet functions. Apply this inequality to each coordinate slice of ψ, integrate over the other two coordinates, and use $E\int|\psi|^2 = (\hbar^2/2m)\int|\nabla\psi|^2$. The same lower bound follows, with equality for the state above.

9.4.6 The quantum mechanical angular momentum operator is given by $\mathbf{L} = -i(\mathbf{r}\times\nabla)$. Show that

$$\mathbf{L}\cdot\mathbf{L}\psi = l(l+1)\psi$$

leads to the associated Legendre equation.

Hint. Section 8.3 and Exercise 8.3.1 may be helpful.

Solution.

In the given units $L_i = -i\epsilon_{ijk}x_j\partial_k$. The derivative in the first operator must also act on the coordinate in the second. Contracting the two epsilon symbols gives

$$\begin{aligned}
L_iL_i\psi &= -(\delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl})x_j\partial_k(x_l\partial_m\psi)\\
&= -\sum_{j,k} x_j\partial_k(x_j\partial_k\psi) + \sum_{j,k} x_j\partial_k(x_k\partial_j\psi)\\
&= -\sum_j x_j\partial_j\psi - r^2\nabla^2\psi + 3\sum_j x_j\partial_j\psi + \sum_{j,k} x_jx_k\partial_k\partial_j\psi\\
&= 2\mathbf{r}\cdot\nabla\psi - r^2\nabla^2\psi + \sum_{j,k} x_jx_k\partial_j\partial_k\psi.
\end{aligned}$$

If $D = \mathbf{r}\cdot\nabla = r\partial_r$, then $D^2\psi = D\psi + \sum_{j,k} x_jx_k\partial_j\partial_k\psi$. Therefore

$$\mathbf{L}^2 = -r^2\nabla^2 + D^2 + D = -r^2\nabla^2 + r^2\partial_r^2 + 2r\partial_r = -\Delta_\Omega.$$

The Instructor's Manual, p. 114, prints $L^2 = +\Delta_\Omega$; the sign must be negative for the stated $L_i = -i(\mathbf{r}\times\nabla)_i$. For example, $\Delta_\Omega\cos\theta = -2\cos\theta$, whereas $L^2\cos\theta = 2\cos\theta$. The eigenvalue equation is consequently

$$\frac{1}{\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta\psi) + \frac{1}{\sin^2\theta}\partial_\varphi^2\psi + l(l+1)\psi = 0.$$

Separate an angular mode as $\Theta(\theta)\Phi(\varphi)$. Setting $\Phi'' = -m^2\Phi$ leaves

$$\frac{1}{\sin\theta}(\sin\theta\,\Theta')' + \left[l(l+1) - \frac{m^2}{\sin^2\theta}\right]\Theta = 0.$$

Put $u = \cos\theta$ and $\Theta(\theta) = P(u)$. Then $\Theta' = -\sin\theta P'$ and

$$\frac{1}{\sin\theta}(\sin\theta\,\Theta')' = \frac{1}{\sin\theta}\frac{d}{d\theta}[-(1-u^2)P'] = (1-u^2)P'' - 2uP'.$$

Thus

$$\boxed{(1-u^2)P'' - 2uP' + \left[l(l+1) - \frac{m^2}{1-u^2}\right]P = 0},$$

the associated Legendre equation. Radial factors are unrestricted by this angular eigenvalue equation alone.

9.4.7 The 1-D Schrödinger wave equation for a particle in a potential field $V = \frac{1}{2}kx^2$ is

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \frac{1}{2}kx^2\psi = E\psi(x).$$

(a) Defining

$$a = \left(\frac{mk}{\hbar^2}\right)^{1/4}, \quad \lambda = \frac{2E}{\hbar}\left(\frac{m}{k}\right)^{1/2},$$

and setting $\xi = ax$, show that

$$\frac{d^2\psi(\xi)}{d\xi^2} + (\lambda - \xi^2)\psi(\xi) = 0.$$

(b) Substituting

$$\psi(\xi) = y(\xi)e^{-\xi^2/2},$$

show that $y(\xi)$ satisfies the Hermite differential equation.

Solution.

Assume the harmonic-oscillator parameters $m, k, \hbar$ are positive. **(a)** Since $\xi = ax$, $d/dx = a\,d/d\xi$ and $x^2 = \xi^2/a^2$. Substitution gives

$$-\frac{\hbar^2 a^2}{2m}\psi_{\xi\xi} + \frac{k\xi^2}{2a^2}\psi = E\psi.$$

Multiply by $-2m/(\hbar^2a^2)$ and move all terms to the left:

$$\psi_{\xi\xi} + \left[\frac{2mE}{\hbar^2a^2} - \frac{mk}{\hbar^2a^4}\xi^2\right]\psi = 0.$$

The definition $a^4 = mk/\hbar^2$ makes the second coefficient one, while $a^2 = \sqrt{mk}/\hbar$ gives

$$\frac{2mE}{\hbar^2a^2} = \frac{2E}{\hbar}\sqrt{\frac{m}{k}} = \lambda.$$

Hence $\boxed{\psi_{\xi\xi} + (\lambda - \xi^2)\psi = 0}$.

(b) Differentiating $\psi = ye^{-\xi^2/2}$ twice gives

$$\psi' = e^{-\xi^2/2}(y' - \xi y), \qquad \psi'' = e^{-\xi^2/2}[y'' - 2\xi y' + (\xi^2 - 1)y].$$

The terms $+\xi^2 y$ and $-\xi^2 y$ cancel upon substitution, leaving

$$\boxed{y'' - 2\xi y' + (\lambda - 1)y = 0}.$$

This is Hermite's equation with parameter $2\alpha = \lambda - 1$. When $\alpha = n$ is a nonnegative integer, the polynomial solutions in Exercise 8.3.3 give $\psi = H_n(\xi)e^{-\xi^2/2}$ and $E = \hbar\sqrt{k/m}(n + \frac{1}{2})$.

9.5 Laplace and Poisson Equations

9.5.1 Verify that the following are solutions of Laplace's equation:

(a) $\psi_1 = 1/r$, $r \neq 0$, (b) $\psi_2 = \dfrac{1}{2r} \ln \dfrac{r+z}{r-z}$, $x^2 + y^2 > 0$ (exclude the entire z-axis).

Solution.

(a) For a radial function in three dimensions, $\nabla^2 f = r^{-2}(r^2 f')'$. At $r > 0$, with $f = 1/r$,

$$f' = -r^{-2}, \qquad (r^2 f')' = (-1)' = 0, \qquad \boxed{\nabla^2(1/r) = 0 \quad (r > 0)}.$$

This is a local statement away from the singularity. Its nonzero flux through a sphere, $\int \partial_r(1/r)\, dS = -4\pi$, is why it cannot be extended as a source-free solution through the origin.

(b) The domain must exclude the whole z-axis, where the logarithm is singular; $r \neq 0$ alone would not suffice. With $z = r\cos\theta$ and $0 < \theta < \pi$,

$$\psi_2 = \frac{g(\theta)}{r}, \qquad g(\theta) = \frac{1}{2} \log \frac{1+\cos\theta}{1-\cos\theta} = \log \cot \frac{\theta}{2}.$$

The radial and azimuthal contributions vanish:

$$\partial_r(r^2 \partial_r \psi_2) = \partial_r[-g(\theta)] = 0, \qquad \partial_\varphi^2 \psi_2 = 0.$$

For the remaining angular term, direct differentiation gives

$$g' = -\frac{\sin\theta}{2}\left(\frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta}\right) = -\frac{1}{\sin\theta}, \quad (\sin\theta\, g')' = (-1)' = 0.$$

Consequently

$$\boxed{\nabla^2 \psi_2 = \frac{1}{r^3 \sin\theta}(\sin\theta\, g')' = 0 \quad \text{where } x^2 + y^2 > 0}.$$

No claim of harmonicity across the excluded axis follows from this calculation.

9.5.2 If Ψ is a solution of Laplace's equation, $\nabla^2 \Psi = 0$, show that $\partial\Psi/\partial z$ is also a solution.

Solution.

Where the third derivatives exist and are continuous, interchange of mixed derivatives gives

$$\begin{aligned} \nabla^2(\partial_z \Psi) &= \partial_x^2 \partial_z \Psi + \partial_y^2 \partial_z \Psi + \partial_z^3 \Psi \\ &= \partial_z(\partial_x^2 \Psi + \partial_y^2 \Psi + \partial_z^2 \Psi) = \partial_z(\nabla^2 \Psi) = 0. \end{aligned}$$

Thus $\partial_z \Psi$ solves Laplace's equation on the same open region. The conclusion also holds in the distributional sense, where partial derivatives commute without classical differentiability assumptions. Boundary values generally change on differentiation; the assertion concerns the PDE, not preservation of prescribed boundary data.

9.5.3 Show that an argument based on Eq. (9.88) can be used to prove that the Laplace and Poisson equations with Dirichlet boundary conditions have unique solutions on a bounded region with a piecewise smooth boundary, for solutions regular enough for Green's identity and continuous up to the boundary.

Solution.

Use the bounded region and regularity assumptions of the text's uniqueness argument. An unbounded region requires an appropriate condition at infinity as well. Let ψ_1, ψ_2 satisfy the same Poisson equation $\nabla^2 \psi_i = f$ on a bounded region Ω, with the same Dirichlet values on its entire boundary. Their difference $u = \psi_1 - \psi_2$ satisfies

$$\nabla^2 u = 0 \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega.$$

The product rule followed by the divergence theorem gives Green's identity, including complex conjugation if the functions are complex:

$$\int_{\partial\Omega} u^* \partial_n u \, dS = \int_\Omega \nabla \cdot (u^* \nabla u) \, dV = \int_\Omega \left(|\nabla u|^2 + u^* \nabla^2 u \right) dV.$$

The boundary integral is zero because $u = 0$ there, and the Laplacian term is zero in the interior. It follows that

$$\int_\Omega |\nabla u|^2 \, dV = 0.$$

The integrand is nonnegative, so $\nabla u = 0$ (everywhere for a classical smooth solution). Thus u is constant on each connected component, and its zero boundary values make each constant zero. Hence $\psi_1 = \psi_2$. Taking $f = 0$ proves the Laplace case as well. This proves uniqueness when a solution exists; it does not assert existence for arbitrary boundary data. For the need for a condition at infinity, $u = z$ and $u = 0$ are distinct harmonic functions in the half-space $z > 0$ with the same zero values on $z = 0$.

9.6 Wave Equation

Solve the wave equation, Eq. (9.89), subject to the indicated conditions.

9.6.1 Determine $\psi(x, t)$ given that at $t = 0$ $\psi_0(x) = \sin x$ and $\partial\psi(x)/\partial t = \cos x$.

Solution.

Take $c > 0$, the speed in $\psi_{tt} = c^2 \psi_{xx}$. To incorporate both initial data, write $\psi = F(x - ct) + G(x + ct)$. For initial displacement f and velocity v,

$$F + G = f, \qquad -cF' + cG' = v.$$

Differentiate the first equation and solve these two linear equations:

$$F' = \tfrac{1}{2}(f' - v/c), \qquad G' = \tfrac{1}{2}(f' + v/c).$$

Integrating and combining the two traveling waves yields d'Alembert's formula

$$\psi(x, t) = \frac{f(x - ct) + f(x + ct)}{2} + \frac{1}{2c} \int_{x-ct}^{x+ct} v(s) \, ds.$$

For $f(x) = \sin x$, $v(x) = \cos x$, the integral is $\sin(x+ct) - \sin(x-ct) = 2\cos x \sin ct$. The first term is $\sin x \cos ct$, so

$$\boxed{\psi(x,t) = \sin x \cos ct + \frac{\cos x}{c} \sin ct}.$$

At $t = 0$ this is $\sin x$, and $\psi_t = -c \sin x \sin ct + \cos x \cos ct$ becomes $\cos x$. Each term obeys $\psi_{tt} = c^2 \psi_{xx}$, completing the check.

9.6.2 Determine $\psi(x,t)$ given that at $t = 0$ $\psi_0(x) = \delta(x)$ (Dirac delta function) and the initial time derivative of ψ is zero.

Solution.

With initial velocity zero, d'Alembert's formula, interpreted for distributions, gives

$$\boxed{\psi(x,t) = \tfrac{1}{2}\delta(x-ct) + \tfrac{1}{2}\delta(x+ct)}.$$

The manual prints $\delta(x-ct)$ twice. The second pulse must have argument $x+ct$ to obtain zero initial velocity. For any smooth compactly supported test function φ,

$$\langle \psi(\cdot,t), \varphi \rangle = \tfrac{1}{2}[\varphi(ct) + \varphi(-ct)].$$

At $t = 0$ this is $\varphi(0)$, while its time derivative is $\frac{c}{2}[\varphi'(ct) - \varphi'(-ct)]$, which is zero at 0. The second time derivative is $\frac{c^2}{2}[\varphi''(ct) + \varphi''(-ct)]$, exactly the action of $c^2 \partial_x^2 \psi$ on the same test function. Thus both the PDE and the initial data hold distributionally. The unit impulse splits into equal masses moving with speeds $+c$ and $-c$.

9.6.3 Determine $\psi(x,t)$ given that at $t = 0$ $\psi_0(x)$ is a single square-wave pulse as defined below, and the initial time derivative of ψ is zero.

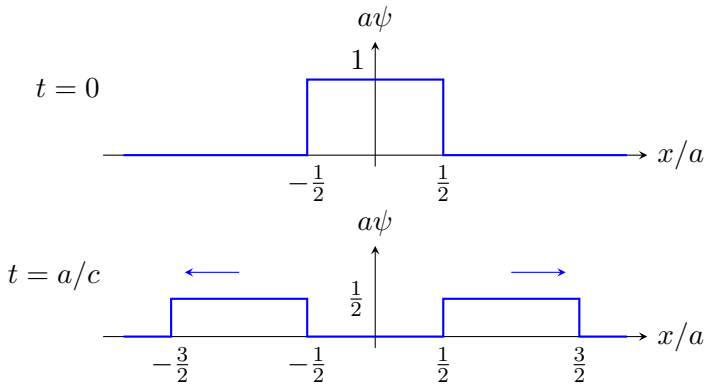
$$\psi_0(x) = 0, \ |x| > a/2, \quad \psi_0(x) = 1/a, \ |x| < a/2.$$

Solution.

Take $a, c > 0$ and write $f(x) = a^{-1}\mathbf{1}_{\{|x|<a/2\}}$, where $\mathbf{1}$ denotes an indicator function. Since the initial velocity is zero,

$$\boxed{\psi(x,t) = \frac{1}{2a}\left[\mathbf{1}_{\{|x-ct|<a/2\}} + \mathbf{1}_{\{|x+ct|<a/2\}}\right]}.$$

The right-moving pulse occupies $(ct - a/2, ct + a/2)$ and the left-moving pulse occupies $(-ct - a/2, -ct + a/2)$; each has height $1/(2a)$ and integral $1/2$. At $t = 0$ they coincide to reproduce f. For $0 < t < a/(2c)$ their overlap has height $1/a$; for $t > a/(2c)$ the pulses are disjoint. Endpoint values may be assigned the half-jump convention; they do not affect the associated distribution. As for any traveling profile, each term solves the wave equation weakly, and their time derivatives cancel initially.



9.6.4 Determine $\psi(x,t)$ given that at $t = 0$ $\psi_0 = 0$ for all x, but $\partial\psi/\partial t = \sin(x)$.

Solution.

Here $f = 0$ and $v(x) = \sin x$. The initial-displacement terms in d'Alembert's formula vanish, leaving

$$\begin{aligned}\psi(x,t) &= \frac{1}{2c}\int_{x-ct}^{x+ct} \sin s\, ds \\ &= \frac{1}{2c}[\cos(x-ct) - \cos(x+ct)] = \boxed{\frac{\sin x \sin ct}{c}}.\end{aligned}$$

Direct checks give $\psi(x,0) = 0$, $\psi_t(x,0) = \sin x$, and $\psi_{tt} = -c\sin x \sin ct = c^2\psi_{xx}$.

9.7 Heat-Flow, or Diffusion PDE

9.7.1 For a homogeneous spherical solid with constant thermal diffusivity, K, and no heat sources, the equation of heat conduction becomes

$$\frac{\partial T(r,t)}{\partial t} = K\nabla^2 T(r,t).$$

Assume a solution of the form

$$T = R(r)\mathcal{T}(t)$$

and separate variables. Show that the radial equation may take on the standard form

$$r^2\frac{d^2R}{dr^2} + 2r\frac{dR}{dr} + \alpha^2 r^2 R = 0,$$

and that $\sin\alpha r/r$ and $\cos\alpha r/r$ are its solutions. For a linearly independent pair take $\alpha \neq 0$; also give the independent pair at $\alpha = 0$.

Solution.

Spherical symmetry removes the angular derivatives, so

$$R(r)\mathcal{T}'(t) = K\mathcal{T}(t)\frac{1}{r^2}(r^2R')'.$$

Divide by $KR\mathcal{T}$ and equate the independent-variable expressions to a separation constant, conventionally $-\alpha^2$:

$$\frac{\mathcal{T}'}{K\mathcal{T}} = \frac{(r^2R')'}{r^2R} = -\alpha^2.$$

This gives the decaying time factor and radial equation

$$\mathcal{T} = Ce^{-K\alpha^2 t}, \qquad \boxed{r^2R'' + 2rR' + \alpha^2 r^2 R = 0}.$$

To solve the latter, let $u = rR$. Then

$$R' = \frac{u'}{r} - \frac{u}{r^2}, \qquad R'' = \frac{u''}{r} - \frac{2u'}{r^2} + \frac{2u}{r^3},$$

so $r^2R'' + 2rR' + \alpha^2 r^2 R = r(u'' + \alpha^2 u)$. For $\alpha \neq 0$,

$$\boxed{R = A\frac{\sin \alpha r}{r} + B\frac{\cos \alpha r}{r}}.$$

These two functions form a basis only for $\alpha \neq 0$. At $\alpha = 0$, the independent radial solutions are 1 and $1/r$. If the solid contains its center and the temperature is finite there, the cosine term (or $1/r$ term) must be absent. The sine term is regular because $\sin \alpha r/r = \alpha - \alpha^3 r^2/6 + \cdots$. The choice $-\alpha^2$ describes the usual decaying modes for $K > 0$; allowed values depend on the surface boundary condition, which was not specified.

9.7.2 Separate variables in the thermal diffusion equation of Exercise 9.7.1 in circular cylindrical coordinates. Assume that you can neglect end effects and take $T = T(\rho, t)$.

Solution.

With no dependence on z or φ, the cylindrical Laplacian is $\rho^{-1}\partial_\rho(\rho\partial_\rho)$. Set $T(\rho, t) = R(\rho)\mathcal{T}(t)$. The PDE becomes

$$R\mathcal{T}' = K\mathcal{T}\frac{1}{\rho}(\rho R')', \qquad \frac{\mathcal{T}'}{K\mathcal{T}} = \frac{(\rho R')'}{\rho R} = -\alpha^2.$$

Therefore

$$\boxed{\mathcal{T} = Ce^{-K\alpha^2 t}, \qquad \rho^2 R'' + \rho R' + \alpha^2\rho^2 R = 0}.$$

For nonzero α put $s = \alpha\rho$; then $s^2R_{ss} + sR_s + s^2R = 0$, the order-zero Bessel equation. Its solution is $R = AJ_0(\alpha\rho) + BY_0(\alpha\rho)$. If the axis is included, finiteness there selects J_0 since Y_0 has a logarithmic singularity. For $\alpha = 0$, integrate $(\rho R')' = 0$ directly: $\rho R' = B$ and $R = A + B\log\rho$, again leaving only a constant on the axis. A radial boundary condition would determine the allowed α; none is supplied here.

9.7.3 Solve the PDE

$$\frac{\partial\psi}{\partial t} = a^2\frac{\partial^2\psi}{\partial x^2},$$

to obtain $\psi(x, t)$ for a rod of infinite extent (in both the $+x$ and $-x$ directions), with a heat pulse at time $t = 0$ that corresponds to $\psi_0(x) = A\delta(x)$.

Solution.

Let $D = a^2 > 0$. Use the Fourier transform convention

$$\widehat{\psi}(k, t) = \int_{\mathbb{R}} \psi(x, t)e^{-ikx}\,dx, \qquad \psi(x, t) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{\psi}(k, t)e^{ikx}\,dk.$$

Integration by parts (or distributional differentiation) gives $\partial_t \widehat{\psi} = -Dk^2 \widehat{\psi}$ and $\widehat{\psi}(k,0) = A$. Solving this scalar ODE yields $\widehat{\psi} = Ae^{-Dk^2 t}$. For $t > 0$, put $I(x) = \int_{\mathbb{R}} e^{-Dtk^2} e^{ikx}\, dk$. Its derivative, using $ke^{-Dtk^2} = -(2Dt)^{-1}\partial_k e^{-Dtk^2}$, is

$$I'(x) = -\frac{i}{2Dt}\int_{\mathbb{R}} \partial_k(e^{-Dtk^2})e^{ikx}\, dk = -\frac{x}{2Dt} I(x).$$

The boundary term vanishes, and $I(0) = \sqrt{\pi/(Dt)}$ from the Gaussian integral. Thus $I(x) = \sqrt{\pi/(Dt)}e^{-x^2/(4Dt)}$ and

$$\boxed{\psi(x,t) = \frac{A}{\sqrt{4\pi Dt}} e^{-x^2/(4Dt)} \quad (t > 0)}.$$

In terms of a, the denominator is $2|a|\sqrt{\pi t}$. Writing $2a\sqrt{\pi t}$ presumes $a > 0$; the diffusivity itself is a^2. Direct differentiation gives

$$\psi_t = \left(-\frac{1}{2t} + \frac{x^2}{4Dt^2}\right)\psi, \qquad D\psi_{xx} = \left(-\frac{1}{2t} + \frac{x^2}{4Dt^2}\right)\psi.$$

The total heat is $\int \psi\, dx = A$. To check the initial impulse, substitute $x = 2\sqrt{Dt}\, u$ in its action on a test function:

$$\int \psi(x,t)\varphi(x)\, dx = \frac{A}{\sqrt{\pi}} \int e^{-u^2} \varphi(2\sqrt{Dt}\, u)\, du \longrightarrow A\varphi(0).$$

Dominated convergence applies because φ is bounded and the Gaussian is integrable. This verifies the initial condition as a distribution, rather than assigning a finite value to $\delta(0)$. If $a = 0$, the PDE is $\psi_t = 0$ and the initial distribution $A\delta(x)$ remains unchanged.

9.7.4 Solve the same PDE as in Exercise 9.7.3 for a rod of length L, with positive diffusivity $a^2 > 0$ and $L > 0$, with position on the rod given by the variable x, with the two ends of the rod at $x = 0$ and $x = L$ kept (at all times t) at the respective temperatures $T = 1$ and $T = 0$, and with the rod initially at $T(x) = 0$, for $0 < x \le L$.

Solution.

Take $D = a^2 > 0$ and $L > 0$. The steady solution satisfying the two endpoint temperatures is $S(x) = 1 - x/L$, since $S'' = 0$, $S(0) = 1$, $S(L) = 0$. Write $T = S + u$. Then

$$u_t = Du_{xx}, \quad u(0,t) = u(L,t) = 0, \quad u(x,0) = \frac{x}{L} - 1 \quad (0 < x < L),$$

Separate its zero-boundary modes: $X'' = -\kappa^2 X$, $X(0) = X(L) = 0$ give $X_n = \sin(n\pi x/L)$, $\kappa_n = n\pi/L$, and the time factors are $e^{-D\kappa_n^2 t}$. Hence

$$u(x,t) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{L} e^{-D(n\pi/L)^2 t}.$$

Sine orthogonality gives

$$\begin{aligned} b_n &= \frac{2}{L}\int_0^L \left(\frac{x}{L} - 1\right)\sin\frac{n\pi x}{L}\, dx \\ &= \frac{2}{L}\left[-\frac{(x/L - 1)\cos(n\pi x/L)}{n\pi/L} + \frac{\sin(n\pi x/L)}{L(n\pi/L)^2}\right]_0^L = -\frac{2}{n\pi}. \end{aligned}$$

Therefore

$$\boxed{T(x,t) = 1 - \frac{x}{L} - \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\sin\frac{n\pi x}{L}e^{-D(n\pi/L)^2 t}}.$$

For every $t > 0$ the Gaussian decay in n makes the series and all needed differentiated series converge uniformly, so termwise verification of the heat equation is valid. At $x = 0, L$ all sine terms vanish, giving the required endpoint values. For $0 < x < L$, its initial limit is zero because

$$\sum_{n=1}^{\infty}\frac{\sin(n\theta)}{n} = \frac{\pi - \theta}{2} \quad (0 < \theta < 2\pi), \qquad \theta = \frac{\pi x}{L}.$$

For example, this identity follows by taking the imaginary part of $\sum_{n\geq 1} r^n e^{in\theta}/n = -\log(1 - re^{i\theta})$ and letting $r \uparrow 1$; $\arg(1 - e^{i\theta}) = (\theta - \pi)/2$. The initial limit is justified without assuming uniform convergence at $x = 0$: for fixed $0 < \theta < \pi$, partial sums of $\sin(n\theta)$ are bounded by $1/|\sin(\theta/2)|$. Summation by parts with the decreasing coefficients $e^{-\eta n^2}/n$ bounds the tail starting at $N + 1$ by $1/[(N + 1)|\sin(\theta/2)|$, uniformly for $\eta \geq 0$. Take $t \downarrow 0$ in the finite initial portion and then let $N \to \infty$. The long-time limit is $S(x)$.

The initial interior temperature tends to zero as $x \downarrow 0$, whereas the imposed endpoint is one. There is consequently a corner discontinuity at $(0, 0)$; the initial limit is taken at fixed interior x, as the question specifies, not jointly with $x \to 0$. For numerical evaluation with $\eta = D\pi^2 t/L^2 > 0$, truncation after $N \geq 1$ terms has the uniform error bound

$$|R_N| \leq \frac{2}{\pi}\sum_{n>N}\frac{e^{-\eta n^2}}{n} \leq \frac{2}{\pi}\int_N^{\infty}\frac{e^{-\eta s^2}}{s}\,ds \leq \frac{e^{-\eta N^2}}{\pi\eta N^2}.$$

The last inequality uses $1/s \leq s/N^2$ for $s \geq N$ and integrates $se^{-\eta s^2}$ exactly.

10 Chapter 10. Green's Functions

Problems

10.1 One-Dimensional Problems

10.1.1 Show that

$$G(x,t) = \begin{cases} x, & 0 \le x < t, \\ t, & t < x \le 1, \end{cases}$$

is the Green's function for the operator $L = -d^2/dx^2$ and the boundary conditions $y(0) = 0$, $y'(1) = 0$.

Solution.

For fixed $0 < t < 1$, write the proposed kernel as $G(x,t) = \min(x,t)$. It is continuous at $x = t$, and its two derivatives there are

$$G_x(t^-,t) = 1, \qquad G_x(t^+,t) = 0.$$

Thus $[G_x]_{t^-}^{t^+} = -1$. A continuous, piecewise smooth function has a delta function in its second derivative with coefficient equal to this derivative jump. Consequently

$$-G_{xx} = \delta(x-t).$$

Away from t, both branches are linear, so there is no ordinary part of $-G_{xx}$. The endpoint conditions also hold: $G(0,t) = 0$ and $G_x(1,t) = 0$.

To check the resulting solution directly, let f be continuous and split the integral at $t = x$:

$$y(x) = \int_0^1 G(x,t) f(t)\,dt = \int_0^x t f(t)\,dt + x\int_x^1 f(t)\,dt.$$

Differentiation gives

$$\begin{aligned} y'(x) &= xf(x) + \int_x^1 f(t)\,dt - xf(x) = \int_x^1 f(t)\,dt, \\ y''(x) &= -f(x). \end{aligned}$$

Hence $-y'' = f$, $y(0) = 0$, and $y'(1) = 0$. Finally, the difference of two solutions would be $ax + b$; its endpoint conditions force $b = 0$ and $a = 0$. Thus this kernel gives the unique solution.

10.1.2 Find the Green's function for

(a) $Ly(x) = \dfrac{d^2y(x)}{dx^2} + y(x), \quad y(0) = 0, \quad y'(1) = 0.$

(b) $Ly(x) = \dfrac{d^2y(x)}{dx^2} - y(x), \quad y(x)$ finite for $-\infty < x < \infty$, meaning bounded as $x \to \pm\infty$.

Solution.

(a) For $y'' + y = 0$, choose a left solution and a right solution

$$u(x) = \sin x, \qquad v(x) = \cos(1-x).$$

They satisfy $u(0)=0$ and $v'(1)=0$, respectively. Continuity at $x=t$ is built into the form

$$G(x,t)=C\begin{cases}u(x)v(t), & x<t,\\ u(t)v(x), & x>t.\end{cases}$$

The derivative jump must be $+1$ because the coefficient of y'' is 1. The needed Wronskian is

$$\begin{aligned}u(t)v'(t)-u'(t)v(t)&=\sin t\sin(1-t)-\cos t\cos(1-t)\\&=-\cos\big(t+(1-t)\big)=-\cos 1.\end{aligned}$$

Therefore $C(-\cos 1)=1$, and

$$\boxed{G(x,t)=-\frac{1}{\cos 1}\begin{cases}\sin x\cos(1-t), & 0\le x\le t,\\ \sin t\cos(1-x), & t\le x\le 1.\end{cases}}$$

Both branches solve $G_{xx}+G=0$ away from the source, are continuous at the source, have derivative jump 1, and satisfy the specified endpoint conditions. These facts prove $(\partial_x^2+1)G=\delta(x-t)$. There is no zero mode: $h(0)=0$ gives $h=A\sin x$, and $h'(1)=A\cos 1=0$ gives $A=0$ since $\cos 1\neq 0$.

(b) "Finite for $-\infty<x<\infty$" must here mean bounded as $x\to\pm\infty$. Finiteness only at each finite point would allow both e^x and e^{-x} and would not select a unique Green's function. The bounded homogeneous solution on the left is proportional to e^x; on the right it is proportional to e^{-x}. Impose continuity by writing

$$G(x,t)=C\begin{cases}e^{x-t}, & x<t,\\ e^{t-x}, & x>t.\end{cases}$$

Its derivative jump is $-C-C=-2C$, so $-2C=1$ gives

$$\boxed{G(x,t)=-\frac{1}{2}e^{-|x-t|}.}$$

It decays at both infinities, solves $G_{xx}-G=0$ for $x\neq t$, and has derivative jump 1. Thus $y(x)=\int_{\mathbb{R}}G(x,t)f(t)\,dt$ solves $y''-y=f$ whenever the integral is defined, for example for bounded continuous f. In that example $|y(x)|\le\|f\|_\infty\int_{\mathbb{R}}e^{-|x-t|}/2\,dt=\|f\|_\infty$. No nonzero combination Ae^x+Be^{-x} is bounded at both infinities, which proves uniqueness in the bounded class.

10.1.3 Show that the function $y(x)$ defined by Eq. (10.26) satisfies the initial-value problem defined by Eq. (10.24) and its initial conditions $y(0)=y'(0)=0$.

Solution.

The referenced formulas, Eqs. (10.24) and (10.26) on textbook pp. 453–454, are

$$y''+y=f(x),\qquad y(x)=\int_0^x\sin(x-t)f(t)\,dt.$$

For continuous f, the Leibniz rule includes both an endpoint term and a derivative of the integrand. Applying it once gives

$$\begin{aligned}y'(x)&=\sin(x-x)f(x)+\int_0^x\cos(x-t)f(t)\,dt\\&=\int_0^x\cos(x-t)f(t)\,dt.\end{aligned}$$

Apply the rule a second time:

$$\begin{aligned} y''(x) &= \cos(x-x)f(x) - \int_0^x \sin(x-t)f(t)\,dt \\ &= f(x) - y(x). \end{aligned}$$

This proves $y'' + y = f$. At $x = 0$, both displayed integrals have zero length, so $y(0) = y'(0) = 0$.

For forward evolution $x \geq 0$ with source times $t \geq 0$, the equivalent causal kernel is $G(x,t) = \Theta(x-t)\sin(x-t)$. The displayed Volterra integral itself also solves the initial-value problem for negative x when its integral is given the usual orientation. The kernel vanishes for $x < t$, is continuous with value zero at $x = t$, and its first derivative jumps from 0 to 1. Thus $(\partial_x^2 + 1)G = \delta(x-t)$. The difference of any two solutions is $A\sin x + B\cos x$; its zero initial position and velocity force $A = B = 0$, so the verified solution is unique.

10.1.4 Find the Green's function for the equation

$$-\frac{d^2y}{dx^2} - \frac{y}{4} = f(x),$$

with boundary conditions $y(0) = y(\pi) = 0$.

$$\textit{ANS. } G(x,t) = \begin{cases} 2\sin(x/2)\cos(t/2), & 0 \leq x < t, \\ 2\cos(x/2)\sin(t/2), & t < x \leq \pi. \end{cases}$$

Solution.

For $y'' + y/4 = 0$, take

$$u(x) = \sin(x/2), \qquad v(x) = \cos(x/2).$$

Here $u(0) = 0$ and $v(\pi) = 0$. A continuous kernel with the required endpoint behavior is $Cu(x)v(t)$ for $x < t$ and $Cu(t)v(x)$ for $x > t$. Its derivative jump is

$$\begin{aligned} G_x(t^+,t) - G_x(t^-,t) &= C\left[-\frac{1}{2}\sin^2(t/2) - \frac{1}{2}\cos^2(t/2)\right] \\ &= -\frac{C}{2}. \end{aligned}$$

Because the operator is $-\partial_x^2 - 1/4$, this jump must be -1. Therefore $C = 2$ and

$$\boxed{G(x,t) = \begin{cases} 2\sin(x/2)\cos(t/2), & 0 \leq x \leq t, \\ 2\cos(x/2)\sin(t/2), & t \leq x \leq \pi. \end{cases}}$$

The branches solve the homogeneous equation off the diagonal; continuity and the jump prove $(-\partial_x^2 - 1/4)G = \delta(x-t)$.

For a direct check of the source sign, put

$$I(x) = \int_0^x \sin(t/2)f(t)\,dt, \qquad J(x) = \int_x^\pi \cos(t/2)f(t)\,dt.$$

Then $y = 2\cos(x/2)I + 2\sin(x/2)J$. Since $I' = \sin(x/2)f$ and $J' = -\cos(x/2)f$, the two endpoint terms cancel in the first derivative:

$$y' = -\sin(x/2)I + \cos(x/2)J.$$

Differentiating again gives

$$y'' = -\tfrac{1}{2}\cos(x/2)I - \tfrac{1}{2}\sin(x/2)J - \left[\sin^2(x/2) + \cos^2(x/2)\right]f = -\frac{y}{4} - f.$$

Also $I(0) = J(\pi) = 0$ gives the two Dirichlet conditions. A homogeneous solution satisfying $y(0) = 0$ is $A\sin(x/2)$, and its value at π is A, so the homogeneous boundary problem has only the zero solution.

10.1.5 Construct the Green's function for

$$x^2\frac{d^2y}{dx^2} + x\frac{dy}{dx} + (k^2x^2 - 1)y = 0,$$

subject to the boundary conditions $y(0) = 0$, $y(1) = 0$. Use the operator exactly as printed with ordinary dt source integration. Construct its inverse for nonresonant real k, treat $k = 0$ separately, and explain the obstruction at a positive zero of $J_1(k)$.

Solution.

The normalization of the operator matters at this singular endpoint. The Instructor's Manual, p. 119, first divides the printed ODE by x. Its symmetric kernel is the Green's function of that divided operator. For the operator exactly as printed, with an ordinary dt source integral, the kernel has an additional factor $1/t$. An ordinary inverse also requires a nonresonant parameter. Both conventions are specified below. Write

$$D = x^2\frac{d^2}{dx^2} + x\frac{d}{dx} + k^2x^2 - 1, \qquad L = \frac{1}{x}D = \frac{d}{dx}\left(x\frac{d}{dx}\right) + k^2x - \frac{1}{x}.$$

First take $k > 0$ with $J_1(k) \neq 0$, and let $0 < t < 1$. On either side of the source, $Dy = 0$ is Bessel's equation of order one. Its independent solutions are $J_1(kx)$ and $Y_1(kx)$. Their small-x behavior is

$$J_1(kx) \sim \frac{kx}{2}, \qquad Y_1(kx) \sim -\frac{2}{\pi kx}.$$

Thus the left homogeneous solution obeying $y(0) = 0$ is $u(x) = J_1(kx)$. Choose the right solution

$$v(x) = Y_1(kx) - \frac{Y_1(k)}{J_1(k)}J_1(kx), \qquad v(1) = 0.$$

To determine the normalization, define $W = uv' - u'v$, with derivatives with respect to x. Dividing the homogeneous ODE by x^2 gives $y'' = -y'/x - (k^2 - x^{-2})y$. Hence

$$W' = uv'' - u''v = -\frac{1}{x}(uv' - u'v) = -\frac{W}{x}, \qquad W = \frac{C}{x}.$$

The multiple of J_1 in v does not change W. Using the displayed small-x forms of J_1, Y_1 gives

$$W \sim \frac{kx}{2}\frac{2}{\pi kx^2} - \frac{k}{2}\left(-\frac{2}{\pi kx}\right) = \frac{2}{\pi x}.$$

Therefore the exact Wronskian is $W(x) = 2/(\pi x)$.

For $LG_L = \delta(x - t)$, the jump condition is $t[G_{L,x}]_{t^-}^{t^+} = 1$. Set $G_L = Cu(x_<)v(x_>)$, where $x_< = \min(x, t)$ and $x_> = \max(x, t)$. Its jump is $CW(t)$, giving $t\,C\,2/(\pi t) = 1$, or $C = \pi/2$. Consequently

$$\boxed{G_L(x,t) = \frac{\pi}{2}J_1(kx_<)\left[Y_1(kx_>) - \frac{Y_1(k)}{J_1(k)}J_1(kx_>)\right].}$$

This is the symmetric kernel in the manual. Since $D = xL$,

$$D_x G_L(x,t) = x\delta(x-t) = t\delta(x-t).$$

Thus the answer for the printed operator is

$$\boxed{G_D(x,t) = \frac{G_L(x,t)}{t}, \qquad D_x G_D(x,t) = \delta(x-t).}$$

In particular $[G_{D,x}] = 1/t^2$, exactly as required by the leading coefficient x^2. The two equivalent source formulas are

$$Dy = F \quad \Longrightarrow \quad y(x) = \int_0^1 G_D(x,t)F(t)\,dt = \int_0^1 G_L(x,t)\frac{F(t)}{t}\,dt.$$

For each interior t, both kernels are continuous at $x = t$, vanish at $x = 1$, and tend to zero as $x \downarrow 0$.

The singular endpoint also restricts allowable sources if the integrated solution is required to vanish at zero. Sources supported inside $(0,1)$ are sufficient; more generally $F(x) = O(x^\alpha)$ for $\alpha > 0$ is sufficient. The small-x kernel estimates give terms bounded by constants times $x^{-1}\int_0^x |F(t)|dt$ and $x\int_x^1 |F(t)|t^{-2}dt$. Up to constant multipliers, these bounds are respectively

$$\frac{1}{x}\int_0^x t^\alpha\,dt = \frac{x^\alpha}{\alpha+1}, \qquad x\int_x^1 t^{\alpha-2}\,dt = \begin{cases} \dfrac{x - x^\alpha}{\alpha - 1}, & \alpha \neq 1, \\ -x\log x, & \alpha = 1. \end{cases}$$

For every $\alpha > 0$ both tend to zero, including the logarithmic case. An arbitrary nonzero constant source at zero is not covered by this boundary-value inverse.

At a positive root $J_1(k) = 0$, the nonzero function $u = J_1(kx)$ satisfies both endpoint conditions. The denominator vanishes for a reason: the homogeneous problem has a zero mode. Green's identity gives for an admissible solution of $Dy = F$

$$\int_0^1 u(x)\frac{F(x)}{x}\,dx = \int_0^1 (uLy - yLu)\,dx = \big[x(uy' - u'y)\big]_0^1 = 0.$$

A general source does not satisfy this compatibility condition, and a compatible solution is not unique because one can add cu. Therefore no ordinary inverse Green's function exists at these resonances.

The case $k = 0$ must be treated separately, rather than counted as a positive Bessel zero. Now the homogeneous solutions are x and $1/x$. Choose $u = x$, $v = x - 1/x$; then $W = 2/x$. The kernels are

$$\boxed{G_{L,0}(x,t) = \frac{1}{2}x_<\left(x_> - \frac{1}{x_>}\right), \qquad G_{D,0}(x,t) = \frac{1}{2t}x_<\left(x_> - \frac{1}{x_>}\right).}$$

These are also the $k \downarrow 0$ limits of the previous expressions. No nonzero multiple of x vanishes at 1, so this case is nonresonant. For real negative k, replace it by $|k|$ throughout, since the printed operator depends only on k^2.

10.1.6 Given that

$$L = (1 - x^2)\frac{d^2}{dx^2} - 2x\frac{d}{dx}$$

and that $G(\pm 1, t)$ remains finite, show that no Green's function can be constructed by the techniques of this section.

Note. The solutions to $L = 0$ needed for the regions $x < t$ and $x > t$ are linearly dependent.

Solution.

Put the operator in divergence form:

$$Ly = \frac{d}{dx}\big[(1 - x^2)y'(x)\big].$$

For a homogeneous solution, one integration gives $(1 - x^2)y' = C$, and a second gives

$$y(x) = D + \frac{C}{2}\log\frac{1 + x}{1 - x}.$$

The logarithm diverges at either endpoint. Thus a homogeneous solution finite at -1 is a constant, and a homogeneous solution finite at $+1$ is also a constant.

For an interior source $-1 < t < 1$, the proposed Green's function would therefore have to be

$$G(x, t) = \begin{cases} A(t), & x < t, \\ B(t), & x > t. \end{cases}$$

It must be continuous at t, because a discontinuity in G would produce a delta derivative in LG, which is absent from the required source. So $A(t) = B(t)$. Both ordinary derivatives are zero, and hence

$$\big[(1 - x^2)G_x(x, t)\big]_{t^-}^{t^+} = 0.$$

But integrating $LG = \delta(x - t)$ across t requires this flux jump to be 1. This contradiction rules out an ordinary Green's function under the specified finite-endpoint conditions.

The obstruction is the constant zero mode $L1 = 0$. For a regular source f, endpoint finiteness of a solution to $Ly = f$ forces the endpoint fluxes $(1 - x^2)y'$ to vanish: a nonzero limiting flux would integrate to the logarithmic divergence displayed above. Consequently

$$\int_{-1}^{1} f(x)\,dx = \big[(1 - x^2)y'(x)\big]_{-1}^{1} = 0.$$

A delta source has integral 1 and violates that condition. Even when a source is compatible, an arbitrary additive constant remains. A Green's function with the constant mode projected out would solve a modified source equation, not $LG = \delta$ requested here.

10.1.7 Find the Green's function for

$$\frac{d^2\psi}{dt^2} + k\frac{d\psi}{dt} = f(t),$$

subject to the initial conditions $\psi(0) = \psi'(0) = 0$, and solve this ODE for $t > 0$ given $f(t) = \exp(-t)$.

Solution.

Use u for the source time and t for the observation time. For $t < u$, the zero initial data require the causal kernel to vanish. For $t > u$, the homogeneous equation is $G_{tt} + kG_t = 0$. When $k \neq 0$

its solution is $A(u) + B(u)e^{-kt}$. Continuity at $t = u$ requires $A + Be^{-ku} = 0$, so it can be written $C(u)(1 - e^{-k(t-u)})$. Its first derivative at u^+ is $kC(u)$, while that at u^- is zero. The coefficient of ψ'' is one; therefore the jump must be one, and $C(u) = 1/k$. We obtain

$$\boxed{G(t,u) = \Theta(t-u)\frac{1 - e^{-k(t-u)}}{k} \quad (k \neq 0).}$$

For $k = 0$, integrate $G_{tt} = 0$ on each side and use the same matching conditions, or take the limit:

$$\boxed{G(t,u) = \Theta(t-u)(t-u) \quad (k = 0).}$$

For either expression G is continuous with value zero at the source and $[G_t] = 1$, proving $(\partial_t^2 + k\partial_t)G = \delta(t-u)$.

The manual's kernel on pp. 119–120 is normalized for the multiplied equation $(e^{kt}\psi')' = e^{kt}f$. It is $e^{-ku}G(t,u)$ and must be integrated against $e^{ku}f(u)$. The kernel above acts directly on the original source $f(u)$. Also, the apparent singularities at $k = 0$ and $k = 1$ in the final algebraic formula are removable and need the cases given below.

For $f(u) = e^{-u}$ and $k \neq 0, 1$, the convolution is

$$\begin{aligned}
\psi(t) &= \frac{1}{k}\int_0^t \left(1 - e^{-k(t-u)}\right) e^{-u}\, du \\
&= \frac{1}{k}\left[1 - e^{-t} - e^{-kt}\int_0^t e^{(k-1)u}\, du\right] \\
&= \frac{1}{k}\left[1 - e^{-t} - \frac{e^{-t} - e^{-kt}}{k-1}\right] \\
&= \boxed{\frac{(k-1) - ke^{-t} + e^{-kt}}{k(k-1)}}.
\end{aligned}$$

Its derivatives are

$$\psi'(t) = \frac{e^{-t} - e^{-kt}}{k-1}, \qquad \psi''(t) = \frac{-e^{-t} + ke^{-kt}}{k-1},$$

so $\psi'' + k\psi' = e^{-t}$ and $\psi(0) = \psi'(0) = 0$. For the exceptional values, evaluate the integrals directly:

$$\begin{aligned}
k = 1: \quad \psi(t) &= \int_0^t (e^{-u} - e^{-t})\, du = \boxed{1 - (1+t)e^{-t}}, \\
k = 0: \quad \psi(t) &= \int_0^t (t-u)e^{-u}\, du = \boxed{t - 1 + e^{-t}}.
\end{aligned}$$

For $k = 1$, the derivatives are te^{-t} and $(1-t)e^{-t}$; for $k = 0$, they are $1 - e^{-t}$ and e^{-t}. These verify the ODE and zero initial data in both cases. There is no boundary condition at infinite time, so a negative real k is also permitted in the causal formula despite the possible growth of the response.

10.1.8 Take $k > 0$ and time dependence $e^{-i\omega\tau}$, with $\omega > 0$. Verify that the Green's function

$$G(x,x') = -\frac{i}{2k}\exp\left(ik|x - x'|\right)$$

yields an outgoing wave solution to the ODE

$$\left(\frac{d^2}{dx^2} + k^2\right)\psi(x) = g(x).$$

Note. Compare with Example 10.1.3.

Solution.

Take $k > 0$ and the time convention $e^{-i\omega\tau}$ used in Example 10.1.3. Its Eq. (10.28), textbook p. 454, omits k inside the exponential; the expression in this exercise and Table 10.1 has the correct factor $ik|x - x'|$. Let $s = x - x'$. Away from $s = 0$, each branch is proportional to e^{iks} or e^{-iks}, and hence solves $G_{xx} + k^2 G = 0$. Both branches have the common value $-i/(2k)$ at $s = 0$. Their first derivatives are

$$G_x(x, x') = \begin{cases} -\frac{1}{2}e^{ik(x'-x)}, & x < x', \\ \frac{1}{2}e^{ik(x-x')}, & x > x'. \end{cases}$$

Thus $G_x(x'^+, x') - G_x(x'^-, x') = 1$. Continuity rules out a delta derivative, and this jump gives

$$(\partial_x^2 + k^2)G(x, x') = \delta(x - x').$$

For a continuous compactly supported source g, define

$$F(x) = \int_{-\infty}^{x} e^{-ikt} g(t)\, dt, \qquad H(x) = \int_{x}^{\infty} e^{ikt} g(t)\, dt.$$

The convolution has the explicit form

$$\psi(x) = -\frac{i}{2k}\left[e^{ikx}F(x) + e^{-ikx}H(x)\right].$$

Because $F' = e^{-ikx}g$ and $H' = -e^{ikx}g$, their endpoint contributions cancel in the first derivative. A second differentiation gives

$$\psi'(x) = \frac{1}{2}\left[e^{ikx}F(x) - e^{-ikx}H(x)\right],$$
$$\psi''(x) = \frac{ik}{2}\left[e^{ikx}F(x) + e^{-ikx}H(x)\right] + g(x).$$

Since $k^2\psi = -ik[e^{ikx}F + e^{-ikx}H]/2$, their sum is g.

If the source is supported in $[a, b]$, then for $x > b$ only the first term remains: $\psi = C_+ e^{ikx}$. For $x < a$ only the second remains: $\psi = C_- e^{-ikx}$. With the time factor $e^{-i\omega\tau}$, constant phase on the right obeys $kx - \omega\tau =$ constant and moves to increasing x; on the left it obeys $-kx - \omega\tau =$ constant and moves to decreasing x. Both waves travel away from the source, which verifies the outgoing condition.

10.1.9 Construct the 1-D Green's function for the modified Helmholtz equation, with $k > 0$ in the displayed answer,

$$\left(\frac{d^2}{dx^2} - k^2\right)\psi(x) = f(x).$$

The boundary conditions are that the Green's function must vanish for $x \to \infty$ and $x \to -\infty$.

ANS. $G(x_1, x_2) = -\dfrac{1}{2k}\exp\left(-k|x_1 - x_2|\right).$

Solution.

The printed formula assumes $k > 0$. For a real nonzero parameter of either sign, set $\kappa = |k|$; the operator contains $k^2 = \kappa^2$, while decay requires the positive root. For $x < t$, decay as $x \to -\infty$

selects $e^{\kappa x}$; for $x > t$, decay as $x \to +\infty$ selects $e^{-\kappa x}$. Continuity can be imposed by a single coefficient:

$$G(x,t) = C \begin{cases} e^{\kappa(x-t)}, & x < t, \\ e^{-\kappa(x-t)}, & x > t. \end{cases}$$

The jump of its derivative is $-\kappa C - \kappa C = -2\kappa C$. The coefficient of ψ'' is one, so this must equal one. Hence

$$\boxed{G(x,t) = -\frac{1}{2\kappa} e^{-\kappa|x-t|}.}$$

It is continuous at $x = t$, solves $G_{xx} - \kappa^2 G = 0$ off the diagonal, and decays at both infinities. Its derivative limits are $-1/2$ and $+1/2$, so $(\partial_x^2 - k^2)G = \delta(x-t)$. Therefore

$$\psi(x) = -\frac{1}{2\kappa}\left[e^{-\kappa x}\int_{-\infty}^{x} e^{\kappa t} f(t)\,dt + e^{\kappa x}\int_{x}^{\infty} e^{-\kappa t} f(t)\,dt\right]$$

is the required solution for, for example, a continuous compactly supported source. A homogeneous solution $Ae^{\kappa x} + Be^{-\kappa x}$ can decay at both ends only if $A = B = 0$, proving uniqueness.

For $k = 0$, no Green's function for d^2/dx^2 can vanish at both infinities. Each branch would be affine; decay at its corresponding infinity would make that branch zero, leaving no derivative jump. The divergent constant in the $\kappa \downarrow 0$ expansion

$$G(x,t) = -\frac{1}{2\kappa} + \frac{1}{2}|x-t| + O(\kappa)$$

reflects that obstruction. Subtracting the constant would yield the Laplace fundamental solution, but it would not satisfy decay at infinity.

10.1.10 From the eigenfunction expansion of the Green's function show that

(a) $$\frac{2}{\pi^2}\sum_{n=1}^{\infty} \frac{\sin n\pi x \,\sin n\pi t}{n^2} = \begin{cases} x(1-t), & 0 \le x < t, \\ t(1-x), & t < x \le 1. \end{cases}$$

(b) $$\frac{2}{\pi^2}\sum_{n=0}^{\infty} \frac{\sin(n+\frac{1}{2})\pi x \,\sin(n+\frac{1}{2})\pi t}{(n+\frac{1}{2})^2} = \begin{cases} x, & 0 \le x < t, \\ t, & t < x \le 1. \end{cases}$$

Solution.

In both parts use $L = -d^2/dx^2$ on $[0,1]$. For a normalized eigenfunction ϕ_n, $L\phi_n = \lambda_n \phi_n$, the inverse has coefficients $1/\lambda_n$:

$$G(x,t) = \sum_n \frac{\phi_n(x)\phi_n(t)}{\lambda_n}.$$

There is no complex conjugation to display here because the chosen eigenfunctions are real.

(a) With $\phi(0) = \phi(1) = 0$, the equation $-\phi'' = \lambda\phi$ has positive eigenvalues. Indeed, integration by parts gives

$$\lambda \int_0^1 |\phi|^2 dx = \int_0^1 |\phi'|^2 dx,$$

and equality at $\lambda = 0$ would make ϕ a constant vanishing at the endpoints. Write $\lambda = q^2$ with $q > 0$. The condition at zero removes the cosine term, and $\sin q = 0$ gives $q = n\pi$, $n \ge 1$. Since

$$\int_0^1 \sin^2(n\pi x)\,dx = \left[\frac{x}{2} - \frac{\sin(2n\pi x)}{4n\pi}\right]_0^1 = \frac{1}{2},$$

we have $\phi_n(x) = \sqrt{2}\sin(n\pi x)$ and $\lambda_n = n^2\pi^2$. The eigenfunction formula is therefore

$$G_D(x,t) = \frac{2}{\pi^2}\sum_{n=1}^{\infty}\frac{\sin(n\pi x)\sin(n\pi t)}{n^2}.$$

Construct the same kernel directly. The left and right homogeneous solutions are x and $1-x$, and continuity suggests $Cx_<(1-x_>)$. Its derivative jump is $-Ct - C(1-t) = -C$. Requiring $-G_{xx} = \delta$ gives $C = 1$. Thus

$$\boxed{\frac{2}{\pi^2}\sum_{n=1}^{\infty}\frac{\sin(n\pi x)\sin(n\pi t)}{n^2} = \begin{cases} x(1-t), & x \le t, \\ t(1-x), & t \le x. \end{cases}}$$

(b) Now impose $\phi(0) = 0$, $\phi'(1) = 0$. The same energy identity shows that the eigenvalues are positive: its boundary term vanishes also for these mixed conditions. Again $\phi = A\sin(qx)$, while the second endpoint condition gives $q\cos q = 0$. Thus

$$q_n = (n+\tfrac{1}{2})\pi, \qquad \lambda_n = (n+\tfrac{1}{2})^2\pi^2, \qquad n = 0, 1, 2, \ldots.$$

The normalization is unchanged because $\sin(2q_n) = 0$: $\int_0^1 \sin^2(q_n x)dx = 1/2$. Hence

$$G_M(x,t) = \frac{2}{\pi^2}\sum_{n=0}^{\infty}\frac{\sin((n+\frac{1}{2})\pi x)\sin((n+\frac{1}{2})\pi t)}{(n+\frac{1}{2})^2}.$$

Exercise 10.1.1 constructed the unique kernel for precisely these conditions, $G_M(x,t) = \min(x,t)$. It follows that

$$\boxed{\frac{2}{\pi^2}\sum_{n=0}^{\infty}\frac{\sin((n+\frac{1}{2})\pi x)\sin((n+\frac{1}{2})\pi t)}{(n+\frac{1}{2})^2} = \begin{cases} x, & x \le t, \\ t, & t \le x. \end{cases}}$$

For completeness, the identification does not require differentiating either series twice term by term. For each fixed interior t, integrate by parts on $(0,t)$ and $(t,1)$, using $-G_{xx} = \delta(x-t)$ and the matching endpoint conditions. Because each closed kernel is affine on either side of t, integration by parts twice gives

$$\begin{aligned}\int_0^1 G(-\phi_n'')\,dx &= \big[-G\phi_n' + G_x\phi_n\big]_0^{t^-} + \big[-G\phi_n' + G_x\phi_n\big]_{t^+}^1 \\ &= \big[G_x(t^-,t) - G_x(t^+,t)\big]\phi_n(t) = \phi_n(t).\end{aligned}$$

Here the outer-endpoint terms vanish; the terms with $\phi_n'(t)$ cancel because G is continuous, and the derivative jump is -1. Since $-\phi_n'' = \lambda_n\phi_n$, this gives

$$\lambda_n\int_0^1 G(x,t)\phi_n(x)\,dx = \int_0^1 G(x,t)(-\phi_n''(x))\,dx = \phi_n(t).$$

Thus the closed kernels have exactly the stated Fourier coefficients. Completeness gives equality in L^2. Both series converge absolutely and uniformly by comparison with $\sum n^{-2}$ or $\sum(n+1/2)^{-2}$, so their sums are continuous. Equality therefore holds everywhere, including $x = t$ and the endpoints by continuity.

10.1.11 Derive an integral equation corresponding to

$$y''(x) - y(x) = 0, \quad y(1) = 1, \quad y(-1) = 1,$$

(a) by integrating twice.

(b) by forming the Green's function.

$$\textit{ANS. } y(x) = 1 - \int_{-1}^{1} K(x,t)\, y(t)\, dt,$$
$$K(x,t) = \begin{cases} \frac{1}{2}(1-x)(t+1), & x > t, \\ \frac{1}{2}(1-t)(x+1), & x < t. \end{cases}$$

Solution.

(a) Rewrite the ODE as $y'' = y$, and set $c = y'(-1)$. Integrating first to s and then to x gives

$$y'(s) = c + \int_{-1}^{s} y(t)\, dt, \qquad y(x) = 1 + c(x+1) + \int_{-1}^{x}\int_{-1}^{s} y(t)\, dt\, ds.$$

Interchanging the order on the triangle $-1 \le t \le s \le x$ turns the double integral into $\int_{-1}^{x}(x-t)y(t)dt$. At $x = 1$,

$$1 = 1 + 2c + \int_{-1}^{1}(1-t)y(t)\, dt, \qquad c = -\frac{1}{2}\int_{-1}^{1}(1-t)y(t)\, dt.$$

Substitution yields

$$y(x) = 1 + \int_{-1}^{x}(x-t)y(t)\, dt - \frac{x+1}{2}\int_{-1}^{1}(1-t)y(t)\, dt.$$

Split the last integral at $t = x$. On $t < x$, its combined coefficient is

$$(x-t) - \frac{x+1}{2}(1-t) = \frac{2x - 2t - x - 1 + xt + t}{2} = -\frac{(1-x)(t+1)}{2}.$$

On $t > x$, the coefficient is $-(x+1)(1-t)/2$. We obtain

$$\boxed{y(x) = 1 - \int_{-1}^{1} K(x,t)y(t)\, dt, \quad K(x,t) = \begin{cases} \frac{1}{2}(1-x)(t+1), & t \le x, \\ \frac{1}{2}(1-t)(x+1), & x \le t. \end{cases}}$$

(b) Homogenize the boundary conditions by setting $u = y - 1$. Then $u(-1) = u(1) = 0$, and the original equation gives $-u'' = -y$. Construct the Green's function of $-d^2/dx^2$ on $[-1, 1]$: the left and right homogeneous solutions are $x + 1$ and $1 - x$. The continuous kernel $C(x_< + 1)(1 - x_>)$ has derivative jump

$$-C(t+1) - C(1-t) = -2C.$$

Setting that jump to -1 gives $C = 1/2$, which is exactly K above. Thus

$$u(x) = \int_{-1}^{1} K(x,t)(-y(t))\, dt,$$

and adding 1 recovers the same integral equation.

Conversely, $K(\pm 1, t) = 0$ makes any solution of that integral equation satisfy $y(\pm 1) = 1$. Applying $-d^2/dx^2$ gives $-y'' = -y$, or $y'' - y = 0$. As an explicit check, write $y = A\cosh x + B\sinh x$. The two endpoint values give $A\cosh 1 + B\sinh 1 = 1$ and $A\cosh 1 - B\sinh 1 = 1$, so

$$\boxed{y(x) = \frac{\cosh x}{\cosh 1}.}$$

Direct integration of the two branches of K gives $\int_{-1}^{1} K(x,t)\cosh t\, dt = \cosh 1 - \cosh x$, verifying the integral equation with this solution.

10.1.12 The general second-order linear ODE with constant coefficients is

$$y''(x) + a_1 y'(x) + a_2 y(x) = 0.$$

Given the boundary conditions $y(0) = y(1) = 0$, integrate twice and develop the integral equation

$$y(x) = \int_0^1 K(x,t)\, y(t)\, dt,$$

with

$$K(x,t) = \begin{cases} a_2 t(1-x) + a_1(x-1), & t < x, \\ a_2 x(1-t) + a_1 x, & x < t. \end{cases}$$

Note that $K(x,t)$ is symmetric and continuous if $a_1 = 0$. How is this related to self-adjointness of the ODE? For this self-adjointness discussion take a_1, a_2 real, and distinguish the ordinary unit-weight inner product from a weighted one.

Solution.

Let $c = y'(0)$ and use $y(0) = 0$. One integration of $y'' = -a_1 y' - a_2 y$ gives

$$y'(x) = c - a_1 y(x) - a_2 \int_0^x y(t)\, dt.$$

Integrate once more. Reversing the order in the double integral gives

$$y(x) = cx - a_1 \int_0^x y(t)\, dt - a_2 \int_0^x (x-t) y(t)\, dt.$$

The second boundary condition supplies

$$0 = c - a_1 \int_0^1 y(t)\, dt - a_2 \int_0^1 (1-t) y(t)\, dt.$$

Eliminate c from the preceding expression:

$$\begin{aligned} y(x) = a_1 x \int_0^1 y(t)\, dt - a_1 \int_0^x y(t)\, dt \\ + a_2 x \int_0^1 (1-t) y(t)\, dt - a_2 \int_0^x (x-t) y(t)\, dt. \end{aligned}$$

For $t < x$, the coefficient of $y(t)$ is

$$a_1(x-1) + a_2\big[x(1-t) - (x-t)\big] = a_1(x-1) + a_2 t(1-x).$$

For $t > x$, it is $a_1 x + a_2 x(1-t)$. Hence

$$\boxed{y(x) = \int_0^1 K(x,t)y(t)\,dt, \qquad K(x,t) = \begin{cases} a_2 t(1-x) + a_1(x-1), & t < x, \\ a_2 x(1-t) + a_1 x, & t > x. \end{cases}}$$

The value of K at the single point $t = x$ does not affect the integral.

The reverse implication can also be checked without assuming the ODE. Let the integral on the right be $F(x)$ and split it at $t = x$. The first branch at the diagonal minus the second branch there is $-a_1$, so the Leibniz rule gives

$$F'(x) = -a_1 y(x) + \int_0^x (-a_2 t + a_1)y(t)\,dt + \int_x^1 \big[a_2(1-t) + a_1\big]y(t)\,dt.$$

A second differentiation gives

$$F'' = -a_1 y' + \big[(-a_2 x + a_1) - (a_2(1-x) + a_1)\big]y = -a_1 y' - a_2 y.$$

Thus $y = F$ implies the original ODE. Also $K(0,t) = K(1,t) = 0$, so the integral equation incorporates both boundary conditions.

The self-adjointness discussion assumes real a_1, a_2 and the unit-weight inner product. A nonzero real a_1 does not make the ODE intrinsically incapable of a self-adjoint formulation; it requires an integrating factor, or equivalently a weighted inner product. For real coefficients and zero Dirichlet data, integration by parts gives

$$\left(\frac{d^2}{dx^2} + a_1\frac{d}{dx} + a_2\right)^\dagger = \frac{d^2}{dx^2} - a_1\frac{d}{dx} + a_2.$$

So the displayed operator is formally self-adjoint in unit weight precisely when $a_1 = 0$. In this case

$$K(x,t) = a_2 x_<(1 - x_>),$$

which is continuous and symmetric. When $a_1 \neq 0$, the kernel jumps by a_1 as t passes from below to above x and is not symmetric. Multiplication of the ODE by $p(x) = e^{a_1 x}$ instead gives

$$(py')' + a_2 py = 0.$$

This differential expression is self-adjoint with the Dirichlet domain; the original expression is symmetric in the weighted scalar product $\int_0^1 p(x)\phi(x)^*\psi(x)dx$.

The homogeneous integral equation need not have a nonzero solution for arbitrary coefficients. To identify its exceptional parameters, put $y = e^{-a_1 x/2}v$. Then

$$y' = e^{-a_1 x/2}(v' - a_1 v/2), \qquad y'' = e^{-a_1 x/2}(v'' - a_1 v' + a_1^2 v/4),$$

and substitution reduces the ODE to

$$v'' + (a_2 - a_1^2/4)v = 0, \qquad v(0) = v(1) = 0.$$

For real coefficients, nonzero solutions exist exactly when $a_2 - a_1^2/4 = n^2\pi^2$, $n = 1, 2, \ldots$; they are $y = Ce^{-a_1 x/2}\sin(n\pi x)$. Otherwise the only solution of either formulation is $y = 0$.

10.1.13 Transform the ODE

$$\frac{d^2y(r)}{dr^2} - k^2 y(r) + V_0 \frac{e^{-r}}{r} y(r) = 0$$

and the boundary conditions $y(0) = y(\infty) = 0$ into an integral equation of the form

$$y(r) = -V_0 \int_0^\infty G(r,t)\, \frac{e^{-t}}{t}\, y(t)\, dt.$$

The quantities V_0 and k^2 are constants. Take $k > 0$ (or, more generally, $\Re k > 0$ for the displayed decaying kernel), and the regular branch $y(r) = O(r)$ at the origin. The ODE is derived from the Schrödinger wave equation with a mesonic potential:

$$G(r,t) = \begin{cases} -\dfrac{1}{k} e^{-kt} \sinh kr, & 0 \le r < t, \\ -\dfrac{1}{k} e^{-kr} \sinh kt, & t < r < \infty. \end{cases}$$

Solution.

The decaying kernel in the question assumes k is the positive square root of a positive k^2 (more generally $\Re k > 0$). The singular endpoint is interpreted using the regular solution $y(r) = O(r)$ as $r \downarrow 0$. The manual's final sentence on p. 120, claiming that the question's G is minus the Green's function, is incorrect for the chapter's convention $LG = +\delta$. Isolate the free operator and source:

$$L_0 = \frac{d^2}{dr^2} - k^2, \qquad L_0 y(r) = -V_0 \frac{e^{-r}}{r} y(r).$$

For $L_0 u = 0$, choose $u(r) = \sinh(kr)$, which vanishes at zero, and $v(r) = e^{-kr}$, which decays at infinity. Their Wronskian is

$$\begin{aligned} uv' - u'v &= -k \sinh(kr) e^{-kr} - k \cosh(kr) e^{-kr} \\ &= -k e^{-kr} \big[\sinh(kr) + \cosh(kr) \big] = -k. \end{aligned}$$

The continuous kernel $Cu(r_<)v(r_>)$ has derivative jump $-kC$. For $L_0 G = \delta(r - t)$ the jump must be 1, so $C = -1/k$:

$$\boxed{G(r,t) = -\frac{1}{k} \sinh(kr_<) e^{-kr_>}, \qquad r_< = \min(r,t), \quad r_> = \max(r,t).}$$

It satisfies $G(0,t) = 0$ and $G(\infty,t) = 0$. The matching calculation shows directly that it is the Green's function with a positive unit delta source. Equivalently, the method-of-images form is

$$G(r,t) = -\frac{1}{2k} \left[e^{-k|r-t|} - e^{-k(r+t)} \right].$$

The second term cancels the first at $r = 0$ and is homogeneous as a function of positive r.

Apply this inverse to the source $-V_0 e^{-t} y(t)/t$:

$$\boxed{y(r) = -V_0 \int_0^\infty G(r,t) \frac{e^{-t}}{t}\, y(t)\, dt.}$$

Thus the negative sign outside the integral and the negative sign in G are both required. In split form, the same equation reads

$$y(r) = \frac{V_0}{k}\left[e^{-kr}\int_0^r \sinh(kt)\frac{e^{-t}}{t}y(t)\,dt + \sinh(kr)\int_r^\infty e^{-kt}\frac{e^{-t}}{t}y(t)\,dt\right].$$

On the regular branch $y(t) = O(t)$, the source factor $y(t)/t$ is bounded near zero. For example, substitution of $y = At + Bt^2 + \cdots$ into the ODE gives $2B + V_0 A = 0$, so $y(t) = At - (V_0 A/2)t^2 + \cdots$. At infinity the exponential factors ensure convergence for a decaying y.

Conversely, apply L_0 to the integral equation. The delta identity gives $L_0 y = -V_0 e^{-r} y/r$, and the kernel enforces the endpoint conditions. There is no omitted homogeneous term: $Ae^{kr} + Be^{-kr}$ can decay at infinity only if $A = 0$, and its value at zero then forces $B = 0$. The integral equation is therefore equivalent to the original boundary problem within this regular, decaying class; it is a homogeneous eigenvalue equation for the allowed parameters, not an assertion that all V_0, k admit a nonzero solution.

The limit $k \downarrow 0$ gives $G(r,t) \to -\min(r,t)$, which approaches $-t$, not zero, as $r \to \infty$. Accordingly there is no inverse of d^2/dr^2 satisfying these same two zero boundary values for an arbitrary source: its right homogeneous branch would have to be identically zero. This explains why the decay assumption on the free parameter is needed.

10.2 Problems in Two and Three Dimensions

10.2.1 Show that the fundamental Green's function for the 1-D Laplace equation, $|x_1 - x_2|/2$, is consistent with the form found in Example 10.1.1.

Solution.

Write $x = x_1$, $t = x_2$. The fundamental solution $G_0(x,t) = |x-t|/2$ has derivative $-1/2$ for $x < t$ and $+1/2$ for $x > t$. Its derivative jump is 1, so

$$\partial_x^2 G_0(x,t) = \delta(x-t).$$

To impose zero Dirichlet values at $x = 0, 1$, add a homogeneous solution in x, $h(x,t) = a(t)x + b(t)$. The endpoint equations are

$$0 = G_0(0,t) + b(t) = t/2 + b(t),$$
$$0 = G_0(1,t) + a(t) + b(t) = (1-t)/2 + a(t) - t/2.$$

Hence $b = -t/2$, $a = t - 1/2$, and

$$G_D(x,t) = \frac{1}{2}|x-t| + xt - \frac{x+t}{2}.$$

This added affine function has zero second derivative and cannot change the source normalization. Evaluating the two branches gives

$$x < t: \quad G_D = \frac{t-x}{2} + xt - \frac{x+t}{2} = -x(1-t),$$
$$x > t: \quad G_D = \frac{x-t}{2} + xt - \frac{x+t}{2} = -t(1-x).$$

Thus

$$\boxed{G_D(x,t) = -\begin{cases} x(1-t), & x \le t, \\ t(1-x), & t \le x. \end{cases}}$$

The operator in Example 10.1.1 is $-d^2/dx^2$, whereas the Laplace operator here is $+d^2/dx^2$. Their Green's functions therefore differ by a minus sign. Consistency requires retaining that operator sign, as also shown in Example 10.2.1, textbook p. 464. Indeed $\partial_x^2 G_D = \delta$, while $(-\partial_x^2)(-G_D) = \delta$; the latter positive kernel is exactly the one in Example 10.1.1.

10.2.2 Show that if

$$L\psi(\mathbf{r}) \equiv \nabla \cdot [p(\mathbf{r})\nabla\psi(\mathbf{r})] + q(\mathbf{r})\psi(\mathbf{r}),$$

then L is Hermitian for $p(\mathbf{r})$ and $q(\mathbf{r})$ real, assuming common homogeneous Dirichlet boundary conditions, sufficient regularity for Green's identity, and a vanishing boundary form at infinity if the region is unbounded, and that the scalar product is an integral over that region with unit weight.

Solution.

For an operator domain, the Dirichlet conditions used here are homogeneous: both test functions vanish at the boundary. Nonzero prescribed boundary data must first be subtracted using a fixed extension. On an unbounded region the corresponding boundary form at infinity must also vanish. Let Ω have a smooth boundary, with sufficiently regular p, q, ϕ, ψ for the integrations below, and take p, q real. The unit-weight scalar product is

$$\langle \phi, \psi \rangle = \int_\Omega \phi^* \psi \, dV.$$

The product rule gives

$$\phi^* \nabla \cdot (p\nabla\psi) = \nabla \cdot (p\phi^* \nabla\psi) - p\nabla\phi^* \cdot \nabla\psi.$$

Integrate and use the divergence theorem:

$$\begin{aligned} \langle \phi, L\psi \rangle &= \int_{\partial\Omega} p\phi^* \partial_n \psi \, dS - \int_\Omega p\nabla\phi^* \cdot \nabla\psi \, dV + \int_\Omega q\phi^* \psi \, dV \\ &= -\int_\Omega p\nabla\phi^* \cdot \nabla\psi \, dV + \int_\Omega q\phi^* \psi \, dV, \end{aligned}$$

because $\phi = 0$ on $\partial\Omega$. Reality of the coefficients gives $(L\phi)^* = \nabla \cdot (p\nabla\phi^*) + q\phi^*$. Repeating the integration by parts with ψ outside the divergence yields

$$\begin{aligned} \langle L\phi, \psi \rangle &= \int_{\partial\Omega} p\psi \partial_n \phi^* \, dS - \int_\Omega p\nabla\psi \cdot \nabla\phi^* \, dV + \int_\Omega q\phi^* \psi \, dV \\ &= -\int_\Omega p\nabla\psi \cdot \nabla\phi^* \, dV + \int_\Omega q\phi^* \psi \, dV. \end{aligned}$$

The remaining volume integrals agree term by term, so

$$\boxed{\langle \phi, L\psi \rangle = \langle L\phi, \psi \rangle.}$$

Equivalently, their difference is the boundary form

$$\int_{\partial\Omega} p \left(\phi^* \partial_n \psi - \psi \partial_n \phi^* \right) dS = 0.$$

This proves the Hermitian (symmetric) property on the stated common domain. If "self-adjoint" is used in its stronger operator-theoretic sense, a domain realization must also be specified; the usual smooth, bounded, uniformly elliptic Dirichlet realization is one such setting.

10.2.3 Show that the terms $+k^2$ in the Helmholtz operator and $-k^2$ in the modified Helmholtz operator do not affect the behavior of $G(\mathbf{r}_1, \mathbf{r}_2)$ in the immediate vicinity of the singular point $\mathbf{r}_1 = \mathbf{r}_2$. Specifically, show that

$$\lim_{\varepsilon \downarrow 0} \int_{|\mathbf{r}_2 - \mathbf{r}_1| < \varepsilon} k^2 G(\mathbf{r}_1, \mathbf{r}_2)\, d^3 r_2 = 0.$$

Solution.

The printed limit, including its value -1, is incorrect for a shrinking neighborhood of the source (textbook p. 466). The integration region must be specified as a ball of radius ε, and the local limit is 0. The manual, p. 121, computes a different, whole-space Fourier integral, formally giving $+1/k^2$ for its Helmholtz denominator $k^2 - p^2$. It does not establish the printed local claim. Fix $\mathbf{r}_1$ and put $s = |\mathbf{r}_2 - \mathbf{r}_1|$. The three-dimensional outgoing and modified kernels are respectively

$$G_H(s) = -\frac{e^{iks}}{4\pi s}, \qquad G_M(s) = -\frac{e^{-ks}}{4\pi s}, \qquad k > 0.$$

Their Taylor expansions at $s = 0$ start with

$$G_H(s) = -\frac{1}{4\pi s} - \frac{ik}{4\pi} + O(s), \qquad G_M(s) = -\frac{1}{4\pi s} + \frac{k}{4\pi} + O(s).$$

Both therefore have the same leading singularity as the Laplace kernel. Since $d^3 r_2 = 4\pi s^2 ds$ for a radial integral,

$$\begin{aligned} \int_{s<\varepsilon} k^2 G(s)\, d^3 r_2 &= 4\pi k^2 \int_0^\varepsilon \left[-\frac{1}{4\pi s} + O(1) \right] s^2\, ds \\ &= -\frac{k^2 \varepsilon^2}{2} + O(\varepsilon^3). \end{aligned}$$

Consequently the corrected assertion is

$$\boxed{\lim_{\varepsilon \downarrow 0} \int_{|\mathbf{r}_2 - \mathbf{r}_1| < \varepsilon} k^2 G(\mathbf{r}_1, \mathbf{r}_2)\, d^3 r_2 = 0.}$$

The same conclusion holds for $-k^2 G$.

The exact radial integrals independently show the limit. For Helmholtz,

$$\begin{aligned} I_H(\varepsilon) &= -k^2 \int_0^\varepsilon s e^{iks}\, ds \\ &= -k^2 \left[e^{iks} \left(\frac{s}{ik} + \frac{1}{k^2} \right) \right]_0^\varepsilon \\ &= e^{ik\varepsilon}(ik\varepsilon - 1) + 1 \longrightarrow 0. \end{aligned}$$

For modified Helmholtz,

$$\begin{aligned} I_M(\varepsilon) &= -k^2 \int_0^\varepsilon s e^{-ks}\, ds \\ &= \Big[(1 + ks) e^{-ks} \Big]_0^\varepsilon = (1 + k\varepsilon) e^{-k\varepsilon} - 1 \longrightarrow 0. \end{aligned}$$

Integrating the defining equation over the ball now gives

$$\int_{s=\varepsilon} \nabla G \cdot d\mathbf{S} \pm \int_{s<\varepsilon} k^2 G \, dV = 1.$$

The second integral tends to zero, while the flux tends to one. Therefore it is the Laplacian term that supplies the delta singularity; the zeroth-order $\pm k^2$ term cannot change its coefficient. For comparison, taking the entire space for the modified kernel means $\varepsilon \to \infty$, and gives $I_M \to -1$. That global value must not be substituted for the shrinking-ball limit.

10.2.4 Show that

$$-\frac{\exp(ik|\mathbf{r}_1 - \mathbf{r}_2|)}{4\pi|\mathbf{r}_1 - \mathbf{r}_2|}$$

satisfies the appropriate criteria and therefore is a Green's function for the Helmholtz equation. Use $k > 0$ and the outgoing convention with time factor $e^{-i\omega\tau}$, $\omega > 0$.

Solution.

Let $s = |\mathbf{r}_1 - \mathbf{r}_2|$ and take $k > 0$, with time dependence $e^{-i\omega\tau}$. For a radial function $G(s)$ at $s > 0$,

$$\nabla^2 G = G'' + \frac{2}{s} G'.$$

Write $G = F(s)/s$. Then

$$G' = \frac{F'}{s} - \frac{F}{s^2}, \qquad G'' = \frac{F''}{s} - \frac{2F'}{s^2} + \frac{2F}{s^3},$$

so the first-derivative terms cancel and

$$(\nabla^2 + k^2)G = \frac{F'' + k^2 F}{s}.$$

The homogeneous radial solutions are consequently $G = (Ae^{iks} + Be^{-iks})/s$. The $e^{iks - i\omega\tau}$ phase moves outward, whereas $e^{-iks - i\omega\tau}$ moves inward. Thus the outgoing condition selects $B = 0$.

For $G = Ae^{iks}/s$, the flux through a sphere of radius ε centered on the source is

$$4\pi\varepsilon^2 G'(\varepsilon) = 4\pi A e^{ik\varepsilon}(ik\varepsilon - 1) \longrightarrow -4\pi A.$$

Exercise 10.2.3 shows that the integral of $k^2 G$ over that small ball vanishes. A positive unit delta source therefore requires $-4\pi A = 1$, giving

$$\boxed{G(\mathbf{r}_1, \mathbf{r}_2) = -\frac{e^{ik|\mathbf{r}_1 - \mathbf{r}_2|}}{4\pi|\mathbf{r}_1 - \mathbf{r}_2|}.}$$

It solves the homogeneous Helmholtz equation away from the source, is smooth there, and has a locally integrable $1/s$ singularity. Its outgoing Sommerfeld condition is explicit:

$$s(\partial_s - ik)G = \frac{e^{iks}}{4\pi s} \longrightarrow 0.$$

To verify the delta identity against an arbitrary smooth compactly supported test function η, apply Green's second identity outside a ball of radius ε around the source. The outer boundary

contributes zero, while the outward normal of this punctured region at the inner sphere points in the direction $-\hat{\mathbf{s}}$. Since $(\nabla^2 + k^2)G = 0$ there, the identity reads

$$\int_{s>\varepsilon} G(\nabla^2 + k^2)\eta \, dV = \int_{s=\varepsilon} [-G\,\partial_s\eta + \eta\,\partial_s G]\, dS.$$

The first surface term is $O(\varepsilon)$, because $G = O(1/\varepsilon)$ and the sphere has area $O(\varepsilon^2)$. In the second term, $\eta = \eta(\mathbf{r}_2) + O(\varepsilon)$ and the flux tends to one, so its limit is $\eta(\mathbf{r}_2)$. This proves, with its sign and normalization,

$$(\nabla_1^2 + k^2)G(\mathbf{r}_1, \mathbf{r}_2) = \delta^{(3)}(\mathbf{r}_1 - \mathbf{r}_2).$$

Convolution with a compactly supported source therefore produces the outgoing solution of the inhomogeneous Helmholtz equation.

10.2.5 Find the Green's function for the 3-D Helmholtz equation, Exercise 10.2.4, when the wave is a standing wave. Find the customary equal average of outgoing and incoming kernels, and state the remaining freedom to add a regular standing homogeneous solution.

Solution.

With $s = |\mathbf{r}_1 - \mathbf{r}_2|$, the outgoing and incoming Green's functions are

$$G_{\text{out}} = -\frac{e^{iks}}{4\pi s}, \qquad G_{\text{in}} = -\frac{e^{-iks}}{4\pi s}.$$

For real k, they are complex conjugates. Each solves $(\nabla^2 + k^2)G = \delta^{(3)}$; its small-sphere flux tends to 1. The equal average therefore still has a unit, rather than doubled, delta source:

$$\begin{aligned} G_{\text{st}} &= \frac{1}{2}(G_{\text{out}} + G_{\text{in}}) \\ &= -\frac{e^{iks} + e^{-iks}}{8\pi s} = \boxed{-\frac{\cos(ks)}{4\pi s}}. \end{aligned}$$

Multiplying by $e^{-i\omega\tau}$ and taking the real part gives $-\cos(ks)\cos(\omega\tau)/(4\pi s)$, a standing radial wave. Equivalently, the outgoing and incoming amplitudes are equal, so their opposite radial fluxes cancel. Directly, $\operatorname{Im}(G_{\text{st}}^* \partial_s G_{\text{st}}) = 0$ because G_{st} is real.

Its derivative and source flux are

$$G'_{\text{st}} = \frac{k\sin(ks)}{4\pi s} + \frac{\cos(ks)}{4\pi s^2}, \qquad 4\pi s^2 G'_{\text{st}} = ks\sin(ks) + \cos(ks) \longrightarrow 1.$$

Also $F(s) = sG_{\text{st}} = -\cos(ks)/(4\pi)$ satisfies $F'' + k^2 F = 0$. These checks establish the same differential operator and source as in Exercise 10.2.4.

A standing-wave condition alone does not uniquely specify the regular homogeneous part. The answer above uses the customary equal average of the outgoing and incoming Green's functions. For any real constant C, adding $C\sin(ks)/s$ gives another real radial standing-wave kernel with the same delta source; an additional phase convention or boundary condition is needed to exclude it. Indeed $\sin(ks)/s = k - k^3 s^2/6 + \cdots$ is smooth at the origin, its small-sphere flux tends to zero, and it satisfies the homogeneous Helmholtz equation everywhere. Thus it does not alter the singular normalization.

10.2.6 Verify that the formula given for the 3-D Green's function of the modified Helmholtz equation in Table 10.1 is correct when the boundary conditions of the problem are that G vanish at infinity. Use the positive decay root $k > 0$ in the table and also identify its $k \downarrow 0$ limit.

Solution.

The three-dimensional entry in Table 10.1, textbook p. 463, is

$$\boxed{G(\mathbf{r}_1, \mathbf{r}_2) = -\frac{e^{-ks}}{4\pi s}, \qquad s = |\mathbf{r}_1 - \mathbf{r}_2|.}$$

Here $k > 0$ is the decay constant. For a real nonzero parameter of either sign, replace k by $|k|$. The limit $k = 0$ gives the decaying three-dimensional Laplace kernel $-1/(4\pi s)$. For $s > 0$, put $F = sG = -e^{-ks}/(4\pi)$. The radial calculation of Exercise 10.2.4 gives

$$(\nabla^2 - k^2)G = \frac{F'' - k^2 F}{s} = 0.$$

The kernel decays at infinity, and its leading singularity is $-1/(4\pi s)$.

Check the entire source normalization on a sphere of arbitrary radius $R > 0$, not just in a limiting expansion. The derivative is

$$G'(s) = \frac{e^{-ks}}{4\pi}\left(\frac{k}{s} + \frac{1}{s^2}\right),$$

so its flux is

$$\Phi(R) = 4\pi R^2 G'(R) = (1 + kR)e^{-kR}.$$

The integral of G over the ball is

$$\begin{aligned} I(R) &= 4\pi \int_0^R G(s) s^2 \, ds = -\int_0^R s e^{-ks} \, ds \\ &= \left[e^{-ks}\left(\frac{s}{k} + \frac{1}{k^2}\right)\right]_0^R = \frac{(1 + kR)e^{-kR} - 1}{k^2}. \end{aligned}$$

Therefore

$$\int_{s<R} (\nabla^2 - k^2)G \, dV = \Phi(R) - k^2 I(R) = (1 + kR)e^{-kR} - \left[(1 + kR)e^{-kR} - 1\right] = 1.$$

In particular $\Phi(R) \to 1$ and $k^2 I(R) \to 0$ as $R \downarrow 0$. The test-function argument from Exercise 10.2.4 applies without change to the singularity, proving

$$(\nabla_1^2 - k^2)G = \delta^{(3)}(\mathbf{r}_1 - \mathbf{r}_2).$$

For uniqueness in the class of kernels with that singularity and decay, the difference h of two such solutions is regular everywhere, vanishes at infinity, and obeys $\nabla^2 h = k^2 h$. Apply this to its real part: a positive maximum would be attained at a finite point and would have $\nabla^2 h \le 0$, contradicting $k^2 h > 0$; a negative minimum gives the opposite contradiction. Hence the real part is zero, and the same argument applies to the imaginary part. This establishes uniqueness for $k > 0$ under these conditions.

10.2.7 An electrostatic potential (mks units) is

$$\varphi(\mathbf{r}) = Z\,\frac{e^{-ar}}{4\pi\epsilon_0 r}, \qquad a > 0.$$

Reconstruct the electrical charge distribution that will produce this potential. Note that $\varphi(\mathbf{r})$ vanishes exponentially for large r, showing that the net charge is zero.

$$ANS.\ \ \rho(\mathbf{r}) = Z\,\delta(\mathbf{r}) - \frac{Za^2e^{-ar}}{4\pi r}.$$

Solution.

The exponential-decay and zero-net-charge conclusion assumes $a > 0$. At $a = 0$ the potential is Coulombic and its net charge is Z, not zero. Use Poisson's equation in SI units,

$$\nabla^2\varphi = -\frac{\rho}{\epsilon_0}, \qquad \rho = -\epsilon_0\nabla^2\varphi.$$

First compute away from the origin. For $h(r) = e^{-ar}/r$,

$$h' = -e^{-ar}\left(\frac{a}{r} + \frac{1}{r^2}\right), \qquad r^2h' = -(ar+1)e^{-ar},$$

and therefore

$$\frac{d}{dr}(r^2h') = -ae^{-ar} + a(ar+1)e^{-ar} = a^2re^{-ar}.$$

Thus $\nabla^2 h = r^{-2}(r^2h')' = a^2e^{-ar}/r$ for $r > 0$. This ordinary calculation omits the point singularity. Its coefficient is fixed by the flux through a small sphere:

$$\lim_{R\downarrow 0} 4\pi R^2 h'(R) = -4\pi \lim_{R\downarrow 0}(aR+1)e^{-aR} = -4\pi.$$

The integral of a^2h over that ball tends to zero, so the full distributional identity is

$$\nabla^2\frac{e^{-ar}}{r} = a^2\frac{e^{-ar}}{r} - 4\pi\delta^{(3)}(\mathbf{r}).$$

Insert $\varphi = Zh/(4\pi\epsilon_0)$ into Poisson's equation:

$$\boxed{\rho(\mathbf{r}) = Z\delta^{(3)}(\mathbf{r}) - \frac{Za^2e^{-ar}}{4\pi r}.}$$

The first term is a point charge Z at the origin; the second is a spherically symmetric cloud of opposite sign. Its $1/r$ density singularity is integrable because the radial volume element contains r^2dr.

Compute its total charge explicitly. Integration by parts gives

$$\int_0^\infty re^{-ar}\,dr = \Big[-\frac{r}{a}e^{-ar}\Big]_0^\infty + \frac{1}{a}\int_0^\infty e^{-ar}\,dr = \frac{1}{a^2}.$$

Consequently

$$Q_{\text{cloud}} = 4\pi\int_0^\infty r^2\left(-\frac{Za^2e^{-ar}}{4\pi r}\right)dr = -Z, \qquad Q_{\text{total}} = Z - Z = 0.$$

As an independent check, the enclosed charge at a finite radius is

$$Q(R) = Z - Za^2 \int_0^R r e^{-ar}\, dr = Z(1 + aR)e^{-aR}.$$

The electric field obtained by differentiating the supplied potential is

$$E_r = -\varphi'(r) = \frac{Ze^{-ar}}{4\pi\epsilon_0}\left(\frac{a}{r} + \frac{1}{r^2}\right),$$

and Gauss' law gives $4\pi\epsilon_0 R^2 E_r(R) = Q(R)$ exactly. It tends to Z as $R \downarrow 0$ and to zero as $R \to \infty$, confirming both the point charge and the screening cloud.

11 Chapter 11. Complex Variable Theory

Problems

11.2 Cauchy-Riemann Conditions

11.2.1 Show whether or not the function $f(z) = \Re(z) = x$ is analytic.

Solution.

Here $u = x$ and $v = 0$, so $u_x = 1$ whereas $v_y = 0$. The necessary Cauchy–Riemann equation $u_x = v_y$ fails at every point. Equivalently, for real h, $[f(z+h) - f(z)]/h = 1$, whereas $[f(z+ih) - f(z)]/(ih) = 0$. These direction-dependent limits disagree. Thus f is nowhere complex differentiable and analytic on no open set.

11.2.2 Having shown that the real part $u(x, y)$ and the imaginary part $v(x, y)$ of an analytic function $w(z)$ each satisfy Laplace's equation, show that neither $u(x, y)$ nor $v(x, y)$ can have either a maximum or a minimum in the interior of any region in which $w(z)$ is analytic and nonconstant, with the region connected. Any stationary points of the components are saddles.

Solution.

Exclude the constant case: a constant harmonic function has both non-strict maxima and minima everywhere. For nonconstant analytic w, expand at an interior point:

$$w(z_0 + h) = w(z_0) + a_m h^m + O(h^{m+1}), \qquad a_m \neq 0, \quad m \geq 1.$$

Writing $a_m = |a_m| e^{i\alpha}$ and $h = re^{i\theta}$ gives

$$u(z_0 + h) - u(z_0) = |a_m| r^m \cos(\alpha + m\theta) + O(r^{m+1}).$$

Choose $\theta = -\alpha/m$ and $(\pi - \alpha)/m$. For small $r > 0$ the leading terms dominate the remainder and have opposite signs. Thus nearby values of u lie both above and below $u(z_0)$. Choosing $\alpha + m\theta = \pi/2, 3\pi/2$ proves the same for v. If all Taylor coefficients vanish, w is locally constant, hence constant on its connected domain by the identity theorem. At a stationary point the nonconstant components have saddle behavior; at other points they have a nonzero gradient.

11.2.3 Find the analytic function

$$w(z) = u(x, y) + iv(x, y)$$

(a) if $u(x, y) = x^3 - 3xy^2$, (b) if $v(x, y) = e^{-y} \sin x$.

Solution.

(a) Cauchy–Riemann gives $v_y = u_x = 3x^2 - 3y^2$. Integrating in y gives $v = 3x^2 y - y^3 + g(x)$. Then $v_x = -u_y = 6xy$ forces $g' = 0$. Since $(x + iy)^3 = x^3 - 3xy^2 + i(3x^2 y - y^3)$,

$$\boxed{w(z) = z^3 + iC, \qquad C \in \mathbb{R}}.$$

(b) Here $u_x = v_y = -e^{-y} \sin x$, so $u = e^{-y} \cos x + g(y)$. The equation $u_y = -v_x = -e^{-y} \cos x$ gives $g' = 0$. Therefore

$$\boxed{w(z) = e^{iz} + C, \qquad C \in \mathbb{R}},$$

using $e^{iz} = e^{-y}(\cos x + i \sin x)$. The constants preserve the specified real or imaginary component.

11.2.4 If there is a common connected region in which $w_1 = u(x,y) + iv(x,y)$ and $w_2 = w_1^* = u(x,y) - iv(x,y)$ are both analytic, prove that $u(x,y)$ and $v(x,y)$ are constants.

Solution.

For w_1, $u_x = v_y$ and $u_y = -v_x$. For $w_2 = u - iv$, $u_x = -v_y$ and $u_y = v_x$. Adding and subtracting yields $u_x = u_y = v_x = v_y = 0$. Along every path in a connected component, $du = u_x dx + u_y dy = 0$ and $dv = v_x dx + v_y dy = 0$, so both functions are constant there. The two resulting complex constants are conjugates. The manual further writes $w_1 = w_2$, but equality requires $v = 0$. For example, $w_1 = 1 + i$ and $w_2 = 1 - i$ are both analytic and satisfy the hypotheses without being equal.

11.2.5 Starting from $f(z) = 1/(x+iy)$, show that $1/z$ is analytic in the entire finite z plane except at the point $z = 0$. This extends our discussion of the analyticity of z^n to negative integer powers n.

Solution.

Rationalizing the denominator gives $u = x/(x^2+y^2)$ and $v = -y/(x^2+y^2)$. Set $D = x^2+y^2 > 0$. Then

$$u_x = \frac{y^2 - x^2}{D^2} = v_y, \qquad u_y = -\frac{2xy}{D^2} = -v_x.$$

These partial derivatives are continuous off the origin, proving analyticity there. The derivative is

$$f' = u_x + iv_x = \frac{y^2 - x^2 + 2ixy}{D^2} = -\frac{(x-iy)^2}{D^2} = \boxed{-z^{-2}}.$$

At zero there is a pole. Products of $1/z$ consequently give analytic negative integer powers on $\mathbb{C} \setminus \{0\}$.

11.2.6 Show that, assuming real differentiability of u, v at the point, the Cauchy-Riemann equations imply that the derivative $f'(z)$ has the same value for $dz = a\,dx + ib\,dy$ (with neither a nor b zero) as it has for $dz = dx$.

Solution.

Assume real differentiability of u, v at the point; continuous first partial derivatives nearby are sufficient. For $h = a\,dx + ib\,dy$, Cauchy–Riemann gives

$$\begin{aligned} df &= (u_x + iv_x)a\,dx + (u_y + iv_y)b\,dy \\ &= (u_x + iv_x)a\,dx + (-v_x + iu_x)b\,dy \\ &= (u_x + iv_x)(a\,dx + ib\,dy). \end{aligned}$$

The remainder in this real differential is $o(|h|)$. Dividing by $h \neq 0$ and taking the limit yields $\boxed{f' = u_x + iv_x}$ independently of direction. Cauchy–Riemann equalities at an isolated point, without real differentiability, are not alone sufficient for this conclusion. For a concrete reason for the hypothesis, set

$$f(x+iy) = \begin{cases} xy/\sqrt{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0). \end{cases}$$

Here $v = 0$, and both partial derivatives of u at zero are zero because u vanishes on each axis, so Cauchy–Riemann holds there. Along the real axis the difference quotient is zero. Along $h = t(1+i)$ with $t > 0$, however,

$$\frac{f(h)}{h} = \frac{t/\sqrt{2}}{t(1+i)} = \frac{1-i}{2\sqrt{2}} \neq 0.$$

Thus real differentiability cannot be omitted.

11.2.7 Using $f(re^{i\theta}) = R(r,\theta)e^{i\Theta(r,\theta)}$, in which $R(r,\theta)$ and $\Theta(r,\theta)$ are differentiable real functions of r and θ, show that the Cauchy-Riemann conditions in polar coordinates become on a local patch with $r > 0$ and $f \neq 0$

$$\text{(a)}\ \frac{\partial R}{\partial r} = \frac{R}{r}\frac{\partial \Theta}{\partial \theta}, \qquad \text{(b)}\ \frac{1}{r}\frac{\partial R}{\partial \theta} = -R\frac{\partial \Theta}{\partial r}.$$

Hint. Set up the derivative first with δz radial and then with δz tangential.

Solution.

On a regular polar patch $r > 0$, use a differentiable phase for $f = Re^{i\Theta}$. A radial displacement gives

$$f'_r = e^{-i\theta}\partial_r f = e^{i(\Theta-\theta)}(R_r + iR\Theta_r).$$

A tangential displacement is $dz = ire^{i\theta}d\theta$, giving

$$f'_\theta = \frac{e^{-i\theta}}{ir}\partial_\theta f = e^{i(\Theta-\theta)}\left(\frac{R\Theta_\theta}{r} - i\frac{R_\theta}{r}\right).$$

Cancel the common phase and equate real and imaginary parts:

$$\boxed{R_r = \frac{R}{r}\Theta_\theta, \qquad \frac{1}{r}R_\theta = -R\Theta_r}.$$

A smooth argument is a local choice away from zeros of f; it is not generally defined at a zero or at $r = 0$.

11.2.8 As an extension of Exercise 11.2.7 show that on a nonzero analytic patch with $r > 0$, $\Theta(r,\theta)$ satisfies the 2-D Laplace equation in polar coordinates,

$$\frac{\partial^2 \Theta}{\partial r^2} + \frac{1}{r}\frac{\partial \Theta}{\partial r} + \frac{1}{r^2}\frac{\partial^2 \Theta}{\partial \theta^2} = 0.$$

Solution.

Use a local nonzero analytic branch, so $R > 0$ and a smooth single-valued phase Θ is available. Put $L = \ln R$. The equations in 11.2.7 become $L_r = \Theta_\theta/r$ and $L_\theta = -r\Theta_r$. Their mixed derivatives are

$$L_{r\theta} = \Theta_{\theta\theta}/r, \qquad L_{\theta r} = -\Theta_r - r\Theta_{rr}.$$

They agree, so division by r gives

$$\boxed{\Theta_{rr} + \frac{1}{r}\Theta_r + \frac{1}{r^2}\Theta_{\theta\theta} = 0}.$$

11.2.9 For each of the following functions $f(z)$, find $f'(z)$ and identify the maximal region within which $f(z)$ is analytic.

$$\text{(a)}\ f(z) = \frac{\sin z}{z}, \qquad \text{(d)}\ f(z) = e^{-1/z},$$

$$\text{(b)}\ f(z) = \frac{1}{z^2+1}, \qquad \text{(e)}\ f(z) = z^2 - 3z + 2,$$

(c) $f(z) = \dfrac{1}{z(z+1)}$, (f) $f(z) = \tan(z)$, (g) $f(z) = \tanh(z)$.

Solution.

Quotient and chain rules give the following derivatives. Domains listed are in the finite complex plane:

	$f'(z)$	domain
(a)	$(z\cos z - \sin z)/z^2$	$\mathbb{C}$ after setting $f(0) = 1$
(b)	$-2z/(z^2+1)^2$	$\mathbb{C} \setminus \{i, -i\}$
(c)	$-(2z+1)/[z^2(z+1)^2]$	$\mathbb{C} \setminus \{0, -1\}$
(d)	$e^{-1/z}/z^2$	$\mathbb{C} \setminus \{0\}$
(e)	$2z - 3$	$\mathbb{C}$
(f)	$1/\cos^2 z$	$\mathbb{C} \setminus \{\pi/2 + n\pi : n \in \mathbb{Z}\}$
(g)	$1/\cosh^2 z$	$\mathbb{C} \setminus \{i(\pi/2 + n\pi) : n \in \mathbb{Z}\}$

For (a), $\sin z/z = 1 - z^2/3! + z^4/5! - \cdots$ gives $f'(0) = 0$. The literal quotient is undefined at zero; the entire function is its removable extension. In (b),(c) the denominator zeros do not cancel; (d) has an essential singularity at zero. For (f), $\cos z = 0$ is equivalent to $e^{2iz} = -1$, giving its listed real sequence. For (g), $e^{2z} = -1$ gives the listed imaginary sequence.

11.2.10 For what complex values do each of the following functions $f(z)$ have a derivative after choosing single-valued branches (and specifying any extension at a branch point)?

(a) $f(z) = z^{3/2}$, (b) $f(z) = z^{-3/2}$, (c) $f(z) = \tan^{-1}(z)$, (d) $f(z) = \tanh^{-1}(z)$.

Solution.

These expressions require branches. Derivatives below are local analytic derivatives away from branch points; a chosen cut is a further boundary of that branch. For the powers choose $-\pi < \arg z < \pi$. On the slit plane,

$$(z^{3/2})' = \tfrac{3}{2} z^{1/2}, \qquad (z^{-3/2})' = -\tfrac{3}{2} z^{-5/2}.$$

Other cuts allow local analytic branches through every nonzero point. The origin is a branch point of both powers, and the second diverges there. For (a), the pointwise derivative at zero does exist if a single-valued selection is defined with $f(0) = 0$: its value is zero. The manual's exclusion of that derivative is incorrect; this does not give analyticity on a neighborhood of the branch point. Indeed $|f(h)/h| = |h|^{1/2} \to 0$, independently of the square-root sign.

Near zero choose the inverse-function branches

$$\arctan z = \frac{\operatorname{Log}(1+iz) - \operatorname{Log}(1-iz)}{2i}, \qquad \operatorname{arctanh} z = \frac{\operatorname{Log}(1+z) - \operatorname{Log}(1-z)}{2}.$$

Differentiating the logarithms gives

$$(\arctan z)' = \tfrac{1}{2}\left(\frac{1}{1+iz} + \frac{1}{1-iz}\right) = \boxed{\frac{1}{1+z^2}},$$

$$(\operatorname{arctanh} z)' = \tfrac{1}{2}\left(\frac{1}{1+z} + \frac{1}{1-z}\right) = \boxed{\frac{1}{1-z^2}}.$$

Local branches have derivatives away from $\pm i$ and ± 1, respectively. Standard cuts run outward along the imaginary axis from $\pm i$ for (c), and along the real axis from ± 1 for (d).

11.2.11 Two-dimensional irrotational fluid flow is conveniently described by a complex potential $f(z) = u(x, y) + iv(x, y)$. We label the real part, $u(x, y)$, the *velocity potential*, and the imaginary part, $v(x, y)$, the *stream function*. The fluid velocity $\mathbf{V}$ is given by $\mathbf{V} = \nabla u$. If $f(z)$ is analytic:

(a) Show that $df/dz = V_x - iV_y$.

(b) Show that $\nabla \cdot \mathbf{V} = 0$ (no sources or sinks).

(c) Show that $\nabla \times \mathbf{V} = 0$ (irrotational, nonturbulent flow).

Solution.

Write $\mathbf{V} = (u_x, u_y, 0)$. **(a)** A real increment gives $f' = u_x + iv_x$. Since $v_x = -u_y$, $\boxed{f' = V_x - iV_y}$. **(b)** Differentiating $u_x = v_y$ in x and $u_y = -v_x$ in y gives $u_{xx} + u_{yy} = v_{yx} - v_{xy} = 0$. Thus $\nabla \cdot \mathbf{V} = 0$. **(c)** The only possibly nonzero curl component is $\partial_x u_y - \partial_y u_x = 0$ by equality of mixed derivatives. The field is incompressible and irrotational on its analytic region. The first occurrence of $u(x, v)$ in the printed problem is a typographical error; the potential depends on the coordinates x, y.

11.2.12 The function $f(z)$ is analytic. Show that the ordinary complex derivative with respect to the conjugate variable exists at z_0 exactly when $f'(z_0) = 0$. Deduce that it exists throughout a connected region exactly when f is constant. Distinguish this from the Wirtinger partial derivative.

Hint. For the Wirtinger interpretation, use the chain rule and take $x = (z+z^*)/2$, $y = (z-z^*)/2i$.

Note. This result emphasizes that our analytic function $f(z)$ is not just a complex function of two real variables x and y. It is a function of the complex variable $x + iy$.

Solution.

Distinguish the ordinary complex derivative in the conjugate variable from the Wirtinger partial derivative. The latter exists and is zero for every analytic f. The former can exist at isolated critical points, but exists throughout a connected region only for a constant function. For the ordinary interpretation examine

$$\frac{f(z_0 + h) - f(z_0)}{\overline{h}} = f'(z_0)\frac{h}{\overline{h}} + o(1).$$

Real increments give $f'(z_0)$ and purely imaginary increments give $-f'(z_0)$. The limit therefore exists only if $f'(z_0) = 0$. This condition is also sufficient because the remaining term tends to zero. For example $f(z) = z^2$ has this derivative zero at $z_0 = 0$. If the condition holds throughout a connected region, $f' = 0$ there and f is constant.

The chain rule in the hint instead defines the Wirtinger derivative:

$$\partial_{\bar{z}} f = \tfrac{1}{2}(\partial_x + i\partial_y)(u + iv) = \tfrac{1}{2}[(u_x - v_y) + i(v_x + u_y)] = \boxed{0}.$$

This is exactly the Cauchy–Riemann condition, not a condition that f be constant.

11.3 Cauchy's Integral Theorem

11.3.1 Show that $\displaystyle\int_{z_1}^{z_2} f(z)\,dz = -\int_{z_2}^{z_1} f(z)\,dz$, with the same path traversed in reverse.

Solution.

Parametrize the specified path by $\gamma(t)$, $0 \le t \le 1$, from z_1 to z_2. Its reverse is $\widetilde{\gamma}(t) = \gamma(1-t)$:

$$\int_{\widetilde{C}} f\,dz = -\int_0^1 f(\gamma(1-t))\gamma'(1-t)\,dt = -\int_0^1 f(\gamma(s))\gamma'(s)\,ds.$$

Here $s = 1 - t$ reverses the limits. The last member equals $-\int_C f\,dz$, proving the identity. The reverse integral follows the same geometric path in the opposite direction; two different paths need not give this relation.

11.3.2 Prove that

$$\left|\int_C f(z)\,dz\right| \le |f|_{\max} \cdot L,$$

where f is continuous on the piecewise smooth finite-length contour C, $|f|_{\max}$ is the maximum value of $|f(z)|$ along the contour C and L is the length of the contour.

Solution.

For a piecewise smooth parametrization $\gamma : [a, b] \to C$,

$$\left|\int_C f\,dz\right| = \left|\int_a^b f(\gamma(t))\gamma'(t)\,dt\right| \le \int_a^b |f(\gamma(t))|\,|\gamma'(t)|\,dt.$$

Bound $|f(\gamma(t))|$ by $M = |f|_{\max}$ and use $L = \int_a^b |\gamma'(t)|dt$: $\boxed{|\int_C f\,dz| \le ML}$. Continuity on the compact contour guarantees a maximum; a finite supremum suffices for the same argument.

11.3.3 Show that the integral

$$\int_{3+4i}^{4-3i} (4z^2 - 3iz)\,dz$$

has the same value on the two paths: (a) the straight line connecting the integration limits, and (b) an arc on the circle $|z| = 5$.

Solution.

The polynomial has the entire primitive $P(z) = 4z^3/3 - 3iz^2/2$. Along each smooth segment, $dP(\gamma(t))/dt = (4\gamma(t)^2 - 3i\gamma(t))\gamma'(t)$. Thus the straight path $\gamma(t) = a + t(b-a)$ and any circular arc $\gamma(\theta) = 5e^{i\theta}$ both give $P(b) - P(a)$, where $a = 3 + 4i$, $b = 4 - 3i$. The endpoint powers are

$$a^2 = -7 + 24i, \quad b^2 = 7 - 24i, \quad a^3 = -117 + 44i, \quad b^3 = -44 - 117i.$$

Hence

$$\begin{aligned} P(b) - P(a) &= \tfrac{4}{3}(73 - 161i) - \tfrac{3i}{2}(14 - 48i) \\ &= \frac{292 - 644i}{3} - 72 - 21i = \boxed{\frac{76 - 707i}{3}}. \end{aligned}$$

The chain rule verifies this evaluation for both requested parametrizations, including either choice of connecting circular arc.

11.3.4 Let $F(z) = \displaystyle\int_{\pi(1+i)}^{z} \cos 2\zeta \, d\zeta$. Show that $F(z)$ is independent of the path connecting the limits of integration, and evaluate $F(\pi i)$.

Solution.

The integrand has the entire primitive $\frac{1}{2}\sin 2\zeta$. Thus every path gives

$$F(z) = \tfrac{1}{2}[\sin 2z - \sin(2\pi + 2\pi i)].$$

At $z = \pi i$, periodicity $\sin(w + 2\pi) = \sin w$ makes the two terms cancel. Therefore $\boxed{F(\pi i) = 0}$, and the primitive proves path independence throughout the plane.

11.3.5 Evaluate $\displaystyle\oint_C (x^2 - iy^2)\, dz$, where the integration is (a) clockwise around the unit circle, (b) on a square with vertices at $\pm 1 \pm i$. Explain why the results of parts (a) and (b) are or are not identical.

Solution.

Expand in real differentials:

$$(x^2 - iy^2)(dx + i\, dy) = (x^2 dx + y^2 dy) + i(x^2 dy - y^2 dx).$$

The real part is $d[(x^3 + y^3)/3]$, whose closed integral is zero. **(a)** For clockwise traversal set $x = \cos t, y = -\sin t$, $0 \le t \le 2\pi$. The imaginary part becomes $(-\cos^3 t + \sin^3 t)dt$. Each odd power has zero integral over a full period, so the answer is 0. **(b)** Traverse the square counterclockwise from $-1 - i$. On a horizontal edge $dz = dx$; on a vertical edge $dz = i\, dy$. The four contributions are

$$\begin{aligned}
I_{\text{bottom}} &= \int_{-1}^{1} (x^2 - i)\, dx = \frac{2}{3} - 2i, \\
I_{\text{right}} &= \int_{-1}^{1} (1 - iy^2) i\, dy = \frac{2}{3} + 2i, \\
I_{\text{top}} &= \int_{1}^{-1} (x^2 - i)\, dx = -\frac{2}{3} + 2i, \\
I_{\text{left}} &= \int_{1}^{-1} (1 - iy^2) i\, dy = -\frac{2}{3} - 2i.
\end{aligned}$$

Opposite edges cancel pairwise, giving $\boxed{0}$. Reversing the orientation reverses all four signs and still gives zero. These equal answers follow from symmetry, not analyticity: $u_x = 2x$, $v_y = -2y$ fail Cauchy–Riemann on every open set.

11.3.6 Verify that

$$\int_0^{1+i} z^*\, dz$$

depends on the path by evaluating the integral for the two paths shown in Fig. 11.7. Recall that $f(z) = z^*$ is not an analytic function of z and that Cauchy's integral theorem therefore does not apply.

Solution.

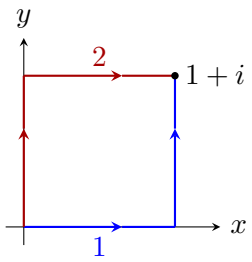
In the actual figure, path 1 is $0 \to 1 \to 1 + i$ and path 2 is $0 \to i \to 1 + i$. Along path 1,

$$I_1 = \int_0^1 x\, dx + \int_0^1 (1 - iy) i\, dy = \frac{1}{2} + i + \frac{1}{2} = \boxed{1 + i}.$$

Along the first segment of path 2, $z = iy$, so $dz = i\,dy$:

$$I_2 = \int_0^1 (-iy)i\,dy + \int_0^1 (x - i)\,dx = \frac{1}{2} + \frac{1}{2} - i = \boxed{1 - i}.$$

The manual's second-path calculation omits the factor i in $dz = i\,dy$ and prints an incorrect second value. The difference $2i$ establishes path dependence.



11.3.7 Show that

$$\oint_C \frac{dz}{z^2 + z} = 0,$$

in which the contour C is a counterclockwise circle defined by $|z| = R > 1$.

Hint. Direct use of the Cauchy integral theorem is illegal. The integral may be evaluated by expanding into partial fractions and then treating the two terms individually. This yields 0 for $R > 1$ and $2\pi i$ for $0 < R < 1$.

Solution.

Partial fractions give $1/[z(z + 1)] = 1/z - 1/(z + 1)$. For the positively oriented circle $R > 1$, both poles lie inside. The two terms therefore contribute $2\pi i$ and $-2\pi i$, giving $\boxed{0}$. For $0 < R < 1$ only the origin lies inside, giving $2\pi i$. At $R = 1$ the path hits -1, so the ordinary integral is undefined. The zero-integral theorem cannot be applied directly over the singularities, which is why the decomposition is useful.

11.4 Cauchy's Integral Formula

Unless explicitly stated otherwise, closed contours occurring in these exercises are to be understood as traversed in the mathematically positive (counterclockwise) direction.

11.4.1 Show that

$$\frac{1}{2\pi i} \oint z^{m-n-1}\,dz, \quad m \text{ and } n \text{ integers}$$

(with the contour encircling the origin once), is a representation of the Kronecker δ_{mn}.

Solution.

On $z = Re^{i\theta}$, $0 \le \theta \le 2\pi$,

$$\frac{1}{2\pi i} \oint z^{m-n-1}\,dz = \frac{R^{m-n}}{2\pi} \int_0^{2\pi} e^{i(m-n)\theta}\,d\theta.$$

For $m = n$ this is 1. For the nonzero integer $k = m - n$, the integral is $[e^{ik\theta}/(ik)]_0^{2\pi} = 0$. Thus the result is $\boxed{\delta_{mn}}$. Contour deformation gives the same value for any contour avoiding zero with positive winding number one.

11.4.2 Evaluate

$$\oint_C \frac{dz}{z^2-1},$$

where C is the circle $|z-1|=1$.

Solution.

The poles at $1, -1$ have distances $0, 2$ from the center 1. Only 1 is inside the specified unit-radius circle. Using

$$\frac{1}{z^2-1} = \frac{1}{2}\left(\frac{1}{z-1} - \frac{1}{z+1}\right),$$

the first term contributes πi and the second integrates to zero. Therefore $\boxed{\oint_C dz/(z^2-1) = \pi i}$.

The manual's answer 0 is incorrect for this contour; textbook p. 491 confirms $|z-1|=1$, enclosing only one pole.

11.4.3 Assuming that $f(z)$ is analytic on and within a closed contour C and that the point z_0 is within C, show that

$$\oint_C \frac{f'(z)}{z-z_0}\,dz = \oint_C \frac{f(z)}{(z-z_0)^2}\,dz.$$

Solution.

Along the contour, the single-valued function $f(z)/(z-z_0)$ obeys

$$d\left(\frac{f(z)}{z-z_0}\right) = \left[\frac{f'(z)}{z-z_0} - \frac{f(z)}{(z-z_0)^2}\right] dz.$$

Its closed integral is zero. Moving one term to the opposite side proves the stated equality. Both integrals also equal $2\pi i f'(z_0)$ by Cauchy's formulas.

11.4.4 You know that $f(z)$ is analytic on and within a closed contour C, with z_0 inside C and n a nonnegative integer. You suspect that the nth derivative $f^{(n)}(z_0)$ is given by

$$f^{(n)}(z_0) = \frac{n!}{2\pi i}\oint_C \frac{f(z)}{(z-z_0)^{n+1}}\,dz.$$

Using mathematical induction (Section 1.4), prove that this expression is correct.

Solution.

The base case $n=0$ is Cauchy's integral formula. Assume the formula holds for n. The point z_0 is a positive distance from C; on a small neighborhood of z_0, denominators stay bounded away from zero, allowing differentiation under this finite contour integral. Since

$$\partial_{z_0}(z-z_0)^{-n-1} = -(n+1)(z-z_0)^{-n-2}(-1) = (n+1)(z-z_0)^{-n-2},$$

we obtain

$$f^{(n+1)}(z_0) = \frac{n!}{2\pi i}\oint_C \frac{(n+1)f(z)}{(z-z_0)^{n+2}}\,dz = \frac{(n+1)!}{2\pi i}\oint_C \frac{f(z)}{(z-z_0)^{n+2}}\,dz.$$

This is the induction step, proving the result for every $n \geq 0$.

11.4.5

(a) A function $f(z)$ is analytic within a closed contour C (and continuous on C). If $f(z) \neq 0$ within C and $|f(z)| \geq m > 0$ on C, show that

$$|f(z)| \geq m$$

for all points within C.

Hint. Consider $w(z) = 1/f(z)$.

(b) If $f(z) = 0$ within the contour C, show that the foregoing result does not hold and that it is possible to have $|f(z)| = 0$ at one or more points in the interior with $|f(z)| > 0$ over the entire bounding contour. Cite a specific example of an analytic function that behaves this way.

Solution.

As printed on p. 491, (a) is a maximum-modulus statement that does not require $f \neq 0$. Thus (b)'s claim that its conclusion fails at zeros is false. The hint suggests the intended minimum-modulus principle. We resolve both statements. For the **original upper-bound version**, continuity gives a maximum of $|f|$ on the closed bounded region. If it exceeded M, it would occur at an interior point z_0. For a nonconstant function write $f(z_0 + h) = f(z_0) + a_m h^m + O(h^{m+1})$, $a_m \neq 0$. If $f(z_0) \neq 0$, choose the direction of h so that $a_m h^m$ is a positive real multiple of $f(z_0)$. For small h the modulus then increases, a contradiction. If the maximum is attained at a zero, its value is zero and the function is identically zero. Constants satisfy the boundary bound as well. Hence $\boxed{|f| \leq M}$ throughout, whether or not zeros occur.

For the **corrected lower bound in (a)**, assume f has no zero on the closed region and $|f| \geq m > 0$ on C. Then $1/f$ is analytic and obeys $|1/f| \leq 1/m$ on the boundary. The upper-bound result gives $|1/f| \leq 1/m$ inside, hence $|f| \geq m$. For **(b)**, $f(z) = z$ on the unit disk has boundary modulus 1 and $f(0) = 0$. This disproves a positive interior *lower* bound without nonvanishing, while still satisfying the printed upper bound.

11.4.6 Evaluate

$$\oint_C \frac{e^{iz}}{z^3}\, dz,$$

for the contour a square with sides of length $a > 1$, centered at $z = 0$.

Solution.

Let $g = e^{iz}$. The square surrounds zero once, so

$$\oint_C \frac{e^{iz}}{z^3}\, dz = \frac{2\pi i}{2!} g''(0) = \pi i[-e^{iz}]_{z=0} = \boxed{-\pi i}.$$

Any positive side length suffices; $a > 1$ is stronger than needed.

11.4.7 Evaluate

$$\oint_C \frac{\sin^2 z - z^2}{(z-a)^3}\, dz,$$

where the contour encircles the point $z = a$.

Solution.

Set $g(z) = \sin^2 z - z^2$. Then $g' = \sin 2z - 2z$ and $g'' = 2\cos 2z - 2$. Cauchy's derivative formula gives

$$\oint_C \frac{g(z)}{(z-a)^3}\,dz = \frac{2\pi i}{2!}g''(a) = \boxed{2\pi i(\cos 2a - 1)}.$$

The formula still applies when a numerator zero lowers or removes the apparent pole.

11.4.8 Evaluate

$$\oint_C \frac{dz}{z(2z+1)},$$

for the contour the unit circle.

Solution.

Decompose

$$\frac{1}{z(2z+1)} = \frac{1}{z} - \frac{2}{2z+1} = \frac{1}{z} - \frac{1}{z+1/2}.$$

Both poles are inside the unit circle. Their contributions, $2\pi i$ and $-2\pi i$, cancel, giving $\boxed{0}$.

11.4.9 Evaluate

$$\oint_C \frac{f(z)}{z(2z+1)^2}\,dz,$$

for the contour the unit circle, assuming f is analytic on a neighborhood of the closed disk.

Hint. Make a partial fraction expansion.

Solution.

Assume f is analytic on and within the unit circle; additional singularities would supply additional contributions. Write $1/[z(2z+1)^2] = A/z + B/(2z+1) + D/(2z+1)^2$. Clearing denominators yields $1 = A(2z+1)^2 + Bz(2z+1) + Dz$. Constant, quadratic, and linear coefficients give, respectively, $A = 1$, $4A + 2B = 0$, $4A + B + D = 0$; hence $B = D = -2$. Therefore

$$\frac{f(z)}{z(2z+1)^2} = \frac{f(z)}{z} - \frac{f(z)}{z+1/2} - \frac{f(z)/2}{(z+1/2)^2}.$$

Cauchy's formula for values and first derivatives now gives

$$\boxed{2\pi i f(0) - 2\pi i f(-1/2) - \pi i f'(-1/2)}.$$

11.5 Laurent Expansion

11.5.1 Develop the Taylor expansion of $\ln(1+z)$.

ANS. $\displaystyle\sum_{n=1}^{\infty}(-1)^{n-1}\frac{z^n}{n}$.

Solution.

Choose the logarithm analytic near $z=0$ with value zero there. For $|z|<1$, the geometric series gives $1/(1+z)=\sum_{n=0}^{\infty}(-1)^n z^n$. It converges uniformly on every smaller closed disk, allowing termwise integration from zero to z:

$$\boxed{\ln(1+z)=\sum_{n=0}^{\infty}\frac{(-1)^n z^{n+1}}{n+1}=\sum_{n=1}^{\infty}(-1)^{n-1}\frac{z^n}{n}}.$$

Both sides vanish at zero, fixing the integration constant. The nearest singularity is $z=-1$, so the Taylor radius is one.

11.5.2 Derive the binomial expansion

$$(1+z)^m=1+mz+\frac{m(m-1)}{1\cdot 2}z^2+\cdots=\sum_{n=0}^{\infty}\binom{m}{n}z^n$$

for m, any real number. The expansion is convergent for $|z|<1$. Why?

Solution.

Choose $(1+z)^m=\exp[m\,\mathrm{Log}(1+z)]$ with value 1 at zero. Repeated differentiation gives

$$\frac{d^n}{dz^n}(1+z)^m=m(m-1)\cdots(m-n+1)(1+z)^{m-n}.$$

Set $z=0$ and divide by $n!$ to obtain

$$\boxed{(1+z)^m=\sum_{n=0}^{\infty}\frac{m(m-1)\cdots(m-n+1)}{n!}z^n}.$$

The empty product for $n=0$ is one. For a nonterminating series, the absolute successive-term ratio is $|z|\,|m-n|/(n+1)\to|z|$, proving convergence for $|z|<1$. The disk also avoids the singularity at -1. For nonnegative integer m the series terminates and is valid everywhere; for other real m the Taylor radius is one.

11.5.3 A function $f(z)$ is analytic on and within the unit circle. Also, $|f(z)|<1$ for $|z|\le 1$ and $f(0)=0$. Show that $|f(z)|<|z|$ for $0<|z|\le 1$.

Hint. One approach is to show that $f(z)/z$ is analytic and then to express $[f(z_0)/z_0]^n$ by the Cauchy integral formula. Finally, consider absolute magnitudes and take the nth root. This exercise is sometimes called Schwarz's theorem.

Solution.

The strict inequality excludes $z=0$, where both sides are zero. It holds for $0<|z|\le 1$. The series $f(z)=a_1z+a_2z^2+\cdots$ shows that $g(z)=f(z)/z$ extends analytically to zero by $g(0)=a_1$.

Compactness of the unit circle and the strict hypothesis give $|g(\zeta)| = |f(\zeta)| \le M < 1$ there. For $|z_0| < 1$, Cauchy's formula applied to g^n gives

$$g(z_0)^n = \frac{1}{2\pi i} \oint_{|\zeta|=1} \frac{g(\zeta)^n}{\zeta - z_0}\, d\zeta, \qquad |g(z_0)|^n \le \frac{M^n}{1 - |z_0|}.$$

Take nth roots and let $n \to \infty$ to get $|g(z_0)| \le M$. This also holds on the boundary. Thus $|f(z)| \le M|z| < |z|$ for nonzero z, with equality $0 = 0$ at zero.

11.5.4 If $f(z)$ is a real function of the complex variable $z = x + iy$, that is, $f(x) = f^*(x)$ for real x near zero, and the Laurent expansion about the origin, $f(z) = \sum a_n z^n$ on $0 < |z| < R$, has $a_n = 0$ for $n < -N$, show that all of the coefficients a_n are real.

Hint. Show that $z^N f(z)$ is analytic (via Morera's theorem, Section 11.4).

Solution.

Here *real* means real-valued for real arguments near the Laurent center, not throughout an open complex region. The finite principal part makes

$$g(z) = z^N f(z) = \sum_{j=0}^{\infty} a_{j-N} z^j$$

analytic after its removable extension to zero. For nonzero real x near zero, $g(x) = x^N f(x)$ is real, and continuity gives a real value at zero. Each analytic derivative at zero may be computed along the real line; derivatives of this real-valued restriction are real. Hence $a_{j-N} = g^{(j)}(0)/j! \in \mathbb{R}$ for every $j \ge 0$. The remaining coefficients vanish and are real as well.

11.5.5 Prove that the Laurent expansion of a given function about a given point is unique; that is, if

$$f(z) = \sum_{n=-N}^{\infty} a_n (z - z_0)^n = \sum_{n=-N}^{\infty} b_n (z - z_0)^n,$$

with both series representing f on a common annulus, show that $a_n = b_n$ for all n.

Hint. Use the Cauchy integral formula.

Solution.

Choose a circle $|\zeta - z_0| = r$ in the common convergence annulus. Uniform convergence there permits termwise integration:

$$\begin{aligned} \frac{1}{2\pi i} \oint f(\zeta)(\zeta - z_0)^{-k-1}\, d\zeta &= \sum_n a_n \frac{1}{2\pi i} \oint (\zeta - z_0)^{n-k-1}\, d\zeta \\ &= \sum_n a_n \delta_{nk} = a_k. \end{aligned}$$

The second series gives b_k for the identical integral. Thus $\boxed{a_k = b_k}$ for every integer k. The proof also works for an infinite principal part on a common annulus.

11.5.6 Obtain the Laurent expansion of e^z/z^2 about $z = 0$.

Solution.

Divide the everywhere-convergent exponential series by z^2:

$$\boxed{\frac{e^z}{z^2} = \sum_{n=0}^{\infty} \frac{z^{n-2}}{n!} = \frac{1}{z^2} + \frac{1}{z} + \frac{1}{2} + \frac{z}{6} + \frac{z^2}{24} + \cdots}.$$

The annulus is $0 < |z| < \infty$. The pole has order two and residue one.

11.5.7 Obtain the Laurent expansion of $ze^z/(z-1)$ about $z = 1$.

Solution.

Put $w = z - 1$. Then

$$\frac{ze^z}{z-1} = e\left(\frac{1}{w} + 1\right) \sum_{n=0}^{\infty} \frac{w^n}{n!}.$$

The w^{-1} coefficient is e; for $m \geq 0$ the w^m coefficient is $e[1/(m+1)! + 1/m!] = e(m+2)/(m+1)!$. Consequently

$$\boxed{\frac{ze^z}{z-1} = \frac{e}{z-1} + e \sum_{m=0}^{\infty} \frac{m+2}{(m+1)!}(z-1)^m}, \qquad 0 < |z-1| < \infty.$$

The first regular terms are $2e + \frac{3}{2}e(z-1) + \frac{2}{3}e(z-1)^2$.

11.5.8 Obtain the Laurent expansion of $(z-1)e^{1/z}$ about $z = 0$.

Solution.

Multiplication of the exponential series gives

$$(z-1)e^{1/z} = \sum_{n=0}^{\infty} \frac{z^{1-n}}{n!} - \sum_{n=0}^{\infty} \frac{z^{-n}}{n!}.$$

The leading term is z, the constants cancel, and the coefficient of z^{-k} for $k \geq 1$ is $1/(k+1)! - 1/k! = -k/(k+1)!$. Thus

$$\boxed{(z-1)e^{1/z} = z - \sum_{k=1}^{\infty} \frac{k}{(k+1)!} z^{-k} = z - \frac{1}{2z} - \frac{1}{3z^2} - \frac{1}{8z^3} - \cdots}.$$

It converges for $0 < |z| < \infty$; its infinite principal part makes zero an essential singularity.

11.6 Singularities

11.6.1 As an example of an essential singularity consider $e^{1/z}$ as z approaches zero. For any complex number z_0, $z_0 \neq 0$, show that

$$e^{1/z} = z_0$$

has an infinite number of solutions.

Solution.

Write $z_0 = \rho e^{i\alpha}$ with $\rho > 0$, choosing one argument. All solutions of $e^w = z_0$ have $w = \ln\rho + i(\alpha + 2\pi n)$, $n \in \mathbb{Z}$. Therefore

$$\boxed{z_n = \frac{1}{\ln\rho + i(\alpha + 2\pi n)}}$$

whenever the denominator is nonzero. Substitution gives $e^{1/z_n} = \rho e^{i\alpha} = z_0$. At most one integer has a zero denominator, and all remaining denominators are distinct. Thus infinitely many distinct solutions remain, with $z_n \to 0$ as $|n| \to \infty$. For $z_0 = 1$, simply omit $n = 0$. The manual argues from roots of truncated exponential polynomials. Passing to the limit does not by itself establish distinct roots of the original equation. The explicit logarithmic preimages above supply that missing argument.

11.6.2 Show that the function

$$w(z) = (z^2 - 1)^{1/2}$$

is single-valued if we make branch cuts on the real axis for $x \geq 1$ and for $x \leq -1$.

Solution.

Remove both rays including their endpoints: $D = \mathbb{C} \setminus ((-\infty, -1] \cup [1, \infty))$. For $z = x + iy$, $1 - z^2$ is nonpositive real only if $xy = 0$ and $1 - x^2 + y^2 \leq 0$. The choice $x = 0$ gives $1 + y^2 > 0$; otherwise $y = 0$, $|x| \geq 1$, exactly the removed rays. Thus the principal square root of $1 - z^2$ is analytic on D, and the explicit branch

$$\boxed{w(z) = i\sqrt{1 - z^2}}$$

is single-valued and satisfies $w^2 = z^2 - 1$. Its negative is the other branch; one reference value selects its sign. The branch points ± 1 themselves are excluded from the analytic domain, not merely the open rays. The manual instead discusses a cut joining -1 to 1. That is the different branch domain in Example 11.6.4, not the two outward rays specified here.

11.6.3 A function $f(z)$ can be represented by

$$f(z) = \frac{f_1(z)}{f_2(z)},$$

in which $f_1(z)$ and $f_2(z)$ are analytic. The denominator, $f_2(z)$, vanishes at $z = z_0$, showing that $f(z)$ has a pole at $z = z_0$. However, $f_1(z_0) \neq 0$, $f_2'(z_0) \neq 0$. Show that a_{-1}, the coefficient of $(z - z_0)^{-1}$ in a Laurent expansion of $f(z)$ at $z = z_0$, is given by

$$a_{-1} = \frac{f_1(z_0)}{f_2'(z_0)}.$$

Solution.

With $h = z - z_0$, Taylor expansion gives

$$f_1(z) = f_1(z_0) + O(h), \quad f_2(z) = f_2'(z_0)h[1 + O(h)].$$

The last factor has an analytic reciprocal near zero because its constant term is one. Hence

$$f(z) = \frac{f_1(z_0) + O(h)}{f_2'(z_0)h}[1 + O(h)] = \frac{f_1(z_0)}{f_2'(z_0)}h^{-1} + O(1).$$

Therefore $\boxed{a_{-1} = f_1(z_0)/f_2'(z_0)}$. The assumptions make this coefficient nonzero and the pole simple.

11.6.4 Determine a unique branch for the function of Exercise 11.6.2 that will cause the value it yields for $f(i)$ to be the same as that found for $f(i)$ in Example 11.6.4. Although Exercise 11.6.2 and Example 11.6.4 describe the same multivalued function, the specific values assigned for various z will not agree everywhere, due to the difference in the location of the branch cuts. Identify the portions of the complex plane where both these descriptions do and do not agree, and characterize the differences.

Solution.

The segment-cut branch of Example 11.6.4 is positive for real $z > 1$. At $z = i$, the arguments of $i + 1$ and $i - 1$ are $\pi/4$ and $3\pi/4$ and their moduli are both $\sqrt{2}$. Its value is

$$(\sqrt{2})^{1/2}(\sqrt{2})^{1/2}e^{i(\pi/4+3\pi/4)/2} = \boxed{i\sqrt{2}}.$$

The outward-cut branch with this value is $F(z) = i\sqrt{1 - z^2}$ from 11.6.2. Since its domain is connected, the reference value fixes its sign.

Call the segment-cut branch G. Where both are defined, $F^2 = G^2 = z^2 - 1$ and neither vanishes; thus F/G is continuous and takes only ± 1. Their common domain has the upper and lower half-planes as its two components. Equality at i implies $F = G$ in the upper half-plane. At $-i$, the segment-cut continuation gives $G(-i) = -i\sqrt{2}$, while $F(-i) = i\sqrt{2}$; hence $F = -G$ in the lower half-plane. These statements specify precisely their agreement and sign difference. The manual gives $f(i) = 2i$. Each factor contributes modulus $2^{1/4}$, not $\sqrt{2}$, so their product has modulus $\sqrt{2}$. Also $(i\sqrt{2})^2 = -2 = i^2 - 1$, whereas $(2i)^2 = -4$.

11.6.5 Find all singularities of

$$z^{-1/3} + \frac{z^{-1/4}}{(z-3)^3} + (z-2)^{1/2},$$

and identify their types (e.g., second-order branch point, fifth-order pole, ...). Include any singularities at the point at infinity.

Note. A branch point is of nth order if it requires n, but no fewer, circuits around the point to restore the original value.

Solution.

The second numerator is $z^{-1/4}$ on textbook p. 507; the transcription $z - 1/4$ loses the fractional power. At zero,

$$\frac{z^{-1/4}}{(z-3)^3} = -\frac{1}{27}z^{-1/4}(1 - z/3)^{-3}.$$

The first two terms therefore acquire multipliers $e^{-2\pi iN/3}$ and $e^{-2\pi iN/4}$ after N circuits. Their distinct leading fractional powers cannot cancel. Both return to their initial values iff N is divisible by 3 and 4; the least such positive integer is 12. Zero is a branch point of order 12 in the question's convention. At 3, the chosen local numerator is analytic and nonzero, giving a third-order pole. At 2, only $(z-2)^{1/2}$ branches, with order 2. For infinity put $w = 1/z$:

$$f(1/w) = w^{1/3} + \frac{w^{13/4}}{(1-3w)^3} + w^{-1/2}(1-2w)^{1/2}.$$

The analytic factors have nonzero constant terms, so circuit counts must be divisible by $3, 4, 2$; their least common multiple is 12. Infinity is an order-12 branch point, not an isolated meromorphic pole. There are no other finite singularities.

11.6.6 The function $F(z) = \ln(z^2+1)$ is made single-valued by straight-line branch cuts from $(x,y) = (0,-1)$ to $(-\infty,-1)$ and from $(0,+1)$ to $(0,+\infty)$. See Fig. 11.17. If $F(0) = -2\pi i$, find the value of $F(i-2)$.

Solution.

Factor $z^2+1 = (z-i)(z+i)$. For the upward cut from i, choose $-3\pi/2 < \arg(z-i) < \pi/2$; for the leftward cut from $-i$, choose $-\pi < \arg(z+i) < \pi$. At zero these arguments are $-\pi/2, \pi/2$ and both moduli are one. The required branch is therefore

$$F(z) = \ln|z-i| + \ln|z+i| + i[\arg(z-i) + \arg(z+i) - 2\pi].$$

At $z = -2+i$, the factors are -2 and $-2+2i$. Their arguments in these ranges are $-\pi$ and $3\pi/4$; their moduli are 2 and $2\sqrt{2}$. Thus

$$\boxed{F(i-2) = \ln(4\sqrt{2}) - 9\pi i/4}.$$

Exponentiation verifies $e^F = 4-4i = (-2+i)^2+1$.

11.6.7 Show that negative numbers have logarithms in the complex plane. In particular, find $\ln(-1)$.

ANS. $\mathrm{Log}(-1) = i\pi$ (principal value).

Solution.

For $r > 0$, $-r = re^{i(\pi+2\pi n)}$, $n \in \mathbb{Z}$. All logarithms are consequently

$$\boxed{\ln(-r) = \ln r + i(2n+1)\pi}.$$

In particular $\ln(-1) = i(2n+1)\pi$. The printed $i\pi$ is the principal value, or one chosen branch value, rather than the complete multivalued answer.

11.6.8 For noninteger m, show that the binomial expansion of Exercise 11.5.2 holds only for a suitably defined branch of the function $(1+z)^m$. Show how the z-plane is cut. Explain why $|z| < 1$ may be taken as the circle of convergence for the expansion of this branch, in light of the cut you have chosen.

Solution.

Choose $(1+z)^m = \exp[m\,\mathrm{Log}(1+z)]$ with $-\pi < \arg(1+z) < \pi$. The cut is $(-\infty,-1]$ and the value at zero is 1. This analytic branch has the Taylor coefficients derived in 11.5.2, and the open unit disk avoids the cut. For noninteger real m, a circuit about -1 multiplies the value by $e^{2\pi im} \neq 1$. Thus -1 is a genuine branch point, distance one from the Taylor center. The radius is exactly one; the branch point prevents a larger convergence disk.

11.6.9 The Taylor expansion of Exercises 11.5.2 and 11.6.8 is not suitable for branches other than the one suitably defined branch of the function $(1+z)^m$ for noninteger m. (Note that other branches cannot have the same Taylor expansion since they must be distinguishable.) Using the same branch cut of the earlier exercises for all other branches, find the corresponding Taylor expansions, detailing the phase assignments and Taylor coefficients.

Solution.

For the same cut, let

$$\operatorname{Log}_k(1+z) = \ln|1+z| + i\arg(1+z) + 2\pi ik, \quad k \in \mathbb{Z},$$

where the displayed argument lies in $(-\pi, \pi)$. Then $(1+z)_k^m = e^{2\pi imk}(1+z)_0^m$, so

$$\boxed{(1+z)_k^m = e^{2\pi imk}\sum_{n=0}^{\infty}\binom{m}{n}z^n, \qquad |z|<1}.$$

Every coefficient is multiplied by the same phase; the constant coefficient already distinguishes the branches. For $m = p/q$ in lowest terms, k modulo q gives q distinct branches. For irrational m, all integer k give distinct ones.

11.6.10

(a) Develop a Laurent expansion of $f(z) = [z(z-1)]^{-1}$ about the point $z = 1$ valid for small values of $|z-1|$. Specify the exact range over which your expansion holds. This is an analytic continuation of the infinite series in Eq. (11.49).

(b) Determine the Laurent expansion of $f(z)$ about $z = 1$ but for $|z-1|$ large.

Hint. Make a partial fraction decomposition of this function and use the geometric series.

Solution.

Put $w = z-1$, so $f = 1/[w(1+w)] = 1/w - 1/(1+w)$. **(a)** For $|w|<1$, geometric expansion gives

$$\boxed{f = \frac{1}{w} - 1 + w - w^2 + \cdots = \sum_{n=-1}^{\infty}(-1)^{n+1}w^n, \qquad 0<|w|<1}.$$

The pole at $w = 0$ excludes the center; the pole at $w = -1$ sets the outer radius. **(b)** For $|w|>1$ use inverse powers:

$$\boxed{f = w^{-2}(1+w^{-1})^{-1} = \sum_{n=0}^{\infty}(-1)^n w^{-n-2}, \qquad |w|>1}.$$

The annuli differ, so uniqueness of Laurent coefficients on a fixed annulus is not contradicted. The manual writes the outer series as $\sum_{n=-2}^{\infty}(-1)^n(z-1)^{-n}$. Its lower index should be 2; otherwise it begins with growing powers and cannot equal a function that decays as $(z-1)^{-2}$ at infinity.

11.6.11

(a) Given $f_1(z) = \int_0^\infty e^{-zt}\,dt$ (with t real), show that the domain in which $f_1(z)$ exists (and is analytic) is $\Re(z) > 0$.

(b) Show that $f_2(z) = 1/z$ equals $f_1(z)$ over $\Re(z) > 0$ and is therefore an analytic continuation of $f_1(z)$ over the entire z-plane except for $z = 0$.

(c) Expand $1/z$ about the point $z = -i$. You will have

$$f_3(z) = \sum_{n=0}^{\infty} a_n (z+i)^n.$$

What is the domain of this formula for $f_3(z)$?

$$\textit{ANS. } \frac{1}{z} = i\sum_{n=0}^{\infty} i^{-n}(z+i)^n, \quad |z+i| < 1.$$

Solution.

(a) For $z \neq 0$,

$$\int_0^T e^{-zt}\,dt = (1 - e^{-zT})/z.$$

If $\Re z > 0$, the exponential tends to zero; convergence is locally uniform, giving analyticity. If $\Re z < 0$, the exponential grows. If $\Re z = 0$, $z \neq 0$, it oscillates without a limit. For $z = 0$, the integral equals T. Thus ordinary convergence occurs exactly on $\Re z > 0$. **(b)** There the limit is $1/z$, whose analytic domain is $\mathbb{C} \setminus \{0\}$. The continuation has no negative-integer singularities; the manual's mention of those points is extraneous to this function. **(c)** Set $w = z + i$, so

$$\boxed{\frac{1}{z} = \frac{i}{1+iw} = i\sum_{n=0}^{\infty}(-i)^n w^n = i\sum_{n=0}^{\infty} i^{-n}(z+i)^n, \quad |z+i| < 1}.$$

The pole at zero is distance one from the center $-i$, fixing the exact convergence radius. The manual expands about $+i$ in part (c), while the question requests $-i$. The expansion here uses $z + i$ and has constant coefficient $1/(-i) = i$.

11.7 Calculus of Residues

11.7.1 Determine the nature of the singularities of each of the following functions and evaluate the residues ($a > 0$).

(a) $\dfrac{1}{z^2 + a^2}$. (b) $\dfrac{1}{(z^2 + a^2)^2}$. (c) $\dfrac{z^2}{(z^2 + a^2)^2}$.

(d) $\dfrac{\sin 1/z}{z^2 + a^2}$. (e) $\dfrac{ze^{iz}}{z^2 + a^2}$. (f) $\dfrac{ze^{iz}}{z^2 - a^2}$.

(g) $\dfrac{e^{iz}}{z^2 - a^2}$. (h) $\dfrac{z^{-k}}{z+1}$, $0 < k < 1$.

Hint. For the point at infinity, use the transformation $w = 1/z$ for $|z| \to \infty$. For the residue, transform $f(z)\,dz$ into $g(w)\,dw$ and look at the behavior of $g(w)$.

Solution.

Infinity corresponds to $w = 1/z \to 0$, hence to $|z| \to \infty$, not to $|z| \to 0$ as the printed hint says. The nature of the singularity at infinity is the nature of $f(1/w)$ at zero, whereas its residue is

$$\operatorname{Res}_\infty f = \operatorname{Res}_{w=0}\left[-\frac{f(1/w)}{w^2}\right].$$

These are different transformations: the second includes the differential.

(a)–(c) Factor $z^2 + a^2 = (z - ia)(z + ia)$. For (a), cancellation of the vanishing factor gives

$$\operatorname{Res}_{ia} \frac{1}{z^2 + a^2} = \frac{1}{2ia}, \qquad \operatorname{Res}_{-ia} \frac{1}{z^2 + a^2} = -\frac{1}{2ia}.$$

Both poles are simple. For (b), they are double, so

$$\operatorname{Res}_{ia} \frac{1}{(z^2 + a^2)^2} = \frac{d}{dz}(z + ia)^{-2}\bigg|_{ia} = -\frac{2}{(2ia)^3} = \frac{1}{4ia^3}.$$

Replacing ia by $-ia$ gives $-1/(4ia^3)$. For (c), the double-pole formula instead gives

$$\frac{d}{dz}\frac{z^2}{(z + ia)^2}\bigg|_{ia} = \frac{2z(z + ia) - 2z^2}{(z + ia)^3}\bigg|_{ia} = \frac{-2a^2}{-8ia^3} = \frac{1}{4ia}.$$

The residue at $-ia$ is $-1/(4ia)$. At infinity the three functions become

$$\frac{w^2}{1 + a^2w^2}, \quad \frac{w^4}{(1 + a^2w^2)^2}, \quad \frac{w^2}{(1 + a^2w^2)^2}.$$

Thus infinity is removable, with a zero of order $2, 4, 2$, respectively. None of the transformed differentials contains w^{-1}, so all three residues at infinity are zero.

(d) Zero is essential, since multiplication of $\sin(1/z)$ by a nonzero analytic function cannot remove its infinitely many negative Laurent powers. The other finite singularities are simple:

$$\operatorname{Res}_{ia} f = \frac{\sin(-i/a)}{2ia} = -\frac{\sinh(1/a)}{2a}, \qquad \operatorname{Res}_{-ia} f = \frac{\sin(i/a)}{-2ia} = -\frac{\sinh(1/a)}{2a}.$$

To obtain the residue at zero directly, multiply the convergent Laurent series on $0 < |z| < a$:

$$\frac{\sin(1/z)}{z^2 + a^2} = \sum_{j,k \ge 0} \frac{(-1)^{j+k} z^{2j-2k-1}}{a^{2j+2}(2k + 1)!}.$$

A term has exponent -1 exactly when $j = k$. Consequently

$$\boxed{\operatorname{Res}_0 f = \sum_{k=0}^{\infty} \frac{1}{a^{2k+2}(2k + 1)!} = \frac{\sinh(1/a)}{a}}.$$

At infinity $f(1/w) = w^2 \sin w/(1 + a^2w^2)$ has a zero of order three, and its residue at infinity is zero.

(e)–(g) If D has a simple zero p, the residue of N/D is $N(p)/D'(p)$. Applying this rule gives

	finite poles	residues in the same order	$\operatorname{Res}_\infty f$
(e)	$ia,\ -ia$	$e^{-a}/2,\ e^{a}/2$	$-\cosh a$
(f)	$a,\ -a$	$e^{ia}/2,\ e^{-ia}/2$	$-\cos a$
(g)	$a,\ -a$	$e^{ia}/(2a),\ -e^{-ia}/(2a)$	$-i\sin a/a$

For example, the first two rows follow from $pe^{ip}/(2p) = e^{ip}/2$, and the third from $e^{ip}/(2p)$. All these finite poles are simple. Each function has an essential singularity at infinity, since $e^{i/w}$ remains after multiplication by a nonzero rational factor. Nevertheless its residue there is defined. The residue theorem on a large circle, together with the definition above, gives $\operatorname{Res}_\infty f = -\sum_{\text{finite } p} \operatorname{Res}_p f$. The sums are $(e^{-a}+e^{a})/2$, $(e^{ia}+e^{-ia})/2$, and $(e^{ia}-e^{-ia})/(2a)$, which yield the displayed answers. In (d), the manual gives positive residues at $\pm ia$; both must be $-\sinh(1/a)/(2a)$, as the displayed substitution shows. It reverses the paired exponents in (e), giving $e^{\pm a}/2$ at $\pm ia$. Its (f) incorrectly uses poles $\pm ia$ and residues $-ie^{\mp a}/2$, although the denominator is z^2-a^2. In (g) it reverses the exponential signs as well. Substitution into $N(p)/D'(p)$ gives the table above, with each residue paired to its pole.

(h) The nonintegral power produces branch points at zero and infinity: continuation once around zero multiplies z^{-k} by $e^{-2\pi ik}$, while $f(1/w) = w^{k+1}/(1+w)$ is also multivalued at $w=0$. A rational $k = p/q$ in lowest terms gives q sheets; irrational k gives infinitely many. On any local branch at -1 there is a simple pole with

$$\boxed{\operatorname{Res}_{-1} f = (-1)^{-k} = e^{-ik(2m+1)\pi}}.$$

For the branch $0 < \arg z < 2\pi$, take $m = 0$. The residue depends on the branch. Ordinary Laurent residues at the branch points zero and infinity are not defined.

11.7.2 Evaluate the residues at $z = 0$ and $z = -1$ of $\pi\cot\pi z/z(z+1)$.

Solution.

Use $\pi\cot\pi z = z^{-1} - (\pi^2/3)z + O(z^3)$. At zero,

$$\frac{\pi\cot\pi z}{z(z+1)} = \left(\frac{1}{z^2} - \frac{\pi^2}{3} + O(z^2)\right)(1 - z + z^2 - \cdots) = \frac{1}{z^2} - \frac{1}{z} + O(1).$$

At $z = -1$, put $w = z+1$ and use the period one of $\cot\pi z$:

$$\frac{\pi\cot\pi z}{z(z+1)} = \frac{w^{-1}+O(w)}{w(w-1)} = -\frac{1}{w^2}(1 + w + w^2 + \cdots) + O(1) = -\frac{1}{w^2} - \frac{1}{w} + O(1).$$

Thus both points are double poles and

$$\boxed{\operatorname{Res}_0 f = \operatorname{Res}_{-1} f = -1}.$$

11.7.3 The classical definition of the exponential integral $\operatorname{Ei}(x)$ for $x > 0$ is the Cauchy principal value integral

$$\operatorname{Ei}(x) = \int_{-\infty}^{x} \frac{e^t}{t}\,dt,$$

where the integration range is cut at $t = 0$. Show that this definition yields a convergent result for positive x.

Solution.

Fix $x > 0$ and choose $0 < \delta < x$. Split the negative side at $-x$ and change $t = -u$ in the interval $[-x, -\delta]$:

$$\int_{-\infty}^{-\delta} \frac{e^t}{t}\,dt + \int_{\delta}^{x} \frac{e^t}{t}\,dt = \int_{-\infty}^{-x} \frac{e^t}{t}\,dt - \int_{\delta}^{x} \frac{e^{-u}}{u}\,du + \int_{\delta}^{x} \frac{e^u}{u}\,du$$
$$= \int_{-\infty}^{-x} \frac{e^t}{t}\,dt + \int_{\delta}^{x} \frac{e^u - e^{-u}}{u}\,du.$$

The first integral converges absolutely by exponential decay. In the second, $e^u - e^{-u} = 2u + O(u^3)$, so the integrand has the finite limit 2 at zero. Therefore

$$\boxed{\mathrm{Ei}(x) = \int_{-\infty}^{-x} \frac{e^t}{t}\,dt + \int_{0}^{x} \frac{2\sinh u}{u}\,du}$$

is finite. The cancellation uses equal distances δ on the two sides of the pole, precisely the principal-value prescription.

11.7.4 Writing a Cauchy principal value integral to deal with the singularity at $x = 1$, show that, if $0 < p < 1$,

$$\int_0^\infty \frac{x^{-p}}{x-1}\,dx = \pi \cot p\pi.$$

Solution.

For the printed denominator $x - 1$, the answer is $+\pi\cot(\pi p)$, not $-\pi\cot(\pi p)$. The manual derives its negative answer for the different denominator $1 - x$. Write the intended integral as

$$I = \lim_{\delta \downarrow 0} \left(\int_0^{1-\delta} + \int_{1+\delta}^{\infty} \right) \frac{x^{-p}}{x-1}\,dx.$$

The endpoint estimates are $O(x^{-p})$ at zero and $O(x^{-p-1})$ at infinity, so $0 < p < 1$ makes both endpoints integrable. Only $x = 1$ requires a principal value.

Use $F(z) = z^{-p}/(z-1)$ with $0 < \arg z < 2\pi$ and a keyhole contour about the positive real axis. On the upper bank $z^{-p} = x^{-p}$; on the lower bank it equals qx^{-p}, where $q = e^{-2\pi i p}$. The straight portions, directed outward above and inward below, give $(1-q)I$. Exclude the boundary pole at 1 by semicircles into the domain. The upper semicircle is clockwise and contributes $-i\pi$; the lower is also clockwise and contributes $-i\pi q$. These limits follow by substituting $z - 1 = \delta e^{i\theta}$; the analytic remainder has integral $O(\delta)$. The origin circle is $O(\epsilon^{1-p})$ and the outer circle is $O(R^{-p})$, so both vanish. No pole remains inside the indented contour, hence

$$(1-q)I - i\pi(1+q) = 0.$$

Finally

$$1 + q = 2e^{-\pi i p}\cos \pi p, \qquad 1 - q = 2ie^{-\pi i p}\sin \pi p,$$

and therefore

$$\boxed{⨍_0^\infty \frac{x^{-p}}{x-1}\,dx = i\pi \frac{1+q}{1-q} = \pi \cot \pi p}.$$

For example, $p = 1/4$ gives $+\pi$, which also distinguishes the two signs.

11.7.5 Explain why Eq. (11.88) is not directly applicable to the product expansion of $\sin z$. Show how the expansion, Eq. (11.89), can be obtained by expanding instead $\sin z/z$.

Solution.

The product formula quoted in the text contains the factors $f(0)$ and $\exp[zf'(0)/f(0)]$. Since $\sin 0 = 0$, direct substitution would both annihilate the product and divide by zero. Instead let

$$g(z) = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases}$$

The Taylor expansion $g(z) = 1 - z^2/3! + z^4/5! - \cdots$ gives $g(0) = 1$ and $g'(0) = 0$. Its zeros are the nonzero points $\pm n\pi$, each simple. To justify the product for this specific function, use the cotangent pole expansion (11.81): on $|z| < \pi$,

$$\frac{g'(z)}{g(z)} = \cot z - \frac{1}{z} = \sum_{n=1}^{\infty} \frac{2z}{z^2 - n^2\pi^2} = \sum_{n=1}^{\infty} \frac{d}{dz} \operatorname{Log}\left(1 - \frac{z^2}{n^2\pi^2}\right).$$

The removable value at zero is understood. On every smaller closed disk the summands are bounded by a constant times n^{-2}, so the series converges uniformly and may be integrated from zero to z. Choose the analytic logarithms that vanish at zero; since g has no zero in this disk and $g(0) = 1$, integration gives

$$\operatorname{Log} g(z) = \sum_{n=1}^{\infty} \operatorname{Log}\left(1 - \frac{z^2}{n^2\pi^2}\right).$$

Exponentiating yields the paired product. Equivalently, the canonical factors for opposite zeros are

$$g(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z}{n\pi}\right) e^{z/(n\pi)} \left(1 + \frac{z}{n\pi}\right) e^{-z/(n\pi)}.$$

Pairing the positive and negative zeros cancels the exponentials and multiplies the two linear factors:

$$\boxed{\sin z = z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2\pi^2}\right)}.$$

The paired product converges uniformly on compact sets because $\sum_{n\geq 1} |z|^2/(n^2\pi^2)$ does. It defines an entire function, so its equality to g on $|z| < \pi$ extends to the whole plane by the identity theorem. Multiplication by z proves the displayed formula, including its simple zero and derivative one at zero. The statement preceding (11.88) needs additional restrictions: zeros and the two values $f(0), f'(0)$ do not determine an arbitrary entire function. For example, e^{z^2} has no zeros, $f(0) = 1$, and $f'(0) = 0$, while that formula would give 1. The logarithmic-derivative argument above establishes the sine product without this unjustified generalization.

11.7.6 Starting from the observations

1. $f(z) = a_n z^n$ has n zeros counted with multiplicity, where $a_n \neq 0$ and $n \geq 1$, and
2. for sufficiently large $|R|$, $\left|\sum_{m=0}^{n-1} a_m R^m\right| < |a_n R^n|$,

use Rouché's theorem to prove the fundamental theorem of algebra (namely that every polynomial of degree n has n roots).

Solution.

The first comparison function must be the monomial $a_n z^n$, as in the textbook, not the full polynomial whose zeros are being proved to exist. Zeros are counted with multiplicity. Let

$$P(z) = a_n z^n + \sum_{m=0}^{n-1} a_m z^m, \qquad a_n \neq 0, \quad S = \sum_{m=0}^{n-1} |a_m|.$$

Choose $R > \max(1, S/|a_n|)$. At every point of $|z| = R$,

$$\left| \sum_{m=0}^{n-1} a_m z^m \right| \leq \sum_{m=0}^{n-1} |a_m| R^m \leq S R^{n-1} < |a_n| R^n = |a_n z^n|.$$

Rouché's theorem says that P and $a_n z^n$ have the same number of zeros inside this circle. The monomial has its sole zero at zero, of multiplicity n, so P has exactly n zeros there. The same inequality holds for every larger radius. Any additional finite zero would lie within some such larger circle and increase the count, which is impossible. Thus P has exactly n complex roots with multiplicity.

11.7.7 Using Rouché's theorem, show that all the zeros of $F(z) = z^6 - 4z^3 + 10$ lie between the circles $|z| = 1$ and $|z| = 2$.

Solution.

On $|z| = 1$ compare F with the constant 10:

$$|z^6 - 4z^3| \leq 1 + 4 = 5 < 10.$$

Rouché's theorem gives no zeros in $|z| < 1$ and the strict inequality also excludes a zero on the circle. On $|z| = 2$, compare with z^6:

$$|-4z^3 + 10| \leq 4(2^3) + 10 = 42 < 2^6 = 64.$$

There are therefore six zeros in $|z| < 2$, counted with multiplicity, and none on its boundary. A degree-six polynomial has exactly these six zeros, so all lie in $1 < |z| < 2$. As an independent check, setting $w = z^3$ yields $w^2 - 4w + 10 = 0$, hence $w = 2 \pm i\sqrt{6}$. Both values have modulus $\sqrt{10}$, so every root satisfies $|z| = 10^{1/6}$, strictly between one and two. The manual claims $|z^6 + 10| > |-4z^3|$ for all points inside $|z| = 2$. It fails when $z^3 = i\sqrt{10}$: the left side is zero and the right side is positive. The comparisons above apply on the two boundary circles, as required by Rouché's theorem.

11.7.8 Derive the pole expansions of $\sec z$ and $\csc z$ given in Eqs. (11.82) and (11.83).

Solution.

A simple pole of $1/h$ at a simple zero a of h has residue $1/h'(a)$. Thus, with $a_n = (n + \frac{1}{2})\pi$,

$$\operatorname{Res}_{a_n} \sec z = (-1)^{n+1}, \quad \operatorname{Res}_{-a_n} \sec z = (-1)^n, \qquad \operatorname{Res}_{n\pi} \csc z = (-1)^n.$$

To justify a pole expansion, integrate $F(\zeta)/(\zeta - z)$ on expanding rectangles, with z fixed inside. For $\sec \zeta$, use vertical sides $\Re\zeta = \pm N\pi$ and horizontal sides $\Im\zeta = \pm N\pi$. On a vertical side $|\sec(N\pi + iy)| = \operatorname{sech} y$, and $|\zeta - z|^{-1} = O(N^{-1})$; its integral is $O(N^{-1})$ because $\int_{\mathbb{R}} \operatorname{sech} y \, dy < \infty$.

On a horizontal side $|\cos(x+iN\pi)| \geq \sinh N\pi$, so its integral tends to zero exponentially. For $\csc\zeta$, choose vertical sides $\Re\zeta = \pm(N+\frac{1}{2})\pi$ and the same horizontal heights. There $|\csc(\pm(N+\frac{1}{2})\pi+iy)| = \operatorname{sech} y$, and the estimates are identical.

The residue theorem therefore gives, in the limit,

$$0 = F(z) + \sum_a \frac{\operatorname{Res}_a F}{a-z}, \qquad F(z) = \sum_a \frac{\operatorname{Res}_a F}{z-a}.$$

For $\sec z$ pair the poles a_n and $-a_n$:

$$\frac{(-1)^{n+1}}{z-a_n} + \frac{(-1)^n}{z+a_n} = \frac{2(-1)^n a_n}{a_n^2 - z^2}.$$

For $\csc z$ separate the pole at zero and pair $\pm n\pi$:

$$\frac{(-1)^n}{z-n\pi} + \frac{(-1)^n}{z+n\pi} = \frac{2(-1)^n z}{z^2 - n^2\pi^2}.$$

Consequently, off their poles,

$$\boxed{\sec z = \pi \sum_{n=0}^{\infty} \frac{(-1)^n(2n+1)}{(n+\frac{1}{2})^2\pi^2 - z^2}}, \qquad \boxed{\csc z = \frac{1}{z} + 2z \sum_{n=1}^{\infty} \frac{(-1)^n}{z^2 - n^2\pi^2}}.$$

The second paired series is absolutely locally uniformly convergent. In the first, the leading term is the convergent alternating series $2(-1)^n/a_n$, and subtracting that leading term leaves a locally uniform $O(n^{-3})$ series. This specifies the pairing and convergence needed for the pole expansion.

11.7.9 Given that $f(z) = (z^2-3z+2)/z$, apply a partial fraction decomposition to f'/f and show directly that $\oint_C f'(z)/f(z)\,dz = 2\pi i(N_f - P_f)$, where N_f and P_f are, respectively, the numbers of zeros and poles encircled by C (including their multiplicities), for a simple positively oriented contour avoiding $0, 1, 2$.

Solution.

Factor the numerator and take its logarithmic derivative algebraically:

$$f(z) = \frac{(z-1)(z-2)}{z}, \qquad \frac{f'}{f} = \frac{1}{z-1} + \frac{1}{z-2} - \frac{1}{z}.$$

For a positively oriented simple closed contour through none of $0, 1, 2$, each of the first two fractions integrates to $2\pi i$ if its pole is inside and to zero otherwise. The last contributes $-2\pi i$ if zero is inside. Hence

$$\boxed{\oint_C \frac{f'}{f}\,dz = 2\pi i\big(\mathbf{1}_{\{1 \text{ inside}\}} + \mathbf{1}_{\{2 \text{ inside}\}} - \mathbf{1}_{\{0 \text{ inside}\}}\big) = 2\pi i(N_f - P_f)}.$$

For a general closed curve the same formula uses winding-number weighted counts; the unweighted interpretation assumes a simple positively oriented contour avoiding all zeros and poles.

11.7.10 For a shrinking semicircle about a simple pole, the limiting contribution is one-half the full-circle integral, with compatible orientation. Show, by a specific example, that

$$\int_{\text{Semicircle}} f(z)\,dz = \frac{1}{2}\oint_{\text{Circle}} f(z)\,dz$$

does not necessarily hold if the integral encircles a pole of higher order.

Hint. Try $f(z) = z^{-2}$.

Solution.

Take $f(z) = z^{-2}$ and the upper semicircle $z = Re^{i\theta}$, $0 \le \theta \le \pi$. Since $dz = iRe^{i\theta}\,d\theta$,

$$\int_{\text{upper arc}} \frac{dz}{z^2} = \frac{i}{R}\int_0^{\pi} e^{-i\theta}\,d\theta = -\frac{1}{R}[e^{-i\theta}]_0^{\pi} = \frac{2}{R}.$$

On the full circle the same computation gives $-R^{-1}[e^{-i\theta}]_0^{2\pi} = 0$, equivalently the residue is zero. Thus $2/R \neq \frac{1}{2}(0)$ for every $R > 0$; the semicircle integral even diverges as the radius shrinks. The cancellation on the other half of the circle is essential for a pole of order two.

11.7.11 A function $f(z)$ is analytic along the real axis except for a third-order pole at $z = x_0$. The Laurent expansion about $z = x_0$ has the form

$$f(z) = \frac{a_{-3}}{(z - x_0)^3} + \frac{a_{-1}}{z - x_0} + g(z),$$

with $g(z)$ analytic at $z = x_0$. Assume the integrals converge separately at both infinite endpoints. Show that the Cauchy principal value technique is applicable, in the sense that

(a) $\lim_{\delta \to 0}\left\{\int_{-\infty}^{x_0-\delta} f(x)\,dx + \int_{x_0+\delta}^{\infty} f(x)\,dx\right\}$ is finite.

(b) $\lim_{\delta \downarrow 0}\int_{C_{x_0}} f(z)\,dz = \pm i\pi a_{-1}$,

where C_{x_0} denotes a semicircle of radius δ about $z = x_0$.

Solution.

The hypothesis controls the singularity at x_0, but does not ensure convergence at infinity. Part (a) needs convergent tails, or a separately specified principal value there. For instance $f(z) = (z - x_0)^{-3} + 1$ satisfies the local hypothesis and has divergent tails. Also, the equality in (b) is a limit as the semicircle shrinks. Assume the tails in (a) converge, and choose $h > 0$ within the Laurent expansion's disk. Put $t = x - x_0$. The singular terms are odd, so for $j = 1, 3$ and every $0 < \delta < h$,

$$\int_{-h}^{-\delta} t^{-j}\,dt + \int_{\delta}^{h} t^{-j}\,dt = 0$$

(the substitution $t = -u$ in the first integral proves the equality). Meanwhile continuity of g gives

$$\lim_{\delta \downarrow 0}\left(\int_{x_0-h}^{x_0-\delta} g(x)\,dx + \int_{x_0+\delta}^{x_0+h} g(x)\,dx\right) = \int_{x_0-h}^{x_0+h} g(x)\,dx.$$

Adding the two nonsingular convergent tails proves local principal-value existence and thus the qualified statement (a).

On a semicircle set $z - x_0 = \delta e^{i\theta}$. Then

$$\int \frac{a_{-3}\,dz}{(z-x_0)^3} = ia_{-3}\delta^{-2}\int_{\theta_1}^{\theta_2} e^{-2i\theta}\,d\theta = 0, \qquad \theta_2 - \theta_1 = \pm\pi,$$

because the endpoint values of $e^{-2i\theta}$ are equal. The simple pole contributes $ia_{-1}(\theta_2-\theta_1) = \pm i\pi a_{-1}$. Finally the analytic part is bounded by $\pi\delta \max_{|z-x_0|=\delta}|g(z)| \to 0$. Therefore

$$\boxed{\lim_{\delta\downarrow 0}\int_{C_{x_0}} f(z)\,dz = \pm i\pi a_{-1}},$$

with plus for counterclockwise traversal and minus for clockwise traversal.

11.7.12 The unit step function is defined as (compare Exercise 1.15.13)

$$u(s-a) = \begin{cases} 0, & s < a \\ 1, & s > a. \end{cases}$$

Show that $u(s)$ has the integral representations

(a) $u(s) = \lim_{\epsilon\to 0^+} \dfrac{1}{2\pi i}\displaystyle\int_{-\infty}^{\infty} \frac{e^{ixs}}{x - i\epsilon}\,dx.$

(b) $u(s) = \dfrac{1}{2} + \dfrac{1}{2\pi i}\,⨍_{-\infty}^{\infty} \dfrac{e^{ixs}}{x}\,dx.$

Note. The parameter s is real and nonzero; the symmetric convention at $s = 0$ is discussed below.

Solution.

The published erratum removes the principal-value symbol in (a): for fixed $\epsilon > 0$ the pole is off the real axis. For $s \neq 0$ the ordinary oscillatory improper integral exists. At $s = 0$ its two tails do not converge separately; a symmetric limit assigns $u(0) = 1/2$, while the problem's definition leaves $u(0)$ unspecified. For $s > 0$, close the contour in the upper half-plane. The exponential has modulus $e^{-s\,\Im z}$, so Jordan's estimate makes the large semicircle contribution tend to zero. The sole pole is $z = i\epsilon$ and has residue $e^{-\epsilon s}$. Thus

$$\frac{1}{2\pi i}\int_{\mathbb{R}} \frac{e^{ixs}}{x - i\epsilon}\,dx = e^{-\epsilon s} \longrightarrow 1.$$

For $s < 0$, close below, where the exponential decays. There are no poles and the clockwise contour gives zero. These are exactly the two prescribed step values.

For (b), evaluate $J(s) = ⨍_{\mathbb{R}} e^{ixs}/x\,dx$ directly. For $s > 0$ close above and indent the origin into the upper half-plane. The small clockwise arc has limit $-i\pi$; the indented domain contains no pole. Hence $J(s) - i\pi = 0$. For $s < 0$, close below; the lower indentation is counterclockwise and contributes $+i\pi$, so $J(s) + i\pi = 0$. Consequently

$$J(s) = i\pi\ \operatorname{sgn} s, \qquad \boxed{\frac{1}{2} + \frac{J(s)}{2\pi i} = \frac{1 + \operatorname{sgn} s}{2} = u(s) \quad (s \neq 0)}.$$

For completeness the connection between the prescriptions is visible in

$$\frac{1}{x - i\epsilon} = \frac{x}{x^2+\epsilon^2} + i\frac{\epsilon}{x^2+\epsilon^2}.$$

The second term has integral $i\pi$ in the limit: put $x = \epsilon t$, then use dominated convergence with $(1+t^2)^{-1}$. The first tends to the principal-value term (pair the two sides near zero and use oscillatory convergence in the tails). At $s = 0$, the symmetric integral of the first term is zero and that of the second is $i\pi$, giving the optional value $1/2$ in both representations.

11.8 Evaluation of Definite Integrals

11.8.1 Generalizing Example 11.8.1, for real a, b, show that

$$\int_0^{2\pi} \frac{d\theta}{a \pm b\cos\theta} = \int_0^{2\pi} \frac{d\theta}{a \pm b\sin\theta} = \frac{2\pi}{(a^2-b^2)^{1/2}}, \quad \text{for } a > |b|.$$

What happens if $|b| > |a|$?

Solution.

Take a, b real with $a > |b|$, and first consider $a + b\cos\theta$. If $b = 0$, the integral is $2\pi/a$. Otherwise set $z = e^{i\theta}$, $d\theta = dz/(iz)$ and $D = \sqrt{a^2 - b^2} > 0$. Then

$$I = \frac{2}{i}\oint_{|z|=1} \frac{dz}{bz^2 + 2az + b}.$$

The roots of the denominator are

$$r_+ = \frac{-a + D}{b} = -\frac{b}{a + D}, \qquad r_- = \frac{-a - D}{b}, \qquad r_+ r_- = 1.$$

Since $|r_+| = |b|/(a + D) < 1$, only r_+ is inside. The residue of the rational integrand there is

$$\frac{1}{2br_+ + 2a} = \frac{1}{2D}.$$

The residue theorem gives

$$\boxed{I = \frac{2}{i}(2\pi i)\frac{1}{2D} = \frac{2\pi}{\sqrt{a^2 - b^2}}}.$$

Changing b to $-b$ leaves the result unchanged. Shifting θ by $\pi/2$ turns sine into cosine, and a full-period integral is unchanged by that shift, proving the other two formulas.

If $|b| > |a|$, the denominator vanishes at angles with $\cos\theta = -a/b$ (or the corresponding sine values). Its derivative there is nonzero because $|\cos\theta| < 1$. Locally the reciprocal therefore has the form $c/(\theta - \theta_0) + O(1)$, $c \neq 0$. The ordinary improper integrals then do not exist. An analytic continuation of the displayed square root is not their value; any principal-value prescription must be specified separately.

11.8.2 Show that

$$\int_0^{\pi} \frac{d\theta}{(a + \cos\theta)^2} = \frac{\pi a}{(a^2 - 1)^{3/2}}, \quad a > 1.$$

Solution.

For $a > 1$, Exercise 11.8.1 and the substitution $\theta \mapsto 2\pi - \theta$ give

$$J(a) = \int_0^{\pi} \frac{d\theta}{a + \cos\theta} = \frac{\pi}{\sqrt{a^2 - 1}}.$$

On any closed interval of parameters lying in $(1, \infty)$, the denominator is bounded away from zero. We may therefore differentiate under the finite integral:

$$J'(a) = -\int_0^{\pi} \frac{d\theta}{(a + \cos\theta)^2} = -\pi a(a^2 - 1)^{-3/2}.$$

Multiplying by -1 proves

$$\boxed{\int_0^{\pi} \frac{d\theta}{(a + \cos\theta)^2} = \frac{\pi a}{(a^2 - 1)^{3/2}}}.$$

11.8.3 Show that

$$\int_0^{2\pi} \frac{d\theta}{1-2t\cos\theta+t^2} = \frac{2\pi}{1-t^2}, \quad \text{for } |t|<1.$$

What happens if $|t|>1$? What happens if $|t|=1$?

Solution.

For $t \neq 0$ write

$$1-2t\cos\theta+t^2 = (1-tz)(1-t/z), \qquad z=e^{i\theta}.$$

Thus

$$I(t) = \frac{1}{i}\oint_{|z|=1} \frac{dz}{(1-tz)(z-t)}.$$

The two poles are $z=t$ and $z=1/t$. If $|t|<1$, only t is inside and its residue is $1/(1-t^2)$. If $|t|>1$, only $1/t$ is inside and its residue is $1/[-t(1/t-t)] = 1/(t^2-1)$. Therefore

$$\boxed{I(t) = \begin{cases} 2\pi/(1-t^2), & |t|<1, \\ 2\pi/(t^2-1), & |t|>1. \end{cases}}$$

The case $t=0$ directly gives 2π, consistent with the first branch. These calculations also hold for complex t off the unit circle.

For real t with $|t|=1$, the denominator is $4\sin^2(\theta/2)$ at $t=1$ and $4\cos^2(\theta/2)$ at $t=-1$. Each has a quadratic zero, and the integral diverges to $+\infty$. If complex t is admitted, the other points $|t|=1$ put simple poles on the integration path instead. Those ordinary improper integrals are also undefined; they need not diverge to positive infinity.

11.8.4 Evaluate

$$\int_0^{2\pi} \frac{\cos 3\theta\, d\theta}{5-4\cos\theta}.$$

ANS. $\pi/12$.

Solution.

Because $\sin 3\theta/(5-4\cos\theta)$ changes sign under $\theta \mapsto 2\pi-\theta$, its integral vanishes. We may thus replace $\cos 3\theta$ by $e^{3i\theta}$ without changing the desired integral. With $z=e^{i\theta}$,

$$I = \frac{1}{i}\oint_{|z|=1} \frac{z^3\, dz}{-2z^2+5z-2}.$$

The denominator is $-2(z-\frac{1}{2})(z-2)$, so only $z=1/2$ lies inside. Its residue is

$$\left.\frac{z^3}{5-4z}\right|_{z=1/2} = \frac{1/8}{3} = \frac{1}{24}.$$

It follows that

$$\boxed{I = \frac{1}{i}(2\pi i)\frac{1}{24} = \frac{\pi}{12}}.$$

11.8.5 With the calculus of residues, show that

$$\int_0^\pi \cos^{2n}\theta\, d\theta = \pi\frac{(2n)!}{2^{2n}(n!)^2} = \pi\frac{(2n-1)!!}{(2n)!!}, \quad n = 0, 1, 2, \ldots.$$

The double factorial notation is defined in Eq. (1.76).

Hint. $\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta}) = \frac{1}{2}(z + z^{-1}), \quad |z| = 1.$

Solution.

The even power has period π, so twice the required integral is its integral over a full period. Set $z = e^{i\theta}$:

$$2I_n = \frac{1}{i\,2^{2n}}\oint_{|z|=1}\frac{(z+z^{-1})^{2n}}{z}\,dz = \frac{1}{i\,2^{2n}}\oint_{|z|=1}\frac{(z^2+1)^{2n}}{z^{2n+1}}\,dz.$$

Expand the numerator, $(z^2+1)^{2n} = \sum_{j=0}^{2n}\binom{2n}{j}z^{2j}$. The z^{-1} term occurs when $2j-(2n+1) = -1$, or $j = n$, and has coefficient $\binom{2n}{n}$. Hence

$$2I_n = \frac{2\pi}{2^{2n}}\binom{2n}{n}, \qquad \boxed{I_n = \pi\frac{(2n)!}{2^{2n}(n!)^2}}.$$

Finally $(2n)!! = 2^n n!$ and $(2n-1)!! = (2n)!/(2^n n!)$, so their ratio equals the factorial ratio above. For $n = 0$ the empty-product convention $(-1)!! = 0!! = 1$ gives $I_0 = \pi$ as required.

11.8.6 Verify that simplification of the expression in Eq. (11.112) yields the result given in Eq. (11.113).

Solution.

In the cited example, $I = \int_0^\infty \ln x/(1+x^3)\,dx$. Put $q = e^{2\pi i/3} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$. Equation (11.112), as printed in the textbook, reads

$$(1-q)I - \frac{2\pi i}{3}q\,\frac{2\pi}{3\sqrt{3}} = (2\pi i)\frac{\pi i}{9}q^{-1}.$$

Move the second term to the right and use $i^2 = -1$:

$$(1-q)I = -\frac{2\pi^2}{9}q^{-1} + \frac{4\pi^2 i}{9\sqrt{3}}q.$$

Since $q^{-1} = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$, the two terms are

$$\frac{\pi^2}{9} + \frac{i\pi^2\sqrt{3}}{9}, \qquad -\frac{2\pi^2}{9} - \frac{2i\pi^2}{9\sqrt{3}}.$$

Their sum is $-\pi^2/9 + i\pi^2/(9\sqrt{3}) = -(2\pi^2/27)(\frac{3}{2} - i\frac{\sqrt{3}}{2})$. But $1 - q = \frac{3}{2} - i\frac{\sqrt{3}}{2} \neq 0$. Canceling this factor gives

$$\boxed{I = -\frac{2\pi^2}{27}},$$

which is Eq. (11.113).

11.8.7 Complete the details of Example 11.8.8 by verifying that there is no contribution to the contour integral from either the small or the large circles of the contour, and that Eq. (11.115) simplifies to the result given as (11.116).

Solution.

Use $F(z) = z^p/(1+z^2)$ with $0 < \arg z < 2\pi$, and let the keyhole run outward along the upper bank and inward along the lower bank. This orientation reverses the incorrectly drawn lower arrows in the textbook's Fig. 11.26, as specified in the published errata. On the small circle $|z| = \epsilon < 1$,

$$\left| \int_{\text{small}} F(z)\, dz \right| \le \frac{2\pi \epsilon^{p+1}}{1-\epsilon^2} \longrightarrow 0.$$

On the large circle $|z| = R > 1$,

$$\left| \int_{\text{large}} F(z)\, dz \right| \le \frac{2\pi R^{p+1}}{R^2-1} = O(R^{p-1}) \longrightarrow 0.$$

Thus the two arc contributions vanish for $-1 < p < 1$, in particular for the stated $0 < p < 1$.

The upper bank contributes I and the lower bank $-e^{2\pi i p} I$. The residues at i and $-i$, with arguments $\pi/2$ and $3\pi/2$ on this branch, are

$$\frac{e^{ip\pi/2}}{2i}, \qquad -\frac{e^{3ip\pi/2}}{2i}.$$

It follows that

$$(1 - e^{2\pi i p}) I = \pi (e^{ip\pi/2} - e^{3ip\pi/2}) = -2\pi i e^{ip\pi} \sin(p\pi/2).$$

Also $1 - e^{2\pi i p} = -2ie^{ip\pi} \sin p\pi$. For $0 < p < 1$ these factors are nonzero; using $\sin p\pi = 2\sin(p\pi/2)\cos(p\pi/2)$ gives

$$\boxed{I = \int_0^\infty \frac{x^p}{1+x^2}\, dx = \frac{\pi}{2\cos(p\pi/2)}}.$$

The same result extends throughout $-1 < p < 1$, with $p = 0$ obtained by continuity or directly from $\int_0^\infty dx/(1+x^2) = \pi/2$.

11.8.8 Evaluate

$$\int_{-\infty}^{\infty} \frac{\cos bx - \cos ax}{x^2}\, dx, \quad a > b > 0.$$

ANS. $\pi(a-b)$.

Solution.

Consider

$$F(z) = \frac{e^{ibz} - e^{iaz}}{z^2} = \frac{i(b-a)}{z} + \frac{a^2-b^2}{2} + O(z).$$

Close the real line above and indent the origin by a small clockwise upper semicircle. The large arc is bounded by $2\pi/R$ because $|e^{iaz}|, |e^{ibz}| \le 1$ there, so it vanishes. The small arc contributes $-i\pi[i(b-a)] = \pi(b-a)$. There are no poles inside the indented contour, hence

$$⨍_{\mathbb{R}} F(x)\, dx + \pi(b-a) = 0.$$

Taking real parts removes the simple pole: the real integrand $(\cos bx - \cos ax)/x^2$ tends to $(a^2-b^2)/2$ at zero and is absolutely integrable at infinity. Thus it needs no principal value, and

$$\boxed{\int_{\mathbb{R}} \frac{\cos bx - \cos ax}{x^2}\, dx = \pi(a-b)}.$$

The same proof allows $b = 0$, since the large-arc estimate still holds.

11.8.9 Prove that

$$\int_{-\infty}^{\infty} \frac{\sin^2 x}{x^2}\,dx = \pi.$$

Hint. $\sin^2 x = \frac{1}{2}(1-\cos 2x)$.

Solution.

Use $\sin^2 x = (1-\cos 2x)/2$ and the $b=0$, $a=2$ case just proved:

$$\int_{\mathbb{R}} \frac{\sin^2 x}{x^2}\,dx = \frac{1}{2}\int_{\mathbb{R}} \frac{1-\cos 2x}{x^2}\,dx = \frac{1}{2}(2\pi) = \boxed{\pi}.$$

The origin is removable with limiting value one, and the tails are bounded by x^{-2}, so this is an ordinary absolutely convergent integral. The actual textbook prints the limits $-\infty$ to ∞ but gives $\pi/2$. The published errata also state $\pi/2$; that is the half-line value $\int_0^\infty (\sin x/x)^2\,dx$. Evenness and the contour calculation above prove that the full-line value is π. The manual's worked solution on p. 138 correctly gives π for the full line and $\pi/2$ for the half-line. Its separate errata list on p. 3, like the printed textbook answer, incorrectly assigns the half-line value to this full-line question.

11.8.10 Show that

$$\int_0^{\infty} \frac{x\sin x}{x^2+1}\,dx = \frac{\pi}{2e}.$$

Solution.

Integrate $F(z) = ze^{iz}/(z^2+1)$ over an upper semicircle. Its rational factor is $O(R^{-1})$ on the large arc, and the exponential decays in the upper half-plane, so Jordan's estimate makes that arc vanish. The only enclosed pole is i:

$$\operatorname{Res}_i F = \left.\frac{ze^{iz}}{z+i}\right|_{z=i} = \frac{e^{-1}}{2}.$$

Therefore

$$\int_{\mathbb{R}} \frac{xe^{ix}}{x^2+1}\,dx = \frac{\pi i}{e}.$$

The imaginary part $x\sin x/(x^2+1)$ is even, so its full-line integral is twice the required one:

$$\boxed{\int_0^{\infty} \frac{x\sin x}{x^2+1}\,dx = \frac{\pi}{2e}}.$$

The tails converge by Dirichlet's test (the amplitude $x/(x^2+1)$ decreases to zero for $x>1$), although they are not absolutely convergent.

11.8.11 A quantum mechanical calculation of a transition probability leads to the function $f(t,\omega) = 2(1-\cos\omega t)/\omega^2$. Show that

$$\int_{-\infty}^{\infty} f(t,\omega)\,d\omega = 2\pi|t|, \qquad t\in\mathbb{R}.$$

Solution.

For a general real time parameter the answer is $2\pi|t|$. The printed $2\pi t$ assumes $t \geq 0$. If $t = 0$ the integrand is zero. If $t \neq 0$, put $y = |t|\omega$; then $d\omega = dy/|t|$ and $\cos \omega t = \cos y$. Thus

$$\begin{aligned}\int_{\mathbb{R}} \frac{2(1-\cos \omega t)}{\omega^2}\, d\omega &= 2|t| \int_{\mathbb{R}} \frac{1-\cos y}{y^2}\, dy \\ &= 2|t|\,\pi = \boxed{2\pi|t|},\end{aligned}$$

where Exercise 11.8.8 with $a = 1, b = 0$ evaluates the last integral. At $\omega = 0$ the integrand extends continuously with value t^2; the tails are $O(\omega^{-2})$, so the substitution is justified in an absolutely convergent integral.

11.8.12 Show that ($a > 0$, $k \in \mathbb{R}$):

(a) $\displaystyle\int_{-\infty}^{\infty} \frac{\cos x}{x^2+a^2}\, dx = \frac{\pi}{a} e^{-a}.$

How is the right side modified if $\cos x$ is replaced by $\cos kx$?

(b) $\displaystyle\int_{-\infty}^{\infty} \frac{x \sin x}{x^2+a^2}\, dx = \pi e^{-a}.$

How is the right side modified if $\sin x$ is replaced by $\sin kx$?

Solution.

Let $k > 0$. Closing above for $F(z) = e^{ikz}/(z^2+a^2)$ encloses only ia, whose residue is $e^{-ka}/(2ia)$. The rational factor is $O(R^{-2})$, so the large arc vanishes. Taking real parts gives

$$\int_{\mathbb{R}} \frac{\cos kx}{x^2+a^2}\, dx = 2\pi i \frac{e^{-ka}}{2ia} = \frac{\pi}{a} e^{-ka}.$$

Cosine is even in k, and $k = 0$ gives π/a by the elementary arctangent integral. Hence

$$\boxed{\text{(a)} \quad \int_{\mathbb{R}} \frac{\cos kx}{x^2+a^2}\, dx = \frac{\pi}{a} e^{-a|k|} \qquad (k \in \mathbb{R}).}$$

For (b), integrate $G(z) = ze^{ikz}/(z^2+a^2)$. For $k > 0$ Jordan's estimate again removes the upper arc, and $\operatorname{Res}_{ia} G = e^{-ka}/2$. Taking imaginary parts yields $\int_{\mathbb{R}} x \sin kx/(x^2+a^2)\, dx = \pi e^{-ka}$. For $k < 0$ the integrand changes sign under $k \mapsto -k$, so

$$\boxed{\text{(b)} \quad \int_{\mathbb{R}} \frac{x \sin kx}{x^2+a^2}\, dx = \begin{cases} \pi\ \operatorname{sgn}(k) e^{-a|k|}, & k \neq 0, \\ 0, & k = 0. \end{cases}}$$

Setting $k = 1$ recovers both stated formulas. The second formula has a jump in its limits as $k \to 0$; the integral at $k = 0$ is exactly zero. Differentiation of the first formula across $k = 0$ would therefore be invalid. For nonzero k, convergence of the second integral's tails follows from Dirichlet's test, not absolute convergence.

11.8.13 Use the contour shown (Fig. 11.28) with $R \to \infty$ to prove that

$$\int_{-\infty}^{\infty} \frac{\sin x}{x}\, dx = \pi.$$

Solution.

Use exactly the rectangular contour of Fig. 11.28: its vertical sides are $\Re z = \pm R$, its top is $\Im z = R$, and its bottom follows the real axis except for a clockwise semicircular indentation of radius ϵ at zero. Integrate $F(z) = e^{iz}/z$ counterclockwise around this indented domain. The pole at zero has been excluded, so the contour integral is zero.

On either vertical side, $z = \pm R + iy$, $0 \le y \le R$, and

$$\left| \int_{\text{side}} F(z)\, dz \right| \le \frac{1}{R} \int_0^R e^{-y}\, dy \le \frac{1}{R}.$$

On the top, $|z| \ge R$, $|e^{iz}| = e^{-R}$ and the length is $2R$, so the bound is $2e^{-R}$. All three contributions vanish. On the small arc $e^{iz}/z = 1/z + i + O(z)$; its integral tends to $-i\pi$, since the argument decreases from π to zero. Consequently

$$⨍_{\mathbb{R}} \frac{e^{ix}}{x}\, dx - i\pi = 0.$$

Taking imaginary parts gives

$$\boxed{\int_{-\infty}^{\infty} \frac{\sin x}{x}\, dx = \pi}.$$

Here $\sin x/x \to 1$ at zero and each infinite tail converges by Dirichlet's test; the real integral itself does not need a principal-value qualification.

11.8.14 In the quantum theory of atomic collisions, we encounter the integral

$$I = \int_{-\infty}^{\infty} \frac{\sin t}{t} e^{ipt}\, dt,$$

in which p is real. Show that

$$I = 0, \quad |p| > 1$$
$$I = \pi, \quad |p| < 1.$$

What happens if $p = \pm 1$?

Solution.

Use $\sin t = (e^{it} - e^{-it})/(2i)$ to write

$$I(p) = \frac{1}{2i} \left\{ ⨍_{\mathbb{R}} \frac{e^{i(p+1)t}}{t}\, dt - ⨍_{\mathbb{R}} \frac{e^{i(p-1)t}}{t}\, dt \right\}.$$

The pole terms at zero cancel in their difference, because $\sin t/t$ is continuous there. For any real nonzero q, the upper or lower indented contour gives $⨍_{\mathbb{R}} e^{iqt}/t\, dt = i\pi \operatorname{sgn} q$. Thus, for $p \ne \pm 1$,

$$\boxed{I(p) = \frac{\pi}{2}\{\operatorname{sgn}(p+1) - \operatorname{sgn}(p-1)\} = \begin{cases} \pi, & |p| < 1, \\ 0, & |p| > 1. \end{cases}}$$

Both frequencies are then nonzero, so each tail of the original integral converges by Dirichlet's test.

At $p = \pm 1$ the ordinary improper complex integral does not exist: its two imaginary tails diverge logarithmically. A symmetric principal value at infinity, or Abel regularization, gives $\pi/2$. Indeed for $p = 1$,

$$\frac{\sin t}{t} e^{it} = \frac{\sin 2t}{2t} + i \frac{\sin^2 t}{t}.$$

The real part integrates to $\pi/2$. The imaginary part is odd, so it cancels on symmetric intervals, but

$$\int_1^R \frac{\sin^2 t}{t}\, dt = \frac{1}{2} \ln R - \frac{1}{2} \int_1^R \frac{\cos 2t}{t}\, dt$$

diverges as $\frac{1}{2} \ln R$, since the second term has a finite limit. For $p = -1$ its sign is reversed. This proves both the failure of ordinary convergence and the symmetric value at the endpoints.

11.8.15 Show that

$$\int_0^\infty \frac{dx}{(x^2 + a^2)^2} = \frac{\pi}{4a^3}, \quad a > 0.$$

Solution.

The function $(z^2 + a^2)^{-2}$ is $O(R^{-4})$ on a large upper semicircle, so that arc contributes $O(R^{-3})$. Its sole enclosed pole is the double pole at ia, with residue

$$\frac{d}{dz}(z + ia)^{-2}\bigg|_{ia} = -\frac{2}{(2ia)^3} = \frac{1}{4ia^3}.$$

Thus

$$\int_{\mathbb{R}} \frac{dx}{(x^2 + a^2)^2} = 2\pi i \frac{1}{4ia^3} = \frac{\pi}{2a^3}.$$

The integrand is even, giving

$$\boxed{\int_0^\infty \frac{dx}{(x^2 + a^2)^2} = \frac{\pi}{4a^3}}.$$

11.8.16 Evaluate

$$\int_{-\infty}^\infty \frac{x^2}{1 + x^4}\, dx.$$

ANS. $\pi/\sqrt{2}$.

Solution.

The roots of $1 + z^4$ are $e^{i\pi/4}$, $e^{3i\pi/4}$, $e^{5i\pi/4}$, $e^{7i\pi/4}$. Only the first two lie above the real axis. At a root p the residue of $z^2/(1 + z^4)$ is $p^2/(4p^3) = 1/(4p)$. Their sum is

$$\frac{1}{4}\left(e^{-i\pi/4} + e^{-3i\pi/4}\right) = \frac{1}{4}\left(\frac{1 - i}{\sqrt{2}} + \frac{-1 - i}{\sqrt{2}}\right) = -\frac{i}{2\sqrt{2}}.$$

The integrand is $O(R^{-2})$, so its large semicircle integral vanishes. The residue theorem therefore yields

$$\boxed{\int_{\mathbb{R}} \frac{x^2}{1 + x^4}\, dx = 2\pi i \left(-\frac{i}{2\sqrt{2}}\right) = \frac{\pi}{\sqrt{2}}}.$$

11.8.17 Evaluate

$$\int_0^\infty \frac{x^p \ln x}{x^2+1}\,dx, \quad 0 < p < 1.$$

ANS. $\dfrac{\pi^2 \sin(\pi p/2)}{4\cos^2(\pi p/2)}$.

Solution.

From Exercise 11.8.7,

$$J(p) = \int_0^\infty \frac{x^p}{1+x^2}\,dx = \frac{\pi}{2\cos(\pi p/2)}, \qquad -1 < p < 1.$$

Fix p in this interval and choose $\eta > 0$ so that $-1 < p-\eta < p+\eta < 1$. Near zero the differentiated integrand is bounded by $x^{p-\eta}|\ln x|$, and near infinity by $x^{p+\eta-2}\ln x$. Both bounds are integrable. Differentiation under the integral is therefore valid:

$$\int_0^\infty \frac{x^p \ln x}{1+x^2}\,dx = J'(p) = \frac{\pi}{2}\frac{(\pi/2)\sin(\pi p/2)}{\cos^2(\pi p/2)}$$
$$= \boxed{\frac{\pi^2 \sin(\pi p/2)}{4\cos^2(\pi p/2)}}.$$

This includes the stated range $0 < p < 1$.

11.8.18 Evaluate

$$\int_0^\infty \frac{(\ln x)^2}{1+x^2}\,dx,$$

(a) by appropriate series expansion of the integrand to obtain

$$4\sum_{n=0}^{\infty}(-1)^n(2n+1)^{-3},$$

(b) and by contour integration to obtain

$$\frac{\pi^3}{8}.$$

Hint. $x \to z = e^t$. Try the contour shown in Fig. 11.29, letting $R \to \infty$.

Solution.

Let I denote the required integral. **(a)** In its part from one to infinity set $x = 1/u$. Then $dx = -du/u^2$, $(\ln x)^2 = (\ln u)^2$, and $1 + x^2 = (1+u^2)/u^2$, so this part equals the part from zero to one. Thus

$$I = 2\int_0^1 \frac{(\ln x)^2}{1+x^2}\,dx = 2\sum_{n=0}^{\infty}(-1)^n \int_0^1 x^{2n}(\ln x)^2\,dx.$$

To evaluate the coefficient put $x = e^{-u}$:

$$\int_0^1 x^{2n}(\ln x)^2\,dx = \int_0^\infty u^2 e^{-(2n+1)u}\,du = \frac{2}{(2n+1)^3}.$$

For $\alpha = 2n+1 > 0$, the two integrations by parts are

$$\int_0^\infty u^2 e^{-\alpha u}\,du = \left[-\frac{u^2 e^{-\alpha u}}{\alpha}\right]_0^\infty + \frac{2}{\alpha}\int_0^\infty u e^{-\alpha u}\,du$$
$$= \frac{2}{\alpha}\left(\left[-\frac{u e^{-\alpha u}}{\alpha}\right]_0^\infty + \frac{1}{\alpha}\int_0^\infty e^{-\alpha u}\,du\right) = \frac{2}{\alpha^3}.$$

Each boundary term vanishes because exponential decay dominates the polynomial at infinity, and the polynomial is zero at the origin. Since $\sum 2/(2n+1)^3$ converges, the integrals of the absolute values of the series terms are summable, justifying the interchange even at the endpoint. Therefore

$$\boxed{I = 4\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^3}}.$$

(b) Set $x = e^t$. Because $e^t/(1+e^{2t}) = 1/(2\cosh t)$, we have

$$I = \int_{-\infty}^{\infty} \frac{t^2}{2\cosh t}\,dt.$$

Integrate $F(z) = z^2/(2\cosh z)$ around the rectangle with vertices $-R, R, R+i\pi, -R+i\pi$, as in Fig. 11.29. On the vertical sides $|\cosh(\pm R + iy)| \geq \sinh R$ and $|z|^2 \leq R^2 + \pi^2$; their integrals are $O(R^2 e^{-R})$. On the top $\cosh(t+i\pi) = -\cosh t$, and its direction is from R to $-R$. After adding bottom and top and passing to the limit, the contour integral is

$$\int_{\mathbb{R}} \frac{t^2 + (t+i\pi)^2}{2\cosh t}\,dt = 2I - \frac{\pi^2}{2}\int_{\mathbb{R}} \operatorname{sech} t\,dt.$$

The term $2i\pi t$ integrates to zero by oddness. Also, with $u = e^t$,

$$\int_{\mathbb{R}} \operatorname{sech} t\,dt = 2\int_0^\infty \frac{du}{1+u^2} = \pi.$$

The only pole in the rectangle is $z = i\pi/2$, with residue

$$\frac{(i\pi/2)^2}{2\sinh(i\pi/2)} = \frac{-\pi^2/4}{2i} = \frac{i\pi^2}{8}.$$

Hence $2I - \pi^3/2 = 2\pi i(i\pi^2/8) = -\pi^3/4$. Solving gives

$$\boxed{I = \frac{\pi^3}{8}}, \qquad \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^3} = \frac{\pi^3}{32}.$$

11.8.19 Prove that

$$\int_0^\infty \frac{\ln(1+x^2)}{1+x^2}\,dx = \pi \ln 2.$$

Solution.

Use the principal logarithm and consider separately

$$F_+(z) = \frac{\operatorname{Log}(z+i)}{z^2+1}, \qquad F_-(z) = \frac{\operatorname{Log}(z-i)}{z^2+1}.$$

The logarithm in F_+ has its cut on the ray $\{x - i : x \le 0\}$, below the real axis. Close its contour above; the pole at i has residue $\operatorname{Log}(2i)/(2i)$. The logarithm in F_- has its cut above the real axis; close below, clockwise, enclosing the pole at $-i$ with residue $\operatorname{Log}(-2i)/(-2i)$. The large arcs in both cases are $O((\ln R)/R)$ and vanish. Therefore

$$\int_{\mathbb{R}} F_+(x)\,dx = \pi \operatorname{Log}(2i), \qquad \int_{\mathbb{R}} F_-(x)\,dx = \pi \operatorname{Log}(-2i).$$

For real x, the two logarithms are conjugates, so their sum is $\ln(x^2+1)$. Also $\operatorname{Log}(2i) = \ln 2 + i\pi/2$ and $\operatorname{Log}(-2i) = \ln 2 - i\pi/2$. Adding the two equations yields

$$\int_{\mathbb{R}} \frac{\ln(1+x^2)}{1+x^2}\,dx = 2\pi \ln 2.$$

Evenness then gives

$$\boxed{\int_0^\infty \frac{\ln(1+x^2)}{1+x^2}\,dx = \pi \ln 2}.$$

11.8.20 Show that

$$\int_0^\infty \frac{x^a}{(x+1)^2}\,dx = \frac{\pi a}{\sin \pi a},$$

where $-1 < a < 1$, with the quotient interpreted as its limit 1 at $a = 0$.

Hint. Use the contour shown in Fig. 11.26, noting that $z = 0$ is a branch point and the positive x-axis can be chosen to be a cut line.

Solution.

For $a \neq 0$ use $F(z) = z^a/(1+z)^2$ on $0 < \arg z < 2\pi$ and a keyhole contour about the positive real axis. The small circle is bounded by $2\pi\epsilon^{a+1}/(1-\epsilon)^2$, and the large circle by $2\pi R^{a+1}/(R-1)^2$. Both vanish exactly when $-1 < a < 1$. The upper and lower banks contribute $(1 - e^{2\pi i a})I$.

The enclosed pole at -1 is double. Since $\arg(-1) = \pi$,

$$\operatorname{Res}_{-1} F = \frac{d}{dz} z^a \bigg|_{-1} = a e^{i\pi(a-1)} = -a e^{i\pi a}.$$

It follows that

$$(1 - e^{2\pi i a})I = -2\pi i a e^{i\pi a}.$$

Using $1 - e^{2\pi i a} = -2i e^{i\pi a} \sin \pi a$ gives

$$\boxed{I = \frac{\pi a}{\sin \pi a} \qquad (-1 < a < 1,\ a \neq 0)}.$$

At $a = 0$ the displayed quotient means its removable limit, equal to one. Directly, $\int_0^\infty (1+x)^{-2}\,dx = [-(1+x)^{-1}]_0^\infty = 1$. This also verifies the continuous value at zero.

11.8.21 Show that

$$\int_{-\infty}^\infty \frac{x^2\,dx}{x^4 - 2x^2\cos 2\theta + 1} = \frac{\pi}{2|\sin\theta|} = \frac{\pi}{2^{1/2}(1-\cos 2\theta)^{1/2}}.$$

Assume θ is real and $\sin\theta \neq 0$. Exercise 11.8.16 is a special case of this result.

Solution.

The formula with $\sin\theta$ in the denominator requires $\sin\theta > 0$, for example $0 < \theta < \pi$. For any real $\theta \notin \pi\mathbb{Z}$, the general answer is $\pi/(2|\sin\theta|)$. At $\theta \in \pi\mathbb{Z}$ the integral diverges. First take $0 < \theta < \pi$, $\theta \neq \pi/2$, and factor

$$D(z) = (z^2 - e^{2i\theta})(z^2 - e^{-2i\theta}).$$

The upper poles are $u = e^{i\theta}$ and $v = -e^{-i\theta}$. Since $D'(z) = 4z(z^2 - \cos 2\theta)$,

$$\operatorname{Res}_u \frac{z^2}{D(z)} = \frac{u}{4i\sin 2\theta}, \qquad \operatorname{Res}_v \frac{z^2}{D(z)} = -\frac{v}{4i\sin 2\theta} = \frac{e^{-i\theta}}{4i\sin 2\theta}.$$

Their sum simplifies to

$$\frac{2\cos\theta}{4i(2\sin\theta\cos\theta)} = \frac{1}{4i\sin\theta}.$$

The large upper arc vanishes because the rational function is $O(R^{-2})$. Hence

$$I = 2\pi i\,\frac{1}{4i\sin\theta} = \frac{\pi}{2\sin\theta}.$$

At $\theta = \pi/2$ the poles coalesce, but the integral can be evaluated directly:

$$\frac{x^2}{(x^2+1)^2} = \frac{1}{x^2+1} - \frac{1}{(x^2+1)^2}, \qquad I = \pi - \frac{\pi}{2} = \frac{\pi}{2}.$$

This agrees with the formula. In general the denominator depends on $\sin^2\theta$ because $D(x) = (x^2-1)^2 + 4x^2\sin^2\theta$; reducing to $0 < \theta < \pi$ gives the absolute value. Finally

$$\boxed{I = \frac{\pi}{2|\sin\theta|} = \frac{\pi}{\sqrt{2}\,\sqrt{1-\cos 2\theta}}}.$$

If $\sin\theta = 0$, the real poles at ± 1 behave like positive multiples of $(x \mp 1)^{-2}$ and cause divergence. Choosing $\theta = \pi/4$ recovers Exercise 11.8.16.

11.8.22 For real $n > 1$, show that

$$\int_0^\infty \frac{dx}{1+x^n} = \frac{\pi/n}{\sin(\pi/n)}.$$

Hint. Try the contour shown in Fig. 11.30, with $\theta = 2\pi/n$.

Solution.

The convergence condition is $n > 1$ (for real n); it is necessary and was omitted from the statement. Choose a branch of z^n with argument in the sector $0 < \arg z < \alpha = 2\pi/n$, and integrate $1/(1+z^n)$ over the sector boundary, with small and large circular arcs of radii ϵ and R. The small arc is $O(\epsilon)$, and the large one is $O(R^{1-n})$, so both vanish for $n > 1$. On the outgoing real ray the integral is I, while on the incoming ray $z = re^{i\alpha}$ it is $-e^{i\alpha}I$, since $z^n = r^n e^{2\pi i} = r^n$.

There is one pole in the sector, $z_0 = e^{i\pi/n}$, with

$$\operatorname{Res}_{z_0} \frac{1}{1+z^n} = \frac{1}{nz_0^{n-1}} = \frac{z_0}{nz_0^n} = -\frac{e^{i\pi/n}}{n}.$$

Thus

$$(1 - e^{2\pi i/n})I = -\frac{2\pi i}{n} e^{i\pi/n}.$$

Since $1 - e^{2i\beta} = -2ie^{i\beta} \sin \beta$, we obtain

$$\boxed{\int_0^\infty \frac{dx}{1 + x^n} = \frac{\pi}{n \sin(\pi/n)} \qquad (n > 1)}.$$

The branch argument also covers nonintegral real $n > 1$; the small circle excludes its branch point at zero. For $0 < n \le 1$ the tail is comparable to x^{-n} and diverges; for $n \le 0$ it does not even tend to zero fast enough to be integrable. The manual loses a minus sign when replacing z_0^{1-n}/n by an exponential. Since $z_0^n = -1$, the residue is $-z_0/n = -e^{i\pi/n}/n$, not $+e^{i\pi/n}/n$. Its positive final integral agrees with the correctly signed derivation above.

11.8.23

(a) Show that

$$f(z) = z^4 - 2z^2 \cos 2\theta + 1$$

has zeros at $e^{i\theta}$, $e^{-i\theta}$, $-e^{i\theta}$, and $-e^{-i\theta}$.

(b) For real θ with $\sin \theta \neq 0$, show that

$$\int_{-\infty}^\infty \frac{dx}{x^4 - 2x^2 \cos 2\theta + 1} = \frac{\pi}{2|\sin \theta|} = \frac{\pi}{2^{1/2}(1 - \cos 2\theta)^{1/2}}.$$

Exercise 11.8.22 ($n = 4$) is a special case of this result.

Solution.

(a) Since $2 \cos 2\theta = e^{2i\theta} + e^{-2i\theta}$,

$$\begin{aligned} z^4 - 2z^2 \cos 2\theta + 1 &= (z^2 - e^{2i\theta})(z^2 - e^{-2i\theta}) \\ &= (z - e^{i\theta})(z + e^{i\theta})(z - e^{-i\theta})(z + e^{-i\theta}). \end{aligned}$$

The four listed numbers are exactly the zeros, with repetitions counted when they coincide.

(b) On the positive half-line put $x = 1/u$. Then

$$dx = -\frac{du}{u^2}, \qquad x^4 - 2x^2 \cos 2\theta + 1 = \frac{1 - 2u^2 \cos 2\theta + u^4}{u^4},$$

so

$$\int_0^\infty \frac{dx}{x^4 - 2x^2 \cos 2\theta + 1} = \int_0^\infty \frac{u^2\, du}{u^4 - 2u^2 \cos 2\theta + 1}.$$

Both integrands are even, and Exercise 11.8.21 evaluates twice the latter integral. Consequently

$$\boxed{\int_{\mathbb{R}} \frac{dx}{x^4 - 2x^2 \cos 2\theta + 1} = \frac{\pi}{2|\sin \theta|} = \frac{\pi}{\sqrt{2}\sqrt{1 - \cos 2\theta}}}.$$

As in 11.8.21, the printed expression without the absolute value requires $\sin \theta > 0$, and $\sin \theta = 0$ is divergent. At $\theta = \pi/4$, the half-line value is $\pi/(2\sqrt{2})$, the $n = 4$ case of Exercise 11.8.22.

11.8.24 Show that

$$\int_0^\infty \frac{x^{-a}}{x+1}\,dx = \frac{\pi}{\sin a\pi},$$

where $0 < a < 1$.

Hint. You have a branch point and you will need a cut line. Try the contour shown in Fig. 11.26.

Solution.

Use $F(z) = z^{-a}/(1+z)$ with $0 < \arg z < 2\pi$. On a keyhole contour, the small-circle bound is $2\pi\epsilon^{1-a}/(1-\epsilon)$ and the large-circle bound is $2\pi R^{1-a}/(R-1)$. The first tends to zero if $a < 1$ and the second if $a > 0$. The two straight banks give $(1 - e^{-2\pi i a})I$. At the only pole, $z = -1$,

$$\operatorname{Res}_{-1} F = e^{-i\pi a}.$$

Therefore

$$(1 - e^{-2\pi i a})I = 2\pi i e^{-i\pi a}.$$

Using $1 - e^{-2\pi i a} = 2ie^{-i\pi a}\sin\pi a$ yields

$$\boxed{\int_0^\infty \frac{x^{-a}}{1+x}\,dx = \frac{\pi}{\sin\pi a} \qquad (0 < a < 1)}.$$

The same inequalities also prove ordinary convergence at both real endpoints.

11.8.25 Show that

$$\int_0^\infty \frac{\cosh bx}{\cosh x}\,dx = \frac{\pi}{2\cos(\pi b/2)}, \quad |b| < 1, \quad b \in \mathbb{R}.$$

Hint. Choose a contour that encloses one zero of $\cosh z$, hence one pole of $1/\cosh z$.

Solution.

For real $|b| < 1$, integrate $F(z) = e^{bz}/\cosh z$ around the rectangle $-R, R, R + i\pi, -R + i\pi$. On either vertical side, $|\cosh(\pm R + iy)| \geq \sinh R$, so the contributions are $O(e^{-(1-|b|)R})$. Let

$$J = \int_{\mathbb{R}} \frac{e^{bx}}{\cosh x}\,dx.$$

On the top, $F(x + i\pi) = -e^{i\pi b}F(x)$; the reversed direction therefore gives $+e^{i\pi b}J$. The one pole inside is $i\pi/2$:

$$\operatorname{Res}_{i\pi/2} F = \frac{e^{i\pi b/2}}{\sinh(i\pi/2)} = \frac{e^{i\pi b/2}}{i}.$$

The residue theorem gives

$$(1 + e^{i\pi b})J = 2\pi e^{i\pi b/2}.$$

Now $1 + e^{i\pi b} = 2e^{i\pi b/2}\cos(\pi b/2)$, so $J = \pi/\cos(\pi b/2)$. Pairing x with $-x$ in J,

$$J = \int_0^\infty \frac{e^{bx} + e^{-bx}}{\cosh x}\,dx = 2\int_0^\infty \frac{\cosh bx}{\cosh x}\,dx.$$

Thus

$$\boxed{\int_0^\infty \frac{\cosh bx}{\cosh x}\,dx = \frac{\pi}{2\cos(\pi b/2)}}.$$

The printed hint calls this a pole of $\cosh z$. That function is entire; its simple zero at $i\pi/2$ produces the pole of the reciprocal used here.

11.8.26 Show that

$$\int_0^\infty \cos(t^2)\,dt = \int_0^\infty \sin(t^2)\,dt = \frac{\sqrt{\pi}}{2\sqrt{2}}.$$

Hint. Try the contour shown in Fig. 11.30, with $\theta = \pi/4$.

Note. These are the Fresnel integrals for the special case of infinity as the upper limit. For the general case of a varying upper limit, asymptotic expansions of the Fresnel integrals are the topic of Exercise 12.6.1.

Solution.

Integrate the entire function e^{iz^2} around the sector $0 \le \arg z \le \pi/4$, $|z| \le R$. The real segment contributes $I_R = \int_0^R e^{it^2}\,dt$. On the returning ray put $z = e^{i\pi/4}t$; then $iz^2 = -t^2$ and its contribution is $-e^{i\pi/4}\int_0^R e^{-t^2}\,dt$. On the circular arc $z = Re^{i\theta}$,

$$|e^{iz^2}| = e^{-R^2\sin 2\theta}.$$

For $0 \le \theta \le \pi/4$, concavity of sine gives $\sin 2\theta \ge 4\theta/\pi$, so

$$\left|\int_{\text{arc}} e^{iz^2}\,dz\right| \le R\int_0^{\pi/4} e^{-4R^2\theta/\pi}\,d\theta \le \frac{\pi}{4R} \longrightarrow 0.$$

Cauchy's theorem now implies

$$\lim_{R\to\infty} I_R = e^{i\pi/4}\int_0^\infty e^{-t^2}\,dt.$$

For clarity, the Gaussian value follows by squaring its integral and using polar coordinates in the first quadrant:

$$\left(\int_0^\infty e^{-t^2}\,dt\right)^2 = \int_0^{\pi/2}\int_0^\infty e^{-r^2} r\,dr\,d\theta = \frac{\pi}{2}\cdot\frac{1}{2} = \frac{\pi}{4}.$$

The integral is positive, so it is $\sqrt{\pi}/2$. Since $e^{i\pi/4} = (1+i)/\sqrt{2}$, taking real and imaginary parts gives

$$\boxed{\int_0^\infty \cos(t^2)\,dt = \int_0^\infty \sin(t^2)\,dt = \frac{\sqrt{\pi}}{2\sqrt{2}}}.$$

The vanishing arc estimate proves convergence of these improper oscillatory integrals as well as their values.

11.8.27 Show that

$$\int_0^1 \frac{1}{(x^2-x^3)^{1/3}}\,dx = 2\pi/\sqrt{3}.$$

Hint. Try the contour shown in Fig. 11.31.

Solution.

The real integrand is $x^{-2/3}(1-x)^{-1/3}$, with its positive real cube root. Both endpoint exponents exceed -1, so it is integrable. To use the dogbone contour of Fig. 11.31, define on $\mathbb{C}\setminus[0,1]$

$$F(z) = \frac{e^{i\pi/3}}{z}\left(1-\frac{1}{z}\right)^{-1/3},$$

where the power in parentheses uses its principal logarithm. The map $1-1/z$ meets the nonpositive real axis exactly on the removed segment, so this branch is analytic on the contour's domain. It satisfies $F(z)^3 = 1/[z^2(1-z)]$.

For $0 < x < 1$, approaching from above makes $\arg(1 - 1/z) \to \pi$, whereas approaching from below makes it tend to $-\pi$. Thus

$$F_+(x) = x^{-2/3}(1-x)^{-1/3}, \qquad F_-(x) = e^{2\pi i/3} F_+(x).$$

Orient the outer circle counterclockwise; the inner dogbone then goes from zero to one above the cut and from one to zero below it. The endpoint circles contribute $O(\epsilon^{1/3})$ at zero and $O(\epsilon^{2/3})$ at one, hence vanish. If I is the desired integral, the two banks contribute $(1 - e^{2\pi i/3})I$. At infinity, $F(z) = e^{i\pi/3}/z + O(z^{-2})$, so the outer circle tends to $2\pi i e^{i\pi/3}$. There are no poles in the remaining domain, and Cauchy's theorem gives

$$(1 - e^{2\pi i/3})I + 2\pi i e^{i\pi/3} = 0.$$

Since $1 - e^{2\pi i/3} = -i\sqrt{3}\, e^{i\pi/3}$,

$$\boxed{I = \frac{2\pi}{\sqrt{3}}}.$$

Independently, $x = t/(1+t)$ transforms I into $\int_0^\infty t^{-2/3}/(1+t)\, dt$, the $a = 2/3$ case of 11.8.24.

11.8.28 Evaluate

$$\int_{-\infty}^{\infty} \frac{\tan^{-1} ax\, dx}{x(x^2 + b^2)},$$

for a and b positive, with $ab < 1$. Explain why the integrand does not have a singularity at $x = 0$.

Hint. Try the contour shown in Fig. 11.32, and use Eq. (1.137) to represent $\tan^{-1} az$. After cancellation, the integrals on segments B and B' combine to give an elementary integral.

Solution.

Let

$$A(z) = \arctan(az) = \frac{1}{2i}\{\mathrm{Log}(1 + iaz) - \mathrm{Log}(1 - iaz)\}, \qquad F(z) = \frac{A(z)}{z(z^2 + b^2)},$$

using the branches continued from $A(0) = 0$. The Taylor expansion $A(z) = az + O(z^3)$ gives $F(z) = a/b^2 + O(z^2)$ at zero, so the apparent singularity is removable.

Use the upper contour of Fig. 11.32, with the slit from i/a to $i\infty$ excluded. The assumption $ab < 1$ places the pole ib below the branch point. The small circle around i/a has integral $O(\rho|\ln\rho|) \to 0$, and the large arc vanishes because $A(z)$ is bounded at large $|z|$ on each bank and $F(z) = O(|z|^{-3})$. The enclosed residue is

$$\mathrm{Res}_{ib}\, F = \frac{A(ib)}{(ib)(2ib)} = -\frac{i}{4b^2} \ln\frac{1+ab}{1-ab},$$

where $A(ib) = \frac{i}{2}\ln[(1+ab)/(1-ab)]$ follows by substituting $z = ib$ into the logarithms.

For $z = iy$, $y > 1/a$, the first logarithm has arguments $+\pi$ on the right bank and $-\pi$ on the left bank, while the second has no jump. Hence $A_{\text{right}} - A_{\text{left}} = \pi$. The right bank is traversed downward and the left bank upward, so their combined contribution is

$$-i\int_{1/a}^{\infty} \frac{\pi\, dy}{iy(b^2 - y^2)} = \pi \int_{1/a}^{\infty} \frac{dy}{y(y^2 - b^2)}.$$

The elementary antiderivative is

$$\int \frac{dy}{y(y^2 - b^2)} = \frac{1}{2b^2} \ln\left(1 - \frac{b^2}{y^2}\right),$$

as direct differentiation verifies. Its definite value is $-\ln(1-a^2b^2)/(2b^2)$. Calling the real-axis integral I, the residue theorem gives

$$I - \frac{\pi}{2b^2}\ln(1-a^2b^2) = \frac{\pi}{2b^2}\ln\frac{1+ab}{1-ab}.$$

Combine the logarithms, whose arguments are positive under $ab<1$:

$$\boxed{I = \frac{\pi}{2b^2}\ln\left[\frac{1+ab}{1-ab}(1-a^2b^2)\right] = \frac{\pi}{b^2}\ln(1+ab)}.$$

The tails are $O(|x|^{-3})$ and the origin is removable, so this real integral is absolutely convergent.

11.9 Evaluation of Sums

11.9.1 Show that if $f(z)$ is analytic at $z=z_0$ and $g(z)$ has a simple pole at $z=z_0$ with residue b_0, then $f(z)g(z)$ has at most a simple pole at $z=z_0$, with residue $f(z_0)b_0$.

Solution.

Write the Taylor and Laurent expansions, respectively:

$$f(z) = f(z_0) + f'(z_0)(z-z_0) + O((z-z_0)^2), \qquad g(z) = \frac{b_0}{z-z_0} + b_1 + O(z-z_0).$$

Multiplication gives

$$f(z)g(z) = \frac{f(z_0)b_0}{z-z_0} + \{f(z_0)b_1 + f'(z_0)b_0\} + O(z-z_0).$$

The coefficient of $(z-z_0)^{-1}$ is therefore

$$\boxed{\operatorname{Res}_{z_0}(fg) = f(z_0)b_0}.$$

The claim that the product also has a simple pole requires $f(z_0)\neq 0$. If $f(z_0)=0$, the pole is canceled and the product has a removable singularity, with residue zero. For example, $f(z)=z-z_0$ and $g(z)=1/(z-z_0)$ give the constant one.

11.9.2 Show that $\cot z$ has magnitude $O(1)$ away from fixed neighborhoods of its poles. Deduce that $I_N = \oint_{C_N} f(z)\pi\cot(\pi z)\,dz \to 0$ on $C_N : |z| = N+\frac{1}{2}$ if $(N+\frac{1}{2})\max_{C_N}|f(z)| \to 0$.

Solution.

For $z=x+iy$, the identities

$$|\sin z|^2 = \sin^2 x + \sinh^2 y, \qquad |\cos z|^2 = \cos^2 x + \sinh^2 y$$

follow by expanding sine and cosine into real and imaginary parts. Hence

$$|\cot z|^2 = \frac{\cos^2 x + \sinh^2 y}{\sin^2 x + \sinh^2 y}.$$

If $|y|\geq\delta>0$, then

$$|\cot z|^2 \leq \frac{1+\sinh^2 y}{\sinh^2 y} = \coth^2|y| \leq \coth^2\delta.$$

In the remaining strip $|y| \le \delta$, impose a fixed positive distance from the poles $\pi\mathbb{Z}$. Periodicity reduces x to $[-\pi/2, \pi/2]$; the resulting closed set is compact and contains no zero of $\sin z$. The continuous quotient is therefore bounded there too. Thus $\cot z$ is uniformly bounded away from fixed neighborhoods of its poles. Here *order one* means $O(1)$, not a nonzero lower bound: $\cot z$ has zeros at $(n+\frac{1}{2})\pi$ even for arbitrarily large $|z|$. The exclusion of the poles must have a fixed positive width.

For the text's circles C_N, with radius $R_N = N + \frac{1}{2}$,

$$|z - n| \ge \big||z| - |n|\big| \ge \tfrac{1}{2} \qquad (z \in C_N,\ n \in \mathbb{Z}).$$

Consequently $|\pi \cot \pi z| \le M$ on every C_N, with one constant M independent of N. If $R_N \max_{C_N} |f(z)| \to 0$, the estimation lemma gives

$$|I_N| = \left| \oint_{C_N} f(z)\pi \cot \pi z \, dz \right| \le 2\pi R_N M \max_{C_N} |f(z)| \longrightarrow 0.$$

The decay of $zf(z)$ used here is uniform on the contours. Directional or pointwise decay alone does not supply this bound.

11.9.3 Evaluate $\dfrac{1}{1^3} - \dfrac{1}{3^3} + \dfrac{1}{5^3} - \cdots$.

Solution.

Exercise 11.8.18 established both

$$\int_0^\infty \frac{(\ln x)^2}{1+x^2}\,dx = 4\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^3} \quad \text{and} \quad \int_0^\infty \frac{(\ln x)^2}{1+x^2}\,dx = \frac{\pi^3}{8}.$$

Dividing the second value by the factor four in the first identity gives

$$\boxed{1 - \frac{1}{3^3} + \frac{1}{5^3} - \cdots = \frac{\pi^3}{32}}.$$

That earlier derivation justified the series interchange by absolute integrability and evaluated the integral by the specified rectangular contour, so no unproved summation formula is needed here.

11.9.4 Evaluate $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$.

Solution.

Resolve the summand into partial fractions:

$$\frac{1}{n(n+2)} = \frac{1}{2}\left(\frac{1}{n} - \frac{1}{n+2}\right).$$

For a finite upper limit N, every term from $1/3$ through $1/N$ cancels between the two sums:

$$S_N = \frac{1}{2}\left(1 + \frac{1}{2} - \frac{1}{N+1} - \frac{1}{N+2}\right).$$

Taking $N \to \infty$ gives

$$\boxed{\sum_{n=1}^{\infty} \frac{1}{n(n+2)} = \frac{3}{4}}.$$

The summands are $O(n^{-2})$, which also confirms convergence before performing the cancellation.

11.9.5 Evaluate $\displaystyle\sum_{n=-\infty}^{\infty} \frac{(-1)^n}{(n+a)^2}$, where a is real and not an integer.

Solution.

Integrate

$$F(z) = \frac{\pi \csc \pi z}{(z+a)^2}$$

on expanding rectangles with vertical sides at half-integers and horizontal sides at imaginary parts $\pm N$. On a vertical side $|\csc \pi z| = \operatorname{sech}(\pi \Im z) \le 1$; on the horizontal sides it decays exponentially. Also $(z+a)^{-2} = O(N^{-2})$ on the boundary for fixed real a. Thus the boundary integral tends to zero. At each integer n, the residue is $(-1)^n/(n+a)^2$ because the residue of $\pi \csc \pi z$ is $(-1)^n$.

Since a is not an integer, the remaining pole at $-a$ is double and distinct from all the integer poles. Its residue is

$$\frac{d}{dz}(\pi \csc \pi z)\bigg|_{z=-a} = -\pi^2 \csc(-\pi a)\cot(-\pi a) = -\pi^2 \csc \pi a \cot \pi a.$$

The residue theorem and the vanishing boundary integral give

$$0 = \sum_{n\in\mathbb{Z}} \frac{(-1)^n}{(n+a)^2} - \pi^2 \csc \pi a \cot \pi a.$$

Therefore

$$\boxed{\sum_{n=-\infty}^{\infty} \frac{(-1)^n}{(n+a)^2} = \pi^2 \csc \pi a \cot \pi a}.$$

The series is absolutely convergent, so the limiting rectangular ordering does not restrict the meaning of the sum.

11.9.6

(a) Using a method based on contour integration, evaluate $\displaystyle\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2}$.

(b) Check your work by relating your answer to an appropriate expression involving zeta functions.

Solution.

(a) Integrate $F(z) = \pi \tan(\pi z)/z^2$ over the square with sides $\Re z = \pm N$, $\Im z = \pm N$, where N is a positive integer. On the vertical sides $\tan(\pi(N+iy)) = i \tanh \pi y$, so its modulus is at most one. On the horizontal sides its modulus is bounded by $\coth \pi N$. Since $|z| \ge N$, the contour integral is $O(N^{-1})$ and vanishes.

The expansion $\tan \pi z = \pi z + O(z^3)$ gives the residue π^2 at zero. Each other pole is $p = n + \frac{1}{2}$, $n \in \mathbb{Z}$. Because

$$\operatorname{Res}_p(\pi \tan \pi z) = \frac{\pi \sin \pi p}{-\pi \sin \pi p} = -1,$$

the residue of F is $-1/p^2$. Summing residues yields

$$\sum_{n\in\mathbb{Z}} \frac{1}{(n+\frac{1}{2})^2} = \pi^2.$$

Pairing positive and negative half-integers and factoring the two gives

$$\sum_{n\in\mathbb{Z}} \frac{1}{(n+\frac{1}{2})^2} = 2\sum_{n=0}^{\infty} \frac{4}{(2n+1)^2} = 8\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2}.$$

Thus

$$\boxed{\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{8}}.$$

(b) Split the defining zeta series into its odd and even parts:

$$\zeta(2) = \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} + \sum_{n=1}^{\infty} \frac{1}{(2n)^2} = \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} + \frac{1}{4}\zeta(2).$$

The odd sum is $(3/4)\zeta(2)$. Using $\zeta(2) = \pi^2/6$ gives $(3/4)(\pi^2/6) = \pi^2/8$, in agreement with the contour result; conversely that result supplies the same value of $\zeta(2)$. The discussion on textbook p. 545 assigns residue $+1$ to $\pi\tan(\pi z)$. The quotient calculation above shows that each residue is -1; this sign is required in the contour sum.

11.9.7 Show that $\dfrac{1}{\cosh(\pi/2)} - \dfrac{1}{3\cosh(3\pi/2)} + \dfrac{1}{5\cosh(5\pi/2)} - \cdots = \dfrac{\pi}{8}$.

Solution.

Let

$$S = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)\cosh((2n+1)\pi/2)}, \qquad F(z) = \frac{1}{z\cos z\cosh z}.$$

Integrate F over the square with sides $\Re z = \pm R$, $\Im z = \pm R$, choosing $R = N\pi$. On a vertical side, $|\cos(\pm R + iy)| = \cosh y$ and $|\cosh(\pm R + iy)| \geq \sinh R$, while $|z| \geq R$. Its integral is bounded by

$$\frac{1}{R\sinh R}\int_{-R}^{R} \operatorname{sech} y\, dy \leq \frac{\pi}{R\sinh R}.$$

On a horizontal side the roles of cosine and hyperbolic cosine interchange, giving the same bound. Thus the square integral tends to zero.

The residue at zero is one. Put $a_n = (n+\frac{1}{2})\pi$, $n \geq 0$. At the positive real poles,

$$\operatorname{Res}_{a_n} F = \frac{1}{-a_n \sin a_n \cosh a_n} = \frac{(-1)^{n+1}}{a_n \cosh a_n}.$$

The residue at $-a_n$ is the same, as direct substitution shows. At the positive imaginary poles,

$$\operatorname{Res}_{ia_n} F = \frac{1}{(ia_n)\cos(ia_n)\sinh(ia_n)} = \frac{1}{(ia_n)\cosh a_n\, i(-1)^n} = \frac{(-1)^{n+1}}{a_n\cosh a_n},$$

and again the negative imaginary pole has the same residue. The four poles for each n therefore contribute

$$4\frac{(-1)^{n+1}}{a_n\cosh a_n} = -\frac{8}{\pi}\frac{(-1)^n}{(2n+1)\cosh((2n+1)\pi/2)}.$$

Their series is absolutely convergent by exponential decay. The residue theorem in the limit now gives $0 = 2\pi i(1 - 8S/\pi)$, and hence

$$\boxed{S = \frac{\pi}{8}}.$$

11.9.8 For $-\pi \le \varphi \le +\pi$, show that $\displaystyle\sum_{n=1}^{\infty}(-1)^n\frac{\sin n\varphi}{n^3} = \frac{\varphi}{12}(\varphi^2-\pi^2)$.

Solution.

The original series converges absolutely and uniformly in real φ, since $|\sin n\varphi|/n^3 \le 1/n^3$. To justify the differentiations, first insert $0<r<1$:

$$S_r(\varphi) = \sum_{n=1}^{\infty}\frac{(-1)^n r^n \sin n\varphi}{n^3}.$$

The differentiated series are then uniformly convergent, and the logarithmic power series gives

$$S_r''(\varphi) = -\sum_{n=1}^{\infty}\frac{(-1)^n r^n \sin n\varphi}{n} = \Im\,\mathrm{Log}(1+re^{i\varphi}).$$

The real part of $1+re^{i\varphi}$ is positive, so this imaginary part is bounded in absolute value by $\pi/2$. For $-\pi<\varphi<\pi$,

$$1+e^{i\varphi} = 2\cos(\varphi/2)e^{i\varphi/2}, \qquad \lim_{r\uparrow 1} S_r''(\varphi) = \frac{\varphi}{2}.$$

For $0 \le \varphi \le \pi$, integrate twice, using $S_r(0)=0$:

$$S_r(\varphi) = C_r\varphi + \int_0^{\varphi}(\varphi-t)\Im\,\mathrm{Log}(1+re^{it})\,dt.$$

Also $S_r(\pi)=0$, term by term, so

$$C_r = -\frac{1}{\pi}\int_0^{\pi}(\pi-t)\Im\,\mathrm{Log}(1+re^{it})\,dt \longrightarrow -\frac{1}{\pi}\int_0^{\pi}\frac{(\pi-t)t}{2}\,dt = -\frac{\pi^2}{12}.$$

The bounded argument justifies dominated convergence in both integrals. The uniform limit $S_r \to S$ therefore satisfies

$$S(\varphi) = -\frac{\pi^2\varphi}{12} + \int_0^{\varphi}\frac{(\varphi-t)t}{2}\,dt = -\frac{\pi^2\varphi}{12}+\frac{\varphi^3}{12}.$$

Both sides are odd, so the formula extends to negative φ. Continuity includes the endpoints, where both sides vanish:

$$\boxed{\sum_{n=1}^{\infty}\frac{(-1)^n\sin n\varphi}{n^3} = \frac{\varphi}{12}(\varphi^2-\pi^2), \quad -\pi\le\varphi\le\pi}.$$

11.10 Miscellaneous Topics

11.10.1 A function $f(z) = u(x, y) + iv(x, y)$ satisfies the conditions for the Schwarz reflection principle. Show that

(a) u is an even function of y. (b) v is an odd function of y.

Solution.

The Schwarz reflection principle gives $f(\bar{z}) = \overline{f(z)}$ on the reflected domain. Writing $z = x + iy$ and equating real and imaginary parts,

$$u(x, -y) + iv(x, -y) = u(x, y) - iv(x, y).$$

Thus

$$\boxed{u(x, -y) = u(x, y)}, \qquad \boxed{v(x, -y) = -v(x, y)}.$$

The real part is even in y and the imaginary part is odd in y, at all pairs of points related by the reflection.

11.10.2 A function $f(z)$ can be expanded in a Laurent series about the origin with the coefficients a_n real. Show that the complex conjugate of this function of z is the same function of the complex conjugate of z; that is,

$$f^*(z) = f(z^*).$$

Verify this explicitly for

(a) $f(z) = z^n$, n an integer. (b) $f(z) = \sin z$.

If $f(z) = iz$ ($a_1 = i$), show that the foregoing statement does not hold.

Solution.

On the Laurent series' annulus of convergence,

$$f(z) = \sum_{n=-\infty}^{\infty} a_n z^n, \qquad a_n \in \mathbb{R}.$$

The annulus is invariant under complex conjugation. Conjugating its convergent partial sums and then taking the limit gives

$$\overline{f(z)} = \sum_{n=-\infty}^{\infty} \overline{a_n}\, \bar{z}^{\,n} = \sum_{n=-\infty}^{\infty} a_n \bar{z}^{\,n} = f(\bar{z}).$$

For $f(z) = z^n$ this follows directly from conjugation of products if $n \geq 0$ and of reciprocals if $n < 0$ (the latter require $z \neq 0$). For the sine function,

$$\sin(x + iy) = \sin x \cosh y + i \cos x \sinh y.$$

Replacing y by $-y$ changes only the sign of its imaginary part, so $\sin(\bar{z}) = \overline{\sin z}$ explicitly. In contrast, for $f(z) = iz$,

$$\overline{f(z)} = -i\bar{z}, \qquad f(\bar{z}) = i\bar{z}.$$

They disagree for every $z \neq 0$, showing why the reality of the coefficients is essential. The manual's sine series omits both alternating signs and the odd powers. The correct series is $\sin z = \sum_{n=0}^{\infty} (-1)^n z^{2n+1}/(2n+1)!$, whose real coefficients agree with the direct Cartesian verification above.

11.10.3 The function $f(z)$ is analytic in a domain that is connected, invariant under complex conjugation, and includes a real interval. When z is real ($z = x$), $f(x)$ is pure imaginary.

(a) Show that
$$f(z^*) = -[f(z)]^*.$$

(b) For the specific case $f(z) = iz$, develop the Cartesian forms of $f(z)$, $f(z^*)$, and $f^*(z)$. Do not quote the general result of part (a).

Solution.

(a) Define $g(z) = -if(z)$. If $f(x)$ is pure imaginary on the real axis, then $g(x)$ is real there. Schwarz reflection gives
$$g(\bar{z}) = \overline{g(z)}, \qquad -if(\bar{z}) = i\,\overline{f(z)}.$$
Dividing by $-i$ proves
$$\boxed{f(\bar{z}) = -\overline{f(z)}}.$$
This identity is understood on the reflected continuation, or on a conjugation-invariant connected domain containing the real interval. For an arbitrary nonsymmetric original domain, one can assert it on the common component connected to that interval; $f(\bar{z})$ need not even be defined at every z of the original domain.

(b) Compute the three expressions separately from $z = x + iy$:
$$f(z) = i(x + iy) = -y + ix, \qquad f(\bar{z}) = i(x - iy) = y + ix, \qquad \overline{f(z)} = -y - ix.$$
The second is the negative of the third, directly confirming (a) without invoking it.

11.10.4 How do circles centered on the origin in the z-plane transform for

(a) $w_1(z) = z + \dfrac{1}{z}$, (b) $w_2(z) = z - \dfrac{1}{z}$, for $z \neq 0$?

What happens when $|z| \to 1$?

Solution.

Parameterize a circle by $z = re^{i\theta}$, $r > 0$. Then $z^{-1} = r^{-1}(\cos\theta - i\sin\theta)$. **(a)** For $w_1 = u + iv = z + z^{-1}$,
$$u = (r + r^{-1})\cos\theta, \qquad v = (r - r^{-1})\sin\theta.$$
If $r \neq 1$, eliminate θ to obtain the ellipse
$$\boxed{\frac{u^2}{(r + r^{-1})^2} + \frac{v^2}{(r - r^{-1})^2} = 1}.$$
Its horizontal semiaxis is $r + r^{-1}$ and its vertical semiaxis is $|r - r^{-1}|$. For $r < 1$ the sign of the sine coefficient reverses the direction of traversal, but not the geometric ellipse.

(b) For $w_2 = z - z^{-1}$ the coordinates are
$$u = (r - r^{-1})\cos\theta, \qquad v = (r + r^{-1})\sin\theta,$$
giving the ellipse
$$\boxed{\frac{u^2}{(r - r^{-1})^2} + \frac{v^2}{(r + r^{-1})^2} = 1 \qquad (r \neq 1)}.$$

Here the major axis is vertical. As $r \to 1$, one semiaxis in each case tends to zero. At $r = 1$ the parameterizations, rather than the divided equations, give $w_1 = 2\cos\theta$ and $w_2 = 2i\sin\theta$. The images are the segments $[-2, 2]$ and $[-2i, 2i]$, respectively, each traced twice during a full circuit of the unit circle.

11.10.5 What part of the z-plane corresponds to the interior of the unit circle in the w-plane if

(a) $w = \dfrac{z-1}{z+1}$? (b) $w = \dfrac{z-i}{z+i}$?

Solution.

(a) The condition $|w| < 1$ is $|z-1| < |z+1|$. Squaring and substituting $z = x + iy$ gives

$$(x-1)^2 + y^2 < (x+1)^2 + y^2 \quad \Longleftrightarrow \quad -2x < 2x \quad \Longleftrightarrow \quad x > 0.$$

Thus the disk corresponds exactly to the open right half-plane. The pole $z = -1$ is outside it; the imaginary axis maps to its boundary.

(b) Similarly,

$$|z-i| < |z+i| \quad \Longleftrightarrow \quad x^2 + (y-1)^2 < x^2 + (y+1)^2 \quad \Longleftrightarrow \quad y > 0.$$

The disk corresponds to the open upper half-plane. The pole $z = -i$ lies outside it, and the real axis maps to the boundary. The inverse formulas $z = (1+w)/(1-w)$ and $z = i(1+w)/(1-w)$ show that every point of $|w| < 1$ is attained, since $w = 1$ is not in the open disk.

11.10.6

(a) Writing $z = x + iy$, $w = u + iv$, show that if $w = 1/z$, the circle in the xy plane defined by $(x-a)^2 + (y-b)^2 = r^2$ transforms into $(u-A)^2 + (v-B)^2 = R^2$ when $r > 0$ and $a^2 + b^2 \neq r^2$. Determine the image line when the circle passes through zero, omitting zero from the original circle.

(b) Does the center of the circle in the z plane transform into the center of the corresponding circle in the w plane?

Solution.

Inversion maps a circle through the origin to a straight line. The claimed circle-to-circle form requires $a^2 + b^2 - r^2 \neq 0$; the origin itself is excluded from the map. Because $z = 1/w = (u - iv)/(u^2 + v^2)$,

$$x = \frac{u}{u^2+v^2}, \qquad y = -\frac{v}{u^2+v^2}.$$

Expand the original circle as $x^2 + y^2 - 2ax - 2by + a^2 + b^2 - r^2 = 0$ and multiply by $u^2 + v^2$. With $D = a^2 + b^2 - r^2$, the result is

$$1 - 2au + 2bv + D(u^2 + v^2) = 0.$$

If $D \neq 0$, divide by D and complete both squares:

$$\left(u - \frac{a}{D}\right)^2 + \left(v + \frac{b}{D}\right)^2 = \frac{a^2+b^2}{D^2} - \frac{1}{D} = \frac{r^2}{D^2}.$$

Therefore

$$\boxed{A = \frac{a}{D}, \qquad B = -\frac{b}{D}, \qquad R = \frac{r}{|D|}}.$$

If $D = 0$, the equation is instead the line $\boxed{2au - 2bv = 1}$.

For (b), when $a + ib \neq 0$ the image of the original center is

$$\frac{1}{a + ib} = \frac{a - ib}{a^2 + b^2},$$

whereas the center of the image circle is $(a - ib)/(a^2 + b^2 - r^2)$. For a genuine circle $r > 0$ these are different whenever both are defined. If the original center is zero, its image is undefined even though the image circle has center zero. Thus a circle's center does not in general map to its image's center. The manual writes the signed quantity $r/(r^2 - a^2 - b^2)$ for the radius. Its square gives the correct circle equation, but the geometric radius must be nonnegative: $R = r/|r^2 - a^2 - b^2|$.

11.10.7 Assume that a curve in the xy plane passes through point z_0 in the direction $dz = e^{i\theta} ds$, where s indicates arc length on the curve. Then, if $w = f(z)$, with $f(z)$ analytic at $z = z_0$, we have $dw = (dw/dz)\, dz = f'(z) e^{i\theta} ds$, where dw is in the direction the mapping of the xy curve passes through $w_0 = f(z_0)$ in the w plane. Use this observation to prove that if $f'(z_0) \neq 0$, the angle at which two curves intersect in the z plane is the same (both in magnitude and direction) as the angle of intersection of their mappings in the w plane.

Solution.

Since $f'(z_0) \neq 0$, write it uniquely in modulus-argument form $f'(z_0) = \rho e^{i\alpha}$ with $\rho > 0$ and α defined modulo 2π. Let the two oriented tangents be

$$dz_1 = e^{i\theta_1} ds_1, \qquad dz_2 = e^{i\theta_2} ds_2, \qquad ds_1, ds_2 > 0.$$

Their images satisfy

$$dw_j = f'(z_0) dz_j = \rho e^{i(\alpha + \theta_j)} ds_j, \qquad j = 1, 2.$$

Thus each tangent rotates by the same angle α and each infinitesimal length is multiplied by the same positive factor ρ. The oriented angle between the image tangents is consequently

$$(\alpha + \theta_2) - (\alpha + \theta_1) = \theta_2 - \theta_1 \pmod{2\pi}.$$

It equals the original oriented angle, preserving both magnitude and sense. The condition $f'(z_0) \neq 0$ ensures that neither image tangent collapses to zero, so its direction is defined.

12 Chapter 12. Further Topics in Analysis

Problems

12.1 Orthogonal Polynomials

12.1.1 Starting from the Rodrigues formula in Table 12.1 for the Hermite polynomials H_n, derive the generating function for the H_n given in that table.

Solution.

The convention in Table 12.1 is

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}.$$

Taylor's theorem applied to the entire function $f(x) = e^{-x^2}$ gives

$$f(x-t) = \sum_{n=0}^{\infty} \frac{(-t)^n}{n!} f^{(n)}(x).$$

Multiply by e^{x^2} and insert the Rodrigues expression term by term:

$$\sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!} = e^{x^2} e^{-(x-t)^2} = \boxed{e^{2xt-t^2}}.$$

The Taylor series converges for every finite complex t, so the generating function is an identity, not only a formal expansion. As a normalization check, its first three coefficients are $1, 2x, 2x^2-1$, corresponding to $H_0 = 1$, $H_1 = 2x$, $H_2 = 4x^2 - 2$ after multiplying by $n!$.

12.1.2

(a) Starting from the Laguerre ODE,

$$xy'' + (1-x)y' + \lambda y = 0,$$

obtain the Rodrigues formula for its polynomial solutions $L_n(x)$.

(b) From the Rodrigues formula, scaled as in Table 12.1, derive the generating function for the $L_n(x)$ given in that table.

Solution.

(a) If y is a polynomial of degree n with leading coefficient $a_n \neq 0$, the coefficient of x^n in the ODE is $(\lambda - n)a_n$. Thus $\lambda = n$. The weight needed to put the equation in self-adjoint form satisfies

$$(xw)' = (1-x)w, \quad xw' + w = w - xw, \quad w' = -w,$$

so choose $w = e^{-x}$. The Rodrigues construction with $p = x$ gives

$$\boxed{L_n(x) = \frac{e^x}{n!} \frac{d^n}{dx^n}(x^n e^{-x})}.$$

We can verify this construction directly. Put $F = x^n e^{-x}$, so $xF' = (n-x)F$. Differentiate $n+1$ times. Since both x and $n-x$ are linear, the product rule leaves

$$xF^{(n+2)} + (n+1)F^{(n+1)} = (n-x)F^{(n+1)} - (n+1)F^{(n)}.$$

With $G = F^{(n)}$ this becomes $xG'' + (x+1)G' + (n+1)G = 0$. Now $G = n!e^{-x}L_n$, so $G' = n!e^{-x}(L_n' - L_n)$ and $G'' = n!e^{-x}(L_n'' - 2L_n' + L_n)$. Substitution cancels the common factor and gives $xL_n'' + (1-x)L_n' + nL_n = 0$, exactly the required ODE.

Leibniz's rule also shows that the result is a polynomial:

$$L_n(x) = \sum_{k=0}^{n} \binom{n}{k} \frac{(-x)^k}{k!}.$$

Indeed, differentiating x^n $n-k$ times gives $n!x^k/k!$, while the remaining k derivatives of e^{-x} give $(-1)^k e^{-x}$. This fixes the scale $L_n(0) = 1$ and gives leading coefficient $(-1)^n/n!$.

(b) Sum that explicit polynomial formula:

$$\begin{aligned}\sum_{n=0}^{\infty} L_n(x)t^n &= \sum_{k=0}^{\infty} \frac{(-x)^k}{k!} \sum_{n=k}^{\infty} \binom{n}{k} t^n \\ &= \sum_{k=0}^{\infty} \frac{(-x)^k}{k!} \frac{t^k}{(1-t)^{k+1}} \\ &= \boxed{\frac{1}{1-t} \exp\left(-\frac{xt}{1-t}\right)}, \qquad |t| < 1.\end{aligned}$$

The inner sum follows by differentiating $(1-t)^{-1}$ k times and multiplying by $t^k/k!$. Interchanging the sums is justified because the corresponding absolute sum is at most $(1-|t|)^{-1}\exp[|x||t|/(1-|t|)] < \infty$.

12.1.3 Carry out in detail the steps needed to confirm that the $(n+1)$-fold differentiation of Eq. (12.10) leads to Eq. (12.12).

Solution.

Set $F = wp^n$ and $A = (n-1)p' + q$. Equation (12.10) is $pF' = AF$. Because p has degree at most two, $p^{(j)} = 0$ for $j \geq 3$; because A is at most linear, $A^{(j)} = 0$ for $j \geq 2$. Leibniz's formula therefore gives, with every surviving term displayed,

$$\begin{aligned}D^{n+1}(pF') &= pF^{(n+2)} + (n+1)p'F^{(n+1)} + \frac{n(n+1)}{2}p''F^{(n)}, \\ D^{n+1}(AF) &= AF^{(n+1)} + (n+1)A'F^{(n)}.\end{aligned}$$

The last term in the first line has coefficient zero if $n = 0$. Bring the right side to the left and divide by nonzero w. The coefficient of $F^{(n+1)}/w$ is

$$(n+1)p' - [(n-1)p' + q] = 2p' - q.$$

The coefficient of $F^{(n)}/w = y_n$ is

$$\begin{aligned}\frac{n(n+1)}{2}p'' - (n+1)[(n-1)p'' + q'] &= \frac{(n+1)(2-n)}{2}p'' - (n+1)q' \\ &= -\left[\frac{n^2-n-2}{2}p'' + (n+1)q'\right].\end{aligned}$$

Hence

$$\boxed{\frac{p}{w}F^{(n+2)} + \frac{2p' - q}{w}F^{(n+1)} - \left[\frac{n^2 - n - 2}{2}p'' + (n+1)q'\right] y_n = 0},$$

which is Eq. (12.12).

12.1.4 Confirm the algebraic steps that convert Eq. (12.12) into Eq. (12.16).

Solution.

Write $G = D^n(wp^n) = wy_n$ and

$$S = \frac{w'}{w} = \frac{q - p'}{p}.$$

Two product differentiations give

$$\frac{G'}{w} = y_n' + Sy_n, \qquad \frac{G''}{w} = y_n'' + 2Sy_n' + (S' + S^2)y_n.$$

Put $A = (n^2 - n - 2)p''/2 + (n+1)q'$. Substitution into Eq. (12.12) gives

$$py_n'' + [2pS + 2p' - q]y_n' + [pS' + pS^2 + (2p' - q)S - A]y_n = 0.$$

The first-derivative coefficient simplifies to $2(q - p') + 2p' - q = q$. In the last coefficient,

$$pS^2 + (2p' - q)S = S[(q - p') + 2p' - q] = p'S,$$

so

$$pS' + p'S - A = (pS)' - A = q' - p'' - A = -\frac{n(n-1)}{2}p'' - nq'.$$

Thus

$$\boxed{py_n'' + qy_n' - \left[\frac{n(n-1)}{2}p'' + nq'\right] y_n = 0},$$

exactly Eq. (12.16), with eigenvalue $\lambda_n = -n(n-1)\alpha - n\mu$ when $p'' = 2\alpha$, $q' = \mu$.

The source's intermediate Eqs. (12.13)–(12.15), on textbook p. 553, contain algebraic errors. Equation (12.16) is correct. The consistent intermediate identities are the following:

$$\begin{aligned}
\frac{p}{w}G'' &= py_n'' + \frac{2(q - p')}{w}G' - \left[p'' - q' + \frac{q(q - p')}{p}\right] y_n, \\
0 &= py_n'' + \frac{q}{w}G' - \left[\frac{n(n-1)}{2}p'' + nq' + \frac{q(q - p')}{p}\right] y_n, \\
\frac{q}{w}G' &= qy_n' + \frac{q(q - p')}{p}y_n.
\end{aligned}$$

For the first identity, eliminate y_n' from G''/w using $G'/w = y_n' + Sy_n$; then $p(S' - S^2) = q' - p'' - q(q - p')/p$. The other two follow by substitution. The opposite signs printed in (12.14) and (12.15) would cancel each other, but neither intermediate identity is valid with those signs.

12.1.5 Given the following integral representations, in which $n \in \mathbb{Z}$ and the contours encircle the origin once counterclockwise but no other singular points, derive the corresponding generating functions:

(a) Bessel functions:

$$J_n(x) = \frac{1}{2\pi i} \oint e^{(x/2)(t-1/t)} t^{-n-1}\, dt.$$

(b) Modified Bessel functions:

$$I_n(x) = \frac{1}{2\pi i} \oint e^{(x/2)(t+1/t)} t^{-n-1}\, dt.$$

Solution.

Both integrands are analytic in the punctured t-plane, so their Laurent coefficients on $0 < |t| < \infty$ are extracted by the given positively oriented contour integrals. Uniqueness of the Laurent expansion therefore gives

$$\boxed{e^{(x/2)(t-1/t)} = \sum_{n=-\infty}^{\infty} J_n(x) t^n}, \qquad \boxed{e^{(x/2)(t+1/t)} = \sum_{n=-\infty}^{\infty} I_n(x) t^n}.$$

For $x \neq 0$, the essential singularity at zero requires the full integer range rather than a Taylor series. At $x = 0$ both generating functions are one, so only the $n = 0$ coefficient survives. To exhibit the coefficients directly, multiply the two exponential series:

$$e^{xt/2} e^{\mp x/(2t)} = \sum_{j,k \geq 0} \frac{(x/2)^{j+k} (\mp 1)^k}{j!k!} t^{j-k}.$$

For $n \geq 0$, the terms with $j - k = n$ have $j = k + n$ and give

$$J_n(x) = \sum_{k \geq 0} \frac{(-1)^k (x/2)^{n+2k}}{k!(n+k)!}, \quad I_n(x) = \sum_{k \geq 0} \frac{(x/2)^{n+2k}}{k!(n+k)!}.$$

The negative coefficients similarly give $J_{-n} = (-1)^n J_n$ and $I_{-n} = I_n$. On any compact annulus in t, the product of exponential series converges absolutely and uniformly, justifying collection of coefficients and contour integration term by term.

12.1.6 Expand the generating function for the Legendre polynomials, $(1 - 2tz + t^2)^{-1/2}$, in powers of t. Assume that t is small. Collect the coefficients of t^0, t^1, and t^2.

ANS. $a_0 = P_0(z) = 1, \quad a_1 = P_1(z) = z, \quad a_2 = P_2(z) = \frac{1}{2}(3z^2 - 1).$

Solution.

Use $(1 + u)^{-1/2} = 1 - u/2 + 3u^2/8 + O(u^3)$ with $u = -2tz + t^2$. To retain only powers through t^2,

$$-\frac{u}{2} = zt - \frac{t^2}{2}, \qquad \frac{3u^2}{8} = \frac{3}{8}(4z^2t^2 + O(t^3)) = \frac{3}{2}z^2t^2 + O(t^3).$$

Consequently

$$(1 - 2tz + t^2)^{-1/2} = 1 + zt + \frac{3z^2 - 1}{2} t^2 + O(t^3).$$

Thus $\boxed{P_0 = 1,\ P_1 = z,\ P_2 = (3z^2 - 1)/2}$. Here z is fixed and the branch is chosen to equal one at $t = 0$; the expansion is valid in a sufficiently small disk about zero.

12.1.7 The set of Chebyshev polynomials usually denoted $U_n(x)$ has the generating-function formula

$$\frac{1}{1-2xt+t^2} = \sum_{n=0}^{\infty} U_n(x)t^n.$$

Derive a recurrence formula (for integer $n \geq 0$) connecting three U_n of consecutive n, using $U_{-1}(x) = 0$ at $n = 0$.

Solution.

Multiply the generating-function identity by its denominator:

$$(1-2xt+t^2)\sum_{n=0}^{\infty} U_n(x)t^n = 1.$$

The constant term gives $U_0 = 1$, and the coefficient of t gives $U_1 - 2xU_0 = 0$, so $U_1 = 2x$. At power t^{n+1} for $n \geq 1$,

$$U_{n+1} - 2xU_n + U_{n-1} = 0.$$

Hence

$$\boxed{U_{n+1} = 2xU_n - U_{n-1} \quad (n \geq 1)}.$$

To write this recurrence at $n = 0$ as requested, define $U_{-1} = 0$; otherwise the three-term formula starts at $n = 1$ and the $n = 0$ case is the separate initial equation $U_1 = 2xU_0$. The recurrence gives $U_2 = 4x^2 - 1$ and $U_3 = 8x^3 - 4x$, which also follow by direct expansion of the rational generating function.

12.2 Bernoulli Numbers

12.2.1 Verify the identities, Eqs. (12.32) and (12.46).

Solution.

For Eq. (12.32), multiply the numerator and denominator involving e^{-t} by e^t:

$$\frac{-t}{e^{-t}-1} = \frac{te^t}{e^t-1} = \frac{t(e^t-1)+t}{e^t-1} = t + \frac{t}{e^t-1}.$$

Subtracting t gives

$$\boxed{\frac{t}{e^t-1} = \frac{-t}{e^{-t}-1} - t}.$$

The first equality already proves Eq. (12.46):

$$\boxed{\frac{te^t}{e^t-1} = \frac{-t}{e^{-t}-1}}.$$

These identities hold away from the denominator's zeros and extend through the removable point $t = 0$ by continuity; both quotients there have limiting value one.

12.2.2 Show that the first Bernoulli polynomials are

$$B_0(s) = 1$$
$$B_1(s) = s - \tfrac{1}{2}$$
$$B_2(s) = s^2 - s + \tfrac{1}{6}.$$

Note that $B_n(0) = B_n$, the Bernoulli number.

Solution.

The defining generating function is $te^{st}/(e^t - 1) = \sum_{n\geq 0} B_n(s)t^n/n!$. First invert the series $(e^t-1)/t = 1+t/2+t^2/6+O(t^3)$. If its inverse is $1+at+bt^2+O(t^3)$, the product requires $a+1/2 = 0$ and $b + a/2 + 1/6 = 0$. Thus $a = -1/2$, $b = 1/12$. Multiply by $e^{st} = 1 + st + s^2t^2/2 + O(t^3)$:

$$\frac{te^{st}}{e^t - 1} = 1 + \left(s - \frac{1}{2}\right)t + \left(\frac{s^2}{2} - \frac{s}{2} + \frac{1}{12}\right)t^2 + O(t^3).$$

Comparing with $B_0 + B_1 t + B_2 t^2/2! + \cdots$ yields

$$\boxed{B_0(s) = 1, \quad B_1(s) = s - \frac{1}{2}, \quad B_2(s) = s^2 - s + \frac{1}{6}}.$$

At $s = 0$ the generating function is the Bernoulli-number generating function itself, proving $B_n(0) = B_n$ for every n.

12.2.3 Show that

$$\tan x = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}2^{2n}(2^{2n} - 1)B_{2n}}{(2n)!}x^{2n-1}, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}.$$

Hint. $\tan x = \cot x - 2\cot 2x$.

Solution.

First verify the hinted identity:

$$\cot x - 2\cot 2x = \frac{\cos x}{\sin x} - \frac{\cos^2 x - \sin^2 x}{\sin x \cos x} = \frac{\sin x}{\cos x} = \tan x.$$

The identity extends through zero by continuity after the poles cancel. From the Bernoulli generating function,

$$\cot x = i + \frac{2i}{e^{2ix} - 1} = \frac{1}{x} + \sum_{n=1}^{\infty} \frac{(-1)^n 2^{2n} B_{2n}}{(2n)!}x^{2n-1}, \quad |x| < \pi.$$

Here the $B_1 = -1/2$ term cancels the initial i, and all odd Bernoulli numbers beyond B_1 vanish. Substitute this series at x and at $2x$:

$$\cot x - 2\cot 2x = \left(\frac{1}{x} - \frac{2}{2x}\right) + \sum_{n\geq 1} \frac{(-1)^n 2^{2n} B_{2n}}{(2n)!}[1 - 2^{2n}]x^{2n-1}$$
$$= \boxed{\sum_{n=1}^{\infty} \frac{(-1)^{n-1}2^{2n}(2^{2n} - 1)B_{2n}}{(2n)!}x^{2n-1}}.$$

Both cotangent expansions converge for $|x| < \pi/2$. The nearest poles of $\tan x$ are at $\pm\pi/2$, so this is also its exact Taylor radius. The first terms are $x + x^3/3 + 2x^5/15 + \cdots$.

12.3 Euler-Maclaurin Integration Formula

12.3.1 The Euler-Maclaurin integration formula may be used for the evaluation of finite series:

$$\sum_{m=1}^{n} f(m) = \int_1^n f(x)\,dx + \frac{1}{2}f(1) + \frac{1}{2}f(n) + \frac{B_2}{2!}\left[f'(n) - f'(1)\right] + \cdots .$$

Show that

(a) $\displaystyle\sum_{m=1}^{n} m = \tfrac{1}{2}n(n+1)$.

(b) $\displaystyle\sum_{m=1}^{n} m^2 = \tfrac{1}{6}n(n+1)(2n+1)$.

(c) $\displaystyle\sum_{m=1}^{n} m^3 = \tfrac{1}{4}n^2(n+1)^2$.

(d) $\displaystyle\sum_{m=1}^{n} m^4 = \tfrac{1}{30}n(n+1)(2n+1)(3n^2+3n-1)$.

Solution.

For each polynomial, the Euler–Maclaurin correction sequence terminates because sufficiently high derivatives vanish. Use $B_2 = 1/6$, $B_4 = -1/30$.

(a) For $f(x) = x$, the first-derivative difference is $1 - 1 = 0$, so

$$\sum_{m=1}^{n} m = \frac{n^2-1}{2} + \frac{1+n}{2} = \boxed{\frac{n(n+1)}{2}}.$$

(b) For $f(x) = x^2$, $f' = 2x$ and higher odd derivatives vanish. Hence

$$\begin{aligned}\sum_{m=1}^{n} m^2 &= \frac{n^3-1}{3} + \frac{1+n^2}{2} + \frac{1}{12}(2n-2)\\ &= \frac{2n^3+3n^2+n}{6} = \boxed{\frac{n(n+1)(2n+1)}{6}}.\end{aligned}$$

(c) For $f(x) = x^3$, $f' = 3x^2$ and $f^{(3)} = 6$, whose endpoint difference is zero. Thus

$$\begin{aligned}\sum_{m=1}^{n} m^3 &= \frac{n^4-1}{4} + \frac{1+n^3}{2} + \frac{1}{12}(3n^2-3)\\ &= \frac{n^4+2n^3+n^2}{4} = \boxed{\frac{n^2(n+1)^2}{4}}.\end{aligned}$$

(d) For $f(x) = x^4$, $f' = 4x^3$, $f^{(3)} = 24x$, and $f^{(5)} = 0$. Retain both nonzero endpoint corrections:

$$\begin{aligned}\sum_{m=1}^{n} m^4 &= \frac{n^5-1}{5} + \frac{1+n^4}{2} + \frac{1}{12}(4n^3-4) - \frac{1}{720}(24n-24)\\ &= \frac{6n^5+15n^4+10n^3-n}{30}\\ &= \boxed{\frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}}.\end{aligned}$$

The last factorization follows from $(n+1)(2n+1)(3n^2+3n-1) = 6n^4+15n^3+10n^2-1$.

12.3.2 The Euler-Maclaurin integration formula provides a way of calculating the Euler-Mascheroni constant γ to high accuracy. Using $f(x) = 1/x$ in Eq. (12.57) (with interval $[1, n]$) and the definition of γ, Eq. (1.13), we obtain

$$\gamma = \sum_{s=1}^{n} s^{-1} - \ln n - \frac{1}{2n} + \sum_{k=1}^{N} \frac{B_{2k}}{(2k)n^{2k}} + R_{n,N}.$$

Using double-precision arithmetic, calculate the approximation for $N = 1, 2, \ldots$ at $n = 1000$, as in the printed answer below.

Note. See D. E. Knuth, Euler's constant to 1271 places. *Math. Comput.* **16**: 275 (1962).

ANS. For $n = 1000$, $N = 2$ $\gamma \approx 0.5772\,1566\,4901$ (truncated, rather than rounded, to twelve decimal places).

Solution.

The printed finite-N equality omits the remainder $R_{n,N}$; it is restored above. Put $H_n = \sum_{s=1}^{n} 1/s$ and define the computable approximation

$$A_{n,N} = H_n - \ln n - \frac{1}{2n} + \sum_{k=1}^{N} \frac{B_{2k}}{2k\, n^{2k}}.$$

To see the signs, apply Euler–Maclaurin to the tail from n to M. Since $f^{(2k-1)}(x) = -(2k-1)!/x^{2k}$, it gives

$$H_M - H_n = \ln(M/n) + \frac{1}{2M} - \frac{1}{2n} + \sum_{k=1}^{N} \frac{B_{2k}}{2k}(n^{-2k} - M^{-2k}) + \text{remainder}.$$

Let $M \to \infty$ and use $H_M - \ln M \to \gamma$. This yields $\gamma = A_{n,N} + R_{n,N}$. A rigorous signed remainder bound, derived in Exercise 12.6.6, is

$$0 < (-1)^N R_{n,N} < \frac{|B_{2N+2}|}{(2N+2)n^{2N+2}}.$$

Using IEEE double arithmetic, $n = 1000$, an accurately summed harmonic sum, and then accurately summing the corrections, gives

N	$A_{1000,N}$	last correction
1	0.5772156649015416	$8.3333333333 \times 10^{-8}$
2	0.5772156649015333	$-8.3333333333 \times 10^{-15}$
3	0.5772156649015333	$3.9682539683 \times 10^{-21}$
4	0.5772156649015333	$-4.1666666667 \times 10^{-27}$
5	0.5772156649015333	$7.5757575758 \times 10^{-33}$
6	0.5772156649015333	$-2.1092796093 \times 10^{-38}$

The exact truncation error at $N = 2$ is below $1/(252\,1000^6) = 3.9682539683 \times 10^{-21}$. The remaining roughly 4×10^{-16} difference from $\gamma = 0.5772156649015328606\ldots$ is double-arithmetic rounding in the harmonic sum and subtraction. More corrections cannot recover those already lost digits. To twelve decimal places the result is $\boxed{\gamma = 0.577215664902}$, consistent with the textbook's truncated display $0.5772\,1566\,4901$.

12.4 Dirichlet Series

12.4.1 From $B_{2n} = (-1)^{n-1}\dfrac{2(2n)!}{(2\pi)^{2n}}\zeta(2n)$, show that

(a) $\zeta(2) = \dfrac{\pi^2}{6}$, (d) $\zeta(8) = \dfrac{\pi^8}{9450}$,

(b) $\zeta(4) = \dfrac{\pi^4}{90}$, (e) $\zeta(10) = \dfrac{\pi^{10}}{93{,}555}$,

(c) $\zeta(6) = \dfrac{\pi^6}{945}$,

Solution.

Rearrange the stated identity before inserting numbers:

$$\zeta(2n) = \frac{(-1)^{n-1}B_{2n}(2\pi)^{2n}}{2(2n)!}.$$

With $B_2 = 1/6$, $B_4 = -1/30$, $B_6 = 1/42$, $B_8 = -1/30$, $B_{10} = 5/66$, the five evaluations are

$$\begin{aligned}
\zeta(2) &= \frac{(1/6)4\pi^2}{2\cdot 2} = \boxed{\pi^2/6},\\
\zeta(4) &= \frac{(1/30)16\pi^4}{2\cdot 24} = \boxed{\pi^4/90},\\
\zeta(6) &= \frac{(1/42)64\pi^6}{2\cdot 720} = \boxed{\pi^6/945},\\
\zeta(8) &= \frac{(1/30)256\pi^8}{2\cdot 40320} = \boxed{\pi^8/9450},\\
\zeta(10) &= \frac{(5/66)1024\pi^{10}}{2\cdot 3628800} = \boxed{\pi^{10}/93555}.
\end{aligned}$$

Each alternating sign cancels the sign of the corresponding Bernoulli number, leaving the positive values required by the defining zeta series.

12.4.2 The integral

$$\int_0^1 \frac{dx}{x}\,[\ln(1-x)]^2$$

appears in the fourth-order correction to the magnetic moment of the electron. Show that it equals $2\zeta(3)$.

Hint. Let $1 - x = e^{-t}$.

Solution.

Let $1-x=e^{-t}$, so $x=1-e^{-t}$, $dx=e^{-t}dt$, and $\ln(1-x)=-t$. The endpoints $x=0,1$ become $t=0,\infty$, giving

$$I=\int_0^\infty \frac{t^2e^{-t}}{1-e^{-t}}dt.$$

For $t>0$, $e^{-t}/(1-e^{-t})=\sum_{m=1}^\infty e^{-mt}$. All summands are nonnegative, so Tonelli's theorem permits interchange:

$$I=\sum_{m=1}^\infty \int_0^\infty t^2e^{-mt}dt.$$

Put $u=mt$ in each integral. Two integrations by parts give $\int_0^\infty u^2e^{-u}du=2\int_0^\infty ue^{-u}du=2$. Consequently

$$\boxed{I=2\sum_{m=1}^\infty \frac{1}{m^3}=2\zeta(3)}.$$

The finite value also proves convergence of the original improper integral, including its logarithmic endpoint singularity.

12.4.3

(a) Show that

$$\int_0^\infty \frac{(\ln z)^2}{1+z^2}\,dz=4\left(1-\frac{1}{3^3}+\frac{1}{5^3}-\frac{1}{7^3}+\cdots\right).$$

(b) By contour integration show that the integral, equivalently four times the parenthesized series, evaluates to $\pi^3/8$.

Solution.

(a) In the part from 1 to ∞, set $z=1/t$. Since $dz/(1+z^2)=-dt/(1+t^2)$ and $(\ln z)^2=(\ln t)^2$,

$$I=2\int_0^1 \frac{(\ln t)^2}{1+t^2}dt.$$

Expand the denominator geometrically. The absolute sum of the integrals is finite, because

$$\int_0^1 t^{2k}(\ln t)^2dt=\int_0^\infty u^2e^{-(2k+1)u}du=\frac{2}{(2k+1)^3}.$$

Here $t=e^{-u}$ was used. Thus termwise integration gives

$$\boxed{I=4\sum_{k=0}^\infty \frac{(-1)^k}{(2k+1)^3}=4\beta(3)}.$$

(b) Integrate $F(z)=(\operatorname{Log} z)^2/(1+z^2)$ round an upper semicircle, indenting the origin, using $0<\arg z<\pi$ inside the contour. The large arc is $O((\ln R)^2/R)$ and the small arc is $O(\epsilon|\ln\epsilon|^2)$; both vanish. The sole enclosed pole is $z=i$, where

$$\operatorname{Res}_{z=i}F=\frac{(i\pi/2)^2}{2i}=\frac{i\pi^2}{8},\qquad \oint F\,dz=-\frac{\pi^3}{4}.$$

On the negative real edge, $z = -t$ and $\operatorname{Log} z = \ln t + i\pi$. The two real-axis contributions therefore add to

$$2I + 2i\pi J - \pi^2 K, \qquad J = \int_0^\infty \frac{\ln t}{1+t^2}dt, \quad K = \int_0^\infty \frac{dt}{1+t^2} = \frac{\pi}{2}.$$

The substitution $t = 1/u$ gives $J = -J$, so $J = 0$. Equating to the residue result yields $2I - \pi^3/2 = -\pi^3/4$, hence

$$\boxed{I = \pi^3/8}, \qquad \boxed{\beta(3) = \pi^3/32}.$$

The value $\pi^3/8$ refers to the integral, or to four times the series in parentheses in (a). The unmultiplied alternating odd-cube series has value $\pi^3/32$.

12.4.4 Show that Catalan's constant, $\beta(2)$, may be written as

$$\beta(2) = 2\sum_{k=1}^{\infty}(4k-3)^{-2} - \frac{\pi^2}{8}.$$

Hint. $\pi^2 = 6\zeta(2)$.

Solution.

Separate the positive odd integers into their two residue classes:

$$A = \sum_{k\geq 1}(4k-3)^{-2}, \qquad B = \sum_{k\geq 1}(4k-1)^{-2}.$$

Absolute convergence permits grouping. By definition $\beta(2) = A - B$, while

$$A + B = \sum_{j\geq 0}(2j+1)^{-2} = \zeta(2) - \sum_{j\geq 1}(2j)^{-2} = \frac{3}{4}\zeta(2) = \frac{\pi^2}{8}.$$

Eliminate $B = \pi^2/8 - A$ to obtain

$$\boxed{\beta(2) = 2\sum_{k=1}^{\infty}(4k-3)^{-2} - \frac{\pi^2}{8}}.$$

12.4.5 Show that

(a) $\displaystyle\int_0^1 \frac{\ln(1+x)}{x}\,dx = \frac{1}{2}\zeta(2)$, (b) $\displaystyle\lim_{a\to 1^-}\int_0^a \frac{\ln(1-x)}{x}\,dx = -\zeta(2)$.

Note that the integrand in part (b) diverges for $a = 1$ but that the integral is convergent.

Solution.

(a) For $0 \leq x < 1$, expand $\ln(1+x)/x = \sum_{k\geq 1}(-1)^{k+1}x^{k-1}/k$. The sum of the integrals of the absolute values is $\sum_{k\geq 1} 1/k^2 < \infty$, which justifies termwise integration up to the endpoint. Therefore

$$\int_0^1 \frac{\ln(1+x)}{x}dx = \sum_{k\geq 1}\frac{(-1)^{k+1}}{k^2} = \zeta(2) - 2\sum_{j\geq 1}\frac{1}{(2j)^2} = \boxed{\frac{1}{2}\zeta(2)}.$$

(b) For $a < 1$, the logarithm series is uniformly convergent on $[0, a]$ after removing the harmless limit at zero, giving

$$\int_0^a \frac{\ln(1-x)}{x}dx = -\sum_{k=1}^{\infty}\frac{a^k}{k^2}.$$

Dominated convergence, with dominating summable sequence $1/k^2$, permits $a \uparrow 1$, so

$$\boxed{\lim_{a\to 1^-}\int_0^a \frac{\ln(1-x)}{x}dx = -\zeta(2)}.$$

The printed right-hand side in (b) has the wrong sign. The integral is negative since $\ln(1-x)/x < 0$ for $0 < x < 1$. The integrand diverges logarithmically at one, but its absolute integral is $\zeta(2)$ and is finite.

12.4.6

(a) Show that the equation $\ln 2 = \sum_{s=1}^{\infty}(-1)^{s+1}s^{-1}$, Eq. (1.53), may be rewritten as

$$\ln 2 = \sum_{s=2}^{n} 2^{-s}\zeta(s) + \sum_{p=1}^{\infty}(2p)^{-n-1}\left[1 - \frac{1}{2p}\right]^{-1}.$$

Hint. Take the terms in pairs.

(b) Calculate $\ln 2$ to six significant figures.

Solution.

(a) Pair neighboring terms of the convergent alternating harmonic series:

$$\ln 2 = \sum_{p=1}^{\infty}\left(\frac{1}{2p-1} - \frac{1}{2p}\right) = \sum_{p=1}^{\infty}\frac{(2p)^{-2}}{1-(2p)^{-1}}.$$

For $r = (2p)^{-1}$ the finite geometric identity is

$$\frac{r^2}{1-r} = \sum_{s=2}^{n} r^s + \frac{r^{n+1}}{1-r}.$$

Sum it over p. The first sum has finitely many s, and all terms are nonnegative, giving exactly

$$\boxed{\ln 2 = \sum_{s=2}^{n} 2^{-s}\zeta(s) + \sum_{p=1}^{\infty}\frac{(2p)^{-n-1}}{1-(2p)^{-1}}}.$$

The usual nonempty first sum takes $n \geq 2$; the $n = 1$ case also holds if that sum is interpreted as empty.

(b) The remainder R_n is positive and, since $1 - (2p)^{-1} \geq 1/2$, obeys

$$0 < R_n \leq 2\sum_{p\geq 1}(2p)^{-n-1} = 2^{-n}\zeta(n+1).$$

Take $n = 24$. Direct summation of the 23 zeta terms gives

$$S_{24} = 0.6931471209552993497\ldots, \qquad R_{24} < 5.9604646552 \times 10^{-8}.$$

Both ends of the interval $S_{24} < \ln 2 < S_{24} + R_{24}^{\text{bound}}$ round to the same six-significant-figure value:

$$\boxed{\ln 2 = 0.693147 \quad \text{(six significant figures)}}.$$

The error bound comes from the derived remainder, independently of any precomputed logarithm value.

12.4.7

(a) Show that the equation $\pi/4 = \sum_{s=1}^{\infty}(-1)^{s+1}(2s-1)^{-1}$, Eq. (12.72), may be rewritten as

$$\frac{\pi}{4} = 1 - 2\sum_{s=1}^{n} 4^{-2s}\zeta(2s) - 2\sum_{p=1}^{\infty}(4p)^{-2n-2}\left[1 - \frac{1}{(4p)^2}\right]^{-1}.$$

(b) Calculate $\pi/4$ to six significant figures.

Solution.

(a) Separate the first term of the arctangent series, then pair the rest:

$$\frac{\pi}{4} = 1 + \sum_{p=1}^{\infty}\left(-\frac{1}{4p-1} + \frac{1}{4p+1}\right) = 1 - 2\sum_{p=1}^{\infty}\frac{1}{(4p)^2-1}.$$

With $r = (4p)^{-2}$,

$$\frac{1}{(4p)^2-1} = \frac{r}{1-r} = \sum_{s=1}^{n} r^s + \frac{r^{n+1}}{1-r}.$$

Summing over p yields

$$\boxed{\frac{\pi}{4} = 1 - 2\sum_{s=1}^{n} 4^{-2s}\zeta(2s) - 2\sum_{p=1}^{\infty}\frac{(4p)^{-2n-2}}{1-(4p)^{-2}}}.$$

(b) The omitted correction has negative sign. Its magnitude is bounded using $1 - (4p)^{-2} \geq 15/16$:

$$0 < R_n := 2\sum_{p\geq 1}\frac{(4p)^{-2n-2}}{1-(4p)^{-2}} \leq \frac{32}{15}\,4^{-2n-2}\zeta(2n+2).$$

For $n = 6$, the finite part and bound are

$$S_6 = 0.7853981713451978431\ldots, \qquad R_6 < 7.9477727265 \times 10^{-9}.$$

Thus $S_6 - R_6^{\text{bound}} < \pi/4 < S_6$, whose endpoints both round to

$$\boxed{\pi/4 = 0.785398 \quad \text{(six significant figures)}}.$$

12.5 Infinite Products

12.5.1 Assume $a_n \geq 0$, no factor vanishes, and the factors are eventually positive. Product convergence means a finite nonzero limit. Using

$$\ln \prod_{n=1}^{\infty} (1 \pm a_n) = \sum_{n=1}^{\infty} \ln(1 \pm a_n)$$

and the Maclaurin expansion of $\ln(1 \pm a_n)$, show that the infinite product $\prod_{n=1}^{\infty}(1 \pm a_n)$ converges or diverges with the infinite series $\sum_{n=1}^{\infty} a_n$.

Solution.

The unqualified statement is false for signed or complex a_n. It is valid in its usual comparison-test form when $a_n \geq 0$ and the factors are eventually positive. A convergent infinite product here means a finite, nonzero limit, with no zero factor. Discarding finitely many initial nonzero factors only multiplies all later partial products by a fixed nonzero constant. We therefore apply real logarithms to the eventually positive tail. For that tail, convergence of the product is equivalent to convergence of its logarithmic series. If $a_n \not\to 0$, a nonzero product limit is already impossible, since the ratio of two successive partial products would not tend to one. Otherwise, eventually $0 \leq a_n \leq 1/2$. Integration of $1/(1+t)$ and $1/(1-t)$ on $[0, a_n]$ gives

$$\frac{a_n}{2} \leq \ln(1 + a_n) \leq a_n, \qquad a_n \leq -\ln(1 - a_n) \leq 2a_n.$$

The comparison test now proves the stated equivalence for both choices of sign. If $\sum a_n$ diverges, the plus-product grows without bound and the minus-product tends to zero.

To see why the hypotheses matter, take $a_n = (-1)^n/\sqrt{n+1}$. Its series converges by the alternating test, and every $1 + a_n$ is positive. But the logarithm expansion gives

$$\ln(1 + a_n) = a_n - \frac{1}{2}a_n^2 + O(|a_n|^3).$$

Here $\sum a_n^2 = \sum 1/(n+1) = \infty$, while $\sum |a_n|^3 < \infty$. Consequently the logarithmic partial sums tend to $-\infty$, and $\prod(1 + a_n)$ tends to zero, not a nonzero limit.

A useful corrected version for signed or complex terms is: if $a_n \to 0$, $\sum |a_n|^2 < \infty$, and no factor vanishes, then $\prod(1 + a_n)$ converges to a nonzero limit exactly when $\sum a_n$ converges. Indeed $\log(1 + a_n) - a_n = O(|a_n|^2)$, using the logarithm branch near one; its error series is absolutely convergent. To justify necessity as well as sufficiency, suppose the partial products $P_n \to L \neq 0$. Choose a disk about L avoiding zero and one analytic logarithm ℓ there. For large n, $\ell(P_n) - \ell(P_{n-1}) = \log(1 + a_n)$: both sides tend to zero, and their difference is an integer multiple of $2\pi i$. The logarithmic tail telescopes to a convergent limit. Conversely, a convergent logarithmic series exponentiates to a nonzero product. Subtracting the absolutely convergent error proves the claim about $\sum a_n$. The minus case follows by replacing a_n by $-a_n$.

12.5.2 An infinite product appears in the form

$$\prod_{n=1}^{\infty} \left(\frac{1 + a/n}{1 + b/n} \right),$$

where a and b are constants not equal to negative integers, so all factors are defined and nonzero. Show that this infinite product converges to a finite nonzero limit only if $a = b$.

Solution.

Use the standard finite, nonzero meaning of product convergence, and assume every denominator and numerator factor is nonzero. If a zero limit were counted as convergence, the asserted "only if" would fail. For fixed real or complex a, b, the tail has a logarithm branch near one. Its expansion is

$$\log \frac{1+a/n}{1+b/n} = \frac{a-b}{n} + O(n^{-2}).$$

The error series converges absolutely, so for the partial product P_N,

$$\log P_N = (a-b)H_N + C + o(1) = (a-b)\log N + C_1 + o(1).$$

The finitely many initial factors only alter the nonzero multiplicative constant. Thus $P_N = KN^{a-b}(1+o(1))$, $K \neq 0$. If $\Re(a-b) > 0$, its modulus tends to infinity; if $\Re(a-b) < 0$, it tends to zero. If $a - b = i\beta$ with real $\beta \neq 0$, its phase $\beta \ln N$ has no limit modulo 2π: integers nearest $e^{2\pi k/|\beta|}$ and $e^{(2\pi k+\pi)/|\beta|}$ give opposite limiting phases. Hence no finite nonzero limit exists unless $a = b$. When $a = b$, every defined factor equals one, so

$$\boxed{\text{the nonzero product converges exactly when } a = b.}$$

For a concrete zero-limit exception, $a = 0, b = 1$ gives $P_N = \prod_{n=1}^{N} n/(n+1) = 1/(N+1) \to 0$. If a is a negative integer a numerator factor is zero; if b is a negative integer the written product is undefined at that factor.

12.5.3 Show that the infinite product representations of $\sin x$ and $\cos x$ are consistent with the identity $2 \sin x \cos x = \sin 2x$.

Solution.

The product formulas are

$$\sin x = x \prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2\pi^2}\right), \quad \cos x = \prod_{n=1}^{\infty} \left(1 - \frac{4x^2}{(2n-1)^2\pi^2}\right).$$

In the sine product rewrite $x^2/(n^2\pi^2) = 4x^2/[(2n)^2\pi^2]$. The two products then supply the even and odd positive integers:

$$\begin{aligned} 2\sin x \cos x &= 2x \prod_{n\geq 1} \left(1 - \frac{4x^2}{(2n)^2\pi^2}\right) \prod_{n\geq 1} \left(1 - \frac{4x^2}{(2n-1)^2\pi^2}\right) \\ &= 2x \prod_{m=1}^{\infty} \left(1 - \frac{(2x)^2}{m^2\pi^2}\right) = \boxed{\sin 2x}. \end{aligned}$$

The deviations of the factors from one are bounded on compact x sets by a constant times $1/n^2$. Their absolute summability justifies interleaving the factors. At zeros of a factor the equality extends by continuity, so it holds for all complex x.

12.5.4 Determine the limit to which $\displaystyle\prod_{n=2}^{\infty}\left(1+\frac{(-1)^n}{n}\right)$ converges.

Solution.

Pair the factor with even index $2k$ with the following odd index:

$$\left(1+\frac{1}{2k}\right)\left(1-\frac{1}{2k+1}\right)=\frac{2k+1}{2k}\frac{2k}{2k+1}=1.$$

To ensure that pairing has not hidden a subsequence issue, let $P_N=\prod_{n=2}^{N}[1+(-1)^n/n]$. Then exactly

$$P_{2M+1}=1, \qquad P_{2M}=\frac{2M+1}{2M}.$$

Both subsequences tend to one. Therefore the complete sequence of partial products converges and

$$\boxed{\prod_{n=2}^{\infty}\left(1+\frac{(-1)^n}{n}\right)=1}.$$

12.5.5 Show that $\displaystyle\prod_{n=2}^{\infty}\left[1-\frac{2}{n(n+1)}\right]=\frac{1}{3}$.

Solution.

Factor each term before taking any limit:

$$1-\frac{2}{n(n+1)}=\frac{n^2+n-2}{n(n+1)}=\frac{(n-1)(n+2)}{n(n+1)}.$$

The finite product telescopes in two separate pieces:

$$\begin{aligned}P_N&=\left(\prod_{n=2}^{N}\frac{n-1}{n}\right)\left(\prod_{n=2}^{N}\frac{n+2}{n+1}\right)\\&=\left(\frac{1}{2}\frac{2}{3}\cdots\frac{N-1}{N}\right)\left(\frac{4}{3}\frac{5}{4}\cdots\frac{N+2}{N+1}\right)\\&=\frac{1}{N}\frac{N+2}{3}.\end{aligned}$$

Taking $N\to\infty$ proves $\boxed{P_\infty=1/3}$.

12.5.6 Prove that $\displaystyle\prod_{n=2}^{\infty}\left(1-\frac{1}{n^2}\right)=\frac{1}{2}$.

Solution.

Use $1-1/n^2=(n-1)(n+1)/n^2$. For a finite upper limit,

$$P_N=\left(\prod_{n=2}^{N}\frac{n-1}{n}\right)\left(\prod_{n=2}^{N}\frac{n+1}{n}\right)=\frac{1}{N}\frac{N+1}{2}=\frac{N+1}{2N}.$$

Both telescoping factors are exact for every $N\geq 2$. Thus $\boxed{\lim_{N\to\infty}P_N=1/2}$.

12.5.7 Verify the Euler identity $\prod_{p=1}^{\infty}(1+z^p)=\prod_{q=1}^{\infty}(1-z^{2q-1})^{-1}$, $|z|<1$.

Solution.

For $|z|<1$, all factors $1\pm z^n$ are nonzero and $\sum_{n\geq 1}|z|^n<\infty$. The corresponding products and their logarithmic tails converge absolutely, permitting the following factorization and regrouping:

$$\prod_{p=1}^{\infty}(1+z^p)=\frac{\prod_{p=1}^{\infty}(1-z^{2p})}{\prod_{p=1}^{\infty}(1-z^p)}.$$

Split the denominator into even and odd indices:

$$\prod_{p\geq 1}(1-z^p)=\prod_{q\geq 1}(1-z^{2q})\prod_{q\geq 1}(1-z^{2q-1}).$$

The even-index product is nonzero, so it cancels legitimately, leaving

$$\boxed{\prod_{p=1}^{\infty}(1+z^p)=\prod_{q=1}^{\infty}(1-z^{2q-1})^{-1}}.$$

The same argument is uniform on $|z|\leq r<1$. No boundary claim at $|z|=1$ follows from it.

12.5.8 Show that $\prod_{r=1}^{\infty}(1+x/r)e^{-x/r}$ converges for all finite x (except for the zeros of $1+x/r$).

Hint. Write the nth factor as $1+a_n$. Here convergence means a nonzero product; at a negative integer x the partial products instead have the zero limit.

Solution.

Write the rth factor as $1+a_r$ with $u=x/r$:

$$\begin{aligned}(1+u)e^{-u}&=(1+u)\left(1-u+\frac{u^2}{2}-\frac{u^3}{6}+\cdots\right)\\&=1-\frac{u^2}{2}+\frac{u^3}{3}+O(u^4).\end{aligned}$$

Hence $a_r=-x^2/(2r^2)+O(r^{-3})$ for fixed finite complex x, and $\sum|a_r|<\infty$. Once $|a_r|<1/2$, the convergent logarithm series has the bound

$$|\log(1+a_r)|\leq\sum_{k\geq 1}\frac{|a_r|^k}{k}\leq 2|a_r|.$$

The logarithmic tail therefore converges absolutely, proving a finite nonzero product whenever no factor vanishes. The exceptional points are exactly $x=-1,-2,\ldots$, since exponentials never vanish.

The estimate is uniform for $|x|\leq R$ after finitely many factors: $|a_r|\leq C_R/r^2$. Thus the product even defines an entire function by locally uniform convergence. At the excluded points it still converges as a sequence of partial products, with value zero; they are excluded only from the convention of a nonzero convergent infinite product.

12.5.9 Derive the formula, for $0 < x < \pi$ with real logarithms,

$$\ln \sin x = \ln x + \sum a_n x^n,$$

giving the explicit form for the coefficients a_n.

Hint. $d(\ln \sin x)/dx = \cot x$.

Solution.

It is convenient to remove the known logarithmic singularity first:

$$\frac{d}{dx} \ln \frac{\sin x}{x} = \cot x - \frac{1}{x}.$$

The Bernoulli expansion of the right side is

$$\cot x - \frac{1}{x} = \sum_{k=1}^{\infty} \frac{(-1)^k 2^{2k} B_{2k}}{(2k)!} x^{2k-1}, \qquad |x| < \pi.$$

Integrate term by term from zero, using $\sin x/x \to 1$ to fix the integration constant to zero:

$$\boxed{\ln \frac{\sin x}{x} = \sum_{k=1}^{\infty} \frac{(-1)^k 2^{2k} B_{2k}}{2k(2k)!} x^{2k}}.$$

Therefore

$$\boxed{a_{2k} = \frac{(-1)^k 2^{2k} B_{2k}}{2k(2k)!} = -\frac{\zeta(2k)}{k\pi^{2k}}, \qquad a_{2k-1} = 0 \quad (k \geq 1)}.$$

The constant coefficient is $a_0 = 0$ by the normalization at zero. In particular, $\ln(\sin x/x) = -x^2/6 - x^4/180 - x^6/2835 - \cdots$. For real logarithms the question's form $\ln \sin x = \ln x + \cdots$ uses $0 < x < \pi$. For $-\pi < x < 0$, write $\ln|\sin x| = \ln|x| + \cdots$. In complex form, the quotient has an analytic logarithm on $|x| < \pi$ normalized to zero at the origin; any separate logarithms must use consistent branches.

12.5.10 Using the infinite product representations of $\sin z$, show that for $|z| < \pi$

$$z \cot z = 1 - 2 \sum_{m,n=1}^{\infty} \left(\frac{z}{n\pi}\right)^{2m},$$

and hence that the Bernoulli numbers are given by the formula

$$B_{2n} = (-1)^{n-1} \frac{2(2n)!}{(2\pi)^{2n}} \zeta(2n).$$

This is an alternate route to Eq. (12.38).

Hint. The result of Exercise 12.5.9 will be helpful.

Solution.

Take the logarithmic derivative of the sine product on a compact subset avoiding its zeros. Local uniform convergence of the derivative series follows from its $O(n^{-2})$ terms. Thus

$$\cot z = \frac{1}{z} + \sum_{n=1}^{\infty} \frac{-2z}{n^2\pi^2 - z^2}, \qquad z \cot z = 1 - 2\sum_{n=1}^{\infty} \frac{z^2}{n^2\pi^2 - z^2}.$$

The requested double power series is restricted to $|z| < \pi$; it is not a representation for arbitrary z away from the poles. In that disk each denominator can be expanded geometrically:

$$\frac{z^2}{n^2\pi^2 - z^2} = \sum_{m=1}^{\infty} \left(\frac{z}{n\pi}\right)^{2m}.$$

For $|z| \le r < \pi$ the double absolute sum is bounded by

$$\sum_{n\ge 1} \frac{r^2/(n\pi)^2}{1 - r^2/(n\pi)^2} \le \frac{r^2\zeta(2)}{\pi^2[1 - r^2/\pi^2]} < \infty,$$

so the two summations may be interchanged. Hence

$$\boxed{z\cot z = 1 - 2\sum_{m,n\ge 1} (z/(n\pi))^{2m}} = 1 - 2\sum_{m\ge 1} \frac{\zeta(2m)}{\pi^{2m}} z^{2m}.$$

Independently, the Bernoulli generating function gives

$$z\cot z = 1 + \sum_{m\ge 1} \frac{(-1)^m 2^{2m} B_{2m}}{(2m)!} z^{2m}.$$

Equate the coefficients of z^{2m} and solve for B_{2m}:

$$\boxed{B_{2m} = (-1)^{m-1} \frac{2(2m)!}{(2\pi)^{2m}} \zeta(2m)}.$$

12.6 Asymptotic Series

12.6.1 Integrating by parts, develop asymptotic expansions as $x \to +\infty$ of the Fresnel integrals

$$\text{(a) } C(x) = \int_0^x \cos\frac{\pi u^2}{2}\, du, \quad \text{(b) } s(x) = \int_0^x \sin\frac{\pi u^2}{2}\, du.$$

These integrals appear in the analysis of a knife-edge diffraction pattern.

Solution.

Take $x \to +\infty$ and write $S(x) = s(x)$ for the sine integral in the question. Put $\theta = \pi x^2/2$. Rotating the Gaussian integration ray through $\pi/4$ gives

$$\int_0^{\infty} e^{i\pi u^2/2} du = e^{i\pi/4} \int_0^{\infty} e^{-\pi t^2/2} dt = \frac{1+i}{2}.$$

The rotation is justified by an arc estimate: on $u = Re^{i\phi}$, $|e^{i\pi u^2/2}| = e^{-\pi R^2 \sin(2\phi)/2}$; its integral over $0 \le \phi \le \pi/4$, including the small region next to $\phi = 0$, is bounded using $\sin 2\phi \ge 4\phi/\pi$:

$$R\int_0^{\pi/4} e^{-\pi R^2 \sin(2\phi)/2} d\phi \le R\int_0^{\infty} e^{-2R^2\phi} d\phi = \frac{1}{2R}.$$

Therefore

$$C(x) + iS(x) = \frac{1+i}{2} - T_0(x), \qquad T_m(x) = \int_x^{\infty} u^{-2m} e^{i\pi u^2/2} du.$$

Since $d(e^{i\pi u^2/2}) = i\pi u e^{i\pi u^2/2} du$, integration by parts gives

$$\begin{aligned} T_m &= \left[\frac{e^{i\pi u^2/2}}{i\pi u^{2m+1}}\right]_x^\infty + \frac{2m+1}{i\pi}\int_x^\infty u^{-2m-2} e^{i\pi u^2/2} du \\ &= \frac{ie^{i\theta}}{\pi x^{2m+1}} - \frac{i(2m+1)}{\pi} T_{m+1}. \end{aligned}$$

Repeated substitution therefore gives the finite identity

$$T_0 = \frac{ie^{i\theta}}{\pi x}\sum_{k=0}^{N-1} \frac{(-i)^k (2k-1)!!}{(\pi x^2)^k} + \frac{(-i)^N (2N-1)!!}{\pi^N} T_N,$$

where $(-1)!! = 1$. The same integration-by-parts identity, followed by an absolute bound on T_{N+1}, gives

$$|T_N| \le \frac{1}{\pi x^{2N+1}} + \frac{2N+1}{\pi}\int_x^\infty u^{-2N-2} du = \frac{2}{\pi x^{2N+1}}.$$

Thus the remainder is of the order of the next inverse power for every fixed N, proving asymptotic validity.

Separate the even and odd k contributions by defining

$$\begin{aligned} f(x) &\sim \frac{1}{\pi x}\left[1 - \frac{3}{\pi^2 x^4} + \frac{105}{\pi^4 x^8} - \cdots\right], \\ g(x) &\sim \frac{1}{\pi^2 x^3}\left[1 - \frac{15}{\pi^2 x^4} + \frac{945}{\pi^4 x^8} - \cdots\right]. \end{aligned}$$

Then $T_0 \sim e^{i\theta}(g + if)$, so the requested real expansions are

$$\boxed{C(x) \sim \frac{1}{2} + f(x)\sin\theta - g(x)\cos\theta}, \qquad \boxed{S(x) \sim \frac{1}{2} - f(x)\cos\theta - g(x)\sin\theta}.$$

For example $C = 1/2 + \sin\theta/(\pi x) - \cos\theta/(\pi^2 x^3) + O(x^{-5})$, with the corresponding sine-integral formula obtained from the second box. For large negative x, use the oddness of both integrals rather than the positive-infinity constants $1/2$.

12.6.2 Rederive the asymptotic expansions of $\mathrm{Ci}(x)$ and $\mathrm{si}(x)$ as $x \to +\infty$ by repeated integration by parts.

Hint. $\mathrm{Ci}(x) + i\,\mathrm{si}(x) = -\displaystyle\int_x^\infty \frac{e^{it}}{t}\,dt.$

Solution.

Use the definitions in the hint, so $\mathrm{si}(x) = \mathrm{Si}(x) - \pi/2$, and take $x \to +\infty$. Set

$$J_m(x) = \int_x^\infty \frac{e^{it}}{t^{m+1}} dt.$$

Integration by parts with $dv = e^{it}dt$ gives

$$J_m = \left[\frac{e^{it}}{it^{m+1}}\right]_x^\infty + \frac{m+1}{i} J_{m+1} = \frac{ie^{ix}}{x^{m+1}} - i(m+1)J_{m+1}.$$

Consequently

$$\mathrm{Ci}(x) + i\,\mathrm{si}(x) = -J_0 = -\frac{ie^{ix}}{x}\sum_{k=0}^{N-1}\frac{(-i)^k k!}{x^k} - (-i)^N N! J_N.$$

A further integration by parts bounds the remainder integral by

$$|J_N| \le x^{-N-1} + (N+1)\int_x^\infty t^{-N-2}dt = 2x^{-N-1}.$$

The residual after N displayed terms is therefore $O(x^{-N-1})$. Collect the real even/odd subsequences:

$$A(x) \sim 1 - \frac{2!}{x^2} + \frac{4!}{x^4} - \cdots, \qquad B(x) \sim 1 - \frac{3!}{x^2} + \frac{5!}{x^4} - \cdots.$$

Since $-J_0 \sim e^{ix}[-iA/x - B/x^2]$, multiplication by $\cos x + i\sin x$ yields

$$\boxed{\mathrm{Ci}(x) \sim \frac{\sin x}{x}A(x) - \frac{\cos x}{x^2}B(x)},$$

$$\boxed{\mathrm{si}(x) \sim -\frac{\cos x}{x}A(x) - \frac{\sin x}{x^2}B(x)}.$$

These are fixed-order asymptotic truncations; the factorial coefficients do not define convergent infinite series at a fixed finite x.

12.6.3 Derive the asymptotic expansion of the Gauss error function as $x \to +\infty$

$$\begin{aligned}\mathrm{erf}(x) &= \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}\,dt \\ &\approx 1 - \frac{e^{-x^2}}{\sqrt{\pi}\,x}\left(1 - \frac{1}{2x^2} + \frac{1\cdot 3}{2^2x^4} - \frac{1\cdot 3\cdot 5}{2^3x^6} + \cdots + (-1)^n\frac{(2n-1)!!}{2^n x^{2n}}\right).\end{aligned}$$

Hint. $\mathrm{erf}(x) = 1 - \mathrm{erfc}(x) = 1 - \dfrac{2}{\sqrt{\pi}}\displaystyle\int_x^\infty e^{-t^2}\,dt.$

Normalized so that $\mathrm{erf}(\infty) = 1$, this function plays an important role in probability theory. It may be expressed in terms of the Fresnel integrals (Exercise 12.6.1), the incomplete gamma functions (Section 13.6), or the confluent hypergeometric functions (Section 18.5).

Solution.

For $x > 0$, let

$$I_m(x) = \int_x^\infty t^{-2m}e^{-t^2}dt, \qquad \mathrm{erfc}(x) = \frac{2}{\sqrt{\pi}}I_0(x).$$

Use $d(e^{-t^2}) = -2te^{-t^2}dt$ in integration by parts:

$$I_m = \left[-\frac{e^{-t^2}}{2t^{2m+1}}\right]_x^\infty - \frac{2m+1}{2}\int_x^\infty t^{-2m-2}e^{-t^2}dt = \frac{e^{-x^2}}{2x^{2m+1}} - \frac{2m+1}{2}I_{m+1}.$$

After N repetitions,

$$I_0 = \frac{e^{-x^2}}{2x}\sum_{k=0}^{N-1}\frac{(-1)^k(2k-1)!!}{2^k x^{2k}} + \frac{(-1)^N(2N-1)!!}{2^N}I_N.$$

All I_m are positive, and the recurrence implies $0 < I_N < e^{-x^2}/(2x^{2N+1})$. The remainder in erfc has sign $(-1)^N$ and magnitude less than its first omitted term:

$$|R_N| < \frac{e^{-x^2}}{\sqrt{\pi}x}\frac{(2N-1)!!}{2^N x^{2N}}.$$

Finally erf $= 1 -$ erfc gives

$$\boxed{\operatorname{erf}(x) \sim 1 - \frac{e^{-x^2}}{\sqrt{\pi}x}\left[1 - \frac{1}{2x^2} + \frac{3}{4x^4} - \frac{15}{8x^6} + \cdots\right]}.$$

The term with index $k = n$ is $(-1)^n(2n-1)!!/(2^n x^{2n})$, as requested. The remainder bound proves the expansion for $x \to +\infty$. For $x \to -\infty$ use $\operatorname{erf}(-x) = -\operatorname{erf}(x)$.

12.6.4 The asymptotic expressions for the various Bessel functions, Section 14.6, contain the series

$$P_\nu(z) \sim 1 + \sum_{n=1}^{\infty}(-1)^n \frac{\prod_{s=1}^{2n}[4\nu^2 - (2s-1)^2]}{(2n)!(8z)^{2n}},$$
$$Q_\nu(z) \sim \sum_{n=1}^{\infty}(-1)^{n+1} \frac{\prod_{s=1}^{2n-1}[4\nu^2 - (2s-1)^2]}{(2n-1)!(8z)^{2n-1}}.$$

Show that these are asymptotic expansions for fixed ν as $z = x \to +\infty$; distinguish terminating half-integer orders from divergent cases. Then specify a complex sector of validity.

Solution.

Divergence at fixed z does not by itself prove that a series is asymptotic to a particular function. Also, for half-integer ν the products terminate, so the blanket divergence argument in the manual has exceptions. We first prove the asymptotic statement for fixed real ν and $z = x \to +\infty$, then give its complex extension on a specified sector. Put

$$\mu = \nu^2 - \frac{1}{4}, \qquad A_k = \frac{\prod_{s=1}^{k}[4\nu^2 - (2s-1)^2]}{k!8^k}, \quad A_0 = 1.$$

The outgoing Bessel solution in Section 14.6 is written

$$H_\nu^{(1)}(x) = \sqrt{\frac{2}{\pi x}}e^{i(x - \nu\pi/2 - \pi/4)}w(x), \qquad w = P_\nu + iQ_\nu.$$

Substituting $H = x^{-1/2}e^{ix}$ times a constant times w into $x^2H'' + xH' + (x^2 - \nu^2)H = 0$ gives

$$w'' + 2iw' - \frac{\mu}{x^2}w = 0, \qquad w(x) \to 1.$$

Only the leading outgoing normalization $w \to 1$ from Section 14.6 is used here; the higher coefficients and their remainder will be proved. For $w \sim \sum_{k\ge 0} c_k x^{-k}$, the coefficient of x^{-k-2} is

$$[k(k+1) - \mu]c_k - 2i(k+1)c_{k+1} = 0.$$

Hence

$$c_{k+1} = \frac{i[4\nu^2 - (2k+1)^2]}{8(k+1)}c_k, \qquad c_k = i^k A_k.$$

The real even powers and imaginary odd powers are exactly the two series in the question, including their signs.

Here is a remainder argument. For an integrable forcing f, define

$$(Kf)(x) = \frac{1}{2i}\int_x^\infty [e^{2i(t-x)} - 1]f(t)\,dt.$$

Differentiating under the integral (the boundary kernel is zero) gives

$$(Kf)' = -\int_x^\infty e^{2i(t-x)}f(t)\,dt, \qquad (Kf)'' = f(x) + 2i\int_x^\infty e^{2i(t-x)}f(t)\,dt.$$

Thus $(Kf)'' + 2i(Kf)' = f$, and $Kf \to 0$ at infinity. Apply this to $f = \mu w/x^2$, which is integrable because $w \to 1$. The difference $w - 1 - Kf$ solves $d'' + 2id' = 0$, so $d = A + Be^{-2ix}$. Since $d \to 0$ on the positive axis, $A = B = 0$. This proves the Volterra equation

$$w(x) = 1 + \frac{\mu}{2i}\int_x^\infty [e^{2i(t-x)} - 1]\frac{w(t)}{t^2}dt.$$

Its kernel has absolute value at most 1 after division by $2i$. On $x \geq R > |\mu|$, its operator T has sup-norm at most $|\mu|/R < 1$, so the bounded outgoing solution is uniquely obtained by successive substitution. Let $S_N = \sum_{k=0}^{N-1} c_k x^{-k}$. The recurrence cancels every lower power, leaving precisely

$$r_N := S_N'' + 2iS_N' - \mu S_N/x^2 = [N(N-1) - \mu]c_{N-1}x^{-N-1}.$$

Then $E_N = w - S_N$ satisfies

$$E_N = T[E_N] - \frac{1}{2i}\int_x^\infty [e^{2i(t-x)} - 1]r_N(t)dt.$$

The explicit last integral is $O(x^{-N})$. In the weighted norm $\|E\|_N = \sup_{x\geq R} x^N|E(x)|$,

$$x^N|T[E](x)| \leq |\mu|\|E\|_N x^N \int_x^\infty t^{-N-2}dt \leq \frac{|\mu|}{(N+1)R}\|E\|_N.$$

The same contraction argument constructs $E_N = O(x^{-N})$; uniqueness identifies $S_N + E_N$ with w. This is the required asymptotic remainder estimate for every fixed truncation. Taking real and imaginary parts proves the P_ν, Q_ν expansions; retaining one extra term when necessary gives the next even or odd inverse-power remainder.

For nonterminating cases the ratio of two consecutive P terms is

$$\left|\frac{[4\nu^2 - (4n+1)^2][4\nu^2 - (4n+3)^2]}{(2n+1)(2n+2)(8x)^2}\right| \sim \frac{n^2}{x^2},$$

and the Q ratio has the same growth. Thus they diverge for fixed x, while their finite truncations have the proved large-x meaning. If $\nu = \pm(j + 1/2)$, $j = 0, 1, \ldots$, a numerator factor vanishes at $s = j + 1$ and all later coefficients are zero; the resulting finite expansions remain valid, and in fact solve the transformed equation exactly. These terminating cases satisfy the usual fixed-order remainder definition. They are excluded only if one imposes the textbook's extra divergence convention, Eq. (12.98); see Exercise 12.6.5.

The proof also supplies a complex sector, rather than assuming that an expansion on the real axis automatically extends to all directions. For $\Re z \geq 0$, integrate along $t = z + u$, $u \geq 0$. Then

$$|e^{2iu} - 1| \leq 2, \qquad |z + u| \geq (|z| + u)/\sqrt{2}.$$

Thus the unweighted operator norm on $|z| \geq R$ is at most $2|\mu|/R$. In the weighted norm its bound is

$$\frac{|\mu|\, 2^{(N+2)/2}}{(N+1)R},$$

and the residual integral is bounded by $|[N(N-1)-\mu]c_{N-1}|\, 2^{(N+1)/2}|z|^{-N}/N$. Taking R sufficiently large gives the same contraction and $O(|z|^{-N})$ remainder, uniformly in this closed right half-plane. The incoming solution obeys $w_-'' - 2iw_-' - \mu w_-/z^2 = 0$; its kernel uses $e^{-2iu} - 1$, with the same bound, and its coefficients are $(-i)^k A_k$. Therefore $P = (w + w_-)/2$ selects the even terms and $Q = (w - w_-)/(2i)$ selects the odd terms. These analytic solutions agree with the Hankel normalizations on the positive axis and hence throughout the sector. This argument also works for each fixed complex ν, since only $|\mu|$ enters the bounds. Wider continuation sectors require the corresponding Hankel branches; no expansion across an unspecified branch or Stokes boundary is being asserted.

12.6.5 For $x > 1$,

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n \frac{1}{x^{n+1}}.$$

Test this series to see if it is an asymptotic series.

Solution.

The answer depends on the stated convention. With the textbook's partial sum through x^{-m}, $m \geq 1$, finite geometric summation gives

$$s_m(x) = \sum_{k=1}^{m} (-1)^{k-1} x^{-k} = \frac{1 - (-1/x)^m}{1+x}, \qquad R_m(x) = \frac{(-1)^m}{x^m(1+x)}.$$

Consequently

$$x^m R_m(x) = \frac{(-1)^m}{1+x}.$$

For each fixed m this tends to zero as $x \to +\infty$, satisfying Eq. (12.97). For each fixed $x > 1$, however, its absolute value is the constant $1/(1+x)$; it does not diverge as $m \to \infty$. Thus the extra requirement in Eq. (12.98) is not met:

not an asymptotic series under the textbook's convention.

This agrees with the manual. Its convergence follows independently from the geometric ratio $|-1/x| < 1$.

Terminology qualification: the usual Poincaré definition uses the fixed-order condition (12.97) without the additional divergence requirement (12.98). The book itself acknowledges this alternative in footnote 5 on p.583. Under that definition the same series is both convergent and asymptotic; the manual's answer is correct under the book's explicitly stricter convention. In that usual notation,

$$\frac{1}{1+x} \sim \frac{1}{x} - \frac{1}{x^2} + \frac{1}{x^3} - \cdots, \qquad x \to +\infty.$$

The exact remainder above proves the claim under either convention without inferring asymptotic behavior merely from convergence or divergence.

12.6.6 Derive the following Bernoulli-number asymptotic series for the Euler-Mascheroni constant, defined in Eq. (1.13):

$$\gamma \sim \sum_{s=1}^{n} s^{-1} - \ln n - \frac{1}{2n} + \sum_{k=1}^{\infty} \frac{B_{2k}}{(2k)n^{2k}}.$$

Here n plays the role of x.

Hint. Apply the Euler-Maclaurin integration formula to $f(x) = x^{-1}$ over the interval $[1, n]$ for $N = 1, 2, \ldots$.

Solution.

Write $H_n = \sum_{s=1}^{n} 1/s$. Applying Euler–Maclaurin to the finite tail $\sum_{s=n+1}^{M} 1/s$, with $f^{(2k-1)}(x) = -(2k-1)!x^{-2k}$, gives

$$H_M - H_n = \ln(M/n) + \frac{1}{2M} - \frac{1}{2n} + \sum_{k=1}^{N} \frac{B_{2k}}{2k}(n^{-2k} - M^{-2k}) + \text{remainder}.$$

Let $M \to \infty$ and use the definition of γ. This produces

$$\gamma = H_n - \ln n - \frac{1}{2n} + \sum_{k=1}^{N} \frac{B_{2k}}{2k\, n^{2k}} + R_{n,N}.$$

To prove the asymptotic nature, rather than merely write formal corrections, we now derive a signed bound for $R_{n,N}$. Define

$$K(n) = \int_0^{\infty} e^{-nt} \left[\frac{1}{t} - \frac{1}{e^t - 1} \right] dt.$$

This integral is finite: the bracket tends to $1/2$ at zero and is $O(1/t)$ at infinity. Its consecutive difference is

$$\begin{aligned} K(n) - K(n+1) &= \int_0^{\infty} \frac{e^{-nt} - e^{-(n+1)t}}{t} dt - \int_0^{\infty} e^{-(n+1)t} dt \\ &= \ln(1 + 1/n) - \frac{1}{n+1}. \end{aligned}$$

The first equality used $(1 - e^{-t})/(e^t - 1) = e^{-t}$; the logarithmic integral follows by integrating $\int_n^{n+1} e^{-at} da$ first in t. This is also the consecutive difference of $H_n - \ln n - \gamma$. The bracket in $K(n)$ is bounded, say by C, so $|K(n)| \le C \int_0^{\infty} e^{-nt} dt = C/n \to 0$. Both sequences therefore tend to zero as $n \to \infty$, so they are equal. Thus, with $q(t) = 1/(e^t - 1) - 1/t + 1/2$,

$$\gamma = H_n - \ln n - \frac{1}{2n} + \int_0^{\infty} e^{-nt} q(t) dt.$$

The cotangent partial fractions, evaluated at $it/2$, give

$$q(t) = \frac{1}{2} \coth(t/2) - \frac{1}{t} = 2t \sum_{j=1}^{\infty} \frac{1}{(2\pi j)^2 + t^2}.$$

Expand each denominator by a *finite* geometric identity through t^{2N-2}. With $a_j = 2\pi j$, this yields

$$q(t) = \sum_{k=1}^{N} \frac{B_{2k}}{(2k)!} t^{2k-1} + (-1)^N 2t^{2N+1} \sum_{j=1}^{\infty} \frac{1}{a_j^{2N}(a_j^2 + t^2)}.$$

The coefficients follow from the zeta/Bernoulli identity, and the last sum is positive. Since $\int_0^\infty e^{-nt}t^m dt = m!/n^{m+1}$, integration gives the stated correction terms and the rigorous bound

$$\boxed{0 < (-1)^N R_{n,N} < \frac{|B_{2N+2}|}{(2N+2)n^{2N+2}}}.$$

Indeed, replace $(a_j^2+t^2)^{-1}$ by a_j^{-2} in the positive remainder and use $2(2N+1)!\zeta(2N+2)/(2\pi)^{2N+2} = |B_{2N+2}|/(2N+2)$. For each fixed N this remainder is smaller by two powers of n than the last correction, proving the requested asymptotic series. The coefficients grow factorially, so taking infinitely many terms at fixed n is not justified.

12.6.7 Develop an asymptotic series for

$$\int_0^\infty \frac{e^{-xv}}{1+v^2}\,dv.$$

Take x to be real and positive.

$$\textit{ANS.}\ \frac{1}{x} - \frac{2!}{x^3} + \frac{4!}{x^5} - \cdots + \frac{(-1)^n(2n)!}{x^{2n+1}}.$$

Solution.

The published errata (Instructor's Manual, 6 April 2012, textbook p. 585) replace the original denominator $(1+v^2)^2$ by $1+v^2$. The corrected integral in the question now matches its printed answer. We give both versions below to make the change explicit. For the corrected integral $I(x)$, use the exact finite identity

$$\frac{1}{1+v^2} = \sum_{k=0}^{N}(-1)^k v^{2k} + \frac{(-1)^{N+1}v^{2N+2}}{1+v^2}.$$

Multiply by e^{-xv} and integrate for $x>0$. With $u=xv$,

$$\int_0^\infty e^{-xv}v^{2k}dv = x^{-2k-1}\int_0^\infty e^{-u}u^{2k}du = \frac{(2k)!}{x^{2k+1}},$$

where repeated integration by parts evaluates the last integral. Consequently

$$\boxed{I(x) \sim \sum_{k=0}^{\infty}\frac{(-1)^k(2k)!}{x^{2k+1}} = \frac{1}{x} - \frac{2!}{x^3} + \frac{4!}{x^5} - \cdots}.$$

The actual remainder after $k=N$ is

$$R_N = (-1)^{N+1}\int_0^\infty \frac{e^{-xv}v^{2N+2}}{1+v^2}dv, \qquad 0 < (-1)^{N+1}R_N < \frac{(2N+2)!}{x^{2N+3}}.$$

This proves the asymptotic statement without interchanging an infinite geometric series outside its disk of convergence.

For comparison, the *original* squared-denominator integral has

$$\int_0^\infty \frac{e^{-xv}}{(1+v^2)^2}dv \sim \boxed{\sum_{k=0}^{\infty}\frac{(-1)^k(k+1)(2k)!}{x^{2k+1}}} = \frac{1}{x} - \frac{4}{x^3} + \frac{72}{x^5} - \cdots.$$

Its coefficients follow from $(1+w)^{-2} = \sum(-1)^k(k+1)w^k$. More precisely, the finite remainder is $(-1)^{N+1}w^{N+1}[(N+2)+(N+1)w]/(1+w)^2$. For $w=v^2\ge 0$ its magnitude is at most $(N+2)v^{2N+2}$, so integration bounds the original integral's remainder by $(N+2)(2N+2)!/x^{2N+3}$. The factor $k+1$ shows exactly why the original problem and printed answer were inconsistent.

12.7 Method of Steepest Descents

We present here a rather small number of exercises on the method of steepest descents. Several additional exercises appear elsewhere in this book, in particular in Section 14.6, where the technique is applied to the contour integral representations of Bessel functions.

12.7.1 Let F be nonconstant and analytic on a connected open region. Derive the correct circle mean of $|F|^2$ about z_0 from the Taylor coefficients of F. Prove that $|F|$ has no interior local maximum and no positive interior local minimum. In particular, if F never vanishes, it has no interior modulus extremum.

Solution.

Both the asserted mean-value identity for $|F|^2$ and the unqualified absence of minima are false. For $F(z) = z$ at $z_0 = 0$, the circle average is r^2, whereas $|F(0)|^2 = 0$ is a minimum. The nonconstant, nonzero case used in the preceding text needs a corrected argument. Expand in a disk about z_0 lying in the analytic region:

$$F(z_0 + re^{i\theta}) = \sum_{n=0}^{\infty} a_n r^n e^{in\theta}.$$

Uniform convergence on a smaller circle permits multiplication and integration term by term. Since $(2\pi)^{-1} \int_0^{2\pi} e^{i(n-m)\theta} d\theta = \delta_{nm}$,

$$\begin{aligned} \frac{1}{2\pi} \int_0^{2\pi} |F(z_0 + re^{i\theta})|^2 d\theta &= \sum_{n,m \geq 0} a_n \overline{a_m} r^{n+m} \delta_{nm} \\ &= \boxed{|F(z_0)|^2 + \sum_{n=1}^{\infty} |a_n|^2 r^{2n}}. \end{aligned}$$

At least one a_n, $n \geq 1$, is nonzero: otherwise the Taylor series would make F constant near z_0, and the identity theorem would make it constant throughout the connected region. Thus the mean strictly exceeds the center value on every sufficiently small circle. A local maximum would make every value on that circle no larger than the center, a contradiction. Therefore $|F|$ has no interior local maximum unless F is constant.

If $F(z_0) \neq 0$, continuity gives a smaller disk where F never vanishes. Apply the same maximum argument to the analytic function $1/F$. A local minimum of $|F|$ would be a local maximum of $|1/F|$, again impossible for nonconstant F. Thus a nonconstant analytic function has no positive interior local minimum; zeros are allowed minima. A constant function is the usual exception to both statements.

For comparison, the true mean-value identities are $\langle F \rangle = F(z_0)$, and, on a zero-free disk, $\langle \log |F| \rangle = \log |F(z_0)|$. The latter follows by taking the real part of the mean-value formula for a local analytic logarithm. It directly proves the no-extremum assertion in the zero-free setting relevant to the saddle-point discussion.

12.7.2 Find the steepest path and leading asymptotic expansion as $s \to +\infty$ for the Fresnel integrals $\displaystyle\int_0^s \cos x^2 \, dx$, $\displaystyle\int_0^s \sin x^2 \, dx$.

Hint. Use $\displaystyle\int_0^1 e^{itz^2} \, dz$.

Solution.

Put $t = s^2 > 0$ and scale $x = sz$:

$$\int_0^s e^{ix^2}dx = sJ(t), \qquad J(t) = \int_0^1 e^{itz^2}dz.$$

For $f(z) = iz^2$, the saddle is $z = 0$, with $f''(0) = 2i$. Writing $z = re^{i\theta}$ gives $\Re(itz^2) = -tr^2\sin 2\theta$ and $\Im(itz^2) = tr^2\cos 2\theta$. The descending ray from zero is therefore $\theta = \pi/4$; on it the exponential becomes e^{-tr^2}. The endpoint $z = 1$ has a descending constant-phase path

$$z = \sqrt{1+iv}, \quad 0 \le v < \infty, \qquad itz^2 = it - tv, \quad dz = \frac{i\,dv}{2\sqrt{1+iv}}.$$

Both paths approach the same direction at infinity. Deforming the finite interval to their difference is legitimate because the integrand is entire and the joining segment at infinity has exponentially vanishing integral. Hence the exact contour decomposition is

$$J(t) = e^{i\pi/4}\int_0^\infty e^{-tr^2}dr - \frac{i}{2}e^{it}\int_0^\infty \frac{e^{-tv}}{\sqrt{1+iv}}dv.$$

The first integral is the endpoint half-Gaussian $\sqrt{\pi}/(2\sqrt{t})$. For the second, $(1+iv)^{-1/2} = 1 - iv/2 - 3v^2/8 + O(v^3)$ near zero. Termwise integration of any finite Taylor truncation gives

$$J(t) = \frac{e^{i\pi/4}\sqrt{\pi}}{2\sqrt{t}} - \frac{ie^{it}}{2t} - \frac{e^{it}}{4t^2} + O(t^{-3}).$$

To justify the error, split the Laplace integral at a fixed small $v = \delta$: its Taylor remainder contributes $O(\int_0^\infty v^2 e^{-tv}dv) = O(t^{-3})$ at the order displayed, and its remote tail is exponentially small. Multiplying by s and taking real and imaginary parts yields

$$\boxed{\int_0^s \cos x^2 dx} \boxed{= \sqrt{\frac{\pi}{8}} + \frac{\sin s^2}{2s} - \frac{\cos s^2}{4s^3} + O(s^{-5}),}$$

$$\boxed{\int_0^s \sin x^2 dx} \boxed{= \sqrt{\frac{\pi}{8}} - \frac{\cos s^2}{2s} - \frac{\sin s^2}{4s^3} + O(s^{-5}).}$$

The saddle is an endpoint, so it contributes half of a full Gaussian integral. The manual's limiting value is twice the correct $\sqrt{\pi/8}$.

12.7.3 Show that the formula

$$\Gamma(1+s) \approx \sqrt{2\pi s}\, s^s e^{-s}$$

holds for complex values of s (with $\Re(s)$ large and positive).

Hint. This involves assigning a phase to s and then demanding that $\Im[sf(z)]$ be constant in the vicinity of the saddle point.

Solution.

Use the principal $\operatorname{Log} s$ for $\Re s > 0$. Starting with $\Gamma(1+s) = \int_0^\infty e^{-u}u^s du$, set $u = sz$. The resulting ray can be rotated to the positive z axis: the rotation stays inside $|\arg u| < \pi/2$, the exponential decays at infinity, and u^s is integrable at zero since $\Re s > 0$. Along the deformed rays $\operatorname{Log} u = \operatorname{Log} s + \operatorname{Log} z$, so $u^s = s^s z^s$ with consistent branches. Thus

$$\Gamma(1+s) = s^{s+1}\int_0^\infty e^{sf(z)}dz, \qquad f(z) = \operatorname{Log} z - z.$$

The saddle is $z_0 = 1$, because $f' = 1/z - 1$; there $f(1) = -1$ and $f''(1) = -1$. Write $s = \rho e^{i\phi}$. Near the saddle,

$$sf(z) = -s - \frac{s}{2}(z-1)^2 + \frac{s}{3}(z-1)^3 + \cdots .$$

To make the quadratic decrement real and negative, choose

$$z - 1 = e^{-i\phi/2}\frac{v}{\sqrt{\rho}} = \frac{v}{\sqrt{s}}, \qquad v \in \mathbb{R}.$$

This is the local steepest path and keeps $\Im[sf(z)]$ constant to quadratic order. Since $dz = dv/\sqrt{s}$, its leading Gaussian contribution is

$$\int e^{sf(z)}dz \sim \frac{e^{-s}}{\sqrt{s}}\int_{-\infty}^{\infty} e^{-v^2/2}dv = \frac{\sqrt{2\pi}e^{-s}}{\sqrt{s}}.$$

Therefore, with the same branch for all powers,

$$\boxed{\Gamma(1+s) \sim \sqrt{2\pi s}\, s^s e^{-s}}.$$

The factor $e^{-i\phi/2}$ from the path is essential; replacing the Gaussian width by $1/\sqrt{|s|}$ would lose that phase.

For completeness, the approximation remains valid throughout the stated limit $\Re s \to +\infty$, even if the phase changes with s. A useful uniform remainder follows from Euler–Maclaurin. The Euler limit for $\Gamma(s)$ gives

$$\log\Gamma(s) = \lim_{M\to\infty}\left[\log M! + s\log M - \sum_{k=0}^{M}\operatorname{Log}(s+k)\right].$$

Apply Euler–Maclaurin to $h(t) = \operatorname{Log}(s+t)$:

$$\sum_{k=0}^{M} h(k) = \int_0^M h(t)dt + \frac{h(0)+h(M)}{2} + \frac{h'(M)-h'(0)}{12} - \frac{1}{2}\int_0^M \widetilde{B}_2(t)h''(t)dt,$$

where $\widetilde{B}_2(t) = B_2(\{t\})$ is periodic. Using $h' = 1/(s+t)$, $h'' = -1/(s+t)^2$, and $\int h = (s+t)\operatorname{Log}(s+t) - (s+t)$, the finite sum is

$$\begin{aligned}\sum_{k=0}^{M}\operatorname{Log}(s+k) = {}& (s+M+\tfrac{1}{2})\operatorname{Log}(s+M) - M \\ & - (s-\tfrac{1}{2})\operatorname{Log} s + \frac{1}{12(s+M)} - \frac{1}{12s} \\ & + \frac{1}{2}\int_0^M \frac{\widetilde{B}_2(t)}{(s+t)^2}dt.\end{aligned}$$

The real Gaussian calculation gives $\log M! = (M+\frac{1}{2})\log M - M + \frac{1}{2}\log(2\pi) + o(1)$. After inserting both formulas in the Euler limit, the remaining M-dependent logarithms are

$$(M+s+\tfrac{1}{2})\big[\log M - \operatorname{Log}(M+s)\big] = -(M+s+\tfrac{1}{2})\operatorname{Log}(1+s/M) \longrightarrow -s$$

for each fixed s with $\Re s > 0$; also $1/(M+s) \to 0$. Adding $\operatorname{Log} s$ to pass from $\Gamma(s)$ to $\Gamma(1+s)$ therefore yields

$$\log\Gamma(1+s) = (s+\tfrac{1}{2})\operatorname{Log} s - s + \tfrac{1}{2}\ln 2\pi + R(s), \qquad R(s) = \frac{1}{12s} - \frac{1}{2}\int_0^\infty \frac{\widetilde{B}_2(t)}{(s+t)^2}dt.$$

Since $|\widetilde{B}_2| \le 1/6$ and $|s+t| \ge \Re s + t$,

$$|R(s)| \le \frac{1}{12|s|} + \frac{1}{12\Re s} \le \frac{1}{6\Re s} \longrightarrow 0.$$

Exponentiation proves that the ratio to the boxed expression tends to one for the entire requested right-half-plane limit.

12.8 Dispersion Relations

12.8.1 Assume that the function $f(z)$ satisfies the conditions for the dispersion relations. In addition, assume that $f(z) = f^*(z^*)$, i.e., that it meets the conditions of the Schwarz reflection principle, Eq. (11.127). Show that $f(z)$ is identically zero.

Solution.

The dispersion assumptions are analyticity in the upper half-plane and across the real axis, with sufficient decay for the large arc to vanish. On the real axis the additional reflection condition gives $f(x) = f(x)^*$, so $v(x) = \Im f(x) = 0$. The first Hilbert-transform equation is

$$u(x_0) = \frac{1}{\pi}\,\mathrm{PV}\int_{-\infty}^{\infty} \frac{v(x)}{x - x_0}dx.$$

Its integrand vanishes identically, hence $u(x_0) = 0$ for every real x_0 too. Thus $f = 0$ on the real axis. Because that axis lies in a region through which f is analytic, the identity theorem implies

$$\boxed{f(z) \equiv 0}$$

on the connected analytic upper-half-plane domain. Equivalently, Schwarz reflection would extend f to an entire function with decay in both half-planes, and Liouville's theorem gives the same result.

12.8.2 For $f(z)$ such that we may replace the closed contour of the Cauchy integral formula by an integral over the real axis we have

$$f(x_0) = \frac{1}{2\pi i}\left\{\int_{-\infty}^{x_0-\delta} \frac{f(x)}{x - x_0}\,dx + \int_{x_0+\delta}^{\infty} \frac{f(x)}{x - x_0}\,dx\right\} + \frac{1}{2\pi i}\int_C \frac{f(x)}{x - x_0}\,dx.$$

Here we take C to be a small semicircle about x_0 in the lower half-plane. Show that as $\delta \downarrow 0$ the formula for $f(x_0)$ reduces to

$$f(x_0) = \frac{1}{\pi i}\mathrm{PV}\int_{-\infty}^{\infty} \frac{f(x)}{x - x_0}\,dx,$$

which is Eq. (12.117).

Solution.

Parameterize the lower semicircle from its left endpoint to its right endpoint as

$$z = x_0 + \delta e^{i\theta}, \quad -\pi \le \theta \le 0, \qquad dz = i\delta e^{i\theta}d\theta.$$

Its orientation is counterclockwise around the lower half of the small circle, so

$$\int_C \frac{f(z)}{z-x_0}dz = i\int_{-\pi}^{0} f(x_0+\delta e^{i\theta})d\theta \longrightarrow i\pi f(x_0).$$

The two real integrals together tend to their Cauchy principal value. Taking $\delta \to 0$ in the formula in the question therefore gives

$$f(x_0) = \frac{1}{2\pi i}\,\mathrm{PV}\int_{-\infty}^{\infty}\frac{f(x)}{x-x_0}dx + \frac{1}{2}f(x_0).$$

Subtract the half contribution and multiply by two:

$$\boxed{f(x_0) = \frac{1}{\pi i}\,\mathrm{PV}\int_{-\infty}^{\infty}\frac{f(x)}{x-x_0}dx}.$$

The real-axis integral in the requested formula must be read as a Cauchy principal value; an ordinary improper integral at the pole need not exist. The source uses a cut integral sign for this convention.

12.8.3

(a) The function $f(z) = e^{iz}$ does not vanish at the endpoints of the range of $\arg z$, 0 and π. Show, with the help of Jordan's lemma, Eq. (11.102), that Eq. (12.116) still holds.

(b) For $f(z) = e^{iz}$ verify by direct integration the dispersion relations, Eq. (12.117) or Eqs. (12.118).

Solution.

The printed lower endpoint `a` is 0: the contour is the upper semicircle $0 \le \arg z \le \pi$. (a) For z_0 fixed in the upper half-plane, put $z = Re^{i\theta}$ on the large semicircle, $0 \le \theta \le \pi$. Then

$$\left|\int_{\mathrm{arc}} \frac{e^{iz}}{z-z_0}dz\right| \le \frac{R}{R-|z_0|}\int_0^{\pi} e^{-R\sin\theta}d\theta.$$

On $0 \le \theta \le \pi/2$, $\sin\theta \ge 2\theta/\pi$; symmetry therefore bounds the last integral by $2\int_0^{\pi/2} e^{-2R\theta/\pi}d\theta \le \pi/R$. The arc contribution is at most $\pi/(R-|z_0|) \to 0$, despite the nonvanishing endpoint values. Cauchy's formula consequently gives

$$\boxed{e^{iz_0} = \frac{1}{2\pi i}\int_{-\infty}^{\infty}\frac{e^{ix}}{x-z_0}dx}, \qquad \Im z_0 > 0.$$

(b) Set $t = x - x_0$. The symmetric principal value is

$$\mathrm{PV}\int_{-\infty}^{\infty}\frac{e^{ix}}{x-x_0}dx = e^{ix_0}\left[\mathrm{PV}\int_{-\infty}^{\infty}\frac{\cos t}{t}dt + i\int_{-\infty}^{\infty}\frac{\sin t}{t}dt\right].$$

The cosine integral is zero by oddness. To evaluate the sine integral, first insert $e^{-\epsilon t}$ on $t > 0$. If $A(b) = \int_0^{\infty} e^{-\epsilon t}\sin(bt)dt/t$, then $A'(b) = \epsilon/(\epsilon^2+b^2)$ and $A(0) = 0$, so $A(1) = \arctan(1/\epsilon) \to \pi/2$. The damping can be removed uniformly in the tail: integration by parts against a bounded primitive of $\sin t$ gives $|\int_A^{\infty} e^{-\epsilon t}\sin t\,dt/t| \le 2/A$ for every $\epsilon \ge 0$. On $[0, A]$, dominated convergence applies. Thus the whole-line sine integral is π, and

$$\mathrm{PV}\int_{-\infty}^{\infty}\frac{e^{ix}}{x-x_0}dx = i\pi e^{ix_0}.$$

Dividing by $i\pi$ verifies Eq. (12.117). Separating components also gives

$$\frac{1}{\pi}\,\mathrm{PV}\int\frac{\sin x}{x-x_0}dx=\cos x_0,\qquad -\frac{1}{\pi}\,\mathrm{PV}\int\frac{\cos x}{x-x_0}dx=\sin x_0,$$

with both integrals over the whole real line, verifying both equations in (12.118).

12.8.4 With $f(x)=u(x)+iv(x)$ and $f(x)=f^*(-x)$, assume the dispersion relations. For (a), require $v\in C^1$ and $v^{(j)}(x)=O(x^{-2-\epsilon-j})$; for (b), require $u\in C^1$ and $u^{(j)}(x)=O(x^{-1-\epsilon-j})$, $j=0,1$, with $0<\epsilon<1$. Show that as $x_0\to+\infty$,

(a) $u(x_0)=-\dfrac{2}{\pi x_0^2}\displaystyle\int_0^\infty xv(x)\,dx+o(x_0^{-2})$,

(b) $v(x_0)=\dfrac{2}{\pi x_0}\displaystyle\int_0^\infty u(x)\,dx+o(x_0^{-1})$.

In quantum mechanics relations of this form are often called *sum rules*.

Solution.

The crossing condition implies $u(-x)=u(x)$ and $v(-x)=-v(x)$. Split the Hilbert integrals into positive and negative halves, substitute $x\mapsto -x$ on the latter, and combine denominators. With $a=x_0>0$,

$$u(a)=\frac{2}{\pi}\,\mathrm{PV}\int_0^\infty\frac{xv(x)}{x^2-a^2}dx,\qquad v(a)=-\frac{2a}{\pi}\,\mathrm{PV}\int_0^\infty\frac{u(x)}{x^2-a^2}dx.$$

The displayed moments and the passage to large a require additional decay/tail control. Crossing and the ordinary dispersion assumptions alone do not suffice. Also, a zero leading moment gives a little-o conclusion, not an equivalence to zero. For example $f(z)=1/(1-iz)$ is analytic above the real axis, decays, and satisfies crossing, but $v(x)=x/(1+x^2)$ makes $\int_0^\infty xv(x)dx$ divergent.

Here are sufficient assumptions and a justified expansion. For (a), assume v, v' are continuous and, for some $0<\epsilon<1$, $v^{(j)}(x)=O(x^{-2-\epsilon-j})$, $j=0,1$. For (b), assume instead $u^{(j)}(x)=O(x^{-1-\epsilon-j})$, $j=0,1$. Each condition is imposed on the function in its corresponding integral. The exact kernel decompositions give

$$u(a)=-\frac{2}{\pi a^2}\int_0^\infty xv(x)dx+\frac{2}{\pi a^2}\,\mathrm{PV}\int_0^\infty\frac{x^3v(x)}{x^2-a^2}dx,$$
$$v(a)=\frac{2}{\pi a}\int_0^\infty u(x)dx-\frac{2}{\pi a}\,\mathrm{PV}\int_0^\infty\frac{x^2u(x)}{x^2-a^2}dx.$$

Both final integrals tend to zero. To see the control near the moving pole, put $h=x^3v$ in (a) or $h=x^2u$ in (b), and split the range at $a/2, 3a/2$. On the middle interval write $g(x)=h(x)/(x+a)$, so the principal value is

$$\mathrm{PV}\int_{a/2}^{3a/2}\frac{g(x)}{x-a}dx=\int_{a/2}^{3a/2}\frac{g(x)-g(a)}{x-a}dx.$$

The constant term cancels by symmetry; $g'=O(a^{-1-\epsilon})$ bounds this integral by $O(a^{-\epsilon})$. On the lower interval, $|x^2-a^2|\geq 3a^2/4$ and $h(x)=O(x^{1-\epsilon})$ at large x, so its contribution is $O(a^{-\epsilon})+$

$O(a^{-2})$. On the upper interval the denominator is bounded below by a constant times x^2, giving $O(\int_{3a/2}^{\infty} x^{-1-\epsilon}dx) = O(a^{-\epsilon})$. It follows that

$$\boxed{u(a) = -\frac{2}{\pi a^2}\int_0^{\infty} xv(x)dx + o(a^{-2})},$$

$$\boxed{v(a) = \frac{2}{\pi a}\int_0^{\infty} u(x)dx + o(a^{-1})}.$$

When the respective moment is nonzero, these imply exactly the two originally printed $\sim$ relations. When it vanishes, a further moment is needed to find the first nonzero term.

12.8.5

(a) Given the integral equation (valid for all real x_0)

$$\frac{1}{1+x_0^2} = \frac{1}{\pi}\text{PV}\int_{-\infty}^{\infty}\frac{u(x)}{x-x_0}\,dx,$$

use Hilbert transforms to determine $u(x_0)$.

(b) Verify that the $u(x_0)$ found as your answer to part (a) actually satisfies the integral equation.

(c) From $f(z)|_{y=0} = u(x) + iv(x)$, replace x by z and determine $f(z)$. Verify that the conditions for the Hilbert transforms are satisfied.

(d) Are the crossing conditions satisfied?

ANS. (a) $u(x_0) = \dfrac{x_0}{1+x_0^2}$, (c) $f(z) = (z+i)^{-1}$.

Solution.

The principal-value symbol restores the cut integral sign in the source; it is necessary at $x = x_0$. All integrals below have that meaning. Use the convention

$$(Hg)(a) = \frac{1}{\pi}\,\text{PV}\int_{-\infty}^{\infty}\frac{g(x)}{x-a}dx, \qquad u = Hv, \quad v = -Hu.$$

(a) The given equation is $Hu = 1/(1+a^2)$, so $v(a) = -1/(1+a^2)$ and $u = Hv$. To evaluate the latter transform, use

$$\frac{1}{(1+x^2)(x-a)} = \frac{1}{1+a^2}\left[\frac{1}{x-a} - \frac{x+a}{1+x^2}\right].$$

The principal value of $1/(x-a)$ on the whole real line is zero; $x/(1+x^2)$ contributes zero by symmetric odd integration; and $\int dx/(1+x^2) = \pi$. Therefore

$$u(a) = -\frac{1}{\pi}\,\text{PV}\int\frac{dx}{(1+x^2)(x-a)} = \boxed{\frac{a}{1+a^2}}.$$

Within the decaying Hilbert-transform class, inversion $H^2 = -1$ fixes this solution uniquely.

(b) Check the original transform directly with another decomposition:

$$\frac{x}{(1+x^2)(x-a)} = \frac{1}{1+a^2}\left[\frac{a}{x-a} + \frac{1-ax}{1+x^2}\right].$$

Its integral is $\pi/(1+a^2)$ by the same three evaluations. Division by π reproduces the given equation, including its positive sign.

(c) On the real axis,

$$f(x) = u + iv = \frac{x-i}{x^2+1} = \frac{1}{x+i}.$$

Thus $\boxed{f(z) = 1/(z+i)}$. Its only pole is at $-i$, below the real axis, so it is analytic in and above the real axis. Also $f(z) = O(1/z)$ there. For the Cauchy kernel $f(z)/(z-z_0)$ the large arc is $O(1/R)$, verifying the needed dispersion condition.

(d) The standard crossing conditions are not satisfied: u is odd, whereas v is even. In fact

$$f(-x) = \frac{1}{-x+i} = -\frac{1}{x-i} = -f(x)^*,$$

with a minus sign instead of the requested $f(-x) = f(x)^*$.

12.8.6 Assume the full-frequency dispersion hypotheses for $f(\omega) = n^2(\omega) - 1$, including analyticity above the real axis and high-frequency decay. In (a), constancy means on the entire real frequency axis.

(a) If the real part of the complex index of refraction (squared) is constant (no optical dispersion), show that the imaginary part is zero (no absorption).

(b) Conversely, if there is absorption, show that there must be dispersion. In other words, if the imaginary part of $n^2 - 1$ is not zero, show that the real part of $n^2 - 1$ is not constant.

Solution.

The conclusion uses the dispersion hypotheses on $f(\omega) = n^2(\omega) - 1$, including its decay at high frequency, and constancy over the entire frequency axis. Constancy over only a finite frequency interval is not enough for this argument. Write $f = u + iv$. (a) If the real part of n^2 is constant, then $u = \Re(n^2 - 1) = K$ is constant. Since $f(\omega) \to 0$ at high frequency, $K = 0$. The second Hilbert-transform equation then gives

$$v(\omega_0) = -\frac{1}{\pi}\,\mathrm{PV}\int_{-\infty}^{\infty} \frac{u(\omega)}{\omega - \omega_0} d\omega = 0.$$

Thus $\boxed{\Im n^2 = 0}$, meaning no absorption under the stated model. The constant real value of n^2 is in fact one because of the high-frequency normalization.

(b) This is the contrapositive of (a). If v is not identically zero and u were constant, the preceding argument would force $v = 0$, a contradiction. Hence

$$\boxed{\Im(n^2 - 1) \not\equiv 0 \quad \Longrightarrow \quad \Re(n^2 - 1) \text{ is not constant}}.$$

This asserts the existence of dispersion somewhere in the full spectrum; it does not demand that its size be appreciable at every frequency.

12.8.7 Given $u(x) = x/(x^2+1)$ and $v(x) = -1/(x^2+1)$, show by direct evaluation of each integral that

$$\int_{-\infty}^{\infty} |u(x)|^2\,dx = \int_{-\infty}^{\infty} |v(x)|^2\,dx.$$

ANS. $\displaystyle\int_{-\infty}^{\infty} |u(x)|^2\,dx = \int_{-\infty}^{\infty} |v(x)|^2\,dx = \frac{\pi}{2}.$

Solution.

Use $x = \tan\theta$, $dx = \sec^2\theta\,d\theta$, with $-\pi/2 < \theta < \pi/2$. For the real functions in the question,

$$\int_{-\infty}^{\infty} |u|^2 dx = \int_{-\pi/2}^{\pi/2} \frac{\tan^2\theta}{\sec^4\theta} \sec^2\theta\,d\theta = \int_{-\pi/2}^{\pi/2} \sin^2\theta\,d\theta,$$

$$\int_{-\infty}^{\infty} |v|^2 dx = \int_{-\pi/2}^{\pi/2} \frac{1}{\sec^4\theta} \sec^2\theta\,d\theta = \int_{-\pi/2}^{\pi/2} \cos^2\theta\,d\theta.$$

With $\sin^2\theta = (1-\cos 2\theta)/2$ and $\cos^2\theta = (1+\cos 2\theta)/2$, the oscillatory endpoint term $\sin 2\theta/4$ vanishes at both limits. Each integral is therefore

$$\boxed{\pi/2}.$$

This directly checks the Parseval equality without assuming it.

12.8.8 Take $u(x) = \delta(x)$, a delta function, and assume that the Hilbert transform equations hold. Interpret both displayed kernel integrals as distributions: replace each reciprocal $1/q$ by $q/(q^2+\epsilon^2)$, use the same $\epsilon > 0$ for both factors, integrate, and then take $\epsilon \downarrow 0$ against smooth compactly supported test functions.

(a) Show that

$$\delta(w) = \frac{1}{\pi^2}\int_{-\infty}^{\infty} \frac{dy}{y(y-w)}.$$

(b) With changes of variables $w = s - t$ and $x = s - y$, transform the δ representation of part (a) into

$$\delta(s-t) = \frac{1}{\pi^2}\int_{-\infty}^{\infty} \frac{dx}{(x-s)(x-t)}.$$

Note. The δ function is discussed in Section 1.11.

Solution.

These formulas are distributional kernel identities. They are not ordinary improper integrals, nor does the pointwise principal value at coincident poles exist. A consistent regularization is essential. With $u(x) = \delta(x)$, the second Hilbert relation gives

$$v(y) = -\frac{1}{\pi}\,\mathrm{PV}\int \frac{\delta(x)}{x-y}dx = \frac{1}{\pi}\,\mathrm{PV}\,\frac{1}{y}.$$

The first relation $u = Hv$ therefore formally gives the expression in (a). To specify and prove its meaning, regularize both kernels by the same positive ϵ and define

$$D_\epsilon(w) = \frac{1}{\pi^2}\int_{-\infty}^{\infty} \frac{y(y-w)}{(y^2+\epsilon^2)((y-w)^2+\epsilon^2)}dy.$$

For real $w \neq 0$, close the rational integrand in the upper half-plane. The arc vanishes as $O(1/R)$, and the two residues before division by π^2 are

$$\frac{-w + i\epsilon}{2w(w - 2i\epsilon)}, \qquad \frac{w + i\epsilon}{2w(w + 2i\epsilon)}.$$

Their sum is $-i\epsilon/(w^2 + 4\epsilon^2)$, so

$$\boxed{D_\epsilon(w) = \frac{2\epsilon}{\pi(w^2 + 4\epsilon^2)}}.$$

The formula extends to $w = 0$ by continuity; directly the integral there is $1/(2\pi\epsilon)$, the same value. For every smooth compactly supported test function φ, substitute $w = 2\epsilon t$:

$$\int_{-\infty}^{\infty} D_\epsilon(w)\varphi(w)dw = \int_{-\infty}^{\infty} \frac{\varphi(2\epsilon t)}{\pi(1 + t^2)} dt \longrightarrow \varphi(0).$$

Dominated convergence applies because φ is bounded and the Cauchy density has integral one. Therefore $D_\epsilon \to \delta$ in the distributional sense, establishing the precise meaning of

$$\boxed{\delta(w) = \frac{1}{\pi^2} \int \frac{dy}{y(y - w)}}.$$

For $w \neq 0$ the separately interpreted principal values would give zero, while at $w = 0$ the integral of $1/y^2$ diverges; this is why a pointwise calculation cannot replace the distributional limit.

(b) Set $w = s - t$ and $x = s - y$. Then $y = s - x$, $y - w = t - x$, and $dy = -dx$. Reversing the limits removes the minus sign, while

$$y(y - w) = (s - x)(t - x) = (x - s)(x - t).$$

Thus

$$\boxed{\delta(s - t) = \frac{1}{\pi^2} \int_{-\infty}^{\infty} \frac{dx}{(x - s)(x - t)}}$$

with the same regularization: its finite-ϵ kernel is $2\epsilon/[\pi((s - t)^2 + 4\epsilon^2)]$. This also supplies the consistent kernel identity used in the Parseval calculation.

13 Chapter 13. Gamma Function

Problems

13.1 Definitions, Properties

13.1.1 Derive the recurrence relations

$$\Gamma(z+1) = z\Gamma(z)$$

from the Euler integral, Eq. (13.5),

$$\Gamma(z) = \int_0^\infty e^{-t} t^{z-1}\, dt.$$

The Euler integral is used for $\Re z > 0$; the recurrence then extends meromorphically.

Solution.

Initially let $\Re z > 0$, so both Euler integrals and the boundary terms below converge. Integrate $\Gamma(z+1) = \int_0^\infty t^z e^{-t} dt$ by parts with $u = t^z$ and $dv = e^{-t}dt$:

$$\Gamma(z+1) = \left[-t^z e^{-t}\right]_0^\infty + z\int_0^\infty t^{z-1} e^{-t} dt.$$

At zero, $|t^z| = t^{\Re z} \to 0$; at infinity the exponential outweighs the power. Hence

$$\boxed{\Gamma(z+1) = z\Gamma(z)}.$$

Repeated application gives $\Gamma(z) = \Gamma(z+N)/[z(z+1)\cdots(z+N-1)]$ with $\Re(z+N) > 0$. This extends the identity to the meromorphic gamma function; at poles it is interpreted by meromorphic continuation, not as multiplication of finite values. In particular $\Gamma(1) = \int_0^\infty e^{-t} dt = 1$ and $\Gamma(N+1) = N!$ for every nonnegative integer N.

13.1.2 In a power-series solution for the Legendre functions of the second kind we encounter the expression

$$\frac{(n+1)(n+2)(n+3)\cdots(n+2s-1)(n+2s)}{2\cdot 4\cdot 6\cdot 8\cdots(2s-2)(2s)\cdot(2n+3)(2n+5)(2n+7)\cdots(2n+2s+1)},$$

in which s is a positive integer.

(a) Rewrite this expression in terms of factorials.

(b) Rewrite this expression using Pochhammer symbols; see Eq. (1.72).

Solution.

(a) For the factorials of integers take $n \geq 0$ integral. Write the numerator as $(n+2s)!/n!$, the even product as $2^s s!$, and the odd product as

$$\frac{(2n+2s+1)!!}{(2n+1)!!} = \frac{(2n+2s+1)!}{2^{n+s}(n+s)!}\frac{2^n n!}{(2n+1)!}.$$

Multiplying the two denominator factors and then dividing the numerator by them gives

$$\boxed{\frac{(n+2s)!(n+s)!(2n+1)!}{s!(n!)^2(2n+2s+1)!}}.$$

For nonintegral n, factorial notation means $q! = \Gamma(q+1)$, and the equality holds wherever the original denominator is nonzero, with any removable gamma quotients interpreted by limits.

(b) By definition $(q)_s = q(q+1)\cdots(q+s-1)$. The numerator is $(n+1)_{2s}$, and the two denominator products are $2^s(1)_s$ and $2^s(n+\frac{3}{2})_s$. Thus

$$\boxed{\frac{(n+1)_{2s}}{4^s s!(n+\frac{3}{2})_s} = \frac{(\frac{n+1}{2})_s(\frac{n+2}{2})_s}{s!(n+\frac{3}{2})_s}}.$$

The last equality follows by separating the $2s$ consecutive numerator factors into the odd and even positions; each pair supplies a factor 4. These finite-product forms also cover complex n away from zeros of the original denominator.

13.1.3 Show that $\Gamma(z)$ may be written

$$\Gamma(z) = 2\int_0^\infty e^{-t^2} t^{2z-1}\,dt, \quad \Re(z) > 0,$$
$$\Gamma(z) = \int_0^1 \left[\ln\left(\frac{1}{t}\right)\right]^{z-1} dt, \quad \Re(z) > 0.$$

Solution.

Start from $\Gamma(z) = \int_0^\infty e^{-u}u^{z-1}du$, with $\Re z > 0$. Put $u = t^2$; since $du = 2t\,dt$ and the limits remain zero and infinity,

$$\boxed{\Gamma(z) = 2\int_0^\infty e^{-t^2} t^{2z-1}\,dt}.$$

For the second representation put $u = -\log t$. Then $t = e^{-u}$, $du = -dt/t$, and $e^{-u} = t$; the endpoints $u = 0, \infty$ correspond to $t = 1, 0$. Therefore

$$\Gamma(z) = \int_1^0 t(-\log t)^{z-1}\frac{-dt}{t} = \boxed{\int_0^1 [\log(1/t)]^{z-1}\,dt}.$$

All powers use the real logarithm of their positive bases. The condition $\Re z > 0$ controls the original lower endpoint and, after substitution, the corresponding endpoint of each transformed integral.

13.1.4 In a Maxwellian distribution the fraction of particles of mass m with speed between v and $v + dv$ is

$$\frac{dN}{N} = 4\pi\left(\frac{m}{2\pi kT}\right)^{3/2}\exp\left(-\frac{mv^2}{2kT}\right)v^2\,dv,$$

where N is the total number of particles, k is Boltzmann's constant, and T is the absolute temperature. The average or expectation value of v^n is defined as $\langle v^n\rangle = N^{-1}\int v^n\,dN$. Show that

$$\langle v^n\rangle = \left(\frac{2kT}{m}\right)^{n/2}\frac{\Gamma(\frac{n+3}{2})}{\Gamma(\frac{3}{2})}.$$

This is an extension of Example 13.1.1, in which the distribution was in kinetic energy $E = mv^2/2$, with $dE = mv\,dv$.

Assume $m, k, T > 0$ and $\Re n > -3$.

Solution.

Assume $m, k, T > 0$. The moment requires $\Re n > -3$ for convergence at zero. Set $b = m/(2kT) > 0$. The normalized speed density is $(4/\sqrt{\pi})b^{3/2}v^2e^{-bv^2}$, so

$$\langle v^n\rangle = \frac{4b^{3/2}}{\sqrt{\pi}}\int_0^\infty v^{n+2}e^{-bv^2}\,dv.$$

With $u = bv^2$, $v = (u/b)^{1/2}$ and $dv = \frac{1}{2}b^{-1/2}u^{-1/2}du$, this becomes

$$\begin{aligned}\langle v^n\rangle &= \frac{4b^{3/2}}{\sqrt{\pi}}\frac{1}{2}b^{-(n+3)/2}\int_0^\infty e^{-u}u^{(n+1)/2}\,du\\ &= \frac{2}{\sqrt{\pi}}b^{-n/2}\Gamma\left(\frac{n+3}{2}\right).\end{aligned}$$

Using $\Gamma(3/2) = \frac{1}{2}\Gamma(1/2) = \sqrt{\pi}/2$, we obtain

$$\boxed{\langle v^n\rangle = \left(\frac{2kT}{m}\right)^{n/2}\frac{\Gamma((n+3)/2)}{\Gamma(3/2)}}.$$

At $n = 0$ this equals one, checking normalization; at $n = 2$ it gives $\langle v^2\rangle = 3kT/m$. The convergence restriction follows also directly from v^{n+2} near $v = 0$.

13.1.5 By transforming the integral into a gamma function, show that

$$-\int_0^1 x^k \ln x\,dx = \frac{1}{(k+1)^2}, \quad k > -1.$$

Solution.

Put $u = -(k+1)\log x$, which is positive for $0 < x < 1$ since $k > -1$. Then

$$x = e^{-u/(k+1)}, \quad dx = -\frac{1}{k+1}e^{-u/(k+1)}du, \quad -\log x = \frac{u}{k+1}.$$

After reversing the transformed limits, the factors x^k and the exponential in dx combine to e^{-u}, giving

$$-\int_0^1 x^k \log x\,dx = \frac{1}{(k+1)^2}\int_0^\infty ue^{-u}\,du = \frac{\Gamma(2)}{(k+1)^2} = \boxed{\frac{1}{(k+1)^2}}.$$

The restriction $k > -1$ ensures convergence at zero and makes the change of variable valid with these limits.

13.1.6 Show that

$$\int_0^\infty e^{-x^4}\,dx = \Gamma\left(\frac{5}{4}\right).$$

Solution.

With $t = x^4$, we have $dx = \frac{1}{4}t^{-3/4}dt$. Thus

$$\int_0^\infty e^{-x^4}dx = \frac{1}{4}\int_0^\infty e^{-t}t^{1/4-1}dt = \frac{1}{4}\Gamma\left(\frac{1}{4}\right) = \boxed{\Gamma\left(\frac{5}{4}\right)},$$

where the last equality is the recurrence from Exercise 13.1.1. The transformed power $t^{-3/4}$ is integrable at zero.

13.1.7 Show that

$$\lim_{x\to 0}\frac{\Gamma(ax)}{\Gamma(x)} = \frac{1}{a}.$$

Take a fixed $a \neq 0$.

Solution.

The parameter a is fixed and nonzero. Use the recurrence to remove the two poles near $x = 0$:

$$\frac{\Gamma(ax)}{\Gamma(x)} = \frac{\Gamma(1+ax)/(ax)}{\Gamma(1+x)/x} = \frac{1}{a}\frac{\Gamma(1+ax)}{\Gamma(1+x)}.$$

The gamma function is continuous and nonzero near 1, with $\Gamma(1) = 1$. Consequently

$$\boxed{\lim_{x\to 0}\frac{\Gamma(ax)}{\Gamma(x)} = \frac{1}{a}}.$$

This argument also works for complex $a \neq 0$ and a complex limit $x \to 0$. For $a = 0$ the numerator would be $\Gamma(0)$ throughout, so the original ratio would not be defined.

13.1.8 Locate the poles of $\Gamma(z)$. Show that they are simple poles and determine the residues.

Solution.

The Euler integral is analytic for $\Re z > 0$; on compact subsets, its derivatives are dominated by integrable functions of the form $e^{-t}t^{\sigma-1}|\log t|^j$ near zero and infinity. For any point outside that half-plane, shift far enough to use

$$\Gamma(z) = \frac{\Gamma(z+N+1)}{z(z+1)\cdots(z+N)}.$$

Its only possible singularities are $z = 0, -1, -2, \ldots$. At $z = -n$ choose $N = n$ and take the residue:

$$\begin{aligned}\lim_{z\to -n}(z+n)\Gamma(z) &= \frac{\Gamma(1)}{(-n)(-n+1)\cdots(-1)} \\ &= \boxed{\frac{(-1)^n}{n!}} \qquad (n = 0, 1, 2, \ldots).\end{aligned}$$

For $n = 0$ the denominator product is empty and equals one. The denominator has exactly one vanishing factor, and the displayed residue is nonzero. Hence each listed singularity is a simple pole, and the shifted formula proves that there are no others.

13.1.9 Show that the equation $\Gamma(x) = k$, $k \neq 0$, has an infinite number of real roots.

The level k is real.

Solution.

Here k must be a nonzero real number. A nonreal k cannot equal $\Gamma(x)$ at a real nonsingular x. On the interval $(-N, -N+1)$, with $N \geq 1$, repeated recurrence shows that $\Gamma(x)$ has the fixed sign $(-1)^N$. Its residues at the endpoints give

$$\Gamma(x) \longrightarrow (-1)^N \infty \quad \text{as } x \downarrow -N \text{ or } x \uparrow -N+1.$$

At the midpoint, recurrence from $\Gamma(1/2) = \sqrt{\pi}$ gives

$$\Gamma\left(\frac{1}{2} - N\right) = \frac{(-1)^N 2^N \sqrt{\pi}}{(2N-1)!!}.$$

Its magnitude tends to zero, since the ratio of consecutive magnitudes is $1/(N+1/2) \to 0$. Choose arbitrarily large N of the parity making $(-1)^N = \operatorname{sgn} k$ and with midpoint magnitude less than $|k|$. The function is continuous between its poles, and its magnitude tends to infinity at either endpoint. The intermediate value theorem therefore gives a root of $\Gamma(x) = k$ on each side of the midpoint. Infinitely many disjoint intervals have the required parity, proving infinitely many real roots.

13.1.10 Show that, for integer s,

(a) $$\int_0^\infty x^{2s+1} \exp(-ax^2)\, dx = \frac{s!}{2a^{s+1}}.$$

(b) $$\int_0^\infty x^{2s} \exp(-ax^2)\, dx = \frac{\Gamma(s+\frac{1}{2})}{2a^{s+1/2}} = \frac{(2s-1)!!}{2^{s+1}a^s}\sqrt{\frac{\pi}{a}}.$$

These Gaussian integrals are of major importance in statistical mechanics.

Take $a > 0$ and $s = 0, 1, 2, \ldots$.

Solution.

For these convergent real Gaussian integrals, take $a > 0$ and integer $s \geq 0$. In both cases put $u = ax^2$, so $dx = \frac{1}{2}a^{-1/2}u^{-1/2}du$. **(a)** Since $x^{2s+1} = a^{-s-1/2}u^{s+1/2}$,

$$\int_0^\infty x^{2s+1} e^{-ax^2} dx = \frac{1}{2a^{s+1}} \int_0^\infty u^s e^{-u} du = \boxed{\frac{s!}{2a^{s+1}}}.$$

(b) With $x^{2s} = a^{-s}u^s$ instead,

$$\int_0^\infty x^{2s} e^{-ax^2} dx = \frac{\Gamma(s+1/2)}{2a^{s+1/2}}.$$

Repeated recurrence gives

$$\Gamma\left(s + \frac{1}{2}\right) = \frac{2s-1}{2}\frac{2s-3}{2}\cdots\frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{(2s-1)!!}{2^s}\sqrt{\pi}.$$

Thus

$$\boxed{\int_0^\infty x^{2s} e^{-ax^2} dx = \frac{(2s-1)!!}{2^{s+1}a^s}\sqrt{\frac{\pi}{a}}}.$$

The convention $(-1)!! = 1$ includes $s = 0$. Negative integer s would make the lower endpoint divergent in both parts.

13.1.11 Express the coefficient of the nth term of the expansion of $(1+x)^{1/2}$ in powers of x

(a) in terms of factorials of integers,

(b) in terms of the double factorial (!!) functions.

$$\textit{ANS. } a_n = (-1)^{n+1}\frac{(2n-3)!}{2^{2n-2}n!(n-2)!} = (-1)^{n+1}\frac{(2n-3)!!}{(2n)!!}, \quad n = 2, 3, \ldots.$$

Solution.

The coefficient of x^n is the binomial coefficient

$$a_n = \binom{1/2}{n} = \frac{(1/2)(1/2-1)\cdots(1/2-n+1)}{n!}.$$

The first two coefficients are $a_0 = 1$ and $a_1 = 1/2$. For $n \geq 2$, pull a factor $1/2$ from each numerator term and count the $n-1$ negative factors:

$$a_n = \frac{(-1)^{n-1}(2n-3)!!}{2^n n!} = \boxed{(-1)^{n+1}\frac{(2n-3)!!}{(2n)!!}}.$$

Here $(2n)!! = 2^n n!$. To express the odd double factorial using ordinary factorials, separate the even factors of $(2n-3)!$:

$$(2n-3)! = (2n-3)!!\, 2^{n-2}(n-2)!.$$

Therefore

$$\boxed{a_n = (-1)^{n+1}\frac{(2n-3)!}{2^{2n-2}n!(n-2)!} \quad (n \geq 2)}.$$

For example $a_2 = -1/8$ and $a_3 = 1/16$, in agreement with direct multiplication of the first few binomial factors.

13.1.12 Express the coefficient of the nth term of the expansion of $(1+x)^{-1/2}$ in powers of x

(a) in terms of the factorials of integers,

(b) in terms of the double factorial (!!) functions.

$$\textit{ANS. } a_n = (-1)^n\frac{(2n)!}{2^{2n}(n!)^2} = (-1)^n\frac{(2n-1)!!}{(2n)!!}, \quad n = 1, 2, 3, \ldots.$$

Solution.

For $n \geq 1$,

$$a_n = \binom{-1/2}{n} = \frac{(-1/2)(-3/2)\cdots(-(2n-1)/2)}{n!} = \frac{(-1)^n(2n-1)!!}{2^n n!}.$$

Since $(2n)!! = 2^n n!$, the double-factorial result is

$$\boxed{a_n = (-1)^n\frac{(2n-1)!!}{(2n)!!}}.$$

Splitting $(2n)!$ into its even and odd factors gives $(2n)! = (2n)!!(2n-1)!! = 2^n n!(2n-1)!!$, so

$$\boxed{a_n = (-1)^n\frac{(2n)!}{2^{2n}(n!)^2}}.$$

The coefficient $a_0 = 1$ is also covered if $(-1)!! = 0!! = 1$. The first terms are $1 - x/2 + 3x^2/8 - 5x^3/16 + \cdots$.

13.1.13 The Legendre polynomial P_n may be written as

$$P_n(\cos\theta) = 2\frac{(2n-1)!!}{(2n)!!}\Bigg\{\cos n\theta + \frac{1}{1}\cdot\frac{n}{2n-1}\cos(n-2)\theta + \frac{1\cdot 3}{1\cdot 2}\frac{n(n-1)}{(2n-1)(2n-3)}\cos(n-4)\theta + \frac{1\cdot 3\cdot 5}{1\cdot 2\cdot 3}\frac{n(n-1)(n-2)}{(2n-1)(2n-3)(2n-5)}\cos(n-6)\theta + \cdots\Bigg\}.$$

Let $n = 2s+1$. Then the above can be written

$$P_n(\cos\theta) = P_{2s+1}(\cos\theta) = \sum_{m=0}^{s} a_m \cos(2m+1)\theta.$$

Find a_m in terms of factorials and double factorials.

Solution.

Let k count downward steps of two from the highest harmonic n. Reading the coefficient of $\cos(n-2k)\theta$ from the given series, including its prefactor, gives

$$c_k = 2\frac{(2n-1)!!}{(2n)!!}\frac{(2k-1)!!}{k!}\frac{n!}{(n-k)!}\frac{(2n-2k-1)!!}{(2n-1)!!}.$$

Cancel $(2n-1)!!$ and use $(2n)!! = 2^n n!$ to obtain

$$c_k = 2^{1-n}\frac{(2k-1)!!(2n-2k-1)!!}{k!(n-k)!}.$$

Now $n = 2s+1$ and $n-2k = 2m+1$ imply $k = s-m$ and $n-k = s+m+1$. Hence

$$\boxed{a_m = \frac{(2s-2m-1)!!(2s+2m+1)!!}{2^{2s}(s-m)!(s+m+1)!} \quad (0 \le m \le s)}.$$

The convention $(-1)!! = 1$ includes the highest harmonic $m = s$. Finally apply $(2q-1)!! = (2q)!/(2^q q!)$ to both factors. Their powers of two multiply to 2^{2s+1}, giving

$$\boxed{a_m = \frac{(2s-2m)!(2s+2m+2)!}{2^{4s+1}[(s-m)!]^2[(s+m+1)!]^2}}.$$

For $s = 1$ this gives $P_3(\cos\theta) = (3\cos\theta + 5\cos 3\theta)/8$, which also follows by inserting $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ in $P_3(x) = (5x^3 - 3x)/2$.

13.1.14

(a) Show that $\Gamma(\frac{1}{2}-n)\Gamma(\frac{1}{2}+n) = (-1)^n\pi$, where n is an integer.

(b) Express $\Gamma(\frac{1}{2}+n)$ and $\Gamma(\frac{1}{2}-n)$ separately in terms of $\pi^{1/2}$ and a double factorial function.

ANS. $\Gamma(\frac{1}{2}+n)=\dfrac{(2n-1)!!}{2^n}\pi^{1/2}$.

The displayed double-factorial answer uses $n\ge 0$ and $(-1)!!=1$. For negative integer n, interchange the two half-integer formulas derived below.

Solution.

First take $n=N\ge 0$. Recurrence upward from 1/2 yields

$$\Gamma\left(\frac{1}{2}+N\right)=\Gamma\left(\frac{1}{2}\right)\prod_{j=1}^{N}\left(j-\frac{1}{2}\right)=\boxed{\frac{(2N-1)!!}{2^N}\sqrt{\pi}}.$$

Recurrence downward gives

$$\Gamma\left(\frac{1}{2}-N\right)=\frac{\Gamma(1/2)}{\prod_{j=1}^{N}(1/2-j)}=\boxed{\frac{(-1)^N2^N}{(2N-1)!!}\sqrt{\pi}}.$$

The products are empty when $N=0$, consistently with $(-1)!!=1$. Their multiplication cancels the double factorial and power of two:

$$\boxed{\Gamma(1/2-N)\Gamma(1/2+N)=(-1)^N\pi}.$$

For a negative integer $n=-N$, interchange the two boxed individual formulas. The product is unchanged and $(-1)^{-N}=(-1)^N$, so the product identity holds for every integer n without introducing undefined negative odd double factorials.

13.1.15 Show that if $\Gamma(x+iy)=u+iv$, then $\Gamma(x-iy)=u-iv$.

This is a special case of the Schwarz reflection principle, Section 11.10.

Take real x,y and exclude poles of the gamma functions.

Solution.

For real $x>0$ and real y, the Euler integral is absolutely convergent and $t^{x+iy-1}=\exp[(x+iy-1)\log t]$ with real $\log t$. Complex conjugation may therefore pass under the integral:

$$\overline{\Gamma(x+iy)}=\int_0^\infty e^{-t}\overline{t^{x+iy-1}}dt=\int_0^\infty e^{-t}t^{x-iy-1}dt=\Gamma(x-iy).$$

For any other nonsingular $z=x+iy$, choose an integer N with $\Re(z+N)>0$ and use

$$\Gamma(z)=\frac{\Gamma(z+N)}{\prod_{j=0}^{N-1}(z+j)}.$$

Conjugating numerator and denominator and applying the proven half-plane result gives $\overline{\Gamma(z)}=\Gamma(\bar z)$ there as well. Thus if $\Gamma(x+iy)=u+iv$,

$$\boxed{\Gamma(x-iy)=u-iv}.$$

At a pole neither side is a finite complex value; the equality is understood as the corresponding meromorphic reflection relation.

13.1.16 Prove that $|\Gamma(\alpha+i\beta)| = |\Gamma(\alpha)| \prod_{n=0}^{\infty} \left[1+\frac{\beta^2}{(\alpha+n)^2}\right]^{-1/2}$.

This equation has been useful in calculations of beta decay theory.

Take real α, β with $\alpha \notin \{0, -1, -2, \ldots\}$.

Solution.

Assume real α, β and $\alpha \notin \{0, -1, -2, \ldots\}$, so every displayed factor and $\Gamma(\alpha)$ are defined. The Euler product-limit representation is

$$\Gamma(z) = \lim_{N\to\infty} \frac{N!\, N^z}{z(z+1)\cdots(z+N)}.$$

Since $|N^{\alpha+i\beta}| = N^{\alpha}$, division of its absolute value at $z = \alpha+i\beta$ by that at $z = \alpha$ gives

$$\begin{aligned}\frac{|\Gamma(\alpha+i\beta)|}{|\Gamma(\alpha)|} &= \lim_{N\to\infty} \prod_{j=0}^{N} \frac{|\alpha+j|}{|\alpha+j+i\beta|} \\ &= \prod_{j=0}^{\infty} \left(1+\frac{\beta^2}{(\alpha+j)^2}\right)^{-1/2}.\end{aligned}$$

This product converges to a positive value: for sufficiently large j, its logarithm in magnitude is bounded by a constant times j^{-2}, and the finitely many preceding factors are nonzero. Multiplying by $|\Gamma(\alpha)|$ gives the requested identity. The argument applies to negative nonintegral α as well as positive α, using the continued product-limit representation. At a nonpositive integer α the written expression would contain an undefined pole times a zero factor; it must not be used literally there.

13.1.17 Show that for n, a positive integer,

$$|\Gamma(n+ib+1)| = \left(\frac{\pi b}{\sinh \pi b}\right)^{1/2} \prod_{s=1}^{n} (s^2+b^2)^{1/2}.$$

Take real b; at $b = 0$, interpret $\pi b/\sinh(\pi b)$ by its limit, one.

Solution.

Take real b. For $b \neq 0$, reflection and recurrence give

$$\Gamma(ib)\Gamma(1-ib) = \frac{\pi}{\sin(\pi i b)}, \qquad \Gamma(1+ib) = ib\Gamma(ib).$$

Because $\sin(i\pi b) = i\sinh(\pi b)$ and the two factors $\Gamma(1 \pm ib)$ are conjugates, this yields

$$|\Gamma(1+ib)|^2 = \frac{ib\pi}{i\sinh \pi b} = \frac{\pi b}{\sinh \pi b}.$$

Apply recurrence n more times:

$$\Gamma(n+1+ib) = \left[\prod_{s=1}^{n} (s+ib)\right] \Gamma(1+ib).$$

Taking the absolute value and using $|s+ib| = \sqrt{s^2+b^2}$ gives

$$\boxed{|\Gamma(n+1+ib)| = \left(\frac{\pi b}{\sinh \pi b}\right)^{1/2} \prod_{s=1}^{n} \sqrt{s^2+b^2}}.$$

At $b = 0$, the quotient $\pi b/\sinh \pi b$ is interpreted by its limit 1. The formula then gives $n! = \Gamma(n+1)$, as it should. Its square root is the nonnegative real one; the quotient is positive for either sign of nonzero real b.

13.1.18 Show that for all real values of x and y, $|\Gamma(x)| \geq |\Gamma(x+iy)|$.

Exclude $x = 0, -1, -2, \ldots$, where the displayed $\Gamma(x)$ is undefined.

Solution.

Exclude $x = 0, -1, -2, \ldots$, where $\Gamma(x)$ is not a finite defined value. For other real x and real y, Exercise 13.1.16 gives

$$\frac{|\Gamma(x+iy)|}{|\Gamma(x)|} = \prod_{n=0}^{\infty} \left(1 + \frac{y^2}{(x+n)^2}\right)^{-1/2}.$$

Each factor is positive and at most one, and the product converges. Every partial product is therefore at most one, as is its limit. Consequently

$$\boxed{|\Gamma(x+iy)| \leq |\Gamma(x)|}.$$

Equality holds at $y = 0$. If $y \neq 0$, already the $n = 0$ factor is strictly smaller than one and all later factors are at most one, so the inequality is strict. This proof includes negative nonintegral x, where the original Euler integral cannot be used directly.

13.1.19 Show that $\left|\Gamma(\frac{1}{2} + iy)\right|^2 = \dfrac{\pi}{\cosh \pi y}$.

Take real y.

Solution.

Put $z = 1/2 + iy$ in the reflection formula. Then $1 - z = 1/2 - iy = \bar{z}$, so Exercise 13.1.15 gives

$$\left|\Gamma\left(\frac{1}{2} + iy\right)\right|^2 = \Gamma(z)\Gamma(1-z) = \frac{\pi}{\sin \pi z}.$$

The denominator is

$$\sin\left(\frac{\pi}{2} + i\pi y\right) = \cos(i\pi y) = \frac{e^{\pi y} + e^{-\pi y}}{2} = \cosh \pi y.$$

Hence

$$\boxed{\left|\Gamma\left(\frac{1}{2} + iy\right)\right|^2 = \frac{\pi}{\cosh \pi y}}.$$

For real y the denominator is positive, and at $y = 0$ the formula reduces to $\Gamma(1/2)^2 = \pi$.

13.1.20 The probability density associated with the normal distribution of statistics is given by

$$f(x) = \frac{1}{\sigma(2\pi)^{1/2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right],$$

with $(-\infty, \infty)$ for the range of x. Show that

(a) $\langle x \rangle$, the mean value of x, is equal to μ,

(b) the standard deviation $(\langle x^2 \rangle - \langle x \rangle^2)^{1/2}$ is given by σ.

Assume real μ and $\sigma > 0$.

Solution.

Take real μ and $\sigma > 0$, as required for the stated probability density. Set $u = (x-\mu)/\sigma$, so $x = \mu + \sigma u$ and $f(x)dx = (2\pi)^{-1/2} e^{-u^2/2} du$. The Gaussian integral gives

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-u^2/2} du = 1, \qquad \int_{-\infty}^{\infty} u e^{-u^2/2} du = 0$$

(the second integrand is odd). Integration by parts also gives

$$\int_{-\infty}^{\infty} u^2 e^{-u^2/2} du = \left[-u e^{-u^2/2}\right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} e^{-u^2/2} du = \sqrt{2\pi}.$$

(a) Expanding the first moment now yields $\langle x \rangle = \mu \cdot 1 + \sigma \cdot 0 = \boxed{\mu}$. **(b)** Expanding $(\mu + \sigma u)^2$ yields

$$\langle x^2 \rangle = \mu^2 + 2\mu\sigma \cdot 0 + \sigma^2, \qquad \boxed{\sqrt{\langle x^2 \rangle - \langle x \rangle^2} = \sqrt{\sigma^2} = \sigma}.$$

The final equality uses the positive scale convention.

13.1.21 For the gamma distribution

$$f(x) = \begin{cases} \dfrac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}, & x > 0, \\ 0, & x \le 0, \end{cases}$$

show that

(a) $\langle x \rangle$, the mean value of x, is equal to $\alpha\beta$,

(b) σ^2, its variance, defined as $\langle x^2 \rangle - \langle x \rangle^2$, has the value $\alpha\beta^2$.

Assume real $\alpha, \beta > 0$.

Solution.

A gamma probability distribution requires $\alpha > 0$ and $\beta > 0$. For any moment exponent q with $\Re q > -\alpha$, substitute $x = \beta u$, $dx = \beta du$:

$$\begin{aligned} \langle x^q \rangle &= \frac{1}{\beta^\alpha \Gamma(\alpha)} \int_0^{\infty} x^{\alpha+q-1} e^{-x/\beta} dx \\ &= \frac{\beta^q}{\Gamma(\alpha)} \int_0^{\infty} u^{\alpha+q-1} e^{-u} du = \frac{\beta^q \Gamma(\alpha+q)}{\Gamma(\alpha)}. \end{aligned}$$

At $q = 0$ this is one, checking normalization. **(a)** With $q = 1$ and the gamma recurrence,

$$\boxed{\langle x \rangle = \frac{\beta\Gamma(\alpha+1)}{\Gamma(\alpha)} = \alpha\beta}.$$

(b) With $q = 2$, $\langle x^2 \rangle = \beta^2 \alpha(\alpha+1)$, so

$$\boxed{\sigma^2 = \alpha(\alpha+1)\beta^2 - (\alpha\beta)^2 = \alpha\beta^2}.$$

13.1.22 The wave function of a particle scattered by a Coulomb potential is $\psi(r,\theta)$. Given that at the origin the wave function becomes

$$\psi(0) = e^{-\pi\gamma/2}\Gamma(1+i\gamma),$$

where $\gamma > 0$ is a dimensionless parameter, show that

$$|\psi(0)|^2 = \frac{2\pi\gamma}{e^{2\pi\gamma}-1}.$$

Solution.

The real exponential factor has squared magnitude $e^{-\pi\gamma}$. Using Exercise 13.1.17 with the empty product ($n = 0$),

$$|\psi(0)|^2 = e^{-\pi\gamma}|\Gamma(1+i\gamma)|^2 = e^{-\pi\gamma}\frac{\pi\gamma}{\sinh\pi\gamma}.$$

Write $\sinh\pi\gamma = (e^{\pi\gamma} - e^{-\pi\gamma})/2$ and multiply numerator and denominator by $e^{\pi\gamma}$:

$$\boxed{|\psi(0)|^2 = \frac{2\pi\gamma}{e^{2\pi\gamma}-1}}.$$

The limit as $\gamma \downarrow 0$ is one, agreeing with $\Gamma(1) = 1$ in the original wave function.

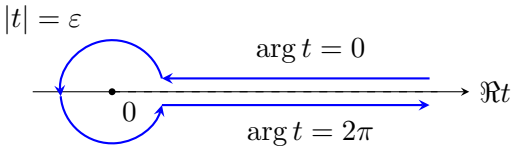
13.1.23 Derive the contour integral representation of Eq. (13.31),

$$2i\Gamma(\nu+1)\sin\nu\pi = \int_C e^{-t}e^{-\pi i\nu}t^\nu\,dt.$$

Use $0 < \arg t < 2\pi$: C runs inward on the upper lip of the positive real cut, counterclockwise around a fixed circle about zero, and outward on the lower lip. At negative integer ν, interpret the gamma–sine product by continuation.

Solution.

Use the branch $t^\nu = \exp[\nu(\log|t| + i\arg t)]$ with $0 < \arg t < 2\pi$, cut along the positive real axis. The contour starts at positive infinity above the cut, runs inward to a small circle of radius ε, traverses that circle counterclockwise, then returns below the cut to positive infinity. The published erratum replaces the ambiguous $(-t)^\nu$ by $e^{-\pi i\nu}t^\nu$ with this continuous branch convention.



The two lips of the cut are displaced in the sketch for visibility. Initially assume $\Re\nu > -1$. On the upper lip $t = x$, $t^\nu = x^\nu$, but the direction is inward. Its contribution is

$$I_{\text{upper}} = -e^{-\pi i\nu}\int_\varepsilon^\infty e^{-x}x^\nu dx.$$

On the lower lip the continuous argument is 2π, so $t^\nu = e^{2\pi i\nu}x^\nu$ and the outward contribution is

$$I_{\text{lower}} = e^{\pi i\nu}\int_\varepsilon^\infty e^{-x}x^\nu dx.$$

On the small circle put $t = \varepsilon e^{i\theta}$, so $dt = i\varepsilon e^{i\theta} d\theta$ and $t^\nu = \varepsilon^\nu e^{i\nu\theta}$. For fixed ν its integral in magnitude is at most a constant times $\varepsilon^{\Re\nu+1} e^\varepsilon$, which tends to zero. Thus, as $\varepsilon \downarrow 0$,

$$\int_C e^{-t} e^{-\pi i\nu} t^\nu dt = (e^{\pi i\nu} - e^{-\pi i\nu}) \int_0^\infty e^{-x} x^\nu dx$$
$$= \boxed{2i \sin(\pi\nu)\Gamma(\nu+1)}.$$

For $\Re\nu \le -1$, keep a nonzero circle radius; the individual ray integrals cannot be extended to zero separately. For fixed ε the complete contour integral is entire in ν: the circle is compact and the ray tails are dominated uniformly on compact parameter sets by an exponential times a fixed power. Deformation within the slit plane makes it independent of ε. Therefore the identity continues analytically, with

$$2i\Gamma(\nu+1)\sin\pi\nu = -\frac{2\pi i}{\Gamma(-\nu)}.$$

Only nonnegative integer ν makes the integrand entire in t and gives zero. At negative integers the integral is nonzero; the text's unqualified statement about integer ν needs this restriction. Indeed, for $\nu = -m$, $m = 1, 2, \ldots$, the integrand is $(-1)^m e^{-t} t^{-m}$. The coefficient of t^{-1} is $(-1)^m(-1)^{m-1}/(m-1)! = -1/(m-1)!$, so the counterclockwise contour gives $-2\pi i/(m-1)!$, agreeing with the continued formula.

13.2 Digamma and Polygamma Functions

13.2.1 For "small" values of x,

$$\ln\Gamma(x+1) = -\gamma x + \sum_{n=2}^{\infty} (-1)^n \frac{\zeta(n)}{n} x^n,$$

where γ is the Euler-Mascheroni constant and $\zeta(n)$ the Riemann zeta function. For what values of x does this series converge?

ANS. $-1 < x \le 1$.

Note that if $x = 1$, we obtain

$$\gamma = \sum_{n=2}^{\infty} (-1)^n \frac{\zeta(n)}{n},$$

a series for the Euler-Mascheroni constant. The convergence of this series is exceedingly slow. For actual computation of γ, other, indirect, approaches are far superior (see Exercise 12.3.2).

Solution.

Write $\zeta(n) = 1 + d_n$, where

$$0 < d_n = \sum_{k=2}^{\infty} k^{-n} \le 2^{-n} + \int_2^\infty t^{-n} dt = 2^{-n}\left(1 + \frac{2}{n-1}\right).$$

Thus the series is the sum of $\sum_{n=2}^\infty (-x)^n/n$, whose radius is one, and an absolutely convergent correction for $|x| < 2$. It converges absolutely for $|x| < 1$. At $x = 1$ the first series is alternating harmonic, while at $x = -1$ it is a positive harmonic tail; the correction converges at both endpoints. For $|x| > 1$ the terms $\zeta(n)x^n/n$ do not tend to zero. Hence the real convergence set is exactly

$$\boxed{-1 < x \le 1}.$$

To identify the endpoint sum, let $x \uparrow 1$ in the interior identity and use Abel's theorem for the convergent power series. Since $\Gamma(2) = 1$, its logarithm tends to zero, giving

$$\gamma = \sum_{n=2}^{\infty} (-1)^n \frac{\zeta(n)}{n}.$$

The leading $1/n$ behavior explains the slow convergence. For complex x, the same decomposition gives conditional convergence on $|x| = 1$ except at $x = -1$.

13.2.2 Show that the series expansion of $\ln \Gamma(x+1)$ (Exercise 13.2.1) may be written as

(a) $$\ln \Gamma(x+1) = \frac{1}{2} \ln \left(\frac{\pi x}{\sin \pi x} \right) - \gamma x - \sum_{n=1}^{\infty} \frac{\zeta(2n+1)}{2n+1} x^{2n+1},$$

(b) $$\ln \Gamma(x+1) = \frac{1}{2} \ln \left(\frac{\pi x}{\sin \pi x} \right) - \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right) + (1-\gamma)x - \sum_{n=1}^{\infty} [\zeta(2n+1) - 1] \frac{x^{2n+1}}{2n+1}.$$

Determine the range of convergence of each of these expressions.

Solution.

Use the real logarithm initially on $-1 < x < 1$ (with removable quotients at $x = 0$). Split the original series into even and odd powers:

$$\ln \Gamma(1+x) = -\gamma x + \sum_{n=1}^{\infty} \frac{\zeta(2n) x^{2n}}{2n} - \sum_{n=1}^{\infty} \frac{\zeta(2n+1) x^{2n+1}}{2n+1}.$$

Reflection and recurrence give $\Gamma(1+x)\Gamma(1-x) = \pi x / \sin \pi x$. Adding the series at x and $-x$ cancels the odd terms, so

$$\sum_{n=1}^{\infty} \frac{\zeta(2n) x^{2n}}{2n} = \frac{1}{2} \ln \frac{\pi x}{\sin \pi x}.$$

(a) Substitution gives exactly the first requested form. Its remaining odd series has radius one; at $x = 1$ its positive terms behave as $1/(2n+1)$, and at $x = -1$ their negatives do. It therefore converges for $\boxed{-1 < x < 1}$ only. In particular, the finite limit of the complete expression at $x = 1$ requires cancellation of divergent pieces; direct endpoint substitution in the separated terms is invalid.

(b) Split $\zeta(2n+1) = 1 + [\zeta(2n+1) - 1]$ and use

$$\sum_{n=1}^{\infty} \frac{x^{2n+1}}{2n+1} = \frac{1}{2} \ln \frac{1+x}{1-x} - x.$$

This follows by integrating the geometric series for $(1-x^2)^{-1}$ from zero to x and removing the linear term. It produces the second displayed form, including $(1-\gamma)x$. Now $\zeta(2n+1) - 1 \sim 2^{-2n-1}$, so the remaining series has radius two. At $x = \pm 2$ its terms again form a signed harmonic tail, and it diverges. Thus this series alone converges on $\boxed{-2 < x < 2}$, but the expression as written with separate real logarithms has the domain $-1 < x < 1$. If those logarithms are first combined, their contribution is

$$\frac{1}{2} \ln \frac{\pi x (1-x)}{(1+x) \sin \pi x}.$$

This has a removable limit $\frac{1}{2}\ln(1/2)$ at $x = 1$ and is real on $-1 < x < 2$. Consequently the combined form extends the identity across $x = 1$ to that larger interval. This continuation must be distinguished from the domain of the separate logarithms.

13.2.3 Verify that for n, a positive integer, the following two forms of the digamma function are equal to each other:

$$\psi(n+1) = \sum_{j=1}^{n} \frac{1}{j} - \gamma \quad \text{and} \quad \psi(n+1) = \sum_{j=1}^{\infty} \frac{n}{j(n+j)} - \gamma.$$

Solution.

For a fixed positive integer n, partial fractions give

$$\frac{n}{j(n+j)} = \frac{1}{j} - \frac{1}{j+n}.$$

Sum only to a finite N before canceling:

$$\begin{aligned} \sum_{j=1}^{N} \frac{n}{j(n+j)} &= H_N - (H_{N+n} - H_n) \\ &= H_n - \sum_{j=N+1}^{N+n} \frac{1}{j}. \end{aligned}$$

The final sum lies between zero and $n/(N+1)$, hence tends to zero as $N \to \infty$. Therefore

$$\boxed{\sum_{j=1}^{\infty} \frac{n}{j(n+j)} = H_n}, \qquad \psi(n+1) = H_n - \gamma,$$

and the two representations agree. This argument avoids subtracting two divergent harmonic series separately.

13.2.4 Show that $\psi(z+1)$ has the series expansion

$$\psi(z+1) = -\gamma + \sum_{n=2}^{\infty} (-1)^n \zeta(n) z^{n-1}.$$

The expansion holds for $|z| < 1$.

Solution.

On every closed disk $|z| \le r < 1$, the Maclaurin series for $\ln\Gamma(1+z)$ may be differentiated term by term. For example, the derivative terms have absolute values at most $\zeta(2)r^{n-1}$, whose sum converges. Since $\psi(1+z) = d\ln\Gamma(1+z)/dz$, differentiation gives

$$\boxed{\psi(1+z) = -\gamma + \sum_{n=2}^{\infty} (-1)^n \zeta(n) z^{n-1}}, \qquad |z| < 1.$$

The domain $|z| < 1$ is needed: on $|z| = 1$ the summands have magnitudes $\zeta(n) \to 1$, so none of that boundary belongs to the convergence set. The closest pole of the expanded function is at $z = -1$, consistent with radius one.

13.2.5 For a power-series expansion of $\ln\Gamma(z+1)$, AMS-55 (see Additional Readings for the reference) lists

$$\ln\Gamma(z+1) = -\ln(1+z) + z(1-\gamma) + \sum_{n=2}^{\infty}(-1)^n\,[\zeta(n)-1]\,\frac{z^n}{n}.$$

(a) Show that this agrees with Eq. (13.44) for $|z|<1$.

(b) What is the range of convergence of this new expression?

Solution.

(a) For $|z|<1$,

$$\log(1+z) = z - \sum_{n=2}^{\infty}(-1)^n\frac{z^n}{n}.$$

Add this to the series for $\log\Gamma(1+z)$. Combining the linear coefficients and then each higher coefficient gives

$$\log\Gamma(1+z) + \log(1+z) = (1-\gamma)z + \sum_{n=2}^{\infty}(-1)^n\frac{\zeta(n)-1}{n}z^n.$$

Moving $\log(1+z)$ to the right proves the claimed agreement.

(b) Set $e_n = \zeta(n) - 1 - 2^{-n}$. The estimate

$$0 < e_n \le 3^{-n}\left(1 + \frac{3}{n-1}\right)$$

shows that the new series is $\sum_{n=2}^{\infty}(-z/2)^n/n$ plus a correction absolutely convergent for $|z|<3$. Thus its radius is two; on $|z|=2$ it converges except at $z=-2$, by the Dirichlet test for the leading series. Its real convergence set is $\boxed{-2 < z \le 2}$.

Convergence of the series is distinct from the domain of the complete logarithmic identity: with ordinary real logarithms that identity requires $z > -1$, and hence holds for $-1 < z \le 2$. It is undefined at $z=-1$. More precisely, the analytic series equals $\log\Gamma(2+z) - (1-\gamma)z$ on $|z|<2$, with the logarithm continued from zero. Subtracting $\log(1+z)$ gives the requested identity on a slit disk, for example $|z|<2$ with $z \notin (-2,-1]$, using compatible analytic branches. The endpoint $z=2$ follows by Abel's theorem and continuity.

13.2.6 Show that

$$\frac{1}{2}\ln\left(\frac{\pi z}{\sin\pi z}\right) = \sum_{n=1}^{\infty}\frac{\zeta(2n)}{2n}z^{2n}, \quad |z|<1.$$

Hint. Use Eqs. (13.23) and (13.44).

Solution.

Write the expansion in Eq. (13.44) at z and $-z$ and add:

$$\begin{aligned}\log\Gamma(1+z) + \log\Gamma(1-z) &= \sum_{k=2}^{\infty}\frac{(-1)^k\zeta(k)}{k}[z^k + (-z)^k] \\ &= 2\sum_{n=1}^{\infty}\frac{\zeta(2n)}{2n}z^{2n}.\end{aligned}$$

The linear terms and all odd powers cancel. Reflection, $\Gamma(z)\Gamma(1-z) = \pi/\sin\pi z$, followed by $z\Gamma(z) = \Gamma(1+z)$, gives

$$\Gamma(1+z)\Gamma(1-z) = \frac{\pi z}{\sin\pi z}.$$

In $|z| < 1$ the quotient has no zero or pole, with value one at zero. Choose its analytic logarithm to vanish there. The sum of the two gamma logarithms has that same value and exponential, so there is no additional $2\pi i$ constant. Dividing by two gives

$$\boxed{\frac{1}{2}\log\frac{\pi z}{\sin\pi z} = \sum_{n=1}^{\infty}\frac{\zeta(2n)}{2n}z^{2n}}, \qquad |z| < 1.$$

The corrected hint points to the power series actually needed.

13.2.7 Write out a Weierstrass infinite-product definition of $\ln\Gamma(z+1)$. Without differentiating, show that this leads directly to the Maclaurin expansion of $\ln\Gamma(z+1)$, Eq. (13.44).

Solution.

Weierstrass' product, after using $\Gamma(1+z) = z\Gamma(z)$, is

$$\frac{1}{\Gamma(1+z)} = e^{\gamma z}\prod_{k=1}^{\infty}\left(1+\frac{z}{k}\right)e^{-z/k}.$$

Taking the analytic logarithm that is zero at $z = 0$ gives

$$\log\Gamma(1+z) = -\gamma z + \sum_{k=1}^{\infty}\left[\frac{z}{k} - \log\left(1+\frac{z}{k}\right)\right].$$

For $|z| < 1$, expand each logarithm. Its linear term cancels:

$$\frac{z}{k} - \log\left(1+\frac{z}{k}\right) = \sum_{n=2}^{\infty}\frac{(-1)^n z^n}{nk^n}.$$

The double series is absolutely convergent, since

$$\sum_{n=2}^{\infty}\sum_{k=1}^{\infty}\frac{|z|^n}{nk^n} \le \zeta(2)\sum_{n=2}^{\infty}|z|^n < \infty.$$

Interchange the sums and identify $\sum_{k=1}^{\infty}k^{-n}$:

$$\boxed{\log\Gamma(1+z) = -\gamma z + \sum_{n=2}^{\infty}(-1)^n\frac{\zeta(n)}{n}z^n}.$$

No differentiation of the product was used.

13.2.8 Derive the difference relation for the polygamma function,

$$\psi^{(m)}(z+2) = \psi^{(m)}(z+1) + (-1)^m \frac{m!}{(z+1)^{m+1}}, \quad m = 0, 1, 2, \ldots.$$

Solution.

Differentiate $\Gamma(z+2) = (z+1)\Gamma(z+1)$ logarithmically at regular arguments. The result is

$$\psi(z+2) = \frac{1}{z+1} + \psi(z+1).$$

This already proves the case $m = 0$. For $m \geq 1$, repeated ordinary differentiation gives

$$\frac{d^m}{dz^m}(z+1)^{-1} = (-1)(-2)\cdots(-m)(z+1)^{-m-1} = \frac{(-1)^m m!}{(z+1)^{m+1}}.$$

Hence

$$\boxed{\psi^{(m)}(z+2) = \psi^{(m)}(z+1) + \frac{(-1)^m m!}{(z+1)^{m+1}}}, \qquad m = 0, 1, 2, \ldots.$$

These are identities of meromorphic functions; a displayed value at a pole is not a finite number. For integer positive arguments, the formulas recursively reduce to the values at one.

13.2.9 The Pochhammer symbol $(a)_n$ is defined (for integral n) as

$$(a)_n = a(a+1)\cdots(a+n-1), \quad (a)_0 = 1.$$

(a) Express $(a)_n$ in terms of factorials.

(b) Find $(d/da)(a)_n$ in terms of $(a)_n$ and digamma functions.

ANS. $\dfrac{d}{da}(a)_n = (a)_n\,[\psi(a+n) - \psi(a)]$.

(c) Show that

$$(a)_{n+k} = (a+n)_k \cdot (a)_n.$$

The finite-product indices n, k are nonnegative integers. Products involving singular gamma or digamma factors are evaluated by their polynomial continuation.

Solution.

The finite-product definition here uses nonnegative integers n and, in part (c), k. **(a)** Repeated gamma recurrence gives

$$\boxed{(a)_n = \frac{\Gamma(a+n)}{\Gamma(a)} = \frac{(a+n-1)!}{(a-1)!}}.$$

The factorials have their generalized meaning. The finite product is a polynomial in a and defines any removable value of this quotient at gamma poles.

(b) At points where all factors are nonzero, logarithmic differentiation of the finite product gives

$$\frac{d}{da}(a)_n = (a)_n \sum_{j=0}^{n-1} \frac{1}{a+j}.$$

Iterating $\psi(a+1)-\psi(a)=1/a$ identifies the finite sum:

$$\boxed{\frac{d}{da}(a)_n=(a)_n[\psi(a+n)-\psi(a)]}.$$

At exceptional arguments use the equivalent polynomial formula

$$\frac{d}{da}(a)_n=\sum_{j=0}^{n-1}\prod_{\substack{0\le\ell<n\\ \ell\neq j}}(a+\ell).$$

For example, at $a=-j$ with $0\le j<n$ its value is $(-1)^j j!(n-1-j)!$, even though the separated digamma factors have poles. For $n=0$ the empty product is one and its derivative is zero.

(c) Split the product after its first n factors:

$$\begin{aligned}(a)_{n+k}&=\left[\prod_{j=0}^{n-1}(a+j)\right]\left[\prod_{j=n}^{n+k-1}(a+j)\right]\\&=\boxed{(a)_n(a+n)_k}.\end{aligned}$$

This polynomial proof also includes zeros and empty products.

13.2.10 Verify the following special values of the ψ form of the digamma and polygamma functions:

$$\psi(1)=-\gamma,\quad \psi^{(1)}(1)=\zeta(2),\quad \psi^{(2)}(1)=-2\zeta(3).$$

Solution.

From the convergent digamma representation,

$$\psi(1+z)=-\gamma+\sum_{k=1}^{\infty}\left(\frac{1}{k}-\frac{1}{k+z}\right),$$

putting $z=0$ gives $\boxed{\psi(1)=-\gamma}$. Differentiation on compact sets away from the poles gives

$$\psi^{(1)}(1+z)=\sum_{k=1}^{\infty}\frac{1}{(k+z)^2},\qquad \psi^{(2)}(1+z)=-2\sum_{k=1}^{\infty}\frac{1}{(k+z)^3}.$$

Both differentiated sums converge absolutely and locally uniformly. Therefore at $z=0$,

$$\boxed{\psi^{(1)}(1)=\sum_{k=1}^{\infty}k^{-2}=\zeta(2)},\qquad \boxed{\psi^{(2)}(1)=-2\sum_{k=1}^{\infty}k^{-3}=-2\zeta(3)}.$$

13.2.11 Verify:

(a) $\displaystyle\int_0^{\infty}e^{-r}\ln r\,dr=-\gamma.$

(b) $\displaystyle\int_0^{\infty}re^{-r}\ln r\,dr=1-\gamma.$

(c) $\displaystyle\int_0^\infty r^n e^{-r}\ln r\,dr = (n-1)! + n\int_0^\infty r^{n-1}e^{-r}\ln r\,dr, \quad n = 1, 2, 3, \ldots.$

Hint. These may be verified by integration by parts, or by differentiating the Euler integral formula for $\Gamma(n+1)$ with respect to n.

Solution.

For s in a compact subset of $\Re s > 0$, differentiation of Euler's integral is justified by the integrable bounds $r^{\varepsilon-1}|\log r|$ near zero and a polynomial times $e^{-r}|\log r|$ at infinity. Hence

$$\Gamma'(s) = \int_0^\infty r^{s-1}e^{-r}\log r\,dr = \Gamma(s)\psi(s).$$

(a) At $s = 1$, $\Gamma(1) = 1$ and $\psi(1) = -\gamma$, so the integral is $\boxed{-\gamma}$.

(c) Write $I_n = \int_0^\infty r^n e^{-r}\log r\,dr$. For integers $n \geq 1$, integrate by parts with $u = r^n\log r$, $dv = e^{-r}dr$. Since $r^n\log r \to 0$ at zero and the exponential removes the upper boundary term,

$$\begin{aligned} I_n &= \left[-r^n e^{-r}\log r\right]_0^\infty + \int_0^\infty e^{-r}(nr^{n-1}\log r + r^{n-1})dr \\ &= \boxed{nI_{n-1} + (n-1)!}. \end{aligned}$$

(b) In particular, $I_1 = I_0 + 0! = \boxed{1-\gamma}$. The general closed form is $I_n = n!(H_n - \gamma)$, as is checked by this recurrence and $I_0 = -\gamma$. Integration by parts at $n = 0$ would have a divergent boundary term, which is why part (a) was evaluated before using this recurrence.

13.2.12 Dirac relativistic wave functions for hydrogen involve factors such as $\Gamma[2(1-\alpha^2Z^2)^{1/2}+1]$ where α, the fine structure constant, is approximated here by 1/137 and Z is the atomic number. Expand $\Gamma[2(1-\alpha^2Z^2)^{1/2}+1]$ in a series of powers of α^2Z^2.

Solution.

Let $q = \alpha^2Z^2$ and expand about $q = 0$, using the square-root branch that equals one there. The binomial series gives

$$2\sqrt{1-q}+1 = 3 + \delta, \qquad \delta = -q - \frac{q^2}{4} - \frac{q^3}{8} + O(q^4).$$

Set $p = \psi(3)$, $r = \psi^{(1)}(3)$ and $s = \psi^{(2)}(3)$. The recurrences in Exercise 13.2.8 yield

$$p = \frac{3}{2} - \gamma, \qquad r = \zeta(2) - 1 - \frac{1}{4} = \zeta(2) - \frac{5}{4}, \qquad s = -2\zeta(3) + 2 + \frac{1}{4} = \frac{9}{4} - 2\zeta(3).$$

Differentiating $\Gamma' = \Gamma\psi$ twice more gives

$$\Gamma'' = \Gamma(\psi^2 + \psi'), \qquad \Gamma^{(3)} = \Gamma(\psi^3 + 3\psi\psi' + \psi'').$$

Since $\Gamma(3) = 2$, Taylor expansion in δ is

$$\Gamma(3+\delta) = 2\left[1 + p\delta + \frac{p^2+r}{2}\delta^2 + \frac{p^3+3pr+s}{6}\delta^3 + O(\delta^4)\right].$$

Now $\delta^2 = q^2 + q^3/2 + O(q^4)$ and $\delta^3 = -q^3 + O(q^4)$. Substitution and collection give

$$\begin{aligned} \Gamma(2\sqrt{1-q}+1) = 2 - 2pq + \left(p^2 + r - \frac{p}{2}\right)q^2 \\ + \left[-\frac{p}{4} + \frac{p^2+r}{2} - \frac{p^3+3pr+s}{3}\right]q^3 + O(q^4). \end{aligned}$$

In particular, the first three terms in the original variables are

$$\boxed{2 + (2\gamma - 3)\alpha^2 Z^2 + \left[\gamma^2 - \frac{5}{2}\gamma + \zeta(2) + \frac{1}{4}\right]\alpha^4 Z^4 + O(\alpha^6 Z^6)}.$$

Higher terms follow from the same two Taylor expansions. The expansion is local at $q = 0$ and converges for $|q| < 1$; there the chosen square root has positive real part, so its gamma argument encounters no gamma pole. Physically, a short truncation requires $\alpha^2 Z^2$ small.

13.2.13 The quantum mechanical description of a particle in a Coulomb field requires a knowledge of the argument of $\Gamma(z)$ when z is complex. Determine the argument of $\Gamma(1 + ib)$ for small, real b.

Solution.

Choose the argument continuously from $b = 0$, where $\Gamma(1) = 1$ and the argument is zero. For $|b| < 1$, substitute $z = ib$ in

$$\log\Gamma(1 + z) = -\gamma z + \sum_{n=2}^{\infty}(-1)^n \frac{\zeta(n)}{n} z^n.$$

Even powers of ib are real. For the odd powers $n = 2m + 1$, $(-1)^{2m+1} i^{2m+1} = (-1)^{m+1} i$. Therefore

$$\boxed{\arg\Gamma(1 + ib) = -\gamma b + \frac{\zeta(3)}{3} b^3 - \frac{\zeta(5)}{5} b^5 + \frac{\zeta(7)}{7} b^7 - \cdots}.$$

The leading approximation is $-\gamma b$, with an error of order b^3. This is the continuous local phase (and agrees with the principal phase for sufficiently small b); arbitrary arguments differ by 2π multiples. Its oddness in b also follows from $\Gamma(1 - ib) = \overline{\Gamma(1 + ib)}$.

13.2.14 Using digamma and polygamma functions, sum the series

(a) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$, (b) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n^2 - 1}$.

Note. You can use Exercise 13.2.8 to calculate the needed digamma functions.

Solution.

Use the digamma identity

$$\psi(1 + z) + \gamma = \sum_{j=1}^{\infty} \frac{z}{j(j + z)}.$$

(a) At $z = 1$ the requested sum is $\psi(2) + \gamma$. The recurrence $\psi(2) = \psi(1) + 1 = -\gamma + 1$ gives

$$\boxed{\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1}.$$

(b) Put $j = n - 1$; then $n^2 - 1 = j(j + 2)$ and

$$\sum_{n=2}^{\infty} \frac{1}{n^2 - 1} = \sum_{j=1}^{\infty} \frac{1}{j(j+2)} = \frac{1}{2}[\psi(3) + \gamma]$$
$$= \frac{1}{2}\left(1 + \frac{1}{2}\right) = \boxed{\frac{3}{4}}.$$

As a separate check, the finite partial sums are $1 - 1/(N+1)$ in (a) and $\frac{1}{2}[1 + 1/2 - 1/N - 1/(N+1)]$ in (b), respectively; their limits give the same answers.

13.2.15 Show that

$$\sum_{n=1}^{\infty} \frac{1}{(n+a)(n+b)} = \frac{1}{(b-a)} \left[\psi(1+b) - \psi(1+a)\right],$$

where $a \neq b$, and neither a nor b is a negative integer. It is of some interest to compare this summation with the corresponding integral,

$$\int_1^{\infty} \frac{dx}{(x+a)(x+b)} = \frac{1}{b-a} \left[\ln(1+b) - \ln(1+a)\right].$$

The relation between $\psi(x)$ and $\ln x$ is made explicit in the analysis leading to Stirling's formula.

For the displayed ordinary real improper integral and real logarithms, require real $a, b > -1$ as well as $a \neq b$. The series has the wider domain stated above.

Solution.

For the stated admissible a, b, every denominator is nonzero and the summands are $O(n^{-2})$, so the series converges absolutely. Subtract the two convergent digamma representations:

$$\begin{aligned}\psi(1+b) - \psi(1+a) &= \sum_{n=1}^{\infty} \left(\frac{1}{n+a} - \frac{1}{n+b}\right) \\ &= (b-a) \sum_{n=1}^{\infty} \frac{1}{(n+a)(n+b)}.\end{aligned}$$

Division by $b - a$ proves

$$\boxed{\sum_{n=1}^{\infty} \frac{1}{(n+a)(n+b)} = \frac{\psi(1+b) - \psi(1+a)}{b-a}}.$$

The removable case $b = a$ is $\sum_{n=1}^{\infty} (n+a)^{-2} = \psi^{(1)}(1+a)$.

The accompanying real improper integral requires stronger conditions, for example real $a, b > -1$. Merely excluding negative integers does not prevent an interior pole: $a = -3/2$, $b = 0$ gives one at $x = 3/2$. Under $a, b > -1$, partial fractions give

$$\begin{aligned}\int_1^R \frac{dx}{(x+a)(x+b)} &= \frac{1}{b-a} \Big[\ln(x+a) - \ln(x+b)\Big]_1^R \\ &= \frac{1}{b-a} \left[\ln \frac{R+a}{R+b} + \ln(1+b) - \ln(1+a)\right].\end{aligned}$$

The first logarithm tends to zero, proving the quoted integral. When $b \to a > -1$, its value tends to $1/(1+a)$, whereas the series tends to the trigamma value above.

13.3 The Beta Function

13.3.1 Verify the following beta function identities:

(a) $B(a,b) = B(a+1,b) + B(a,b+1)$,

(b) $B(a,b) = \dfrac{a+b}{b} B(a,b+1)$,

(c) $B(a,b) = \dfrac{b-1}{a} B(a+1,b-1)$,

(d) $B(a,b)B(a+b,c) = B(b,c)B(a,b+c)$.

Solution.

Start where the beta integrals converge; in part (c) take $\Re b > 1$ as well as $\Re a > 0$. The results then extend as meromorphic identities wherever the displayed values are defined.

(a) Insert $1 = t + (1-t)$ into the defining integral:

$$\begin{aligned} B(a,b) &= \int_0^1 t^{a-1}(1-t)^{b-1}[t + (1-t)]dt \\ &= \boxed{B(a+1,b) + B(a,b+1)}. \end{aligned}$$

(b) Gamma recurrence gives

$$B(a,b+1) = \frac{\Gamma(a)b\Gamma(b)}{(a+b)\Gamma(a+b)} = \frac{b}{a+b} B(a,b),$$

so $\boxed{B(a,b) = \dfrac{a+b}{b} B(a,b+1)}$.

(c) Use $\Gamma(a+1) = a\Gamma(a)$ and $\Gamma(b-1) = \Gamma(b)/(b-1)$:

$$B(a+1,b-1) = \frac{a\Gamma(a)\Gamma(b)}{(b-1)\Gamma(a+b)} = \frac{a}{b-1} B(a,b).$$

Rearranging gives the requested identity. When one of the separated factors has a pole, an equivalent continued identity must be interpreted by a limit, not by multiplying undefined values.

(d) Multiplication cancels the intermediate gamma factor:

$$\begin{aligned} B(a,b)B(a+b,c) &= \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \frac{\Gamma(a+b)\Gamma(c)}{\Gamma(a+b+c)} \\ &= \frac{\Gamma(a)\Gamma(b)\Gamma(c)}{\Gamma(a+b+c)} \\ &= \frac{\Gamma(b)\Gamma(c)}{\Gamma(b+c)} \frac{\Gamma(a)\Gamma(b+c)}{\Gamma(a+b+c)} \\ &= \boxed{B(b,c)B(a,b+c)}. \end{aligned}$$

13.3.2

(a) Show that

$$\int_{-1}^{1}(1-x^2)^{1/2}x^{2n}\,dx=\begin{cases}\pi/2, & n=0\\ \pi\dfrac{(2n-1)!!}{(2n+2)!!}, & n=1,2,3,\ldots.\end{cases}$$

(b) Show that

$$\int_{-1}^{1}(1-x^2)^{-1/2}x^{2n}\,dx=\begin{cases}\pi, & n=0,\\ \pi\dfrac{(2n-1)!!}{(2n)!!}, & n=1,2,3,\ldots.\end{cases}$$

Solution.

Both integrands are even. In twice the integral from zero to one, put $u=x^2$, so $2\,dx=u^{-1/2}du$.
(a) This gives

$$\begin{aligned}\int_{-1}^{1}(1-x^2)^{1/2}x^{2n}dx&=\int_0^1 u^{n-1/2}(1-u)^{1/2}du\\ &=B\left(n+\frac{1}{2},\frac{3}{2}\right)=\frac{\Gamma(n+1/2)\Gamma(3/2)}{\Gamma(n+2)}\\ &=\frac{[(2n-1)!!\sqrt{\pi}/2^n](\sqrt{\pi}/2)}{(n+1)!}=\boxed{\pi\frac{(2n-1)!!}{(2n+2)!!}}.\end{aligned}$$

Here $(2n+2)!!=2^{n+1}(n+1)!$. With $(-1)!!=1$ this includes $n=0$, giving $\pi/2$.

(b) The same substitution changes the second integral to

$$\begin{aligned}B\left(n+\frac{1}{2},\frac{1}{2}\right)&=\frac{\Gamma(n+1/2)\Gamma(1/2)}{\Gamma(n+1)}\\ &=\frac{(2n-1)!!\pi}{2^n n!}=\boxed{\pi\frac{(2n-1)!!}{(2n)!!}}.\end{aligned}$$

This also includes $n=0$, because $0!!=1$. The endpoint powers $\pm1/2$ are greater than -1, so both improper integrals exist.

13.3.3 Show that

$$\int_{-1}^{1}(1-x^2)^n\,dx=\frac{2(2n)!!}{(2n+1)!!},\quad n=0,1,2,\ldots.$$

Solution.

Evenness followed by $u=x^2$ yields

$$\int_{-1}^{1}(1-x^2)^n dx=2\int_0^1(1-x^2)^n dx=\int_0^1 u^{-1/2}(1-u)^n du=B\left(\frac{1}{2},n+1\right).$$

Now $\Gamma(n+3/2)=(2n+1)!!\sqrt{\pi}/2^{n+1}$, so

$$B\left(\frac{1}{2},n+1\right)=\frac{\sqrt{\pi}\,n!}{(2n+1)!!\sqrt{\pi}/2^{n+1}}=\boxed{\frac{2(2n)!!}{(2n+1)!!}}.$$

At $n=0$ this is two, the length of the interval, checking the empty-product convention.

13.3.4 Evaluate $\displaystyle\int_{-1}^{1}(1+x)^a(1-x)^b\,dx$ in terms of the beta function.

ANS. $2^{a+b+1}B(a+1,b+1)$.

Require $\Re a, \Re b > -1$ for this improper integral.

Solution.

The improper integral requires $\Re a > -1$ and $\Re b > -1$; the bases inside the interval are positive. Put $u = (1+x)/2$, so $x = 2u-1$, $dx = 2du$, $1+x = 2u$ and $1-x = 2(1-u)$. The limits $x = -1, 1$ become $u = 0, 1$. Therefore

$$\begin{aligned}\int_{-1}^{1}(1+x)^a(1-x)^b dx &= 2^{a+b+1}\int_0^1 u^a(1-u)^b du\\ &= \boxed{2^{a+b+1}B(a+1,b+1)}\\ &= 2^{a+b+1}\frac{\Gamma(a+1)\Gamma(b+1)}{\Gamma(a+b+2)}.\end{aligned}$$

The factor 2^{a+b+1} includes the Jacobian two. Outside the stated convergence domain the gamma expression may continue meromorphically, but that continuation is not the original convergent improper integral.

13.3.5 Show, by means of the beta function, that

$$\int_t^z \frac{dx}{(z-x)^{1-\alpha}(x-t)^\alpha} = \frac{\pi}{\sin\pi\alpha}, \quad 0<\alpha<1.$$

Take real $t < z$.

Solution.

Take real $t < z$ for the displayed real integral. Set $u = (x-t)/(z-t)$. Then

$$x - t = (z-t)u, \quad z - x = (z-t)(1-u), \quad dx = (z-t)du.$$

The denominator has total power $(1-\alpha)+\alpha = 1$, so its factor $z-t$ cancels the Jacobian. Hence

$$\begin{aligned}\int_t^z \frac{dx}{(z-x)^{1-\alpha}(x-t)^\alpha} &= \int_0^1 u^{-\alpha}(1-u)^{\alpha-1}du\\ &= B(1-\alpha,\alpha) = \Gamma(1-\alpha)\Gamma(\alpha)\\ &= \boxed{\frac{\pi}{\sin\pi\alpha}}.\end{aligned}$$

The conditions $1-\alpha > 0$ and $\alpha > 0$ are exactly the two endpoint convergence requirements. Reflection gives the last equality; the interval length has canceled completely.

13.3.6 Show that the Dirichlet integral

$$\iint x^p y^q\,dx\,dy = \frac{p!\,q!}{(p+q+2)!} = \frac{B(p+1,q+1)}{p+q+2},$$

where the range of integration is the triangle bounded by the positive x- and y-axes and the line $x + y = 1$.

The integral requires $\Re p, \Re q > -1$; factorials outside nonnegative integers mean gamma functions.

Solution.

The integral converges for $\Re p, \Re q > -1$; ordinary integer factorials correspond to $p, q \geq 0$. Describe the triangle by $0 \leq x \leq 1$ and $0 \leq y \leq 1 - x$. Integrate with respect to y first:

$$\begin{aligned}\int_0^1 \int_0^{1-x} x^p y^q \, dy \, dx &= \frac{1}{q+1} \int_0^1 x^p (1-x)^{q+1} dx \\ &= \frac{B(p+1, q+2)}{q+1} \\ &= \frac{\Gamma(p+1)\Gamma(q+2)}{(q+1)\Gamma(p+q+3)} \\ &= \boxed{\frac{\Gamma(p+1)\Gamma(q+1)}{\Gamma(p+q+3)}}.\end{aligned}$$

Since $\Gamma(q+2) = (q+1)\Gamma(q+1)$, the inner-integral factor has canceled. Recurrence in the remaining denominator gives

$$\frac{\Gamma(p+1)\Gamma(q+1)}{\Gamma(p+q+3)} = \boxed{\frac{B(p+1, q+1)}{p+q+2}} = \frac{p!q!}{(p+q+2)!}$$

for nonnegative integers. At $p = q = 0$ the value is $1/2$, the area of the triangle.

13.3.7 Show that

$$\int_0^\infty \int_0^\infty e^{-(x^2+y^2+2xy\cos\theta)} \, dx \, dy = \frac{\theta}{2\sin\theta}.$$

What are the limits on θ?

Hint. Consider oblique xy-coordinates.

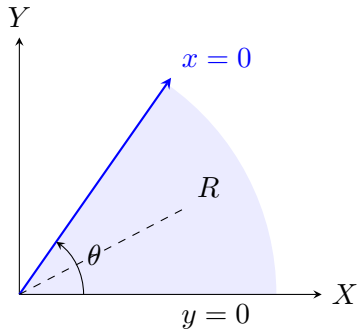
ANS. $-\pi < \theta < \pi$ for the unreduced formula, with value $1/2$ at $\theta = 0$ by continuity. The integral itself converges for every real θ except odd multiples of π.

Solution.

First take $0 < \theta < \pi$. Introduce Cartesian variables

$$X = x + y\cos\theta, \qquad Y = y\sin\theta.$$

Then $X^2 + Y^2 = x^2 + y^2 + 2xy\cos\theta$ and $\partial(X, Y)/\partial(x, y) = \sin\theta > 0$. The original positive quadrant maps onto the sector between the positive X-axis ($y = 0$) and the ray at angle θ ($x = 0$).



In polar coordinates $X = R\cos\phi$, $Y = R\sin\phi$,

$$I(\theta) = \frac{1}{\sin\theta}\int_0^\theta d\phi \int_0^\infty e^{-R^2} R\, dR$$
$$= \frac{\theta}{\sin\theta}\left[-\frac{1}{2}e^{-R^2}\right]_0^\infty = \boxed{\frac{\theta}{2\sin\theta}}.$$

Both the original integral and this quotient are even in θ, so the result extends to $-\pi < \theta < 0$. At zero use $u = x + y$, $v = x$, where $0 \le v \le u$:

$$I(0) = \int_0^\infty e^{-u^2}\int_0^u dv\, du = \int_0^\infty u e^{-u^2} du = \frac{1}{2},$$

the removable limit of the formula.

The stated interval $-\pi < \theta < \pi$ is the range for the unreduced formula $\theta/(2\sin\theta)$, including its removable value at zero. For unrestricted real θ, the integral is periodic and converges except at odd multiples of π. Indeed, if $c = \cos\theta > -1$, the quadratic form on $x, y \ge 0$ is at least $\min(1, 1 + c)(x^2 + y^2)$, so a convergent Gaussian dominates it. If $c = -1$, the integrand is $e^{-(x-y)^2}$; on $x \ge 1$, $x \le y \le x + 1$ it is at least e^{-1}, giving infinite integral. In general reduce to $\vartheta = \arccos(\cos\theta) \in [0, \pi]$ and use $\vartheta/(2\sin\vartheta)$, with the endpoint conventions just established.

13.3.8 Evaluate (using the beta function)

(a) $$\int_0^{\pi/2} \cos^{1/2}\theta\, d\theta = \frac{(2\pi)^{3/2}}{16[\Gamma(5/4)]^2},$$

(b) $$\int_0^{\pi/2} \cos^n\theta\, d\theta = \int_0^{\pi/2} \sin^n\theta\, d\theta = \frac{\sqrt{\pi}\,[(n-1)/2]!}{2(n/2)!}$$
$$= \begin{cases} \dfrac{(n-1)!!}{n!!} & \text{for } n \text{ odd}, \\ \dfrac{\pi}{2}\cdot\dfrac{(n-1)!!}{n!!} & \text{for } n \text{ even}. \end{cases}$$

The general integral requires $\Re n > -1$; its odd/even forms use nonnegative integer n and $(-1)!! = 0!! = 1$.

Solution.

For $u = \sin^2\theta$, the limits become zero and one and

$$du = 2\sin\theta\cos\theta\, d\theta, \qquad d\theta = \frac{du}{2\sqrt{u(1-u)}}, \qquad \cos^n\theta = (1-u)^{n/2}.$$

Thus the common form is

$$I_n = \int_0^{\pi/2} \cos^n\theta\, d\theta = \frac{1}{2}\int_0^1 u^{-1/2}(1-u)^{(n-1)/2} du = \frac{\sqrt{\pi}\,\Gamma((n+1)/2)}{2\Gamma((n+2)/2)}.$$

This integral formula requires $\Re n > -1$; the odd/even factorial forms below use nonnegative integer n. Replacing θ by $\pi/2 - \theta$ proves the same value for $\int_0^{\pi/2}\sin^n\theta\, d\theta$.

(a) At $n = 1/2$, $I_{1/2} = \sqrt{\pi}\,\Gamma(3/4)/[2\Gamma(5/4)]$. Reflection and recurrence give

$$\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}, \qquad \Gamma(1/4) = 4\Gamma(5/4).$$

Thus

$$\boxed{I_{1/2} = \frac{\pi\sqrt{2\pi}}{8[\Gamma(5/4)]^2} = \frac{(2\pi)^{3/2}}{16[\Gamma(5/4)]^2}}.$$

(b) The gamma expression is precisely $\sqrt{\pi}\,[(n-1)/2]!/[2(n/2)!]$. For $n = 2m+1$,

$$I_{2m+1} = \frac{\sqrt{\pi}\,m!}{2\Gamma(m+3/2)} = \frac{2^m m!}{(2m+1)!!} = \boxed{\frac{(2m)!!}{(2m+1)!!}}.$$

For $n = 2m$,

$$I_{2m} = \frac{\sqrt{\pi}\,\Gamma(m+1/2)}{2m!} = \frac{\pi}{2}\frac{(2m-1)!!}{2^m m!} = \boxed{\frac{\pi}{2}\frac{(2m-1)!!}{(2m)!!}}.$$

The case $m = 0$ in the even formula gives $I_0 = \pi/2$.

13.3.9 Evaluate $\displaystyle\int_0^1 (1-x^4)^{-1/2}\,dx$ as a beta function.

ANS. $\dfrac{[\Gamma(5/4)]^2 \cdot 4}{(2\pi)^{1/2}} = 1.311028777.$

Solution.

Put $u = x^4$, so $dx = \frac{1}{4}u^{-3/4}du$. The endpoint exponents $-3/4$ and $-1/2$ are both integrable, and

$$I = \frac{1}{4}\int_0^1 u^{-3/4}(1-u)^{-1/2}du = \boxed{\frac{1}{4}B\left(\frac{1}{4},\frac{1}{2}\right)} = \frac{\Gamma(1/4)\sqrt{\pi}}{4\Gamma(3/4)}.$$

Use $\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$ to remove $\Gamma(3/4)$, then $\Gamma(1/4) = 4\Gamma(5/4)$:

$$I = \frac{[\Gamma(1/4)]^2}{4\sqrt{2\pi}} = \boxed{\frac{4[\Gamma(5/4)]^2}{\sqrt{2\pi}}} = 1.311028777146\ldots.$$

For an independent numerical quadrature one can also put $x = \sin\phi$; cancellation of $\cos\phi$ gives the nonsingular integral $\int_0^{\pi/2}(1+\sin^2\phi)^{-1/2}d\phi$.

13.3.10 Using beta functions, show that the integral representation

$$J_\nu(z) = \frac{2}{\pi^{1/2}\Gamma(\nu+\frac{1}{2})}\left(\frac{z}{2}\right)^\nu \int_0^{\pi/2} \sin^{2\nu}\theta\cos(z\cos\theta)\,d\theta, \quad \Re(\nu) > -\tfrac{1}{2},$$

reduces to the Bessel series

$$J_\nu(z) = \sum_{s=0}^{\infty}(-1)^s \frac{1}{s!\,\Gamma(s+\nu+1)}\left(\frac{z}{2}\right)^{2s+\nu},$$

thereby confirming its validity.

Solution.

Expand the cosine, keeping the same branch of $(z/2)^\nu$ throughout:

$$\cos(z\cos\theta) = \sum_{s=0}^{\infty}\frac{(-1)^s z^{2s}\cos^{2s}\theta}{(2s)!}.$$

For fixed z, the absolute sum is bounded by $e^{|z|}$. The remaining weight has magnitude $\sin^{2\Re\nu}\theta$, which is integrable at zero exactly when $\Re\nu > -1/2$. Dominated convergence therefore permits integration term by term. The beta integral for each term is

$$\int_0^{\pi/2} \sin^{2\nu}\theta\cos^{2s}\theta\,d\theta = \frac{1}{2}B\left(\nu+\frac{1}{2}, s+\frac{1}{2}\right) = \frac{\Gamma(\nu+1/2)\Gamma(s+1/2)}{2\Gamma(s+\nu+1)}.$$

Insert this in the proposed representation and cancel its 2 and $\Gamma(\nu+1/2)$ factors:

$$J_\nu(z) = \left(\frac{z}{2}\right)^\nu \sum_{s=0}^{\infty} \frac{(-1)^s z^{2s}}{\sqrt{\pi}(2s)!}\frac{\Gamma(s+1/2)}{\Gamma(s+\nu+1)}.$$

Finally,

$$\Gamma\left(s+\frac{1}{2}\right) = \frac{(2s)!\sqrt{\pi}}{4^s s!}, \qquad \frac{\Gamma(s+1/2)}{\sqrt{\pi}(2s)!} = \frac{1}{4^s s!}.$$

Thus

$$\boxed{J_\nu(z) = \sum_{s=0}^{\infty} \frac{(-1)^s}{s!\Gamma(s+\nu+1)}\left(\frac{z}{2}\right)^{2s+\nu}},$$

the defining Bessel series. For noninteger ν both representations use a chosen branch of z^ν; a value at zero is taken only when the corresponding limit exists.

13.3.11 Given the associated Legendre function, defined in Chapter 15,

$$P_m^m(x) = (2m-1)!!\,(1-x^2)^{m/2},$$

show that

(a) $\displaystyle\int_{-1}^{1}[P_m^m(x)]^2\,dx = \frac{2}{2m+1}(2m)!, \quad m = 0, 1, 2, \ldots,$

(b) $\displaystyle\int_{-1}^{1}[P_m^m(x)]^2\frac{dx}{1-x^2} = 2\cdot(2m-1)!, \quad m = 1, 2, 3, \ldots.$

Solution.

Square the stated expression: $[P_m^m(x)]^2 = [(2m-1)!!]^2(1-x^2)^m$. **(a)** Exercise 13.3.3 gives

$$\begin{aligned}\int_{-1}^{1}[P_m^m(x)]^2dx &= [(2m-1)!!]^2\frac{2(2m)!!}{(2m+1)!!}\\ &= \frac{2(2m-1)!!(2m)!!}{2m+1} = \boxed{\frac{2(2m)!}{2m+1}}.\end{aligned}$$

We used $(2m+1)!! = (2m+1)(2m-1)!!$ and $(2m-1)!!(2m)!! = (2m)!$. The case $m = 0$ gives two.

(b) Division by $1-x^2$ reduces the exponent to $m-1$. For $m \geq 1$ the same beta evaluation yields

$$\begin{aligned}\int_{-1}^{1}[P_m^m(x)]^2\frac{dx}{1-x^2} &= [(2m-1)!!]^2\frac{2(2m-2)!!}{(2m-1)!!}\\ &= 2(2m-1)!!(2m-2)!! = \boxed{2(2m-1)!}.\end{aligned}$$

At $m = 0$ the second integral diverges at both endpoints, explaining its different lower limit on m. An alternative associated-Legendre convention introduces $(-1)^m$, which has no effect on either squared integral.

13.3.12 Show that, for integers p and q,

(a) $\displaystyle\int_0^1 x^{2p+1}(1-x^2)^{-1/2}\,dx = \frac{(2p)!!}{(2p+1)!!}$,

(b) $\displaystyle\int_0^1 x^{2p}(1-x^2)^{q}\,dx = \frac{(2p-1)!!\,(2q)!!}{(2p+2q+1)!!}$.

Take $p \geq 0$ in (a), and $p, q \geq 0$ in (b).

Solution.

For the stated integer parameters, require $p \geq 0$ in (a), and $p, q \geq 0$ in (b). Negative integer values produce nonintegrable endpoint powers. Use $u = x^2$, $dx = du/(2\sqrt{u})$. **(a)** Since $x^{2p+1}dx = u^p du/2$,

$$\int_0^1 x^{2p+1}(1-x^2)^{-1/2}dx = \frac{1}{2}B\left(p+1,\frac{1}{2}\right) = \frac{p!\sqrt{\pi}}{2\Gamma(p+3/2)}$$
$$= \frac{p!\sqrt{\pi}\,2^{p+1}}{2(2p+1)!!\sqrt{\pi}} = \boxed{\frac{(2p)!!}{(2p+1)!!}}.$$

(b) Here $x^{2p}dx = u^{p-1/2}du/2$, so

$$\int_0^1 x^{2p}(1-x^2)^{q}dx = \frac{1}{2}B\left(p+\frac{1}{2},q+1\right) = \frac{\Gamma(p+1/2)q!}{2\Gamma(p+q+3/2)}$$
$$= \frac{[(2p-1)!!\sqrt{\pi}/2^p]q!}{2[(2p+2q+1)!!\sqrt{\pi}/2^{p+q+1}]}$$
$$= \boxed{\frac{(2p-1)!!(2q)!!}{(2p+2q+1)!!}}.$$

These formulas include $p = 0$ or $q = 0$ with the usual empty products $(-1)!! = 0!! = 1$.

13.3.13 A particle of mass m moving in a symmetric potential that is well described by $V(x) = A|x|^n$ has a total energy $\frac{1}{2}m(dx/dt)^2 + V(x) = E$. Solving for dx/dt and integrating we find that the period of motion is

$$\tau = 2\sqrt{2m}\int_0^{x_{\max}} \frac{dx}{(E-Ax^n)^{1/2}},$$

where $x_{\max}$ is a classical turning point given by $Ax_{\max}^n = E$. Show that

$$\tau = \frac{2}{n}\sqrt{\frac{2\pi m}{E}}\left(\frac{E}{A}\right)^{1/n}\frac{\Gamma(1/n)}{\Gamma(1/n+\frac{1}{2})}.$$

Assume $m, A, E, n > 0$.

Solution.

For the bounded oscillation considered here, take $m, A, E > 0$ and $n > 0$. The positive turning point is $x_{\max} = (E/A)^{1/n}$. On the outward quarter-cycle, energy conservation gives

$$\frac{dx}{dt} = \sqrt{\frac{2}{m}}\sqrt{E-Ax^n}, \qquad \frac{\tau}{4} = \sqrt{\frac{m}{2}}\int_0^{x_{\max}}\frac{dx}{\sqrt{E-Ax^n}},$$

which explains the factor in the question. Set $u = Ax^n/E = (x/x_{\max})^n$. Then

$$x = x_{\max}u^{1/n}, \quad dx = \frac{x_{\max}}{n}u^{1/n-1}du, \quad (E - Ax^n)^{-1/2} = E^{-1/2}(1-u)^{-1/2}.$$

Both endpoints become zero and one, respectively. Consequently

$$\begin{aligned}\tau &= \frac{2}{n}\sqrt{\frac{2m}{E}}\left(\frac{E}{A}\right)^{1/n}\int_0^1 u^{1/n-1}(1-u)^{-1/2}du \\ &= \frac{2}{n}\sqrt{\frac{2m}{E}}\left(\frac{E}{A}\right)^{1/n}B\left(\frac{1}{n},\frac{1}{2}\right) \\ &= \boxed{\frac{2}{n}\sqrt{\frac{2\pi m}{E}}\left(\frac{E}{A}\right)^{1/n}\frac{\Gamma(1/n)}{\Gamma(1/n+1/2)}}.\end{aligned}$$

As a check, $n = 2$ and $A = m\omega^2/2$ give $\tau = \pi\sqrt{2m/A} = 2\pi/\omega$, the harmonic-oscillator period, independent of E.

13.3.14 Referring to Exercise 13.3.13,

(a) Determine the limit as $n \to \infty$ of

$$\frac{2}{n}\sqrt{\frac{2\pi m}{E}}\left(\frac{E}{A}\right)^{1/n}\frac{\Gamma(1/n)}{\Gamma(1/n+\frac{1}{2})}.$$

(b) Find $\lim_{n\to\infty}\tau$ from the behavior of the integrand $(E - Ax^n)^{-1/2}$.

(c) Investigate the behavior of the physical system (potential well) as $n \to \infty$. Obtain the period from inspection of this limiting physical system.

For a dimensionful family, write $V_n(x) = A(|x|/L)^n$ with fixed $A, L > 0$; the displayed formula uses length units $L = 1$.

Solution.

(a) Use the recurrence before taking the limit:

$$\frac{\Gamma(1/n)}{n\Gamma(1/n+1/2)} = \frac{\Gamma(1+1/n)}{\Gamma(1/n+1/2)} \longrightarrow \frac{\Gamma(1)}{\Gamma(1/2)} = \frac{1}{\sqrt{\pi}}.$$

Also $(E/A)^{1/n} \to 1$ for fixed positive numerical E/A. Hence the period written in the question tends to

$$\boxed{\tau_\infty = 2\sqrt{\frac{2m}{E}}}.$$

When comparing dimensionful potentials at different n, fix a length L and write $V_n(x) = A(|x|/L)^n$ with fixed energy scale A. The question uses length units with $L = 1$; the dimensionful limiting period is $2L\sqrt{2m/E}$.

(b) To justify passing to the limit despite the moving turning point, set $x = x_{\max}u$:

$$\tau_n = 2\sqrt{\frac{2m}{E}}\,x_{\max}\int_0^1 (1-u^n)^{-1/2}du.$$

For $0 \le u < 1$, this integrand tends to one. For $n \ge 1$, $u^n \le u$, so

$$0 < (1-u^n)^{-1/2} \le (1-u)^{-1/2}, \qquad \int_0^1 (1-u)^{-1/2}du = 2.$$

Dominated convergence gives an integral limit of one, and $x_{\max} = L(E/A)^{1/n} \to L$. This proves the same limit without ignoring the endpoint singularity.

(c) For $|x| < L$, $V_n(x) \to 0$; for $|x| > L$, $V_n(x) \to \infty$. The limiting motion is free flight between reflecting walls at $\pm L$, with speed $v = \sqrt{2E/m}$. A round trip covers distance $4L$, so

$$\boxed{\tau_\infty = \frac{4L}{v} = 2L\sqrt{\frac{2m}{E}}}.$$

The isolated value $V_n(\pm L) = A$ does not affect the limiting travel time derived in (b).

13.3.15 Show that

$$\int_0^\infty \frac{\sinh^\alpha x}{\cosh^\beta x}\,dx = \frac{1}{2}B\left(\frac{\alpha+1}{2}, \frac{\beta-\alpha}{2}\right), \quad -1 < \alpha < \beta.$$

Hint. Let $\sinh^2 x = u$.

Solution.

The endpoint behavior explains the given restrictions: near zero $\sinh^\alpha x/\cosh^\beta x \sim x^\alpha$, whereas at infinity it is asymptotic to $2^{\beta-\alpha}e^{(\alpha-\beta)x}$. Thus for real parameters convergence requires $\alpha > -1$ and $\beta > \alpha$.

Set $u = \sinh^2 x$. Since $\cosh^2 x = 1 + u$,

$$du = 2\sinh x \cosh x\,dx = 2\sqrt{u(1+u)}\,dx.$$

It follows that

$$\int_0^\infty \frac{\sinh^\alpha x}{\cosh^\beta x}dx = \frac{1}{2}\int_0^\infty u^{(\alpha-1)/2}(1+u)^{-(\beta+1)/2}du.$$

To recognize the beta integral, put $v = u/(1+u)$, $du = dv/(1-v)^2$. In general this shows

$$\int_0^\infty u^{p-1}(1+u)^{-p-q}du = \int_0^1 v^{p-1}(1-v)^{q-1}dv = B(p,q).$$

Here $p = (\alpha+1)/2$, $q = (\beta-\alpha)/2$, both positive. Therefore

$$\boxed{\int_0^\infty \frac{\sinh^\alpha x}{\cosh^\beta x}dx = \frac{1}{2}B\left(\frac{\alpha+1}{2}, \frac{\beta-\alpha}{2}\right)}.$$

For complex parameters the corresponding absolute-convergence conditions are $\Re\alpha > -1$ and $\Re(\beta-\alpha) > 0$.

13.3.16 The beta distribution of probability theory has a probability density

$$f(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1},$$

with x restricted to the interval $(0,1)$. Show that

(a) $\langle x \rangle$, the mean value, is $\dfrac{\alpha}{\alpha+\beta}$.

(b) σ^2, its variance, is $\langle x^2 \rangle - \langle x \rangle^2 = \dfrac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$.

Assume real $\alpha, \beta > 0$.

Solution.

A beta probability density requires real $\alpha, \beta > 0$. Its prefactor is $1/B(\alpha,\beta)$. Consequently

$$\langle x^q \rangle = \frac{B(\alpha+q,\beta)}{B(\alpha,\beta)} = \frac{\Gamma(\alpha+q)\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\alpha+\beta+q)}, \qquad \Re q > -\alpha.$$

At $q = 0$ this equals one and verifies normalization. **(a)** For $q = 1$, apply gamma recurrence to numerator and denominator:

$$\langle x \rangle = \frac{[\alpha\Gamma(\alpha)]\Gamma(\alpha+\beta)}{\Gamma(\alpha)[(\alpha+\beta)\Gamma(\alpha+\beta)]} = \boxed{\frac{\alpha}{\alpha+\beta}}.$$

(b) For $q = 2$ the same cancellation gives

$$\langle x^2 \rangle = \frac{\alpha(\alpha+1)\Gamma(\alpha)\Gamma(\alpha+\beta)}{\Gamma(\alpha)(\alpha+\beta)(\alpha+\beta+1)\Gamma(\alpha+\beta)} = \frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)}.$$

Subtract the squared mean using a common denominator:

$$\begin{aligned}
\sigma^2 &= \frac{\alpha(\alpha+1)(\alpha+\beta) - \alpha^2(\alpha+\beta+1)}{(\alpha+\beta)^2(\alpha+\beta+1)} \\
&= \frac{\alpha[(\alpha^2+\alpha\beta+\alpha+\beta) - (\alpha^2+\alpha\beta+\alpha)]}{(\alpha+\beta)^2(\alpha+\beta+1)} \\
&= \boxed{\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}}.
\end{aligned}$$

13.3.17 From

$$\lim_{n\to\infty} \frac{\displaystyle\int_0^{\pi/2} \sin^{2n}\theta\, d\theta}{\displaystyle\int_0^{\pi/2} \sin^{2n+1}\theta\, d\theta} = 1,$$

derive the Wallis formula for π:

$$\frac{\pi}{2} = \frac{2\cdot 2}{1\cdot 3}\cdot\frac{4\cdot 4}{3\cdot 5}\cdot\frac{6\cdot 6}{5\cdot 7}\cdots.$$

Solution.

Write $I_k = \int_0^{\pi/2} \sin^k \theta \, d\theta$. Exercise 13.3.8 gives

$$I_{2n} = \frac{\pi}{2} \frac{(2n-1)!!}{(2n)!!}, \qquad I_{2n+1} = \frac{(2n)!!}{(2n+1)!!}.$$

Therefore

$$\frac{I_{2n}}{I_{2n+1}} = \frac{\pi}{2} \frac{(2n-1)!!(2n+1)!!}{[(2n)!!]^2}.$$

Define the finite product

$$P_n = \prod_{j=1}^{n} \frac{(2j)^2}{(2j-1)(2j+1)} = \frac{[(2n)!!]^2}{(2n-1)!!(2n+1)!!}.$$

The previous ratio is $(\pi/2)/P_n$. Taking its stated limit of one gives

$$\boxed{\frac{\pi}{2} = \lim_{n\to\infty} P_n = \frac{2\cdot 2}{1\cdot 3}\frac{4\cdot 4}{3\cdot 5}\frac{6\cdot 6}{5\cdot 7}\cdots}.$$

For completeness, the limit used here also follows without a gamma asymptotic formula: since $0 \le \sin\theta \le 1$, $I_{2n+1} \le I_{2n} \le I_{2n-1}$. The already evaluated odd integrals satisfy $I_{2n-1}/I_{2n+1} = (2n+1)/(2n)$, so

$$1 \le \frac{I_{2n}}{I_{2n+1}} \le 1 + \frac{1}{2n} \longrightarrow 1.$$

This proves convergence of the infinite product as well as its value.

13.4 Stirling's Series

13.4.1 Rewrite Stirling's series to give $\Gamma(z+1)$ instead of $\ln\Gamma(z+1)$.

ANS. $\Gamma(z+1) \sim \sqrt{2\pi}\, z^{z+1/2} e^{-z} \left(1 + \frac{1}{12z} + \frac{1}{288z^2} - \frac{139}{51{,}840z^3} + \cdots\right)$.

Interpret this as an asymptotic expansion as $|z| \to \infty$ in a fixed sector $|\arg z| \le \pi - \delta$, $\delta > 0$, with consistent logarithm branches.

Solution.

Use Stirling's logarithmic expansion on the positive real axis, or on a fixed closed sector $|\arg z| \le \pi - \delta$ with a consistent logarithm branch:

$$\log\Gamma(z+1) = \frac{1}{2}\log(2\pi) + \left(z + \frac{1}{2}\right)\log z - z + \frac{1}{12z} - \frac{1}{360z^3} + O(z^{-5}).$$

Let $u = 1/(12z) - 1/(360z^3) + O(z^{-5})$ and exponentiate. Through order z^{-3},

$$\begin{aligned} e^u &= 1 + u + \frac{u^2}{2} + \frac{u^3}{6} + O(z^{-4}) \\ &= 1 + \frac{1}{12z} + \frac{1}{2\cdot 12^2 z^2} + \left(-\frac{1}{360} + \frac{1}{6\cdot 12^3}\right) z^{-3} + O(z^{-4}). \end{aligned}$$

The cubic coefficient is $-1/360 + 1/10368 = (-144+5)/51840 = -139/51840$. Thus

$$\boxed{\Gamma(z+1) \sim \sqrt{2\pi}\, z^{z+1/2} e^{-z} \left(1 + \frac{1}{12z} + \frac{1}{288z^2} - \frac{139}{51840z^3} + \cdots\right)}.$$

This is an asymptotic expansion, not a convergent infinite power series in $1/z$. A finite truncation and its remainder order are meaningful; the ellipsis does not license summing infinitely many terms at a fixed z.

13.4.2 Use Stirling's formula to estimate 52!, the number of possible rearrangements of cards in a standard deck of playing cards.

Solution.

The leading Stirling estimate is

$$52! \approx \sqrt{2\pi \cdot 52} \left(\frac{52}{e}\right)^{52}.$$

Compute its logarithm first:

$$L_0 = \frac{1}{2}\ln(2\pi) + \frac{105}{2}\ln 52 - 52 = 156.3592337587296\ldots.$$

Exponentiation gives

$$\boxed{52! \approx e^{L_0} = 8.05290203839 \times 10^{67} \approx 8.1 \times 10^{67}}.$$

The first logarithmic correction gives

$$e^{L_0 + 1/(12\cdot 52)} = 8.06581767642 \times 10^{67}.$$

For comparison, direct integer multiplication $1 \cdot 2 \cdots 52$ gives $52! = 8.065817517094\ldots \times 10^{67}$. The leading estimate has relative error $-0.00160126096\ldots$, about -0.1601%; including $1/(12\cdot 52)$ reduces the relative error to 1.97534×10^{-8}. The size and sign are consistent with the next logarithmic term $-1/(360 \cdot 52^3)$.

13.4.3 Show that the constants C_1 and C_2 in Stirling's formula have the respective values zero and $\frac{1}{2}\ln 2\pi$ by using the logarithm of the Legendre duplication formula and that of the recurrence formula.

Solution.

Let $F(z) = \ln\Gamma(z+1)$ for large positive z. Before the constants are determined, the integrated expansion is

$$F(z) = C_2 + \left(z + \frac{1}{2}\right)\ln z + (C_1 - 1)z + O(z^{-1}).$$

The recurrence gives $F(z) - F(z-1) = \ln z$. Substituting and canceling C_2 and the common linear terms yields

$$0 = \left(z - \frac{1}{2}\right)\ln\frac{z}{z-1} + C_1 - 1 + O(z^{-1}).$$

Since $\ln[z/(z-1)] = -\ln(1 - 1/z) = 1/z + O(z^{-2})$, the first term tends to one. Therefore $\boxed{C_1 = 0}$.

Now use duplication in the form

$$F(z) + F\left(z - \frac{1}{2}\right) = \frac{1}{2}\ln\pi - 2z\ln 2 + F(2z).$$

The needed shifted expansion is

$$\begin{aligned} F\left(z - \frac{1}{2}\right) &= C_2 + z\ln\left(z - \frac{1}{2}\right) - z + \frac{1}{2} + O(z^{-1}) \\ &= C_2 + z\ln z + z\left[-\frac{1}{2z} + O(z^{-2})\right] - z + \frac{1}{2} + O(z^{-1}) \\ &= C_2 + z\ln z - z + O(z^{-1}). \end{aligned}$$

Consequently the left side of duplication is $2C_2 + (2z + 1/2)\ln z - 2z + O(z^{-1})$, while its right side is

$$C_2 + \left(2z + \frac{1}{2}\right)\ln z - 2z + \frac{1}{2}\ln\pi + \frac{1}{2}\ln 2 + O(z^{-1}).$$

Canceling the common terms and taking the limit gives

$$\boxed{C_2 = \frac{1}{2}\ln(2\pi)}.$$

The corrected instruction supplies the recurrence needed to fix C_1, in addition to the duplication formula used for C_2.

13.4.4 Without using Stirling's series show that

(a) $\ln(n!) < \displaystyle\int_1^{n+1} \ln x\,dx$, (b) $\ln(n!) > \displaystyle\int_1^{n} \ln x\,dx$; n is an integer ≥ 2.

Note that the arithmetic mean of these two integrals gives a good approximation for Stirling's series.

Solution.

The logarithm is strictly increasing. **(a)** On each interval $[j, j+1]$, $\ln x > \ln j$ except at its left endpoint. Integrating over a unit interval and summing $j = 1, \ldots, n$ gives

$$\boxed{\ln(n!) = \sum_{j=1}^{n} \ln j < \sum_{j=1}^{n} \int_j^{j+1} \ln x\,dx = \int_1^{n+1} \ln x\,dx}.$$

(b) On $[j-1, j]$ for $j = 2, \ldots, n$, $\ln x < \ln j$ except at the right endpoint. Therefore

$$\boxed{\int_1^{n} \ln x\,dx = \sum_{j=2}^{n} \int_{j-1}^{j} \ln x\,dx < \sum_{j=2}^{n} \ln j = \ln(n!)}.$$

Evaluating the elementary antiderivative $x\ln x - x$ gives the explicit two-sided bound

$$n\ln n - n + 1 < \ln(n!) < (n+1)\ln(n+1) - n.$$

The arithmetic mean of these bounds is

$$\frac{n\ln n + (n+1)\ln(n+1) - 2n + 1}{2} = \left(n + \frac{1}{2}\right)\ln n - n + 1 + O(n^{-1}),$$

which reproduces Stirling's growing terms without assuming that formula. It does not determine Stirling's exact constant term.

13.4.5 Test for convergence

$$\sum_{p=0}^{\infty}\left[\frac{\Gamma(p+\frac{1}{2})}{p!}\right]^2\frac{2p+1}{2p+2}=\pi\sum_{p=0}^{\infty}\frac{(2p-1)!!\,(2p+1)!!}{(2p)!!\,(2p+2)!!}.$$

This series arises in an attempt to describe the magnetic field created by and enclosed by a current loop.

Solution.

Let

$$a_p=\left[\frac{\Gamma(p+1/2)}{\Gamma(p+1)}\right]^2\frac{2p+1}{2p+2}>0.$$

Stirling's formula gives the gamma quotient

$$\frac{\Gamma(p+1/2)}{\Gamma(p+1)}=p^{-1/2}\left[1-\frac{1}{8p}+O(p^{-2})\right].$$

This is also the $a=-1/2$, $b=0$ specialization of the detailed quotient expansion in Exercise 13.4.6. Since $(2p+1)/(2p+2)\to 1$, it follows that

$$\lim_{p\to\infty}p\,a_p=1.$$

By limit comparison with $\sum_{p=1}^{\infty}1/p$,

$$\boxed{\sum_{p=0}^{\infty}a_p \text{ diverges to } +\infty}.$$

The finite $p=0$ term is $\pi/2$ and does not alter convergence. For the second form, use $\Gamma(p+1/2)/p!=\sqrt{\pi}(2p-1)!!/(2p)!!$ and absorb $(2p+1)/(2p+2)$ into the corresponding double factorials. This proves that it is the same positive series. A ratio tending to one alone would not establish its divergence; the harmonic comparison above does.

13.4.6 Show that $\lim_{x\to\infty}x^{b-a}\dfrac{\Gamma(x+a+1)}{\Gamma(x+b+1)}=1$.

Solution.

Take fixed a,b and $x\to+\infty$. Complex a,b are also allowed with logarithms continued in the large right half-plane. For either fixed parameter c, Stirling's formula gives

$$\log\Gamma(x+c+1)=\left(x+c+\frac{1}{2}\right)\log(x+c)-(x+c)+\frac{1}{2}\log(2\pi)+\frac{1}{12(x+c)}+O(x^{-3}).$$

Expand $\log(x+c)=\log x+c/x-c^2/(2x^2)+O(x^{-3})$. Multiplication by $x+c+1/2$ and cancellation of the constant c against $-(x+c)$ gives

$$\begin{aligned}\log\Gamma(x+c+1)=&\left(x+\frac{1}{2}\right)\log x-x+\frac{1}{2}\log(2\pi)+c\log x\\&+\frac{c(c+1)/2+1/12}{x}+O(x^{-2}).\end{aligned}$$

Subtract the $c=b$ expression from the $c=a$ expression and add $(b-a)\log x$:

$$\log\left[x^{b-a}\frac{\Gamma(x+a+1)}{\Gamma(x+b+1)}\right]=\frac{(a-b)(a+b+1)}{2x}+O(x^{-2})\longrightarrow 0.$$

Exponentiating proves

$$\boxed{\lim_{x\to\infty}x^{b-a}\frac{\Gamma(x+a+1)}{\Gamma(x+b+1)}=1}.$$

The calculation also gives the first relative correction $(a-b)(a+b+1)/(2x)$, useful for checking the rate of approach.

13.4.7 Show that $\lim_{n\to\infty}\dfrac{(2n-1)!!}{(2n)!!}n^{1/2}=\pi^{-1/2}$.

Solution.

Express the double factorials in gamma form:

$$(2n-1)!!=\frac{2^n\Gamma(n+1/2)}{\sqrt{\pi}},\qquad (2n)!!=2^n\Gamma(n+1).$$

Their normalized ratio is consequently

$$\frac{(2n-1)!!}{(2n)!!}\sqrt{n}=\frac{1}{\sqrt{\pi}}\,n^{1/2}\frac{\Gamma(n+1/2)}{\Gamma(n+1)}.$$

Exercise 13.4.6 with $a=-1/2$ and $b=0$ shows that the final factor tends to one, so

$$\boxed{\lim_{n\to\infty}\frac{(2n-1)!!}{(2n)!!}\sqrt{n}=\pi^{-1/2}}.$$

More precisely it equals $\pi^{-1/2}[1-1/(8n)+O(n^{-2})]$, approaching the limit from below for sufficiently large n.

13.4.8 A set of N distinguishable particles is assigned to states ψ_i, $i=1,2,\ldots,M$. If the numbers of particles in the various states are $n_1,n_2,\ldots,n_M$ (with $M\ll N$), the number of ways this can be done is

$$W=\frac{N!}{n_1!\,n_2!\cdots n_M!}.$$

The entropy associated with this assignment is $S=k\ln W$, where k is Boltzmann's constant. In the limit $N\to\infty$, with $n_i=p_iN$ (so p_i is the fraction of the particles in state i), find S as a function of N and the p_i.

(a) In the limit of large N, find the entropy associated with an arbitrary set of n_i. Is the entropy an extensive function of the system size (i.e., is it proportional to N)?

(b) Find the set of p_i that maximize S.

Hint. Remember that $\sum_i p_i=1$ and that this is a constrained maximization (see Section 22.3).

Note. These formulas correspond to classical, or Boltzmann, statistics.

Take M fixed and the fractions p_i fixed as $N\to\infty$. Extensivity refers to the leading term; $M\ll N$ alone is insufficient if M grows with N.

Solution.

Use $0! = 1$ and $0 \ln 0 = 0$ for unoccupied states. **(a)** With fixed M and fixed fractions p_i, $\sum_i n_i = N$ and $n_i = Np_i$. The leading Stirling terms give

$$\begin{aligned}\frac{S}{k} &= \ln N! - \sum_i \ln n_i! \\ &= N \ln N - N - \sum_i (n_i \ln n_i - n_i) + O(\ln N) \\ &= N \ln N - \sum_i N p_i (\ln N + \ln p_i) + O(\ln N) \\ &= -N \sum_i p_i \ln p_i + O(\ln N).\end{aligned}$$

Thus

$$\boxed{S \sim -Nk \sum_{i=1}^{M} p_i \ln p_i}$$

when the entropy per particle is nonzero; more generally the additive estimate in the preceding line includes the zero-entropy case. This leading term is extensive.

If q fractions are positive, retaining the half-logarithm terms makes the finite-size correction explicit:

$$\frac{S}{k} = -N \sum_{p_i>0} p_i \ln p_i - \frac{q-1}{2} \ln(2\pi N) - \frac{1}{2} \sum_{p_i>0} \ln p_i + O(N^{-1}).$$

Here every occupied n_i tends to infinity; zero fractions were omitted before applying Stirling's formula. Extensivity is the leading large-N statement for fixed state fractions and a fixed number of states. The exact finite-N entropy has smaller corrections. The condition $M \ll N$ alone does not ensure extensivity if M also grows: for example $M = \sqrt{N}$, $p_i = 1/M$ gives a leading $(k/2)N \ln N$.

(b) Maximize $H = -\sum_i p_i \ln p_i$ subject to $\sum_i p_i = 1$. In the interior of the probability simplex, introduce $\mathcal{L} = H - \lambda(\sum_i p_i - 1)$. Then

$$\frac{\partial \mathcal{L}}{\partial p_i} = -\ln p_i - 1 - \lambda = 0, \qquad p_i = e^{-1-\lambda}.$$

All probabilities are equal, and normalization gives

$$\boxed{p_i = \frac{1}{M} \quad (i = 1, \ldots, M)}, \qquad \boxed{S_{\max} = Nk \ln M + O(\ln N)}.$$

The Hessian of H is $-\operatorname{diag}(1/p_i)$, strictly negative on every nonzero tangent displacement, so this is the unique maximum. A boundary with only $q < M$ occupied states has maximum $\ln q < \ln M$ and cannot improve it.

For a finite integer assignment, if $n_i \geq n_j + 2$, moving one particle from i to j changes the multiplicity by $W_{\text{new}}/W_{\text{old}} = n_i/(n_j + 1) > 1$. Thus the exact maximizing counts differ by at most one; they are exactly uniform when M divides N, and tend to the probabilities above otherwise.

13.5 Riemann Zeta Function

13.5.1 Show that the symmetrical functional relation

$$\Gamma\left(\frac{z}{2}\right)\pi^{-z/2}\zeta(z) = \Gamma\left(\frac{1-z}{2}\right)\pi^{-(1-z)/2}\zeta(1-z)$$

follows from the equation

$$\left(e^{2\pi iz}-1\right)\zeta(z) = \zeta(1-z)\frac{(2\pi)^z}{\Gamma(z)}\left(e^{3\pi iz/2}-e^{\pi iz/2}\right).$$

Solution.

Initially avoid integer z and poles of the gamma factors so that the following divisions are valid. Factor the exponentials:

$$e^{2\pi iz}-1 = 2ie^{\pi iz}\sin\pi z, \qquad e^{3\pi iz/2}-e^{\pi iz/2} = 2ie^{\pi iz}\sin\frac{\pi z}{2}.$$

Cancel $2ie^{\pi iz}$ in the supplied relation to obtain

$$\sin\pi z\,\zeta(z) = \frac{(2\pi)^z}{\Gamma(z)}\sin\frac{\pi z}{2}\,\zeta(1-z).$$

Use reflection twice:

$$\sin\pi z = \frac{\pi}{\Gamma(z)\Gamma(1-z)}, \qquad \sin\frac{\pi z}{2} = \frac{\pi}{\Gamma(z/2)\Gamma(1-z/2)}.$$

After cancellation this gives

$$\zeta(z) = \frac{(2\pi)^z\Gamma(1-z)}{\Gamma(z/2)\Gamma(1-z/2)}\,\zeta(1-z).$$

Duplication with $w=(1-z)/2$ reads

$$\Gamma\left(\frac{1-z}{2}\right)\Gamma\left(1-\frac{z}{2}\right) = 2^z\sqrt{\pi}\,\Gamma(1-z).$$

Substitute for $\Gamma(1-z)$ and cancel $\Gamma(1-z/2)$ and 2^z:

$$\zeta(z) = \pi^{z-1/2}\frac{\Gamma((1-z)/2)}{\Gamma(z/2)}\zeta(1-z).$$

Multiplication by $\Gamma(z/2)\pi^{-z/2}$ yields

$$\boxed{\Gamma\left(\frac{z}{2}\right)\pi^{-z/2}\zeta(z) = \Gamma\left(\frac{1-z}{2}\right)\pi^{-(1-z)/2}\zeta(1-z)}.$$

The two meromorphic functions therefore agree by continuation; at exceptional points their limits, rather than the canceled zero or pole factors separately, determine the identity.

13.5.2 Prove that

$$\int_0^\infty \frac{x^n e^x\,dx}{(e^x-1)^2} = n!\,\zeta(n).$$

Assuming n to be real, show that each side of the equation diverges if $n = 1$. Hence the preceding equation carries the condition $n > 1$. Integrals such as this appear in the quantum theory of transport effects: thermal and electrical conductivity.

Solution.

Since $d[(e^x-1)^{-1}]/dx = -e^x/(e^x-1)^2$, integration by parts gives

$$\begin{aligned} I_n &= \int_0^\infty \frac{x^n e^x}{(e^x-1)^2}dx \\ &= \left[-\frac{x^n}{e^x-1}\right]_0^\infty + n\int_0^\infty \frac{x^{n-1}}{e^x-1}dx. \end{aligned}$$

At infinity the boundary term vanishes exponentially. At zero $x^n/(e^x-1) \sim x^{n-1}$, so it vanishes for real $n > 1$. Expanding $(e^x-1)^{-1} = \sum_{k=1}^\infty e^{-kx}$ and integrating the nonnegative terms then gives

$$I_n = n\sum_{k=1}^\infty \int_0^\infty x^{n-1}e^{-kx}dx = n\sum_{k=1}^\infty \frac{\Gamma(n)}{k^n} = \boxed{\Gamma(n+1)\zeta(n) = n!\zeta(n)}.$$

For a noninteger n, the notation $n!$ means $\Gamma(n+1)$. The original integrand behaves as x^{n-2} near zero. Thus $n = 1$ gives a logarithmically divergent integral, matching the pole of $\zeta(1)$; the real integral converges exactly for $n > 1$. For complex n the same argument, with absolute convergence, holds for $\Re n > 1$.

13.5.3 The Bloch-Grüneisen approximation for the resistance in a monovalent metal at absolute temperature T is

$$\rho = C\frac{T^5}{\Theta^6}\int_0^{\Theta/T} \frac{x^5\,dx}{(e^x-1)(1-e^{-x})},$$

where Θ is the Debye temperature characteristic of the metal.

(a) For $T \to \infty$, show that

$$\rho \approx \frac{C}{4}\cdot\frac{T}{\Theta^2}.$$

(b) For $T \to 0$, show that

$$\rho \approx 5!\,\zeta(5)\,C\frac{T^5}{\Theta^6}.$$

Solution.

Take $T, \Theta > 0$ and set $y = \Theta/T$. First rewrite

$$\frac{1}{(e^x-1)(1-e^{-x})} = \frac{e^x}{(e^x-1)^2}.$$

(a) For high temperature, $y \to 0$. Differentiating the local expansion $(e^x-1)^{-1} = x^{-1} - 1/2 + x/12 - x^3/720 + \cdots$ gives

$$\frac{e^x}{(e^x-1)^2} = \frac{1}{x^2} - \frac{1}{12} + O(x^2).$$

Multiply by x^5 and integrate:

$$\int_0^y \frac{x^5 e^x}{(e^x-1)^2}dx = \frac{y^4}{4} - \frac{y^6}{72} + O(y^8).$$

Hence

$$\rho = C\frac{T^5}{\Theta^6}\left[\frac{1}{4}\left(\frac{\Theta}{T}\right)^4 - \frac{1}{72}\left(\frac{\Theta}{T}\right)^6 + O\left(\frac{\Theta^8}{T^8}\right)\right]$$
$$= \boxed{\frac{C}{4}\frac{T}{\Theta^2}}\left[1 - \frac{1}{18}\left(\frac{\Theta}{T}\right)^2 + O\left(\frac{\Theta^4}{T^4}\right)\right].$$

(b) For low temperature, $y \to \infty$. The positive integral tends to the complete one from Exercise 13.5.2 at $n = 5$:

$$\int_0^\infty \frac{x^5 e^x}{(e^x-1)^2}dx = 5!\zeta(5).$$

Thus

$$\boxed{\rho \sim 5!\zeta(5)C\frac{T^5}{\Theta^6}}.$$

The omitted tail is exponentially small. Explicitly, for $x \geq y > 0$, $e^x/(e^x-1)^2 \leq e^{-x}/(1-e^{-y})^2$, so its integral is at most

$$\frac{e^{-y}}{(1-e^{-y})^2}\left(y^5 + 5y^4 + 20y^3 + 60y^2 + 120y + 120\right).$$

The polynomial follows by five integrations by parts of $\int_y^\infty x^5 e^{-x}dx$ and proves the limiting replacement.

13.5.4 Derive the following expansion of the Debye function for $n \geq 1$:

$$\int_0^x \frac{t^n\,dt}{e^t-1} = x^n\left[\frac{1}{n} - \frac{x}{2(n+1)} + \sum_{k=1}^\infty \frac{B_{2k}x^{2k}}{(2k+n)(2k)!}\right], \quad |x| < 2\pi.$$

The complete integral $(0,\infty)$ equals $n!\,\zeta(n+1)$ (Exercise 13.5.6).

For complex x, take integer $n \geq 1$. For real $n \geq 1$, the real-power version uses $0 < x < 2\pi$; complex continuation needs a specified power branch.

Solution.

The Bernoulli generating function is analytic for $|t| < 2\pi$:

$$\frac{t}{e^t-1} = 1 - \frac{t}{2} + \sum_{k=1}^\infty \frac{B_{2k}}{(2k)!}t^{2k}.$$

Its nearest nonremovable poles are at $t = \pm 2\pi i$. Multiply by t^{n-1} and integrate from zero to x. For integer $n \geq 1$ uniform convergence on every smaller disk justifies termwise integration, giving

$$\int_0^x \frac{t^n}{e^t-1}dt = \frac{x^n}{n} - \frac{x^{n+1}}{2(n+1)} + \sum_{k=1}^\infty \frac{B_{2k}x^{n+2k}}{(n+2k)(2k)!}$$
$$= \boxed{x^n\left[\frac{1}{n} - \frac{x}{2(n+1)} + \sum_{k=1}^\infty \frac{B_{2k}x^{2k}}{(2k+n)(2k)!}\right]}.$$

Every denominator $n+2k$ comes from integrating a monomial; the lower-limit terms vanish. The same calculation is valid for real $n>0$ and $0<x<2\pi$, with the integrable factor t^{n-1}; for complex x and noninteger n a consistent power branch is needed. The full integral to infinity is instead $\Gamma(n+1)\zeta(n+1)$, as derived in Exercise 13.5.6.

13.5.5 The equilibrium radiation energy per unit volume of a blackbody is given by

$$u=\frac{8\pi k^4T^4}{c^3h^3}\int_0^\infty\frac{x^3}{e^x-1}\,dx.$$

Show that the integral in this expression is equal to $3!\,\zeta(4)$. The final result is the Stefan-Boltzmann law.

Solution.

For $x>0$ the geometric series gives $(e^x-1)^{-1}=\sum_{m=1}^\infty e^{-mx}$. The terms are nonnegative, so integrate them by monotone convergence. In each integral put $t=mx$:

$$\begin{aligned}\int_0^\infty\frac{x^3}{e^x-1}dx&=\sum_{m=1}^\infty\int_0^\infty x^3e^{-mx}dx\\&=\sum_{m=1}^\infty\frac{1}{m^4}\int_0^\infty t^3e^{-t}dt=\boxed{3!\zeta(4)}=\frac{\pi^4}{15}.\end{aligned}$$

Here $\Gamma(4)=3!=6$ and $\zeta(4)=\pi^4/90$. Substitution gives

$$\boxed{u=\frac{8\pi^5k^4}{15c^3h^3}T^4}.$$

The supplied formula describes radiation energy per unit volume, not total radiated energy or power per unit area. For isotropic equilibrium radiation, the outgoing flux through a surface is

$$j^\star=\frac{cu}{4\pi}\int_0^{2\pi}d\phi\int_0^{\pi/2}\cos\theta\sin\theta\,d\theta=\frac{cu}{4}.$$

Thus the Stefan–Boltzmann flux law is

$$\boxed{j^\star=\sigma T^4,\qquad\sigma=\frac{2\pi^5k^4}{15c^2h^3}}.$$

This also distinguishes the two quantities' different powers of c.

13.5.6 As a generalization of the result in Exercise 13.5.5, show that

$$\int_0^\infty\frac{x^s\,dx}{e^x-1}=s!\,\zeta(s+1),\quad\Re(s)>0.$$

Solution.

Let $\sigma=\Re s>0$ and use $x^s=e^{s\ln x}$ for $x>0$. Near zero the integrand is $O(x^{\sigma-1})$ in absolute value, and at infinity it has exponential decay. Moreover,

$$\sum_{n=1}^\infty\int_0^\infty|x^se^{-nx}|dx=\Gamma(\sigma+1)\sum_{n=1}^\infty n^{-\sigma-1}<\infty.$$

Thus the geometric expansion may be integrated term by term:

$$\begin{aligned}\int_0^\infty \frac{x^s}{e^x-1}dx &= \sum_{n=1}^\infty \int_0^\infty x^s e^{-nx}dx \\ &= \sum_{n=1}^\infty n^{-s-1}\int_0^\infty t^s e^{-t}dt \\ &= \boxed{\Gamma(s+1)\zeta(s+1) = s!\zeta(s+1)}.\end{aligned}$$

The substitution in the second line is $t = nx$, including its Jacobian $dx = dt/n$. The condition $\Re s > 0$ is exactly what makes both the endpoint and the absolutely summed integrals converge.

13.5.7 Prove that

$$\int_0^\infty \frac{x^s\,dx}{e^x+1} = s!\,(1-2^{-s})\,\zeta(s+1), \quad \Re(s) > 0.$$

Exercises 13.5.6 and 13.5.7 give the Mellin integral transform of $1/(e^x \pm 1)$; this transform is defined in Eq. (20.9).

Solution.

The algebraic identity

$$\frac{1}{e^x+1} = \frac{1}{e^x-1} - \frac{2}{e^{2x}-1}$$

follows by taking a common denominator. For $\Re s > 0$ both integrals on the right converge separately, and Exercise 13.5.6 gives

$$\begin{aligned}\int_0^\infty \frac{x^s}{e^x+1}dx &= \Gamma(s+1)\zeta(s+1) - 2\int_0^\infty \frac{x^s}{e^{2x}-1}dx \\ &= \Gamma(s+1)\zeta(s+1) - 2\cdot 2^{-s-1}\int_0^\infty \frac{t^s}{e^t-1}dt \\ &= \boxed{\Gamma(s+1)(1-2^{-s})\zeta(s+1)}.\end{aligned}$$

In the second integral $t = 2x$, so the power and Jacobian give 2^{-s-1}.

The original Fermi integral actually converges for $\Re s > -1$, because its denominator tends to two at zero. Analytic continuation therefore extends the answer to that half-plane, with a removable product at $s = 0$:

$$(1-2^{-s})\zeta(s+1) \longrightarrow \ln 2, \qquad \int_0^\infty \frac{dx}{e^x+1} = \left[-\ln(1+e^{-x})\right]_0^\infty = \ln 2.$$

The extension does not permit separating the two divergent Bose integrals when $-1 < \Re s \le 0$.

13.5.8 The neutrino energy density (Fermi distribution) in the early history of the universe is given by

$$\rho_\nu = \frac{4\pi}{h^3}\int_0^\infty \frac{x^3}{\exp(x/kT)+1}\,dx.$$

Show that

$$\rho_\nu = \frac{7\pi^5}{30h^3}(kT)^4.$$

Use $kT > 0$ and the convention $c = 1$. With x an energy in conventional units, insert c^{-3} in both formulas.

Solution.

For positive kT, put $t = x/(kT)$, so $x^3dx = (kT)^4t^3dt$. Then

$$\rho_\nu = \frac{4\pi(kT)^4}{h^3}\int_0^\infty \frac{t^3}{e^t+1}dt.$$

At $s = 3$ the preceding exercise gives

$$\int_0^\infty \frac{t^3}{e^t+1}dt = 3!(1-2^{-3})\zeta(4) = 6\cdot\frac{7}{8}\cdot\frac{\pi^4}{90} = \frac{7\pi^4}{120}.$$

Multiplying by the prefactor yields

$$\boxed{\rho_\nu = \frac{7\pi^5}{30h^3}(kT)^4}.$$

As an energy density with x an energy, the supplied formula uses $c = 1$. Restoring conventional length/time units requires a factor c^{-3} on both the stated integral and its answer. Indeed a massless momentum satisfies $x = pc$, so the radial phase-space factor p^2dp becomes x^2dx/c^3; one additional factor x supplies the energy.

13.5.9 Prove that

$$\psi^{(n)}(z) = (-1)^{n+1}\int_0^\infty \frac{t^n e^{-zt}}{1-e^{-t}}\,dt, \quad \Re(z) > 0.$$

The derivative order is an integer $n \geq 1$.

Solution.

Require an integer $n \geq 1$ as well as $\Re z > 0$. For $n = 0$ the written integral diverges as $\int_0 dt/t$ and cannot equal the finite digamma function. For $t > 0$, expand $(1-e^{-t})^{-1} = \sum_{p=0}^\infty e^{-pt}$. Absolute convergence justifies termwise integration, because

$$\sum_{p=0}^\infty \int_0^\infty t^n e^{-(\Re z+p)t}dt = n!\sum_{p=0}^\infty (\Re z+p)^{-n-1} < \infty.$$

Therefore

$$\begin{aligned}(-1)^{n+1}\int_0^\infty \frac{t^n e^{-zt}}{1-e^{-t}}dt &= (-1)^{n+1}\sum_{p=0}^\infty \int_0^\infty t^n e^{-(z+p)t}dt \\ &= (-1)^{n+1}n!\sum_{p=0}^\infty \frac{1}{(z+p)^{n+1}} \\ &= \boxed{\psi^{(n)}(z)}.\end{aligned}$$

The second equality is the scaled gamma integral; the last is the differentiated digamma series. The near-zero behavior t^{n-1} also directly verifies the required condition on n.

13.5.10 Show that $\zeta(s)$ is analytic in the entire finite complex plane except at $s = 1$, where it has a simple pole with a residue of $+1$.

Hint. The contour integral representation will be useful.

Solution.

Use the positive-real-axis cut and continuous argument $0 < \arg t < 2\pi$ from Exercise 13.1.23. Let C_ε run inward along the upper lip, counterclockwise around the circle $|t| = \varepsilon$, then outward along the lower lip, where $0 < \varepsilon < 2\pi$. Define

$$H(s) = \int_{C_\varepsilon} \frac{t^{s-1}}{e^t - 1} dt.$$

Keeping the circle radius fixed makes H entire in s: the compact circle contains no pole on its path, and the two ray tails decay exponentially and uniformly for s in a compact set. The poles $t = \pm 2\pi i, \pm 4\pi i, \ldots$ lie outside the small circle.

For $\Re s > 1$ the circle can shrink to zero: its contribution is $O(\varepsilon^{\Re s - 1})$. The upper ray contributes $-\Gamma(s)\zeta(s)$ and the lower ray contributes $e^{2\pi i(s-1)}\Gamma(s)\zeta(s)$. Thus

$$H(s) = (e^{2\pi i s} - 1)\Gamma(s)\zeta(s).$$

Use $e^{2\pi i s} - 1 = 2ie^{\pi i s} \sin \pi s$ and gamma reflection to rewrite this as

$$\boxed{\zeta(s) = \frac{e^{-\pi i s}\Gamma(1-s)}{2\pi i} H(s)}.$$

The right side is meromorphic on the whole finite plane and agrees with the original zeta function on a nonempty open set. It therefore supplies its meromorphic continuation. Since H is entire, its only possible poles are the poles of $\Gamma(1 - s)$, namely $s = 1, 2, 3, \ldots$.

At an integer $m \geq 2$, the two ray contributions cancel. On the small circle the integrand is single-valued and analytic at zero, because

$$\frac{t^{m-1}}{e^t - 1} = t^{m-2}\left(1 - \frac{t}{2} + O(t^2)\right).$$

Hence its circle integral is zero and $H(m) = 0$. This zero cancels the possible simple pole of $\Gamma(1 - s)$, proving regularity at every $m \geq 2$.

At $s = 1$ the circle integrand is $(e^t - 1)^{-1} = t^{-1} - 1/2 + O(t)$, so the counterclockwise contour gives $H(1) = 2\pi i$. Writing $s = 1 + \eta$,

$$\Gamma(1 - s) = \Gamma(-\eta) = -\frac{1}{\eta} + O(1), \qquad e^{-\pi i s} = -1 + O(\eta).$$

Substitution yields

$$\boxed{\zeta(s) = \frac{1}{s-1} + O(1)},$$

so the pole is simple and its residue is $+1$. At nonpositive integers $\Gamma(1 - s)$ is regular, giving no additional poles. For example $H(0) = -\pi i$ from the coefficient $-1/(2t)$, which gives the finite value $\zeta(0) = -1/2$.

The alternating-series quotient alone does not prove regularity throughout $\Re s > 0$: its denominator $1 - 2^{1-s}$ vanishes also at $s = 1 + 2\pi i k/\ln 2$ for nonzero integers k. Cancellation at those points must be justified. The entire-contour argument above establishes that all such points are regular and leaves only the pole at one.

13.6 Other Related Functions

13.6.1 Show that $\gamma(a,x) = e^{-x}\sum_{n=0}^{\infty}\frac{(a-1)!}{(a+n)!}x^{a+n}$

(a) by repeatedly integrating by parts,

(b) by transforming it into Eq. (13.76).

Take $x > 0$ and $\Re a > 0$ for the defining lower-gamma integral; factorials mean gamma functions.

Solution.

For the defining lower incomplete-gamma integral, take $x > 0$ and $\Re a > 0$; factorials here mean gamma functions and do not require integral a. **(a)** Integrate $e^{-t}t^{a-1}$ by parts, integrating the power and differentiating the exponential:

$$\gamma(a,x) = \left[\frac{t^a e^{-t}}{a}\right]_0^x + \frac{1}{a}\int_0^x t^a e^{-t}dt = \frac{x^a e^{-x}}{a} + \frac{\gamma(a+1,x)}{a}.$$

Repeat this step N times. With $(a)_N = a(a+1)\cdots(a+N-1)$,

$$\gamma(a,x) = e^{-x}\sum_{n=0}^{N-1}\frac{x^{a+n}}{(a)_{n+1}} + \frac{\gamma(a+N,x)}{(a)_N}.$$

Let $\sigma = \Re a > 0$. Since $|(a)_N| \geq (\sigma)_N$, the final remainder has the bound

$$\left|\frac{\gamma(a+N,x)}{(a)_N}\right| \leq \frac{x^{\sigma+N}}{(\sigma+N)(\sigma)_N} = \frac{\Gamma(\sigma)x^{\sigma+N}}{\Gamma(\sigma+N+1)} \longrightarrow 0.$$

The ratio of successive bounds is $x/(\sigma+N+1) \to 0$. Therefore

$$\boxed{\gamma(a,x) = e^{-x}\sum_{n=0}^{\infty}\frac{\Gamma(a)}{\Gamma(a+n+1)}x^{a+n}}.$$

(b) To transform this into Eq. (13.76), use

$$\frac{\Gamma(a)}{\Gamma(a+n+1)} = \frac{1}{n!}\int_0^1 u^{a-1}(1-u)^n du.$$

The exponential series converges uniformly on $0 \leq u \leq 1$, and $u^{\Re a-1}$ is integrable. Thus

$$\begin{aligned}
e^{-x}\sum_{n=0}^{\infty}\frac{\Gamma(a)x^{a+n}}{\Gamma(a+n+1)} &= x^a\int_0^1 u^{a-1}e^{-x}\sum_{n=0}^{\infty}\frac{[x(1-u)]^n}{n!}du \\
&= x^a\int_0^1 u^{a-1}e^{-xu}du \\
&= x^a\sum_{m=0}^{\infty}\frac{(-x)^m}{m!}\int_0^1 u^{a+m-1}du \\
&= \boxed{x^a\sum_{m=0}^{\infty}\frac{(-1)^m x^m}{m!(a+m)}}.
\end{aligned}$$

This is exactly the lower incomplete-gamma series in Eq. (13.76). Both series converge for every finite positive x in the stated parameter domain, although they need not be equally convenient numerically.

13.6.2 Show that

(a) $\dfrac{d^m}{dx^m}\left[x^{-a}\gamma(a,x)\right] = (-1)^m x^{-a-m}\gamma(a+m,x)$,

(b) $\dfrac{d^m}{dx^m}\left[e^x\gamma(a,x)\right] = e^x\dfrac{\Gamma(a)}{\Gamma(a-m)}\gamma(a-m,x)$.

Take $x>0$ and integer $m\geq 0$. The direct integral proofs use $\Re a>0$ in (a) and $\Re a>m$ in (b); beyond these domains continue the whole products meromorphically.

Solution.

Let m be a nonnegative integer and $x>0$. **(a)** For $\Re a>0$, put $t=xu$ in the defining integral:

$$x^{-a}\gamma(a,x)=\int_0^1 u^{a-1}e^{-xu}du.$$

Differentiation under this finite integral is justified by the integrable weight $u^{\Re a-1}$. Each derivative supplies $-u$, so

$$\begin{aligned}\frac{d^m}{dx^m}[x^{-a}\gamma(a,x)] &= (-1)^m\int_0^1 u^{a+m-1}e^{-xu}du\\ &= \boxed{(-1)^m x^{-a-m}\gamma(a+m,x)}.\end{aligned}$$

(b) Initially take $\Re a>m$ so every lower gamma integral appearing below is defined. The first product derivative is

$$\frac{d}{dx}[e^x\gamma(a,x)] = e^x\gamma(a,x)+x^{a-1}.$$

The manual's intermediate derivative includes an extra e^{-x} in the second term; it cancels against e^x. The lower recurrence gives $\gamma(a,x)=(a-1)\gamma(a-1,x)-x^{a-1}e^{-x}$. Substitution cancels the power term:

$$\frac{d}{dx}[e^x\gamma(a,x)] = (a-1)e^x\gamma(a-1,x).$$

Apply the same formula with a replaced successively by $a-1,\ldots,a-m+1$:

$$\boxed{\frac{d^m}{dx^m}[e^x\gamma(a,x)] = e^x(a-1)(a-2)\cdots(a-m)\gamma(a-m,x) = e^x\frac{\Gamma(a)}{\Gamma(a-m)}\gamma(a-m,x)}.$$

Outside the initial integral domain, the entire product on the right must be interpreted by meromorphic continuation. A zero coefficient cannot be multiplied literally by a pole of $\gamma(a-m,x)$. For example, when $a=m=1$ the left side is $d(e^x-1)/dx=e^x$, not zero. The continued product has precisely this finite value. At $m=0$ both identities reduce to themselves.

13.6.3 Show that $\gamma(a,x)$ and $\Gamma(a,x)$ satisfy the recurrence relations

(a) $\gamma(a+1,x) = a\,\gamma(a,x)-x^a e^{-x}$,

(b) $\Gamma(a+1,x) = a\,\Gamma(a,x)+x^a e^{-x}$.

Take $x > 0$. The lower-integral proof starts at $\Re a > 0$ and then continues meromorphically; the upper-integral identity holds for every complex a.

Solution.

(a) For $x > 0$, $\Re a > 0$, integrate $\gamma(a+1, x) = \int_0^x t^a e^{-t} dt$ by parts with $u = t^a$, $dv = e^{-t} dt$:

$$\begin{aligned}\gamma(a+1, x) &= \left[-t^a e^{-t}\right]_0^x + a \int_0^x t^{a-1} e^{-t} dt \\ &= \boxed{a\gamma(a, x) - x^a e^{-x}}.\end{aligned}$$

The lower boundary term vanishes because $\Re a > 0$.

(b) The same integration by parts over $[x, \infty)$ gives

$$\begin{aligned}\Gamma(a+1, x) &= \left[-t^a e^{-t}\right]_x^\infty + a \int_x^\infty t^{a-1} e^{-t} dt \\ &= \boxed{a\Gamma(a, x) + x^a e^{-x}}.\end{aligned}$$

For fixed $x > 0$ this upper-integral proof works for every complex a, because zero is outside the interval and the exponential controls infinity. Adding the two relations cancels the endpoint terms and recovers $\Gamma(a+1) = a\Gamma(a)$. The lower relation also continues meromorphically in a.

13.6.4 Show that the asymptotic expansion (for large x) of the incomplete gamma function $\Gamma(a, x)$ has the form

$$\Gamma(a, x) \sim x^{a-1} e^{-x} \sum_{n=0}^{\infty} \frac{\Gamma(a)}{\Gamma(a-n)} \cdot \frac{1}{x^n},$$

and that the above expression is equivalent to

$$\Gamma(a, x) \sim x^{a-1} e^{-x} \sum_{n=0}^{\infty} (a-n)_n \frac{1}{x^n}.$$

Take fixed a and $x \to +\infty$. Interpret $\Gamma(a)/\Gamma(a-n)$ as the polynomial $(a-1)\cdots(a-n)$ at gamma poles.

Solution.

For fixed a and $x \to +\infty$, integrate the upper gamma integral by parts in the direction of decreasing parameter:

$$\Gamma(a, x) = x^{a-1} e^{-x} + (a-1)\Gamma(a-1, x).$$

After N steps this is the exact identity

$$\Gamma(a, x) = x^{a-1} e^{-x} \sum_{n=0}^{N-1} \frac{c_n(a)}{x^n} + c_N(a)\Gamma(a-N, x), \qquad c_n(a) = \prod_{j=1}^{n} (a-j), \quad c_0 = 1.$$

The remainder has the required order. Indeed, putting $t = x + u$ gives

$$\Gamma(a-N, x) = e^{-x} x^{a-N-1} \int_0^\infty e^{-u} (1 + u/x)^{a-N-1} du.$$

For $x \geq 1$ its integral is bounded in absolute value by $\int_0^\infty e^{-u}(1+u)^{\max(\Re a - N - 1, 0)} du < \infty$. Thus the remainder is $O(e^{-x} x^{\Re a - N - 1})$, establishing an asymptotic expansion rather than an unjustified integration of an infinite binomial series:

$$\boxed{\Gamma(a,x) \sim x^{a-1} e^{-x} \sum_{n=0}^{\infty} \frac{c_n(a)}{x^n}}.$$

Gamma recurrence and reversal of the finite product give

$$c_n(a) = \frac{\Gamma(a)}{\Gamma(a-n)} = (a-n)(a-n+1)\cdots(a-1) = (a-n)_n,$$

which proves both requested forms. At gamma poles the coefficient is the finite polynomial $c_n(a)$, not an undefined quotient of gamma values. For example $a = 0$ gives $c_n(0) = (-1)^n n!$. If a is a positive integer, the products eventually vanish and the expansion terminates exactly. Otherwise the ratio of successive term magnitudes behaves as n/x for fixed x, so the infinite series diverges even though every fixed truncation is asymptotic.

13.6.5 A series expansion of the incomplete beta function yields

$$B_x(p,q) = x^p \left\{ \frac{1}{p} + \frac{1-q}{p+1} x + \frac{(1-q)(2-q)}{2!\,(p+2)} x^2 + \cdots \right.$$
$$\left. + \frac{(1-q)(2-q)\cdots(n-q)}{n!\,(p+n)} x^n + \cdots \right\}.$$

Given that $0 \leq x \leq 1$, $p > 0$, and $q > 0$, test this series for convergence. What happens at $x = 1$?

Solution.

Let the coefficient after removing x^p be

$$a_n = \frac{(1-q)_n}{n!(p+n)}, \qquad n = 0, 1, \ldots.$$

If q is a positive integer, $(1-q)_n = 0$ for $n \geq q$, so the series is a finite polynomial times x^p and convergence is immediate. Otherwise no such termination occurs, and

$$\frac{a_{n+1} x^{n+1}}{a_n x^n} = x \, \frac{n+1-q}{n+1} \frac{p+n}{p+n+1} \longrightarrow x.$$

It follows that the series converges absolutely for $0 \leq x < 1$. In the manual's reverse ratio (term $n-1$)/(term n), the factor is $1/x$, not x; its stated large-n limit must be $1/x$.

At $x = 1$, use the gamma quotient for the coefficients:

$$a_n = \frac{\Gamma(n+1-q)}{\Gamma(1-q)\Gamma(n+1)(p+n)} \sim \frac{n^{-q-1}}{\Gamma(1-q)}.$$

Since $q > 0$, comparison with the convergent positive series $\sum n^{-q-1}$ proves *absolute* convergence at the endpoint, regardless of the finite initial pattern of signs. In fact $\sum |a_n| < \infty$, so the series

is uniformly convergent on $0 \le x \le 1$ after multiplication by x^p. The integral defining $B_x(p,q)$ is continuous as $x \uparrow 1$ for $p, q > 0$, hence

$$\boxed{B_1(p,q) = \sum_{n=0}^{\infty} \frac{(1-q)_n}{n!(p+n)} = B(p,q) = \frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}}.$$

Thus the full specified interval, including $x = 1$, belongs to the convergence set. At $x = 0$ its value is zero because $p > 0$.

13.6.6 Using the definitions of the various functions, show that

(a) $\operatorname{si}(x) = \dfrac{1}{2i}\,[E_1(ix) - E_1(-ix)]$,

(b) $\operatorname{Ci}(x) = -\dfrac{1}{2}\,[E_1(ix) + E_1(-ix)]$,

(c) $E_1(ix) = -\operatorname{Ci}(x) + i\,\operatorname{si}(x)$.

Take $x > 0$, principal E_1, and $\operatorname{si}(x) = \operatorname{Si}(x) - \pi/2$.

Solution.

Take $x > 0$ and the principal branch of E_1. The real sine integral here is $\operatorname{si}(x) = \operatorname{Si}(x) - \pi/2$. For $\varepsilon > 0$ the scaled defining integral is

$$E_1((\varepsilon \pm i)x) = \int_1^{\infty} \frac{e^{-(\varepsilon \pm i)xu}}{u}\,du = \int_x^{\infty} \frac{e^{-(\varepsilon \pm i)t}}{t}\,dt.$$

Let $\varepsilon \downarrow 0$. The oscillatory sine and cosine tails converge by the Dirichlet test. To justify equality with the Abel limits, set $g_\varepsilon(t) = e^{-\varepsilon t}/t$. An integration by parts of $\int_R^{\infty} g_\varepsilon(t) e^{\pm it}dt$ bounds its magnitude by

$$g_\varepsilon(R) + \int_R^{\infty} |g_\varepsilon'(t)|dt = 2g_\varepsilon(R) \le \frac{2}{R}.$$

This bound is uniform for $\varepsilon \ge 0$; on $[x, R]$ ordinary dominated convergence applies. First let $\varepsilon \downarrow 0$, then $R \to \infty$, to obtain the ordinary improper integrals. Using the definitions

$$\operatorname{si}(x) = -\int_x^{\infty} \frac{\sin t}{t}dt, \qquad \operatorname{Ci}(x) = -\int_x^{\infty} \frac{\cos t}{t}dt,$$

we obtain

$$\boxed{E_1(ix) = -\operatorname{Ci}(x) + i\,\operatorname{si}(x)}, \qquad E_1(-ix) = -\operatorname{Ci}(x) - i\,\operatorname{si}(x).$$

This proves (c). Subtracting the two identities and dividing by $2i$ proves (a); adding and multiplying by $-1/2$ proves (b). The limiting procedure also specifies the contour at infinity, which cannot be changed silently in a complex exponential integral.

13.6.7 The potential produced by a $1s$ hydrogen electron is given by

$$V(r) = \frac{q}{4\pi\epsilon_0 a_0}\left[\frac{1}{2r}\,\gamma(3, 2r) + \Gamma(2, 2r)\right].$$

(a) For $r \ll 1$, show that

$$V(r) = \frac{q}{4\pi\epsilon_0 a_0}\left[1 - \frac{2}{3}r^2 + \cdots\right].$$

(b) For $r \gg 1$, show that

$$V(r) \sim \frac{q}{4\pi\epsilon_0 a_0} \cdot \frac{1}{r}.$$

Here r is expressed in units of a_0, the Bohr radius.

Note. $V(r)$ is illustrated in Fig. 13.8.

Solution.

Write $V(r) = [q/(4\pi\epsilon_0 a_0)]F(r)$. For positive integer parameters the incomplete gamma formulas give

$$\gamma(3, 2r) = 2\left[1 - e^{-2r}(1 + 2r + 2r^2)\right], \qquad \Gamma(2, 2r) = e^{-2r}(1 + 2r).$$

Insert these expressions before taking either limit:

$$\begin{aligned} F(r) &= \frac{1 - e^{-2r}(1 + 2r + 2r^2)}{r} + e^{-2r}(1 + 2r) \\ &= \boxed{\frac{1 - (1 + r)e^{-2r}}{r}}. \end{aligned}$$

(a) For small positive r, multiplication of the exponential Taylor series gives

$$(1 + r)e^{-2r} = 1 - r + \frac{2}{3}r^3 - \frac{2}{3}r^4 + \frac{2}{5}r^5 + O(r^6).$$

Thus the numerator's apparent zero-over-zero singularity is removable:

$$F(r) = 1 - \frac{2}{3}r^2 + \frac{2}{3}r^3 - \frac{2}{5}r^4 + O(r^5), \qquad \boxed{V(r) = \frac{q}{4\pi\epsilon_0 a_0}\left[1 - \frac{2}{3}r^2 + O(r^3)\right]}.$$

In particular $F(0) = 1$ and $F'(0) = 0$.

(b) The exact expression also gives

$$F(r) = \frac{1}{r} - \left(1 + \frac{1}{r}\right)e^{-2r} = \frac{1}{r}[1 - (1 + r)e^{-2r}].$$

Therefore

$$\boxed{V(r) \sim \frac{q}{4\pi\epsilon_0 a_0 r} \quad (r \to \infty)}.$$

This is the leading Coulomb potential; its relative correction is the explicitly displayed exponentially small term.

13.6.8 The potential produced by a $2p$ hydrogen electron can be shown to be

$$\begin{aligned} V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{24a_0}&\left[\frac{1}{r}\gamma(5, r) + \Gamma(4, r)\right] \\ &- \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{120a_0}\left[\frac{1}{r^3}\gamma(7, r) + r^2\Gamma(2, r)\right] P_2(\cos\theta). \end{aligned}$$

Here r is expressed in units of a_0, the Bohr radius. $P_2(\cos\theta)$ is a Legendre polynomial (Section 15.1).

(a) For $r \ll 1$, show that

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{a_0} \left[\frac{1}{4} - \frac{1}{120} r^2 P_2(\cos\theta) + \cdots \right].$$

(b) For $r \gg 1$, show that

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{a_0 r} \left[1 - \frac{6}{r^2} P_2(\cos\theta) + \cdots \right].$$

Solution.

Factor out $C = q/(4\pi\epsilon_0 a_0)$ and write $V = C[A_0(r) - A_2(r)P_2(\cos\theta)]$, where

$$A_0 = \frac{\gamma(5,r)/r + \Gamma(4,r)}{24}, \qquad A_2 = \frac{\gamma(7,r)/r^3 + r^2\Gamma(2,r)}{120}.$$

(a) Integrating the exponential power series gives

$$\gamma(j,r) = \frac{r^j}{j} - \frac{r^{j+1}}{j+1} + O(r^{j+2}), \qquad \Gamma(j,r) = (j-1)! - \gamma(j,r).$$

Consequently

$$24A_0 = \frac{r^4}{5} + O(r^5) + 6 - \frac{r^4}{4} + O(r^5) = 6 - \frac{r^4}{20} + O(r^5),$$
$$120A_2 = \frac{r^4}{7} + O(r^5) + r^2\left[1 - \frac{r^2}{2} + O(r^3)\right] = r^2 - \frac{5}{14}r^4 + O(r^5).$$

Divide by the normalizing constants and substitute:

$$\boxed{V = C\left[\frac{1}{4} - \frac{r^2}{120}P_2(\cos\theta) + r^4\left(-\frac{1}{480} + \frac{P_2(\cos\theta)}{336}\right) + O(r^5)\right]}.$$

This includes the requested leading terms and identifies the first omitted orders.

(b) At integer j, $\Gamma(j,r) = (j-1)!e^{-r}\sum_{k=0}^{j-1} r^k/k!$. Write $S_j(r) = \sum_{k=0}^{j} r^k/k!$. Substituting the finite upper-gamma formulas and $\gamma(j,r) = (j-1)! - \Gamma(j,r)$ gives

$$A_0 = \frac{1}{r} - e^{-r}\left[\frac{S_4(r)}{r} - \frac{S_3(r)}{4}\right], \qquad A_2 = \frac{6}{r^3} - e^{-r}\left[\frac{6S_6(r)}{r^3} - \frac{r^2(1+r)}{120}\right].$$

For the first bracket, $S_4/r = 1/r + 1 + r/2 + r^2/6 + r^3/24$ and $S_3/4 = 1/4 + r/4 + r^2/8 + r^3/24$. Their difference is $1/r + 3/4 + r/4 + r^2/24$. For the second bracket,

$$\frac{6S_6}{r^3} = \frac{6}{r^3} + \frac{6}{r^2} + \frac{3}{r} + 1 + \frac{r}{4} + \frac{r^2}{20} + \frac{r^3}{120};$$

subtracting $r^2/120 + r^3/120$ leaves the r^2 coefficient $1/20 - 1/120 = 1/24$ and cancels the r^3 term. Thus the exact expressions are

$$A_0 = \frac{1}{r} - e^{-r}\left(\frac{1}{r} + \frac{3}{4} + \frac{r}{4} + \frac{r^2}{24}\right),$$
$$A_2 = \frac{6}{r^3} - e^{-r}\left(\frac{6}{r^3} + \frac{6}{r^2} + \frac{3}{r} + 1 + \frac{r}{4} + \frac{r^2}{24}\right).$$

Thus

$$\boxed{V = \frac{C}{r}\left[1 - \frac{6}{r^2}P_2(\cos\theta) + O(r^3 e^{-r})\right]}.$$

The monopole and quadrupole coefficients are fixed by $\Gamma(5) = 24$ and $\Gamma(7) = 720$, respectively; $720/120 = 6$ explains the second coefficient.

13.6.9 Prove that the exponential integral has the expansion

$$\int_x^\infty \frac{e^{-t}}{t}\,dt = -\gamma - \ln x - \sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n \cdot n!},$$

where γ is the Euler-Mascheroni constant.

Take $x > 0$ for this real improper integral and logarithm.

Solution.

Take $x > 0$. Using $E_1(x) = \Gamma(0, x)$, subtract the lower incomplete-gamma series from $\Gamma(a)$ and let $a \to 0$:

$$E_1(x) = \lim_{a\to 0}\left[\Gamma(a) - \frac{x^a}{a}\right] - \sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n\,n!}.$$

The series with $n \geq 1$ is locally uniformly convergent in a near zero, because its factorial denominator dominates all powers of a fixed x. In the singular pair, use $a\Gamma(a) = \Gamma(1 + a)$:

$$\lim_{a\to 0} \frac{\Gamma(1+a) - x^a}{a} = \Gamma'(1) - \ln x = \Gamma(1)\psi(1) - \ln x = -\gamma - \ln x.$$

The evaluation subscript in the textbook's Eq. (13.82) must be $a = 0$, not $a = 1$; the limit being evaluated is $a \to 0$. Therefore

$$\boxed{E_1(x) = \int_x^\infty \frac{e^{-t}}{t}\,dt = -\gamma - \ln x - \sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n\,n!}}.$$

The ratio of successive absolute series terms is $xn/(n+1)^2 \to 0$, so the series converges for every finite x. As an independent differential check,

$$\frac{d}{dx}\left[-\gamma - \ln x - \sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n\,n!}\right] = -\frac{1}{x} - \frac{e^{-x} - 1}{x} = -\frac{e^{-x}}{x},$$

which is the derivative of the defining integral. For complex x, use the principal $\operatorname{Log} x$ on the plane cut along the negative real axis; zero remains a logarithmic singularity.

13.6.10 Show that $E_1(z)$ may be written as

$$E_1(z) = e^{-z}\int_0^\infty \frac{e^{-zt}}{1+t}\,dt.$$

Show also that we must impose the condition $|\arg z| \leq \pi/2$ and $z \neq 0$; convergence on the imaginary axis is conditional.

Solution.

Initially let $\Re z > 0$. The scaled exponential integral is

$$E_1(z) = \int_1^\infty \frac{e^{-zu}}{u}\, du.$$

For positive real z this follows directly from $v = zu$ in $\int_z^\infty e^{-v} dv/v$; both sides are analytic in the right half-plane, so the equality holds there by analytic continuation. Set $u = 1 + t$, with $du = dt$, to obtain

$$\boxed{E_1(z) = e^{-z} \int_0^\infty \frac{e^{-zt}}{1+t} dt}.$$

If $\Re z > 0$, the integral is absolutely convergent. If $z = ib$, $b \neq 0$, the primitive of e^{-ibt} is bounded and $(1+t)^{-1}$ decreases to zero, so the Dirichlet test gives conditional convergence. Taking the limit from $\Re z > 0$ identifies this value with the principal E_1 on the imaginary axis.

If $\Re z < 0$, even conditional convergence fails. For a fixed $L > 0$ chosen so that $1 - e^{-zL} \neq 0$, the integral over $[R, R + L]$ satisfies

$$\int_R^{R+L} \frac{e^{-zt}}{1+t} dt = \frac{e^{-zR}}{R} \left[\frac{1 - e^{-zL}}{z} + O(R^{-1}) \right].$$

Its magnitude grows exponentially, contradicting the Cauchy criterion for convergence of an improper integral. Therefore

$$\boxed{|\arg z| \leq \frac{\pi}{2}, \qquad z \neq 0}$$

is the convergence domain of this particular representation. The exclusion $z = 0$ is essential, and the boundary $|\arg z| = \pi/2$ is conditional rather than absolute convergence. The function itself has analytic continuation beyond this integral's convergence sector.

13.6.11 Related to the exponential integral by a simple change of variable is the function

$$E_n(x) = \int_1^\infty \frac{e^{-xt}}{t^n}\, dt.$$

Show that $E_n(x)$ satisfies the recurrence relation

$$E_{n+1}(x) = \frac{1}{n} e^{-x} - \frac{x}{n} E_n(x), \quad n = 1, 2, 3, \cdots.$$

Take $x > 0$ for the displayed recurrence. At $x = 0$, use the limiting interpretation when $n = 1$.

Solution.

For $x > 0$, integrate $E_{n+1}(x)$ by parts with $u = e^{-xt}$ and $dv = t^{-n-1} dt$, so $du = -xe^{-xt} dt$ and $v = -t^{-n}/n$:

$$\begin{aligned} E_{n+1}(x) &= \left[-\frac{e^{-xt}}{nt^n} \right]_1^\infty - \frac{x}{n} \int_1^\infty \frac{e^{-xt}}{t^n} dt \\ &= \boxed{\frac{e^{-x}}{n} - \frac{x}{n} E_n(x)}, \qquad n = 1, 2, \ldots. \end{aligned}$$

The upper boundary term vanishes and the lower one is $+e^{-x}/n$. When $n > 1$, the identity also holds at $x = 0$ using the finite values of both integrals. For $n = 1$, $E_1(0)$ is divergent, so the extension must instead use the limit $xE_1(x) \to 0$, established by Exercise 13.6.9. It then gives $E_2(0) = 1$ without assigning a value to $0 \cdot E_1(0)$.

13.6.12 With $E_n(x)$ as defined in Exercise 13.6.11, show that for $n > 1$,

$$E_n(0) = 1/(n-1).$$

Solution.

At $x = 0$ the exponential is one. For $n > 1$ evaluate the improper integral through a finite upper limit:

$$E_n(0) = \lim_{R\to\infty} \int_1^R t^{-n} dt = \lim_{R\to\infty} \frac{R^{1-n} - 1}{1-n} = \boxed{\frac{1}{n-1}}.$$

The exponent $1 - n$ is negative, making $R^{1-n} \to 0$. At $n = 1$ the integral is $\ln R \to \infty$, while for $n < 1$ it diverges as a positive power of R. Thus the stated real condition $n > 1$ is necessary as well as sufficient. Dominated convergence also shows $E_n(x) \to E_n(0)$ as $x \downarrow 0$ for these n, since $0 \le e^{-xt}t^{-n} \le t^{-n}$.

13.6.13 Develop the following power-series expansions:

(a) $\operatorname{si}(x) = -\dfrac{\pi}{2} + \displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)(2n+1)!}$,

(b) $\operatorname{Ci}(x) = \gamma + \ln x + \displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{2n(2n)!}$.

Use $x > 0$ for the displayed real logarithm; the sine-integral series also has its finite limit at zero.

Solution.

(a) For $x > 0$, use the convergent Dirichlet integral $\int_0^\infty \sin t\, dt/t = \pi/2$ to write

$$\operatorname{si}(x) = -\frac{\pi}{2} + \int_0^x \frac{\sin t}{t} dt.$$

For example its constant follows by Abel regularization: $J(\varepsilon) = \int_0^\infty e^{-\varepsilon t} \sin t\, dt/t$ satisfies $J'(\varepsilon) = -1/(1+\varepsilon^2)$ and $J(\infty) = 0$, hence $J(\varepsilon) = \arctan(1/\varepsilon)$ and $J(0^+) = \pi/2$. Now the sine series, divided by t, is uniformly convergent on every finite interval. Integrating term by term gives

$$\boxed{\operatorname{si}(x) = -\frac{\pi}{2} + \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)(2n+1)!}}.$$

The apparent value at zero is the finite limit $-\pi/2$.

(b) Exercise 13.6.6 gives $\operatorname{Ci}(x) = -[E_1(ix) + E_1(-ix)]/2$. Insert the series from Exercise 13.6.9. The logarithms satisfy

$$\operatorname{Log}(ix) = \ln x + i\pi/2, \qquad \operatorname{Log}(-ix) = \ln x - i\pi/2,$$

so their imaginary constants cancel. Odd powers cancel as well, while $(ix)^{2n} + (-ix)^{2n} = 2(-1)^n x^{2n}$. Consequently

$$\boxed{\operatorname{Ci}(x) = \gamma + \ln x + \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{2n(2n)!}}.$$

Both displayed power series have infinite radius of convergence; Ci itself retains the explicitly separated logarithmic singularity. The formulas for real logarithms here use $x > 0$; complex continuation requires the same chosen logarithm branch.

13.6.14 An analysis of a center-fed linear antenna leads to the expression

$$\int_0^x \frac{1-\cos t}{t}\,dt.$$

Show that this is equal to $\gamma + \ln x - \mathrm{Ci}(x)$.

Take $x > 0$ in the displayed real-logarithm identity; extend continuously at zero.

Solution.

For $x > 0$, Exercise 13.6.13 gives

$$\mathrm{Ci}(x) = \gamma + \ln x + \int_0^x \frac{\cos t - 1}{t} dt.$$

The integral is finite at its lower endpoint because $(1-\cos t)/t = t/2 + O(t^3)$. Moving the integral to the other side proves

$$\boxed{\int_0^x \frac{1-\cos t}{t} dt = \gamma + \ln x - \mathrm{Ci}(x)}.$$

One can also check the identity by differentiation: $\mathrm{Ci}'(x) = \cos x/x$, so the derivative of the right side is exactly $(1-\cos x)/x$; its limit at zero is zero because $\mathrm{Ci}(x) - \gamma - \ln x \to 0$. For negative real x, the original integral is even in x, and its real form is $\gamma + \ln|x| - \mathrm{Ci}(|x|)$. At zero the continuous value is zero.

13.6.15 Using the relation

$$\Gamma(a) = \gamma(a,x) + \Gamma(a,x),$$

show that if $\gamma(a,x)$ satisfies the relations of Exercise 13.6.2, then $\Gamma(a,x)$ must satisfy the same relations.

Take $x > 0$ and integer $m \geq 0$; interpret the gamma quotient as the finite polynomial $(a-1)\cdots(a-m)$.

Solution.

Initially take $x > 0$, $m \geq 0$ integral and $\Re a > m$, so all complete and lower gamma factors used separately below are regular. Write $\Gamma(a,x) = \Gamma(a) - \gamma(a,x)$.

(a) Direct differentiation of the complete-gamma term gives

$$\frac{d^m}{dx^m}[x^{-a}\Gamma(a)] = (-1)^m (a)_m x^{-a-m}\Gamma(a) = (-1)^m x^{-a-m}\Gamma(a+m).$$

Subtract the corresponding lower-gamma derivative from Exercise 13.6.2(a):

$$\begin{aligned}\frac{d^m}{dx^m}[x^{-a}\Gamma(a,x)] &= (-1)^m x^{-a-m}[\Gamma(a+m) - \gamma(a+m,x)] \\ &= \boxed{(-1)^m x^{-a-m}\Gamma(a+m,x)}.\end{aligned}$$

(b) Every derivative of $e^x\Gamma(a)$ equals $e^x\Gamma(a)$. Put $P_m(a) = (a-1)\cdots(a-m)$, so $\Gamma(a) = P_m(a)\Gamma(a-m)$. Subtraction of Exercise 13.6.2(b) then gives

$$\begin{aligned}\frac{d^m}{dx^m}[e^x\Gamma(a,x)] &= e^x\Gamma(a) - e^x P_m(a)\gamma(a-m,x) \\ &= e^x P_m(a)[\Gamma(a-m) - \gamma(a-m,x)] \\ &= \boxed{e^x \frac{\Gamma(a)}{\Gamma(a-m)}\Gamma(a-m,x)}.\end{aligned}$$

For fixed $x > 0$, upper incomplete gamma functions are entire in their parameter. Thus both identities extend to every complex a, with the quotient in (b) interpreted as the polynomial $P_m(a)$ at exceptional gamma arguments. For instance $a = 1$, $m = 1$ gives zero on both sides of (b), since $e^x\Gamma(1, x) = 1$.

13.6.16 For $x > 0$, show that

$$\int_x^\infty \frac{t^n\,dt}{e^t - 1} = \sum_{k=1}^\infty e^{-kx}\left[\frac{x^n}{k} + \frac{nx^{n-1}}{k^2} + \frac{n(n-1)x^{n-2}}{k^3} + \cdots + \frac{n!}{k^{n+1}}\right].$$

Take $n = 0, 1, 2, \ldots$ so the bracket terminates.

Solution.

The terminating bracket requires a nonnegative integer n. For $t \geq x > 0$, expand the positive geometric series:

$$\frac{1}{e^t - 1} = \sum_{k=1}^\infty e^{-kt}.$$

Tonelli's theorem permits interchange of sum and integral. In each term set $t = x + u$ and expand the finite binomial:

$$\begin{aligned}\int_x^\infty \frac{t^n}{e^t - 1}dt &= \sum_{k=1}^\infty e^{-kx}\int_0^\infty (x+u)^n e^{-ku}du \\ &= \sum_{k=1}^\infty e^{-kx}\sum_{j=0}^n \binom{n}{j}x^{n-j}\int_0^\infty u^j e^{-ku}du.\end{aligned}$$

Substitution $v = ku$ gives $\int_0^\infty u^j e^{-ku}du = j!/k^{j+1}$. Cancel $j!$ against the binomial coefficient to obtain

$$\boxed{\int_x^\infty \frac{t^n}{e^t - 1}dt = \sum_{k=1}^\infty e^{-kx}\sum_{j=0}^n \frac{n!}{(n-j)!}\frac{x^{n-j}}{k^{j+1}}}.$$

Writing the finite inner sum out gives exactly the bracket in the question. For fixed $x > 0$ its outer sum converges absolutely by its exponential factor. In the boundary case $n = 0$ it is $\sum_{k=1}^\infty e^{-kx}/k = -\ln(1 - e^{-x})$, also obtained by direct integration of $(e^t - 1)^{-1}$.

14 Chapter 14. Bessel Functions

Problems

14.1 Bessel Functions of the First Kind, $J_\nu(x)$

14.1.1 From the product of the generating functions $g(x,t)g(x,-t)$, show that

$$1 = [J_0(x)]^2 + 2[J_1(x)]^2 + 2[J_2(x)]^2 + \cdots$$

and therefore that $|J_0(x)| \le 1$ and $|J_n(x)| \le 1/\sqrt{2}$, $n = 1, 2, 3, \ldots$.

Hint. Use uniqueness of power series, (Section 1.2).

Take x real for the two inequalities.

Solution.

The inequalities require real x. The identity of squares also holds for complex x, but those squares are then not nonnegative. First obtain the negative-order relation from the generating function:

$$g(x,1/t) = g(x,-t), \qquad \sum_{m=-\infty}^{\infty} J_{-m}(x)t^m = \sum_{m=-\infty}^{\infty} (-1)^m J_m(x)t^m.$$

Uniqueness of Laurent coefficients gives $J_{-m} = (-1)^m J_m$. Now multiply the two absolutely convergent Laurent expansions:

$$1 = g(x,t)g(x,-t) = \sum_{m,n\in\mathbb{Z}} J_m(x)J_n(x)(-1)^n t^{m+n}.$$

The coefficient of t^0 comes from $n = -m$. Thus

$$\begin{aligned} 1 &= \sum_{m\in\mathbb{Z}} (-1)^{-m} J_m J_{-m} = \sum_{m\in\mathbb{Z}} J_m^2 \\ &= J_0^2 + \sum_{m=1}^{\infty} (J_m^2 + J_{-m}^2) = \boxed{J_0^2 + 2\sum_{m=1}^{\infty} J_m^2}. \end{aligned}$$

For real x, every summand is nonnegative. Dropping all terms but J_0^2 gives $J_0^2 \le 1$; retaining only $2J_n^2$ gives $2J_n^2 \le 1$. Taking nonnegative square roots proves the two bounds.

14.1.2 Using a generating function $g(x,t) = g(u+v,t) = g(u,t)g(v,t)$, show that

(a) $J_n(u+v) = \sum_{s=-\infty}^{\infty} J_s(u)J_{n-s}(v)$,

(b) $J_0(u+v) = J_0(u)J_0(v) + 2\sum_{s=1}^{\infty} J_s(u)J_{-s}(v)$.

These are addition theorems for the Bessel functions.

Solution.

The exponent is linear in its first argument, so

$$e^{(u+v)(t-t^{-1})/2} = e^{u(t-t^{-1})/2} e^{v(t-t^{-1})/2}.$$

Insert the Laurent series on both sides:

$$\sum_{n\in\mathbb{Z}} J_n(u+v)t^n = \sum_{s,r\in\mathbb{Z}} J_s(u)J_r(v)t^{s+r}.$$

(a) To obtain t^n on the right, set $r = n - s$. Coefficient comparison gives

$$\boxed{J_n(u+v) = \sum_{s=-\infty}^{\infty} J_s(u)J_{n-s}(v).}$$

(b) Set $n = 0$ and separate $s = 0$ from the pairs $s, -s$:

$$J_0(u+v) = J_0(u)J_0(v) + \sum_{s=1}^{\infty}\big[J_s(u)J_{-s}(v) + J_{-s}(u)J_s(v)\big].$$

Because $J_{-s}(z) = (-1)^s J_s(z)$, the two terms inside each bracket are equal. Hence

$$\boxed{J_0(u+v) = J_0(u)J_0(v) + 2\sum_{s=1}^{\infty} J_s(u)J_{-s}(v).}$$

Absolute convergence on any circle $0 < |t| < \infty$ justifies the product and coefficient collection; alternatively they follow by multiplying the two convergent exponential series defining the generating function.

14.1.3 Using only the generating function

$$e^{(x/2)(t-1/t)} = \sum_{n=-\infty}^{\infty} J_n(x)t^n$$

and not the explicit series form of $J_n(x)$, show that $J_n(x)$ has odd or even parity according to whether n is odd or even, that is,

$$J_n(x) = (-1)^n J_n(-x).$$

Solution.

Changing both signs leaves the exponent unchanged:

$$g(-x,-t) = \exp\left[\frac{-x}{2}\left(-t + \frac{1}{t}\right)\right] = g(x,t).$$

Expand these equal functions in powers of t:

$$\sum_{n\in\mathbb{Z}} J_n(x)t^n = \sum_{n\in\mathbb{Z}} J_n(-x)(-t)^n = \sum_{n\in\mathbb{Z}} (-1)^n J_n(-x)t^n.$$

For every integer n, equality of Laurent coefficients gives

$$\boxed{J_n(x) = (-1)^n J_n(-x).}$$

Thus even integer orders give even functions of x, and odd integer orders give odd functions. No explicit coefficient series for J_n was used.

14.1.4 Use the basic recurrence formulas, Eqs. (14.7) and (14.8), to prove the following formulas:

(a) $\dfrac{d}{dx}\big[x^n J_n(x)\big] = x^n J_{n-1}(x)$,

(b) $\dfrac{d}{dx}\big[x^{-n} J_n(x)\big] = -x^{-n} J_{n+1}(x)$,

(c) $J_n(x) = J'_{n+1} + \dfrac{n+1}{x} J_{n+1}(x)$.

Solution.

The two given recurrences are

$$J_{n-1} + J_{n+1} = \frac{2n}{x} J_n, \qquad J_{n-1} - J_{n+1} = 2J'_n.$$

Adding and subtracting them, respectively, gives

$$J'_n + \frac{n}{x} J_n = J_{n-1}, \qquad J'_n - \frac{n}{x} J_n = -J_{n+1}.$$

(a) The product rule and the first identity yield

$$\frac{d}{dx}(x^n J_n) = x^n \left(J'_n + \frac{n}{x} J_n\right) = \boxed{x^n J_{n-1}}.$$

(b) Using the second identity instead,

$$\frac{d}{dx}(x^{-n} J_n) = x^{-n} \left(J'_n - \frac{n}{x} J_n\right) = \boxed{-x^{-n} J_{n+1}}.$$

(c) Replace n by $n+1$ in the first identity:

$$\boxed{J_n = J'_{n+1} + \frac{n+1}{x} J_{n+1}.}$$

The manipulations hold at $x \neq 0$; any expression with a removable singularity at zero is extended there by its series limit.

14.1.5 Derive the Jacobi-Anger expansion

$$e^{i\rho\cos\varphi} = \sum_{m=-\infty}^{\infty} i^m J_m(\rho) e^{im\varphi}.$$

This is an expansion of a plane wave in a series of cylindrical waves.

Solution.

In the generating function, put $x = \rho$ and $t = ie^{i\varphi}$. Since $t^{-1} = -ie^{-i\varphi}$,

$$t - t^{-1} = i(e^{i\varphi} + e^{-i\varphi}) = 2i\cos\varphi.$$

Therefore

$$\begin{aligned} e^{i\rho\cos\varphi} &= \sum_{m=-\infty}^{\infty} J_m(\rho)(ie^{i\varphi})^m \\ &= \boxed{\sum_{m=-\infty}^{\infty} i^m J_m(\rho) e^{im\varphi}}. \end{aligned}$$

The factors $e^{im\varphi}$ are the angular cylindrical modes; their radial coefficients are $J_m(\rho)$.

14.1.6 Show that

(a) $\cos x = J_0(x) + 2\sum_{n=1}^{\infty}(-1)^n J_{2n}(x)$,

(b) $\sin x = 2\sum_{n=0}^{\infty}(-1)^n J_{2n+1}(x)$.

Solution.

Set $\varphi = 0$, $\rho = x$ in Exercise 14.1.5:

$$e^{ix} = \sum_{m\in\mathbb{Z}} i^m J_m(x).$$

Pair the terms of orders m and $-m$. Since $i^{-m}J_{-m} = i^{-m}(-1)^m J_m = i^m J_m$,

$$e^{ix} = J_0 + 2\sum_{m=1}^{\infty} i^m J_m = J_0 + 2\sum_{n=1}^{\infty}(-1)^n J_{2n} + 2i\sum_{n=0}^{\infty}(-1)^n J_{2n+1}.$$

For real x, comparison of real and imaginary parts gives

$$\boxed{\cos x = J_0(x) + 2\sum_{n=1}^{\infty}(-1)^n J_{2n}(x)}, \qquad \boxed{\sin x = 2\sum_{n=0}^{\infty}(-1)^n J_{2n+1}(x)}.$$

Both sides are entire functions, so the identities also extend to complex x.

14.1.7 To help remove the generating function from the realm of magic, show that it can be derived from the recurrence relation, Eq. (14.7).

Hint.

(a) Assume a generating function of the form

$$g(x,t) = \sum_{m=-\infty}^{\infty} J_m(x)t^m.$$

(b) Multiply Eq. (14.7) by t^n and sum over n.

(c) Rewrite the preceding result as

$$\left(t + \frac{1}{t}\right) g(x,t) = \frac{2t}{x}\frac{\partial g(x,t)}{\partial t}.$$

(d) Integrate and adjust the "constant" of integration (a function of x) so that the coefficient of the zeroth power, t^0, is $J_0(x)$ as given by Eq. (14.6).

Solution.

Start with the proposed Laurent series $g = \sum_m J_m(x)t^m$. Multiply $J_{n-1} + J_{n+1} = 2nJ_n/x$ by t^n and sum. Shifting the indices on the left gives

$$\sum_n J_{n-1}t^n = tg, \qquad \sum_n J_{n+1}t^n = t^{-1}g,$$

while the right side is $(2t/x)\partial_t g$. Thus, for fixed $x \neq 0$,

$$\partial_t g = \frac{x}{2}(1 + t^{-2})g.$$

Solving this first-order equation gives

$$g(x,t) = C(x) \exp\left[\frac{x}{2}(t - t^{-1})\right].$$

To find C, multiply the two elementary exponential series:

$$e^{xt/2}e^{-x/(2t)} = \sum_{r,s\geq 0} \frac{(-1)^s(x/2)^{r+s}}{r!s!}t^{r-s}.$$

Only $r = s$ contributes to t^0. Its coefficient is

$$\sum_{s=0}^{\infty} \frac{(-1)^s(x/2)^{2s}}{(s!)^2} = J_0(x).$$

The coefficient of t^0 in g is consequently $C(x)J_0(x)$, which must equal $J_0(x)$. This yields $C(x) = 1$ wherever $J_0(x) \neq 0$. At an isolated zero of J_0, matching that one coefficient alone does not determine C. Continuity in x of the Bessel family fixes the same value there; the recurrence by itself leaves an overall normalization undetermined. The limit at $x = 0$ likewise gives $g(0,t) = 1$. Hence

$$\boxed{g(x,t) = e^{(x/2)(t-t^{-1})}.}$$

14.1.8 Show, by direct differentiation, that

$$J_\nu(x) = \sum_{s=0}^{\infty} \frac{(-1)^s}{s!\Gamma(s+\nu+1)}\left(\frac{x}{2}\right)^{\nu+2s}$$

satisfies the two recurrence relations

$$J_{\nu-1}(x) + J_{\nu+1}(x) = \frac{2\nu}{x}J_\nu(x),$$
$$J_{\nu-1}(x) - J_{\nu+1}(x) = 2J'_\nu(x),$$

and Bessel's differential equation

$$x^2J''_\nu(x) + xJ'_\nu(x) + (x^2 - \nu^2)J_\nu(x) = 0.$$

Solution.

Use a fixed branch of x^ν (in particular $x > 0$ for real ν), and write

$$J_\nu(x) = \sum_{s=0}^{\infty} a_s x^{\nu+2s}, \qquad a_s = \frac{(-1)^s}{2^{\nu+2s}s!\Gamma(s+\nu+1)}.$$

The convergent series may be differentiated term by term on compact sets away from a branch point. First,

$$J'_\nu + \frac{\nu}{x}J_\nu = \sum_{s=0}^{\infty} 2(\nu+s)a_s x^{\nu+2s-1}$$
$$= \sum_{s=0}^{\infty} \frac{(-1)^s x^{\nu-1+2s}}{2^{\nu-1+2s}s!\Gamma(s+\nu)} = J_{\nu-1},$$

where $\Gamma(s+\nu+1) = (s+\nu)\Gamma(s+\nu)$ was used. On subtracting rather than adding the ν/x term, the $s=0$ term vanishes. Put $s = r+1$ in the remaining sum:

$$J'_\nu - \frac{\nu}{x}J_\nu = \sum_{s=1}^{\infty} 2sa_s x^{\nu+2s-1}$$
$$= -\sum_{r=0}^{\infty} \frac{(-1)^r x^{\nu+1+2r}}{2^{\nu+1+2r}r!\Gamma(r+\nu+2)} = -J_{\nu+1}.$$

Adding the last two identities gives $J_{\nu-1} - J_{\nu+1} = 2J'_\nu$; subtracting gives $J_{\nu-1} + J_{\nu+1} = 2\nu J_\nu/x$.

For the differential equation, the coefficient contributed by $x^2J''_\nu + xJ'_\nu - \nu^2 J_\nu$ at $x^{\nu+2s}$ is

$$\big[(\nu+2s)(\nu+2s-1) + (\nu+2s) - \nu^2\big]a_s = 4s(s+\nu)a_s.$$

The term $x^2 J_\nu$ contributes a_{s-1} for $s \geq 1$. The coefficient ratio is

$$\frac{a_s}{a_{s-1}} = -\frac{1}{4s(s+\nu)},$$

so $4s(s+\nu)a_s + a_{s-1} = 0$. The initial coefficient ($s=0$) already vanishes. Every power therefore cancels, proving

$$\boxed{x^2J''_\nu + xJ'_\nu + (x^2-\nu^2)J_\nu = 0.}$$

If a reciprocal gamma factor vanishes at an exceptional integer order, the identities are interpreted by continuation in ν, avoiding spurious division by a zero factor.

14.1.9 Prove that

$$\frac{\sin x}{x} = \int_0^{\pi/2} J_0(x\cos\theta)\cos\theta\, d\theta, \qquad \frac{1-\cos x}{x} = \int_0^{\pi/2} J_1(x\cos\theta)\, d\theta.$$

Hint. The definite integral

$$\int_0^{\pi/2} \cos^{2s+1}\theta\, d\theta = \frac{2\cdot 4\cdot 6\cdots(2s)}{1\cdot 3\cdot 5\cdots(2s+1)}$$

may be useful.

Solution.

The supplied cosine integral can be written in factorial form:

$$I_s = \int_0^{\pi/2} \cos^{2s+1}\theta\, d\theta = \frac{(2s)!!}{(2s+1)!!} = \frac{2^{2s}(s!)^2}{(2s+1)!}.$$

Expand J_0, multiply by $\cos\theta$, and integrate each term:

$$\int_0^{\pi/2} J_0(x\cos\theta)\cos\theta\,d\theta = \sum_{s=0}^{\infty} \frac{(-1)^s x^{2s}}{2^{2s}(s!)^2} I_s$$
$$= \sum_{s=0}^{\infty} \frac{(-1)^s x^{2s}}{(2s+1)!} = \boxed{\frac{\sin x}{x}}.$$

For J_1, the power of $\cos\theta$ is again $2s+1$:

$$\int_0^{\pi/2} J_1(x\cos\theta)\,d\theta = \sum_{s=0}^{\infty} \frac{(-1)^s x^{2s+1}}{2^{2s+1}s!(s+1)!} I_s$$
$$= \sum_{s=0}^{\infty} \frac{(-1)^s x^{2s+1}}{2(s+1)(2s+1)!}$$
$$= \sum_{s=0}^{\infty} \frac{(-1)^s x^{2s+1}}{(2s+2)!} = \boxed{\frac{1-\cos x}{x}}.$$

For fixed x, the Bessel series converge uniformly on this finite integration interval, justifying the termwise integrations. At $x = 0$, the two apparent quotients have their continuous values 1 and 0, which also follow immediately from the integrands.

14.1.10 Derive

$$J_n(x) = (-1)^n x^n \left(\frac{1}{x}\frac{d}{dx}\right)^n J_0(x).$$

Hint. Try mathematical induction (Section 1.4).
Here n is a nonnegative integer; at $x = 0$ use the continuous limit.

Solution.

Let $D = x^{-1}d/dx$ and take $n = 0, 1, 2, \ldots$. For $n = 0$, the claimed identity is simply $J_0 = J_0$. Assume that $J_n = (-1)^n x^n D^n J_0$. Then

$$x^{-n} J_n = (-1)^n D^n J_0.$$

Differentiate this equation and use Exercise 14.1.4(b):

$$-x^{-n} J_{n+1} = (-1)^n \frac{d}{dx}(D^n J_0) = (-1)^n x D^{n+1} J_0.$$

Multiplying by $-x^n$ gives

$$J_{n+1} = (-1)^{n+1} x^{n+1} D^{n+1} J_0.$$

This is the required induction step, so

$$\boxed{J_n(x) = (-1)^n x^n \left(\frac{1}{x}\frac{d}{dx}\right)^n J_0(x)}.$$

The proof is for $x \neq 0$. The even power series of J_0 shows that the apparent singularities of the final expression are removable at zero.

14.1.11 Show that between any two consecutive zeros of $J_n(x)$ there is one and only one zero of $J_{n+1}(x)$.

Hint. Equations (14.10) and (14.11) may be useful.

Use consecutive strictly positive zeros, excluding the origin.

Solution.

Use consecutive positive zeros, excluding the origin. For integer $n \geq 1$, the origin is also a zero of J_n; the assertion would be false on the interval between zero and its first positive zero. Let $a < b$ be consecutive positive zeros of J_n. The function $f = x^{-n}J_n$ also vanishes at a, b. Rolle's theorem gives at least one $c \in (a, b)$ for which $f'(c) = 0$. By Exercise 14.1.4(b),

$$f' = -x^{-n}J_{n+1},$$

so $J_{n+1}(c) = 0$.

Suppose there were two such zeros $c < d$ inside (a, b). Apply Rolle's theorem to $g = x^{n+1}J_{n+1}$ on $[c, d]$. Exercise 14.1.4(a) gives

$$g' = x^{n+1}J_n,$$

so g' would vanish at some point of (c, d). That would be a zero of J_n between its consecutive zeros a, b, a contradiction. There is therefore exactly one zero of J_{n+1} in (a, b).

For strict interlacing there are no shared positive zeros. If both $J_n(a)$ and $J_{n+1}(a)$ vanished, the recurrence $J_n' = nJ_n/x - J_{n+1}$ would give $J_n'(a) = 0$. At a positive point, Bessel's ODE is regular; zero value and zero derivative force the identically zero solution, contrary to the nonzero defining series. The same argument shows that each positive zero is simple.

14.1.12 An analysis of antenna radiation patterns for a system with a circular aperture involves the equation

$$g(u) = \int_0^1 f(r)J_0(ur)r\,dr.$$

If $f(r) = 1 - r^2$, show that

$$g(u) = \frac{2}{u^2}J_2(u).$$

Solution.

For $u \neq 0$, put $x = ur$, so $r\,dr = x\,dx/u^2$:

$$g(u) = \frac{1}{u^2}\int_0^u \left(1 - \frac{x^2}{u^2}\right) xJ_0(x)\,dx.$$

The derivative identity $(xJ_1)' = xJ_0$ allows integration by parts:

$$\begin{aligned} g(u) &= \frac{1}{u^2}\left[\left(1 - \frac{x^2}{u^2}\right) xJ_1(x)\right]_0^u + \frac{2}{u^4}\int_0^u x^2J_1(x)\,dx \\ &= \frac{2}{u^4}\left[x^2J_2(x)\right]_0^u = \boxed{\frac{2J_2(u)}{u^2}}. \end{aligned}$$

The first endpoint expression vanishes at $x = u$ because of its prefactor and at $x = 0$ because $xJ_1(x) = O(x^2)$. The second identity uses $(x^2J_2)' = x^2J_1$ and $x^2J_2(x) = O(x^4)$. At $u = 0$, direct integration gives $g(0) = \int_0^1 (r - r^3)dr = 1/4$; the answer has the same continuous limit because $J_2(u) = u^2/8 + O(u^4)$.

14.1.13 The differential cross section in a nuclear scattering experiment is given by $d\sigma/d\Omega = |f(\theta)|^2$. An approximate treatment leads to

$$f(\theta) = \frac{-ik}{2\pi} \int_0^{2\pi} \int_0^R \exp[ik\rho \sin\theta \sin\varphi] \rho \, d\rho \, d\varphi.$$

Here θ is an angle through which the scattered particle is scattered. R is the nuclear radius. Show that

$$\frac{d\sigma}{d\Omega} = (\pi R^2) \frac{1}{\pi} \left[\frac{J_1(kR\sin\theta)}{\sin\theta} \right]^2.$$

Solution.

Put $b = k \sin\theta$. The order-zero angular representation is

$$\frac{1}{2\pi} \int_0^{2\pi} e^{ib\rho\sin\varphi} \, d\varphi = J_0(b\rho).$$

Performing that integral first therefore gives

$$f(\theta) = -ik \int_0^R \rho J_0(b\rho) \, d\rho.$$

For $b \neq 0$, change variable to $x = b\rho$ and integrate $xJ_0(x) = (xJ_1(x))'$:

$$\int_0^R \rho J_0(b\rho) \, d\rho = \frac{1}{b^2} \left[xJ_1(x) \right]_0^{bR} = \frac{R}{b} J_1(bR).$$

Consequently

$$f(\theta) = -iR \frac{J_1(kR\sin\theta)}{\sin\theta}.$$

For real k, R, θ, the quotient is real, and $|-i|^2 = 1$. Thus

$$\boxed{\frac{d\sigma}{d\Omega} = R^2 \left[\frac{J_1(kR\sin\theta)}{\sin\theta} \right]^2 = (\pi R^2) \frac{1}{\pi} \left[\frac{J_1(kR\sin\theta)}{\sin\theta} \right]^2.}$$

The forward and backward axial directions are understood by continuity: $J_1(z) = z/2 + O(z^3)$ gives $f \to -ikR^2/2$ and $d\sigma/d\Omega \to k^2R^4/4$. The apparent division by $\sin\theta = 0$ is therefore removable.

14.1.14 A set of functions $C_n(x)$ satisfies the recurrence relations

$$C_{n-1}(x) - C_{n+1}(x) = \frac{2n}{x} C_n(x),$$
$$C_{n-1}(x) + C_{n+1}(x) = 2C_n'(x).$$

(a) What linear second-order ODE does the $C_n(x)$ satisfy?

(b) By a change of variable transform your ODE into Bessel's equation. This suggests that $C_n(x)$ may be expressed in terms of Bessel functions of transformed argument.

Solution.

(a) Adding and subtracting the supplied recurrences gives

$$C_{n-1} = C_n' + \frac{n}{x}C_n, \qquad C_{n+1} = C_n' - \frac{n}{x}C_n.$$

Differentiate the second identity:

$$C_{n+1}' = C_n'' - \frac{n}{x}C_n' + \frac{n}{x^2}C_n.$$

The first identity with n replaced by $n+1$ also says

$$C_{n+1}' = C_n - \frac{n+1}{x}C_{n+1} = C_n - \frac{n+1}{x}\left(C_n' - \frac{n}{x}C_n\right).$$

Equate these two expressions, move everything to the left, and collect coefficients:

$$C_n'' + \frac{1}{x}C_n' - \left(1 + \frac{n^2}{x^2}\right)C_n = 0.$$

Multiplication by x^2 gives the modified Bessel equation

$$\boxed{x^2C_n'' + xC_n' - (x^2 + n^2)C_n = 0.}$$

(b) Put $z = ix$ and $C_n(x) = Y(z)$. Then $C_n' = iY'$, $C_n'' = -Y''$, and $x^2 = -z^2$. Substituting these into the boxed ODE gives

$$\boxed{z^2Y'' + zY' + (z^2 - n^2)Y = 0},$$

The manual's displayed ODE on p. 197 has x^2C_n'; the correct first-derivative term is xC_n', as the derivation above shows. This is Bessel's equation of order n. Its solutions can consequently be expressed using Bessel functions with imaginary argument. For example $I_n(x) = i^{-n}J_n(ix)$ is the regular modified Bessel function; the phase depending on n makes it satisfy the supplied recurrences.

14.1.15

(a) Show by direct differentiation and substitution that

$$J_\nu(x) = \frac{1}{2\pi i}\int_C e^{(x/2)(t-1/t)}t^{-\nu-1}\,dt$$

(this is the Schlaefli integral representation of J_ν), and that the equivalent equation,

$$J_\nu(x) = \frac{1}{2\pi i}\left(\frac{x}{2}\right)^\nu \int_C e^{s-x^2/4s}s^{-\nu-1}\,ds,$$

both satisfy Bessel's equation. C is the contour shown in Fig. 14.5. The negative real axis is the cut line.

Hint. This exercise is aimed at providing details of the discussion that starts at Eq. (14.38).

(b) Show that the first integral (with n an integer) may be transformed into

$$J_n(x) = \frac{1}{2\pi}\int_0^{2\pi} e^{i(x\sin\theta - n\theta)}\,d\theta = \frac{i^{-n}}{2\pi}\int_0^{2\pi} e^{i(x\cos\theta + n\theta)}\,d\theta.$$

Take $x > 0$ initially. The open contour starts below the negative-real cut at $-\infty$, circles zero counterclockwise, and returns above the cut; use $-\pi < \arg t < \pi$.

Solution.

For noninteger ν, the contour in Fig. 14.5 is an open Hankel contour: it starts at $-\infty$ below the cut, circles zero counterclockwise, and returns to $-\infty$ above the cut. The earlier transcription used $\oint$; the open $\int_C$ has now been restored, as in the textbook. Take $x > 0$ and $-\pi < \arg t < \pi$ initially. The two unbounded ends then decay exponentially. Write

$$f(t,x) = e^{x(t-t^{-1})/2} t^{-\nu-1}, \quad S = t - t^{-1}, \quad T = t + t^{-1}.$$

(a) Differentiation with respect to x gives $f_x = Sf/2$ and $f_{xx} = S^2 f/4$. Consequently the Bessel operator applied under the integral has integrand

$$x^2 f_{xx} + x f_x + (x^2 - \nu^2) f = f\left(\frac{x^2 T^2}{4} + \frac{xS}{2} - \nu^2\right),$$

where $S^2 + 4 = T^2$. This is a total derivative. Indeed, since $tT' = S$,

$$\begin{aligned} \frac{d}{dt}\left[e^{xS/2} t^{-\nu}\left(\nu + \frac{xT}{2}\right)\right] &= f\left[\left(\frac{xT}{2} - \nu\right)\left(\nu + \frac{xT}{2}\right) + \frac{xS}{2}\right] \\ &= f\left(\frac{x^2 T^2}{4} + \frac{xS}{2} - \nu^2\right). \end{aligned}$$

Its integral is zero because the bracketed primitive tends to zero at both ends of the contour. Therefore $F(x) = (2\pi i)^{-1} \int_C f\, dt$ satisfies $x^2 F'' + xF' + (x^2 - \nu^2)F = 0$.

In the same integral put $s = xt/2$. Then

$$\frac{x}{2}(t - t^{-1}) = s - \frac{x^2}{4s}, \qquad t^{-\nu-1} dt = (x/2)^{\nu} s^{-\nu-1} ds,$$

which gives the second representation. Positive dilation changes neither the cut nor the contour's winding around zero; it may be deformed back to a fixed contour without crossing a singularity. This proves its equivalence and hence its differential equation.

For completeness the normalization is also fixed, rather than merely identifying an unspecified solution of the ODE. Expand the second exponential and use the Hankel formula for the reciprocal gamma function:

$$\begin{aligned} F(x) &= (x/2)^{\nu} \sum_{r=0}^{\infty} \frac{(-x^2/4)^r}{r!} \frac{1}{2\pi i} \int_C e^s s^{-\nu-r-1}\, ds \\ &= \sum_{r=0}^{\infty} \frac{(-1)^r (x/2)^{\nu+2r}}{r!\Gamma(\nu + r + 1)} = J_\nu(x). \end{aligned}$$

The contour can have a fixed positive distance from zero, so the series is uniformly dominated there by $e^{x^2/(4|s|)}$, and the negative-real tails decay exponentially. These facts justify the termwise integration and differentiation. Other admissible x follow by analytic continuation on a fixed branch.

(b) For integer n the two banks of the cut cancel, leaving a counterclockwise circle. Set $t = e^{i\theta}$, so $t - t^{-1} = 2i \sin\theta$ and $t^{-n-1} dt = i e^{-in\theta} d\theta$:

$$\boxed{J_n(x) = \frac{1}{2\pi} \int_0^{2\pi} e^{i(x \sin\theta - n\theta)}\, d\theta}.$$

Now set $\theta = \pi/2 - \phi$. The phase becomes $x\cos\phi + n\phi - n\pi/2$. Reversing the integration limits and using the 2π periodicity for integer n yields

$$\boxed{J_n(x) = \frac{i^{-n}}{2\pi}\int_0^{2\pi} e^{i(x\cos\phi + n\phi)}\,d\phi}.$$

14.1.16 The contour C in Exercise 14.1.15 is deformed to the path $-\infty$ to -1, unit circle $e^{-i\pi}$ to $e^{i\pi}$, and finally -1 to $-\infty$. Show that

$$J_\nu(x) = \frac{1}{\pi}\int_0^\pi \cos(\nu\theta - x\sin\theta)\,d\theta - \frac{\sin\nu\pi}{\pi}\int_0^\infty e^{-\nu\theta - x\sinh\theta}\,d\theta.$$

This is Bessel's integral.

Hint. The negative values of the variable of integration u must be represented in a manner consistent with the presence of the branch cut, for example, by writing $u = te^{\pm i\pi}$.

Take $x > 0$ initially and use the cut convention of Exercise 14.1.15.

Solution.

The hint's $e^{\pm ix}$ must be $e^{\pm i\pi}$: these are the arguments of the two banks of the negative real axis. Use the open contour and branch specified in Exercise 14.1.15, with $x > 0$ initially. Split its integral into a lower ray, the unit circle, and an upper ray. On the lower ray put $t = e^{u-i\pi}$ with $u : \infty \to 0$. On the upper ray use $t = e^{u+i\pi}$ with $u : 0 \to \infty$. In both cases $t - t^{-1} = -2\sinh u$, whereas

$$t^{-\nu-1}dt = e^{-\nu u}e^{+i\pi\nu}du \quad \text{(lower)}, \qquad t^{-\nu-1}dt = e^{-\nu u}e^{-i\pi\nu}du \quad \text{(upper)}.$$

Thus their combined contribution is

$$\frac{1}{2\pi i}\left[-e^{i\pi\nu} + e^{-i\pi\nu}\right]\int_0^\infty e^{-\nu u - x\sinh u}\,du = -\frac{\sin\pi\nu}{\pi}\int_0^\infty e^{-\nu u - x\sinh u}\,du.$$

On the circle $t = e^{i\theta}$, $-\pi \le \theta \le \pi$, and the contribution is

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{i(x\sin\theta - \nu\theta)}d\theta = \frac{1}{\pi}\int_0^\pi \cos(\nu\theta - x\sin\theta)d\theta.$$

Here the sine of the phase is odd and its cosine is even. Adding the three parts gives exactly the stated formula. For integer $\nu = n$, the ray contribution vanishes because $\sin\pi n = 0$.

14.1.17

(a) Show that

$$J_\nu(x) = \frac{2}{\pi^{1/2}\Gamma(\nu + \frac{1}{2})}\left(\frac{x}{2}\right)^\nu \int_0^{\pi/2} \cos(x\sin\theta)\cos^{2\nu}\theta\,d\theta,$$

where $\nu > -\frac{1}{2}$.

Hint. Here is a chance to use series expansion and term-by-term integration. The formulas of Section 13.3 will prove useful.

(b) Transform the integral in part (a) into

$$\begin{aligned} J_\nu(x) &= \frac{1}{\pi^{1/2}\Gamma(\nu+\frac{1}{2})}\left(\frac{x}{2}\right)^\nu \int_0^\pi \cos(x\cos\theta)\sin^{2\nu}\theta\,d\theta \\ &= \frac{1}{\pi^{1/2}\Gamma(\nu+\frac{1}{2})}\left(\frac{x}{2}\right)^\nu \int_0^\pi e^{\pm ix\cos\theta}\sin^{2\nu}\theta\,d\theta \\ &= \frac{1}{\pi^{1/2}\Gamma(\nu+\frac{1}{2})}\left(\frac{x}{2}\right)^\nu \int_{-1}^1 e^{\pm ipx}(1-p^2)^{\nu-1/2}\,dp. \end{aligned}$$

These are alternate integral representations of $J_\nu(x)$.

Solution.

(a) Expand the cosine and integrate each term. For real $\nu > -1/2$, the endpoint weight is integrable; uniform convergence of the cosine series on the finite interval justifies the operation. The beta integral gives

$$\begin{aligned} \int_0^{\pi/2} \sin^{2r}\theta\cos^{2\nu}\theta\,d\theta &= \frac{1}{2}B(r+\tfrac{1}{2},\nu+\tfrac{1}{2}) \\ &= \frac{\Gamma(r+\frac{1}{2})\Gamma(\nu+\frac{1}{2})}{2\Gamma(r+\nu+1)}. \end{aligned}$$

Since $\Gamma(r+1/2) = \sqrt{\pi}(2r)!/(4^r r!)$, the proposed right-hand side becomes

$$\begin{aligned} \frac{2(x/2)^\nu}{\sqrt{\pi}\Gamma(\nu+1/2)}\sum_{r=0}^\infty \frac{(-1)^r x^{2r}}{(2r)!}\frac{\Gamma(r+1/2)\Gamma(\nu+1/2)}{2\Gamma(r+\nu+1)} &= \sum_{r=0}^\infty \frac{(-1)^r (x/2)^{\nu+2r}}{r!\Gamma(r+\nu+1)} \\ &= J_\nu(x). \end{aligned}$$

The factor x^ν is taken on the same branch as J_ν.

(b) Replacing θ by $\pi/2-\theta$ first changes the integrand to $\cos(x\cos\theta)\sin^{2\nu}\theta$. Reflection about $\pi/2$ leaves this integrand unchanged, so

$$2\int_0^{\pi/2}\cos(x\cos\theta)\sin^{2\nu}\theta\,d\theta = \int_0^\pi \cos(x\cos\theta)\sin^{2\nu}\theta\,d\theta.$$

This gives the first requested form. The integral of $\sin(x\cos\theta)\sin^{2\nu}\theta$ is zero: the substitution $\theta \mapsto \pi-\theta$ reverses its sign. We may consequently replace the cosine by either $e^{ix\cos\theta}$ or $e^{-ix\cos\theta}$. Finally $p = \cos\theta$ gives

$$dp = -\sin\theta\,d\theta, \qquad \sin^{2\nu}\theta\,d\theta = -(1-p^2)^{\nu-1/2}dp.$$

Reversing the limits $1, -1$ yields the final integral in the question. The same proof works for complex ν with $\Re\nu > -1/2$.

14.1.18 Given that C is the contour in Fig. 14.5,

(a) From

$$J_\nu(x) = \frac{1}{2\pi i}\left(\frac{x}{2}\right)^\nu \int_C t^{-\nu-1}e^{t-x^2/4t}\,dt$$

derive the recurrence relation

$$J_\nu'(x) = \frac{\nu}{x}J_\nu(x) - J_{\nu+1}(x).$$

(b) From

$$J_\nu(x) = \frac{1}{2\pi i}\int_C t^{-\nu-1}e^{(x/2)(t-1/t)}\,dt$$

derive the recurrence relation

$$J'_\nu(x) = \frac{1}{2}\big[J_{\nu-1}(x) - J_{\nu+1}(x)\big].$$

The integrals are over the open Hankel contour of Exercise 14.1.15, with its fixed branch and $x > 0$ initially.

Solution.

Use a fixed open Hankel contour, the branch convention of Exercise 14.1.15, and initially $x > 0$. Its exponentially decaying tails justify differentiating under the integral.

(a) Both the prefactor and the exponential depend on x:

$$\begin{aligned} J'_\nu(x) &= \frac{\nu}{x}J_\nu(x) + \frac{(x/2)^\nu}{2\pi i}\int_C t^{-\nu-1}\left(-\frac{x}{2t}\right)e^{t-x^2/(4t)}dt \\ &= \frac{\nu}{x}J_\nu(x) - \frac{(x/2)^{\nu+1}}{2\pi i}\int_C t^{-(\nu+1)-1}e^{t-x^2/(4t)}dt \\ &= \boxed{\frac{\nu}{x}J_\nu(x) - J_{\nu+1}(x)}. \end{aligned}$$

(b) Differentiating the other representation gives

$$\begin{aligned} J'_\nu(x) &= \frac{1}{4\pi i}\int_C \left[t^{-\nu} - t^{-\nu-2}\right]e^{x(t-t^{-1})/2}dt \\ &= \frac{1}{2}\left[\frac{1}{2\pi i}\int_C t^{-(\nu-1)-1}e^{x(t-t^{-1})/2}dt - \frac{1}{2\pi i}\int_C t^{-(\nu+1)-1}e^{x(t-t^{-1})/2}dt\right] \\ &= \boxed{\tfrac{1}{2}[J_{\nu-1}(x) - J_{\nu+1}(x)]}. \end{aligned}$$

The identities then continue to the domains of the Bessel functions.

14.1.19 Show that the recurrence relation

$$J'_n(x) = \frac{1}{2}\big[J_{n-1}(x) - J_{n+1}(x)\big]$$

follows directly from differentiation of

$$J_n(x) = \frac{1}{\pi}\int_0^\pi \cos(n\theta - x\sin\theta)\,d\theta.$$

The order n in this trigonometric integral is an integer.

Solution.

For integer n, differentiating the finite-interval integral gives

$$J'_n(x) = \frac{1}{\pi}\int_0^\pi \sin(n\theta - x\sin\theta)\sin\theta\,d\theta.$$

Set $A = n\theta - x\sin\theta$. The elementary subtraction identity $\cos(A-\theta) - \cos(A+\theta) = 2\sin A\sin\theta$ therefore gives

$$\begin{aligned} 2J_n'(x) &= \frac{1}{\pi}\int_0^\pi \{\cos[(n-1)\theta - x\sin\theta] - \cos[(n+1)\theta - x\sin\theta]\}\,d\theta \\ &= J_{n-1}(x) - J_{n+1}(x). \end{aligned}$$

Dividing by two proves the recurrence. The restriction to integer order matters: for noninteger order the extra ray integral from Exercise 14.1.16 must also be differentiated.

14.1.20 Evaluate

$$\int_0^\infty e^{-ax}J_0(bx)\,dx, \qquad a, b > 0.$$

Actually the results hold for $a \ge 0$, $-\infty < b < \infty$. This is a Laplace transform of J_0.

Hint. Either an integral representation of J_0 or a series expansion will be helpful.

For real $a \ge 0$ and b, exclude $(a,b) = (0,0)$; the $a = 0$, $b \ne 0$ integral is conditionally convergent.

Solution.

For $a > 0$ and real b, use the integer-order integral representation:

$$\begin{aligned} L(a,b) &= \frac{1}{\pi}\int_0^\pi \int_0^\infty e^{-ax}\cos(bx\sin\theta)\,dx\,d\theta \\ &= \frac{1}{\pi}\int_0^\pi \frac{a}{a^2 + b^2\sin^2\theta}\,d\theta. \end{aligned}$$

The interchange is justified by the integrable bound e^{-ax}. The inner Laplace integral is the real part of $\int_0^\infty e^{-(a-ib\sin\theta)x}dx = (a - ib\sin\theta)^{-1}$. Reflect the remaining integrand about $\pi/2$ and put $u = \tan\theta$:

$$\begin{aligned} L(a,b) &= \frac{2a}{\pi}\int_0^{\pi/2} \frac{d\theta}{a^2 + b^2\sin^2\theta} = \frac{2a}{\pi}\int_0^\infty \frac{du}{a^2 + (a^2+b^2)u^2} \\ &= \frac{2}{\pi\sqrt{a^2+b^2}}\left[\arctan\frac{u\sqrt{a^2+b^2}}{a}\right]_0^\infty = \boxed{\frac{1}{\sqrt{a^2+b^2}}}. \end{aligned}$$

The claimed extension to $a \ge 0$ excludes $(a,b) = (0,0)$, where the integrand is 1 and the integral diverges. At $a = 0$, $b \ne 0$, the integral converges conditionally and equals $1/|b|$. To justify that boundary value, $J_0(bx) = J_0(|b|x)$ and its large-x expansion is an oscillating cosine times $x^{-1/2}$ plus an $O(x^{-3/2})$ remainder. The first term converges by integration by parts and the remainder absolutely. Abel's theorem for this convergent improper integral then gives its value as $\lim_{a\downarrow 0} L(a,b) = 1/|b|$.

14.1.21 Using the symmetries of the trigonometric functions, confirm that for integer n,

$$\frac{1}{2\pi}\int_0^{2\pi} \cos(x\sin\theta - n\theta)\,d\theta = \frac{1}{\pi}\int_0^\pi \cos(x\sin\theta - n\theta)\,d\theta.$$

Solution.

Let $F(\theta) = \cos(x\sin\theta - n\theta)$. For integer n,

$$\begin{aligned} F(2\pi - \theta) &= \cos[-x\sin\theta - 2\pi n + n\theta] \\ &= \cos[-(x\sin\theta - n\theta)] = F(\theta). \end{aligned}$$

In the integral from π to 2π, substitute $u = 2\pi - \theta$. Then

$$\int_{\pi}^{2\pi} F(\theta)d\theta = \int_{0}^{\pi} F(2\pi - u)du = \int_{0}^{\pi} F(u)du.$$

Hence the integral over $[0, 2\pi]$ is twice that over $[0, \pi]$. Division by 2π proves the stated equality.

14.1.22

(a) Plot the intensity, Φ^2 of Eq. (14.25), as a function of $(\sin\alpha/\lambda)$ along a diameter of the circular diffraction pattern. Locate the first two minima.

(b) Estimate the fraction of the total light intensity that falls within the central maximum.

Hint. $[J_1(x)]^2/x$ may be written as a derivative and the area integral of the intensity integrated by inspection.

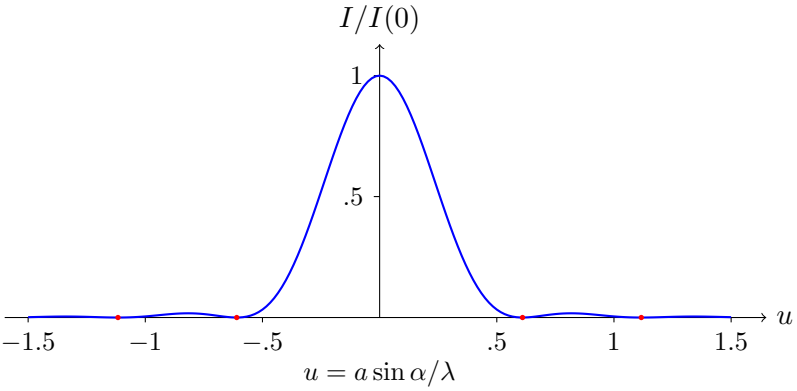
For the total-intensity estimate use the paraxial diffraction plane and an aperture large compared with the wavelength.

Solution.

(a) Write $q = \sin\alpha/\lambda$ and $u = aq$. Equation (14.25), normalized by its central intensity, is

$$\boxed{\frac{I(q)}{I(0)} = \left[\frac{2J_1(2\pi aq)}{2\pi aq}\right]^2}, \qquad I(0)/I(0) = 1.$$

The function is even, so the diameter profile is symmetric:



The noncentral minima occur at the nonzero zeros of J_1. The first two positive zeros, found from $J_1(j) = 0$, are

$$j_{11} = 3.8317059702, \qquad j_{12} = 7.0155866698.$$

Hence their positions on either side of the center are

$$\boxed{\frac{\sin\alpha}{\lambda} = \pm\frac{0.6098349456}{a}, \qquad \frac{\sin\alpha}{\lambda} = \pm\frac{1.1165652972}{a}}.$$

The point $q = 0$ is a maximum, not a zero: the apparent $0/0$ there has limit one. Physical angles require $|\sin\alpha| \le 1$, so a given minimum is accessible only if its displayed value satisfies that bound.

(b) In the small-angle diffraction plane the annular area element is proportional to $q\,dq$. Put $x = 2\pi aq$; the enclosed energy is therefore proportional to $\int_0^X J_1(x)^2\,dx/x$. Using $J_0' = -J_1$ and $J_1' = J_0 - J_1/x$,

$$\frac{d}{dx}(J_0^2 + J_1^2) = 2J_0(-J_1) + 2J_1(J_0 - J_1/x) = -\frac{2J_1^2}{x},$$
$$\int_0^X \frac{J_1(x)^2}{x}dx = \frac{1}{2}[1 - J_0(X)^2 - J_1(X)^2].$$

Both squared Bessel functions tend to zero as $X \to \infty$, so the total radial integral is $1/2$. At the first dark ring, $X = j_{11}$ and $J_1(X) = 0$. The central fraction is consequently

$$\boxed{1 - J_0(j_{11})^2 = 0.8377848692\ldots \simeq 83.8\%}.$$

This is the usual paraxial, large-aperture estimate ($a/\lambda \gg 1$). The manual, p. 201, says a/λ is small when extending the pattern to infinity; the regime is reversed. Likewise the text on p. 650 calls $2\pi a/\lambda$ small while giving its value as 57120, which is large. For a subwavelength aperture, the first dark ring may not exist at any physical angle, and the 83.8% plane-area estimate is not applicable.

14.1.23 The fraction of light incident on a circular aperture (normal incidence) that is transmitted is given by

$$T = 2\int_0^{2ka} J_2(x)\frac{dx}{x} - \frac{1}{2ka}\int_0^{2ka} J_2(x)\,dx.$$

Here a is the radius of the aperture and k is the wave number, $2\pi/\lambda$. Show that

(a) $T = 1 - \dfrac{1}{ka}\displaystyle\sum_{n=0}^{\infty} J_{2n+1}(2ka)$,

(b) $T = 1 - \dfrac{1}{2ka}\displaystyle\int_0^{2ka} J_0(x)\,dx$.

Solution.

Put $A = 2ka > 0$. The recurrence from Exercise 14.1.4 gives $[J_1(x)/x]' = -J_2(x)/x$, while $J_1(x)/x \to 1/2$ at zero. Thus

$$2\int_0^A \frac{J_2(x)}{x}\,dx = -2\left[\frac{J_1(x)}{x}\right]_0^A = 1 - \frac{2J_1(A)}{A}.$$

Also $2J_1' = J_0 - J_2$, so

$$\int_0^A J_2(x)dx = \int_0^A J_0(x)dx - 2J_1(A),$$

because $J_1(0) = 0$. Substitution into the defining formula for T cancels the two terms containing $J_1(A)$ and proves part (b):

$$\boxed{T = 1 - \frac{1}{A}\int_0^A J_0(x)dx}.$$

For part (a), successively use $J_{2r} = 2J_{2r+1}' + J_{2r+2}$. After N substitutions and integration,

$$\int_0^A J_0(x)dx = 2\sum_{r=0}^{N-1} J_{2r+1}(A) + \int_0^A J_{2N}(x)dx.$$

The series definition bounds $|J_m(x)|$ for $0 \le x \le A$ by $(A/2)^m e^{A^2/4}/m!$. Thus the last integral tends to zero as $N \to \infty$, and

$$\boxed{T = 1 - \frac{1}{ka} \sum_{r=0}^{\infty} J_{2r+1}(2ka)}.$$

At $ka = 0$ the expressions with denominators have their continuous limit $T = 0$, as follows from $A^{-1} \int_0^A J_0 \to 1$.

14.1.24 The amplitude $U(\rho, \varphi, t)$ of a vibrating circular membrane of radius a satisfies the wave equation

$$\nabla^2 U \equiv \frac{\partial^2 U}{\partial \rho^2} + \frac{1}{\rho}\frac{\partial U}{\partial \rho} + \frac{1}{\rho^2}\frac{\partial^2 U}{\partial \varphi^2} = \frac{1}{v^2}\frac{\partial^2 U}{\partial t^2}.$$

Here v is the phase velocity of the wave, determined by the properties of the membrane.

(a) Show that a physically relevant solution is

$$U(\rho, \varphi, t) = J_m(k\rho)\big(c_1 e^{im\varphi} + c_2 e^{-im\varphi}\big)\big(b_1 e^{i\omega t} + b_2 e^{-i\omega t}\big).$$

(b) From the Dirichlet boundary condition $J_m(ka) = 0$, find the allowable values of k.

The separated wave satisfies $\omega = vk$; the two angular factors coincide for $m = 0$.

Solution.

(a) Write $U = P(\rho)\Phi(\varphi)T(t)$ and divide the wave equation by $P\Phi T$. The result is

$$\frac{P''}{P} + \frac{P'}{\rho P} + \frac{\Phi''}{\rho^2 \Phi} = \frac{T''}{v^2 T} = -k^2.$$

Therefore $T'' + v^2 k^2 T = 0$. Defining $\omega = vk$ gives $T = b_1 e^{i\omega t} + b_2 e^{-i\omega t}$. Multiplication of the remaining equation by ρ^2 separates the angle:

$$\frac{\Phi''}{\Phi} = -m^2, \qquad \rho^2 P'' + \rho P' + (k^2\rho^2 - m^2)P = 0.$$

Single-valuedness under $\varphi \mapsto \varphi + 2\pi$ requires integer m, so $\Phi = c_1 e^{im\varphi} + c_2 e^{-im\varphi}$. With $z = k\rho$ the radial equation is Bessel's equation. Finiteness at the center excludes Y_m, leaving $P = J_m(k\rho)$ up to a constant. Multiplying the three factors proves the displayed solution. Its parameters must satisfy $\omega = vk$; they are not independent. For $m = 0$ the two angular exponentials are the same constant and represent only one angular mode.

(b) Let α_{mn} be the nth positive zero of J_m, $m = 0, 1, \ldots$, $n = 1, 2, \ldots$. The fixed-edge condition gives

$$ka = \alpha_{mn}, \qquad \boxed{k_{mn} = \frac{\alpha_{mn}}{a}, \qquad \omega_{mn} = \frac{v\alpha_{mn}}{a}}.$$

The negative-m radial solutions duplicate these since $J_{-m} = (-1)^m J_m$. For $m > 0$ the two independent real angular factors are $\cos m\varphi$ and $\sin m\varphi$. A zero value of k gives a regular harmonic radial function proportional to ρ^m (a constant for $m = 0$); the edge condition then forces its coefficient to vanish, so it supplies no additional normal mode.

14.1.25 Example 14.1.2 describes the TM modes of electromagnetic cavity oscillation. To obtain the transverse electric (TE) modes, we set $E_z = 0$ and work from the z component of the magnetic induction B:

$$\nabla^2 B_z + \alpha^2 B_z = 0$$

with boundary conditions

$$B_z(0) = B_z(l) = 0 \quad \text{and} \quad \left.\frac{\partial B_z}{\partial \rho}\right|_{\rho=a} = 0.$$

Show that the TE resonant frequencies are given by

$$\omega_{mnp} = c\sqrt{\frac{\beta_{mn}^2}{a^2} + \frac{p^2\pi^2}{l^2}}, \quad p = 1, 2, 3, \ldots,$$

and identify the quantities β_{mn}.

(Here $\alpha^2 = \omega^2/c^2$, with ω the angular frequency; the height of the cavity is l.)

The TE fields must also obey $\nabla \cdot B = 0$ and the perfectly conducting side condition $B_\rho = 0$; these exclude a spurious constant transverse scalar mode.

Solution.

Write $B_z = P(\rho)\Phi(\varphi)Z(z)$ with harmonic time factor $e^{-i\omega t}$. Separation gives

$$\Phi'' + m^2\Phi = 0, \quad Z'' + g^2 Z = 0, \quad \rho^2 P'' + \rho P' + (q^2\rho^2 - m^2)P = 0, \qquad q^2 = \frac{\omega^2}{c^2} - g^2.$$

Periodicity makes m an integer. The end conditions $Z(0) = Z(l) = 0$ give $Z = \sin(p\pi z/l)$ and $g = p\pi/l$, $p = 1, 2, \ldots$. Regularity at the axis gives $P = J_m(q\rho)$ for $q > 0$. The side condition is

$$P'(a) = qJ_m'(qa) = 0.$$

Thus, with β_{mn} the nth *positive zero of* J_m',

$$q = \frac{\beta_{mn}}{a}, \qquad \boxed{\omega_{mnp} = c\sqrt{\frac{\beta_{mn}^2}{a^2} + \frac{p^2\pi^2}{l^2}}}, \quad m \geq 0,\ n \geq 1,\ p \geq 1.$$

The manual, p. 202, mistakenly identifies β_{mn} as a zero of J_m; it must be a zero of J_m'. Its separated sine also needs the factor z, and its oscillatory time factor needs i: $\sin(p\pi z/l)$ and $e^{\pm i\omega t}$.

The scalar Helmholtz equation and the displayed scalar boundary conditions alone also admit a constant transverse profile at $q = 0$. Excluding it for a TE electromagnetic mode requires Maxwell's equations, not merely the scalar equation. For a Neumann transverse eigenfunction, integration by parts gives

$$q^2 \int_D |P\Phi|^2 \, dA = \int_D |\nabla_\perp (P\Phi)|^2 \, dA \geq 0.$$

If $q = 0$, the profile is constant over the disk. For a perfectly conducting side wall $B_\rho = 0$; integrating $\nabla \cdot B = 0$ over a thin disk-shaped slab therefore gives $d(\int_D B_z \, dA)/dz = 0$. The end condition $B_z = 0$ makes this flux zero, which rules out a nonzero constant transverse profile. Hence only the positive derivative zeros represent TE modes.

14.1.26 A conducting cylinder can accommodate traveling electromagnetic waves; when used for this purpose it is called a wave guide. The equations describing traveling waves are the same as those of Example 14.1.2, but there is no boundary condition on E_z at $z = 0$ or $z = h$ other than that its z dependence be oscillatory. For each TM mode (values of m and j of Example 14.1.2), there is a minimum frequency that can be transmitted through a wave guide of radius a. Explain why this is so, and give a formula for the cutoff frequencies.

Solution.

Take a separated traveling mode with axial wave number k_z:

$$E_z = J_m(\alpha_{mj}\rho/a)e^{im\varphi}e^{ik_z z - i\omega t}, \qquad J_m(\alpha_{mj}) = 0, \quad \alpha_{mj} > 0.$$

The radial and angular factors have transverse Laplacian eigenvalue $-\alpha_{mj}^2/a^2$, and the axial derivative supplies $-k_z^2$. Thus the Helmholtz equation becomes

$$-\frac{\alpha_{mj}^2}{a^2} - k_z^2 + \frac{\omega^2}{c^2} = 0, \qquad k_z^2 = \frac{\omega^2}{c^2} - \frac{\alpha_{mj}^2}{a^2}.$$

Below $c\alpha_{mj}/a$, k_z is imaginary and the axial field is evanescent. Above that value k_z is real and nonzero, so traveling waves exist. The cutoff is therefore

$$\boxed{\omega_{c,mj} = \frac{c\alpha_{mj}}{a}, \qquad f_{c,mj} = \frac{c\alpha_{mj}}{2\pi a}}.$$

At equality $k_z = 0$; this is the onset of propagation, with axial group velocity $d\omega/dk_z = c^2 k_z/\omega = 0$. Strictly propagating modes require $\omega > \omega_c$. The manual's ordinary-frequency formula on p. 203 omits the factor c. The dispersion relation above also shows the dimensionally necessary factor explicitly.

14.1.27 Plot the three lowest TM and the three lowest TE angular resonant frequencies, ω_{mnp}, as a function of the ratio radius/length (a/l) for $0 \le a/l \le 1.5$.

Hint. Try plotting ω^2 (in units of c^2/a^2) vs. $(a/l)^2$. Why this choice?

Count the $\pm m$ degeneracy as one mode family. The endpoint $a/l = 0$ is the infinite-length limit.

Solution.

Use the dimensionless coordinates suggested in the hint:

$$r = a/l, \qquad X = r^2, \qquad Y = (a\omega/c)^2.$$

Each individual mode is a straight line,

$$Y_{mnp}^{\mathrm{TM}} = \alpha_{mn}^2 + p^2\pi^2 X, \quad p \ge 0; \qquad Y_{mnp}^{\mathrm{TE}} = \beta_{mn}^2 + p^2\pi^2 X, \quad p \ge 1.$$

The requested angular frequencies follow from $\omega = (c/a)\sqrt{Y}$ and $r = \sqrt{X}$; $0 \le r \le 1.5$ means $0 \le X \le 2.25$. The three lowest mode families change as the aspect ratio changes. The two angular polarizations for $m > 0$ have the same frequency and are counted as one family here; at crossings the plotted families coincide. The point $r = 0$ is an infinite-length limit, not a finite cavity, and the axial indices then become degenerate.

For TM modes the lowest transverse roots are

$$A = \alpha_{01}^2 = 5.783185963, \quad B = \alpha_{11}^2 = 14.68197064, \quad C = \alpha_{21}^2 = 26.37461643.$$

To find the first three modes, it suffices to sort A, $A+\pi^2X$, $A+4\pi^2X$, B, and C. Higher p on the first transverse branch cannot precede its first three members; higher p on either of the other branches exceeds the first branch with that same p. Higher transverse eigenvalues cannot improve the first three. The crossover equations and results are

$$A+4\pi^2r_1^2=B,\quad A+\pi^2r_2^2=B,\quad A+\pi^2r_3^2=C,$$

$$(r_1,r_2,r_3)=(0.4747724183,\ 0.9495448366,\ 1.444419644).$$

Thus the ordering, with indices written as (m,n,p), is

r interval	lowest	second	third
$0<r<r_1$	TM_{010}	TM_{011}	TM_{012}
$r_1<r<r_2$	TM_{010}	TM_{011}	TM_{110}
$r_2<r<r_3$	TM_{010}	TM_{110}	TM_{011}
$r_3<r\le 1.5$	TM_{010}	TM_{110}	TM_{210}

For TE modes let

$$D=\beta_{11}^2=3.389957717,\quad E=\beta_{21}^2=9.328363213,\quad F=\beta_{01}^2=14.68197064.$$

Here the candidates are $D+\pi^2X$, $D+4\pi^2X$, $D+9\pi^2X$, $E+\pi^2X$, and $F+\pi^2X$. The crossovers satisfy

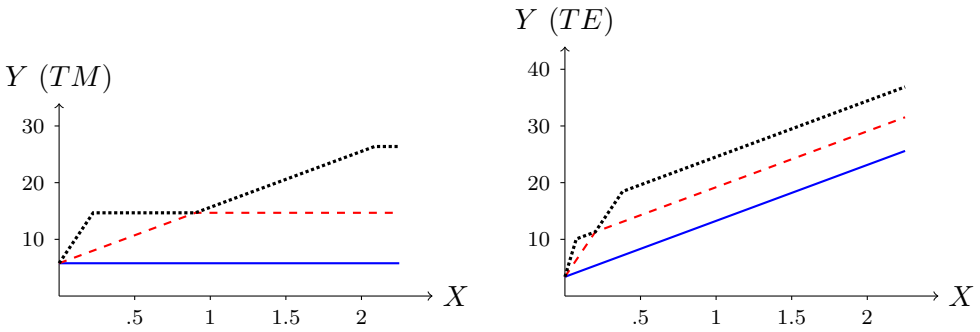
$$D+9\pi^2s_1^2=E+\pi^2s_1^2,\quad D+4\pi^2s_2^2=E+\pi^2s_2^2,$$

$$D+4\pi^2s_3^2=F+\pi^2s_3^2,\qquad (s_1,s_2,s_3)=(0.2742458463,\ 0.4478415918,\ 0.6175543472).$$

This gives

r interval	lowest	second	third
$0<r<s_1$	TE_{111}	TE_{112}	TE_{113}
$s_1<r<s_2$	TE_{111}	TE_{112}	TE_{211}
$s_2<r<s_3$	TE_{111}	TE_{211}	TE_{112}
$s_3<r\le 1.5$	TE_{111}	TE_{211}	TE_{011}

These orderings give the following plots of the three lowest frequencies in squared coordinates. Blue solid is the lowest, red dashed the second, and black dotted the third:



The transverse ordering used above can also be checked without surveying indefinitely many roots. In the radial Rayleigh quotient, increasing m^2 increases the nonnegative term $m^2 \int |P|^2 d\rho/\rho$, so each radial eigenvalue increases with m. For TM, $\alpha_{02} = 5.520078 > \alpha_{21}$ and $\alpha_{31} = 6.380162 > \alpha_{21}$ exclude the remaining branches. For TE, the zero Neumann eigenvalue is excluded as in Exercise 14.1.25; $\beta_{31} = 4.201189 > \beta_{01}$ and $\beta_{12} = 5.331443 > \beta_{01}$ exclude higher positive branches. Each fixed-m sequence increases with its radial index. In all these labels $n = 1$ means the first *positive* root; in particular $\beta_{01} = 3.831706$, not zero.

14.1.28 Show that the integral

$$\int_0^a x^m J_n(x)\, dx, \quad m \geq n \geq 0,$$

(a) is integrable for $m + n$ odd in terms of Bessel functions and powers of x, i.e., is expressible as linear combinations of $a^p J_q(a)$;

(b) may be reduced for $m + n$ even to integrated terms plus $\displaystyle\int_0^a J_0(x)\, dx$.

Here m, n are nonnegative integers with $m \geq n$.

Solution.

The parity language presumes nonnegative integer m, n. Define $I_{m,n}(a) = \int_0^a x^m J_n(x) dx$. Since $[x^{n+1} J_{n+1}(x)]' = x^{n+1} J_n(x)$, integration by parts gives

$$\begin{aligned} I_{m,n} &= \int_0^a x^{m-n-1}\, d[x^{n+1} J_{n+1}(x)] \\ &= a^m J_{n+1}(a) - (m - n - 1) I_{m-1,n+1}(a). \end{aligned} \tag{*}$$

The lower boundary term is zero because $x^m J_{n+1}(x) = O(x^{m+n+1})$. Every repetition decreases the power by one, increases the order by one, and leaves their sum $N = m + n$ unchanged.

(a) If N is odd, then $m - n$ is odd. After $r = (m - n - 1)/2$ repetitions the remaining indices satisfy $m - r = (n + r) + 1$. At that stage the coefficient of the remaining integral in (*) is zero. Thus the procedure ends with the antiderivative $I_{j+1,j} = a^{j+1} J_{j+1}(a)$, and all earlier terms are constants times $a^{m-r} J_{n+r+1}(a)$. For example,

$$I_{n+3,n} = a^{n+3} J_{n+1}(a) - 2a^{n+2} J_{n+2}(a).$$

This supplies a finite constructive expression of the required type.

(b) If N is even, each coefficient $m - n - 1 - 2r$ in (*) is odd and cannot vanish. Continue until the power becomes zero. All the terms already integrated have the desired form, and only a constant multiple of $I_{0,N}$ remains. For even $N = 2h$, the recurrence $J_{2j} = J_{2j-2} - 2J'_{2j-1}$ gives

$$\begin{aligned} I_{0,2h}(a) &= I_{0,2h-2}(a) - 2J_{2h-1}(a) \\ &= \boxed{\int_0^a J_0(x) dx - 2\sum_{j=1}^{h} J_{2j-1}(a)}. \end{aligned}$$

All odd-order Bessel functions vanish at zero, so no omitted lower boundary constants occur. Inserting this into the repeated formula (*) proves the requested reduction.

14.1.29 Show that

$$\int_0^{\alpha_{0n}} \left(1 - \frac{y}{\alpha_{0n}}\right) J_0(y) y \, dy = \frac{1}{\alpha_{0n}} \int_0^{\alpha_{0n}} J_0(y) \, dy.$$

Here α_{0n} is the nth zero of $J_0(y)$. This relation is useful (see Exercise 14.2.9): The expression on the right is easier and quicker to evaluate, and is much more accurate. Taking the difference of two terms in the expression on the left leads to a large relative error.

Solution.

Set $A = \alpha_{0n} > 0$, so $J_0(A) = 0$. First use $[yJ_1(y)]' = yJ_0(y)$:

$$\begin{aligned}\int_0^A (1 - y/A) y J_0(y) dy &= [(1 - y/A) y J_1(y)]_0^A + \frac{1}{A} \int_0^A y J_1(y) dy \\ &= \frac{1}{A} \int_0^A y J_1(y) dy.\end{aligned}$$

The upper boundary vanishes because $1 - A/A = 0$, and the lower boundary because $yJ_1(y) = O(y^2)$. Now $J_1 = -J_0'$ and a second integration by parts gives

$$\begin{aligned}\frac{1}{A} \int_0^A y J_1(y) dy &= -\frac{1}{A} [y J_0(y)]_0^A + \frac{1}{A} \int_0^A J_0(y) dy \\ &= \boxed{\frac{1}{A} \int_0^A J_0(y) dy}.\end{aligned}$$

The last boundary term is zero by the defining root condition. The identity avoids separately evaluating the two oscillatory moments $\int yJ_0$ and $A^{-1} \int y^2 J_0$ and then subtracting them; the smaller final quantity is computed directly.

14.2 Orthogonality

14.2.1 Show that

$$\int_0^a J_\nu(kx) J_\nu(k'x) x \, dx = \frac{a[k' J_\nu(ka) J_\nu'(k'a) - k J_\nu'(ka) J_\nu(k'a)]}{(k^2 - k'^2)},$$

where $J_\nu'(ka) = \dfrac{d}{d(kx)} J_\nu(kx)\Big|_{x=a}$, and that

$$\int_0^a [J_\nu(kx)]^2 x \, dx = \frac{a^2}{2} \left\{ [J_\nu'(ka)]^2 + \left(1 - \frac{\nu^2}{k^2 a^2}\right) [J_\nu(ka)]^2 \right\}, \qquad \nu > -1.$$

These two integrals are usually called the first and second Lommel integrals.

Take real $\nu > -1$, $a > 0$, and $k, k' > 0$. The first displayed quotient requires $k^2 \neq k'^2$; its coincident limit is the second integral.

Solution.

Let $y(x) = J_\nu(kx)$ and $z(x) = J_\nu(k'x)$, initially with $k, k' > 0$, $k^2 \neq k'^2$. Their equations in self-adjoint form are

$$(xy')' + (k^2 x - \nu^2/x) y = 0, \qquad (xz')' + (k'^2 x - \nu^2/x) z = 0.$$

Multiply the second by y, the first by z, and subtract:

$$\frac{d}{dx}\{x(yz' - y'z)\} = (k^2 - k'^2)xyz.$$

Integrate first from $\epsilon > 0$ to a, so no individually divergent origin terms have been introduced. To take $\epsilon \to 0$, write

$$y = Ax^\nu \left[1 - \frac{k^2x^2}{4(\nu+1)} + O(x^4)\right], \quad z = Bx^\nu \left[1 - \frac{k'^2x^2}{4(\nu+1)} + O(x^4)\right].$$

The leading powers cancel in the bracket, leaving

$$x(yz' - y'z) = \frac{AB(k^2 - k'^2)}{2(\nu+1)}x^{2\nu+2} + O(x^{2\nu+4}) \longrightarrow 0 \qquad (\nu > -1).$$

At a, the chain rule gives $y' = kJ'_\nu(ka)$ and $z' = k'J'_\nu(k'a)$. Hence

$$\boxed{\int_0^a xyz\,dx = \frac{a[k'J_\nu(ka)J'_\nu(k'a) - kJ'_\nu(ka)J_\nu(k'a)]}{k^2 - k'^2}}.$$

For the second integral let $k' \to k$ and apply l'Hôpital's rule. Writing $J = J_\nu(ka)$ and using primes for argument derivatives,

$$\int_0^a xJ_\nu(kx)^2dx = \frac{a[J(J' + kaJ'') - ka(J')^2]}{-2k}.$$

Bessel's equation at ka says $kaJ'' = -J' - (ka - \nu^2/(ka))J$. Therefore

$$J(J' + kaJ'') - ka(J')^2 = -ka\left[(J')^2 + \left(1 - \frac{\nu^2}{k^2a^2}\right)J^2\right],$$

which gives

$$\boxed{\int_0^a xJ_\nu(kx)^2dx = \frac{a^2}{2}\left[J'_\nu(ka)^2 + \left(1 - \frac{\nu^2}{k^2a^2}\right)J_\nu(ka)^2\right]}.$$

The condition $\nu > -1$ also makes the integrand $O(x^{2\nu+1})$ integrable. Coincident wave numbers use the second formula; any zero-parameter value must be taken as an existing limit, not inserted into the displayed denominators.

14.2.2

(a) If $\beta_{\nu m}$ is the mth positive zero of J'_ν, so that $(d/d\rho)J_\nu(\beta_{\nu m}\rho/a)|_{\rho=a} = 0$, show that the Bessel functions are orthogonal over the interval $[0, a]$ with an orthogonality integral

$$\int_0^a J_\nu\left(\beta_{\nu m}\frac{\rho}{a}\right)J_\nu\left(\beta_{\nu n}\frac{\rho}{a}\right)\rho\,d\rho = 0, \qquad m \neq n, \quad \nu > -1.$$

(b) Derive the corresponding normalization integral ($m = n$).

ANS. (b) $\dfrac{a^2}{2}\left(1-\dfrac{\nu^2}{\beta_{\nu m}^2}\right)[J_\nu(\beta_{\nu m})]^2$, $\nu > -1$.

These formulas refer to positive derivative roots. For $\nu = 0$ the additional constant mode has norm $a^2/2$.

Solution.

Interpret $\beta_{\nu m}$ as a positive argument satisfying $J'_\nu(\beta_{\nu m}) = 0$, so the derivative with respect to ρ vanishes at $\rho = a$. Put $k = \beta_{\nu m}/a$ and $k' = \beta_{\nu n}/a$ in the first Lommel integral. For $m \neq n$ the denominator is nonzero, while both terms in its numerator contain a vanishing derivative. Thus

$$\boxed{\int_0^a \rho J_\nu(\beta_{\nu m}\rho/a)J_\nu(\beta_{\nu n}\rho/a)d\rho = 0 \quad (m \neq n)}.$$

For $m = n$, use the second Lommel integral instead. The squared derivative term is zero, and $k^2a^2 = \beta_{\nu m}^2$, giving

$$\boxed{N_{\nu m} = \frac{a^2}{2}\left(1-\frac{\nu^2}{\beta_{\nu m}^2}\right)J_\nu(\beta_{\nu m})^2}.$$

The origin terms vanish for $\nu > -1$ by the calculation in Exercise 14.2.1. The integral defining $N_{\nu m}$ is positive, so this also implies $\beta_{\nu m} > |\nu|$ for every positive derivative zero under consideration. For $\nu = 0$ the Neumann problem has the additional constant mode $J_0(0\rho/a) = 1$, of norm $a^2/2$. It is not one of the positive roots, and the displayed expression containing ν^2/β^2 must not be used as $0/0$. Its orthogonality to the positive modes follows from $\int_0^a \rho J_0(\beta\rho/a)d\rho = a^2J_1(\beta)/\beta = 0$.

14.2.3 Verify that the orthogonality equation, Eq. (14.44), and the normalization equation, Eq. (14.46), hold for $\nu > -1$.

Hint. Using power-series expansions, examine the behavior of Eq. (14.43) as $\rho \to 0$.

Solution.

The two equations to be checked are the Dirichlet orthogonality and normalization formulas

$$\int_0^a \rho J_\nu(\alpha_{\nu m}\rho/a)J_\nu(\alpha_{\nu n}\rho/a)d\rho = 0 \quad (m \neq n),$$

$$\int_0^a \rho J_\nu(\alpha_{\nu n}\rho/a)^2 d\rho = \frac{a^2}{2}J_{\nu+1}(\alpha_{\nu n})^2.$$

The only possible issue when $-1 < \nu < 0$ is the singular-looking origin. Put $k = \alpha_{\nu m}/a$, $k' = \alpha_{\nu n}/a$. The series begin

$$J_\nu(k\rho) = \frac{(k/2)^\nu}{\Gamma(\nu+1)}\rho^\nu\left[1-\frac{k^2\rho^2}{4(\nu+1)}+O(\rho^4)\right].$$

In the boundary expression $\rho[J_\nu(k\rho)\partial_\rho J_\nu(k'\rho) - \partial_\rho J_\nu(k\rho)J_\nu(k'\rho)]$, the terms of order $\rho^{2\nu}$ cancel exactly. The next term is

$$\frac{(kk'/4)^\nu(k^2-k'^2)}{2(\nu+1)\Gamma(\nu+1)^2}\rho^{2\nu+2},$$

which vanishes for precisely the stated range $\nu > -1$. At $\rho = a$ both function values vanish for distinct Dirichlet roots, so the first Lommel formula gives orthogonality. For a repeated root, the second formula reduces to $a^2J'_\nu(\alpha_{\nu n})^2/2$. The recurrence $J'_\nu = \nu J_\nu/x - J_{\nu+1}$ yields $J'_\nu(\alpha_{\nu n}) = -J_{\nu+1}(\alpha_{\nu n})$, proving normalization. All relevant integrals converge because their origin integrands behave as $\rho^{2\nu+1}$.

14.2.4 From Eq. (14.44), develop a proof that $J_\nu(z)$, $\nu > -1$ has no complex roots (with a nonzero imaginary part).

Hint.

(a) Use the series form of $J_\nu(z)$ to exclude pure imaginary roots.

(b) Assume $\alpha_{\nu m}$ to be complex and take $\alpha_{\nu n}$ to be $\alpha_{\nu m}^*$.

Solution.

Take real $\nu > -1$ and consider nonzero roots on a fixed branch. The factor z^ν can be removed by defining the entire even function

$$F_\nu(z) = \Gamma(\nu+1)(2/z)^\nu J_\nu(z) = \sum_{r=0}^{\infty} \frac{(-1)^r \Gamma(\nu+1)(z^2/4)^r}{r!\Gamma(\nu+r+1)}.$$

(a) If $z = iy$ with real $y \neq 0$, every term of $F_\nu(iy)$ is positive, because $(-1)^r(iy)^{2r} = y^{2r}$ and all gamma arguments $\nu + r + 1$ are positive. Thus $F_\nu(iy) > 0$, excluding nonzero pure imaginary roots.

(b) The series coefficients are real, so if z_0 is a root then z_0^* is also a root. Suppose both its real and imaginary parts are nonzero. Then $z_0^2 \neq (z_0^*)^2$. Apply the first Lommel identity on $0 \le x \le 1$ with $k = z_0$, $k' = z_0^*$. The upper boundary term vanishes because both arguments are roots; the lower boundary vanishes by the same $x^{2\nu+2}$ estimate as before. With conjugate branches for the two factors, this says

$$0 = \int_0^1 x J_\nu(z_0 x) J_\nu(z_0^* x)\,dx = \int_0^1 x |J_\nu(z_0 x)|^2\,dx.$$

But the integrand is nonnegative and the series shows that the function is not identically zero on $(0,1)$, so the integral is strictly positive. This contradiction excludes every remaining nonreal possibility. Therefore all nonzero zeros are real. The corrected reference is Eq. (14.44), the orthogonality identity; the original reference to (11.49) is unrelated.

14.2.5

(a) In the series expansion

$$f(\rho) = \sum_{m=1}^{\infty} c_{\nu m} J_\nu\left(\alpha_{\nu m}\frac{\rho}{a}\right), \quad 0 \le \rho \le a, \quad \nu > -1,$$

with $J_\nu(\alpha_{\nu m}) = 0$, show that the coefficients are given by

$$c_{\nu m} = \frac{2}{a^2[J_{\nu+1}(\alpha_{\nu m})]^2} \int_0^a f(\rho) J_\nu\left(\alpha_{\nu m}\frac{\rho}{a}\right)\rho\,d\rho.$$

(b) In the series expansion

$$f(\rho) = f_{\text{extra}}(\rho) + \sum_{m=1}^{\infty} d_{\nu m} J_\nu\left(\beta_{\nu m}\frac{\rho}{a}\right), \quad 0 \le \rho \le a, \quad \nu > -1,$$

with $(d/d\rho)J_\nu(\beta_{\nu m}\rho/a)\big|_{\rho=a} = 0$, show that the coefficients are given by

$$d_{\nu m} = \frac{2}{a^2(1-\nu^2/\beta_{\nu m}^2)[J_\nu(\beta_{\nu m})]^2} \int_0^a f(\rho) J_\nu\left(\beta_{\nu m}\frac{\rho}{a}\right)\rho\,d\rho.$$

Use real $\nu > -1$ and convergence in $L^2((0,a),\rho\,d\rho)$; the roots are positive. In (b), include $f_{\text{extra}} = 0$ for $\nu > 0$, a constant mode for $\nu = 0$, and the mode proportional to $I_\nu(\kappa\rho/a)$ with $I'_\nu(\kappa) = 0$ for $-1 < \nu < 0$. Its coefficient is determined below.

Solution.

Use the weighted inner product $\langle g, h\rangle = \int_0^a g(\rho)^* h(\rho)\rho\,d\rho$. The coefficient formulas hold whenever the stated expansions converge in this weighted L^2 space.

(a) Multiply the Dirichlet expansion by $J_\nu(\alpha_{\nu n}\rho/a)\rho$ and integrate. Orthogonality removes every term except $m = n$:

$$\int_0^a f(\rho)J_\nu(\alpha_{\nu n}\rho/a)\rho d\rho = c_{\nu n}\frac{a^2}{2}J_{\nu+1}(\alpha_{\nu n})^2.$$

Division gives the asserted $c_{\nu n}$.

(b) Repeat with $J_\nu(\beta_{\nu n}\rho/a)\rho$ instead. The Neumann orthogonality and norm from Exercise 14.2.2 give

$$\int_0^a f(\rho)J_\nu(\beta_{\nu n}\rho/a)\rho d\rho = d_{\nu n}\frac{a^2}{2}\left(1 - \frac{\nu^2}{\beta_{\nu n}^2}\right)J_\nu(\beta_{\nu n})^2.$$

Solving for $d_{\nu n}$ proves the second displayed coefficient formula for each positive root.

Positive derivative roots alone do not give an expansion of an arbitrary function throughout the stated range $\nu > -1$. For $\nu = 0$ one must add the constant mode with coefficient $d_{00} = 2a^{-2}\int_0^a f(\rho)\rho\,d\rho$. For $-1 < \nu < 0$ one must also include a negative-eigenvalue mode $I_\nu(\kappa\rho/a)$, where $I'_\nu(\kappa) = 0$, $\kappa > 0$. These omissions do not change the coefficient formulas for the positive-root terms. Here is a direct check of the latter qualification. For $-1 < \nu < 0$, $I_\nu(x) > 0$ by its series, its derivative is negative near zero ($I_\nu \sim (x/2)^\nu/\Gamma(\nu+1)$), and positive for large x. Thus it has a positive derivative zero. If $w = xI'_\nu/I_\nu$, the modified Bessel equation gives $xw' = x^2 + \nu^2 - w^2$. At every zero of w its derivative is positive, so it crosses zero only once. The corresponding radial function has eigenvalue $-\kappa^2/a^2$ and is orthogonal to all positive-eigenvalue modes by the same boundary-bracket argument. For clarity, derive its norm rather than assuming a modified Lommel formula. Put $y(x) = I_\nu(x)$ and

$$F(x) = \frac{1}{2}\{x^2[y(x)^2 - y'(x)^2] + \nu^2 y(x)^2\}.$$

Differentiation gives

$$F' = x(y^2 - y'^2) + (x^2 + \nu^2)yy' - x^2y'y''.$$

The modified equation says $x^2y'' = (x^2 + \nu^2)y - xy'$. Substitution cancels the two yy' terms and the two xy'^2 terms, leaving $F' = xy^2$. Near zero, the leading $x^{2\nu}$ contributions cancel in F, so $F = O(x^{2\nu+2}) \to 0$ for $\nu > -1$. Now set $x = \kappa\rho/a$ and use $I'_\nu(\kappa) = 0$:

$$N_- = \frac{a^2}{\kappa^2}\int_0^\kappa xI_\nu(x)^2\,dx = \frac{a^2}{\kappa^2}[F(\kappa) - F(0)].$$

Thus its norm is

$$N_- = \frac{a^2}{2}\left(1 + \frac{\nu^2}{\kappa^2}\right)I_\nu(\kappa)^2,$$

so its coefficient is $N_-^{-1}\int_0^a f(\rho)I_\nu(\kappa\rho/a)\rho d\rho$. At the Dirichlet endpoint $\rho = a$, a series of zero-valued basis functions is zero; nonzero prescribed endpoint values are understood as interior limits or in L^2, rather than equality at that single point.

14.2.6 A right circular cylinder has an electrostatic potential of $\psi(\rho,\varphi)$ on both ends. The potential on the curved cylindrical surface is zero. Find the potential at all interior points.

Hint. Choose your coordinate system and adjust your z dependence to exploit the symmetry of your potential.

Assume bounded, piecewise smooth end data, and seek the bounded harmonic interior solution. At an incompatible rim, impose the face limits away from the junction.

Solution.

Let the cylinder have radius a and length l, with its ends at $z=\pm l/2$. Denote the sought interior potential by V and the identical prescribed end potential by $\psi(\rho,\varphi)$. Separation of $\nabla^2 V=0$ gives the angular factors $e^{im\varphi}$, regular radial factors $J_{|m|}(k\rho)$, and axial equation $Z''-k^2Z=0$. The side condition forces $k=\alpha_{|m|n}/a$, where $J_{|m|}(\alpha_{|m|n})=0$. Identical end data are even in z; hence use $\cosh(kz)$ rather than the odd $\sinh(kz)$. Normalize the axial factor to one on both ends:

$$\boxed{V(\rho,\varphi,z)=\sum_{m=-\infty}^{\infty}\sum_{n=1}^{\infty}A_{mn}J_{|m|}(\alpha_{|m|n}\rho/a)e^{im\varphi}\frac{\cosh(\alpha_{|m|n}z/a)}{\cosh(\alpha_{|m|n}l/(2a))}}.$$

At either end this is the double Fourier–Bessel expansion of ψ. Multiply it by $e^{-im\varphi}J_{|m|}(\alpha_{|m|n}\rho/a)\rho$ and integrate over the disk. The angular norm is 2π and the radial norm is $a^2J_{|m|+1}(\alpha_{|m|n})^2/2$. Therefore

$$\boxed{A_{mn}=\frac{\displaystyle\int_0^{2\pi}\int_0^a\psi(\rho,\varphi)e^{-im\varphi}J_{|m|}(\alpha_{|m|n}\rho/a)\rho\,d\rho\,d\varphi}{\pi a^2J_{|m|+1}(\alpha_{|m|n})^2}}.$$

Every term satisfies Laplace's equation, the grounded side, and reflection symmetry; the coefficient projection supplies both end conditions. For real data $A_{-m,n}=A_{mn}^*$, making the potential real. For compatible boundary data and finite-energy solutions, the difference $U=V_1-V_2$ has zero boundary values. Green's identity gives $\int|\nabla U|^2d^3x=0$, hence $U=0$.

A discontinuous rim need not give a finite-energy field, so that energy argument should not be applied to each such field without qualification. Uniqueness still holds among bounded harmonic solutions with the specified face limits. To see this, let Γ be the two rim circles and introduce the positive harmonic barrier

$$B(\mathbf{x})=\int_\Gamma\frac{d\ell_{\mathbf{y}}}{|\mathbf{x}-\mathbf{y}|}.$$

It is finite in the interior, harmonic there, and diverges to $+\infty$ at the rim: the local line integral is comparable to $\int_{-\delta}^{\delta}(d^2+s^2)^{-1/2}ds=2\,\mathrm{arsinh}(\delta/d)\to\infty$. For any $\epsilon>0$, both $U-\epsilon B$ and $-U-\epsilon B$ have nonpositive boundary upper limits, including the rim because U is bounded. The maximum principle gives $|U(\mathbf{x})|\le\epsilon B(\mathbf{x})$ in the interior. Letting $\epsilon\downarrow 0$ proves $U=0$. The series therefore supplies the bounded interior solution and, at incompatible rims, the face limits away from the junction.

14.2.7 A function $f(x)$ is expressed as a Bessel series:

$$f(x)=\sum_{n=1}^{\infty}a_nJ_m(\alpha_{mn}x),$$

with α_{mn} the nth root of J_m. Prove the Parseval relation,

$$\int_0^1[f(x)]^2x\,dx=\frac{1}{2}\sum_{n=1}^{\infty}a_n^2[J_{m+1}(\alpha_{mn})]^2.$$

Let $m > -1$ be real, let α_{mn} enumerate the positive zeros, and assume convergence to f in $L^2((0,1), x\,dx)$.

Solution.

First take a finite partial sum $S_N(x) = \sum_{n=1}^{N} a_n J_m(\alpha_{mn}x)$. Expanding its square, then using orthogonality and the Dirichlet norm, gives

$$\int_0^1 S_N(x)^2 x\,dx = \sum_{n=1}^{N}\sum_{j=1}^{N} a_n a_j \int_0^1 J_m(\alpha_{mn}x)J_m(\alpha_{mj}x)x\,dx$$
$$= \frac{1}{2}\sum_{n=1}^{N} a_n^2 J_{m+1}(\alpha_{mn})^2.$$

To pass to an infinite series, the expansion must converge to f in $L^2((0,1), x\,dx)$; merely writing a formal series does not justify Parseval's equality. Under that assumption, the Cauchy–Schwarz inequality gives

$$\left|\int_0^1 (f^2 - S_N^2)x\,dx\right| \le \|f - S_N\|\,\|f + S_N\| \longrightarrow 0.$$

The finite identity therefore tends to the formula in the question. For real f and coefficients this is the usual Parseval identity. For complex functions, the norm version is

$$\int_0^1 |f(x)|^2 x\,dx = \frac{1}{2}\sum_{n=1}^{\infty} |a_n|^2 J_{m+1}(\alpha_{mn})^2;$$

the bilinear square identity also follows by the same limiting argument. Completeness of the Dirichlet Bessel basis identifies the limit with f, rather than merely its orthogonal projection.

14.2.8 Prove that

$$\sum_{n=1}^{\infty} (\alpha_{mn})^{-2} = \frac{1}{4(m+1)}.$$

Hint. Expand x^m in a Bessel series and apply the Parseval relation.
Here $m > -1$ is real and α_{mn} is the nth positive zero of J_m.

Solution.

Expand $f(x) = x^m$ on $0 < x < 1$ in the Dirichlet Bessel basis. For $m > -1$, $\int_0^1 |x^m|^2 x\,dx = 1/[2(m+1)]$ is finite. The coefficient formula gives

$$a_n = \frac{2}{J_{m+1}(\alpha_{mn})^2}\int_0^1 x^{m+1} J_m(\alpha_{mn}x)dx.$$

Set $\alpha = \alpha_{mn}$. The recurrence implies

$$\frac{d}{dx}[x^{m+1}J_{m+1}(\alpha x)] = \alpha x^{m+1} J_m(\alpha x).$$

Its lower boundary vanishes as x^{2m+2}; hence

$$\int_0^1 x^{m+1} J_m(\alpha x)dx = \frac{J_{m+1}(\alpha)}{\alpha}, \qquad a_n = \frac{2}{\alpha_{mn} J_{m+1}(\alpha_{mn})}.$$

Insert these coefficients into Parseval's relation:

$$\frac{1}{2(m+1)} = \frac{1}{2}\sum_{n=1}^{\infty} \frac{4J_{m+1}(\alpha_{mn})^2}{\alpha_{mn}^2 J_{m+1}(\alpha_{mn})^2} = 2\sum_{n=1}^{\infty} \alpha_{mn}^{-2}.$$

Thus

$$\boxed{\sum_{n=1}^{\infty} \frac{1}{\alpha_{mn}^2} = \frac{1}{4(m+1)}}.$$

The expansion is used in the weighted L^2 sense. At $x = 1$ every basis function is zero whereas $x^m = 1$; that single endpoint has zero measure and does not affect the integral or Parseval's identity.

14.2.9 A right circular cylinder of length l and radius a has on its end caps a potential

$$\psi\Big|_{z=\pm\frac{l}{2}} = 100\left(1 - \frac{\rho}{a}\right).$$

The potential on the curved surface (the side) is zero. Using the Bessel series from Exercise 14.2.6, calculate the electrostatic potential for $\rho/a = 0.0(0.2)1.0$ and $z/l = 0.0(0.1)0.5$. Take $a/l = 0.5$.

Hint. From Exercise 14.1.29 you have

$$\int_0^{\alpha_{0n}} \left(1 - \frac{y}{\alpha_{0n}}\right) J_0(y)\, y\, dy.$$

Show that this equals

$$\frac{1}{\alpha_{0n}} \int_0^{\alpha_{0n}} J_0(y)\, dy.$$

Numerical evaluation of this latter form rather than the former is both faster and more accurate.

Note. For $\rho/a = 0.0$ and $z/l = 0.5$ the convergence is slow, 20 terms giving only 98.4 rather than 100.

Check value. For $\rho/a = 0.4$ and $z/l = 0.3$, $\psi = 24.559$.

Solution.

The end data are independent of φ, so only $m = 0$ remains in Exercise 14.2.6. Put $r = \rho/a$, $s = z/l$, and $\alpha_n = \alpha_{0n}$. Since $l/(2a) = 1$,

$$\boxed{V(r,s) = \sum_{n=1}^{\infty} A_n J_0(\alpha_n r) \frac{\cosh(2\alpha_n s)}{\cosh \alpha_n}}, \qquad A_n = \frac{200}{J_1(\alpha_n)^2} \int_0^1 (1-r) J_0(\alpha_n r) r\, dr.$$

With $y = \alpha_n r$, the last integral is

$$\frac{1}{\alpha_n^2} \int_0^{\alpha_n} (1 - y/\alpha_n) y J_0(y) dy = \frac{1}{\alpha_n^3} \int_0^{\alpha_n} J_0(y) dy,$$

where the final equality is proved by the two integrations by parts in Exercise 14.1.29. Consequently the stable coefficients are

$$\boxed{A_n = \frac{200}{\alpha_n^3 J_1(\alpha_n)^2} \int_0^{\alpha_n} J_0(y) dy}.$$

The resulting potentials are:

$\rho/a \backslash z/l$	0.0	0.1	0.2	0.3	0.4	0.5
0.0	14.108	15.805	21.353	32.341	52.640	100.000
0.2	13.290	14.881	20.061	30.213	48.293	80.000
0.4	10.989	12.287	16.485	24.559	38.270	60.000
0.6	7.617	8.504	11.359	16.786	25.844	40.000
0.8	3.747	4.179	5.568	8.204	12.648	20.000
1.0	0.000	0.000	0.000	0.000	0.000	0.000

The last column is the prescribed boundary value $100(1-r)$; the last row is the grounded side. For the interior entries, the integrals were accumulated between consecutive zeros of J_0, and 120 modes were summed. Comparing 80 and 120 modes changed every interior entry by less than 10^{-20}, comfortably below the 0.001 displayed precision. This convergence comparison is a numerical check, not a universal truncation-error theorem. The printed check value 24.558 is not correctly rounded to three decimals: the converged value is $V(0.4, 0.3) = 24.5585639709\ldots$, hence 24.559. At the top center, a 20-mode sum is $98.424595978\ldots$; prescribing the exact boundary value avoids reporting its slowly convergent truncation as the boundary potential. Symmetry gives $V(r,-s) = V(r,s)$ for the lower half of the cylinder.

14.3 Neumann Functions, Bessel Functions of the Second Kind

14.3.1 Prove that the Neumann functions Y_n (with n an integer) satisfy the recurrence relations

$$Y_{n-1}(x) + Y_{n+1}(x) = \frac{2n}{x} Y_n(x),$$
$$Y_{n-1}(x) - Y_{n+1}(x) = 2Y_n'(x).$$

Hint. These relations may be proved by differentiating the recurrence relations for J_ν or by using the limit form of Y_ν but not dividing everything by zero.

Solution.

Begin with noninteger ν, where the defining denominator is nonzero. Abbreviate $C = \cos\pi\nu$ and $S = \sin\pi\nu$. Since $\cos\pi(\nu \pm 1) = -C$ and $\sin\pi(\nu \pm 1) = -S$,

$$Y_{\nu-1} = \frac{CJ_{\nu-1} + J_{-\nu+1}}{S}, \qquad Y_{\nu+1} = \frac{CJ_{\nu+1} + J_{-\nu-1}}{S}.$$

Add these and apply the J recurrence at both orders ν and $-\nu$:

$$Y_{\nu-1} + Y_{\nu+1} = \frac{C(2\nu J_\nu/x) - 2\nu J_{-\nu}/x}{S} = \frac{2\nu}{x}\frac{CJ_\nu - J_{-\nu}}{S} = \frac{2\nu}{x} Y_\nu.$$

Subtracting instead gives

$$Y_{\nu-1} - Y_{\nu+1} = \frac{C(J_{\nu-1} - J_{\nu+1}) + (J_{-\nu+1} - J_{-\nu-1})}{S}$$
$$= \frac{2CJ_\nu' - 2J_{-\nu}'}{S} = 2Y_\nu'.$$

Now let $\nu \to n$. The defining removable limits give $Y_\nu \to Y_n$, $Y_{\nu\pm1} \to Y_{n\pm1}$, and the same limits hold after x differentiation away from the branch point $x = 0$. Both recurrences therefore hold for integer n, with no division by $\sin\pi n = 0$.

14.3.2 Show that for integer n

$$Y_{-n}(x) = (-1)^n Y_n(x).$$

Solution.

For noninteger ν, solve the definition for the opposite-order first-kind function:

$$J_{-\nu} = \cos \pi\nu \, J_\nu - \sin \pi\nu \, Y_\nu.$$

Apply the definition once more with order $-\nu$:

$$\begin{aligned} Y_{-\nu} &= \frac{\cos \pi\nu \, J_{-\nu} - J_\nu}{-\sin \pi\nu} \\ &= \frac{(\cos^2 \pi\nu - 1)J_\nu - \cos \pi\nu \sin \pi\nu \, Y_\nu}{-\sin \pi\nu} \\ &= \sin \pi\nu \, J_\nu + \cos \pi\nu \, Y_\nu. \end{aligned}$$

Taking the removable limit $\nu \to n \in \mathbb{Z}$ gives

$$\boxed{Y_{-n}(x) = (-1)^n Y_n(x)}.$$

This also includes $n = 0$, where the equality is immediate.

14.3.3 Show that

$$Y_0'(x) = -Y_1(x).$$

Solution.

Set $n = 0$ in the derivative recurrence:

$$2Y_0' = Y_{-1} - Y_1.$$

Exercise 14.3.2 gives $Y_{-1} = -Y_1$, and hence

$$2Y_0' = -2Y_1, \qquad \boxed{Y_0' = -Y_1}.$$

The manual's last equation in this exercise, p. 209, prints J_1, J_0' where Y_1, Y_0' are required. The latter are the functions used in the recurrence above.

14.3.4 If X and Z are any two solutions of Bessel's equation, show that

$$X_\nu(x)Z_\nu'(x) - X_\nu'(x)Z_\nu(x) = \frac{A_\nu}{x},$$

in which A_ν may depend on ν but is independent of x. This is a special case of Exercise 7.6.11.

Solution.

Define $W = X_\nu Z_\nu' - X_\nu' Z_\nu$ on an interval not containing $x = 0$. Differentiation cancels the mixed terms:

$$W' = X_\nu Z_\nu'' - X_\nu'' Z_\nu.$$

Both functions satisfy $y'' = -y'/x - (1 - \nu^2/x^2)y$. Substitution therefore gives

$$\begin{aligned} W' &= X_\nu \left[-\frac{Z'_\nu}{x} - \left(1 - \frac{\nu^2}{x^2}\right) Z_\nu \right] - \left[-\frac{X'_\nu}{x} - \left(1 - \frac{\nu^2}{x^2}\right) X_\nu \right] Z_\nu \\ &= -\frac{1}{x}(X_\nu Z'_\nu - X'_\nu Z_\nu) = -\frac{W}{x}. \end{aligned}$$

Thus $(xW)' = W + xW' = 0$, proving

$$\boxed{W(x) = \frac{A_\nu}{x}}.$$

The constant depends on the chosen solutions and their order, but not on x. It is zero for linearly dependent solutions.

14.3.5 Verify the Wronskian formulas

$$\begin{aligned} J_\nu(x)J_{-\nu+1}(x) + J_{-\nu}(x)J_{\nu-1}(x) &= \frac{2\sin\nu\pi}{\pi x}, \\ J_\nu(x)Y'_\nu(x) - J'_\nu(x)Y_\nu(x) &= \frac{2}{\pi x}. \end{aligned}$$

Solution.

For noninteger ν, the derivative recurrences give

$$J_{1-\nu} = -J'_{-\nu} - \frac{\nu}{x}J_{-\nu}, \qquad J_{\nu-1} = J'_\nu + \frac{\nu}{x}J_\nu.$$

Multiplying and adding cancels the terms proportional to ν/x:

$$J_\nu J_{1-\nu} + J_{-\nu}J_{\nu-1} = -J_\nu J'_{-\nu} + J'_\nu J_{-\nu} = -W[J_\nu, J_{-\nu}].$$

The manual's corresponding line on p. 209 has a minus sign before the second product; it must be plus, as the cancellation above shows. By Exercise 14.3.4, $W = A/x$. The leading series terms are

$$J_\nu = \frac{(x/2)^\nu}{\Gamma(\nu+1)}[1 + O(x^2)], \qquad J_{-\nu} = \frac{(x/2)^{-\nu}}{\Gamma(1-\nu)}[1 + O(x^2)].$$

Their derivatives give

$$A = -\frac{2\nu}{\Gamma(\nu+1)\Gamma(1-\nu)} = -\frac{2}{\Gamma(\nu)\Gamma(1-\nu)} = -\frac{2\sin\pi\nu}{\pi}.$$

Therefore

$$\boxed{J_\nu J_{1-\nu} + J_{-\nu}J_{\nu-1} = \frac{2\sin\pi\nu}{\pi x}}.$$

Finally $Y_\nu = (\cos\pi\nu\, J_\nu - J_{-\nu})/\sin\pi\nu$, so bilinearity of the Wronskian gives

$$W[J_\nu, Y_\nu] = -\frac{W[J_\nu, J_{-\nu}]}{\sin\pi\nu} = \boxed{\frac{2}{\pi x}}.$$

Both formulas extend to integer orders by their removable order limits. In the first, both sides then vanish because J_{-n} is proportional to J_n; the second remains nonzero.

14.3.6 As an alternative to letting x approach zero in the evaluation of the Wronskian constant, we may invoke the uniqueness of power-series expansions. The coefficient of x^{-1} in the series expansion of $u_\nu(x)v'_\nu(x) - u'_\nu(x)v_\nu(x)$ is then A_ν. Show by series expansion that the coefficients of x^0 and x^1 of $J_\nu(x)J'_{-\nu}(x) - J'_\nu(x)J_{-\nu}(x)$ are each zero.

Solution.

For noninteger ν, write the first two series terms as

$$J_\nu = Ax^\nu(1 - ax^2 + O(x^4)), \quad J_{-\nu} = Bx^{-\nu}(1 - bx^2 + O(x^4)),$$

where

$$A = \frac{2^{-\nu}}{\Gamma(\nu+1)}, \quad B = \frac{2^\nu}{\Gamma(1-\nu)}, \quad a = \frac{1}{4(\nu+1)}, \quad b = \frac{1}{4(1-\nu)}.$$

All powers in either product $J_\nu J'_{-\nu}$ or $J'_\nu J_{-\nu}$ are x^{2r-1} with integer $r \geq 0$. Consequently an x^0 term cannot occur. For x^1, differentiate the two displayed series and multiply:

$$J_\nu J'_{-\nu} = AB\left[-\frac{\nu}{x} + \{a\nu - b(2-\nu)\}x + O(x^3)\right],$$
$$J'_\nu J_{-\nu} = AB\left[\frac{\nu}{x} + \{-b\nu - a(\nu+2)\}x + O(x^3)\right].$$

Subtracting, the x coefficient is

$$2AB[(\nu+1)a + (\nu-1)b] = 2AB\left[\frac{1}{4} - \frac{1}{4}\right] = 0.$$

Thus both requested coefficients vanish, leaving $-2\nu AB/x + O(x^3)$ at this order. At integer $\nu = n$ the negative-order relation makes the entire Wronskian identically zero, so the same coefficient conclusion holds there without using intermediate gamma factors at their poles.

14.3.7

(a) By differentiating and substituting into Bessel's ODE for $\nu = 0$, show that $\displaystyle\int_0^\infty \cos(x\cosh t)\,dt$ is a solution.

Hint. Rearrange the final integral to $\displaystyle\int_0^\infty \frac{d}{dt}\left[x\sin(x\cosh t)\sinh t\right]dt$.

(b) Show that $\displaystyle Y_0(x) = -\frac{2}{\pi}\int_0^\infty \cos(x\cosh t)\,dt$ is linearly independent of $J_0(x)$.

Take $x > 0$. Justify differentiation through an upper-half-plane convergence factor before taking its real-axis boundary limit.

Solution.

For real $x > 0$, the integral is conditionally convergent. Its formal x derivatives are not convergent ordinary improper integrals, and the proposed boundary term $x\sin(x\cosh t)\sinh t$ does not vanish as $t \to \infty$. The manual's unregularized integration by parts on p. 211 therefore needs the limiting argument below.

(a) Begin in the upper half-plane, with

$$H(z) = \int_0^\infty e^{iz\cosh t}dt, \qquad \Im z > 0.$$

The factor $e^{-\Im z\cosh t}$ makes the integral and all local z derivatives absolutely convergent. Direct differentiation gives

$$\begin{aligned} z^2H'' + zH' + z^2H &= \int_0^\infty [-z^2\sinh^2 t + iz\cosh t]e^{iz\cosh t}dt \\ &= \int_0^\infty \frac{d}{dt}\left[iz\sinh t\, e^{iz\cosh t}\right]dt = 0. \end{aligned}$$

At $t = 0$ the bracket is zero, and at infinity exponential damping overcomes $\sinh t$. Thus H solves the order-zero Bessel equation. Its coefficients are analytic near every positive real x, so its solutions continue analytically across that regular part of the axis. Equivalently, write H locally as a linear combination of the analytic solutions J_0 and Y_0 and continue those functions.

Now put $u = \cosh t$. The real part of the boundary value is

$$F(x) = \int_1^\infty \frac{\cos(xu)}{\sqrt{u^2-1}}du = \int_0^\infty \cos(x\cosh t)dt.$$

The singularity at $u = 1$ is integrable and the tail converges by Dirichlet's test. Abel convergence therefore identifies it with $\lim_{\epsilon\downarrow 0}\Re H(x+i\epsilon)$. The analytic continuation just established makes this limit a smooth solution:

$$\boxed{x^2F'' + xF' + x^2F = 0 \quad (x > 0)}.$$

(b) To compare it with J_0, split off an elementary logarithmic integral:

$$F(x) = \int_1^\infty \frac{\cos(xu)}{u}\,du + \int_1^\infty \cos(xu)\left(\frac{1}{\sqrt{u^2-1}} - \frac{1}{u}\right)du.$$

The first is $-\operatorname{Ci}(x) = -\gamma - \ln x + o(1)$. The difference in parentheses is absolutely integrable; dominated convergence gives for the second integral

$$\lim_{R\to\infty}[\operatorname{arcosh} R - \ln R] = \ln 2.$$

Consequently

$$-\frac{2}{\pi}F(x) = \frac{2}{\pi}[\ln(x/2) + \gamma] + o(1).$$

This diverges logarithmically as $x \downarrow 0$, whereas $J_0(x)$ tends to one. They are linearly independent. The logarithmic coefficient and constant also agree with the defining series of Y_0; a difference of two such solutions would be a multiple of J_0 tending to zero, so that multiple is zero. Hence $Y_0(x) = -2F(x)/\pi$ with the stated normalization.

14.3.8 Verify the expansion formula for $Y_n(x)$ given in Eq. (14.61).

Hint. Start from Eq. (14.60) and perform the indicated differentiations on the power-series expansions of J_ν and $J_{-\nu}$. The digamma functions ψ arise from the differentiation of the gamma function. You will need the identity (not derived in this book) $\lim_{z\to -n}\psi(z)/\Gamma(z) = (-1)^{n-1}n!$, where n is a positive integer.

Take a nonnegative integer n and $x > 0$ initially; a complex continuation uses a consistent logarithm branch.

Solution.

Let $L = \ln(x/2)$, take integer $n \geq 0$, and use a fixed branch of the logarithm. Equation (14.60) reads

$$Y_n = \frac{1}{\pi}\left[\partial_\nu J_\nu|_n - (-1)^n\,\partial_\nu J_{-\nu}|_n\right].$$

The first order derivative of the series is

$$\partial_\nu J_\nu|_n = J_n L - \sum_{r=0}^{\infty} \frac{(-1)^r (x/2)^{n+2r} \psi(n+r+1)}{r!(n+r)!}. \tag{1}$$

For the second, the power $(x/2)^{-\nu}$ contributes $-L$, while the derivative of $1/\Gamma(r-\nu+1)$ contributes $+\psi(r-\nu+1)/\Gamma(r-\nu+1)$. Thus

$$\partial_\nu J_{-\nu}|_n = -J_{-n} L + \sum_{r=0}^{\infty} \frac{(-1)^r (x/2)^{2r-n}}{r!} \left.\frac{\psi(z)}{\Gamma(z)}\right|_{z=r-n+1}. \tag{2}$$

The values at gamma poles are limits. To derive them, set $z = -q+\epsilon$, $q = 0, 1, \ldots$. The simple pole of gamma gives

$$\frac{1}{\Gamma(-q+\epsilon)} = (-1)^q q!\epsilon + O(\epsilon^2), \quad \frac{\psi}{\Gamma} = -\left(\frac{1}{\Gamma}\right)' \longrightarrow (-1)^{q+1} q!.$$

Split (2) at $r = n$. For $r < n$, $q = n-r-1$, and its summands become $(-1)^n (n-r-1)!(x/2)^{2r-n}/r!$. For $r \geq n$, put $r = n+k$ and use $\Gamma(k+1) = k!$. Since $J_{-n} = (-1)^n J_n$, this yields

$$\begin{aligned}\partial_\nu J_{-\nu}|_n = (-1)^{n+1} J_n L &+ (-1)^n \sum_{r=0}^{n-1} \frac{(n-r-1)!}{r!} (x/2)^{2r-n} \\ &+ (-1)^n \sum_{k=0}^{\infty} \frac{(-1)^k (x/2)^{n+2k} \psi(k+1)}{k!(n+k)!}.\end{aligned}$$

Substitute this and (1) into the order-limit formula. The logarithms add, and both sums are subtracted:

$$\boxed{Y_n(x) = \frac{2}{\pi} J_n(x) \ln(x/2) - \frac{1}{\pi} \sum_{r=0}^{n-1} \frac{(n-r-1)!}{r!} (x/2)^{2r-n}} \\ \boxed{-\frac{1}{\pi} \sum_{k=0}^{\infty} \frac{(-1)^k [\psi(k+1) + \psi(n+k+1)]}{k!(n+k)!} (x/2)^{n+2k}.}$$

This is Eq. (14.61). When $n = 0$ the finite sum is empty; negative integer orders follow from $Y_{-n} = (-1)^n Y_n$. The gamma-pole limit above also supplies $q = 0$, needed by the last finite-sum term.

14.3.9 If Bessel's ODE (with solution J_ν) is differentiated with respect to ν, one obtains

$$x^2 \frac{d^2}{dx^2}\left(\frac{\partial J_\nu}{\partial \nu}\right) + x \frac{d}{dx}\left(\frac{\partial J_\nu}{\partial \nu}\right) + (x^2 - \nu^2)\frac{\partial J_\nu}{\partial \nu} = 2\nu J_\nu.$$

Use the above equation to show that $Y_n(x)$ is a solution to Bessel's ODE.

Hint. Equation (14.60) will be useful.

Solution.

Denote the Bessel operator by $L_\nu = x^2\partial_x^2 + x\partial_x + x^2 - \nu^2$. Differentiate $L_\nu J_\nu = 0$ with respect to order:

$$L_\nu(\partial_\nu J_\nu) - 2\nu J_\nu = 0.$$

Because $J_{-\nu}$ satisfies the same equation with the same ν^2, differentiating $L_\nu J_{-\nu} = 0$ likewise gives

$$L_\nu(\partial_\nu J_{-\nu}) = 2\nu J_{-\nu}.$$

At $\nu = n$, apply L_n to Eq. (14.60):

$$\begin{aligned} L_n Y_n &= \frac{1}{\pi} L_n \left[\partial_\nu J_\nu|_n - (-1)^n \, \partial_\nu J_{-\nu}|_n\right] \\ &= \frac{2n}{\pi}[J_n - (-1)^n J_{-n}] \\ &= \frac{2n}{\pi}[J_n - (-1)^{2n} J_n] = 0. \end{aligned}$$

Thus $\boxed{x^2 Y_n'' + x Y_n' + (x^2 - n^2) Y_n = 0}$, including $n = 0$, for which both inhomogeneous terms already vanish.

14.3.10 For the case $m = 0$, $a = 1$, and $b = 2$, the coaxial wave-guide TM boundary conditions become $f(\lambda) = 0$, with

$$f(x) = \frac{J_0(2x)}{Y_0(2x)} - \frac{J_0(x)}{Y_0(x)}.$$

This function is plotted in Fig. 14.9.

(a) Calculate $f(x)$ for $x = 0.0(0.1)10.0$ and plot $f(x)$ vs. x to find the approximate location of the roots.

(b) Call a root-finding program to determine the first three roots to higher precision.

ANS. (b) 3.1230, 6.2734, 9.4182.

Note. The higher roots can be expected to appear at intervals whose length approaches π. Why? AMS-55 (see Additional Readings) gives an approximate formula for the roots. The function $g(x) = J_0(x)Y_0(2x) - J_0(2x)Y_0(x)$ is much better behaved than the $f(x)$ previously discussed.

Tabulate $x = 0$ as undefined and use the values $0.1, 0.2, \ldots, 10.0$ elsewhere. The limiting zero at the origin is not an annular eigenvalue.

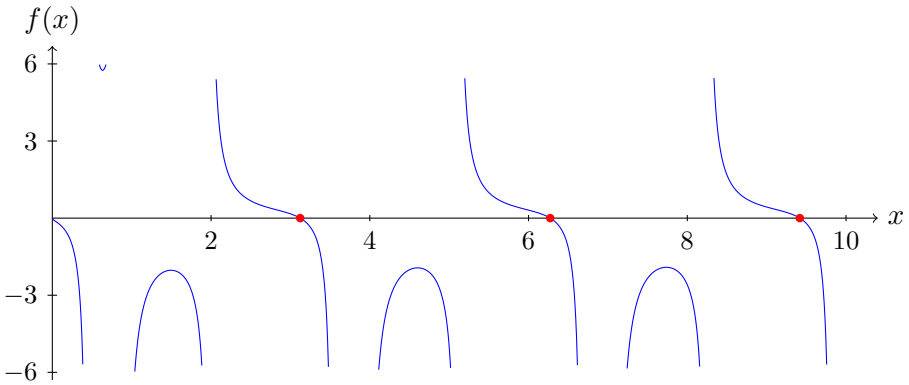
Solution.

At $x = 0$ the quoted ratios are undefined. Their difference has limiting value zero, but this is not an eigenvalue of the annular Dirichlet problem. Poles of either denominator must also be distinguished from roots; a sign change across a pole is not a zero.

(a) The requested grid values, rounded to four decimals, are below. The origin entry is undefined; its right-hand limit is zero.

x	$f(x)$	x	$f(x)$	x	$f(x)$	x	$f(x)$
0.0	undef.	2.6	0.5341	5.2	5.2300	7.8	-1.9419
0.1	-0.2656	2.7	0.4305	5.3	2.2930	7.9	-2.1253
0.2	-0.6690	2.8	0.3441	5.4	1.4109	8.0	-2.5934
0.3	-1.7451	2.9	0.2613	5.5	0.9937	8.1	-3.8402
0.4	-8.1648	3.0	0.1673	5.6	0.7530	8.2	-10.3590
0.5	10.7813	3.1	0.0388	5.7	0.5959	8.3	9.6855
0.6	5.8986	3.2	-0.1741	5.8	0.4827	8.4	2.9820
0.7	6.2993	3.3	-0.6075	5.9	0.3925	8.5	1.6783
0.8	10.8328	3.4	-1.8044	6.0	0.3110	8.6	1.1349
0.9	-142.7633	3.5	-9.5528	6.1	0.2248	8.7	0.8408
1.0	-8.2314	3.6	11.3692	6.2	0.1151	8.8	0.6571
1.1	-4.2257	3.7	6.8360	6.3	-0.0542	8.9	0.5298
1.2	-2.9376	3.8	8.0074	6.4	-0.3728	9.0	0.4326
1.3	-2.3652	3.9	18.3403	6.5	-1.1445	9.1	0.3496
1.4	-2.1021	4.0	-22.6755	6.6	-4.2622	9.2	0.2677
1.5	-2.0284	4.1	-6.4402	6.7	28.1279	9.3	0.1711
1.6	-2.1260	4.2	-3.7568	6.8	7.4939	9.4	0.0333
1.7	-2.4670	4.3	-2.7318	6.9	7.4278	9.5	-0.2061
1.8	-3.3644	4.4	-2.2430	7.0	12.9089	9.6	-0.7246
1.9	-6.8081	4.5	-2.0077	7.1	-70.6154	9.7	-2.3274
2.0	23.0048	4.6	-1.9344	7.2	-8.1293	9.8	-21.9636
2.1	3.6951	4.7	-2.0067	7.3	-4.2476	9.9	9.2367
2.2	1.8835	4.8	-2.2745	7.4	-2.9405	10.0	7.0840
2.3	1.2180	4.9	-2.9403	7.5	-2.3396		
2.4	0.8780	5.0	-4.9933	7.6	-2.0424		
2.5	0.6728	5.1	-45.1412	7.7	-1.9218		

The plot separates each interval between poles; vertical values outside $[-6, 6]$ are clipped. Red points mark the first three roots.



The grid brackets them in $(3.1, 3.2)$, $(6.2, 6.3)$, and $(9.4, 9.5)$, but root finding should use the continuous determinant

$$g(x) = J_0(x)Y_0(2x) - J_0(2x)Y_0(x), \qquad f(x) = -\frac{g(x)}{Y_0(x)Y_0(2x)}.$$

(b) Solving $g(x) = 0$ inside those intervals gives

$$\boxed{\lambda_1 = 3.1230309196, \qquad \lambda_2 = 6.2734357140, \qquad \lambda_3 = 9.4182075423}.$$

At these roots both denominators in f are nonzero, and direct substitution verifies both $g(\lambda_j)=0$ and $f(\lambda_j)=0$. The corresponding radial eigenfunction

$$R_j(\rho)=J_0(\lambda_j\rho)Y_0(\lambda_j)-Y_0(\lambda_j\rho)J_0(\lambda_j)$$

vanishes at $\rho=1$ identically and at $\rho=2$ by $g=0$. At zero parameter the radial equation has solution $A+B\ln\rho$; both boundary values zero force $A=B=0$. Consistently, $g(0^+)=2\ln 2/\pi\neq 0$, although $f(0^+)=0$ because each Y_0 grows logarithmically.

For large x, substitute $J_0(x)\sim\sqrt{2/(\pi x)}\cos(x-\pi/4)$ and $Y_0(x)\sim\sqrt{2/(\pi x)}\sin(x-\pi/4)$. The sine subtraction identity gives

$$g(x)=\frac{\sqrt{2}}{\pi x}\{\sin[(2x-\pi/4)-(x-\pi/4)]+O(x^{-1})\}=\frac{\sqrt{2}}{\pi x}\{\sin x+O(x^{-1})\}.$$

To justify the ordering of the roots as well as the spacing, set $u(\rho)=\sqrt{\rho}\,R(\rho)$. Differentiating $R=\rho^{-1/2}u$ in $R''+\rho^{-1}R'+\lambda^2R=0$ gives

$$-u''-\frac{u}{4\rho^2}=\lambda^2u,\qquad u(1)=u(2)=0.$$

On $[1,2]$, $-1/4\le -1/(4\rho^2)\le -1/16$. Compare its Rayleigh quotient with that of $-u''$, whose Dirichlet eigenvalues on an interval of length one are $(j\pi)^2$. The minimum–maximum characterization therefore gives, for every $j\ge 1$,

$$\sqrt{(j\pi)^2-\tfrac{1}{4}}\le\lambda_j\le\sqrt{(j\pi)^2-\tfrac{1}{16}}.$$

These disjoint intervals contain the three computed roots for $j=1,2,3$, so they are indeed the first three. The same bounds give $\lambda_j=j\pi+O(j^{-1})$, hence $\lambda_{j+1}-\lambda_j\to\pi$.

14.4 Hankel Functions

14.4.1 Verify the Wronskian formulas

(a) $J_\nu(x)H_\nu^{(1)\prime}(x)-J_\nu'(x)H_\nu^{(1)}(x)=\dfrac{2i}{\pi x}$,

(b) $J_\nu(x)H_\nu^{(2)\prime}(x)-J_\nu'(x)H_\nu^{(2)}(x)=-\dfrac{2i}{\pi x}$,

(c) $Y_\nu(x)H_\nu^{(1)\prime}(x)-Y_\nu'(x)H_\nu^{(1)}(x)=-\dfrac{2}{\pi x}$,

(d) $Y_\nu(x)H_\nu^{(2)\prime}(x)-Y_\nu'(x)H_\nu^{(2)}(x)=-\dfrac{2}{\pi x}$,

(e) $H_\nu^{(1)}(x)H_\nu^{(2)\prime}(x)-H_\nu^{(1)\prime}(x)H_\nu^{(2)}(x)=-\dfrac{4i}{\pi x}$,

(f) $H_\nu^{(2)}(x)H_{\nu+1}^{(1)}(x)-H_\nu^{(1)}(x)H_{\nu+1}^{(2)}(x)=\dfrac{4}{i\pi x}$,

(g) $J_{\nu-1}(x)H_\nu^{(1)}(x)-J_\nu(x)H_{\nu-1}^{(1)}(x)=\dfrac{2}{i\pi x}$.

Solution.

Write $J = J_\nu$, $Y = Y_\nu$, $H^{(1)} = J + iY$, $H^{(2)} = J - iY$, and $W[J, Y] = JY' - J'Y = 2/(\pi x)$. Expand each requested expression:

$$\begin{aligned}
\text{(a)}\quad & W[J, H^{(1)}] = J(J' + iY') - J'(J + iY) = iW[J, Y] = \frac{2i}{\pi x}, \\
\text{(b)}\quad & W[J, H^{(2)}] = J(J' - iY') - J'(J - iY) = -iW[J, Y] = -\frac{2i}{\pi x}, \\
\text{(c)}\quad & W[Y, H^{(1)}] = YJ' - Y'J + i(YY' - Y'Y) = -W[J, Y] = -\frac{2}{\pi x}, \\
\text{(d)}\quad & W[Y, H^{(2)}] = YJ' - Y'J - i(YY' - Y'Y) = -\frac{2}{\pi x}, \\
\text{(e)}\quad & W[H^{(1)}, H^{(2)}] = (J + iY)(J' - iY') - (J' + iY')(J - iY) \\
& \qquad = -2i(JY' - J'Y) = -\frac{4i}{\pi x}.
\end{aligned}$$

For (f), the derivative recurrence gives $H^{(j)}_{\nu+1} = \nu H^{(j)}_\nu / x - H^{(j)\prime}_\nu$. Substitute in the left side; its two ν/x products cancel:

$$\begin{aligned}
H^{(2)}_\nu H^{(1)}_{\nu+1} - H^{(1)}_\nu H^{(2)}_{\nu+1} &= -H^{(2)}_\nu H^{(1)\prime}_\nu + H^{(1)}_\nu H^{(2)\prime}_\nu \\
&= W[H^{(1)}, H^{(2)}] = -\frac{4i}{\pi x} = \frac{4}{i\pi x}.
\end{aligned}$$

For (g), use $J_{\nu-1} = J'_\nu + \nu J_\nu / x$ and $H^{(1)}_{\nu-1} = H^{(1)\prime}_\nu + \nu H^{(1)}_\nu / x$:

$$J_{\nu-1} H^{(1)}_\nu - J_\nu H^{(1)}_{\nu-1} = J'_\nu H^{(1)}_\nu - J_\nu H^{(1)\prime}_\nu = -\frac{2i}{\pi x} = \frac{2}{i\pi x}.$$

This verifies all seven signs with the same Wronskian convention.

14.4.2 Show that the integral forms

$$\text{(a)}\quad \frac{1}{i\pi} \int_{0C_1}^{\infty e^{i\pi}} e^{(x/2)(t-1/t)} \frac{dt}{t^{\nu+1}} = H^{(1)}_\nu(x),$$

$$\text{(b)}\quad \frac{1}{i\pi} \int_{\infty e^{-i\pi} C_2}^{0} e^{(x/2)(t-1/t)} \frac{dt}{t^{\nu+1}} = H^{(2)}_\nu(x)$$

satisfy Bessel's ODE. The contours C_1 and C_2 are shown in Fig. 14.10.

Take $x > 0$ and $-\pi < \arg t < \pi$. Approach zero from positive t and the negative-real endpoints on their indicated banks.

Solution.

Take $x > 0$ and the branch $-\pi < \arg t < \pi$, with limiting values on either bank of the negative axis. Put $f = e^{x(t-t^{-1})/2} t^{-\nu-1}$. The computation in Exercise 14.1.15 gives the exact identity

$$(x^2 \partial_x^2 + x\partial_x + x^2 - \nu^2) f = \frac{\partial}{\partial t} Q(t, x),$$

$$Q(t, x) = e^{x(t-t^{-1})/2} t^{-\nu} \left[\nu + \frac{x}{2}(t + t^{-1}) \right].$$

We now check the different endpoints of these contours. On the positive axis as $t \downarrow 0$, the exponential is $e^{-x/(2t)+O(t)}$; it dominates every power of t^{-1} and makes $Q \to 0$. On either bank $t = Re^{\pm i\pi}$ as $R \to \infty$, it is $e^{-xR/2+O(R^{-1})}$ and again $Q \to 0$. These estimates also justify differentiation under the integrals on compact positive x intervals.

(a) Integrating the identity from 0^+ along C_1 to the upper negative-real infinity gives $[Q]_{0^+}^{\infty e^{i\pi}} = 0$. After multiplication by $1/(\pi i)$, the first integral therefore satisfies Bessel's equation.

(b) Along C_2 the orientation is from the lower negative-real infinity to 0^+. Both endpoint values still vanish, so the second integral satisfies the same equation. These endpoint estimates establish the ODE, while Exercises 14.4.3–14.4.4 establish that their normalizations are precisely $J_\nu \pm iY_\nu$.

14.4.3 Show that the substitution $t = e^{i\pi}/s$ into Eq. (14.90) for $H_\nu^{(1)}(x)$ not only produces the integrand for the similar integral representation of $H_{-\nu}^{(1)}(x)$ but that the contour in s is the same geometric contour up to deformation, but with reversed orientation.

Solution.

Choose a convenient representative of C_1: the positive segment $0 < t \le 1$, the upper unit semicircle, and the upper bank of the negative ray from -1 to $-\infty$. This can be deformed to the contour in Fig. 14.10 without crossing the cut or the origin. Under $s = e^{i\pi}/t = -1/t$,

$$t : 0^+ \longmapsto s : \infty e^{i\pi}, \qquad t : i \longmapsto s : i, \qquad t : \infty e^{i\pi} \longmapsto s : 0^+.$$

The two rays are exchanged. On the semicircle, $t = e^{i\theta}$ maps to $s = e^{i(\pi-\theta)}$, so that arc is also unchanged as a set but traversed backwards. Thus the entire transformed contour is C_1 with reversed orientation.

On this branch, $\log t = i\pi - \log s$, so

$$t - t^{-1} = s - s^{-1}, \qquad t^{-\nu-1}dt = -e^{-i\pi\nu}s^{\nu-1}ds.$$

The minus sign from the differential cancels the reversed contour. Therefore

$$\begin{aligned} H_\nu^{(1)}(x) &= \frac{e^{-i\pi\nu}}{\pi i}\int_{C_1} e^{x(s-s^{-1})/2}s^{\nu-1}ds \\ &= \boxed{e^{-i\pi\nu}H_{-\nu}^{(1)}(x)}. \end{aligned}$$

For any other admissible representative, "the same contour" means the same deformable contour class with the same branch and endpoints; the representative just chosen makes the reversal explicit.

14.4.4 Using the integrals and contours given in Exercise 14.4.2, show that

$$\frac{1}{2i}\left[H_\nu^{(1)}(x) - H_\nu^{(2)}(x)\right] = Y_\nu(x).$$

Solution.

Temporarily denote the two contour integrals by A_ν and B_ν, without using their algebraic Hankel definitions. Joining C_2 to C_1 gives the Schlaefli contour: the paths approaching the positive origin can be joined through the exponentially decaying region. Its prefactor is half the Hankel prefactor, so

$$2J_\nu = A_\nu + B_\nu. \tag{1}$$

Exercise 14.4.3 gives $A_{-\nu} = e^{i\pi\nu} A_\nu$. The same inversion on the lower contour, now with $t = e^{-i\pi}/s$, gives $B_{-\nu} = e^{-i\pi\nu} B_\nu$. Applying (1) at order $-\nu$ therefore yields

$$2J_{-\nu} = e^{i\pi\nu} A_\nu + e^{-i\pi\nu} B_\nu. \tag{2}$$

For noninteger ν, insert these into the definition of Y_ν:

$$\begin{aligned} Y_\nu &= \frac{\cos\pi\nu\,(A_\nu + B_\nu) - e^{i\pi\nu} A_\nu - e^{-i\pi\nu} B_\nu}{2\sin\pi\nu} \\ &= \frac{-i\sin\pi\nu\, A_\nu + i\sin\pi\nu\, B_\nu}{2\sin\pi\nu} = \boxed{\frac{A_\nu - B_\nu}{2i}}. \end{aligned}$$

The limit in order proves the integer case. Combining this result with (1) gives $A_\nu = J_\nu + iY_\nu$ and $B_\nu = J_\nu - iY_\nu$, verifying both identifications from the contours themselves.

14.4.5 Show that the integrals in Exercise 14.4.2 may be transformed to yield

(a) $H_\nu^{(1)}(x) = \dfrac{1}{\pi i} \displaystyle\int_{C_3} e^{x\sinh\gamma - \nu\gamma}\, d\gamma,$

(b) $H_\nu^{(2)}(x) = \dfrac{1}{\pi i} \displaystyle\int_{C_4} e^{x\sinh\gamma - \nu\gamma}\, d\gamma,$

where C_3 and C_4 are the contours in Fig. 14.11.

Take $x > 0$ initially and retain the upper/lower logarithm branches and the orientations of C_1, C_2.

Solution.

Put $t = e^\gamma$ using a continuous logarithm on each contour. Then

$$\frac{dt}{t} = d\gamma, \qquad t - t^{-1} = 2\sinh\gamma, \qquad t^{-\nu} = e^{-\nu\gamma}.$$

The transformed integrand is consequently

$$e^{x(t-t^{-1})/2} t^{-\nu-1} dt = e^{x\sinh\gamma - \nu\gamma} d\gamma.$$

(a) For C_1, positive $t \downarrow 0$ corresponds to $\gamma \to -\infty$ on the real axis, and the upper negative ray corresponds to $\gamma \to +\infty + i\pi$. A representative is

$$C_3: \quad -\infty \;\longrightarrow\; 0 \;\longrightarrow\; i\pi \;\longrightarrow\; +\infty + i\pi.$$

Thus $H_\nu^{(1)} = (\pi i)^{-1} \int_{C_3} e^{x\sinh\gamma - \nu\gamma} d\gamma$.

(b) The lower negative ray of C_2 begins at $\gamma = +\infty - i\pi$, and the positive-origin endpoint is $\gamma = -\infty$. Its orientation is

$$C_4: \quad +\infty - i\pi \;\longrightarrow\; -i\pi \;\longrightarrow\; 0 \;\longrightarrow\; -\infty.$$

This gives the second requested integral with its stated sign. The integrand is entire in γ, so these piecewise paths may be smoothed to those in Fig. 14.11. For $x > 0$, it decays at $\Re\gamma \to -\infty$ on height zero and at $\Re\gamma \to +\infty$ on heights $\pm\pi$, validating the endpoint deformations.

14.4.6

(a) Transform $H_0^{(1)}(x)$, Eq. (14.90), into

$$H_0^{(1)}(x) = \frac{1}{i\pi}\int_C e^{ix\cosh s}\,ds,$$

where the contour C runs from $-\infty - i\pi/2$ through the origin of the s-plane to $\infty + i\pi/2$.

(b) Justify rewriting $H_0^{(1)}(x)$ as

$$H_0^{(1)}(x) = \frac{2}{i\pi}\int_0^{\infty+i\pi/2} e^{ix\cosh s}\,ds.$$

(c) Verify that this integral representation actually satisfies Bessel's differential equation. (The $i\pi/2$ in the upper limit is not essential. It serves as a convergence factor. We can replace it by $ia\pi/2$ and take the limit $a \to 0$.)

Take $x > 0$ and keep a positive imaginary contour height, or a positive imaginary part of x, during differentiation. Take the damping limit only afterward.

Solution.

(a) In the order-zero contour formula set $t = ie^s = e^{s+i\pi/2}$. Directly,

$$t - t^{-1} = i(e^s + e^{-s}) = 2i\cosh s, \qquad dt/t = ds.$$

The initial positive-origin endpoint maps to $s = -\infty - i\pi/2$, the point $t = i$ maps to $s = 0$, and the final upper negative infinity maps to $s = +\infty + i\pi/2$. Therefore

$$\boxed{H_0^{(1)}(x) = \frac{1}{\pi i}\int_C e^{ix\cosh s} ds}.$$

(b) Choose the contour to consist of a path C_+ from 0 to $\infty + i\pi/2$ and its reflected path from $-\infty - i\pi/2$ to 0. On the reflected half put $u = -s$. Because $\cosh(-u) = \cosh u$, reversing its limits cancels $ds = -du$, so its integral equals the integral on C_+. Hence

$$\boxed{H_0^{(1)}(x) = \frac{2}{\pi i}\int_0^{\infty+i\pi/2} e^{ix\cosh s} ds}.$$

(c) For $x > 0$, choose a contour whose final ray has fixed height η, $0 < \eta \le \pi/2$. On that ray

$$|e^{ix\cosh(R+i\eta)}| = e^{-x\sinh R\sin\eta},$$

which decays faster than the hyperbolic factors from any two derivatives. Differentiation under the integral is now legitimate:

$$\begin{aligned}(x^2\partial_x^2 + x\partial_x + x^2)e^{ix\cosh s} &= [-x^2\sinh^2 s + ix\cosh s]e^{ix\cosh s}\\ &= \frac{d}{ds}[ix\sinh s\, e^{ix\cosh s}].\end{aligned}$$

The lower boundary is zero since $\sinh 0 = 0$; the upper is zero by the damping estimate. Thus the integral solves the Bessel ODE. The real-axis form is a boundary limit, not a license to discard an undamped boundary term after differentiating. Take $\eta \downarrow 0$ only after the convergent-contour calculation, or use $x + i\epsilon$ and Abel convergence as in Exercise 14.3.7.

14.4.7 From

$$H_0^{(1)}(x) = \frac{2}{i\pi}\int_0^\infty e^{ix\cosh s}\,ds$$

show that

(a) $J_0(x) = \dfrac{2}{\pi}\displaystyle\int_0^\infty \sin(x\cosh s)\,ds,$

(b) $J_0(x) = \dfrac{2}{\pi}\displaystyle\int_1^\infty \frac{\sin(xt)}{\sqrt{t^2-1}}\,dt.$

This last result is a Fourier sine transform.

These ordinary oscillatory integrals hold for $x > 0$; the value at zero is obtained after integration by a limit.

Solution.

These real integrals are for $x > 0$. The sine-transform formula cannot be extended to $x < 0$ unchanged, since its right side is odd in x whereas J_0 is even. At $x = 0$ the limit must be taken after integration. For $x > 0$, expand the exponential and retain its prefactor:

$$\begin{aligned} H_0^{(1)}(x) &= -\frac{2i}{\pi}\int_0^\infty [\cos(x\cosh s) + i\sin(x\cosh s)]ds \\ &= \frac{2}{\pi}\int_0^\infty \sin(x\cosh s)ds - \frac{2i}{\pi}\int_0^\infty \cos(x\cosh s)ds. \end{aligned}$$

(a) Since $H_0^{(1)} = J_0 + iY_0$ and both J_0, Y_0 are real on the positive axis, its real part is

$$\boxed{J_0(x) = \frac{2}{\pi}\int_0^\infty \sin(x\cosh s)ds}.$$

(b) Put $t = \cosh s$. The limits become $1, \infty$, and $dt = \sinh s\,ds = \sqrt{t^2-1}\,ds$. Thus

$$\boxed{J_0(x) = \frac{2}{\pi}\int_1^\infty \frac{\sin(xt)}{\sqrt{t^2-1}}dt}.$$

The square-root singularity at $t = 1$ is integrable and the oscillatory $1/t$ tail converges by Dirichlet's test. These facts also identify the ordinary improper integral with its Abel limit.

14.4.8 From $H_0^{(1)}(x) = \dfrac{2}{i\pi}\displaystyle\int_0^\infty e^{ix\cosh s}\,ds$ (see Exercises 14.4.5 and 14.4.6), show that

(a) $Y_0(x) = -\dfrac{2}{\pi}\displaystyle\int_0^\infty \cos(x\cosh s)\,ds,$

(b) $Y_0(x) = -\dfrac{2}{\pi}\displaystyle\int_1^\infty \frac{\cos(xt)}{\sqrt{t^2-1}}\,dt.$

These are the integral representations in Eq. (14.63). This last result is a Fourier cosine transform.

Take $x > 0$ for both ordinary oscillatory integrals.

Solution.

For $x > 0$, multiplication by $2/(i\pi) = -2i/\pi$ gives

$$H_0^{(1)}(x) = \frac{2}{\pi}\int_0^\infty \sin(x\cosh s)ds - \frac{2i}{\pi}\int_0^\infty \cos(x\cosh s)ds.$$

(a) Equating imaginary parts with $J_0 + iY_0$ therefore yields

$$\boxed{Y_0(x) = -\frac{2}{\pi}\int_0^\infty \cos(x\cosh s)ds}.$$

(b) With $t = \cosh s$, use $ds = dt/\sqrt{t^2-1}$ and the limits $t = 1, \infty$:

$$\boxed{Y_0(x) = -\frac{2}{\pi}\int_1^\infty \frac{\cos(xt)}{\sqrt{t^2-1}}dt}.$$

As in Exercise 14.4.7, the finite endpoint is integrable and the tail converges conditionally. Thus taking the imaginary part of the Abel boundary value is consistent with the displayed ordinary improper integrals. In particular the prefactor changes the cosine integral's sign; omitting $1/i$ would interchange the first- and second-kind identifications.

14.5 Modified Bessel Functions, $I_\nu(x)$ and $K_\nu(x)$

14.5.1 Show that $e^{(x/2)(t+1/t)} = \sum_{n=-\infty}^{\infty} I_n(x)t^n$, thus generating modified Bessel functions, $I_n(x)$.

Solution.

Start from $\exp[X(T - T^{-1})/2] = \sum_{n\in\mathbb{Z}} J_n(X)T^n$ and put $X = ix$, $T = -it$. Since $T^{-1} = i/t$,

$$\frac{ix}{2}(-it - i/t) = \frac{x}{2}(t + t^{-1}), \qquad J_n(ix)(-it)^n = i^{-n}J_n(ix)t^n = I_n(x)t^n.$$

Thus, for $t \neq 0$,

$$\boxed{e^{x(t+t^{-1})/2} = \sum_{n=-\infty}^{\infty} I_n(x)t^n}.$$

The equality $(-i)^n = i^{-n}$ holds for all integers n. Absolute convergence on compact annuli follows either from the original generating series or from the product of the two exponential series $e^{xt/2}e^{x/(2t)}$.

14.5.2 Verify the following identities

(a) $1 = I_0(x) + 2\sum_{n=1}^{\infty}(-1)^n I_{2n}(x)$,

(b) $e^x = I_0(x) + 2\sum_{n=1}^{\infty} I_n(x)$,

(c) $e^{-x} = I_0(x) + 2\sum_{n=1}^{\infty}(-1)^n I_n(x)$,

(d) $\cosh x = I_0(x) + 2\sum_{n=1}^{\infty} I_{2n}(x)$,

(e) $\sinh x = 2\sum_{n=1}^{\infty} I_{2n-1}(x)$.

Solution.

Invariance of the generating function under $t \mapsto 1/t$ gives $I_{-n} = I_n$. Pair positive and negative orders:

$$e^{x(t+t^{-1})/2} = I_0 + \sum_{n=1}^{\infty} I_n (t^n + t^{-n}).$$

(a) Put $t = i$. Odd-n pairs cancel, while for $n = 2j$, $i^{2j} + i^{-2j} = 2(-1)^j$. Since $i + i^{-1} = 0$,

$$\boxed{1 = I_0 + 2\sum_{j=1}^{\infty} (-1)^j I_{2j}}.$$

(b) Put $t = 1$; all pairs equal two, so

$$\boxed{e^x = I_0 + 2\sum_{n\geq 1} I_n}.$$

(c) Put $t = -1$; each pair is $2(-1)^n$, giving

$$\boxed{e^{-x} = I_0 + 2\sum_{n\geq 1} (-1)^n I_n}.$$

(d) Add (b) and (c) and divide by two. Odd orders cancel and even orders double, so

$$\boxed{\cosh x = I_0 + 2\sum_{n\geq 1} I_{2n}}.$$

(e) Subtract (c) from (b) and divide by two. The constant and even orders cancel, giving

$$\boxed{\sinh x = 2\sum_{n\geq 1} I_{2n-1}}.$$

All these substitutions are on the unit circle, where the generating series converges absolutely.

14.5.3

(a) From the generating function of Exercise 14.5.1 show that

$$I_n(x) = \frac{1}{2\pi i} \oint e^{(x/2)(t+1/t)} \frac{dt}{t^{n+1}}.$$

(b) For $n = \nu$, not an integer, show that the preceding integral representation may be generalized to

$$I_\nu(x) = \frac{1}{2\pi i} \int_C e^{(x/2)(t+1/t)} \frac{dt}{t^{\nu+1}}.$$

The contour C is the same as that for $J_\nu(x)$ (Fig. 14.5).

In (b) use $x > 0$ initially and the open Hankel contour with $-\pi < \arg t < \pi$.

Solution.

(a) Divide the Laurent expansion from Exercise 14.5.1 by t^{n+1} and integrate counterclockwise around any circle centered at zero. The integral of t^{m-n-1} is $2\pi i$ if $m = n$ and zero otherwise. Thus coefficient extraction gives

$$\boxed{I_n(x) = \frac{1}{2\pi i} \oint e^{x(t+t^{-1})/2} t^{-n-1} dt}.$$

The exponential has an essential singularity at zero, but its Laurent coefficient is extracted exactly as for any isolated singularity.

For noninteger order in (b), the contour is the open Hankel contour of Fig. 14.5, not an ordinary closed circle. Use the two banks of the negative-real cut and a fixed circle about zero, with $-\pi < \arg t < \pi$; take $x > 0$ initially. Put $s = xt/2$. The proposed integral becomes

$$\frac{(x/2)^\nu}{2\pi i} \int_C e^{s + x^2/(4s)} s^{-\nu-1} ds.$$

Expand only the second exponential and use the reciprocal-gamma Hankel integral:

$$\begin{aligned} \frac{(x/2)^\nu}{2\pi i} \sum_{r=0}^{\infty} \frac{(x^2/4)^r}{r!} \int_C e^s s^{-\nu-r-1} ds &= \sum_{r=0}^{\infty} \frac{(x/2)^{\nu+2r}}{r!\Gamma(\nu + r + 1)} \\ &= \boxed{I_\nu(x)}. \end{aligned}$$

A fixed-radius contour stays away from zero, so the expanded factor is uniformly bounded; its tails have the decaying factor e^s. This justifies termwise integration. The equality then continues to compatible complex x branches.

14.5.4 For $\nu > -\frac{1}{2}$ show that $I_\nu(z)$ may be represented by

$$\begin{aligned} I_\nu(z) &= \frac{1}{\pi^{1/2}\Gamma(\nu + \frac{1}{2})} \left(\frac{z}{2}\right)^\nu \int_0^\pi e^{\pm z\cos\theta} \sin^{2\nu}\theta \, d\theta \\ &= \frac{1}{\pi^{1/2}\Gamma(\nu + \frac{1}{2})} \left(\frac{z}{2}\right)^\nu \int_{-1}^1 e^{\pm zp} (1-p^2)^{\nu-1/2} \, dp \\ &= \frac{2}{\pi^{1/2}\Gamma(\nu + \frac{1}{2})} \left(\frac{z}{2}\right)^\nu \int_0^{\pi/2} \cosh(z\cos\theta) \sin^{2\nu}\theta \, d\theta. \end{aligned}$$

Use one consistent branch of z^ν and of $I_\nu(z)$; at zero use an existing limit.

Solution.

For $\nu > -1/2$ the endpoint weight is integrable. Expand $e^{z\cos\theta}$ in its uniformly convergent series. Reflection $\theta \mapsto \pi - \theta$ cancels every odd power of $\cos\theta$, leaving

$$\int_0^\pi e^{z\cos\theta} \sin^{2\nu}\theta d\theta = \sum_{r=0}^{\infty} \frac{z^{2r}}{(2r)!} \int_0^\pi \cos^{2r}\theta \sin^{2\nu}\theta d\theta.$$

The remaining integral is $B(r + 1/2, \nu + 1/2)$, so multiplication by the prescribed prefactor gives

$$\begin{aligned} \frac{(z/2)^\nu}{\sqrt{\pi}\Gamma(\nu + 1/2)} \sum_{r=0}^{\infty} \frac{z^{2r}}{(2r)!} \frac{\Gamma(r + 1/2)\Gamma(\nu + 1/2)}{\Gamma(r + \nu + 1)} &= \sum_{r=0}^{\infty} \frac{(z/2)^{\nu+2r}}{r!\Gamma(r + \nu + 1)} \\ &= I_\nu(z), \end{aligned}$$

where $\Gamma(r+1/2)/[\sqrt{\pi}(2r)!] = 1/(4^r r!)$. Changing z to $-z$ inside the exponential does not change any surviving even term, proving both signs in the first integral.

With $p = \cos\theta$, $dp = -\sin\theta d\theta$, and hence

$$\sin^{2\nu}\theta d\theta = -(1-p^2)^{\nu-1/2}dp.$$

Reversing the limits gives the second representation. Finally the reflection symmetry averages the two exponentials to $\cosh(z\cos\theta)$, whose integral over $[0,\pi]$ is twice its integral over $[0,\pi/2]$. This proves the third form. All factors z^ν use the same branch as $I_\nu(z)$.

14.5.5 The cylindrical cavity depicted in Fig. 14.4 has radius a and height h. For this exercise, the end caps $z = 0$ and h are at zero potential, while the cylindrical wall $\rho = a$ has a potential of functional form $V = V(\varphi, z)$.

(a) Show that the electrostatic potential $\Phi(\rho,\varphi,z)$ has the functional form

$$\Phi(\rho,\varphi,z) = \sum_{m=0}^{\infty}\sum_{n=1}^{\infty} I_m(k_n\rho)\,(a_{mn}\sin m\varphi + b_{mn}\cos m\varphi)\sin k_n z,$$

where $k_n = n\pi/h$.

(b) Show that the coefficients a_{mn} and b_{mn} are given by

$$\left\{\begin{array}{c} a_{mn} \\ b_{mn} \end{array}\right\} = \frac{2-\delta_{m0}}{\pi h\, I_m(k_n a)}\int_0^{2\pi}\int_0^h V(\varphi,z)\left\{\begin{array}{c} \sin m\varphi \\ \cos m\varphi \end{array}\right\}\sin k_n z\, dz\, d\varphi.$$

Hint. Expand $V(\varphi,z)$ as a double series and use the orthogonality of the trigonometric functions. Set $a_{0n} = 0$ by convention, since the zero angular sine contributes nothing.

Solution.

(a) Set $\Phi = P(\rho)F(\varphi)Z(z)$ in Laplace's equation. The separated angular and axial equations and boundary conditions give

$$F'' + m^2 F = 0, \quad F = a\sin m\varphi + b\cos m\varphi, \quad m = 0, 1, \ldots,$$
$$Z'' + k_n^2 Z = 0, \quad Z(0) = Z(h) = 0, \quad Z = \sin k_n z, \quad k_n = n\pi/h, \quad n \geq 1.$$

The radial equation is then

$$\rho^2 P'' + \rho P' - (k_n^2\rho^2 + m^2)P = 0.$$

Its regular solution is $I_m(k_n\rho)$; K_m is singular at the axis. Adding all separated solutions gives precisely the double series in part (a).

(b) Evaluate it at $\rho = a$, multiply by the desired sine or cosine in φ and by $\sin k_n z$, and integrate over the wall. Distinct indices vanish by orthogonality. The axial norm is

$$\int_0^h \sin^2(n\pi z/h)dz = h/2.$$

For $m > 0$ both angular norms are π; for $m = 0$ the constant cosine has norm 2π. Dividing by the product of norms and by $I_m(k_n a)$ gives

$$\boxed{\begin{pmatrix} a_{mn} \\ b_{mn} \end{pmatrix} = \frac{2-\delta_{m0}}{\pi h I_m(k_n a)}\int_0^{2\pi}\int_0^h V(\varphi,z)\begin{pmatrix} \sin m\varphi \\ \cos m\varphi \end{pmatrix}\sin k_n z\, dz\, d\varphi}.$$

The height is h throughout, as in the published erratum; the original l in the coefficient formula must be replaced by h. Also $\sin 0\varphi = 0$, so a_{0n} is not determined by orthogonality and has no effect on the solution. The displayed formula makes the convenient choice $a_{0n} = 0$.

14.5.6 Verify that $K_\nu(x)$ as defined in Eq. (14.106) is equivalent to

$$K_\nu(x) = \frac{\pi}{2}\,\frac{I_{-\nu}(x) - I_\nu(x)}{\sin \nu\pi}$$

and from this show that

$$K_\nu(x) = K_{-\nu}(x).$$

Take $x > 0$ initially; at integer order interpret the quotient by its removable order limit.

Solution.

For noninteger ν, put $S = \sin \pi\nu$ and $C = \cos \pi\nu$. The definitions imply

$$\begin{aligned} H_\nu^{(1)}(ix) &= J_\nu(ix) + i\frac{CJ_\nu(ix) - J_{-\nu}(ix)}{S} \\ &= \frac{i}{S}[e^{-i\pi\nu}J_\nu(ix) - J_{-\nu}(ix)], \end{aligned}$$

because $S + iC = ie^{-i\pi\nu}$. Using the branch-compatible relations $J_\nu(ix) = i^\nu I_\nu(x)$ and $J_{-\nu}(ix) = i^{-\nu}I_{-\nu}(x)$,

$$\begin{aligned} K_\nu(x) &= \frac{\pi}{2} i^{\nu+1} H_\nu^{(1)}(ix) \\ &= \frac{\pi}{2S}\left[i^{2\nu+2}e^{-i\pi\nu}I_\nu - i^2 I_{-\nu}\right] = \boxed{\frac{\pi}{2}\frac{I_{-\nu} - I_\nu}{\sin \pi\nu}}. \end{aligned}$$

Changing ν to $-\nu$ reverses the signs of both numerator and denominator, so

$$\boxed{K_{-\nu} = K_\nu}.$$

At integer order both quotients are understood through their removable limits in ν, which preserves the same equality.

14.5.7 Show that $K_\nu(x)$ satisfies the following recurrence relations:

$$\begin{aligned} K_{\nu-1}(x) - K_{\nu+1}(x) &= -\frac{2\nu}{x}K_\nu(x), \\ K_{\nu-1}(x) + K_{\nu+1}(x) &= -2K_\nu'(x). \end{aligned}$$

Note. These differ from the recurrence relations for I_ν.

Solution.

Let $c = \pi/(2\sin \pi\nu)$ and first take noninteger ν. Since the sine changes sign when the order changes by one,

$$K_{\nu-1} = -c(I_{-\nu+1} - I_{\nu-1}), \qquad K_{\nu+1} = -c(I_{-\nu-1} - I_{\nu+1}).$$

For the difference, the I recurrence at orders ν and $-\nu$ gives

$$\begin{aligned} K_{\nu-1} - K_{\nu+1} &= -c[(I_{-\nu+1} - I_{-\nu-1}) - (I_{\nu-1} - I_{\nu+1})] \\ &= -c\left[\frac{2\nu}{x}I_{-\nu} - \frac{2\nu}{x}I_\nu\right] = \boxed{-\frac{2\nu}{x}K_\nu}. \end{aligned}$$

For the sum, use $I_{\mu-1} + I_{\mu+1} = 2I_\mu'$:

$$\begin{aligned} K_{\nu-1} + K_{\nu+1} &= -c[(I_{-\nu+1} + I_{-\nu-1}) - (I_{\nu-1} + I_{\nu+1})] \\ &= -2c(I_{-\nu}' - I_\nu') = \boxed{-2K_\nu'}. \end{aligned}$$

Taking order limits gives the same relations for integer ν. The opposite signs relative to the I recurrences arise from the sign reversal of $\sin \pi\nu$, not from a change in the ODE.

14.5.8 If $\mathcal{K}_\nu = e^{\nu\pi i}K_\nu$, show that $\mathcal{K}_\nu$ satisfies the same recurrence relations as I_ν.

Solution.

The neighboring phase factors satisfy $e^{i\pi(\nu-1)} = e^{i\pi(\nu+1)} = -e^{i\pi\nu}$. Multiplying the K recurrences by this common negative phase therefore gives

$$\mathcal{K}_{\nu-1} - \mathcal{K}_{\nu+1} = -e^{i\pi\nu}(K_{\nu-1} - K_{\nu+1}) = \frac{2\nu}{x}e^{i\pi\nu}K_\nu = \boxed{\frac{2\nu}{x}\mathcal{K}_\nu},$$
$$\mathcal{K}_{\nu-1} + \mathcal{K}_{\nu+1} = -e^{i\pi\nu}(K_{\nu-1} + K_{\nu+1}) = 2e^{i\pi\nu}K'_\nu = \boxed{2\mathcal{K}'_\nu}.$$

The order-dependent phase is constant with respect to x, so it passes through the derivative. These are exactly the recurrences for I_ν.

14.5.9 Show that when K_0 is evaluated from its series expansion about $x = 0$, the formula given as Eq. (14.110) only follows if a specific branch of its logarithmic term is chosen.

Solution.

For positive x, the small-argument Hankel expansion is

$$H_0^{(1)}(z) = 1 + \frac{2i}{\pi}[\ln(z/2) + \gamma] + O(z^2 \ln z).$$

Insert $z = ix$ in $K_0(x) = (\pi i/2)H_0^{(1)}(ix)$:

$$\begin{aligned} K_0(x) &= \frac{\pi i}{2} - \ln(ix/2) - \gamma + O(x^2 \ln x) \\ &= -\ln(x/2) - \gamma - 2\pi ik + O(x^2 \ln x), \end{aligned}$$

where $\ln(ix/2) = \ln(x/2) + i\pi/2 + 2\pi ik$. Thus the real expression in Eq. (14.110) requires

$$\boxed{\arg(ix) = \pi/2, \qquad k = 0}.$$

On this branch the $+\pi i/2$ prefactor contribution cancels the imaginary part of the logarithm, leaving $K_0 = -\ln x - \gamma + \ln 2 + \cdots$. The full logarithmic series has coefficient $-I_0(x)$; selecting another logarithm branch changes the full function by $-2\pi ikI_0(x)$, not merely its leading constant. This is why the branch is part of the definition of the real positive-axis function K_0.

14.5.10 For $\nu > -\frac{1}{2}$ show that $K_\nu(z)$ may be represented by

$$\begin{aligned} K_\nu(z) &= \frac{\pi^{1/2}}{\Gamma(\nu + \frac{1}{2})}\left(\frac{z}{2}\right)^\nu \int_0^\infty e^{-z\cosh t}\sinh^{2\nu} t\, dt, \qquad -\frac{\pi}{2} < \arg z < \frac{\pi}{2} \\ &= \frac{\pi^{1/2}}{\Gamma(\nu + \frac{1}{2})}\left(\frac{z}{2}\right)^\nu \int_1^\infty e^{-zp}(p^2 - 1)^{\nu-1/2}\, dp. \end{aligned}$$

Solution.

Let $\nu > -1/2$ and initially $z > 0$. Denote the second proposed expression by $F(z)$, and put $c = \sqrt{\pi}/[2^\nu\Gamma(\nu + 1/2)]$. Thus

$$F(z) = cz^\nu \int_1^\infty e^{-zp}(p^2 - 1)^{\nu-1/2} dp.$$

Both endpoints are integrable: at $p=1$ the power is $\nu-1/2>-1$, and at infinity the exponential dominates. Differentiation under the integral gives

$$\begin{aligned} z^2F''+zF'-(z^2+\nu^2)F &= cz^\nu\int_1^\infty e^{-zp}(p^2-1)^{\nu-1/2}[z^2(p^2-1)-(2\nu+1)zp]dp \\ &= -cz^{\nu+1}\int_1^\infty \frac{d}{dp}[e^{-zp}(p^2-1)^{\nu+1/2}]dp = 0. \end{aligned}$$

The boundary at $p=1$ vanishes because $\nu+1/2>0$. Thus F solves the modified Bessel equation.

Its scale is fixed by the large-positive-z limit. Put $p=1+u/z$; after collecting powers,

$$F(z)=\sqrt{\frac{\pi}{2z}}\frac{e^{-z}}{\Gamma(\nu+1/2)}\int_0^\infty e^{-u}u^{\nu-1/2}\left(1+\frac{u}{2z}\right)^{\nu-1/2}du.$$

Dominated convergence makes the integral tend to $\Gamma(\nu+1/2)$, so $F(z)\sim\sqrt{\pi/(2z)}e^{-z}$. This is the normalization of K_ν from its Hankel definition:

$$\frac{\pi}{2}i^{\nu+1}\sqrt{\frac{2}{\pi iz}}e^{i(iz-\pi\nu/2-\pi/4)}=\sqrt{\frac{\pi}{2z}}e^{-z}.$$

Here the leading Hankel asymptotic form, established independently in Section 14.6, is used only to fix the multiplicative constant. Two decaying modified-Bessel solutions are proportional: their Wronskian is A/z, and exponential decay of the functions and their derivatives forces $A=0$. The common leading coefficient then gives $F=K_\nu$.

Finally, substitute $p=\cosh t$, $dp=\sinh t\,dt$:

$$(p^2-1)^{\nu-1/2}dp=\sinh^{2\nu}t\,dt,$$

which gives the first representation. Absolute convergence and analyticity extend both formulas to $\Re z>0$, equivalently $-\pi/2<\arg z<\pi/2$, with a consistent branch of z^ν.

14.5.11 Show that $I_\nu(x)$ and $K_\nu(x)$ satisfy the Wronskian relation

$$I_\nu(x)K_\nu'(x)-I_\nu'(x)K_\nu(x)=-\frac{1}{x}.$$

Solution.

Use the definitions at argument $w=ix$:

$$I_\nu(x)=i^{-\nu}J_\nu(w),\qquad K_\nu(x)=\frac{\pi}{2}i^{\nu+1}H_\nu^{(1)}(w).$$

Each x derivative supplies an additional factor i from $dw/dx=i$. Thus

$$\begin{aligned} I_\nu K_\nu'-I_\nu'K_\nu &= \frac{\pi}{2}i^{-\nu}i^{\nu+1}i[J_\nu(w)H_\nu^{(1)\prime}(w)-J_\nu'(w)H_\nu^{(1)}(w)] \\ &= -\frac{\pi}{2}\frac{2i}{\pi w}=-\frac{i}{ix}=\boxed{-\frac{1}{x}}. \end{aligned}$$

The derivative primes inside the brackets refer to the argument w, whereas those on I,K refer to x; keeping that distinction is essential for the sign and normalization.

14.5.12 Verify that the coefficient in the axial Green's function of Eq. (14.117) is -1.

Use the cylindrical unit-source convention, whose radial equation has source $\delta(\rho_1-\rho_2)/\rho_1$. The inconsistent source factors in Eqs. (14.114) and (14.116) are corrected below.

Solution.

The coefficient -1 is correct for the cylindrical point-source normalization. On p. 684, Eq. (14.114) must have ρ_1^{-1}, not ρ_1^{-2}, multiplying the three coordinate delta functions. Equation (14.116) must have $\delta(\rho_1-\rho_2)/\rho_1$ on its right, not the unweighted delta function printed there. Indeed the cylindrical volume element is $\rho_1 d\rho_1 d\varphi_1 dz_1$, so a unit point source is $\delta(\rho_1-\rho_2)\delta(\varphi_1-\varphi_2)\delta(z_1-z_2)/\rho_1$. After the angular and axial expansions, put $r=\rho_1$, $b=\rho_2>0$. The correct radial equation is

$$\left[\frac{d^2}{dr^2}+\frac{1}{r}\frac{d}{dr}-\frac{m^2}{r^2}-k^2\right]g(r,b)=\frac{\delta(r-b)}{r},\qquad k>0.$$

Write a continuous candidate $g=AI_m(kr_<)K_m(kr_>)$. Its derivative jump is

$$\begin{aligned}g'(b^+)-g'(b^-)&=Ak[I_m(kb)K_m'(kb)-I_m'(kb)K_m(kb)]\\&=Ak\left(-\frac{1}{kb}\right)=-\frac{A}{b}.\end{aligned}$$

Multiplying the ODE by r and integrating across b requires $[rg']_{b^-}^{b^+}=1$, so $-A=1$. Hence

$$\boxed{g(r,b)=-I_m(kr_<)K_m(kr_>)}.$$

It is regular at the axis and decays at infinity. If one instead uses the literally printed unweighted radial delta, the jump must be one and the coefficient would be $A=-b$, not -1. The printed ρ_1^{-2} source would require $A=-1/b$.

The proof uses $k>0$. For integer $|m|>0$, its $k\downarrow 0$ limit is $-(r_</r_>)^{|m|}/(2|m|)$. For $m=0$ the limit has an additive logarithmic divergence; the zero-wave-number radial equation has no solution both regular at the axis and decaying at infinity. It must be handled as a limiting Fourier component, not by substituting $I_0(0)K_0(0)$ into the formula.

14.5.13 If $r=(x^2+y^2)^{1/2}$, prove that

$$\frac{1}{r}=\frac{2}{\pi}\int_0^\infty \cos(xt)K_0(yt)\,dt.$$

This is a Fourier cosine transform of K_0.

Take real x and $y>0$. For negative y, use $K_0(|y|t)$; $y=0$ is only a limit.

Solution.

For the ordinary real integral take $y>0$ and real x. If negative y is to be allowed, replace $K_0(yt)$ by $K_0(|y|t)$. The case $y=0$ is a limit of the transform, not a pointwise-defined integral of $K_0(0)$. Insert the convergent representation $K_0(yt)=\int_0^\infty e^{-yt\cosh u}du$:

$$\int_0^\infty \cos(xt)K_0(yt)dt=\int_0^\infty\int_0^\infty \cos(xt)e^{-yt\cosh u}\,dt\,du.$$

The absolute double integral is bounded by $\int_0^\infty du/(y\cosh u)=\pi/(2y)$, so changing the order is justified. The inner elementary Laplace integral is

$$\int_0^\infty e^{-yt\cosh u}\cos(xt)dt=\frac{y\cosh u}{y^2\cosh^2 u+x^2}.$$

Now put $v = \sinh u$, $dv = \cosh u\, du$:

$$\begin{aligned}\int_0^\infty \cos(xt)K_0(yt)dt &= y\int_0^\infty \frac{dv}{y^2v^2 + x^2 + y^2}\\ &= \frac{1}{\sqrt{x^2+y^2}}\left[\arctan\frac{yv}{\sqrt{x^2+y^2}}\right]_0^\infty\\ &= \frac{\pi}{2\sqrt{x^2+y^2}}.\end{aligned}$$

Multiplying by $2/\pi$ proves the formula. For $x \neq 0$, the limit $y \downarrow 0$ of this evaluated transform is $1/|x|$.

14.5.14 Derive the integral representation

$$I_n(x) = \frac{1}{\pi}\int_0^\pi e^{x\cos\theta}\cos(n\theta)\, d\theta,$$

where n is an integer.

Hint. Start with the corresponding integral representation of $J_n(x)$. Equation (14.112) is a special case of this representation.

Solution.

For integer n, take the coefficient-extraction contour from Exercise 14.5.3 to be the unit circle $t = e^{i\theta}$, $-\pi \le \theta \le \pi$. Since $dt/t = i\, d\theta$,

$$\begin{aligned}I_n(x) &= \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{x\cos\theta}e^{-in\theta}d\theta\\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{x\cos\theta}[\cos n\theta - i\sin n\theta]d\theta.\end{aligned}$$

The sine term is odd and integrates to zero; the cosine term is even. Therefore

$$\boxed{I_n(x) = \frac{1}{\pi}\int_0^\pi e^{x\cos\theta}\cos(n\theta)d\theta}.$$

This holds for negative integers too, because both $I_{-n} = I_n$ and $\cos(-n\theta) = \cos(n\theta)$. The published qualification that n is an integer is necessary: noninteger order has the additional branch-cut contribution, so an ordinary closed-circle coefficient extraction is unavailable.

14.5.15 Show that

$$K_0(z) = \int_0^\infty e^{-z\cosh t}\, dt$$

satisfies the modified Bessel equation. How can you establish that this form is linearly independent of $I_0(z)$?

Take $\Re z > 0$ for the integral and its differentiated forms.

Solution.

Take $\Re z > 0$, which ensures absolute convergence and allows two derivatives under the integral. With $F(z) = \int_0^\infty e^{-z\cosh t}dt$,

$$\begin{aligned}z^2F'' + zF' - z^2F &= \int_0^\infty [z^2\sinh^2 t - z\cosh t]e^{-z\cosh t}dt\\ &= -\int_0^\infty \frac{d}{dt}[z\sinh t\, e^{-z\cosh t}]dt = 0.\end{aligned}$$

At zero the bracket vanishes because $\sinh 0 = 0$; at infinity the exponential dominates the hyperbolic factor. Hence F solves the order-zero modified Bessel equation. Exercise 14.5.10 at $\nu = 0$ has prefactor one and identifies its normalization as $F = K_0$.

To prove independence from I_0, take positive real z and fix any $T > 0$. Positivity gives

$$F(z) \geq \int_0^T e^{-z\cosh t} dt \longrightarrow T \quad (z \downarrow 0).$$

Since T is arbitrary, $F(z) \to \infty$, whereas $I_0(z) = 1 + O(z^2)$. No constant multiple of I_0 can therefore equal F, so the two solutions are linearly independent.

14.5.16 The cylindrical cavity of Exercise 14.5.5 has along the cylinder walls the potential walls:

$$V(z) = \begin{cases} 100\,\dfrac{z}{h}, & 0 \leq \dfrac{z}{h} \leq 1/2, \\ 100\left(1 - \dfrac{z}{h}\right), & 1/2 \leq \dfrac{z}{h} \leq 1. \end{cases}$$

With the radius-height ratio $a/h = 0.5$, calculate the potential for $z/h = 0.1(0.1)0.5$ and $\rho/a = 0.0(0.2)1.0$.

Check value. For $z/h = 0.3$ and $\rho/a = 0.8$, $V = 26.396$.

Solution.

The boundary data are independent of φ, so only the $m = 0$ cosine modes of Exercise 14.5.5 survive. Put $r = \rho/a$, $s = z/h$. Write the wall data as $V(s) = \sum_{n\geq 1} B_n \sin n\pi s$, where

$$B_n = 2\int_0^1 V(s)\sin n\pi s\, ds.$$

Reflection $s \mapsto 1 - s$ leaves V unchanged and multiplies the sine by $(-1)^{n+1}$. Therefore even-n coefficients vanish. For odd n the two halves add, and integration by parts gives

$$\begin{aligned} B_n &= 400\int_0^{1/2} s\sin n\pi s\, ds \\ &= 400\left[-\frac{s\cos n\pi s}{n\pi} + \frac{\sin n\pi s}{n^2\pi^2}\right]_0^{1/2} = \frac{400\sin(n\pi/2)}{n^2\pi^2}. \end{aligned}$$

The final expression also gives zero for even n. Since $a/h = 1/2$, the potential is

$$\boxed{\Phi(r,s) = \frac{400}{\pi^2}\sum_{n=1}^{\infty} \frac{\sin(n\pi/2)\sin(n\pi s)}{n^2}\,\frac{I_0(n\pi r/2)}{I_0(n\pi/2)}}.$$

It vanishes on $s = 0, 1$, stays regular at $r = 0$, and at $r = 1$ reproduces the triangular wall potential, whose maximum is 50. The numerical results are

$\rho/a \backslash z/h$	0.1	0.2	0.3	0.4	0.5
0.0	7.118	13.657	19.006	22.551	23.798
0.2	7.262	13.954	19.461	23.138	24.437
0.4	7.682	14.836	20.843	24.958	26.441
0.6	8.345	16.261	23.180	28.212	30.130
0.8	9.165	18.088	26.396	33.252	36.364
1.0	10.000	20.000	30.000	40.000	50.000

The wall row uses the prescribed boundary values. The interior entries sum the modes through $n = 160$; comparison with $n = 100$ changes them by less than 10^{-14}. The series is evaluated through the ratio of the two I_0 values, which prevents a large coefficient and a small normalizing factor from being stored separately. The requested check is $\Phi(0.8, 0.3) = 26.3955684295\ldots \simeq 26.396$, and the symmetry $\Phi(r, 1-s) = \Phi(r, s)$ determines the other half of the cylinder.

14.6 Asymptotic Expansions

14.6.1 Determine the asymptotic dependence of the modified Bessel functions $I_\nu(x)$, given

$$I_\nu(x) = \frac{1}{2\pi i}\int_C e^{(x/2)(t+1/t)}\frac{dt}{t^{\nu+1}}.$$

The contour starts and ends at $t = -\infty$, encircling the origin in a positive sense. There are two saddle points. Only the one at $z = +1$ contributes significantly to the asymptotic form.

Find the leading form as $x \to +\infty$ with ν fixed; use the open contour and branch of Exercise 14.5.3.

Solution.

Take $x \to +\infty$ with order ν fixed. The rapidly varying exponent is $w(t) = x(t + t^{-1})/2$, with

$$w'(t) = \frac{x}{2}(1 - t^{-2}), \qquad w''(t) = xt^{-3}.$$

Thus the saddles are $t = \pm 1$, not $\pm x$. The manual, p. 219, prints $t = \pm x$ in identifying the saddles; the differentiated exponent gives $t = \pm 1$. Their exponential factors are e^x and e^{-x}, so the positive saddle supplies the dominant positive-real-axis asymptotic form. Near it, put $t = 1 + iu$ along the upward local descent direction:

$$w(1 + iu) = x - \frac{xu^2}{2} + O(xu^3), \quad t^{-\nu-1} = 1 + O(u), \quad dt = i\,du.$$

The contributing width is $u = O(x^{-1/2})$. Keeping the leading terms and extending the local Gaussian integral to the real line,

$$\begin{aligned} I_\nu(x) &\sim \frac{1}{2\pi i}e^x i\int_{-\infty}^{\infty} e^{-xu^2/2}du \\ &= \frac{e^x}{2\pi}\sqrt{\frac{2\pi}{x}} = \boxed{\frac{e^x}{\sqrt{2\pi x}}}. \end{aligned}$$

The next relative terms are powers of $1/x$ for fixed order; this leading approximation is not uniform when ν grows with x. The branch-cut contour is understood as in Exercise 14.5.3.

14.6.2 Determine the asymptotic dependence of the modified Bessel function of the second kind, $K_\nu(x)$, by using

$$K_\nu(x) = \frac{1}{2}\int_0^\infty e^{(-x/2)(s+1/s)}\frac{ds}{s^{1-\nu}}.$$

Find the leading form as $x \to +\infty$ with ν fixed.

Solution.

For $x \to +\infty$ and fixed ν, write the given integral as

$$K_\nu(x) = \frac{1}{2}\int_0^\infty e^{w(s)} s^{\nu-1} ds, \qquad w(s) = -\frac{x}{2}(s + s^{-1}).$$

Since $s + s^{-1} \geq 2$ for positive s, its unique real maximum is at $s = 1$. In detail,

$$w'(s) = -\frac{x}{2}(1 - s^{-2}), \quad w''(1) = -x, \quad w(1) = -x.$$

Put $s = 1 + u$. Then $w = -x - xu^2/2 + O(xu^3)$ and $s^{\nu-1} = 1 + O(u)$. Only a neighborhood of width $x^{-1/2}$ contributes at leading order, so the endpoints can be extended to the full Gaussian line in the local variable:

$$K_\nu(x) \sim \frac{1}{2}e^{-x}\int_{-\infty}^\infty e^{-xu^2/2} du = \frac{1}{2}e^{-x}\sqrt{\frac{2\pi}{x}}$$
$$= \boxed{\sqrt{\frac{\pi}{2x}}e^{-x}}.$$

Both original endpoints are exponentially suppressed; no contour deformation is needed for positive real x.

14.6.3 Verify that the integral representations

$$I_n(z) = \frac{1}{\pi}\int_0^\pi e^{z\cos t}\cos(nt)\, dt,$$
$$K_\nu(z) = \int_0^\infty e^{-z\cosh t}\cosh(\nu t)\, dt, \qquad \Re(z) > 0,$$

satisfy the modified Bessel equation by direct substitution into that equation (the index n is an integer). How can you check the normalization?

Solution.

For the I integral put $f(t) = e^{z\cos t}$ and $c(t) = \cos nt$. Direct differentiation gives

$$f'' = (-z\cos t + z^2\sin^2 t)f, \qquad c'' = -n^2 c.$$

Here primes on f, c denote t derivatives. Applying the modified Bessel operator with respect to z under the integral therefore gives

$$(z^2\partial_z^2 + z\partial_z - z^2 - n^2)I_n = \frac{1}{\pi}\int_0^\pi [-f''c + fc'']dt$$
$$= \frac{1}{\pi}[fc' - f'c]_0^\pi = 0.$$

At both endpoints $f' = -z\sin t\, f = 0$, and $c' = -n\sin nt = 0$ because n is an integer. This explains the published integer qualification; it is not needed merely for convergence, but for the boundary terms.

For the K integral use $f(t) = e^{-z\cosh t}$, $c(t) = \cosh \nu t$. Now $f'' = (-z\cosh t + z^2 \sinh^2 t)f$ and $c'' = \nu^2 c$, so

$$(z^2\partial_z^2 + z\partial_z - z^2 - \nu^2)K_\nu = \int_0^\infty [f''c - fc'']dt$$
$$= [f'c - fc']_0^\infty = 0.$$

At zero both first derivatives vanish. At infinity, $\Re z > 0$ gives double-exponential damping, stronger than the fixed-order factors $\cosh \nu t$ and $\sinh \nu t$.

To check normalization, for integer $n \geq 1$ the first nonzero term of the I integral comes from $z^n \cos^n t/n!$. The coefficient of $\cos nt$ in $\cos^n t$ is 2^{1-n}, and $\int_0^\pi \cos^2 nt\, dt = \pi/2$. Thus the leading term is $z^n/(2^n n!)$, exactly the defining I_n coefficient. For $n = 0$ its value at zero is one; negative orders agree by $I_{-n} = I_n$. For K, the dominant large-positive-z region is $t = 0$, where $\cosh t = 1 + t^2/2 + \cdots$ and $\cosh \nu t = 1 + \cdots$. Hence

$$K_\nu(z) \sim e^{-z} \int_0^\infty e^{-zt^2/2} dt = \sqrt{\frac{\pi}{2z}} e^{-z},$$

which fixes the standard decaying normalization.

14.6.4

(a) Show that when K_ν is defined by Eq. (14.128),

$$\frac{dK_0(z)}{dz} = -K_1(z).$$

(b) Show that the indefinite integral of $-K_1(x)$ as defined by Eq. (14.128) has in the limit of small z the value $-\ln z + C$, and therefore, by comparison with Eq. (14.110), that K_0 as defined by Eq. (14.128) has the correct normalization.

To identify the full K_0 solution, also determine the constant C or require decay at infinity; the logarithmic coefficient alone permits addition of I_0.

Solution.

(a) The two instances of Eq. (14.128) are

$$K_0(z) = \int_1^\infty \frac{e^{-zp}}{\sqrt{p^2-1}} dp, \qquad K_1(z) = z\int_1^\infty e^{-zp}\sqrt{p^2-1}\, dp.$$

For $\Re z > 0$, differentiate the first and use $d\sqrt{p^2-1}/dp = p/\sqrt{p^2-1}$:

$$K_0'(z) = -\int_1^\infty e^{-zp}\, d\sqrt{p^2-1}$$
$$= -[e^{-zp}\sqrt{p^2-1}]_1^\infty - z\int_1^\infty e^{-zp}\sqrt{p^2-1} dp = \boxed{-K_1(z)}.$$

The boundary bracket vanishes at both ends.

(b) Set $u = zp$ in the K_1 integral for positive z:

$$K_1(z) = \frac{1}{z}\int_z^\infty e^{-u}\sqrt{u^2-z^2}\, du.$$

As $z \downarrow 0$, the integrand is bounded by ue^{-u} and tends to it; its integral tends to $\Gamma(2) = 1$. Thus $K_1(z) \sim 1/z$ and an antiderivative of $-K_1$ has leading logarithmic term $-\ln z$. That leading coefficient fixes the scale of the singular solution but does not by itself exclude adding a multiple of I_0. To identify the full normalization, fix the constant or the decaying behavior as well. The constant can be found directly from the K_0 integral:

$$K_0(z) = \int_1^\infty \frac{e^{-zp}}{p}\,dp + \int_1^\infty e^{-zp}\left(\frac{1}{\sqrt{p^2-1}} - \frac{1}{p}\right)dp.$$

The first integral is $E_1(z) = -\ln z - \gamma + o(1)$. The second tends, by absolute integrability, to

$$\int_1^\infty \left(\frac{1}{\sqrt{p^2-1}} - \frac{1}{p}\right)dp = \lim_{R\to\infty}(\operatorname{arcosh} R - \ln R) = \ln 2.$$

Therefore

$$\boxed{K_0(z) = -\ln z + \ln 2 - \gamma + o(1)},$$

which reproduces both the logarithmic scale and the constant in Eq. (14.110).

14.6.5 Verify that Eq. (14.132) can be rearranged to the form given as Eq. (14.133).

Solution.

The coefficient of $(2z)^{-r}/r!$ in Eq. (14.132) is

$$\frac{\Gamma(\nu + r + 1/2)}{\Gamma(\nu - r + 1/2)}.$$

For $r \geq 1$, apply the gamma recurrence $2r$ times and pair factors equidistant from ν:

$$\begin{aligned}\frac{\Gamma(\nu + r + 1/2)}{\Gamma(\nu - r + 1/2)} &= \prod_{j=1}^{r}\left(\nu + j - \tfrac{1}{2}\right)\left(\nu - j + \tfrac{1}{2}\right)\\ &= 4^{-r}\prod_{j=1}^{r}[4\nu^2 - (2j-1)^2].\end{aligned}$$

Since $4^r(2z)^r = (8z)^r$, the expansion is

$$\boxed{K_\nu(z) \sim \sqrt{\frac{\pi}{2z}}e^{-z}\sum_{r=0}^{\infty}\frac{\prod_{j=1}^{r}[4\nu^2 - (2j-1)^2]}{r!(8z)^r}}.$$

The empty product at $r = 0$ is one. Its first three terms are

$$1 + \frac{4\nu^2 - 1}{8z} + \frac{(4\nu^2 - 1)(4\nu^2 - 9)}{2!(8z)^2} + \cdots,$$

as in Eq. (14.133). At exceptional half-integer orders the product form also makes the terminating factors explicit. In general this is an asymptotic series, not a convergent infinite sum at a fixed z.

14.6.6

(a) Show that

$$y(z) = z^\nu \int e^{-zt}(t^2-1)^{\nu-1/2}\,dt$$

satisfies the modified Bessel equation, provided the contour is chosen so that

$$e^{-zt}(t^2-1)^{\nu+1/2}$$

has the same value at the initial and final points of the contour.

(b) Verify that the contours shown in Fig. 14.14 are suitable for this problem.

Require convergence of the integral and its differentiated forms. For the two endpoint contours in (b), take $\Re\nu > -1/2$; the unbounded contour additionally needs $\Re z > 0$.

Solution.

(a) Put $F(z) = \int_C e^{-zt}(t^2-1)^{\nu-1/2}dt$, so $y = z^\nu F$. The product rule gives

$$y' = \nu z^{\nu-1}F + z^\nu F', \qquad y'' = \nu(\nu-1)z^{\nu-2}F + 2\nu z^{\nu-1}F' + z^\nu F''.$$

Substitution cancels the terms proportional to $\nu^2 F$:

$$\begin{aligned} z^2y'' + zy' - (z^2+\nu^2)y &= z^{\nu+2}(F''-F) + (2\nu+1)z^{\nu+1}F' \\ &= z^{\nu+1}\int_C e^{-zt}(t^2-1)^{\nu-1/2}[z(t^2-1) - (2\nu+1)t]dt \\ &= -z^{\nu+1}\int_C \frac{d}{dt}[e^{-zt}(t^2-1)^{\nu+1/2}]dt. \end{aligned}$$

Thus matching the two endpoint values of the bracket makes the ODE residual zero. Differentiation under the integral requires a fixed contour with convergent derivatives, as in the examples below; a formal endpoint condition alone does not ensure the integral exists.

(b) In Fig. 14.14, contour (1) joins $t = 1$ and positive-real infinity, and contour (2) joins $t = -1$ and $t = 1$ on a fixed bank of the cut. Their orientations change the solution's sign but not the ODE test. For these endpoint contours take $\Re\nu > -1/2$. The unbounded contour (1) also requires $\Re z > 0$ as drawn; without these conditions the stated suitability is not automatic. Under these assumptions, at $t = \pm 1$ the bracket vanishes like $(t \mp 1)^{\nu+1/2}$, and the integrand is locally integrable like $(t \mp 1)^{\nu-1/2}$. On contour (1), exponential decay makes the bracket and the differentiated integrands vanish at infinity. On contour (2), both endpoint bracket values are zero, with a consistent branch of $(t^2-1)^{\nu-1/2}$ along the upper arc. Hence both contours satisfy the required conditions.

14.6.7 Use the asymptotic expansions to verify the following Wronskian formulas:

(a) $J_\nu(x)J_{-\nu-1}(x) + J_{-\nu}(x)J_{\nu+1}(x) = -2\sin\nu\pi/\pi x$,

(b) $J_\nu(x)Y_{\nu+1}(x) - J_{\nu+1}(x)Y_\nu(x) = -2/\pi x$,

(c) $J_\nu(x)H^{(2)}_{\nu-1}(x) - J_{\nu-1}(x)H^{(2)}_\nu(x) = 2/i\pi x$,

(d) $I_\nu(x)K'_\nu(x) - I'_\nu(x)K_\nu(x) = -1/x$,

(e) $I_\nu(x)K_{\nu+1}(x) + I_{\nu+1}(x)K_\nu(x) = 1/x$.

Solution.

For fixed real order and $x \to +\infty$, put $A = \sqrt{2/(\pi x)}$, $\chi = x - \nu\pi/2 - \pi/4$. Then

$$J_\nu \sim A\cos\chi, \quad Y_\nu \sim A\sin\chi, \quad H_\nu^{(2)} \sim Ae^{-i\chi},$$

$$I_\nu \sim \frac{e^x}{\sqrt{2\pi x}}, \qquad K_\nu \sim \sqrt{\frac{\pi}{2x}}e^{-x}, \qquad I_\nu' \sim I_\nu, \quad K_\nu' \sim -K_\nu.$$

To exhibit the Wronskians explicitly, use $J_{-\nu-1} = J_{-\nu}' - \nu J_{-\nu}/x$, $J_{\nu+1} = \nu J_\nu/x - J_\nu'$, $Y_{\nu+1} = \nu Y_\nu/x - Y_\nu'$, and $H_{\nu-1}^{(2)} = H_\nu^{(2)\prime} + \nu H_\nu^{(2)}/x$. The ν/x products cancel in each pair, leaving

$$\text{(a) } W[J_\nu, J_{-\nu}], \qquad \text{(b) } -W[J_\nu, Y_\nu], \qquad \text{(c) } W[J_\nu, H_\nu^{(2)}].$$

Part (d) is already a Wronskian and (e) is its negative. Abel's identity makes each exactly a constant divided by x. Thus computing its leading coefficient at infinity determines the exact identity, not just an approximation.

(a) The phases for orders $-\nu$, $-\nu-1$, and $\nu+1$ are $\chi+\nu\pi$, $\chi+\nu\pi+\pi/2$, and $\chi-\pi/2$. Consequently

$$\begin{aligned} J_\nu J_{-\nu-1} + J_{-\nu}J_{\nu+1} &\sim A^2[-\cos\chi\sin(\chi+\nu\pi) + \cos(\chi+\nu\pi)\sin\chi] \\ &= -A^2\sin\nu\pi = \boxed{-\frac{2\sin\nu\pi}{\pi x}}. \end{aligned}$$

(b) Since $Y_{\nu+1} \sim -A\cos\chi$ and $J_{\nu+1} \sim A\sin\chi$,

$$J_\nu Y_{\nu+1} - J_{\nu+1}Y_\nu \sim -A^2(\cos^2\chi + \sin^2\chi) = \boxed{-\frac{2}{\pi x}}.$$

The Y notation here is the published correction to N.

(c) The order-$\nu-1$ phase is $\chi+\pi/2$, so

$$\begin{aligned} J_\nu H_{\nu-1}^{(2)} - J_{\nu-1}H_\nu^{(2)} &\sim A^2e^{-i\chi}[-i\cos\chi + \sin\chi] \\ &= -iA^2 = \boxed{\frac{2}{i\pi x}}. \end{aligned}$$

(d) The product of the two leading modified functions is $I_\nu K_\nu \sim 1/(2x)$. Therefore

$$I_\nu K_\nu' - I_\nu' K_\nu \sim -2I_\nu K_\nu = \boxed{-\frac{1}{x}}.$$

(e) Changing the fixed order by one does not alter these leading factors, giving $I_\nu K_{\nu+1} + I_{\nu+1}K_\nu \sim 1/x$. To confirm its exact Wronskian status, substitute $K_{\nu+1} = -K_\nu' + \nu K_\nu/x$ and $I_{\nu+1} = I_\nu' - \nu I_\nu/x$. The order terms cancel, leaving $-W[I_\nu, K_\nu]$. Thus it is exactly $\boxed{1/x}$.

14.6.8 Verify that the Green's function for the 2-D Helmholtz equation (operator $\nabla^2 + k^2$) with outgoing-wave boundary conditions is

$$G(\boldsymbol{\rho}_1, \boldsymbol{\rho}_2) = -\frac{i}{4} H_0^{(1)}(k|\boldsymbol{\rho}_1 - \boldsymbol{\rho}_2|).$$

Hint. $H_0^{(1)}(k\rho)$ is known to be an outgoing-wave solution to the homogeneous Helmholtz equation. Use $k > 0$, time dependence $e^{-i\omega t}$, and $(\nabla^2 + k^2)G = +\delta^{(2)}$. The displayed minus sign uses this positive-source convention.

Solution.

The originally printed $+iH_0^{(1)}/4$ has source $-\delta^{(2)}$. With the book's Chapter 10 convention $LG = +\delta$, the coefficient must instead be $-i/4$; the manual, p. 223, derives this minus sign. Both conventions are common, so the source sign must be stated explicitly. Let $r = |\boldsymbol{\rho}_1 - \boldsymbol{\rho}_2|$, $k > 0$, and $G = CH_0^{(1)}(kr)$. For $r > 0$, the radial Helmholtz equation is Bessel's equation, so $(\nabla^2 + k^2)G = 0$ there. Near the source,

$$H_0^{(1)}(kr) = 1 + \frac{2i}{\pi}[\ln(kr/2) + \gamma] + O(r^2 \ln r), \quad G'(r) = \frac{2iC}{\pi r} + O(r \ln r).$$

Integrate the operator over a disk of radius ϵ and use the divergence theorem:

$$\int_{r<\epsilon} (\nabla^2 + k^2)G \, d^2r = 2\pi\epsilon G'(\epsilon) + 2\pi k^2 \int_0^\epsilon G(r) r \, dr$$
$$\longrightarrow 4iC.$$

The last integral tends to zero as $O(\epsilon^2|\ln\epsilon|)$. A unit positive delta source therefore requires $4iC = 1$:

$$\boxed{G_+(\boldsymbol{\rho}_1, \boldsymbol{\rho}_2) = -\frac{i}{4} H_0^{(1)}(kr), \qquad (\nabla^2 + k^2)G_+ = \delta^{(2)}(\boldsymbol{\rho}_1 - \boldsymbol{\rho}_2)}.$$

Changing the sign gives the expression originally printed in the question and a negative unit source. Finally the large-r form $H_0^{(1)}(kr) \sim \sqrt{2/(\pi kr)}e^{i(kr-\pi/4)}$ satisfies $(\partial_r - ik)G = O(r^{-3/2})$, the outgoing condition for time dependence $e^{-i\omega t}$.

14.6.9 From the asymptotic form of $K_\nu(z)$, Eq. (14.134), derive the asymptotic form of $H_\nu^{(1)}(z)$, Eq. (14.138). Note particularly the phase, $(\nu + \frac{1}{2})\pi/2$.

Solution.

The definition gives $H_\nu^{(1)}(z) = (2/\pi)i^{-\nu-1}K_\nu(-iz)$. Take $z > 0$ first and rotate clockwise to $-iz = ze^{-i\pi/2}$, fixing the square-root and power branches. Equation (14.134) then gives

$$K_\nu(-iz) \sim \sqrt{\frac{\pi}{2(-iz)}} e^{iz}[P_\nu(z) + iQ_\nu(z)],$$

because $i(-iz) = z$. On this branch $(-iz)^{-1/2} = z^{-1/2}e^{i\pi/4}$. Combining the prefactors,

$$\frac{2}{\pi} i^{-\nu-1} \sqrt{\frac{\pi}{2z}} e^{i\pi/4} = \sqrt{\frac{2}{\pi z}} e^{-i(\nu+1)\pi/2 + i\pi/4} = \sqrt{\frac{2}{\pi z}} e^{-i(\nu+1/2)\pi/2}.$$

Hence

$$\boxed{H_\nu^{(1)}(z) \sim \sqrt{\frac{2}{\pi z}} e^{i[z-(\nu+1/2)\pi/2]}[P_\nu(z) + iQ_\nu(z)]}.$$

The additional $\pi/4$ is the phase of the rotated square root; omitting it would give an incorrect outgoing-wave phase. Extension to complex z uses this same continuation: the sector $-3\pi/2 < \arg(-iz) < 3\pi/2$ becomes $-\pi < \arg z < 2\pi$. The asymptotic estimates hold on closed subsectors away from their boundary rays, with order fixed.

14.6.10 Apply Stokes' method for obtaining an asymptotic expansion for the Hankel function $H_\nu^{(1)}$ as follows:

(a) Replace the Bessel function in Bessel's equation by $x^{-1/2}y(x)$ and show that $y(x)$ satisfies

$$y''(x) + \left(1 - \frac{\nu^2 - \frac{1}{4}}{x^2}\right) y(x) = 0.$$

(b) Develop a power-series solution with negative powers of x starting with the assumed form

$$y(x) \sim e^{ix} \sum_{n=0}^{\infty} a_n x^{-n}.$$

Obtain the recurrence relation giving a_{n+1} in terms of a_n. Check your result against the asymptotic series, Eq. (14.138).

(c) From Eq. (14.125), determine the initial coefficient, a_0.

The inverse-power ansatz in (b) is an asymptotic expansion at positive infinity for fixed ν, rather than a generally convergent series.

Solution.

(a) Put $F = x^{-1/2}y$. Its derivatives are

$$F' = x^{-1/2}y' - \tfrac{1}{2}x^{-3/2}y, \qquad F'' = x^{-1/2}y'' - x^{-3/2}y' + \tfrac{3}{4}x^{-5/2}y.$$

Substituting in $x^2F'' + xF' + (x^2 - \nu^2)F = 0$ cancels the y' terms and gives

$$x^{3/2}y'' + x^{3/2}y + (1/4 - \nu^2)x^{-1/2}y = 0.$$

After division by $x^{3/2}$, $\boxed{y'' + [1 - (\nu^2 - 1/4)/x^2]y = 0}$.

(b) Write $y = e^{ix}h$, $h \sim \sum_{n\geq 0} a_n x^{-n}$, and put $\mu = \nu^2 - 1/4$. Since $y'' + y = e^{ix}(h'' + 2ih')$, the equation becomes

$$h'' + 2ih' - \mu x^{-2}h = 0.$$

The coefficient of x^{-n-2} is

$$[n(n+1) - \mu]a_n - 2i(n+1)a_{n+1} = 0.$$

Thus

$$\boxed{a_{n+1} = \frac{i[4\nu^2 - (2n+1)^2]}{8(n+1)} a_n}, \qquad a_n = a_0 \frac{i^n}{8^n n!} \prod_{j=1}^{n} [4\nu^2 - (2j-1)^2].$$

Even $n = 2r$ have factor $i^{2r} = (-1)^r$ and build P_ν; odd $n = 2r+1$ have $i^{2r+1} = i(-1)^r$ and build iQ_ν. This reproduces Eq. (14.138) term by term.

(c) The leading Hankel form is $H_\nu^{(1)}(x) \sim \sqrt{2/(\pi x)}e^{i(x - \nu\pi/2 - \pi/4)}$. Since $H = x^{-1/2}y$,

$$\boxed{a_0 = \sqrt{\frac{2}{\pi}} e^{-i(\nu+1/2)\pi/2}}.$$

The formal inverse-power series is generally asymptotic; for half-integer order a factor in the recurrence can terminate it.

14.6.11 Using the method of steepest descents, evaluate the second Hankel function given by

$$H_\nu^{(2)}(t) = \frac{1}{\pi i}\int_{C_2} e^{(t/2)(z-1/z)}\frac{dz}{z^{\nu+1}},$$

with contour C_2 as shown in Fig. 14.10.

ANS. $H_\nu^{(2)}(t) \approx \sqrt{\dfrac{2}{\pi t}}\, e^{-i(t-\pi/4-\nu\pi/2)}$.

Take $t \to +\infty$ with order ν fixed.

Solution.

Take positive $t \to \infty$, with ν fixed. Here $w(z) = t(z - z^{-1})/2$, $g(z) = z^{-\nu-1}$, and

$$w'(z) = \frac{t}{2}(1 + z^{-2}), \qquad w''(z) = -tz^{-3}.$$

The lower contour goes through $z_0 = -i$, where $w(z_0) = -it$ and $w''(z_0) = it$. Its oriented descent direction is $e^{i\pi/4}$: writing $z = -i + e^{i\pi/4}u$ gives

$$w(z) = -it - \frac{tu^2}{2} + O(tu^3), \qquad dz = e^{i\pi/4}du.$$

On the lower branch $\arg(-i) = -\pi/2$, so $g(-i) = e^{i(\nu+1)\pi/2}$. The Gaussian approximation is

$$\begin{aligned} H_\nu^{(2)}(t) &\sim \frac{1}{\pi i}e^{i(\nu+1)\pi/2}e^{-it}e^{i\pi/4}\sqrt{\frac{2\pi}{t}} \\ &= \boxed{\sqrt{\frac{2}{\pi t}}e^{-i(t-\nu\pi/2-\pi/4)}}. \end{aligned}$$

The factor $1/i = e^{-i\pi/2}$ was included in the last phase simplification. This is the incoming-wave counterpart of the upper-contour asymptotic expression.

14.6.12

(a) In applying the method of steepest descents to the Hankel function $H_\nu^{(1)}(t)$, show that $w(z,t)$, which appears in Eq. (14.121), satisfies

$$\Re[w(z,t)] < \Re[w(z_0,t)] = 0$$

for z on the contour C_1 (Fig. 14.10) but away from the point $z = z_0 = i$.

(b) For general values of $z = re^{i\theta}$, show that

$$\Re[w(z,t)] > 0 \quad \text{for } 0 < r < 1, \quad \begin{cases} \pi/2 < \theta \le \pi \\ -\pi \le \theta < -\pi/2 \end{cases}$$

and

$$\Re[w(z,t)] > 0 \quad \text{for } r > 1, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}.$$

Your demonstration verifies that the distribution of the sign of w is as shown schematically in Fig. 14.15.

(c) Explain why the contour C_1 (Fig. 14.10) cannot be deformed to go through both saddle points, and why it may not go through the saddle point at $-i$ if it is to end at $z = -\infty$ with argument $+\pi$.

Take $t > 0$. In (a) and (c) use an admissible descending representative of C_1; the claimed restriction concerns steepest descent, not arbitrary analytic detours through regions of exponential growth.

Solution.

For $t > 0$ and $z = re^{i\theta}$, direct separation into real and imaginary parts gives

$$\boxed{\Re w = \frac{t}{2}(r - r^{-1})\cos\theta, \qquad \Im w = \frac{t}{2}(r + r^{-1})\sin\theta}.$$

(a) An explicit upper descent contour is

$$r = \tan(\theta/2), \qquad 0 < \theta < \pi.$$

It starts at 0^+, passes through i, and ends at infinity on the upper bank of the negative axis. Since $r + r^{-1} = 2/\sin\theta$ and $r - r^{-1} = -2\cot\theta$,

$$\Im w = t, \qquad \Re w = -\frac{t\cos^2\theta}{\sin\theta} \le 0.$$

Equality occurs only at $\theta = \pi/2$, namely $z = i$. Thus the saddle is the unique maximum of the exponential modulus on this admissible representative of C_1.

(b) If $r < 1$, the factor $r - r^{-1}$ is negative. Therefore $\Re w > 0$ precisely in the left half-plane, $\pi/2 < \theta \le \pi$ or $-\pi \le \theta < -\pi/2$. If $r > 1$, that factor is positive, so $\Re w > 0$ in the right half-plane $-\pi/2 < \theta < \pi/2$. The original second inequality printed $\Re w < 0$; it is corrected to $\Re w > 0$ in the statement. The lower angular endpoint in the first pair must also be $-\pi/2$, as marked in the statement. Both corrected signs agree with Fig. 14.15. The remaining open regions have $\Re w < 0$; the unit circle and imaginary axis have $\Re w = 0$.

(c) A descending path leaving 0^+ lies in the right half of the unit disk, where $\Re w < 0$. Its passage through $+i$ connects to the upper part of the exterior left half-plane, which reaches the upper negative-real endpoint. Passage through $-i$ instead connects to the lower exterior left half-plane and the lower bank of the cut. To move from that lower branch to the upper endpoint one must cross the cut, or leave the decaying regions through a region with $\Re w > 0$. A single steepest path also has constant $\Im w$, whereas the two saddles have $\Im w = +t$ and $-t$. The claimed impossibility is about an admissible descent contour for this saddle approximation. An arbitrary analytic contour can be deformed to detour through both saddles while entering growth regions; Cauchy's theorem does not forbid that. Such a detour cannot be evaluated by retaining only independent decaying Gaussian neighborhoods of the two saddles.

14.6.13 Calculate the first 15 partial sums of $P_0(x)$ and $Q_0(x)$, Eqs. (14.135) and (14.136). Let x vary from 4 to 10 in unit steps. Determine the number of terms to be retained for maximum accuracy and the accuracy achieved as a function of x. Specifically, how small may x be without raising the error above 3×10^{-6}?

ANS. $x_{\min} = 6$.

Measure absolute error separately in P_0 and Q_0 on the stated integer grid; the quoted minimum is a grid minimum.

Solution.

For order zero define the exact coefficients recursively by

$$a_0 = 1, \qquad a_{j+1} = -\frac{(2j+1)^2}{8(j+1)} a_j.$$

The N-term partial sums requested are

$$P_N(x) = \sum_{r=0}^{N-1} \frac{(-1)^r a_{2r}}{x^{2r}}, \qquad Q_N(x) = \sum_{r=0}^{N-1} \frac{(-1)^r a_{2r+1}}{x^{2r+1}}.$$

For example $P_2 = 1 - 9/(128x^2)$ and $Q_2 = -1/(8x) + 225/(3072x^3)$. The first fifteen P_N are:

$N\backslash x$	4	5	6	7	8	9	10
1	1.0	1.0	1.0	1.0	1.0	1.0	1.0
2	0.99560546875	0.9971875	0.998046875	0.99856505102	0.99890136719	0.99913194444	0.999296875
3	0.99604356289	0.99736694336	0.99813341211	0.9986117616	0.99892874807	0.9991490382	0.99930809021
4	0.99590379203	0.99733030327	0.99812114142	0.99860689542	0.99892656415	0.99914796093	0.99930751771
5	0.99599647456	0.99734585282	0.99812475775	0.99860794906	0.99892692619	0.99914810204	0.99930757845
6	0.99589155404	0.99733458706	0.99812293827	0.99860755959	0.99892682373	0.99914807048	0.99930756745
7	0.99607263834	0.99734703108	0.99812433395	0.99860777908	0.99892686794	0.99914808124	0.99930757049
8	0.99562993071	0.99732756059	0.99812281746	0.99860760386	0.99892684092	0.99914807605	0.9993075693
9	0.99708580943	0.99736853993	0.99812503394	0.99860779201	0.99892686314	0.99914807942	0.99930756992
10	0.99088757806	0.99725688252	0.99812083998	0.99860753045	0.99892683949	0.99914807658	0.9993075695
11	1.0240553954	0.99763928142	0.99813081449	0.99860798749	0.99892687112	0.99914807958	0.99930756986
12	0.8061441789	0.99603138045	0.99810168909	0.998607007	0.99892681917	0.99914807569	0.99930756948
13	2.5306366164	1.0041750657	0.99820412944	0.99860954066	0.99892692196	0.99914808177	0.99930756996
14	-13.649436996	0.95527379385	0.99777695255	0.99860177835	0.99892668085	0.9991480705	0.99930756923
15	163.94765864	1.2987965369	0.99986087045	0.9986295992	0.99892734245	0.99914809495	0.99930757051

The first fifteen Q_N are:

$N\backslash x$	4	5	6	7	8	9	10
1	-0.03125	-0.025	-0.020833333333	-0.017857142857	-0.015625	-0.013888888889	-0.0125
2	-0.03010559082	-0.0244140625	-0.020494249132	-0.017643608783	-0.015481948853	-0.013788419496	-0.012426757812
3	-0.030327375978	-0.024486737061	-0.020523455408	-0.017657121486	-0.015488879639	-0.01379226559	-0.012429028893
4	-0.03022192386	-0.024464622149	-0.020517283542	-0.017655023566	-0.015488055794	-0.013791904365	-0.01242885612
5	-0.0303149282	-0.02447710498	-0.020519702798	-0.017655627738	-0.015488237443	-0.013791967295	-0.0124288805
6	-0.030183479479	-0.024465813621	-0.020518183117	-0.017655348909	-0.015488173259	-0.013791949726	-0.012428874987
7	-0.030455541237	-0.024480770374	-0.020519581033	-0.017655537349	-0.01548820647	-0.013791956909	-0.012428876813
8	-0.029679880565	-0.02445347924	-0.020517809689	-0.01765536192	-0.015488182799	-0.013791952864	-0.01242887598
9	-0.032594314255	-0.024519106453	-0.02052076772	-0.017655577152	-0.015488205034	-0.013791955866	-0.012428876481
10	-0.018638099211	-0.024317976197	-0.02051447217	-0.017655240606	-0.015488178415	-0.013791953026	-0.012428876097
11	-0.10160699937	-0.025083229231	-0.020531106254	-0.017655893912	-0.015488217977	-0.013791956361	-0.012428876462
12	0.49794492096	-0.021544099364	-0.020477683307	-0.017654352383	-0.015488146505	-0.013791951601	-0.01242887604
13	-4.6776880071	-0.041097087738	-0.020682649385	-0.017658697607	-0.015488300751	-0.013791959718	-0.012428876623
14	47.926278172	0.086091498118	-0.019756769828	-0.017644276747	-0.015487908821	-0.013791943422	-0.012428875675
15	-573.85493274	-0.87606832271	-0.024620742001	-0.017699935591	-0.015489066979	-0.013791981471	-0.012428877467

To measure accuracy, recover the exact amplitude functions from the two Bessel functions, with $\chi = x - \pi/4$:

$$P(x) = \sqrt{\frac{\pi x}{2}}[J_0(x)\cos\chi + Y_0(x)\sin\chi],$$

$$Q(x) = \sqrt{\frac{\pi x}{2}}[Y_0(x)\cos\chi - J_0(x)\sin\chi].$$

These formulas invert the J_0, Y_0 amplitude-phase formulas and provide references independent of truncating the asymptotic series. Minimizing the absolute error separately among the fifteen sums gives:

x	best N_P	$\lvert P_{N_P} - P\rvert$	best N_Q	$\lvert Q_{N_Q} - Q\rvert$	relative Q error
4	5	4.40308e-05	4	4.36604e-05	1.44258e-03
5	6	5.39862e-06	5	5.36775e-06	2.19345e-04
6	7	6.72951e-07	6	6.70164e-07	3.26609e-05
7	8	8.48751e-08	7	8.46089e-08	4.79223e-06
8	9	1.07992e-08	8	1.07726e-08	6.95539e-07
9	10	1.38345e-09	9	1.38072e-09	1.00110e-07
10	11	1.78200e-10	10	1.77910e-10	1.43143e-08

The optimum is $N_P = x + 1$, $N_Q = x$ for these seven integer arguments. Extra terms eventually make the errors larger: the coefficient ratio grows quadratically in the order before division by the index, so at fixed x the term magnitudes eventually grow. The least-term region is near coefficient index $j \simeq 2x$, which explains the observed truncation counts.

The supplied answer $x_{\min} = 6$ uses absolute errors in both P and Q, and the integer grid specified in the question. If "error" means relative error in both amplitude functions, the corresponding minimum on this grid is 8, because Q is small. No threshold for all real x between grid points is established by the seven sampled arguments alone. Under the absolute-error interpretation, at $x = 5$ both optimal errors exceed 5.3×10^{-6}, whereas at $x = 6$ both are below 6.8×10^{-7}. Thus the requested grid threshold is $\boxed{x_{\min} = 6}$. The error table uses the full precision of the computed sums, rather than the rounded values in the two partial-sum tables.

14.7 Spherical Bessel Functions

Unless a problem explicitly states otherwise, the spherical index n (and m) is a nonnegative integer. Singular formulas are initially for nonzero argument; a value at zero requires its regular limit.

14.7.1 Show how one can obtain Eq. (14.162) starting from Eq. (14.161).

Solution.

Put $\nu = n + \frac{1}{2}$ in Eq. (14.161). The coefficient of i^s/x^s in $P_\nu(x) + iQ_\nu(x)$ is

$$a_s = \frac{\prod_{j=1}^{s}[4\nu^2 - (2j-1)^2]}{s!8^s}.$$

For this half-integer order each factor is

$$(2n+1)^2 - (2j-1)^2 = 4(n+j)(n-j+1).$$

For $0 \le s \le n$ the two products are

$$\prod_{j=1}^{s}(n+j) = \frac{(n+s)!}{n!}, \qquad \prod_{j=1}^{s}(n-j+1) = \frac{n!}{(n-s)!},$$

and the factor at $j = n+1$ is zero. Thus the expansion terminates:

$$P_{n+1/2}(x) + iQ_{n+1/2}(x) = \sum_{s=0}^{n} \frac{i^s(n+s)!}{s!(2x)^s(n-s)!}.$$

Since $e^{-i(n+1)\pi/2} = (-i)^{n+1}$, Eq. (14.161) becomes

$$\boxed{h_n^{(1)}(x) = (-i)^{n+1}\frac{e^{ix}}{x}\sum_{s=0}^{n}\frac{i^s(n+s)!}{s!(2x)^s(n-s)!}.}$$

Finally, $(2r)!! = 2^r r!$ gives

$$\frac{(2n+2s)!!}{(2n-2s)!!} = 2^{2s}\frac{(n+s)!}{(n-s)!}.$$

Consequently the same sum is $\sum_{s=0}^{n} i^s(2n+2s)!!/[s!(8x)^s(2n-2s)!!]$, the other form in Eq. (14.162). The finite expression satisfies the spherical Bessel equation and has the outgoing normalization $(-i)^{n+1}e^{ix}/x$; it is exact for $x \neq 0$, not merely a large-x approximation.

14.7.2 Show that if

$$y_n(x) = \sqrt{\frac{\pi}{2x}}\, Y_{n+1/2}(x),$$

it automatically equals

$$(-1)^{n+1}\sqrt{\frac{\pi}{2x}}\, J_{-n-1/2}(x).$$

Solution.

For noninteger ν, the defining relation for Y_ν is

$$Y_\nu(x) = \frac{\cos(\pi\nu)J_\nu(x) - J_{-\nu}(x)}{\sin(\pi\nu)}.$$

At $\nu = n + \frac{1}{2}$, with integer n, $\cos(\pi\nu) = 0$ and $\sin(\pi\nu) = (-1)^n$. Hence

$$Y_{n+1/2}(x) = -(-1)^n J_{-n-1/2}(x) = (-1)^{n+1}J_{-n-1/2}(x).$$

Multiplication by $\sqrt{\pi/(2x)}$ gives the asserted formula. For complex x, use the same branch of $x^{1/2}$ in both expressions; their resulting spherical functions agree by analytic continuation.

14.7.3 Derive the trigonometric-polynomial forms of $j_n(z)$ and $y_n(z)$[3]:

$$j_n(z) = \frac{1}{z}\sin\left(z - \frac{n\pi}{2}\right)\sum_{s=0}^{[n/2]}\frac{(-1)^s(n+2s)!}{(2s)!(2z)^{2s}(n-2s)!}$$
$$+\frac{1}{z}\cos\left(z - \frac{n\pi}{2}\right)\sum_{s=0}^{[(n-1)/2]}\frac{(-1)^s(n+2s+1)!}{(2s+1)!(2z)^{2s+1}(n-2s-1)!},$$

$$y_n(z) = \frac{(-1)^{n+1}}{z}\cos\left(z + \frac{n\pi}{2}\right)\sum_{s=0}^{[n/2]}\frac{(-1)^s(n+2s)!}{(2s)!(2z)^{2s}(n-2s)!}$$
$$-\frac{(-1)^{n+1}}{z}\sin\left(z + \frac{n\pi}{2}\right)\sum_{s=0}^{[(n-1)/2]}\frac{(-1)^s(n+2s+1)!}{(2s+1)!(2z)^{2s+1}(n-2s-1)!}.$$

Solution.

Separate the even and odd indices in the finite sum of Exercise 14.7.1. Write

$$P = \sum_{s=0}^{\lfloor n/2\rfloor}\frac{(-1)^s(n+2s)!}{(2s)!(2z)^{2s}(n-2s)!},$$
$$Q = \sum_{s=0}^{\lfloor (n-1)/2\rfloor}\frac{(-1)^s(n+2s+1)!}{(2s+1)!(2z)^{2s+1}(n-2s-1)!}.$$

[3]The upper summation limit $[n/2]$ means the largest integer that does not exceed $n/2$.

An empty sum is zero, so $Q = 0$ when $n = 0$. With $\chi = z - n\pi/2$ the prefactor is $(-i)^{n+1}e^{iz} = \sin\chi - i\cos\chi$. Therefore

$$h_n^{(1)}(z) = \frac{1}{z}\{P\sin\chi + Q\cos\chi + i[-P\cos\chi + Q\sin\chi]\}.$$

For positive real z, P, Q are real and $h_n^{(1)} = j_n + iy_n$; separating its two parts gives

$$\boxed{j_n = \frac{P\sin\chi + Q\cos\chi}{z}, \qquad y_n = \frac{-P\cos\chi + Q\sin\chi}{z}.}$$

These finite analytic identities then extend to complex $z \neq 0$. To put the second one in the question's phase convention, use

$$\cos\chi = (-1)^n\cos(z + n\pi/2), \qquad \sin\chi = (-1)^n\sin(z + n\pi/2).$$

Thus

$$\boxed{y_n(z) = \frac{(-1)^{n+1}}{z}\{P\cos(z + n\pi/2) - Q\sin(z + n\pi/2)\}.}$$

There are two corrections to the printed formulas. The power in the odd sum for j_n is $2s + 1$, as in the published errata. In the displayed phase convention for y_n, the sign before its second sum must be minus. The actual textbook, p. 710, prints plus. For $n = 1$, $P = 1$, $Q = 1/z$; the correct answer is $y_1 = -\sin z/z - \cos z/z^2$. The printed plus would change the last sign and fail the spherical Bessel equation. The same substitution gives $j_1 = \sin z/z^2 - \cos z/z$, which also checks the corrected odd power.

14.7.4 Use the integral representation of $J_\nu(x)$,

$$J_\nu(x) = \frac{1}{\pi^{1/2}\Gamma(\nu + \frac{1}{2})}\left(\frac{x}{2}\right)^\nu \int_{-1}^{1} e^{\pm ixp}(1 - p^2)^{\nu - 1/2}\, dp,$$

to show that the spherical Bessel functions $j_n(x)$ are expressible in terms of trigonometric functions; that is, for example,

$$j_0(x) = \frac{\sin x}{x}, \qquad j_1(x) = \frac{\sin x}{x^2} - \frac{\cos x}{x}.$$

Solution.

Insert $\nu = n + \frac{1}{2}$ and multiply by $\sqrt{\pi/(2x)}$. Since $\Gamma(n + 1) = n!$,

$$j_n(x) = \frac{x^n}{2^{n+1}n!}\int_{-1}^{1} e^{ixp}(1 - p^2)^n\, dp.$$

Expand the polynomial and introduce

$$S(x) = \int_{-1}^{1} e^{ixp}dp = \frac{2\sin x}{x}.$$

Differentiation under this finite-interval integral gives $S^{(2r)}(x) = (-1)^r\int_{-1}^{1} p^{2r}e^{ixp}dp$. The two factors $(-1)^r$ cancel in the binomial expansion, so

$$\boxed{j_n(x) = \frac{x^n}{2^{n+1}n!}\sum_{r=0}^{n}\binom{n}{r}S^{(2r)}(x).}$$

Every derivative of $\sin x/x$ is a finite combination of $\sin x, \cos x$ multiplied by integer negative powers of x. This proves the requested statement for every nonnegative integer n. In particular,

$$j_0 = S/2 = \frac{\sin x}{x}.$$

For $n = 1$,

$$S'' = -\frac{2\sin x}{x} - \frac{4\cos x}{x^2} + \frac{4\sin x}{x^3},$$

whence

$$j_1 = \frac{x}{4}(S + S'') = \frac{\sin x}{x^2} - \frac{\cos x}{x}.$$

The apparent singularities at $x = 0$ cancel, as is already clear from the original integral multiplied by x^n.

14.7.5

(a) Derive the recurrence relations

$$f_{n-1}(x) + f_{n+1}(x) = \frac{2n+1}{x} f_n(x),$$
$$n f_{n-1}(x) - (n+1) f_{n+1}(x) = (2n+1) f_n'(x)$$

satisfied by the spherical Bessel functions $j_n(x)$, $y_n(x)$, $h_n^{(1)}(x)$, and $h_n^{(2)}(x)$.

(b) Show, from these two recurrence relations, that the spherical Bessel function $f_n(x)$ satisfies the differential equation

$$x^2 f_n''(x) + 2x f_n'(x) + \big[x^2 - n(n+1)\big] f_n(x) = 0.$$

Solution.

(a) For any cylindrical solution F_ν, set $f_n = \sqrt{\pi/(2x)} F_{n+1/2}$. Its two cylindrical recurrences give, respectively,

$$f_{n-1} + f_{n+1} = \frac{2n+1}{x} f_n, \qquad f_n' = \frac{1}{2}(f_{n-1} - f_{n+1}) - \frac{f_n}{2x}.$$

The last term arises from differentiating $x^{-1/2}$. Replace f_n/x by $(f_{n-1} + f_{n+1})/(2n+1)$:

$$\begin{aligned}(2n+1) f_n' &= \frac{2n+1}{2}(f_{n-1} - f_{n+1}) - \frac{1}{2}(f_{n-1} + f_{n+1}) \\ &= n f_{n-1} - (n+1) f_{n+1}.\end{aligned}$$

This also yields the useful equivalent pair

$$f_n' = f_{n-1} - \frac{n+1}{x} f_n = \frac{n}{x} f_n - f_{n+1}.$$

All four functions in the question inherit these relations by linearity from their cylindrical counterparts.

(b) Differentiate $f_{n+1} = n f_n/x - f_n'$:

$$f_{n+1}' = \frac{n}{x} f_n' - \frac{n}{x^2} f_n - f_n''.$$

Use the first equivalent derivative identity at order $n+1$:

$$f'_{n+1} = f_n - \frac{n+2}{x} f_{n+1} = f_n - \frac{n(n+2)}{x^2} f_n + \frac{n+2}{x} f'_n.$$

Equating the right sides and moving all terms to one side gives

$$f''_n + \frac{2}{x} f'_n + \left(1 - \frac{n(n+1)}{x^2}\right) f_n = 0.$$

Multiplication by x^2 proves the required equation.

14.7.6 Prove by mathematical induction (Section 1.4) that

$$j_n(x) = (-1)^n x^n \left(\frac{1}{x}\frac{d}{dx}\right)^n \left(\frac{\sin x}{x}\right)$$

for n, an arbitrary nonnegative integer.

Solution.

Let $D = x^{-1} d/dx$. The assertion is true for $n = 0$ because $j_0 = \sin x/x$. Assume it is true at n. The derivative recurrence in Exercise 14.7.5 gives

$$j_{n+1} = \frac{n}{x} j_n - j'_n = -x^n \frac{d}{dx}(x^{-n} j_n).$$

Insert the induction hypothesis before differentiating:

$$\begin{aligned} j_{n+1} &= -x^n \frac{d}{dx}\{(-1)^n D^n j_0\} \\ &= (-1)^{n+1} x^{n+1} D(D^n j_0) = (-1)^{n+1} x^{n+1} D^{n+1} j_0. \end{aligned}$$

This is the assertion for $n+1$, completing the induction. The displayed differential expression is initially interpreted at $x \neq 0$; its continuous limit at the origin is supplied by the power series of j_n.

14.7.7 From the discussion of orthogonality of the spherical Bessel functions, show that a Wronskian relation for $j_n(x)$ and $y_n(x)$ is

$$j_n(x) y'_n(x) - j'_n(x) y_n(x) = \frac{1}{x^2}.$$

Solution.

The spherical equation is self-adjoint in the form

$$(x^2 f'_n)' + [x^2 - n(n+1)] f_n = 0.$$

Multiply the equation for y_n by j_n, the equation for j_n by y_n, and subtract. The undifferentiated terms cancel:

$$j_n (x^2 y'_n)' - y_n (x^2 j'_n)' = \frac{d}{dx}\{x^2 (j_n y'_n - j'_n y_n)\} = 0.$$

Thus $x^2 W$ is a constant. To determine it, use the leading series at the origin,

$$j_n \sim \frac{x^n}{(2n+1)!!}, \qquad y_n \sim -(2n-1)!!\, x^{-n-1},$$

where $(-1)!! = 1$. Their leading contribution to the Wronskian is

$$\begin{aligned} W &\sim \frac{(n+1)(2n-1)!!}{(2n+1)!!}\frac{1}{x^2} + \frac{n(2n-1)!!}{(2n+1)!!}\frac{1}{x^2} \\ &= \frac{(2n+1)(2n-1)!!}{(2n+1)!!}\frac{1}{x^2} = \frac{1}{x^2}. \end{aligned}$$

Because $x^2 W$ is exactly constant, the asymptotic computation fixes the exact result $\boxed{W[j_n, y_n] = x^{-2}}$. The second function is y_n, correcting the printed n_n in accordance with the published errata.

14.7.8 Verify

$$h_n^{(1)}(x) h_n^{(2)\prime}(x) - h_n^{(1)\prime}(x) h_n^{(2)}(x) = -\frac{2i}{x^2}.$$

Solution.

Write $j = j_n$, $y = y_n$. Bilinearity and antisymmetry of the Wronskian give

$$\begin{aligned} W[h_n^{(1)}, h_n^{(2)}] &= W[j + iy, j - iy] \\ &= W[j, j] - iW[j, y] + iW[y, j] + W[y, y] \\ &= -iW[j, y] - iW[j, y] = -2iW[j, y]. \end{aligned}$$

The two same-function Wronskians vanish. Substitution of Exercise 14.7.7 yields $\boxed{-2i/x^2}$, including its sign.

14.7.9 Verify Poisson's integral representation of the spherical Bessel function,

$$j_n(z) = \frac{z^n}{2^{n+1} n!} \int_0^\pi \cos(z \cos\theta) \sin^{2n+1}\theta \, d\theta.$$

Solution.

Expand the cosine; its series is uniformly convergent on the finite θ interval for every fixed complex z. Its rth term requires

$$\begin{aligned} \int_0^\pi \cos^{2r}\theta \sin^{2n+1}\theta \, d\theta &= \int_{-1}^1 p^{2r}(1-p^2)^n dp \\ &= \int_0^1 u^{r-1/2}(1-u)^n du \\ &= \frac{\Gamma(r+1/2) n!}{\Gamma(n+r+3/2)}. \end{aligned}$$

In the second line $u = p^2$ was used after doubling the integral on $0 < p < 1$. Since $\Gamma(r+1/2) = \sqrt{\pi}(2r)!/(4^r r!)$, the proposed right side becomes

$$\begin{aligned} \frac{z^n}{2^{n+1} n!} \sum_{r=0}^\infty \frac{(-1)^r z^{2r}}{(2r)!} \frac{\Gamma(r+1/2) n!}{\Gamma(n+r+3/2)} &= \frac{\sqrt{\pi} z^n}{2^{n+1}} \sum_{r=0}^\infty \frac{(-1)^r z^{2r}}{4^r r! \Gamma(n+r+3/2)} \\ &= \sqrt{\frac{\pi}{2z}} J_{n+1/2}(z) = j_n(z). \end{aligned}$$

This proves the formula. At $z = 0$ it has its regular limiting value, 1 for $n = 0$ and 0 for $n > 0$.

14.7.10 A well-known integral representation for $K_\nu(x)$ has the form

$$K_\nu(x) = \frac{2^\nu \Gamma(\nu + \frac{1}{2})}{\sqrt{\pi}\, x^\nu} \int_0^\infty \frac{\cos xt}{(t^2+1)^{\nu+1/2}}\, dt.$$

Starting from this formula, show that

$$k_n(x) = \frac{2^{n+2}(n+1)!}{\pi x^{n+1}} \int_0^\infty \frac{k^2 j_0(kx)}{(k^2+1)^{n+2}}\, dk.$$

Take $x > 0$ and nonnegative integer n. The starting cosine integral converges for real $\nu > -1/2$ (absolutely when $\nu > 0$), and the required order is $\nu = n + 1/2$.

Solution.

The book uses

$$k_n(x) = \sqrt{\frac{2}{\pi x}} K_{n+1/2}(x),$$

so its normalization differs from that of j_n. Inserting $\nu = n + \frac{1}{2}$ in the given integral gives

$$k_n(x) = \frac{2^{n+1} n!}{\pi x^{n+1}} \int_0^\infty \frac{\cos(xt)}{(1+t^2)^{n+1}} dt.$$

Integrate this last integral by parts, taking the cosine as the derivative of $\sin(xt)/x$:

$$\begin{aligned}\int_0^\infty \frac{\cos(xt)}{(1+t^2)^{n+1}} dt &= \left[\frac{\sin(xt)}{x(1+t^2)^{n+1}}\right]_0^\infty + \frac{2(n+1)}{x} \int_0^\infty \frac{t \sin(xt)}{(1+t^2)^{n+2}} dt \\ &= 2(n+1) \int_0^\infty \frac{t^2 j_0(xt)}{(1+t^2)^{n+2}} dt.\end{aligned}$$

The boundary term is zero at both ends for $x > 0$, $n \geq 0$. Multiplying the prefactor by $2(n+1)$ and renaming t as k gives exactly

$$\boxed{k_n(x) = \frac{2^{n+2}(n+1)!}{\pi x^{n+1}} \int_0^\infty \frac{k^2 j_0(kx)}{(1+k^2)^{n+2}} dk.}$$

Both integrals are absolutely convergent for these parameters.

14.7.11 Show that

$$\int_0^\infty J_\mu(x) J_\nu(x) \frac{dx}{x} = \frac{2}{\pi} \frac{\sin[(\mu-\nu)\pi/2]}{\mu^2 - \nu^2}, \qquad \mu + \nu > 0.$$

Take real orders. At $\mu = \nu > 0$, interpret the quotient by continuity as $1/(2\nu)$.

Solution.

Take real orders with $\mu + \nu > 0$. The two Bessel equations in self-adjoint form are

$$(xJ'_\alpha)' + \left(x - \frac{\alpha^2}{x}\right) J_\alpha = 0.$$

Multiply the μ equation by J_ν, the ν equation by J_μ, and subtract. The result is

$$\frac{d}{dx}\{x(J_\nu J'_\mu - J'_\nu J_\mu)\} = \frac{\mu^2 - \nu^2}{x} J_\mu J_\nu.$$

Integrate on $[\epsilon, R]$. At the lower endpoint the bracket is $O(\epsilon^{\mu+\nu})$, which tends to zero; when a leading coefficient vanishes at a negative integer order, the actual power only improves this estimate. At the upper endpoint put $\chi_\alpha = x - \alpha\pi/2 - \pi/4$ and use

$$J_\alpha = \sqrt{\frac{2}{\pi x}}[\cos\chi_\alpha + O(x^{-1})], \quad J'_\alpha = \sqrt{\frac{2}{\pi x}}[-\sin\chi_\alpha + O(x^{-1})].$$

It follows that

$$\begin{aligned} x(J_\nu J'_\mu - J'_\nu J_\mu) &= \frac{2}{\pi}[-\cos\chi_\nu \sin\chi_\mu + \sin\chi_\nu \cos\chi_\mu] + O(x^{-1}) \\ &\longrightarrow \frac{2}{\pi}\sin\frac{(\mu-\nu)\pi}{2}. \end{aligned}$$

The integrand is integrable at zero and $O(x^{-2})$ at infinity, so both limiting operations are justified. For $\mu^2 \neq \nu^2$,

$$\boxed{\int_0^\infty J_\mu J_\nu \frac{dx}{x} = \frac{2}{\pi}\frac{\sin[(\mu-\nu)\pi/2]}{\mu^2-\nu^2}.}$$

At equal orders the printed quotient is $0/0$ and must be interpreted by continuity. Here $\mu = \nu > 0$ and its limit is $1/(2\nu)$. The other vanishing denominator, $\mu = -\nu$, is excluded by $\mu + \nu > 0$. Indeed, dividing numerator and denominator by $\mu - \nu$ gives the limit $(2/\pi)(\pi/2)/(2\nu) = 1/(2\nu)$. Dominated convergence is valid in a sufficiently small neighborhood of any fixed positive ν, using the same origin and infinity bounds.

14.7.12 Derive Eq. (14.190):

$$\int_{-\infty}^{\infty} j_m(x) j_n(x)\, dx = 0, \qquad \begin{cases} m \neq n, \\ m, n \geq 0. \end{cases}$$

Solution.

The power series gives $j_n(-x) = (-1)^n j_n(x)$. The product is integrable at the origin and decays as $O(x^{-2})$ at infinity. If $m + n$ is odd, its integral over the whole real line therefore vanishes by oddness.

If $m + n$ is even, the product is even. On the positive half-line, use Exercise 14.7.11 with $\mu = m + \frac{1}{2}$, $\nu = n + \frac{1}{2}$:

$$\begin{aligned} \int_0^\infty j_m(x) j_n(x) dx &= \frac{\pi}{2}\int_0^\infty J_{m+1/2}(x) J_{n+1/2}(x)\frac{dx}{x} \\ &= \frac{\sin[(m-n)\pi/2]}{(m+1/2)^2 - (n+1/2)^2} \\ &= \frac{\sin[(m-n)\pi/2]}{(m-n)(m+n+1)}. \end{aligned}$$

For distinct nonnegative integers of the same parity, $m - n$ is a nonzero even integer: the numerator is zero and the denominator is nonzero. Doubling this zero proves the whole-line result. The denominator in this half-line intermediate formula is $(m-n)(m+n+1)$, not $m^2 - n^2$ as in the manual, p. 228. The final orthogonality conclusion is unaffected.

14.7.13 Derive Eq. (14.191):

$$\int_{-\infty}^{\infty} \left[j_n(x)\right]^2 dx = \frac{\pi}{2n+1}.$$

Solution.

Use the equal-order limit established in Exercise 14.7.11, now with $\nu = n + \frac{1}{2} > 0$:

$$\int_0^{\infty} \frac{J_{n+1/2}(x)^2}{x} dx = \frac{1}{2(n+1/2)} = \frac{1}{2n+1}.$$

Since $j_n^2 = \pi J_{n+1/2}^2/(2x)$ on $x > 0$,

$$\int_0^{\infty} j_n(x)^2 dx = \frac{\pi}{2(2n+1)}.$$

The square is even, regardless of the parity of n. Its integral over the whole line is twice the last expression, hence $\boxed{\pi/(2n+1)}$.

14.7.14 The Fresnel integrals (Fig. 14.19 and Exercise 12.7.2) occurring in diffraction theory are given by

$$x(t) = \sqrt{\frac{\pi}{2}}\, C\left(\sqrt{\frac{2}{\pi}}\, t\right) = \int_0^t \cos(v^2)\, dv,$$
$$y(t) = \sqrt{\frac{\pi}{2}}\, S\left(\sqrt{\frac{2}{\pi}}\, t\right) = \int_0^t \sin(v^2)\, dv.$$

Show that these integrals may be expanded in series of spherical Bessel functions as follows:

$$x(\sqrt{s}) = \frac{1}{2}\int_0^s j_{-1}(u) u^{1/2}\, du = s^{1/2} \sum_{n=0}^{\infty} j_{2n}(s),$$

$$y(\sqrt{s}) = \frac{1}{2}\int_0^s j_0(u) u^{1/2}\, du = s^{1/2} \sum_{n=0}^{\infty} j_{2n+1}(s).$$

Hint. To establish the equality of the integral and the sum, you may wish to work with their derivatives. The spherical Bessel analogs of Eqs. (14.8) and (14.12) may be helpful.

Use the standard $C(q) = \int_0^q \cos(\pi u^2/2)\, du$ and $S(q) = \int_0^q \sin(\pi u^2/2)\, du$. In the subsequent formulas set $s = t^2 \geq 0$.

Solution.

Use capital X, Y temporarily to distinguish the two functions from their arguments. With the standard Fresnel convention $C(q) = \int_0^q \cos(\pi u^2/2) du$ and $S(q) = \int_0^q \sin(\pi u^2/2) du$, the substitution $u = \sqrt{2/\pi}\, v$ gives

$$X(t) = \sqrt{\frac{\pi}{2}} C\left(\sqrt{\frac{2}{\pi}}\, t\right), \qquad Y(t) = \sqrt{\frac{\pi}{2}} S\left(\sqrt{\frac{2}{\pi}}\, t\right).$$

Next put $s = t^2$ for $t \geq 0$, and define $F(s) = X(\sqrt{s})$, $H(s) = Y(\sqrt{s})$. In the defining integrals set $u = v^2$, so $dv = du/(2\sqrt{u})$:

$$F(s) = \frac{1}{2}\int_0^s \frac{\cos u}{\sqrt{u}}du = \frac{1}{2}\int_0^s \sqrt{u}\, j_{-1}(u)du,$$
$$H(s) = \frac{1}{2}\int_0^s \frac{\sin u}{\sqrt{u}}du = \frac{1}{2}\int_0^s \sqrt{u}\, j_0(u)du,$$

where $j_{-1}(u) = \cos u/u$ and $j_0(u) = \sin u/u$. The Fresnel arguments printed on p. 711 are $\sqrt{\pi/2}\, t$; the correct arguments are $\sqrt{2/\pi}\, t$. Also the functions in the subsequent s-integrals are $X(\sqrt{s})$ and $Y(\sqrt{s})$, not the same functions evaluated at s. The printed second integral has a stray x after $u^{1/2}$, which must be deleted.

To prove the series, form finite sums

$$E_N(s) = \sqrt{s}\sum_{n=0}^{N} j_{2n}(s) = \sqrt{\frac{\pi}{2}}\sum_{n=0}^{N} J_{2n+1/2}(s),$$

$$O_N(s) = \sqrt{s}\sum_{n=0}^{N} j_{2n+1}(s) = \sqrt{\frac{\pi}{2}}\sum_{n=0}^{N} J_{2n+3/2}(s).$$

Use $2J'_\alpha = J_{\alpha-1} - J_{\alpha+1}$. Consecutive terms cancel, leaving the explicit terminal terms

$$E'_N(s) = \frac{1}{2}\sqrt{\frac{\pi}{2}}[J_{-1/2}(s) - J_{2N+3/2}(s)],$$
$$O'_N(s) = \frac{1}{2}\sqrt{\frac{\pi}{2}}[J_{1/2}(s) - J_{2N+5/2}(s)].$$

The series for J_α bounds the terminal term on $0 \leq s \leq R$ by $(R/2)^\alpha e^{R^2/4}/\Gamma(\alpha+1)$ for these positive orders, which tends to zero. Integrate these finite identities from zero to s; both sums vanish at zero. Their limits are therefore $F(s)$ and $H(s)$, respectively. This establishes

$$\boxed{F(s) = \sqrt{s}\sum_{n=0}^{\infty} j_{2n}(s), \qquad H(s) = \sqrt{s}\sum_{n=0}^{\infty} j_{2n+1}(s).}$$

Equivalently, in the original parameter,

$$\boxed{X(t) = t\sum_{n=0}^{\infty} j_{2n}(t^2), \qquad Y(t) = t\sum_{n=0}^{\infty} j_{2n+1}(t^2).}$$

These last formulas also extend to negative real t by oddness.

14.7.15 A hollow sphere of radius a (Helmholtz resonator) contains standing sound waves. Find the minimum positive frequency of oscillation in terms of the radius a and the velocity of sound v. The sound waves satisfy the wave equation

$$\nabla^2\psi = \frac{1}{v^2}\frac{\partial^2\psi}{\partial t^2}$$

and the boundary condition $\dfrac{\partial\psi}{\partial r} = 0$, $r = a$.

The spatial part of this PDE is the same as the PDE discussed in Example 14.7.1, but here we have a Neumann boundary condition, in contrast to the Dirichlet boundary condition of that example.

ANS. $\nu_{\min} = 0.3313\, v/a, \quad \lambda_{\max} = 3.018\, a.$

Solution.

Separate a time-harmonic mode as $\psi = R_l(r)Y_{lm}(\theta,\phi)e^{-i\omega t}$ and set $k = \omega/v$. The radial equation is

$$(r^2 R_l')' + [k^2 r^2 - l(l+1)]R_l = 0.$$

Regularity at the center selects $R_l(r) = j_l(kr)$ for $k > 0$. The rigid-wall condition is $j_l'(ka) = 0$. A spatially constant Neumann mode has $k = \omega = 0$. The quoted minimum means the lowest *positive* oscillation frequency; the zero mode must be excluded explicitly.

For $l = 1$ differentiation of $j_1(x) = \sin x/x^2 - \cos x/x$ gives

$$j_1'(x) = \frac{F(x)}{x^3}, \qquad F(x) = (x^2 - 2)\sin x + 2x\cos x.$$

Differentiate the numerator completely:

$$F' = 2x\sin x + (x^2 - 2)\cos x + 2\cos x - 2x\sin x = x^2\cos x.$$

Thus F increases from $F(0) = 0$ to $x = \pi/2$, then decreases strictly to $F(\pi) = -2\pi < 0$. It has exactly one zero in $(\pi/2, \pi)$ and none earlier. Numerical solution gives

$$x_1 = ka = 2.0815759778181006\ldots.$$

To establish that this is the global first positive eigenvalue, first consider $l = 0$. The derivative numerator is $x\cos x - \sin x$, with derivative $-x\sin x < 0$ on $(0,\pi)$. It starts at zero, hence j_0' has no positive root below π. For every $l \geq 2$, multiply the radial equation by $\overline{R}_l$ and integrate, using the regular origin and Neumann endpoint:

$$k^2 = \frac{\int_0^a [r^2|R_l'|^2 + l(l+1)|R_l|^2]dr}{\int_0^a r^2|R_l|^2 dr} \geq \frac{l(l+1)}{a^2} \geq \frac{6}{a^2}.$$

Since $x_1 < \sqrt{6}$, no higher angular order can lower this frequency. The $l = 1$ root is consequently the required one. Converting angular frequency to cycles per unit time,

$$\boxed{\nu_{\min} = \frac{x_1}{2\pi}\frac{v}{a} = 0.3312931063\ldots\frac{v}{a}, \qquad \lambda_{\max} = \frac{2\pi a}{x_1} = 3.0184751237\ldots a.}$$

At $k = 0$ the regular radial solutions are r^l; their Neumann condition permits only $l = 0$, confirming the constant zero mode.

14.7.16

(a) Show that the parity of $i_n(x)$ (the behavior under $x \to -x$) is $(-1)^n$.

(b) Show that $k_n(x)$ has no definite parity.

Solution.

(a) From the series of $I_{n+1/2}$,

$$i_n(x) = \sqrt{\frac{\pi}{2x}} I_{n+1/2}(x) = \frac{\sqrt{\pi} x^n}{2^{n+1}} \sum_{s=0}^{\infty} \frac{x^{2s}}{4^s s! \Gamma(n+s+3/2)}.$$

The series after x^n contains only even powers and is entire. Hence $i_n(-x) = (-1)^n i_n(x)$. This argument also handles the square-root branches: their product has the displayed integer power series, which supplies the continuation through the origin.

(b) The finite half-integer K formula, with the book's normalization, is

$$k_n(x) = \frac{e^{-x}}{x} \sum_{s=0}^{n} \frac{(n+s)!}{s!(2x)^s (n-s)!}.$$

Its continuation to $-x$ is

$$k_n(-x) = -\frac{e^{x}}{x} \sum_{s=0}^{n} \frac{(-1)^s (n+s)!}{s!(2x)^s (n-s)!}.$$

For positive $x \to \infty$ both finite sums tend to 1, so $k_n(-x)/k_n(x) \sim -e^{2x}$. This ratio is neither identically $+1$ nor identically -1. Therefore k_n is neither even nor odd for any nonnegative integer n. Already $k_0(x) = e^{-x}/x$ exhibits the failure.

14.7.17 Show that the Wronskian of the spherical modified Bessel functions is given by

$$i_n(x) k_n'(x) - i_n'(x) k_n(x) = -\frac{1}{x^2}.$$

Solution.

Keep the two normalization factors distinct:

$$i_n = a(x) I_\nu(x), \quad k_n = b(x) K_\nu(x), \quad \nu = n + \tfrac{1}{2}, \quad a = \sqrt{\frac{\pi}{2x}}, \quad b = \sqrt{\frac{2}{\pi x}}.$$

Differentiate each product and subtract:

$$\begin{aligned} i_n k_n' - i_n' k_n &= a I_\nu (b' K_\nu + b K_\nu') - (a' I_\nu + a I_\nu') b K_\nu \\ &= ab (I_\nu K_\nu' - I_\nu' K_\nu) + (ab' - a' b) I_\nu K_\nu. \end{aligned}$$

Here $a'/a = b'/b = -1/(2x)$, so the last coefficient is zero, and $ab = 1/x$. Exercise 14.5.11 supplies $W[I_\nu, K_\nu] = -1/x$. Consequently

$$\boxed{W[i_n, k_n] = \frac{1}{x}\left(-\frac{1}{x}\right) = -\frac{1}{x^2}.}$$

Using $\sqrt{\pi/(2x)}$ for both functions would introduce an extra factor $\pi/2$; the result here uses precisely the book's Eq. (14.194) definition of k_n.

15 Chapter 15. Legendre Functions

Problems

15.1 Legendre Polynomials

15.1.1 Derive the Legendre ODE by manipulation of the Legendre polynomial recurrence relations. Suggested starting point: Eqs. (15.24) and (15.25).

Solution.

The two needed recurrences are

$$P'_{n-1} = -nP_n + xP'_n, \qquad (1-x^2)P'_n = nP_{n-1} - nxP_n.$$

Differentiate the second, including both product derivatives:

$$(1-x^2)P''_n - 2xP'_n = nP'_{n-1} - nP_n - nxP'_n.$$

Insert the first recurrence on the right:

$$n(-nP_n + xP'_n) - nP_n - nxP'_n = -n(n+1)P_n.$$

Moving this term to the left gives

$$\boxed{(1-x^2)P''_n - 2xP'_n + n(n+1)P_n = 0}.$$

For $n = 0$, $P_0 = 1$ verifies the equation directly, so no undefined P_{-1} is needed at that endpoint.

15.1.2 Derive the following closed formula for the Legendre polynomials $P_n(x)$,

$$P_n(x) = \sum_{k=0}^{[n/2]} (-1)^k \frac{(2n-2k)!}{2^n k!(n-k)!(n-2k)!}\, x^{n-2k},$$

where $[n/2]$ stands for the integer part of $n/2$.

Hint. Further expand Eq. (15.12) and rearrange the resulting double sum.

Solution.

Start in a sufficiently small disk about $t = 0$, where the binomial series converges:

$$(1-2xt+t^2)^{-1/2} = \sum_{j=0}^{\infty} \binom{-1/2}{j} (-2xt+t^2)^j.$$

Choose k copies of t^2 from the j factors. The other $j-k$ copies contribute $(-2xt)^{j-k}$, so the resulting term is

$$\binom{-1/2}{j}\binom{j}{k}(-2x)^{j-k}t^{j+k}.$$

To obtain t^n, set $j = n-k$. The condition $0 \le k \le j$ becomes $0 \le k \le \lfloor n/2 \rfloor$. Therefore

$$P_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{-1/2}{n-k}\binom{n-k}{k}(-2x)^{n-2k}.$$

Use $\binom{-1/2}{j} = (-1)^j(2j)!/[2^{2j}(j!)^2]$. The sign becomes $(-1)^{n-k+n-2k} = (-1)^k$ and the power of two is $2^{n-2k-2(n-k)} = 2^{-n}$. The remaining factorials give

$$\boxed{P_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k \frac{(2n-2k)!}{2^n k!(n-k)!(n-2k)!} x^{n-2k}}.$$

Although the generating series was used locally in t, the coefficient is a polynomial identity and holds for every finite complex x.

15.1.3 By differentiation and direct substitution of the series form given in Exercise 15.1.2, show that $P_n(x)$ satisfies the Legendre ODE. Note that there is no restriction on x. We may have any x, $-\infty < x < \infty$, and indeed any z in the entire finite complex plane.

Solution.

Write $P_n = \sum_{k=0}^{\lfloor n/2 \rfloor} A_k x^{n-2k}$, with A_k as in Exercise 15.1.2. The Legendre operator acts on a monomial as

$$L_n[x^j] = j(j-1)x^{j-2} + [n(n+1) - j(j+1)]x^j.$$

At power x^{n-2k}, the second term comes from A_k, while the first comes from A_{k-1}. Its total coefficient is

$$(n-2k+2)(n-2k+1)A_{k-1} + 2k(2n-2k+1)A_k,$$

because $n(n+1) - (n-2k)(n-2k+1) = 2k(2n-2k+1)$. Direct cancellation of factorials in the formula for A_k gives

$$\frac{A_k}{A_{k-1}} = -\frac{(n-2k+2)(n-2k+1)}{2k(2n-2k+1)}.$$

Thus every displayed coefficient is zero. The leading coefficient ($k = 0$) is multiplied by $n(n+1) - n(n+1) = 0$. At the bottom, the last monomial is either 1 or x, whose second derivative is zero, so no lower power remains. Hence $L_n[P_n] = 0$ identically. All manipulations concern finite polynomials, which proves the claim on the entire finite complex plane.

15.1.4 The shifted Legendre polynomials, designated by the symbol $P_n^*(x)$ (where the asterisk does not mean complex conjugate) are orthogonal with unit weight on $[0, 1]$, with normalization integral $\langle P_n^* | P_n^* \rangle = 1/(2n+1)$. The P_n^* through $n = 6$ are shown in Table 15.2.

Table 15.2 Shifted Legendre Polynomials

$P_0^*(x) = 1$
$P_1^*(x) = 2x - 1$
$P_2^*(x) = 6x^2 - 6x + 1$
$P_3^*(x) = 20x^3 - 30x^2 + 12x - 1$
$P_4^*(x) = 70x^4 - 140x^3 + 90x^2 - 20x + 1$
$P_5^*(x) = 252x^5 - 630x^4 + 560x^3 - 210x^2 + 30x - 1$
$P_6^*(x) = 924x^6 - 2772x^5 + 3150x^4 - 1680x^3 + 420x^2 - 42x + 1$

(a) Find the recurrence relation satisfied by the P_n^*.

(b) Show that all the coefficients of the P_n^* are integers.

Hint. Look at the closed formula in Exercise 15.1.2.

Solution.

The listed normalization and signs are obtained by $P_n^*(x) = P_n(2x-1)$. Indeed, $y = 2x-1$ gives

$$\int_0^1 P_n^* P_m^* dx = \frac{1}{2}\int_{-1}^1 P_n(y)P_m(y)dy = \frac{\delta_{nm}}{2n+1}.$$

The manual's last expression on p. 232 retains an extra factor $1/2$. The change of variable contributes $1/2$, but the unshifted norm contributes $2/(2n+1)$, leaving $1/(2n+1)$ as stated correctly in the textbook. (a) Replace the argument in $(n+1)P_{n+1} = (2n+1)yP_n - nP_{n-1}$:

$$\boxed{(n+1)P_{n+1}^*(x) = (2n+1)(2x-1)P_n^*(x) - nP_{n-1}^*(x)}.$$

The initial polynomials are $P_0^* = 1$, $P_1^* = 2x-1$.

(b) The recurrence alone involves division by $n+1$, so an explicit coefficient formula is more useful for proving integrality. Transform Rodrigues's formula, using $D_y = \frac{1}{2}D_x$ and $y^2 - 1 = 4x(x-1)$:

$$P_n^*(x) = \frac{1}{n!}D_x^n[x^n(x-1)^n].$$

Expand $(x-1)^n = \sum_{k=0}^n (-1)^{n-k}\binom{n}{k}x^k$ and differentiate:

$$P_n^*(x) = \sum_{k=0}^n (-1)^{n-k}\binom{n}{k}\frac{(n+k)!}{n!k!}x^k = \boxed{\sum_{k=0}^n (-1)^{n-k}\binom{n}{k}\binom{n+k}{k}x^k}.$$

Every coefficient is a signed product of integers.

15.1.5 Given the series

$$\alpha_0 + \alpha_2\cos^2\theta + \alpha_4\cos^4\theta + \alpha_6\cos^6\theta = a_0P_0 + a_2P_2 + a_4P_4 + a_6P_6,$$

where the arguments of the P_n are $\cos\theta$, express the coefficients α_i as a column vector $\boldsymbol{\alpha}$ and the coefficients a_i as a column vector $\mathbf{a}$ and determine the matrices A and B such that

$$\mathsf{A}\boldsymbol{\alpha} = \mathbf{a} \quad \text{and} \quad \mathsf{B}\mathbf{a} = \boldsymbol{\alpha}.$$

Check your computation by showing that $\mathsf{AB} = \mathbf{1}$ (unit matrix). Repeat for the odd case

$$\alpha_1\cos\theta + \alpha_3\cos^3\theta + \alpha_5\cos^5\theta + \alpha_7\cos^7\theta = a_1P_1 + a_3P_3 + a_5P_5 + a_7P_7.$$

Note. $P_n(\cos\theta)$ and $\cos^n\theta$ are tabulated in terms of each other in AMS-55 (see Additional Readings for the complete reference).

Solution.

Put $x = \cos\theta$ and order the even columns as $\boldsymbol{\alpha} = (\alpha_0, \alpha_2, \alpha_4, \alpha_6)^T$, $\mathbf{a} = (a_0, a_2, a_4, a_6)^T$. The coefficients of P_0, P_2, P_4, P_6 in ascending powers of x give the columns of B_e:

$$\boxed{\mathsf{B}_e = \begin{pmatrix} 1 & -1/2 & 3/8 & -5/16 \\ 0 & 3/2 & -15/4 & 105/16 \\ 0 & 0 & 35/8 & -315/16 \\ 0 & 0 & 0 & 231/16 \end{pmatrix}}.$$

Solving the triangular equations from the last row upward gives

$$a_6 = \frac{16}{231}\alpha_6, \quad a_4 = \frac{8}{35}\alpha_4 + \frac{24}{77}\alpha_6, \quad a_2 = \frac{2}{3}\alpha_2 + \frac{4}{7}\alpha_4 + \frac{10}{21}\alpha_6,$$

and $a_0 = \alpha_0 + \alpha_2/3 + \alpha_4/5 + \alpha_6/7$. Thus

$$\boxed{\mathsf{A}_e = \begin{pmatrix} 1 & 1/3 & 1/5 & 1/7 \\ 0 & 2/3 & 4/7 & 10/21 \\ 0 & 0 & 8/35 & 24/77 \\ 0 & 0 & 0 & 16/231 \end{pmatrix}}.$$

The diagonal products are one. The six upper off-diagonal entries of $\mathsf{A}_e\mathsf{B}_e$ are, explicitly,

$$-\tfrac{1}{2} + \tfrac{1}{2}, \quad \tfrac{3}{8} - \tfrac{5}{4} + \tfrac{7}{8}, \quad -\tfrac{5}{16} + \tfrac{35}{16} - \tfrac{63}{16} + \tfrac{33}{16},$$
$$-\tfrac{5}{2} + \tfrac{5}{2}, \quad \tfrac{35}{8} - \tfrac{45}{4} + \tfrac{55}{8}, \quad -\tfrac{9}{2} + \tfrac{9}{2}.$$

All vanish, and entries below the diagonal vanish by triangularity. Hence $\mathsf{A}_e\mathsf{B}_e = 1$.

For the odd case use the order $1, 3, 5, 7$. The same coefficient comparison and back substitution give

$$\boxed{\mathsf{B}_o = \begin{pmatrix} 1 & -3/2 & 15/8 & -35/16 \\ 0 & 5/2 & -35/4 & 315/16 \\ 0 & 0 & 63/8 & -693/16 \\ 0 & 0 & 0 & 429/16 \end{pmatrix}}, \qquad \boxed{\mathsf{A}_o = \begin{pmatrix} 1 & 3/5 & 3/7 & 1/3 \\ 0 & 2/5 & 4/9 & 14/33 \\ 0 & 0 & 8/63 & 8/39 \\ 0 & 0 & 0 & 16/429 \end{pmatrix}}.$$

Here the six off-diagonal products are

$$-\tfrac{3}{2} + \tfrac{3}{2}, \quad \tfrac{15}{8} - \tfrac{21}{4} + \tfrac{27}{8}, \quad -\tfrac{35}{16} + \tfrac{189}{16} - \tfrac{297}{16} + \tfrac{143}{16},$$
$$-\tfrac{7}{2} + \tfrac{7}{2}, \quad \tfrac{63}{8} - \tfrac{77}{4} + \tfrac{91}{8}, \quad -\tfrac{11}{2} + \tfrac{11}{2},$$

again zero, with diagonal entries one. Therefore $\mathsf{A}_o\mathsf{B}_o = 1$. In both cases $\mathbf{a} = \mathsf{A}\boldsymbol{\alpha}$ and $\boldsymbol{\alpha} = \mathsf{B}\mathbf{a}$.

15.1.6 By differentiating the generating function $g(x,t)$ with respect to t, multiplying by $2t$, and then adding $g(x,t)$, show that inside the generating function's Taylor disk (in particular, for $-1 \le x \le 1$ and $|t| < 1$)

$$\frac{1-t^2}{(1-2tx+t^2)^{3/2}} = \sum_{n=0}^{\infty}(2n+1)P_n(x)t^n.$$

This result is useful in calculating the charge induced on a grounded metal sphere by a nearby point charge.

Solution.

Write $D = 1 - 2xt + t^2$ and $g = D^{-1/2}$. Then

$$g_t = -\frac{1}{2}D^{-3/2}(-2x + 2t) = (x - t)D^{-3/2}.$$

Consequently

$$2tg_t + g = \frac{2t(x-t) + D}{D^{3/2}} = \frac{1-t^2}{(1-2xt+t^2)^{3/2}}.$$

Termwise differentiation of the generating series in its convergence disk gives $2tg_t + g = \sum_{n\ge0}(2n + 1)P_n(x)t^n$. Equating the two expressions proves the identity. In particular, for real $-1 \le x \le 1$ it holds for $|t| < 1$.

15.1.7

(a) Derive Eq. (15.26),

$$(1-x^2)P_n'(x) = (n+1)xP_n(x) - (n+1)P_{n+1}(x).$$

(b) Write out the relation of Eq. (15.26) to preceding equations in symbolic form analogous to the symbolic forms for Eqs. (15.22) to (15.25).

Solution.

(a) Begin with Eq. (15.25) and eliminate nP_{n-1} using the three-term recurrence:

$$nP_{n-1} = (2n+1)xP_n - (n+1)P_{n+1}.$$

It follows that

$$(1-x^2)P_n' = [(2n+1)-n]xP_n - (n+1)P_{n+1} = \boxed{(n+1)(xP_n - P_{n+1})}.$$

This elimination uses $n \geq 1$. For $n = 0$, $P_0 = 1$ and $P_1 = x$ give $0 = x - x$ directly, so no P_{-1} is required. The manual drops the factor x from $(n+1)xP_n$ in its last line for part (a), p. 233. The textbook equation and the derivation above retain it. (b) To express the result in the same symbolic style as the source, use $(15.24)_{n\mapsto n+1}$ and $x(15.23)$:

$$P_n' = -(n+1)P_{n+1} + xP_{n+1}', \qquad xP_{n+1}' = (n+1)xP_n + x^2P_n'.$$

Substitute the second equation into the first and collect P_n'. Thus the indicated combination is

$$\boxed{(15.24)_{n\mapsto n+1} + x(15.23) \implies (15.26)},$$

where the notation means eliminating their common xP_{n+1}' term.

15.1.8 Prove that

$$P_n'(1) = \frac{d}{dx}P_n(x)\Big|_{x=1} = \frac{1}{2}n(n+1).$$

Solution.

Since P_n is a polynomial, substitute $x = 1$ directly into its ODE:

$$(1-1)P_n''(1) - 2P_n'(1) + n(n+1)P_n(1) = 0.$$

The generating function at $x = 1$ is $(1-t)^{-1} = \sum_{n\geq 0} t^n$, so $P_n(1) = 1$. Therefore

$$\boxed{P_n'(1) = \frac{n(n+1)}{2}}.$$

There is no singular limiting derivative here: the factor $1-x^2$ multiplies a finite polynomial derivative. The formula also gives $P_0'(1) = 0$.

15.1.9 Show that $P_n(\cos\theta) = (-1)^n P_n(-\cos\theta)$ by use of the recurrence relation relating P_n, P_{n+1}, and P_{n-1} and your knowledge of P_0 and P_1.

Solution.

The base cases are $P_0(-x) = 1 = P_0(x)$ and $P_1(-x) = -x = -P_1(x)$. Assume $P_n(-x) = (-1)^n P_n(x)$ and $P_{n-1}(-x) = (-1)^{n-1} P_{n-1}(x)$. Evaluate the three-term recurrence at $-x$:

$$\begin{aligned}(n+1)P_{n+1}(-x) &= -(2n+1)x(-1)^n P_n(x) - n(-1)^{n-1}P_{n-1}(x)\\ &= (-1)^{n+1}[(2n+1)xP_n(x) - nP_{n-1}(x)]\\ &= (-1)^{n+1}(n+1)P_{n+1}(x).\end{aligned}$$

Divide by $n+1$ and apply induction. Thus $P_n(-x) = (-1)^n P_n(x)$ for every $n \geq 0$. Putting $x = \cos\theta$ gives the requested form.

15.1.10 From Eq. (15.27) write out the coefficient of t^2 in terms of $\cos n\theta$, $n \leq 2$. This coefficient is $P_2(\cos\theta)$.

Solution.

Equation (15.27) factors the generating function as

$$(1 - te^{i\theta})^{-1/2}(1 - te^{-i\theta})^{-1/2}.$$

The coefficient of t^2 receives exactly three contributions:

$$\frac{3}{8}e^{2i\theta} + \left(\frac{1}{2}e^{i\theta}\right)\left(\frac{1}{2}e^{-i\theta}\right) + \frac{3}{8}e^{-2i\theta}.$$

Since $e^{2i\theta} + e^{-2i\theta} = 2\cos 2\theta$,

$$\boxed{P_2(\cos\theta) = \frac{1}{4} + \frac{3}{4}\cos 2\theta = \frac{1}{2}(3\cos^2\theta - 1)}.$$

The only angular frequencies are 0 and 2.

15.1.11 Derive, for integers $n \geq 1$, the recurrence relation

$$(1 - x^2)P_n'(x) = nP_{n-1}(x) - nxP_n(x)$$

from the Legendre polynomial generating function.

Solution.

Let $g = (1 - 2xt + t^2)^{-1/2} = \sum P_n(x)t^n$. The generating series of the proposed right-hand side is

$$\begin{aligned}\sum_{n\geq 1}(nP_{n-1} - nxP_n)t^n &= t\sum_{j\geq 0}(j+1)P_j t^j - xtg_t\\ &= tg + t(t-x)g_t.\end{aligned}$$

On the other hand, $g_x = tD^{-3/2}$ and $g_t = (x-t)D^{-3/2}$, so

$$tg + t(t-x)g_t = \frac{t[D - (x-t)^2]}{D^{3/2}} = \frac{t(1-x^2)}{D^{3/2}} = (1-x^2)g_x.$$

Compare coefficients of t^n:

$$\boxed{(1-x^2)P_n' = nP_{n-1} - nxP_n}, \qquad n \geq 1.$$

The $n = 0$ coefficient is the separate identity $P_0' = 0$.

15.1.12 Evaluate $\displaystyle\int_0^1 P_n(x)\,dx$.

ANS. $n = 2s$, 1 for $s = 0$, 0 for $s > 0$;
$n = 2s+1$, $P_{2s}(0)/(2s+2) = (-1)^s(2s-1)!!/(2s+2)!!$.

Hint. Use a recurrence relation to replace $P_n(x)$ by derivatives and then integrate by inspection. Alternatively, you can integrate the generating function.

Solution.

For $n \geq 1$, use $P'_{n+1} - P'_{n-1} = (2n+1)P_n$ and integrate:

$$I_n := \int_0^1 P_n(x)dx = \frac{[P_{n+1} - P_{n-1}]_0^1}{2n+1} = \frac{P_{n-1}(0) - P_{n+1}(0)}{2n+1},$$

because both endpoint values at one equal one. If $n = 2s > 0$, both polynomials in the numerator are odd and vanish at zero, so $I_{2s} = 0$. Separately, $I_0 = 1$. If $n = 2s+1$, the recurrence at $x = 0$ gives

$$P_{2s+2}(0) = -\frac{2s+1}{2s+2}P_{2s}(0).$$

Thus $I_{2s+1} = P_{2s}(0)/(2s+2)$. Starting with $P_0(0) = 1$ and repeating the same recurrence yields $P_{2s}(0) = (-1)^s(2s-1)!!/(2s)!!$, whence

$$\boxed{I_{2s+1} = (-1)^s\frac{(2s-1)!!}{(2s+2)!!}}.$$

Use $(-1)!! = 0!! = 1$; at $s = 0$ the formula gives $I_1 = 1/2$. The denominator shown in dark red in the answer applies the published correction for textbook p. 723 in the Instructor's Manual errata dated 6 April 2012.

15.1.13 Show that each term in the summation

$$\sum_{r=[n/2]+1}^{n} \left(\frac{d}{dx}\right)^n \frac{(-1)^r n!}{r!\,(n-r)!}\,x^{2n-2r}$$

vanishes (r and n integral). Here $[n/2]$ is the largest integer $\leq n/2$.

Solution.

For every integer $r \geq \lfloor n/2 \rfloor + 1$, one has $2r > n$, so the exponent $d = 2n - 2r$ satisfies $0 \leq d < n$ when $r \leq n$. The derivative of a monomial of degree d is zero after $d+1$ derivatives, and hence

$$D^n x^{2n-2r} = 0.$$

The scalar factorial coefficient cannot change this result. Therefore each individual summand vanishes, not only their sum. For $n = 0$ the indicated index set is empty, with sum zero.

15.1.14 Show that $\displaystyle\int_{-1}^1 x^m P_n(x)\,dx = 0$ when m, n are integers with $0 \leq m < n$.

Hint. Use Rodrigues formula or expand x^m in Legendre polynomials.

Solution.

Take nonnegative integers $m < n$. Rodrigues's formula gives

$$I = \frac{1}{2^n n!}\int_{-1}^1 x^m D^n(x^2-1)^n\,dx.$$

The polynomial $(x^2-1)^n = (x-1)^n(x+1)^n$ and all its derivatives of orders $0,\dots,n-1$ vanish at both endpoints. Every boundary term from n integrations by parts is therefore zero. Moving all derivatives onto x^m leaves

$$I = \frac{(-1)^n}{2^n n!}\int_{-1}^{1}(D^n x^m)(x^2-1)^n\,dx = 0,$$

because $D^n x^m = 0$. This proves the required orthogonality.

15.1.15 Show that

$$\int_{-1}^{1} x^n P_n(x)\,dx = \frac{2n!}{(2n+1)!!}.$$

Note. You are expected to use the Rodrigues formula and integrate by parts, but also see if you can get the result from Eq. (15.14) by inspection.

Solution.

The supplied transcription had $2^{n+1}n!/(2n+1)!!$. The actual textbook p. 724 has $2n!/(2n+1)!!$, which is the correct value; for example the integral at $n=1$ is $2/3$, not $4/3$. Move the n Rodrigues derivatives onto x^n as in Exercise 15.1.14:

$$I_n = \frac{(-1)^n}{2^n n!}\int_{-1}^{1} n!(x^2-1)^n\,dx = \frac{1}{2^n}\int_{-1}^{1}(1-x^2)^n\,dx.$$

Put $u = x^2$ on the positive half of the even integral:

$$\int_{-1}^{1}(1-x^2)^n dx = \int_0^1 u^{-1/2}(1-u)^n\,du = \frac{\Gamma(1/2)n!}{\Gamma(n+3/2)}.$$

Using $\Gamma(n+3/2) = (2n+1)!!\sqrt{\pi}/2^{n+1}$ yields

$$\boxed{I_n = \frac{2n!}{(2n+1)!!} = \frac{2^{n+1}(n!)^2}{(2n+1)!}}.$$

For the alternative inspection argument, the leading coefficient of P_n is $(2n-1)!!/n!$, so $x^n = n!P_n/(2n-1)!!+$ a polynomial of degree below n. Orthogonality and $\int P_n^2 = 2/(2n+1)$ give the same result.

15.1.16 Show that

$$\int_{-1}^{1} x^{2r}P_{2n}(x)\,dx = \frac{2^{2n+1}(2r)!\,(r+n)!}{(2r+2n+1)!\,(r-n)!}, \qquad r,n\in\mathbb{Z},\quad r\ge n\ge 0.$$

Solution.

For integers $r \ge n \ge 0$, apply Rodrigues to P_{2n} and integrate by parts $2n$ times. The endpoint terms vanish because $(x^2-1)^{2n}$ has zeros of order $2n$ at both ends:

$$I = \frac{(2r)!}{2^{2n}(2n)!(2r-2n)!}\int_{-1}^{1} x^{2r-2n}(1-x^2)^{2n}dx.$$

With $u = x^2$, the even integral becomes

$$B(r-n+\tfrac{1}{2},2n+1) = \frac{\Gamma(r-n+1/2)(2n)!}{\Gamma(r+n+3/2)}.$$

For a nonnegative integer j, $\Gamma(j+1/2) = (2j)!\sqrt{\pi}/(4^j j!)$. Substitution gives

$$I = \frac{(2r)!}{2^{2n}(2r-2n)!}\frac{(2r-2n)!\sqrt{\pi}}{4^{r-n}(r-n)!}\frac{4^{r+n+1}(r+n+1)!}{(2r+2n+2)!\sqrt{\pi}}$$
$$= \boxed{\frac{2^{2n+1}(2r)!(r+n)!}{(2r+2n+1)!(r-n)!}}.$$

The last step cancels $2(r+n+1)$ against the last factorial factor.

15.1.17 As a generalization of Exercises 15.1.15 and 15.1.16, show that, for nonnegative integers r, the Legendre expansions of x^s are

(a) $x^{2r} = \displaystyle\sum_{n=0}^{r} \frac{2^{2n}(4n+1)(2r)!\,(r+n)!}{(2r+2n+1)!\,(r-n)!}\,P_{2n}(x), \quad s = 2r,$

(b) $x^{2r+1} = \displaystyle\sum_{n=0}^{r} \frac{2^{2n+1}(4n+3)(2r+1)!\,(r+n+1)!}{(2r+2n+3)!\,(r-n)!}\,P_{2n+1}(x), \quad s = 2r+1.$

Solution.

For a polynomial of degree s, the expansion terminates at P_s, because the nonzero leading coefficients of $P_0, \ldots, P_s$ make them a basis for polynomials of degree at most s. Orthogonality gives

$$a_j = \frac{2j+1}{2}\int_{-1}^{1} x^s P_j(x)dx.$$

Parity eliminates all j of parity opposite to s.

(a) With $s = 2r$, $j = 2n$, use Exercise 15.1.16 and multiply by $(4n+1)/2$:

$$\boxed{x^{2r} = \sum_{n=0}^{r} \frac{2^{2n}(4n+1)(2r)!(r+n)!}{(2r+2n+1)!(r-n)!}P_{2n}(x)}.$$

(b) For clarity, evaluate the odd moment separately. Moving $2n+1$ Rodrigues derivatives onto x^{2r+1} gives

$$\int_{-1}^{1} x^{2r+1}P_{2n+1}(x)dx = \frac{(2r+1)!}{2^{2n+1}(2n+1)!(2r-2n)!}B(r-n+\tfrac{1}{2}, 2n+2).$$

The two signs from integration by parts and $(x^2-1)^{2n+1}$ cancel. Evaluating the beta function by the same half-integer gamma formula as before gives

$$\int_{-1}^{1} x^{2r+1}P_{2n+1}(x)dx = \frac{2^{2n+2}(2r+1)!(r+n+1)!}{(2r+2n+3)!(r-n)!}.$$

Multiplication by $(4n+3)/2$ therefore yields

$$\boxed{x^{2r+1} = \sum_{n=0}^{r} \frac{2^{2n+1}(4n+3)(2r+1)!(r+n+1)!}{(2r+2n+3)!(r-n)!}P_{2n+1}(x)}.$$

15.1.18 In numerical work (for e.g., the Gauss-Legendre quadrature), it is useful to establish that $P_n(x)$ has n real zeros in the interior of $[-1, 1]$. Show that this is so.

Hint. Rolle's theorem shows that the first derivative of $(x^2 - 1)^n$ has one zero in the interior of $[-1, 1]$. Extend this argument to the second, third, and ultimately the nth derivative.

Solution.

To connect the proposed Rolle argument with Rodrigues's formula, start with $(x^2 - 1)^n$, not the printed $(x^2 - 1)^{2n}$. The nth derivative of the latter is not proportional to P_n. Let $F(x) = (x^2 - 1)^n$, $n \geq 1$. Its endpoints are zeros of multiplicity n. We prove inductively that $F^{(k)}$ has at least k distinct zeros strictly between -1 and 1, for $0 \leq k \leq n$. The assertion for $k = 0$ is empty. If it holds for $k - 1 < n$, then $F^{(k-1)}$ still vanishes at both endpoints, because their multiplicities are $n - k + 1 \geq 1$. Its $k - 1$ interior zeros, together with the two endpoints, determine k disjoint open intervals. Rolle's theorem supplies a zero of $F^{(k)}$ in each interval. At $k = n$, Rodrigues gives $F^{(n)} = 2^n n! P_n$, so P_n has at least n distinct interior zeros. Its degree is exactly n, so these are all its zeros and each is simple. For $n = 0$ the statement means no zeros, consistent with $P_0 = 1$.

15.2 Orthogonality

15.2.1 Using a Rodrigues formula, show that the $P_n(x)$ are orthogonal and that

$$\int_{-1}^{1} [P_n(x)]^2 \, dx = \frac{2}{2n + 1}.$$

Hint. Integrate by parts.

Solution.

Suppose $m < n$. Rodrigues's formula and n integrations by parts give

$$\int_{-1}^{1} P_m P_n \, dx = \frac{(-1)^n}{2^n n!} \int_{-1}^{1} P_m^{(n)}(x)(x^2 - 1)^n \, dx = 0.$$

Every boundary term vanishes because $(x^2 - 1)^n$ and its first $n - 1$ derivatives vanish at ± 1; the remaining derivative of P_m is zero. Symmetry covers $m > n$.

For $m = n$, the leading coefficient $c_n = (2n)!/[2^n (n!)^2]$ gives $P_n^{(n)} = n! c_n$. The same integrations by parts yield

$$\int_{-1}^{1} P_n^2 dx = \frac{c_n}{2^n} \int_{-1}^{1} (1 - x^2)^n dx.$$

The beta integral in Exercise 15.1.15 is $2^{2n+1}(n!)^2/(2n + 1)!$. Therefore

$$\int_{-1}^{1} P_n^2 dx = \frac{(2n)!}{2^{2n}(n!)^2} \frac{2^{2n+1}(n!)^2}{(2n + 1)!} = \boxed{\frac{2}{2n + 1}}.$$

Together these establish $\int P_m P_n = 2\delta_{mn}/(2n + 1)$.

15.2.2 You have constructed a set of orthogonal functions by the Gram-Schmidt process (Section 5.2), taking $u_n(x) = x^n$, $n = 0, 1, 2, \ldots$, in increasing order with $w(x) = 1$ and an interval $-1 \leq x \leq 1$. Prove that the nth such function constructed in this way is proportional to $P_n(x)$.

Hint. Use mathematical induction (Section 1.4).

Solution.

Without normalization, let the Gram–Schmidt output be

$$g_n = x^n - \sum_{j=0}^{n-1} \frac{\langle g_j, x^n \rangle}{\langle g_j, g_j \rangle} g_j.$$

Its leading coefficient is one, since every subtracted polynomial has degree below n. For $n = 0$, $g_0 = 1 = P_0$. Assume the earlier g_j are proportional to P_j. Both g_n and P_n are orthogonal to all polynomials of degree less than n: this follows from the construction for g_n and from Rodrigues's formula for P_n. Put $c_n = (2n)!/[2^n(n!)^2]$, the leading coefficient of P_n. Then $h = g_n - P_n/c_n$ has degree below n and is orthogonal to all such polynomials, including itself. Thus $\int_{-1}^{1} h(x)^2 dx = 0$, so the continuous polynomial h is zero. Consequently

$$\boxed{g_n = \frac{2^n (n!)^2}{(2n)!} P_n}.$$

Any optional nonzero normalization in Gram–Schmidt changes only this scalar factor. This completes the induction.

15.2.3 Expand the Dirac delta function $\delta(x)$ in a series of Legendre polynomials using the interval $-1 \le x \le 1$, with convergence understood in the distributional sense.

Solution.

The coefficient of P_n is obtained by pairing the delta distribution with the polynomial:

$$a_n = \frac{2n+1}{2} \langle \delta, P_n \rangle = \frac{2n+1}{2} P_n(0).$$

Odd n vanish. For $n = 2s$, use $P_{2s}(0) = (-1)^s (2s-1)!!/(2s)!!$:

$$\boxed{\delta(x) = \sum_{s=0}^{\infty} (-1)^s \frac{4s+1}{2} \frac{(2s-1)!!}{(2s)!!} P_{2s}(x)}.$$

This is a distributional expansion, not a pointwise function series for the delta distribution. To verify it, let ϕ be smooth on $[-1, 1]$, with Legendre coefficients $c_n = (2n+1) \int \phi P_n/2$. Pairing the proposed series with ϕ gives $\sum c_n P_n(0) = \phi(0)$. Here is the convergence justification. Set $L = -D[(1-x^2)D]$, so $LP_n = n(n+1)P_n$. For $n \ge 1$, transfer L from P_n to ϕ by two integrations by parts. Both boundary terms contain $1 - x^2$ and vanish. Repeating this k times gives

$$c_n = \frac{2n+1}{2[n(n+1)]^k} \int_{-1}^{1} (L^k \phi)(x) P_n(x)\, dx.$$

Since $|P_n(x)| \le 1$ on this interval, $|c_n| \le (2n+1)\|L^k \phi\|_\infty / [n(n+1)]^k = O(n^{1-2k})$. Taking $k = 2$ makes $\sum |c_n|$ converge, so $\sum c_n P_n$ converges uniformly. Its L^2 limit is ϕ by polynomial completeness, and continuity then gives equality at every point. Evaluation at zero therefore proves the required delta action.

15.2.4 Verify the Dirac delta function expansions as distributions on $[-1,1]$

$$\delta(1-x)=\sum_{n=0}^{\infty}\frac{2n+1}{2}P_n(x),$$
$$\delta(1+x)=\sum_{n=0}^{\infty}(-1)^n\frac{2n+1}{2}P_n(x).$$

These expressions appear in a resolution of the Rayleigh plane wave expansion (Exercise 15.2.24) into incoming and outgoing spherical waves.

Note. Assume that the entire Dirac delta function is covered when integrating over $[-1,1]$.

Solution.

Use the full endpoint-mass convention specified in the question. Orthogonal projection gives

$$a_n^{(+)}=\frac{2n+1}{2}P_n(1)=\frac{2n+1}{2},\qquad a_n^{(-)}=\frac{2n+1}{2}P_n(-1)=(-1)^n\frac{2n+1}{2}.$$

Therefore the two proposed expansions follow. For a direct distributional verification, expand a smooth test function as $\phi=\sum c_nP_n$. Pairing the first series with ϕ gives $\sum c_n=\phi(1)$, while pairing the second gives $\sum(-1)^nc_n=\phi(-1)$. These are precisely the actions of $\delta(1-x)$ and $\delta(1+x)$, respectively. The uniform convergence proved in Exercise 15.2.3 justifies both endpoint evaluations. The endpoint convention matters: inserting a half-weight convention would change both coefficients by a factor of two. These are distributional identities, with no assertion of pointwise convergence.

15.2.5 Neutrons (mass 1) are being scattered by a nucleus of mass A ($A>1$). In the center-of-mass system the scattering is isotropic. Then, in the laboratory system the average of the cosine of the angle of deflection of the neutron is

$$\langle\cos\psi\rangle=\frac{1}{2}\int_0^{\pi}\frac{A\cos\theta+1}{(A^2+2A\cos\theta+1)^{1/2}}\sin\theta\,d\theta.$$

Show, by expansion of the denominator, that $\langle\cos\psi\rangle=2/(3A)$.

Solution.

Set $x=\cos\theta$, so $\sin\theta\,d\theta=-dx$, and factor out $A>1$ from the square root:

$$\langle\cos\psi\rangle=\frac{1}{2}\int_{-1}^{1}(x+A^{-1})(1+2x/A+A^{-2})^{-1/2}dx.$$

The generating function at $t=-1/A$ gives

$$(1+2x/A+A^{-2})^{-1/2}=\sum_{n=0}^{\infty}(-A^{-1})^nP_n(x).$$

The series converges uniformly on $[-1,1]$, so it may be integrated termwise. As $1=P_0$ and $x=P_1$, orthogonality kills every term except $n=0$ in the constant part and $n=1$ in the x part:

$$\langle\cos\psi\rangle=\frac{1}{2}\left[\frac{1}{A}\int_{-1}^{1}P_0^2dx-\frac{1}{A}\int_{-1}^{1}P_1^2dx\right]=\frac{1}{2}\left(\frac{2}{A}-\frac{2}{3A}\right)=\boxed{\frac{2}{3A}}.$$

15.2.6 A particular function $f \in L^2[-1,1]$ is expanded in a Legendre series over this same interval, converging in L^2. Show that the expansion is unique.

Solution.

In the usual $L^2[-1,1]$ interpretation of a Legendre expansion, suppose $f = \sum a_n P_n = \sum b_n P_n$, both sums converging in L^2. Taking the inner product with a fixed P_m is continuous in that norm, by the Cauchy–Schwarz inequality. Hence

$$\int_{-1}^{1} f P_m dx = \frac{2a_m}{2m+1} = \frac{2b_m}{2m+1},$$

so $a_m = b_m$ for every m. The same proof applies to any stronger convergence that permits these integrations. Thus the unique coefficients are

$$\boxed{a_m = \frac{2m+1}{2} \int_{-1}^{1} f(x) P_m(x) dx}.$$

A convergence class is needed to justify extracting a coefficient from an infinite series; the argument above uses the standard square-integrable Fourier–Legendre setting.

15.2.7 A real function $f \in L^2[-1,1]$ is expanded in its L^2-convergent Legendre series $f(x) = \sum_{n=0}^{\infty} a_n P_n(x)$. Show that

$$\int_{-1}^{1} [f(x)]^2 \, dx = \sum_{n=0}^{\infty} \frac{2a_n^2}{2n+1}.$$

This is a statement that the Legendre polynomials form a complete set.

Solution.

For a finite partial sum $S_N = \sum_{n=0}^{N} a_n P_n$, orthogonality gives, with the cross terms written explicitly,

$$\int_{-1}^{1} S_N^2 dx = \sum_{n,m=0}^{N} a_n a_m \int_{-1}^{1} P_n P_m dx = \sum_{n=0}^{N} \frac{2a_n^2}{2n+1}.$$

For a real square-integrable f, its Legendre projections converge in L^2. One way to establish completeness is that polynomials approximate continuous functions uniformly on the compact interval, continuous functions are dense in L^2, and the P_n span all polynomials. Moreover,

$$\left| \int (f^2 - S_N^2) \right| \leq \|f - S_N\|_2 (\|f\|_2 + \|S_N\|_2) \longrightarrow 0.$$

Taking the limit proves

$$\boxed{\int_{-1}^{1} f^2 dx = \sum_{n=0}^{\infty} \frac{2a_n^2}{2n+1}}.$$

The norm identity for complex f uses $|f|^2$ and $|a_n|^2$. The printed bilinear identity with squares also holds for complex L^2 functions, but is then not the nonnegative norm formula.

15.2.8

(a) For

$$f(x) = \begin{cases} +1, & 0 < x < 1, \\ -1, & -1 < x < 0, \end{cases}$$

show that

$$\int_{-1}^{1} [f(x)]^2 \, dx = 2 \sum_{n=0}^{\infty} (4n+3) \left[\frac{(2n-1)!!}{(2n+2)!!} \right]^2 .$$

(b) By testing the series, prove that it is convergent.

(c) The value of the integral in part (a) is 2. Check the rate at which the series converges by summing its first 10 terms.

Solution.

(a) The function is odd, so only odd Legendre coefficients occur. For $j = 2n+1$,

$$a_{2n+1} = \frac{4n+3}{2} \int_{-1}^{1} \operatorname{sgn}(x) P_{2n+1}(x) dx = (4n+3) \int_{0}^{1} P_{2n+1}(x) dx.$$

Exercise 15.1.12 gives $a_{2n+1} = (-1)^n (4n+3)(2n-1)!!/(2n+2)!!$. Insert this in Parseval:

$$\int_{-1}^{1} f^2 dx = 2 \sum_{n=0}^{\infty} (4n+3) \left[\frac{(2n-1)!!}{(2n+2)!!} \right]^2 .$$

(b) Write

$$\frac{(2n-1)!!}{(2n+2)!!} = \frac{1}{2(n+1)} \frac{\binom{2n}{n}}{4^n}.$$

Stirling's formula gives $\binom{2n}{n}/4^n \sim (\pi n)^{-1/2}$. Thus the entire summand, including its leading factor 2, is asymptotic to $2/(\pi n^2)$. Comparison with $\sum n^{-2}$ proves convergence independently of invoking Parseval. It also predicts a tail of order $1/N$.

(c) The first ten terms ($n = 0, \ldots, 9$), evaluated by exact rational arithmetic, give

$$\boxed{S_{10} = \frac{66586053015}{34359738368} = 1.9379091977316420525\ldots}.$$

The missing amount is $2 - S_{10} = 0.062090802268357 9475\ldots$, or about 3.105% of the integral. The slow $1/N$ tail is visible even though all terms are positive. The manual, p. 237, prints 1.943 . . . for ten terms. That is instead the eleven-term value $S_{11} = 267102370903/137438953472 = 1.943425529338128\ldots$. The requested ten terms run from $n = 0$ through $n = 9$.

15.2.9 Prove that

$$\int_{-1}^{1} x(1-x^2) P_n' P_m' \, dx = \frac{2n(n^2-1)}{4n^2-1} \, \delta_{m,n-1} + \frac{2n(n+2)(n+1)}{(2n+1)(2n+3)} \, \delta_{m,n+1}.$$

Solution.

Let I_{nm} denote the integral. The boundary term vanishes because $x(1-x^2)P_n' P_m$ is zero at $x = \pm 1$. Integrating by parts gives

$$I_{nm} = -\int_{-1}^{1} P_m \left[(1-x^2) P_n' + x\{(1-x^2)P_n'\}' \right] dx.$$

The ODE says $\{(1-x^2)P_n'\}' = -n(n+1)P_n$. Also $(1-x^2)P_n' = nP_{n-1} - nxP_n$. Consequently

$$I_{nm} = \int_{-1}^{1} P_m[n(n+2)xP_n - nP_{n-1}]dx.$$

Replace xP_n by $[(n+1)P_{n+1} + nP_{n-1}]/(2n+1)$. The coefficient of P_{n-1} becomes $n[n(n+2) - (2n+1)]/(2n+1) = n(n^2-1)/(2n+1)$, and that of P_{n+1} is $n(n+1)(n+2)/(2n+1)$. Orthogonality now yields

$$\boxed{I_{nm} = \frac{2n(n^2-1)}{(2n-1)(2n+1)}\delta_{m,n-1} + \frac{2n(n+1)(n+2)}{(2n+1)(2n+3)}\delta_{m,n+1}}.$$

For $n = 0$ the original derivative is zero; for $n = 1$ the first coefficient is zero. Thus the boundary index cases are included.

15.2.10 The coincidence counting rate, $W(\theta)$, in a gamma-gamma angular correlation experiment has the form

$$W(\theta) = \sum_{n=0}^{\infty} a_{2n} P_{2n}(\cos\theta).$$

Show that data in the range $\pi/2 \le \theta \le \pi$ can, in principle, define the function $W(\theta)$ (and permit a determination of the coefficients a_{2n}). This means that although data in the range $0 \le \theta < \pi/2$ may be useful as a check, they are not essential.

Solution.

Every P_{2n} is even, so with $x = \cos\theta$ the stated expansion implies $W(-x) = W(x)$, equivalently $W(\pi - \theta) = W(\theta)$. Data for $\pi/2 \le \theta \le \pi$ give $W(x)$ on $[-1, 0]$; reflection supplies its values on $[0, 1]$. The coefficient formula makes this explicit:

$$a_{2n} = \frac{4n+1}{2}\int_{-1}^{1} W(x)P_{2n}(x)dx = (4n+1)\int_{-1}^{0} W(x)P_{2n}(x)dx.$$

Changing back to θ gives

$$\boxed{a_{2n} = (4n+1)\int_{\pi/2}^{\pi} W(\theta)P_{2n}(\cos\theta)\sin\theta\, d\theta}.$$

Thus ideal knowledge on the stated half-range fixes every coefficient and the full even function. The other half-range tests the assumed even symmetry and improves experimental precision.

15.2.11 A conducting sphere of radius r_0 is placed in an initially uniform electric field, E_0. Show the following:

(a) The induced surface charge density is $\sigma = 3\varepsilon_0 E_0 \cos\theta$,

(b) The induced electric dipole moment is $P = 4\pi r_0^3 \varepsilon_0 E_0$.

Note. The induced electric dipole moment can be calculated either from the surface charge [part (a)] or by noting that the final electric field E is the result of superimposing a dipole field on the original uniform field.

Solution.

Choose the z axis along $\mathbf{E}_0$ and take the initially uncharged conductor. The imposed potential is $-E_0 r\cos\theta$. The decaying dipole harmonic is $A\cos\theta/r^2$, so outside the sphere

$$V = -E_0 r\cos\theta + \frac{A}{r^2}\cos\theta.$$

A conducting surface is equipotential. Its $\cos\theta$ coefficient must vanish at $r = r_0$, giving $A = E_0 r_0^3$. Therefore

$$E_r = -\partial_r V = E_0\left(1 + \frac{2r_0^3}{r^3}\right)\cos\theta.$$

The field inside the conductor is zero, so the normal jump condition gives

$$\boxed{\sigma = \varepsilon_0 E_r(r_0) = 3\varepsilon_0 E_0\cos\theta}.$$

It integrates to zero, as required for induced charge on a neutral sphere. The dipole moment has only a z component by axial symmetry:

$$\begin{aligned} p_z &= \int_{\text{surface}} (r_0\cos\theta)\sigma\, dA \\ &= 6\pi\varepsilon_0 E_0 r_0^3 \int_0^\pi \cos^2\theta\sin\theta\, d\theta \\ &= 6\pi\varepsilon_0 E_0 r_0^3\left(\frac{2}{3}\right) = \boxed{4\pi\varepsilon_0 E_0 r_0^3}. \end{aligned}$$

The coefficient of the induced potential independently gives the same answer, since a dipole has $V_{\text{dip}} = p_z\cos\theta/(4\pi\varepsilon_0 r^2)$.

15.2.12 Obtain as a Legendre expansion the electrostatic potential of the circular ring of Example 15.2.3, for points (r,θ) with $r < a$.

Solution.

Let $C = q/(4\pi\varepsilon_0)$ and a be the ring radius, as in the source example. Inside the sphere $r < a$ there is no charge, and the axisymmetric potential is regular at the origin. Hence it has the form $\psi = \sum_{n\ge 0} b_n r^n P_n(\cos\theta)$. On the positive axis every ring element is the same distance away:

$$\psi(z,0) = \frac{C}{\sqrt{a^2+z^2}} = \frac{C}{a}\sum_{s=0}^{\infty}(-1)^s\frac{(2s-1)!!}{(2s)!!}\left(\frac{z}{a}\right)^{2s}.$$

Since $P_n(1) = 1$, equality of these power-series coefficients gives $b_{2s} = C(-1)^s(2s-1)!!/[a^{2s+1}(2s)!!]$ and $b_{2s+1} = 0$. Therefore

$$\boxed{\psi(r,\theta) = \frac{q}{4\pi\varepsilon_0 a}\sum_{s=0}^{\infty} P_{2s}(0)\left(\frac{r}{a}\right)^{2s} P_{2s}(\cos\theta)}, \qquad r < a.$$

The series converges uniformly on every $r \le \rho < a$, and gives $\psi(0) = C/a$. Its axis values and regular solid harmonics determine all coefficients uniquely.

15.2.13 Calculate the electric field produced by the charged conducting ring of Example 15.2.3 for

(a) $r > a$, (b) $r < a$.

Solution.

Put $C = q/(4\pi\varepsilon_0)$, $c_s = P_{2s}(0)$, and $x = \cos\theta$. There is no azimuthal dependence, so $E_\varphi = 0$. For any potential $\psi(r, x)$, $E_r = -\psi_r$ and $E_\theta = (\sin\theta/r)\psi_x$.

(a) The exterior ring potential is $\psi = C\sum c_s a^{2s} r^{-2s-1} P_{2s}(x)$. Termwise differentiation gives

$$\boxed{E_r = \frac{C}{r^2}\sum_{s=0}^{\infty}(2s+1)c_s\left(\frac{a}{r}\right)^{2s} P_{2s}(x)},$$

$$\boxed{E_\theta = \frac{C\sin\theta}{r^2}\sum_{s=0}^{\infty} c_s\left(\frac{a}{r}\right)^{2s} P'_{2s}(x)}, \qquad r > a.$$

(b) Differentiate the interior result of Exercise 15.2.12:

$$\boxed{E_r = -\frac{C}{a^2}\sum_{s=1}^{\infty} 2s\, c_s\left(\frac{r}{a}\right)^{2s-1} P_{2s}(x)},$$

$$\boxed{E_\theta = \frac{C\sin\theta}{a^2}\sum_{s=1}^{\infty} c_s\left(\frac{r}{a}\right)^{2s-1} P'_{2s}(x)}, \qquad r < a.$$

The $s = 0$ term is a constant and contributes no field. Both interior components approach zero at the origin, interpreted as the single Cartesian field $\mathbf{E}(0) = 0$. Uniform convergence on compact subregions justifies the derivatives; the singular charged ring itself is excluded.

15.2.14 As an extension of Example 15.2.3, find the potential $\psi(r, \theta)$ produced by a charged conducting disk, Fig. 15.4, for $r > a$, where a is the radius of the disk. The charge density σ (on each side of the disk) is

$$\sigma(\rho) = \frac{q}{4\pi a(a^2 - \rho^2)^{1/2}}, \qquad \rho^2 = x^2 + y^2.$$

Hint. The definite integral you get can be evaluated as a beta function, Section 13.3. For more details see section 5.03 of Smythe in Additional Readings.

$$\textit{ANS. } \psi(r,\theta) = \frac{q}{4\pi\varepsilon_0 r}\sum_{l=0}^{\infty}(-1)^l\frac{1}{2l+1}\left(\frac{a}{r}\right)^{2l} P_{2l}(\cos\theta).$$

Solution.

Both faces contribute. A circular annulus of radius ρ and width $d\rho$ carries

$$dq = 2\sigma(\rho)\,2\pi\rho\,d\rho = \frac{q}{a}\frac{\rho\,d\rho}{\sqrt{a^2-\rho^2}}.$$

Its integral from zero to a is q, confirming the normalization. Apply the exterior ring expansion to each annulus:

$$\psi = \frac{1}{4\pi\varepsilon_0 r}\sum_{l=0}^{\infty}\frac{P_{2l}(0)P_{2l}(\cos\theta)}{r^{2l}}\int_0^a \rho^{2l}\,dq.$$

For $r > a$ the geometrically dominated series can be integrated termwise, even with the integrable edge singularity in σ. Using $u = \rho^2/a^2$,

$$\int_0^a \rho^{2l}\, dq = \frac{q}{a}\int_0^a \frac{\rho^{2l+1}}{\sqrt{a^2-\rho^2}}d\rho$$
$$= \frac{qa^{2l}}{2}B(l+1,\tfrac{1}{2}) = qa^{2l}\frac{2^{2l}(l!)^2}{(2l+1)!}.$$

Multiply by $P_{2l}(0) = (-1)^l(2l)!/[2^{2l}(l!)^2]$; all factors cancel except $(-1)^l/(2l+1)$. Hence

$$\boxed{\psi(r,\theta) = \frac{q}{4\pi\varepsilon_0 r}\sum_{l=0}^{\infty}\frac{(-1)^l}{2l+1}\left(\frac{a}{r}\right)^{2l}P_{2l}(\cos\theta)}.$$

15.2.15 The hemisphere defined by $r = a$, $0 \le \theta < \pi/2$, has an electrostatic potential $+V_0$. The hemisphere $r = a$, $\pi/2 < \theta \le \pi$ has an electrostatic potential $-V_0$. Show that the potential at interior points is

$$V = V_0\sum_{n=0}^{\infty}\frac{4n+3}{2n+2}\left(\frac{r}{a}\right)^{2n+1}P_{2n}(0)P_{2n+1}(\cos\theta)$$
$$= V_0\sum_{n=0}^{\infty}(-1)^n\frac{(4n+3)(2n-1)!!}{(2n+2)!!}\left(\frac{r}{a}\right)^{2n+1}P_{2n+1}(\cos\theta).$$

Hint. You need Exercise 15.1.12.

Solution.

Let $x = \cos\theta$. The boundary potential is $V(a,x) = V_0 \operatorname{sgn} x$, except for the unspecified equatorial jump. Its Legendre coefficients are those of Exercise 15.2.8:

$$b_{2n} = 0, \qquad b_{2n+1} = V_0(4n+3)\int_0^1 P_{2n+1}(x)dx = V_0\frac{4n+3}{2n+2}P_{2n}(0).$$

Regularity at the origin selects r^l, while the other radial solution r^{-l-1} is excluded. To match each boundary coefficient at $r = a$, multiply it by $(r/a)^l$. Thus

$$\boxed{V(r,\theta) = V_0\sum_{n=0}^{\infty}\frac{4n+3}{2n+2}P_{2n}(0)\left(\frac{r}{a}\right)^{2n+1}P_{2n+1}(\cos\theta)}.$$

Substituting the double-factorial value of $P_{2n}(0)$ gives the second series in the question. For $r < a$ it converges locally uniformly and is harmonic. Its boundary limit is $\pm V_0$ away from the equator and the symmetric value zero at the jump. This also shows $V(0) = 0$ and the reflection antisymmetry.

15.2.16 A conducting sphere of radius a is divided into two electrically separate hemispheres by a thin insulating barrier at its equator. The top hemisphere is maintained at a potential V_0, and the bottom hemisphere at $-V_0$. Take $V \to 0$ at infinity and the ideal limit of zero barrier width.

(a) Show that the electrostatic potential exterior to the two hemispheres is

$$V(r,\theta) = V_0\sum_{s=0}^{\infty}(-1)^s(4s+3)\frac{(2s-1)!!}{(2s+2)!!}\left(\frac{a}{r}\right)^{2s+2}P_{2s+1}(\cos\theta).$$

(b) Calculate the electric charge density σ on the outside surface. Interpret the boundary derivative by its radial Abel limit. Show that the pole densities are finite although the unregularized series diverges there; identify the equatorial singularity that produces the infinite ideal capacitance.

ANS. (b) With $\rho = a/r$, away from the equator,

$$\sigma = \varepsilon_0 \lim_{r\downarrow a} E_r = \frac{\varepsilon_0 V_0}{a} \lim_{\rho\uparrow 1} \sum_{s=0}^{\infty} (-1)^s (4s+3) \frac{(2s-1)!!}{(2s)!!} \rho^{2s+3} P_{2s+1}(\cos\theta).$$

Solution.

(a) The boundary coefficients are $b_{2s+1} = V_0(-1)^s(4s+3)(2s-1)!!/(2s+2)!!$ and the even ones are zero. The exterior solution that vanishes at infinity uses r^{-l-1}, so

$$\boxed{V = V_0 \sum_{s=0}^{\infty} (-1)^s (4s+3) \frac{(2s-1)!!}{(2s+2)!!} \left(\frac{a}{r}\right)^{2s+2} P_{2s+1}(\cos\theta)}, \quad r > a.$$

(b) Differentiate first at $r > a$, where this is legitimate:

$$E_r = \frac{V_0}{a} \sum_{s=0}^{\infty} (-1)^s (4s+3) \frac{(2s-1)!!}{(2s)!!} \left(\frac{a}{r}\right)^{2s+3} P_{2s+1}(\cos\theta).$$

Consequently

$$\boxed{\sigma(\theta) = \frac{\varepsilon_0 V_0}{a} \lim_{\rho\uparrow 1} \sum_{s=0}^{\infty} (-1)^s (4s+3) \frac{(2s-1)!!}{(2s)!!} \rho^{2s+3} P_{2s+1}(\cos\theta)}.$$

This is the radial, or Abel, boundary limit. The printed answer is missing $1/a$. Moreover, ordinary series divergence at the poles does not mean the physical charge density is infinite there. The physical singularity and infinite capacitance occur at the equatorial potential jump.

For an explicit pole check, the spherical Poisson kernel from Exercise 15.1.6 gives, on the positive axis,

$$\frac{V(r,0)}{V_0} = \frac{\rho(1-\rho^2)}{2} \int_{-1}^{1} \frac{\operatorname{sgn} u}{(1-2\rho u+\rho^2)^{3/2}} du = 1 - \frac{1-\rho^2}{\sqrt{1+\rho^2}}, \quad \rho = \frac{a}{r}.$$

The antiderivative of the integrand without the sign is $[\rho\sqrt{1-2\rho u+\rho^2}]^{-1}$, which gives the last equality. Since $-\partial_r = (\rho^2/a)\partial_\rho$, differentiation at $\rho = 1$ yields

$$\boxed{\sigma(0) = \frac{\sqrt{2}\varepsilon_0 V_0}{a}, \qquad \sigma(\pi) = -\frac{\sqrt{2}\varepsilon_0 V_0}{a}}.$$

The unsmoothed series has terms of magnitude proportional to $\sqrt{s}$ at those poles, so an ordinary termwise boundary sum cannot give these finite values.

To locate the actual divergence, let Ω be a unit vector on the north hemisphere. Split the boundary data into V_0 on the whole sphere minus $2V_0$ on the south. Its exterior Poisson integral is

$$\frac{V}{V_0} = \rho - \frac{\rho(1-\rho^2)}{2\pi} \int_{\text{south}} \frac{d\Omega'}{|\rho\Omega - \Omega'|^3}.$$

At a point strictly north of the equator the integral and its radial derivative remain finite as $\rho \uparrow 1$. The derivative of $\rho(1-\rho^2)$ is -2 there, whereas the coefficient of the derivative of the integral is zero. Using $-\partial_r = (\rho^2/a)\partial_\rho$ therefore gives

$$\sigma(\Omega) = \frac{\varepsilon_0 V_0}{a}\left[1 + \frac{1}{\pi}\int_{\text{south}} \frac{d\Omega'}{|\Omega - \Omega'|^3}\right].$$

For small angular distance $h > 0$ north of the equator, local tangential coordinates reduce the leading singular part to

$$\int_0^\infty \int_{-\infty}^\infty \frac{dX\, dY}{[X^2 + (Y+h)^2]^{3/2}} = \int_0^\infty \frac{2\, dY}{(Y+h)^2} = \frac{2}{h}.$$

To justify this local approximation, fix a small angular patch of radius δ about the closest equatorial point. The spherical area element and distance in that patch differ from their tangent-plane values by multiplicative factors tending to one as $\delta \to 0$. The integral outside the patch stays bounded for fixed δ. Inside, substitute $X = h\xi$, $Y = h\eta$: the area contributes h^2 and the denominator h^3. After multiplying by h, the patch expands to the whole half-plane and its integral tends to 2. The multiplicative bounds tend to one as $\delta \to 0$, proving

$$\sigma \sim \frac{2\varepsilon_0 V_0}{\pi a h} \qquad (h \downarrow 0).$$

An equatorial strip has area $2\pi a^2\, dh$ to leading order, so its charge grows as $4\varepsilon_0 V_0 a \int dh/h$. The hemisphere charge, and hence its ratio to the voltage difference $2V_0$, diverges logarithmically. This establishes the infinite ideal capacitance at the equator.

15.2.17 By writing $\varphi_s(x) = \sqrt{(2s+1)/2}\, P_s(x)$, a Legendre polynomial is renormalized to unity. Explain how $|\varphi_s\rangle\langle\varphi_s|$ acts as a projection operator. In particular, show that if $|f\rangle = \sum_n a'_n|\varphi_n\rangle$, then

$$|\varphi_s\rangle\langle\varphi_s|f\rangle = a'_s|\varphi_s\rangle.$$

Solution.

The normalization gives $\langle\varphi_s, \varphi_n\rangle = \delta_{sn}$. For $\Pi_s = |\varphi_s\rangle\langle\varphi_s|$,

$$\Pi_s|f\rangle = |\varphi_s\rangle \sum_n a'_n\langle\varphi_s|\varphi_n\rangle = \boxed{a'_s|\varphi_s\rangle}.$$

It is a projection because

$$\Pi_s^2 = |\varphi_s\rangle\langle\varphi_s|\varphi_s\rangle\langle\varphi_s| = \Pi_s, \qquad \Pi_s^\dagger = \Pi_s.$$

Its image is the one-dimensional span of φ_s, and the discarded remainder is orthogonal to that span: $\langle\varphi_s, (1-\Pi_s)f\rangle = 0$. In the x representation its kernel is $\Pi_s(x,y) = \varphi_s(x)\varphi_s(y)^*$.

15.2.18 Expand x^8 as a Legendre series. Determine the Legendre coefficients from Eq. (15.40),

$$a_m = \frac{2m+1}{2}\int_{-1}^1 x^8 P_m(x)\, dx.$$

Check your values against AMS-55, Table 22.9. (For the complete reference, see Additional Readings.) This illustrates the expansion of a simple function $f(x)$.

Hint. Gaussian quadrature can be used to evaluate the integral.

Solution.

Parity makes every odd coefficient zero. Since x^8 has degree eight, coefficients above a_8 also vanish by Exercise 15.1.14. For $m = 2n$, $0 \le n \le 4$, the exact moment formula gives

$$a_{2n} = \frac{4n+1}{2}\int_{-1}^{1} x^8 P_{2n}(x)\,dx = \frac{2^{2n}(4n+1)8!(4+n)!}{(2n+9)!(4-n)!}.$$

Substitute $n = 0, 1, 2, 3, 4$ and cancel the factorials:

m	0	2	4	6	8
a_m	1/9	40/99	48/143	64/495	128/6435

Thus

$$\boxed{x^8 = \frac{715P_0 + 2600P_2 + 2160P_4 + 832P_6 + 128P_8}{6435}}.$$

Expansion of the five polynomials cancels all powers below eight and leaves coefficient one on x^8, providing a separate exact check. The x^8 column of AMS-55, Table 22.9, p. 798 lists denominator 6435 and numerators $715, 2600, 2160, 832, 128$ in its P_0, P_2, P_4, P_6, P_8 rows, respectively. These agree with all five coefficients derived above.

15.2.19 Calculate and tabulate the electrostatic potential created by a ring of charge, Example 15.2.3, for $r/a = 1.5(0.5)5.0$ and $\theta = 0°(15°)90°$. Carry terms through $P_{22}(\cos\theta)$.

Note. The convergence of your series will be slow for $r/a = 1.5$. Truncating the series at P_{22} limits you to about four-significant-figure accuracy.

Check value. For $r/a = 2.5$ and $\theta = 60°$, $\psi = 0.40272(q/4\pi\varepsilon_0 a)$.

Solution.

Set $R = r/a$ and report the dimensionless potential $F = 4\pi\varepsilon_0 a\,\psi/q$. The required truncation is

$$F_{22}(R,\theta) = \sum_{s=0}^{11}(-1)^s \frac{\binom{2s}{s}}{4^s}\frac{P_{2s}(\cos\theta)}{R^{2s+1}}.$$

For each angle start $P_0 = 1$, $P_1 = \cos\theta$ and use $P_{n+1} = [(2n+1)\cos\theta\,P_n - nP_{n-1}]/(n+1)$ up to P_{22}. Summing the displayed twelve terms at 65-digit working precision gives the following values; six decimals describe the truncation, not a uniform six-decimal accuracy claim for the exact potential.

R	0°	15°	30°	45°	60°	75°	90°
1.5	0.554696	0.560829	0.579888	0.613868	0.665263	0.729928	0.768045
2.0	0.447214	0.450895	0.461951	0.480116	0.503626	0.526419	0.536591
2.5	0.371391	0.373649	0.380258	0.390536	0.402722	0.413298	0.417623
3.0	0.316228	0.317677	0.321844	0.328102	0.335155	0.340944	0.343220
3.5	0.274721	0.275694	0.278459	0.282517	0.286950	0.290473	0.291829
4.0	0.242536	0.243216	0.245133	0.247901	0.250864	0.253173	0.254050
4.5	0.216930	0.217422	0.218801	0.220769	0.222847	0.224443	0.225045
5.0	0.196116	0.196483	0.197504	0.198952	0.200464	0.201615	0.202046

For example, $F_{22}(2.5, 60°) = 0.402721525118444\ldots$. The printed check value uses the wrong length in its prefactor: 0.40272 multiplies $q/(4\pi\varepsilon_0 a)$, not $q/(4\pi\varepsilon_0 r)$. With the latter prefactor the coefficient

is $RF = 1.00680381279611\ldots$. A simple rigorous truncation bound follows from $|P_n| \le 1$ and $\binom{2s}{s}/4^s \le 1$:

$$|F - F_{22}| \le \sum_{s=12}^{\infty} R^{-2s-1} = \frac{R^{-25}}{1 - R^{-2}}.$$

The numerical verification also compares every grid point with direct Coulomb-ring integration; the bound controls the omitted infinite tail.

15.2.20 Calculate and tabulate the electrostatic potential created by a charged disk (Exercise 15.2.14), for $r/a = 1.5(0.5)5.0$ and $\theta = 0°(15°)90°$. Carry terms through $P_{22}(\cos\theta)$.

Check value. For $r/a = 2.0$ and $\theta = 15°$, $\psi = 0.46638(q/4\pi\varepsilon_0 a)$.

Solution.

Use the same normalization $F = 4\pi\varepsilon_0 a\,\psi/q$ and $R = r/a$. Exercise 15.2.14 gives the twelve-term approximation

$$F_{22}(R,\theta) = \sum_{s=0}^{11} \frac{(-1)^s}{2s+1} \frac{P_{2s}(\cos\theta)}{R^{2s+1}}.$$

The same three-term polynomial recurrence used in Exercise 15.2.19 evaluates all the needed polynomials; only the multipole weights change. The requested grid, computed at 65-digit working precision, is

R	0°	15°	30°	45°	60°	75°	90°
1.5	0.588001	0.592890	0.607783	0.633185	0.668769	0.708689	0.729727
2.0	0.463648	0.466381	0.474451	0.487268	0.503036	0.517445	0.523599
2.5	0.380506	0.382123	0.386792	0.393878	0.401998	0.408801	0.411517
3.0	0.321751	0.322766	0.325659	0.329924	0.334617	0.338380	0.339837
3.5	0.278300	0.278973	0.280872	0.283620	0.286569	0.288874	0.289752
4.0	0.244979	0.245445	0.246752	0.248620	0.250592	0.252108	0.252680
4.5	0.218669	0.219005	0.219940	0.221264	0.222647	0.223699	0.224093
5.0	0.197396	0.197644	0.198335	0.199308	0.200314	0.201074	0.201358

At the check point, $F_{22}(2, 15°) = 0.466380536818770\ldots$. Again the printed numerical coefficient multiplies $q/(4\pi\varepsilon_0 a)$. If the prefactor is instead $q/(4\pi\varepsilon_0 r)$, its coefficient is $RF = 0.932761073637541\ldots$. For all the tabulated points,

$$|F - F_{22}| \le \sum_{s=12}^{\infty} \frac{R^{-2s-1}}{2s+1} \le \frac{R^{-25}}{25(1 - R^{-2})}.$$

The last bound separates truncation from the six-decimal display rounding. The independent numerical check uses the disk potential obtained by integrating the Coulomb ring kernel over its prescribed charge density.

15.2.21 Calculate the first five (nonvanishing) coefficients in the Legendre series expansion of $f(x) = 1 - |x|$, evaluating the coefficients in the series by numerical integration. Actually these coefficients can be obtained in closed form. Compare your coefficients with those listed in Exercise 18.4.26.

ANS. $a_0 = 0.5000$, $a_2 = -0.6250$, $a_4 = 0.1875$, $a_6 = -0.1016$, $a_8 = 0.0664$.

Solution.

Since f is even, $a_{2n+1} = 0$, while

$$a_{2n} = (4n+1)\int_0^1 (1-x)P_{2n}(x)\,dx.$$

Numerical quadrature on the smooth interval $[0,1]$ gives

n_{degree}	numerical coefficient	exact coefficient
0	0.5000000000	$1/2$
2	-0.6250000000	$-5/8$
4	0.1875000000	$3/16$
6	-0.1015625000	$-13/128$
8	0.0664062500	$17/256$

Here numerical integration, rather than rounding a preassigned answer, was used; the supplementary program reproduces it at high precision.

For an independent closed form, $a_0 = \int_0^1 (1-x)dx = 1/2$. For $n \geq 1$, $\int_0^1 P_{2n} = 0$, so $a_{2n} = -(4n+1)\int_0^1 xP_{2n}dx$. Use the xP_{2n} recurrence and Exercise 15.1.12:

$$\begin{aligned}\int_0^1 xP_{2n}dx &= \frac{(2n+1)I_{2n+1} + 2nI_{2n-1}}{4n+1} \\ &= \frac{P_{2n}(0)}{4n+1}\left[\frac{2n+1}{2n+2} - \frac{2n}{2n-1}\right] \\ &= -\frac{P_{2n}(0)}{(2n-1)(2n+2)}.\end{aligned}$$

Thus

$$\boxed{a_{2n} = (-1)^n(4n+1)\frac{(2n-3)!!}{(2n+2)!!}}, \quad n \geq 1.$$

The source formula for $|x|$ in Exercise 18.4.26 has constant $1/2$ and the opposite nonconstant coefficients, exactly as $1-|x|$ requires.

15.2.22 Calculate and tabulate the exterior electrostatic potential created by the two charged hemispheres of Exercise 15.2.16, for $r/a = 1.5(0.5)5.0$ and $\theta = 0°(15°)90°$. Carry terms through $P_{23}(\cos\theta)$.

Check value. For $r/a = 2.0$ and $\theta = 45°$, $V = 0.27066V_0$.

Solution.

Write $R = r/a$ and $G = V/V_0$. Keep $s = 0, \ldots, 11$ in the exterior hemisphere solution:

$$G_{23}(R,\theta) = \sum_{s=0}^{11}(-1)^s \frac{(4s+3)\binom{2s}{s}}{4^s(2s+2)}\frac{P_{2s+1}(\cos\theta)}{R^{2s+2}}.$$

The polynomial recurrence is evaluated through P_{23}, then the twelve contributions are summed at 65-digit working precision.

R	0°	15°	30°	45°	60°	75°	90°
1.5	0.537744	0.532551	0.515181	0.478972	0.406992	0.260331	0.000000
2.0	0.329180	0.323560	0.305432	0.270657	0.211743	0.120305	0.000000
2.5	0.220080	0.215255	0.200089	0.172575	0.129952	0.070921	0.000000
3.0	0.156726	0.152795	0.140654	0.119398	0.088122	0.047211	0.000000
3.5	0.116968	0.113786	0.104077	0.087472	0.063803	0.033839	0.000000
4.0	0.090491	0.087897	0.080046	0.066831	0.048383	0.025503	0.000000
4.5	0.072020	0.069879	0.063440	0.052724	0.037977	0.019937	0.000000
5.0	0.058643	0.056853	0.051498	0.042657	0.030616	0.016027	0.000000

The equatorial values are exactly zero because all retained polynomials are odd. The check point gives

$$G_{23}(2, 45^\circ) = 0.270656664912169\ldots,$$

which rounds to the stated 0.27066. Since $\binom{2s}{s}/4^s \le 1$ and $(4s+3)/(2s+2) < 2$,

$$|G - G_{23}| \le 2\sum_{s=12}^{\infty} R^{-2s-2} = \frac{2R^{-26}}{1-R^{-2}}.$$

The reported decimal values are the requested finite truncations. Higher-order summation with its own bounded tail and direct Poisson integrals provide independent numerical checks.

15.2.23

(a) Given $f(x) = 2.0$, $|x| < 0.5$ and $f(x) = 0$, $0.5 < |x| < 1.0$, expand $f(x)$ in a Legendre series and calculate the coefficients a_n through a_{80} (analytically).

(b) Evaluate $\displaystyle\sum_{n=0}^{80} a_n P_n(x)$ for $x = 0.400(0.005)0.600$. Plot your results.

Note. This illustrates the Gibbs phenomenon of Section 19.3 and the danger of trying to calculate with a series expansion in the vicinity of a discontinuity.

Solution.

(a) The coefficient formula and the two nonzero intervals give

$$a_n = (2n+1)\int_{-1/2}^{1/2} P_n(x)dx.$$

Every odd coefficient is zero by parity, and $a_0 = 1$. For even $n \ge 2$, integrate $(2n+1)P_n = P'_{n+1} - P'_{n-1}$ and use oddness of the endpoint polynomials:

$$\boxed{a_n = 2[P_{n+1}(1/2) - P_{n-1}(1/2)] \quad (n = 2, 4, \ldots, 80)}.$$

This is an exact analytic formula for all the requested coefficients. For an expression involving only integers and powers of two, use

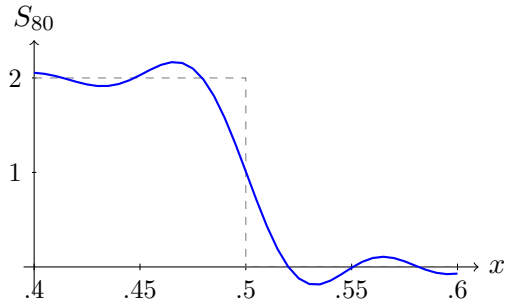
$$P_j(1/2) = \sum_{k=0}^{\lfloor j/2 \rfloor} \frac{(-1)^k (2j-2k)!}{2^{2j-2k} k!(j-k)!(j-2k)!}.$$

All 41 nonzero coefficients were calculated as exact rational numbers from these polynomial values. Their decimal displays are:

n	a_n	n	a_n
0	1.000000000	42	0.115695743
2	-1.875000000	44	-0.429575813
4	1.054687500	46	0.309464967
6	0.266601562	48	0.108557716
8	-0.982086182	50	-0.403296282
10	0.663539886	52	0.291080519
12	0.203865767	54	0.102595591
14	-0.751645535	56	-0.381317431
16	0.524470681	58	0.275626635
18	0.171100741	60	0.097518496
20	-0.632362136	62	-0.362580937
22	0.447317591	64	0.262399140
24	0.150272358	66	0.093127210
26	-0.556355477	68	-0.346360592
28	0.396552617	70	0.250910070
30	0.135556642	72	0.089280100
32	-0.502500382	74	-0.332139273
34	0.359902602	76	0.240809175
36	0.124454578	78	0.085873358
38	-0.461771439	80	-0.319537401
40	0.331841571		

(b) For each specified rational x, compute $P_0 = 1$, $P_1 = x$, then $P_{n+1} = [(2n+1)xP_n - nP_{n-1}]/(n+1)$ through $n = 80$, and sum $S_{80} = \sum a_n P_n$. Exact rational arithmetic avoids cancellation error; only the final display is rounded.

x	$S_{80}(x)$	x	$S_{80}(x)$
0.400	2.055021750	0.505	0.710618291
0.405	2.043444528	0.510	0.430206382
0.410	2.020966606	0.515	0.189030075
0.415	1.990922807	0.520	0.002436961
0.420	1.958725538	0.525	-0.121105466
0.425	1.931073999	0.530	-0.181043732
0.430	1.914789974	0.535	-0.184550845
0.435	1.915445056	0.540	-0.145134072
0.440	1.936002637	0.545	-0.080266216
0.445	1.975722662	0.550	-0.008445837
0.450	2.029559995	0.555	0.053799979
0.455	2.088226666	0.560	0.094441298
0.460	2.138990947	0.565	0.107554625
0.465	2.167165489	0.570	0.093742176
0.470	2.158112203	0.575	0.059315097
0.475	2.099484848	0.580	0.014435104
0.480	1.983362297	0.585	-0.029385877
0.485	1.807911778	0.590	-0.061902155
0.490	1.578269431	0.595	-0.076321894
0.495	1.306431978	0.600	-0.070616739
0.500	1.010104145		



The dashed line is the original step function; the solid line connects all 41 requested computed values. On this grid the largest value is $S_{80}(0.465) = 2.167165489\ldots$ and the smallest is $S_{80}(0.535) = -0.184550845\ldots$. At the jump, $S_{80}(0.5) = 1.010104145\ldots$ approaches the midpoint value one as the truncation order grows. The finite overshoot near the jump illustrates the Gibbs phenomenon; it is not a failure of the exact coefficient calculation.

15.2.24 A plane wave may be expanded in a series of spherical waves by the Rayleigh equation,

$$e^{ikr\cos\gamma} = \sum_{n=0}^{\infty} a_n j_n(kr) P_n(\cos\gamma).$$

Show that $a_n = i^n(2n+1)$.

Hint.

1. Use the orthogonality of the P_n to solve for $a_n j_n(kr)$.
2. Differentiate n times with respect to (kr) and set $r = 0$ to eliminate the r-dependence.
3. Evaluate the remaining integral by Exercise 15.1.15.

Note. This problem may also be treated by noting that both sides of the equation satisfy the Helmholtz equation. The equality can be established by showing that the solutions have the same behavior at the origin and also behave alike at large distances.

Solution.

Put $z = kr$ and $x = \cos\gamma$. Orthogonality in the proposed expansion gives

$$a_n j_n(z) = \frac{2n+1}{2} \int_{-1}^{1} e^{izx} P_n(x)\,dx.$$

Expand e^{izx} at $z = 0$. All powers x^j with $j < n$ have zero integral against P_n by Exercise 15.1.14. The first surviving term is

$$\frac{2n+1}{2}\frac{(iz)^n}{n!}\int_{-1}^{1} x^n P_n(x)\,dx = \frac{(2n+1)i^n z^n}{(2n+1)!!}.$$

Here the corrected moment is $2n!/(2n+1)!!$. Since $j_n(z) = z^n/(2n+1)!! + O(z^{n+2})$, comparison gives

$$\boxed{a_n = i^n(2n+1)}.$$

Equivalently, differentiating exactly n times at zero cancels the common $n!/(2n+1)!!$ factor. There is no division by j_n at one of its nonzero roots. The full identity is verified by the recurrence argument in Exercise 15.2.25 and the integral argument in Exercise 15.2.27.

15.2.25 Verify the Rayleigh equation of Exercise 15.2.24 by starting with the following steps:

(a) Differentiate with respect to (kr) to establish

$$\sum_n a_n j_n'(kr) P_n(\cos\gamma) = i \sum_n a_n j_n(kr) \cos\gamma \, P_n(\cos\gamma).$$

(b) Use a recurrence relation to replace $\cos\gamma \, P_n(\cos\gamma)$ by a linear combination of P_{n-1} and P_{n+1}.

(c) Use a recurrence relation to replace j_n' by a linear combination of j_{n-1} and j_{n+1}.

Solution.

Again put $z = kr$, $x = \cos\gamma$, and $F = \sum a_n j_n(z) P_n(x)$. (a) If $F = e^{izx}$, then $F_z = ixF$, giving the displayed differentiated equation.

(b) Use $xP_n = [(n+1)P_{n+1} + nP_{n-1}]/(2n+1)$. The coefficient of P_m on the right becomes

$$i\left[\frac{m\,a_{m-1}}{2m-1} j_{m-1} + \frac{(m+1)a_{m+1}}{2m+3} j_{m+1}\right].$$

The first term is absent for $m = 0$.

(c) The spherical Bessel recurrence is

$$j_m' = \frac{m}{2m+1} j_{m-1} - \frac{m+1}{2m+1} j_{m+1},$$

including $j_0' = -j_1$ separately. Comparing coefficients on both sides requires

$$\frac{a_m}{2m+1} = \frac{i\,a_{m-1}}{2m-1}, \qquad -\frac{a_m}{2m+1} = \frac{i\,a_{m+1}}{2m+3}.$$

These are the same recurrence, $a_{m+1} = i(2m+3)a_m/(2m+1)$. Starting with $a_0 = 1$ gives $a_n = i^n(2n+1)$.

For sufficiency, insert these coefficients in F. The spherical Bessel power series supplies a concrete uniform bound: for $|z| \le M$,

$$|j_n(z)| \le \sum_{k=0}^{\infty} \frac{M^{n+2k}}{2^k k!(2n+2k+1)!!} \le \frac{M^n}{(2n+1)!!} e^{M^2/2}.$$

The ratio of consecutive bounds after multiplication by $2n+1$ tends to zero. Thus $\sum(2n+1)|j_n(z)|$ converges uniformly on that disk. The derivative recurrence above gives the same conclusion for $\sum(2n+1)|j_n'(z)|$, with $j_0' = -j_1$ handled separately. Together with $|P_n(x)| \le 1$, these bounds justify differentiating F term by term for $-1 \le x \le 1$. The calculation above proves $F_z = ixF$. At zero, $j_0(0) = 1$ and $j_n(0) = 0$ for $n > 0$, so $F(0, x) = 1$. Solving this first-order equation yields $F(z, x) = e^{izx}$, verifying the Rayleigh identity for all finite z.

15.2.26 From Exercise 15.2.24 show that

$$j_n(kr) = \frac{1}{2i^n} \int_{-1}^{1} e^{ikr\mu} P_n(\mu)\, d\mu.$$

This means that (apart from a constant factor) the spherical Bessel function $j_n(kr)$ is an integral transform of the Legendre polynomial $P_n(\mu)$.

Solution.

Multiply the Rayleigh expansion by $P_m(\mu)$ and integrate:

$$\int_{-1}^{1} e^{iz\mu} P_m(\mu)\,d\mu = \sum_{n\ge 0} i^n (2n+1) j_n(z) \frac{2\delta_{nm}}{2n+1} = 2i^m j_m(z).$$

Uniform convergence for $\mu \in [-1,1]$ and fixed z justifies the interchange. Divide by $2i^m$ and rename the index:

$$\boxed{j_n(z) = \frac{1}{2i^n} \int_{-1}^{1} e^{iz\mu} P_n(\mu)\,d\mu}.$$

The integral is entire in z, so the formula also holds for complex z and at zero, with the usual removable values of j_n.

15.2.27 Rewriting the formula of Exercise 15.2.26 as

$$j_n(z) = \frac{1}{2}(-i)^n \int_0^{\pi} e^{iz\cos\theta} P_n(\cos\theta)\sin\theta\,d\theta, \qquad n = 0, 1, 2, \ldots,$$

verify it by transforming the right-hand side into

$$\frac{z^n}{2^{n+1}n!} \int_0^{\pi} \cos(z\cos\theta)\sin^{2n+1}\theta\,d\theta$$

and using Exercise 14.7.9.

Solution.

Put $x = \cos\theta$, reverse the limits, and insert Rodrigues:

$$I = \frac{(-i)^n}{2} \int_{-1}^{1} e^{izx} P_n(x)\,dx = \frac{(-i)^n}{2^{n+1}n!} \int_{-1}^{1} e^{izx} D^n (x^2-1)^n\,dx.$$

After n integrations by parts, all boundary terms vanish and the exponential contributes $(-iz)^n$. Since $(-i)^n(-i)^n = (-1)^n$, it combines with $(x^2-1)^n = (-1)^n(1-x^2)^n$ to give

$$I = \frac{z^n}{2^{n+1}n!} \int_{-1}^{1} e^{izx}(1-x^2)^n\,dx.$$

The sine part is odd in x and integrates to zero, even for complex z. Returning to θ gives exactly

$$I = \frac{z^n}{2^{n+1}n!} \int_0^{\pi} \cos(z\cos\theta)\sin^{2n+1}\theta\,d\theta.$$

To verify the spherical Bessel value directly, expand the cosine and integrate each term:

$$\int_{-1}^{1} x^{2k}(1-x^2)^n\,dx = B(k+\tfrac{1}{2}, n+1) = \frac{\Gamma(k+1/2)n!}{\Gamma(k+n+3/2)}.$$

Therefore

$$I = \sum_{k=0}^{\infty} \frac{(-1)^k z^{n+2k}}{2^k k!(2n+2k+1)!!} = j_n(z).$$

In obtaining the coefficient, use $\Gamma(k+1/2) = (2k)!\sqrt{\pi}/(4^k k!)$ and the analogous half-integer formula in the denominator. This reproduces the Poisson representation from Exercise 14.7.9 without assuming it.

15.3 Physical Interpretation of Generating Function

15.3.1 Develop the electrostatic potential for the array of charges shown in Fig. 15.7. This is a linear electric quadrupole.

Solution.

The actual Figure 15.7 has charges $q, -2q, q$ at $z = -a, 0, a$. Coulomb's law therefore gives, at every point away from these charges,

$$V = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{\sqrt{r^2+a^2+2ar\cos\theta}} - \frac{2}{r} + \frac{1}{\sqrt{r^2+a^2-2ar\cos\theta}}\right].$$

For $r > a$, expand the two displaced charges with the generating function. Their nth coefficients add to $1+(-1)^n$. All odd terms vanish, and the $n=0$ term cancels the central charge:

$$\boxed{V = \frac{2q}{4\pi\varepsilon_0 r}\sum_{s=1}^{\infty}\left(\frac{a}{r}\right)^{2s} P_{2s}(\cos\theta)}.$$

In particular,

$$V = \frac{2qa^2}{4\pi\varepsilon_0 r^3}P_2(\cos\theta) + O(a^4/r^5) = \frac{qa^2(3\cos^2\theta - 1)}{4\pi\varepsilon_0 r^3} + O(a^4/r^5).$$

The total charge and dipole moment are zero, while $\sum q_i z_i^2 = 2qa^2 \neq 0$, identifying the leading quadrupole.

15.3.2 Calculate the electrostatic potential of the array of charges shown in Fig. 15.8. Here is an example of two equal but oppositely directed quadrupoles. The quadrupole contributions cancel. The octopole terms do not cancel.

Solution.

Figure 15.8 has charges $-q, 2q, -2q, q$ at $-2a, -a, a, 2a$. Thus the exact potential is

$$V = \frac{q}{4\pi\varepsilon_0}\left[-\frac{1}{|\mathbf{r}+2a\hat{\mathbf{z}}|} + \frac{2}{|\mathbf{r}+a\hat{\mathbf{z}}|} - \frac{2}{|\mathbf{r}-a\hat{\mathbf{z}}|} + \frac{1}{|\mathbf{r}-2a\hat{\mathbf{z}}|}\right].$$

For $r > 2a$, the coefficient of $a^n P_n/r^{n+1}$ is

$$-(-2)^n + 2(-1)^n - 2 + 2^n = [1-(-1)^n](2^n - 2).$$

It is zero for every even n and also for $n = 1$. Putting $n = 2s+1$ leaves

$$\boxed{V = \frac{2q}{4\pi\varepsilon_0 r}\sum_{s=1}^{\infty}(2^{2s+1}-2)\left(\frac{a}{r}\right)^{2s+1} P_{2s+1}(\cos\theta)}.$$

The leading term is

$$\boxed{V_{\text{oct}} = \frac{12qa^3}{4\pi\varepsilon_0 r^4}P_3(\cos\theta)}.$$

This also follows from the moments: $\mu_0 = \mu_1 = \mu_2 = 0$ and $\mu_3 = q[8-(-8)-2+2(-1)]a^3 = 12qa^3$.

15.3.3 Show that the electrostatic potential produced by a charge q at $z = a$ for $r < a$ is

$$\varphi(r) = \frac{q}{4\pi\varepsilon_0 a} \sum_{n=0}^{\infty} \left(\frac{r}{a}\right)^n P_n(\cos\theta).$$

Solution.

The source-point distance is $|\mathbf{r} - a\hat{\mathbf{z}}| = \sqrt{a^2 + r^2 - 2ar\cos\theta}$. For $0 \le r < a$, factor out a and use the generating function with $t = r/a$:

$$\varphi = \frac{q}{4\pi\varepsilon_0 a}\left[1 - 2\frac{r}{a}\cos\theta + \left(\frac{r}{a}\right)^2\right]^{-1/2} = \boxed{\frac{q}{4\pi\varepsilon_0 a} \sum_{n=0}^{\infty} \left(\frac{r}{a}\right)^n P_n(\cos\theta)}.$$

The condition $r/a < 1$ ensures convergence uniformly on angular directions for each fixed smaller radius. At $r = 0$ the value is $q/(4\pi\varepsilon_0 a)$, with all higher terms vanishing.

15.3.4 Using $\mathbf{E} = -\nabla\varphi$, determine the components of the electric field corresponding to the (pure) electric dipole potential,

$$\varphi(r) = \frac{2aq\, P_1(\cos\theta)}{4\pi\varepsilon_0 r^2}.$$

Here it is assumed that $r \gg a$.

$$\textit{ANS. } E_r = +\frac{4aq\cos\theta}{4\pi\varepsilon_0 r^3}, \quad E_\theta = +\frac{2aq\sin\theta}{4\pi\varepsilon_0 r^3}, \quad E_\varphi = 0.$$

Solution.

Since $P_1(\cos\theta) = \cos\theta$, put $C = 2aq/(4\pi\varepsilon_0)$ and write $\varphi = Cr^{-2}\cos\theta$. The spherical gradient gives

$$E_r = -\partial_r\varphi = 2Cr^{-3}\cos\theta, \qquad E_\theta = -\frac{1}{r}\partial_\theta\varphi = Cr^{-3}\sin\theta, \qquad E_\varphi = -\frac{1}{r\sin\theta}\partial_\varphi\varphi = 0.$$

Substituting C yields all three stated components. For a pair of charges separated by $2a$ these are its leading $r \gg a$ terms; for an ideal point dipole of fixed moment $2aq$ they are exact at $r > 0$.

15.3.5 Operating in spherical polar coordinates, show that for $r > 0$

$$\frac{\partial}{\partial z}\left[\frac{P_l(\cos\theta)}{r^{l+1}}\right] = -(l+1)\frac{P_{l+1}(\cos\theta)}{r^{l+2}}.$$

This is the key step in the mathematical argument that the derivative of one multipole leads to the next higher multipole.

Hint. Compare with Exercise 3.10.28.

Solution.

At fixed Cartesian x, y, write $\rho = \sqrt{x^2 + y^2}$, $r = \sqrt{\rho^2 + z^2}$ and $u = z/r = \cos\theta$. Then

$$\frac{\partial r}{\partial z} = u, \qquad \frac{\partial u}{\partial z} = \frac{1 - u^2}{r}.$$

Hence

$$\partial_z\left[r^{-l-1}P_l(u)\right] = r^{-l-2}\left[-(l+1)uP_l(u) + (1-u^2)P_l'(u)\right].$$

Exercise 15.1.7 gives $(1-u^2)P_l' = (l+1)(uP_l - P_{l+1})$. The uP_l terms cancel, leaving

$$\boxed{\partial_z\left[\frac{P_l(\cos\theta)}{r^{l+1}}\right] = -(l+1)\frac{P_{l+1}(\cos\theta)}{r^{l+2}}}, \qquad r > 0.$$

This chain-rule calculation is equivalent to $\partial_z = \cos\theta\,\partial_r - (\sin\theta/r)\partial_\theta$.

15.3.6 A point electric dipole of strength $p^{(1)}$ is placed at $z = a$; a second point electric dipole of equal but opposite strength is at the origin. Keeping the product $p^{(1)}a$ constant, let $a \to 0$. Show that this results in a point electric quadrupole.

Hint. Exercise 15.3.5 (when proved) will be helpful.

Solution.

Define $F_1(\mathbf{r}) = P_1(\cos\theta)/r^2$. The positive z dipole at $a\hat{\mathbf{z}}$ and its negative at zero have combined potential

$$V_a(\mathbf{r}) = \frac{p^{(1)}}{4\pi\varepsilon_0}[F_1(\mathbf{r} - a\hat{\mathbf{z}}) - F_1(\mathbf{r})].$$

At each fixed point away from the origin, Taylor expansion gives $F_1(\mathbf{r}-a\hat{\mathbf{z}})-F_1(\mathbf{r}) = -a\,\partial_z F_1+O(a^2)$. Exercise 15.3.5 evaluates $\partial_z F_1 = -2P_2/r^3$. Holding $K = p^{(1)}a$ fixed, the error is $O(Ka) \to 0$, so

$$\boxed{V_{\text{quad}} = \frac{2K}{4\pi\varepsilon_0 r^3}P_2(\cos\theta)}.$$

In the convention $V_{\text{quad}} = p^{(2)}P_2/(4\pi\varepsilon_0 r^3)$, the resulting quadrupole strength is $p^{(2)} = 2ap^{(1)}$. The dipole terms cancel, while a nonzero $r^{-3}P_2$ term remains.

15.3.7 A point electric octupole may be constructed by placing a point electric quadrupole (pole strength $p^{(2)}$ in the z-direction) at $z = a$ and an equal but opposite point electric quadrupole at $z = 0$ and then letting $a \to 0$, subject to $p^{(2)}a =$ constant. Find the electrostatic potential corresponding to a point electric octupole. Show from the construction of the point electric octupole that the corresponding potential may be obtained by differentiating the point quadrupole potential.

Solution.

Use the same strength convention: $V^{(2)} = p^{(2)}P_2(\cos\theta)/(4\pi\varepsilon_0 r^3)$. For a positive quadrupole displaced to $z = a$ and its negative at zero,

$$V_a^{(3)} = \frac{p^{(2)}}{4\pi\varepsilon_0}\left[F_2(\mathbf{r} - a\hat{\mathbf{z}}) - F_2(\mathbf{r})\right], \qquad F_2 = \frac{P_2(\cos\theta)}{r^3}.$$

The Taylor difference is $-a\partial_z F_2 + O(a^2)$, and $\partial_z F_2 = -3P_3/r^4$. Therefore, keeping $K_2 = ap^{(2)}$ fixed,

$$\boxed{V^{(3)} = \frac{3K_2}{4\pi\varepsilon_0 r^4}P_3(\cos\theta) = \frac{3K_2}{8\pi\varepsilon_0 r^4}(5\cos^3\theta - 3\cos\theta)}.$$

Thus $p^{(3)} = 3ap^{(2)}$ in this convention, and the construction is precisely $V^{(3)} = -a\,\partial_z V^{(2)}$ in the limiting sense. The sign is set by putting the positive source at $+a$. The manual, p. 245, instead writes $48a^3q$ for the numerator of the octupole moment, although this exercise supplies $p^{(2)}$, not a charge q. In the stated strength convention the moment is $3ap^{(2)}$. The manual's form would agree only after the additional, unstated assignment $p^{(2)} = 16a^2q$.

15.3.8 A point charge q is in the interior of a hollow conducting sphere of radius r_0. The charge q is displaced a distance $0 < a < r_0$ from the center of the sphere. If the conducting sphere is grounded, show that the potential in the interior produced by q and the distributed induced charge is the same as that produced by q and its image charge q'. The image charge is at a distance $a' = r_0^2/a$ from the center, collinear with q and the origin (Fig. 15.9).

Hint. Calculate the electrostatic potential for $a < r_0 < a'$. Show that the potential vanishes for $r = r_0$ if we take $q' = -qr_0/a$.

Solution.

For $0 < a < r_0$, place $q' = -qr_0/a$ at $a' = r_0^2/a > r_0$. Consider

$$V(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\left[\frac{q}{|\mathbf{r} - a\hat{\mathbf{z}}|} - \frac{qr_0/a}{|\mathbf{r} - a'\hat{\mathbf{z}}|}\right].$$

On $r = r_0$, the two distances satisfy

$$|\mathbf{r} - a'\hat{\mathbf{z}}|^2 = r_0^2 + \frac{r_0^4}{a^2} - \frac{2r_0^3}{a}\cos\theta = \frac{r_0^2}{a^2}|\mathbf{r} - a\hat{\mathbf{z}}|^2.$$

Their positive square roots therefore differ by r_0/a and the two potential terms cancel exactly.

The Legendre version of the same cancellation holds for $a < r < r_0$:

$$V = \frac{1}{4\pi\varepsilon_0}\sum_{n=0}^{\infty}\left[\frac{qa^n}{r^{n+1}} + \frac{q'r^n}{(a')^{n+1}}\right]P_n(\cos\theta).$$

At $r = r_0$, inserting q', a' makes each bracket zero. The image term is harmonic everywhere inside the sphere, because its source is outside; the real charge has exactly the required Coulomb singularity. The difference from any other grounded solution is harmonic, including across the canceled source singularity, and zero on the boundary. Uniqueness for the Dirichlet problem proves the result throughout the interior, not only in the expansion annulus. For $a = 0$, take the limit directly: $V = q(1/r - 1/r_0)/(4\pi\varepsilon_0)$.

15.4 Associated Legendre Equation

15.4.1 Apply the Frobenius method to Eq. (15.72) to obtain Eq. (15.73) and verify that the numerator of that equation becomes zero if $\lambda = l(l+1)$ and $j = l - m$.

Solution.

Write U for the polynomial factor in Eq. (15.72):

$$(1 - x^2)U'' - 2(m+1)xU' + [\lambda - m(m+1)]U = 0.$$

For $U = \sum_{j\geq 0} a_j x^{j+k}$, the lowest power x^{k-2} has coefficient $k(k-1)a_0$, so $k = 0$ or 1. The next power, x^{k-1}, has coefficient $k(k+1)a_1$. Thus $k = 1$ forces $a_1 = 0$, whereas $k = 0$ leaves a_0, a_1 independent; these are the even and odd branches at the ordinary point $x = 0$. The coefficient of x^{j+k} is

$$(j+k+2)(j+k+1)a_{j+2} + \{\lambda - m(m+1) - (j+k)(j+k-1) - 2(m+1)(j+k)\}a_j.$$

Set it to zero:

$$a_{j+2} = \frac{(j+k)^2 + (2m+1)(j+k) - \lambda + m(m+1)}{(j+k+1)(j+k+2)}a_j.$$

For $k = 0$ this is exactly Eq. (15.73). If $\lambda = l(l+1)$, the numerator factors as

$$(j+m)(j+m+1) - l(l+1) = (j+m-l)(j+m+l+1),$$

which vanishes at $j = l - m$. The parity of $l - m$ selects the even or odd power series. For nonnegative integers $m \leq l$, that branch terminates in a polynomial of degree $l - m$.

15.4.2 Starting from the entries for P_2^2 and P_2^1 in Table 15.3, apply a recurrence formula to obtain P_2^0 (which is P_2), P_2^{-1}, and P_2^{-2}. Compare your results with the value of P_2 from Table 15.1 and with values of P_2^{-1} and P_2^{-2} obtained by applying Eq. (15.81) to entries from Table 15.3.

Solution.

Put $s = \sqrt{1-x^2}$ on $-1 < x < 1$. The supplied entries are $P_2^2 = 3s^2$ and $P_1^1 = -s$. Equation (15.88) with $l = m = 1$ gives $P_2^1 = 3xP_1^1 - 2P_0^1 = -3xs$, because $P_0^1 = -sP_0' = 0$. Now use the fixed-degree recurrence

$$P_2^{m+1} + \frac{2mx}{s}P_2^m + [6 - m(m-1)]P_2^{m-1} = 0.$$

At $m = 1$, this gives $3s^2 - 6x^2 + 6P_2^0 = 0$, hence

$$\boxed{P_2^0 = \frac{3x^2-1}{2}}.$$

At $m = 0$, $P_2^1 + 6P_2^{-1} = 0$, so

$$\boxed{P_2^{-1} = \frac{1}{2}xs}.$$

At $m = -1$, $P_2^0 - 2xP_2^{-1}/s + 4P_2^{-2} = 0$. Substituting the two results gives $4P_2^{-2} = x^2 - (3x^2-1)/2 = s^2/2$, so

$$\boxed{P_2^{-2} = \frac{1}{8}(1-x^2)}.$$

Independently, Eq. (15.81) predicts $P_2^{-1} = -(1!/3!)P_2^1 = xs/2$ and $P_2^{-2} = (0!/4!)P_2^2 = s^2/8$, in agreement. The endpoint values follow by continuity.

15.4.3 Prove that, for integers $0 \le m \le l$ and $-1 < x < 1$,

$$P_l^{-m}(x) = (-1)^m \frac{(l-m)!}{(l+m)!} P_l^m(x),$$

where $P_l^m(x)$ is defined by

$$P_l^m(x) = \frac{(-1)^m}{2^l l!}(1-x^2)^{m/2}\frac{d^{l+m}}{dx^{l+m}}(x^2-1)^l.$$

Hint. One approach is to apply Leibniz's formula to $(x+1)^l(x-1)^l$.

Solution.

Let $F = (x-1)^l(x+1)^l$ and take integers $0 \le m \le l$. Leibniz's rule gives

$$D^{l-m}F = \sum_{j=0}^{l-m} \frac{(l-m)!(l!)^2(x-1)^{l-j}(x+1)^{m+j}}{j!(l-m-j)!(l-j)!(m+j)!}.$$

For $D^{l+m}F$, the number of derivatives on the first factor must lie between m and l. Write that number as $m + j$:

$$D^{l+m}F = \sum_{j=0}^{l-m} \frac{(l+m)!(l!)^2(x-1)^{l-m-j}(x+1)^{j}}{(m+j)!(l-j)!(l-m-j)!j!}.$$

Multiplication of the second sum by $(x^2-1)^m$ makes its two powers exactly those of the first sum. Their coefficients then differ only by $(l-m)!/(l+m)!$. Thus

$$D^{l-m}F = \frac{(l-m)!}{(l+m)!}(x^2-1)^m D^{l+m}F.$$

Insert this identity into the Rodrigues definition with order $-m$:

$$\begin{aligned} P_l^{-m} &= \frac{(-1)^m}{2^l l!}(1-x^2)^{-m/2}\frac{(l-m)!}{(l+m)!}(-1)^m(1-x^2)^m D^{l+m}F \\ &= \boxed{(-1)^m\frac{(l-m)!}{(l+m)!}P_l^m}. \end{aligned}$$

On $(-1,1)$ choose the positive square root of $1-x^2$. The identity also extends with any consistently chosen complex branch.

15.4.4 Show that

$$\begin{aligned} P_{2l}^1(0) &= 0, \\ P_{2l+1}^1(0) &= (-1)^{l+1}\frac{(2l+1)!!}{(2l)!!}, \end{aligned}$$

by each of these three methods:

(a) Use of recurrence relations,

(b) Expansion of the generating function,

(c) Rodrigues formula.

Solution.

(a) The fixed-order recurrence at $x=0$, with $m=1$, is

$$lP_{l+1}^1(0) = -(l+1)P_{l-1}^1(0), \qquad l \geq 1.$$

The starting values are $P_0^1(0)=0$ and $P_1^1(0)=-1$. Every even-index value is consequently zero. Repeatedly using $l=2,4,\ldots,2s$ for the odd subsequence gives

$$P_{2s+1}^1(0) = (-1)^{s+1}\frac{3\cdot 5\cdots(2s+1)}{2\cdot 4\cdots 2s} = \boxed{(-1)^{s+1}\frac{(2s+1)!!}{(2s)!!}}.$$

(b) Differentiate the ordinary generating function in x and use $P_l^1 = -\sqrt{1-x^2}P_l'$:

$$\sum_{l\geq 0} P_l^1(0)t^l = -t(1+t^2)^{-3/2} = -\sum_{s\geq 0}(-1)^s\frac{(2s+1)!!}{2^s s!}t^{2s+1}.$$

There are no even powers, and $2^s s! = (2s)!!$, yielding the same values.

(c) Rodrigues, or its explicit polynomial expansion, shows that P_{2s} has no linear term. The coefficient of x in P_{2s+1} is

$$(-1)^s\frac{(2s+2)!}{2^{2s+1}s!(s+1)!} = (-1)^s\frac{(2s+1)!!}{(2s)!!}.$$

At zero, $P_n^1 = -P_n'$ is the negative of this coefficient. This proves both formulas by the third requested method.

15.4.5 Evaluate $P_l^m(0)$ for integers $1 \le m \le l$.

$$ANS.\ P_l^m(0) = \begin{cases} (-1)^{(l+m)/2}\dfrac{(l+m-1)!!}{(l-m)!!}, & l+m \text{ even}, \\ 0, & l+m \text{ odd}. \end{cases}$$

Solution.

For integers $1 \le m \le l$, $P_l^m(0) = (-1)^m P_l^{(m)}(0)$. Only a term x^m in the polynomial P_l can survive this evaluation. The powers in P_l are $l, l-2, \ldots$, so no such term exists when $l+m$ is odd. If $l+m$ is even, put $k = (l-m)/2$ in the closed polynomial formula. Differentiating x^m contributes $m!$, giving

$$P_l^m(0) = (-1)^{(l+m)/2}\frac{(l+m)!}{2^l[(l-m)/2]![(l+m)/2]!}.$$

Split the factorials into even and odd factors:

$$\frac{(l+m)!}{2^{(l+m)/2}[(l+m)/2]!} = (l+m-1)!!, \qquad 2^{(l-m)/2}[(l-m)/2]! = (l-m)!!.$$

Hence the stated nonzero value is

$$\boxed{P_l^m(0) = (-1)^{(l+m)/2}\frac{(l+m-1)!!}{(l-m)!!}}.$$

For integer $m > l$, the defining derivative is zero; the factorial answer is intended for the usual range $m \le l$.

15.4.6 Starting from the potential of a finite dipole, Eq. (15.65), verify the formulas for the electric field components given as Eqs. (15.130) and (15.131).

Solution.

Equation (15.65) is the convergent finite-dipole expansion

$$V = \frac{2q}{4\pi\varepsilon_0}\sum_{s=0}^{\infty}\frac{a^{2s+1}}{r^{2s+2}}P_{2s+1}(\cos\theta), \qquad r > a.$$

Radial differentiation gives

$$E_r = \frac{q}{\pi\varepsilon_0}\sum_{s\ge 0}(s+1)\frac{a^{2s+1}}{r^{2s+3}}P_{2s+1} = \boxed{\frac{qa}{\pi\varepsilon_0 r^3}\left[P_1 + 2(a/r)^2 P_3 + 3(a/r)^4 P_5 + \cdots\right]}.$$

For the angular derivative, $dP_l(\cos\theta)/d\theta = -\sin\theta\, P_l'(\cos\theta) = P_l^1(\cos\theta)$. Therefore

$$E_\theta = -\frac{1}{r}\partial_\theta V = \boxed{-\frac{qa}{2\pi\varepsilon_0 r^3}\left[P_1^1 + (a/r)^2 P_3^1 + (a/r)^4 P_5^1 + \cdots\right]}.$$

These are Eqs. (15.130) and (15.131), with $E_\varphi = 0$. The opposite signs of P_l^1 and $\sin\theta\, P_l'$ are included in the convention, so the leading E_θ points in the positive θ direction for positive qa.

15.4.7 Show that

$$P_l^l(\cos\theta) = (-1)^l(2l-1)!!\,\sin^l\theta, \qquad l = 0, 1, 2, \ldots, \quad 0 \le \theta \le \pi.$$

Solution.

The Rodrigues definition at $m = l$ involves $D^{2l}(x^2-1)^l$. Only the leading monomial x^{2l} survives this derivative, with value $(2l)!$. Thus

$$P_l^l(x) = \frac{(-1)^l(2l)!}{2^l l!}(1-x^2)^{l/2} = (-1)^l(2l-1)!!(1-x^2)^{l/2}.$$

For $x = \cos\theta$ and $0 \le \theta \le \pi$, $\sqrt{1-x^2} = \sin\theta \ge 0$, so

$$\boxed{P_l^l(\cos\theta) = (-1)^l(2l-1)!!\,\sin^l\theta}.$$

The convention $(-1)!! = 1$ includes $l = 0$.

15.4.8 Derive the associated Legendre recurrence relation, for $-1 < x < 1$ and integer orders $-l \le m \le l$, $l \ge 1$, with a zero-coefficient term omitted when its order falls outside this range,

$$P_l^{m+1}(x) + \frac{2mx}{(1-x^2)^{1/2}}P_l^m(x) + \big[l(l+1) - m(m-1)\big]P_l^{m-1}(x) = 0.$$

Solution.

Differentiate the ordinary Legendre equation $m-1$ times, for $m \ge 1$, and put $u = D^{m-1}P_l$. The product rule gives

$$(1-x^2)u'' - 2mxu' + [l(l+1) - m(m-1)]u = 0.$$

For example, the u' coefficient is $-2(m-1)x - 2x = -2mx$; the u subtraction is $(m-1)(m-2) + 2(m-1) = m(m-1)$. Multiply this equation by $(-1)^{m+1}(1-x^2)^{(m-1)/2}$. Its three terms become, respectively,

$$P_l^{m+1}, \qquad \frac{2mx}{\sqrt{1-x^2}}P_l^m, \qquad [l(l+1) - m(m-1)]P_l^{m-1}.$$

Their sum is zero, proving the recurrence. For the extension to negative orders, denote the displayed left-hand side by R_m and write $c_k = (-1)^k(l-k)!/(l+k)!$, so $P_l^{-k} = c_k P_l^k$. For $1 \le k < l$, the adjacent ratios are

$$c_{k-1} = -(l+k)(l-k+1)c_k, \qquad (l-k)(l+k+1)c_{k+1} = -c_k.$$

Substitution into R_{-k} consequently gives

$$R_{-k} = -c_k\left\{P_l^{k+1} + \frac{2kx}{\sqrt{1-x^2}}P_l^k + [l(l+1) - k(k-1)]P_l^{k-1}\right\} = -c_k R_k = 0.$$

At $m = 0$ the recurrence is $P_l^1 + l(l+1)P_l^{-1} = 0$, which is exactly the negative-order formula. At $m = l$, Rodrigues's formula gives $P_l^{l+1} = 0$; at $m = -l$ the coefficient of the out-of-range order $-l-1$ is zero and that term is omitted. The same two-term identity remains. The $l = 0$ case is simply $P_0^1 = 0$. At $x = \pm 1$ the separate quotients can be singular: the recurrence is interpreted as the limit of their combined expression from $-1 < x < 1$.

15.4.9 Develop a recurrence relation that will yield $P_l^1(x)$, for $l \geq 1$ and $-1 < x < 1$, as

$$P_l^1(x) = f_1(x,l)P_l(x) + f_2(x,l)P_{l-1}(x).$$

Follow either of the procedures (a) or (b):

(a) Derive a recurrence relation of the preceding form. Give $f_1(x,l)$ and $f_2(x,l)$ explicitly.

(b) Find the appropriate recurrence relation in print.

(1) Give the source.

(2) Verify the recurrence relation.

ANS. (a) $P_l^1(x) = \dfrac{lx}{(1-x^2)^{1/2}}P_l - \dfrac{l}{(1-x^2)^{1/2}}P_{l-1}.$

Solution.

Use the derivation option (a). The defining relation is $P_l^1 = -\sqrt{1-x^2}\,P_l'$, while the ordinary recurrence gives $(1-x^2)P_l' = lP_{l-1} - lxP_l$. Divide the latter by $\sqrt{1-x^2}$ and reverse the sign:

$$\boxed{P_l^1(x) = \frac{lx}{\sqrt{1-x^2}}P_l(x) - \frac{l}{\sqrt{1-x^2}}P_{l-1}(x)}, \qquad -1 < x < 1, \quad l \geq 1.$$

Thus $f_1 = lx/\sqrt{1-x^2}$ and $f_2 = -l/\sqrt{1-x^2}$. The numerator cancels at the endpoints; the associated function there is recovered by a limit. For $l = 0$, $P_0^1 = 0$ directly.

15.4.10 Show that, for $0 \leq \theta \leq \pi$, $\sin\theta \dfrac{d}{d\cos\theta}P_n(\cos\theta) = -P_n^1(\cos\theta)$.

Solution.

For $0 \leq \theta \leq \pi$, set $x = \cos\theta$ in $P_n^1(x) = -\sqrt{1-x^2}\,dP_n/dx$. Because the square root is $\sin\theta$, this gives

$$\boxed{\sin\theta \frac{d}{d\cos\theta}P_n(\cos\theta) = -P_n^1(\cos\theta)}.$$

The minus sign is the published erratum for textbook p. 754, listed in the Instructor's Manual dated 6 April 2012. It is required by the book's Condon–Shortley definition. For instance $n = 1$ gives $\sin\theta = -(-\sin\theta)$.

15.4.11 Show that for nonnegative integer degrees and admissible orders $|m| \leq \min(l, l')$ in part (a)

(a) $$\int_0^\pi \left(\frac{dP_l^m}{d\theta}\frac{dP_{l'}^m}{d\theta} + \frac{m^2 P_l^m P_{l'}^m}{\sin^2\theta}\right)\sin\theta\,d\theta = \frac{2l(l+1)}{2l+1}\frac{(l+m)!}{(l-m)!}\delta_{ll'},$$

(b) $$\int_0^\pi \left(\frac{P_l^1}{\sin\theta}\frac{dP_{l'}^1}{d\theta} + \frac{P_{l'}^1}{\sin\theta}\frac{dP_l^1}{d\theta}\right)\sin\theta\,d\theta = 0.$$

These integrals occur in the theory of scattering of electromagnetic waves by spheres.

Solution.

(a) Put $u = P_l^m(\cos\theta)$, $v = P_{l'}^m(\cos\theta)$. The self-adjoint angular equation for u is

$$(\sin\theta\,u')' + \left[l(l+1)\sin\theta - \frac{m^2}{\sin\theta}\right]u = 0.$$

Multiply by v and integrate. Integration by parts yields

$$\int_0^\pi \left(u'v' + \frac{m^2uv}{\sin^2\theta}\right)\sin\theta\,d\theta = l(l+1)\int_0^\pi uv\sin\theta\,d\theta + [\sin\theta\,u'v]_0^\pi.$$

For admissible integer orders, the boundary term is zero: for $m = 0$, $u' = O(\sin\theta)$; for $|m| > 0$, both functions are $O(\sin^{|m|}\theta)$, so the product in that term is $O(\sin^{2|m|}\theta)$. The fixed-m orthogonality and normalization give

$$\boxed{\int_0^\pi \left(u'v' + \frac{m^2uv}{\sin^2\theta}\right)\sin\theta\,d\theta = \frac{2l(l+1)}{2l+1}\frac{(l+m)!}{(l-m)!}\delta_{ll'}}.$$

(b) With $u = P_l^1$, $v = P_{l'}^1$, the apparent denominators cancel the integration weight. The integrand becomes $uv' + vu' = (uv)'$:

$$\int_0^\pi (uv' + vu')d\theta = [uv]_0^\pi = 0,$$

because both order-one functions vanish at the two poles.

15.4.12 As a repeat of Exercise 15.2.9, show, using associated Legendre functions, that for integers $n \geq 2$ and $m \geq 0$

$$\int_{-1}^1 x(1-x^2)P_n'(x)P_m'(x)\,dx = \frac{n+1}{2n+1}\frac{2}{2n-1}\frac{n!}{(n-2)!}\,\delta_{m,n-1} + \frac{n}{2n+1}\frac{2}{2n+3}\frac{(n+2)!}{n!}\,\delta_{m,n+1}.$$

Solution.

Since $P_n^1 = -\sqrt{1-x^2}P_n'$, the integrand is $xP_n^1P_m^1$. The fixed-order recurrence gives

$$(2n+1)xP_n^1 = (n+1)P_{n-1}^1 + nP_{n+1}^1.$$

Multiplying by P_m^1 and integrating leaves only two norms:

$$I_{nm} = \frac{n+1}{2n+1}\frac{2(n-1)n}{2n-1}\delta_{m,n-1} + \frac{n}{2n+1}\frac{2(n+1)(n+2)}{2n+3}\delta_{m,n+1}.$$

Here $\int_{-1}^1 (P_j^1)^2 dx = 2j(j+1)/(2j+1)$ was used. Writing $n(n-1) = n!/(n-2)!$ and $(n+1)(n+2) = (n+2)!/n!$ gives exactly the displayed formula. The factorial form of its first term assumes $n \geq 2$. The polynomial form above remains meaningful at $n = 0, 1$ and gives the correct zero term, without a negative factorial.

15.4.13 Evaluate $\displaystyle\int_0^\pi \sin^2\theta\,P_n^1(\cos\theta)\,d\theta$.

Solution.

Insert $P_n^1(\cos\theta) = -\sin\theta\,P_n'(\cos\theta)$ and then put $x = \cos\theta$:

$$I_n = -\int_0^\pi \sin^3\theta\,P_n'(\cos\theta)d\theta = -\int_{-1}^1 (1-x^2)P_n'(x)dx.$$

Integration by parts has a zero endpoint term, so

$$I_n = -2\int_{-1}^{1} xP_n(x)dx.$$

Because $x = P_1$ and its squared norm is $2/3$,

$$\boxed{I_n = -\frac{4}{3}\,\delta_{n1}}, \qquad n = 0, 1, 2, \ldots.$$

In particular, $P_1^1 = -\sin\theta$ directly gives $-\int_0^\pi \sin^3\theta\, d\theta = -4/3$. The manual's answer on p. 251 has the opposite sign, $+4\delta_{n1}/3$. The negative sign follows from the book's $P_1^1 = -\sin\theta$ convention and the direct $n = 1$ integral.

15.4.14 The associated Legendre function $P_l^m(x)$ satisfies the self-adjoint ODE

$$(1-x^2)\frac{d^2P_l^m(x)}{dx^2} - 2x\frac{dP_l^m(x)}{dx} + \left[l(l+1) - \frac{m^2}{1-x^2}\right]P_l^m(x) = 0.$$

From the differential equations for $P_l^m(x)$ and $P_l^k(x)$ show that for integer orders $0 \le m, k \le l$ with $k \ne m$,

$$\int_{-1}^{1} P_l^m(x)P_l^k(x)\frac{dx}{1-x^2} = 0.$$

Solution.

The eigenvalue that is compared is m^2, so the required condition is $k^2 \ne m^2$, or distinct nonnegative orders. If $k = -m \ne 0$, the two functions are proportional and the stated zero integral is false. Let $u = P_l^m$, $v = P_l^k$, with nonnegative integer orders $m, k \le l$ and $m \ne k$. In self-adjoint form their equations are

$$[(1-x^2)u']' + \left[l(l+1) - \frac{m^2}{1-x^2}\right]u = 0$$

and the analogous equation for v. Multiply the first by v, the second by u, and subtract:

$$\frac{d}{dx}[(1-x^2)(vu' - uv')] + \frac{k^2-m^2}{1-x^2}uv = 0.$$

At least one order is positive. The endpoint products in the bracket vanish: a positive-order function is $(1-x^2)^{m/2}$ times a polynomial, and the combined weighted Wronskian is $O((1-x^2)^{(m+k)/2})$ or smaller. Also $uv/(1-x^2) = O((1-x^2)^{(m+k)/2-1})$, which is integrable because $m+k > 0$. Integration therefore gives

$$(k^2-m^2)\int_{-1}^{1} \frac{P_l^m P_l^k}{1-x^2}dx = 0.$$

Division by the nonzero eigenvalue difference proves the result. As a concrete excluded case, $P_1^1 = -\sqrt{1-x^2}$ and $P_1^{-1} = \sqrt{1-x^2}/2$ give integral -1, not zero.

15.4.15 Determine the vector potential and the magnetic induction field of a magnetic quadrupole by differentiating the magnetic dipole potential.

$$\textit{ANS.}\quad \mathbf{A}_{MQ} = -\frac{\mu_0}{2}(Ia^2)(dz)\frac{P_2^1(\cos\theta)}{r^3}\,\hat{\mathbf{e}}_\varphi + \text{higher-order terms},$$
$$\mathbf{B}_{MQ} = \mu_0(Ia^2)(dz)\left[\frac{3P_2(\cos\theta)}{r^4}\hat{\mathbf{e}}_r - \frac{P_2^1(\cos\theta)}{r^4}\hat{\mathbf{e}}_\theta + \cdots\right].$$

This corresponds to placing a current loop of radius a at $z \to dz$ and an oppositely directed current loop at $z \to -dz$. The vector potential and magnetic induction field of a point quadrupole are given by the leading terms in these expansions if we take the limit $dz \to 0$, $a \to 0$, and $I \to \infty$ subject to $Ia^2dz =$ constant.

Solution.

Take a positive current loop at $+h$ and the opposite loop at $-h$, where $h = dz$ in the question. A small loop at the origin has

$$A_\varphi^{(D)} = \frac{\mu_0 Ia^2}{4}\frac{\sin\theta}{r^2} = -\frac{\mu_0 Ia^2}{4}\frac{P_1^1(\cos\theta)}{r^2}.$$

Translation along the z axis does not rotate $\hat{\mathbf{e}}_\varphi$. The difference of the two translated potentials is $-2h\,\partial_z A_\varphi^{(D)} + O(h^3)$. For fixed cylindrical radius ρ, $\sin\theta/r^2 = \rho/(\rho^2+z^2)^{3/2}$, so

$$\partial_z(\sin\theta/r^2) = -3\rho z/r^5 = \frac{P_2^1(\cos\theta)}{r^3}.$$

Thus, writing $C = \mu_0 Ia^2 h/2$,

$$\boxed{\mathbf{A}_{MQ} = -C\frac{P_2^1(\cos\theta)}{r^3}\hat{\mathbf{e}}_\varphi}.$$

The axisymmetric curl has components $B_r = (r\sin\theta)^{-1}\partial_\theta(\sin\theta A_\varphi)$ and $B_\theta = -r^{-1}\partial_r(rA_\varphi)$. The Legendre ODE gives $\partial_\theta(\sin\theta P_2^1) = -6\sin\theta P_2$. Consequently

$$B_r = \frac{6C}{r^4}P_2, \qquad B_\theta = -\frac{2C}{r^4}P_2^1, \qquad B_\varphi = 0.$$

These are exactly the stated coefficients after substituting C. For fixed observation points away from the origin, the next translation term is $O(Ia^2h^3) = O((Ia^2h)h^2)$, and the next finite-radius loop term is $O(Ia^4h) = O((Ia^2h)a^2)$. Both vanish in the simultaneous point limit at fixed Ia^2h. The final sentence of the problem calls this a point dipole. This construction and its r^{-4} field describe a point quadrupole.

15.4.16 A single circular wire loop of radius a carries a constant current I. Model the wire as a line of zero cross section; choose $I > 0$ counterclockwise when seen from $+z$, with the flux normal along $+z$.

(a) Find the magnetic induction $\mathbf{B}$ for $r < a$, $\theta = \pi/2$.

(b) Calculate the integral of the magnetic flux $(\mathbf{B}\cdot d\boldsymbol{\sigma})$ over the area of the current loop, that is,

$$\int_0^a r\,dr\int_0^{2\pi} d\varphi\, B_z\left(r, \theta = \frac{\pi}{2}\right).$$

ANS. ∞.

The Earth is within such a ring current, in which I approximates millions of amperes arising from the drift of charged particles in the Van Allen belt.

Solution.

Let $\rho = r/a < 1$ in the equatorial plane. To obtain an interior expansion, first use the exact field on the axis:

$$B_z(z) = \frac{\mu_0 I}{2a}(1 + z^2/a^2)^{-3/2} = \frac{\mu_0 I}{2a}\sum_{n=0}^{\infty}(-1)^n \frac{(2n+1)!!}{(2n)!!}(z/a)^{2n}.$$

In the source-free ball, write $\mathbf{B} = -\nabla\Psi$ and match this axis series with regular solid harmonics. Integration in z determines

$$\Psi = -\frac{\mu_0 I}{2}\sum_{n=0}^{\infty}(-1)^n \frac{(2n-1)!!}{(2n)!!}\left(\frac{r}{a}\right)^{2n+1} P_{2n+1}(\cos\theta).$$

Indeed $-\partial_z\Psi$ on the positive axis reproduces the preceding series term by term; regular axial harmonic expansions are unique.

(a) At the equator $P_{2n+1}(0) = 0$, so $B_r = 0$. Differentiating Ψ in θ, and using Exercise 15.4.4, gives

$$\boxed{\mathbf{B}(r, \pi/2) = \frac{\mu_0 I}{2a}\sum_{n=0}^{\infty}(2n+1)\left[\frac{\binom{2n}{n}}{4^n}\right]^2 \rho^{2n}\,\hat{\mathbf{z}}}.$$

Here $\hat{\mathbf{z}} = -\hat{\mathbf{e}}_\theta$ on the equator. The first terms are $\mu_0 I[1 + \frac{3}{4}\rho^2 + \frac{45}{64}\rho^4 + \cdots]/(2a)$.

(b) All coefficients $c_n = (2n+1)[\binom{2n}{n}/4^n]^2$ are positive for positive I. Tonelli's theorem permits integration to the edge as an extended nonnegative integral:

$$\Phi = 2\pi\int_0^a rB_z\,dr = \pi\mu_0 I a\sum_{n=0}^{\infty}\frac{c_n}{2n+2}.$$

Stirling's formula gives $c_n \to 2/\pi$, so the summand is asymptotic to $1/(\pi n)$. The harmonic comparison proves

$$\boxed{\Phi = +\infty \quad (I > 0)}$$

for an ideal zero-thickness wire, with the sign reversed if the current or surface orientation is reversed. For $r < a$ the field itself is finite; the divergence occurs only when the flux integral reaches the wire.

15.4.17 The vector potential $\mathbf{A}$ of a magnetic dipole, dipole moment $\mathbf{m}$, is given by $\mathbf{A}(\mathbf{r}) = (\mu_0/4\pi)(\mathbf{m}\times\mathbf{r}/r^3)$. Show by direct computation that the magnetic induction $\mathbf{B} = \nabla\times\mathbf{A}$ is given, for $r > 0$, by

$$\mathbf{B} = \frac{\mu_0}{4\pi}\frac{3\hat{\mathbf{r}}(\hat{\mathbf{r}}\cdot\mathbf{m}) - \mathbf{m}}{r^3}.$$

Solution.

The formula describes points $r > 0$. Put $\mathbf{F} = \mathbf{r}/r^3$. For constant $\mathbf{m}$, the vector product rule gives

$$\nabla\times(\mathbf{m}\times\mathbf{F}) = \mathbf{m}(\nabla\cdot\mathbf{F}) - (\mathbf{m}\cdot\nabla)\mathbf{F}.$$

The required derivatives can be computed componentwise:

$$\partial_j F_i = \frac{\delta_{ij}}{r^3} - \frac{3r_i r_j}{r^5}, \qquad \nabla \cdot \mathbf{F} = \frac{3}{r^3} - \frac{3r^2}{r^5} = 0.$$

Therefore

$$[(\mathbf{m} \cdot \nabla)\mathbf{F}]_i = \frac{m_i}{r^3} - \frac{3r_i(\mathbf{m} \cdot \mathbf{r})}{r^5}.$$

Multiplying its negative by $\mu_0/(4\pi)$ yields

$$\boxed{\mathbf{B} = \frac{\mu_0}{4\pi r^3}[3\hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \mathbf{m}) - \mathbf{m}]}.$$

The ordinary-function field excludes $r = 0$. Its distributional curl contains an additional contact term, not a value assigned to r^{-3} at the origin. To find that term explicitly, integrate $-F_i \partial_j \phi$ outside a sphere of radius ϵ and integrate by parts. The inner-boundary contribution tends to

$$\phi(0) \int_{S^2} \hat{r}_i \hat{r}_j \, d\Omega = \frac{4\pi}{3} \delta_{ij} \phi(0).$$

The off-diagonal integrals vanish by reflection; the three diagonal integrals are equal and sum to $\int d\Omega = 4\pi$. Therefore

$$\partial_j F_i = \mathrm{PV}\left(\frac{\delta_{ij}}{r^3} - \frac{3r_i r_j}{r^5}\right) + \frac{4\pi}{3} \delta_{ij} \delta(\mathbf{r}),$$

where the principal value uses spherical exclusions. Its trace is $\nabla \cdot \mathbf{F} = 4\pi\delta(\mathbf{r})$. In the curl formula, the two source terms combine as $\mu_0(4\pi - 4\pi/3)\mathbf{m}\delta/(4\pi) = (2\mu_0/3)\mathbf{m}\delta(\mathbf{r})$, in addition to the principal value of the displayed field.

15.4.18

(a) Show that in the point dipole limit the magnetic induction field of the current loop becomes

$$B_r(r, \theta) = \frac{\mu_0 m}{2\pi r^3} P_1(\cos\theta),$$
$$B_\theta(r, \theta) = -\frac{\mu_0 m}{4\pi r^3} P_1^1(\cos\theta),$$

with $m = I\pi a^2$ and $r > 0$.

(b) Compare these results with the magnetic induction of the point magnetic dipole of Exercise 15.4.17. Take $\mathbf{m} = \hat{\mathbf{z}} m$.

Solution.

The leading small-loop vector potential is $A_\varphi = \mu_0 I a^2 \sin\theta/(4r^2) = \mu_0 m \sin\theta/(4\pi r^2)$, where $m = I\pi a^2$. Its curl gives

$$B_r = \frac{1}{r\sin\theta} \partial_\theta(\sin\theta A_\varphi) = \frac{\mu_0 m}{2\pi r^3} \cos\theta,$$
$$B_\theta = -\frac{1}{r} \partial_r(r A_\varphi) = \frac{\mu_0 m}{4\pi r^3} \sin\theta.$$

Thus

$$\boxed{B_r = \frac{\mu_0 m}{2\pi r^3} P_1(\cos\theta), \qquad B_\theta = -\frac{\mu_0 m}{4\pi r^3} P_1^1(\cos\theta)}.$$

The printed B_θ denominator 2π must be 4π. Its value as written is too large by a factor of two. For part (b), insert $\mathbf{m} = m\hat{\mathbf{z}}$ into Exercise 15.4.17 and use $\hat{\mathbf{z}} = \cos\theta\,\hat{\mathbf{e}}_r - \sin\theta\,\hat{\mathbf{e}}_\theta$. Then

$$3m\cos\theta\,\hat{\mathbf{e}}_r - m\hat{\mathbf{z}} = 2m\cos\theta\,\hat{\mathbf{e}}_r + m\sin\theta\,\hat{\mathbf{e}}_\theta,$$

which agrees with both corrected components.

15.4.19 A uniformly charged spherical shell is rotating with constant angular velocity.

(a) Calculate the magnetic induction $\mathbf{B}$ along the axis of rotation outside the sphere.

(b) Using the vector potential series of Example 15.4.2, find $\mathbf{A}$ and then $\mathbf{B}$ for all points outside the sphere.

Solution.

Let the shell radius be a, its uniform surface charge density be σ, and its angular velocity be $\omega\hat{\mathbf{z}}$. A strip at polar angle α has charge $dq = 2\pi\sigma a^2 \sin\alpha\, d\alpha$ and current $dI = (\omega/2\pi)dq = \sigma\omega a^2 \sin\alpha\, d\alpha$. Its circular radius is $a\sin\alpha$ and its center is $a\cos\alpha$.

(a) Add the standard axial field of each strip:

$$B_z(z) = \frac{\mu_0\sigma\omega a^4}{2}\int_{-1}^{1}\frac{1-u^2}{(z^2+a^2-2azu)^{3/2}}du, \qquad z > a.$$

Set $t = z^2 + a^2 - 2azu$. The integral becomes

$$\frac{1}{8a^3z^3}\int_{(z-a)^2}^{(z+a)^2}[4a^2z^2 - (z^2+a^2-t)^2]t^{-3/2}dt.$$

An antiderivative of the numerator times $t^{-3/2}$ is

$$2(z^2-a^2)^2t^{-1/2} + 4(z^2+a^2)t^{1/2} - \frac{2}{3}t^{3/2}.$$

Since $z > a$, the endpoint square roots are $z+a$ and $z-a$. The difference of the antiderivative values is

$$\begin{aligned}&2(z^2-a^2)^2\left(\frac{1}{z+a}-\frac{1}{z-a}\right) + 8a(z^2+a^2) - \frac{2}{3}[(z+a)^3-(z-a)^3]\\ &\quad = -4a(z^2-a^2) + 8a(z^2+a^2) - 4az^2 - \frac{4}{3}a^3 = \frac{32a^3}{3}.\end{aligned}$$

Multiplication by $1/(8a^3z^3)$ leaves $4/(3z^3)$. Therefore

$$\boxed{\mathbf{B}(z\hat{\mathbf{z}}) = \frac{2\mu_0\sigma\omega a^4}{3|z|^3}\hat{\mathbf{z}}}, \qquad |z| > a,$$

where reflection symmetry supplies the negative axis.

(b) Use the same decaying vector-potential expansion as the loop example:

$$A_\varphi = \sum_{l\geq 1} c_l r^{-l-1} P_l^1(\cos\theta).$$

Its radial curl is $B_r = -\sum l(l+1)c_l r^{-l-2}P_l(\cos\theta)$. On the positive axis, $P_l(1) = 1$, and part (a) has only z^{-3}. Uniqueness of the power series therefore gives $c_1 = -\mu_0\sigma\omega a^4/3$ and $c_l = 0$ for $l > 1$. Consequently

$$\boxed{A_\varphi = -\frac{\mu_0\sigma\omega a^4}{3r^2}P_1^1 = \frac{\mu_0\sigma\omega a^4}{3r^2}\sin\theta},$$

$$\boxed{B_r = \frac{2\mu_0\sigma\omega a^4}{3r^3}\cos\theta, \qquad B_\theta = \frac{\mu_0\sigma\omega a^4}{3r^3}\sin\theta, \qquad B_\varphi = 0}.$$

This is exactly a dipole with $m = 4\pi\sigma\omega a^4/3 = Q\omega a^2/3$, $Q = 4\pi a^2\sigma$. The manual's part (a), p. 252, has an extra factor 3 in its denominator: its displayed coefficient is 2/9 instead of 2/3. This also conflicts with its own radial field on the positive axis. In its expressions in terms of P_1^1, both A_φ and B_θ require a minus sign. The printed positive signs conflict with $P_1^1 = -\sin\theta$ and with the printed positive radial field.

15.4.20 In the liquid-drop model of the nucleus, a spherical nucleus is subjected to small deformations. Consider a sphere of radius r_0 that is deformed so that its new surface is given by

$$r = r_0\big[1 + \alpha_2 P_2(\cos\theta)\big].$$

Find the area of the deformed sphere through terms of order α_2^2.

Hint.

$$dA = \left[r^2 + \left(\frac{dr}{d\theta}\right)^2\right]^{1/2} r\sin\theta\, d\theta\, d\varphi.$$

ANS. $A = 4\pi r_0^2\left[1 + \frac{4}{5}\alpha_2^2 + O\big(\alpha_2^3\big)\right]$.

Note. The area element dA follows from noting that the line element ds for fixed φ is given by

$$ds = (r^2 d\theta^2 + dr^2)^{1/2} = \left[r^2 + \left(\frac{dr}{d\theta}\right)^2\right]^{1/2} d\theta.$$

Solution.

Put $\alpha = \alpha_2$, $P = P_2(\cos\theta)$ and $\dot P = dP_2(\cos\theta)/d\theta$. Then $r = r_0(1 + \alpha P)$ and $r_\theta = r_0\alpha\dot P$. For small α the radius is positive, so the area integrand is

$$\begin{aligned} r\sqrt{r^2 + r_\theta^2} &= r^2\sqrt{1 + (r_\theta/r)^2} \\ &= r^2 + \frac{1}{2}r_\theta^2 + O(\alpha^3) \\ &= r_0^2[1 + 2\alpha P + \alpha^2(P^2 + \tfrac{1}{2}\dot P^2)] + O(\alpha^3). \end{aligned}$$

The first-order integral vanishes because $\int_{-1}^{1} P_2 dx = 0$. The norm gives $\int_{-1}^{1} P_2^2 dx = 2/5$. Also $\dot P = -3\cos\theta\sin\theta$, whence

$$\int_0^\pi \dot P^2 \sin\theta\, d\theta = 9\int_{-1}^{1}(x^2 - x^4)dx = 9\left(\frac{2}{3} - \frac{2}{5}\right) = \frac{12}{5}.$$

Integrating over φ now gives

$$A = 2\pi r_0^2\left[2 + \alpha^2\left(\frac{2}{5} + \frac{6}{5}\right) + O(\alpha^3)\right] = \boxed{4\pi r_0^2[1 + \tfrac{4}{5}\alpha_2^2 + O(\alpha_2^3)]}.$$

The reference radius is kept as specified in the given surface equation.

15.5 Spherical Harmonics

15.5.1 Show that the parity of $Y_l^m(\theta,\varphi)$ is $(-1)^l$. Note the disappearance of any m dependence.
Hint. For the parity operation in spherical polar coordinates, see Exercise 3.10.25.

Solution.

Spatial parity sends $\mathbf{r}$ to $-\mathbf{r}$, so $(\theta,\varphi)\mapsto(\pi-\theta,\varphi+\pi)$. Rodrigues's formula gives $P_l^m(-x)=(-1)^{l+m}P_l^m(x)$ for integer admissible orders; negative m follow from the factorial relation. The azimuthal factor changes by $e^{im(\varphi+\pi)}=(-1)^m e^{im\varphi}$. The normalization constant is unchanged. Multiplying the two signs,

$$\boxed{Y_l^m(\pi-\theta,\varphi+\pi)=(-1)^{l+2m}Y_l^m(\theta,\varphi)=(-1)^l Y_l^m(\theta,\varphi)}.$$

Thus it is spatial parity, rather than θ reflection alone, that removes the m dependence.

15.5.2 Prove that $Y_l^m(0,\varphi)=\left(\dfrac{2l+1}{4\pi}\right)^{1/2}\delta_{m0}$.

Solution.

For $m>0$, $P_l^m(x)=(-1)^m(1-x^2)^{m/2}P_l^{(m)}(x)$. The derivative is a finite polynomial, so at $x=1$ the factor $(1-x^2)^{m/2}$ forces $P_l^m(1)=0$. The relation between positive and negative orders gives the same zero for $m<0$. For $m=0$, $P_l(1)=1$, the factorial normalization ratio is one, and the azimuthal exponential is one. Therefore

$$\boxed{Y_l^m(0,\varphi)=\sqrt{\frac{2l+1}{4\pi}}\delta_{m0}}.$$

The result is independent of the otherwise undefined azimuthal coordinate at the pole.

15.5.3 In the theory of Coulomb excitation of nuclei we encounter $Y_l^m(\pi/2,0)$. Show that

$$\begin{aligned}Y_l^m\left(\frac{\pi}{2},0\right)&=\left(\frac{2l+1}{4\pi}\right)^{1/2}\frac{[(l-m)!\,(l+m)!]^{1/2}}{(l-m)!!\,(l+m)!!}\,(-1)^{(l+m)/2},\quad l+m\text{ even},\\&=0,\qquad l+m\text{ odd}.\end{aligned}$$

Solution.

The book's definition, Eq. (15.137), is

$$Y_l^m(\theta,\varphi)=\sqrt{\frac{2l+1}{4\pi}\frac{(l-m)!}{(l+m)!}}\,P_l^m(\cos\theta)e^{im\varphi},$$

with the Condon–Shortley sign already in P_l^m. At the point in question, $x=0$ and the exponential equals one. If $l+m$ is odd, Exercise 15.4.5 gives zero. If $l+m$ is even, substitution gives

$$Y_l^m(\tfrac{\pi}{2},0)=(-1)^{(l+m)/2}\sqrt{\frac{2l+1}{4\pi}\frac{(l-m)!}{(l+m)!}}\,\frac{(l+m-1)!!}{(l-m)!!}.$$

Use $(l+m)! = (l+m)!!(l+m-1)!!$ to rewrite the magnitude:

$$\boxed{Y_l^m(\tfrac{\pi}{2},0) = (-1)^{(l+m)/2}\sqrt{\frac{2l+1}{4\pi}}\,\frac{\sqrt{(l-m)!(l+m)!}}{(l-m)!!(l+m)!!}}.$$

The negative-order identity extends the result to $-l \le m < 0$ as well. The printed phase $(-1)^{(l-m)/2}$ differs by $(-1)^m$ from the phase required by the book's own definition. For $l = m = 1$, $Y_1^1(\pi/2, 0) = -\sqrt{3/(8\pi)}$, which directly disproves the printed positive sign. The manual obtains the correct $(l+m)/2$ phase before claiming agreement with the printed result.

15.5.4 The orthogonal azimuthal functions yield a useful representation of the Dirac delta function. Show that as a distribution on the circle

$$\delta(\varphi_1 - \varphi_2) = \frac{1}{2\pi}\sum_{m=-\infty}^{\infty} e^{im(\varphi_1-\varphi_2)}.$$

Note. This formula assumes that φ_1 and φ_2 are restricted to $0 \le \varphi < 2\pi$. Without this restriction there will be additional delta-function contributions at intervals of 2π in $\varphi_1 - \varphi_2$.

Solution.

On the circle, the normalized functions are $\phi_m(\varphi) = e^{im\varphi}/\sqrt{2\pi}$. For a smooth periodic f, its Fourier coefficients are $c_m = (2\pi)^{-1}\int_0^{2\pi} f(\varphi_2)e^{-im\varphi_2}d\varphi_2$. Pair the proposed kernel with f:

$$\int_0^{2\pi}\frac{1}{2\pi}\sum_{m\in\mathbb{Z}} e^{im(\varphi_1-\varphi_2)}f(\varphi_2)d\varphi_2 = \sum_m c_m e^{im\varphi_1} = f(\varphi_1).$$

Smoothness makes this test-function Fourier series converge, proving the periodic delta identity. One explicit regularization is to insert $\rho^{|m|}$:

$$\frac{1}{2\pi}\sum_m \rho^{|m|}e^{imu} = \frac{1-\rho^2}{2\pi(1-2\rho\cos u+\rho^2)}, \quad 0<\rho<1.$$

It is nonnegative, has total integral one on the circle, and tends uniformly to zero away from $u = 0$ modulo 2π; hence it tends to the periodic delta as $\rho \uparrow 1$. On the real line the same distribution is $\sum_{k\in\mathbb{Z}}\delta(u-2\pi k)$, in agreement with the note.

15.5.5 Derive the spherical harmonic closure relation as an identity of distributions for spherical area measure, using $0 < \theta_1 < \pi$ in its coordinate form and the periodic delta in azimuth:

$$\begin{aligned}\sum_{l=0}^{\infty}\sum_{m=-l}^{+l} Y_l^m(\theta_1,\varphi_1)\big[Y_l^m(\theta_2,\varphi_2)\big]^* &= \frac{1}{\sin\theta_1}\,\delta(\theta_1-\theta_2)\,\delta(\varphi_1-\varphi_2)\\ &= \delta(\cos\theta_1-\cos\theta_2)\,\delta(\varphi_1-\varphi_2).\end{aligned}$$

Solution.

Let $d\Omega = \sin\theta\, d\theta\, d\varphi$ and define

$$K(\Omega_1,\Omega_2) = \sum_{l,m} Y_l^m(\Omega_1)Y_l^m(\Omega_2)^*.$$

For a smooth function $f = \sum f_{lm} Y_l^m$, with $f_{lm} = \int Y_l^m(\Omega)^* f(\Omega) d\Omega$,

$$\int K(\Omega_1, \Omega_2) f(\Omega_2) d\Omega_2 = \sum_{l,m} f_{lm} Y_l^m(\Omega_1) = f(\Omega_1).$$

Thus K is the delta distribution for spherical area measure.

For completeness of the basis used here, first expand in Fourier modes in φ. At fixed m, the polar factors are $(1-x^2)^{|m|/2}$ times polynomials of successive degrees in $x = \cos\theta$. If an L^2 function is orthogonal to every such factor, its product with $(1-x^2)^{|m|/2}$ is orthogonal to every polynomial. Polynomial density makes that product zero almost everywhere, and then the original function is zero almost everywhere. This proves completeness of each polar family and hence of the combined spherical harmonics.

The distributional passage to the infinite kernel can be made without assuming pointwise convergence of an arbitrary L^2 series. Let K_N retain degrees $l \le N$, so integration against f gives its orthogonal projection $S_N f$. For every smooth test function h,

$$\left| \int h^*(\Omega_1) \left[\int K_N(\Omega_1, \Omega_2) f(\Omega_2) d\Omega_2 - f(\Omega_1) \right] d\Omega_1 \right| \le \|h\|_2 \|S_N f - f\|_2 \longrightarrow 0.$$

Thus the limiting kernel acts as the identity when tested in both variables, which is precisely the delta kernel. For smooth f this agrees with the evaluation formula used above.

The coordinate kernel that reproduces the same test-function action is

$$\boxed{K = \frac{\delta(\theta_1 - \theta_2)\delta_{2\pi}(\varphi_1 - \varphi_2)}{\sin\theta_1} = \delta(\cos\theta_1 - \cos\theta_2)\delta_{2\pi}(\varphi_1 - \varphi_2)}.$$

Indeed $d(\cos\theta) = -\sin\theta\, d\theta$, so the Jacobian cancels. The closure formula is a distributional identity. Its polar coordinate form is used for $0 < \theta_1 < \pi$; at a pole the invariant spherical delta, rather than division by a zero sine, specifies its meaning.

15.5.6 In some circumstances it is desirable to replace the imaginary exponential of our spherical harmonic by sine or cosine. Morse and Feshbach (see Additional Readings) define

$$\overset{e}{Y}_{ml} = P_l^m(\cos\theta)\cos m\varphi, \qquad m \ge 0,$$
$$\overset{o}{Y}_{ml} = P_l^m(\cos\theta)\sin m\varphi, \qquad m > 0,$$

and their normalization integrals are

$$\int_0^{2\pi}\int_0^{\pi} \left[\overset{e \text{ or } o}{Y}_{mn}(\theta,\varphi) \right]^2 \sin\theta\, d\theta\, d\varphi = \frac{4\pi}{2(2n+1)} \frac{(n+m)!}{(n-m)!}, \qquad 1 \le m \le n,$$
$$= \frac{4\pi}{2n+1}, \qquad m = 0,\ n \ge 1 \quad \text{(cosine family)},$$
$$= 4\pi, \qquad n = 0.$$

These spherical harmonics are often named according to the patterns of their positive and negative regions on the surface of a sphere: zonal harmonics for $m = 0$, sectoral harmonics for $m = n$, and tesseral harmonics for $0 < m < n$. For $\overset{e}{Y}_{mn}$, $n = 4$, $m = 0, 2, 4$, indicate on a diagram of a hemisphere (one diagram for each spherical harmonic) the regions in which the spherical harmonic is positive.

Solution.

First separate the two normalization integrals. The polar one is $2(n+m)!/[(2n+1)(n-m)!]$. For $m>0$, both $\int_0^{2\pi}\cos^2 m\varphi\,d\varphi$ and its sine counterpart equal π, giving the stated $2\pi(n+m)!/[(2n+1)(n-m)!]$. For $m=0$, the cosine factor is identically one and its integral is 2π, so

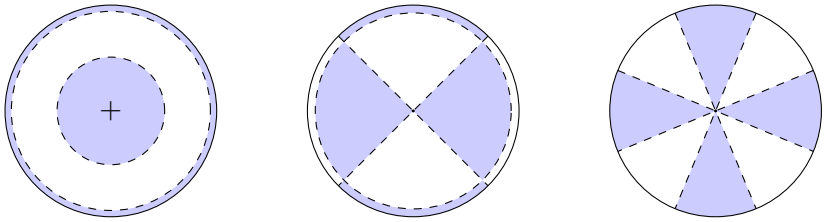
$$\boxed{\|Y^e_{0n}\|^2=\frac{4\pi}{2n+1}}.$$

The printed normalization for $n\geq 1$ is valid for $m>0$. For the included $m=0$ harmonics it is smaller than the correct norm by a factor of two. The separate $n=0$ value 4π is correct.

For $n=4$, the three real functions are

$$Y^e_{04}=\frac{1}{8}(35x^4-30x^2+3),\quad Y^e_{24}=\frac{15}{2}(1-x^2)(7x^2-1)\cos 2\varphi,\quad Y^e_{44}=105(1-x^2)^2\cos 4\varphi,$$

where $x=\cos\theta$. The manual's first polynomial on p. 253 prints $35\cos^2\theta$ in the leading term. It must be $35\cos^4\theta$, as in P_4 above; the squared power would incorrectly reduce this degree-four polynomial to degree two. On the north hemisphere, viewed from above, put $\rho=\sin\theta$, so its orthographic projection is the unit disk. In the diagrams blue regions are positive; white regions are negative, and boundary curves are zeros.



$m=0$ (zonal) $m=2$ (tesseral) $m=4$ (sectoral)

For $m=0$, the two latitude zeros have $x^2=(15\pm 2\sqrt{30})/35$; the polar cap and the equatorial band are positive. For $m=2$, the latitude zero is $x=1/\sqrt{7}$ and the meridians are $\varphi=\pi/4+j\pi/2$. Inside that latitude circle the sign is that of $\cos 2\varphi$; outside it the sign reverses. For $m=4$ there are no interior latitude zeros, and positivity is $\cos 4\varphi>0$, namely sectors centered at $\varphi=j\pi/2$ of half-width $\pi/8$. The pole is also a zero for $m=2,4$.

15.5.7 Suppose $f(r,\theta,\varphi)$ has the regular Laplace series below, converging uniformly on the closed ball under consideration:

$$f(r,\theta,\varphi)=\sum_{l,m}a_{lm}r^l Y_l^m(\theta,\varphi).$$

Letting $\langle\cdots\rangle_{\text{sphere}}$ denote the average over a sphere centered on the origin, show that

$$\big\langle f(r,\theta,\varphi)\big\rangle_{\text{sphere}}=f(0,0,0).$$

Solution.

Use $Y_0^0=1/\sqrt{4\pi}$ and orthogonality:

$$\int Y_l^m\,d\Omega=\sqrt{4\pi}\int Y_l^m(Y_0^0)^*d\Omega=\sqrt{4\pi}\,\delta_{l0}\delta_{m0}.$$

Averaging the regular solid-harmonic expansion over a sphere of radius r therefore leaves only its constant term:

$$\langle f\rangle_{\text{sphere}} = \frac{1}{4\pi}\sum_{l,m} a_{lm} r^l \int Y_l^m \, d\Omega = \frac{a_{00}}{\sqrt{4\pi}}.$$

At the origin every $r^l Y_l^m$ with $l > 0$ tends to zero, whereas the $l = 0$ term has the same value $a_{00}/\sqrt{4\pi}$. Thus

$$\boxed{\langle f\rangle_{\text{sphere}} = f(0,0,0)}.$$

The termwise average is justified on spheres inside the convergence region of the stated regular Laplace expansion. This is the mean-value property for the assumed regular harmonic expansion. It does not hold for an arbitrary function; for example, $f = r^2$ has sphere average r^2 but $f(0) = 0$.

15.6 Legendre Functions of the Second Kind

15.6.1 Use the real branches specified below, with $x \in \mathbb{R} \setminus \{-1, 1\}$ and integer $l \geq 0$. Show that if l is even, $Q_l(-x) = -Q_l(x)$, and that if l is odd, $Q_l(-x) = Q_l(x)$.

Solution.

Use the real Legendre functions with

$$Q_0(x) = \frac{1}{2}\ln\left|\frac{1+x}{1-x}\right|, \qquad Q_1(x) = xQ_0(x) - 1, \qquad x \neq \pm 1.$$

Replacing x by $-x$ inverts the logarithm's positive absolute argument, so Q_0 is odd and Q_1 is even. Assume $Q_n(-x) = (-1)^{n+1}Q_n(x)$ and $Q_{n-1}(-x) = (-1)^n Q_{n-1}(x)$. Then

$$\begin{aligned}(n+1)Q_{n+1}(-x) &= -(2n+1)x(-1)^{n+1}Q_n(x) - n(-1)^n Q_{n-1}(x)\\ &= (-1)^{n+2}[(2n+1)xQ_n(x) - nQ_{n-1}(x)]\\ &= (-1)^{n+2}(n+1)Q_{n+1}(x).\end{aligned}$$

Induction proves

$$\boxed{Q_l(-x) = (-1)^{l+1}Q_l(x)}.$$

In particular even degree gives an odd function and odd degree an even function. The assertion refers to these consistent real/Ferrers branches; arbitrary independently chosen logarithm branches would introduce additional multiples of P_l.

15.6.2 Show that, for nonnegative integers p and the real analytic Q_l branch at $x = 0$,

(a) $$Q_{2p}(x) = (-1)^p 2^{2p}\sum_{s=0}^{p}(-1)^s \frac{(p+s)!\,(p-s)!}{(2s+1)!\,(2p-2s)!}\,x^{2s+1} + 2^{2p}\sum_{s=p+1}^{\infty}\frac{(p+s)!\,(2s-2p)!}{(2s+1)!\,(s-p)!}\,x^{2s+1}, \qquad |x| < 1,$$

(b) $$Q_{2p+1}(x) = (-1)^{p+1}2^{2p}\sum_{s=0}^{p}(-1)^s \frac{(p+s)!\,(p-s)!}{(2s)!\,(2p-2s+1)!}\,x^{2s} + 2^{2p+1}\sum_{s=p+1}^{\infty}\frac{(p+s)!\,(2s-2p-2)!}{(2s)!\,(s-p-1)!}\,x^{2s}, \qquad |x| < 1.$$

Solution.

Use the real analytic branch at $x = 0$. The Legendre equation gives the coefficient recurrence for any Taylor solution:

$$(j+2)(j+1)a_{j+2} = [j(j+1) - l(l+1)]a_j.$$

Its initial scales follow from $Q_0 = x + \cdots$, $Q_1 = -1 + \cdots$ and the degree recurrence:

$$Q'_{2p}(0) = (-1)^p \frac{(2p)!!}{(2p-1)!!}, \qquad Q_{2p+1}(0) = (-1)^{p+1} \frac{(2p)!!}{(2p+1)!!}.$$

For example, the values for the odd-degree branch obey $(2p+1)Q_{2p+1}(0) = -2pQ_{2p-1}(0)$. For the even-degree derivative, $(1-x^2)Q'_{2p} = 2p(Q_{2p-1} - xQ_{2p})$ gives

$$Q'_{2p}(0) = 2pQ_{2p-1}(0) = (-1)^p \frac{2p(2p-2)!!}{(2p-1)!!} = (-1)^p \frac{(2p)!!}{(2p-1)!!} \quad (p \geq 1).$$

The case $p = 0$ comes from Q_0.

(a) Write $Q_{2p} = \sum_{s\geq 0} C_s x^{2s+1}$. The Taylor recurrence is

$$\frac{C_{s+1}}{C_s} = \frac{(s+p+1)(2s-2p+1)}{(s+1)(2s+3)}, \qquad C_0 = (-1)^p 2^{2p} \frac{(p!)^2}{(2p)!}.$$

For $0 \leq s \leq p$, its product gives

$$C_s = (-1)^{p+s} 2^{2p} \frac{(p+s)!(p-s)!}{(2s+1)!(2p-2s)!}.$$

To check the product explicitly, the factorial ratio between adjacent terms is

$$-\frac{p+s+1}{p-s} \frac{(2p-2s)(2p-2s-1)}{(2s+3)(2s+2)} = \frac{(p+s+1)(2s-2p+1)}{(s+1)(2s+3)}.$$

At the transition, $C_p = 2^{2p}/(2p+1)$ and $C_{p+1} = 2^{2p}/[(p+1)(2p+3)]$. For $s \geq p+1$, the all-positive continuation is

$$C_s = 2^{2p} \frac{(p+s)!(2s-2p)!}{(2s+1)!(s-p)!}.$$

Its adjacent ratio reduces to the same recurrence by canceling $2(s-p+1)$ against $s-p+1$. These formulas give exactly the two sums in part (a).

(b) Put $Q_{2p+1} = \sum_{s\geq 0} D_s x^{2s}$. This time

$$\frac{D_{s+1}}{D_s} = \frac{(s+p+1)(2s-2p-1)}{(s+1)(2s+1)}, \qquad D_0 = (-1)^{p+1} 2^{2p} \frac{(p!)^2}{(2p+1)!}.$$

For $0 \leq s \leq p$, repeated multiplication gives

$$D_s = (-1)^{p+1+s} 2^{2p} \frac{(p+s)!(p-s)!}{(2s)!(2p-2s+1)!}.$$

Its ratio is $-(p+s+1)(2p-2s+1)(2p-2s)/[(p-s)(2s+2)(2s+1)]$, equal to the required ratio. The transition has $D_p = -2^{2p}$ and $D_{p+1} = 2^{2p}/(p+1)$. For the positive tail,

$$D_s = 2^{2p+1} \frac{(p+s)!(2s-2p-2)!}{(2s)!(s-p-1)!}, \qquad s \geq p+1.$$

Canceling consecutive factorials again gives the same recurrence. The initial value, all within-range ratios and both transition values therefore establish the two requested sums without leaving a missing index case. Their term ratios tend to x^2, proving convergence for $|x| < 1$; uniqueness of the Taylor solution identifies them with Q_l.

15.6.3

(a) Use integer $l \geq 0$ and the exterior Q_l branch decaying at infinity, on $|x| > 1$. Starting with the assumed form

$$Q_l(x) = \sum_{j=0}^{\infty} b_{lj} x^{k-j},$$

show that

$$Q_l(x) = b_{l0} x^{-l-1} \sum_{s=0}^{\infty} \frac{(l+s)!\,(l+2s)!\,(2l+1)!}{s!\,(l!)^2 (2l+2s+1)!}\, x^{-2s}.$$

(b) The standard choice of b_{l0} is

$$b_{l0} = \frac{2^l (l!)^2}{(2l+1)!},$$

leading to the final result

$$Q_l(x) = x^{-l-1} \sum_{s=0}^{\infty} \frac{(l+2s)!}{(2s)!!\,(2l+2s+1)!!}\, x^{-2s}.$$

Show that this choice of b_{l0} brings this negative power-series form of $Q_n(x)$ into agreement with the closed-form solutions.

Solution.

(a) For a monomial x^α, the Legendre operator is

$$L_l[x^\alpha] = \alpha(\alpha-1)x^{\alpha-2} + [l(l+1) - \alpha(\alpha+1)]x^\alpha.$$

The leading term of the proposed series therefore requires $k(k+1) = l(l+1)$, giving $k = l$ or $k = -l-1$. The solution that decays at infinity uses the latter root. At the next power, x^{-l-2}, the coefficient is $[l(l+1) - (l+1)(l+2)]b_{l1} = -2(l+1)b_{l1}$. It follows that $b_{l1} = 0$, and the recurrence then makes every odd j vanish. Set $c_s = b_{l,2s}$. At power $x^{-l-1-2s}$ the equation gives

$$2s(2l+2s+1)c_s = (l+2s)(l+2s-1)c_{s-1}.$$

Hence

$$\frac{c_s}{c_0} = \prod_{j=1}^{s} \frac{(l+2j)(l+2j-1)}{2j(2l+2j+1)} = \frac{(l+s)!(l+2s)!(2l+1)!}{s!(l!)^2(2l+2s+1)!}.$$

For the last equality, the numerator product is $(l+2s)!/l!$, while the denominator product is $s!\,l!(2l+2s+1)!/[(l+s)!(2l+1)!]$. This proves the requested series, convergent for $|x| > 1$.

(b) The normalization can be checked against the closed logarithmic solutions for every degree by the integral

$$R_l(z) = \frac{1}{2}\int_{-1}^{1} \frac{P_l(t)}{z-t} dt, \qquad |z| > 1.$$

Directly, $R_0 = \frac{1}{2}\,\mathrm{Log}[(z+1)/(z-1)]$ with the branch vanishing at infinity, and $R_1 = zR_0 - 1$. For $l \geq 1$, multiply the polynomial degree recurrence by $1/[2(z-t)]$ and use $t/(z-t) = z/(z-t) - 1$ and $\int P_l = 0$. This gives

$$(l+1)R_{l+1} = (2l+1)zR_l - lR_{l-1}.$$

Thus R_l agrees with the standard closed forms Q_l by induction. Expand $1/(z-t) = \sum_{j\geq 0} t^j z^{-j-1}$ under the finite integral. All moments with $j < l$ vanish, so its leading coefficient is

$$b_{l0} = \frac{1}{2}\int_{-1}^{1} t^l P_l(t)dt = \frac{l!}{(2l+1)!!} = \boxed{\frac{2^l (l!)^2}{(2l+1)!}}.$$

Multiplying the part (a) coefficient by this value and splitting even and odd factorials gives

$$\boxed{Q_l(z) = z^{-l-1}\sum_{s=0}^{\infty} \frac{(l+2s)!}{(2s)!!(2l+2s+1)!!} z^{-2s}}, \qquad |z| > 1.$$

The integral and series specify a consistent exterior logarithm branch.

15.6.4

(a) Using the recurrence relations, prove (independent of the Wronskian relation) that, for integers $n \geq 1$ on a common consistent branch away from $x = \pm 1$,

$$n\big[P_n(x)Q_{n-1}(x) - P_{n-1}(x)Q_n(x)\big] = P_1(x)Q_0(x) - P_0(x)Q_1(x).$$

(b) By direct substitution show that the right-hand side of this equation equals 1.

Solution.

Let $D_n = P_n Q_{n-1} - P_{n-1}Q_n$. For $n \geq 2$, both families obey

$$nF_n = (2n-1)xF_{n-1} - (n-1)F_{n-2}.$$

Substitute this for P_n and Q_n:

$$\begin{aligned} nD_n &= [(2n-1)xP_{n-1} - (n-1)P_{n-2}]Q_{n-1} \\ &\quad - P_{n-1}[(2n-1)xQ_{n-1} - (n-1)Q_{n-2}] \\ &= (n-1)[P_{n-1}Q_{n-2} - P_{n-2}Q_{n-1}] \\ &= (n-1)D_{n-1}. \end{aligned}$$

Repeating gives $nD_n = D_1$, with $n = 1$ already included. Finally $P_0 = 1$, $P_1 = x$, and $Q_1 = xQ_0 - 1$, so

$$D_1 = xQ_0 - (xQ_0 - 1) = 1.$$

Therefore

$$\boxed{n[P_nQ_{n-1} - P_{n-1}Q_n] = 1}.$$

The argument uses only degree recurrences and starting values, as requested, and is valid on any common consistent branch away from the singular endpoints.

16 Chapter 16. Angular Momentum

Problems

Throughout this chapter use $\hbar = 1$, as in the textbook. Scalar spherical harmonics are normalized with the Condon–Shortley phase; their indices satisfy $L \in \{0, 1, \ldots\}$ and $-L \le M \le L$. Terms beyond a ladder endpoint mean zero, rather than an undefined harmonic. Abstract spin multiplets may also have half-integral L; in that case $L \pm M$ and $2L$ are nonnegative integers.

16.1 Angular Momentum Operators

16.1.1 The quantum mechanical angular momentum operators $L_x \pm iL_y$ in 3-D physical space are given by

$$L_x + iL_y = e^{i\varphi}\left(\frac{\partial}{\partial\theta} + i\cot\theta\,\frac{\partial}{\partial\varphi}\right),$$
$$L_x - iL_y = -e^{-i\varphi}\left(\frac{\partial}{\partial\theta} - i\cot\theta\,\frac{\partial}{\partial\varphi}\right).$$

Show that

(a) $(L_x + iL_y)Y_L^M(\theta,\varphi) = \sqrt{(L-M)(L+M+1)}\,Y_L^{M+1}(\theta,\varphi)$,

(b) $(L_x - iL_y)Y_L^M(\theta,\varphi) = \sqrt{(L+M)(L-M+1)}\,Y_L^{M-1}(\theta,\varphi)$.

Solution.

Use $\hbar = 1$ and the Condon–Shortley normalization

$$Y_L^M = N_{LM}P_L^M(\cos\theta)e^{iM\varphi}, \qquad N_{LM} = \sqrt{\frac{2L+1}{4\pi}\frac{(L-M)!}{(L+M)!}}.$$

Put $P = P_L^M(\cos\theta)$, $A = P_L^{M+1}(\cos\theta)$ and $B = (L+M)(L-M+1)P_L^{M-1}(\cos\theta)$. The associated-Legendre recurrences (15.87) and (15.91), with $d/d\theta = -\sin\theta\, d/d(\cos\theta)$, read

$$-M\cot\theta\,P = \frac{A+B}{2}, \qquad \frac{dP}{d\theta} = \frac{A-B}{2}.$$

Adding them, or subtracting the second from the first, gives

$$(\partial_\theta - M\cot\theta)P = A, \qquad (-\partial_\theta - M\cot\theta)P = B.$$

The first identity also follows directly by differentiating $P_L^M = (-1)^M\sin^M\theta\, d^M P_L(x)/dx^M$, $x = \cos\theta$, for $M \ge 0$.

Since $\partial_\varphi e^{iM\varphi} = iMe^{iM\varphi}$,

$$L_+Y_L^M = N_{LM}P_L^{M+1}e^{i(M+1)\varphi} = \frac{N_{LM}}{N_{L,M+1}}Y_L^{M+1}.$$

The normalization ratio is

$$\frac{N_{LM}^2}{N_{L,M+1}^2} = \frac{(L-M)!(L+M+1)!}{(L+M)!(L-M-1)!} = (L-M)(L+M+1),$$

proving (a). Similarly,

$$L_- Y_L^M = (L+M)(L-M+1)\frac{N_{LM}}{N_{L,M-1}} Y_L^{M-1},$$

and

$$\frac{N_{LM}^2}{N_{L,M-1}^2} = \frac{1}{(L+M)(L-M+1)}.$$

Taking positive square roots establishes

$$\boxed{L_\pm Y_L^M = \sqrt{(L \mp M)(L \pm M + 1)}\, Y_L^{M\pm 1}}.$$

These formulas hold for integer $-L \le M \le L$ with the endpoint understood as annihilation: $L_+ Y_L^L = L_- Y_L^{-L} = 0$. Indeed $Y_L^{\pm L}$ is proportional to $\sin^L\theta\, e^{\pm iL\varphi}$, and direct substitution gives zero. No harmonic outside the permitted M range is needed.

16.1.2 With $L_\pm$ given by

$$L_\pm = L_x \pm iL_y = \pm e^{\pm i\varphi}\left(\frac{\partial}{\partial\theta} \pm i\cot\theta\,\frac{\partial}{\partial\varphi}\right),$$

show that

(a) $Y_l^m = \sqrt{\dfrac{(l+m)!}{(2l)!\,(l-m)!}}\,(L_-)^{l-m} Y_l^l$,

(b) $Y_l^m = \sqrt{\dfrac{(l-m)!}{(2l)!\,(l+m)!}}\,(L_+)^{l+m} Y_l^{-l}$.

Solution.

Each application of L_- changes m to $m-1$ with the coefficient proved in 16.1.1. Starting at $m = l$, after k steps we therefore have

$$(L_-)^k Y_l^l = \left[\prod_{j=0}^{k-1}(2l-j)(j+1)\right]^{1/2} Y_l^{l-k}.$$

The two products are $\prod_{j=0}^{k-1}(2l-j) = (2l)!/(2l-k)!$ and $\prod_{j=0}^{k-1}(j+1) = k!$. Set $k = l-m$:

$$(L_-)^{l-m} Y_l^l = \sqrt{\frac{(2l)!(l-m)!}{(l+m)!}}\, Y_l^m.$$

Dividing by this positive coefficient proves

$$\boxed{Y_l^m = \sqrt{\frac{(l+m)!}{(2l)!(l-m)!}}\,(L_-)^{l-m} Y_l^l}.$$

For raising from $m = -l$, the successive squared coefficients are again $(2l - j)(j + 1)$, because at step j the magnetic index is $-l + j$. Thus

$$(L_+)^k Y_l^{-l} = \sqrt{\frac{(2l)!\,k!}{(2l-k)!}}\, Y_l^{-l+k}.$$

Putting $k = l + m$ and solving for the final harmonic gives

$$\boxed{Y_l^m = \sqrt{\frac{(l-m)!}{(2l)!(l+m)!}}\,(L_+)^{l+m} Y_l^{-l}}.$$

For $k = 0$ the product is empty and equals one, so both extreme states are included.

16.1.3 Using the known forms of L_+ and L_- (Exercise 16.1.2), show that

$$\int \left[Y_L^M\right]^* L_- \left(L_+ Y_L^M\right) d\Omega = \int \left(L_+ Y_L^M\right)^* \left(L_+ Y_L^M\right) d\Omega.$$

Here $d\Omega$ is the element of solid angle $(\sin\theta\, d\theta\, d\varphi)$, and the integration is over the entire angular space.

Solution.

We prove the needed adjoint identity by integration by parts rather than assuming it. Let u, v be smooth single-valued functions on the sphere, with φ-periodic coordinate expressions. Since

$$L_- v = -e^{-i\varphi} v_\theta + i e^{-i\varphi} \cot\theta\, v_\varphi,$$

we have

$$\langle u, L_- v\rangle = \int_0^{2\pi} \int_0^{\pi} \left[-u^* e^{-i\varphi} \sin\theta\, v_\theta + i u^* e^{-i\varphi} \cos\theta\, v_\varphi\right] d\theta\, d\varphi.$$

The θ boundary term vanishes because of $\sin\theta$ at the poles; the φ boundary term vanishes by periodicity. After moving both derivatives onto their coefficients, the remaining integrand is

$$v\left[e^{-i\varphi} \sin\theta\, u_\theta^* + e^{-i\varphi} \cos\theta\, u^* - e^{-i\varphi} \cos\theta\, u^* - i e^{-i\varphi} \cos\theta\, u_\varphi^*\right].$$

The middle two terms cancel, leaving

$$\langle u, L_- v\rangle = \int v\, e^{-i\varphi} (u_\theta^* - i \cot\theta\, u_\varphi^*)\, d\Omega = \langle L_+ u, v\rangle.$$

Take $u = Y_L^M$ and $v = L_+ Y_L^M$, both regular spherical harmonics or zero. This is exactly the requested equality. Moreover both sides equal $\boxed{(L - M)(L + M + 1)}$ by normalization and 16.1.1, including zero at the upper endpoint.

16.1.4

(a) Show that $\mathbf{J}^2 = \dfrac{1}{2}\left[J_+ J_- + J_- J_+\right] + J_z^2$.

(b) Use the result from part (a) and the explicit formulas for L_+ and L_- from Exercise 16.1.2 to verify that all the spherical harmonics with $l = 2$ are eigenfunctions of $\mathbf{L}^2$ with eigenvalue $l(l + 1) = 6$.

Solution.

(a) Preserve operator order while expanding:

$$J_+J_- = J_x^2 + J_y^2 + iJ_yJ_x - iJ_xJ_y,$$
$$J_-J_+ = J_x^2 + J_y^2 + iJ_xJ_y - iJ_yJ_x.$$

Their sum is $2J_x^2 + 2J_y^2$ because the mixed terms cancel. Consequently

$$\boxed{\mathbf{J}^2 = \tfrac{1}{2}(J_+J_- + J_-J_+) + J_z^2}.$$

(b) For an arbitrary smooth test function f, direct product differentiation in the given ladder operators yields

$$L_+L_-f = -f_{\theta\theta} - \cot\theta f_\theta - \cot^2\theta f_{\varphi\varphi} - if_\varphi,$$
$$L_-L_+f = -f_{\theta\theta} - \cot\theta f_\theta - \cot^2\theta f_{\varphi\varphi} + if_\varphi.$$

For example, writing $c = \cot\theta$ and using $c' = -\csc^2\theta$, the two parts of L_+L_-f before cancellation are

$$e^{i\varphi}\partial_\theta(L_-f) = -f_{\theta\theta} + ic'f_\varphi + icf_{\theta\varphi},$$
$$e^{i\varphi}ic\,\partial_\varphi(L_-f) = -cf_\theta - icf_{\theta\varphi} + c^2if_\varphi - c^2f_{\varphi\varphi}.$$

The mixed derivatives cancel and $i(c'+c^2) = -i$. Interchanging the ladder signs changes the last first-order term from $-if_\varphi$ to $+if_\varphi$. The manual (printed p. 257) writes the difference $(L_+L_- - L_-L_+)/2$ when extracting the common second-order terms. It must use the sum. The half-difference is $L_z = -i\partial_\varphi$, not $\mathbf{L}^2 - L_z^2$. Since $L_z^2f = -f_{\varphi\varphi}$, (a) gives

$$\mathbf{L}^2f = -f_{\theta\theta} - \cot\theta f_\theta - \csc^2\theta f_{\varphi\varphi}.$$

Write $s = \sin\theta, c = \cos\theta$. The five $l = 2$ harmonics are nonzero constant multiples of $s^2e^{\pm2i\varphi}$, $sc\,e^{\pm i\varphi}$, and $3c^2 - 1$. The operator acts on $F(\theta)e^{im\varphi}$ as $[-F'' - \cot\theta F' + m^2F/s^2]e^{im\varphi}$. Explicitly,

F	m^2	F', F''	$-F'' - (c/s)F' + m^2F/s^2$
s^2	4	$2sc,\ 2(c^2 - s^2)$	$-2c^2 + 2s^2 - 2c^2 + 4 = 6s^2$
sc	1	$c^2 - s^2,\ -4sc$	$4sc - c^3/s + cs + c/s = 6sc$
$3c^2 - 1$	0	$-6sc,\ -6(c^2 - s^2)$	$12c^2 - 6s^2 = 6(3c^2 - 1)$

Both signs of m have the same m^2, so the first two rows each verify two harmonics. Multiplying by their normalization constants does not change the eigenvalue: $\boxed{\mathbf{L}^2Y_2^m = 6Y_2^m}$ for all five m.

16.1.5 Derive the following relations for normalized eigenstates of the Hermitian angular-momentum algebra in one irreducible multiplet, with relative phases chosen so the ladder coefficients are positive. Do not assume coordinate formulas for ψ_L^M:

(a) $\psi_L^M(\theta,\varphi) = \sqrt{\dfrac{(L+M)!}{(2L)!\,(L-M)!}}\,(L_-)^{L-M}\psi_L^L(\theta,\varphi),$

(b) $\psi_L^M(\theta,\varphi) = \sqrt{\dfrac{(L-M)!}{(2L)!\,(L+M)!}}\,(L_+)^{L+M}\psi_L^{-L}(\theta,\varphi).$

Solution.

The numerical coefficients require normalized states in a single angular-momentum multiplet and a consistent ladder phase convention. Eigenvalue equations alone leave arbitrary magnitudes, phases, and possible degeneracy labels. Assume the Hermitian angular-momentum algebra, and choose states with $\mathbf{L}^2\psi_L^M = L(L+1)\psi_L^M$ and $L_z\psi_L^M = M\psi_L^M$. The commutators $[L_z, L_\pm] = \pm L_\pm$ and $[\mathbf{L}^2, L_\pm] = 0$ show that $L_\pm\psi_L^M$ has the same L and magnetic index $M \pm 1$. Within a fixed irreducible multiplet it is therefore proportional to $\psi_L^{M\pm1}$.

Using $L_\pm^\dagger = L_\mp$ and $[L_+, L_-] = 2L_z$ in the identity of 16.1.4 gives

$$L_-L_+ = \mathbf{L}^2 - L_z^2 - L_z, \qquad L_+L_- = \mathbf{L}^2 - L_z^2 + L_z.$$

Thus normalization determines the squared proportionality factors:

$$\|L_+\psi_L^M\|^2 = L(L+1) - M^2 - M = (L-M)(L+M+1),$$
$$\|L_-\psi_L^M\|^2 = L(L+1) - M^2 + M = (L+M)(L-M+1).$$

Choose the relative phases so these factors have positive square roots. Iteration now gives, without any coordinate formula,

$$(L_-)^k\psi_L^L = \sqrt{\prod_{j=0}^{k-1}(2L-j)(j+1)}\,\psi_L^{L-k} = \sqrt{\frac{(2L)!k!}{(2L-k)!}}\,\psi_L^{L-k}.$$

Setting $k = L - M$ and inverting the coefficient yields (a). Starting at $-L$ and raising gives the identical product; setting $k = L + M$ yields (b):

$$\boxed{\psi_L^M = \sqrt{\frac{(L+M)!}{(2L)!(L-M)!}}(L_-)^{L-M}\psi_L^L},$$

$$\boxed{\psi_L^M = \sqrt{\frac{(L-M)!}{(2L)!(L+M)!}}(L_+)^{L+M}\psi_L^{-L}}.$$

The argument also applies to abstract half-integral multiplets: $2L, L \pm M$ are still nonnegative integers. Ordinary scalar spherical harmonics are the integral-L realization.

16.1.6 Derive the operator equations for integers $n \geq 0$, on $0 < \theta < \pi$, with the derivatives in $\cos\theta$ taken at fixed φ:

$$(L_+)^nY_L^M(\theta,\varphi) = (-1)^n e^{in\varphi}\sin^{n+M}\theta\,\frac{d^n\sin^{-M}\theta\,Y_L^M(\theta,\varphi)}{(d\cos\theta)^n},$$
$$(L_-)^nY_L^M(\theta,\varphi) = e^{-in\varphi}\sin^{n-M}\theta\,\frac{d^n\sin^{M}\theta\,Y_L^M(\theta,\varphi)}{(d\cos\theta)^n}.$$

Hint. Try mathematical induction (Section 1.4).

Solution.

All derivatives with respect to $x = \cos\theta$ below hold φ fixed. A function $g_m = F(\theta)e^{im\varphi}$ satisfies

$$\begin{aligned} L_+g_m &= e^{i\varphi}(\partial_\theta - m\cot\theta)g_m = e^{i\varphi}\sin^m\theta\,\partial_\theta(\sin^{-m}\theta\,g_m) \\ &= -e^{i\varphi}\sin^{m+1}\theta\,\partial_x(\sin^{-m}\theta\,g_m), \\ L_-g_m &= -e^{-i\varphi}(\partial_\theta + m\cot\theta)g_m = -e^{-i\varphi}\sin^{-m}\theta\,\partial_\theta(\sin^{m}\theta\,g_m) \\ &= e^{-i\varphi}\sin^{1-m}\theta\,\partial_x(\sin^{m}\theta\,g_m). \end{aligned}$$

These establish both claims for $n = 1$; $n = 0$ is the identity.

Assume the raising formula at n: $g_{M+n} = (-1)^n e^{in\varphi} \sin^{M+n}\theta\, \partial_x^n(\sin^{-M}\theta\, Y_L^M)$. It has magnetic index $M + n$. Applying the first identity and canceling the two opposite powers of sine inside the derivative gives

$$L_+ g_{M+n} = (-1)^{n+1} e^{i(n+1)\varphi} \sin^{M+n+1}\theta\, \partial_x^{n+1}(\sin^{-M}\theta\, Y_L^M).$$

This is the $n + 1$ formula.

For lowering, the induction hypothesis is $g_{M-n} = e^{-in\varphi} \sin^{n-M}\theta\, \partial_x^n(\sin^{M}\theta\, Y_L^M)$. Use the second identity with $m = M - n$; its factor $\sin^m\theta$ cancels $\sin^{n-M}\theta$, leaving

$$L_- g_{M-n} = e^{-i(n+1)\varphi} \sin^{n+1-M}\theta\, \partial_x^{n+1}(\sin^{M}\theta\, Y_L^M).$$

Thus both stated equations hold for every integer $n \geq 0$. They are coordinate expressions on $0 < \theta < \pi$; the resulting harmonics extend regularly to the poles.

16.1.7 Show, using $(L_-)^n$, that

$$Y_L^{-M}(\theta,\varphi) = (-1)^M \left[Y_L^M(\theta,\varphi)\right]^*.$$

Solution.

First let $M \geq 0$. Iterating the raising coefficient from zero to M gives

$$Y_L^M = c_M (L_+)^M Y_L^0, \qquad c_M = \sqrt{\frac{(L-M)!}{(L+M)!}}.$$

Indeed the squared product is $\prod_{j=0}^{M-1}(L-j)(L+j+1) = (L+M)!/(L-M)!$. The differential forms show, for any differentiable f,

$$(L_+ f)^* = e^{-i\varphi}(\partial_\theta - i\cot\theta\,\partial_\varphi) f^* = -L_- f^*.$$

Because Y_L^0 is real, repeated conjugation therefore gives

$$(Y_L^M)^* = (-1)^M c_M (L_-)^M Y_L^0.$$

Lowering from zero to $-M$ has the same squared coefficient product $(L+M)!/(L-M)!$, so $c_M (L_-)^M Y_L^0 = Y_L^{-M}$. Consequently

$$\boxed{Y_L^{-M} = (-1)^M (Y_L^M)^*}.$$

Replacing M by $-M$ and conjugating proves the identical statement for negative M, since $(-1)^{-M} = (-1)^M$ for integral M. The manual (printed p. 258) says to start with Y_0^0. For a fixed degree L the starting function must be Y_L^0. Indeed both ladders annihilate Y_0^0, so they cannot generate any nonconstant harmonic from it.

16.1.8 Verify by explicit calculation that

(a) $L_+ Y_1^0(\theta,\varphi) = -\sqrt{\dfrac{3}{4\pi}}\sin\theta\, e^{i\varphi} = \sqrt{2}\, Y_1^1(\theta,\varphi)$,

(b) $L_- Y_1^0(\theta,\varphi) = +\sqrt{\dfrac{3}{4\pi}}\sin\theta\, e^{-i\varphi} = \sqrt{2}\, Y_1^{-1}(\theta,\varphi)$.

The signs have the indicated values because the spherical harmonics were defined to be consistent with the results obtained using the ladder operators L_+ and L_- (Condon–Shortley phase).

Solution.

The relevant normalized harmonics are

$$Y_1^0 = \sqrt{\frac{3}{4\pi}}\cos\theta, \qquad Y_1^1 = -\sqrt{\frac{3}{8\pi}}\sin\theta\, e^{i\varphi}, \qquad Y_1^{-1} = \sqrt{\frac{3}{8\pi}}\sin\theta\, e^{-i\varphi}.$$

Since Y_1^0 is independent of φ, the azimuthal derivatives in $L_\pm$ vanish. Direct differentiation gives

$$L_+ Y_1^0 = e^{i\varphi}\sqrt{\frac{3}{4\pi}}(-\sin\theta) = -\sqrt{\frac{3}{4\pi}}\sin\theta\, e^{i\varphi} = \boxed{\sqrt{2}\, Y_1^1},$$

$$L_- Y_1^0 = -e^{-i\varphi}\sqrt{\frac{3}{4\pi}}(-\sin\theta) = \sqrt{\frac{3}{4\pi}}\sin\theta\, e^{-i\varphi} = \boxed{\sqrt{2}\, Y_1^{-1}}.$$

The minus sign in Y_1^1 is precisely the phase needed to make both ladder coefficients positive.

16.2 Angular Momentum Coupling

16.2.1 Derive recursion relations for Clebsch–Gordan coefficients. Use them to calculate $C(11J\,|\,m_1 m_2 M)$ for $J = 0, 1, 2$.

Hint. Use the known matrix elements of $J_+ = J_{1+} + J_{2+}$, J_{i+}, and $\mathbf{J}^2 = (\mathbf{J}_1 + \mathbf{J}_2)^2$, etc.

Solution.

Write $|m_1, m_2\rangle = |j_1 m_1\rangle|j_2 m_2\rangle$ and

$$|JM\rangle = \sum_{m_1+m_2=M} C_M(m_1, m_2)|m_1, m_2\rangle.$$

Define $a_i^\pm(m) = \sqrt{(j_i \mp m)(j_i \pm m + 1)}$. Apply $J_+ = J_{1+} + J_{2+}$ to this expansion and collect the coefficient of the *target* state $|m_1, m_2\rangle$:

$$\sqrt{(J-M)(J+M+1)}\, C_{M+1}(m_1, m_2) = a_1^+(m_1-1)C_M(m_1-1, m_2) + a_2^+(m_2-1)C_M(m_1, m_2-1).$$

Similarly, applying J_- gives

$$\sqrt{(J+M)(J-M+1)}\, C_{M-1}(m_1, m_2) = a_1^-(m_1+1)C_M(m_1+1, m_2) + a_2^-(m_2+1)C_M(m_1, m_2+1).$$

Coefficients outside the allowed magnetic ranges are zero. A recursion within a fixed M follows from

$$\mathbf{J}^2 = \mathbf{J}_1^2 + \mathbf{J}_2^2 + 2J_{1z}J_{2z} + J_{1+}J_{2-} + J_{1-}J_{2+}:$$

$$\begin{aligned}[J(J+1) - j_1(j_1+1) - j_2(j_2+1) - 2m_1m_2]C_M(m_1, m_2) \\ = a_1^+(m_1-1)a_2^-(m_2+1)C_M(m_1-1, m_2+1) \\ + a_1^-(m_1+1)a_2^+(m_2-1)C_M(m_1+1, m_2-1).\end{aligned}$$

Together with normalization and phase, these determine the coefficients.

For $j_1 = j_2 = 1$, choose positive lowering coefficients and a positive coefficient of the largest m_1 in each highest-M state, consistent with the text's Condon–Shortley convention. Start with $|2,2\rangle = |1,1\rangle$. The first lowering gives

$$2|2,1\rangle = \sqrt{2}(|1,0\rangle + |0,1\rangle).$$

The next gives

$$\sqrt{6}|2,0\rangle = |1,-1\rangle + 2|0,0\rangle + |-1,1\rangle.$$

Lowering this numerator gives $3\sqrt{2}(|0,-1\rangle + |-1,0\rangle)$; after dividing by its existing $\sqrt{6}$ and the next ladder factor $\sqrt{6}$, $|2,-1\rangle = (|0,-1\rangle + |-1,0\rangle)/\sqrt{2}$. The final lowering gives $2|2,-2\rangle = 2|-1,-1\rangle$. Thus the complete quintet is

M	$\lvert 2, M\rangle$
2	$\lvert 1,1\rangle$
1	$(\lvert 1,0\rangle + \lvert 0,1\rangle)/\sqrt{2}$
0	$(\lvert 1,-1\rangle + 2\lvert 0,0\rangle + \lvert -1,1\rangle)/\sqrt{6}$
-1	$(\lvert 0,-1\rangle + \lvert -1,0\rangle)/\sqrt{2}$
-2	$\lvert -1,-1\rangle$

The normalized vector at $M = 1$ orthogonal to the quintet is $|1,1\rangle_{\text{coupled}} = (|1,0\rangle - |0,1\rangle)/\sqrt{2}$. Its two lowerings are

$$J_-|1,1\rangle_{\text{coupled}} = |1,-1\rangle - |-1,1\rangle = \sqrt{2}|1,0\rangle_{\text{coupled}},$$

$$J_-|1,0\rangle_{\text{coupled}} = |0,-1\rangle - |-1,0\rangle = \sqrt{2}|1,-1\rangle_{\text{coupled}}.$$

Hence the triplet is

M	$\lvert 1, M\rangle_{\text{coupled}}$
1	$(\lvert 1,0\rangle - \lvert 0,1\rangle)/\sqrt{2}$
0	$(\lvert 1,-1\rangle - \lvert -1,1\rangle)/\sqrt{2}$
-1	$(\lvert 0,-1\rangle - \lvert -1,0\rangle)/\sqrt{2}$

Finally write the singlet as $u|1,-1\rangle + v|0,0\rangle + w|-1,1\rangle$. Orthogonality to the $J = 2$ and $J = 1$ states gives $u + 2v + w = 0$ and $u - w = 0$, so $w = u, v = -u$. Normalization and the chosen sign give

$$\boxed{|0,0\rangle_{\text{coupled}} = \frac{|1,-1\rangle - |0,0\rangle + |-1,1\rangle}{\sqrt{3}}}.$$

Every Clebsch–Gordan coefficient requested is the coefficient of its product ket in these nine normalized states; all unlisted coefficients are zero. Their dimensions $5 + 3 + 1 = 9$ exhaust the product basis.

16.2.2 Defining $\big(Y_l\chi\big)^M_J$ by the formula

$$\big(Y_l\chi\big)^M_J = \sum C\big(l\tfrac{1}{2}J \,\big|\, m_l m_s M\big) Y_l^{m_l}\chi_{m_s},$$

where $\chi_{\pm 1/2}$ are the spin up and down eigenfunctions of $\sigma_3 = \sigma_z$, show that $\big(Y_l\chi\big)^M_J$ is a J, M eigenfunction.

Solution.

Let $\chi_+ = \chi_{1/2}$ and $\chi_- = \chi_{-1/2}$, so that $S_z\chi_\pm = \pm\chi_\pm/2$ with $\mathbf{S} = \boldsymbol{\sigma}/2$. Every nonzero Clebsch–Gordan term has $m_l + m_s = M$; therefore $J_z = L_z + S_z$ acting on the sum gives M times that sum.

To check $\mathbf{J}^2$ explicitly, consider the fixed-M product basis

$$u = Y_l^{M-1/2}\chi_+, \qquad v = Y_l^{M+1/2}\chi_-,$$

omitting a vector if its orbital magnetic index is outside $[-l, l]$. In this basis,

$$\mathbf{J}^2 = \mathbf{L}^2 + \mathbf{S}^2 + 2L_zS_z + L_+S_- + L_-S_+ = \left[l(l+1) + \tfrac{1}{4}\right] I + \begin{pmatrix} M & q \\ q & -M \end{pmatrix},$$

where $K = l + \frac{1}{2}$ and $q = \sqrt{K^2 - M^2}$. For example, $L_+S_-u = qv$, $L_-S_+v = qu$; the diagonal terms are $l(l+1) + 3/4 \pm (M \mp 1/2)$. The square of the last matrix is $(M^2 + q^2)I = K^2 I$, so its eigenvalues are $+K, -K$.

The corresponding normalized eigenvectors are

$$\Psi_+ = \sqrt{\frac{K+M}{2K}}\,u + \sqrt{\frac{K-M}{2K}}\,v, \qquad \Psi_- = -\sqrt{\frac{K-M}{2K}}\,u + \sqrt{\frac{K+M}{2K}}\,v.$$

Indeed $q\sqrt{K-M} = (K-M)\sqrt{K+M}$ and $q\sqrt{K+M} = (K+M)\sqrt{K-M}$. The two components of the matrix times Ψ_+ are consequently $[M+K-M]\sqrt{K+M}$ and $[K+M-M]\sqrt{K-M}$, divided by $\sqrt{2K}$; these are the components of $K\Psi_+$. For Ψ_- the components are $[-M+K+M]\sqrt{K-M}$ and $[-K+M-M]\sqrt{K+M}$, divided by $\sqrt{2K}$, which equal $-K\Psi_-$. The complete eigenvalues are thus

$$l(l+1) + \tfrac{1}{4} + K = (l+\tfrac{1}{2})(l+\tfrac{3}{2}), \qquad l(l+1) + \tfrac{1}{4} - K = (l-\tfrac{1}{2})(l+\tfrac{1}{2}).$$

These are $J(J+1)$ for $J = l + 1/2$ and $J = l - 1/2$. Their displayed coefficients are precisely the normalized Clebsch–Gordan coefficients in the stated coupling convention. At the extreme $M = \pm(l + 1/2)$ only the permitted basis vector survives. The lower multiplet exists only for $l \geq 1$. Consequently

$$\boxed{\mathbf{J}^2(Y_l\chi)_J^M = J(J+1)(Y_l\chi)_J^M, \qquad J_z(Y_l\chi)_J^M = M(Y_l\chi)_J^M}.$$

16.2.3 Find the (j, m) states of a p electron ($l = 1$), in which the orbital angular momentum of the electron is coupled to its spin angular momentum ($s = 1/2$) to form states whose conventional labelings are ${}^2p_{1/2}$ and ${}^2p_{3/2}$. The notation is of the general form ${}^{2s+1}(\text{symbol})_j$, where "symbol" is that indicating the l value (i.e., s, p, ...).

Solution.

Write $p_m = Y_1^m$ and $\alpha = \chi_{1/2}$, $\beta = \chi_{-1/2}$. The tensor products are normalized and mutually orthogonal. Since $1 \otimes \frac{1}{2} = \frac{3}{2} \oplus \frac{1}{2}$, there are four ${}^2p_{3/2}$ states and two ${}^2p_{1/2}$ states.

The highest state is $|\frac{3}{2}, \frac{3}{2}\rangle = p_1\alpha$. Use $L_-p_1 = \sqrt{2}p_0$, $L_-p_0 = \sqrt{2}p_{-1}$, $S_-\alpha = \beta$, $S_-\beta = 0$:

$$J_-(p_1\alpha) = \sqrt{2}p_0\alpha + p_1\beta.$$

Divide by $\sqrt{3}$, the $J = 3/2$ ladder coefficient, to get $|\frac{3}{2}, \frac{1}{2}\rangle$ below. Lower its numerator once more:

$$J_-(\sqrt{2}p_0\alpha + p_1\beta) = 2p_{-1}\alpha + 2\sqrt{2}p_0\beta.$$

Dividing by the previous $\sqrt{3}$ and the next ladder coefficient 2 gives $|\frac{3}{2},-\frac{1}{2}\rangle$. The last lowering is $J_-(p_{-1}\alpha+\sqrt{2}p_0\beta)/\sqrt{3}=\sqrt{3}p_{-1}\beta$. Therefore

$$\boxed{\begin{aligned}|\tfrac{3}{2},\tfrac{3}{2}\rangle &= p_1\alpha,\\ |\tfrac{3}{2},\tfrac{1}{2}\rangle &= \sqrt{\tfrac{2}{3}}p_0\alpha+\sqrt{\tfrac{1}{3}}p_1\beta,\\ |\tfrac{3}{2},-\tfrac{1}{2}\rangle &= \sqrt{\tfrac{1}{3}}p_{-1}\alpha+\sqrt{\tfrac{2}{3}}p_0\beta,\\ |\tfrac{3}{2},-\tfrac{3}{2}\rangle &= p_{-1}\beta.\end{aligned}}$$

For the remaining $M=1/2$ state, orthogonality to the quartet requires $\sqrt{2}u+v=0$ in $up_0\alpha+vp_1\beta$. Normalization gives $|u|^2+|v|^2=1$; choosing the coefficient of $p_1\beta$ positive fixes $u=-1/\sqrt{3}$, $v=\sqrt{2/3}$. Its lowering coefficient for $J=1/2$ is one:

$$\begin{aligned}J_-\left(-\frac{p_0\alpha}{\sqrt{3}}+\sqrt{\frac{2}{3}}p_1\beta\right) &= -\sqrt{\frac{2}{3}}p_{-1}\alpha-\frac{p_0\beta}{\sqrt{3}}+\frac{2}{\sqrt{3}}p_0\beta\\ &= -\sqrt{\frac{2}{3}}p_{-1}\alpha+\frac{p_0\beta}{\sqrt{3}}.\end{aligned}$$

Thus

$$\boxed{\begin{aligned}|\tfrac{1}{2},\tfrac{1}{2}\rangle &= -\sqrt{\tfrac{1}{3}}p_0\alpha+\sqrt{\tfrac{2}{3}}p_1\beta,\\ |\tfrac{1}{2},-\tfrac{1}{2}\rangle &= -\sqrt{\tfrac{2}{3}}p_{-1}\alpha+\sqrt{\tfrac{1}{3}}p_0\beta.\end{aligned}}$$

The manual's displayed doublet changes the relative phase between its two M states and does not obey $J_-|\frac{1}{2},\frac{1}{2}\rangle=|\frac{1}{2},-\frac{1}{2}\rangle$. The explicit lowering above fixes that phase. A common overall sign change of both states would be harmless. The superscript 2 is $2s+1$, and the letter p records $l=1$.

16.2.4 Repeat Exercise 16.2.3 for $l=1$, $s=3/2$. Apply the conventional labels to the j,m states.

Solution.

Here $1\otimes\frac{3}{2}=\frac{5}{2}\oplus\frac{3}{2}\oplus\frac{1}{2}$. Because $2s+1=4$ and $l=1$, the labels are ${}^4p_{5/2}$, ${}^4p_{3/2}$, and ${}^4p_{1/2}$, with $6,4,2$ states, respectively. Write $p_+=Y_1^1$, $p_0=Y_1^0$, $p_-=Y_1^{-1}$ and let χ_μ be the normalized spin-3/2 state. The elementary lowering actions are

$$L_-p_+=\sqrt{2}p_0,\quad L_-p_0=\sqrt{2}p_-,\quad L_-p_-=0,$$

$$S_-\chi_{3/2}=\sqrt{3}\chi_{1/2},\quad S_-\chi_{1/2}=2\chi_{-1/2},\quad S_-\chi_{-1/2}=\sqrt{3}\chi_{-3/2},\quad S_-\chi_{-3/2}=0.$$

These specify every matrix element used below.

For a compact complete table, express a state of given M by its coefficient column in the ordered product basis

$$B_M=(p_+\chi_{M-1},\ p_0\chi_M,\ p_-\chi_{M+1}),\qquad |JM\rangle=B_Mc_M^{(J)}.$$

A nonexistent spin state is omitted, and its coefficient is zero. Let $\sigma(\mu)=\sqrt{(3/2+\mu)(5/2-\mu)}$ for allowed μ; set $\sigma=0$ outside the spin range. Applying $J_-=L_-+S_-$ and collecting the three target coefficients gives the explicit rule

$$\sqrt{(J+M)(J-M+1)}\,c_{M-1}^{(J)}=\begin{pmatrix}\sigma(M-1) & 0 & 0\\ \sqrt{2} & \sigma(M) & 0\\ 0 & \sqrt{2} & \sigma(M+1)\end{pmatrix}c_M^{(J)}.\qquad(*)$$

Thus the rows below give full expansions, not only their multiplicities.

For $J = 5/2$, start with $c_{5/2} = (1, 0, 0)^T$. The first lowering gives $(\sqrt{3}, \sqrt{2}, 0)^T$ and has norm $\sqrt{5}$. The next gives

$$\frac{1}{\sqrt{5}}(2\sqrt{3},\ \sqrt{6}+\sqrt{6},\ 2)^T;$$

dividing by the next factor $2\sqrt{2}$ gives the $M = 1/2$ row. For the remaining steps the factors in (*) are $3, 2\sqrt{2}, \sqrt{5}$. For example the middle step has numerator

$$\left(\frac{3}{\sqrt{10}},\ 3\sqrt{\frac{3}{5}},\ 3\sqrt{\frac{3}{10}}\right)^T,$$

which, divided by 3, gives the $M = -1/2$ row. The complete sextet is

M	c_+	c_0	c_-
5/2	1	0	0
3/2	$\sqrt{3/5}$	$\sqrt{2/5}$	0
1/2	$\sqrt{3/10}$	$\sqrt{3/5}$	$\sqrt{1/10}$
−1/2	$\sqrt{1/10}$	$\sqrt{3/5}$	$\sqrt{3/10}$
−3/2	0	$\sqrt{2/5}$	$\sqrt{3/5}$
−5/2	0	0	1

$({}^4p_{5/2})$.

For the penultimate step, the nonzero numerator components are $1/\sqrt{5}+3/\sqrt{5} = 4/\sqrt{5}$ and $\sqrt{6/5}+\sqrt{6/5} = 2\sqrt{6/5}$; division by $2\sqrt{2}$ gives the indicated row. The last numerator is $2/\sqrt{5}+3/\sqrt{5} = \sqrt{5}$, leaving coefficient one.

At $M = 3/2$, the highest state of the $J = 3/2$ multiplet must be orthogonal to $(\sqrt{3/5}, \sqrt{2/5}, 0)^T$. A positive first coefficient fixes it to $(\sqrt{2/5}, -\sqrt{3/5}, 0)^T$. Its first lowering has numerator

$$(2\sqrt{2/5},\ 2/\sqrt{5}-3/\sqrt{5},\ -\sqrt{6/5})^T.$$

Divide by $\sqrt{3}$ to obtain the next row. The next lowering has numerator $(2\sqrt{2/5},\ 2/\sqrt{15},\ -2\sqrt{8/15})^T$ and factor 2. The final lowering has nonzero components $3/\sqrt{5}$ and $-\sqrt{6/5}$ and factor $\sqrt{3}$. Consequently

M	c_+	c_0	c_-
3/2	$\sqrt{2/5}$	$-\sqrt{3/5}$	0
1/2	$\sqrt{8/15}$	$-1/\sqrt{15}$	$-\sqrt{2/5}$
−1/2	$\sqrt{2/5}$	$1/\sqrt{15}$	$-\sqrt{8/15}$
−3/2	0	$\sqrt{3/5}$	$-\sqrt{2/5}$

$({}^4p_{3/2})$.

For the remaining highest state at $M = 1/2$, write $c = (A, B, C)^T$. Orthogonality to the two existing rows gives

$$\sqrt{3}A + \sqrt{6}B + C = 0, \qquad 2\sqrt{2}A - B - \sqrt{6}C = 0.$$

Substituting $C = -\sqrt{3}A - \sqrt{6}B$ in the second equation yields $5\sqrt{2}A + 5B = 0$, so $B = -\sqrt{2}A$ and $C = \sqrt{3}A$. Normalization gives $6A^2 = 1$; choose $A = 1/\sqrt{6}$. Applying (*) with its factor one produces

$$\left(\frac{1}{\sqrt{2}},\ \frac{1}{\sqrt{3}}-\frac{2}{\sqrt{3}},\ -\sqrt{\frac{2}{3}}+\sqrt{\frac{3}{2}}\right)^T = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{6}}\right)^T.$$

Thus the doublet is

M	c_+	c_0	c_-
$1/2$	$1/\sqrt{6}$	$-1/\sqrt{3}$	$1/\sqrt{2}$
$-1/2$	$1/\sqrt{2}$	$-1/\sqrt{3}$	$1/\sqrt{6}$

$({}^4p_{1/2})$.

Each row has norm one; rows of the same M but different J are orthogonal by the construction and the lowering algebra. The dimension $6+4+2=12=3\cdot 4$ confirms that every product-basis state is included. A common sign change within a complete multiplet changes no physics; the tables use the same ladder phase convention as 16.2.1. The manual's $J=3/2, M=-1/2$ row contains an invalid spin label on its p_+ term. That label must be $-3/2$, since $1+(-3/2)=-1/2$. The corrected table above obeys $m_l+m_s=M$ in every entry. Its $J=3/2$ multiplet differs from the manual by one common, harmless overall sign. Its intermediate lowering work on printed p. 261 also has two p_0 labels that must be p_-: the term with spin $-1/2$ in $J=5/2, M=-3/2$, and the term with spin $+1/2$ in $J=1/2, M=-1/2$. The sums m_l+m_s fix both labels.

16.2.5 A deuterium atom consists of a proton, a neutron, and an electron. Each of these particles has spin 1/2. The coupling of these three spins can produce J values of 3/2 and 1/2. We consider here only states with no orbital angular momentum.

(a) Show that these J, M states consist of one quartet ($J=3/2$) and two linearly independent doublets ($J=1/2$).

Hint. Make a vector-model diagram.

(b) One way to analyze this problem is to couple the spins of the proton and neutron to form a nuclear triplet or singlet, and then to couple the resultant nuclear spin to the electron spin. Find the states that are obtained in this way (designate the single-particle states p_α, p_β, n_α, n_β, e_α, e_β).

(c) Another way to analyze this problem is to couple the spins of the proton and electron to form an atomic triplet or singlet, and then to couple that resultant to the neutron spin. Find the states that result from this coupling scheme.

(d) Show that the coupling schemes of parts (b) and (c) span the same Hilbert space.

Note. The actual interaction energies among these angular momenta cause the scheme of part (b) to be the better way of treating this problem (the triplet nuclear state is substantially the more stable), and the system actually looks like a spin-1 deuterium nucleus plus an electron.

Solution.

Use the fixed tensor order proton–neutron–electron. The particles are distinct, so reordering their tensor factors while writing a coupling scheme introduces no fermionic exchange sign.

(a) The product space has dimension $2^3=8$. Its numbers of states at $M=3/2, 1/2, -1/2, -3/2$ are respectively $1, 3, 3, 1$. The unique highest state generates one $J=3/2$ quartet, removing one state from each row. The remainder has two states at each of $M=\pm 1/2$, hence two independent $J=1/2$ doublets. Equivalently, the coupling tree is

$$(\tfrac{1}{2}\otimes\tfrac{1}{2})\otimes\tfrac{1}{2}=(1\oplus 0)\otimes\tfrac{1}{2}=(\tfrac{3}{2}\oplus\tfrac{1}{2})\oplus\tfrac{1}{2}.$$

(b) Couple p and n first. Their triplet and singlet are

$$t_+=p_\alpha n_\alpha,\qquad t_0=\frac{p_\alpha n_\beta+p_\beta n_\alpha}{\sqrt{2}},\qquad t_-=p_\beta n_\beta,\qquad s=\frac{p_\alpha n_\beta-p_\beta n_\alpha}{\sqrt{2}}.$$

Applying the $1 \otimes 1/2$ formulas of 16.2.3 gives the quartet

$$Q_{3/2} = t_+ e_\alpha = p_\alpha n_\alpha e_\alpha,$$
$$Q_{1/2} = \sqrt{\tfrac{2}{3}} t_0 e_\alpha + \sqrt{\tfrac{1}{3}} t_+ e_\beta,$$
$$Q_{-1/2} = \sqrt{\tfrac{1}{3}} t_- e_\alpha + \sqrt{\tfrac{2}{3}} t_0 e_\beta,$$
$$Q_{-3/2} = t_- e_\beta = p_\beta n_\beta e_\beta,$$

and two doublets, denoted T_M for nuclear triplet and S_M for nuclear singlet:

$$T_{1/2} = \sqrt{\tfrac{2}{3}} t_+ e_\beta - \sqrt{\tfrac{1}{3}} t_0 e_\alpha, \qquad S_{1/2} = s e_\alpha,$$
$$T_{-1/2} = \sqrt{\tfrac{1}{3}} t_0 e_\beta - \sqrt{\tfrac{2}{3}} t_- e_\alpha, \quad S_{-1/2} = s e_\beta.$$

To display all product coefficients and compare the next scheme, define

$$A = p_\alpha n_\alpha e_\beta, \quad B = p_\alpha n_\beta e_\alpha, \quad C = p_\beta n_\alpha e_\alpha,$$
$$D = p_\alpha n_\beta e_\beta, \quad E = p_\beta n_\alpha e_\beta, \quad F = p_\beta n_\beta e_\alpha.$$

Substitution of t_0, s then gives

M	Q_M	T_M	S_M
$1/2$	$(A+B+C)/\sqrt{3}$	$(2A-B-C)/\sqrt{6}$	$(B-C)/\sqrt{2}$
$-1/2$	$(D+E+F)/\sqrt{3}$	$(D+E-2F)/\sqrt{6}$	$(D-E)/\sqrt{2}$

together with the two extreme quartet states already written. For instance $J_- A = D + E$, $J_- B = D+F$ and $J_- C = E+F$. Consequently $J_-(2A-B-C) = D+E-2F$ and $J_-(B-C) = D-E$, explicitly checking the doublet ladder phases.

(c) Replace the first pair by pe:

$$t'_+ = p_\alpha e_\alpha, \quad t'_0 = (p_\alpha e_\beta + p_\beta e_\alpha)/\sqrt{2}, \quad t'_- = p_\beta e_\beta, \quad s' = (p_\alpha e_\beta - p_\beta e_\alpha)/\sqrt{2}.$$

Couple these to n with the same coefficients:

$$T'_{1/2} = \sqrt{\tfrac{2}{3}} t'_+ n_\beta - \sqrt{\tfrac{1}{3}} t'_0 n_\alpha, \qquad S'_{1/2} = s' n_\alpha,$$
$$T'_{-1/2} = \sqrt{\tfrac{1}{3}} t'_0 n_\beta - \sqrt{\tfrac{2}{3}} t'_- n_\alpha, \quad S'_{-1/2} = s' n_\beta.$$

Reordering the displayed factors back to pne gives

M	T'_M	S'_M
$1/2$	$(2B-A-C)/\sqrt{6}$	$(A-C)/\sqrt{2}$
$-1/2$	$(D-2E+F)/\sqrt{6}$	$(D-F)/\sqrt{2}$

and the same four Q_M: the quartet is totally symmetric under permuting its spin labels.

(d) For $M = 1/2$, the coefficient dot products give

$$\langle T_M, T'_M \rangle = (-2-2+1)/6 = -\tfrac{1}{2}, \qquad \langle S_M, T'_M \rangle = (2+1)/\sqrt{12} = \tfrac{\sqrt{3}}{2}.$$

The corresponding products with S'_M are $\sqrt{3}/2$ and $1/2$. Using the second rows of the tables gives exactly the same four numbers at $M = -1/2$. Thus, for both magnetic values,

$$\boxed{\begin{pmatrix} T'_M \\ S'_M \end{pmatrix} = \begin{pmatrix} -1/2 & \sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{pmatrix} \begin{pmatrix} T_M \\ S_M \end{pmatrix}}.$$

The real matrix squares to the identity and has orthonormal rows. It therefore changes basis in the same two-dimensional doublet space at each M. Together with the unchanged quartet, it is a unitary change of basis in the full eight-dimensional Hilbert space. The manual repeats the relative doublet-phase mismatch from 16.2.3 in its triplet–electron coupling. The expansions here use $J_-T_{1/2} = T_{-1/2}$ and $J_-T'_{1/2} = T'_{-1/2}$, so the same recoupling matrix applies to both M values.

16.3 Spherical Tensors

16.3.1 For $l = 1$, Eq. (16.52) becomes

$$Y_1^m(\theta', \varphi') = \sum_{m'=-1}^{1} D^1_{m'm}(\alpha, \beta, \gamma)\, Y_1^{m'}(\theta, \varphi).$$

Rewrite these spherical harmonics in Cartesian form. Show that the resulting Cartesian coordinate equations are equivalent to the Euler rotation matrix $\mathsf{A}(\alpha, \beta, \gamma)$, Eq. (3.37).

Solution.

Use the passive rotations of Eq. (3.37): first α about z, then β about the intermediate y, then γ about the new z. Define

$$Z(t) = \begin{pmatrix} \cos t & \sin t & 0 \\ -\sin t & \cos t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad B(\beta) = \begin{pmatrix} \cos\beta & 0 & -\sin\beta \\ 0 & 1 & 0 \\ \sin\beta & 0 & \cos\beta \end{pmatrix}.$$

Thus the Cartesian column transforms by $\mathbf{r}' = A\mathbf{r}$, where $A = Z(\gamma)B(\beta)Z(\alpha)$.

The $l = 1$ harmonics are linear combinations of the Cartesian components of $\hat{\mathbf{r}}$:

$$\mathcal{Y} = \begin{pmatrix} Y_1^{-1} \\ Y_1^0 \\ Y_1^1 \end{pmatrix} = \sqrt{\frac{3}{4\pi}}\, Q\hat{\mathbf{r}}, \qquad Q = \begin{pmatrix} 1/\sqrt{2} & -i/\sqrt{2} & 0 \\ 0 & 0 & 1 \\ -1/\sqrt{2} & -i/\sqrt{2} & 0 \end{pmatrix}.$$

Direct row products give $QQ^\dagger = I$. In the index convention of Eq. (16.52), $Y_1^m(\Omega') = \sum_{m'} D^1_{m'm} Y_1^{m'}(\Omega)$; consequently the matrix acting on this column is $(D^1)^T$, and

$$\mathcal{Y}' = \sqrt{\frac{3}{4\pi}} QA\hat{\mathbf{r}} = QAQ^\dagger \mathcal{Y}, \qquad \boxed{(D^1)^T = QAQ^\dagger, \quad A = Q^\dagger (D^1)^T Q}.$$

The transpose here follows from the displayed summation indices.

For completeness the elementary spherical-basis matrices are

$$QZ(t)Q^\dagger = E(t) = \mathrm{diag}(e^{it}, 1, e^{-it}),$$

$$QB(\beta)Q^\dagger = H(\beta) = \begin{pmatrix} (1+c_\beta)/2 & -s_\beta/\sqrt{2} & (1-c_\beta)/2 \\ s_\beta/\sqrt{2} & c_\beta & -s_\beta/\sqrt{2} \\ (1-c_\beta)/2 & s_\beta/\sqrt{2} & (1+c_\beta)/2 \end{pmatrix},$$

where $c_t = \cos t$, $s_t = \sin t$. For example the first diagonal entry of H is $c_\beta/2 + 1/2$, and its first-row middle entry is $-s_\beta/\sqrt{2}$; the other entries follow from the same row-column products. Hence $(D^1)^T = E(\gamma)H(\beta)E(\alpha)$. Converting this back to Cartesian coordinates gives

$$B(\beta)Z(\alpha) = \begin{pmatrix} c_\beta c_\alpha & c_\beta s_\alpha & -s_\beta \\ -s_\alpha & c_\alpha & 0 \\ s_\beta c_\alpha & s_\beta s_\alpha & c_\beta \end{pmatrix},$$

and the final multiplication yields

$$\boxed{A = \begin{pmatrix} c_\gamma c_\beta c_\alpha - s_\gamma s_\alpha & c_\gamma c_\beta s_\alpha + s_\gamma c_\alpha & -c_\gamma s_\beta \\ -s_\gamma c_\beta c_\alpha - c_\gamma s_\alpha & -s_\gamma c_\beta s_\alpha + c_\gamma c_\alpha & s_\gamma s_\beta \\ s_\beta c_\alpha & s_\beta s_\alpha & c_\beta \end{pmatrix}}.$$

This is exactly Eq. (3.37), with the same passive-angle convention.

16.3.2 In proving the addition theorem, we assumed that $Y_l^k(\theta_1, \varphi_1)$ could be expanded in a series of $Y_l^m(\theta_2, \varphi_2)$, in which m varied from $-l$ to $+l$ but l was held fixed. What arguments can you develop to justify summing only over the upper index, m, and not over the lower index, l?

Hints. Examine the homogeneous harmonic polynomials $r^l Y_l^m$. Their restrictions to the sphere have terms of the form $\cos^{l-p}\theta \sin^p\theta e^{im\varphi}$, or $x^{l-p-s}y^p z^s/r^l$. Alternatively examine the rotation invariance of the angular Laplacian. The original angular hint omits the azimuthal factor; homogeneity alone also needs the harmonicity condition to exclude lower-degree traces.

Solution.

Let R be the fixed rotation, and let $U_R f(\hat{\mathbf{r}}) = f(R\hat{\mathbf{r}})$ denote its action on functions. A rotation preserves r, dot products, and the Cartesian Laplacian. The identity

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial}{\partial r}\right) - \frac{\mathbf{L}^2}{r^2}$$

therefore implies $U_R \mathbf{L}^2 = \mathbf{L}^2 U_R$. Expand the rotated harmonic in the complete orthonormal spherical-harmonic basis:

$$U_R Y_l^m = \sum_{k=0}^{\infty} \sum_{q=-k}^{k} a_{kq} Y_k^q.$$

Applying $\mathbf{L}^2$ in the two orders gives

$$l(l+1) \sum_{kq} a_{kq} Y_k^q = \sum_{kq} k(k+1) a_{kq} Y_k^q.$$

Multiply by $Y_{k'}^{q'*}$ and integrate:

$$\big[k'(k'+1) - l(l+1)\big] a_{k'q'} = 0.$$

For nonnegative integers k', l, the factor is zero only for $k' = l$. All other coefficients vanish, proving

$$\boxed{U_R Y_l^m = \sum_{q=-l}^{l} a_{lq} Y_l^q}.$$

The polynomial hint gives the same conclusion when its harmonic condition is retained. The solid harmonic $r^l Y_l^m$ is a homogeneous harmonic polynomial of degree l. Linear substitution by a rotation preserves its degree, and rotation invariance of ∇^2 preserves its being harmonic. It therefore remains in the space spanned by $r^l Y_l^{-l}, \ldots, r^l Y_l^l$. Homogeneity alone would not suffice: a general homogeneous polynomial may contain traces such as r^2 times a lower-degree harmonic. Harmonicity excludes those traces.

16.3.3 An atomic electron with angular momentum l and magnetic quantum number m has a wave function

$$\psi(r,\theta,\varphi) = f(r)Y_l^m(\theta,\varphi).$$

Show that the sum of the electron densities in a given complete shell is spherically symmetric; that is, $\sum_{m=-l}^{l}\psi^*(r,\theta,\varphi)\psi(r,\theta,\varphi)$ is independent of θ and φ.

Solution.

All members of the specified shell have the same radial factor. Consequently

$$\sum_{m=-l}^{l}\psi_m^*\psi_m = |f(r)|^2\sum_{m=-l}^{l}|Y_l^m(\theta,\varphi)|^2.$$

Apply the addition theorem to two identical directions. Their separation angle is zero, so $\cos\gamma = 1$, and $P_l(1) = 1$:

$$1 = P_l(1) = \frac{4\pi}{2l+1}\sum_{m=-l}^{l}Y_l^{m*}(\theta,\varphi)Y_l^m(\theta,\varphi).$$

Thus

$$\boxed{\sum_{m=-l}^{l}\psi_m^*\psi_m = \frac{2l+1}{4\pi}|f(r)|^2}.$$

This depends only on r. Equal occupancy of all magnetic states is the complete-shell condition used by the stated sum; a common spin-occupancy factor merely multiplies this radial density.

16.3.4 The potential energy of an electron at point $\mathbf{r}_e$ in the field of Z protons at points $\mathbf{r}_p$ is

$$\Phi = -\frac{e^2}{4\pi\varepsilon_0}\sum_{p=1}^{Z}\frac{1}{|\mathbf{r}_e-\mathbf{r}_p|}.$$

Show that for r_e larger than all r_p, this may be written as

$$\Phi = -\frac{e^2}{4\pi\varepsilon_0 r_e}\sum_{p=1}^{Z}\sum_{L,M}\left(\frac{r_p}{r_e}\right)^L\frac{4\pi}{2L+1}Y_L^M(\theta_p,\varphi_p)^*Y_L^M(\theta_e,\varphi_e).$$

How should Φ be written for $r_e < r_p$?

Solution.

The displayed expression contains the charge product $(-e)(+e) = -e^2$, so it is an interaction energy. The electric potential of the protons alone has coefficient $+e/(4\pi\varepsilon_0)$. For each proton define γ_p as the angle between $\mathbf{r}_e$ and $\mathbf{r}_p$. With $t = r_p/r_e < 1$,

$$\frac{1}{|\mathbf{r}_e-\mathbf{r}_p|} = \frac{1}{r_e}(1-2t\cos\gamma_p+t^2)^{-1/2} = \frac{1}{r_e}\sum_{L=0}^{\infty}t^L P_L(\cos\gamma_p).$$

The addition theorem is

$$P_L(\cos\gamma_p) = \frac{4\pi}{2L+1}\sum_{M=-L}^{L}Y_L^{M*}(\Omega_p)Y_L^M(\Omega_e).$$

Substitution into each term of the proton sum gives exactly the first requested expansion.

For $r_e < r_p$, factor out r_p instead. The same angular factor then gives

$$\boxed{\Phi = -\frac{e^2}{4\pi\varepsilon_0}\sum_{p=1}^{Z}\frac{1}{r_p}\sum_{L=0}^{\infty}\left(\frac{r_e}{r_p}\right)^L\frac{4\pi}{2L+1}\sum_{M=-L}^{L}Y_L^{M*}(\Omega_p)Y_L^M(\Omega_e)}$$

when the electron lies inside every proton radius. More generally, the expansion can be used term by term, even when some protons are inside r_e and others outside:

$$\boxed{\Phi = -\frac{e^2}{4\pi\varepsilon_0}\sum_{p=1}^{Z}\sum_{L=0}^{\infty}\frac{r_{<p}^L}{r_{>p}^{L+1}}\frac{4\pi}{2L+1}\sum_{M=-L}^{L}Y_L^{M*}(\Omega_p)Y_L^M(\Omega_e)},$$

where $r_{<p} = \min(r_e, r_p)$ and $r_{>p} = \max(r_e, r_p)$. The strictly unequal radii assumed above make each generating series convergent. Reversing which radius is larger changes only the radial factor, not the angular addition theorem.

16.3.5 Two protons are uniformly distributed within the same spherical volume. If the coordinates of one element of charge are $(r_1, \theta_1, \varphi_1)$ and the coordinates of the other are $(r_2, \theta_2, \varphi_2)$ and r_{12} is the distance between them, the element of repulsion energy will be given by

$$d\psi = \rho^2\frac{d\tau_1\, d\tau_2}{r_{12}} = \rho^2\frac{r_1^2 dr_1 \sin\theta_1\, d\theta_1\, d\varphi_1\, r_2^2 dr_2 \sin\theta_2\, d\theta_2\, d\varphi_2}{r_{12}},$$

where

$$\rho = \frac{\text{charge}}{\text{volume}} = \frac{3e}{4\pi R^3} \qquad \text{and} \qquad r_{12}^2 = r_1^2 + r_2^2 - 2r_1 r_2\cos\gamma.$$

Here ρ is the charge density and γ is the angle between $\mathbf{r}_1$ and $\mathbf{r}_2$. Calculate the total electrostatic energy (of repulsion) of the two protons. This calculation is used in accounting for the mass difference in "mirror" nuclei, such as O^{15} and N^{15}.

ANS. $\dfrac{6}{5}\dfrac{e^2}{R}$.

Solution.

The angular average of the Coulomb expansion retains only $L = 0$. Indeed $\int P_L(\hat{\mathbf{r}}_1 \cdot \hat{\mathbf{r}}_2)\, d\Omega_2 = 4\pi\delta_{L0}$, so

$$\int d\Omega_1 \int d\Omega_2 \frac{1}{r_{12}} = \frac{16\pi^2}{\max(r_1, r_2)}.$$

The diagonal $r_1 = r_2$ has zero measure in the remaining double radial integral. Split its square into the two equal triangles:

$$\begin{aligned} U &= 16\pi^2\rho^2\int_0^R\int_0^R \frac{r_1^2 r_2^2}{\max(r_1, r_2)}\, dr_2\, dr_1 \\ &= 32\pi^2\rho^2\int_0^R r_1\, dr_1\int_0^{r_1} r_2^2\, dr_2 \\ &= \frac{32\pi^2\rho^2}{3}\int_0^R r_1^4\, dr_1 = \frac{32\pi^2\rho^2 R^5}{15}. \end{aligned}$$

Substitute $\rho^2 = 9e^2/(16\pi^2 R^6)$:

$$\boxed{U = \frac{32\cdot 9}{15\cdot 16}\frac{e^2}{R} = \frac{6}{5}\frac{e^2}{R}}.$$

There is no factor 1/2: this is the mutual energy of two distinct charge distributions, and each pair with one element from each distribution is counted once. The problem uses a Coulomb constant of one; in SI units the result is multiplied by $1/(4\pi\varepsilon_0)$.

16.3.6 Each of the two $1s$ electrons in helium may be described by a hydrogenic wave function

$$\psi(r) = \left(\frac{Z^3}{\pi a_0^3}\right)^{1/2} e^{-Zr/a_0}$$

in the absence of the other electron. Here Z, the atomic number, is 2. The symbol a_0 is the Bohr radius, $\hbar^2/me^2$. Find the mutual potential energy of the two electrons, given by

$$\int \psi^*(\mathbf{r}_1)\psi^*(\mathbf{r}_2)\frac{e^2}{|\mathbf{r}_1 - \mathbf{r}_2|}\,\psi(\mathbf{r}_1)\psi(\mathbf{r}_2)\,d^3r_1\,d^3r_2.$$

ANS. $\dfrac{5e^2Z}{8a_0}$.

Solution.

Put $\lambda = 2Z/a_0$ and $N = Z^3/(\pi a_0^3)$, so $|\psi(r)|^2 = Ne^{-\lambda r}$. Its normalization is

$$4\pi N \int_0^\infty r^2 e^{-\lambda r}\,dr = 4\pi N\,\frac{2}{\lambda^3} = 1.$$

The angular integration is the same as in 16.3.5. Splitting the radial quadrant into $r_2 < r_1$ and $r_1 < r_2$ gives

$$U = 32\pi^2 e^2 N^2 \int_0^\infty r_1 e^{-\lambda r_1}\,dr_1 \int_0^{r_1} r_2^2 e^{-\lambda r_2}\,dr_2.$$

Set $x = \lambda r_2$, $y = \lambda r_1$ and reverse the order:

$$U = \frac{32\pi^2 e^2 N^2}{\lambda^5} \int_0^\infty x^2 e^{-x} \left(\int_x^\infty y e^{-y}\,dy\right) dx.$$

Integration by parts gives

$$\int_x^\infty y e^{-y} dy = [-ye^{-y}]_x^\infty + \int_x^\infty e^{-y} dy = (x+1)e^{-x}.$$

For $I_k = \int_0^\infty x^k e^{-2x} dx$ and $k \geq 1$,

$$I_k = \left[-\frac{x^k e^{-2x}}{2}\right]_0^\infty + \frac{k}{2}\int_0^\infty x^{k-1} e^{-2x} dx = \frac{k}{2} I_{k-1}.$$

Since $I_0 = 1/2$, this gives $I_1 = 1/4$, $I_2 = 1/4$, and $I_3 = 3/8$, or generally $I_k = k!/2^{k+1}$. Hence the dimensionless integral is

$$\int_0^\infty (x^3 + x^2) e^{-2x}\,dx = \frac{3!}{2^4} + \frac{2!}{2^3} = \frac{3}{8} + \frac{1}{4} = \frac{5}{8}.$$

The prefactor simplifies without an approximation:

$$\frac{32\pi^2 e^2}{(2Z/a_0)^5} \frac{Z^6}{\pi^2 a_0^6} = \frac{e^2 Z}{a_0}.$$

Thus

$$\boxed{U = \frac{5e^2 Z}{8a_0} = \frac{5e^2}{4a_0} \quad (Z = 2)}.$$

This evaluates the stated independent hydrogenic-orbital integral; it uses the same Coulomb-unit convention as the problem.

16.3.7 The probability of finding a $1s$ hydrogen electron in a volume element $r^2 dr \sin\theta\, d\theta\, d\varphi$ is

$$\frac{1}{\pi a_0^3}\, e^{-2r/a_0}\, r^2 dr \sin\theta\, d\theta\, d\varphi,$$

where r is the distance of the electron from the nucleus. Find the electrostatic potential of this charge distribution at points $\mathbf{r}_1$, where you may not assume that $\mathbf{r}_1$ is on the polar axis of your coordinate system. Calculate the potential from

$$V(\mathbf{r}_1) = -\frac{e}{4\pi\varepsilon_0} \int \frac{\rho(\mathbf{r}_2)}{r_{12}}\, d^3 r_2,$$

where $r_{12} = |\mathbf{r}_1 - \mathbf{r}_2|$ and $\rho(r) = e^{-2r/a_0}/(\pi a_0^3)$ is the probability density. Expand $1/r_{12}$. Apply the Legendre polynomial addition theorem and show that the angular dependence of $V(\mathbf{r}_1)$ drops out.

ANS. $V(r_1) = -\dfrac{e}{4\pi\varepsilon_0}\dfrac{1}{2r_1}\left[\gamma\left(3, \dfrac{2r_1}{a_0}\right) + \dfrac{2r_1}{a_0}\Gamma\left(2, \dfrac{2r_1}{a_0}\right)\right]$, where γ and Γ are incomplete gamma functions, Eq. (13.73).

Solution.

The supplied transcription incorrectly puts both incomplete gamma functions under $1/(2r_1)$. The actual textbook answer (printed p. 807) has coefficients $1/(2r_1)$ and $1/a_0$, respectively. The latter is restored below and in the answer line. The textbook's instruction "Expand r_{12}" must instead say "Expand $1/r_{12}$", the Coulomb kernel in the integral. Write $R = r_1$ and $r = r_2$. Expand the inverse separation:

$$\frac{1}{r_{12}} = \sum_{l=0}^{\infty} \frac{r_<^l}{r_>^{l+1}} \frac{4\pi}{2l+1} \sum_{m=-l}^{l} Y_l^m(\Omega_1) Y_l^{m*}(\Omega_2).$$

The density is independent of Ω_2. Because $\int Y_l^{m*}\, d\Omega_2 = \sqrt{4\pi}\delta_{l0}\delta_{m0}$ and $Y_0^0(\Omega_1) = 1/\sqrt{4\pi}$, its angular integral is $4\pi/r_>$. Thus no choice of polar axis is needed, and

$$V(R) = -\frac{e}{4\pi\varepsilon_0}\frac{4}{a_0^3}\left[\frac{1}{R}\int_0^R r^2 e^{-2r/a_0}\, dr + \int_R^\infty r e^{-2r/a_0}\, dr\right].$$

Put $q = 2R/a_0$, $t = 2r/a_0$. By the definitions $\gamma(s,q) = \int_0^q t^{s-1}e^{-t}dt$ and $\Gamma(s,q) = \int_q^\infty t^{s-1}e^{-t}dt$,

$$\boxed{V(R) = -\frac{e}{4\pi\varepsilon_0}\left[\frac{\gamma(3,q)}{2R} + \frac{\Gamma(2,q)}{a_0}\right]}.$$

For clarity, the required integrations by parts are

$$\begin{aligned}
\int_0^q te^{-t}dt &= [-te^{-t}]_0^q + \int_0^q e^{-t}dt = 1 - (q+1)e^{-q},\\
\gamma(3,q) = \int_0^q t^2 e^{-t}dt &= [-t^2e^{-t}]_0^q + 2\int_0^q te^{-t}dt\\
&= 2 - e^{-q}(q^2+2q+2),\\
\Gamma(2,q) = \int_q^\infty te^{-t}dt &= [-te^{-t}]_q^\infty + \int_q^\infty e^{-t}dt = e^{-q}(q+1).
\end{aligned}$$

Since $q/(2R) = 1/a_0$, substitution cancels the q^2 and q radial contributions to leave

$$\boxed{V(R) = -\frac{e}{4\pi\varepsilon_0}\left[\frac{1}{R} - e^{-2R/a_0}\left(\frac{1}{R} + \frac{1}{a_0}\right)\right]}.$$

For example, $(1-e^{-2R/a_0})/R \to 2/a_0$ as $R \to 0$, so $V(0) = -e/(4\pi\varepsilon_0 a_0)$. At large R the exponential disappears and the potential is that of charge $-e$, as required.

16.3.8 A hydrogen electron in a $2p$ orbital has a charge distribution

$$\rho = -\frac{e}{64\pi a_0^5}\, r^2 e^{-r/a_0} \sin^2\theta,$$

where $a_0 = \hbar^2/me^2$ is the Bohr radius, and r is the distance between the electron and the nucleus. Find the electrostatic electron–nucleus interaction energy, treating the nucleus as a point charge $+e$ at the origin. Also find the electric potential of the electron distribution at an arbitrary field point.

Solution.

For the hydrogen atom, the stated electrostatic energy is interpreted as the electron–nucleus interaction. The manual instead displays a position-dependent electric potential; both quantities are obtained here. A classical self-energy $\frac{1}{2}\int \rho V\, d^3r$ would be a different quantity, so a second charge is not required for every notion of electrostatic energy.

Let the field point be (R, Θ, Φ) and write $a = a_0$. The identity $\sin^2\theta = \frac{2}{3}[1 - P_2(\cos\theta)]$ gives

$$\rho(r,\theta) = -\frac{e}{96\pi a^5} r^2 e^{-r/a}[1 - P_2(\cos\theta)].$$

Its total charge is

$$-\frac{e}{96\pi a^5}\,(4\pi)\int_0^\infty r^4 e^{-r/a} dr = -\frac{e}{24a^5}(4!a^5) = -e.$$

Only the monopole and quadrupole survive. The addition theorem and orthogonality give, for $l = 0, 2$,

$$\int \frac{P_l(\cos\theta)}{|\mathbf{R}-\mathbf{r}|}\, d\Omega = \frac{4\pi}{2l+1}\frac{r_<^l}{r_>^{l+1}} P_l(\cos\Theta).$$

To see the factor, write $P_l(\cos\theta) = \sqrt{4\pi/(2l+1)}\,Y_l^0(\Omega)$ in the Coulomb expansion: the angular integral selects its $l, 0$ term. Consequently

$$V(R,\Theta) = -\frac{e}{4\pi\varepsilon_0}\frac{1}{24a^5}\left[I_0(R) - \frac{P_2(\cos\Theta)}{5} I_2(R)\right],$$

where splitting each radial integral at $r = R$ gives

$$I_0(R) = \frac{1}{R}\int_0^R r^4 e^{-r/a} dr + \int_R^\infty r^3 e^{-r/a} dr,$$
$$I_2(R) = \frac{1}{R^3}\int_0^R r^6 e^{-r/a} dr + R^2 \int_R^\infty r e^{-r/a} dr.$$

Thus, with $q = R/a$,

$$\boxed{V(R,\Theta) = -\frac{e}{4\pi\varepsilon_0}\left\{\frac{1}{24}\left[\frac{\gamma(5,q)}{R} + \frac{\Gamma(4,q)}{a}\right] - \frac{P_2(\cos\Theta)}{120}\left[\frac{a^2\gamma(7,q)}{R^3} + \frac{R^2\Gamma(2,q)}{a^3}\right]\right\}.}$$

The manual's potential on printed p. 265 lacks the overall factor $-e$ required by the given electron charge density. The displayed result restores both the charge and its negative sign. A test charge q_{test} has energy $q_{\text{test}}V(R,\Theta)$.

For the hydrogen nucleus, $q_{\text{test}} = +e$ and $R = 0$. Since $e^{-r/a} = 1 + O(r)$ near the origin, the interior contributions are

$$\frac{1}{R}\int_0^R r^4 e^{-r/a} dr = \frac{R^4}{5} + O(R^5), \qquad \frac{1}{R^3}\int_0^R r^6 e^{-r/a} dr = \frac{R^4}{7} + O(R^5).$$

Both vanish; the exterior quadrupole contribution is $R^2 \int_R^\infty r e^{-r/a} dr = O(R^2)$ and also vanishes. The surviving contribution in the braces is $\Gamma(4,0)/(24a) = 3!/(24a) = 1/(4a)$. Therefore

$$\boxed{U_{\text{electron–nucleus}} = eV(0) = -\frac{e^2}{4\pi\varepsilon_0}\frac{1}{4a_0}}.$$

Equivalently one may integrate the nuclear potential directly:

$$U = \int \rho \frac{e}{4\pi\varepsilon_0 r} d^3r = -\frac{e^2}{4\pi\varepsilon_0}\frac{1}{24a^5}\int_0^\infty r^3 e^{-r/a} dr = -\frac{e^2}{4\pi\varepsilon_0}\frac{1}{4a}.$$

In the Coulomb units used in the question's $a_0 = \hbar^2/(me^2)$, set $1/(4\pi\varepsilon_0) = 1$: the energy is $-e^2/(4a_0)$. This is the electrostatic interaction energy; the kinetic energy is not part of the stated electrostatic calculation.

16.3.9 Interpret the delta and both sums as distributions with respect to $d\Omega = \sin\theta\, d\theta\, d\varphi$.

(a) As a Laplace series and as an example of Eq. (5.27), show that

$$\delta(\Omega_1 - \Omega_2) = \sum_{l=0}^{\infty}\sum_{m=-l}^{l} Y_l^m(\theta_2,\varphi_2)^* Y_l^m(\theta_1,\varphi_1).$$

(b) Show also that this same representation of the Dirac delta function may be written as

$$\delta(\Omega_1 - \Omega_2) = \sum_{l=0}^{\infty} \frac{2l+1}{4\pi} P_l(\cos\gamma),$$

and identify γ. Now, if you can justify equating the summations over l term by term, you have an alternate derivation of the spherical harmonic addition theorem.

Solution.

These infinite sums represent a distribution on the sphere, with measure $d\Omega = \sin\theta\, d\theta\, d\varphi$; they are not ordinary pointwise convergent formulas for a function.

(a) For a smooth test function f, let $f_{lm} = \int Y_l^{m*}(\Omega_2) f(\Omega_2)\, d\Omega_2$. Completeness gives $f(\Omega_1) = \sum_{lm} f_{lm} Y_l^m(\Omega_1)$. Therefore

$$\int \left[\sum_{lm} Y_l^{m*}(\Omega_2) Y_l^m(\Omega_1)\right] f(\Omega_2)\, d\Omega_2 = \sum_{lm} f_{lm} Y_l^m(\Omega_1) = f(\Omega_1).$$

This is precisely the defining action of $\delta(\Omega_1 - \Omega_2)$, proving (a) in the distributional sense.

(b) Define the finite-dimensional projection kernel

$$K_l(\Omega_1,\Omega_2) = \sum_{m=-l}^{l} Y_l^m(\Omega_1) Y_l^{m*}(\Omega_2).$$

A common rotation of the two directions mixes their m components by the same unitary matrix: rotations preserve $d\Omega$ and hence the harmonic inner product. In the sum, the two matrices multiply to the identity. Thus K_l is unchanged by a common rotation. Choose one taking Ω_2 to the north pole. The other direction then has polar angle γ, their angular separation. At the pole,

$$Y_l^m(0,\varphi) = 0 \quad (m \neq 0), \qquad Y_l^0(0,\varphi) = \sqrt{\frac{2l+1}{4\pi}}.$$

It follows directly, without assuming the addition theorem, that

$$K_l(\Omega_1,\Omega_2) = Y_l^0(\gamma,0)\sqrt{\frac{2l+1}{4\pi}} = \boxed{\frac{2l+1}{4\pi}P_l(\cos\gamma)},$$

where

$$\cos\gamma = \cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2\cos(\varphi_1-\varphi_2).$$

Summing these kernels gives the stated Legendre representation of the delta distribution.

The term-by-term identification is justified by orthogonal projection, rather than by subtracting two arbitrary infinite sums. Apply the degree-l projector in the first angular variable. It annihilates all K_k with $k \neq l$, since $\int Y_l^{m*}Y_k^q\,d\Omega = \delta_{lk}\delta_{mq}$, and leaves K_l unchanged. Thus each degree block is uniquely determined. The boxed identity is the addition theorem. The manual (printed p. 266) says that reconstructing the arbitrary test function yields $Y_l^m(\Omega_1)$. Its preceding sum instead yields $f(\Omega_1)$, as the delta's defining action requires. A single harmonic is recovered only when f is that harmonic.

16.3.10 Verify the following, with $L \geq 0$ and $|M| \leq L$; in (d) also require $L \geq 1$ and $|M+1| \leq L-1$:

(a) $$\int Y_L^M(\theta,\varphi)Y_0^0(\theta,\varphi)Y_L^{M*}(\theta,\varphi)\,d\Omega = \frac{1}{\sqrt{4\pi}},$$

(b) $$\int Y_L^M Y_1^0 Y_{L+1}^{M*}\,d\Omega = \sqrt{\frac{3}{4\pi}\frac{(L+M+1)(L-M+1)}{(2L+1)(2L+3)}},$$

(c) $$\int Y_L^M Y_1^1 Y_{L+1}^{M+1*}\,d\Omega = \sqrt{\frac{3}{8\pi}\frac{(L+M+1)(L+M+2)}{(2L+1)(2L+3)}},$$

(d) $$\int Y_L^M Y_1^1 Y_{L-1}^{M+1*}\,d\Omega = -\sqrt{\frac{3}{8\pi}\frac{(L-M)(L-M-1)}{(2L-1)(2L+1)}}.$$

These integrals were used in an investigation of the angular correlation of internal conversion electrons.

Solution.

Let

$$N_{LM} = \sqrt{\frac{2L+1}{4\pi}\frac{(L-M)!}{(L+M)!}}, \qquad Y_L^M = N_{LM}P_L^M(\cos\theta)e^{iM\varphi}.$$

All integrals use normalized harmonics and $d\Omega$. Each harmonic appearing in an integral must have nonnegative degree and $|m| \leq l$. In particular, (d) is an integral of the displayed harmonics only when $L \geq 1$ and $|M+1| \leq L-1$; absent terms in recurrence formulas are omitted.

(a) Since $Y_0^0 = 1/\sqrt{4\pi}$,

$$\int Y_L^M Y_0^0 Y_L^{M*} d\Omega = \frac{1}{\sqrt{4\pi}} \int |Y_L^M|^2 d\Omega = \boxed{\frac{1}{\sqrt{4\pi}}}.$$

(b) Put $x = \cos\theta$. Equation (15.88) gives

$$x P_L^M = \frac{L-M+1}{2L+1} P_{L+1}^M + \frac{L+M}{2L+1} P_{L-1}^M.$$

For the upper term, the normalization ratio is

$$\frac{N_{LM}}{N_{L+1,M}} = \sqrt{\frac{2L+1}{2L+3}\frac{L+M+1}{L-M+1}}.$$

Multiplying by $(L-M+1)/(2L+1)$ therefore gives

$$A_+ = \sqrt{\frac{(L+1)^2 - M^2}{(2L+1)(2L+3)}}.$$

For the lower term,

$$\frac{N_{LM}}{N_{L-1,M}} = \sqrt{\frac{2L+1}{2L-1}\frac{L-M}{L+M}}, \qquad A_- = \sqrt{\frac{L^2-M^2}{(2L-1)(2L+1)}}.$$

At an endpoint the zero coefficient removes the nonexistent term. Thus $\cos\theta\, Y_L^M = A_+ Y_{L+1}^M + A_- Y_{L-1}^M$. Multiply by Y_{L+1}^{M*}, integrate, and use $Y_1^0 = \sqrt{3/(4\pi)}\cos\theta$:

$$\boxed{\int Y_L^M Y_1^0 Y_{L+1}^{M*} d\Omega = \sqrt{\frac{3}{4\pi}\frac{(L+M+1)(L-M+1)}{(2L+1)(2L+3)}}}.$$

(c),(d) Equation (15.89) is

$$\sqrt{1-x^2}\, P_L^M = \frac{P_{L-1}^{M+1} - P_{L+1}^{M+1}}{2L+1}.$$

The required ratios, including the azimuthal factor $e^{i\varphi}$, are

$$\frac{N_{LM}}{(2L+1)N_{L+1,M+1}} = \sqrt{\frac{(L+M+1)(L+M+2)}{(2L+1)(2L+3)}} = B_+,$$
$$\frac{N_{LM}}{(2L+1)N_{L-1,M+1}} = \sqrt{\frac{(L-M)(L-M-1)}{(2L-1)(2L+1)}} = B_-.$$

For example the upper factorial quotient cancels $(L-M)!$ and leaves $(L+M+2)!/(L+M)! = (L+M+2)(L+M+1)$; the lower quotient similarly leaves $(L-M)(L-M-1)$. Hence

$$\sin\theta\, e^{i\varphi} Y_L^M = -B_+ Y_{L+1}^{M+1} + B_- Y_{L-1}^{M+1}.$$

Because $Y_1^1 = -\sqrt{3/(8\pi)}\sin\theta\, e^{i\varphi}$, orthogonality now selects each coefficient:

$$\boxed{\begin{aligned} \int Y_L^M Y_1^1 Y_{L+1}^{M+1*} d\Omega &= \sqrt{\frac{3}{8\pi}\frac{(L+M+1)(L+M+2)}{(2L+1)(2L+3)}}, \\ \int Y_L^M Y_1^1 Y_{L-1}^{M+1*} d\Omega &= -\sqrt{\frac{3}{8\pi}\frac{(L-M)(L-M-1)}{(2L-1)(2L+1)}}. \end{aligned}}$$

16.3.11 For nonnegative integers L, N, show that

$$\text{(a)}\ \int_{-1}^{1} xP_L(x)P_N(x)\,dx = \begin{cases} \dfrac{2(L+1)}{(2L+1)(2L+3)}, & N = L+1, \\ \dfrac{2L}{(2L-1)(2L+1)}, & N = L-1, \\ 0, & \text{all other } N \geq 0. \end{cases}$$

$$\text{(b)}\ \int_{-1}^{1} x^2P_L(x)P_N(x)\,dx = \begin{cases} \dfrac{2(L+1)(L+2)}{(2L+1)(2L+3)(2L+5)}, & N = L+2, \\ \dfrac{2(2L^2+2L-1)}{(2L-1)(2L+1)(2L+3)}, & N = L, \\ \dfrac{2L(L-1)}{(2L-3)(2L-1)(2L+1)}, & N = L-2, \\ 0, & \text{all other } N \geq 0. \end{cases}$$

Solution.

Take L, N to be nonnegative integers. Write the three-term recurrence as

$$xP_L = a_L P_{L+1} + b_L P_{L-1}, \qquad a_L = \frac{L+1}{2L+1}, \quad b_L = \frac{L}{2L+1}.$$

At $L = 0$ this is $xP_0 = P_1$; the negative-degree term is omitted. Orthogonality is $\int_{-1}^{1} P_j P_N\,dx = 2\delta_{jN}/(2N+1)$.

(a) Multiplying the recurrence by P_N and integrating gives

$$\int_{-1}^{1} xP_L P_N\,dx = a_L \frac{2\delta_{N,L+1}}{2L+3} + b_L \frac{2\delta_{N,L-1}}{2L-1}.$$

This gives respectively $2(L+1)/[(2L+1)(2L+3)]$ and $2L/[(2L-1)(2L+1)]$.

(b) Apply the recurrence a second time:

$$\begin{aligned} x^2P_L &= a_L(a_{L+1}P_{L+2} + b_{L+1}P_L) + b_L(a_{L-1}P_L + b_{L-1}P_{L-2}) \\ &= A_L P_{L+2} + B_L P_L + C_L P_{L-2}, \end{aligned}$$

where the coefficients are

$$A_L = \frac{(L+1)(L+2)}{(2L+1)(2L+3)}, \qquad C_L = \frac{L(L-1)}{(2L+1)(2L-1)},$$

$$\begin{aligned} B_L &= \frac{(L+1)^2}{(2L+1)(2L+3)} + \frac{L^2}{(2L+1)(2L-1)} \\ &= \frac{(L+1)^2(2L-1) + L^2(2L+3)}{(2L+1)(2L-1)(2L+3)} \\ &= \frac{(2L+1)(2L^2+2L-1)}{(2L+1)(2L-1)(2L+3)} = \frac{2L^2+2L-1}{(2L-1)(2L+3)}. \end{aligned}$$

For $L = 0, 1$, direct application of $xP_0 = P_1$ removes every negative-degree term; the surviving formulas still give these same A_L, B_L. Integration against P_N multiplies the three coefficients by $2/(2L+5)$, $2/(2L+1)$, and $2/(2L-3)$, respectively. These are exactly the three stated answers. Both displayed case lists need a final case 0 for every other nonnegative integer N. A case with a negative value of $N = L-1$ or $N = L-2$ is absent.

16.3.12 Since $xP_n(x)$ is a polynomial (of degree $n+1$), it may be represented by the Legendre series

$$xP_n(x) = \sum_{s=0}^{\infty} a_s P_s(x).$$

(a) Show that $a_s = 0$ for $s < n-1$ and $s > n+1$.

(b) For $n \geq 1$, calculate a_{n-1}, a_n, and a_{n+1} and reproduce the recurrence relation, Eq. (15.18). Handle $n = 0$ separately, since a_{-1} is not an expansion coefficient.

Note. This argument may be put in a general form to demonstrate the existence of a three-term recurrence relation for any of our complete sets of orthogonal polynomials:

$$x\varphi_n = a_{n+1}\varphi_{n+1} + a_n\varphi_n + a_{n-1}\varphi_{n-1}.$$

Solution.

Since the Legendre polynomials of degrees $0, \ldots, n+1$ form a basis of the polynomials of degree at most $n+1$, the expansion of xP_n is finite and $a_s = 0$ for $s > n+1$. For a coefficient with $s < n-1$, use the inner product:

$$a_s = \frac{2s+1}{2}\int_{-1}^{1} xP_n(x)P_s(x)\,dx = \frac{2s+1}{2}\int_{-1}^{1} P_n(x)[xP_s(x)]\,dx.$$

The bracket has degree $s+1 < n$. It is a combination of $P_0, \ldots, P_{n-1}$, each orthogonal to P_n; hence $a_s = 0$. Only degrees $n-1, n, n+1$ can occur.

The middle coefficient vanishes because $x[P_n(x)]^2$ is odd:

$$a_n = \frac{2n+1}{2}\int_{-1}^{1} x[P_n(x)]^2 dx = 0.$$

To find the upper coefficient without assuming the desired recurrence, use Rodrigues' formula. The highest term in $(x^2-1)^n$ is x^{2n}, whose nth derivative is $(2n)!x^n/n!$. Thus the leading coefficient of P_n is

$$\kappa_n = \frac{(2n)!}{2^n(n!)^2}.$$

Matching the coefficient of x^{n+1} in $xP_n = a_{n+1}P_{n+1} + a_{n-1}P_{n-1}$ gives

$$a_{n+1} = \frac{\kappa_n}{\kappa_{n+1}} = \frac{2(n+1)^2}{(2n+2)(2n+1)} = \frac{n+1}{2n+1}.$$

For $n \geq 1$, evaluate at $x = 1$, where every $P_j(1) = 1$:

$$1 = a_{n+1} + a_{n-1}, \qquad a_{n-1} = 1 - \frac{n+1}{2n+1} = \frac{n}{2n+1}.$$

Therefore

$$\boxed{(2n+1)xP_n = (n+1)P_{n+1} + nP_{n-1} \quad (n \geq 1)}.$$

For $n = 0$ the separate identity is $xP_0 = P_1$.

The same degree and inner-product argument works for orthogonal polynomials with any positive weight on their interval: multiplication by x is symmetric in that inner product, so only the neighboring degrees and the original degree can appear. The middle coefficient need not vanish unless the weight and interval supply the parity symmetry used here.

16.4 Vector Spherical Harmonics

16.4.1 Construct the vector spherical harmonics with total $J = 0, M = 0$ and total $J = 1, M = 0$, for every allowed orbital degree L. The notation is $\mathbf{Y}_{JLM}$: the first index is total angular momentum, and the second is orbital angular momentum.

ANS. $\mathbf{Y}_{010} = -\hat{\mathbf{r}}(4\pi)^{-1/2}$
$\mathbf{Y}_{000} = \mathbf{0}$ (formal zero only)
$\mathbf{Y}_{120} = -\hat{\mathbf{r}}(2\pi)^{-1/2}\cos\theta - \hat{\boldsymbol{\theta}}(8\pi)^{-1/2}\sin\theta$
$\mathbf{Y}_{110} = \hat{\boldsymbol{\varphi}}\, i\,(3/8\pi)^{1/2}\sin\theta$
$\mathbf{Y}_{100} = \hat{\mathbf{r}}(4\pi)^{-1/2}\cos\theta - \hat{\boldsymbol{\theta}}(4\pi)^{-1/2}\sin\theta.$

Solution.

Use the spherical Cartesian basis

$$\mathbf{e}_+ = -\frac{\mathbf{e}_x + i\mathbf{e}_y}{\sqrt{2}}, \qquad \mathbf{e}_0 = \mathbf{e}_z, \qquad \mathbf{e}_- = \frac{\mathbf{e}_x - i\mathbf{e}_y}{\sqrt{2}}.$$

The first index of $\mathbf{Y}_{JLM}$ is the total angular momentum; the second is the orbital degree. Its definition is

$$\mathbf{Y}_{JLM} = \sum_{m,q} C(L\,1\,J \mid m\,q\,M) Y_L^m \mathbf{e}_q.$$

The triangle condition $|L-1| \le J \le L+1$ excludes $J = L = 0$. The printed $\mathbf{Y}_{000} = 0$ is a useful formal zero, not a normalized vector harmonic. There is also no negative-degree orbital harmonic.

For $J = M = 0$, only $L = 1$ is allowed. The singlet coefficients from 16.2.1 give

$$\mathbf{Y}_{010} = \frac{1}{\sqrt{3}}(Y_1^1\mathbf{e}_- - Y_1^0\mathbf{e}_0 + Y_1^{-1}\mathbf{e}_+).$$

With $\mathbf{e}_\rho = \cos\varphi\,\mathbf{e}_x + \sin\varphi\,\mathbf{e}_y$, the $m = \pm 1$ terms combine to $-\sqrt{3/(4\pi)}\sin\theta\,\mathbf{e}_\rho$. The $m = 0$ term is $-\sqrt{3/(4\pi)}\cos\theta\,\mathbf{e}_z$. Since $\hat{\mathbf{r}} = \sin\theta\,\mathbf{e}_\rho + \cos\theta\,\mathbf{e}_z$,

$$\boxed{\mathbf{Y}_{010} = -\frac{\hat{\mathbf{r}}}{\sqrt{4\pi}}}.$$

For $J = 1, M = 0, L = 0$, coupling a scalar to spin one is immediate:

$$\boxed{\mathbf{Y}_{100} = Y_0^0\mathbf{e}_0 = \frac{\cos\theta\,\hat{\mathbf{r}} - \sin\theta\,\hat{\boldsymbol{\theta}}}{\sqrt{4\pi}}}.$$

For $L = 1$, the antisymmetric coefficients give

$$\mathbf{Y}_{110} = \frac{1}{\sqrt{2}}(Y_1^1\mathbf{e}_- - Y_1^{-1}\mathbf{e}_+).$$

Insert $Y_1^{\pm 1} = \mp\sqrt{3/(8\pi)}\sin\theta\,e^{\pm i\varphi}$. The Cartesian bracket is

$$-e^{i\varphi}(\mathbf{e}_x - i\mathbf{e}_y) + e^{-i\varphi}(\mathbf{e}_x + i\mathbf{e}_y) = 2i(-\sin\varphi\,\mathbf{e}_x + \cos\varphi\,\mathbf{e}_y),$$

so

$$\boxed{\mathbf{Y}_{110} = i\sqrt{\frac{3}{8\pi}}\sin\theta\,\hat{\boldsymbol{\varphi}}}.$$

Finally consider $L=2$. To obtain the $J=1$ coefficients, first write its highest $M=1$ state in the product basis $(m,q)=(2,-1),(1,0),(0,1)$ with coefficients (a,b,c). Its raising must vanish. The coefficients of the two $M=2$ products yield

$$\sqrt{2}a+2b=0, \qquad \sqrt{2}b+\sqrt{6}c=0.$$

Thus $b=-a/\sqrt{2}$, $c=a/\sqrt{6}$, and $a^2+b^2+c^2=5a^2/3=1$ fixes $(a,b,c)=(\sqrt{3/5},-\sqrt{3/10},1/\sqrt{10})$. Lowering once and dividing by $\sqrt{2}$ gives

$$\begin{aligned} C(2\,1\,1\mid 1,-1,0)&=(2a+\sqrt{2}b)/\sqrt{2}=\sqrt{3/10},\\ C(2\,1\,1\mid 0,0,0)&=(\sqrt{6}b+\sqrt{2}c)/\sqrt{2}=-\sqrt{2/5},\\ C(2\,1\,1\mid -1,1,0)&=\sqrt{6}c/\sqrt{2}=\sqrt{3/10}. \end{aligned}$$

Consequently

$$\mathbf{Y}_{120}=\sqrt{\frac{3}{10}}(Y_2^1\mathbf{e}_-+Y_2^{-1}\mathbf{e}_+)-\sqrt{\frac{2}{5}}Y_2^0\mathbf{e}_z.$$

Use $Y_2^{\pm1}=\mp\sqrt{15/(8\pi)}\sin\theta\cos\theta e^{\pm i\varphi}$ and $Y_2^0=\sqrt{5/(16\pi)}(3\cos^2\theta-1)$:

$$\mathbf{Y}_{120}=-\frac{1}{\sqrt{8\pi}}\left[3\sin\theta\cos\theta\,\mathbf{e}_\rho+(3\cos^2\theta-1)\mathbf{e}_z\right].$$

The substitutions $\mathbf{e}_\rho=\sin\theta\,\hat{\mathbf{r}}+\cos\theta\,\hat{\boldsymbol{\theta}}$ and $\mathbf{e}_z=\cos\theta\,\hat{\mathbf{r}}-\sin\theta\,\hat{\boldsymbol{\theta}}$ give radial coefficient $3\sin^2\theta\cos\theta+(3\cos^2\theta-1)\cos\theta=2\cos\theta$ and polar coefficient $3\sin\theta\cos^2\theta-(3\cos^2\theta-1)\sin\theta=\sin\theta$. Therefore

$$\boxed{\mathbf{Y}_{120}=-\frac{\cos\theta}{\sqrt{2\pi}}\hat{\mathbf{r}}-\frac{\sin\theta}{\sqrt{8\pi}}\hat{\boldsymbol{\theta}}}.$$

Together with the formal $\mathbf{Y}_{000}=0$, these reproduce all five displayed expressions.

16.4.2 Use argument inversion $\hat{\mathbf{r}}\mapsto-\hat{\mathbf{r}}$ with a fixed Cartesian basis, and include only allowed index triples. Verify that the parity of $\mathbf{Y}_{L\,L+1\,M}$ is $(-1)^{L+1}$, that of $\mathbf{Y}_{L\,L\,M}$ is $(-1)^L$, and that of $\mathbf{Y}_{L\,L-1\,M}$ is $(-1)^{L+1}$. What happened to the M-dependence of the parity?

Hint. $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\varphi}}$ have odd parity; $\hat{\boldsymbol{\theta}}$ has even parity (compare with Exercise 3.10.25).

Solution.

Under inversion of the direction, $(\theta,\varphi)\mapsto(\pi-\theta,\varphi+\pi)$. For integer m, the associated Legendre and azimuthal factors give

$$P_l^m(-\cos\theta)=(-1)^{l+m}P_l^m(\cos\theta), \qquad e^{im(\varphi+\pi)}=(-1)^m e^{im\varphi}.$$

Their product is $(-1)^{l+2m}=(-1)^l$ times the original harmonic. Thus $Y_l^m(-\hat{\mathbf{r}})=(-1)^lY_l^m(\hat{\mathbf{r}})$, independently of m.

In the fixed Cartesian spherical basis, $\mathbf{e}_q$ is independent of the field direction. Every term in

$$\mathbf{Y}_{JLM}(\hat{\mathbf{r}})=\sum_{m,q}C(L\,1\,J\mid m\,q\,M)Y_L^m(\hat{\mathbf{r}})\mathbf{e}_q$$

has the same orbital degree L. Therefore

$$\mathbf{Y}_{JLM}(-\hat{\mathbf{r}})=(-1)^L\mathbf{Y}_{JLM}(\hat{\mathbf{r}}).$$

Applying this to the three orbital degrees $L+1, L, L-1$ gives

$$\boxed{\begin{aligned}\mathbf{Y}_{L,L+1,M}(-\hat{\mathbf{r}}) &= (-1)^{L+1}\mathbf{Y}_{L,L+1,M},\\ \mathbf{Y}_{L,L,M}(-\hat{\mathbf{r}}) &= (-1)^{L}\mathbf{Y}_{L,L,M},\\ \mathbf{Y}_{L,L-1,M}(-\hat{\mathbf{r}}) &= (-1)^{L+1}\mathbf{Y}_{L,L-1,M}.\end{aligned}}$$

The last case is present only when $L \geq 1$. The M dependence disappears because the two scalar parity factors contain $2m$, an even integer.

This is the argument-parity convention in the question: the Cartesian component functions are evaluated at $-\hat{\mathbf{r}}$. If one instead defines parity on a physical polar-vector field by $(\mathcal{P}\mathbf{F})(\mathbf{r}) = -\mathbf{F}(-\mathbf{r})$, that operation includes an additional intrinsic minus sign. In the moving spherical basis the same calculation distributes the argument parity among the components and the basis vectors: $\hat{\mathbf{r}}, \hat{\boldsymbol{\varphi}}$ are odd and $\hat{\boldsymbol{\theta}}$ is even, as stated in the hint.

16.4.3 Verify the orthonormality of the vector spherical harmonics $\mathbf{Y}_{JLM_J}$ under the Hermitian inner product $\int \mathbf{Y}^* \cdot \mathbf{Y}'\, d\Omega$, for allowed indices.

Solution.

The textbook's Eq. (16.89) omits the complex conjugate on one vector harmonic. Orthonormality uses the Hermitian product $\int \mathbf{Y}^* \cdot \mathbf{Y}'\, d\Omega$. An unconjugated product cannot express it: already $\mathbf{e}_+ \cdot \mathbf{e}_+ = 0$, whereas $\mathbf{e}_+^* \cdot \mathbf{e}_+ = 1$. The spherical Cartesian basis obeys $\mathbf{e}_q^* \cdot \mathbf{e}_{q'} = \delta_{qq'}$. Write $C_{mq}^{JM} = C(L\,1\,J \mid m\,q\,M)$. Expanding both vector harmonics before integrating gives

$$\begin{aligned}&\int \mathbf{Y}^*_{JLM} \cdot \mathbf{Y}_{J'L'M'}\, d\Omega\\ &\quad= \sum_{m,q}\sum_{m',q'} (C_{mq}^{JM})^* C(L'\,1\,J' \mid m'\,q'\,M')(\mathbf{e}_q^* \cdot \mathbf{e}_{q'}) \int Y_L^{m*} Y_{L'}^{m'}\, d\Omega\\ &\quad= \delta_{LL'} \sum_{m,q} C(L\,1\,J \mid m\,q\,M)^* C(L\,1\,J' \mid m\,q\,M').\end{aligned}$$

The remaining sum is the inner product of two normalized coupled states expressed in the orthonormal product basis. The Clebsch–Gordan change of basis is unitary, so the sum is $\delta_{JJ'}\delta_{MM'}$. Therefore, for all allowed indices,

$$\boxed{\int \mathbf{Y}^*_{JLM} \cdot \mathbf{Y}_{J'L'M'}\, d\Omega = \delta_{JJ'}\delta_{LL'}\delta_{MM'}}.$$

This does not assign norm one to the excluded formal zero $\mathbf{Y}_{000}$.

16.4.4 Jackson's *Classical Electrodynamics* (see Additional Readings) defines $\mathbf{Y}_{LLM}$ by the equation

$$\mathbf{Y}_{LLM}(\theta,\varphi) = \frac{1}{\sqrt{L(L+1)}}\,\mathbf{L}\,Y_L^M(\theta,\varphi),$$

in which the angular momentum operator $\mathbf{L}$ is given by

$$\mathbf{L} = -i(\mathbf{r} \times \nabla).$$

Show that this definition agrees with Eq. (16.88) for $L \geq 1$ and $|M| \leq L$.

Solution.

The displayed definition requires $L \geq 1$. At $L = 0$ its denominator vanishes, and the formal $\mathbf{Y}_{000} = 0$ is not a normalized member of this family. In the spherical Cartesian basis of 16.4.1, express the vector operator in terms of $L_\pm = L_x \pm iL_y$:

$$\mathbf{L} = \mathbf{e}_x L_x + \mathbf{e}_y L_y + \mathbf{e}_0 L_z = \frac{\mathbf{e}_- L_+ - \mathbf{e}_+ L_-}{\sqrt{2}} + \mathbf{e}_0 L_z.$$

Applying the scalar ladder formulas gives

$$\begin{aligned}\frac{\mathbf{L}Y_L^M}{\sqrt{L(L+1)}} = &\sqrt{\frac{(L-M)(L+M+1)}{2L(L+1)}}\, Y_L^{M+1}\mathbf{e}_- \\ &+ \frac{M}{\sqrt{L(L+1)}} Y_L^M \mathbf{e}_0 \\ &- \sqrt{\frac{(L+M)(L-M+1)}{2L(L+1)}}\, Y_L^{M-1}\mathbf{e}_+.\end{aligned}$$

These are the three Clebsch–Gordan coefficients for coupling orbital L and vector spin 1 to total $J = L$.

One may establish their phase and total J directly. At $M = L$ the two nonzero coefficients are $-1/\sqrt{L+1}$ on $(m, q) = (L-1, 1)$ and $\sqrt{L/(L+1)}$ on $(L, 0)$. Their norm squared is one. Raising gives the coefficient

$$-\frac{\sqrt{2L}}{\sqrt{L+1}} + \sqrt{\frac{L}{L+1}}\sqrt{2} = 0$$

on $(L, 1)$. Thus this is a highest state of total $J = L$, with positive coefficient at the largest orbital m.

The subsequent lowering uses the vector transformation law. On any scalar f, the total lowering generator acting on $\mathbf{L}f$ satisfies

$$J_-(\mathbf{L}f) = \mathbf{L}(L_- f).$$

To exhibit the cancellation, the Cartesian spin-one matrices are $(S_i)_{jk} = -i\epsilon_{ijk}$, whereas $[L_i, L_j] = i\epsilon_{ijk}L_k$. Therefore, for each Cartesian i,

$$J_i(\mathbf{L}f) - \mathbf{L}(L_i f) = \sum_{j,k} \mathbf{e}_j\, i\epsilon_{ijk} L_k f + \sum_{j,k} \mathbf{e}_j(-i\epsilon_{ijk}) L_k f = 0.$$

Taking the $x - iy$ combination proves the stated lowering identity. It follows that lowering the displayed M expression gives $\sqrt{(L+M)(L-M+1)}$ times the expression at $M-1$. The normalized highest state and these positive lowering factors uniquely generate the $J = L$ multiplet. Thus its coefficients are exactly those of Eq. (16.88), and

$$\boxed{\mathbf{Y}_{LLM} = \frac{\mathbf{L}Y_L^M}{\sqrt{L(L+1)}}}.$$

16.4.5 For $L \geq 1$, show that

$$\sum_{M=-L}^{L} \mathbf{Y}^*_{LLM}(\theta,\varphi) \cdot \mathbf{Y}_{LLM}(\theta,\varphi) = \frac{2L+1}{4\pi}.$$

Hint. One way is to use Exercise 16.4.4 with $\mathbf{L}$ expanded in Cartesian coordinates and to apply raising and lowering operators.

Solution.

This identity holds for $L \geq 1$. The book's formal $\mathbf{Y}_{000} = 0$ would give zero on the left for $L = 0$, whereas the stated right side would be $1/(4\pi)$. Use 16.4.4. Since Cartesian basis vectors are orthonormal,

$$\begin{aligned}\sum_M \mathbf{Y}^*_{LLM} \cdot \mathbf{Y}_{LLM} &= \frac{1}{L(L+1)} \sum_M \left(|L_x Y_L^M|^2 + |L_y Y_L^M|^2 + |L_z Y_L^M|^2\right) \\ &= \frac{1}{L(L+1)} \sum_M \left(\frac{|L_+ Y_L^M|^2 + |L_- Y_L^M|^2}{2} + M^2 |Y_L^M|^2 \right).\end{aligned}$$

The second equality follows by expanding $L_x = (L_+ + L_-)/2$ and $L_y = (L_+ - L_-)/(2i)$: their cross terms cancel. Now collect the coefficient of a fixed $|Y_L^k|^2$. The raising term comes from $M = k-1$ and the lowering term from $M = k+1$. Including the L_z term, the coefficient is

$$\begin{aligned}&\frac{1}{2}(L-k+1)(L+k) + \frac{1}{2}(L+k+1)(L-k) + k^2 \\ &\quad = \frac{1}{2}[L(L+1) - k^2 + k] + \frac{1}{2}[L(L+1) - k^2 - k] + k^2 \\ &\quad = L(L+1).\end{aligned}$$

At $k = \pm L$, the out-of-range source term has coefficient zero, so the same collection is valid at both endpoints. Hence

$$\sum_M \mathbf{Y}^*_{LLM} \cdot \mathbf{Y}_{LLM} = \sum_{k=-L}^{L} |Y_L^k|^2 = \boxed{\frac{2L+1}{4\pi}},$$

where the last step is the scalar addition theorem at coincident directions.

16.4.6 For $L \geq 1$ and $|M| \leq L$, show that

$$\int \mathbf{Y}^*_{LLM} \cdot \left(\hat{\mathbf{r}} \times \mathbf{Y}_{LLM}\right) d\Omega = 0.$$

The integrand represents an interference term in electromagnetic radiation that contributes to angular distributions but not to total intensity.

Solution.

The original, unconjugated version has a stronger but different property: its integrand is identically zero. For any vector $\mathbf{A}$, including one with complex components, the bilinear scalar triple product is

$$\mathbf{A} \cdot (\hat{\mathbf{r}} \times \mathbf{A}) = \sum_{ijk} \epsilon_{ijk} A_i \hat{r}_j A_k = 0.$$

Interchanging i and k changes ϵ_{ijk} to its negative but leaves the commuting product A_iA_k unchanged. The sum therefore equals its own negative and is zero at every point. Taking $\mathbf{A} = \mathbf{Y}_{LLM}$ proves the original integral.

The source has no complex conjugate in this exercise. Accordingly its following statement about a nonzero angular interference term cannot describe the integrand as printed. The physically relevant Hermitian version, with the first $\mathbf{Y}_{LLM}$ conjugated, can be nonzero pointwise; its angular integral also vanishes, as shown next.

For $L \geq 1$ put $N = \sqrt{L(L+1)}$ and $Y = Y_L^M$. Using $\mathbf{L} = -i\hat{\boldsymbol{\varphi}}\,\partial_\theta + i\hat{\boldsymbol{\theta}}(\sin\theta)^{-1}\partial_\varphi$ and $\partial_\varphi Y = iMY$ gives

$$A_\theta = -\frac{M}{N\sin\theta}Y, \qquad A_\varphi = -\frac{i}{N}\partial_\theta Y, \qquad A_r = 0.$$

Since $\hat{\mathbf{r}} \times \hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\varphi}}$ and $\hat{\mathbf{r}} \times \hat{\boldsymbol{\varphi}} = -\hat{\boldsymbol{\theta}}$,

$$\begin{aligned}
\mathbf{A}^* \cdot (\hat{\mathbf{r}} \times \mathbf{A}) &= -A_\theta^* A_\varphi + A_\varphi^* A_\theta \\
&= -\frac{iM}{L(L+1)\sin\theta}\left[Y^*\partial_\theta Y + (\partial_\theta Y^*)Y\right] \\
&= -\frac{iM}{L(L+1)\sin\theta}\partial_\theta |Y|^2.
\end{aligned}$$

The modulus is independent of φ, so integration gives

$$\int \mathbf{A}^* \cdot (\hat{\mathbf{r}} \times \mathbf{A})\, d\Omega = -\frac{2\pi i M}{L(L+1)}\left[|Y_L^M(\theta,0)|^2\right]_{\theta=0}^{\theta=\pi} = 0.$$

For $M = 0$ the prefactor is zero; for $M \neq 0$ the spherical harmonic vanishes at both poles. For example $L = M = 1$ gives $|Y_1^1|^2 = 3\sin^2\theta/(8\pi)$ and hence

$$-\frac{i}{2\sin\theta}\frac{d}{d\theta}\left(\frac{3\sin^2\theta}{8\pi}\right) = -\frac{3i}{8\pi}\cos\theta,$$

which is nonzero away from the equator but integrates to zero. Thus the corrected Hermitian interpretation has the angular behavior described in the prose.

17 Chapter 17. Group Theory

Problems

17.1 Introduction to Group Theory

17.1.1 The Vierergruppe (German: four-membered group) is a group different from the C_4 group introduced in Example 17.1.4. The Vierergruppe has the multiplication table shown in Table 17.2. Determine whether this group is cyclic and whether it is abelian.

Solution.

The diagonal of Table 17.2 gives

$$I^2 = A^2 = B^2 = C^2 = I.$$

Thus the powers of a nonidentity element g alternate $I, g, I, g, \ldots$, generating only two elements. The powers of I generate only I. No element generates all four, so the group is $\boxed{\text{not cyclic}}$.

The table is symmetric across its diagonal. More explicitly,

$$AB = BA = C, \qquad BC = CB = A, \qquad CA = AC = B,$$

and products with I commute automatically. These exhaust all pairs of distinct elements; an element commutes with itself. Hence the Vierergruppe is $\boxed{\text{abelian}}$. It is isomorphic to $C_2 \times C_2$, for example under $I \mapsto (0,0)$, $A \mapsto (1,0)$, $B \mapsto (0,1)$, $C \mapsto (1,1)$ with componentwise addition modulo two.

17.1.2

(a) Show that the permutations of n distinct objects satisfy the group postulates.

(b) Construct the multiplication table for the permutations of three objects, giving each permutation a name of some sort. (Suggestion: Use I for the permutation that leaves the order unchanged.)

(c) Show that this permutation group (named S_3) is isomorphic with D_3 and identify corresponding operations. Is your identification unique?

Solution.

(a) A permutation is a bijection of $\{1, \ldots, n\}$ to itself; multiplication means composition, with the right-hand permutation acting first. The composition of two bijections is a bijection, proving closure. For every label j, $[(pq)r](j) = p(q(r(j))) = [p(qr)](j)$, proving associativity. The identity is $I(j) = j$. For each p, every target k has a unique preimage; define $p^{-1}(k)$ to be that preimage. Then $pp^{-1} = p^{-1}p = I$ and p^{-1} is another permutation. All four postulates hold.

(b) For three labels use cycle notation

$$r = (123), \qquad s = (23), \qquad r^2 = (132), \quad sr = (13), \quad sr^2 = (12).$$

Here (123) moves label 1 to 2, 2 to 3, and 3 to 1. The elementary relations are $r^3 = s^2 = I$ and $srs = r^{-1}$. They reduce any product to one of the six listed elements. For example, $r(sr) = rsr = s$

because $rs = sr^{-1}$. The complete table (row times column) is

	I	r	r^2	s	sr	sr^2
I	I	r	r^2	s	sr	sr^2
r	r	r^2	I	sr^2	s	sr
r^2	r^2	I	r	sr	sr^2	s
s	s	sr	sr^2	I	r	r^2
sr	sr	sr^2	s	r^2	I	r
sr^2	sr^2	s	sr	r	r^2	I

.

(c) Number the triangle vertices as in Fig. 17.1. A counterclockwise 120° rotation has vertex action (123); the 180° rotation about the axis through vertex 1 fixes that vertex and interchanges 2 and 3. Hence an isomorphism is

$$\boxed{I \leftrightarrow I, \quad C_3 \leftrightarrow r, \quad C_3^2 \leftrightarrow r^2, \quad C_2 \leftrightarrow s, \quad C_2' \leftrightarrow sr, \quad C_2'' \leftrightarrow sr^2}.$$

The table just constructed is Table 17.1 with these substitutions, so every product is preserved. Equivalently the geometric action on the three vertices is injective, and both groups have six elements. The identification is not unique: C_3 can be assigned either 3-cycle and C_2 any of the three transpositions; the relations then determine the other images. These $2 \cdot 3 = 6$ choices correspond to the possible vertex relabelings.

17.1.3 *Rearrangement theorem*: Given a group of distinct elements $(I, a, b, \ldots, n)$, show that the set of products $(aI, a^2, ab, ac, \ldots, an)$ reproduces all the group elements in a new order.

Solution.

Fix $a \in \mathcal{G}$ and consider $L_a(g) = ag$. Closure ensures its values lie in $\mathcal{G}$. If $L_a(b) = L_a(c)$, then

$$ab = ac \quad \Longrightarrow \quad a^{-1}(ab) = a^{-1}(ac) \quad \Longrightarrow \quad b = c.$$

Thus no two distinct elements yield the same product. Furthermore, for every $h \in \mathcal{G}$, the element $g = a^{-1}h$ lies in $\mathcal{G}$ and satisfies $L_a(g) = a(a^{-1}h) = h$. Therefore L_a is also onto. It is a bijection with inverse $L_{a^{-1}}$. The list $(aI, a^2, ab, \ldots)$ consequently contains each group element exactly once. This proof works for infinite groups as well as for the finite list in the question.

17.1.4 A group $\mathcal{G}$ has a subgroup $\mathcal{H}$ with elements h_i. Let x be a fixed element of the original group $\mathcal{G}$ and not a member of $\mathcal{H}$. The transform

$$xh_ix^{-1}, \qquad i = 1, 2, \ldots$$

generates a conjugate subgroup $x\mathcal{H}x^{-1}$. Show that this conjugate subgroup satisfies each of the four group postulates and therefore is a group.

Solution.

Let $\mathcal{K} = x\mathcal{H}x^{-1}$. For $h, k \in \mathcal{H}$,

$$(xhx^{-1})(xkx^{-1}) = xh(x^{-1}x)kx^{-1} = x(hk)x^{-1} \in \mathcal{K},$$

because $hk \in \mathcal{H}$. This proves closure. Associativity is inherited from multiplication in $\mathcal{G}$; explicitly both associations of three elements reduce to $x(hk\ell)x^{-1}$.

The identity is $xIx^{-1} = I$, which belongs to $\mathcal{K}$. The inverse of xhx^{-1} is $xh^{-1}x^{-1} \in \mathcal{K}$:

$$(xhx^{-1})(xh^{-1}x^{-1}) = xhh^{-1}x^{-1} = I,$$

and the reverse product is also I. Thus $\mathcal{K}$ satisfies all four group postulates.

In fact $h \mapsto xhx^{-1}$ is a group isomorphism from $\mathcal{H}$ to $\mathcal{K}$: the product formula proves the homomorphism property, and conjugation by x^{-1} is its inverse. The condition $x \notin \mathcal{H}$ is unnecessary for this conclusion; it only makes a potentially different conjugate subgroup of interest.

17.1.5

(a) A particular group is abelian. A second group is created by replacing g_i by g_i^{-1} for each element in the original group. Show that the two groups are isomorphic.

Note. This means showing that if $ab = c$, then $a^{-1}b^{-1} = c^{-1}$.

(b) Continuing part (a), show that the second group is also abelian.

Solution.

(a) First recall the inverse-of-product identity:

$$(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = I, \qquad (b^{-1}a^{-1})(ab) = I.$$

Hence $(ab)^{-1} = b^{-1}a^{-1}$. In an abelian group a^{-1} and b^{-1} commute, so

$$f(ab) = (ab)^{-1} = b^{-1}a^{-1} = a^{-1}b^{-1} = f(a)f(b), \qquad f(g) = g^{-1}.$$

Also $f(f(g)) = g$, making f bijective. It is therefore an isomorphism.

(b) Every pair in the set of inverses has the form a^{-1}, b^{-1}, and

$$a^{-1}b^{-1} = (ab)^{-1} = (ba)^{-1} = b^{-1}a^{-1}.$$

Thus the second group is abelian. In fact it is the same set $\mathcal{G}$, because inversion permutes a group's elements. The noteworthy conclusion is that inversion is an automorphism when the group is abelian; in a general group it reverses the multiplication order instead.

17.1.6 Consider a cubic crystal consisting of identical atoms at $\mathbf{r} = (la, ma, na)$, with l, m, and n taking on all integral values.

(a) Show that each Cartesian axis is a fourfold symmetry axis.

(b) The cubic point group will consist of all operations (rotations, reflections, inversion) that leave the simple cubic crystal invariant and that do not move the atom at $l = m = n = 0$. From a consideration of the permutation of the positive and negative coordinate axes, predict how many elements this cubic group will contain.

Solution.

(a) A 90° counterclockwise rotation about z takes

$$(la, ma, na) \longmapsto (-ma, la, na).$$

The new coordinates again have integer coefficients. Repeating the rotation gives $(-la, -ma, na)$, then $(ma, -la, na)$, then the original point. The same argument applies cyclically to x and y. These

axes have exactly fourfold rotational symmetry: the four nearest lattice points in their perpendicular coordinate plane are at angular separations 90°, so a smaller positive rotation cannot permute them.

(b) A point symmetry fixing the origin preserves distances, so it permutes the six nearest lattice points $\{\pm a\mathbf{e}_x, \pm a\mathbf{e}_y, \pm a\mathbf{e}_z\}$. The image of $\mathbf{e}_x$ has six choices. Orthogonality leaves four choices for the image of $\mathbf{e}_y$, and then two for $\mathbf{e}_z$. Thus there are

$$\boxed{6 \cdot 4 \cdot 2 = 48}$$

possible signed permutation matrices. Every one really is a symmetry: it merely permutes the integers (l, m, n) and changes some signs. No other point symmetry is possible because the images of three orthonormal coordinate vectors determine the entire orthogonal transformation. Exactly 24 have determinant $+1$ and are rotations; multiplying by inversion pairs these with the 24 determinant--1 operations. The full point group is commonly denoted O_h.

17.1.7 A plane is covered with regular hexagons, as shown in Fig. 17.3.

(a) Determine the rotational symmetry of an axis perpendicular to the plane through the common vertex of three hexagons (A). That is, if the axis has n-fold symmetry, show (with careful explanation) what n is.

(b) Repeat part (a) for an axis perpendicular to the plane through the geometric center of one hexagon (B).

(c) Find all the different kinds of axes within the plane of hexagons about which a 180° rotation is a symmetry element (this corresponds to turning the plane over by rotation about that axis).

Solution.

(a) At a vertex A, three edges leave at angular separations 120°. A rotation preserving the tiling must permute these three rays, so its angle is a multiple of 120°. Rotation through 120° indeed permutes the three incident hexagons and extends to the whole regular tiling. One way to see the extension is that reflection or rotation takes each regular hexagon to the neighboring hexagon determined by its edge; continuing across edges determines all tiles. Thus the axis through A is $\boxed{\text{threefold}}$.

(b) At a center B, the six vertices are equally spaced. A 60° rotation permutes the hexagon and its six neighboring tiles, and hence the entire tiling by the same edge-continuation argument. Any symmetry fixing B must permute its six nearest vertices, excluding a smaller positive angle. The axis through B is $\boxed{\text{sixfold}}$.

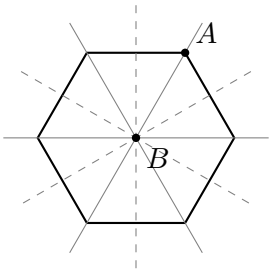
(c) A 180° rotation about a line in the plane acts, on that plane, as reflection in the line. The two kinds are

$$\boxed{\begin{array}{l}\text{lines through opposite vertices of a hexagon,}\\ \text{lines through midpoints of opposite edges of a hexagon.}\end{array}}$$

There are three directions of each kind through a given center. Their lattice translates give all the corresponding axes throughout the plane; an axis of the first kind also contains some tile edges.

To check completeness, take side length one and a hexagon with vertices $(\pm 1, 0)$, $(\pm 1/2, \pm\sqrt{3}/2)$. Its centers form the triangular lattice generated by $(3/2, \sqrt{3}/2)$ and $(0, \sqrt{3})$. Reflection must permute the six shortest center-to-center vectors. This restricts its direction to the six directions spaced by 30°. For a horizontal axis $y = d$, reflection of the center at the origin requires $(0, 2d)$ to be a center, so $d = k\sqrt{3}/2$. Each such line contains some centers and is of the opposite-vertex type. For

a vertical axis $x = d$, the corresponding condition is $(2d, 0)$ in the center lattice, which requires $d = 3k/2$. These are the opposite-edge-midpoint axes. Rotations by 60° produce the remaining directions. No additional offsets or directions are possible.



Solid lines join opposite vertices; dashed lines join opposite edge midpoints. Both sets are reflection axes.

17.2 Representation of Groups

17.2.1 For any representation K of a group, and for any group element a, show that

$$\left[U^K(a)\right]^{-1} = U^K\left(a^{-1}\right).$$

Solution.

A matrix representation maps the identity to the identity matrix and preserves products. Hence

$$U^K(a)U^K(a^{-1}) = U^K(aa^{-1}) = U^K(I) = I,$$

and also

$$U^K(a^{-1})U^K(a) = U^K(a^{-1}a) = I.$$

Thus $U^K(a^{-1})$ is both a right and left inverse of $U^K(a)$. By uniqueness of the matrix inverse,

$$\boxed{[U^K(a)]^{-1} = U^K(a^{-1})}.$$

No assumption of a faithful representation or of unitarity is needed.

17.2.2 Show that these four matrices form a representation of the Vierergruppe, whose multiplication table is in Table 17.2.

$$\mathsf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathsf{A} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathsf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \mathsf{C} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.$$

Solution.

The given matrices obey $A = -I$ and $C = -B$. Direct multiplication gives

$$B^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^2 = I,$$

so $A^2 = (-I)^2 = I$ and $C^2 = (-B)^2 = I$ as well. The products of distinct nonidentity matrices are

$$AB = BA = -B = C, \qquad AC = CA = B, \qquad BC = CB = -B^2 = -I = A.$$

Multiplication by I leaves every matrix unchanged. These statements give all 16 products:

	I	A	B	C
I	I	A	B	C
A	A	I	C	B
B	B	C	I	A
C	C	B	A	I

.

The table is exactly Table 17.2; therefore the four matrices form a representation of the Vierergruppe. Since all four are different, this particular representation is faithful.

17.2.3 Show that the matrices I, A, B, and C of Exercise 17.2.2 are reducible. Reduce them.

Note. This means transforming B and C to diagonal form (by the same unitary transformation).

Solution.

The eigenvectors of B are

$$v_+ = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, \quad Bv_+ = v_+, \qquad v_- = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}, \quad Bv_- = -v_-.$$

Let $Q = (v_+\ v_-) = 2^{-1/2}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}$. Its columns are orthonormal, so $Q^\dagger Q = I$. Because $C = -B$, the same eigenvectors diagonalize C:

$$Q^\dagger B Q = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}, \qquad Q^\dagger C Q = \begin{pmatrix}-1 & 0\\ 0 & 1\end{pmatrix}.$$

The scalar matrices $I, A = -I$ are unchanged. Thus the transformed representation is a direct sum of two one-dimensional representations, with characters in the order (I, A, B, C)

$$\boxed{(1,-1,1,-1) \quad \text{and} \quad (1,-1,-1,1)}.$$

Each line span$\{v_\pm\}$ is invariant under every group element, proving reducibility and supplying the reduction.

17.2.4

(a) Once you have a matrix representation of any group, a 1-D representation can be obtained by taking the determinants of the matrices. Show that the multiplicative relations are preserved in this determinant representation.

(b) Use determinants to obtain a 1-D representation of D_3 from the 2×2 representation in Eq. (17.2).

Solution.

(a) If $U(ab) = U(a)U(b)$, the determinant product theorem gives

$$\chi_{\det}(ab) = \det U(ab) = \det U(a)\det U(b) = \chi_{\det}(a)\chi_{\det}(b).$$

Also $\chi_{\det}(I) = 1$ and $\chi_{\det}(a^{-1}) = 1/\chi_{\det}(a)$. Every $U(a)$ is invertible, so no determinant is zero. The determinants therefore define a one-dimensional representation.

(b) In Eq. (17.2), the matrices of C_3 and C_3^2 have determinant $(-1/2)^2 + (\sqrt{3}/2)^2 = 1$. The C_2 matrix has determinant -1; either other twofold matrix has determinant $(-1/2)(1/2) - (\sqrt{3}/2)^2 = -1$. Thus

g	I	C_3	C_3^2	C_2	C_2'	C_2''
$\chi_{\det}(g)$	1	1	1	-1	-1	-1

This is the A_2 representation of D_3. Products of two flips have determinant $+1$, products of a flip with a rotation have determinant -1, and products of rotations have determinant $+1$, as required by the group table.

17.2.5 Show that the cyclic group of n objects, C_n, may be represented by r^m, $m = 0, 1, 2, \ldots, n-1$. Here r is the image of an abstract generator, given by

$$r = \exp(2\pi i s/n).$$

The parameter s takes on the values $s = 1, 2, 3, \ldots, n$, with each value of s yielding a different 1-D (irreducible) representation of C_n.

Take $n \geq 1$ and complex representations; the scalar image need not have order n.

Solution.

Let g be the abstract generator of C_n, so $g^n = I$. For each integer $s \in \{1, \ldots, n\}$ set

$$r_s = e^{2\pi i s/n}, \qquad U_s(g^m) = r_s^m.$$

Because $r_s^n = e^{2\pi i s} = 1$, reducing m modulo n does not change this value. For arbitrary integers j, k,

$$U_s(g^j g^k) = r_s^{j+k} = r_s^j r_s^k = U_s(g^j) U_s(g^k).$$

Thus this is a representation; it is irreducible since its space is one-dimensional.

The representations are distinct: equivalence in one dimension cannot change the scalar $U_s(g)$, and the n numbers $e^{2\pi i s/n}$ are distinct for these s. They are all the irreducible representations. Indeed any complex representation matrix of g satisfies $U(g)^n = I$, so its eigenvalues are nth roots of unity; each eigenline is invariant under every power of g. An irreducible representation must therefore have dimension one.

The scalar r_s is the image of a generator, not necessarily an element of order n. Its order is $n/\gcd(s, n)$. The representation is faithful only when $\gcd(s, n) = 1$; $s = n$ gives the trivial representation. To check the order, $r_s^k = 1$ is equivalent to n dividing sk; the smallest positive such k is $n/\gcd(s, n)$.

17.2.6 Develop the irreducible 2×2 matrix representation of the group of rotations (including those that turn it over) that transform a square into itself. Give the group multiplication table.

Note. This group has the name D_4 (see Fig. 17.5).

Solution.

Take r to be a counterclockwise 90° rotation and s a 180° turn about the x axis in the plane of the square. Their action on in-plane coordinates is

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Thus $R^4 = S^2 = I$ and $SRS = R^{-1}$. The eight matrices are

g	I	r	r^2	r^3
$U(g)$	$\begin{pmatrix}1&0\\0&1\end{pmatrix}$	$\begin{pmatrix}0&-1\\1&0\end{pmatrix}$	$\begin{pmatrix}-1&0\\0&-1\end{pmatrix}$	$\begin{pmatrix}0&1\\-1&0\end{pmatrix}$

,

g	s	rs	r^2s	r^3s
$U(g)$	$\begin{pmatrix}1&0\\0&-1\end{pmatrix}$	$\begin{pmatrix}0&1\\1&0\end{pmatrix}$	$\begin{pmatrix}-1&0\\0&1\end{pmatrix}$	$\begin{pmatrix}0&-1\\-1&0\end{pmatrix}$

.

The first four preserve the displayed side of the square; the last four turn it over. All are real orthogonal.

Every element is $r^i s^\epsilon$, $0 \le i < 4$, $\epsilon = 0, 1$. Since $sr^j = r^{-j}s$, the product rule is

$$(r^i s^\epsilon)(r^j s^\eta) = r^{\,i+(-1)^\epsilon j} s^{\,\epsilon+\eta},$$

with exponents reduced modulo four and two. This gives the full table:

	I	r	r^2	r^3	s	rs	r^2s	r^3s
I	I	r	r^2	r^3	s	rs	r^2s	r^3s
r	r	r^2	r^3	I	rs	r^2s	r^3s	s
r^2	r^2	r^3	I	r	r^2s	r^3s	s	rs
r^3	r^3	I	r	r^2	r^3s	s	rs	r^2s
s	s	r^3s	r^2s	rs	I	r^3	r^2	r
rs	rs	s	r^3s	r^2s	r	I	r^3	r^2
r^2s	r^2s	rs	s	r^3s	r^2	r	I	r^3
r^3s	r^3s	r^2s	rs	s	r^3	r^2	r	I

.

To prove irreducibility even over $\mathbb{C}$, a proper nonzero invariant subspace would be a line. It would have to be an eigenline of R, whose distinct eigenvalues are $i, -i$ with eigenvectors $(1, -i)^T$ and $(1, i)^T$. But S interchanges these lines. Neither is invariant under the whole group, so the two-dimensional representation is irreducible.

17.3 Symmetry and Physics

17.3.1 Consider a quantum mechanics problem with D_3 symmetry, with the threefold symmetry axis taken as the z direction, and with orbitals $\varphi(\mathbf{r} - \mathbf{R}_j)$ located at the vertices of an equilateral triangle. This is the same system geometry as in Example 17.3.2, but in the present problem φ will no longer be chosen to have spherical symmetry.

Given that $\varphi(\mathbf{r}) = (z/r)f(r)$ (so φ has the symmetry of a p orbital oriented along the symmetry axis), construct linear combinations of the φ that are bases for irreducible representations of D_3, for each basis indicating its representation.

Solution.

Write $\varphi_j(\mathbf{r}) = \varphi(\mathbf{r} - \mathbf{R}_j)$, with the vertices numbered cyclically. Use active operations: $r = C_3$ sends $\varphi_1 \to \varphi_2 \to \varphi_3 \to \varphi_1$. A twofold rotation $s = C_2$ about the in-plane axis through vertex 1 also reverses z. Since each local orbital is proportional to z, its transformation is

$$s\varphi_1 = -\varphi_1, \qquad s\varphi_2 = -\varphi_3, \qquad s\varphi_3 = -\varphi_2.$$

Thus the sum

$$\boxed{\psi_{A_2} = \frac{\varphi_1 + \varphi_2 + \varphi_3}{\sqrt{3}}}$$

is unchanged by r and changes sign under s. Its characters on $(I, 2C_3, 3C_2)$ are $(1, 1, -1)$, identifying A_2.

Choose two independent zero-sum combinations

$$u = \frac{2\varphi_1 - \varphi_2 - \varphi_3}{\sqrt{6}}, \qquad v = \frac{\varphi_2 - \varphi_3}{\sqrt{2}}.$$

For example

$$ru = \frac{-\varphi_1 + 2\varphi_2 - \varphi_3}{\sqrt{6}} = -\frac{1}{2}u + \frac{\sqrt{3}}{2}v,$$

and the same substitution gives

$$rv = -\frac{\sqrt{3}}{2}u - \frac{1}{2}v, \qquad su = -u, \qquad sv = v.$$

Hence the coefficient matrices in the ordered basis (u, v) are

$$D_E(r) = \begin{pmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{pmatrix}, \qquad D_E(s) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Their traces are -1 and 0, so the characters are $(2, -1, 0)$. The rotation eigenlines are interchanged by the flip, making this two-dimensional space irreducible. Therefore

$$\boxed{\text{span}\{\varphi_1, \varphi_2, \varphi_3\} = A_2 \oplus E, \qquad E \text{ has basis } (u, v)}.$$

The manual's $-I_2$ for a twofold operation in this E space has the wrong trace, -2. Every such operation has eigenvalues $+1, -1$ and trace zero, as the direct transformations show. The square-root factors normalize the coefficient vectors. If spatially overlapping orbitals are used, their actual Hilbert-space norms must be obtained from their overlap integrals; the symmetry assignments do not depend on those common scale choices.

17.4 Discrete Groups

17.4.1 The Vierergruppe has the multiplication table shown in Table 17.2.

(a) Divide its elements into classes.

(b) Using the class information, determine for the Vierergruppe its number of inequivalent irreducible representations and their dimensions.

(c) Construct a character table for the Vierergruppe.

Solution.

(a) The group is abelian, so $gag^{-1} = agg^{-1} = a$ for every a, g. Its conjugacy classes are therefore

$$\boxed{\{I\}, \quad \{A\}, \quad \{B\}, \quad \{C\}}.$$

(b) The number of inequivalent complex irreducible representations equals the number of classes, hence is four. Their positive integer dimensions satisfy $d_1^2 + d_2^2 + d_3^2 + d_4^2 = |\mathcal{G}| = 4$. Every term is at least one, so each $d_j = 1$.

(c) In a one-dimensional representation set $\chi(A) = a$, $\chi(B) = b$. Since $A^2 = B^2 = I$, both a, b are ± 1. The relation $C = AB$ fixes $\chi(C) = ab$. The four independent choices of a, b give

	I	A	B	C
A_1	1	1	1	1
A_2	1	1	−1	−1
A_3	1	−1	1	−1
A_4	1	−1	−1	1

.

Each row obeys the multiplication table. Its squared norm is four, and two distinct rows have two equal and two opposite entries, giving inner product zero. Thus the table also satisfies character orthogonality and contains all four irreducibles.

17.4.2 The group D_3 may be discussed as a permutation group of three objects. Operation C_3, for instance, moves vertex 1 to the position formerly occupied by vertex 3; likewise vertex 2 moves to the original position of vertex 1 and vertex 3 moves to the original position of vertex 2. So this shuffling could be described as the permutation of $(1, 2, 3)$ to $(2, 3, 1)$. Using now letters a, b, c to avoid notational confusion, this permutation $(abc) \to (bca)$ corresponds to the matrix equation

$$C_3 \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} b \\ c \\ a \end{pmatrix},$$

thereby identifying a 3×3 representation of the operation C_3.

(a) Develop analogous 3×3 representations for the other elements of D_3.

(b) Reduce your 3×3 representation to the direct sum of a 1×1 and a 2×2 representation. Note: This 3×3 representation must be reducible or Eq. (17.10) would be violated.

Here the name C_3 follows the displayed label-column matrix. The vertex-motion description has been reversed to agree with it; the original convention mismatch is explained below.

Solution.

The original prose used an active vertex map $1 \to 2 \to 3 \to 1$, which sends a column of labels at fixed positions from (a, b, c) to (c, a, b). The displayed matrix instead gives (b, c, a), the inverse reordering. We retain the displayed matrix for this exercise; using the active convention would interchange the labels C_3 and C_3^2.

(a) Put

$$R = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \qquad S = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Here S interchanges the second and third entries. Direct multiplication gives $R^3 = S^2 = I$ and $SRS = R^{-1}$, so R, S generate a representation of D_3. The six matrices, named consistently with

the multiplication table $C_2C_3 = C_2'$, are

$$U(I) = I_3, \quad U(C_3) = R, \qquad\qquad U(C_3^2) = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

$$U(C_2) = S, \quad U(C_2') = SR = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad U(C_2'') = SR^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

(b) All permutations preserve the sum of the three components. Consequently

$$e_0 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

is fixed, and its orthogonal complement is invariant. Choose an orthonormal basis there:

$$e_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}, \qquad e_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \qquad Q = (e_0\ e_1\ e_2).$$

For example

$$Re_1 = -\tfrac{1}{2}e_1 - \tfrac{\sqrt{3}}{2}e_2, \qquad Re_2 = \tfrac{\sqrt{3}}{2}e_1 - \tfrac{1}{2}e_2, \qquad Se_1 = e_1, \quad Se_2 = -e_2.$$

Thus

$$Q^\dagger RQ = 1 \oplus \begin{pmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{pmatrix}, \qquad Q^\dagger SQ = 1 \oplus \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Every other group matrix is a product of R, S, so the same $1 + 2$ block structure holds for all six matrices. The scalar block is A_1; the two-dimensional block has characters $(2, -1, 0)$ and is E. Hence

$$\boxed{U = A_1 \oplus E}.$$

17.4.3 The group named D_4 has a fourfold axis of symmetry, and twofold axes in four directions perpendicular to the fourfold axis. See Fig. 17.5. D_4 has the following classes (the numbers preceding the class descriptors indicate the number of elements in the class): I, $2C_4$, C_2, $2C_2'$, $2C_2''$. The twofold axes marked with primes are in the plane of fourfold symmetry.

(a) Find the number and dimensions of the irreducible representations.

(b) Given that all the characters of the representations of dimension 1 are ± 1 and that $C_2 = C_4^2$, use the orthogonality conditions to construct a complete character table for D_4.

Solution.

(a) There are five classes, so there are five irreducible representations. Their dimensions have squares summing to eight. No dimension can exceed two. If k dimensions are two and the other $5 - k$ are one, then

$$4k + (5 - k) = 8 \quad \Longrightarrow \quad k = 1.$$

Thus there are four one-dimensional representations and one two-dimensional representation.

(b) Write the group as $r^4 = s^2 = I$, $srs = r^{-1}$, as in 17.2.6. Its classes are

$$\{I\}, \quad \{r, r^3\}, \quad \{r^2\}, \quad \{s, r^2 s\}, \quad \{rs, r^3 s\}.$$

In one dimension, $a = U(r)$ and $b = U(s)$ are scalars. The relation $bab = a^{-1}$ and $b^2 = 1$ imply $a = a^{-1}$, so $a, b = \pm 1$. The class characters are therefore $(1, a, 1, b, ab)$. Their four choices give the first four rows below.

For the remaining representation, $\chi_E(I) = 2$. Column orthogonality between a nonidentity column g and the identity column requires

$$\sum_K \chi_K(I)^* \chi_K(g) = 0.$$

For r, r^3, the first four entries sum to zero, so $2\chi_E(r) = 0$. The same reasoning gives zero on both flip classes. For r^2, the four scalar entries sum to four, so $4 + 2\chi_E(r^2) = 0$, giving $\chi_E(r^2) = -2$. Thus the complete table is

D_4	I	$2C_4$	C_2	$2C_2'$	$2C_2''$
A_1	1	1	1	1	1
A_2	1	1	1	-1	-1
B_1	1	-1	1	1	-1
B_2	1	-1	1	-1	1
E	2	0	-2	0	0

.

Using class weights $(1, 2, 1, 2, 2)$, each row has squared norm eight and distinct rows have inner product zero. For example $\langle E, E\rangle = 4 + 4 = 8$, while $\langle E, A_1\rangle = 2 - 2 = 0$. This verifies the required orthogonality; the E matrices in 17.2.6 realize its last row.

17.4.4 The eight functions $\pm x^3$, $\pm x^2 y$, $\pm xy^2$, $\pm y^3$ span a reducible function space for D_4, with C_4 a 90° counterclockwise rotation in the xy plane, $C_2 = C_4^2$, $C_2' = (x \to -x,\ y \to y)$, $C_2'' = (x \to y,\ y \to x)$, and the remaining members of D_4 are additional members of the classes containing the above operations. Find the characters of the reducible representation on this span, and find the direct sum of irreducible representations of which it consists.

Also distinguish the formal permutation representation on eight independent signed labels, which is the interpretation used by the manual.

Solution.

As actual functions, f and $-f$ are linearly dependent. The eight listed signed functions are not an eight-dimensional basis: their function space has dimension four. The manual's eight-dimensional answer describes a different, formal permutation space whose eight signed labels are treated as independent. We give both interpretations explicitly.

For the actual cubic functions take the independent basis

$$f_1 = x^3, \quad f_2 = x^2 y, \quad f_3 = xy^2, \quad f_4 = y^3.$$

Use the substitution $r : (x, y) \mapsto (y, -x)$ for the quarter-turn action on functions. Its inverse convention gives the same characters. The transformed basis is

$$r(f_1, f_2, f_3, f_4) = (f_4, -f_3, f_2, -f_1).$$

Thus no diagonal coefficient occurs and $\chi(r) = 0$. The half-turn sends every cubic to its negative, so $\chi(r^2) = -4$. For $s : (x, y) \mapsto (-x, y)$,

$$s(f_1, f_2, f_3, f_4) = (-f_1, f_2, -f_3, f_4),$$

whose trace is zero. The diagonal reflection $(x, y) \mapsto (y, x)$ exchanges $f_1 \leftrightarrow f_4$ and $f_2 \leftrightarrow f_3$, again with trace zero. Therefore

$$\chi_{\text{cubic}} = (4, 0, -4, 0, 0) \quad \text{on } (I, 2C_4, C_2, 2C_2', 2C_2'').$$

The multiplicity formula gives

$$n_{A_1} = n_{A_2} = n_{B_1} = n_{B_2} = \frac{4-4}{8} = 0, \qquad n_E = \frac{4 \cdot 2 + (-4)(-2)}{8} = 2.$$

Hence the function-space answer is

$$\boxed{\mathcal{P}_3 = E \oplus E}.$$

Explicit invariant pairs are (x^3, y^3) and (x^2y, xy^2). A quarter-turn has eigenvalues $i, -i$ in each pair and a reflection interchanges the corresponding eigenlines, proving each copy irreducible.

For the formal eight-label interpretation, introduce independent symbols $e_{j,+}, e_{j,-}$ representing the labels $+f_j, -f_j$. A group operation permutes these eight symbols, so its character counts fixed labels. The identity fixes eight. The quarter-turn and diagonal reflection fix none. The half-turn exchanges each + label with its − partner and fixes none. The reflection $x \mapsto -x$ fixes the four labels $\pm x^2y, \pm y^3$. Thus

$$\chi_{\text{formal}} = (8, 0, 0, 4, 0),$$

and

$$\boxed{\mathcal{V}_{\text{formal}} = 2A_1 \oplus 2B_1 \oplus 2E}.$$

Indeed its multiplicities are $(8 + 2 \cdot 4)/8 = 2$ for A_1, B_1, $(8 - 2 \cdot 4)/8 = 0$ for A_2, B_2, and $(8 \cdot 2)/8 = 2$ for E. The map $e_{j,\pm} \mapsto \pm f_j$ has kernel spanned by $e_{j,+} + e_{j,-}$, of dimension four. Its quotient is the actual cubic function space. This explains why the two legitimate spaces have different characters and dimensions.

17.4.5 The group C_{4v} has a fourfold symmetry axis in the z direction, reflection symmetries (σ_v) about the xz and yz planes, and additional reflection symmetries (σ_d, d = dihedral) about planes that contain the z axis but are 45° from the x and y axes. See Fig. 17.6. The character table for C_{4v} follows.

	I	$2C_4$	C_2	$2\sigma_v$	$2\sigma_d$
A_1	1	1	1	1	1
A_2	1	1	1	−1	−1
B_1	1	−1	1	1	−1
B_2	1	−1	1	−1	1
E	2	0	−2	0	0

(a) Construct the matrix representing one member of each class of C_{4v} using as a basis a p_z orbital at each of the points $(x, y) = (a, 0)$, $(0, a)$, $(-a, 0)$, $(0, -a)$, and therefrom extract the characters of the reducible representation for which these p_z orbitals form a basis. A p_z orbital has functional form $(z/r)f(r)$.

(b) Determine the irreducible representations contained in our reducible p_z representation.

(c) Form those linear combinations of our p_z functions that are bases for each of the irreducible representations found in part (b).

The printed reference in (c) says (a); the irreducible summands are determined in (b).

Solution.

Let f_1, f_2, f_3, f_4 denote the p_z orbitals centered, in order, at $+x, +y, -x, -y$. A C_{4v} operation leaves z unchanged, so the local p_z shape incurs no sign change; only the centers are permuted. We use matrices whose jth column gives the image of f_j.

(a) Choose $r = C_4$ counterclockwise, $s = \sigma_v$ reflection in the xz plane, and $d = \sigma_d$ reflection in $x = y$. Representatives of the five classes are

$$P_I = I_4, \qquad P_r = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad P_{r^2} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix},$$

$$P_s = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad P_d = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

The traces are

$$\boxed{\chi = (4, 0, 0, 2, 0)}.$$

For example s fixes the two orbitals on the x axis, while r, r^2, d fix none.

(b) With class sizes $(1, 2, 1, 2, 2)$, $n_K = 8^{-1} \sum_{\text{classes}} |C| \chi_K^* \chi$ gives

$$n_{A_1} = n_{B_1} = \frac{4 + 2(2)}{8} = 1, \qquad n_{A_2} = n_{B_2} = \frac{4 - 2(2)}{8} = 0, \qquad n_E = \frac{2(4)}{8} = 1.$$

Thus $\boxed{\Gamma = A_1 \oplus B_1 \oplus E}$.

(c) Suitable combinations are

$$\boxed{\begin{aligned} A_1 &: \quad a = \tfrac{1}{2}(f_1 + f_2 + f_3 + f_4), \\ B_1 &: \quad b = \tfrac{1}{2}(f_1 - f_2 + f_3 - f_4), \\ E &: \quad e_x = (f_1 - f_3)/\sqrt{2}, \qquad e_y = (f_2 - f_4)/\sqrt{2}. \end{aligned}}$$

Substituting the center permutations gives

$$ra = a, \quad sa = a, \qquad rb = -b, \quad sb = b,$$

$$re_x = e_y, \quad re_y = -e_x, \qquad se_x = e_x, \quad se_y = -e_y, \qquad de_x = e_y, \quad de_y = e_x.$$

These generator actions identify the stated representations. The four coefficient vectors are mutually orthonormal and hence independent; their actual function norms, if orbitals overlap, are handled as explained in 17.3.1.

17.4.6 Using the notation and geometry of Exercise 17.4.5, repeat that exercise for the eight-member basis consisting of a p_x and a p_y orbital at each of the points $(x, y) = (a, 0)$, $(0, a)$, $(-a, 0)$, $(0, -a)$.

Solution.

Write $x_j = p_x(\mathbf{r} - \mathbf{R}_j)$ and $y_j = p_y(\mathbf{r} - \mathbf{R}_j)$, with the four sites ordered as in 17.4.5. All x_j have positive lobe along positive x, and all y_j along positive y. The ordered basis is

$$(x_1, y_1, x_2, y_2, x_3, y_3, x_4, y_4).$$

Unlike p_z, these local orbitals rotate and reflect in their two-component orientation space.

(a) If P_g is the site matrix from 17.4.5 and O_g the matrix acting on (p_x, p_y), then

$$D(g)_{(k,b),(j,a)} = (P_g)_{kj}(O_g)_{ba}, \qquad \boxed{D(g) = P_g \otimes O_g}.$$

This formula determines every entry of the requested 8×8 matrices. Explicitly, put

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad H = I_2.$$

In the following block matrices each entry is 2×2:

$$D(I) = I_8, \qquad D(r) = \begin{pmatrix} 0 & 0 & 0 & R \\ R & 0 & 0 & 0 \\ 0 & R & 0 & 0 \\ 0 & 0 & R & 0 \end{pmatrix}, \qquad D(r^2) = \begin{pmatrix} 0 & 0 & -H & 0 \\ 0 & 0 & 0 & -H \\ -H & 0 & 0 & 0 \\ 0 & -H & 0 & 0 \end{pmatrix},$$

$$D(s) = \begin{pmatrix} S & 0 & 0 & 0 \\ 0 & 0 & 0 & S \\ 0 & 0 & S & 0 \\ 0 & S & 0 & 0 \end{pmatrix}, \qquad D(d) = \begin{pmatrix} 0 & T & 0 & 0 \\ T & 0 & 0 & 0 \\ 0 & 0 & 0 & T \\ 0 & 0 & T & 0 \end{pmatrix}.$$

For example $rx_1 = y_2$, $ry_1 = -x_2$, $sx_1 = x_1$, and $sy_1 = -y_1$, which explain the orientation signs. The trace of a tensor product is the product of traces, so

$$\chi_D = \chi_{\text{site}}\chi_{\text{orientation}} = (4,0,0,2,0)(2,0,-2,0,0) = \boxed{(8,0,0,0,0)}.$$

(b) Only the identity contributes to the multiplicity sum: $n_K = 8^{-1}(8)\chi_K(I) = \dim K$. Therefore

$$\boxed{\Gamma = A_1 \oplus A_2 \oplus B_1 \oplus B_2 \oplus E \oplus E}.$$

(c) The following four combinations give the scalar representations:

$$\boxed{\begin{aligned} A_1 &: \quad a_1 = (x_1 + y_2 - x_3 - y_4)/2, \\ A_2 &: \quad a_2 = (y_1 - x_2 - y_3 + x_4)/2, \\ B_1 &: \quad b_1 = (x_1 - y_2 - x_3 + y_4)/2, \\ B_2 &: \quad b_2 = (y_1 + x_2 - y_3 - x_4)/2. \end{aligned}}$$

They arise by taking the site pair $(f_1 - f_3, f_2 - f_4)$ and combining it with the orientation pair (p_x, p_y) as a scalar product, antisymmetric product, difference of diagonal products, and symmetric off-diagonal product. One can also check directly that their eigenvalues under the two generators are

	a_1	a_2	b_1	b_2
r	1	1	-1	-1
s	1	-1	1	-1

.

Since $d = rs$, its eigenvalues are the corresponding products, so these are exactly the four character-table rows.

The two independent E pairs are

$$\boxed{\begin{aligned} u_x &= (x_1 + x_2 + x_3 + x_4)/2, \quad u_y = (y_1 + y_2 + y_3 + y_4)/2, \\ v_x &= (x_1 - x_2 + x_3 - x_4)/2, \quad v_y = (y_1 - y_2 + y_3 - y_4)/2. \end{aligned}}$$

The first pair is the site A_1 combination times the orientation pair; the second uses the site B_1 combination. Their generator matrices are respectively (R, S) and $(-R, S)$. Both have characters $(2, 0, -2, 0, 0)$ and have no common invariant line. The eight coefficient columns displayed above have norm one and pairwise zero inner products, so they form a complete basis. As in 17.3.1, coefficient normalization is distinct from spatial normalization when local orbitals overlap.

17.5 Direct Products

17.5.1 The group C_{4v} has eight elements, corresponding to the rotational and reflection symmetries of a square that cannot be turned over. See Fig. 17.6. Symmetry rotations about the z axis are denoted C_4, C_2, C_4'. Reflections relative to the xz and yz planes are named σ_v and σ_v'; those at 45° relative to the xz and yz planes are called σ_d and σ_d' (d indicates "dihedral"). The character table for C_{4v} is in Table 17.4.

(a) Find the direct sum of irreducible representations of C_{4v} corresponding to the direct product $E \otimes E$.

(b) A basis for E (in the context of Fig. 17.5) consists of the two functions $\varphi_1 = x$, $\varphi_2 = y$. Apply a few of the group operations to this basis and verify the entries for E in the character table.

(c) Assume now that we have two sets of variables, x_1, y_1 and x_2, y_2, and we form the direct-product basis x_1x_2, x_1y_2, y_1x_2, y_1y_2. Determine how the direct-product basis functions can be combined to form bases for each of the irreducible representations in the direct sum corresponding to $E \otimes E$.

Solution.

(a) For matrices A, B, $\mathrm{tr}(A \otimes B) = (\mathrm{tr}\, A)(\mathrm{tr}\, B)$. The product character is consequently

$$\chi_{E\otimes E} = (2, 0, -2, 0, 0)^2 = (4, 0, 4, 0, 0).$$

For each of the four one-dimensional irreducibles, $\chi(I) = \chi(C_2) = 1$, so its multiplicity is $(4+4)/8 = 1$. For E itself it is $[2(4) + (-2)(4)]/8 = 0$. Thus

$$\boxed{E \otimes E = A_1 \oplus A_2 \oplus B_1 \oplus B_2}.$$

(b) Take $rx = y$, $ry = -x$, $sx = x$, $sy = -y$, and $dx = y$, $dy = x$. In the ordered basis (x, y) their coefficient matrices are

$$D(r) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad D(r^2) = -I_2, \quad D(s) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad D(d) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Their traces, together with $\mathrm{tr}\, I_2 = 2$, are $(2, 0, -2, 0, 0)$ as in the E row.

(c) The product basis has independent variables for particles 1 and 2. Its four convenient combinations are

$$\boxed{\begin{array}{ll} A_1: & a_1 = (x_1x_2 + y_1y_2)/\sqrt{2}, \\ A_2: & a_2 = (x_1y_2 - y_1x_2)/\sqrt{2}, \\ B_1: & b_1 = (x_1x_2 - y_1y_2)/\sqrt{2}, \\ B_2: & b_2 = (x_1y_2 + y_1x_2)/\sqrt{2}. \end{array}}$$

Apply r to each factor: $x_1x_2 \to y_1y_2$, $y_1y_2 \to x_1x_2$, $x_1y_2 \to -y_1x_2$, and $y_1x_2 \to -x_1y_2$. Thus the four eigenvalues of r are $(1, 1, -1, -1)$. Under s a product acquires one minus sign for each y, so the eigenvalues are $(1, -1, 1, -1)$. Under d the two variables are exchanged, giving $(1, -1, -1, 1)$. These reproduce the four claimed one-dimensional characters. Their independent coefficient vectors span the full four-dimensional tensor-product space.

17.6 Symmetric Group

17.6.1

(a) The objects $(abcd)$ are permuted to $(dacb)$. Write out a 4×4 matrix representation of this one permutation.

Hint: Compare with Exercise 17.4.2.

(b) Is the permutation $(abdc) \to (dacb)$ odd or even?

(c) Is either of the permutations in (a) and (b) a possible member of the D_4 group, which was the subject of Exercise 17.4.3? Why or why not?

Keep the two printed inputs distinct. In (c), distinguish the standard cyclic square labeling $1, 2, 3, 4$ from an embedding obtained by relabeling its vertices.

Solution.

(a) The target column selects entries $4, 1, 3, 2$ of the original column:

$$\boxed{P = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}}, \qquad P \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \begin{pmatrix} d \\ a \\ c \\ b \end{pmatrix}.$$

As a map from old positions to new positions, this is $(1\,2\,4)$ with position 3 fixed. It is a 3-cycle, has order three, and is even: a 3-cycle is a product of two transpositions.

(b) The printed input changes from $(abcd)$ in (a) to $(abdc)$ in (b). These are different permutations. The manual's answer *even* applies to (a), not to (b) as printed. For the literal input (a, b, d, c), the target (d, a, c, b) selects entries $3, 1, 4, 2$. Its matrix is

$$Q = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

Its position cycle is $(1\,2\,4\,3)$, which is a product of three transpositions. Thus the answer to (b) as printed is

$$\boxed{\text{odd}, \qquad \det Q = -1}.$$

Equivalently the row-selection list $(3, 1, 4, 2)$ has three inversions, also giving $(-1)^3 = -1$.

(c) For the permutation in (a), the answer is unambiguously $\boxed{\text{no}}$: its order is three, while D_4 has only elements of orders one, two, and four. In particular its order cannot divide $|D_4| = 8$.

The original reference in (c) was ambiguous. For the literal permutation of (b), the geometric labeling must be specified. With slots $1, 2, 3, 4$ placed in that cyclic order around the square, Q

is not a symmetry: it sends the adjacent pair $(1,2)$ to $(2,4)$, a diagonal pair. But a four-cycle is compatible with an abstract D_4 embedding. If the cyclic order of vertices is chosen to be $1,2,4,3$, then $Q = (1\,2\,4\,3)$ is a quarter-turn. This distinction resolves the ambiguity without silently changing the input in (b).

17.6.2

(a) The permutation group of four objects, S_4, has $4! = 24$ elements. Treating the four elements of the cyclic group, C_4, as permutations, set up a 4×4 matrix representation of C_4. Note that C_4 is a subgroup of S_4.

(b) How do you know that this 4×4 matrix representation of C_4 must be reducible?

Solution.

Let R cyclically shift a column of four entries to the left. The four matrices are

$$U(I) = I_4, \qquad U(C_4) = R = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

$$U(C_4^2) = R^2 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad U(C_4^3) = R^3 = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Four shifts restore the original column, so $R^4 = I_4$. The products satisfy $R^j R^k = R^{j+k \pmod 4}$, exactly the cyclic-group law. All four are permutation matrices, so they form a subgroup of the full permutation group S_4 (the original note writes P_4; this denotes the permutation group S_4 just introduced).

For reducibility, the vector $(1,1,1,1)^T$ is fixed by all four matrices, giving a proper invariant line. More fully, let $\omega = i$ and define

$$v_k = \frac{1}{2} \begin{pmatrix} 1 \\ \omega^k \\ \omega^{2k} \\ \omega^{3k} \end{pmatrix}, \qquad k = 0, 1, 2, 3.$$

Cyclic shifting gives $Rv_k = \omega^k v_k$; the last component works because $\omega^{4k} = 1$. Their inner products are

$$v_j^\dagger v_k = \frac{1}{4} \sum_{m=0}^{3} \omega^{(k-j)m} = \begin{cases} 1, & j = k, \\ 0, & j \neq k, \end{cases}$$

by the finite geometric-series formula. Thus $V = (v_0\, v_1\, v_2\, v_3)$ is unitary and

$$\boxed{V^\dagger R^m V = \operatorname{diag}(1, i^m, (-1)^m, (-i)^m)}.$$

The representation reduces to the four one-dimensional irreducible representations of C_4.

17.6.3 The permutation group of four objects, S_4, has five classes.

(a) Determine the number of elements in each class of S_4 and identify one element of each class as a product of cycles.

(b) Two of the irreducible representations of S_4 are of dimension 1 (and are usually denoted A_1 and A_2). Noting that permutations can be classified as even or odd, find the characters of A_1 and A_2.

Hint. Set up a character table and fill in the A_1 and A_2 lines.

(c) One irreducible representation of S_4 (usually denoted E) is of dimension 2. Determine the dimensions of all the irreducible representations of S_4 other than A_1, A_2, and E.

(d) Complete the character table of S_4.

Hint. Only the even permutations have nonzero characters in the E representation.

Solution.

(a) Conjugation relabels a permutation's cycles and preserves their lengths. Conversely two permutations with the same cycle lengths are conjugate by a suitable relabeling. The five possibilities and their sizes are

cycle type	example	class size
$1+1+1+1$	I	1
$2+1+1$	(12)	$\binom{4}{2} = 6$
$2+2$	$(12)(34)$	$\frac{1}{2}\binom{4}{2} = 3$
$3+1$	(123)	$\binom{4}{3}(3-1)! = 8$
4	(1234)	$(4-1)! = 6$

For two transpositions, dividing by two removes the ordering of the two pairs. For a cycle of length k, cyclically shifting its notation describes the same cycle, leaving $k!/k = (k-1)!$ choices. The sizes sum to 24.

(b) The trivial character is $(1, 1, 1, 1, 1)$. A cycle of length k is a product of $k-1$ transpositions, so the signs of the five classes are $(1, -1, 1, 1, -1)$. These give the A_1, A_2 rows below.

(c) There are five irreducibles. The known dimensions $1, 1, 2$ account for $1 + 1 + 4 = 6$ of the squared-dimension sum. The other dimensions d, e obey $d^2 + e^2 = 24 - 6 = 18$. The only positive integer solution is $d = e = 3$.

(d) A useful independent construction supplies the remaining characters. The ordinary four-dimensional permutation representation has character equal to its number of fixed letters:

$$(4, 2, 0, 1, 0).$$

Removing its invariant all-ones line leaves a three-dimensional representation with character

$$\chi_{T_2} = (3, 1, -1, 0, -1).$$

Its squared character norm is $9 + 6(1) + 3(1) + 8(0) + 6(1) = 24$, proving irreducibility. Multiplying by the sign representation gives a second irreducible,

$$\chi_{T_1} = (3, -1, -1, 0, 1).$$

This fixes our names T_1, T_2; interchanging their names is an equally valid convention.

To construct E, act on the three pair partitions

$$12|34, \qquad 13|24, \qquad 14|23.$$

The respective numbers of fixed partitions for the five classes are $(3,1,3,0,1)$. For example (12) fixes $12|34$ and exchanges the other two; $(12)(34)$ fixes all three; a 3-cycle cycles all three. A 4-cycle fixes $13|24$ and exchanges the other two. Remove the all-ones invariant line from this three-dimensional permutation space to obtain $\chi_E = (2,0,2,-1,0)$. Its norm is $4+3(4)+8(1)=24$, again proving irreducibility.

The complete table is therefore

S_4	I	$6(12)$	$3(12)(34)$	$8(123)$	$6(1234)$
A_1	1	1	1	1	1
A_2	1	-1	1	1	-1
E	2	0	2	-1	0
T_1	3	-1	-1	0	1
T_2	3	1	-1	0	-1

.

Using the displayed class sizes, distinct rows are orthogonal; for instance $\langle T_1, T_2\rangle = 9-6+3-6=0$ and $\langle E, A_1\rangle = 2+6-8=0$. All five rows have norm squared 24, and their squared dimensions sum to 24, as required.

17.7 Continuous Groups

17.7.1 Determine three $SU(2)$ subgroups of $SU(3)$.

Solution.

Every $SU(2)$ matrix can be written

$$W = \begin{pmatrix} a & b \\ -b^* & a^* \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

To see that this describes every such matrix, let its first row be (a,b), of norm one. Every unit row orthogonal to it has the form $e^{i\delta}(-b^*, a^*)$. The determinant is then $e^{i\delta}(|a|^2+|b|^2) = e^{i\delta}$; requiring determinant one gives $e^{i\delta}=1$. Conversely the displayed matrix has orthonormal rows and determinant $|a|^2+|b|^2=1$. Embed this block into each of the three coordinate pairs:

$$\boxed{\begin{aligned} H_{12} &= \left\{ \begin{pmatrix} a & b & 0 \\ -b^* & a^* & 0 \\ 0 & 0 & 1 \end{pmatrix} \right\}, \\ H_{13} &= \left\{ \begin{pmatrix} a & 0 & b \\ 0 & 1 & 0 \\ -b^* & 0 & a^* \end{pmatrix} \right\}, \\ H_{23} &= \left\{ \begin{pmatrix} 1 & 0 & 0 \\ 0 & a & b \\ 0 & -b^* & a^* \end{pmatrix} \right\}. \end{aligned}}$$

Each set has the stated constraint on a, b. Products and adjoints act only in the chosen 2×2 block, so each set is a subgroup. Each embedding is injective and preserves multiplication, making these subgroups isomorphic to $SU(2)$. They fix, respectively, the third, second, and first basis vector.

For explicit generators, let E_{ij} have a single one in row i, column j. In the (i,j) block take

$$T_x = \frac{E_{ij}+E_{ji}}{2}, \qquad T_y = \frac{E_{ij}-E_{ji}}{2i}, \qquad T_z = \frac{E_{ii}-E_{jj}}{2}.$$

Using $E_{ab}E_{cd} = \delta_{bc}E_{ad}$, put $A = E_{ij}$, $B = E_{ji}$, and $D = E_{ii} - E_{jj}$. Then

$$[A,B] = D, \qquad [D,A] = 2A, \qquad [D,B] = -2B.$$

Substitution displays all three commutators:

$$\begin{aligned}
[T_x,T_y] &= \frac{[A+B,A-B]}{4i} = \frac{-2D}{4i} = i\,\frac{D}{2} = iT_z,\\
[T_y,T_z] &= \frac{[A-B,D]}{4i} = \frac{-2(A+B)}{4i} = i\,\frac{A+B}{2} = iT_x,\\
[T_z,T_x] &= \frac{[D,A+B]}{4} = \frac{A-B}{2} = i\,\frac{A-B}{2i} = iT_y.
\end{aligned}$$

In terms of the Gell-Mann matrices, the three triples are

$$\left(\frac{\lambda_1}{2},\frac{\lambda_2}{2},\frac{\lambda_3}{2}\right), \quad \left(\frac{\lambda_4}{2},\frac{\lambda_5}{2},\frac{\lambda_3+\sqrt{3}\lambda_8}{4}\right), \quad \left(\frac{\lambda_6}{2},\frac{\lambda_7}{2},\frac{-\lambda_3+\sqrt{3}\lambda_8}{4}\right).$$

Equations (17.62) and (17.63) omit a factor $1/2$ in their linear combinations. Direct addition gives $\sqrt{3}\lambda_8 - \lambda_3 = \mathrm{diag}(0,2,-2)$ and $\sqrt{3}\lambda_8 + \lambda_3 = \mathrm{diag}(2,0,-2)$. Their displayed diagonal matrices therefore equal half those combinations. The generator triples above include the needed factors.

17.7.2 Prove that the matrices $U(n)$ (unitary matrices of order n) form a group, and that $SU(n)$ (those with determinant unity) form a subgroup of $U(n)$.

Solution.

Let U, V be $n \times n$ unitary matrices: $U^\dagger U = UU^\dagger = I_n$ and likewise for V. Their product is unitary because

$$(UV)^\dagger(UV) = V^\dagger U^\dagger UV = V^\dagger V = I_n,$$

and similarly $(UV)(UV)^\dagger = I_n$. Thus $U(n)$ is closed. Matrix multiplication is associative, I_n is a unitary identity, and the inverse of U is $U^\dagger$, itself unitary: $(U^\dagger)^\dagger U^\dagger = UU^\dagger = I_n$. This proves all group axioms for $U(n)$.

For $SU(n)$, impose $\det U = 1$ in addition. If $U, V \in SU(n)$, then

$$\det(UV) = \det U \, \det V = 1,$$

so the product is in $SU(n)$. The identity has determinant one, and $\det U^{-1} = (\det U)^{-1} = 1$, so the inverse is in $SU(n)$. Associativity is inherited from $U(n)$. Hence $\boxed{SU(n) \le U(n)}$. For comparison, unitarity alone gives $|\det U|^2 = \det(U^\dagger U) = 1$; restricting this unit-modulus determinant to the value one is precisely the special-unitary condition.

17.7.3 Using Eq. (17.56) for the matrix elements of $SU(2)$ corresponding to rotations about the coordinate axes, find the matrix corresponding to a rotation defined by Euler angles (α, β, γ). The Euler angles are defined in Section 3.4.

Solution.

With the Euler convention of Section 3.4, the successive passive coordinate rotations are about z, then the intermediate y, then the final z. Equation (17.56) therefore prescribes

$$U(\alpha,\beta,\gamma) = e^{i\gamma\sigma_3/2} e^{i\beta\sigma_2/2} e^{i\alpha\sigma_3/2}.$$

Since $\sigma_j^2 = I$, separating even and odd powers of the exponential gives $e^{it\sigma_j/2} = I\cos(t/2) + i\sigma_j \sin(t/2)$. In particular

$$Z(t) = \begin{pmatrix} e^{it/2} & 0 \\ 0 & e^{-it/2} \end{pmatrix}, \qquad Y(\beta) = \begin{pmatrix} \cos(\beta/2) & \sin(\beta/2) \\ -\sin(\beta/2) & \cos(\beta/2) \end{pmatrix}.$$

First multiply on the right:

$$Y(\beta)Z(\alpha) = \begin{pmatrix} c\,e^{i\alpha/2} & s\,e^{-i\alpha/2} \\ -s\,e^{i\alpha/2} & c\,e^{-i\alpha/2} \end{pmatrix}, \qquad c = \cos(\beta/2),\ s = \sin(\beta/2).$$

Left multiplication by $Z(\gamma)$ multiplies its first row by $e^{i\gamma/2}$ and its second by $e^{-i\gamma/2}$. Hence

$$\boxed{U(\alpha,\beta,\gamma) = \begin{pmatrix} e^{i(\alpha+\gamma)/2}\cos(\beta/2) & e^{i(\gamma-\alpha)/2}\sin(\beta/2) \\ -e^{i(\alpha-\gamma)/2}\sin(\beta/2) & e^{-i(\alpha+\gamma)/2}\cos(\beta/2) \end{pmatrix}}.$$

Its determinant is $c^2 + s^2 = 1$ and its columns are orthonormal, so it is in $SU(2)$. This specifies the lift obtained from the stated exponential product. Changing a single Euler angle by 2π changes its sign; the two matrices $U, -U$ correspond to the same spatial rotation, consistent with the double cover of $SO(3)$ by $SU(2)$.

17.7.4 For a product of three quarks, the member of $SU(3)$ representation **10** with $(I_3, Y) = \left(+\frac{3}{2}, 1\right)$ is uuu.

(a) Apply operators in the root diagram for $SU(3)$, Fig. 17.8, to obtain all the remaining members of the decuplet comprising the representation **10**.

(b) The two representations **8** can be chosen to have for $I_3 = \frac{1}{2}$, $Y = 1$ the respective members $\psi_1\left(\frac{1}{2}, 1\right) = (ud - du)u$ and $\psi_2\left(\frac{1}{2}, 1\right) = 2uud - udu - duu$. Briefly explain why this choice is possible.

(c) Using the operators in the root diagram and the above $\psi_1\left(\frac{1}{2}, 1\right)$, find expressions for $\psi_1\left(-\frac{1}{2}, 1\right)$, $\psi_1(-1, 0)$, $\psi_1(1, 0)$, $\psi_1\left(-\frac{1}{2}, -1\right)$, and $\psi_1\left(\frac{1}{2}, -1\right)$.

(d) Taking each of the six ψ_1 functions you now have, apply an operator that will convert it into $\psi_1(0, 0)$. Show that you obtain exactly two linearly independent $\psi_1(0, 0)$, thereby justifying the claim that the ψ_1 are an octet at the points shown in Fig. 17.10.

(e) Show that the octet built starting from $\psi_2\left(\frac{1}{2}, 1\right)$ is linearly independent from that built from ψ_1.

(f) Find the wave function $\psi(0,0)$ that is linearly independent of all the $\psi(0,0)$ functions found in parts (a)–(e). It is the sole member of the representation **1**.

Use the ordered flavor tensor space with u, d, s orthonormal, and the root convention $I_-u = d$, $U_-d = s$, $V_-u = s$ from Fig. 17.8. The conflicting U, V matrix labels in Eq. (17.67) are corrected below.

Solution.

The supplied transcription has $I_3 = 2/3$ for uuu. The actual textbook (printed p. 861) correctly has $I_3 = 3/2$: each u contributes $1/2$. This value is restored in the question. Also, the matrices labeled $U_\pm$ and $V_\pm$ in Eq. (17.67) are interchanged relative to Eqs. (17.64)–(17.66) and Fig. 17.8. We use the consistent root-diagram convention below.

Use an ordered tensor product, so $uds = u \otimes d \otimes s$; the 27 ordered words are orthonormal. The single-quark weights and root actions are

$$u : (1/2, 1/3), \qquad d : (-1/2, 1/3), \qquad s : (0, -2/3),$$

$$I_-u = d, \quad I_+d = u, \qquad U_-d = s, \quad U_+s = d, \qquad V_-u = s, \quad V_+s = u,$$

with all unlisted single-quark actions zero. On a word the operator acts on all three slots:

$$T(abc) = (Ta)bc + a(Tb)c + ab(Tc).$$

Thus the total generators are sums of the three single-slot generators, not products of them. A word with occupation numbers (n_u, n_d, n_s) has

$$I_3 = (n_u - n_d)/2, \qquad Y = (n_u + n_d - 2n_s)/3.$$

(a) The decuplet. Let $\mathcal{S}(aab) = aab + aba + baa$ for $a \neq b$, and

$$\mathcal{S}(uds) = uds + usd + dus + dsu + sud + sdu.$$

The ten normalized states are

I_3	Y	state
$3/2$	1	uuu
$1/2$	1	$\mathcal{S}(uud)/\sqrt{3}$
$-1/2$	1	$\mathcal{S}(ddu)/\sqrt{3}$
$-3/2$	1	ddd
1	0	$\mathcal{S}(uus)/\sqrt{3}$
0	0	$\mathcal{S}(uds)/\sqrt{6}$
-1	0	$\mathcal{S}(dds)/\sqrt{3}$
$1/2$	-1	$\mathcal{S}(ssu)/\sqrt{3}$
$-1/2$	-1	$\mathcal{S}(ssd)/\sqrt{3}$
0	-2	sss

To derive every coefficient, a symmetric word sum with occupations n_a has $N = 3!/(n_u!n_d!n_s!)$ distinct words. A root operator changing a into b produces each target word $n_b + 1$ times. The ratio of target and source word counts is $N'/N = n_a/(n_b + 1)$. Therefore its action on normalized sums is

$$T_{b\leftarrow a} D_{n_u,n_d,n_s} = \sqrt{n_a(n_b+1)}\, D_{\text{new occupations}}.$$

For example $I_- uuu = duu + udu + uud = \sqrt{3} D_{uud}$. Successive I_- operations down the top row have coefficients $\sqrt{3}, 2, \sqrt{3}$. The remaining rows follow from

$$V_- D_{uuu} = \sqrt{3} D_{uus}, \quad I_- D_{uus} = \sqrt{2} D_{uds}, \quad I_- D_{uds} = \sqrt{2} D_{dds},$$
$$V_- D_{uus} = 2 D_{uss}, \qquad I_- D_{uss} = D_{dss}, \qquad V_- D_{uss} = \sqrt{3} D_{sss}.$$

Here the subscript denotes occupations, so $D_{uss} = D_{ssu}$. These paths reach all ten states listed. The root actions preserve complete permutation symmetry; the symmetric three-quark space has $\binom{3+3-1}{3} = 10$ occupation states, so none is omitted.

(b) Choosing the two octet highest states. At weight $(1/2, 1)$ the ordered basis is (uud, udu, duu). The decuplet coefficient vector is $(1, 1, 1)/\sqrt{3}$. The two proposed vectors have coefficients

$$a = (0, 1, -1), \qquad b = (2, -1, -1).$$

Their sums are zero, so both are orthogonal to the decuplet. Also $a \cdot b = -1 + 1 = 0$ and their squared norms are $2, 6$. They span the two-dimensional remaining weight space. Each is annihilated by I_+ because its coefficients sum to zero, and by U_+, V_+ because it contains no strange quark. They can therefore be taken as the two highest states. The constructions below exhibit all eight states generated from each, thereby identifying the two octets directly.

(c) The six nonzero weights of the first octet. Use unnormalized states to keep every replacement visible. Starting with $A = udu - duu$, the following path gives all requested weights:

state	(I_3, Y)	word expression and operation
A	$(1/2, 1)$	$udu - duu$
B	$(-1/2, 1)$	$I_- A = udd - dud$
C	$(-1, 0)$	$V_- B = sdd - dsd$
D	$(-1/2, -1)$	$U_- C = sds - dss$
E	$(1/2, -1)$	$I_+ D = sus - uss$
F	$(1, 0)$	$V_+ E = suu - usu$

.

For clarity the first and third steps are

$$I_-(udu - duu) = (ddu + udd) - (ddu + dud) = udd - dud,$$

$$U_-(sdd - dsd) = (ssd + sds) - (ssd + dss) = sds - dss.$$

The remaining steps are equally explicit:

$$V_-(udd - dud) = sdd - dsd,$$
$$I_+(sds - dss) = sus - uss,$$
$$V_+(sus - uss) = (uus + suu) - (uus + usu) = suu - usu.$$

Every listed vector has squared norm two. Dividing them by $\sqrt{2}$ produces normalized nonzero-weight states.

(d) The two zero-weight states. Approach weight $(0, 0)$ from each of the six preceding weights. Define

$$Z_1 = sdu + uds - dsu - dus, \qquad Z_2 = usd + uds - sud - dus.$$

The complete list of results is

operation	result	in the (Z_1, Z_2) basis
V_-A	$sdu + uds - dsu - dus$	Z_1
U_-B	$usd + uds - sud - dus$	Z_2
I_+C	$sud + sdu - usd - dsu$	$Z_1 - Z_2$
V_+D	$uds + sdu - dus - dsu$	Z_1
U_+E	$dus + sud - uds - usd$	$-Z_2$
I_-F	$sdu + sud - dsu - usd$	$Z_1 - Z_2$

.

For example V_-A changes either u in each term of $udu - duu$, giving the four terms in Z_1; all six rows are obtained by the same slot-by-slot rule. The inner products are

$$\langle Z_1, Z_1\rangle = \langle Z_2, Z_2\rangle = 4, \qquad \langle Z_1, Z_2\rangle = 2.$$

Their Gram determinant is $16 - 4 = 12 > 0$, so they are independent, while the table proves that all six incoming results are in their span. One orthonormal zero-weight pair is

$$\boxed{z_1 = Z_1/2, \qquad z_2 = (2Z_2 - Z_1)/\sqrt{12}}.$$

Indeed the second numerator is orthogonal to Z_1 and has norm squared $4(4) + 4 - 4(2) = 12$. The six distinct nonzero weights and these two independent zero weights supply exactly eight states.

(e) Independence of the second octet. Interchange slots 1 and 2, calling this unitary operation P_{12}. The first highest state is odd under it: $P_{12}(udu - duu) = -(udu - duu)$. The second is even: $P_{12}(2uud - udu - duu) = 2uud - udu - duu$. Every total root operator commutes with P_{12}, because it sums identical actions on all three slots. Hence all states generated from the first are odd and all from the second are even. These eigenspaces are orthogonal: if $P_{12}v = -v$ and $P_{12}w = w$, then $\langle v, w\rangle = \langle P_{12}v, P_{12}w\rangle = -\langle v, w\rangle = 0$.

For an explicit check that the second space also has eight states, use the same path I_-, V_-, U_-, I_+, V_+ as in (c):

(I_3, Y)	ψ_2
$(1/2, 1)$	$2uud - udu - duu$
$(-1/2, 1)$	$dud + udd - 2ddu$
$(-1, 0)$	$dsd + sdd - 2dds$
$(-1/2, -1)$	$2ssd - sds - dss$
$(1/2, -1)$	$2ssu - sus - uss$
$(1, 0)$	$usu + suu - 2uus$

.

For instance lowering the first row gives $2(dud+udd)-(ddu+udd)-(ddu+dud) = dud+udd-2ddu$. Applying V_- to its first row and U_- to its second yields

$$T_1 = 2sud + 2usd - sdu - uds - dsu - dus,$$
$$T_2 = sud + dus + usd + uds - 2sdu - 2dsu.$$

Their Gram matrix is $\begin{pmatrix} 12 & 6 \\ 6 & 12 \end{pmatrix}$ with determinant $108 > 0$, giving two more independent states at zero weight. The six outer states have norm squared six; a normalized central pair is $T_1/\sqrt{12}$ and $(2T_2 - T_1)/6$.

For completeness, the slot-symmetric subspace under P_{12} has dimension $6 \cdot 3 = 18$. Its completely symmetric decuplet takes ten dimensions, leaving eight. This orthogonal complement is invariant

under $SU(3)$, because both the slot exchange and full symmetrization commute with $U \otimes U \otimes U$. The second construction supplies eight independent vectors there and hence exhausts that invariant complement. The first octet lies in the nine-dimensional slot-antisymmetric space; the remaining one-dimensional component is found next.

(f) The singlet. There are six ordered words at zero weight, in the order

$$(uds, usd, dus, dsu, sud, sdu).$$

The decuplet supplies the all-ones vector; the two octets supply Z_1, Z_2, T_1, T_2. These five vectors are independent: Z_1, Z_2 have odd P_{12} parity, while the other three have even parity; T_1, T_2 are independent and each has coefficient sum zero, so neither their span nor any of its nonzero vectors is the all-ones line. The normalized vector orthogonal to all five is

$$\boxed{\psi_{\mathbf{1}} = \frac{uds - usd - dus + dsu + sud - sdu}{\sqrt{6}}}.$$

Its coefficients are $(1, -1, -1, 1, 1, -1)/\sqrt{6}$. Their sum is zero, their dot products with Z_1, Z_2 are zero, and orthogonality to T_1, T_2 follows also from opposite P_{12} parity. Its squared norm is one.

Finally this is an invariant state, not merely an unused vector: it is the fully antisymmetric tensor. Under $U \otimes U \otimes U$ it is multiplied by $\det U = 1$ for $U \in SU(3)$. Equivalently each root operator produces pairs of opposite terms and annihilates it. Thus it spans the trivial representation. The first octet starts orthogonal to this invariant singlet and stays orthogonal under the unitary group action. It therefore fills the eight-dimensional complement of the singlet within the nine-dimensional slot-antisymmetric space.

One can also verify irreducibility directly from the displayed states. In either octet all nonzero outer weights other than the starting state have a nonzero raising image: use I_+ on B, C, V_+ on D, U_+ on E, F. At zero weight the replacement rules give

$$I_+(aZ_1 + bZ_2) = (a - b)F, \qquad V_+(aZ_1 + bZ_2) = (2a + b)A.$$

Both vanish only if $a = b = 0$. The same equations hold with Z_j replaced by T_j and A, F by the second-octet states. Thus the common kernel of I_+, U_+, V_+ in each octet is exactly its one-dimensional highest-state line. In the decuplet the occupation formula shows that any d can be raised by I_+ and any s by V_+; only uuu is killed by all raising operators. Every nonzero invariant summand of a finite-dimensional unitary representation has a highest state. The commuting Hermitian operators Y, I_3 preserve that summand and admit a simultaneous eigenbasis there. Choose a weight vector of largest Y, then largest I_3 among that Y eigenspace. Both U_+, V_+ would increase Y, while I_+ would increase I_3 at the same Y; therefore all three annihilate the chosen vector. Two nonzero orthogonal summands would give two independent common-kernel vectors, which the preceding calculations exclude. All three constructed multiplets are therefore irreducible.

The dimensions and the explicit independent states give

$$\boxed{\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = \mathbf{10} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{1}}, \qquad 10 + 8 + 8 + 1 = 27.$$

17.8 Lorentz Group

17.8.1 Show that in $3+1$ dimensions (this means three spatial dimension plus time), a boost in the xy plane at an angle θ from the x direction has, in coordinates (x^0, x^1, x^2, x^3), the generator

$$\mathsf{S} = i \begin{pmatrix} 0 & \cos\theta & \sin\theta & 0 \\ \cos\theta & 0 & 0 & 0 \\ \sin\theta & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Use $x^0 = ct$, metric $\mathrm{diag}(1,-1,-1,-1)$, and $U(\rho) = e^{i\rho S}$ for the passive boost.

Solution.

Let $x^0 = ct$ and $\mathbf{n} = (\cos\theta, \sin\theta, 0)$. For an infinitesimal passive boost of rapidity $\delta\rho$, the frame velocity is $c\,\delta\rho\,\mathbf{n}$ to first order. Thus

$$\delta x = -\delta\rho\cos\theta\, x^0, \qquad \delta y = -\delta\rho\sin\theta\, x^0, \qquad \delta z = 0.$$

Write the most general linear first-order time change as $\delta x^0 = -\delta\rho(ax + by + dz + ex^0)$. Preservation of $(x^0)^2 - x^2 - y^2 - z^2$ to first order gives

$$2x^0\delta x^0 - 2x\delta x - 2y\delta y = 2x^0\delta\rho\big[(\cos\theta - a)x + (\sin\theta - b)y - dz - ex^0\big] = 0.$$

Because this polynomial vanishes for every x, y, z, x^0, each coefficient vanishes: $a = \cos\theta$, $b = \sin\theta$, $d = e = 0$. Hence

$$x' = (I_4 - \delta\rho\, B)x + O(\delta\rho^2), \qquad B = \begin{pmatrix} 0 & \cos\theta & \sin\theta & 0 \\ \cos\theta & 0 & 0 & 0 \\ \sin\theta & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The book defines a generator by $U(\delta\rho) = I_4 + i\delta\rho\, S + O(\delta\rho^2)$. Comparing first-order coefficients, $iS = -B$, so

$$\boxed{S = iB},$$

which is the requested generator. With $\eta = \mathrm{diag}(1,-1,-1,-1)$ one also checks $B^T\eta + \eta B = 0$, the infinitesimal Lorentz condition.

17.8.2

(a) Show that the generator in Exercise 17.8.1 produces a Lorentz transformation matrix for rapidity ρ given by

$$U(\rho;\theta) = \begin{pmatrix} \cosh\rho & -\cos\theta\sinh\rho & -\sin\theta\sinh\rho & 0 \\ -\cos\theta\sinh\rho & \sin^2\theta + \cos^2\theta\cosh\rho & \cos\theta\sin\theta(\cosh\rho - 1) & 0 \\ -\sin\theta\sinh\rho & \cos\theta\sin\theta(\cosh\rho - 1) & \cos^2\theta + \sin^2\theta\cosh\rho & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Note. This transformation matrix is symmetric. All single boosts (in any spatial direction) have symmetric transformation matrices.

(b) Verify that the transformation matrix of part (a) is consistent with (1) rotating the spatial coordinates to align the boost direction with a coordinate axis, (2) performing a boost in the direction of that axis using Eq. (17.70), and (3) rotating back to the original coordinate system.

Solution.

Put $c_\theta = \cos\theta$, $s_\theta = \sin\theta$, and let B be the real matrix in 17.8.1, so $S = iB$. Direct multiplication gives

$$B^2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & c_\theta^2 & c_\theta s_\theta & 0 \\ 0 & c_\theta s_\theta & s_\theta^2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad B^3 = B,$$

using $c_\theta^2 + s_\theta^2 = 1$. Thus every positive odd power is B and every positive even power is B^2. Separating the exponential series,

$$\begin{aligned} U(\rho;\theta) = e^{i\rho S} = e^{-\rho B} &= I_4 - B\left(\rho + \frac{\rho^3}{3!} + \cdots\right) + B^2\left(\frac{\rho^2}{2!} + \frac{\rho^4}{4!} + \cdots\right) \\ &= I_4 - B\sinh\rho + B^2(\cosh\rho - 1). \end{aligned}$$

Writing $C = \cosh\rho$, $H = \sinh\rho$ and inserting the entries yields

$$\boxed{U(\rho;\theta) = \begin{pmatrix} C & -c_\theta H & -s_\theta H & 0 \\ -c_\theta H & s_\theta^2 + c_\theta^2 C & c_\theta s_\theta (C-1) & 0 \\ -s_\theta H & c_\theta s_\theta (C-1) & c_\theta^2 + s_\theta^2 C & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}}.$$

For example the xx entry is $1 + c_\theta^2(C-1) = s_\theta^2 + c_\theta^2 C$. The matrix is symmetric. Along the time and $\mathbf{n}$ directions it is the 2×2 block $\begin{pmatrix} C & -H \\ -H & C \end{pmatrix}$; on the two perpendicular directions it is the identity. The relation $C^2 - H^2 = 1$ consequently gives both $U^T\eta U = \eta$ and $\det U = 1$.

(b) Rotate the spatial coordinates so that $\mathbf{n}$ becomes the x direction:

$$R = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & c_\theta & s_\theta & 0 \\ 0 & -s_\theta & c_\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad U_x = \begin{pmatrix} C & -H & 0 & 0 \\ -H & C & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

First applying R, then U_x, then $R^{-1} = R^T$ gives $R^T U_x R$. The first multiplication is

$$U_x R = \begin{pmatrix} C & -Hc_\theta & -Hs_\theta & 0 \\ -H & Cc_\theta & Cs_\theta & 0 \\ 0 & -s_\theta & c_\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Left multiplication by R^T forms c_θ times row 2 minus s_θ times row 3 for the new x row, and s_θ times row 2 plus c_θ times row 3 for the new y row. These are exactly the two middle rows of the boxed matrix. The other rows are unchanged. This verifies the rotation–boost–inverse-rotation construction.

17.8.3 Obtain the Lorentz transformation matrix for a boost of finite amount ρ' in the x direction followed by a finite boost ρ'' in the y direction, with both rapidities real and nonzero. Show that there are no values of ρ and θ that can bring this transformation to the form given in Exercise 17.8.2.

Solution.

Write $C' = \cosh\rho'$, $H' = \sinh\rho'$ and similarly C'', H'' for ρ''. Since the x boost acts first, the product is $U_y(\rho'')U_x(\rho')$:

$$U_yU_x = \begin{pmatrix} C'' & 0 & -H'' & 0 \\ 0 & 1 & 0 & 0 \\ -H'' & 0 & C'' & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} C' & -H' & 0 & 0 \\ -H' & C' & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$= \boxed{\begin{pmatrix} C''C' & -C''H' & -H'' & 0 \\ -H' & C' & 0 & 0 \\ -H''C' & H''H' & C'' & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}}.$$

The spatial xy entry is zero, whereas the yx entry is $H''H'$. For real nonzero ρ', ρ'', this product is nonzero. The matrix is therefore not symmetric. Every single boost in 17.8.2 is symmetric, so no choice of ρ, θ can produce this two-boost matrix.

The impossibility statement requires both rapidities to be nonzero. If $\rho'' = 0$, the result is the single x boost, with $\theta = 0$ and $\rho = \rho'$. If $\rho' = 0$, it is the single y boost, with $\theta = \pi/2$ and $\rho = \rho''$. If both vanish, the result is the identity.

17.9 Lorentz Covariance of Maxwell's Equations

17.9.1 Apply the Lorentz transformation of Eq. (17.80) to $F^{\mu\lambda}$ as given in Eq. (17.79). Verify that the result is a matrix F' whose elements confirm the results given in Eqs. (17.82) and (17.83).

Take a real boost velocity $|v| < c$, with the metric and passive convention of Exercise 17.8.1.

Solution.

Factor out the common ε_0 in Eq. (17.79) and use $\beta = v/c$, $\gamma = (1-\beta^2)^{-1/2} > 0$, so $\gamma^2(1-\beta^2) = 1$. The transformation is $F' = UFU^T$; the particular boost matrix is symmetric, so this is UFU, as in Eq. (17.81). The first multiplication gives

$$\frac{UF}{\varepsilon_0} = \begin{pmatrix} -\beta\gamma E_z & -\gamma E_x + \beta\gamma cB_y & -\gamma E_y - \beta\gamma cB_x & -\gamma E_z \\ E_x & 0 & -cB_z & cB_y \\ E_y & cB_z & 0 & -cB_x \\ \gamma E_z & \beta\gamma E_x - \gamma cB_y & \beta\gamma E_y + \gamma cB_x & \beta\gamma E_z \end{pmatrix}.$$

Right multiplication by U mixes only columns 0 and 3. The six independent entries are therefore

$$\begin{aligned} F'^{01}/\varepsilon_0 &= -\gamma(E_x - \beta cB_y), & F'^{02}/\varepsilon_0 &= -\gamma(E_y + \beta cB_x), \\ F'^{03}/\varepsilon_0 &= (\beta^2\gamma^2 - \gamma^2)E_z = -E_z, & F'^{12}/\varepsilon_0 &= -cB_z, \\ F'^{13}/\varepsilon_0 &= \gamma(cB_y - \beta E_x), & F'^{23}/\varepsilon_0 &= -\gamma(cB_x + \beta E_y). \end{aligned}$$

For instance F'^{13}/ε_0 is row 1 of UF/ε_0 dotted with column 3 of U, namely $E_x(-\beta\gamma) + (cB_y)\gamma$. The remaining off-diagonal entries are their negatives because $(UFU^T)^T = UF^TU^T = -UFU^T$; every diagonal entry is zero.

Define

$$X = \gamma(E_x - \beta c B_y), \quad Y = \gamma(E_y + \beta c B_x), \quad P = \gamma(c B_x + \beta E_y), \quad Q = \gamma(c B_y - \beta E_x).$$

The entire transformed matrix is

$$F' = \varepsilon_0 \begin{pmatrix} 0 & -X & -Y & -E_z \\ X & 0 & -cB_z & Q \\ Y & cB_z & 0 & -P \\ E_z & -Q & P & 0 \end{pmatrix}.$$

Reading it in the same pattern as the original field tensor gives

$$\boxed{\begin{array}{lll} E'_x = \gamma(E_x - vB_y), & E'_y = \gamma(E_y + vB_x), & E'_z = E_z, \\ B'_x = \gamma(B_x + vE_y/c^2), & B'_y = \gamma(B_y - vE_x/c^2), & B'_z = B_z. \end{array}}$$

These are precisely Eqs. (17.82) and (17.83), including their signs.

17.9.2 Confirm that the generalization of Eqs. (17.82) and (17.83) to a boost corresponding to an arbitrary velocity $\mathbf{v}$ with $|\mathbf{v}| < c$ is properly given by Eq. (17.84).

At $\mathbf{v} = 0$, interpret the direction-dependent projections through the continuous zero-velocity limit.

Solution.

For $0 < |\mathbf{v}| < c$, put $\mathbf{n} = \mathbf{v}/|\mathbf{v}|$ and $\gamma = (1 - |\mathbf{v}|^2/c^2)^{-1/2}$. Split the fields into parallel and perpendicular parts:

$$\mathbf{E}_\parallel = (\mathbf{E}\cdot\mathbf{n})\mathbf{n}, \quad \mathbf{E}_\perp = \mathbf{E} - \mathbf{E}_\parallel, \qquad \mathbf{B}_\parallel = (\mathbf{B}\cdot\mathbf{n})\mathbf{n}, \quad \mathbf{B}_\perp = \mathbf{B} - \mathbf{B}_\parallel.$$

Choose a proper spatial rotation taking $\mathbf{n}$ to the z axis. In those coordinates 17.9.1 says that the longitudinal components do not change, while the transverse electric components are $\gamma(E_x - vB_y, E_y + vB_x)$. These are exactly $\gamma(\mathbf{E}_\perp + \mathbf{v}\times\mathbf{B})$. Likewise the transverse magnetic components are $\gamma(\mathbf{B}_\perp - \mathbf{v}\times\mathbf{E}/c^2)$. Proper rotations preserve dot and cross products, so these vector statements remain valid after rotating back.

Adding the unchanged longitudinal parts gives

$$\begin{aligned} \mathbf{E}' &= \mathbf{E}_\parallel + \gamma(\mathbf{E} - \mathbf{E}_\parallel + \mathbf{v}\times\mathbf{B}) \\ &= \boxed{\gamma(\mathbf{E} + \mathbf{v}\times\mathbf{B}) + (1-\gamma)\mathbf{E}_\parallel}, \\ \mathbf{B}' &= \mathbf{B}_\parallel + \gamma(\mathbf{B} - \mathbf{B}_\parallel - \mathbf{v}\times\mathbf{E}/c^2) \\ &= \boxed{\gamma(\mathbf{B} - \mathbf{v}\times\mathbf{E}/c^2) + (1-\gamma)\mathbf{B}_\parallel}. \end{aligned}$$

These are Eq. (17.84), with $\mathbf{E}_\parallel = \mathbf{E}_v$, $\mathbf{B}_\parallel = \mathbf{B}_v$.

At $\mathbf{v} = 0$ the direction $\hat{\mathbf{v}}$ is undefined; the equations mean their continuous limit, $\mathbf{E}' = \mathbf{E}$, $\mathbf{B}' = \mathbf{B}$. An equivalent form makes this endpoint explicit. Since $\gamma^2 - 1 = \gamma^2 v^2/c^2$,

$$\frac{\gamma - 1}{v^2} = \frac{\gamma^2}{(\gamma+1)c^2}.$$

Thus, for all $|\mathbf{v}| < c$, including zero,

$$\boxed{\begin{aligned}\mathbf{E}' &= \gamma(\mathbf{E} + \mathbf{v} \times \mathbf{B}) - \frac{\gamma^2}{\gamma + 1}\frac{\mathbf{v}(\mathbf{v} \cdot \mathbf{E})}{c^2}, \\ \mathbf{B}' &= \gamma(\mathbf{B} - \mathbf{v} \times \mathbf{E}/c^2) - \frac{\gamma^2}{\gamma + 1}\frac{\mathbf{v}(\mathbf{v} \cdot \mathbf{B})}{c^2}.\end{aligned}}$$

Both the longitudinal invariance and the z-boost component formulas follow immediately from this expression.

18 Chapter 18. More Special Functions

Problems

Unless an exercise says otherwise, polynomial degrees and derivative orders in this chapter are nonnegative integers. The associated Laguerre index k is a fixed nonnegative integer. Real and complex extensions are stated separately where needed.

18.1 Hermite Functions

18.1.1 Assume that the Hermite polynomials are known to be solutions of the Hermite ODE, Eq. (18.1). Assume further that the recurrence relation, Eq. (18.3), and the values of $H_n(0)$ are also known. Given the existence of a generating function

$$g(x,t) = \sum_{n=0}^{\infty} H_n(x)\,\frac{t^n}{n!},$$

(a) Differentiate $g(x,t)$ with respect to x and using the recurrence relation develop a first-order PDE for $g(x,t)$.

(b) Integrate with respect to x, holding t fixed.

(c) Evaluate $g(0,t)$ using the known values of $H_n(0)$.

(d) Finally, show that $g(x,t) = \exp(-t^2 + 2tx)$.

Solution.

The manual prints $g(x,0) = f(t)$ in part (c); the evaluation is $g(0,t) = f(t)$.

First derive the derivative identity from precisely the stated data. Write $D = d/dx$ and $L_n = D^2 - 2xD + 2n$. The ODE gives

$$L_n H_{n+1} = -2H_{n+1}, \qquad L_n H_{n-1} = 2H_{n-1},$$

while the product rule gives

$$L_n(xH_n) = xL_nH_n + 2H_n' - 2xH_n = 2H_n' - 2xH_n.$$

Apply L_n to the known identity $H_{n+1} - 2xH_n + 2nH_{n-1} = 0$:

$$\begin{aligned} 0 &= -2H_{n+1} - 4H_n' + 4xH_n + 4nH_{n-1} \\ &= -2(2xH_n - 2nH_{n-1}) - 4H_n' + 4xH_n + 4nH_{n-1} \\ &= 8nH_{n-1} - 4H_n'. \end{aligned}$$

Thus $H_n' = 2nH_{n-1}$ for $n \geq 1$; $H_0' = 0$ because $H_0 = 1$. We have not assumed the derivative formula separately.

(a) Differentiate the proposed generating series, initially as a formal power series:

$$\frac{\partial g}{\partial x} = \sum_{n=1}^{\infty} 2nH_{n-1}(x)\frac{t^n}{n!} = 2t\sum_{m=0}^{\infty} H_m(x)\frac{t^m}{m!} = \boxed{2tg}.$$

(b) Holding t fixed, $\partial_x(e^{-2tx}g) = 0$, so $g(x,t) = e^{2tx}f(t)$. (c) At $x = 0$, the known values are $H_{2m}(0) = (-1)^m(2m)!/m!$ and $H_{2m+1}(0) = 0$. Therefore

$$f(t) = g(0,t) = \sum_{m=0}^{\infty} \frac{(-1)^m(2m)!}{m!}\frac{t^{2m}}{(2m)!} = \sum_{m=0}^{\infty}\frac{(-t^2)^m}{m!} = e^{-t^2}.$$

(d) Substitution gives

$$\boxed{g(x,t) = e^{2tx-t^2}}.$$

The resulting exponential is entire in t for each x, so its coefficient identity also establishes convergence of the initially formal generating series everywhere.

18.1.2 In developing the properties of the Hermite polynomials, start at a number of different points, such as:

1. Hermite's ODE, Eq. (18.1),
2. Rodrigues's formula, Eq. (18.8),
3. Integral representation, Eq. (18.10),
4. Generating function, Eq. (18.2),
5. Gram–Schmidt construction of a complete set of orthogonal polynomials over $(-\infty,\infty)$ with a weighting factor of $\exp(-x^2)$ (Section 5.2).

Outline how you can go from any one of these starting points to all the other points. In the ODE and Gram–Schmidt constructions, select the degree-n polynomial with leading coefficient 2^n.

Solution.

To identify a unique Hermite polynomial, supplement the ODE or Gram–Schmidt construction by degree n and leading coefficient 2^n. The ODE alone also has nonpolynomial solutions, and orthogonality alone does not fix the scale. Here is a closed chain, with both directions to the fifth starting point.

ODE $\longrightarrow$ Rodrigues. Let $w = e^{-x^2}$. Differentiating $w' + 2xw = 0$ exactly $n+1$ times gives

$$w^{(n+2)} + 2xw^{(n+1)} + 2(n+1)w^{(n)} = 0.$$

Put $w^{(n)} = (-1)^n w R_n$. Since $w'/w = -2x$ and $w''/w = 4x^2 - 2$, division by $(-1)^n w$ gives

$$\begin{aligned} 0 &= R_n'' - 4xR_n' + (4x^2-2)R_n + 2x(R_n' - 2xR_n) + 2(n+1)R_n \\ &= R_n'' - 2xR_n' + 2nR_n. \end{aligned}$$

Differentiation of w repeatedly produces w times a polynomial; its highest term is $(-2x)^n$. Thus R_n has degree n and leading coefficient 2^n. Two such polynomial solutions differ by a polynomial of degree $m < n$; applying L_n to its leading term cx^m gives $2(n-m)cx^m$, which cannot vanish unless $c = 0$. Hence

$$H_n = R_n = (-1)^n e^{x^2}\frac{d^n}{dx^n}e^{-x^2}.$$

Rodrigues $\longrightarrow$ generating function. Taylor's theorem for the entire function w gives

$$\sum_{n=0}^{\infty} H_n(x)\frac{t^n}{n!} = e^{x^2}\sum_{n=0}^{\infty}\frac{(-t)^n}{n!}w^{(n)}(x) = e^{x^2}w(x-t) = e^{-t^2+2xt}.$$

Generating function $\longleftrightarrow$ contour integral. Cauchy's coefficient formula on a positively oriented circle about zero is

$$H_n(x) = \frac{n!}{2\pi i} \oint_{|z|=R} \frac{e^{-z^2+2xz}}{z^{n+1}}\, dz.$$

Conversely, multiply by $t^n/n!$ and sum for $|t| < R$. The geometric series converges uniformly on that circle, so

$$\sum_{n=0}^{\infty} H_n(x)\frac{t^n}{n!} = \frac{1}{2\pi i} \oint \frac{e^{-z^2+2xz}}{z-t}\, dz = e^{-t^2+2xt}.$$

Any fixed t is covered by choosing $R > |t|$.

Generating function $\longrightarrow$ ODE. For $g = e^{-t^2+2xt}$, direct differentiation gives

$$g_{xx} - 2xg_x + 2tg_t = [4t^2 - 4xt + 2t(2x - 2t)]g = 0.$$

Its $t^n/n!$ coefficient is $H_n'' - 2xH_n' + 2nH_n = 0$. This closes the first four starting points.

Rodrigues $\longleftrightarrow$ Gram–Schmidt. For every polynomial q of degree less than n, integration by parts n times yields

$$\int_{-\infty}^{\infty} wH_n q\, dx = (-1)^n \int_{-\infty}^{\infty} w^{(n)} q\, dx = \int_{-\infty}^{\infty} wq^{(n)}\, dx = 0.$$

Every boundary term is a polynomial times e^{-x^2} and vanishes. For $q = H_n$, its nth derivative is $2^n n!$, so the same calculation gives

$$\int_{-\infty}^{\infty} wH_n^2\, dx = 2^n n!\sqrt{\pi}.$$

Therefore Rodrigues produces the orthogonal polynomials generated by Gram–Schmidt from $1, x, x^2, \ldots$, with the prescribed leading coefficients. Conversely, two degree-n polynomials with that leading coefficient and orthogonality to all lower degrees have difference d of degree below n. Orthogonality gives $\int wd^2 = 0$, hence $d = 0$. Thus Gram–Schmidt recovers Rodrigues and enters the preceding chain.

Completeness in $L^2(\mathbb{R}, w\,dx)$ can also be justified. If f is orthogonal to every polynomial, set $F(z) = \int_{\mathbb{R}} f(x)w(x)e^{zx}\, dx$. Cauchy–Schwarz bounds this integral and all its differentiated integrals uniformly on compact z sets by Gaussian moments, so F is entire. All $F^{(n)}(0)$ vanish, implying $F = 0$. In particular the Fourier transform of the integrable function fw is zero; Fourier uniqueness gives $fw = 0$ almost everywhere, hence $f = 0$.

18.1.3 Prove that $\big|H_n(x)\big| \le \big|H_n(ix)\big|$. Take x real.

Solution.

The assertion is for real x. It is false for arbitrary complex x: for $n = 2$, $x = i$, the two moduli are 6 and 2. The finite series is

$$H_n(x) = n! \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{(-1)^k (2x)^{n-2k}}{k!(n-2k)!}.$$

For real x, the triangle inequality gives

$$|H_n(x)| \le n! \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{(2|x|)^{n-2k}}{k!(n-2k)!}.$$

On the other hand $(-1)^k i^{n-2k} = i^n$, so

$$H_n(ix) = i^n n! \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{(2x)^{n-2k}}{k!(n-2k)!}.$$

All nonzero powers on the right have the same parity. For $x > 0$ all terms are nonnegative; for $x < 0$ they all have the sign $(-1)^n$. Its modulus is consequently exactly the positive sum in the bound. At $x = 0$ the conclusion follows directly from the constant terms. Thus $\boxed{|H_n(x)| \le |H_n(ix)|}$ for every real x.

18.1.4 Rewrite the series form of $H_n(x)$, Eq. (18.9), as an ascending power series.

$$\textit{ANS. } H_{2n}(x) = (-1)^n \sum_{s=0}^{n} (-1)^s (2x)^{2s} \frac{(2n)!}{(2s)!\,(n-s)!},$$

$$H_{2n+1}(x) = (-1)^n \sum_{s=0}^{n} (-1)^s (2x)^{2s+1} \frac{(2n+1)!}{(2s+1)!\,(n-s)!}.$$

Solution.

Start with $H_N(x) = N! \sum_{k=0}^{\lfloor N/2 \rfloor} (-1)^k (2x)^{N-2k}/[k!(N-2k)!]$. For $N = 2n$, put $s = n - k$. As k goes from 0 to n, s goes from n to 0, and reversing the finite sum gives

$$H_{2n}(x) = (2n)! \sum_{s=0}^{n} \frac{(-1)^{n-s}(2x)^{2s}}{(n-s)!(2s)!}$$

$$= \boxed{(-1)^n \sum_{s=0}^{n} \frac{(-1)^s (2n)!(2x)^{2s}}{(2s)!(n-s)!}}.$$

For $N = 2n + 1$ the same change gives $N - 2k = 2s + 1$, hence

$$\boxed{H_{2n+1}(x) = (-1)^n \sum_{s=0}^{n} \frac{(-1)^s (2n+1)!(2x)^{2s+1}}{(2s+1)!(n-s)!}}.$$

The even formula needs $(-1)^s$, not the textbook's $(-1)^{2s}$. For example the corrected $n = 1$ formula gives $H_2(x) = 4x^2 - 2$, directly checking the relative sign.

18.1.5

(a) Expand x^{2r} in a series of even-order Hermite polynomials.

(b) Expand x^{2r+1} in a series of odd-order Hermite polynomials.

$$\textit{ANS. (a) } x^{2r} = \frac{(2r)!}{2^{2r}} \sum_{n=0}^{r} \frac{H_{2n}(x)}{(2n)!\,(r-n)!}$$

$$\text{(b) } x^{2r+1} = \frac{(2r+1)!}{2^{2r+1}} \sum_{n=0}^{r} \frac{H_{2n+1}(x)}{(2n+1)!\,(r-n)!}, \qquad r = 0, 1, 2, \ldots.$$

Hint. Use a Rodrigues representation and integrate by parts.

Solution.

Because H_n has degree n and leading coefficient 2^n, the polynomials $H_0, \ldots, H_m$ form a basis for polynomials of degree at most m. Write $x^m = \sum_{n=0}^{m} c_n H_n(x)$. Orthogonality gives

$$c_n = \frac{1}{2^n n! \sqrt{\pi}} \int_{-\infty}^{\infty} x^m e^{-x^2} H_n(x)\, dx.$$

By Rodrigues and n integrations by parts, with all Gaussian boundary terms zero,

$$\int_{-\infty}^{\infty} x^m e^{-x^2} H_n(x)\, dx = \frac{m!}{(m-n)!} \int_{-\infty}^{\infty} x^{m-n} e^{-x^2}\, dx.$$

Odd powers integrate to zero. For $m - n = 2j$, substitution $u = x^2$ gives

$$\int_{-\infty}^{\infty} x^{2j} e^{-x^2}\, dx = \Gamma(j + \tfrac{1}{2}) = \frac{(2j)!\sqrt{\pi}}{4^j j!}.$$

Consequently

$$c_n = \frac{m!}{2^n 4^j n! j!} = \frac{m!}{2^m n! j!}, \qquad n = m - 2j.$$

(a) Take $m = 2r$, $n = 2k$ and $j = r - k$:

$$\boxed{x^{2r} = \frac{(2r)!}{2^{2r}} \sum_{k=0}^{r} \frac{H_{2k}(x)}{(2k)!(r-k)!}}.$$

(b) Take $m = 2r + 1$, $n = 2k + 1$ and again $j = r - k$:

$$\boxed{x^{2r+1} = \frac{(2r+1)!}{2^{2r+1}} \sum_{k=0}^{r} \frac{H_{2k+1}(x)}{(2k+1)!(r-k)!}}.$$

These finite identities include $r = 0$, giving $1 = H_0$ and $x = H_1/2$.

18.1.6 Show that

(a) $$\int_{-\infty}^{\infty} H_n(x) \exp\left[-\frac{x^2}{2}\right] dx = \begin{cases} \sqrt{2\pi}\, n!/(n/2)!, & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases}$$

(b) $$\int_{-\infty}^{\infty} x H_n(x) \exp\left[-\frac{x^2}{2}\right] dx = \begin{cases} 0, & n \text{ even}, \\ \sqrt{2\pi}\, \dfrac{(n+1)!}{((n+1)/2)!}, & n \text{ odd}. \end{cases}$$

Solution.

(a) Multiply the generating function by $e^{-x^2/2}$ and integrate. For real t, completion of the square gives

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{t^n}{n!} \int_{-\infty}^{\infty} e^{-x^2/2} H_n(x)\, dx &= \int_{-\infty}^{\infty} e^{-x^2/2 + 2tx - t^2}\, dx \\ &= e^{t^2} \int_{-\infty}^{\infty} e^{-(x-2t)^2/2}\, dx \\ &= \sqrt{2\pi}\, e^{t^2} = \sqrt{2\pi} \sum_{k=0}^{\infty} \frac{t^{2k}}{k!}. \end{aligned}$$

Coefficient comparison gives

$$\boxed{\int_{-\infty}^{\infty} e^{-x^2/2} H_n(x)\,dx = \begin{cases} \sqrt{2\pi}\, n!/(n/2)!, & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases}}$$

The interchange is justified on every bounded t set by a Gaussian bound $C_R e^{-x^2/2+2R|x|}$; differentiating at $t = 0$ gives the same coefficient calculation without a formal interchange.

(b) Inserting x and putting $y = x - 2t$ gives

$$\sum_{n=0}^{\infty} \frac{t^n}{n!} \int_{-\infty}^{\infty} x e^{-x^2/2} H_n(x)\,dx = e^{t^2} \int_{-\infty}^{\infty} (y + 2t) e^{-y^2/2}\,dy$$
$$= 2t\sqrt{2\pi}\, e^{t^2} = \sqrt{2\pi} \sum_{k=0}^{\infty} \frac{2t^{2k+1}}{k!}.$$

The odd Gaussian moment is zero. Thus even n give zero, and for $n = 2k + 1$ the answer is $\sqrt{2\pi}\, 2(2k + 1)!/k!$. Since $2(2k + 1)!/k! = (2k + 2)!/(k + 1)!$, this is

$$\boxed{\int_{-\infty}^{\infty} x e^{-x^2/2} H_n(x)\,dx = \begin{cases} 0, & n \text{ even}, \\ \sqrt{2\pi}\,(n + 1)!/((n + 1)/2)!, & n \text{ odd}. \end{cases}}$$

Both square roots implement the published correction of 2π to $\sqrt{2\pi}$. The case $n = 0$ in (a) is the usual Gaussian integral and already detects the original normalization error.

18.1.7

(a) Using the Cauchy integral formula, develop an integral representation of $H_n(x)$ based on Eq. (18.2) with the contour enclosing the point $z = -x$.

$$ANS.\ H_n(x) = \frac{n!}{2\pi i}\, e^{x^2} \oint \frac{e^{-z^2}}{(z + x)^{n+1}}\,dz.$$

(b) Show by direct substitution that this result satisfies the Hermite equation.

Solution.

(a) Cauchy's coefficient formula applied to the generating function gives

$$H_n(x) = \frac{n!}{2\pi i} \oint_{|w|=R} \frac{e^{-w^2+2xw}}{w^{n+1}}\,dw.$$

Set $z = w - x$. Then $w = z + x$ and $-w^2 + 2xw = x^2 - z^2$. The positively oriented circle now encloses $z = -x$, and hence

$$\boxed{H_n(x) = \frac{n!}{2\pi i} e^{x^2} \oint \frac{e^{-z^2}}{(z + x)^{n+1}}\,dz}.$$

Any positively oriented simple contour enclosing $-x$ gives the same result, since this is the only finite singularity of the integrand.

(b) Use the original w contour, which stays fixed as x varies; its integrand is entire in x, so differentiation under the finite contour integral is valid. Put $E = e^{-w^2+2xw}$. Then

$$H_n' = \frac{n!}{2\pi i}\oint \frac{2wE}{w^{n+1}}\,dw, \qquad H_n'' = \frac{n!}{2\pi i}\oint \frac{4w^2E}{w^{n+1}}\,dw.$$

Combining all three terms of the ODE gives

$$\begin{aligned} H_n'' - 2xH_n' + 2nH_n &= \frac{n!}{2\pi i}\oint \frac{(4w^2 - 4xw + 2n)E}{w^{n+1}}\,dw \\ &= -\frac{2n!}{2\pi i}\oint \frac{d}{dw}\left(\frac{E}{w^n}\right)dw = 0. \end{aligned}$$

Indeed the derivative inside is $(-2w+2x)E/w^n - nE/w^{n+1}$, giving precisely the displayed numerator when multiplied by -2. The integral of this derivative is zero because its primitive is single-valued on the closed contour, even though it can have a pole inside. This proves the equation directly, including $n = 0$.

18.2 Applications of Hermite Functions

18.2.1 Prove that $\left(2x - \dfrac{d}{dx}\right)^n 1 = H_n(x)$.

Hint. Check out the cases $n = 0$ and $n = 1$ and then use mathematical induction (Section 1.4).

Solution.

Let $A = 2x - D$, where $D = d/dx$ acts on everything to its right. The two recurrence identities give

$$AH_n = 2xH_n - H_n' = 2xH_n - 2nH_{n-1} = H_{n+1}.$$

Now $A^0 1 = 1 = H_0$ and $A1 = 2x = H_1$. If $A^n 1 = H_n$, then

$$A^{n+1}1 = A(A^n 1) = AH_n = H_{n+1}.$$

Mathematical induction therefore proves $\boxed{(2x - D)^n 1 = H_n(x)}$ for every nonnegative integer n. The operator power means repeated composition; x and D do not commute, since $D(xf) = f + xDf$.

18.2.2 Show that $\displaystyle\int_{-\infty}^{\infty} x^m e^{-x^2} H_n(x)\,dx = 0$ for m an integer, $0 \le m \le n-1$.

Solution.

The manual's polynomial expansion must start at H_0, not H_1; its final integral is zero. For example, the constant polynomial cannot be expanded using only $H_1, \ldots, H_m$.

For integers $0 \le m < n$, Rodrigues gives

$$\begin{aligned} \int_{-\infty}^{\infty} x^m e^{-x^2} H_n(x)\,dx &= (-1)^n \int_{-\infty}^{\infty} x^m \frac{d^n}{dx^n} e^{-x^2}\,dx \\ &= \int_{-\infty}^{\infty} e^{-x^2} \frac{d^n}{dx^n} x^m\,dx = \boxed{0}. \end{aligned}$$

There are n integrations by parts: their $(-1)^n$ cancels the Rodrigues sign. Every boundary term is a polynomial times a derivative of the Gaussian and vanishes at both infinities. Finally $D^n x^m = 0$ because $n > m$. For $n = 0$ no integer m meets the stated range, so there is no exceptional missing case.

18.2.3 The transition probability between two oscillator states m and n depends on

$$\int_{-\infty}^{\infty} xe^{-x^2} H_n(x) H_m(x)\, dx.$$

Show that this integral equals $\pi^{1/2} 2^{n-1} n!\, \delta_{m,n-1} + \pi^{1/2} 2^n (n+1)!\, \delta_{m,n+1}$. This result shows that such transitions can occur only between states of adjacent energy levels, $m = n \pm 1$.

Hint. Multiply the generating function, Eq. (18.2), by itself using two different sets of variables (x, s) and (x, t). Alternatively, the factor x may be eliminated by the recurrence relation, Eq. (18.3).

Solution.

The recurrence relation can be solved for multiplication by x:

$$xH_n = \tfrac{1}{2} H_{n+1} + nH_{n-1}.$$

Use $\int_{\mathbb{R}} e^{-x^2} H_j H_m\, dx = \sqrt{\pi}\, 2^j j!\, \delta_{jm}$. Multiplying the recurrence by $e^{-x^2} H_m$ and integrating yields

$$\begin{aligned} I_{nm} &= \tfrac{1}{2}\sqrt{\pi}\, 2^{n+1}(n+1)!\, \delta_{m,n+1} + n\sqrt{\pi}\, 2^{n-1}(n-1)!\, \delta_{m,n-1} \\ &= \boxed{\sqrt{\pi}\, 2^{n-1} n!\, \delta_{m,n-1} + \sqrt{\pi}\, 2^n (n+1)!\, \delta_{m,n+1}}. \end{aligned}$$

For $n = 0$, omit the lowering term before writing its factorial: $xH_0 = H_1/2$. The final formula still applies because $\delta_{m,-1} = 0$ for nonnegative m. Thus the position matrix element vanishes unless $m = n \pm 1$. Normalization of the wave functions changes its nonzero magnitude but not these zeros, giving the stated adjacent-level selection rule.

18.2.4 Show that $\displaystyle\int_{-\infty}^{\infty} x^2 e^{-x^2} H_n(x) H_n(x)\, dx = \pi^{1/2} 2^n n! \left(n + \frac{1}{2}\right)$. This integral occurs in the calculation of the mean-square displacement of our quantum oscillator.

Hint. Use the recurrence relation, Eq. (18.3), and the orthogonality integral.

Solution.

Square $xH_n = H_{n+1}/2 + nH_{n-1}$ and integrate with Gaussian weight:

$$\begin{aligned} \int_{\mathbb{R}} x^2 e^{-x^2} H_n^2\, dx &= \frac{1}{4}\int_{\mathbb{R}} e^{-x^2} H_{n+1}^2\, dx + n\int_{\mathbb{R}} e^{-x^2} H_{n+1} H_{n-1}\, dx + n^2 \int_{\mathbb{R}} e^{-x^2} H_{n-1}^2\, dx \\ &= \frac{1}{4}\sqrt{\pi}\, 2^{n+1}(n+1)! + n^2\sqrt{\pi}\, 2^{n-1}(n-1)! \\ &= \sqrt{\pi}\, 2^{n-1} n!\,[(n+1) + n] \\ &= \boxed{\sqrt{\pi}\, 2^n n!\, \left(n + \tfrac{1}{2}\right)}. \end{aligned}$$

The mixed term is zero by orthogonality. At $n = 0$ the terms with H_{-1} are omitted at the first line, leaving $\sqrt{\pi}/2$; this agrees with the final formula. Dividing by the norm $\sqrt{\pi} 2^n n!$ gives $\langle x^2 \rangle_n = n + 1/2$ in the dimensionless oscillator coordinate.

18.2.5 Evaluate

$$\int_{-\infty}^{\infty} x^2 e^{-x^2} H_n(x) H_m(x)\, dx$$

in terms of n and m and appropriate Kronecker delta functions.

ANS. $2^{n-1}\pi^{1/2}(2n+1) n!\, \delta_{nm} + 2^n \pi^{1/2} (n+2)!\, \delta_{n+2,m} + 2^{n-2}\pi^{1/2} n!\, \delta_{n-2,m}$.

Solution.

Apply the multiplication recurrence twice, keeping each coefficient:

$$\begin{aligned} x^2 H_n &= \tfrac{1}{2} x H_{n+1} + n x H_{n-1} \\ &= \tfrac{1}{2}\left(\tfrac{1}{2} H_{n+2} + (n+1) H_n\right) + n\left(\tfrac{1}{2} H_n + (n-1) H_{n-2}\right) \\ &= \tfrac{1}{4} H_{n+2} + \tfrac{2n+1}{2} H_n + n(n-1) H_{n-2}. \end{aligned}$$

Multiplication by $e^{-x^2} H_m$ and integration therefore gives

$$\begin{aligned} \int_{\mathbb{R}} x^2 e^{-x^2} H_n H_m \, dx &= \tfrac{1}{4}\sqrt{\pi} 2^{n+2} (n+2)! \, \delta_{m,n+2} \\ &\quad + \tfrac{2n+1}{2}\sqrt{\pi} 2^n n! \, \delta_{mn} \\ &\quad + n(n-1)\sqrt{\pi} 2^{n-2} (n-2)! \, \delta_{m,n-2} \\ &= \boxed{\begin{aligned} &\sqrt{\pi} 2^{n-1} (2n+1) n! \, \delta_{mn} \\ &+ \sqrt{\pi} 2^n (n+2)! \, \delta_{m,n+2} + \sqrt{\pi} 2^{n-2} n! \, \delta_{m,n-2}. \end{aligned}} \end{aligned}$$

For $n = 0, 1$ the last Hermite term is absent; the last delta in the boxed formula is zero for all allowed $m \geq 0$, so that form covers these cases without a negative factorial. Setting $m = n$ recovers Exercise 18.2.4.

18.2.6 Show that $\displaystyle\int_{-\infty}^{\infty} x^r e^{-x^2} H_n(x) H_{n+p}(x) \, dx = \begin{cases} 0, & p > r \\ 2^n \pi^{1/2} (n+r)!, & p = r, \end{cases}$ with n, p, and r nonnegative integers.

Hint. Apply the multiplication recurrence, Eq. (18.3), r times, to $x^r H_n$.

Solution.

Multiplication by x raises a Hermite index by one with coefficient $1/2$, or lowers it by one with coefficient equal to the current index: $x H_j = H_{j+1}/2 + j H_{j-1}$. Iterating this relation r times gives

$$x^r H_n = 2^{-r} H_{n+r} + \text{a polynomial of degree at most } n + r - 2.$$

To see why the highest coefficient is exactly 2^{-r}, the only way to reach index $n + r$ in r steps is to raise the index at every step; each contributes $1/2$. Any lowering step decreases the final index by two relative to that highest index. This also follows by matching leading terms: $x^r H_n$ has leading term $2^n x^{n+r}$, whereas H_{n+r} has leading term $2^{n+r} x^{n+r}$.

If $p > r$, the polynomial $x^r H_n$ has degree below $n + p$ and is orthogonal to H_{n+p}. If $p = r$, only the displayed highest Hermite term survives the integral. Consequently

$$\boxed{\int_{-\infty}^{\infty} x^r e^{-x^2} H_n H_{n+p} \, dx = \begin{cases} 0, & p > r, \\ 2^{-r}\sqrt{\pi} 2^{n+r} (n+r)! = 2^n \sqrt{\pi} (n+r)!, & p = r. \end{cases}}$$

The case $r = 0, p = 0$ is exactly the normalization integral. All indices are nonnegative integers as required by the question.

18.2.7 With $\psi_n(x) = \dfrac{e^{-x^2/2}H_n(x)}{(2^n n!\,\pi^{1/2})^{1/2}}$, verify that

$$a\,\psi_n(x) = \frac{x+ip}{\sqrt{2}}\,\psi_n(x) = \frac{1}{\sqrt{2}}\left(x + \frac{d}{dx}\right)\psi_n(x) = n^{1/2}\psi_{n-1}(x),$$
$$a^\dagger\psi_n(x) = \frac{x-ip}{\sqrt{2}}\,\psi_n(x) = \frac{1}{\sqrt{2}}\left(x - \frac{d}{dx}\right)\psi_n(x) = (n+1)^{1/2}\psi_{n+1}(x).$$

Note. The usual quantum mechanical operator approach establishes these raising and lowering properties before the form of $\psi_n(x)$ is known. Use dimensionless $p = -i\,d/dx$; at $n = 0$ the lowering formula means $a\psi_0 = 0$.

Solution.

The manual's two reduced ladder expressions also need $e^{-x^2/2}$ in place of e^{-x^2}; its recurrence uses $2xH_n - 2nH_{n-1} = H_{n+1}$.

In the stated dimensionless variables $p = -iD$, so $ip = D$. Thus $a = (x + D)/\sqrt{2}$ and $a^\dagger = (x - D)/\sqrt{2}$. These are the corrected signs and operator actions in the published erratum. Set $N_n = (2^n n!\sqrt{\pi})^{-1/2}$, so $\psi_n = N_n e^{-x^2/2}H_n$. The product rule gives

$$D\psi_n = N_n e^{-x^2/2}(H_n' - xH_n).$$

For lowering, the xH_n terms cancel:

$$\begin{aligned} a\psi_n &= \frac{N_n}{\sqrt{2}}e^{-x^2/2}H_n' = \sqrt{2}\,nN_n e^{-x^2/2}H_{n-1} \\ &= \sqrt{2}\,n\frac{N_n}{N_{n-1}}\psi_{n-1} = \sqrt{2}\,n\frac{1}{\sqrt{2n}}\psi_{n-1} = \boxed{\sqrt{n}\,\psi_{n-1}} \qquad (n \geq 1). \end{aligned}$$

For $n = 0$, $H_0' = 0$, so $a\psi_0 = 0$ directly; no function ψ_{-1} is needed. For raising,

$$\begin{aligned} a^\dagger\psi_n &= \frac{N_n}{\sqrt{2}}e^{-x^2/2}(2xH_n - H_n') \\ &= \frac{N_n}{\sqrt{2}}e^{-x^2/2}H_{n+1} = \frac{N_n}{\sqrt{2}N_{n+1}}\psi_{n+1} \\ &= \frac{\sqrt{2(n+1)}}{\sqrt{2}}\psi_{n+1} = \boxed{\sqrt{n+1}\,\psi_{n+1}}. \end{aligned}$$

The normalization ratios, not only the polynomial recurrences, produce the square-root coefficients.

18.2.8

(a) Verify the operator identity

$$x - ip = x - \frac{d}{dx} = -\exp\left(\frac{x^2}{2}\right)\frac{d}{dx}\exp\left(-\frac{x^2}{2}\right).$$

(b) The normalized simple harmonic oscillator wave function is

$$\psi_n(x) = (\pi^{1/2}2^n n!)^{-1/2}\exp\left(-\frac{x^2}{2}\right)H_n(x).$$

Show that this may be written as

$$\psi_n(x) = (\pi^{1/2}2^n n!)^{-1/2}\left(x - \frac{d}{dx}\right)^n\exp\left(-\frac{x^2}{2}\right).$$

Note. This corresponds to an n-fold application of the raising operator of Exercise 18.2.7. Use the same dimensionless momentum $p = -i\,d/dx$.

Solution.

(a) An operator identity must act on a function. For any differentiable f, the product rule gives

$$\begin{aligned} -e^{x^2/2}D(e^{-x^2/2}f) &= -e^{x^2/2}\left[-xe^{-x^2/2}f + e^{-x^2/2}f'\right] \\ &= xf - f' = (x - D)f. \end{aligned}$$

Since $p = -iD$, $x - ip = x - D$, proving the identity as written with the published correction $x - ip$.

(b) Let M denote multiplication by $e^{x^2/2}$, and M^{-1} multiplication by $e^{-x^2/2}$. Part (a) says $x - D = M(-D)M^{-1}$. Consecutive inverse multiplication operators cancel under composition, so induction gives

$$(x - D)^n = M(-D)^n M^{-1}.$$

Apply this identity to $e^{-x^2/2}$:

$$\begin{aligned} (x - D)^n e^{-x^2/2} &= e^{x^2/2}(-1)^n D^n e^{-x^2} \\ &= e^{x^2/2}e^{-x^2}H_n(x) = e^{-x^2/2}H_n(x), \end{aligned}$$

where the second line is Rodrigues's formula. Therefore

$$\boxed{\psi_n(x) = (\sqrt{\pi}2^n n!)^{-1/2}(x - D)^n e^{-x^2/2}}.$$

Equivalently, $(a^\dagger)^n\psi_0 = \sqrt{n!}\,\psi_n$ follows by multiplying the successive factors $\sqrt{1}, \sqrt{2}, \ldots, \sqrt{n}$. Both derivations include $n = 0$ as the identity operator.

18.3 Laguerre Functions

18.3.1 Show with the aid of the Leibniz formula that the series expansion of $L_n(x)$, Eq. (18.53), follows from the Rodrigues representation, Eq. (18.72).

Solution.

Use Eq. (18.72) with $k = 0$:

$$L_n(x) = \frac{e^x}{n!}D^n(x^n e^{-x}).$$

Leibniz's formula assigns j derivatives to x^n and $n-j$ to the exponential. Since $D^j x^n = n!x^{n-j}/(n-j)!$ and $D^{n-j}e^{-x} = (-1)^{n-j}e^{-x}$, it gives

$$\begin{aligned} L_n(x) &= \frac{e^x}{n!}\sum_{j=0}^{n}\binom{n}{j}\frac{n!}{(n-j)!}x^{n-j}(-1)^{n-j}e^{-x} \\ &= \sum_{j=0}^{n}\frac{(-1)^{n-j}n!}{j!(n-j)!^2}x^{n-j}. \end{aligned}$$

Putting $m = n - j$ reverses the finite sum and yields

$$\boxed{L_n(x) = \sum_{m=0}^{n}\frac{(-1)^m n!}{(n-m)!(m!)^2}x^m}.$$

Its first descending terms are

$$\frac{(-1)^n}{n!}\left[x^n - n^2x^{n-1} + \frac{n^2(n-1)^2}{2!}x^{n-2} - \cdots + (-1)^n n!\right],$$

which is Eq. (18.53). The empty differentiations and single-term sum at $n = 0$ both give $L_0 = 1$.

18.3.2

(a) Using the explicit series form, Eq. (18.53), show that

$$L_n'(0) = -n, \qquad L_n''(0) = \tfrac{1}{2}n(n-1).$$

(b) Repeat without using the explicit series form of $L_n(x)$.

Solution.

(a) In the ascending series, the coefficients of x and x^2 are

$$-\frac{n!}{(n-1)!} = -n, \qquad \frac{n!}{(n-2)!(2!)^2} = \frac{n(n-1)}{4}.$$

A first derivative at zero selects the first coefficient; a second derivative selects twice the second. Thus

$$\boxed{L_n'(0) = -n, \qquad L_n''(0) = \frac{n(n-1)}{2}}.$$

For $n = 0, 1$ the absent higher terms give zero exactly where the formulas predict it.

(b) Use the ODE $xL_n'' + (1-x)L_n' + nL_n = 0$ and the independently known value $L_n(0) = 1$. Setting $x = 0$ gives $L_n'(0) + n = 0$. Differentiate the ODE once:

$$xL_n''' + L_n'' + (1-x)L_n'' - L_n' + nL_n' = 0,$$

or

$$xL_n''' + (2-x)L_n'' + (n-1)L_n' = 0.$$

At zero, $2L_n''(0) + (n-1)(-n) = 0$, giving the second formula. No explicit polynomial series has been used in this second proof.

18.3.3 Derive the normalization relation, Eq. (18.71) for the associated Laguerre polynomials, thereby also confirming Eq. (18.55) for the L_n.

Solution.

For fixed nonnegative integer k, Rodrigues's formula is

$$e^{-x}x^k L_n^k(x) = \frac{1}{n!}D^n(e^{-x}x^{n+k}).$$

Assume first $m \le n$. Integrating by parts n times gives

$$\begin{aligned} I_{mn} &= \int_0^\infty e^{-x}x^k L_n^k L_m^k \, dx \\ &= \frac{(-1)^n}{n!}\int_0^\infty e^{-x}x^{n+k} D^n L_m^k \, dx. \end{aligned}$$

At infinity the boundary terms decay exponentially. At zero, $D^{n-1-j}(e^{-x}x^{n+k}) = O(x^{k+j+1})$ for $0 \le j < n$, so each boundary term also vanishes. If $m < n$, then $D^n L_m^k = 0$. If $m = n$, the leading term of L_n^k is $(-1)^n x^n/n!$, giving $D^n L_n^k = (-1)^n$ and

$$I_{nn} = \frac{1}{n!}\int_0^\infty e^{-x}x^{n+k}\,dx = \frac{\Gamma(n+k+1)}{n!}.$$

Symmetry $I_{mn} = I_{nm}$ supplies the case $m > n$. Therefore

$$\boxed{\int_0^\infty e^{-x}x^k L_n^k(x)L_m^k(x)\,dx = \frac{(n+k)!}{n!}\delta_{mn}}.$$

For $k = 0$ the norm is one, proving Eq. (18.55). The same proof works for real $k > -1$, replacing factorials by gamma functions: the displayed zero-endpoint powers still tend to zero.

18.3.4 Expand x^r in a series of associated Laguerre polynomials $L_n^k(x)$, with k fixed and n ranging from 0 to r when r is a nonnegative integer. For other real $r > -k-1$, use an infinite expansion with the convergence interpretation below.

Hint. The Rodrigues form of $L_n^k(x)$ will be useful.

ANS. $x^r = (r+k)!\,r!\sum_{n=0}^{r}\dfrac{(-1)^n L_n^k(x)}{(n+k)!\,(r-n)!}$, $\quad 0 \le x < \infty$. The displayed finite answer applies to $r = 0, 1, 2, \ldots$. For $r > 0$ the infinite version converges locally uniformly on $[0, \infty)$; for $-k-1 < r < 0$ use its Abel sum on $x > 0$. Weighted L^2 convergence holds precisely when $r > -(k+1)/2$.

Solution.

Take the book's fixed nonnegative integer k. Write $h_n = \Gamma(n+k+1)/n!$ for the norm from Exercise 18.3.3. For a nonnegative integer r, x^r is a finite combination of $L_0^k, \ldots, L_r^k$, since their degrees are distinct and their leading coefficients are nonzero. Its coefficient is

$$c_n = \frac{1}{h_n}\int_0^\infty x^{r+k}e^{-x}L_n^k(x)\,dx.$$

Using Rodrigues and integrating by parts n times gives, for $n \le r$,

$$\begin{aligned} c_n &= \frac{1}{\Gamma(n+k+1)}\int_0^\infty x^r D^n(e^{-x}x^{n+k})\,dx \\ &= \frac{(-1)^n}{\Gamma(n+k+1)}\frac{r!}{(r-n)!}\int_0^\infty e^{-x}x^{r+k}\,dx \\ &= \frac{(-1)^n(r+k)!r!}{(n+k)!(r-n)!}. \end{aligned}$$

For $n > r$ the derivative of x^r is zero, so $c_n = 0$. Every endpoint term is zero: at infinity by the exponential and at zero by a positive power at least $k+1$. Hence

$$\boxed{x^r = (r+k)!r!\sum_{n=0}^{r}\frac{(-1)^n L_n^k(x)}{(n+k)!(r-n)!}}, \qquad r = 0, 1, 2, \ldots.$$

The noninteger extension needs parameter and convergence conditions; a finite upper limit r and ordinary factorials no longer specify it. For real $r > -k-1$ the same integration is valid, even when

intermediate derivatives are singular: every zero-endpoint boundary product has order x^{r+k+1}, and the final integral has power $r+k>-1$. Writing the falling derivative as a polynomial in r gives

$$\boxed{c_n=\frac{\Gamma(r+k+1)(-r)_n}{\Gamma(n+k+1)}}.$$

This form also handles nonnegative integer r without undefined gamma quotients and terminates automatically.

For $r>0$ the resulting infinite series converges absolutely and uniformly on each finite interval $0\le x\le A$, and equals x^r. Here are explicit convergence justifications. The useful bound is

$$|L_n^k(x)|\le e^{x/2}\binom{n+k}{n},\qquad x\ge 0.$$

For $k=0$ it follows from the Gaussian identity

$$e^{-x/2}L_n(x)=\frac{1}{2^n n!\sqrt{\pi}}\int_{\mathbb{R}} e^{-y^2}H_n(y)^2 e^{i\sqrt{2x}\,y}\,dy.$$

Indeed the integral of the nonnegative factor without its phase is one. To verify this identity independently, multiply two Hermite generators and integrate with e^{-y^2+icy}; completion of the square gives $\sqrt{\pi}e^{2uv+ic(u+v)-c^2/4}$. Its u^nv^n coefficient, divided by the Hermite norm, is

$$e^{-c^2/4}\sum_{j=0}^{n}\frac{n!(-c^2/2)^j}{(n-j)!(j!)^2}=e^{-c^2/4}L_n(c^2/2).$$

Set $c=\sqrt{2x}$. For integer $k\ge 1$, multiplication of the ordinary Laguerre generator by $(1-t)^{-k}$ gives

$$L_n^k(x)=\sum_{j=0}^{n}\binom{n-j+k-1}{k-1}L_j(x).$$

The coefficients are positive and sum to $\binom{n+k}{n}$, proving the bound. Consequently

$$|c_nL_n^k(x)|\le\frac{e^{A/2}\Gamma(r+k+1)}{k!}\frac{|(-r)_n|}{n!}.$$

For noninteger $r>0$, the last ratio is $O(n^{-r-1})$ by the gamma-quotient asymptotic; this is summable. For integer r it terminates.

To identify the sum, the Laguerre polynomials are complete in $L^2((0,\infty),x^ke^{-x}dx)$. A brief proof is to take a function f orthogonal to all monomials and form $F(z)=\int_0^\infty f(x)x^ke^{-x}e^{zx}dx$. Cauchy–Schwarz makes F analytic for $\Re z<1/2$, all its derivatives at zero vanish, and the identity theorem makes it zero throughout that half-plane. Fourier uniqueness on $z=i\xi$ then gives $f=0$ almost everywhere. Thus the computed orthogonal expansion converges to x^r in this norm whenever $r>-(k+1)/2$. For $r>0$ the locally uniform convergence already proved identifies its continuous sum at every $x\ge 0$. The case $r=0$ is the single term $L_0^k=1$.

For completeness, negative powers have a precise extension. Put $r=-\beta$ with $0<\beta<k+1$ and $x>0$. The same coefficients give the Abel sum

$$x^{-\beta}=\lim_{q\uparrow 1}\sum_{n=0}^{\infty}c_nq^nL_n^k(x).$$

To prove it rather than assume ordinary pointwise convergence, the beta integral for $(\beta)_n/\Gamma(n+k+1)$ gives, for $0 \le q < 1$,

$$\sum_{n=0}^{\infty} c_n q^n L_n^k(x) = \frac{1}{\Gamma(\beta)} \int_0^1 \frac{u^{\beta-1}(1-u)^{k-\beta}}{(1-qu)^{k+1}} \exp\left(-\frac{xqu}{1-qu}\right) du.$$

The series interchange is uniform because $|qu| \le q < 1$. For $q \ge 1/2$, on $u \ge 1/2$ the exponential bounds the inverse power $(1-qu)^{-k-1}$ uniformly; on $u \le 1/2$ that denominator is bounded directly. Thus an integrable constant multiple of the beta weight dominates the integrand. Dominated convergence and $v = u/(1-u)$ yield

$$\lim_{q\uparrow 1}(\cdots) = \frac{1}{\Gamma(\beta)} \int_0^1 \frac{u^{\beta-1}}{(1-u)^{\beta+1}} e^{-xu/(1-u)} du = \frac{1}{\Gamma(\beta)} \int_0^\infty v^{\beta-1} e^{-xv} dv = x^{-\beta}.$$

If also $\beta < (k+1)/2$, the series has the weighted L^2 interpretation above. For negative powers the endpoint $x = 0$ is not a finite value. If $r \le -k-1$, even the coefficient integrals diverge at zero, so the unrestricted orthogonal-expansion prescription in the question does not apply there.

18.3.5 Expand e^{-ax} in a series of associated Laguerre polynomials $L_n^k(x)$, with k fixed and n ranging from 0 to ∞.

(a) Evaluate directly the coefficients in your assumed expansion.

(b) Develop the desired expansion from the generating function.

ANS. $e^{-ax} = \dfrac{1}{(1+a)^{1+k}} \displaystyle\sum_{n=0}^{\infty} \left(\frac{a}{1+a}\right)^n L_n^k(x), \qquad 0 \le x < \infty.$ Assume $\Re a > -1/2$ for an expansion valid at every $x \ge 0$.

Solution.

For an expansion valid throughout $0 \le x < \infty$, take $\Re a > -1/2$. This is both the weighted L^2 condition and the condition $|a/(1+a)| < 1$ for the generating series.

(a) By orthogonality the coefficient of L_n^k is

$$c_n = \frac{n!}{\Gamma(n+k+1)} \int_0^\infty x^k e^{-(1+a)x} L_n^k(x)\, dx.$$

Rodrigues followed by n integrations by parts gives

$$\begin{aligned} c_n &= \frac{1}{\Gamma(n+k+1)} \int_0^\infty e^{-ax} D^n(e^{-x} x^{n+k})\, dx \\ &= \frac{(-1)^n(-a)^n}{\Gamma(n+k+1)} \int_0^\infty e^{-(1+a)x} x^{n+k}\, dx \\ &= \frac{a^n}{\Gamma(n+k+1)} \frac{\Gamma(n+k+1)}{(1+a)^{n+k+1}} \\ &= \frac{1}{(1+a)^{k+1}} \left(\frac{a}{1+a}\right)^n. \end{aligned}$$

The boundary products vanish at zero as positive powers and at infinity because $\Re(1+a) > 0$. This weaker condition suffices to compute coefficients, but by itself does not ensure the requested series converges.

(b) The polynomial coefficients give

$$L_n^k(x) = \sum_{j=0}^{n} \binom{n+k}{n-j} \frac{(-x)^j}{j!}.$$

Put $n = j + m$ in the generating sum. The binomial series then yields

$$\begin{aligned}
\sum_{n=0}^{\infty} L_n^k(x) t^n &= \sum_{j=0}^{\infty} \frac{(-xt)^j}{j!} \sum_{m=0}^{\infty} \binom{m+j+k}{m} t^m \\
&= \sum_{j=0}^{\infty} \frac{(-xt)^j}{j!} (1-t)^{-j-k-1} \\
&= (1-t)^{-k-1} \exp\!\left(-\frac{xt}{1-t}\right), \qquad |t| < 1.
\end{aligned}$$

For $|t| < 1$ the sum of absolute values is bounded by the same calculation with $|x|, |t|$, so changing the order is justified. Set $t = a/(1+a)$, so $1 - t = 1/(1+a)$ and $t/(1-t) = a$. Division by $(1+a)^{k+1}$ gives

$$\boxed{e^{-ax} = \frac{1}{(1+a)^{k+1}} \sum_{n=0}^{\infty} \left(\frac{a}{1+a}\right)^n L_n^k(x)}.$$

The convergence condition follows from $|1+a|^2 - |a|^2 = 1 + 2\Re a > 0$. The Laguerre bound proved in Exercise 18.3.4 also gives absolute locally uniform convergence. At $a = 0$ only the $n = 0$ term remains. At $\Re a = -1/2$ the terms at $x = 0$ have magnitudes proportional to $\binom{n+k}{n}$ and fail to tend to zero; this prevents including the boundary in a formula asserted for all $x \geq 0$.

18.3.6 Show that $\displaystyle\int_0^{\infty} e^{-x} x^{k+1} L_n^k(x) L_n^k(x)\, dx = \frac{(n+k)!}{n!}\,(2n+k+1)$.

Hint. Note that $xL_n^k = (2n+k+1)L_n^k - (n+k)L_{n-1}^k - (n+1)L_{n+1}^k$.

Solution.

Let $h_n = (n+k)!/n!$. Multiply the supplied recurrence by $x^k e^{-x} L_n^k$ and integrate:

$$\begin{aligned}
\int_0^{\infty} x^{k+1} e^{-x} (L_n^k)^2\, dx &= (2n+k+1) \int_0^{\infty} x^k e^{-x} (L_n^k)^2\, dx \\
&\quad - (n+k) \int_0^{\infty} x^k e^{-x} L_{n-1}^k L_n^k\, dx \\
&\quad - (n+1) \int_0^{\infty} x^k e^{-x} L_{n+1}^k L_n^k\, dx.
\end{aligned}$$

Both mixed integrals vanish by orthogonality, leaving

$$\boxed{\int_0^{\infty} x^{k+1} e^{-x} (L_n^k)^2\, dx = (2n+k+1) \frac{(n+k)!}{n!}}.$$

At $n = 0$ the L_{-1}^k term is absent (equivalently define it as zero); directly the integral is $\Gamma(k+2) = (k+1)k!$, consistent with the formula. The manual's last denominator $k!$ must be $n!$; the question's displayed normalization is correct.

18.3.7 Assume that a particular problem in quantum mechanics has led to the ODE

$$\frac{d^2y}{dx^2} - \left[\frac{k^2-1}{4x^2} - \frac{2n+k+1}{2x} + \frac{1}{4}\right] y = 0$$

for nonnegative integers n, k. Write $y(x)$ as $y(x) = A(x)B(x)C(x)$, with the requirement that

(a) $A(x)$ be a **negative** exponential giving the required asymptotic behavior of $y(x)$, and

(b) $B(x)$ be a **positive** power of x giving the behavior of $y(x)$ for $0 \le x \ll 1$.

Determine $A(x)$ and $B(x)$. Find the relation between $C(x)$ and the associated Laguerre polynomial.

ANS. $A(x) = e^{-x/2}$, $B(x) = x^{(k+1)/2}$, $C(x) = L_n^k(x)$. Select the solution regular at $x = 0$ (excluding the logarithmic branch when $k = 0$).

Solution.

At large positive x, the constant coefficient gives $y'' - y/4 \simeq 0$, with exponential factors $e^{\pm x/2}$. The decaying factor is therefore $A = e^{-x/2}$; powers of x can still multiply it. At zero insert $y \sim x^p$ into the terms of order x^{p-2}:

$$p(p-1) - \frac{k^2-1}{4} = 0, \qquad p = \frac{1 \pm k}{2}.$$

The regular positive-power choice is $p = (k+1)/2$, giving $B = x^{(k+1)/2}$. For $k = 0$ the indicial root is repeated: selecting the regular Frobenius branch also excludes its logarithmic companion. The conditions fix a solution only up to overall normalization.

Set $p = (k+1)/2$ and $g = AB = e^{-x/2}x^p$. Then

$$\frac{g'}{g} = \frac{p}{x} - \frac{1}{2}, \qquad \frac{g''}{g} = \frac{p(p-1)}{x^2} - \frac{p}{x} + \frac{1}{4}.$$

Inserting $y = gC$, dividing by g, and using the indicial equation cancels the x^{-2} and constant terms. What remains is

$$C'' + \left(\frac{2p}{x} - 1\right) C' + \frac{(2n+k+1)/2 - p}{x} C = 0,$$

or

$$xC'' + (k+1-x)C' + nC = 0.$$

This is the associated Laguerre equation. More explicitly, insertion of $C = \sum_{j\ge 0} c_j x^j$ gives

$$(j+1)(j+k+1)c_{j+1} + (n-j)c_j = 0, \quad c_{j+1} = \frac{j-n}{(j+1)(j+k+1)} c_j.$$

For nonnegative integer n it terminates at degree n. Choosing $c_0 = (n+k)!/(n!k!)$ gives exactly $C = L_n^k$; an arbitrary c_0 gives a constant multiple. Thus

$$\boxed{y(x) = \text{constant } e^{-x/2} x^{(k+1)/2} L_n^k(x)}.$$

Its full large-x leading behavior is proportional to $e^{-x/2}x^{n+(k+1)/2}$, since $L_n^k \sim (-1)^n x^n/n!$.

18.3.8 From Eq. (18.84) the normalized radial part of the hydrogenic wave function is

$$R_{nL}(r) = \left[\alpha^3 \frac{(n-L-1)!}{2n(n+L)!}\right]^{1/2} e^{-\alpha r/2}(\alpha r)^L L_{n-L-1}^{2L+1}(\alpha r),$$

in which $\alpha = 2Z/na_0 = 2Zme^2/(4\pi\epsilon_0\hbar^2 n)$. Evaluate

(a) $\langle r \rangle = \int_0^\infty r R_{nL}(r) R_{nL}(r) r^2\, dr,$

(b) $\langle r^{-1} \rangle = \int_0^\infty r^{-1} R_{nL}(r) R_{nL}(r) r^2\, dr.$

The quantity $\langle r \rangle$ is the average displacement of the electron from the nucleus, whereas $\langle r^{-1} \rangle$ is the average of the reciprocal displacement.

ANS. $\langle r \rangle = \dfrac{a_0}{2Z}\left[3n^2 - L(L+1)\right], \qquad \langle r^{-1} \rangle = \dfrac{Z}{n^2 a_0}$. Use $Z > 0$, $a_0 > 0$, $n \geq 1$ and $0 \leq L \leq n-1$, with integer n, L.

Solution.

Equation (18.84) has $e^{-\alpha r/2}$, not the $e^{-\alpha r}$ printed in this exercise. The second expression for α also needs $1/n$. Once R_{nL} is defined as a function of r, the moments use $R_{nL}(r)$, not $R_{nL}(\alpha r)$. The printed answers further assume $Z = 1$; the corrected formulas below retain general Z. Take $Z > 0$, $n = 1, 2, \ldots$, and $L = 0, \ldots, n-1$. Put $q = n - L - 1$, $k = 2L + 1$, $x = \alpha r$ and $h_q = (q+k)!/q!$. The corrected radial probability measure is

$$R_{nL}(r)^2 r^2 dr = \frac{1}{2nh_q} e^{-x} x^{k+1} [L_q^k(x)]^2 dx, \qquad \alpha = \frac{2Z}{na_0}.$$

Since $2q + k + 1 = 2n$, Exercise 18.3.6 shows that its integral is 1, explicitly checking normalization.

(b) Multiplication by $r^{-1} = \alpha/x$ removes the additional power of x. Orthogonality immediately gives

$$\langle r^{-1} \rangle = \frac{\alpha}{2nh_q} \int_0^\infty e^{-x} x^k [L_q^k(x)]^2 dx = \frac{\alpha}{2n} = \boxed{\frac{Z}{n^2 a_0}}.$$

(a) Multiplication by $r = x/\alpha$ instead requires $J_2 = \int_0^\infty e^{-x} x^k [x L_q^k(x)]^2 dx$. Use

$$x L_q^k = (2q + k + 1) L_q^k - (q+k) L_{q-1}^k - (q+1) L_{q+1}^k.$$

Squaring and integrating removes every mixed product. Hence

$$\begin{aligned} \frac{J_2}{h_q} &= (2q+k+1)^2 + (q+k)^2 \frac{h_{q-1}}{h_q} + (q+1)^2 \frac{h_{q+1}}{h_q} \\ &= (2q+k+1)^2 + q(q+k) + (q+1)(q+k+1). \end{aligned}$$

Here $h_{q-1}/h_q = q/(q+k)$ and $h_{q+1}/h_q = (q+k+1)/(q+1)$. For $q = 0$ omit the first neighbor term before taking these ratios; its final contribution $q(q+k)$ is zero. Substituting $q = n - L - 1$ and $k = 2L + 1$ gives

$$\begin{aligned} q(q+k) &= (n-L-1)(n+L) = n^2 - n - L(L+1), \\ (q+1)(q+k+1) &= (n-L)(n+L+1) = n^2 + n - L(L+1), \\ J_2/h_q &= 4n^2 + 2n^2 - 2L(L+1). \end{aligned}$$

Consequently

$$\langle r\rangle = \frac{J_2}{2n\alpha h_q} = \frac{3n^2 - L(L+1)}{n\alpha} = \boxed{\frac{a_0}{2Z}[3n^2 - L(L+1)]}.$$

At $n = 1, L = 0$ these become $3a_0/(2Z)$ and Z/a_0; setting $Z = 1$ gives the two textbook answers.

18.3.9 Derive a recurrence formula for the hydrogen wave function expectation values:

$$\frac{s+2}{n^2}\langle r^{s+1}\rangle - (2s+3)a_0\langle r^s\rangle + \frac{s+1}{4}\Big[(2L+1)^2 - (s+1)^2\Big]a_0^2\langle r^{s-1}\rangle = 0,$$

with $s \geq -2L - 1$.

Hint. Transform Eq. (18.80) into a form analogous to Eq. (18.73). Multiply by $\rho^{s+2}u' - c\rho^{s+1}u$, with $u = \rho\Phi$. Adjust c to cancel terms that do not yield expectation values. Use hydrogen, $Z = 1$, with $n \geq 1$, $0 \leq L \leq n-1$ and real s in the stated range.

Solution.

Use hydrogen, $Z = 1$, and $\rho = \alpha r$ with $\alpha = 2/(na_0)$. Let $u(\rho) = \rho R_{nL}(\rho/\alpha)$; its constant scale does not affect moment ratios. Equation (18.80) is

$$u'' = Q(\rho)u, \qquad Q(\rho) = \frac{1}{4} - \frac{n}{\rho} + \frac{L(L+1)}{\rho^2}.$$

Define

$$M_j = \frac{\int_0^\infty \rho^j u^2 d\rho}{\int_0^\infty u^2 d\rho} = \alpha^j\langle r^j\rangle.$$

The radial function may be chosen real. It behaves as $u = O(\rho^{L+1})$ near zero and as an exponential times a polynomial at infinity.

Put $f = \rho^{s+2}$. Multiply $u'' - Qu = 0$ by $fu' - c\rho^{s+1}u$ and integrate first on a finite interval. The first term produces

$$\int fu'u'' = [fu'^2/2] - \frac{1}{2}\int f'u'^2.$$

For a general multiplier gu the other derivative term is

$$\int guu'' = [guu' - g'u^2/2] - \int gu'^2 + \frac{1}{2}\int g''u^2.$$

Thus the unwanted integral in u'^2 cancels when $g = -f'/2$, or $\boxed{c = (s+2)/2}$. The potential terms satisfy

$$-\int fQuu' + \frac{1}{2}\int f'Qu^2 = [-fQu^2/2] + \frac{1}{2}\int (fQ' + 2f'Q)u^2.$$

Adding these three identities and multiplying by two yields

$$0 = \left[fu'^2 - f'uu' + \frac{f''}{2}u^2 - fQu^2\right]_0^\infty + \int_0^\infty \left(2f'Q + fQ' - \frac{f'''}{2}\right)u^2 d\rho.$$

The zero-endpoint boundary bracket is $O(\rho^{s+2L+2})$; it vanishes under the given condition $s \geq -2L-1$. The lowest moment integrand has order ρ^{s+2L+1} and is integrable there. The upper endpoint vanishes exponentially. These estimates justify taking the limits after the finite-interval integrations by parts.

Now compute the remaining coefficient explicitly:

$$\begin{aligned} 2f'Q + fQ' - \tfrac{1}{2}f''' &= \frac{s+2}{2}\rho^{s+1} - n(2s+3)\rho^{s} \\ &\quad + \left[2(s+1)L(L+1) - \frac{s(s+1)(s+2)}{2}\right]\rho^{s-1} \\ &= \frac{s+2}{2}\rho^{s+1} - n(2s+3)\rho^{s} \\ &\quad + \frac{s+1}{2}\big[(2L+1)^2 - (s+1)^2\big]\rho^{s-1}. \end{aligned}$$

Divide the integral identity by $\int u^2$ and substitute $M_j = \alpha^j \langle r^j \rangle$. Multiplying by $a_0/(n\alpha^s)$ gives, since $\alpha = 2/(na_0)$,

$$\boxed{\begin{aligned} &\frac{s+2}{n^2}\langle r^{s+1}\rangle - (2s+3)a_0\langle r^{s}\rangle \\ &\quad + \frac{s+1}{4}\big[(2L+1)^2 - (s+1)^2\big]a_0^2\langle r^{s-1}\rangle = 0. \end{aligned}}$$

For $s = -1$ this gives $\langle r^{-1}\rangle = 1/(n^2 a_0)$. For $s = 0$, inserting that value gives $\langle r\rangle = a_0[3n^2 - L(L+1)]/2$, checking both moments from Exercise 18.3.8. For a hydrogenic ion replace a_0 by a_0/Z throughout. The proof also permits real $s > -2L-2$, a slightly larger range than the one requested.

18.3.10 Show that $\displaystyle\int_{-\infty}^{\infty} x^n e^{-x^2} H_n(xy)\,dx = \sqrt{\pi}\,n!\,P_n(y)$, where P_n is a Legendre polynomial.

Solution.

The manual drops $1/[2(n+1)y]$ in its integration by parts and e^{-x^2} in the last term of its next recurrence. The subsequent integrals are with respect to x, not y. The finite calculation below also covers $y = 0$ without division by y.

The Hermite sum is finite, so termwise integration has no convergence interchange issue. Insert

$$H_n(xy) = n!\sum_{j=0}^{\lfloor n/2\rfloor} \frac{(-1)^j (2xy)^{n-2j}}{j!(n-2j)!}.$$

After multiplication by x^n, the Gaussian power is $2n-2j$. Its integral is

$$\int_{-\infty}^{\infty} x^{2(n-j)} e^{-x^2}\,dx = \frac{(2n-2j)!\sqrt{\pi}}{4^{n-j}(n-j)!}.$$

Thus, using $2^{n-2j}/4^{n-j} = 2^{-n}$,

$$\int_{-\infty}^{\infty} x^n e^{-x^2} H_n(xy)\,dx = \frac{\sqrt{\pi}\,n!}{2^n}\sum_{j=0}^{\lfloor n/2\rfloor} \frac{(-1)^j (2n-2j)!}{j!(n-j)!(n-2j)!} y^{n-2j}.$$

To identify the polynomial, expand the Legendre Rodrigues formula:

$$\begin{aligned} P_n(y) &= \frac{1}{2^n n!} D_y^n (y^2-1)^n \\ &= \frac{1}{2^n n!}\sum_{j=0}^{n} (-1)^j \binom{n}{j} D_y^n y^{2n-2j} \\ &= \frac{1}{2^n}\sum_{j=0}^{\lfloor n/2\rfloor} \frac{(-1)^j (2n-2j)!}{j!(n-j)!(n-2j)!} y^{n-2j}. \end{aligned}$$

Terms with $2n - 2j < n$ vanish upon differentiation. Comparing the two finite sums proves

$$\boxed{\int_{-\infty}^{\infty} x^n e^{-x^2} H_n(xy)dx = \sqrt{\pi}\, n! P_n(y)}.$$

Both sides are polynomials in y, so the identity also holds for complex y and includes $y = 0$ without division by that parameter.

18.4 Chebyshev Polynomials

18.4.1 By evaluating the generating function for special values of x, verify the special values

$$T_n(1) = 1, \qquad T_n(-1) = (-1)^n, \qquad T_{2n}(0) = (-1)^n, \qquad T_{2n+1}(0) = 0.$$

Solution.

Use the book's type-I generator, whose nonconstant terms carry a factor two:

$$G_T(x,t) = \frac{1-t^2}{1-2xt+t^2} = T_0(x) + 2\sum_{n=1}^{\infty} T_n(x)t^n, \qquad |t| < 1.$$

At the three requested arguments it becomes

$$G_T(1,t) = \frac{1+t}{1-t} = 1 + 2\sum_{n=1}^{\infty} t^n,$$
$$G_T(-1,t) = \frac{1-t}{1+t} = 1 + 2\sum_{n=1}^{\infty} (-1)^n t^n,$$
$$G_T(0,t) = \frac{1-t^2}{1+t^2} = 1 + 2\sum_{j=1}^{\infty} (-1)^j t^{2j}.$$

Equating coefficients, including the constant coefficient $T_0 = 1$, gives

$$\boxed{T_n(1) = 1, \quad T_n(-1) = (-1)^n, \quad T_{2j}(0) = (-1)^j, \quad T_{2j+1}(0) = 0}.$$

All expansions used are ordinary geometric series with $|t| < 1$.

18.4.2 By evaluating the generating function for special values of x, verify the special values

$$U_n(1) = n + 1, \qquad U_n(-1) = (-1)^n(n+1), \qquad U_{2n}(0) = (-1)^n, \qquad U_{2n+1}(0) = 0.$$

Solution.

The type-II generator has no extra factor two:

$$G_U(x,t) = \frac{1}{1-2xt+t^2} = \sum_{n=0}^{\infty} U_n(x)t^n.$$

For $|t| < 1$,

$$G_U(1,t) = (1-t)^{-2} = \sum_{n=0}^{\infty}(n+1)t^n,$$
$$G_U(-1,t) = (1+t)^{-2} = \sum_{n=0}^{\infty}(-1)^n(n+1)t^n,$$
$$G_U(0,t) = (1+t^2)^{-1} = \sum_{j=0}^{\infty}(-1)^j t^{2j}.$$

The first two series follow by differentiating $(1 \mp t)^{-1}$. Comparison gives

$$\boxed{U_n(1) = n+1, \quad U_n(-1) = (-1)^n(n+1), \quad U_{2j}(0) = (-1)^j, \quad U_{2j+1}(0) = 0}.$$

18.4.3 Another Chebyshev generating function is

$$\frac{1-xt}{1-2xt+t^2} = \sum_{n=0}^{\infty} X_n(x)t^n, \qquad |t| < 1.$$

How is $X_n(x)$ related to $T_n(x)$ and $U_n(x)$? Take $-1 \le x \le 1$ for the stated radius $|t| < 1$.

Solution.

Write the proposed generator in two ways:

$$\frac{1-xt}{1-2xt+t^2} = \frac{1}{2}\left[1 + \frac{1-t^2}{1-2xt+t^2}\right] = 1 + \sum_{n=1}^{\infty} T_n(x)t^n,$$
$$\frac{1-xt}{1-2xt+t^2} = (1-xt)\sum_{n=0}^{\infty} U_n(x)t^n$$
$$= U_0(x) + \sum_{n=1}^{\infty}[U_n(x) - xU_{n-1}(x)]t^n.$$

Therefore

$$\boxed{X_n = T_n = U_n - xU_{n-1} \quad (n \ge 1), \qquad X_0 = T_0 = U_0 = 1}.$$

Defining $U_{-1} = 0$ includes $n = 0$ in the first expression. The stated radius $|t| < 1$ applies for real $-1 \le x \le 1$. For other x, restrict t to the distance from zero to the nearest uncancelled pole of the rational generator. The coefficient identities themselves are polynomial identities, so hold for every complex x.

18.4.4 Given

$$(1-x^2)U_n''(x) - 3xU_n'(x) + n(n+2)U_n(x) = 0,$$

show that $V_n(x)$, Eq. (18.106), satisfies

$$(1-x^2)V_n''(x) - xV_n'(x) + n^2V_n(x) = 0,$$

which is Chebyshev's equation. Use $n \geq 1$ and $-1 < x < 1$.

Solution.

On $-1 < x < 1$ put $h = \sqrt{1-x^2}$, $u = U_{n-1}$ and $V_n = hu$, for $n \geq 1$. Since $h' = -x/h$ and $h'' = -1/h^3$,

$$V_n' = hu' - \frac{x}{h}u, \qquad V_n'' = hu'' - \frac{2x}{h}u' - \frac{1}{h^3}u.$$

Substitute these derivatives into the claimed equation:

$$\begin{aligned}(1-x^2)V_n'' - xV_n' + n^2V_n &= h^3u'' - 2xhu' - \frac{u}{h} - xhu' + \frac{x^2u}{h} + n^2hu \\ &= h\big[(1-x^2)u'' - 3xu' + (n^2-1)u\big].\end{aligned}$$

Because $(n-1)((n-1)+2) = n^2-1$, the bracket is exactly the supplied type-II equation at index $n-1$, and is zero. This proves the result. The formulas with h specify the positive root on this real interval; analytic continuation uses a consistent branch.

18.4.5 Show that the Wronskian of $T_n(x)$ and $V_n(x)$ is given by

$$T_n(x)V_n'(x) - T_n'(x)V_n(x) = -\frac{n}{(1-x^2)^{1/2}}.$$

For $n \geq 1$ and $-1 < x < 1$, this verifies that T_n and $V_n = \sin(n \arccos x)$ are independent solutions of Eq. (18.97). If the sine formula is extended to $n = 0$, it gives the zero function. Find an independent zero-order solution.

Solution.

Put $x = \cos\theta$, $0 < \theta < \pi$, so $D_x = -(\sin\theta)^{-1}D_\theta$. For $n \geq 1$,

$$T_n = \cos n\theta, \quad V_n = \sin n\theta, \quad T_n' = \frac{n\sin n\theta}{\sin\theta}, \quad V_n' = -\frac{n\cos n\theta}{\sin\theta}.$$

Thus

$$T_nV_n' - T_n'V_n = -\frac{n(\cos^2 n\theta + \sin^2 n\theta)}{\sin\theta} = \boxed{-\frac{n}{\sqrt{1-x^2}}}.$$

It is nonzero throughout $(-1,1)$, proving independence there. At $n = 0$ this particular family gives $T_0 = 1$ and $\sin(0\theta) = 0$, so its second member has collapsed to zero. The transformed equation is then $d^2y/d\theta^2 = 0$, whose general solution is $A + B\theta$. The missing independent solution is therefore $\arccos x$, equivalently $\arcsin x$ up to a constant and a sign. It also follows from $\lim_{n\to 0}\sin(n\arccos x)/n = \arccos x$, treating the order as continuous in this limit. The independent zero-order solution is distinct from the zero member $V_0 = 0$ of the sine family.

18.4.6 Show that $W_n(x) = (1-x^2)^{-1/2}T_{n+1}(x)$ is a solution of

$$(1-x^2)W_n''(x) - 3xW_n'(x) + n(n+2)W_n(x) = 0.$$

Work on $-1 < x < 1$.

Solution.

Let $h = \sqrt{1-x^2}$ and $T = T_{n+1}$ on $-1 < x < 1$. For $W = T/h$, direct differentiation gives

$$W' = \frac{T'}{h} + \frac{xT}{h^3}, \qquad W'' = \frac{T''}{h} + \frac{2xT'}{h^3} + \frac{(1+2x^2)T}{h^5}.$$

Insert these expressions and collect coefficients:

$$h^2W'' - 3xW' + n(n+2)W = hT'' - \frac{xT'}{h} + \frac{(1+2x^2-3x^2)T}{h^3} + \frac{n(n+2)T}{h}$$
$$= \frac{1}{h}\big[(1-x^2)T'' - xT' + (n+1)^2T\big] = 0.$$

The last bracket is the type-I equation for T_{n+1}. Thus $\boxed{W_n = (1-x^2)^{-1/2}T_{n+1}}$ solves the type-II equation, including $n = 0$. The differential identity is on the open interval; W_n itself is singular at its endpoints.

18.4.7 Evaluate the Wronskian of $U_n(x)$ and $W_n(x) = (1-x^2)^{-1/2}T_{n+1}(x)$. Work on $-1 < x < 1$.

Solution.

Set $a = n+1$, $s = \sin\theta$, $x = \cos\theta$. Then $U_n = \sin(a\theta)/s$ and $W_n = \cos(a\theta)/s$. Their θ derivatives are

$$\frac{dU_n}{d\theta} = \frac{a\cos(a\theta)}{s} - \frac{\sin(a\theta)\cos\theta}{s^2}, \qquad \frac{dW_n}{d\theta} = -\frac{a\sin(a\theta)}{s} - \frac{\cos(a\theta)\cos\theta}{s^2}.$$

In $U_n(dW_n/d\theta) - (dU_n/d\theta)W_n$ the two terms with $\cos\theta$ cancel, leaving $-a/s^2$. Multiplying by $d\theta/dx = -1/s$ gives

$$\boxed{U_nW_n' - U_n'W_n = \frac{n+1}{(1-x^2)^{3/2}}}.$$

It is nonzero for every $n \geq 0$ on $(-1,1)$, so here the second solution remains independent even at the lowest polynomial index.

18.4.8 $V_n(x) = (1-x^2)^{1/2}U_{n-1}(x)$ is not defined for $n = 0$. Show that a second and independent solution of the Chebyshev differential equation for $T_n(x)$ $(n = 0)$ is $V_0(x) = \arccos x$ (or $\arcsin x$).

Solution.

At $n = 0$, the equation is $(1-x^2)y'' - xy' = 0$. Writing $v = y'$ gives

$$\frac{v'}{v} = \frac{x}{1-x^2}, \qquad \ln|v| = -\frac{1}{2}\ln(1-x^2) + \text{constant},$$

so $v = C/\sqrt{1-x^2}$ on $(-1,1)$ and

$$y = A + C\arcsin x = \widetilde{A} + \widetilde{C}\arccos x.$$

The constant solution is $T_0 = 1$. The derivative of $\arccos x$ is $-1/\sqrt{1-x^2}$, so its Wronskian with 1 is nonzero, proving independence. Directly,

$$(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}, \qquad (\arccos x)'' = -\frac{x}{(1-x^2)^{3/2}},$$

and the two terms of the ODE cancel. Calling this solution V_0 is a separate zero-order convention. It must not replace the sine-family value $V_0 = 0$ in recurrences or orthogonality formulas for $V_n = \sin(n\arccos x)$.

18.4.9 Show that $V_n(x)$ satisfies the same three-term recurrence relation as $T_n(x)$, Eq. (18.92). The recurrence holds for $n \geq 2$ with the book's $V_0 = \arccos x$. It also holds at $n = 1$ if instead the sine family is extended by $V_0 = 0$.

Solution.

For the sine family $V_n(x) = \sin(n\theta)$, $x = \cos\theta$, addition of the two angle formulas gives

$$\begin{aligned} V_{n+1} + V_{n-1} &= \sin(n\theta + \theta) + \sin(n\theta - \theta) \\ &= 2\sin(n\theta)\cos\theta = 2xV_n. \end{aligned}$$

Hence $\boxed{V_{n+1} - 2xV_n + V_{n-1} = 0}$. Equivalently, multiply $U_n - 2xU_{n-1} + U_{n-2} = 0$ by $\sqrt{1-x^2}$. At $n = 1$ the recurrence uses $V_0 = 0$, not the independent solution $\arccos x$ introduced in Exercise 18.4.8. With that zero member the recurrence holds for every integer $n \geq 1$; with the separate $\arccos x$ convention it only applies without modification for $n \geq 2$.

18.4.10 Verify the series solutions for $T_n(x)$ and $U_n(x)$, Eqs. (18.114) and (18.115).

Solution.

In the manual's first ODE line, x^2T_n must be n^2T_n. After shifting the derivative sum, its factorial is $(n-2m+2)!$, not $(n-2m)!$. Its evaluation at zero also omits the prefactor $n/2$.

Start with the type-II generator near $t = 0$ and expand geometrically:

$$\begin{aligned} \frac{1}{1-2xt+t^2} &= \sum_{j=0}^{\infty} (2xt - t^2)^j \\ &= \sum_{j=0}^{\infty} \sum_{m=0}^{j} (-1)^m \binom{j}{m} (2x)^{j-m} t^{j+m}. \end{aligned}$$

For a fixed power t^n, $j = n - m$ and $0 \leq m \leq \lfloor n/2 \rfloor$. Its coefficient is therefore

$$\boxed{U_n(x) = \sum_{m=0}^{\lfloor n/2 \rfloor} \frac{(-1)^m (n-m)!}{m!(n-2m)!} (2x)^{n-2m}}.$$

For T_n, use $T_n = U_n - xU_{n-1}$ from Exercise 18.4.3. The coefficient of $(-1)^m(2x)^{n-2m}$ in this difference is

$$\begin{aligned} \frac{(n-m)!}{m!(n-2m)!} - \frac{(n-m-1)!}{2m!(n-2m-1)!} &= \frac{(n-m-1)!}{m!(n-2m)!}\left[n - m - \frac{n-2m}{2}\right] \\ &= \frac{n}{2}\frac{(n-m-1)!}{m!(n-2m)!}. \end{aligned}$$

When $n = 2m$, the second term is absent and the same final expression holds by setting its factor $n - 2m$ to zero before cancellation. Thus for $n \geq 1$,

$$\boxed{T_n(x) = \frac{n}{2} \sum_{m=0}^{\lfloor n/2 \rfloor} \frac{(-1)^m (n-m-1)!}{m!(n-2m)!} (2x)^{n-2m}}.$$

The case $T_0 = 1$ is separate; it must not be evaluated through $0 \cdot (-1)!$. These coefficient extractions prove the two stated series including their normalization.

18.4.11 Transform the series form of $T_n(x)$, Eq. (18.114), into an ascending power series.

$$\textit{ANS. } T_{2n}(x) = (-1)^n n \sum_{m=0}^{n} (-1)^m \frac{(n+m-1)!}{(n-m)!\,(2m)!}\,(2x)^{2m},\ n \geq 1,$$

$$T_{2n+1}(x) = \frac{2n+1}{2} \sum_{m=0}^{n} \frac{(-1)^{m+n}(n+m)!}{(n-m)!\,(2m+1)!}\,(2x)^{2m+1}.$$

Solution.

In the finite series for T_N, change the descending-power index j to $n-m$. For $N = 2n$, with $n \geq 1$, the power is $N - 2j = 2m$, $N - j - 1 = n + m - 1$, and $(-1)^j = (-1)^n(-1)^m$. Thus reversing the finite sum yields

$$\boxed{T_{2n}(x) = (-1)^n n \sum_{m=0}^{n} \frac{(-1)^m (n+m-1)!}{(n-m)!(2m)!}(2x)^{2m}}.$$

For $N = 2n+1$, the corresponding values are $N - 2j = 2m+1$ and $N - j - 1 = n + m$, giving

$$\boxed{T_{2n+1}(x) = \frac{2n+1}{2} \sum_{m=0}^{n} \frac{(-1)^{n+m}(n+m)!}{(n-m)!(2m+1)!}(2x)^{2m+1}}.$$

The even formula starts at $n = 1$; $T_0 = 1$ is retained separately. The odd formula includes $n = 0$, where it gives $T_1 = x$.

18.4.12 Rewrite the series form of $U_n(x)$, Eq. (18.115), as an ascending power series.

$$\textit{ANS. } U_{2n}(x) = (-1)^n \sum_{m=0}^{n} (-1)^m \frac{(n+m)!}{(n-m)!\,(2m)!}\,(2x)^{2m},$$

$$U_{2n+1}(x) = (-1)^n \sum_{m=0}^{n} (-1)^m \frac{(n+m+1)!}{(n-m)!\,(2m+1)!}\,(2x)^{2m+1}.$$

Solution.

Use $U_N = \sum_{j=0}^{\lfloor N/2 \rfloor} (-1)^j (N-j)!(2x)^{N-2j}/[j!(N-2j)!]$. For $N = 2n$ put $j = n - m$, so the numerator factorial becomes $(n+m)!$ and the power becomes $2m$. Hence

$$\boxed{U_{2n}(x) = (-1)^n \sum_{m=0}^{n} \frac{(-1)^m (n+m)!}{(n-m)!(2m)!}(2x)^{2m}}.$$

For $N = 2n+1$ the same substitution gives numerator $(n+m+1)!$ and power $2m+1$, so

$$\boxed{U_{2n+1}(x) = (-1)^n \sum_{m=0}^{n} \frac{(-1)^m (n+m+1)!}{(n-m)!(2m+1)!}(2x)^{2m+1}}.$$

Both include $n = 0$, giving $U_0 = 1$ and $U_1 = 2x$. The sign $(-1)^{n-m}$ was replaced by $(-1)^{n+m}$ because their exponents differ by the even integer $2m$.

18.4.13

(a) From the differential equation for T_n (in self-adjoint form) show that

$$\int_{-1}^{1} \frac{dT_m(x)}{dx}\frac{dT_n(x)}{dx}(1-x^2)^{1/2}\,dx = 0, \qquad m \neq n.$$

(b) Confirm the preceding result by showing that

$$\frac{dT_n(x)}{dx} = nU_{n-1}(x).$$

Solution.

(a) Set $p = \sqrt{1-x^2}$ and $w = 1/p$. The self-adjoint equations are

$$(pT_n')' + n^2 wT_n = 0, \qquad (pT_m')' + m^2 wT_m = 0.$$

Multiplying by T_m, T_n respectively, subtracting and integrating gives

$$[p(T_m T_n' - T_n T_m')]_{-1}^{1} + (n^2 - m^2)\int_{-1}^{1} wT_m T_n\,dx = 0.$$

The boundary term is zero because polynomials and their derivatives are finite there while $p \to 0$. For distinct nonnegative m, n, $n^2 - m^2 \neq 0$, so the integral is zero. Next multiply the first self-adjoint equation by T_m alone and integrate by parts:

$$\int_{-1}^{1} pT_m' T_n'\,dx = [pT_m T_n']_{-1}^{1} + n^2\int_{-1}^{1} wT_m T_n\,dx = \boxed{0}.$$

All endpoint integrals converge since w has only square-root singularities.

(b) With $x = \cos\theta$,

$$T_n'(x) = \frac{-n\sin n\theta}{-\sin\theta} = \boxed{nU_{n-1}(x)} \qquad (n \geq 1).$$

Therefore the derivative integral equals mn times the type-II orthogonality integral for indices $m-1, n-1$, and vanishes for $m \neq n$. If either index is zero, its derivative is zero directly. As a normalization check, the same calculation gives $\int_{-1}^{1} p(T_n')^2 dx = \pi n^2/2$ for $n \geq 1$.

18.4.14 The substitution $x = 2x' - 1$ converts $T_n(x)$ into the shifted Chebyshev polynomials $T_n^*(x')$. Verify that this produces the shifted polynomials shown in Table 18.5 and that they satisfy the orthogonality and normalization relation

$$\int_0^1 T_n^*(x')T_m^*(x')\left[x'(1-x')\right]^{-1/2}dx' = \frac{\delta_{mn}\pi}{2-\delta_{n0}}.$$

Table 18.5 Shifted Type I Chebyshev Polynomials

$T_0^* = 1$
$T_1^* = 2x - 1$
$T_2^* = 8x^2 - 8x + 1$
$T_3^* = 32x^3 - 48x^2 + 18x - 1$
$T_4^* = 128x^4 - 256x^3 + 160x^2 - 32x + 1$
$T_5^* = 512x^5 - 1280x^4 + 1120x^3 - 400x^2 + 50x - 1$
$T_6^* = 2048x^6 - 6144x^5 + 6912x^4 - 3584x^3 + 840x^2 - 72x + 1$

Solution.

Put $y = x'$ and $S_n(y) = T_n(2y-1)$. The recurrence becomes

$$S_{n+1} = (4y-2)S_n - S_{n-1}, \qquad S_0 = 1, \quad S_1 = 2y-1.$$

Successive multiplication and subtraction give

$$\begin{aligned}
S_2 &= (4y-2)(2y-1) - 1 = 8y^2 - 8y + 1,\\
S_3 &= (4y-2)(8y^2-8y+1) - (2y-1)\\
&= 32y^3 - 48y^2 + 18y - 1,\\
S_4 &= (4y-2)(32y^3 - 48y^2 + 18y - 1) - (8y^2 - 8y + 1)\\
&= 128y^4 - 256y^3 + 160y^2 - 32y + 1,\\
S_5 &= (4y-2)S_4 - S_3\\
&= 512y^5 - 1280y^4 + 1120y^3 - 400y^2 + 50y - 1,\\
S_6 &= (4y-2)S_5 - S_4\\
&= 2048y^6 - 6144y^5 + 6912y^4 - 3584y^3 + 840y^2 - 72y + 1.
\end{aligned}$$

For example the y^3 coefficient of S_5 is $4(160) - 2(-256) - 32 = 1120$. The source table's $120y^3$ is therefore $1120y^3$. The integration formula needs a star on both polynomials and the same primed variable throughout. The relation is orthogonality with the stated norms, not unit normalization.

In the integral set $x = 2y-1$, so $dx = 2dy$ and $1 - x^2 = 4y(1-y)$. Therefore

$$\begin{aligned}
\int_0^1 \frac{S_n(y)S_m(y)}{\sqrt{y(1-y)}}\,dy &= \int_{-1}^1 \frac{T_n(x)T_m(x)}{\sqrt{1-x^2}}\,dx\\
&= \int_0^\pi \cos(n\theta)\cos(m\theta)\,d\theta\\
&= \boxed{\frac{\pi\delta_{mn}}{2-\delta_{n0}}}.
\end{aligned}$$

The last line is π for $m = n = 0$, $\pi/2$ for $m = n \geq 1$, and zero for unequal nonnegative indices.

18.4.15 The expansion of a power of x in a Chebyshev series leads to the integral

$$I_{mn} = \int_{-1}^1 x^m T_n(x)\,\frac{dx}{\sqrt{1-x^2}}.$$

(a) Show that this integral vanishes for $m < n$.

(b) Show that this integral vanishes for $m + n$ odd.

Solution.

Assume the power and polynomial indices m, n are nonnegative integers. (a) The polynomials $T_0, \ldots, T_m$ form a basis for all polynomials of degree at most m: $T_0 = 1$ and T_j has nonzero leading coefficient 2^{j-1} for $j \geq 1$. Therefore $x^m = \sum_{j=0}^m c_j T_j$. If $m < n$, orthogonality gives

$$I_{mn} = \sum_{j=0}^m c_j \int_{-1}^1 \frac{T_j T_n}{\sqrt{1-x^2}}dx = \boxed{0},$$

since no index j equals n.

(b) The finite series gives $T_n(-x) = (-1)^n T_n(x)$. Changing x to $-x$ in the integral thus gives $I_{mn} = (-1)^{m+n} I_{mn}$. For odd $m+n$ this says $I_{mn} = -I_{mn}$, so again $\boxed{I_{mn} = 0}$. The weight is integrable at both endpoints, so this symmetry argument applies to ordinary convergent integrals.

18.4.16 Evaluate the integral

$$I_{mn} = \int_{-1}^{1} x^m T_n(x) \, \frac{dx}{\sqrt{1-x^2}}$$

for $m \geq n$ and $m+n$ even by each of two methods:

(a) Replacing $T_n(x)$ by its Rodrigues representation.

(b) Using $x = \cos\theta$ to transform the integral to a form with θ as the variable.

$$\textit{ANS. } I_{mn} = \pi \frac{m!\,(m-n-1)!!}{(m-n)!\,(m+n)!!}, \qquad m \geq n,\ m+n \text{ even}.$$

Solution.

The manual uses the incorrect Rodrigues power $n+1/2$ twice; it must be $n-1/2$. Its final gamma quotient has $((m+n)/2)!$ in the denominator, not $((m-n)/2)!$.

Let $m-n = 2j \geq 0$. (a) The Rodrigues representation gives

$$\frac{T_n(x)}{\sqrt{1-x^2}} = \frac{(-1)^n \sqrt{\pi}}{2^n \Gamma(n+1/2)} D^n (1-x^2)^{n-1/2}.$$

Integrate by parts n times. Each derivative of $(1-x^2)^{n-1/2}$ of order at most $n-1$ tends to zero at both endpoints; this follows from its minimum remaining endpoint exponent $1/2$. Hence

$$\begin{aligned} I_{mn} &= \frac{\sqrt{\pi}}{2^n \Gamma(n+1/2)} \frac{m!}{(m-n)!} \int_{-1}^{1} x^{m-n} (1-x^2)^{n-1/2} dx \\ &= \frac{\sqrt{\pi} m!}{2^n (2j)! \Gamma(n+1/2)} B(j+\tfrac{1}{2}, n+\tfrac{1}{2}) \\ &= \frac{\sqrt{\pi} m! \Gamma(j+1/2)}{2^n (2j)! \Gamma(n+j+1)} \\ &= \boxed{\frac{\pi m!}{2^m j!(n+j)!}}. \end{aligned}$$

The beta integral follows by evenness and $u = x^2$. For $n = 0$ no integration by parts is needed and the same beta calculation applies. Since $(2j-1)!! = (2j)!/(2^j j!)$ and $(m+n)!! = 2^{n+j}(n+j)!$, the answer is equivalently

$$\boxed{I_{mn} = \pi \frac{m!(m-n-1)!!}{(m-n)!(m+n)!!}},$$

with $(-1)!! = 1$ when $m = n$.

(b) Substitute $x = \cos\theta$: $I_{mn} = \int_0^\pi \cos^m\theta \cos n\theta \, d\theta$. The binomial expansion is

$$\cos^m\theta = 2^{-m} \sum_{\ell=0}^{m} \binom{m}{\ell} e^{i(m-2\ell)\theta}.$$

Pair the terms ℓ and $m-\ell$. For $n>0$ the coefficient of $\cos n\theta$ is $2^{1-m}\binom{m}{j}$; multiplication by $\cos n\theta$ and integration multiplies it by $\pi/2$. For $n=0$ the unpaired constant coefficient is $2^{-m}\binom{m}{m/2}$ and its integral is π times that coefficient. Both cases give $\pi 2^{-m}\binom{m}{j} = \pi m!/[2^m j!(n+j)!]$, agreeing with (a).

18.4.17 Establish the following bounds, $-1 \le x \le 1$:

(a) $\big|U_n(x)\big| \le n+1$,

(b) $\left|\dfrac{d}{dx}T_n(x)\right| \le n^2$.

Solution.

(a) For $x=\cos\theta$ with $0<\theta<\pi$, a finite geometric sum gives

$$U_n(\cos\theta) = \frac{\sin((n+1)\theta)}{\sin\theta} = \sum_{j=0}^{n} e^{i(n-2j)\theta}.$$

Indeed summing $e^{in\theta}\sum_{j=0}^{n} e^{-2ij\theta}$ and factoring its numerator and denominator into sines produces the middle expression. There are $n+1$ unit-modulus terms, so $\boxed{|U_n(x)| \le n+1}$. By polynomial continuity the bound also holds at $x=\pm1$, where the values are $n+1$ and $(-1)^n(n+1)$; thus it is sharp.

(b) Exercise 18.4.13 gives $T_n' = nU_{n-1}$ for $n \ge 1$. Applying (a) with index $n-1$ gives

$$\boxed{|T_n'(x)| \le n\cdot n = n^2}, \qquad -1 \le x \le 1.$$

For $n=0$ the derivative is zero, again consistent with the bound. For positive n its magnitude at either endpoint is n^2.

18.4.18

(a) Show that for $-1 \le x \le 1$, $\big|V_n(x)\big| \le 1$.

(b) Show that $W_n(x)$ is unbounded in $-1 \le x \le 1$.

Use $n \ge 1$ for V_n and $n \ge 0$ for W_n; unboundedness refers to approach to the endpoints from inside $(-1,1)$.

Solution.

(a) Use $x=\cos\theta$, $0 \le \theta \le \pi$. Then $V_n(x) = \sin(n\theta)$, so $\boxed{|V_n(x)| \le 1}$. This refers to the sine family ($n \ge 1$, or its zero member $V_0 = 0$), not the separate solution $\arccos x$ at order zero, whose maximum is π.

(b) As $x \to 1^-$, $T_{n+1}(x) \to 1$ and $\sqrt{1-x^2} \sim \sqrt{2(1-x)}$. Thus

$$W_n(x) \sim \frac{1}{\sqrt{2(1-x)}} \longrightarrow +\infty.$$

Similarly, as $x \to -1^+$,

$$W_n(x) \sim \frac{(-1)^{n+1}}{\sqrt{2(1+x)}}.$$

Therefore W_n is unbounded for every nonnegative integer n; it has no finite values at the two endpoints. Possible zeros in the interior cannot remove either endpoint singularity.

18.4.19 Verify the orthogonality-normalization integrals for

(a) $T_m(x)$, $T_n(x)$,

(b) $V_m(x)$, $V_n(x)$,

(c) $U_m(x)$, $U_n(x)$,

(d) $W_m(x)$, $W_n(x)$.

Hint. All these can be converted to trigonometric integrals. For V_m, V_n, take $m, n \geq 1$. The zero-norm entry in Eq. (18.117) uses the optional sine-family extension $V_0 = 0$, not the independent solution $\arccos x$.

Solution.

The second cosine in the manual's part (d) is $\cos((n+1)\theta)$, not a repeated $\cos((m+1)\theta)$.

Put $x = \cos\theta$, $0 < \theta < \pi$; reversing the integration limits gives $dx/\sqrt{1-x^2} = d\theta$ and $\sqrt{1-x^2}\,dx = \sin^2\theta\,d\theta$ in the integrals. The four transformations are

$$\int_{-1}^{1} \frac{T_m T_n}{\sqrt{1-x^2}} dx = \int_0^{\pi} \cos(m\theta)\cos(n\theta) d\theta,$$

$$\int_{-1}^{1} \frac{V_m V_n}{\sqrt{1-x^2}} dx = \int_0^{\pi} \sin(m\theta)\sin(n\theta) d\theta,$$

$$\int_{-1}^{1} \sqrt{1-x^2}\, U_m U_n\, dx = \int_0^{\pi} \sin((m+1)\theta)\sin((n+1)\theta) d\theta,$$

$$\int_{-1}^{1} \sqrt{1-x^2}\, W_m W_n\, dx = \int_0^{\pi} \cos((m+1)\theta)\cos((n+1)\theta) d\theta.$$

For the cosine product use $2\cos a\theta \cos b\theta = \cos((a-b)\theta) + \cos((a+b)\theta)$; for the sine product change the last plus sign to a minus sign. Every nonzero integer-frequency cosine integrates to zero on $[0, \pi]$, whereas the zero-frequency cosine integrates to π. Consequently the four answers are, respectively,

$$\boxed{\frac{\pi\delta_{mn}}{2-\delta_{n0}}}, \qquad \boxed{\frac{\pi}{2}\delta_{mn}(1-\delta_{n0})}, \qquad \boxed{\frac{\pi}{2}\delta_{mn}}, \qquad \boxed{\frac{\pi}{2}\delta_{mn}}.$$

The second formula uses $V_0 = \sin 0 = 0$. Substituting the independent zero-order solution $\arccos x$ would change that normalization and its cross integrals. Although W_n is singular at the endpoints, its weight reduces the displayed integrals to bounded trigonometric integrals, so all four improper integrals converge.

18.4.20 Show whether

(a) $T_m(x)$ and $V_n(x)$ are or are not orthogonal over the interval $[-1, 1]$ with respect to the weighting factor $(1-x^2)^{-1/2}$.

(b) $U_m(x)$ and $W_n(x)$ are or are not orthogonal over the interval $[-1, 1]$ with respect to the weighting factor $(1-x^2)^{1/2}$.

Use $n \geq 1$ in part (a), with $V_n = \sin(n \arccos x)$.

Solution.

For an integer j, define

$$S(j) = \int_0^\pi \sin(j\theta) d\theta = \begin{cases} [1-(-1)^j]/j, & j \neq 0, \\ 0, & j = 0. \end{cases}$$

This formula includes negative integers; in particular $S(-j) = -S(j)$.

(a) With the sine-family convention for V_n,

$$\begin{aligned} A_{mn} &= \int_{-1}^1 \frac{T_m V_n}{\sqrt{1-x^2}} dx = \int_0^\pi \cos(m\theta)\sin(n\theta) d\theta \\ &= \tfrac{1}{2}[S(n+m) + S(n-m)] \\ &= \boxed{\begin{cases} 0, & m+n \text{ even}, \\ \dfrac{2n}{n^2-m^2}, & m+n \text{ odd}. \end{cases}} \end{aligned}$$

The odd-parity case cannot have $m = n$, so its denominator is nonzero. For example $A_{0,1} = 2$. Thus the two families are not mutually orthogonal, although all pairs of equal parity are.

(b) The weight cancels the two sine denominators in $U_m W_n$:

$$\begin{aligned} B_{mn} &= \int_{-1}^1 \sqrt{1-x^2}\, U_m W_n \, dx \\ &= \int_0^\pi \sin((m+1)\theta)\cos((n+1)\theta) d\theta \\ &= \tfrac{1}{2}[S(m+n+2) + S(m-n)] \\ &= \boxed{\begin{cases} 0, & m+n \text{ even}, \\ \dfrac{2(m+1)}{(m+1)^2-(n+1)^2}, & m+n \text{ odd}. \end{cases}} \end{aligned}$$

For example $B_{0,1} = -2/3$, so again there is no mutual orthogonality across the full two families.

18.4.21 Derive

(a) $T_{n+1}(x) + T_{n-1}(x) = 2xT_n(x)$,

(b) $T_{m+n}(x) + T_{m-n}(x) = 2T_m(x)T_n(x)$,

from the "corresponding" cosine identities.

Solution.

For $x = \cos\theta$, use $\cos(A+B) + \cos(A-B) = 2\cos A \cos B$. (a) Set $A = n\theta$, $B = \theta$:

$$T_{n+1}(x) + T_{n-1}(x) = \cos((n+1)\theta) + \cos((n-1)\theta) = \boxed{2xT_n(x)}.$$

(b) Set $A = m\theta$, $B = n\theta$:

$$T_{m+n}(x) + T_{m-n}(x) = \cos((m+n)\theta) + \cos((m-n)\theta) = \boxed{2T_m(x)T_n(x)}.$$

If $m < n$, interpret $T_{m-n} = T_{n-m}$, because cosine is even in its order; one may equivalently write $T_{|m-n|}$ throughout. The identities first hold on $[-1, 1]$, and then for all complex x because their two sides are polynomials agreeing on an interval.

18.4.22 A number of equations relate the two types of Chebyshev polynomials. As examples show that

$$T_n(x) = U_n(x) - xU_{n-1}(x)$$

and

$$(1-x^2)U_n(x) = xT_{n+1}(x) - T_{n+2}(x).$$

The first identity uses $n \geq 1$; the second uses $n \geq 0$.

Solution.

Put $x = \cos\theta$ with $0 < \theta < \pi$. For $n \geq 1$,

$$\begin{aligned} U_n - xU_{n-1} &= \frac{\sin((n+1)\theta) - \cos\theta\sin(n\theta)}{\sin\theta} \\ &= \frac{\cos(n\theta)\sin\theta}{\sin\theta} = \boxed{T_n}. \end{aligned}$$

The sine addition formula gives the second line. For the other identity, expand the cosine of a sum:

$$\begin{aligned} xT_{n+1} - T_{n+2} &= \cos\theta\cos((n+1)\theta) - \cos((n+2)\theta) \\ &= \sin\theta\sin((n+1)\theta) = \boxed{(1-x^2)U_n}. \end{aligned}$$

Continuity supplies the endpoints, and polynomial identity extends both formulas to arbitrary complex x. The second holds for $n \geq 0$; the first also holds for $n = 0$ if $U_{-1} = 0$, in which case it reads $T_0 = U_0 = 1$.

18.4.23 Show that

$$\frac{d}{dx}V_n(x) = -\frac{nT_n(x)}{\sqrt{1-x^2}}$$

(a) using the trigonometric forms of V_n and T_n,

(b) using the Rodrigues representation.

Use $n \geq 1$ and $-1 < x < 1$.

Solution.

(a) For $n \geq 1$, $V_n = \sin(n\theta)$ and $x = \cos\theta$, so

$$V_n'(x) = n\cos(n\theta)\frac{d\theta}{dx} = \boxed{-\frac{nT_n(x)}{\sqrt{1-x^2}}}, \qquad -1 < x < 1.$$

(b) Multiply the Rodrigues formula for U_{n-1} by $\sqrt{1-x^2}$:

$$V_n(x) = \frac{(-1)^{n-1}n\sqrt{\pi}}{2^n\Gamma(n+1/2)}D^{n-1}(1-x^2)^{n-1/2}.$$

Differentiate once, then compare with the type-I Rodrigues formula:

$$\begin{aligned} V_n' &= \frac{(-1)^{n-1}n\sqrt{\pi}}{2^n\Gamma(n+1/2)}D^n(1-x^2)^{n-1/2} \\ &= -n\left[\frac{(-1)^n\sqrt{\pi}}{2^n\Gamma(n+1/2)}D^n(1-x^2)^{n-1/2}\right] = -\frac{nT_n}{\sqrt{1-x^2}}. \end{aligned}$$

The signs and normalization factors agree exactly. At $n = 0$ the sine family has zero derivative, while the separately chosen independent solution $\arccos x$ is not covered by this formula.

18.4.24 Starting with $x = \cos\theta$ and $T_n(\cos\theta) = \cos n\theta$, expand

$$x^k = \left(\frac{e^{i\theta} + e^{-i\theta}}{2}\right)^k$$

and show that

$$x^k = \frac{1}{2^{k-1}}\left[T_k(x) + \binom{k}{1}T_{k-2}(x) + \binom{k}{2}T_{k-4} + \cdots\right],$$

the series in brackets terminating after the term containing T_1 or T_0. (If the series ends with a T_0 term, that term carries an additional factor of 1/2.)

Solution.

For a nonnegative integer k, expand binomially:

$$x^k = 2^{-k}\sum_{j=0}^{k}\binom{k}{j}e^{i(k-2j)\theta}.$$

The terms j and $k-j$ have equal binomial coefficients and opposite frequencies. Each such pair sums to $2\binom{k}{j}\cos((k-2j)\theta) = 2\binom{k}{j}T_{k-2j}(x)$. If k is odd all terms pair. If k is even, $j = k/2$ is an unpaired zero-frequency term. Thus the precise formula is

$$\boxed{x^k = 2^{1-k}\sum_{j=0}^{\lfloor(k-1)/2\rfloor}\binom{k}{j}T_{k-2j}(x) + \begin{cases}2^{-k}\binom{k}{k/2}T_0(x), & k \text{ even},\\ 0, & k \text{ odd}.\end{cases}}$$

The first sum is empty at $k = 0$ and the last term is then 1. Factoring out 2^{1-k} shows that the last T_0 coefficient, when present, has the additional factor 1/2 specified by the published erratum. For example $x^2 = (T_2 + T_0)/2$ and $x^3 = (T_3 + 3T_1)/4$.

18.4.25 Develop the following Chebyshev expansions (for $[-1, 1]$):

(a) $(1-x^2)^{1/2} = \dfrac{2}{\pi}\left[1 - 2\sum_{s=1}^{\infty}(4s^2-1)^{-1}T_{2s}(x)\right]$,

(b) $\begin{cases}+1, & 0 < x \le 1\\ -1, & -1 \le x < 0\end{cases} = \dfrac{4}{\pi}\sum_{s=0}^{\infty}(-1)^s(2s+1)^{-1}T_{2s+1}(x).$

Solution.

For $F(\theta) = f(\cos\theta)$, use the cosine-series convention

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}a_nT_n(x), \qquad a_n = \frac{2}{\pi}\int_0^{\pi}F(\theta)\cos(n\theta)d\theta.$$

(a) Here $F(\theta) = \sin\theta$ on $[0, \pi]$. The constant is $a_0 = 4/\pi$, and reflection $\theta \mapsto \pi - \theta$ makes every odd coefficient zero. For $s \ge 1$, the product-to-sum formula gives

$$\begin{aligned}\int_0^{\pi}\sin\theta\cos(2s\theta)d\theta &= \frac{1}{2}\int_0^{\pi}[\sin((2s+1)\theta) + \sin((1-2s)\theta)]d\theta\\ &= \frac{1}{2s+1} + \frac{1}{1-2s} = -\frac{2}{4s^2-1}.\end{aligned}$$

Thus $a_{2s} = -4/[\pi(4s^2-1)]$ and

$$\boxed{\sqrt{1-x^2} = \frac{2}{\pi}\left[1 - 2\sum_{s=1}^{\infty}\frac{T_{2s}(x)}{4s^2-1}\right]}.$$

The coefficients are absolutely summable and $|T_n(x)| \le 1$ on $[-1,1]$, so the series converges uniformly, including both endpoints. Its cosine coefficients identify the continuous function there.

(b) Now $F = 1$ on $0 \le \theta < \pi/2$ and $F = -1$ on $\pi/2 < \theta \le \pi$. Its average is zero. For $n \ge 1$,

$$\begin{aligned} a_n &= \frac{2}{\pi}\left[\int_0^{\pi/2}\cos(n\theta)\,d\theta - \int_{\pi/2}^{\pi}\cos(n\theta)\,d\theta\right] \\ &= \frac{4}{\pi n}\sin\frac{n\pi}{2}. \end{aligned}$$

This vanishes for even n, and equals $4(-1)^s/[\pi(2s+1)]$ for $n = 2s+1$. Therefore

$$\boxed{\operatorname{sgn} x = \frac{4}{\pi}\sum_{s=0}^{\infty}\frac{(-1)^s}{2s+1}T_{2s+1}(x)}, \qquad -1 \le x \le 1,\ x \ne 0.$$

The usual Fourier convergence argument for a piecewise constant function gives its value at each continuity point, including the endpoints of this cosine expansion. At the jump $x = 0$ the series equals the midpoint value 0. In fact every odd $T_{2s+1}(0)$ is zero, so all partial sums already have that value; the series is not uniformly convergent across the jump.

18.4.26

(a) For the interval $[-1,1]$ show that

$$\begin{aligned} |x| &= \frac{1}{2} + \sum_{s=1}^{\infty}(-1)^{s+1}\frac{(2s-3)!!}{(2s+2)!!}(4s+1)P_{2s}(x) \\ &= \frac{2}{\pi} + \frac{4}{\pi}\sum_{s=1}^{\infty}(-1)^{s+1}\frac{1}{4s^2-1}T_{2s}(x). \end{aligned}$$

(b) Show that the ratio of the coefficient of $T_{2s}(x)$ to that of $P_{2s}(x)$ approaches $(\pi s)^{-1/2}$ as $s \to \infty$. This illustrates the relatively rapid convergence of the Chebyshev series.

Hint. With the Legendre recurrence relations, rewrite $xP_n(x)$ as a linear combination of derivatives. The trigonometric substitution $x = \cos\theta$, $T_n(x) = \cos n\theta$ is most helpful for the Chebyshev part.

Solution.

(a) For the Legendre expansion, parity removes all odd degrees and the constant coefficient is $\frac{1}{2}\int_{-1}^{1}|x|dx = 1/2$. At degree $N = 2s \ge 2$ the coefficient is $(2N+1)J_N$, where $J_N = \int_0^1 xP_N(x)dx$. Multiply the self-adjoint Legendre equation $[(1-x^2)P_N']' + N(N+1)P_N = 0$ by x and integrate:

$$0 = [x(1-x^2)P_N']_0^1 - \int_0^1 (1-x^2)P_N'\,dx + N(N+1)J_N.$$

The boundary term is zero, while another integration by parts gives

$$\int_0^1 (1-x^2)P_N' dx = [(1-x^2)P_N]_0^1 + 2J_N = -P_N(0) + 2J_N.$$

Consequently

$$J_N = -\frac{P_N(0)}{(N-1)(N+2)}.$$

Use $P_{2s}(0) = (-1)^s(2s-1)!!/(2s)!!$ and cancel $2s-1$ and $2s+2$:

$$J_{2s} = (-1)^{s+1}\frac{(2s-3)!!}{(2s+2)!!}.$$

It follows that

$$\boxed{|x| = \frac{1}{2} + \sum_{s=1}^{\infty}(-1)^{s+1}(4s+1)\frac{(2s-3)!!}{(2s+2)!!}P_{2s}(x)}.$$

For the Chebyshev expansion, $F(\theta) = |\cos\theta|$ has $a_0 = 4/\pi$ and zero odd coefficients. For $s \geq 1$,

$$\begin{aligned} a_{2s} &= \frac{4}{\pi}\int_0^{\pi/2} \cos\theta\cos(2s\theta)d\theta \\ &= \frac{2}{\pi}\left[\frac{\sin((2s-1)\pi/2)}{2s-1} + \frac{\sin((2s+1)\pi/2)}{2s+1}\right] \\ &= \frac{2}{\pi}(-1)^{s+1}\left[\frac{1}{2s-1} - \frac{1}{2s+1}\right] = \frac{4}{\pi}\frac{(-1)^{s+1}}{4s^2-1}. \end{aligned}$$

Hence

$$\boxed{|x| = \frac{2}{\pi} + \frac{4}{\pi}\sum_{s=1}^{\infty}\frac{(-1)^{s+1}}{4s^2-1}T_{2s}(x)}.$$

Both expansions are valid on the whole closed interval. In the Legendre case the coefficients are $O(s^{-3/2})$ as shown below, and $|P_n(x)| \leq 1$ on this interval; in the Chebyshev case they are $O(s^{-2})$ and $|T_n(x)| \leq 1$. Thus both are absolutely uniformly convergent, and orthogonal completeness identifies their continuous sums with $|x|$.

(b) Convert the double-factorial ratio to gamma functions:

$$\frac{(2s-3)!!}{(2s+2)!!} = \frac{\Gamma(s-1/2)}{4\sqrt{\pi}\,\Gamma(s+2)} \sim \frac{1}{4\sqrt{\pi}}\,s^{-5/2}.$$

Therefore the Legendre and Chebyshev coefficients satisfy

$$A_s \sim \frac{(-1)^{s+1}}{\sqrt{\pi}\,s^{3/2}}, \qquad B_s \sim \frac{(-1)^{s+1}}{\pi s^2},$$

and their ratio is

$$\boxed{\frac{B_s}{A_s} \sim \frac{1}{\sqrt{\pi s}}}.$$

This is the published exponent correction from -1 to $-1/2$. It compares the large-order coefficients; the two expansions use different orthogonality weights.

18.4.27 Show that

$$\frac{\pi^2}{8} = 1 + 2\sum_{s=1}^{\infty}(4s^2-1)^{-2}.$$

Hint. Apply Parseval's identity (or the completeness relation) to the results of Exercise 18.4.26.

Solution.

Use the uniformly convergent Chebyshev expansion of $|x|$ from Exercise 18.4.26. Orthogonality eliminates the mixed terms upon squaring and integrating with weight $(1-x^2)^{-1/2}$. Thus

$$\begin{aligned}\int_{-1}^{1}\frac{x^2}{\sqrt{1-x^2}}dx &= \left(\frac{2}{\pi}\right)^2\pi + \sum_{s=1}^{\infty}\left(\frac{4}{\pi(4s^2-1)}\right)^2\frac{\pi}{2}\\ &= \frac{4}{\pi} + \frac{8}{\pi}\sum_{s=1}^{\infty}(4s^2-1)^{-2}.\end{aligned}$$

The integral on the left is $\int_0^{\pi}\cos^2\theta\,d\theta = \pi/2$. Multiplication by $\pi/4$ gives

$$\boxed{\frac{\pi^2}{8} = 1 + 2\sum_{s=1}^{\infty}(4s^2-1)^{-2}}.$$

Uniform convergence, or equivalently the weighted L^2 Parseval identity, justifies the limiting squared integral.

18.4.28 Show that

(a) $\cos^{-1}x = \dfrac{\pi}{2} - \dfrac{4}{\pi}\displaystyle\sum_{n=0}^{\infty}\frac{1}{(2n+1)^2}T_{2n+1}(x)$,

(b) $\sin^{-1}x = \dfrac{4}{\pi}\displaystyle\sum_{n=0}^{\infty}\frac{1}{(2n+1)^2}T_{2n+1}(x)$.

Use principal real inverse trigonometric functions on $-1 \le x \le 1$.

Solution.

The manual's intermediate constant coefficient is $\pi/2$, not $\pi^2/2$. Its final expansion has the correct constant.

(a) Put $x = \cos\theta$, $0 \le \theta \le \pi$, so $\arccos x = \theta$. The cosine coefficients of this linear function are

$$a_0 = \frac{2}{\pi}\int_0^{\pi}\theta\,d\theta = \pi,$$

and, for $n \ge 1$, integration by parts gives

$$\begin{aligned}a_n &= \frac{2}{\pi}\int_0^{\pi}\theta\cos(n\theta)d\theta\\ &= \frac{2}{\pi}\left[\frac{\theta\sin(n\theta)}{n} + \frac{\cos(n\theta)}{n^2}\right]_0^{\pi} = \frac{2}{\pi}\frac{(-1)^n-1}{n^2}.\end{aligned}$$

Even coefficients vanish, and for $n = 2j+1$ the result is $-4/[\pi(2j+1)^2]$. Therefore

$$\boxed{\arccos x = \frac{\pi}{2} - \frac{4}{\pi}\sum_{j=0}^{\infty} \frac{T_{2j+1}(x)}{(2j+1)^2}}.$$

The coefficients are absolutely summable. The cosine expansion of the continuous function θ therefore converges uniformly on this interval and is valid at both endpoints as well.

(b) The principal real inverse functions satisfy $\arcsin x + \arccos x = \pi/2$ for $-1 \le x \le 1$. Subtracting the first expansion from $\pi/2$ yields

$$\boxed{\arcsin x = \frac{4}{\pi}\sum_{j=0}^{\infty} \frac{T_{2j+1}(x)}{(2j+1)^2}}.$$

At $x = 0$ all terms are zero; at $x = \pm 1$ the sums give $\pm\pi/2$, using $\sum_{j\ge 0}(2j+1)^{-2} = \pi^2/8$.

18.5 Hypergeometric Functions

18.5.1

(a) For c, an integer, and a and b nonintegral, show that

$$ {}_2F_1(a,b;c;x) \quad \text{and} \quad x^{1-c}\,{}_2F_1(a+1-c,b+1-c;2-c;x)$$

provide only one usable series formula for a solution to the hypergeometric equation (counting coincident formulas once).

(b) What happens if a is an integer, say, $a = -1$, and $c = -2$?

Solution.

(a) Write $F(a,b;c;x) = {}_2F_1(a,b;c;x)$. Its coefficient recurrence, obtained by substituting $y = \sum_{j\ge 0} d_j x^j$ into $x(1-x)y'' + [c-(a+b+1)x]y' - aby = 0$, is

$$(j+1)(j+c)d_{j+1} = (j+a)(j+b)d_j.$$

The two Frobenius exponents at zero are 0 and $1-c$. For nonintegral a, b, neither numerator series terminates. If $c = 1$, the two displayed expressions coincide term for term. If $c = 0, -1, -2, \ldots$, the first series encounters $(c)_j = 0$ with nonzero numerator, while the second has the positive integer denominator parameter $2-c$ and is well defined near zero. If $c = 2, 3, \ldots$, the first series is well defined and the second encounters the nonpositive integer parameter $2-c$. Thus precisely one of these two unmodified formulas is usable (or they are identical when $c = 1$).

This does not mean that the second-order ODE has only one independent solution. Nor does a zero denominator parameter make the hypergeometric series zero, as the paragraph following Eq. (18.121) states: it makes the generic series undefined. On a subinterval avoiding $0, 1$ and zeros of a first solution y_1, reduction of order constructs a second solution:

$$y_2 = y_1 \int^x \frac{s^{-c}(1-s)^{c-a-b-1}}{y_1(s)^2}\,ds.$$

Indeed the ODE's first-derivative coefficient is $c/x + (c-a-b-1)/(1-x)$, so the numerator in this integral is $\exp(-\int p(x)dx)$. Its Wronskian is that nonzero numerator. Initial-value uniqueness then extends the independent pair across any zeros of y_1 to the whole interval avoiding $0, 1$. The resonant Frobenius construction generally introduces a logarithm; the loss is in the displayed pair of formulas, not the dimension of the solution space.

(b) For $a = -1, c = -2$, termination occurs before the zero denominator:

$$y_1 = 1 + \frac{(-1)b}{-2}x = 1 + \frac{b}{2}x, \qquad y_2 = x^3 F(2, b+3; 4; x).$$

The equation becomes

$$x(1-x)y'' + (-2-bx)y' + by = 0.$$

Direct substitution of $y_1' = b/2$, $y_1'' = 0$ gives $(-2-bx)b/2 + b(1+bx/2) = 0$. For the other solution, set $y = x^3 u$ and differentiate: $y' = 3x^2 u + x^3 u'$, $y'' = 6xu + 6x^2 u' + x^3 u''$. After substitution and division by x^3, the equation is

$$x(1-x)u'' + [4 - (b+6)x]u' - 2(b+3)u = 0,$$

which is the equation for $F(2, b+3; 4; x)$. Since $y_1 = 1 + O(x)$ and $y_2 = x^3 + O(x^4)$, they are independent. The coefficient recurrence also shows exactly the extra freedom: $d_1 = bd_0/2$, $d_2 = 0$, and at $j = 2$ both sides vanish, leaving d_3 free. Here $F(-1, b; -2; x)$ means the explicitly terminating polynomial; a generic parameter limit need not select the same solution without a specified limiting prescription.

18.5.2 Find the Legendre, Chebyshev I, and Chebyshev II recurrence relations corresponding to the hypergeometric contiguous function relation given as Eq. (18.126).

Solution.

To avoid confusing the hypergeometric argument with the polynomial variable, put $u = (1-z)/2$ and write the contiguous relation as $AF(a, b; c; u) = BF(a-1, b+1; c; u) + CF(a+1, b-1; c; u)$. Substitution into Eq. (18.126) gives the following coefficients:

(a, b, c)	A	B	C
$(-n, n+1, 1)$	$-2n(n+1)(2n+1)z$	$-2n(n+1)^2$	$-2n^2(n+1)$
$(-n, n, \frac{1}{2})$	$-n(4n^2-1)z$	$-n(4n^2-1)/2$	$-n(4n^2-1)/2$
$(-n, n+2, \frac{3}{2})$	$-(n+1)(2n+1)(2n+3)z$	$-(n+2)(2n+1)(2n+3)/2$	$-n(2n+1)(2n+3)/2$

For example, in the Legendre row $a - b = -2n - 1$, and the expression in braces on the left is

$$1 - n^2 - (n+1)^2 + 4n(n+1)(1-u) = 2n(n+1)(1-2u) = 2n(n+1)z.$$

This gives the first entry and illustrates the substitution used for each row.

For Legendre, the three functions are P_n, P_{n+1}, P_{n-1}. Divide the first row by $-2n(n+1)$ for $n \geq 1$:

$$\boxed{(n+1)P_{n+1}(z) - (2n+1)zP_n(z) + nP_{n-1}(z) = 0}.$$

At $n = 0$ the original contiguous equation degenerates to $0 = 0$; the recurrence separately follows from $P_0 = 1, P_1 = z$.

For Chebyshev I the three functions are T_n, T_{n+1}, T_{n-1}. Divide the second row by $-n(4n^2-1)/2$, for $n \geq 1$, to get

$$\boxed{T_{n+1}(z) - 2zT_n(z) + T_{n-1}(z) = 0}.$$

The initial polynomials are $T_0 = 1, T_1 = z$. If the cosine extension $T_{-1} = T_1$ is adopted, the recurrence also holds at zero.

For Chebyshev II, include the normalization factors:

$$F(-n, n+2; \tfrac{3}{2}; u) = \frac{U_n(z)}{n+1}, \quad F(-n-1, n+3; \tfrac{3}{2}; u) = \frac{U_{n+1}(z)}{n+2}.$$

For $n \geq 1$ the last function is U_{n-1}/n. In the third row the factors $n+1, n+2, n$ therefore cancel individually, leaving

$$-z(2n+1)(2n+3)U_n = -\frac{(2n+1)(2n+3)}{2}(U_{n+1} + U_{n-1}),$$

so

$$\boxed{U_{n+1}(z) - 2zU_n(z) + U_{n-1}(z) = 0}.$$

At $n = 0$ the last contiguous coefficient is zero; with $U_{-1} = 0$ it gives $U_1 = 2zU_0$, consistent with $U_0 = 1$.

18.5.3 Transform the following polynomials into hypergeometric functions of argument x^2:

(a) $T_{2n}(x)$;

(b) $x^{-1}T_{2n+1}(x)$;

(c) $U_{2n}(x)$;

(d) $x^{-1}U_{2n+1}(x)$.

ANS. (a) $T_{2n}(x) = (-1)^n\, {}_2F_1\big(-n, n; \tfrac{1}{2}; x^2\big)$.
(b) $x^{-1}T_{2n+1}(x) = (-1)^n(2n+1)\, {}_2F_1\big(-n, n+1; \tfrac{3}{2}; x^2\big)$.
(c) $U_{2n}(x) = (-1)^n\, {}_2F_1\big(-n, n+1; \tfrac{1}{2}; x^2\big)$.
(d) $x^{-1}U_{2n+1}(x) = (-1)^n(2n+2)\, {}_2F_1\big(-n, n+2; \tfrac{3}{2}; x^2\big)$. Interpret the quotients at $x = 0$ by their removable limits.

Solution.

Take $n = 0, 1, 2, \ldots$ and $t = x^2$. For an even function $Y(x) = f(t)$,

$$Y' = 2xf', \qquad Y'' = 2f' + 4tf''.$$

For an odd function $Y(x) = xf(t)$,

$$Y' = f + 2tf', \qquad Y'' = x(6f' + 4tf'').$$

Substitution into the type-I equation $(1-x^2)Y'' - xY' + N^2Y = 0$ gives, respectively,

$$t(1-t)f'' + (\tfrac{1}{2} - t)f' + \frac{N^2}{4}f = 0,$$

$$t(1-t)f'' + (\tfrac{3}{2} - 2t)f' + \frac{N^2-1}{4}f = 0.$$

With $N = 2n$ in the first and $N = 2n+1$ in the second, comparison with $t(1-t)f'' + [c - (a+b+1)t]f' - abf = 0$ gives $(a, b, c) = (-n, n, \frac{1}{2})$ and $(-n, n+1, \frac{3}{2})$.

For type II, use $(1-x^2)Y''-3xY'+N(N+2)Y=0$ instead. The corresponding equations are

$$t(1-t)f''+(\tfrac{1}{2}-2t)f'+\frac{N(N+2)}{4}f=0,$$

$$t(1-t)f''+(\tfrac{3}{2}-3t)f'+\frac{N(N+2)-3}{4}f=0.$$

Their parameters are $(-n,n+1,\frac{1}{2})$ and $(-n,n+2,\frac{3}{2})$. All four analytic solutions at $t=0$ are uniquely determined by their constant term, because c is positive and the coefficient recurrence has no zero denominator. The needed constants are

$$T_{2n}(0)=U_{2n}(0)=(-1)^n,\quad T'_{2n+1}(0)=(-1)^n(2n+1),\quad U'_{2n+1}(0)=(-1)^n(2n+2).$$

For example, with $x=\cos\theta$, $d/dx=-\csc\theta\,d/d\theta$, these follow at $\theta=\pi/2$ from $T_N=\cos N\theta$ and $U_N=\sin((N+1)\theta)/\sin\theta$. Since $F(a,b;c;0)=1$, the four answers are

$$T_{2n}(x)=\boxed{(-1)^nF(-n,n;\tfrac{1}{2};x^2)},$$

$$\frac{T_{2n+1}(x)}{x}=\boxed{(-1)^n(2n+1)F(-n,n+1;\tfrac{3}{2};x^2)},$$

$$U_{2n}(x)=\boxed{(-1)^nF(-n,n+1;\tfrac{1}{2};x^2)},$$

$$\frac{U_{2n+1}(x)}{x}=\boxed{(-1)^n(2n+2)F(-n,n+2;\tfrac{3}{2};x^2)}.$$

The parameter $-n$ terminates every series. The quotients at $x=0$ are removable and take the derivative values just displayed; these are polynomial identities, so hold also for complex x.

18.5.4 Derive or verify the leading factor in the hypergeometric representations of the Chebyshev functions.

Solution.

In Eqs. (18.133)–(18.135), the hypergeometric argument is $(1-x)/2$, which is zero at $x=1$. Since its series starts with one, the prefactors follow from endpoint normalization:

$$T_n(1)=1,\qquad U_n(1)=n+1,\qquad \lim_{x\to1^-}\frac{V_n(x)}{\sqrt{1-x^2}}=U_{n-1}(1)=n.$$

Thus those three factors are, respectively,

$$\boxed{1,\qquad n+1,\qquad n\sqrt{1-x^2}}.$$

The last expression describes the sine family $V_n=\sqrt{1-x^2}U_{n-1}$ for $n\ge1$; at zero order its continuation is $V_0=0$, not the separate solution $\arccos x$.

For the four x^2 representations in Exercise 18.5.3, evaluate at $x=0$ instead. The even values are $(-1)^n$; for each odd quotient use $\lim_{x\to0}Y(x)/x=Y'(0)$. Differentiating the trigonometric representations gives

$$T'_N(0)=N\sin(N\pi/2),\qquad U'_N(0)=-(N+1)\cos((N+1)\pi/2).$$

At $N=2n+1$ these are $(-1)^n(2n+1)$ and $(-1)^n(2n+2)$. Hence the four leading factors, in order, are

$$\boxed{(-1)^n,\quad (-1)^n(2n+1),\quad (-1)^n,\quad (-1)^n(2n+2)}.$$

Both sets of representations are therefore scaled at a point where the hypergeometric series equals one, with no comparison of infinite series left implicit.

18.5.5 Verify that the Legendre function of the second kind, $Q_\nu(z)$, is given by

$$Q_\nu(z) = \frac{\pi^{1/2}\nu!}{\Gamma\left(\nu+\frac{3}{2}\right)(2z)^{\nu+1}}\,{}_2F_1\left(\frac{\nu}{2}+\frac{1}{2},\frac{\nu}{2}+1;\nu+\frac{3}{2};z^{-2}\right),$$

where $|z|>1$, $|\arg z|<\pi$, and $\nu\neq -1,-2,-3,\cdots$. When $\nu+3/2$ is a nonpositive integer, interpret ${}_2F_1/\Gamma(\nu+3/2)$ as the regularized hypergeometric function, as defined in the solution; use $\nu!=\Gamma(\nu+1)$.

Solution.

Put $r=\nu+1$, $t=z^{-2}$ and $Q=z^{-r}f(t)$. Chain-rule differentiation gives

$$Q'=z^{-r-1}[-rf-2tf'],\qquad Q''=z^{-r-2}[r(r+1)f+(4r+6)tf'+4t^2f''].$$

Substitution into $(1-z^2)Q''-2zQ'+\nu(\nu+1)Q=0$ and division by $-4tz^{-r}$ gives

$$t(1-t)f''+[\nu+\tfrac{3}{2}-(\nu+\tfrac{5}{2})t]f'-\frac{(\nu+1)(\nu+2)}{4}f=0.$$

Thus

$$a=\frac{\nu+1}{2},\quad b=\frac{\nu+2}{2},\quad \boxed{c=\nu+\tfrac{3}{2}},$$

confirming the published correction of the denominator parameter. For nonsingular c, the analytic solution at $t=0$ is $f=AF(a,b;c;t)$; its recurrence is

$$\frac{d_{j+1}}{d_j}=\frac{(j+(\nu+1)/2)(j+(\nu+2)/2)}{(j+1)(j+\nu+3/2)}.$$

To fix A, including for noninteger degree, start with real $\nu>0, z>1$ and the conventional exterior second-kind normalization

$$Q_\nu(z)=\int_0^\infty[z+\sqrt{z^2-1}\cosh u]^{-\nu-1}\,du.$$

This integral representation, with the conversion from Olver's normalization, is recorded in DLMF 14.12.6 and 14.3.10. Multiply by $z^{\nu+1}$ and let $z\to+\infty$. For $z\geq 2$, the integrand is bounded by a constant multiple of $(1+\cosh u)^{-\nu-1}$, so dominated convergence gives

$$\begin{aligned}
A&=\int_0^\infty(1+\cosh u)^{-\nu-1}du\\
&=2^{-\nu}\int_0^\infty \operatorname{sech}^{2\nu+2}v\,dv \qquad (u=2v)\\
&=2^{-\nu-1}\int_0^1 s^{-1/2}(1-s)^\nu ds \qquad (s=\tanh^2 v)\\
&=\frac{\sqrt{\pi}\,\Gamma(\nu+1)}{2^{\nu+1}\Gamma(\nu+3/2)}.
\end{aligned}$$

The decaying Frobenius solution is unique in this initial parameter range; the other infinity exponent is ν. Therefore

$$\boxed{Q_\nu(z)=\frac{\sqrt{\pi}\,\Gamma(\nu+1)}{\Gamma(\nu+3/2)(2z)^{\nu+1}}F\left(\frac{\nu+1}{2},\frac{\nu+2}{2};\nu+\tfrac{3}{2};z^{-2}\right)}.$$

Analytic continuation in degree and in z gives the requested exterior branch, with $(2z)^{\nu+1} = \exp((\nu+1)\operatorname{Log}(2z))$ and $|\arg z| < \pi$. The condition $|z| > 1$ makes $|t| < 1$. At nonnegative integer $\nu = n$, the leading coefficient also reduces to $n!/(2n+1)!!$, agreeing with the ordinary Q_n normalization.

At $\nu = -3/2, -5/2, \ldots$, the displayed product must be read as a regularized limit; its separate hypergeometric factor is undefined. A formula that includes these values directly is

$$Q_\nu(z) = \frac{\sqrt{\pi}\,\Gamma(\nu+1)}{(2z)^{\nu+1}} \sum_{j=0}^{\infty} \frac{((\nu+1)/2)_j((\nu+2)/2)_j}{\Gamma(\nu+3/2+j)j!}\, z^{-2j}.$$

Here $1/\Gamma$ is entire and vanishes at its nonpositive integer arguments. This is precisely $F/\Gamma(c)$, not an instruction to multiply an undefined infinity by zero. The regularized convention is independently consistent with DLMF 14.3.7. The excluded negative integer degrees are the poles specified in the question; no value at those poles is asserted.

18.5.6 The incomplete beta function was defined in Eq. (13.78) as

$$B_x(p,q) = \int_0^x t^{p-1}(1-t)^{q-1}\,dt.$$

Show that

$$B_x(p,q) = p^{-1}x^p\,{}_2F_1(p, 1-q; p+1; x).$$

Initially take $0 \le x < 1$ and $\Re p > 0$; at $x = 1$ also require $\Re q > 0$.

Solution.

First take $0 < x < 1$, $\Re p > 0$ and arbitrary complex q, using real logarithms for positive t and $1-t$. The binomial series is uniformly convergent for $0 \le t \le x$:

$$(1-t)^{q-1} = \sum_{j=0}^{\infty} \frac{(1-q)_j}{j!} t^j.$$

Its product with t^{p-1} is integrable at zero, and termwise integration therefore gives

$$B_x(p,q) = \sum_{j=0}^{\infty} \frac{(1-q)_j}{j!} \frac{x^{p+j}}{p+j}.$$

The finite-product cancellation

$$\frac{(p)_j}{(p+1)_j} = \frac{p}{p+j}$$

holds also at $j = 0$. Insert it into the last series:

$$\boxed{B_x(p,q) = \frac{x^p}{p} \sum_{j=0}^{\infty} \frac{(p)_j(1-q)_j}{(p+1)_j j!} x^j = \frac{x^p}{p} F(p, 1-q; p+1; x)}.$$

At $x = 0$ both sides tend to zero under $\Re p > 0$. At $x = 1$, convergence of the defining integral additionally requires $\Re q > 0$, yielding the complete beta function. For complex x, continue along a path avoiding the branch points $0, 1$, with the same logarithms used in the integral and x^p. Parameter continuation beyond $\Re p > 0$ is a continuation of the function, not an ordinary convergent lower-endpoint integral.

18.5.7 Verify the integral representation

$$ {}_2F_1(a,b;c;z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 t^{b-1}(1-t)^{c-b-1}(1-tz)^{-a}\,dt. $$

What restrictions must be placed on the parameters b and c?

Note. Although the power series used to establish this integral representation is only valid for $|z| < 1$, the principal representation is valid on $\mathbb{C} \setminus [1,\infty)$. Elsewhere specify matching analytic continuation or boundary limits. For nonintegral a, $[1,\infty)$ is the conventional cut.

Hint. The integral is suspiciously like a beta function and can be expanded into a series of beta functions.

ANS. $c > b > 0$ for real parameters; in general $\Re c > \Re b > 0$.

Solution.

For $|z| < 1$, expand the last factor:

$$ (1-tz)^{-a} = \sum_{j=0}^{\infty} \frac{(a)_j}{j!} t^j z^j. $$

Assume $\Re b > 0$ and $\Re(c-b) > 0$. The beta weight is absolutely integrable and the binomial series is uniformly absolutely convergent for $t \in [0,1]$, so integration and summation may be interchanged. The proposed right side becomes

$$ \begin{aligned} \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \sum_{j=0}^{\infty} \frac{(a)_j z^j}{j!} \int_0^1 t^{b+j-1}(1-t)^{c-b-1}dt &= \sum_{j=0}^{\infty} \frac{(a)_j z^j}{j!} \frac{\Gamma(c)\Gamma(b+j)}{\Gamma(b)\Gamma(c+j)} \\ &= \sum_{j=0}^{\infty} \frac{(a)_j (b)_j}{(c)_j j!} z^j = F(a,b;c;z). \end{aligned} $$

Near $t = 0$ and $t = 1$, for $z \neq 1$ off the cut, the additional factor is finite and nonzero. Ordinary endpoint convergence thus requires

$$ \boxed{\Re b > 0, \qquad \Re(c-b) > 0}; \qquad \boxed{c > b > 0 \quad \text{for real } b, c}. $$

There is no restriction on a for this initial domain. Choose the branch of $(1-tz)^{-a}$ continued from $z = 0$. For $z \in \mathbb{C} \setminus [1,\infty)$ the integral is analytic: on every compact subset the additional factor and its derivatives are uniformly bounded against the integrable endpoint weight. The identity theorem therefore extends the representation from the unit disk to this cut plane. On the cut itself a boundary value or another explicit prescription is needed; "general z" does not guarantee an ordinary integral through $t = 1/z$. At $z = 1$, the endpoint powers combine and require $\Re(c-a-b) > 0$ instead at $t = 1$, as used next.

18.5.8 Prove that

$$ {}_2F_1(a,b;c;1) = \frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-a)\Gamma(c-b)}, \qquad c \neq 0, -1, -2, \ldots, \quad \Re(c-a-b) > 0. $$

Hint. Here is a chance to use the integral representation in Exercise 18.5.7.

Solution.

Initially impose $\Re b > 0$, $\Re(c-b) > 0$ and $\Re(c-a-b) > 0$, so Euler's integral remains convergent at $z = 1$. Combining its two factors involving $1 - t$ gives

$$\begin{aligned} F(a,b;c;1) &= \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 t^{b-1}(1-t)^{c-a-b-1}\,dt \\ &= \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \frac{\Gamma(b)\Gamma(c-a-b)}{\Gamma(c-a)} \\ &= \boxed{\frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-a)\Gamma(c-b)}}. \end{aligned}$$

For the nonterminating series, the gamma-quotient asymptotic makes its jth term $O(j^{a+b-c-1})$ in magnitude, with real part understood in the exponent. Thus it converges absolutely when $\Re(c-a-b) > 0$, locally uniformly away from $c = 0, -1, -2, \ldots$. Analytic continuation in the parameters removes the temporary Euler-integral conditions on b and $c - b$ and proves the stated formula throughout

$$c \notin \{0, -1, -2, \ldots\}, \qquad \Re(c-a-b) > 0.$$

For real parameters this is the question's $c > a + b$. A gamma function in a denominator is interpreted through its entire reciprocal, so a zero of that reciprocal is included when the other factors are finite. Terminating polynomials can obey a larger parameter range; Exercise 18.5.12 proves their finite identity without imposing this nonterminating convergence condition.

18.5.9 Prove that

$$ {}_2F_1(a,b;c;x) = (1-x)^{-a}\, {}_2F_1\left(a, c-b; c; \frac{-x}{1-x}\right). $$

Hint. Try an integral representation.

Note. This relation is useful in developing a Rodrigues representation of $T_n(x)$ (see Exercise 18.5.10). Take $c \notin \{0, -1, -2, \ldots\}$ and principal branches on the cut plane $x \notin [1, \infty)$; further continuation must follow matching paths.

Solution.

Assume first $\Re b > 0$, $\Re(c-b) > 0$ and take x close to zero. In Euler's integral for $F(a,b;c;x)$ put $u = 1 - t$. Then $dt = -du$, the limits reverse, and

$$1 - tx = 1 - (1-u)x = (1-x)\left(1 + \frac{xu}{1-x}\right).$$

Consequently

$$\begin{aligned} F(a,b;c;x) &= \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 (1-u)^{b-1} u^{c-b-1} [1 - x + xu]^{-a}\,du \\ &= (1-x)^{-a} \frac{\Gamma(c)}{\Gamma(c-b)\Gamma(b)} \int_0^1 u^{c-b-1}(1-u)^{b-1} \left(1 - u\frac{-x}{1-x}\right)^{-a} du \\ &= \boxed{(1-x)^{-a} F\left(a, c-b; c; \frac{-x}{1-x}\right)}. \end{aligned}$$

The prefactor is unchanged when b and $c - b$ exchange places, which is why the last integral has exactly the required normalization. This proves the identity in a neighborhood of $x = 0$. Continue

both sides along the same path, avoiding the singular points and using $(1-x)^{-a} = \exp[-a\,\mathrm{Log}(1-x)]$ on the chosen branch. It holds on the principal cut plane and on matching analytic continuations; independent branch choices need not agree. Parameter continuation removes the temporary beta-integral restrictions whenever the functions are defined. The singular point $x = 1$ requires a separate limiting analysis.

18.5.10 Derive the Rodrigues representation of $T_n(x)$,

$$T_n(x) = \frac{(-1)^n \pi^{1/2} (1-x^2)^{1/2}}{2^n\,\Gamma\left(n+\frac{1}{2}\right)} \frac{d^n}{dx^n}\left[(1-x^2)^{n-1/2}\right].$$

Hint. One possibility is to use the hypergeometric function relation

$${}_2F_1(a,b;c;z) = (1-z)^{-a}\,{}_2F_1\left(a, c-b; c; \frac{-z}{1-z}\right),$$

with $z = (1-x)/2$. An alternate approach is to develop a first-order differential equation for $y = (1-x^2)^{n-1/2}$. Repeated differentiation of this equation leads to the Chebyshev equation. Derive the identity on $-1 < x < 1$; at $x = \pm 1$ use the polynomial's continuous limits.

Solution.

Take $n = 0, 1, 2, \ldots$ and first $-1 < x < 1$. Set $g = (1-x^2)^{n-1/2}$; direct differentiation gives

$$(1-x^2)g' + (2n-1)xg = 0.$$

Differentiate $n+1$ times. The first factor has only two nonzero derivatives and x only one, so Leibniz's formula gives

$$\begin{aligned} 0 &= (1-x^2)g^{(n+2)} - 2(n+1)xg^{(n+1)} - n(n+1)g^{(n)} \\ &\quad + (2n-1)xg^{(n+1)} + (2n-1)(n+1)g^{(n)} \\ &= (1-x^2)g^{(n+2)} - 3xg^{(n+1)} + (n^2-1)g^{(n)}. \end{aligned}$$

Thus $u = g^{(n)}$ satisfies $(1-x^2)u'' - 3xu' + (n^2-1)u = 0$. With $h = \sqrt{1-x^2}$, differentiation of $R = hu$ gives

$$(1-x^2)R'' - xR' + n^2 R = h[(1-x^2)u'' - 3xu' + (n^2-1)u] = 0.$$

Moreover R is a polynomial. Indeed induction gives

$$g^{(j)} = (1-x^2)^{n-1/2-j} p_j(x), \quad p_0 = 1,$$

$$p_{j+1} = (1-x^2)p_j' - 2(n-1/2-j)xp_j, \qquad \deg p_j \le j.$$

At $j = n$, multiplication by h leaves $R = p_n$. The polynomial solution of this degree-n Chebyshev equation is a scalar multiple of T_n: substitution of powers gives $(k+2)(k+1)a_{k+2} + (n^2-k^2)a_k = 0$, fixing all coefficients from the leading one and forcing the opposite parity to vanish.

It remains to determine the scalar, rather than assume its sign. Write $g = (1-x)^{n-1/2}(1+x)^{n-1/2}$ and use Leibniz at $x \to 1^-$. After multiplying by h, every term with fewer than n derivatives on the first factor vanishes as $(1-x)^{n-j}$. The remaining term gives

$$\begin{aligned} R(1) &= (-1)^n \frac{\Gamma(n+1/2)}{\Gamma(1/2)} 2^{n-1/2}\sqrt{2} \\ &= (-1)^n 2^n \frac{\Gamma(n+1/2)}{\sqrt{\pi}}. \end{aligned}$$

Since $T_n(1) = 1$, division by this value proves

$$\boxed{T_n(x) = \frac{(-1)^n\sqrt{\pi}}{2^n\Gamma(n+1/2)}\sqrt{1-x^2}\,\frac{d^n}{dx^n}(1-x^2)^{n-1/2}}.$$

The published notation correction is the gamma function $\Gamma(n+1/2)$ in the denominator. At $n = 0$ the two square-root factors cancel and give $T_0 = 1$. Although the derivative representation separately has endpoint singularities, its product is the polynomial just proved, so it extends continuously to $x = \pm 1$ and algebraically to complex x.

18.5.11 Show that the summation in Eq. (18.43),

$$\sum_{\nu=0}^{\min(m,n)} \frac{(-m)_\nu(-n)_\nu}{\nu!\left(\dfrac{1-m-n}{2}\right)_\nu}\left(\frac{a^2}{2(a^2-1)}\right)^\nu,$$

can be written as a hypergeometric function. Take $m, n \geq 0$ and $a^2 \neq 1$. Interpret a singular lower hypergeometric parameter by the terminating finite series, not an undefined infinite-series quotient.

Solution.

Let m, n be nonnegative integers and set

$$c = \frac{1-m-n}{2}, \qquad w = \frac{a^2}{2(a^2-1)}.$$

Compare each coefficient with the defining hypergeometric series:

$$F(A, B; C; w) = \sum_{j\geq 0} \frac{(A)_j(B)_j}{(C)_j j!}w^j.$$

The numerator parameters are $A = -m, B = -n$. Their Pochhammer symbols terminate at $j = m$ and $j = n$, respectively, so the finite sum is

$$\boxed{F\left(-m, -n; \frac{1-m-n}{2}; \frac{a^2}{2(a^2-1)}\right)}.$$

Here one must specify what "terminates" means when c is itself a nonpositive integer. If $m+n = 2\ell+1$, then $c = -\ell$ and the first denominator zero occurs at $j = \ell + 1$. But $\min(m, n) \leq \ell$, so every term actually retained has a nonzero denominator. If $m+n$ is even, c is a half-integer and no denominator factor vanishes. Thus the original finite sum is well defined in every degree case. At the singular integer denominator parameters, the boxed notation means this terminating polynomial, not a generic infinite series with later $0/0$ coefficients. For $a^2 \neq 1$ the argument is finite. At $a = \pm 1$ it is undefined as written; if one index is zero the polynomial is identically one and has that removable value. With both indices positive its nonzero highest coefficient gives divergence as $w \to \infty$. Any cancellation with other factors in the full Eq. (18.43) must be taken in that full expression, not in this isolated sum.

18.5.12 Verify that

$$ {}_2F_1(-n, b; c; 1) = \frac{(c-b)_n}{(c)_n}, $$

where n is a nonnegative integer and $(c)_n \neq 0$. Use the terminating series if the lower parameter would make a later, unused coefficient singular.

Hint. Here is a chance to use the contiguous function relation Eq. (18.127) and mathematical induction (Section 1.4). Alternatively, use the integral representation and the beta function.

Solution.

For the finite identity take $n = 0, 1, 2, \ldots$ and initially $\Re b > 0$, $\Re(c-b) > 0$. Euler's integral with first parameter $-n$ gives

$$ \begin{aligned} F(-n, b; c; 1) &= \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 t^{b-1}(1-t)^{c-b-1}(1-t)^n dt \\ &= \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \frac{\Gamma(b)\Gamma(c-b+n)}{\Gamma(c+n)} \\ &= \frac{\Gamma(c-b+n)}{\Gamma(c-b)} \frac{\Gamma(c)}{\Gamma(c+n)} = \boxed{\frac{(c-b)_n}{(c)_n}}. \end{aligned} $$

Because the left side is the finite sum

$$ \sum_{j=0}^{n} (-1)^j \binom{n}{j} \frac{(b)_j}{(c)_j}, $$

both sides are rational functions of b, c. Multiply by $(c)_n$: for every $j \leq n$, the quotient $(c)_n/(c)_j = (c+j)_{n-j}$ is a polynomial. The resulting polynomial identity, proved on an open parameter set, holds identically. Division back gives the formula whenever $(c)_n \neq 0$, without requiring the beta-integral restrictions. This includes singular denominator parameters whose first zero occurs strictly after termination, with the terminating convention of Exercise 18.5.11. At $n = 0$ both empty products and the single-term sum equal one.

The terminating version uses a nonnegative integer n; the published erratum's word "integer" should be understood in that polynomial setting. Negative integer n gives a nonterminating series, for which the convergence conditions of Gauss's formula in Exercise 18.5.8 must instead be imposed. At zeros of $(c)_n$ the literal formula is not defined unless a particular removable limit has been specified.

18.6 Confluent Hypergeometric Functions

18.6.1 Verify the confluent hypergeometric representation of the error function

$$ \operatorname{erf}(x) = \frac{2x}{\pi^{1/2}} M\left(\frac{1}{2}, \frac{3}{2}, -x^2\right). $$

Solution.

By definition $M(a, c, z) = \sum_{j \geq 0} (a)_j z^j / [(c)_j j!]$. The finite products telescope:

$$ \frac{(1/2)_j}{(3/2)_j} = \frac{1/2}{j + 1/2} = \frac{1}{2j+1}, $$

including $j = 0$. Hence

$$\frac{2x}{\sqrt{\pi}} M(\tfrac{1}{2}, \tfrac{3}{2}, -x^2) = \frac{2}{\sqrt{\pi}} \sum_{j=0}^{\infty} \frac{(-1)^j x^{2j+1}}{(2j+1)j!}.$$

On every bounded complex path, the exponential series is uniformly absolutely convergent, so its integral is

$$\operatorname{erf} x = \frac{2}{\sqrt{\pi}} \int_0^x e^{-u^2} du = \frac{2}{\sqrt{\pi}} \sum_{j=0}^{\infty} \frac{(-1)^j}{j!} \int_0^x u^{2j} du.$$

Evaluating the last monomial integral gives the preceding series, proving the identity for every complex x, including its zero value at the origin. The entire integrand makes the integration path irrelevant.

18.6.2 Show that the Fresnel integrals $C(x)$ and $s(x)$ of Exercise 12.6.1 may be expressed in terms of the confluent hypergeometric function as

$$C(x) + i\, s(x) = xM\left(\frac{1}{2}, \frac{3}{2}, \frac{i\pi x^2}{2}\right).$$

Solution.

The definitions give

$$C(x) + is(x) = \int_0^x [\cos(\pi u^2/2) + i \sin(\pi u^2/2)] du = \int_0^x e^{i\pi u^2/2} du.$$

Expand the exponential and integrate term by term on the finite path (uniform absolute convergence justifies this):

$$\begin{aligned} C(x) + is(x) &= \sum_{j=0}^{\infty} \frac{(i\pi/2)^j}{j!} \frac{x^{2j+1}}{2j+1} \\ &= x \sum_{j=0}^{\infty} \frac{(1/2)_j}{(3/2)_j j!} \left(\frac{i\pi x^2}{2}\right)^j \\ &= \boxed{xM\left(\tfrac{1}{2}, \tfrac{3}{2}, \frac{i\pi x^2}{2}\right)}. \end{aligned}$$

The factor $1/(2j+1)$ was converted by the exact product identity in Exercise 18.6.1. Both sides are entire, so the proof includes negative real and complex x, with the same analytic definitions of the Fresnel integrals.

18.6.3 By direct differentiation and substitution verify that

$$y = ax^{-a} \int_0^x e^{-t} t^{a-1}\, dt = ax^{-a}\gamma(a, x)$$

satisfies

$$xy'' + (a + 1 + x)y' + ay = 0.$$

For the defining integral, take $x > 0$ and $\Re a > 0$.

Solution.

Take $x > 0$ and $\Re a > 0$, the domain in which the defining lower-endpoint integral converges. Write $G(x) = \gamma(a, x)$, so $G' = e^{-x}x^{a-1}$, and let $y = ax^{-a}G$. The product rule gives

$$y' = -\frac{a}{x}y + \frac{a}{x}e^{-x}.$$

Differentiate this equality once more and substitute for y':

$$\begin{aligned} y'' &= \frac{a}{x^2}y - \frac{a}{x}y' - \frac{a}{x^2}e^{-x} - \frac{a}{x}e^{-x} \\ &= \frac{a(a+1)}{x^2}y - \frac{a(a+1)}{x^2}e^{-x} - \frac{a}{x}e^{-x}. \end{aligned}$$

Thus the four contributions are

$$\begin{aligned} xy'' &= \frac{a(a+1)}{x}y - \frac{a(a+1)}{x}e^{-x} - ae^{-x}, \\ (a+1)y' &= -\frac{a(a+1)}{x}y + \frac{a(a+1)}{x}e^{-x}, \\ xy' &= -ay + ae^{-x}, \qquad ay = ay. \end{aligned}$$

Every term cancels upon addition, proving

$$\boxed{xy'' + (a+1+x)y' + ay = 0}.$$

Also $\gamma(a, x) = x^a/a + O(x^{a+1})$, so $y(0) = 1$ by continuity. Expanding the integral gives $y = M(a, a+1, -x)$; this supplies analytic continuation with matching branches. For $\Re a \le 0$ the displayed integral is generally not an ordinary convergent integral; any extension uses that analytic continuation. For example its removable continuation at $a = 0$ is 1, not a direct evaluation of 0 times a divergent integral.

18.6.4 Show that the modified Bessel function of the second kind, $K_\nu(x)$, is given by

$$K_\nu(x) = \pi^{1/2}e^{-x}(2x)^\nu\, U\big(\nu + \tfrac{1}{2}, 2\nu + 1, 2x\big).$$

Take $x > 0$; first derive the identity for $\Re\nu > -1/2$, then continue in ν.

Solution.

After $t = 2zy$, the manual's exponential inside the integral is e^{-2zy}, not e^{z-2zy}; its outside factor already includes e^{-z}.

Start with the second-kind Bessel integral for $x > 0$ and $\Re\nu > -1/2$:

$$K_\nu(x) = \frac{\sqrt{\pi}(x/2)^\nu}{\Gamma(\nu + 1/2)} \int_1^\infty e^{-xs}(s^2 - 1)^{\nu - 1/2} ds.$$

Set $s = 1 + 2t$, so $ds = 2dt$, $s^2 - 1 = 4t(1+t)$ and $e^{-xs} = e^{-x}e^{-2xt}$. Then

$$\begin{aligned} K_\nu(x) &= \frac{\sqrt{\pi}(x/2)^\nu e^{-x}\, 2\, 4^{\nu-1/2}}{\Gamma(\nu + 1/2)} \int_0^\infty e^{-2xt} t^{\nu-1/2}(1+t)^{\nu-1/2} dt \\ &= \sqrt{\pi}e^{-x}(2x)^\nu \frac{1}{\Gamma(\nu + 1/2)} \int_0^\infty e^{-2xt} t^{\nu-1/2}(1+t)^{\nu-1/2} dt. \end{aligned}$$

In the integral for $U(a,c,z)$, choose

$$a=\nu+\tfrac{1}{2},\quad c=2\nu+1,\quad z=2x,$$

so that $a-1=c-a-1=\nu-1/2$. Therefore

$$\boxed{K_\nu(x)=\sqrt{\pi}e^{-x}(2x)^\nu U(\nu+\tfrac{1}{2},2\nu+1,2x)}.$$

The proof also works for $\Re x>0$ with consistent powers; continuation in ν gives the remaining orders, and continuation in x uses the conventional cut and the same logarithm in $(2x)^\nu$. The integral restrictions belong to this derivation, not to an extra restriction on the continued identity.

18.6.5 Show that the cosine and sine integrals of Section 13.6 may be expressed in terms of confluent hypergeometric functions as

$$\mathrm{Ci}(x)+i\,\mathrm{si}(x)=-e^{ix}U(1,1,-ix).$$

This relation is useful in numerical computation of $\mathrm{Ci}(x)$ and $\mathrm{si}(x)$ for large values of x. Take $x>0$ and $\mathrm{si}(x)=-\int_x^\infty(\sin t)/t\,dt$.

Solution.

For real $x>0$, the convention in Section 13.6 is

$$\mathrm{Ci}\,x=-\int_x^\infty\frac{\cos t}{t}dt,\qquad \mathrm{si}\,x=-\int_x^\infty\frac{\sin t}{t}dt=\mathrm{Si}\,x-\frac{\pi}{2}.$$

These oscillatory improper integrals converge by integration by parts. For $\varepsilon>0$, the Laplace representation gives

$$U(1,1,\varepsilon-ix)=\int_0^\infty\frac{e^{-(\varepsilon-ix)u}}{1+u}du.$$

Multiplying by $e^{-\varepsilon+ix}$ and setting $v=1+u$ gives

$$e^{-\varepsilon+ix}U(1,1,\varepsilon-ix)=\int_1^\infty\frac{e^{-(\varepsilon-ix)v}}{v}\,dv.$$

Let $\varepsilon\downarrow 0$. The tail is uniformly controlled by integration by parts against e^{ixv}, so the Abel limit is the ordinary oscillatory integral. With $t=xv$ it becomes

$$e^{ix}U(1,1,-ix)=\int_1^\infty\frac{e^{ixv}}{v}\,dv=\int_x^\infty\frac{e^{it}}{t}\,dt.$$

Taking its negative gives

$$\boxed{\mathrm{Ci}\,x+i\,\mathrm{si}\,x=-e^{ix}U(1,1,-ix)}.$$

The value of U is continued from the right half-plane, fixing the branch at $-ix$. Equivalently, the answer is $-E_1(-ix)$. The lowercase si is essential: replacing it by Si would require adding $i\pi/2$.

18.6.6 Verify the confluent hypergeometric form of the Hermite polynomial $H_{2n+1}(x)$, Eq. (18.149), by showing that

(a) $H_{2n+1}(x)/x$ satisfies the confluent hypergeometric equation with $a = -n$, $c = 3/2$ and argument x^2,

(b) $\lim_{x\to 0} \dfrac{H_{2n+1}(x)}{x} = (-1)^n \dfrac{2(2n+1)!}{n!}$.

Solution.

(a) Let $t = x^2$ and $H_{2n+1}(x) = xf(t)$. Then

$$H' = f + 2tf', \qquad H'' = x(6f' + 4tf'').$$

Insert these into $H'' - 2xH' + 2(2n+1)H = 0$:

$$x[6f' + 4tf'' - 2f - 4tf' + (4n+2)f] = 0.$$

For $x \neq 0$, division by $4x$ gives

$$\boxed{tf'' + (\tfrac{3}{2} - t)f' + nf = 0},$$

which is Kummer's equation with $a = -n, c = 3/2$. Since the odd polynomial is divisible by x, f is an actual polynomial in t and the equation extends to $x = 0$ as well.

(b) In the finite Hermite series

$$H_{2n+1}(x) = (2n+1)! \sum_{j=0}^{n} \frac{(-1)^j (2x)^{2n+1-2j}}{j!(2n+1-2j)!},$$

the coefficient of x comes only from $j = n$. It is $(-1)^n 2(2n+1)!/n!$. Therefore

$$\lim_{x\to 0} \frac{H_{2n+1}(x)}{x} = \boxed{(-1)^n \frac{2(2n+1)!}{n!}}.$$

Because $M(-n, 3/2, 0) = 1$ and the analytic solution is uniquely fixed by its constant term, parts (a) and (b) together prove

$$H_{2n+1}(x) = (-1)^n \frac{2(2n+1)!}{n!} x M(-n, \tfrac{3}{2}, x^2).$$

For $n = 0$ this reduces to $H_1 = 2x$ without a limiting exception.

18.6.7 Show that the contiguous confluent hypergeometric function equation

$$(c-a)M(a-1,c,x) + (2a-c+x)M(a,c,x) - aM(a+1,c,x) = 0$$

leads to the associated Laguerre function recurrence relation, Eq. (18.66).

Solution.

Use $a = -n$, $c = k+1$ and

$$M(-n, k+1, x) = A_n L_n^k(x), \qquad A_n = \frac{n!\,k!}{(n+k)!}.$$

For $n \geq 1$, the contiguous equation becomes

$$(n+k+1)A_{n+1}L^k_{n+1} + [x-(2n+k+1)]A_nL^k_n + nA_{n-1}L^k_{n-1} = 0.$$

The two ratios are

$$\frac{A_{n+1}}{A_n} = \frac{n+1}{n+k+1}, \qquad \frac{A_{n-1}}{A_n} = \frac{n+k}{n}.$$

Divide by A_n and cancel the factors multiplying these ratios:

$$\boxed{(n+1)L^k_{n+1} - (2n+k+1-x)L^k_n + (n+k)L^k_{n-1} = 0}.$$

This is Eq. (18.66). At $n = 0$ the original coefficient $-a$ is zero, so its last term is absent. The result is $L^k_1 = (k+1-x)L^k_0$, consistent with $L^k_0 = 1$ and the convention $L^k_{-1} = 0$. No division by $n = 0$ is used. Gamma functions extend this proof from integer $k \geq 0$ to nonsingular parameter values, in particular real $k > -1$.

18.6.8 Verify the Kummer transformations:

(a) $M(a,c,x) = e^x M(c-a,c,-x)$,

(b) $U(a,c,x) = x^{1-c}U(a-c+1,2-c,x)$.

For M, take $c \notin \{0,-1,-2,\ldots\}$; for U and powers of x, use matching branches, initially $|\arg x| < \pi$.

Solution.

(a) Put $g(x) = M(c-a,c,-x)$. Its equation, after changing the sign of the argument, is

$$xg'' + (c+x)g' + (c-a)g = 0.$$

For $y = e^x g$, $y' = e^x(g+g')$ and $y'' = e^x(g+2g'+g'')$. Hence

$$xy'' + (c-x)y' - ay = e^x\{xg'' + (c+x)g' + (c-a)g\} = 0.$$

Both y and $M(a,c,x)$ are analytic at zero with value one. For $c \notin \{0,-1,-2,\ldots\}$, their Taylor recurrence $(j+1)(j+c)d_{j+1} = (j+a)d_j$ uniquely fixes every coefficient. Therefore

$$\boxed{M(a,c,x) = e^x M(c-a,c,-x)}.$$

Exercise 18.6.11 supplies a separate integral proof.

(b) Use the corrected connection formula, initially $c \notin \mathbb{Z}$:

$$U(a,c,x) = \frac{\Gamma(1-c)}{\Gamma(a-c+1)}M(a,c,x) + \frac{\Gamma(c-1)}{\Gamma(a)}x^{1-c}M(a-c+1,2-c,x).$$

It follows from Eq. (18.142) by the reflection identity, with its published replacement $\Gamma(-c) \to \Gamma(2-c)$. For example $\pi/[\sin\pi c\,\Gamma(c)] = \Gamma(1-c)$ and $-\pi/[\sin\pi c\,\Gamma(2-c)] = \Gamma(c-1)$. Now replace (a,c) by $(a-c+1,2-c)$:

$$\begin{aligned} U(a-c+1,2-c,x) = {} & \frac{\Gamma(c-1)}{\Gamma(a)}M(a-c+1,2-c,x) \\ & + \frac{\Gamma(1-c)}{\Gamma(a-c+1)}x^{c-1}M(a,c,x). \end{aligned}$$

Multiplication by x^{1-c} simply exchanges the two original terms. Thus

$$\boxed{U(a,c,x) = x^{1-c}U(a-c+1,2-c,x)}.$$

For integer c, take the limit of the entire combination; the separate gamma terms may have poles which cancel. Parameter continuation includes other removable cases. The powers on both sides use the same logarithm, with the usual principal branch initially $|\arg x| < \pi$.

18.6.9 Prove that

(a) $\dfrac{d^n}{dx^n}\,M(a,c,x) = \dfrac{(a)_n}{(c)_n}\,M(a+n,c+n,x)$,

(b) $\dfrac{d^n}{dx^n}\,U(a,c,x) = (-1)^n(a)_n\,U(a+n,c+n,x)$.

Take $n \geq 0$; for M, exclude $c = 0,-1,-2,\ldots$. For U, differentiate on a fixed branch away from $x = 0$.

Solution.

Let n be a nonnegative integer. (a) Differentiate the entire defining series n times:

$$\begin{aligned}\frac{d^n}{dx^n}M(a,c,x) &= \sum_{j=n}^{\infty}\frac{(a)_j}{(c)_j}\frac{x^{j-n}}{(j-n)!}\\ &= \sum_{r=0}^{\infty}\frac{(a)_{n+r}}{(c)_{n+r}}\frac{x^r}{r!}\\ &= \frac{(a)_n}{(c)_n}\sum_{r=0}^{\infty}\frac{(a+n)_r}{(c+n)_r}\frac{x^r}{r!}.\end{aligned}$$

The last step uses $(a)_{n+r} = (a)_n(a+n)_r$ and its c analogue. Therefore

$$\boxed{\frac{d^n}{dx^n}M(a,c,x) = \frac{(a)_n}{(c)_n}M(a+n,c+n,x)}.$$

The two denominator-parameter labels are c, not b, as in the published erratum. Initially exclude nonpositive integer c; terminating cases with n beyond the polynomial degree have zero derivative, exactly as $(a)_n = 0$ predicts.

(b) For $\Re a > 0$, $\Re x > 0$, differentiate the Laplace integral; exponential decay justifies every finite number of derivatives:

$$\begin{aligned}\frac{d^n}{dx^n}U(a,c,x) &= \frac{(-1)^n}{\Gamma(a)}\int_0^{\infty}e^{-xt}t^{a+n-1}(1+t)^{c-a-1}dt\\ &= (-1)^n\frac{\Gamma(a+n)}{\Gamma(a)}U(a+n,c+n,x)\\ &= \boxed{(-1)^n(a)_nU(a+n,c+n,x)}.\end{aligned}$$

The exponent $c-a-1$ stays unchanged because both parameters increase by n. Analytic continuation supplies the usual cut plane and other parameter values. For $a = -N$, the resulting polynomial has zero derivatives for $n > N$, consistently with the vanishing Pochhammer factor.

18.6.10 Verify the following integral representations:

(a) $M(a,c,x) = \dfrac{\Gamma(c)}{\Gamma(a)\Gamma(c-a)}\displaystyle\int_0^1 e^{xt}t^{a-1}(1-t)^{c-a-1}\,dt, \quad c > a > 0,$

(b) $U(a,c,x) = \dfrac{1}{\Gamma(a)}\displaystyle\int_0^{\infty} e^{-xt}t^{a-1}(1+t)^{c-a-1}\,dt, \quad \Re x > 0,\ a > 0.$

Under what conditions can you accept $\Re x = 0$ in part (b)?

Solution.

(a) For complex parameters the endpoint conditions are $\Re a > 0$, $\Re(c-a) > 0$. Expand e^{xt} and integrate its uniformly convergent series against the beta weight:

$$\begin{aligned}\frac{\Gamma(c)}{\Gamma(a)\Gamma(c-a)}\int_0^1 e^{xt}t^{a-1}(1-t)^{c-a-1}dt &= \frac{\Gamma(c)}{\Gamma(a)\Gamma(c-a)}\sum_{j=0}^{\infty}\frac{x^j}{j!}B(a+j,c-a)\\ &= \sum_{j=0}^{\infty}\frac{\Gamma(c)\Gamma(a+j)}{\Gamma(a)\Gamma(c+j)}\frac{x^j}{j!}\\ &= \boxed{M(a,c,x)}.\end{aligned}$$

This holds for every finite complex x; for real parameters its restrictions are precisely $c > a > 0$.

(b) Denote the proposed integral by $I(a,c,x)$. It converges absolutely for $\Re a > 0$, $\Re x > 0$, with no restriction on c. Differentiation under the integral gives

$$\begin{aligned}xI'' + (c-x)I' - aI &= \frac{1}{\Gamma(a)}\int_0^{\infty} e^{-xt}t^{a-1}(1+t)^{c-a-1}[xt(t+1) - ct - a]dt\\ &= -\frac{1}{\Gamma(a)}\left[e^{-xt}t^a(1+t)^{c-a}\right]_0^{\infty} = 0.\end{aligned}$$

The endpoint terms vanish by $\Re a > 0$ at zero and exponential decay at infinity. This verifies the differential equation, but we must also identify the particular solution and its scale.

For noninteger c, its local basis is $M(a,c,x)$ and $x^{1-c}M(a-c+1,2-c,x)$. Let their coefficients in I be A, B. Initially with real $a > 0, c < 1$, dominated convergence at $x \downarrow 0$ gives

$$A = I(a,c,0) = \frac{1}{\Gamma(a)}\int_0^{\infty} t^{a-1}(1+t)^{c-a-1}dt = \frac{\Gamma(1-c)}{\Gamma(a-c+1)}.$$

Here the substitution $v = t/(1+t)$ reduces the integral to $B(a,1-c)$. To compute B, take instead $1 < c < a+1$ and put $u = xt$:

$$x^{c-1}I = \frac{1}{\Gamma(a)}\int_0^{\infty} e^{-u}u^{a-1}(x+u)^{c-a-1}du \longrightarrow \frac{\Gamma(c-1)}{\Gamma(a)}.$$

The exponent $c-a-1$ is negative in this initial strip, so the integrand is bounded by $e^{-u}u^{c-2}$, an integrable function. The analytic first solution vanishes after multiplication by x^{c-1}; hence this limit is B. The local coefficients continue meromorphically in the parameters away from integer c, so the independently computed A, B give

$$I = \frac{\Gamma(1-c)}{\Gamma(a-c+1)}M(a,c,x) + \frac{\Gamma(c-1)}{\Gamma(a)}x^{1-c}M(a-c+1,2-c,x).$$

This is the corrected defining connection formula for $\boxed{U(a,c,x)}$. Limits include integer c; continuation from positive real x gives the whole right half-plane. As a further scale check, $t = u/x$ in the original integral gives $x^a I \to \Gamma(a)/\Gamma(a) = 1$ as $x \to +\infty$.

The boundary $\Re x = 0$ must distinguish $x = 0$ from a nonzero purely imaginary x. At large t, the amplitude is

$$h(t) = t^{a-1}(1+t)^{c-a-1} = t^{c-2}[1 + O(t^{-1})], \quad h'(t) = O(t^{\Re c - 3}) \text{ in magnitude.}$$

At $x = 0$, the ordinary integral converges exactly when $\Re c < 1$ (still $\Re a > 0$ at zero); it is the beta value above. For $x = iy$, $y \neq 0$, integration by parts on $[T, R]$ gives

$$\int_T^R e^{-iyt} h(t)dt = \left[\frac{e^{-iyt}h(t)}{-iy}\right]_T^R + \frac{1}{iy}\int_T^R e^{-iyt}h'(t)dt.$$

If $\Re c < 2$, the endpoint at infinity vanishes and the last integral is absolutely convergent. The original integral is absolutely convergent for $\Re c < 1$ and conditionally convergent for $1 \leq \Re c < 2$. For $\Re c \geq 2$, the nonvanishing leading oscillatory boundary term prevents an ordinary improper limit; repeated integration by parts separates it from lower powers. A value assigned by $\Re x \downarrow 0$ outside these ranges is an Abel continuation, not that ordinary integral.

18.6.11 From the integral representation of $M(a, c, x)$, Exercise 18.6.10(a), show that

$$M(a, c, x) = e^x M(c - a, c, -x).$$

Hint. Replace the variable of integration t by $1 - s$ to release a factor e^x from the integral. For the integral derivation, take $\Re a > 0$ and $\Re(c - a) > 0$; then continue in the nonsingular parameters.

Solution.

For $\Re a > 0$, $\Re(c - a) > 0$, put $s = 1 - t$ in the Euler integral. Reversing the limits cancels the sign from $dt = -ds$:

$$\begin{aligned} M(a, c, x) &= \frac{\Gamma(c)}{\Gamma(a)\Gamma(c - a)} \int_0^1 e^{x(1-s)}(1 - s)^{a-1}s^{c-a-1}ds \\ &= e^x \frac{\Gamma(c)}{\Gamma(c - a)\Gamma(a)} \int_0^1 e^{-xs}s^{(c-a)-1}(1 - s)^{c-(c-a)-1}ds \\ &= \boxed{e^x M(c - a, c, -x)}. \end{aligned}$$

Both the powers and the two gamma factors have exchanged roles, so no further normalization is needed. The finite-interval integral allows arbitrary complex x. Analytic continuation in a, c then gives the identity wherever M is defined, including its terminating polynomial cases. This is independent of the ODE uniqueness proof in Exercise 18.6.8(a).

18.6.12 From the integral representation of $U(a, c, x)$ in Exercise 18.6.10(b), show that the exponential integral is given by

$$E_1(x) = e^{-x}U(1, 1, x).$$

Hint. Replace the variable of integration t in $E_1(x)$ by $x(1 + s)$. Initially take $x > 0$; both sides then have matching continuation to the principal cut plane.

Solution.

For real $x > 0$ start with $E_1(x) = \int_x^\infty e^{-t}dt/t$. Make exactly the suggested substitution $t = x(1 + s)$:

$$dt = x\,ds, \quad t = x \Rightarrow s = 0, \quad E_1(x) = e^{-x}\int_0^\infty \frac{e^{-xs}}{1 + s}ds.$$

The Laplace integral for U with $a = c = 1$ has $s^{a-1} = 1$, $(1 + s)^{c-a-1} = (1 + s)^{-1}$ and $\Gamma(a) = 1$. Thus

$$\boxed{E_1(x) = e^{-x}U(1, 1, x)}.$$

Both sides are analytic on the right half-plane and continue to the principal cut plane $|\arg x| < \pi$ with the same branch. They have a logarithmic singularity at $x = 0$; that point is not included as a finite value.

18.6.13 From the integral representations of $M(a,c,x)$ and $U(a,c,x)$ in Exercise 18.6.10, develop asymptotic expansions of (a) $M(a,c,x)$, (b) $U(a,c,x)$.

Hint. You can use the technique that was employed with $K_\nu(z)$ in Section 14.6.

ANS. (a) $\dfrac{\Gamma(c)}{\Gamma(a)}\dfrac{e^x}{x^{c-a}}\left[1+\dfrac{(1-a)(c-a)}{1!\,x}+\dfrac{(1-a)(2-a)(c-a)(c-a+1)}{2!\,x^2}+\cdots\right]$,

(b) $\dfrac{1}{x^a}\left[1+\dfrac{a(1+a-c)}{1!\,(-x)}+\dfrac{a(a+1)(1+a-c)(2+a-c)}{2!\,(-x)^2}+\cdots\right]$. Use fixed parameters and $x\to+\infty$. Initially assume $\Re c>\Re a>0$ in (a), $\Re a>0$ in (b); discuss terminating $a=0,-1,-2,\ldots$ separately.

Solution.

We derive Poincaré expansions for $x\to+\infty$, with fixed parameters. A displayed infinite asymptotic series means that each fixed truncation has the stated remainder, not that the whole series converges.

(a) Initially take real $0<a<c$. In the Euler integral set $u=1-t$, obtaining

$$M(a,c,x)=\frac{\Gamma(c)e^x}{\Gamma(a)\Gamma(c-a)}\int_0^1 e^{-xu}u^{c-a-1}(1-u)^{a-1}du.$$

Near $u=0$ the Taylor expansion is

$$(1-u)^{a-1}=\sum_{j=0}^{N-1}\frac{(1-a)_j}{j!}u^j+O(u^N).$$

Choose a fixed $0<\delta<1$. On $[0,\delta]$ the remainder integral is $O(x^{-(c-a+N)})$ by $v=xu$. On $[\delta,1]$ the integrable endpoint weight is multiplied by at most $e^{-\delta x}$, making its contribution exponentially small. Replacing each Taylor-term integral's upper limit by infinity also changes it by an exponentially small amount. Since

$$\int_0^\infty e^{-xu}u^{c-a+j-1}du=\frac{\Gamma(c-a+j)}{x^{c-a+j}},$$

we obtain

$$\boxed{M(a,c,x)=\frac{\Gamma(c)}{\Gamma(a)}e^x x^{a-c}\left[\sum_{j=0}^{N-1}\frac{(1-a)_j(c-a)_j}{j!x^j}+O(x^{-N})\right]}.$$

The first three coefficients are 1, $(1-a)(c-a)$ and $(1-a)(2-a)(c-a)(c-a+1)/2$, as printed. The derivation extends to $\Re a>0$, $\Re(c-a)>0$ with complex parameters and the same fixed positive-real limit.

(b) For $\Re a>0$, expand the Laplace-integral amplitude at zero:

$$(1+t)^{c-a-1}=\sum_{j=0}^{N-1}\frac{(-1)^j(1+a-c)_j}{j!}t^j+O(t^N).$$

The same split at a fixed $\delta>0$ works; the tail grows only as a fixed power and is exponentially suppressed by e^{-xt}. Termwise Laplace integration therefore gives

$$U(a,c,x)=\frac{1}{\Gamma(a)}\sum_{j=0}^{N-1}\frac{(-1)^j(1+a-c)_j}{j!}\frac{\Gamma(a+j)}{x^{a+j}}+O(x^{-\Re a-N})$$

$$=\boxed{x^{-a}\left[\sum_{j=0}^{N-1}\frac{(a)_j(1+a-c)_j}{j!(-x)^j}+O(x^{-N})\right]}.$$

In particular the first correction is $-a(1+a-c)/x$, fixing its sign.

These displayed forms need a limit direction and parameter qualifications; a positive-axis exponential expansion for M cannot be applied unchanged at a terminating a. If $a=-m$, $m=0,1,\ldots$, use the exact polynomial $M(-m,c,x)$, whose leading term is $(-1)^m x^m/(c)_m$ when $(c)_m\neq 0$. The apparent exponential prefactor $1/\Gamma(a)$ then vanishes. The U expansion instead terminates at $j=m$ and gives its exact polynomial, with leading term x^m. To justify this without applying the integral outside its domain, use the connection formula from Exercise 18.6.8: its second term vanishes because $1/\Gamma(-m)=0$, while

$$\frac{\Gamma(1-c)}{\Gamma(1-c-m)}=(-1)^m(c)_m.$$

Thus $U(-m,c,x)=(-1)^m(c)_m M(-m,c,x)$, first for noninteger c and then by continuation. Reversing the finite powers of this polynomial gives the displayed terminating U expansion. On other complex rays one must choose the relevant asymptotic sector; the estimates above specifically establish the positive-real expansions requested by the given integrals.

18.6.14 Show that the Wronskian of the two confluent hypergeometric functions $M(a,c,x)$ and $U(a,c,x)$ is given by

$$MU'-M'U=-\frac{(c-1)!\,e^x}{(a-1)!\,x^c}.$$

What happens if a is 0 or a negative integer? Take $c\notin\{0,-1,-2,\ldots\}$ and $x\neq 0$ on a fixed branch. Interpret factorials by gamma functions, including reciprocal-gamma zeros.

Solution.

Assume initially $c\notin\{0,-1,-2,\ldots\}$ so that M is defined. Write $W=MU'-M'U$. Since both functions satisfy $y''+(c/x-1)y'-(a/x)y=0$, differentiation gives

$$W'=MU''-M''U=(1-c/x)(MU'-M'U)=(1-c/x)W.$$

Integration yields $W=C(a,c)e^x x^{-c}$ on a fixed branch. For real $0<a<c$, Exercise 18.6.13 and the corresponding differentiated Laplace estimates give

$$M\sim\frac{\Gamma(c)}{\Gamma(a)}e^x x^{a-c},\quad M'\sim M,\quad U\sim x^{-a},\quad U'\sim -ax^{-a-1}.$$

Thus MU' is smaller than $M'U$ by a factor $O(x^{-1})$, and

$$\lim_{x\to+\infty}e^{-x}x^cW=-\frac{\Gamma(c)}{\Gamma(a)}.$$

Because this quotient is the constant $C(a,c)$, the result is exact, and parameter continuation gives

$$\boxed{MU'-M'U=-\frac{\Gamma(c)}{\Gamma(a)}e^x x^{-c}}.$$

The factorials in the question are these gamma functions.

If $a=-m$ is zero or a negative integer, $1/\Gamma(a)=0$ and $W=0$. The manual's explanation that M does not exist is incorrect: it exists and terminates. The two solutions have become proportional. Their leading polynomial coefficients are $(-1)^m/(c)_m$ for M and 1 for U, so

$$\boxed{U(-m,c,x)=(-1)^m(c)_m M(-m,c,x)=(-1)^m m!L_m^{c-1}(x)}.$$

The degree recurrence in the ODE uniquely fixes the polynomial from that leading coefficient. At $m=0$ both functions equal one. The ODE still has a second independent solution, but it is not supplied by this now-degenerate pair.

18.6.15 The Coulomb wave equation (radial part of the Schrödinger equation with Coulomb potential) is

$$\frac{d^2y}{dr^2} + \left[1 - \frac{2\eta}{r} - \frac{L(L+1)}{r^2}\right] y = 0.$$

Show that a regular solution $y = F_L(\eta, r)$ is given by

$$F_L(\eta, r) = C_L(\eta) r^{L+1} e^{-ir} M(L + 1 - i\eta, 2L + 2, 2ir).$$

For the physical radial problem, take $L \geq 0$ integral, η real and $r > 0$.

Solution.

Set $z = 2ir$, $g(r) = r^{L+1}e^{-ir}$ and $y = Cg(r)f(z)$. The logarithmic derivatives of g are

$$\frac{g'}{g} = \frac{L+1}{r} - i, \qquad \frac{g''}{g} = \frac{L(L+1)}{r^2} - \frac{2i(L+1)}{r} - 1.$$

Since $dz/dr = 2i$,

$$\frac{y''}{Cg} = -4f'' + 4i\left(\frac{L+1}{r} - i\right) f' + \left[\frac{L(L+1)}{r^2} - \frac{2i(L+1)}{r} - 1\right] f.$$

Adding the potential term cancels $L(L+1)/r^2$ and 1. Use $1/r = 2i/z$ in the remainder to obtain

$$\frac{1}{Cg}\left\{y'' + \left[1 - \frac{2\eta}{r} - \frac{L(L+1)}{r^2}\right] y\right\} = -\frac{4}{z}\{zf'' + (2L + 2 - z)f' - (L + 1 - i\eta)f\}.$$

The braces vanish for $f = M(L + 1 - i\eta, 2L + 2, z)$, proving

$$\boxed{F_L(\eta, r) = C_L(\eta) r^{L+1} e^{-ir} M(L + 1 - i\eta, 2L + 2, 2ir)}.$$

For physical $L = 0, 1, 2, \ldots$, the denominator parameter is positive and $M = 1 + O(r)$ at zero. Thus $F_L \sim C_L r^{L+1}$, the regular Frobenius branch; the other exponent is $-L$. The constant C_L is set by the chosen normalization at large radius and is not fixed by this homogeneous ODE alone.

18.6.16

(a) Show that the radial part of the hydrogen-atom wave function, Eq. (18.81), may be written as

$$e^{-\alpha r/2}(\alpha r)^L L^{2L+1}_{n-L-1}(\alpha r) = \frac{(n+L)!}{(n-L-1)!\,(2L+1)!}\, e^{-\alpha r/2}(\alpha r)^L M(L + 1 - n, 2L + 2, \alpha r).$$

(b) It was assumed previously that the total (kinetic + potential) energy E of the electron was negative. Rewrite the (unnormalized) radial wave function for an unbound hydrogenic electron, $E > 0$.

ANS. $e^{i\alpha r/2}(\alpha r)^L M(L + 1 - in, 2L + 2, -i\alpha r)$, regular continuum wave; see the correction below. This representation provides a powerful alternative technique for the calculation of photoionization and recombination coefficients.

In (a), $n \geq L+1$ with integer $n, L \geq 0$ and $\alpha > 0$. In (b), take $Z > 0$ and real $E > 0$; the continuum Coulomb parameter is not a discrete principal quantum number.

Solution.

(a) Put $N = n - L - 1 \geq 0$, $q = 2L + 1$ in

$$L_N^q(x) = \frac{(N+q)!}{N!q!} M(-N, q+1, x).$$

Then $N + q = n + L$, $-N = L + 1 - n$, $q + 1 = 2L + 2$. Multiplication by $e^{-\alpha r/2}(\alpha r)^L$ yields exactly

$$\boxed{e^{-\alpha r/2}(\alpha r)^L L_{n-L-1}^{2L+1}(\alpha r) = \frac{(n+L)!}{(n-L-1)!(2L+1)!} e^{-\alpha r/2}(\alpha r)^L M(L+1-n, 2L+2, \alpha r)}.$$

All factorials have nonnegative integer arguments in the bound-state range $n \geq L + 1$.

(b) Derive the positive-energy parameters directly to fix their signs. With $u = rR$, the physical radial equation is

$$u'' + \left[\frac{2m_e E}{\hbar^2} + \frac{2Z}{a_0 r} - \frac{L(L+1)}{r^2}\right] u = 0.$$

For $E > 0$, define

$$k = \frac{\sqrt{2m_e E}}{\hbar} > 0, \quad \alpha = 2k, \quad \rho = kr, \quad n = \frac{Z}{ka_0} > 0.$$

Here n is a continuous Coulomb parameter, not a bound-state principal quantum number. Dividing the equation by k^2 gives

$$\frac{d^2u}{d\rho^2} + \left[1 + \frac{2n}{\rho} - \frac{L(L+1)}{\rho^2}\right] u = 0.$$

This is Exercise 18.6.15 with $\eta = -n$. On dividing its regular solution by r and absorbing constants into the arbitrary scale,

$$R_{\text{reg}}(r) \;\propto\; e^{-i\alpha r/2}(\alpha r)^L M(L+1+in, 2L+2, i\alpha r).$$

Kummer's transformation $M(a, c, z) = e^z M(c-a, c, -z)$ gives the form printed in the question:

$$\boxed{R_{\text{reg}}(r) \;\propto\; e^{i\alpha r/2}(\alpha r)^L M(L+1-in, 2L+2, -i\alpha r)}.$$

It is regular, $R_{\text{reg}} = O(r^L)$, at the origin.

The label "outgoing wave" is incorrect for this M solution. It is the regular continuum wave, containing incoming and outgoing components. For real n, r, complex conjugation of the last expression produces the preceding expression, and Kummer's transformation proves they are equal. Thus it can be chosen real and has zero radial current $\text{Im}(R^* R')$, whereas a nonzero purely outgoing wave carries positive outward current.

For comparison, a purely outgoing radial solution, with time factor $e^{-iEt/\hbar}$, is

$$R_{\text{out}}(r) \;\propto\; e^{i\alpha r/2}(\alpha r)^L U(L+1-in, 2L+2, -i\alpha r).$$

The asymptotic $U(a, c, z) \sim z^{-a}$ holds for $|\arg z| \leq 3\pi/2 - \delta$, including this ray; see DLMF 13.7.3. Thus its leading radial form is a nonzero constant times $r^{-1}\exp\{i[kr + n\ln(2kr)]\}$, displaying the outward Coulomb phase. This U solution is generally irregular at the origin; regularity and a purely outgoing condition are different boundary conditions.

18.6.17 Evaluate

(a) $\displaystyle\int_0^\infty \big[M_{k\mu}(x)\big]^2 dx,$

(b) $\displaystyle\int_0^\infty \big[M_{k\mu}(x)\big]^2 \frac{dx}{x},$

(c) $\displaystyle\int_0^\infty \big[M_{k\mu}(x)\big]^2 \frac{dx}{x^{1-a}},$

where $2\mu = 0, 1, 2, \ldots,\quad k - \mu - \frac{1}{2} = 0, 1, 2, \ldots,\quad a > -2\mu - 1.$

ANS. The printed norm and general-moment answers are inconsistent with the definition in Eq. (18.152); the corrected evaluations are shown below.

Solution.

Use the definition in Eq. (18.152), without changing its scale. Put

$$q = 2\mu \in \{0, 1, 2, \ldots\}, \qquad N = k - \mu - \tfrac{1}{2} \in \{0, 1, 2, \ldots\}, \qquad 2k = 2N + q + 1.$$

The terminating Kummer–Laguerre relation gives

$$M_{k\mu}(x) = Ae^{-x/2}x^{(q+1)/2}L_N^q(x), \qquad A = \frac{N!q!}{(N+q)!}.$$

Define the common norm factor

$$B = A^2\frac{(N+q)!}{N!} = \boxed{\frac{N!(q!)^2}{(N+q)!}}.$$

(b) Orthogonality immediately yields, with its normalization shown,

$$\int_0^\infty M_{k\mu}(x)^2\frac{dx}{x} = A^2\int_0^\infty e^{-x}x^q[L_N^q(x)]^2dx = \boxed{B}.$$

(a) The first-moment recurrence in Exercise 18.3.6 gives

$$\int_0^\infty M_{k\mu}(x)^2 dx = A^2(2N+q+1)\frac{(N+q)!}{N!} = \boxed{2kB}.$$

The printed answers omit the factor $N!q!/(N+q)! = 1/\binom{N+q}{N}$ in both (a) and (b). For example, $N = q = 1$ gives $k = 2$, $\mu = 1/2$ and $M_{2,1/2} = xe^{-x/2}(1 - x/2)$. Its two integrals are 2 and 1/2, rather than 4 and 1.

(c) For real $a > -q - 1$, the integral is

$$I_a = A^2\int_0^\infty e^{-x}x^{q+a}[L_N^q(x)]^2dx.$$

The given condition is exact at zero: the Kummer polynomial has constant term one, so the original integrand has leading power x^{q+a}. Infinity is harmless because of e^{-x}. A useful exact single sum

follows by changing the Laguerre parameter. Their generators satisfy

$$\begin{aligned}\sum_{N=0}^{\infty} L_N^q(x)t^N &= (1-t)^{-q-1}e^{-xt/(1-t)} \\ &= (1-t)^a \sum_{j=0}^{\infty} L_j^{q+a}(x)t^j \\ &= \left[\sum_{r=0}^{\infty} \frac{(-a)_r}{r!}t^r\right]\left[\sum_{j=0}^{\infty} L_j^{q+a}(x)t^j\right].\end{aligned}$$

Coefficient comparison gives the finite connection formula

$$L_N^q(x) = \sum_{j=0}^{N} \frac{(-a)_{N-j}}{(N-j)!} L_j^{q+a}(x).$$

Square it and integrate with weight $e^{-x}x^{q+a}$. Since $q+a > -1$, orthogonality applies and all unequal-index terms vanish. Thus the complete answer is

$$\boxed{I_a = \left[\frac{N!q!}{(N+q)!}\right]^2 \sum_{j=0}^{N} \frac{[(-a)_{N-j}]^2}{[(N-j)!]^2} \frac{\Gamma(j+q+a+1)}{j!}}.$$

Every term is finite and nonnegative for the stated real parameters. At $a=0$ only $j=N$ survives, giving $I_0 = B$. At $a=1$ only $j = N, N-1$ survive, giving

$$I_1 = A^2\left[\frac{\Gamma(N+q+2)}{N!} + \frac{\Gamma(N+q+1)}{(N-1)!}\right] = (2N+q+1)B,$$

with the second term omitted for $N=0$. This recovers (a).

The printed general formula $(2k)^a q!$ is also incorrect even when the normalization factor happens to be one. At $N=q=0$, $k=1/2$ and $M_{1/2,0} = x^{1/2}e^{-x/2}$, so $I_a = \Gamma(a+1)$; at $a=2$ this is 2, while the printed answer is 1. More generally the next integer moment is

$$I_2 = B\,[6N(N+q+1) + (q+1)(q+2)],$$

obtained from the three nonzero connection coefficients $1, -2, 1$. It exceeds $(2k)^2 B$ by $B[2N(N+q+1)+q+1] > 0$, so even restoring the missing normalization cannot turn all moments into powers of the first moment.

18.7 Dilogarithm

18.7.1 Prove that the expansion of $\mathrm{Li}_2(z)$, Eq. (18.156), converges everywhere on the circle $|z| = 1$.

Solution.

For every $|z| \le 1$,

$$\left|\frac{z^n}{n^2}\right| \le \frac{1}{n^2}, \qquad \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} < \infty.$$

The Weierstrass comparison test therefore proves absolute and uniform convergence on the whole closed unit disk, in particular at every point of its boundary circle. More explicitly,

$$\left|\sum_{n=N+1}^{\infty} \frac{z^n}{n^2}\right| \le \sum_{n=N+1}^{\infty} \frac{1}{n^2} \le \int_N^{\infty} \frac{dt}{t^2} = \frac{1}{N}.$$

This controls every boundary point with the same remainder bound. The sum is continuous there. Convergence at $z = 1$ does not make it analytic across that point: its derivative $-\log(1-z)/z$ has the stated logarithmic branch singularity.

18.7.2 Use the functional relations, Eqs. (18.161) to (18.163), to find the principal value of $\mathrm{Li}_2(1/2)$.

Solution.

On the principal real branch for $0 < z < 1$, the reflection identity is

$$\mathrm{Li}_2(z) + \mathrm{Li}_2(1-z) = \frac{\pi^2}{6} - \ln z \ln(1-z).$$

At $z = 1/2$ the two dilogarithms are equal and both logarithms are $-\ln 2$. Thus

$$2\,\mathrm{Li}_2(1/2) = \frac{\pi^2}{6} - (\ln 2)^2, \qquad \boxed{\mathrm{Li}_2(1/2) = \frac{\pi^2}{12} - \frac{1}{2}(\ln 2)^2}.$$

As a check using a second supplied identity, set $z = 1/2$ in Eq. (18.163). Its transformed argument is -1, so $\mathrm{Li}_2(1/2) + \mathrm{Li}_2(-1) = -\frac{1}{2}\ln^2(1/2)$. Since $\mathrm{Li}_2(-1) = -\pi^2/12$, the same answer follows.

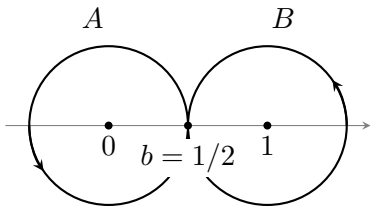
18.7.3 Find all the multiple values of $\mathrm{Li}_2(1/2)$.

Solution.

Let $b = 1/2$ and $F_0 = \pi^2/12 - \frac{1}{2}\ln^2 2$ denote the principal value. To find every value, continue the differential

$$dF = -L_1(z)\frac{dz}{z}, \qquad L_1(z) = \log(1-z),$$

around loops based at b in $\mathbb{C} \setminus \{0, 1\}$. Let A encircle zero once counterclockwise and B encircle one once counterclockwise. They generate all closed paths up to homotopy. The two circles below are valid representatives.



On a sheet where L_1 has increased by $2\pi i n$, traversal of A leaves L_1 unchanged and changes F by

$$\Delta_A F = -2\pi i n \oint_A \frac{dz}{z} = 4\pi^2 n.$$

The principal part contributes zero because $-\log(1-z)/z$ is analytic inside the zero circle. For B, the additive constant $2\pi i n$ contributes zero because B has zero winding about zero. To find the remaining increment, let L_0 be the logarithm of z analytic inside this one-circle neighborhood, with $L_0(1) = 0$. Integration by parts gives

$$\Delta_B F = -[L_1 L_0]_{\text{start}}^{\text{end}} + \oint_B L_0 \, dL_1 = -2\pi i \ln b + 2\pi i L_0(1) = 2\pi i \ln 2.$$

Here L_1 gains $2\pi i$, so B also changes the integer sheet label from n to $n+1$. Inverse loops give the inverse increments. It follows inductively that every word in A, B ends at a value of the form

$$\boxed{F_0 + 2\pi i n \ln 2 + 4\pi^2 m, \qquad m, n \in \mathbb{Z}}.$$

Conversely, follow B, then A, then B^{-1}, then A^{-1}. The two B increments cancel, while A was traversed with sheet label one; this path adds $4\pi^2$ and returns the label to zero. Its repetitions and inverse give every m, after which B^n gives every n. Thus the boxed set is both necessary and sufficient, with no branch values omitted.

The manual's proposed $2n^2\pi^2 - 2\pi i n \ln 2$ correction is not the full branch set and is incorrect for odd n. Substituting a multivalued logarithm into the principal reflection formula while equating its two dilogarithms assigns incompatible branches. For example one loop B gives $F_0 + 2\pi i \ln 2$ with no real shift, directly contradicting that proposed enumeration.

18.7.4 Explain why Eq. (18.161) gives the expected result for $z = 0$ when on the principal branch of the dilogarithm.

Solution.

One cannot substitute $z = 0$ separately into $\ln z$. Take the principal-branch limit instead. For $z \to 0$ off its logarithmic cut,

$$\ln(1-z) = -z + O(z^2), \qquad \ln z \ln(1-z) = -z \ln z + O(z^2 \ln z).$$

Writing $z = re^{i\theta}$ with $|\theta| \le \pi$ gives $|z \ln z| \le r(|\ln r| + \pi) \to 0$; the remainder also vanishes. The dilogarithm series gives $\mathrm{Li}_2(0) = 0$, while its continuous boundary value at one is $\mathrm{Li}_2(1) = \sum n^{-2} = \pi^2/6$. Consequently the two sides of Eq. (18.161) both tend to

$$\boxed{\pi^2/6}.$$

This is a limit of the product, not an assignment of a finite value to $\ln 0$. The principal branch condition also matters: on other sheets the extra logarithmic terms must be continued consistently.

18.7.5 Show that

$$\mathrm{Li}_2\left(\frac{1+z^{-1}}{2}\right) = -\mathrm{Li}_2\left(\frac{1+z}{1-z}\right) - \frac{1}{2}\ln^2\left(\frac{1-z^{-1}}{2}\right).$$

Exclude $z = 0, 1$ and use matching logarithm and dilogarithm branches. Principal values give the identity directly for real $z > 1$ or $z < 0$; continue consistently elsewhere.

Solution.

Set $w = (1 + z^{-1})/2 = (z + 1)/(2z)$, with $z \neq 0, 1$. Two elementary reductions are

$$\frac{w}{w-1} = \frac{(z+1)/(2z)}{(1-z)/(2z)} = \frac{1+z}{1-z}, \qquad 1 - w = \frac{1 - z^{-1}}{2}.$$

Insert them into Eq. (18.163), $\mathrm{Li}_2(w) + \mathrm{Li}_2(w/(w-1)) = -\frac{1}{2}\log^2(1-w)$, and move its second term to the right:

$$\boxed{\mathrm{Li}_2\left(\frac{1+z^{-1}}{2}\right) = -\,\mathrm{Li}_2\left(\frac{1+z}{1-z}\right) - \frac{1}{2}\log^2\left(\frac{1-z^{-1}}{2}\right).}$$

For real $z > 1$ or $z < 0$, both dilogarithm arguments are less than one and the logarithm's argument is positive, so this is an equality of ordinary real principal values. Complex values follow by continuing all three terms along the same path from such a domain.

The formula is not an unrestricted identity between independently selected boundary values on the cut. For $0 < z < 1$, both dilogarithm arguments exceed one, but a small imaginary part of z puts them on opposite sides of their cuts. For example, at $z = 1/3$ the arguments both equal 2. Selecting the same boundary value for both would incorrectly assert $2\,\mathrm{Li}_2(2) = \pi^2/2$, although the boundary values have imaginary parts $\pm\pi \ln 2$. Opposite boundary values, with the matching logarithm, satisfy the continued identity.

18.7.6 The following integral arises in the computation of the electronic energy of the Li atom using a correlated wave function (one that explicitly includes the electron-electron distances as well as the distances of electrons from the nucleus):

$$I = \iiint d^3r_1\, d^3r_2\, d^3r_3\, \frac{e^{-\alpha_1 r_1 - \alpha_2 r_2 - \alpha_3 r_3}}{r_1 r_2 r_3 r_{12} r_{13} r_{23}},$$

where $r_i = |\mathbf{r}_i|$, $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$, and the integrations are over the entire three-dimensional space of each $\mathbf{r}_i$. For convergence of I, we require all $\alpha_j > 0$, but there are no restrictions on their relative magnitudes.

In terms of the auxiliary quantities

$$\zeta_1 = \frac{\alpha_1}{\alpha_2 + \alpha_3}, \qquad \zeta_2 = \frac{\alpha_2}{\alpha_1 + \alpha_3}, \qquad \zeta_3 = \frac{\alpha_3}{\alpha_1 + \alpha_2},$$

this integral has the value

$$I = \frac{32\pi^3}{\alpha_1\alpha_2\alpha_3}\left(-\frac{\pi^2}{2} + \sum_{j=1}^{3}\left[\mathrm{Li}_2(\zeta_j) - \mathrm{Li}_2(-\zeta_j) + \ln\zeta_j \ln\left(\frac{1-\zeta_j}{1+\zeta_j}\right)\right]\right).$$

Rearrange I to a form (first found by Remiddi[4]), in which all terms in the final expression are guaranteed to evaluate to real quantities and can be evaluated as principal values. In the given

[4] E. Remiddi, "Analytic value of the atomic three-electron correlation integral with Slater wave functions," *Physical Review A* **44**, 5492 (1991).

complex expression use continuation from $0 < \zeta_j < 1$, with the logarithm boundary values matching those of $\mathrm{Li}_2(\zeta_j)$. At $\zeta_j = 1$ use the finite limit.

Solution.

For a positive variable ζ, denote one summand by

$$S(\zeta) = \mathrm{Li}_2(\zeta) - \mathrm{Li}_2(-\zeta) + \ln\zeta \log\frac{1-\zeta}{1+\zeta}.$$

Start with $0 < \zeta < 1$, where all terms are real. Reflection and the logarithm of a positive quotient give

$$\mathrm{Li}_2(\zeta) = \frac{\pi^2}{6} - \mathrm{Li}_2(1-\zeta) - \ln\zeta\ln(1-\zeta),$$
$$\log\frac{1-\zeta}{1+\zeta} = \log(1-\zeta) - \ln(1+\zeta).$$

The terms containing $\ln\zeta\log(1-\zeta)$ cancel, leaving

$$S(\zeta) = \frac{\pi^2}{6} - \mathrm{Li}_2(1-\zeta) - \mathrm{Li}_2(-\zeta) - \ln\zeta\ln(1+\zeta).$$

Continue this equality consistently above or below $\zeta = 1$. The imaginary parts in the original combination cancel; the right side is already real for every $\zeta > 0$. At $\zeta = 1$ it has a finite limit, so no $0\log 0$ product needs to be evaluated separately.

Sum over $j = 1, 2, 3$. The three constants $\pi^2/6$ cancel the original $-\pi^2/2$ exactly. Thus the desired form is

$$\boxed{I = -\frac{32\pi^3}{\alpha_1\alpha_2\alpha_3}\sum_{j=1}^{3}\left[\mathrm{Li}_2(-\zeta_j) + \mathrm{Li}_2(1-\zeta_j) + \ln\zeta_j\ln(1+\zeta_j)\right]}.$$

The assumptions $\alpha_j > 0$ imply $\zeta_j > 0$. Consequently $-\zeta_j < 0$, $1-\zeta_j < 1$, and both logarithm arguments are strictly positive. Every term in the final formula therefore has a real principal value, for arbitrary relative magnitudes of the three positive decay parameters. In particular, $\zeta_j = 1$ is harmless because $\mathrm{Li}_2(0) = 0$ and $\ln 1 = 0$.

18.8 Elliptic Integrals

18.8.1 The ellipse $x^2/a^2 + y^2/b^2 = 1$ may be represented parametrically by $x = a\sin\theta$, $y = b\cos\theta$. Show that the length of arc within the first quadrant is

$$a\int_0^{\pi/2}(1 - m\sin^2\theta)^{1/2}\,d\theta = aE(m).$$

Take $a \geq b > 0$, so $0 \leq m = (a^2 - b^2)/a^2 < 1$. The value $m = 1$ is the degenerate limit $b \downarrow 0$.

Solution.

Take $a \geq b > 0$, so the given parameter lies in $0 \leq m < 1$. Differentiating the parametrization gives $dx/d\theta = a\cos\theta$ and $dy/d\theta = -b\sin\theta$. The arc-length element is therefore

$$ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{a^2\cos^2\theta + b^2\sin^2\theta}\,d\theta$$
$$= a\sqrt{1 - \frac{a^2-b^2}{a^2}\sin^2\theta}\,d\theta.$$

As θ increases from 0 to $\pi/2$, the point moves from $(0,b)$ to $(a,0)$ along precisely the first-quadrant arc. Thus its length is

$$\boxed{\ell = a\int_0^{\pi/2}\sqrt{1-m\sin^2\theta}\,d\theta = aE(m)}, \qquad m = 1-b^2/a^2.$$

For $a=b$, $m=0$ and $\ell=\pi a/2$, the quarter-circle length. The limit $b\downarrow 0$ gives $m\uparrow 1$ and $\ell\to a\int_0^{\pi/2}\cos\theta\,d\theta = a$; this is a degenerate limiting ellipse. If $b>a$, interchange the axis labels (or use $\theta\mapsto\pi/2-\theta$) to obtain $bE(1-a^2/b^2)$ in the conventional nonnegative-parameter form.

18.8.2 Derive the series expansion

$$E(m) = \frac{\pi}{2}\left[1-\left(\frac{1}{2}\right)^2\frac{m}{1}-\left(\frac{1\cdot 3}{2\cdot 4}\right)^2\frac{m^2}{3}-\cdots\right].$$

The series is valid for $|m|\le 1$, with the square root continued from $m=0$.

Solution.

For $|m|<1$, expand the integrand uniformly in θ:

$$\sqrt{1-m\sin^2\theta} = 1+\sum_{n=1}^{\infty}(-1)^n\binom{1/2}{n}m^n\sin^{2n}\theta.$$

For $n\ge 1$, direct multiplication of the binomial factors gives

$$(-1)^n\binom{1/2}{n} = -\frac{(2n-3)!!}{(2n)!!},$$

where $(-1)!!=1$. The sine moment is the beta integral

$$\int_0^{\pi/2}\sin^{2n}\theta\,d\theta = \frac{1}{2}B(n+\tfrac12,\tfrac12) = \frac{\pi}{2}\frac{(2n-1)!!}{(2n)!!}.$$

Multiply these factors and use $(2n-3)!! = (2n-1)!!/(2n-1)$:

$$\boxed{E(m) = \frac{\pi}{2}\left[1-\sum_{n=1}^{\infty}\frac{1}{2n-1}\left(\frac{(2n-1)!!}{(2n)!!}\right)^2 m^n\right]}.$$

The first two nonconstant terms are $-(1/2)^2 m$ and $-(1\cdot 3/(2\cdot 4))^2 m^2/3$, which reproduce the requested series. In ordinary powers,

$$E(m) = \frac{\pi}{2}\left(1-\frac{m}{4}-\frac{3m^2}{64}-\frac{5m^3}{256}-\cdots\right).$$

Since $(2n-1)!!/(2n)!! \sim 1/\sqrt{\pi n}$, the absolute coefficients are $O(n^{-2})$. Thus the series is absolutely uniformly convergent for $|m|\le 1$ and extends to the boundary by continuity. In particular $m=1$ gives $E(1)=1$, consistent with its defining integral.

18.8.3 Show that

$$\lim_{m\to 0}\frac{K(m)-E(m)}{m}=\frac{\pi}{4}.$$

Solution.

The original printed exercise has $(K-E)/m$, without an extra factor π in the numerator. That extra factor in the supplied transcription has been removed. The published erratum also makes both arguments explicitly m. Subtract the defining integrands before taking the limit:

$$\begin{aligned}K(m)-E(m)&=\int_0^{\pi/2}\frac{1-(1-m\sin^2\theta)}{\sqrt{1-m\sin^2\theta}}d\theta\\&=m\int_0^{\pi/2}\frac{\sin^2\theta}{\sqrt{1-m\sin^2\theta}}d\theta.\end{aligned}$$

For $|m|$ bounded strictly below one, the integrand is dominated by a constant times $\sin^2\theta$. Dominated convergence gives

$$\boxed{\lim_{m\to 0}\frac{K(m)-E(m)}{m}=\int_0^{\pi/2}\sin^2\theta\,d\theta=\frac{\pi}{4}}.$$

Equivalently the two series start with $K=(\pi/2)(1+m/4+O(m^2))$ and $E=(\pi/2)(1-m/4+O(m^2))$. If the spurious numerator factor π were retained, the limit would instead be $\pi^2/4$, so it cannot accompany the stated right side.

18.8.4 A circular loop of wire in the xy-plane, centered at the origin with radius $a>0$, carries a current I. Given that the vector potential is

$$A_\varphi(\rho,\varphi,z)=\frac{a\mu_0 I}{2\pi}\int_0^{\pi}\frac{\cos\alpha\,d\alpha}{(a^2+\rho^2+z^2-2a\rho\cos\alpha)^{1/2}},$$

show that

$$A_\varphi(\rho,\varphi,z)=\frac{\mu_0 I}{\pi k}\left(\frac{a}{\rho}\right)^{1/2}\left[\left(1-\frac{k^2}{2}\right)K(k^2)-E(k^2)\right],$$

where

$$k^2=\frac{4a\rho}{(a+\rho)^2+z^2}.$$

Note. For extension of this exercise to **B**, see Smythe.[5] Take $\rho>0$ and real z, away from the wire $(\rho,z)=(a,0)$; recover the axis by a limit.

Solution.

Assume $a>0$, $\rho>0$, away from the ideal wire $(\rho,z)=(a,0)$. Set

$$D=(a+\rho)^2+z^2,\quad m=k^2=\frac{4a\rho}{D},\quad 0<m<1.$$

Using $\cos\alpha=2\cos^2(\alpha/2)-1$, the denominator squared becomes

$$a^2+\rho^2+z^2-2a\rho\cos\alpha=D-4a\rho\cos^2(\alpha/2)=D[1-m\cos^2(\alpha/2)].$$

[5] W. R. Smythe, *Static and Dynamic Electricity*, 3rd ed. (McGraw-Hill, New York, 1969), p. 270.

Put $u = \alpha/2$ and then $\theta = \pi/2 - u$:

$$A_\varphi = \frac{a\mu_0 I}{\pi\sqrt{D}} \int_0^{\pi/2} \frac{2\sin^2\theta - 1}{\sqrt{1 - m\sin^2\theta}} d\theta.$$

The difference of the elliptic integrands gives

$$K(m) - E(m) = m \int_0^{\pi/2} \frac{\sin^2\theta}{\sqrt{1 - m\sin^2\theta}} d\theta.$$

Therefore the remaining angular integral is

$$\frac{2}{m}[K(m) - E(m)] - K(m) = \frac{2}{m}\left[(1 - \tfrac{m}{2})K(m) - E(m)\right].$$

Finally

$$\frac{2a}{m\sqrt{D}} = \frac{1}{k}\sqrt{\frac{a}{\rho}},$$

because $k = 2\sqrt{a\rho}/\sqrt{D}$. Substitution yields

$$\boxed{A_\varphi(\rho, \varphi, z) = \frac{\mu_0 I}{\pi k}\sqrt{\frac{a}{\rho}}\left[(1 - \tfrac{k^2}{2})K(k^2) - E(k^2)\right]}.$$

Rotational symmetry explains the absence of φ. At $\rho = 0$ the apparent singular factors are interpreted by a limit: the original integral is zero, so $A_\varphi = 0$ on the axis. At the ideal wire $k^2 = 1$, the logarithmic singularity is real and is not a removable limit. The cancellation near $k = 0$ is also quantified by the next exercise.

18.8.5 An analysis of the magnetic vector potential of a circular current loop leads to the expression

$$f(k^2) = k^{-2}\left[(2 - k^2)K(k^2) - 2E(k^2)\right],$$

where $K(k^2)$ and $E(k^2)$ are the complete elliptic integrals of the first and second kinds. Show that for $k^2 \ll 1$ ($r \gg$ radius of loop)

$$f(k^2) \approx \frac{\pi k^2}{16}.$$

Here $r = \sqrt{\rho^2 + z^2}$ and the loop radius is a. The far-field condition $r \gg a$ is sufficient for $k^2 \ll 1$.

Solution.

Write $m = k^2$. Since the definition divides by m, retain the elliptic series through m^2 (one extra order beyond the desired answer):

$$K(m) = \frac{\pi}{2}\left(1 + \frac{m}{4} + \frac{9m^2}{64} + O(m^3)\right), \quad E(m) = \frac{\pi}{2}\left(1 - \frac{m}{4} - \frac{3m^2}{64} + O(m^3)\right).$$

Then

$$\begin{aligned}(2 - m)K(m) - 2E(m) &= \frac{\pi}{2}\left[2 - \frac{m}{2} + \left(\frac{18}{64} - \frac{1}{4}\right)m^2 - 2 + \frac{m}{2} + \frac{6m^2}{64} + O(m^3)\right] \\ &= \frac{\pi}{2}\left[\frac{m^2}{8} + O(m^3)\right].\end{aligned}$$

The constant and linear terms cancel. Dividing by m gives

$$\boxed{f(k^2) = \frac{\pi k^2}{16} + O(k^4)}.$$

Keeping the next terms $25m^3/256$ in $2K/\pi$ and $-5m^3/256$ in $2E/\pi$ would give $f(m) = \pi m/16 + 3\pi m^2/64 + O(m^3)$. Thus f has the removable value $f(0) = 0$.

For the distant loop field, $D \sim r^2$, $m \sim 4a\rho/r^2$ and the preceding exercise has $A_\varphi = a\mu_0 I f(m)/(\pi\sqrt{D})$. Consequently its leading term is $\mu_0 I a^2 \rho/(4r^3) = \mu_0 I a^2 \sin\theta/(4r^2)$, the magnetic-dipole form. Large r/a is a sufficient way to make m small; it is not the only one, since $m \to 0$ also on approaching the symmetry axis.

18.8.6 Show that

(a) $\dfrac{d\,E(k^2)}{dk} = \dfrac{1}{k}\big(E(k^2) - K(k^2)\big)$,

(b) $\dfrac{d\,K(k^2)}{dk} = \dfrac{E(k^2)}{k(1-k^2)} - \dfrac{K(k^2)}{k}$.

Hint. For part (b) show that

$$E(k^2) = (1-k^2)\int_0^{\pi/2} (1 - k^2\sin^2\theta)^{-3/2}\, d\theta$$

by comparing series expansions. Initially take $0 < k < 1$; both derivatives have removable limits at $k = 0$.

Solution.

Take initially $0 < k < 1$, put $m = k^2$, and use differentiation under the integral on compact subintervals of this range. (a) The chain rule gives

$$\frac{dE(k^2)}{dk} = -k\int_0^{\pi/2} \frac{\sin^2\theta}{\sqrt{1 - k^2\sin^2\theta}}\, d\theta.$$

Since $-k\sin^2\theta = [(1 - k^2\sin^2\theta) - 1]/k$,

$$\boxed{\frac{dE(k^2)}{dk} = \frac{E(k^2) - K(k^2)}{k}}.$$

(b) First verify the corrected hint by the requested series method. Define

$$a_n = \left[\frac{(1/2)_n}{n!}\right]^2, \qquad J(m) = \int_0^{\pi/2} (1 - m\sin^2\theta)^{-3/2} d\theta.$$

The binomial series and the sine moment give

$$J(m) = \frac{\pi}{2}\sum_{n=0}^{\infty} \frac{(3/2)_n (1/2)_n}{(n!)^2} m^n = \frac{\pi}{2}\sum_{n=0}^{\infty} (2n+1) a_n m^n.$$

For $n \geq 1$, $a_n/a_{n-1} = (2n-1)^2/(4n^2)$. Hence the m^n coefficient of $2(1-m)J/\pi$ is

$$(2n+1)a_n - (2n-1)a_{n-1} = a_n\left[(2n+1) - \frac{4n^2}{2n-1}\right] = -\frac{a_n}{2n-1}.$$

The constant is one, so Exercise 18.8.2 identifies the entire series with $2E(m)/\pi$. Thus

$$\boxed{E(m) = (1-m)J(m)}, \qquad |m| < 1.$$

This uses the corrected integrand $1 - k^2 \sin^2\theta$, not $1 - k\sin^2\theta$.

Now differentiate K and rewrite its numerator:

$$\begin{aligned}
\frac{dK(k^2)}{dk} &= k\int_0^{\pi/2} \frac{\sin^2\theta}{(1-k^2\sin^2\theta)^{3/2}}\,d\theta \\
&= \frac{1}{k}\int_0^{\pi/2} \left[\frac{1}{(1-k^2\sin^2\theta)^{3/2}} - \frac{1}{\sqrt{1-k^2\sin^2\theta}}\right] d\theta \\
&= \frac{J(k^2)-K(k^2)}{k} = \boxed{\frac{E(k^2)}{k(1-k^2)} - \frac{K(k^2)}{k}}.
\end{aligned}$$

All elliptic arguments in these formulas are the parameter k^2, as stated in the published erratum. At $k = 0$ both derivatives have the removable value zero: $E(k^2) = \pi/2 - \pi k^2/8 + O(k^4)$ and $K(k^2) = \pi/2 + \pi k^2/8 + O(k^4)$. The singular endpoint $k = 1$ is outside the finite derivative formulas and is approached only as a limit.

19 Chapter 19. Fourier Series

Problems

19.1 General Properties

19.1.1 A real-valued function $f(x)$ (quadratically integrable) is to be represented by a finite Fourier series. A convenient measure of the accuracy of the series is given by the integrated square of the deviation,

$$\Delta_p = \int_0^{2\pi} \left[f(x) - \frac{a_0}{2} - \sum_{n=1}^{p} (a_n \cos nx + b_n \sin nx) \right]^2 dx.$$

Show that the requirement that Δ_p be minimized, that is,

$$\frac{\partial \Delta_p}{\partial a_n} = 0, \qquad \frac{\partial \Delta_p}{\partial b_n} = 0,$$

for all n, leads to choosing a_n and b_n as given in Eqs. (19.2) and (19.3).

Note. Your coefficients a_n and b_n are independent of p. This independence is a consequence of orthogonality and would not hold if we expanded $f(x)$ in a power series. Take the coefficients real as well. For complex data, replace the squared residual by its modulus squared.

Solution.

Write $s_p = a_0/2 + \sum_{n=1}^{p} (a_n \cos nx + b_n \sin nx)$. The squared error as printed is a real least-squares error when f and the coefficients are real. For complex data use $|f - s_p|^2$. All integrals below exist: Cauchy–Schwarz on the finite interval gives $\int |f| \le \sqrt{2\pi}\,(\int |f|^2)^{1/2}$.

For $1 \le m \le p$, differentiate the finite quadratic polynomial:

$$\frac{\partial \Delta_p}{\partial a_m} = -2 \int_0^{2\pi} (f - s_p) \cos mx\, dx = -2 \int_0^{2\pi} f \cos mx\, dx + 2\pi a_m,$$
$$\frac{\partial \Delta_p}{\partial b_m} = -2 \int_0^{2\pi} (f - s_p) \sin mx\, dx = -2 \int_0^{2\pi} f \sin mx\, dx + 2\pi b_m.$$

Here the mixed sine–cosine integrals vanish, while $\int \cos nx \cos mx = \int \sin nx \sin mx = \pi \delta_{nm}$. For the constant coefficient its factor $1/2$ gives instead

$$\frac{\partial \Delta_p}{\partial a_0} = -\int_0^{2\pi} (f - s_p)\, dx = -\int_0^{2\pi} f\, dx + \pi a_0.$$

Setting these derivatives to zero yields

$$\boxed{a_m = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos mx\, dx \quad (m \ge 0), \qquad b_m = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin mx\, dx \quad (m \ge 1).}$$

These stationary values are the unique minimum, not merely a stationary point. If s_p^* has the displayed coefficients and the coefficients are changed by $\alpha_0, \alpha_m, \beta_m$, orthogonality to the residual gives

$$\Delta_p = \Delta_p^* + \frac{\pi}{2} \alpha_0^2 + \pi \sum_{m=1}^{p} (\alpha_m^2 + \beta_m^2).$$

Every added term is nonnegative and vanishes only when all changes vanish. No coefficient formula contains p: adding an orthogonal mode cannot change the projections onto modes already present.

19.1.2 In the analysis of a complex waveform (ocean tides, earthquakes, musical tones, etc.), it might be more convenient to have the Fourier series written as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \alpha_n \cos(nx - \theta_n).$$

Show that this is equivalent to Eq. (19.1) with

$$a_n = \alpha_n \cos\theta_n, \qquad \alpha_n^2 = a_n^2 + b_n^2,$$
$$b_n = \alpha_n \sin\theta_n, \qquad \tan\theta_n = b_n/a_n.$$

Note. The coefficients α_n^2 as a function of n define what is called the *power spectrum*. The importance of α_n^2 lies in their invariance under a shift in the phase θ_n. The waveform and its sine/cosine coefficients are real. Take $\alpha_n \geq 0$; the tangent formula applies only when $a_n \neq 0$. Specify the phase quadrant and the zero-amplitude case.

Solution.

The cosine subtraction formula gives, for each harmonic,

$$\alpha_n \cos(nx - \theta_n) = \alpha_n \cos\theta_n \cos nx + \alpha_n \sin\theta_n \sin nx.$$

Matching the independent sine and cosine coefficients gives $a_n = \alpha_n \cos\theta_n$, $b_n = \alpha_n \sin\theta_n$. Squaring and adding, and then dividing when $a_n \neq 0$, gives

$$\boxed{\alpha_n = \sqrt{a_n^2 + b_n^2}, \qquad \tan\theta_n = b_n/a_n.}$$

We choose the nonnegative amplitude. The tangent alone does not fix the quadrant and is undefined when $a_n = 0$. For nonzero amplitude take $\theta_n = \operatorname{atan2}(b_n, a_n)$ modulo 2π; for $a_n = b_n = 0$ the phase is arbitrary. In particular, when $a_n = 0$ and $b_n \neq 0$, the phase is $\operatorname{sgn}(b_n)\pi/2$ modulo 2π.

Under a shift $x \mapsto x + x_0$, the phase becomes $\theta_n - nx_0$, while the amplitude stays unchanged. Equivalently the coefficient pair is rotated:

$$\begin{pmatrix} a_n' \\ b_n' \end{pmatrix} = \begin{pmatrix} \cos nx_0 & \sin nx_0 \\ -\sin nx_0 & \cos nx_0 \end{pmatrix} \begin{pmatrix} a_n \\ b_n \end{pmatrix}, \qquad (a_n')^2 + (b_n')^2 = a_n^2 + b_n^2.$$

This is the asserted phase invariance of the power spectrum.

19.1.3 A function $f(x)$ is expanded in an exponential Fourier series

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}.$$

If $f(x)$ is real, $f(x) = f^*(x)$, what restriction is imposed on the coefficients c_n?

Solution.

The coefficient formula, with f real, gives

$$c_n^* = \left[\frac{1}{2\pi}\int_{-\pi}^{\pi} f(x) e^{-inx}\, dx\right]^* = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x) e^{inx}\, dx = c_{-n}.$$

Thus

$$\boxed{c_{-n} = c_n^* \quad \text{for every integer } n; \qquad c_0 \in \mathbb{R}.}$$

Conversely, this relation makes every symmetric partial sum real because

$$c_n e^{inx} + c_{-n} e^{-inx} = 2\,\mathrm{Re}(c_n e^{inx}).$$

Consequently its pointwise limit, wherever it exists, is real; its L^2 limit is real almost everywhere as well.

19.1.4 Assuming that f is real and $\displaystyle\int_{-\pi}^{\pi}[f(x)]^2\,dx$ is finite, show that

$$\lim_{m\to\infty} a_m = 0, \qquad \lim_{m\to\infty} b_m = 0.$$

Hint. Integrate $[f(x) - s_n(x)]^2$, where $s_n(x)$ is the nth partial sum, and use Bessel's inequality (Section 5.1). For our finite interval the assumption that $f(x)$ is square integrable ($\displaystyle\int_{-\pi}^{\pi}|f(x)|^2\,dx$ is finite) implies that $\displaystyle\int_{-\pi}^{\pi}|f(x)|\,dx$ is also finite. The converse does not hold.

Solution.

Let s_N be the finite Fourier projection. Expanding the square and using the coefficient integrals gives

$$0 \le \int_{-\pi}^{\pi}(f - s_N)^2\,dx = \int f^2\,dx - 2\int f s_N\,dx + \int s_N^2\,dx,$$

$$\int f s_N\,dx = \int s_N^2\,dx = \frac{\pi}{2}a_0^2 + \pi\sum_{n=1}^{N}(a_n^2 + b_n^2).$$

Therefore

$$\frac{a_0^2}{2} + \sum_{n=1}^{N}(a_n^2 + b_n^2) \le \frac{1}{\pi}\int_{-\pi}^{\pi} f^2\,dx < \infty.$$

The nonnegative partial sums are bounded, so their increments tend to zero: $a_n^2 + b_n^2 \to 0$. Since each square is no larger than the sum,

$$\boxed{a_n \to 0, \qquad b_n \to 0.}$$

No pointwise convergence of the original Fourier series was used. For complex f, use modulus squares throughout, as in 19.1.1, obtaining $|a_n|^2 + |b_n|^2 \to 0$.

The integrability assertion in the hint follows directly from

$$\int_{-\pi}^{\pi}|f|\cdot 1\,dx \le \left(\int |f|^2\,dx\right)^{1/2}\sqrt{2\pi}.$$

For a counterexample to the converse, $f(x) = |x|^{-2/3}$ on $0 < |x| < 1$, and zero elsewhere, has $\int|f| = 6$ but $\int|f|^2 = 2\int_0^1 x^{-4/3}\,dx = \infty$.

19.1.5 Apply the summation technique of this section to show that

$$\sum_{n=1}^{\infty}\frac{\sin nx}{n} = \begin{cases}\dfrac{1}{2}(\pi - x), & 0 < x \le \pi,\\[2ex] -\dfrac{1}{2}(\pi + x), & -\pi \le x < 0.\end{cases}$$

This is the reverse sawtooth wave shown in Fig. 19.5.

Solution.

First insert an Abel factor $0 < r < 1$. Absolute convergence permits taking the imaginary part of the logarithmic power series:

$$S_r(x) = \sum_{n=1}^{\infty}\frac{r^n\sin nx}{n} = \operatorname{Im}\{-\log(1 - re^{ix})\}.$$

Use the principal logarithm. For $0 < x < 2\pi$,

$$1 - e^{ix} = -2ie^{ix/2}\sin(x/2) = 2\sin(x/2)e^{i(x/2-\pi/2)},$$

whose argument is $x/2 - \pi/2 \in (-\pi/2, \pi/2)$. Hence $\lim_{r\uparrow 1} S_r(x) = (\pi - x)/2$. The original series converges for $x \notin 2\pi\mathbb{Z}$ by Dirichlet's test: the partial sums of e^{inx} are bounded by $2/|1 - e^{ix}|$, and $1/n$ decreases to zero. Abel's theorem then identifies its ordinary sum with this limit. Oddness supplies the negative interval:

$$\boxed{\sum_{n=1}^{\infty} \frac{\sin nx}{n} = \begin{cases} (\pi - x)/2, & 0 < x \le \pi, \\ -(\pi + x)/2, & -\pi \le x < 0. \end{cases}}$$

At $x = 0$ every summand is zero; the sum is zero, the mean of the two one-sided limits. At $\pm\pi$ it is also zero. The complete function is 2π-periodic.

19.1.6 Sum the series $\displaystyle\sum_{n=1}^{\infty} (-1)^{n+1} \frac{\sin nx}{n}$ and show that it equals $x/2$ for $-\pi < x < \pi$.

Solution.

For $0 < r < 1$, the alternating logarithmic series gives

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{r^n \sin nx}{n} = \operatorname{Im} \log(1 + re^{ix}).$$

When $-\pi < x < \pi$,

$$1 + e^{ix} = 2\cos(x/2)e^{ix/2}, \qquad \cos(x/2) > 0,$$

so its principal argument is $x/2$. Dirichlet convergence and Abel's theorem, as in 19.1.5, therefore give

$$\boxed{\sum_{n=1}^{\infty} (-1)^{n+1} \frac{\sin nx}{n} = \frac{x}{2}, \qquad -\pi < x < \pi.}$$

The interval is essential: the left side is periodic, whereas $x/2$ is not. At $x = (2k+1)\pi$ its value is zero, not $x/2$. On $((2k-1)\pi, (2k+1)\pi)$ its value is $(x - 2k\pi)/2$.

19.1.7 Sum the trigonometric series $\displaystyle\sum_{n=0}^{\infty} \frac{\sin(2n+1)x}{2n+1}$ and show that it equals

$$\begin{cases} \pi/4, & 0 < x < \pi, \\ -\pi/4, & -\pi < x < 0. \end{cases}$$

Solution.

Subtract the even harmonics from the sum of all harmonics. With S denoting the function in 19.1.5,

$$\sum_{n=0}^{\infty} \frac{\sin(2n+1)x}{2n+1} = S(x) - \frac{1}{2}S(2x).$$

This subtraction is legitimate either for Abel sums before $r \uparrow 1$ or for the convergent ordinary odd/even subseries. For $0 < x < \pi$ both x and $2x$ lie in $(0, 2\pi)$, so

$$S(x) - \frac{1}{2}S(2x) = \frac{\pi - x}{2} - \frac{\pi - 2x}{4} = \frac{\pi}{4}.$$

The series is odd, giving $-\pi/4$ on $(-\pi, 0)$. Thus

$$\boxed{\sum_{n=0}^{\infty} \frac{\sin(2n+1)x}{2n+1} = \begin{cases} \pi/4, & 0 < x < \pi, \\ -\pi/4, & -\pi < x < 0. \end{cases}}$$

At every integer multiple of π, each sine vanishes and the sum is zero, the midpoint value at the jump.

19.1.8 Let $f(z) = \ln(1+z) = \displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1} z^n}{n}$. This series converges to $\ln(1+z)$ for $|z| \le 1$, except at the point $z = -1$.

(a) From the real parts show that

$$\ln\left(2\cos\frac{\theta}{2}\right) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{\cos n\theta}{n}, \qquad -\pi < \theta < \pi.$$

(b) Using a change of variable, transform part (a) into

$$-\ln\left(2\sin\frac{\theta}{2}\right) = \sum_{n=1}^{\infty} \frac{\cos n\theta}{n}, \qquad 0 < \theta < 2\pi.$$

Solution.

(a) Put $z = e^{i\theta}$. For $-\pi < \theta < \pi$ the identity

$$1 + e^{i\theta} = 2\cos(\theta/2) e^{i\theta/2}$$

has a positive real amplitude, so the principal logarithm is

$$\log(1 + e^{i\theta}) = \ln(2\cos(\theta/2)) + i\theta/2.$$

Taking real parts of the convergent logarithmic series gives

$$\boxed{\ln(2\cos(\theta/2)) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{\cos n\theta}{n}.}$$

At the included point $\theta = 0$ this is the convergent alternating harmonic series. The excluded endpoints have the logarithmic singularity.

(b) Replace the angle in (a) by $\theta - \pi$, which belongs to $(-\pi, \pi)$ when $0 < \theta < 2\pi$. Then

$$\cos\frac{\theta - \pi}{2} = \sin\frac{\theta}{2}, \qquad \cos(n\theta - n\pi) = (-1)^n \cos n\theta.$$

Consequently the two alternating factors multiply to -1, yielding

$$\boxed{-\ln(2\sin(\theta/2)) = \sum_{n=1}^{\infty} \frac{\cos n\theta}{n}, \qquad 0 < \theta < 2\pi.}$$

The manual's intermediate exponential identity in (b) has the wrong sign. The correct identity is $2\sin(\theta/2) = ie^{-i\theta/2}(1 - e^{i\theta})$. Its final real-part result agrees with the formula above.

19.1.9

(a) Expand $f(x) = x$ in the interval $(0, 2L)$. Sketch the series you have found (right-hand side of ANS.) over $(-2L, 2L)$.

$$\textit{ANS. } x = L - \frac{2L}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\sin\frac{n\pi x}{L}.$$

(b) Expand $f(x) = x$ as a sine series in the half interval $(0, L)$. Sketch the series you have found (right-hand side of ANS.) over $(-2L, 2L)$.

$$\textit{ANS. } x = \frac{2L}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin\frac{n\pi x}{L}.$$

Solution.

(a) For period $2L$, put $k_n = n\pi/L$. The coefficient normalization on $(0, 2L)$ is $1/L$, so

$$a_0 = \frac{1}{L}\int_0^{2L} x\,dx = 2L.$$

Two elementary antiderivatives are

$$\int x\cos kx\,dx = \frac{x\sin kx}{k} + \frac{\cos kx}{k^2}, \qquad \int x\sin kx\,dx = -\frac{x\cos kx}{k} + \frac{\sin kx}{k^2}.$$

At $k = k_n$, the endpoint sines vanish and the endpoint cosines are one. Consequently $a_n = 0$ and $b_n = -2/k_n = -2L/(n\pi)$. Thus

$$\boxed{x = L - \frac{2L}{\pi}\sum_{n=1}^{\infty}\frac{\sin(n\pi x/L)}{n}, \qquad 0 < x < 2L.}$$

On $(-2L, 0)$ this same series equals $x + 2L$, and at $x = 0$ its value is the jump midpoint L.

(b) A sine series on $(0, L)$ uses the odd extension and the normalization $2/L$. Integration by parts now gives

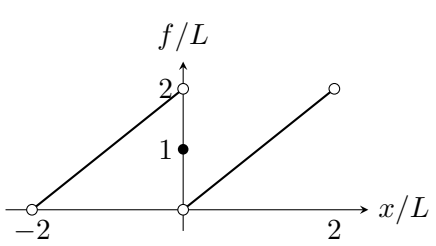
$$b_n = \frac{2}{L}\int_0^L x\sin k_n x\,dx = \frac{2}{L}\left[-\frac{x\cos k_n x}{k_n} + \frac{\sin k_n x}{k_n^2}\right]_0^L = \frac{2L(-1)^{n+1}}{n\pi}.$$

Therefore the corrected answer is

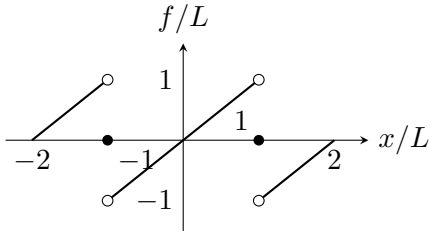
$$\boxed{x = \frac{2L}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin\frac{n\pi x}{L}, \qquad 0 < x < L.}$$

The book's printed answer in (b), with coefficient $4L/\pi$ and only odd harmonics, is incorrect for $f(x) = x$. By 19.1.7 that printed series equals the constant L throughout $(0, L)$. The answer in the question above has been restored in red. The corrected series equals x on $(-L, L)$, $x + 2L$ on $(-2L, -L)$, and $x - 2L$ on $(L, 2L)$; its values at $\pm L$ are zero.

These are the requested sketches, with both axes scaled by L:



(a) Full-interval expansion



(b) Half-interval sine expansion

Open circles are one-sided limits, and filled circles give the series values at the indicated jumps.

19.1.10 In some problems it is convenient to approximate $\sin \pi x$ over the interval $[0, 1]$ by a parabola $ax(1 - x)$, where a is a constant. To get a feeling for the accuracy of this approximation, expand $4x(1 - x)$ in a Fourier sine series $(-1 \le x \le 1)$:

$$f(x) = \begin{cases} 4x(1-x), & 0 \le x \le 1 \\ 4x(1+x), & -1 \le x \le 0 \end{cases} = \sum_{n=1}^{\infty} b_n \sin n\pi x.$$

ANS. $b_n = \dfrac{32}{\pi^3 n^3}$, n odd, $\qquad b_n = 0$, n even.

This approximation is shown in Fig. 19.6.

Solution.

The stated extension is odd, so all cosine coefficients vanish. Put $k = n\pi$; the sine coefficient is

$$b_n = 2\int_0^1 4x(1-x)\sin kx\,dx = 8I_n.$$

Integrating by parts twice, with $\sin k = 0$,

$$\begin{aligned} I_n &= \left[-\frac{x(1-x)\cos kx}{k}\right]_0^1 + \frac{1}{k}\int_0^1 (1-2x)\cos kx\,dx, \\ &= \frac{1}{k}\left[\frac{(1-2x)\sin kx}{k}\right]_0^1 + \frac{2}{k^2}\int_0^1 \sin kx\,dx = \frac{2(1-\cos k)}{k^3}. \end{aligned}$$

It follows that

$$\boxed{b_n = \frac{16[1-(-1)^n]}{\pi^3 n^3} = \begin{cases} 32/(\pi^3 n^3), & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases}}$$

Thus the fundamental sine has coefficient $32/\pi^3$, and the next nonzero coefficient is smaller by a factor 27. The series converges absolutely and uniformly, since its terms are bounded by a convergent multiple of $\sum n^{-3}$. It equals the continuous odd periodic extension, including its zero values at $-1, 0, 1$.

19.1.11 Verify that $\delta(\varphi_1-\varphi_2)=\dfrac{1}{2\pi}\displaystyle\sum_{m=-\infty}^{\infty} e^{im(\varphi_1-\varphi_2)}$ is a Dirac delta function by showing that it satisfies the definition,

$$\int_{-\pi}^{\pi} f(\varphi_1)\,\frac{1}{2\pi}\sum_{m=-\infty}^{\infty} e^{im(\varphi_1-\varphi_2)}\,d\varphi_1 = f(\varphi_2).$$

Hint. Represent $f(\varphi_1)$ by an exponential Fourier series. Use the periodic delta distribution on the circle, with a smooth 2π-periodic test function f; take the series through symmetric finite partial sums.

Solution.

This is the periodic delta distribution on the circle, not a pointwise convergent series of functions. On the real line it is the comb $\delta_{2\pi}(u)=\sum_{k\in\mathbb{Z}}\delta(u-2\pi k)$. The interval endpoints are identified. Take a smooth 2π-periodic test function $f(\varphi)=\sum_{j\in\mathbb{Z}} c_j e^{ij\varphi}$, with $c_j=(2\pi)^{-1}\int_{-\pi}^{\pi} f(\varphi)e^{-ij\varphi}\,d\varphi$. Integration by parts twice shows $c_j=O(j^{-2})$, so its coefficient series is absolutely summable. For a finite symmetric kernel,

$$\begin{aligned}\int_{-\pi}^{\pi} f(\varphi_1)\frac{1}{2\pi}\sum_{m=-N}^{N} e^{im(\varphi_1-\varphi_2)}\,d\varphi_1 &= \sum_{m=-N}^{N} c_{-m}e^{-im\varphi_2}\\ &= \sum_{j=-N}^{N} c_j e^{ij\varphi_2}.\end{aligned}$$

The last expression tends to $f(\varphi_2)$ as $N\to\infty$. This is exactly the defining distributional action, hence

$$\boxed{\delta_{2\pi}(\varphi_1-\varphi_2)=\frac{1}{2\pi}\sum_{m\in\mathbb{Z}} e^{im(\varphi_1-\varphi_2)}.}$$

For φ_2 strictly inside the integration interval, only one member of the delta comb is present. For a piecewise regular function with a jump at φ_2, symmetric Fourier sums give its midpoint value; one must not assert recovery of an arbitrary assigned value at the jump.

19.1.12 Show that integration of the Fourier expansion of $f(x)=x$, $-\pi<x<\pi$, leads to

$$\frac{\pi^2}{12}=\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^2}=1-\frac{1}{4}+\frac{1}{9}-\frac{1}{16}+\cdots.$$

Note. The series for $f(x)=x$ was the subject of Example 19.1.1. Confirm that the change in the defined range from $[0,2\pi]$ to $[-\pi,\pi]$ has no effect on the expansion.

Solution.

From 19.1.6, or from direct integration of its Fourier coefficients,

$$x=2\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n}\sin nx,\qquad -\pi<x<\pi.$$

Integrating from 0 to x gives

$$\frac{x^2}{2}=2\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n^2}(1-\cos nx).$$

The integrated series is absolutely and uniformly convergent; integration can also be justified by integrating the L^2 partial sums, since $|\int_0^x (f - s_N)| \le \sqrt{2\pi}\,\|f - s_N\|_2$ uniformly in x. Average the integrated identity over $(-\pi, \pi)$. Every cosine has zero average, and

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{x^2}{2}\,dx = \frac{1}{2\pi}\left[\frac{x^3}{6}\right]_{-\pi}^{\pi} = \frac{\pi^2}{6}.$$

Consequently

$$\boxed{\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}.}$$

There is no change in the underlying periodic function in the note: Example 19.1.1 defines it as x on $(0, \pi)$ and $x - 2\pi$ on $(\pi, 2\pi)$. Moving that latter interval left by 2π gives x on $(-\pi, 0)$. Integrals over either complete period therefore give exactly these same Fourier coefficients.

19.1.13

(a) Assuming that the Fourier expansion of a real-valued $f(x)$ is uniformly convergent, show that

$$\frac{1}{\pi}\int_{-\pi}^{\pi} \big[f(x)\big]^2 dx = \frac{a_0^2}{2} + \sum_{n=1}^{\infty} \big(a_n^2 + b_n^2\big).$$

This is Parseval's identity. Note that it is a completeness relation for the Fourier expansion.

(b) Given $x^2 = \dfrac{\pi^2}{3} + 4\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n \cos nx}{n^2}$, $-\pi \le x \le \pi$, apply Parseval's identity to obtain $\zeta(4)$ in closed form.

(c) The condition of uniform convergence is not necessary. Show this by applying the Parseval identity to the square wave

$$f(x) = \begin{cases} -1, & -\pi < x < 0 \\ 1, & 0 < x < \pi \end{cases} = \frac{4}{\pi}\sum_{n=1}^{\infty} \frac{\sin(2n-1)x}{2n-1}.$$

Solution.

(a) Orthogonality gives for every finite partial sum

$$\frac{1}{\pi}\int_{-\pi}^{\pi} s_N^2\,dx = \frac{a_0^2}{2} + \sum_{n=1}^{N}(a_n^2 + b_n^2).$$

If $s_N \to f$ uniformly, the s_N and f are bounded and

$$\left|\int (s_N^2 - f^2)\right| \le 2\pi\|s_N - f\|_\infty(\|s_N\|_\infty + \|f\|_\infty) \longrightarrow 0.$$

Taking limits proves the stated identity. The displayed identity is for real functions; the complex version uses modulus squares. Also the manual's first display in 19.1.13(a) is missing the factor $1/\pi$ on its integral: for $f = 1$ its unnormalized left side is 2π, whereas its right side is 2. More generally L^2 convergence suffices, because $|\|s_N\|_2 - \|f\|_2| \le \|s_N - f\|_2$.

(b) For $f = x^2$, $a_0 = 2\pi^2/3$, $a_n = 4(-1)^n/n^2$, and $b_n = 0$. The identity becomes

$$\frac{1}{\pi}\int_{-\pi}^{\pi} x^4\,dx = \frac{2\pi^4}{5} = \frac{1}{2}\left(\frac{2\pi^2}{3}\right)^2 + 16\sum_{n=1}^{\infty}\frac{1}{n^4}.$$

Solving,

$$\boxed{\zeta(4) = \frac{1}{16}\left(\frac{2}{5} - \frac{2}{9}\right)\pi^4 = \frac{\pi^4}{90}.}$$

The given x^2 series is uniformly convergent by comparison with $\sum n^{-2}$, so part (a) applies directly.

(c) The square wave has $\int f^2 = 2\pi$, no cosine terms, and $b_{2j-1} = 4/[\pi(2j-1)]$. Hence Parseval reads

$$2 = \frac{16}{\pi^2}\sum_{j=1}^{\infty}\frac{1}{(2j-1)^2}.$$

To check this independently, integrate the ± 1 square-wave expansion from 0 to π:

$$\pi = \frac{4}{\pi}\sum_{j=1}^{\infty}\frac{1-\cos((2j-1)\pi)}{(2j-1)^2} = \frac{8}{\pi}\sum_{j=1}^{\infty}\frac{1}{(2j-1)^2}.$$

Thus the right side of Parseval is indeed 2. For any fixed partial sum s_N, $s_N(x) \to 0$ as $x \downarrow 0$, whereas the square wave equals one there. Thus $\sup_{0<x<\pi}|s_N(x) - 1| \geq 1$, ruling out uniform convergence. This supplies the requested counterexample to necessity of uniform convergence, while L^2 completeness still justifies Parseval.

19.1.14 Given

$$\varphi_1(x) \equiv \sum_{n=1}^{\infty}\frac{\sin nx}{n} = \begin{cases} -\dfrac{1}{2}(\pi + x), & -\pi \leq x < 0, \\ \dfrac{1}{2}(\pi - x), & 0 < x \leq \pi, \end{cases}$$

show by integrating that

$$\varphi_2(x) \equiv \sum_{n=1}^{\infty}\frac{\cos nx}{n^2} = \begin{cases} \dfrac{1}{4}(\pi + x)^2 - \dfrac{\pi^2}{12}, & -\pi \leq x \leq 0, \\ \dfrac{1}{4}(\pi - x)^2 - \dfrac{\pi^2}{12}, & 0 \leq x \leq \pi. \end{cases}$$

Solution.

The cosine series for φ_2 is absolutely and uniformly convergent. Integrating the sine series for φ_1 from 0 to $x > 0$ is valid because its partial sums converge in L^2 and $|\int_0^x(\varphi_1 - s_N)| \leq \sqrt{\pi}\,\|\varphi_1 - s_N\|_2$. Thus

$$\varphi_2(0) - \varphi_2(x) = \int_0^x \frac{\pi - t}{2}\,dt = \frac{\pi x}{2} - \frac{x^2}{4}.$$

Write $C = \varphi_2(0)$. Evenness then gives, on $[-\pi, \pi]$, $\varphi_2(x) = C - \pi|x|/2 + x^2/4$. Every term in its uniformly convergent cosine series has zero mean, so

$$0 = \int_0^{\pi}\varphi_2(x)\,dx = C\pi - \frac{\pi}{2}\frac{\pi^2}{2} + \frac{1}{4}\frac{\pi^3}{3} = C\pi - \frac{\pi^3}{6}.$$

Therefore $C = \pi^2/6$. Completing the square yields

$$\boxed{\varphi_2(x) = \frac{(\pi - |x|)^2}{4} - \frac{\pi^2}{12}, \qquad -\pi \le x \le \pi,}$$

which gives exactly the two requested branches. Uniform convergence includes $x = 0, \pm\pi$; in particular it also proves $\sum n^{-2} = \pi^2/6$ by setting $x = 0$.

19.1.15 Given

$$\psi_{2s}(x) = \sum_{n=1}^{\infty} \frac{\sin nx}{n^{2s}}, \qquad \psi_{2s+1}(x) = \sum_{n=1}^{\infty} \frac{\cos nx}{n^{2s+1}},$$

develop the following recurrence relations:

(a) $\psi_{2s}(x) = \displaystyle\int_0^x \psi_{2s-1}(t)\, dt,$

(b) $\psi_{2s+1}(x) = \zeta(2s+1) - \displaystyle\int_0^x \psi_{2s}(t)\, dt.$

Note. The functions $\psi_s(x)$ and $\varphi_s(x)$ of this and the preceding exercise are known as *Clausen functions*. In theory they may be used to improve the rate of convergence of a Fourier series. As is often the case, there is the question of how much analytical work we do and how much arithmetic work we demand that a computer do. As computers become steadily more powerful, the balance progressively shifts so that we are doing less and demanding that computers do more. Here $s = 1, 2, \ldots$. When $s = 1$, the first integral is improper at multiples of 2π crossed by the interval.

Solution.

Let s be a positive integer, and use t as the integration variable. For $s \ge 2$, the series for ψ_{2s-1} converges absolutely and uniformly. Thus

$$\int_0^x \psi_{2s-1}(t)\, dt = \sum_{n=1}^{\infty} \frac{1}{n^{2s-1}} \int_0^x \cos nt\, dt = \sum_{n=1}^{\infty} \frac{\sin nx}{n^{2s}} = \boxed{\psi_{2s}(x)}.$$

For every $s \ge 1$, the even-index series is absolutely uniform, giving

$$\begin{aligned} \int_0^x \psi_{2s}(t)\, dt &= \sum_{n=1}^{\infty} \frac{1 - \cos nx}{n^{2s+1}} \\ &= \zeta(2s+1) - \psi_{2s+1}(x). \end{aligned}$$

Rearrangement proves the second recurrence.

The first recurrence at $s = 1$ has an integrable logarithmic singularity at its lower endpoint: $\psi_1(t) = -\ln(2|\sin(t/2)|)$. The integral must be read as improper at every multiple of 2π it crosses; ψ_1 itself is not finite at those points. To justify this base case, insert r^n with $0 < r < 1$. Then

$$\int_0^x -\ln|1 - re^{it}|\, dt = \sum_{n=1}^{\infty} \frac{r^n \sin nx}{n^2}.$$

The right side tends uniformly to $\psi_2(x)$. On any bounded interval,

$$|1 - re^{it}|^2 = (1-r)^2 + 4r\sin^2(t/2)$$

shows, for $r \ge 1/2$, that the absolute logarithm is bounded by $C + |\ln \operatorname{dist}(t, 2\pi\mathbb{Z})|$ near the finitely many singularities. This bound is integrable because $\int_0^\epsilon |\ln t|\, dt = \epsilon(1 - \ln\epsilon)$. Dominated convergence proves the first recurrence also for $s = 1$.

19.1.16 Show that $f(x) = \sum_{n=1}^{\infty} \frac{\cos nx}{n+1}$ may be written as

$$f(x) = \psi_1(x) - \varphi_2(x) + \sum_{n=1}^{\infty} \frac{\cos nx}{n^2(n+1)},$$

where $\psi_1(x)$ and $\varphi_2(x)$ are the Clausen functions defined in Exercises 19.1.14 and 19.1.15. For the finite-valued identity take $x \notin 2\pi\mathbb{Z}$.

Solution.

The elementary coefficient identity is

$$\frac{1}{n} - \frac{1}{n^2} + \frac{1}{n^2(n+1)} = \frac{n(n+1) - (n+1) + 1}{n^2(n+1)} = \frac{1}{n+1}.$$

Multiply by $\cos nx$, sum first to a finite N, and then pass to the limit. The first series is ψ_1, the second is φ_2, and the last is absolutely convergent. Therefore

$$\boxed{f(x) = \psi_1(x) - \varphi_2(x) + \sum_{n=1}^{\infty} \frac{\cos nx}{n^2(n+1)}.}$$

This equality is an ordinary finite-valued identity only for $x \notin 2\pi\mathbb{Z}$. At a multiple of 2π both f and ψ_1 diverge. Away from those points, Dirichlet's test proves convergence of the original series and of ψ_1. The correction series has a simple uniform truncation bound:

$$\left| \sum_{n>N} \frac{\cos nx}{n^2(n+1)} \right| \le \sum_{n>N} \frac{1}{n^3} \le \int_N^{\infty} \frac{dt}{t^3} = \frac{1}{2N^2}.$$

Thus known closed forms for the first two terms leave a rapidly convergent remainder.

19.2 Applications of Fourier Series

19.2.1 Formally expand each sine in the Fourier expansion of a nonzero-height square wave, Eq. (19.26), in powers of x. Show that the coefficients of x^1 form a divergent series. Repeat for the coefficients of x^3.

Note. A power series cannot handle a discontinuity. These infinite coefficients are the result of attempting to beat this basic limitation on power series.

Solution.

Equation (19.26) is

$$f(x) = \frac{h}{2} + \frac{2h}{\pi} \sum_{j=0}^{\infty} \frac{\sin((2j+1)x)}{2j+1}.$$

For each individual term its entire Taylor expansion is

$$\frac{\sin((2j+1)x)}{2j+1} = x - \frac{(2j+1)^2 x^3}{3!} + \frac{(2j+1)^4 x^5}{5!} - \cdots.$$

Retain only the first N harmonics before collecting powers. The linear coefficient is $2hN/\pi$. The cubic coefficient is

$$-\frac{2h}{6\pi}\sum_{j=0}^{N-1}(2j+1)^2 = -\frac{h}{3\pi}\left[4\frac{(N-1)N(2N-1)}{6} + 4\frac{N(N-1)}{2} + N\right] = -\frac{hN(4N^2-1)}{9\pi}.$$

For nonzero h neither coefficient has a finite limit; their magnitudes grow as N and N^3, respectively. Formally, the coefficient of x^{2k+1} would be

$$\frac{2h}{\pi}\frac{(-1)^k}{(2k+1)!}\sum_{j=0}^{\infty}(2j+1)^{2k},$$

and it diverges for every $k \geq 0$. The original request to transform the Fourier series is therefore only formal, not a valid convergent power-series rearrangement. Interchanging the harmonic sum and Taylor sum is precisely the invalid step. A convergent power series on a neighborhood of zero is continuous there, so it cannot reproduce the jump between 0 and h. Away from jumps the square wave is locally constant and has a perfectly valid local constant Taylor series; the obstruction concerns a single series spanning the discontinuity.

19.2.2 Derive the Fourier series expansion of the Dirac delta function $\delta(x)$ in the interval $-\pi < x < \pi$.

(a) What significance can be attached to the constant term?

(b) In what region is this representation valid?

(c) With the identity

$$\sum_{n=1}^{N}\cos nx = \frac{\sin(Nx/2)}{\sin(x/2)}\cos\left[(N+1)\frac{x}{2}\right],$$

show that your Fourier representation of $\delta(x)$ is consistent with Eq. (5.27).

Interpret the expansion distributionally. In the finite-sum identity use the continuous limiting value where $\sin(x/2) = 0$.

Solution.

Apply the coefficient formulas to the distribution on $(-\pi, \pi)$:

$$a_0 = \frac{1}{\pi}\int \delta(x)\,dx = \frac{1}{\pi}, \quad a_n = \frac{\cos 0}{\pi} = \frac{1}{\pi}, \quad b_n = \frac{\sin 0}{\pi} = 0.$$

Thus

$$\boxed{\delta_{2\pi}(x) = \frac{1}{2\pi} + \frac{1}{\pi}\sum_{n=1}^{\infty}\cos nx.}$$

(a) The constant $1/(2\pi)$ is the mean density over one period. All other terms integrate to zero; this term alone ensures unit total mass.

(b) The equality is distributional, and its global extension is periodic: $\delta_{2\pi}(x) = \sum_{k\in\mathbb{Z}}\delta(x-2\pi k)$. It represents the single $\delta(x)$ within $(-\pi, \pi)$, not a single isolated delta on the whole real line. The series cannot be interpreted as an ordinary pointwise function.

(c) The printed finite-sum identity has an incorrect angle: its cosine must be $\cos((N+1)x/2)$, not $\cos((N+1/2)x/2)$. This correction is marked in the question. Indeed, the geometric sum is

$$\sum_{n=1}^{N} e^{inx} = e^{ix}\frac{1-e^{iNx}}{1-e^{ix}} = e^{i(N+1)x/2}\frac{\sin(Nx/2)}{\sin(x/2)}.$$

Taking real parts proves the corrected identity. Adding the constant term and using $2\sin A\cos B = \sin(A+B)+\sin(A-B)$ yields

$$\begin{aligned} 1+2\sum_{n=1}^{N}\cos nx &= \frac{\sin(x/2)+\sin((N+1/2)x)-\sin(x/2)}{\sin(x/2)} \\ &= \frac{\sin((N+1/2)x)}{\sin(x/2)} = D_N(x). \end{aligned}$$

The quotient has the removable value $D_N(0) = 2N+1$. Its normalized integral is one. Moreover $D_N/(2\pi)$ reproduces every Fourier polynomial of degree at most N under integration. For smooth periodic f, the absolutely summable Fourier coefficients then give $\int fD_N/(2\pi) \to f(0)$, as proved in 19.1.11. This is the delta limit.

Specifically Eq. (5.27) is the closure relation $\delta(x-t) = \sum_j \phi_j(x)\phi_j^*(t)$. For the orthonormal functions $1/\sqrt{2\pi}$, $\cos nx/\sqrt{\pi}$, $\sin nx/\sqrt{\pi}$, setting $t=0$ gives exactly the boxed formula. Thus the finite kernel and the general bilinear closure relation agree.

19.2.3 Expand $\delta(x-t)$ in a Fourier series. Compare your result with the bilinear form of Eq. (5.27).

$$ANS.\ \ \delta(x-t) = \frac{1}{2\pi} + \frac{1}{\pi}\sum_{n=1}^{\infty}(\cos nx\cos nt + \sin nx\sin nt)$$

$$= \frac{1}{2\pi} + \frac{1}{\pi}\sum_{n=1}^{\infty}\cos n(x-t).$$ Use the periodic distribution on $(-\pi,\pi)$, with t in that interval, or equivalently work on the circle.

Solution.

For t strictly inside $(-\pi,\pi)$, the delta's sampling rule gives

$$a_0(t) = \frac{1}{\pi}, \qquad a_n(t) = \frac{\cos nt}{\pi}, \qquad b_n(t) = \frac{\sin nt}{\pi}.$$

Substitution gives

$$\begin{aligned} \delta_{2\pi}(x-t) &= \frac{1}{2\pi} + \frac{1}{\pi}\sum_{n=1}^{\infty}(\cos nx\cos nt + \sin nx\sin nt) \\ &= \boxed{\frac{1}{2\pi} + \frac{1}{\pi}\sum_{n=1}^{\infty}\cos n(x-t)}. \end{aligned}$$

The last equality is the cosine subtraction identity, applied first to finite sums. With the normalized sine, cosine and constant basis from 19.2.2, the first line is precisely $\sum_j \phi_j(x)\phi_j^*(t)$ in Eq. (5.27). For example, integrating against $\phi_k(x)$ leaves only the kth orthonormal term and gives $\phi_k(t)$; by the convergent Fourier expansion this extends to smooth test functions. The periodic/distributional and endpoint qualifications of 19.2.2 apply equally to the translated kernel.

19.2.4 Show that integrating the Fourier expansion of the Dirac delta function (Exercise 19.2.2) leads to the Fourier representation of the square wave, Eq. (19.26), with $h = 1$.

Note. Integrating the constant term $(1/2\pi)$ leads to a term $x/2\pi$. What are you going to do with this?

Solution.

Let $H(x)$ be zero for $-\pi < x < 0$ and one for $0 < x < \pi$. Within this open interval, $H' = \delta(x)$ and the symmetric value at zero is $H(0) = 1/2$. Integrating the delta Fourier series from zero, with that symmetric convention, gives

$$H(x) = \frac{1}{2} + \frac{x}{2\pi} + \frac{1}{\pi}\sum_{n=1}^{\infty}\frac{\sin nx}{n}, \qquad -\pi < x < \pi.$$

The term mentioned in the hint must be retained. Expand it using the sawtooth series $x = 2\sum_{n\geq 1}(-1)^{n+1}\sin nx/n$:

$$\begin{aligned} H(x) &= \frac{1}{2} + \frac{1}{\pi}\sum_{n=1}^{\infty}\frac{1+(-1)^{n+1}}{n}\sin nx \\ &= \boxed{\frac{1}{2} + \frac{2}{\pi}\sum_{j=0}^{\infty}\frac{\sin((2j+1)x)}{2j+1}}. \end{aligned}$$

The even terms cancel and the odd terms double, which is Eq. (19.26) with $h = 1$.

There is also a direct way to keep the periodic boundary bookkeeping:

$$\begin{aligned} H(x) &= \int_0^{\pi}\delta_{2\pi}(x-t)\,dt \\ &= \frac{1}{2} + \frac{1}{\pi}\sum_{n=1}^{\infty}\frac{\sin nx - \sin(n(x-\pi))}{n} \\ &= \frac{1}{2} + \frac{1}{\pi}\sum_{n=1}^{\infty}\frac{1-(-1)^n}{n}\sin nx. \end{aligned}$$

The manual's final negative-x branch is printed as 1; it must be 0. Also the derivative of the periodic step is $\delta_{2\pi}(x) - \delta_{2\pi}(x-\pi)$, including the downward boundary jump. Integrating a single periodic delta produces the linear term above, not by itself a periodic primitive. At all jump points the Fourier series takes the midpoint 1/2.

19.2.5 Starting from the Fourier series given as lines 4 and 5 of Table 19.1, show that:

$$\sum_{n=0}^{\infty}\frac{\cos(2n+1)x}{2n+1} = \frac{1}{2}\ln\left|\cot\frac{x}{2}\right|.$$

Exclude $x \in \pi\mathbb{Z}$.

Solution.

On $0 < |x| < \pi$, the two logarithmic series in Table 19.1 read

$$A(x) = \sum_{n=1}^{\infty}\frac{\cos nx}{n} = -\ln(2|\sin(x/2)|), \qquad B(x) = \sum_{n=1}^{\infty}\frac{(-1)^n\cos nx}{n} = -\ln(2\cos(x/2)).$$

In the manual's intermediate first sum, the denominator printed as x must be n, and that sum starts at $n = 1$. Subtracting eliminates every even harmonic:

$$A - B = \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n} \cos nx = 2 \sum_{j=0}^{\infty} \frac{\cos((2j+1)x)}{2j+1}.$$

The corresponding logarithmic difference is $\ln(2\cos(x/2)) - \ln(2|\sin(x/2)|) = \ln|\cot(x/2)|$. Hence

$$\boxed{\sum_{j=0}^{\infty} \frac{\cos((2j+1)x)}{2j+1} = \frac{1}{2} \ln|\cot(x/2)|.}$$

Exclude $x \in \pi\mathbb{Z}$. At even multiples of π the series diverges to $+\infty$; at odd multiples it diverges to $-\infty$. Elsewhere its ordinary convergence follows from Dirichlet's test for $\sum e^{i(2j+1)x}$, whose geometric ratio is $e^{2ix} \neq 1$. Periodicity extends the displayed identity to every other real x.

19.2.6 Develop the Fourier series representation of

$$f(t) = \begin{cases} 0, & -\pi \le \omega t \le 0, \\ \sin \omega t, & 0 \le \omega t \le \pi. \end{cases}$$

This is the output of a simple half-wave rectifier. It is also an approximation of the solar thermal effect that produces "tides" in the atmosphere.

$$\textit{ANS.}\ \ f(t) = \frac{1}{\pi} + \frac{1}{2} \sin \omega t - \frac{2}{\pi} \sum_{n=2,4,6,\ldots} \frac{\cos n\omega t}{n^2 - 1}.$$

Solution.

Put $x = \omega t$ and use the 2π-periodic extension. Only the positive half-period contributes, so

$$a_0 = \frac{1}{\pi} \int_0^{\pi} \sin x \, dx = \frac{2}{\pi}.$$

Orthogonality of sines on $(0, \pi)$ gives

$$b_n = \frac{1}{\pi} \int_0^{\pi} \sin x \sin nx \, dx = \frac{1}{2} \delta_{n1}.$$

For $n \neq 1$, the product-to-sum identity gives

$$\begin{aligned} a_n &= \frac{1}{2\pi} \int_0^{\pi} [\sin((n+1)x) + \sin((1-n)x)] \, dx \\ &= \frac{1}{2\pi} \left[\frac{1 - (-1)^{n+1}}{n+1} + \frac{1 - (-1)^{1-n}}{1-n} \right]. \end{aligned}$$

This vanishes for odd $n > 1$. For even n it is

$$\frac{1}{\pi} \left(\frac{1}{n+1} + \frac{1}{1-n} \right) = -\frac{2}{\pi(n^2 - 1)}.$$

The exceptional coefficient is also zero: $a_1 = (2\pi)^{-1} \int_0^{\pi} \sin 2x \, dx = 0$. Consequently

$$\boxed{f(t) = \frac{1}{\pi} + \frac{1}{2} \sin \omega t - \frac{2}{\pi} \sum_{j=1}^{\infty} \frac{\cos(2j\omega t)}{4j^2 - 1}.}$$

The coefficients of the infinite remainder are $O(j^{-2})$, so the series converges absolutely and uniformly, including the joining points where the rectified waveform is continuous and zero.

19.2.7 A sawtooth wave is given by

$$f(x) = x, \qquad -\pi < x < \pi.$$

Show that

$$f(x) = 2\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin nx.$$

Solution.

Oddness gives $a_0 = a_n = 0$. The sine coefficient is

$$\begin{aligned} b_n &= \frac{2}{\pi}\int_0^{\pi} x \sin nx\, dx \\ &= \frac{2}{\pi}\left[-\frac{x\cos nx}{n} + \frac{\sin nx}{n^2}\right]_0^{\pi} = \frac{2(-1)^{n+1}}{n}. \end{aligned}$$

Thus

$$\boxed{x = 2\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin nx, \qquad -\pi < x < \pi.}$$

The periodic extension is piecewise smooth; its Fourier sum agrees with x at every point of this open interval. At $x = \pm\pi$ the periodic extension jumps from π to $-\pi$, and every sine term vanishes, giving the midpoint zero.

19.2.8 A different sawtooth wave is described by

$$f(x) = \begin{cases} -\dfrac{1}{2}(\pi + x), & -\pi \le x < 0, \\ +\dfrac{1}{2}(\pi - x), & 0 < x \le \pi. \end{cases}$$

Show that $f(x) = \displaystyle\sum_{n=1}^{\infty} (\sin nx/n)$.

Solution.

The function is odd, so only sine terms remain. For $n \ge 1$,

$$\begin{aligned} b_n &= \frac{2}{\pi}\int_0^{\pi} \frac{\pi - x}{2} \sin nx\, dx \\ &= \int_0^{\pi} \sin nx\, dx - \frac{1}{\pi}\int_0^{\pi} x \sin nx\, dx \\ &= \frac{1 - (-1)^n}{n} - \frac{1}{\pi}\left(-\frac{\pi(-1)^n}{n}\right) = \frac{1}{n}. \end{aligned}$$

The manual's last integral in its coefficient calculation prints $x\sin x$; it must be $x\sin nx$. The n-dependence is retained above. Therefore

$$\boxed{f(x) = \sum_{n=1}^{\infty} \frac{\sin nx}{n}}$$

throughout the stated intervals. At zero the two limiting values are $\pm\pi/2$, and the Fourier value is zero. At $\pm\pi$ the function and the series are both zero. This coefficient calculation independently confirms the logarithmic summation in 19.1.5.

19.2.9 A triangular wave (Fig. 19.4) is represented by

$$f(x) = \begin{cases} x, & 0 < x < \pi \\ -x, & -\pi < x < 0. \end{cases}$$

Represent $f(x)$ by a Fourier series.

$$\textit{ANS. } f(x) = \frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1,3,5,\ldots} \frac{\cos nx}{n^2}.$$

Solution.

Here $f(x) = |x|$ is even, so $b_n = 0$. Its constant coefficient is

$$a_0 = \frac{2}{\pi} \int_0^\pi x \, dx = \pi.$$

For $n \geq 1$, integration by parts gives

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^\pi x \cos nx \, dx \\ &= \frac{2}{\pi} \left[\frac{x \sin nx}{n} + \frac{\cos nx}{n^2} \right]_0^\pi = \frac{2[(-1)^n - 1]}{\pi n^2}. \end{aligned}$$

Thus the even coefficients vanish, while each odd one is $-4/(\pi n^2)$:

$$\boxed{|x| = \frac{\pi}{2} - \frac{4}{\pi} \sum_{j=0}^{\infty} \frac{\cos((2j+1)x)}{(2j+1)^2}, \qquad -\pi \leq x \leq \pi.}$$

The periodic triangular wave is continuous, including at the period endpoints, and the series is absolutely uniform by comparison with $\sum n^{-2}$. The cusp in its derivative does not obstruct equality.

19.2.10 Expand

$$f(x) = \begin{cases} 1, & x^2 < x_0^2 \\ 0, & x^2 > x_0^2 \end{cases}$$

in the interval $[-\pi, \pi]$.

Note. This variable-width square wave is of some importance in electronic music. Take x_0 real and include zero width and full width; at a jump state the Fourier midpoint value.

Solution.

Because the definition involves x_0^2, write $\alpha = |x_0|$. First assume $0 < \alpha < \pi$. The waveform is even, giving $b_n = 0$ and

$$a_0 = \frac{2}{\pi} \int_0^\alpha 1 \, dx = \frac{2\alpha}{\pi}, \qquad a_n = \frac{2}{\pi} \int_0^\alpha \cos nx \, dx = \frac{2 \sin n\alpha}{n\pi}.$$

Hence

$$\boxed{f(x) = \frac{\alpha}{\pi} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{\sin n\alpha}{n} \cos nx.}$$

The constant term is the duty fraction: the pulse occupies length 2α of a period 2π. At the two jumps $x = \pm\alpha$ the Fourier value is $1/2$; it is one or zero in the open constant regions.

The usual variable-width formula presumes $0 < |x_0| < \pi$. For $|x_0| \geq \pi$ the function is one almost everywhere in the interval; for $x_0 = 0$ it is zero almost everywhere. The equality points in the original piecewise definition can be assigned arbitrarily without changing any coefficient. One formula for all real x_0 is obtained by replacing α by $\min(|x_0|, \pi)$ above: it reduces to the appropriate constant series in the two limiting cases.

19.2.11 A metal cylindrical tube of radius a is split lengthwise into two nontouching halves. The top half is maintained at a potential $+V$, the bottom half at a potential $-V$. See Fig. 19.9. Separate the variables in Laplace's equation and solve for the electrostatic potential for $r \leq a$. Observe the resemblance between your solution for $r = a$ and the Fourier series for a square wave. Assume an infinitely long tube, negligible slit width and no axial variation. Impose the boundary values away from the two ideal slit lines.

Solution.

Use the ideal infinitely long tube, with translational symmetry along its axis and negligible slit width. Finite end effects or a finite open gap require additional boundary data. At the ideal slit endpoints the two assigned boundary limits disagree, so a single boundary value cannot be imposed there. Take $0 < \varphi < \pi$ to be the top half. In the interior, Laplace's equation is

$$\Phi_{rr} + \frac{1}{r}\Phi_r + \frac{1}{r^2}\Phi_{\varphi\varphi} = 0.$$

With $\Phi = R(r)\Theta(\varphi)$, multiplication by $r^2/(R\Theta)$ separates this into

$$\frac{r^2R'' + rR'}{R} = -\frac{\Theta''}{\Theta} = n^2.$$

Single-valued angular functions have integer n, with $\Theta = \cos n\varphi, \sin n\varphi$. The radial equation has $R = r^n, r^{-n}$ for $n \geq 1$, and $R = 1, \ln r$ for $n = 0$. Boundedness at the axis excludes r^{-n} and $\ln r$, leaving

$$\Phi(r, \varphi) = C_0 + \sum_{n=1}^{\infty}(r/a)^n(C_n \cos n\varphi + D_n \sin n\varphi).$$

The boundary function is odd and has zero mean, so $C_0 = C_n = 0$. Its sine coefficients are

$$D_n = \frac{2}{\pi}\int_0^{\pi} V \sin n\varphi \, d\varphi = \frac{2V}{n\pi}[1 - (-1)^n].$$

Therefore

$$\boxed{\Phi(r, \varphi) = \frac{4V}{\pi}\sum_{j=0}^{\infty}\frac{(r/a)^{2j+1}}{2j+1}\sin((2j+1)\varphi), \qquad r < a.}$$

The series and all differentiated series converge uniformly on compact subdisks, so it is harmonic. As $r \uparrow a$, it becomes the $\pm V$ square-wave series and attains the prescribed values away from the slits. Its radial limit on a slit is the midpoint zero.

For an additional closed-form check put $q = r/a < 1$ and $z = qe^{i\varphi}$. The geometric-series integral gives $\sum_{j\geq 0} z^{2j+1}/(2j+1) = \frac{1}{2}\log[(1+z)/(1-z)]$. Since

$$\frac{1+z}{1-z} = \frac{1 - q^2 + 2iq\sin\varphi}{|1-z|^2},$$

its real part is positive. Thus the principal argument gives

$$\boxed{\Phi(r,\varphi) = \frac{2V}{\pi}\arctan\frac{2ar\sin\varphi}{a^2-r^2}, \qquad r<a.}$$

The limit is $+V$ above and $-V$ below. The bounded harmonic Dirichlet solution is unique: the difference of two such solutions has zero boundary data almost everywhere and its Poisson representation is zero.

19.2.12 A metal cylinder is placed in a (previously) uniform electric field, $\mathbf{E}_0$, with the axis of the cylinder perpendicular to that of the original field.

(a) Find the perturbed electrostatic potential.

(b) Find the induced surface charge on the cylinder as a function of angular position.

Assume an infinitely long neutral cylinder in vacuum. Take its radius to be a and set its constant potential to zero.

Solution.

The usual solution assumes an infinitely long, neutral cylinder in vacuum. The original problem does not specify a net charge; allowing one adds a logarithmic monopole potential. Finite cylinders require end data. Let the radius be a and take the original field to be $E_0\hat{\mathbf{x}}$. Choose the conductor potential as zero, so the far-field potential is $-E_0 r\cos\varphi$.

(a) The separated exterior harmonic functions are $1, \ln r, r^{\pm n}\cos n\varphi, r^{\pm n}\sin n\varphi$. The applied field has only the $n=1$ cosine component. All other angular coefficients vanish because the surface potential is constant and the perturbation decays at infinity. Thus

$$\Phi_{\text{out}} = (-E_0 r + C/r)\cos\varphi.$$

The conductor condition $\Phi(a,\varphi)=0$ fixes $-E_0 a + C/a = 0$, or $C = E_0 a^2$. Hence

$$\boxed{\Phi_{\text{out}}(r,\varphi) = -E_0\left(r - \frac{a^2}{r}\right)\cos\varphi \quad (r\geq a), \qquad \Phi_{\text{in}} = 0.}$$

Each radial term solves the separated Laplace equation, and $\Phi_{\text{out}} + E_0 r\cos\varphi = E_0 a^2\cos\varphi/r \to 0$.

(b) The outward normal field just outside the surface is

$$E_r = -\partial_r\Phi_{\text{out}} = E_0\left(1+\frac{a^2}{r^2}\right)\cos\varphi, \qquad E_r(a^+) = 2E_0\cos\varphi.$$

Since the field inside the metal is zero, the boundary condition gives

$$\boxed{\sigma(\varphi) = \epsilon_0[E_r(a^+) - E_r(a^-)] = 2\epsilon_0 E_0\cos\varphi.}$$

Also $E_\varphi(a) = 0$ and the total induced charge per unit length is $a\int_0^{2\pi}\sigma\,d\varphi = 0$, verifying neutrality. If a net line charge λ is specified instead, the more general solution contains

$$\Phi_{\text{mono}} = -\frac{\lambda}{2\pi\epsilon_0}\ln(r/a), \qquad \sigma_{\text{mono}} = \frac{\lambda}{2\pi a},$$

plus an arbitrary common potential constant. The decaying-perturbation condition used above excludes this monopole.

19.2.13

(a) Find the Fourier series representation of

$$f(x) = \begin{cases} 0, & -\pi < x \le 0 \\ x, & 0 \le x < \pi. \end{cases}$$

(b) From the Fourier expansion show that

$$\frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \cdots .$$

Solution.

(a) All coefficient integrals run only from zero to π:

$$a_0 = \frac{1}{\pi}\int_0^\pi x\,dx = \frac{\pi}{2}.$$

For $n \ge 1$,

$$a_n = \frac{1}{\pi}\left[\frac{x\sin nx}{n} + \frac{\cos nx}{n^2}\right]_0^\pi = \frac{(-1)^n - 1}{\pi n^2},$$
$$b_n = \frac{1}{\pi}\left[-\frac{x\cos nx}{n} + \frac{\sin nx}{n^2}\right]_0^\pi = \frac{(-1)^{n+1}}{n}.$$

Thus

$$\boxed{f(x) = \frac{\pi}{4} - \frac{2}{\pi}\sum_{j=0}^{\infty}\frac{\cos((2j+1)x)}{(2j+1)^2} + \sum_{n=1}^{\infty}\frac{(-1)^{n+1}\sin nx}{n}.}$$

This also follows by splitting $f(x) = (|x| + x)/2$ and combining the triangular and sawtooth expansions. The equality holds on $(-\pi, \pi)$, including zero, where f is continuous. At the period endpoints the series equals the midpoint $\pi/2$ of the jump between 0 and π.

(b) Set $x = 0$. All sine terms vanish and every cosine is one, so

$$0 = \frac{\pi}{4} - \frac{2}{\pi}\sum_{j=0}^{\infty}\frac{1}{(2j+1)^2}.$$

Solving gives

$$\boxed{\sum_{j=0}^{\infty}\frac{1}{(2j+1)^2} = \frac{\pi^2}{8}.}$$

19.2.14 Integrate the Fourier expansion of the unit step function

$$f(x) = \begin{cases} 0, & -\pi < x < 0 \\ 1, & 0 < x < \pi. \end{cases}$$

Show that your integrated series agrees with Exercise 19.2.13.

Solution.

The unit step has the series

$$H(t) = \frac{1}{2} + \frac{2}{\pi}\sum_{j=0}^{\infty} \frac{\sin((2j+1)t)}{2j+1}.$$

Its integral from zero to x is $F(x) = 0$ for negative x and $F(x) = x$ for positive x, as in 19.2.13. Integrating term by term gives

$$F(x) = \frac{x}{2} + \frac{2}{\pi}\sum_{j=0}^{\infty} \frac{1 - \cos((2j+1)x)}{(2j+1)^2}.$$

This step is justified in L^2: integration against the indicator of $(0, x)$ is a bounded linear functional, and the Fourier partial sums converge in L^2. The resulting series is absolutely uniform. Using $\sum_{j\geq 0}(2j+1)^{-2} = \pi^2/8$ yields

$$F(x) = \frac{\pi}{4} + \frac{x}{2} - \frac{2}{\pi}\sum_{j=0}^{\infty} \frac{\cos((2j+1)x)}{(2j+1)^2}.$$

Finally, on $(-\pi, \pi)$, $x/2 = \sum_{n\geq 1}(-1)^{n+1}\sin nx/n$. Substituting it produces exactly the constant, cosine and sine terms in 19.2.13. The original integral is not periodic; replacing its linear term by the sawtooth series supplies the required periodic extension, with midpoint values at the endpoints.

19.2.15 In the interval $(-\pi, \pi)$,

$$\delta_n(x) = \begin{cases} n, & |x| < 1/2n, \\ 0, & |x| > 1/2n. \end{cases}$$

This wave form is the pulse shown in Fig. 19.10.

(a) Expand $\delta_n(x)$ as a Fourier cosine series.

(b) Show that your Fourier series agrees with a Fourier expansion of $\delta(x)$ in the distributional limit as $n \to \infty$.

Solution.

Use a different letter m for the Fourier harmonic. The pulse has half-width $1/(2n)$ and total area $n(1/n) = 1$. It is even, so $b_m = 0$. Its coefficients are

$$a_0 = \frac{1}{\pi}\int_{-1/(2n)}^{1/(2n)} n\,dx = \frac{1}{\pi}, \qquad a_m = \frac{2n}{\pi}\int_0^{1/(2n)} \cos mx\,dx = \frac{2n}{\pi m}\sin\frac{m}{2n}.$$

(a) Therefore

$$\boxed{\delta_n(x) = \frac{1}{2\pi} + \frac{2n}{\pi}\sum_{m=1}^{\infty} \frac{\sin(m/(2n))}{m}\cos mx.}$$

For each fixed n it converges to the pulse away from its two edges, and to $n/2$ at the edges.

(b) At every fixed harmonic m,

$$\frac{2n}{m}\sin\frac{m}{2n}=\frac{\sin(m/(2n))}{m/(2n)}\longrightarrow 1.$$

The limiting coefficients therefore give $1/(2\pi)+(1/\pi)\sum_{m\geq 1}\cos mx$, the delta expansion of 19.2.2. This is a distributional limit; taking the coefficient limit is not justification for a pointwise limit of the infinite series. A direct proof uses any continuous test function near zero:

$$\left|\int f(x)\delta_n(x)\,dx-f(0)\right|=\left|n\int_{-1/(2n)}^{1/(2n)}[f(x)-f(0)]\,dx\right|\leq\sup_{|x|\leq 1/(2n)}|f(x)-f(0)|\longrightarrow 0.$$

This proves the delta limit independently of formal coefficient passage.

19.2.16 Confirm the delta function nature of your Fourier series of Exercise 19.2.15 by showing that for any $f(x)$ that is bounded and measurable in $[-\pi,\pi]$ and continuous at $x=0$,

$$\lim_{n\to\infty}\lim_{M\to\infty}\int_{-\pi}^{\pi}f(x)S_{n,M}(x)\,dx=f(0),$$

Here $S_{n,M}$ is the first M harmonics, including the constant term, of the finite-width pulse δ_n.

Solution.

The displayed integrals require an integrable test function; pointwise finiteness alone does not ensure integrability. A bounded measurable f continuous at zero suffices. The Fourier expansion must first represent the finite-width pulse, followed by the pulse-width limit; no unrestricted interchange of two infinite limits is needed. For fixed n, its Fourier series equals δ_n almost everywhere. If f is bounded, it is in $L^2[-\pi,\pi]$, and the partial sums $S_{n,M}$ satisfy

$$\left|\int f(S_{n,M}-\delta_n)\right|\leq\|f\|_2\,\|S_{n,M}-\delta_n\|_2\longrightarrow 0.$$

Consequently the action of the completed Fourier series is

$$I_n=n\int_{-1/(2n)}^{1/(2n)}f(x)\,dx=\int_{-1/2}^{1/2}f(u/n)\,du.$$

Given $\epsilon>0$, continuity at zero provides $\eta>0$ such that $|f(x)-f(0)|<\epsilon$ when $|x|<\eta$. Choose $n>1/(2\eta)$. Then throughout the last integral $|u/n|<\eta$, and

$$|I_n-f(0)|\leq\int_{-1/2}^{1/2}|f(u/n)-f(0)|\,du\leq\epsilon.$$

Hence

$$\boxed{\lim_{n\to\infty}\lim_{M\to\infty}\int_{-\pi}^{\pi}f(x)S_{n,M}(x)\,dx=f(0).}$$

The same shrinking-interval proof applies directly to any measurable integrable f continuous at zero; no behavior away from that shrinking interval can alter the pulse limit. It does not assert that arbitrary Dirichlet partial sums converge for every function continuous at one point.

19.2.17

(a) Show that the Dirac delta function $\delta(x-a)$, expanded in a Fourier sine series in the half-interval $(0, L)$ $(0 < a < L)$ is given by

$$\delta(x-a) = \frac{2}{L}\sum_{n=1}^{\infty} \sin\frac{n\pi a}{L}\sin\frac{n\pi x}{L}.$$

Note that this series actually describes $-\delta(x+a)+\delta(x-a)$ in the interval $(-L, L)$.

(b) By integrating both sides of the preceding equation from 0 to x, show that the cosine expansion of the square wave

$$f(x) = \begin{cases} 0, & 0 \le x < a \\ 1, & a < x < L, \end{cases}$$

is

$$f(x) = \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\sin\frac{n\pi a}{L} - \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\sin\frac{n\pi a}{L}\cos\frac{n\pi x}{L},$$

for $0 \le x < L$.

(c) Show that the term $\dfrac{2}{\pi}\displaystyle\sum_{n=1}^{\infty}\frac{1}{n}\sin\frac{n\pi a}{L}$ is the average of $f(x)$ on $(0, L)$.

The delta expansion is distributional. At $x = a$ the integrated Fourier series has the midpoint value $1/2$.

Solution.

(a) The sine functions have half-interval norm

$$\int_0^L \sin\frac{n\pi x}{L}\sin\frac{m\pi x}{L}\,dx = \frac{L}{2}\delta_{nm}.$$

This follows by writing the product as half the difference of cosines; for $n = m$ the constant term integrates to $L/2$ and the oscillatory term integrates to zero. Thus the delta coefficient is

$$b_n = \frac{2}{L}\int_0^L \delta(x-a)\sin\frac{n\pi x}{L}\,dx = \frac{2}{L}\sin\frac{n\pi a}{L},$$

which gives the requested expansion

$$\boxed{\delta(x-a) = \frac{2}{L}\sum_{n=1}^{\infty}\sin\frac{n\pi a}{L}\sin\frac{n\pi x}{L}, \qquad 0 < x, a < L.}$$

The equality is distributional. Extending the sine basis to $(-L, L)$ is odd, so the corresponding source is $\delta(x-a)-\delta(x+a)$ there, repeated with period $2L$.

(b) Integrating from zero to x gives the step $H(x-a)$. For each mode,

$$\int_0^x \sin\frac{n\pi t}{L}\,dt = \frac{L}{n\pi}\left(1-\cos\frac{n\pi x}{L}\right).$$

Therefore

$$\boxed{H(x-a) = \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\sin(n\pi a/L)}{n} - \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\sin(n\pi a/L)}{n}\cos\frac{n\pi x}{L}.}$$

An Abel factor justifies the integration before passing to the limiting step; equivalently the integrated coefficients are those of the bounded step, verified directly in 19.2.18. At $x = a$ symmetric Fourier sums give $1/2$; the endpoints have values zero at $x = 0$ and one at $x = L$.

(c) Put $\theta = \pi a/L \in (0, \pi)$ in 19.1.5:

$$\frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\sin n\theta}{n} = \frac{2}{\pi}\frac{\pi-\theta}{2} = 1 - \frac{a}{L}.$$

Independently the step average is $L^{-1}\int_0^L f(x)\,dx = L^{-1}\int_a^L 1\,dx = 1 - a/L$. Thus the first sum is exactly the required average.

19.2.18 Verify the Fourier cosine expansion of the square wave, Exercise 19.2.17(b), by direct calculation of the Fourier coefficients.

Solution.

For a half-interval cosine expansion, write $f(x) = a_0/2 + \sum_{n\geq 1} a_n \cos(n\pi x/L)$. The constant mode has norm L and every nonconstant cosine has norm $L/2$. Accordingly

$$a_0 = \frac{2}{L}\int_a^L 1\,dx = 2\left(1 - \frac{a}{L}\right),$$

and, for $n \geq 1$,

$$a_n = \frac{2}{L}\int_a^L \cos\frac{n\pi x}{L}\,dx = \frac{2}{n\pi}\left[\sin\frac{n\pi x}{L}\right]_a^L = -\frac{2}{n\pi}\sin\frac{n\pi a}{L}.$$

Hence

$$\boxed{f(x) = 1 - \frac{a}{L} - \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\sin(n\pi a/L)}{n}\cos\frac{n\pi x}{L}.}$$

The identity for the first sum in 19.2.17(c) shows exact agreement with that exercise. Its even extension is periodic with period $2L$; it is continuous at $0, L$ and jumps at $\pm a$. Thus the series has the specified step values away from the jumps and the midpoint $1/2$ there.

19.2.19

(a) A string is clamped at both ends $x = 0$ and $x = L$. Assuming small-amplitude vibrations, we find that the amplitude $y(x,t)$ satisfies the wave equation

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y}{\partial t^2}.$$

Here v is the wave velocity. The string is set in vibration by a sharp blow at $x = a$. Hence we have

$$y(x,0) = 0, \qquad \frac{\partial y(x,t)}{\partial t} = Lv_0\delta(x-a) \quad \text{at } t = 0.$$

The constant L is included to compensate for the dimensions (inverse length) of $\delta(x-a)$. With $\delta(x-a)$ given by Exercise 19.2.17(a), solve the wave equation subject to these initial conditions.

$$\textit{ANS. } y(x,t) = \frac{2v_0L}{\pi v}\sum_{n=1}^{\infty}\frac{1}{n}\sin\frac{n\pi a}{L}\sin\frac{n\pi x}{L}\sin\frac{n\pi vt}{L}.$$

(b) Show that the transverse velocity of the string $\partial y(x,t)/\partial t$ is given by

$$\frac{\partial y(x,t)}{\partial t} = 2v_0 \sum_{n=1}^{\infty} \sin\frac{n\pi a}{L} \sin\frac{n\pi x}{L} \cos\frac{n\pi v t}{L}.$$

Take $L, v > 0$ and $0 < a < L$. Interpret the impulsive initial velocity and its propagated series as distributions.

Solution.

(a) Put $k_n = n\pi/L$ and $\Omega_n = vk_n$. The fixed-end condition selects the sine modes, so write $y(x,t) = \sum_{n\geq 1} q_n(t) \sin k_n x$. Substitution and sine orthogonality give

$$q_n'' + \Omega_n^2 q_n = 0, \qquad q_n(t) = A_n \cos\Omega_n t + B_n \sin\Omega_n t.$$

The zero initial displacement forces $A_n = 0$. The sine coefficient of the initial velocity is

$$q_n'(0) = \frac{2}{L}\int_0^L Lv_0\delta(x-a)\sin k_n x\,dx = 2v_0 \sin k_n a.$$

Since $q_n'(0) = \Omega_n B_n$, we obtain

$$\boxed{y(x,t) = \frac{2v_0 L}{\pi v}\sum_{n=1}^{\infty} \frac{\sin(n\pi a/L)\sin(n\pi x/L)\sin(n\pi v t/L)}{n}.}$$

Every mode solves the wave equation and vanishes at both fixed ends.

(b) Differentiating each mode contributes the factor $\Omega_n = n\pi v/L$, cancelling the denominator and prefactor:

$$\boxed{y_t(x,t) = 2v_0 \sum_{n=1}^{\infty} \sin\frac{n\pi a}{L} \sin\frac{n\pi x}{L} \cos\frac{n\pi v t}{L}.}$$

At $t = 0$, the sine closure relation gives exactly $Lv_0\delta(x-a)$. The ideal impulse makes the velocity a distribution. The last series is not asserted to converge as an ordinary velocity function; differentiation and initial conditions are interpreted distributionally.

For an independent propagation description, extend the initial velocity oddly and periodically:

$$G(s) = Lv_0 \sum_{j\in\mathbb{Z}} [\delta(s-a-2jL) - \delta(s+a-2jL)].$$

Then

$$y(x,t) = \frac{1}{2v}\int_{x-vt}^{x+vt} G(s)\,ds, \qquad y_t(x,t) = \frac{1}{2}[G(x+vt) + G(x-vt)].$$

Oddness about both endpoints enforces the clamps. Before either pulse reaches an end, $y = Lv_0/(2v)$ where $|x-a| < vt$, and zero outside; the moving fronts and subsequent reflected images explain the singular velocity and confirm the Fourier solution.

19.2.20 A string, clamped at $x = 0$ and at $x = L$, is vibrating freely. Its motion is described by the wave equation

$$\frac{\partial^2 u(x,t)}{\partial t^2} = v^2 \frac{\partial^2 u(x,t)}{\partial x^2}.$$

Assume a Fourier expansion of the form

$$u(x,t) = \sum_{n=1}^{\infty} b_n(t) \sin \frac{n\pi x}{L}$$

and determine the coefficients $b_n(t)$. The initial conditions are

$$u(x,0) = f(x) \qquad \text{and} \qquad \frac{\partial}{\partial t} u(x,0) = g(x).$$

Note. This is only half the conventional Fourier orthogonality integral interval. However, as long as only the sines are included here, the Sturm–Liouville boundary conditions are still satisfied and the functions are orthogonal.

ANS. $b_n(t) = A_n \cos \dfrac{n\pi v t}{L} + B_n \sin \dfrac{n\pi v t}{L}$,

$A_n = \dfrac{2}{L} \displaystyle\int_0^L f(x) \sin \frac{n\pi x}{L}\, dx$, $B_n = \dfrac{2}{n\pi v} \displaystyle\int_0^L g(x) \sin \frac{n\pi x}{L}\, dx$. Take $L, v > 0$ and real initial data. For the finite-energy solution assume $f \in H_0^1(0,L)$ and $g \in L^2(0,L)$; also state the regularity and endpoint qualifications for a classical solution.

Solution.

Here $H_0^1(0,L)$ means that f and its weak first derivative are square integrable and f has zero endpoint values. Thus the assumptions make the initial displacement-gradient and velocity energies finite. Let $k_n = n\pi/L$ and $\Omega_n = v k_n$, with $L, v > 0$. Substitute the sine expansion into the wave equation:

$$\sum_{n \geq 1} [b_n''(t) + \Omega_n^2 b_n(t)] \sin k_n x = 0.$$

Multiplying by $\sin k_m x$ and integrating from 0 to L isolates $(L/2)[b_m'' + \Omega_m^2 b_m] = 0$. Thus

$$b_n(t) = A_n \cos \Omega_n t + B_n \sin \Omega_n t.$$

At zero, $b_n(0) = A_n$ and $b_n'(0) = \Omega_n B_n$. Project each initial condition onto its sine mode:

$$\boxed{A_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L}\, dx, \qquad B_n = \frac{2}{n\pi v} \int_0^L g(x) \sin \frac{n\pi x}{L}\, dx.}$$

Together with the displayed time dependence these determine the solution. The factor $2/(n\pi v)$ follows from $(2/L)/\Omega_n$; no extra L is missing. The cosine time factor carries the initial displacement, and the sine time factor carries the initial velocity.

For a classical solution the initial data must be regular enough for the differentiations and compatible with the clamps (in particular $f(0) = f(L) = 0$, and $g(0) = g(L) = 0$ when these traces exist). For less regular data the modal formula is interpreted in the corresponding weak sense. For example $f \in H_0^1(0,L)$ and $g \in L^2(0,L)$ give a finite-energy solution. The energy

$$E(t) = \frac{1}{2} \int_0^L (u_t^2 + v^2 u_x^2)\, dx = \frac{L}{4} \sum_{n \geq 1} [(b_n')^2 + \Omega_n^2 b_n^2]$$

is constant, since each summand has derivative $2b_n'(b_n'' + \Omega_n^2 b_n) = 0$.

19.2.21

(a) Let us continue the vibrating string problem in Exercise 19.2.20. We assume now that the presence of a resisting medium will damp the vibrations according to the equation

$$\frac{\partial^2 u(x,t)}{\partial t^2} = v^2 \frac{\partial^2 u(x,t)}{\partial x^2} - k\frac{\partial u(x,t)}{\partial t}.$$

Introduce a Fourier expansion

$$u(x,t) = \sum_{n=1}^{\infty} b_n(t) \sin \frac{n\pi x}{L}$$

and again determine the coefficients $b_n(t)$. Take the initial and boundary conditions to be the same as in Exercise 19.2.20. Take $k \geq 0$ and first consider each mode for which $k < 2n\pi v/L$.

(b) Repeat for modes with $k > 2n\pi v/L$, and include the critical case $k = 2n\pi v/L$. Explain why finite k cannot overdamp every mode.

ANS. (a) $b_n(t) = e^{-kt/2}\big[A_n \cos \omega_n t + B_n \sin \omega_n t\big]$, $\omega_n^2 = \left(\frac{n\pi v}{L}\right)^2 - \left(\frac{k}{2}\right)^2 > 0$,

$$A_n = \frac{2}{L}\int_0^L f(x) \sin \frac{n\pi x}{L}\, dx, \quad B_n = \frac{2}{\omega_n L}\int_0^L g(x) \sin \frac{n\pi x}{L}\, dx + \frac{k}{2\omega_n} A_n.$$

(b) $b_n(t) = e^{-kt/2}\big[A_n \cosh \sigma_n t + B_n \sinh \sigma_n t\big]$, $\sigma_n^2 = \left(\frac{k}{2}\right)^2 - \left(\frac{n\pi v}{L}\right)^2 > 0$,

$$A_n = \frac{2}{L}\int_0^L f(x) \sin \frac{n\pi x}{L}\, dx, \quad B_n = \frac{2}{\sigma_n L}\int_0^L g(x) \sin \frac{n\pi x}{L}\, dx + \frac{k}{2\sigma_n} A_n.$$

Solution.

Assume $k \geq 0$, write $\Omega_n = n\pi v/L$, and let the initial sine coefficients be

$$F_n = \frac{2}{L}\int_0^L f(x) \sin \frac{n\pi x}{L}\, dx, \qquad G_n = \frac{2}{L}\int_0^L g(x) \sin \frac{n\pi x}{L}\, dx.$$

Projection of the damped wave equation gives

$$b_n'' + k b_n' + \Omega_n^2 b_n = 0, \qquad b_n(0) = F_n, \quad b_n'(0) = G_n.$$

To remove the first derivative put $b_n = e^{-kt/2} z_n$. Then

$$b_n' = e^{-kt/2}(z_n' - k z_n/2), \quad b_n'' = e^{-kt/2}(z_n'' - k z_n' + k^2 z_n/4),$$

and cancellation yields

$$z_n'' + (\Omega_n^2 - k^2/4) z_n = 0, \quad z_n(0) = F_n, \quad z_n'(0) = G_n + kF_n/2.$$

(a) Underdamped modes. If $\Omega_n > k/2$, define $\omega_n = \sqrt{\Omega_n^2 - k^2/4} > 0$. The initial conditions give

$$\boxed{b_n = e^{-kt/2}\left[F_n \cos \omega_n t + \frac{G_n + kF_n/2}{\omega_n} \sin \omega_n t\right].}$$

This is the printed formula with $A_n = F_n$ and $B_n = (G_n + kA_n/2)/\omega_n$. The actual textbook's answer in (a), printed p. 957, omits the square on $n\pi v/L$ in ω_n^2. The correct formula is $\omega_n^2 = (n\pi v/L)^2 - (k/2)^2$, already restored in the local question.

(b) Overdamped modes. If $0 < \Omega_n < k/2$, define $\sigma_n = \sqrt{k^2/4 - \Omega_n^2} > 0$. Then

$$\boxed{b_n = e^{-kt/2}\left[F_n \cosh \sigma_n t + \frac{G_n + kF_n/2}{\sigma_n} \sinh \sigma_n t\right].}$$

The two decay exponents $-k/2 \pm \sigma_n$ are negative, since $0 < \sigma_n < k/2$.

Damping is classified mode by mode. No finite k overdamps all infinitely many modes, since Ω_n grows without bound. The critical case $\Omega_n = k/2$, omitted from the original question, also needs a separate formula. Here $z_n'' = 0$, so

$$\boxed{b_n = e^{-kt/2}[F_n + (G_n + kF_n/2)t].}$$

This is also the continuous $\omega_n \to 0$ or $\sigma_n \to 0$ limit. All modes are underdamped if $k < 2\pi v/L$; otherwise modes with $n < kL/(2\pi v)$ are overdamped, equality is critical, and the larger indices remain underdamped. As a check, let $E = \frac{1}{2}\int_0^L (u_t^2 + v^2 u_x^2)\, dx$. Differentiating and integrating the second term by parts gives

$$\begin{aligned} E' &= \int_0^L (u_t u_{tt} + v^2 u_x u_{xt})\, dx \\ &= v^2[u_x u_t]_0^L + \int_0^L u_t(u_{tt} - v^2 u_{xx})\, dx \\ &= -k\int_0^L u_t^2\, dx \le 0 \qquad (k \ge 0). \end{aligned}$$

The boundary term vanishes because the clamped displacement is constant in time, hence $u_t = 0$ at both ends. For smooth data this is a direct calculation; for finite-energy data it follows by limits of finite modal sums. It is consistent with the time dependence of every mode.

19.3 Gibbs Phenomenon

19.3.1 With the partial-sum summation techniques of this section, show that at a discontinuity in $f(x)$ the Fourier series for $f(x)$ takes on the arithmetic mean of the right- and left-hand limits:

$$\lim_{N\to\infty} s_N(x_0) = \tfrac{1}{2}\big[f(x_0 + 0) + f(x_0 - 0)\big].$$

In evaluating $\lim_{r\to\infty} s_r(x_0)$, you may find it convenient to identify part of the integrand as a Dirac delta function. Assume f is 2π-periodic and of bounded variation on a period. The left side is the Fourier-series limit, regardless of the assigned value of $f(x_0)$.

Solution.

The conclusion concerns the Fourier-series value, which need not equal an arbitrarily assigned $f(x_0)$. As in the chapter's Dirichlet conditions, assume a piecewise regular periodic function; bounded variation suffices. The existence of a jump alone is not a general pointwise-convergence hypothesis. The source Eq. (19.29) also omits n from the exponential; the correct finite kernel uses $e^{in(x-t)}$, as in Eq. (19.30). Let

$$m = \frac{f(x_0 + 0) + f(x_0 - 0)}{2}, \qquad D_N(t) = \frac{\sin((N + 1/2)t)}{\sin(t/2)}.$$

The finite partial-sum formula and evenness of the kernel give

$$s_N(x_0) = \frac{1}{2\pi}\int_0^{\pi} [f(x_0+t)+f(x_0-t)]D_N(t)\,dt.$$

Because $D_N = 1 + 2\sum_{j=1}^{N}\cos jt$, its integral on $(0,\pi)$ is exactly π. Subtracting m therefore gives

$$s_N(x_0) - m = \frac{1}{2\pi}\int_0^{\pi} g(t)D_N(t)\,dt, \quad g(t) = f(x_0+t)+f(x_0-t)-2m, \quad g(0+) = 0.$$

We prove that the last integral tends to zero, including its potentially singular neighborhood of zero.

Put $M = N + 1/2$ and $A_N(u) = \int_0^u D_N(t)\,dt$. Split

$$\frac{1}{\sin(t/2)} = \frac{2}{t} + q(t), \qquad A_N(u) = 2\int_0^{Mu}\frac{\sin v}{v}\,dv + \int_0^u q(t)\sin Mt\,dt.$$

Here $q(t) = O(t)$ at zero and is integrable on $[0,\pi]$. The sine integral is uniformly bounded: below $v = 1$ its absolute integral is at most one; beyond one, integration by parts gives

$$\int_1^z \frac{\sin v}{v}\,dv = \Big[-\frac{\cos v}{v}\Big]_1^z - \int_1^z \frac{\cos v}{v^2}\,dv,$$

whose absolute value is at most two. Consequently $|A_N(u)| \le C = 6 + \int_0^{\pi}|q(t)|\,dt$, independently of N, u.

Since g has bounded variation and $g(0+) = 0$, choose a small $\delta > 0$ for which both $|g(\delta)|$ and its variation on $(0,\delta]$ are small. Stieltjes integration by parts gives

$$\left|\int_0^{\delta} g(t)D_N(t)\,dt\right| = \left|g(\delta)A_N(\delta) - \int_{(0,\delta]} A_N(t)\,dg(t)\right| \le C\big(|g(\delta)| + \mathrm{Var}_{(0,\delta]}\,g\big).$$

The variation is the mass of a finite variation measure on $(0,\delta]$; these intervals shrink to the empty set, so that mass tends to zero. Also $g(\delta)\to 0$. Thus the bound tends to zero as $\delta\downarrow 0$, uniformly in N. For fixed δ, the remaining integrand has the form $[g(t)/\sin(t/2)]\sin Mt$ with an integrable bracket on $[\delta,\pi]$. Its integral tends to zero: approximate the bracket in L^1 by a step function, whose finitely many sine integrals are each $O(1/M)$; the approximation error bounds the integral error independently of M. Combining the small-neighborhood bound and the outer limit proves

$$\boxed{\lim_{N\to\infty} s_N(x_0) = \frac{f(x_0+0)+f(x_0-0)}{2}.}$$

The kernel's symmetry is what assigns equal weight to the two limits.

19.3.2 Determine the partial sum, s_n, of the series in Eq. (19.33) by using

(a) $\dfrac{\sin mx}{m} = \displaystyle\int_0^x \cos my\,dy,$

(b) $\displaystyle\sum_{p=1}^{n}\cos(2p-1)y = \frac{\sin 2ny}{2\sin y}.$

Compare the peak limit with Eqs. (19.38) and (19.41).

Solution.

Equation (19.33) represents levels $\pm h/2$. Its first n nonzero harmonics give

$$s_n(x) = \frac{2h}{\pi} \sum_{p=1}^{n} \frac{\sin((2p-1)x)}{2p-1}.$$

There are only finitely many terms, so integrate each cosine and interchange the finite sum with the integral:

$$s_n(x) = \frac{2h}{\pi} \int_0^x \sum_{p=1}^{n} \cos((2p-1)y)\, dy.$$

The identity in the hint follows by telescoping:

$$2 \sin y \cos((2p-1)y) = \sin 2py - \sin(2p-2)y,$$

so its sum is $\sin 2ny$. Consequently the exact partial sum is

$$\boxed{s_n(x) = \frac{h}{\pi} \int_0^x \frac{\sin 2ny}{\sin y}\, dy.}$$

The apparent singularities at multiples of π are removable because the quotient equals twice a finite cosine sum. For $h > 0$, its first positive maximum occurs at $x_n = \pi/(2n)$: the derivative $(h/\pi) \sin 2nx / \sin x$ changes from positive to negative there. Changing variables $t = 2ny$ at that maximum gives

$$s_n(x_n) = \frac{h}{\pi} \int_0^{\pi} \frac{\sin t}{2n \sin(t/(2n))}\, dt \longrightarrow \frac{h}{\pi} \int_0^{\pi} \frac{\sin t}{t}\, dt.$$

The limit follows uniformly after removing $t = 0$, since $(t/(2n))/\sin(t/(2n)) \to 1$ uniformly on $[0, \pi]$. More generally, at $x = z/(2n)$ for fixed z the limiting profile is $(h/\pi) \int_0^z \sin t/t\, dt$.

Eq. (19.40), cited in the original question, is a tail-integral identity, not a partial-sum formula. The exact expression above agrees with the small-x approximation in (19.37) and the peak limit in (19.38). Using (19.40) together with $\int_0^{\infty} \sin t/t\, dt = \pi/2$ gives the peak integral in (19.41). Thus the limiting peak is $(h/2)I$, where I is computed next. Its overshoot is approximately $0.08949h$: about 8.949% of the jump height h, or 17.898% of the positive level $h/2$.

19.3.3

(a) Calculate the value of the Gibbs phenomenon integral

$$I = \frac{2}{\pi} \int_0^{\pi} \frac{\sin t}{t}\, dt$$

by numerical quadrature accurate to 12 significant figures.

(b) Check your result by (1) expanding the integrand as a series, (2) integrating term by term, and (3) evaluating the integrated series. This calls for double-precision calculation.

ANS. $I = 1.178979744472$.

Solution.

(a) Define the removable endpoint by $(\sin t/t)|_{t=0} = 1$. Numerical Gauss–Legendre quadrature on $[0, \pi]$, evaluated with 65-digit arithmetic, gives

$$\boxed{I = 1.17897974447216727023202884582 5\ldots.}$$

Independent tanh–sinh quadrature gives the same displayed digits. To twelve significant figures the result is $\boxed{1.17897974447}$; the book's 1.178979744472 agrees when rounded to twelve decimal places.

(b) The entire power series for the integrand is

$$\frac{\sin t}{t} = \sum_{k=0}^{\infty} \frac{(-1)^k t^{2k}}{(2k+1)!}.$$

It converges uniformly on $[0, \pi]$, so termwise integration gives

$$\begin{aligned} I &= \frac{2}{\pi} \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} \frac{\pi^{2k+1}}{2k+1} \\ &= \boxed{2 \sum_{k=0}^{\infty} \frac{(-1)^k \pi^{2k}}{(2k+1)(2k+1)!}}. \end{aligned}$$

Let $T_k = 2\pi^{2k}/[(2k+1)(2k+1)!]$. Successive magnitudes satisfy

$$\frac{T_{k+1}}{T_k} = \frac{\pi^2(2k+1)}{(2k+3)^2(2k+2)} < 1 \qquad (k \geq 0).$$

To see the monotonicity directly, put $r_k = (2k+1)/[(2k+3)^2(2k+2)]$. Subtracting successive terms gives

$$r_k - r_{k+1} = \frac{16k^3 + 68k^2 + 84k + 23}{2(k+1)(k+2)(2k+3)^2(2k+5)^2} > 0 \qquad (k \geq 0).$$

Thus the largest ratio is $\pi^2 r_0 = \pi^2/18 < 1$. Hence the alternating-series remainder after N terms is at most T_N. Direct evaluation yields

N	sum of terms $k = 0, \ldots, N-1$	bound T_N
4	1.1735737377034521530	5.811×10^{-3}
8	1.1789797154459948275	2.978×10^{-8}
12	1.1789797444721629210	4.406×10^{-15}
16	1.1789797444721672702	5.658×10^{-23}

These bounds concern the mathematical series truncation; arithmetic roundoff must be controlled separately. Summing the sixteen terms using ordinary double-precision arithmetic and compensated summation gives 1.178979744472167, independently meeting the requested twelve-digit accuracy. High-precision series evaluation and both quadrature methods provide the additional cross-checks.

20 Chapter 20. Integral Transforms

Problems

20.2 Fourier Transform

20.2.1

(a) Show that $g(-\omega) = g^*(\omega)$ is a necessary and sufficient condition for $f(x)$ to be real.

(b) Show that $g(-\omega) = -g^*(\omega)$ is a necessary and sufficient condition for $f(x)$ to be pure imaginary.

Note. The condition of part (a) is used in the development of the dispersion relations of Section 12.8.

Use Fourier inversion in the usual L^1 or L^2 setting, and interpret reality almost everywhere; continuous representatives give pointwise statements.

Solution.

Use the convention

$$g(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{i\omega x} dx, \qquad f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(\omega) e^{-i\omega x} d\omega.$$

Conjugating the first formula shows

$$g^*(\omega) = \frac{1}{\sqrt{2\pi}} \int f^*(x) e^{-i\omega x} dx = \mathcal{F}[f^*](-\omega).$$

(a) If $f = f^*$, this is $g(-\omega)$. Conversely, if $g(-\omega) = g^*(\omega)$, conjugate the inverse transform and put $u = -\omega$:

$$\begin{aligned} f^*(x) &= \frac{1}{\sqrt{2\pi}} \int g^*(\omega) e^{i\omega x} d\omega \\ &= \frac{1}{\sqrt{2\pi}} \int g(-\omega) e^{i\omega x} d\omega = \frac{1}{\sqrt{2\pi}} \int g(u) e^{-iux} du = f(x). \end{aligned}$$

(b) Purely imaginary means $f^* = -f$. The conjugated forward formula then gives $g^*(\omega) = -g(-\omega)$. Conversely this last condition, inserted in the preceding inverse calculation, gives $f^*(x) = -f(x)$. For L^1 or L^2 transforms the recovered function is determined almost everywhere; these reality assertions have that meaning. They hold at every point for continuous representatives. Changing a function at an isolated point cannot be detected by its Fourier transform.

20.2.2 The function

$$f(x) = \begin{cases} 1, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$$

is a symmetrical finite step function.

(a) Find $g_c(\omega)$, Fourier cosine transform of $f(x)$.

(b) Taking the inverse cosine transform, show that

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\sin \omega \cos \omega x}{\omega} \, d\omega.$$

(c) From part (b) show that

$$\int_0^\infty \frac{\sin\omega\cos\omega x}{\omega}\,d\omega = \begin{cases} 0, & |x|>1, \\ \dfrac{\pi}{4}, & |x|=1, \\ \dfrac{\pi}{2}, & |x|<1. \end{cases}$$

Solution.

(a) The unitary cosine transform is

$$g_c(\omega) = \sqrt{\frac{2}{\pi}}\int_0^\infty f(x)\cos\omega x\,dx = \sqrt{\frac{2}{\pi}}\int_0^1 \cos\omega x\,dx = \boxed{\sqrt{\frac{2}{\pi}}\frac{\sin\omega}{\omega}}.$$

At zero its continuous value is $\sqrt{2/\pi}$. (b) Substitution in the inverse cosine transform gives

$$f(x) = \sqrt{\frac{2}{\pi}}\int_0^\infty g_c(\omega)\cos\omega x\,d\omega = \frac{2}{\pi}\int_0^\infty \frac{\sin\omega\cos\omega x}{\omega}d\omega.$$

At a jump, inversion gives the mean of the two one-sided limits, so its value at $x=\pm1$ is $1/2$.

(c) To evaluate the integral directly, first note that

$$D_\epsilon(b) = \int_0^\infty e^{-\epsilon u}\frac{\sin bu}{u}du, \qquad D'_\epsilon(b) = \int_0^\infty e^{-\epsilon u}\cos bu\,du = \frac{\epsilon}{\epsilon^2+b^2}.$$

Since $D_\epsilon(0)=0$, integration in b gives $D_\epsilon(b)=\arctan(b/\epsilon)$. Thus $D(b)=(\pi/2)\operatorname{sgn} b$ for $b\neq0$, while $D(0)=0$. Dirichlet convergence makes this Abel limit equal to the ordinary improper integral. Now use

$$\sin\omega\cos\omega x = \tfrac{1}{2}\{\sin[(1+x)\omega]+\sin[(1-x)\omega]\}.$$

The answer is $\frac{\pi}{4}[\operatorname{sgn}(1+x)+\operatorname{sgn}(1-x)]$, with $\operatorname{sgn}0=0$. For $|x|<1$ both signs are positive; for $|x|>1$ they cancel; at $|x|=1$ just one is positive. These are exactly the three stated values.

20.2.3

(a) Show that the Fourier sine and cosine transforms of e^{-at} are

$$g_s(\omega) = \sqrt{\frac{2}{\pi}}\frac{\omega}{\omega^2+a^2}, \qquad g_c(\omega) = \sqrt{\frac{2}{\pi}}\frac{a}{\omega^2+a^2}.$$

Hint. Each of the transforms can be related to the other by integration by parts.

(b) Show that

$$\int_0^\infty \frac{\omega\sin\omega x}{\omega^2+a^2}\,d\omega = \frac{\pi}{2}\,e^{-ax}, \qquad x>0,$$
$$\int_0^\infty \frac{\cos\omega x}{\omega^2+a^2}\,d\omega = \frac{\pi}{2a}\,e^{-ax}, \qquad x>0.$$

These results can also be obtained by contour integration (Exercise 11.8.12).

Assume $a > 0$.

Solution.

Take $a > 0$ for the ordinary exponentially decaying transforms and the answers as printed. Put

$$C = \int_0^\infty e^{-at}\cos\omega t\,dt, \qquad S = \int_0^\infty e^{-at}\sin\omega t\,dt.$$

Integrate by parts using an antiderivative $-e^{-at}/a$ of the exponential. The endpoint terms give

$$C = \frac{1}{a} - \frac{\omega}{a}S, \qquad S = \frac{\omega}{a}C.$$

Substitute the second into the first: $a^2C = a - \omega^2 C$, hence

$$C = \frac{a}{a^2+\omega^2}, \qquad S = \frac{\omega}{a^2+\omega^2}.$$

Multiplication by $\sqrt{2/\pi}$ gives g_c and g_s. For $x > 0$, the cosine and sine inversion formulas therefore read

$$e^{-ax} = \frac{2a}{\pi}\int_0^\infty \frac{\cos\omega x}{a^2+\omega^2}d\omega, \qquad e^{-ax} = \frac{2}{\pi}\int_0^\infty \frac{\omega\sin\omega x}{a^2+\omega^2}d\omega.$$

Solving these equations for the two integrals proves (b). The cosine integral is absolutely convergent. The sine integral converges conditionally for $x > 0$; it follows from inversion of the odd extension of e^{-at} on $t > 0$. At $x = 0$ that odd extension has a jump and its inversion value is zero, so the first formula in (b) must not be extended there by substitution.

20.2.4 Find the Fourier transform of the triangular pulse (Fig. 20.6),

$$f(x) = \begin{cases} h(1-a|x|), & |x| < 1/a, \\ 0, & |x| > 1/a. \end{cases}$$

Note. This function provides another delta sequence with $h = a$ and $a \to \infty$.

Assume $a > 0$, so the pulse has positive width.

Solution.

Take $a > 0$, so that the pulse has the stated positive width. Evenness eliminates the sine part:

$$g(\omega) = \frac{2h}{\sqrt{2\pi}}\int_0^{1/a}(1-ax)\cos\omega x\,dx.$$

For $\omega \neq 0$, use

$$\int x\cos\omega x\,dx = \frac{x\sin\omega x}{\omega} + \frac{\cos\omega x}{\omega^2}.$$

The terms proportional to $\sin(\omega/a)$ cancel, giving

$$\begin{aligned}\int_0^{1/a}(1-ax)\cos\omega x\,dx &= \frac{a[1-\cos(\omega/a)]}{\omega^2} \\ &= \frac{1}{2a}\left[\frac{\sin(\omega/2a)}{\omega/2a}\right]^2.\end{aligned}$$

Thus

$$\boxed{g(\omega) = \frac{h}{a\sqrt{2\pi}}\left[\frac{\sin(\omega/2a)}{\omega/2a}\right]^2}, \qquad g(0) = \frac{h}{a\sqrt{2\pi}}.$$

The area of the pulse is h/a. With $h = a$, substitute $u = ax$ in its action on a continuous test function η:

$$\int f_a(x)\eta(x)dx = \int_{-1}^{1}(1-|u|)\eta(u/a)du \longrightarrow \eta(0)\int_{-1}^{1}(1-|u|)du = \eta(0).$$

This proves the delta-sequence claim, not merely a pointwise concentration of the graph.

20.2.5 Consider the sequence

$$\delta_n(x) = \begin{cases} n, & |x| < 1/2n, \\ 0, & |x| > 1/2n. \end{cases}$$

This is Eq. (1.152). Express $\delta_n(x)$ as a Fourier integral (via the Fourier integral theorem, inverse transform, etc.). Finally, show that we may write

$$\delta(x) = \lim_{n\to\infty}\delta_n(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-ikx}\,dk.$$

The limit and the final Fourier integral are distributional identities, tested against smooth rapidly decreasing functions.

Solution.

The forward transform is

$$\begin{aligned} g_n(k) &= \frac{n}{\sqrt{2\pi}}\int_{-1/(2n)}^{1/(2n)} e^{ikx}dx \\ &= \frac{1}{\sqrt{2\pi}}\frac{\sin(k/2n)}{k/2n}. \end{aligned}$$

Fourier inversion consequently gives

$$\boxed{\delta_n(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{\sin(k/2n)}{k/2n}e^{-ikx}dk},$$

with the midpoint value $n/2$ at either edge of the rectangle. Those two point values do not affect any integral. For a smooth rapidly decreasing test function η,

$$\int \delta_n(x)\eta(x)dx = n\int_{-1/(2n)}^{1/(2n)}\eta(x)dx \longrightarrow \eta(0).$$

Also $|\sin u/u| \le 1$ for real u, and the Fourier transform of η is integrable. Hence dominated convergence permits the limit of the multiplier after pairing with η. Since this multiplier tends to one, the limiting identity is

$$\boxed{\delta(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-ikx}dk}.$$

This last integral is a distributional Fourier identity, not a convergent ordinary integral or a pointwise limit of functions at $x = 0$. The test-function argument supplies its precise meaning.

20.2.6 Using the sequence

$$\delta_n(x) = \frac{n}{\sqrt{\pi}} \exp(-n^2 x^2),$$

show that

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ikx}\, dk.$$

Hint. Remember that $\delta(x)$ is defined in terms of its behavior as part of an integrand.
Interpret the final Fourier integral as a distribution, rather than an ordinary convergent integral.

Solution.

The exponent completes to a square as

$$-n^2x^2 + ikx = -n^2 \left(x - \frac{ik}{2n^2} \right)^2 - \frac{k^2}{4n^2}.$$

Set $z = x - ik/(2n^2)$. The horizontal integration line can be shifted back to the real axis: on its two vertical end segments $z = \pm R + iv$, the modulus is $e^{-n^2R^2+n^2v^2}$, with v in a fixed bounded interval, so those integrals tend to zero. Consequently the remaining Gaussian integral is $\int_{\mathbb{R}} e^{-n^2z^2} dz = \sqrt{\pi}/n$, and

$$\begin{aligned} g_n(k) &= \frac{1}{\sqrt{2\pi}} \frac{n}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-n^2x^2+ikx} dx \\ &= \frac{1}{\sqrt{2\pi}} e^{-k^2/(4n^2)}. \end{aligned}$$

The inverse is therefore the explicit regularized identity

$$\frac{n}{\sqrt{\pi}} e^{-n^2x^2} = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-k^2/(4n^2)} e^{-ikx} dk.$$

For a bounded continuous test function η, put $u = nx$:

$$\int \frac{n}{\sqrt{\pi}} e^{-n^2x^2} \eta(x) dx = \frac{1}{\sqrt{\pi}} \int e^{-u^2} \eta(u/n) du \longrightarrow \eta(0).$$

The Gaussian is an integrable majorant. For a Schwartz test function one can likewise pass to the limit in the Fourier pairing because $0 < e^{-k^2/(4n^2)} \leq 1$ and its transform is integrable. This gives the required delta identity with the factor $1/(2\pi)$. The unregularized exponential integral is understood in this weak, distributional sense; removing the Gaussian does not make it an ordinary improper integral.

20.2.7 The formula

$$\delta(t-x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega(t-x)}\, d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega t} e^{-i\omega x}\, d\omega$$

can be identified as the continuum limit of an eigenfunction expansion. Derive sine and cosine representations of $\delta(t-x)$ that are comparable to the exponential representation just given.

ANS. $\frac{2}{\pi} \int_0^{\infty} \sin \omega t \sin \omega x\, d\omega, \quad \frac{2}{\pi} \int_0^{\infty} \cos \omega t \cos \omega x\, d\omega.$

The two half-line answers below mean distributions for $x, t > 0$. On the full line they require reflected delta terms, as derived in the solution.

Solution.

On a finite interval $(0, L)$ the normalized Dirichlet and Neumann eigenfunctions give, for interior x, t,

$$\delta(t-x) = \frac{2}{L}\sum_{n=1}^{\infty} \sin\frac{n\pi t}{L}\sin\frac{n\pi x}{L},$$

$$\delta(t-x) = \frac{1}{L} + \frac{2}{L}\sum_{n=1}^{\infty} \cos\frac{n\pi t}{L}\cos\frac{n\pi x}{L}.$$

Set $\omega_n = n\pi/L$, so $\Delta\omega = \pi/L$. The factor $2/L = (2/\pi)\Delta\omega$ turns the sums into integrals as $L \to \infty$; the constant Neumann term $1/L$ vanishes. In the weak sense this gives the two quoted kernels.

Their domains can be checked directly from product-to-sum:

$$\frac{2}{\pi}\int_0^{\infty} \sin\omega t \sin\omega x \, d\omega = \delta(t-x) - \delta(t+x),$$
$$\frac{2}{\pi}\int_0^{\infty} \cos\omega t \cos\omega x \, d\omega = \delta(t-x) + \delta(t+x).$$

Here $\delta(u) = \pi^{-1}\int_0^{\infty}\cos\omega u \, d\omega$. The stated sine and cosine answers represent $\delta(t-x)$ on the positive half-line, with $x, t > 0$. On the whole real line they contain the reflected source $\mp\delta(t+x)$ shown above. Endpoint values at zero require the corresponding half-line boundary convention. For a test function supported inside $(0, \infty)$ and fixed $x > 0$, the reflected source has no support, which proves exactly the needed positive-half-line identities.

20.2.8 In a resonant cavity, an electromagnetic oscillation of frequency ω_0 dies out as

$$A(t) = \begin{cases} A_0 e^{-\omega_0 t/2Q} e^{-i\omega_0 t}, & t > 0, \\ 0, & t < 0. \end{cases}$$

The parameter Q is a measure of the ratio of stored energy to energy loss per cycle. Calculate the frequency distribution of the oscillation, $a^*(\omega)a(\omega)$, where $a(\omega)$ is the Fourier transform of $A(t)$.

Note. The larger Q is, the sharper your resonance line will be.

$$ANS.\ \ a^*(\omega)a(\omega) = \frac{|A_0|^2}{2\pi}\frac{1}{(\omega-\omega_0)^2 + (\omega_0/2Q)^2}.$$

Take $\omega_0 > 0$, $Q > 0$. The amplitude A_0 may be complex, so its squared modulus is required in the answer.

Solution.

Put $\gamma = \omega_0/(2Q) > 0$, the amplitude decay rate. Direct transformation of the causal signal gives

$$\begin{aligned} a(\omega) &= \frac{A_0}{\sqrt{2\pi}}\int_0^{\infty} e^{-[\gamma - i(\omega-\omega_0)]t} dt \\ &= \frac{A_0}{\sqrt{2\pi}}\frac{1}{\gamma - i(\omega-\omega_0)}. \end{aligned}$$

The conjugate denominator is $\gamma + i(\omega - \omega_0)$, so their product is $\gamma^2 + (\omega-\omega_0)^2$. Hence

$$\boxed{|a(\omega)|^2 = \frac{|A_0|^2}{2\pi}\frac{1}{(\omega-\omega_0)^2 + (\omega_0/2Q)^2}.}$$

The printed numerator A_0^2 presumes a real initial amplitude; for a complex amplitude it is $|A_0|^2$. Decay requires $\omega_0/Q > 0$, conventionally $\omega_0 > 0$, $Q > 0$. The peak is at ω_0. Half its height occurs at $|\omega - \omega_0| = \gamma$, so the full width at half maximum is $2\gamma = \omega_0/Q$. Increasing Q narrows the line. As a normalization check,

$$\int_{-\infty}^{\infty} |a(\omega)|^2 d\omega = \frac{|A_0|^2}{2\gamma} = \int_0^{\infty} |A(t)|^2 dt.$$

20.2.9 Prove that

$$\frac{\hbar}{2\pi i} \int_{-\infty}^{\infty} \frac{e^{-i\omega t}\, d\omega}{E_0 - i\Gamma/2 - \hbar\omega} = \begin{cases} \exp\left(-\dfrac{\Gamma t}{2\hbar}\right) \exp\left(-i\dfrac{E_0 t}{\hbar}\right), & t > 0, \\ 0, & t < 0. \end{cases}$$

This Fourier integral appears in a variety of problems in quantum mechanics: barrier penetration, scattering, time-dependent perturbation theory, and so on.

(The formula as given assumes that $\Gamma > 0$.)

Hint. Try contour integration.

Solution.

Let

$$\omega_p = \frac{E_0 - i\Gamma/2}{\hbar}, \qquad \hbar > 0, \quad E_0 \in \mathbb{R}, \quad \Gamma > 0.$$

Then the integral is

$$I(t) = -\frac{1}{2\pi i} \int_{-\infty}^{\infty} \frac{e^{-i\omega t}}{\omega - \omega_p} d\omega.$$

The pole lies below the real axis. If $t > 0$, write $\omega = u + iv$; the modulus of the exponential is e^{vt}, so close the contour in the lower half-plane. The closure is clockwise, and Jordan's lemma makes its arc integral vanish. The residue of $e^{-i\omega t}/(\omega - \omega_p)$ is $e^{-i\omega_p t}$; consequently

$$I(t) = -\frac{1}{2\pi i}(-2\pi i) e^{-i\omega_p t} = e^{-\Gamma t/(2\hbar)} e^{-iE_0 t/\hbar}.$$

If $t < 0$, close above instead. There is no pole inside, so the integral is zero. This proves both cases, including the sign from the contour orientation. The manual, p. 334, writes $e^{+i\omega_p t}$ in its residue and final step. The exponent must be negative, as the original integrand shows; the positive sign would produce growth for $t > 0$. The published condition $\Gamma > 0$ is necessary for the causal contour choice. At $t = 0$ the symmetric principal-value inversion gives $1/2$, the mean across the jump. The displayed problem deliberately states only $t \neq 0$. For $\Gamma = 0$, a pole prescription would have to be supplied before the real-axis integral is defined.

20.2.10 Verify that the following are Fourier integral transforms of one another:

(a) $$\begin{cases} \sqrt{\dfrac{2}{\pi}} \cdot \dfrac{1}{\sqrt{a^2 - x^2}}, & |x| < a, \\ 0, & |x| > a, \end{cases}$$ and $J_0(ay)$,

(b) $$\begin{cases} 0, & |x| < a, \\ -\sqrt{\dfrac{2}{\pi}} \dfrac{1}{\sqrt{x^2 - a^2}}, & |x| > a, \end{cases}$$ and $Y_0(a|y|)$,

(c) $\sqrt{\dfrac{\pi}{2}}\,\dfrac{1}{\sqrt{x^2+a^2}}$ and $K_0(a|y|)$.

(d) Can you suggest why $I_0(ay)$ is not included in this list?

(It is helpful to use the integral representation $J_0(x)=\dfrac{2}{\pi}\displaystyle\int_0^1 \frac{\cos xt}{\sqrt{1-t^2}}\,dt$.)

Hint. J_0, Y_0, and K_0 may be transformed most easily by using an exponential representation, reversing the order of integration, and employing the Dirac delta function exponential representation, Eq. (20.20). These cases can be treated equally well as Fourier cosine transforms.

Take $a>0$. In (b),(c), ordinary transform integrals are asserted for $y\neq 0$; the identities also have a tempered-distribution interpretation.

Solution.

Take $a>0$ for these transform pairs. Since all the displayed functions are even, the full unitary transform is $\sqrt{2/\pi}\int_0^\infty f(x)\cos(xy)dx$.

(a) The compactly supported member gives

$$\begin{aligned}\mathcal{F}f(y)&=\frac{2}{\pi}\int_0^a \frac{\cos(xy)}{\sqrt{a^2-x^2}}dx\\&=\frac{2}{\pi}\int_0^1 \frac{\cos(ayu)}{\sqrt{1-u^2}}du=J_0(ay).\end{aligned}$$

The substitution $x=au$ canceled both factors of a. The last equality is the Poisson representation stated in the published erratum.

(b) With the corrected square root, the transform is

$$\mathcal{F}f(y)=-\frac{2}{\pi}\int_a^\infty \frac{\cos(xy)}{\sqrt{x^2-a^2}}dx=-\frac{2}{\pi}\int_1^\infty \frac{\cos(a|y|u)}{\sqrt{u^2-1}}du=Y_0(a|y|),\quad y\neq 0.$$

This is the real oscillatory representation derived in Exercise 14.4.8. Its endpoint singularity is integrable and its infinite tail is conditionally convergent for $y\neq 0$.

(c) The prefactors now multiply to one, giving

$$\mathcal{F}f(y)=\int_0^\infty \frac{\cos(xy)}{\sqrt{x^2+a^2}}dx=K_0(a|y|),\quad y\neq 0.$$

For completeness, this follows by cosine inversion of Exercise 14.5.13:

$$\frac{1}{\sqrt{x^2+a^2}}=\frac{2}{\pi}\int_0^\infty K_0(aq)\cos(xq)dq.$$

With the unitary cosine normalization this says that the cosine transform of $1/\sqrt{x^2+a^2}$ is $\sqrt{2/\pi}K_0(a|y|)$, which is precisely the asserted full-transform relation. Applying the full transform twice reflects the original function: $\mathcal{F}^2f(x)=f(-x)$. All three members here are even, so this also verifies the reverse directions. The logarithmic singularities in (b) and (c) at $y=0$ are not finite point values. Their transform identities hold as tempered distributions and as the indicated ordinary improper integrals away from zero; values at the support edges in (a),(b) are likewise singular.

(d) For $a>0$, $I_0(ay)\sim e^{a|y|}/\sqrt{2\pi a|y|}$. It therefore has neither an ordinary Fourier integral nor a Fourier transform as a tempered distribution represented by this function. This exponential growth is the obstruction to an analogous pair in the present Fourier setting. At $a=0$ the constant $I_0(0)=1$ instead has the distributional transform $\sqrt{2\pi}\delta$, a separate degenerate case.

20.2.11 Show that the right-hand expression has the left-hand expression as its forward Fourier transform, and determine the reverse forward transform:

$$i^n J_n(t) \qquad \text{and} \qquad \begin{cases} \sqrt{\dfrac{2}{\pi}}\, T_n(x)(1-x^2)^{-1/2}, & |x|<1, \\ 0, & |x|>1. \end{cases}$$

$T_n(x)$ is the nth-order Chebyshev polynomial.

(The left-hand-side quantities are the transforms of their right-hand-side counterparts, but the right-hand-side quantities are $(-1)^n$ times the transforms of the left-hand-side expressions.)

Hint. With $T_n(\cos\theta) = \cos n\theta$, the transform of $T_n(x)(1-x^2)^{-1/2}$ leads to an integral representation of $J_n(t)$.

Solution.

Denote the compactly supported right-hand expression by $f_n(x)$. Its transform is

$$\mathcal{F}f_n(t) = \frac{1}{\pi}\int_{-1}^{1} \frac{T_n(x)e^{itx}}{\sqrt{1-x^2}}dx.$$

Set $x = \cos\theta$, where $0 \le \theta \le \pi$; reversing the limits removes the sign of $dx = -\sin\theta\, d\theta$:

$$\mathcal{F}f_n(t) = \frac{1}{\pi}\int_0^{\pi} e^{it\cos\theta}\cos(n\theta)d\theta.$$

For integer $n \ge 0$ the integral is half the full-circle integral, because reflection $\theta \mapsto 2\pi - \theta$ leaves both factors unchanged. The sine part of $e^{in\theta}$ integrates to zero under the same reflection. The integral representation of J_n therefore gives

$$\frac{1}{\pi}\int_0^{\pi} e^{it\cos\theta}\cos n\theta\, d\theta = \frac{1}{2\pi}\int_0^{2\pi} e^{it\cos\theta + in\theta}d\theta = \boxed{i^n J_n(t)}.$$

To check the direction of the transform, apply it again:

$$\mathcal{F}[i^n J_n](x) = f_n(-x) = (-1)^n f_n(x),$$

since $T_n(-x) = (-1)^n T_n(x)$. Thus the inverse transform of $i^n J_n$ is f_n, while its forward transform is $(-1)^n f_n$. This proves, and explains, the published sign correction. The square-root singularities are integrable; reverse inversion can be read in the tempered-distribution sense at the singular edges.

20.2.12 Show that the Fourier exponential transform of

$$f(\mu) = \begin{cases} P_n(\mu), & |\mu| \le 1 \\ 0, & |\mu| > 1 \end{cases}$$

is $i^n\sqrt{2/\pi}\, j_n(\omega)$, where ω is the transform variable (the book prints $(2i^n/2\pi)\, j_n(kr)$). Here $P_n(\mu)$ is a Legendre polynomial and j_n is a spherical Bessel function.

Solution.

The forward transform, with variable ω, is

$$g(\omega) = \frac{1}{\sqrt{2\pi}}\int_{-1}^{1} P_n(\mu)e^{i\omega\mu}d\mu.$$

Use Rodrigues' formula $P_n(\mu) = (2^n n!)^{-1} d^n(\mu^2-1)^n/d\mu^n$. Integrating by parts n times has no endpoint terms: the polynomial $(\mu^2-1)^n$ and its first $n-1$ derivatives vanish at both endpoints. Hence

$$\begin{aligned} g(\omega) &= \frac{(-i\omega)^n}{2^n n!\sqrt{2\pi}} \int_{-1}^{1} (\mu^2-1)^n e^{i\omega\mu} d\mu \\ &= \frac{i^n \omega^n}{2^n n!\sqrt{2\pi}} \int_{-1}^{1} (1-\mu^2)^n e^{i\omega\mu} d\mu. \end{aligned}$$

The spherical Poisson representation of Exercise 14.7.4 is

$$j_n(\omega) = \frac{\omega^n}{2^{n+1} n!} \int_{-1}^{1} (1-\mu^2)^n e^{i\omega\mu} d\mu.$$

Comparison gives

$$\boxed{g(\omega) = \frac{2i^n}{\sqrt{2\pi}} j_n(\omega) = i^n \sqrt{\frac{2}{\pi}}\, j_n(\omega)}.$$

The limit at $\omega = 0$ is automatic. This derivation verifies both the scaling and the transform variable in the marked correction; kr is only a possible later choice of ω.

20.2.13

(a) Show that $f(x) = x^{-1/2}$ is a *self-reciprocal* under both Fourier cosine and sine transforms; that is,

$$\sqrt{\frac{2}{\pi}} \int_0^\infty x^{-1/2} \cos xt\, dx = t^{-1/2},$$
$$\sqrt{\frac{2}{\pi}} \int_0^\infty x^{-1/2} \sin xt\, dx = t^{-1/2}.$$

(b) Use the preceding results to evaluate the Fresnel integrals

$$\int_0^\infty \cos\left(y^2\right) dy \quad \text{and} \quad \int_0^\infty \sin\left(y^2\right) dy.$$

In part (a), take $t > 0$.

Solution.

The differential in the second integral must be dx: the original textbook, p. 979, prints ds. Take $t > 0$; both displayed integrals then converge conditionally. Introduce positive damping before taking the square-root limit:

$$I_\epsilon(t) = \int_0^\infty x^{-1/2} e^{-(\epsilon - it)x} dx = \Gamma(1/2)(\epsilon - it)^{-1/2}.$$

The gamma integral applies because $\Re(\epsilon - it) > 0$. On the principal branch, $\arg(\epsilon - it) \to -\pi/2$, so

$$I(t) = \lim_{\epsilon \downarrow 0} I_\epsilon(t) = \sqrt{\pi}\, t^{-1/2} e^{i\pi/4} = \sqrt{\frac{\pi}{2}}\, t^{-1/2}(1+i).$$

The ordinary integrals converge at zero by the exponent $-1/2$ and at infinity by Dirichlet's test, so Abel convergence identifies this limit with their values. Equating real and imaginary parts and multiplying by $\sqrt{2/\pi}$ proves both self-reciprocal identities. The manual, p. 336, has two further slips: the change $u = -izt$ maps the positive z axis to the negative imaginary axis, and its final factor $t^{1/2}$ must be $t^{-1/2}$. The damped gamma calculation above fixes both the phase and the scaling without an ambiguous contour rotation.

For the Fresnel integrals set $x = y^2$ at $t = 1$. Then $x^{-1/2}dx = 2dy$, giving

$$2\int_0^\infty \cos(y^2)dy = 2\int_0^\infty \sin(y^2)dy = \sqrt{\frac{\pi}{2}}.$$

Thus each is $\boxed{\sqrt{\pi/8}}$. The substitution is first made at a finite upper limit, followed by the convergent improper limit; it does not require absolute convergence.

20.2.14 Show that $\left[\frac{1}{r^2}\right]^T(\mathbf{k}) = \left(\frac{\pi}{2}\right)^{1/2}\frac{1}{k}$.

Here $k = |\mathbf{k}| > 0$. The equality extends distributionally, with a singular zero-frequency value.

Solution.

Let $k = |\mathbf{k}| > 0$ and use a polar axis parallel to $\mathbf{k}$. To make the large-radius integration explicit, first multiply r^{-2} by $e^{-\epsilon r}$, $\epsilon > 0$. Since $d^3r = r^2 dr\, d\Omega$, the radial singularity cancels against the volume factor. The angular integration is

$$\int d\Omega\, e^{ikr\cos\theta} = 2\pi\int_{-1}^{1} e^{ikru}du = 4\pi\frac{\sin kr}{kr}.$$

Thus

$$\begin{aligned}\mathcal{F}\left[\frac{e^{-\epsilon r}}{r^2}\right](\mathbf{k}) &= \frac{4\pi}{(2\pi)^{3/2}k}\int_0^\infty e^{-\epsilon r}\frac{\sin kr}{r}dr \\ &= \sqrt{\frac{2}{\pi}}\frac{1}{k}\arctan\frac{k}{\epsilon}.\end{aligned}$$

The last integral was evaluated in Exercise 20.2.2. Taking $\epsilon \downarrow 0$ gives

$$\boxed{\mathcal{F}[r^{-2}](\mathbf{k}) = \sqrt{\frac{\pi}{2}}\frac{1}{k}}.$$

Here k is the nonnegative magnitude of the wave vector. The formula is for $k > 0$, or is a tempered-distribution identity on all of three-dimensional momentum space. Its value at $k = 0$ is singular, and the original spatial integral is not absolutely convergent at infinity. Both r^{-2} in three dimensions and k^{-1} in three dimensions are locally integrable, so this distributional interpretation is well-defined.

20.2.15 The Fourier transform formulas for a function of two variables are

$$\begin{aligned}F(u,v) &= \frac{1}{2\pi}\iint f(x,y)e^{i(ux+vy)}\,dx\,dy, \\ f(x,y) &= \frac{1}{2\pi}\iint F(u,v)e^{-i(ux+vy)}\,du\,dv,\end{aligned}$$

where the integrations are over the entire xy or uv plane. For $f(x,y) = f\big([x^2+y^2]^{1/2}\big) = f(r)$, show that the zero-order Hankel transforms

$$F(\rho) = \int_0^\infty r f(r) J_0(\rho r)\, dr,$$
$$f(r) = \int_0^\infty \rho F(\rho) J_0(\rho r)\, d\rho,$$

are a special case of the Fourier transforms.

Note. This technique may be generalized to derive the Hankel transforms of order $\nu = 0, \frac{1}{2}, 1, \frac{3}{2}, \ldots$. See the two texts by Sneddon (Additional Readings). It might also be noted that the Hankel transforms of half-integral orders $\nu = \pm\frac{1}{2}$ reduce to Fourier sine and cosine transforms.

Solution.

Put $x = r\cos\theta$, $y = r\sin\theta$, and write $u = \rho\cos\phi$, $v = \rho\sin\phi$. Then $ux + vy = \rho r\cos(\theta-\phi)$ and $dx\,dy = r\,dr\,d\theta$. The forward transform is consequently

$$\begin{aligned} F(u,v) &= \frac{1}{2\pi}\int_0^\infty r f(r) dr \int_0^{2\pi} e^{i\rho r\cos(\theta-\phi)} d\theta \\ &= \int_0^\infty r f(r) J_0(\rho r) dr. \end{aligned}$$

The angular integral is $2\pi J_0(\rho r)$, independent of ϕ, proving that the transform is radial: $F = F(\rho)$. For the inverse, use $du\,dv = \rho\,d\rho\,d\phi$:

$$\begin{aligned} f(x,y) &= \frac{1}{2\pi}\int_0^\infty \rho F(\rho) d\rho \int_0^{2\pi} e^{-i\rho r\cos(\theta-\phi)} d\phi \\ &= \int_0^\infty \rho F(\rho) J_0(\rho r) d\rho, \end{aligned}$$

because J_0 is even. This is the claimed inverse Hankel transform. The factors r and ρ are the two polar Jacobians; the Fourier factors $1/(2\pi)$ cancel the angular integrals. For example, Schwartz radial functions justify the operations directly; the unitary extension acts in the radial spaces with measures $r\,dr$ and $\rho\,d\rho$.

20.2.16 Show that the 3-D Fourier exponential transform of a radially symmetric function may be rewritten as a Fourier sine transform:

$$\frac{1}{(2\pi)^{3/2}}\int_{\mathbb{R}^3} f(r) e^{i\mathbf{k}\cdot\mathbf{r}}\, d^3r = \frac{1}{k}\sqrt{\frac{2}{\pi}}\int_0^\infty r f(r)\sin kr\, dr.$$

Take $k = |\mathbf{k}| > 0$, and use a continuous limit at $k = 0$ when it exists.

Solution.

Choose the z axis along $\mathbf{k}$, with $k = |\mathbf{k}| > 0$. In spherical coordinates,

$$\begin{aligned}
\mathcal{F}f(\mathbf{k}) &= \frac{1}{(2\pi)^{3/2}} \int_0^\infty r^2 f(r) dr \int_0^{2\pi} d\phi \int_0^\pi e^{ikr\cos\theta} \sin\theta \, d\theta \\
&= \frac{2\pi}{(2\pi)^{3/2}} \int_0^\infty r^2 f(r) dr \int_{-1}^1 e^{ikru} du \\
&= \frac{4\pi}{(2\pi)^{3/2}} \int_0^\infty r^2 f(r) \frac{\sin kr}{kr} dr \\
&= \boxed{\frac{1}{k}\sqrt{\frac{2}{\pi}} \int_0^\infty r f(r) \sin kr \, dr}.
\end{aligned}$$

Thus $k\mathcal{F}f(k)$ is the unitary sine transform of $rf(r)$. The marked volume element must be d^3r over all space, as in the published correction. If $\int_0^\infty r^2|f(r)|dr < \infty$, the zero-frequency limit follows from $\sin(kr)/(kr) \to 1$:

$$\mathcal{F}f(0) = \sqrt{\frac{2}{\pi}} \int_0^\infty r^2 f(r) dr.$$

Consequently the displayed $1/k$ does not create a singularity when this finite limit exists.

20.3 Properties of Fourier Transforms

20.3.1 Write the 1-D equivalents of the equations for translation, scale change, sign change, and complex conjugation that were given for 3-D transforms in Eqs. (20.48) to (20.51).

Solution.

Write $g(\omega) = \mathcal{F}f(\omega)$ with the positive forward exponential of Eq. (20.10). The four one-dimensional identities are

$$\begin{aligned}
\mathcal{F}[f(x-X)](\omega) &= e^{i\omega X} g(\omega), \\
\mathcal{F}[f(\alpha x)](\omega) &= \frac{1}{|\alpha|} g(\omega/\alpha), \quad \alpha \in \mathbb{R} \setminus \{0\}, \\
\mathcal{F}[f(-x)](\omega) &= g(-\omega), \\
\mathcal{F}[f^*(-x)](\omega) &= g^*(\omega).
\end{aligned}$$

For translation set $u = x - X$, so $e^{i\omega x} = e^{i\omega X} e^{i\omega u}$. For scaling set $u = \alpha x$: when $\alpha < 0$ the integration limits reverse, changing $1/\alpha$ into $1/|\alpha|$. The third identity is the second at $\alpha = -1$. For the fourth, set $u = -x$ and obtain

$$\frac{1}{\sqrt{2\pi}} \int f^*(-x) e^{i\omega x} dx = \frac{1}{\sqrt{2\pi}} \int f^*(u) e^{-i\omega u} du = g^*(\omega).$$

Equivalently, $\mathcal{F}[f^*](\omega) = g^*(-\omega)$. A scale-change factor without an absolute value presumes a positive scale. The formula above covers either sign; $\alpha = 0$ is not an invertible change of variables. The manual's one-dimensional translation line, p. 337, prints $e^{i\omega t}$ for a shift $f(t-a)$. The phase must contain the fixed displacement, $e^{i\omega a}$, not the integration variable.

20.3.2

(a) Show that by replacement of $\mathbf{r}$ by $\mathbf{r}-\mathbf{R}$ in the formula for the Fourier transform of $f(\mathbf{r})$, one can derive the translation formula, Eq. (20.48).

(b) Using methods similar to those for part (a), establish the formulas for scale change, sign change, and complex conjugation, Eqs. (20.49) to (20.51).

Solution.

Let $g(\mathbf{k}) = (2\pi)^{-3/2}\int f(\mathbf{r})e^{i\mathbf{k}\cdot\mathbf{r}}d^3r$. (a) Put $\mathbf{u} = \mathbf{r}-\mathbf{R}$. Translation has unit Jacobian, and the exponential separates:

$$\begin{aligned}\mathcal{F}[f(\mathbf{r}-\mathbf{R})](\mathbf{k}) &= \frac{1}{(2\pi)^{3/2}}\int f(\mathbf{u})e^{i\mathbf{k}\cdot(\mathbf{u}+\mathbf{R})}d^3u \\ &= \boxed{e^{i\mathbf{k}\cdot\mathbf{R}}g(\mathbf{k})}.\end{aligned}$$

(b) For real nonzero α, set $\mathbf{u} = \alpha\mathbf{r}$. The volume element is $d^3r = |\alpha|^{-3}d^3u$, so

$$\boxed{\mathcal{F}[f(\alpha\mathbf{r})](\mathbf{k}) = |\alpha|^{-3}g(\mathbf{k}/\alpha)}.$$

In particular, at $\alpha = -1$ this gives $\mathcal{F}[f(-\mathbf{r})](\mathbf{k}) = g(-\mathbf{k})$. Finally,

$$\begin{aligned}\mathcal{F}[f^*(-\mathbf{r})](\mathbf{k}) &= \frac{1}{(2\pi)^{3/2}}\int f^*(\mathbf{u})e^{-i\mathbf{k}\cdot\mathbf{u}}d^3u \\ &= \left[\frac{1}{(2\pi)^{3/2}}\int f(\mathbf{u})e^{i\mathbf{k}\cdot\mathbf{u}}d^3u\right]^* = \boxed{g^*(\mathbf{k})}.\end{aligned}$$

The printed α^{-3} in Eq. (20.49) needs $\alpha > 0$. For negative real scaling the correct Jacobian is $|\alpha|^{-3}$, as the sign-change case itself confirms.

20.3.3 Derive Eq. (20.53), the formula for the Fourier transform of $\nabla^2 f(\mathbf{r})$.

Solution.

For a sufficiently decaying differentiable function, integrate by parts in r_j, holding the other coordinates fixed:

$$\begin{aligned}\mathcal{F}[\partial_j f](\mathbf{k}) &= \frac{1}{(2\pi)^{3/2}}\int d^2r_\perp\left([fe^{i\mathbf{k}\cdot\mathbf{r}}]_{r_j=-\infty}^{\infty} - \int f\, ik_j e^{i\mathbf{k}\cdot\mathbf{r}}dr_j\right) \\ &= -ik_j g(\mathbf{k}).\end{aligned}$$

The boundary bracket vanishes. Applying the same calculation to $\partial_j f$ gives $\mathcal{F}[\partial_j^2 f] = (-ik_j)^2 g = -k_j^2 g$. Summing the three coordinate equations yields

$$\boxed{\mathcal{F}[\nabla^2 f](\mathbf{k}) = -(k_x^2+k_y^2+k_z^2)g(\mathbf{k}) = -k^2 g(\mathbf{k})}.$$

Schwartz functions satisfy the required endpoint and interchange conditions. The same identity extends to tempered distributions by defining their derivatives through integration by parts, so singular sources can also be transformed consistently.

20.3.4 Verify Eqs. (20.55) and (20.56), the formulas for the derivatives of 1-D Fourier transforms.

Solution.

Starting from the one-dimensional definition,

$$\begin{aligned}\mathcal{F}[f'](\omega) &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f'(t)e^{i\omega t}dt \\ &= \frac{1}{\sqrt{2\pi}}[f(t)e^{i\omega t}]_{-\infty}^{\infty} - i\omega\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(t)e^{i\omega t}dt \\ &= -i\omega g(\omega).\end{aligned}$$

For n derivatives repeat the operation. Each repetition moves one derivative to the exponential and contributes $-i\omega$:

$$\boxed{\mathcal{F}[f^{(n)}](\omega) = (-i\omega)^n g(\omega)}.$$

More explicitly, the finite-interval integration-by-parts identity has endpoint terms

$$\sum_{j=0}^{n-1}(-i\omega)^j[f^{(n-1-j)}(t)e^{i\omega t}]_{-R}^{R};$$

under the usual decaying-function assumptions each tends to zero as $R \to \infty$. This explains why no initial-value terms occur for this two-sided transform, unlike the one-sided Laplace transform. For distributional derivatives the identity remains valid without incorrectly discarding jump contributions.

20.3.5 Derive the inverse of Eq. (20.56), namely that

$$[t^n f(t)]^T(\omega) = i^{-n}\,\frac{d^n}{d\omega^n}\,g(\omega).$$

Solution.

Differentiate the forward transform with respect to its frequency argument:

$$g'(\omega) = \frac{1}{\sqrt{2\pi}}\int f(t)\frac{\partial}{\partial\omega}e^{i\omega t}dt = i\,\mathcal{F}[tf](\omega).$$

After n derivatives,

$$g^{(n)}(\omega) = \frac{1}{\sqrt{2\pi}}\int (it)^n f(t)e^{i\omega t}dt = i^n\mathcal{F}[t^n f](\omega).$$

Dividing by i^n proves

$$\boxed{\mathcal{F}[t^n f](\omega) = i^{-n}g^{(n)}(\omega)}.$$

For example, $t^n f \in L^1$ justifies these differentiations by dominated convergence, with lower moments controlled by $f \in L^1$ and the highest moment. Schwartz functions satisfy all such conditions. This is the duality between multiplication in direct space and differentiation in frequency space.

20.3.6 The 1-D neutron diffusion equation with a (plane) source is

$$-D\,\frac{d^2\varphi(x)}{dx^2} + K^2 D\varphi(x) = Q\delta(x),$$

where $\varphi(x)$ is the neutron flux, $Q\delta(x)$ is the (plane) source at $x = 0$, and D and K^2 are constants. Apply a Fourier transform. Solve the equation in transform space. Transform your solution back into x-space.

$$\textit{ANS. } \varphi(x) = \frac{Q}{2KD}\,e^{-|Kx|}.$$

For the displayed answer assume $D > 0$, $K > 0$, and seek the whole-line solution bounded at both infinities. These conditions select the decaying solution.

Solution.

For the decaying whole-line solution take $D > 0$ and $K^2 > 0$, and put $\kappa = \sqrt{K^2} > 0$. The printed coefficient $1/K$ presumes $K > 0$. Decay, or boundedness at both infinities, must be part of the boundary specification. The delta transform is $1/\sqrt{2\pi}$ and $\mathcal{F}[\varphi''] = -\omega^2\Phi$. Thus

$$D(\omega^2 + \kappa^2)\Phi(\omega) = \frac{Q}{\sqrt{2\pi}}, \qquad \Phi(\omega) = \frac{Q}{D\sqrt{2\pi}(\omega^2 + \kappa^2)}.$$

Invert, use evenness, and apply Exercise 20.2.3:

$$\begin{aligned}\varphi(x) &= \frac{Q}{2\pi D}\int_{-\infty}^{\infty} \frac{e^{-i\omega x}}{\omega^2 + \kappa^2}\,d\omega \\ &= \frac{Q}{\pi D}\int_0^{\infty} \frac{\cos\omega x}{\omega^2 + \kappa^2}\,d\omega = \boxed{\frac{Q}{2\kappa D}e^{-\kappa|x|}}.\end{aligned}$$

Away from zero, $\varphi'' = \kappa^2\varphi$. The derivative jump is

$$\varphi'(0^+) - \varphi'(0^-) = -\frac{Q}{2D} - \frac{Q}{2D} = -\frac{Q}{D},$$

so $-D\varphi''$ has exactly the source $Q\delta$. A homogeneous addition is $Ae^{\kappa x} + Be^{-\kappa x}$; boundedness at both infinities forces $A = B = 0$. For $K = 0$ and nonzero Q, the formal fundamental solution is $-Q|x|/(2D)$ plus an affine function, and no solution decays at both ends. Substituting $K = 0$ in the printed formula is invalid.

20.4 Fourier Convolution Theorem

20.4.1 Work out the convolution equation corresponding to Eq. (20.71) for

(a) Fourier sine transforms

$$\frac{1}{2}\int_0^{\infty} g(y)\big[f(y+x) + f(y-x)\big]\,dy = \int_0^{\infty} F_s(s)G_s(s)\cos sx\,ds,$$

where f and g are odd functions.

(b) Fourier cosine transforms

$$\frac{1}{2}\int_0^{\infty} g(y)\big[f(y+x) + f(x-y)\big]\,dy = \int_0^{\infty} F_c(s)G_c(s)\cos sx\,ds,$$

where f and g are even functions.

Solution.

Use the unitary half-line transforms, for example $F_s(s) = \sqrt{2/\pi}\int_0^\infty f(u)\sin(su)du$, with the identical normalization in the inverse.

(a) Extend f oddly. Its inverse sine representation and the identity $\sin(A+B)+\sin(A-B) = 2\sin A\cos B$ give

$$f(y+x)+f(y-x) = 2\sqrt{\frac{2}{\pi}}\int_0^\infty F_s(s)\sin(sy)\cos(sx)ds.$$

Multiply by $g(y)/2$ and integrate on $y>0$:

$$\begin{aligned}\frac{1}{2}\int_0^\infty g(y)[f(y+x)+f(y-x)]dy &= \sqrt{\frac{2}{\pi}}\int_0^\infty F_s(s)\cos(sx)\left[\int_0^\infty g(y)\sin(sy)dy\right]ds\\ &= \int_0^\infty F_s(s)G_s(s)\cos(sx)ds.\end{aligned}$$

The factors $\sqrt{2/\pi}$ and $\sqrt{\pi/2}$ cancel.

(b) Extend f evenly. Since $f(x-y) = f(y-x)$, the cosine addition identity similarly gives

$$f(y+x)+f(x-y) = 2\sqrt{\frac{2}{\pi}}\int_0^\infty F_c(s)\cos(sy)\cos(sx)ds.$$

The same multiplication and integration replace the inner integral by $\sqrt{\pi/2}G_c(s)$, giving exactly the second formula. The calculation is justified first for sufficiently regular decaying transforms, and then by the L^2 pairing limit where appropriate; odd/even extensions are essential for arguments that become negative. The manual's intermediate two-delta kernels on p. 339 omit reflected sources. For example, the sine kernel on the full range of its arguments is

$$\begin{aligned}\int_0^\infty \sin(sy)\sin(st)\cos(sx)ds = \frac{\pi}{4}\big[&\delta(t-y-x)+\delta(t-y+x)\\ &-\delta(t+y+x)-\delta(t+y-x)\big].\end{aligned}$$

This follows by applying the product identity and the reflected-source formula in Exercise 20.2.7. The cosine kernel has the same four delta arguments with all plus signs. The reflected terms cannot all be discarded on $t, y > 0$ when x is arbitrary; the odd/even extensions above include them.

20.4.2 Show that for both Fourier sine and Fourier cosine transforms Parseval's relation has the form

$$\int_0^\infty F(t)G(t)\,dt = \int_0^\infty f(y)g(y)\,dy.$$

Solution.

Set $x = 0$ in each of the two formulas just proved. For the sine version the bracket is $f(y)+f(y) = 2f(y)$. For the cosine version it is $f(y)+f(-y) = 2f(y)$ because the extension is even. On either right-hand side $\cos(sx) = 1$. Thus in both cases

$$\boxed{\int_0^\infty f(y)g(y)dy = \int_0^\infty F(s)G(s)ds}.$$

The same result follows by inserting the inverse transform of f into the left-hand side and identifying the transform of g; all normalization factors cancel. For L^2 functions, Cauchy–Schwarz controls both pairings and permits approximation by regular functions. This displayed version is bilinear and is valid also for complex functions. To obtain the norm form set $g = \overline{f}$: the sine and cosine kernels are real, so $G = \overline{F}$, and $\int|f|^2 = \int|F|^2$.

20.4.3

(a) A rectangular pulse is described by

$$f(x) = \begin{cases} 1, & |x| < a, \\ 0, & |x| > a. \end{cases}$$

Show that the Fourier exponential transform is

$$F(t) = \sqrt{\frac{2}{\pi}}\,\frac{\sin at}{t}.$$

This is the single-slit diffraction problem of physical optics. The slit is described by $f(x)$. The diffraction pattern amplitude is given by the Fourier transform $F(t)$.

(b) Use the Parseval relation to evaluate

$$\int_{-\infty}^{\infty} \frac{\sin^2 t}{t^2}\,dt.$$

This integral may also be evaluated by using the calculus of residues (Exercise 11.8.9).

ANS. (b) π.

Take $a > 0$.

Solution.

(a) For $a > 0$, direct integration gives

$$\begin{aligned} F(t) &= \frac{1}{\sqrt{2\pi}} \int_{-a}^{a} e^{itx} dx \\ &= \frac{1}{\sqrt{2\pi}} \frac{e^{iat} - e^{-iat}}{it} = \boxed{\sqrt{\frac{2}{\pi}}\frac{\sin at}{t}}. \end{aligned}$$

At $t = 0$ take the limit $F(0) = a\sqrt{2/\pi}$. (b) Unitary Parseval gives

$$2a = \int_{-a}^{a} 1\,dx = \int_{-\infty}^{\infty} |F(t)|^2 dt = \frac{2}{\pi} \int_{-\infty}^{\infty} \frac{\sin^2 at}{t^2} dt.$$

Set $u = at$, so $dt = du/a$ and $t^{-2} = a^2/u^2$:

$$2a = \frac{2a}{\pi} \int_{-\infty}^{\infty} \frac{\sin^2 u}{u^2} du.$$

Canceling $2a$ gives the required value $\boxed{\pi}$. The apparent singularity at zero is removable and the tails are bounded by u^{-2}, so the final integral is absolutely convergent.

20.4.4 Solve Poisson's equation, $\nabla^2\psi(\mathbf{r}) = -\rho(\mathbf{r})/\varepsilon_0$, by the following sequence of operations:

(a) Take the Fourier transform of both sides of this equation. Solve for the Fourier transform of $\psi(\mathbf{r})$.

(b) Carry out the Fourier inverse transform.

Seek the free-space solution that vanishes at infinity, for sufficiently localized charge. Without this boundary condition, harmonic additions are possible.

Solution.

(a) Let $\widehat{\psi}$ and $\widehat{\rho}$ denote unitary three-dimensional transforms. The transformed equation is

$$-k^2\widehat{\psi}(\mathbf{k}) = -\frac{\widehat{\rho}(\mathbf{k})}{\varepsilon_0}, \qquad \widehat{\psi}(\mathbf{k}) = \frac{\widehat{\rho}(\mathbf{k})}{\varepsilon_0 k^2}.$$

(b) To invert the multiplier, first evaluate the absolutely convergent regulated radial transform, $\eta > 0$:

$$\begin{aligned}\mathcal{F}\left[\frac{e^{-\eta r}}{r}\right](\mathbf{k}) &= \sqrt{\frac{2}{\pi}}\frac{1}{k}\int_0^\infty e^{-\eta r}\sin kr\, dr \\ &= \sqrt{\frac{2}{\pi}}\frac{1}{k^2+\eta^2}.\end{aligned}$$

Letting $\eta \downarrow 0$ in distributions shows that the inverse transform of k^{-2} is $\sqrt{\pi/2}/r$. For our normalization,

$$\mathcal{F}^{-1}[AB](\mathbf{r}) = (2\pi)^{-3/2}\int a(\mathbf{u})b(\mathbf{r}-\mathbf{u})d^3u.$$

Use $a = \rho$ and $b = \sqrt{\pi/2}/r$. Since $\sqrt{\pi/2}/(2\pi)^{3/2} = 1/(4\pi)$,

$$\boxed{\psi(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\int \frac{\rho(\mathbf{u})}{|\mathbf{r}-\mathbf{u}|}d^3u}.$$

This is the free-space solution with vanishing potential at infinity for suitably localized charge. Without a boundary condition, one may add any harmonic function; division by k^2 does not establish uniqueness at the zero mode. For smooth localized charge the Coulomb singularity is locally integrable, and $\nabla^2(1/r) = -4\pi\delta^{(3)}$ verifies the source normalization of the answer.

20.4.5

(a) Given $f(x) = 1 - |x/2|$ for $-2 \le x \le 2$, with $f(x) = 0$ elsewhere, show that the Fourier transform of $f(x)$ is

$$F(t) = \sqrt{\frac{2}{\pi}}\left(\frac{\sin t}{t}\right)^2.$$

(b) Using the Parseval relation, evaluate

$$\int_{-\infty}^{\infty}\left(\frac{\sin t}{t}\right)^4 dt.$$

ANS. (b) $\dfrac{2\pi}{3}$.

Solution.

(a) Let $q(x)$ equal one on $[-1,1]$ and zero elsewhere. In this solution $*$ denotes the unnormalized convolution integral, so the Fourier convolution theorem includes its factor $\sqrt{2\pi}$. The overlap length

of the two intervals in $(q*q)(x) = \int q(y)q(x-y)dy$ is $2-|x|$ for $|x| \le 2$ and zero outside. Thus $f = (q*q)/2$. From Exercise 20.4.3, $Q(t) = \sqrt{2/\pi}\sin t/t$. The unitary convolution theorem gives

$$F(t) = \frac{1}{2}\sqrt{2\pi}\,Q(t)^2 = \boxed{\sqrt{\frac{2}{\pi}}\left(\frac{\sin t}{t}\right)^2}.$$

(b) The direct-space squared norm is

$$\int |f(x)|^2 dx = 2\int_0^2 (1-x/2)^2 dx = 4\int_0^1 u^2 du = \frac{4}{3}.$$

In transform space it is

$$\int |F(t)|^2 dt = \frac{2}{\pi}\int_{-\infty}^{\infty}\left(\frac{\sin t}{t}\right)^4 dt.$$

Equating the norms yields the value $\boxed{(\pi/2)(4/3) = 2\pi/3}$. The manual's Parseval line on p. 341 prints $(\sin t/t)^2$ where it must print $(\sin t/t)^4$: the already squared amplitude $F(t)$ is squared again to form $|F(t)|^2$. The integration differential there is dt.

20.4.6 With $F(t)$ and $G(t)$ the Fourier transforms of $f(x)$ and $g(x)$, respectively, show that

$$\int_{-\infty}^{\infty} \big|f(x)-g(x)\big|^2 dx = \int_{-\infty}^{\infty} \big|F(t)-G(t)\big|^2 dt.$$

If $g(x)$ is an approximation to $f(x)$, the preceding relation indicates that the mean square deviation in t-space is equal to the mean square deviation in x-space.

Assume $f, g \in L^2$, or more generally that the difference and its transform are defined in L^2.

Solution.

Let $h = f - g$. Linearity gives $\mathcal{F}h = F - G$. Parseval's norm identity applied to h is

$$\int |h(x)|^2 dx = \int |\mathcal{F}h(t)|^2 dt,$$

which is exactly the claimed equation. To display the complex cross terms explicitly,

$$|f-g|^2 = |f|^2 + |g|^2 - f\overline{g} - \overline{f}g.$$

The unitary inner-product identity sends each of these four integrals respectively to $|F|^2$, $|G|^2$, $F\overline{G}$, and $\overline{F}G$, whose sum is $|F-G|^2$. The finite mean-square statement requires $f-g \in L^2$ and its L^2 transform; for example $f, g \in L^2$ suffice. It is an equality of integrated squared errors, not pointwise errors at corresponding coordinates.

20.4.7 Use the Parseval relation to evaluate

$$\text{(a)}\ \int_{-\infty}^{\infty}\frac{d\omega}{(\omega^2+a^2)^2}, \qquad \text{(b)}\ \int_{-\infty}^{\infty}\frac{\omega^2\,d\omega}{(\omega^2+a^2)^2}.$$

Hint. Compare Exercise 20.2.3.

ANS. (a) $\dfrac{\pi}{2a^3}$, (b) $\dfrac{\pi}{2a}$.

The displayed answers use $a > 0$; for nonzero real a replace their a by $|a|$.

Solution.

Take $a > 0$ and apply the half-line Parseval identities to $f(x) = e^{-ax}$. Its direct squared norm is

$$\int_0^\infty e^{-2ax}dx = \frac{1}{2a}.$$

(a) From the cosine transform in Exercise 20.2.3,

$$\frac{1}{2a} = \frac{2}{\pi}\int_0^\infty \frac{a^2}{(\omega^2+a^2)^2}d\omega.$$

Divide by $2a^2/\pi$ and double the half-line integral:

$$\boxed{\int_{-\infty}^\infty \frac{d\omega}{(\omega^2+a^2)^2} = \frac{\pi}{2a^3}}.$$

(b) Using the sine transform instead,

$$\frac{1}{2a} = \frac{2}{\pi}\int_0^\infty \frac{\omega^2}{(\omega^2+a^2)^2}d\omega,$$

so the whole-line answer is $\boxed{\pi/(2a)}$. If a is allowed to be negative, the integrands are unchanged and the answers contain $|a|^3$ and $|a|$, respectively. At $a = 0$ both integrals diverge. The printed forms use $a > 0$.

20.4.8 The nuclear form factor $F(\mathbf{k})$ and the charge distribution $\rho(\mathbf{r})$ are 3-D Fourier transforms of each other:

$$F(\mathbf{k}) = \frac{1}{(2\pi)^{3/2}}\int \rho(\mathbf{r})e^{i\mathbf{k}\cdot\mathbf{r}}\,d^3r.$$

If the measured form factor is

$$F(\mathbf{k}) = (2\pi)^{-3/2}\left(1+\frac{k^2}{a^2}\right)^{-1},$$

find the corresponding charge distribution.

$$ANS.\ \ \rho(r) = \frac{a^2}{4\pi}\frac{e^{-ar}}{r}.$$

The parameter a in the displayed answer is the positive scale, $a > 0$.

Solution.

Assume $a > 0$ first. Rewrite the measured transform as

$$F(\mathbf{k}) = \frac{a^2}{(2\pi)^{3/2}}\frac{1}{k^2+a^2}.$$

The radial formula of Exercise 20.2.16 gives

$$\mathcal{F}\left[\frac{e^{-ar}}{r}\right](\mathbf{k}) = \sqrt{\frac{2}{\pi}}\frac{1}{k}\int_0^\infty e^{-ar}\sin kr\,dr = \sqrt{\frac{2}{\pi}}\frac{1}{k^2+a^2}.$$

Therefore multiplication by $a^2/(4\pi)$ produces the given F, because $(4\pi)^{-1}\sqrt{2/\pi} = (2\pi)^{-3/2}$. The charge distribution is

$$\boxed{\rho(r) = \frac{a^2}{4\pi}\frac{e^{-ar}}{r}}.$$

Its total charge is

$$4\pi \int_0^\infty r^2 \rho(r) dr = a^2 \int_0^\infty r e^{-ar} dr = 1,$$

in agreement with $(2\pi)^{3/2} F(0) = 1$. The $1/r$ origin singularity is integrable with the volume element. The scale in the printed decaying exponential is the positive root. For nonzero real a of either sign use $e^{-|a|r}$; the form factor only determines a^2.

20.4.9 Using convolution methods, find an integral whose value is the electrostatic interaction energy between a charge distribution $\rho(\mathbf{r} - \mathbf{A})$ and a unit point charge at $\mathbf{C}$.

Solution.

Let $H(\mathbf{r}) = 1/|\mathbf{r}|$, write $(\rho * H)(\mathbf{v}) = \int \rho(\mathbf{u}) H(\mathbf{v} - \mathbf{u}) d^3u$ for the unnormalized convolution, and use SI normalization, so the potential of unit charge is $H/(4\pi\varepsilon_0)$. The potential at $\mathbf{C}$ of the translated distribution is its interaction energy with the unit point charge:

$$U = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho(\mathbf{r} - \mathbf{A})}{|\mathbf{C} - \mathbf{r}|} d^3r.$$

Putting $\mathbf{u} = \mathbf{r} - \mathbf{A}$ identifies a convolution:

$$\boxed{U = \frac{1}{4\pi\varepsilon_0} (\rho * H)(\mathbf{C} - \mathbf{A}) = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho(\mathbf{u})}{|\mathbf{C} - \mathbf{A} - \mathbf{u}|} d^3u}.$$

If $R(\mathbf{k}) = \mathcal{F}\rho(\mathbf{k})$, then $\mathcal{F}H = \sqrt{2/\pi}/k^2$ and $\mathcal{F}(\rho * H) = (2\pi)^{3/2} R\mathcal{F}H = 4\pi R/k^2$. Inverting yields the equivalent transform-space integral

$$\boxed{U = \frac{1}{\varepsilon_0 (2\pi)^{3/2}} \int \frac{R(\mathbf{k})}{k^2} e^{-i\mathbf{k}\cdot(\mathbf{C}-\mathbf{A})} d^3k}.$$

Both formulas show that only the relative displacement enters. For real charge the frequency-domain integrand has conjugate symmetry, so U is real. These formulas assume the potential integral exists; point-charge singularities are understood through the corresponding distributions. There is no factor 1/2 in the mutual energy of these two specified charge distributions.

20.4.10 With $\psi(\mathbf{r})$ a wave function in ordinary space and $\varphi(\mathbf{p})$ the corresponding momentum function, show that

(a) $$\frac{1}{(2\pi\hbar)^{3/2}} \int \mathbf{r} \psi(\mathbf{r}) e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar} d^3r = i\hbar \nabla_p \varphi(\mathbf{p}),$$

(b) $$\frac{1}{(2\pi\hbar)^{3/2}} \int \mathbf{r}^2 \psi(\mathbf{r}) e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar} d^3r = \left(i\hbar \nabla_p\right)^2 \varphi(\mathbf{p}).$$

(This exercise assumes that the units and scaling of the momentum wave function correspond to the formula $\varphi(\mathbf{p}) = \dfrac{1}{(2\pi\hbar)^{3/2}} \displaystyle\int \psi(\mathbf{r}) e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar} d^3r$.)

Note. ∇_p is the gradient in momentum space:

$$\hat{\mathbf{e}}_x \frac{\partial}{\partial p_x} + \hat{\mathbf{e}}_y \frac{\partial}{\partial p_y} + \hat{\mathbf{e}}_z \frac{\partial}{\partial p_z}.$$

These results may be extended to any positive integer power of $\mathbf{r}$ on the appropriate moment domain, and to an analytic function only when its Maclaurin series can be interchanged with the transform for the wave function in use.

Solution.

Use precisely the momentum-space convention stated in the erratum, whose forward exponential has a negative sign:

$$\varphi(\mathbf{p}) = \frac{1}{(2\pi\hbar)^{3/2}} \int \psi(\mathbf{r}) e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar} d^3r.$$

(a) Differentiation of the exponential gives

$$\nabla_p e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar} = -\frac{i\mathbf{r}}{\hbar} e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar}.$$

Move the derivative under the integral and multiply by $i\hbar$:

$$\boxed{i\hbar\nabla_p\varphi = \frac{1}{(2\pi\hbar)^{3/2}} \int \mathbf{r}\psi(\mathbf{r}) e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar} d^3r}.$$

(b) Apply another component derivative and sum over components:

$$\nabla_p^2 e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar} = -\frac{r_x^2 + r_y^2 + r_z^2}{\hbar^2} e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar}.$$

Since $(i\hbar\nabla_p)^2 = -\hbar^2\nabla_p^2$, this proves

$$\boxed{(i\hbar\nabla_p)^2\varphi = \frac{1}{(2\pi\hbar)^{3/2}} \int r^2\psi(\mathbf{r}) e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar} d^3r}.$$

The exponent printed in part (b) omits the factor i: it reads $-\mathbf{r}\cdot\mathbf{p}/\hbar$. The Fourier kernel must be $e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar}$ in both parts. The same argument works for every polynomial in the coordinate components. Appropriate integrable moments of ψ justify ordinary differentiation; otherwise operator domains or a well-defined distributional interpretation must be specified.

20.4.11 The ordinary space wave function $\psi(\mathbf{r}, t)$ satisfies the time-dependent Schrödinger equation,

$$i\hbar \frac{\partial\psi(\mathbf{r}, t)}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi + V(\mathbf{r})\psi.$$

Show that the corresponding time-dependent momentum wave function satisfies the analogous equation

$$i\hbar \frac{\partial\varphi(\mathbf{p}, t)}{\partial t} = \frac{p^2}{2m}\varphi + V(i\hbar\nabla_p)\varphi.$$

Note. Assume that $V(\mathbf{r})$ may be expressed by a Maclaurin series whose termwise transformation is justified on the chosen wave function (a polynomial potential suffices), and use Exercise 20.4.10. $V(i\hbar\nabla_p)$ is the same function of the variable $i\hbar\nabla_p$ that $V(\mathbf{r})$ is of the variable $\mathbf{r}$.

Solution.

Transform with the negative-exponent momentum convention of Exercise 20.4.10. The time derivative commutes with the spatial transform, producing $i\hbar\partial_t\varphi$. Integrating the spatial Laplacian by parts twice gives

$$\mathcal{F}_p[\nabla_r^2\psi] = -\frac{p^2}{\hbar^2}\varphi,$$

so the kinetic term becomes $p^2\varphi/(2m)$. Write a convergent coordinate expansion with multi-index $\alpha = (\alpha_x, \alpha_y, \alpha_z)$:

$$V(\mathbf{r}) = \sum_{\alpha} v_\alpha \mathbf{r}^\alpha.$$

Exercise 20.4.10, applied component by component, gives

$$\mathcal{F}_p[\mathbf{r}^\alpha \psi] = (i\hbar)^{|\alpha|} \partial_{p_x}^{\alpha_x} \partial_{p_y}^{\alpha_y} \partial_{p_z}^{\alpha_z} \varphi.$$

Consequently

$$\mathcal{F}_p[V\psi] = \sum_{\alpha} v_\alpha (i\hbar\nabla_p)^\alpha \varphi = V(i\hbar\nabla_p)\varphi.$$

Combining the terms gives

$$\boxed{i\hbar \frac{\partial \varphi}{\partial t} = \frac{p^2}{2m}\varphi + V(i\hbar\nabla_p)\varphi}.$$

For a nonpolynomial potential, a local Maclaurin series alone does not justify interchange with an integral over all space. Its action on the chosen wave function must converge and permit that interchange. Polynomial potentials satisfy the algebraic step without an infinite-series issue. When a general potential is treated by its transform instead, the rigorous replacement is the momentum convolution

$$\mathcal{F}_p[V\psi](\mathbf{p}) = \frac{1}{(2\pi\hbar)^{3/2}} \int \widetilde{V}(\mathbf{p} - \mathbf{q})\varphi(\mathbf{q}) d^3q,$$

whenever this convolution, possibly distributional, is defined. The differential expression is its polynomial or convergent-series form.

20.5 Signal-Processing Applications

20.5.1 Find the transfer function $\varphi(\omega)$ for the circuit shown in the left panel of Fig. 20.10. Is this a high-pass, a low-pass, or a more complicated filter?

Solution.

The left circuit has a series resistor R and a capacitor C across the output terminals. With the $e^{i\omega t}$ phasor convention used in Section 20.5, $Z_C = 1/(i\omega C)$. The current is $I = V_{\text{in}}/(R + Z_C)$, and $V_{\text{out}} = IZ_C$. Therefore

$$\boxed{\varphi(\omega) = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{Z_C}{R + Z_C} = \frac{1}{1 + i\omega RC}}.$$

For positive R, C,

$$|\varphi|^2 = \frac{1}{1 + \omega^2 R^2 C^2}, \qquad \arg\varphi = -\tan^{-1}(\omega RC) \quad (\omega \geq 0).$$

The gain decreases from 1 at zero frequency to 0 at high frequency: this is a low-pass filter. The power gain is $1/2$ at $\omega_c = 1/(RC)$. These expressions assume an unloaded output, as in the drawing. The sign of $i\omega$ follows the section's phasor convention; the opposite time convention conjugates the transfer function.

20.5.2 Find the transfer function $\varphi(\omega)$ for the circuit shown in the right panel of Fig. 20.10.

Hint. The potential difference across an inductor is given by $L\,dI/dt$.

Solution.

The intended reference is the right panel of Fig. 20.10, which contains a series inductor L and an output resistor R. Figure 20.11 has a single, different circuit. For a current $Ie^{i\omega t}$, the voltage across the inductor is $L\,dI/dt = i\omega LI$. Kirchhoff's voltage law and the output voltage are consequently

$$V_{\text{in}} = (R + i\omega L)I, \qquad V_{\text{out}} = RI.$$

Dividing gives

$$\boxed{\varphi(\omega) = \frac{R}{R + i\omega L} = \frac{1}{1 + i\omega L/R}}.$$

Its power gain is $[1 + (\omega L/R)^2]^{-1}$, so it is a low-pass filter with half-power frequency R/L, for positive R, L.

20.5.3 Find the transfer function $\varphi(\omega)$ for the circuit shown in Fig. 20.11. This is a band-pass filter.

Hint. Assume the currents in the various parts of the circuit to have the values shown in the figure.

Solution.

Use the currents specified in Fig. 20.11: I_1 flows through R_1, I_2 through the series combination R_2, C_2, and $I_1 + I_2$ through C_1. Put $p = i\omega$. The two node voltages satisfy

$$V_1 = R_1 I_1 = \left(R_2 + \frac{1}{pC_2}\right) I_2, \quad V_{\text{out}} = \frac{I_2}{pC_2}, \quad V_{\text{in}} = V_1 + \frac{I_1 + I_2}{pC_1}.$$

The middle relation gives $I_2 = pC_2 V_{\text{out}}$, and the first then gives

$$V_1 = (1 + pR_2C_2)V_{\text{out}}, \qquad I_1 = \frac{1 + pR_2C_2}{R_1} V_{\text{out}}.$$

Substitute these currents into the input relation and multiply by pR_1C_1:

$$pR_1C_1V_{\text{in}} = \big[pR_1C_1(1 + pR_2C_2) + 1 + pR_2C_2 + pR_1C_2\big]V_{\text{out}}.$$

Thus

$$\boxed{\varphi(\omega) = \frac{i\omega R_1C_1}{1 + i\omega(R_1C_1 + R_1C_2 + R_2C_2) - \omega^2 R_1R_2C_1C_2}}.$$

To check the filter type, write $A = R_1C_1$, $B = R_1R_2C_1C_2$, and $D = R_1C_1 + R_1C_2 + R_2C_2$, all positive. Then

$$|\varphi|^2 = \frac{A^2\omega^2}{(1 - B\omega^2)^2 + D^2\omega^2} = \frac{A^2}{\omega^{-2} + B^2\omega^2 + D^2 - 2B}.$$

It vanishes as $\omega \to 0$ and $\omega \to \infty$. Its maximum occurs when $-2\omega^{-3} + 2B^2\omega = 0$, namely $\omega = 1/\sqrt{B}$, where $|\varphi| = A/D$. It therefore passes an intermediate band of frequencies.

20.5.4 The circuit elements for Exercise 20.5.3 correspond to the successive transfer functions shown in Fig. 20.12. Explain why the transfer function for this exercise is uniformly approximated by the product of the individual unloaded transfer functions when $R_2 \gg R_1$. Explain the loading correction; the original word "only" does not give a necessary condition at each frequency.

Solution.

Taken separately with unloaded outputs, the high-pass and low-pass stages have transfer functions

$$\varphi_1 = \frac{pR_1C_1}{1 + pR_1C_1}, \qquad \varphi_2 = \frac{1}{1 + pR_2C_2}, \qquad p = i\omega.$$

Their product is

$$\varphi_1\varphi_2 = \frac{pR_1C_1}{1 + p(R_1C_1 + R_2C_2) + p^2R_1R_2C_1C_2}.$$

The actual denominator found in Exercise 20.5.3 contains the additional term pR_1C_2. This appears because the second stage draws current from the first: the current through C_1 is $I_1 + I_2$, not I_1. Quantitatively,

$$\frac{I_2}{I_1} = \frac{R_1}{R_2 + 1/(i\omega C_2)}, \qquad \left|\frac{I_2}{I_1}\right| = \frac{R_1}{\sqrt{R_2^2 + 1/(\omega^2C_2^2)}} \le \frac{R_1}{R_2}.$$

Hence $R_2 \gg R_1$ makes the load current small at every real frequency. More explicitly, if $D_0 = (1 + pR_1C_1)(1 + pR_2C_2)$, then

$$\varphi = \frac{\varphi_1\varphi_2}{1 + \eta}, \qquad \eta = \frac{pR_1C_2}{D_0}, \qquad |\eta| \le R_1/R_2.$$

Indeed,

$$|\eta| = \frac{\omega R_1C_2}{\sqrt{1 + \omega^2R_1^2C_1^2}\sqrt{1 + \omega^2R_2^2C_2^2}} \le \frac{R_1}{R_2} \quad (\omega \ge 0),$$

since the first square root is at least 1 and the second at least ωR_2C_2 (the case $\omega = 0$ has $\eta = 0$). For $\delta = R_1/R_2 < 1$, the relative discrepancy, wherever the product is nonzero, therefore satisfies

$$\left|\frac{\varphi}{\varphi_1\varphi_2} - 1\right| = \frac{|\eta|}{|1 + \eta|} \le \frac{|\eta|}{1 - |\eta|} \le \frac{\delta}{1 - \delta}.$$

The quoted limit is a sufficient condition for a uniformly accurate product approximation, not a necessary condition at each frequency. At low frequency the loading may also be small; an ideal buffer between stages makes the product exact. For finite positive elements in the unbuffered circuit, the product is not an identity in frequency.

20.6 Discrete Fourier Transform

20.6.1 Derive the trigonometric forms of discrete orthogonality corresponding to Eq. (20.120):

$$\sum_{k=0}^{N-1} \cos(2\pi pk/N)\sin(2\pi qk/N) = 0$$

$$\sum_{k=0}^{N-1} \cos(2\pi pk/N)\cos(2\pi qk/N) = \begin{cases} N, & p = q = (0 \text{ or } N/2); \\ N/2, & (p+q=N) \text{ or } p=q \text{ but not both}; \\ 0, & \text{otherwise.} \end{cases}$$

$$\sum_{k=0}^{N-1} \sin(2\pi pk/N)\sin(2\pi qk/N) = \begin{cases} N/2, & p = q \text{ and } p+q \neq (0 \text{ or } N); \\ -N/2, & p \neq q \text{ and } p+q=N; \\ 0, & \text{otherwise.} \end{cases}$$

Note. If N is odd, p and q will never have the value $N/2$.
Hint. Consider the use of trigonometric identities such as

$$\sin A \cos B = \frac{1}{2}\big[\sin(A+B) + \sin(A-B)\big].$$

Take integer indices $0 \le p, q < N$ and a positive integer N in these piecewise formulas.

Solution.

For integers m define the discrete selector

$$\Delta_N(m) = \begin{cases} 1, & m \equiv 0 \pmod N, \\ 0, & \text{otherwise.} \end{cases}$$

The geometric series, including its $e^{2\pi im/N} = 1$ case, gives

$$\sum_{k=0}^{N-1} e^{2\pi imk/N} = N\Delta_N(m).$$

Its imaginary part is zero and its real part is $N\Delta_N(m)$. Applying the product identities gives, with $\theta_k = 2\pi k/N$,

$$\sum_k \cos(p\theta_k)\sin(q\theta_k) = \frac{1}{2}\sum_k \{\sin[(q+p)\theta_k] + \sin[(q-p)\theta_k]\} = 0,$$

$$\sum_k \cos(p\theta_k)\cos(q\theta_k) = \frac{N}{2}\{\Delta_N(p-q) + \Delta_N(p+q)\},$$

$$\sum_k \sin(p\theta_k)\sin(q\theta_k) = \frac{N}{2}\{\Delta_N(p-q) - \Delta_N(p+q)\}.$$

These formulas hold for all integer p, q and positive integer N. The displayed piecewise formulas in the question use the representative range $0 \le p, q < N$. Without that range, congruences modulo N must replace the equalities. In this range $\Delta_N(p-q) = 1$ exactly when $p = q$. Also $\Delta_N(p+q) = 1$ when $p + q = 0$ or N. Both selectors equal 1 for $p = q = 0$, and also for $p = q = N/2$ when N is

even. The cosine sum is then N, whereas the sine sum is zero. When exactly one selector is 1, the cosine sum is $N/2$; the sine sum is $N/2$ for $p = q$ and $-N/2$ for $p+q = N$ with $p \neq q$. When neither selector is 1, both sums vanish. These cases prove the corrected formulas, including $N = 1$ and $N = 2$. The final product-to-sum line in the manual's proof on p. 344 has a cosine–cosine product on its left. It must be a sine–sine product to agree with the difference of cosines on its right. The selector derivation above gives that identity directly.

20.6.2 Show in detail how to go from

$$F_p = \frac{1}{N^{1/2}} \sum_{k=0}^{N-1} f_k e^{2\pi i p k/N} \qquad \text{to} \qquad f_k = \frac{1}{N^{1/2}} \sum_{p=0}^{N-1} F_p e^{-2\pi i p k/N}.$$

Solution.

Insert the defining expression for F_p into the proposed inverse, using a different dummy index j:

$$\begin{aligned} \frac{1}{\sqrt{N}} \sum_{p=0}^{N-1} F_p e^{-2\pi i p k/N} &= \frac{1}{N} \sum_{p=0}^{N-1} \sum_{j=0}^{N-1} f_j e^{2\pi i p j/N} e^{-2\pi i p k/N} \\ &= \frac{1}{N} \sum_{j=0}^{N-1} f_j \sum_{p=0}^{N-1} e^{2\pi i p (j-k)/N}. \end{aligned}$$

The inner sum is $N\Delta_N(j-k)$ by the finite geometric series. For $0 \le j, k < N$, the only surviving index is $j = k$. Thus the expression is f_k, which proves

$$\boxed{f_k = \frac{1}{\sqrt{N}} \sum_{p=0}^{N-1} F_p e^{-2\pi i p k/N}}.$$

All sums are finite, so their interchange requires no convergence assumption. The factors $1/N$ in both exponents are essential: without them every exponential is 1 for integer indices, and the transform cannot distinguish the input entries.

20.6.3 The N-membered point sets f_k and F_p are discrete Fourier transforms of each other. Derive the following symmetry relations:

(a) If f_k is real, F_p is Hermitian symmetric; that is, $F_p = F^*_{N-p}$.

(b) If f_k is pure imaginary, then $F_p = -F^*_{N-p}$.

Note. The symmetry of part (a) is an illustration of aliasing. If F_p describes an amplitude at a frequency proportional to p, we necessarily predict an equal amplitude at the frequency proportional to $N - p$.

Indices are understood modulo N, in particular $F_N = F_0$. Equal paired amplitudes in the note mean equal magnitudes, with conjugate phases.

Solution.

Regard the frequency index modulo N, so $F_N = F_0$. Complex conjugation gives

$$\begin{aligned} F^*_{N-p} &= \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} f^*_k e^{-2\pi i (N-p) k/N} \\ &= \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} f^*_k e^{-2\pi i k} e^{2\pi i p k/N} = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} f^*_k e^{2\pi i p k/N}. \end{aligned}$$

If f_k is real, $f_k^* = f_k$, so $F_{N-p}^* = F_p$. If f_k is purely imaginary, $f_k^* = -f_k$, so $F_{N-p}^* = -F_p$. These are precisely (a) and (b). For a real input, the zero-frequency coefficient and, for even N, the coefficient at $p = N/2$ are real because they are their own conjugates. For a purely imaginary input those same coefficients are purely imaginary. In general the paired coefficients have equal magnitudes; their phases are related by conjugation (and by an extra minus sign in part (b)). The index $N - p$ represents the same sampled frequency as $-p$.

20.7 Laplace Transforms

20.7.1 Prove that

$$\lim_{s\to\infty} sf(s) = \lim_{t\to+0} F(t).$$

Hint. Assume that $F(t)$ can be expressed as $F(t) = \sum_{n=0}^{\infty} a_n t^n$.

Let $s \to +\infty$ along the real axis. Assume a finite $F(0+)$ and an exponential bound $|F(t)| \le Me^{ct}$; these hypotheses suffice without an unjustified termwise integration.

Solution.

Take $s \to +\infty$ along the positive real axis. A sufficient hypothesis is that F has the finite right limit $F(0+)$, is locally integrable, and has an exponential bound $|F(t)| \le Me^{ct}$ for $t \ge 0$. In the defining integral put $u = st$:

$$sf(s) = s\int_0^\infty e^{-st}F(t)dt = \int_0^\infty e^{-u}F(u/s)du.$$

For each fixed $u > 0$, $F(u/s) \to F(0+)$. For sufficiently large s, $c/s \le 1/2$, and $|e^{-u}F(u/s)| \le Me^{-u/2}$, which is integrable. Dominated convergence therefore gives

$$\lim_{s\to+\infty} sf(s) = F(0+)\int_0^\infty e^{-u}du = \boxed{F(0+)}.$$

The hint also describes the coefficients: if a Taylor expansion can be integrated termwise, then

$$f(s) = \sum_{n\ge0} \frac{a_n\Gamma(n+1)}{s^{n+1}}, \qquad sf(s) = a_0 + \frac{a_1}{s} + \frac{2a_2}{s^2} + \cdots.$$

Even when this infinite series is only asymptotic, the direct limit proof above remains valid and identifies $a_0 = F(0+)$.

20.7.2 Prove the following precise replacement for the withdrawn exercise:

$$\frac{1}{\pi}\lim_{s\downarrow0} L_t\{\cos xt\}(s) = \delta(x) \quad \text{as distributions in } x.$$

The original formula left the transform variable, approach to zero, and meaning of convergence unspecified; the published errata therefore say to disregard it.

Solution.

The published errata withdraw the statement as ill-defined. It has a useful, precise replacement: let s approach zero through positive real values, take the Laplace transform in t, and interpret convergence as convergence of distributions in x. For $s > 0$, direct integration of $e^{-(s-ix)t}$ gives

$$L_t\{\cos xt\}(s) = \Re\frac{1}{s-ix} = \frac{s}{s^2+x^2}.$$

The resulting functions do not converge to an ordinary function representing a delta: they approach zero at every $x \neq 0$ and diverge at $x = 0$. To prove the correct weak limit, take a smooth test function ϕ with compact support and substitute $x = su$:

$$\int_{-\infty}^{\infty} \frac{s\phi(x)}{\pi(s^2 + x^2)} dx = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\phi(su)}{1 + u^2} du.$$

The integrand is bounded by $\|\phi\|_\infty/(1 + u^2)$, and $\phi(su) \to \phi(0)$. Consequently the limit is

$$\frac{\phi(0)}{\pi} [\tan^{-1} u]_{-\infty}^{\infty} = \phi(0).$$

Thus the repaired assertion is

$$\boxed{\lim_{s \downarrow 0} \frac{1}{\pi} L_t\{\cos xt\}(s) = \delta(x) \quad \text{in the distributional sense.}}$$

The one-sided real approach is part of the assertion; an unspecified complex limit is not the prescription proved here.

20.7.3 Verify that

$$L\left\{\frac{\cos at - \cos bt}{b^2 - a^2}\right\} = \frac{s}{(s^2 + a^2)(s^2 + b^2)}, \qquad a^2 \neq b^2.$$

Solution.

For real a, b and $\Re s > 0$,

$$L\{\cos at\} = \frac{1}{2}\left(\frac{1}{s - ia} + \frac{1}{s + ia}\right) = \frac{s}{s^2 + a^2}.$$

Linearity and a common denominator now yield

$$\begin{aligned} L\left\{\frac{\cos at - \cos bt}{b^2 - a^2}\right\} &= \frac{s}{b^2 - a^2}\left(\frac{1}{s^2 + a^2} - \frac{1}{s^2 + b^2}\right) \\ &= \frac{s}{b^2 - a^2} \frac{(s^2 + b^2) - (s^2 + a^2)}{(s^2 + a^2)(s^2 + b^2)} \\ &= \boxed{\frac{s}{(s^2 + a^2)(s^2 + b^2)}}. \end{aligned}$$

The stipulated $a^2 \neq b^2$ permits the cancellation. For complex frequencies the same algebra applies in a common half-plane of convergence.

20.7.4 Using partial fraction expansions, show that

(a) $L^{-1}\left\{\dfrac{1}{(s + a)(s + b)}\right\} = \dfrac{e^{-at} - e^{-bt}}{b - a}, \quad a \neq b.$

(b) $L^{-1}\left\{\dfrac{s}{(s + a)(s + b)}\right\} = \dfrac{ae^{-at} - be^{-bt}}{a - b}, \quad a \neq b.$

Solution.

(a) Set $1/[(s+a)(s+b)] = A/(s+a) + B/(s+b)$. After multiplication by the denominator, $1 = A(s+b) + B(s+a)$. Setting $s = -a$ and $s = -b$ gives $A = 1/(b-a)$ and $B = 1/(a-b)$. Since $L\{e^{-at}\} = 1/(s+a)$,

$$\boxed{L^{-1}\left\{\frac{1}{(s+a)(s+b)}\right\} = \frac{e^{-at} - e^{-bt}}{b-a}}.$$

(b) Now $s = A(s+b) + B(s+a)$, so the same substitutions give $A = a/(a-b)$, $B = -b/(a-b)$. Hence

$$\boxed{L^{-1}\left\{\frac{s}{(s+a)(s+b)}\right\} = \frac{ae^{-at} - be^{-bt}}{a-b}}.$$

The defining transforms converge for $\Re s > \max(-\Re a, -\Re b)$. Although excluded in the question, the equal-parameter limits are regular:

$$b \to a: \qquad te^{-at}, \qquad (1-at)e^{-at},$$

respectively, obtained by differentiating the numerators with respect to b and the denominators with respect to b.

20.7.5 Using partial fraction expansions, show that for $a^2 \neq b^2$,

(a) $L^{-1}\left\{\dfrac{1}{(s^2+a^2)(s^2+b^2)}\right\} = -\dfrac{1}{a^2-b^2}\left(\dfrac{\sin at}{a} - \dfrac{\sin bt}{b}\right)$.

(b) $L^{-1}\left\{\dfrac{s^2}{(s^2+a^2)(s^2+b^2)}\right\} = \dfrac{1}{a^2-b^2}\{a\sin at - b\sin bt\}$.

For real a, b, use $\sin(at)/a \to t$ if $a = 0$, and the corresponding limit if $b = 0$.

Solution.

For real parameters use $L\{\sin at/a\} = 1/(s^2+a^2)$ in $\Re s > 0$. (a) Equating coefficients in

$$1 = A(s^2+b^2) + B(s^2+a^2)$$

gives $A + B = 0$, $Ab^2 + Ba^2 = 1$. Thus $A = 1/(b^2-a^2)$ and $B = -A$, and inversion gives

$$\boxed{\frac{1}{b^2-a^2}\left(\frac{\sin at}{a} - \frac{\sin bt}{b}\right)}.$$

(b) Equating coefficients in $s^2 = A(s^2+b^2) + B(s^2+a^2)$ gives $A + B = 1$, $Ab^2 + Ba^2 = 0$. Solving, $A = a^2/(a^2-b^2)$, $B = -b^2/(a^2-b^2)$. Therefore

$$\boxed{L^{-1}\left\{\frac{s^2}{(s^2+a^2)(s^2+b^2)}\right\} = \frac{a\sin at - b\sin bt}{a^2-b^2}}.$$

If either parameter is zero, the quotient $\sin(at)/a$ in part (a) means its continuous limit t. The condition $a^2 \neq b^2$ alone does not exclude that case.

20.7.6 Show that

(a) $$\int_0^\infty \frac{\cos s}{s^\nu}\,ds = \frac{\pi}{2\Gamma(\nu)\cos(\nu\pi/2)}, \qquad 0<\nu<1.$$

(b) $$\int_0^\infty \frac{\sin s}{s^\nu}\,ds = \frac{\pi}{2\Gamma(\nu)\sin(\nu\pi/2)}, \qquad 0<\nu<2.$$

Why is ν restricted to $(0,1)$ for (a), to $(0,2)$ for (b)? These integrals may be interpreted as Fourier transforms of $s^{-\nu}$ and as Mellin transforms of $\sin s$ and $\cos s$.

Hint. Replace $s^{-\nu}$ by a Laplace transform integral: $L\{t^{\nu-1}/\Gamma(\nu)\}$. Then integrate with respect to s. The resulting integral can be treated as a beta function (Section 13.3).

Solution.

To avoid confusing the transform parameter with the integration variable, write the latter as x. For $\nu>0$, substitution $v=ux$ into the gamma integral gives

$$x^{-\nu} = \frac{1}{\Gamma(\nu)}\int_0^\infty u^{\nu-1}e^{-ux}du.$$

Insert a convergence factor $e^{-\epsilon x}$, $\epsilon>0$, before changing the order of integration. In the ranges stated in the question the resulting double integral is absolutely integrable: after integrating its absolute value in u, it is $\int_0^\infty e^{-\epsilon x}x^{-\nu}|\cos x|dx$ or its sine counterpart. Thus

$$C_\epsilon = \frac{1}{\Gamma(\nu)}\int_0^\infty \frac{u^{\nu-1}(u+\epsilon)}{(u+\epsilon)^2+1}du,$$
$$S_\epsilon = \frac{1}{\Gamma(\nu)}\int_0^\infty \frac{u^{\nu-1}}{(u+\epsilon)^2+1}du.$$

For $0<\epsilon\le 1$, on $0<u<1$ both integrands are bounded by a constant times $u^{\nu-1}$; for $u\ge 1$ the bounds are $2u^{\nu-2}$ and $u^{\nu-3}$, respectively. Dominated convergence gives

$$C_0 = \frac{1}{\Gamma(\nu)}\int_0^\infty \frac{u^\nu}{1+u^2}du, \qquad S_0 = \frac{1}{\Gamma(\nu)}\int_0^\infty \frac{u^{\nu-1}}{1+u^2}du.$$

For $0<q<2$, put $v=u^2$ and use the beta integral:

$$\int_0^\infty \frac{u^{q-1}}{1+u^2}du = \frac{1}{2}\int_0^\infty \frac{v^{q/2-1}}{1+v}dv = \frac{1}{2}\Gamma(q/2)\Gamma(1-q/2) = \frac{\pi}{2\sin(\pi q/2)}.$$

Take $q=\nu+1$ for the cosine and $q=\nu$ for the sine:

$$\boxed{C_0 = \frac{\pi}{2\Gamma(\nu)\cos(\pi\nu/2)}}, \quad 0<\nu<1; \qquad \boxed{S_0 = \frac{\pi}{2\Gamma(\nu)\sin(\pi\nu/2)}}, \quad 0<\nu<2.$$

At the origin the cosine integrand behaves as $x^{-\nu}$, requiring $\nu<1$, whereas the sine integrand behaves as $x^{1-\nu}$, requiring $\nu<2$. At infinity Dirichlet's test applies for $\nu>0$; for $\nu\le 0$ the ordinary oscillatory improper integral fails to converge. Abel's limiting theorem therefore identifies the regulated limits above with the ordinary improper integrals in precisely the stated ranges. The gamma notation in the corrected statement accommodates noninteger ν.

20.7.7 A function $F(t)$ can be expanded in a Maclaurin series,

$$F(t) = \sum_{n=0}^{\infty} a_n t^n.$$

If this series represents F on the half-line and termwise Laplace integration is justified, then

$$L\{F(t)\} = \int_0^\infty e^{-st} \sum_{n=0}^{\infty} a_n t^n \, dt = \sum_{n=0}^{\infty} a_n \int_0^\infty e^{-st} t^n \, dt.$$

Show that $f(s)$, the Laplace transform of $F(t)$, contains no powers of s greater than s^{-1}. Check your result by calculating $L\{\delta(t)\}$, and comment on this fiasco.

For the impulse comparison use the causal convention in which a delta at $t = 0$ has full unit mass.

Solution.

Whenever termwise integration is justified, $u = st$ gives

$$\int_0^\infty e^{-st} t^n dt = \frac{1}{s^{n+1}} \int_0^\infty e^{-u} u^n du = \frac{n!}{s^{n+1}}, \qquad s > 0.$$

Thus

$$\boxed{f(s) = \sum_{n=0}^{\infty} \frac{a_n n!}{s^{n+1}}},$$

which contains only $s^{-1}, s^{-2}, \ldots$. Existence of a local Maclaurin series does not by itself justify the displayed interchange over the entire half-line. For example, $F(t) = 1/(1+t)$ has a Maclaurin series near zero and a Laplace transform for $s > 0$, but the resulting formal series $\sum_{n\geq 0}(-1)^n n!/s^{n+1}$ diverges for every finite s. It is an asymptotic expansion at infinity, not a convergent identity. With the causal Laplace convention in which a delta impulse at 0 has unit mass, the distributional pairing gives

$$L\{\delta(t)\}(s) = \langle \delta, e^{-st} \rangle = 1.$$

This constant is no contradiction: a delta distribution is not an ordinary function with a Maclaurin series. Its transform lies outside the hypotheses of the argument. The same distinction permits $L\{\delta^{(n)}\} = s^n$ for causal distributions. The convention at the lower endpoint must be specified: integrating a symmetric regularization over $[0, \infty)$ assigns half the mass, whereas the causal 0^- convention used here assigns the full mass.

20.7.8 Show that the Laplace transform of the confluent hypergeometric function $M(a, c, x)$ is

$$L\big\{M(a, c, x)\big\} = \frac{1}{s} \, {}_2F_1\left(a, 1; c; \frac{1}{s}\right).$$

For the ordinary series definitions take $c \notin \{0, -1, -2, \ldots\}$ and $\Re s > 1$; exceptional terminating cases may admit larger domains.

Solution.

Use the defining confluent hypergeometric series

$$M(a, c, t) = \sum_{n=0}^{\infty} \frac{(a)_n}{(c)_n} \frac{t^n}{n!}.$$

For $\Re s > 1$ the series may be integrated absolutely: the sum of the integrals of the absolute terms is

$$\frac{1}{\Re s}\sum_{n=0}^{\infty}\left|\frac{(a)_n}{(c)_n}\right|(\Re s)^{-n},$$

which converges by the ratio test (or terminates). Hence

$$\begin{aligned} L\{M(a,c,t)\}(s) &= \sum_{n=0}^{\infty}\frac{(a)_n}{(c)_n n!}\frac{n!}{s^{n+1}} \\ &= \frac{1}{s}\sum_{n=0}^{\infty}\frac{(a)_n(1)_n}{(c)_n n!}\left(\frac{1}{s}\right)^n = \boxed{\frac{1}{s}\,{}_2F_1\left(a,1;c;\frac{1}{s}\right)}, \end{aligned}$$

since $(1)_n = n!$. The ordinary definitions require $c \notin \{0,-1,-2,\ldots\}$, unless a specified terminating or limiting case removes the singularity. The half-plane $\Re s > 1$ is sufficient for the proof; continuation beyond it must stay within the transform's convergence domain and use consistent analytic branches. Terminating cases can have a larger domain.

20.8 Properties of Laplace Transforms

20.8.1 Use the expression for the transform of a second derivative to obtain the transform of $\cos kt$.

Solution.

Put $F(t) = \cos kt$ and $f(s) = L\{F\}(s)$. The differential identity $F'' = -k^2F$ and initial values $F(0) = 1$, $F'(0) = 0$ give

$$L\{F''\} = s^2 f(s) - sF(0) - F'(0) = s^2 f(s) - s = -k^2 f(s).$$

Solving this algebraic equation,

$$\boxed{L\{\cos kt\}(s) = \frac{s}{s^2+k^2}}, \qquad \Re s > 0 \quad (k \in \mathbb{R}).$$

For $k = 0$ this reduces to $L\{1\} = 1/s$, as required.

20.8.2 A mass m is attached to one end of an unstretched spring, spring constant k (Fig. 20.17). Starting at time $t = 0$, the free end of the spring experiences a constant acceleration a, away from the mass. Using Laplace transforms,

(a) find the position x of m as a function of time.

(b) determine the limiting form of $x(t)$ for small t.

ANS. (a) $x = \frac{1}{2}at^2 - \frac{a}{\omega^2}(1-\cos\omega t)$, $\omega^2 = \frac{k}{m}$,
(b) $x = \frac{a\omega^2}{4!}t^4$, $\omega t \ll 1$.

Use the initially resting preparation $x(0) = x'(0) = 0$, with the driven end initially at rest and $m, k > 0$.

Solution.

Measure x from the initial mass position and subtract the unstretched spring length from the driven endpoint's position. The latter coordinate is $y(t) = at^2/2$, so the spring extension is $y - x$. Newton's law is

$$mx'' = k(y - x), \qquad x'' + \omega^2 x = \frac{a\omega^2 t^2}{2}, \qquad \omega^2 = k/m.$$

The mass is initially at rest at $x = 0$. With $\widehat{x} = L\{x\}$, the transformed equation is

$$(s^2 + \omega^2)\widehat{x} = \frac{a\omega^2}{s^3}, \qquad \widehat{x} = \frac{a\omega^2}{s^3(s^2 + \omega^2)} = \frac{a}{s^3} - \frac{a}{s(s^2 + \omega^2)}.$$

Since $1/[s(s^2 + \omega^2)] = \omega^{-2}[1/s - s/(s^2 + \omega^2)]$, inversion yields

$$\boxed{x(t) = \frac{1}{2}at^2 - \frac{a}{\omega^2}(1 - \cos\omega t)}.$$

Expanding the cosine shows explicitly the cancellation of the quadratic terms:

$$x(t) = \frac{1}{2}at^2 - \frac{a}{\omega^2}\left(\frac{\omega^2 t^2}{2} - \frac{\omega^4 t^4}{24} + \frac{\omega^6 t^6}{720} - \cdots\right) = \boxed{\frac{a\omega^2 t^4}{24} + O(t^6)}.$$

The small-time condition is $|\omega t| \ll 1$. Differentiating the exact answer gives $x'' = a(1 - \cos\omega t)$, which agrees with $\omega^2(y - x)$ and verifies the equation of motion.

20.8.3 Radioactive nuclei decay according to the law

$$\frac{dN}{dt} = -\lambda N,$$

with N the concentration of a given nuclide and λ its particular decay constant. This equation may be interpreted as stating that the rate of decay is proportional to the number of these radioactive nuclei present. They all decay independently.

Consider now a radioactive series of n different nuclides, with Nuclide 1 decaying into Nuclide 2, Nuclide 2 into Nuclide 3, etc., until reaching Nuclide n, which is stable. The concentrations of the various nuclides satisfy the system of ODEs

$$\frac{dN_1}{dt} = -\lambda_1 N_1, \qquad \frac{dN_2}{dt} = \lambda_1 N_1 - \lambda_2 N_2, \qquad \cdots, \qquad \frac{dN_n}{dt} = \lambda_{n-1} N_{n-1}.$$

(a) For the case $n = 3$ find $N_1(t)$, $N_2(t)$, and $N_3(t)$, with $N_1(0) = N_0$ and $N_2(0) = N_3(0) = 0$.

(b) Find an approximate expression for N_2 and N_3, valid for small t when $\lambda_1 \approx \lambda_2$.

(c) Find approximate expressions for N_2 and N_3, valid for large t, when (1) $\lambda_1 \gg \lambda_2$, (2) $\lambda_1 \ll \lambda_2$.

ANS. (a) $N_1(t) = N_0 e^{-\lambda_1 t}$, $N_2(t) = N_0 \dfrac{\lambda_1}{\lambda_2 - \lambda_1}\left(e^{-\lambda_1 t} - e^{-\lambda_2 t}\right)$,

$$N_3(t) = N_0\left(1 - \frac{\lambda_2}{\lambda_2 - \lambda_1}e^{-\lambda_1 t} + \frac{\lambda_1}{\lambda_2 - \lambda_1}e^{-\lambda_2 t}\right).$$

(b) $N_2 \approx N_0 \lambda_1 t$, $N_3 \approx \dfrac{N_0}{2}\lambda_1 \lambda_2 t^2$.

(c) (1) $N_2 \approx N_0 e^{-\lambda_2 t}$

$N_3 \approx N_0\left(1 - e^{-\lambda_2 t}\right)$, $\lambda_1 t \gg 1$.

(2) $N_2 \approx N_0(\lambda_1/\lambda_2)e^{-\lambda_1 t}$,

$N_3 \approx N_0\left(1 - e^{-\lambda_1 t}\right)$, $\lambda_2 t \gg 1$.

The quotient answers in part (a) are written for distinct rates; at equal rates take the continuous limits derived below.

Solution.

(a) Denote Laplace transforms by hats. The initial values and the three rate equations give

$$(s+\lambda_1)\widehat{N}_1 = N_0, \quad (s+\lambda_2)\widehat{N}_2 = \lambda_1\widehat{N}_1, \quad s\widehat{N}_3 = \lambda_2\widehat{N}_2.$$

The first two inversions are

$$\begin{aligned} N_1(t) &= N_0 e^{-\lambda_1 t}, \\ \widehat{N}_2 &= \frac{N_0\lambda_1}{\lambda_2-\lambda_1}\left(\frac{1}{s+\lambda_1} - \frac{1}{s+\lambda_2}\right), \\ N_2(t) &= \frac{N_0\lambda_1}{\lambda_2-\lambda_1}(e^{-\lambda_1 t} - e^{-\lambda_2 t}). \end{aligned}$$

Adding the three original equations gives $(N_1+N_2+N_3)' = 0$, so their sum remains N_0. Subtraction therefore gives

$$\boxed{N_3(t) = N_0\left[1 - \frac{\lambda_2 e^{-\lambda_1 t} - \lambda_1 e^{-\lambda_2 t}}{\lambda_2-\lambda_1}\right]}.$$

The quotient formulas presume $\lambda_1 \neq \lambda_2$. At the removable equal-rate case $\lambda_1 = \lambda_2 = \lambda$, $N_2 = N_0\lambda t e^{-\lambda t}$ and $N_3 = N_0[1-(1+\lambda t)e^{-\lambda t}]$. They follow either from $\widehat{N}_2 = N_0\lambda/(s+\lambda)^2$ or from the limit of the exponential difference.

(b) Expand each exponential and cancel its constant term:

$$\begin{aligned} N_2 &= N_0\lambda_1\left[t - \frac{\lambda_1+\lambda_2}{2}t^2 + O(t^3)\right], \\ N_3 &= N_0\lambda_1\lambda_2\left[\frac{t^2}{2} - \frac{\lambda_1+\lambda_2}{6}t^3 + O(t^4)\right]. \end{aligned}$$

The leading terms are those stated. They hold when $\max(\lambda_1,\lambda_2)t \ll 1$, including equal or nearly equal rates; closeness of the rates is not needed for this small-time expansion.

(c1) If $\lambda_1 \gg \lambda_2$ and $\lambda_1 t \gg 1$, discard $e^{-\lambda_1 t}$ in the exact formulas:

$$N_2 \simeq N_0\frac{\lambda_1}{\lambda_1-\lambda_2}e^{-\lambda_2 t}, \quad N_3 \simeq N_0\left[1 - \frac{\lambda_1}{\lambda_1-\lambda_2}e^{-\lambda_2 t}\right].$$

Since $\lambda_1/(\lambda_1-\lambda_2) = 1+O(\lambda_2/\lambda_1)$, these become $N_0 e^{-\lambda_2 t}$ and $N_0(1-e^{-\lambda_2 t})$. The positive exponent printed for N_2 in this case must be negative. A positive exponent would violate both the rate equation and $0 \le N_2 \le N_0$.

(c2) If $\lambda_1 \ll \lambda_2$ and $\lambda_2 t \gg 1$, discard $e^{-\lambda_2 t}$ instead:

$$N_2 \simeq N_0\frac{\lambda_1}{\lambda_2-\lambda_1}e^{-\lambda_1 t} \simeq N_0\frac{\lambda_1}{\lambda_2}e^{-\lambda_1 t},$$

$$N_3 \simeq N_0\left[1 - \frac{\lambda_2}{\lambda_2-\lambda_1}e^{-\lambda_1 t}\right] \simeq N_0(1-e^{-\lambda_1 t}).$$

The last approximations retain the leading terms in the small rate ratio; they need not resolve a small transient correction to N_3 immediately after the fast decay.

20.8.4 The rate of formation of an isotope in a nuclear reactor is given by

$$\frac{dN_2}{dt} = \varphi\big[\sigma_1 N_1(0) - \sigma_2 N_2(t)\big] - \lambda_2 N_2(t).$$

Here $N_1(0)$ is the concentration of the original isotope (assumed constant), and N_2 is that of the newly formed isotope. The first two terms on the right-hand side describe the production and destruction of the new isotope via neutron absorption; φ is the neutron flux (units $\mathrm{cm^{-2}\,s^{-1}}$); σ_1 and σ_2 (units $\mathrm{cm^2}$) are neutron absorption cross sections. The final term describes the radioactive decay of the new isotope, with decay constant λ_2.

(a) Find the concentration N_2 of the new isotope as a function of time.

(b) For original isotope ^{153}Eu, $\sigma_1 = 400$ barns $= 400\times10^{-24}$ $\mathrm{cm^2}$, $\sigma_2 = 1000$ barns $= 1000\times10^{-24}$ $\mathrm{cm^2}$, and $\lambda_2 = 1.4\times10^{-9}$ $\mathrm{s^{-1}}$. If $N_1(0) = 10^{20}$ and $\varphi = 10^9$ $\mathrm{cm^{-2}\,s^{-1}}$, find N_2, the concentration of ^{154}Eu, after 1 year of continuous irradiation. Is the assumption that N_1 is constant justified?

In (a), retain $N_2(0) = N_{20}$ as an initial datum; for the numerical irradiation in (b), take $N_{20} = 0$.

Solution.

(a) Set $A = \varphi\sigma_1 N_1(0)$ and $b = \varphi\sigma_2 + \lambda_2$. The equation is $N_2' + bN_2 = A$. With $N_2(0) = N_{20}$ its transform obeys

$$(s+b)\widehat{N}_2 = N_{20} + A/s, \quad \widehat{N}_2 = \frac{N_{20}}{s+b} + \frac{A}{b}\left(\frac{1}{s} - \frac{1}{s+b}\right).$$

Therefore

$$\boxed{N_2(t) = N_{20}e^{-bt} + \frac{A}{b}(1 - e^{-bt})}.$$

An initial concentration is needed. For irradiation of the original isotope with none of the new isotope initially present, use $N_{20} = 0$; the formula above also covers a nonzero initial inventory. If $b = 0$, the continuous limit is $N_{20} + At$.

(b) For the stated data,

$$A = (10^9)(400\times10^{-24})(10^{20}) = 4.00\times10^7\ \mathrm{s^{-1}},$$
$$b = (10^9)(1000\times10^{-24}) + 1.4\times10^{-9} = 1.401\times10^{-9}\ \mathrm{s^{-1}}.$$

Taking one year to mean 365 days gives $T = 31\,536\,000$ s. For $N_{20} = 0$,

$$N_2(T) = \frac{4.00\times10^7}{1.401\times10^{-9}}\left(1 - e^{-(1.401\times10^{-9})(31\,536\,000)}\right) = \boxed{1.2340\times10^{15}}.$$

This is 1.2×10^{15} at the precision of the source data. The parent depletion rate is $\varphi\sigma_1 = 4.00\times10^{-13}$ $\mathrm{s^{-1}}$, so the fraction consumed is

$$1 - e^{-\varphi\sigma_1 T} = 1.26143\times10^{-5} = 0.00126143\%.$$

Thus treating N_1 as constant is accurate to about thirteen parts per million over this interval. The calculation uses the same number or concentration units as $N_1(0)$.

20.8.5 In a nuclear reactor ^{135}Xe is formed as both a direct fission product of ^{235}U and by decay of ^{135}I (another fission product), half-life 6.7 hours. The half-life of ^{135}Xe is 9.2 hours. Because ^{135}Xe strongly absorbs thermal neutrons, thereby "poisoning" the nuclear reactor, its concentration is a matter of great interest. The relevant equations are

$$\frac{dN_{\mathrm{I}}}{dt} = \varphi\gamma_{\mathrm{I}}(\sigma_f N_{\mathrm{U}}) - \lambda_{\mathrm{I}} N_{\mathrm{I}},$$
$$\frac{dN_{\mathrm{Xe}}}{dt} = \varphi\big[\gamma_{\mathrm{Xe}}(\sigma_f N_{\mathrm{U}}) - \sigma_{\mathrm{Xe}} N_{\mathrm{Xe}}\big] + \lambda_{\mathrm{I}} N_{\mathrm{I}} - \lambda_{\mathrm{Xe}} N_{\mathrm{Xe}}.$$

Here N_{I}, N_{Xe}, N_{U} are the concentrations of ^{135}I, ^{135}Xe, ^{235}U, with N_{U} assumed to be constant. The neutron flux φ in the reactor causes fission of ^{235}U with cross section σ_f and removes ^{135}Xe by neutron absorption with cross section $\sigma_{\mathrm{Xe}} = 3.5 \times 10^6$ barns $= 3.5 \times 10^{-18}$ cm^2. Neutron absorption by ^{135}I is negligible. The yield of ^{135}I and ^{135}Xe per fission are, respectively, $\gamma_{\mathrm{I}} = 0.060$ and $\gamma_{\mathrm{Xe}} = 0.003$.

(a) Find $N_{\mathrm{Xe}}(t)$ in terms of neutron flux φ and the product $\sigma_f N_{\mathrm{U}}$.

(b) Find $N_{\mathrm{Xe}}(t \to \infty)$.

(c) After N_{Xe} has reached equilibrium, the reactor is shut down: $\varphi = 0$. Find $N_{\mathrm{Xe}}(t)$ following shutdown. Determine when N_{Xe} increases immediately after shutdown; this depends on the pre-shutdown flux. Such an increase may for a few hours interfere with starting the reactor up again.

Hint. The half-life $t_{1/2}$ of a radioactive isotope is the time required for decay of half of the nuclides in a sample. For a decay rate $dN/dt = -\lambda N$, the half-life has the value $t_{1/2} = \ln 2/\lambda$, so λ can be computed as $\lambda = \ln 2/t_{1/2} = 0.693/t_{1/2}$.

In (a), retain initial inventories $N_{\mathrm{I}}(0) = I_0$ and $N_{\mathrm{Xe}}(0) = X_0$, then specialize to $I_0 = X_0 = 0$.

Solution.

Keep the neutron flux and the uranium concentration constant during operation, and abbreviate

$$P = \varphi\sigma_f N_{\mathrm{U}}, \quad \alpha = \lambda_{\mathrm{I}}, \quad \beta = \lambda_{\mathrm{Xe}}, \quad \mu = \beta + \varphi\sigma_{\mathrm{Xe}}, \quad I = N_{\mathrm{I}}, \quad X = N_{\mathrm{Xe}}.$$

The system becomes $I' + \alpha I = P\gamma_{\mathrm{I}}$ and $X' + \mu X = P\gamma_{\mathrm{Xe}} + \alpha I$. The original part (a) did not specify initial inventories. We first keep $I(0) = I_0$, $X(0) = X_0$ arbitrary, and then give the usual initially clean case $I_0 = X_0 = 0$. The equilibrium values, obtained by setting both derivatives to zero, are

$$I_e = \frac{P\gamma_{\mathrm{I}}}{\alpha}, \qquad X_e = \frac{P(\gamma_{\mathrm{I}} + \gamma_{\mathrm{Xe}})}{\mu}.$$

(a) The first transformed equation gives

$$\widehat{I} = \frac{I_e}{s} + \frac{I_0 - I_e}{s + \alpha}, \qquad I(t) = I_e + (I_0 - I_e)e^{-\alpha t}.$$

Subtract X_e from the second equation. Then

$$(s + \mu)L\{X - X_e\} = X_0 - X_e + \frac{\alpha(I_0 - I_e)}{s + \alpha}.$$

Use $1/[(s+\alpha)(s+\mu)] = [1/(s+\alpha) - 1/(s+\mu)]/(\mu - \alpha)$ to obtain

$$\boxed{X(t) = X_e + (X_0 - X_e)e^{-\mu t} + \frac{\alpha(I_0 - I_e)}{\mu - \alpha}(e^{-\alpha t} - e^{-\mu t})}.$$

In particular, for zero initial inventories,

$$\boxed{X(t) = \frac{P(\gamma_{\mathrm{I}} + \gamma_{\mathrm{Xe}})}{\mu}(1 - e^{-\mu t}) - \frac{P\gamma_{\mathrm{I}}}{\mu - \alpha}(e^{-\alpha t} - e^{-\mu t})}.$$

When $\mu = \alpha$, replace $(e^{-\alpha t} - e^{-\mu t})/(\mu - \alpha)$ by $te^{-\alpha t}$. The populations remain finite at this coincident rate. For example, in the initially clean case this gives $X = X_e(1 - e^{-\alpha t}) - P\gamma_{\mathrm{I}} t e^{-\alpha t}$.

(b) Since the rates are positive, the transients vanish, leaving

$$\boxed{N_{\mathrm{Xe}}(\infty) = \frac{\varphi(\gamma_{\mathrm{I}} + \gamma_{\mathrm{Xe}})\sigma_f N_{\mathrm{U}}}{\lambda_{\mathrm{Xe}} + \varphi\sigma_{\mathrm{Xe}}}}.$$

(c) Reset $t = 0$ at shutdown. The initial values are now I_e, X_e, but production and neutron absorption cease. Thus $I = I_e e^{-\alpha t}$ and

$$(s + \beta)\widehat{X} = X_e + \frac{\alpha I_e}{s + \alpha}.$$

Inverting gives

$$\boxed{X(t) = X_e e^{-\beta t} + \frac{\alpha I_e}{\alpha - \beta}(e^{-\beta t} - e^{-\alpha t})}.$$

In the manual's shutdown formula the coefficient of the exponential difference uses $N_{\mathrm{Xe}}(0)$; it must use $N_{\mathrm{I}}(0) = I_e$. The source of xenon after shutdown is αI, as the differential equation directly shows. The initial slope is

$$X'(0) = P\gamma_{\mathrm{I}} - \beta X_e = \frac{P(\gamma_{\mathrm{I}}\varphi\sigma_{\mathrm{Xe}} - \beta\gamma_{\mathrm{Xe}})}{\mu}.$$

A short-term increase is conditional, not automatic: for a positive pre-shutdown flux it occurs exactly when $\varphi\sigma_{\mathrm{Xe}} > \beta\gamma_{\mathrm{Xe}}/\gamma_{\mathrm{I}}$. Here $\alpha = \ln 2/(6.7 \cdot 3600) = 2.87374 \times 10^{-5}$ s^{-1} and $\beta = \ln 2/(9.2 \cdot 3600) = 2.09284 \times 10^{-5}$ s^{-1}. The threshold is consequently

$$\varphi > \frac{\beta}{20\sigma_{\mathrm{Xe}}} = 2.98977 \times 10^{11} \text{ cm}^{-2}\text{s}^{-1}.$$

For an explicit peak time put $A = \alpha I_e/(\alpha - \beta) > 0$. Then $X = (X_e + A)e^{-\beta t} - Ae^{-\alpha t}$ and $X' = 0$ gives

$$t_{\mathrm{peak}} = \frac{1}{\alpha - \beta} \ln \frac{\alpha A}{\beta(X_e + A)}.$$

This is positive exactly under the preceding increase condition; otherwise the maximum after shutdown is at $t = 0$.

20.8.6 Solve Eq. (20.160), which describes a damped simple harmonic oscillator, for $X(0) = X_0$, $X'(0) = 0$, and

(a) $b^2 = 4mk$ (critically damped),

(b) $b^2 > 4mk$ (overdamped).

ANS. (a) $X(t) = X_0 e^{-(b/2m)t}\left(1 + \dfrac{b}{2m}t\right)$.

Solution.

The equation is $mX'' + bX' + kX = 0$. Put $\alpha = b/(2m)$ and $\omega_0^2 = k/m$. Using $X(0) = X_0$ and $X'(0) = 0$ gives

$$(s^2 + 2\alpha s + \omega_0^2)\widehat{X} = (s + 2\alpha)X_0, \quad \widehat{X} = X_0\frac{(s+\alpha)+\alpha}{(s+\alpha)^2 + \omega_0^2 - \alpha^2}.$$

(a) Critical damping means $\omega_0^2 = \alpha^2$, so

$$\widehat{X} = X_0\left(\frac{1}{s+\alpha} + \frac{\alpha}{(s+\alpha)^2}\right), \qquad \boxed{X = X_0 e^{-\alpha t}(1 + \alpha t)}.$$

(b) In the overdamped case define $\kappa = \sqrt{\alpha^2 - \omega_0^2} > 0$. The elementary hyperbolic transform pairs, shifted by $s \mapsto s + \alpha$, give

$$\boxed{X = X_0 e^{-\alpha t}\left(\cosh \kappa t + \frac{\alpha}{\kappa}\sinh \kappa t\right)}.$$

Indeed this is

$$\frac{X_0}{2}\left[\left(1 + \frac{\alpha}{\kappa}\right)e^{(-\alpha+\kappa)t} + \left(1 - \frac{\alpha}{\kappa}\right)e^{(-\alpha-\kappa)t}\right].$$

Both exponents solve the characteristic equation; their coefficients sum to X_0, and their initial derivative is 0. Taking $\kappa \to 0$ gives the critical result because $\sinh \kappa t/\kappa \to t$.

20.8.7 Again solve Eq. (20.160), which describes a damped simple harmonic oscillator, but this time for $X(0) = 0$, $X'(0) = v_0$, and

(a) $b^2 < 4mk$ (underdamped),

(b) $b^2 = 4mk$ (critically damped),

(c) $b^2 > 4mk$ (overdamped).

ANS. (a) $X(t) = \dfrac{v_0}{\omega_1}e^{-(b/2m)t}\sin \omega_1 t$,
(b) $X(t) = v_0 t e^{-(b/2m)t}$.

Solution.

With $\alpha = b/(2m)$, $\omega_0^2 = k/m$, and initial data $X(0) = 0$, $X'(0) = v_0$, transformation of the equation gives

$$(s^2 + 2\alpha s + \omega_0^2)\widehat{X} = v_0, \qquad \widehat{X} = \frac{v_0}{(s+\alpha)^2 + \omega_0^2 - \alpha^2}.$$

(a) If $\omega_1 = \sqrt{\omega_0^2 - \alpha^2} > 0$, the shifted sine transform gives

$$\boxed{X(t) = \frac{v_0}{\omega_1}e^{-\alpha t}\sin \omega_1 t}.$$

(b) At $\omega_0^2 = \alpha^2$ the transform is $v_0/(s+\alpha)^2$, whose inverse is

$$\boxed{X(t) = v_0 t e^{-\alpha t}}.$$

(c) If $\kappa = \sqrt{\alpha^2 - \omega_0^2} > 0$, then

$$\widehat{X} = \frac{v_0}{2\kappa}\left(\frac{1}{s+\alpha-\kappa} - \frac{1}{s+\alpha+\kappa}\right),$$

so

$$\boxed{X(t) = \frac{v_0}{\kappa} e^{-\alpha t}\sinh \kappa t}.$$

All three expressions have $X(0) = 0$, $X'(0) = v_0$. The critical formula is the continuous limit of both others, using $\sin z/z \to 1$ and $\sinh z/z \to 1$.

20.8.8 The motion of a body falling in a resisting medium may be described by

$$m\,\frac{d^2X(t)}{dt^2} = mg - b\,\frac{dX(t)}{dt}$$

when the retarding force is proportional to the velocity. Find $X(t)$ and $dX(t)/dt$ for the initial conditions

$$X(0) = \left.\frac{dX}{dt}\right|_{t=0} = 0.$$

Solution.

Choose the downward direction positive and set $\Gamma = b/m > 0$. The equation is $X'' + \Gamma X' = g$. With both initial values zero,

$$(s^2 + \Gamma s)\widehat{X} = g/s, \qquad \widehat{X} = \frac{g}{s^2(s+\Gamma)}.$$

The partial fractions are

$$\frac{1}{s^2(s+\Gamma)} = \frac{1}{\Gamma s^2} - \frac{1}{\Gamma^2 s} + \frac{1}{\Gamma^2(s+\Gamma)}.$$

Multiplication by $s^2(s+\Gamma)$ verifies the identity: its right numerator is $(s+\Gamma)/\Gamma - s(s+\Gamma)/\Gamma^2 + s^2/\Gamma^2 = 1$. Therefore

$$\boxed{X(t) = \frac{mg}{b}t - \frac{m^2 g}{b^2}(1 - e^{-bt/m})}, \qquad \boxed{X'(t) = \frac{mg}{b}(1 - e^{-bt/m})}.$$

The velocity approaches the terminal value mg/b. Expanding $e^{-\Gamma t} = 1 - \Gamma t + \Gamma^2 t^2/2 - \cdots$ gives the zero-drag limits $X \to gt^2/2$ and $X' \to gt$, and checks that the apparent singularities as $b \to 0$ cancel.

20.8.9 Ringing circuit. In certain electronic devices, resistance, inductance, and capacitance are placed in a circuit as shown in Fig. 20.18. A constant voltage is maintained across the capacitance, keeping it charged. At time $t = 0$ the circuit is disconnected from the voltage source. Find the voltages across each of the elements R, L, and C as a function of time. Assume R to be small.

Hint. By Kirchhoff's laws

$$I_{RL} + I_C = 0 \quad \text{and} \quad E_R + E_L = E_C,$$

where

$$E_R = I_{RL}R, \qquad E_L = L\,\frac{dI_{RL}}{dt}, \qquad E_C = \frac{q_0}{C} + \frac{1}{C}\int_0^t I_C\,dt,$$

q_0 = initial charge of capacitor.

Take $R, L, C > 0$ and $R^2 < 4L/C$. Keep the initial branch current $I_{RL}(0) = I_0$ explicit; if the source was connected long enough for DC steady state, use $I_0 = q_0/(CR)$.

Solution.

The drawing has a series R, L branch in parallel with C. Define $V_0 = q_0/C$, $I(0) = I_0$ for the current through that branch, and use the voltage polarities in the hint. Kirchhoff's laws imply

$$I_C = -I, \qquad E_C' = -I/C, \qquad LI' + RI = E_C.$$

Eliminating I gives

$$LCE_C'' + RCE_C' + E_C = 0, \qquad E_C(0) = V_0, \quad E_C'(0) = -I_0/C.$$

Put $\alpha = R/(2L)$ and $\omega = \sqrt{1/(LC) - \alpha^2} > 0$; "small R" means $R^2 < 4L/C$. Transformation gives

$$\widehat{E}_C(s) = \frac{(s + 2\alpha)V_0 - I_0/C}{(s+\alpha)^2 + \omega^2} = V_0 \frac{s+\alpha}{(s+\alpha)^2 + \omega^2} + \frac{\alpha V_0 - I_0/C}{(s+\alpha)^2 + \omega^2}.$$

Consequently

$$\boxed{E_C = e^{-\alpha t}\left[V_0 \cos \omega t + \frac{\alpha V_0 - I_0/C}{\omega} \sin \omega t\right]}.$$

Differentiate this expression and use $\omega^2 + \alpha^2 = 1/(LC)$. The cosine coefficient of $-CE_C'$ is I_0, while its sine coefficient is $(V_0/L - \alpha I_0)/\omega$. Thus

$$I = e^{-\alpha t}\left[I_0 \cos \omega t + \frac{V_0/L - \alpha I_0}{\omega} \sin \omega t\right],$$

$$\boxed{E_R = Re^{-\alpha t}\left[I_0 \cos \omega t + \frac{V_0/L - \alpha I_0}{\omega} \sin \omega t\right]},$$

$$\boxed{E_L = e^{-\alpha t}\left[(V_0 - RI_0) \cos \omega t + \frac{-\alpha V_0 + (R\alpha - 1/C)I_0}{\omega} \sin \omega t\right]}.$$

The last expression is $E_C - E_R$, and also equals LI'. These formulas display both initial energy stores explicitly.

For the circuit exactly as drawn, if the constant source has been connected long enough to establish a DC steady state, the inductor has zero DC voltage but the resistor does not:

$$\boxed{I_0 = V_0/R}.$$

Substitution gives

$$E_C = V_0 e^{-\alpha t}\left[\cos \omega t + \frac{\alpha - 1/(RC)}{\omega} \sin \omega t\right],$$
$$E_R = V_0 e^{-\alpha t}\left[\cos \omega t + \frac{\alpha}{\omega} \sin \omega t\right],$$
$$E_L = -\frac{V_0}{RC\omega} e^{-\alpha t} \sin \omega t.$$

These have $E_R(0) = E_C(0) = V_0$, $E_L(0) = 0$, as a prior DC steady state requires. The manual instead uses damping $1/(2RC)$ and $E_C(0) = q_0 = 0$. Those are not the equations or initial state of the charged-capacitor circuit in Fig. 20.18. Its damping is $R/(2L)$. An ideal inductor in series with R does not short the capacitor. If a different switching preparation leaves the inductor initially unenergized, set $I_0 = 0$ in the general formulas; this is an additional initial condition, not the DC steady state of the displayed circuit. In that case $E_C = V_0 e^{-\alpha t}(\cos\omega t + \alpha \sin\omega t/\omega)$, $E_R = RV_0 e^{-\alpha t} \sin\omega t/(L\omega)$, and $E_L = V_0 e^{-\alpha t}(\cos\omega t - \alpha \sin\omega t/\omega)$. The $R = 0$ idealization admits this separately specified initial state, but no finite steady DC current V_0/R.

20.8.10 With $J_0(t)$ expressed as a contour integral, apply the Laplace transform operation, reverse the order of integration, and thus show that

$$L\{J_0(t)\} = (s^2 + 1)^{-1/2}, \quad \text{for } s > 0.$$

Solution.

Use the generating-function contour representation

$$J_0(t) = \frac{1}{2\pi i} \oint_{|z|=1} \exp\left[\frac{t}{2}(z - z^{-1})\right] \frac{dz}{z},$$

with counterclockwise orientation. On this contour $z - z^{-1} = 2i \sin\theta$, so the exponential has modulus 1. For real $s > 0$, multiplication by e^{-st} makes the double integral absolutely convergent, and interchange is valid:

$$\begin{aligned} L\{J_0\}(s) &= \frac{1}{2\pi i} \oint \frac{dz}{z} \int_0^\infty e^{-t[s-(z-z^{-1})/2]} dt \\ &= \frac{1}{2\pi i} \oint \frac{2\,dz}{1 + 2sz - z^2}. \end{aligned}$$

Let $D = \sqrt{s^2 + 1} > 0$. The denominator vanishes at $z_\pm = s \pm D$. Here $z_+ > 1$ and $|z_-| = D - s = 1/(D + s) < 1$, so just z_- lies inside. Its residue is

$$\frac{2}{2s - 2z_-} = \frac{1}{D}.$$

The residue theorem therefore gives

$$\boxed{L\{J_0(t)\}(s) = \frac{1}{\sqrt{s^2 + 1}}}, \qquad s > 0.$$

The transform extends analytically to $\Re s > 0$ with the branch whose square root is positive on the positive real axis.

20.8.11 Develop the Laplace transform of $J_n(t)$ from $L\{J_0(t)\}$ by using the Bessel function recurrence relations.

Hint. Here is a chance to use mathematical induction (Section 1.4).

The formula derived by the indicated induction is for integer $n \geq 0$.

Solution.

Write $g_n(s) = L\{J_n(t)\}(s)$, $D = \sqrt{s^2 + 1}$, and $r = D - s$ for $s > 0$. Exercise 20.8.10 gives $g_0 = 1/D$. Since $J_0' = -J_1$ and $J_0(0) = 1$,

$$g_1 = -[sg_0 - J_0(0)] = 1 - s/D = (D - s)/D = r/D.$$

For integer $n \geq 1$, $2J_n' = J_{n-1} - J_{n+1}$ and $J_n(0) = 0$, hence

$$2sg_n = g_{n-1} - g_{n+1}, \qquad g_{n+1} = g_{n-1} - 2sg_n.$$

Assume $g_{n-1} = r^{n-1}/D$ and $g_n = r^n/D$. Then

$$g_{n+1} = \frac{r^{n-1}}{D}(1 - 2sr).$$

The identity

$$r^2 = (D-s)^2 = D^2 - 2sD + s^2 = 1 + 2s^2 - 2sD = 1 - 2s(D-s) = 1 - 2sr$$

makes this r^{n+1}/D. Induction establishes

$$\boxed{L\{J_n(t)\}(s) = \frac{(\sqrt{s^2+1} - s)^n}{\sqrt{s^2+1}}}, \qquad n = 0, 1, 2, \ldots.$$

For negative integer order use $J_{-n} = (-1)^n J_n$; the displayed power formula is intended for nonnegative order. Analytic continuation to $\Re s > 0$ uses the branch specified in Exercise 20.8.10.

20.8.12 A calculation of the magnetic field of a circular current loop in circular cylindrical coordinates leads to the integral

$$\int_0^\infty e^{-kz} k J_1(ka)\, dk, \qquad a > 0, \quad \Re z > 0.$$

Show that this integral is equal to $a/(z^2 + a^2)^{3/2}$.

Solution.

For an ordinary improper integral the originally printed boundary $\Re z = 0$ must be excluded when $a > 0$: the stated integrand has oscillations with amplitude growing like $\sqrt{k}$. The formula holds ordinarily for $\Re z > 0$ and has boundary values in an Abel or distributional sense. Scale Exercise 20.8.10 to obtain, for $a > 0$ and $\Re z > 0$,

$$\int_0^\infty e^{-kz} J_0(ka) dk = \frac{1}{\sqrt{z^2 + a^2}}.$$

Exponential decay permits differentiation with respect to a. Because $J_0'(ka) = -J_1(ka)$,

$$-\int_0^\infty e^{-kz} k J_1(ka) dk = \frac{\partial}{\partial a}(z^2 + a^2)^{-1/2} = -a(z^2 + a^2)^{-3/2}.$$

Therefore

$$\boxed{\int_0^\infty e^{-kz} k J_1(ka) dk = \frac{a}{(z^2 + a^2)^{3/2}}}, \qquad \Re z > 0.$$

The square root is chosen positive for real $z > 0$ and continued throughout the right half-plane. To see the boundary issue directly, at $z = 0$, $kJ_1(ka) \sim \sqrt{2k/(\pi a)}\cos(ka - 3\pi/4)$; its oscillatory primitive does not have a limit. Nevertheless inserting $e^{-\epsilon k}$ and letting $\epsilon \downarrow 0$ gives the Abel value $1/a^2$. For $z = iy$, the analogous limiting formula is singular at $y = \pm a$; it is not a finite ordinary value there. The manual, p. 351, also prints $K_1(ka)$ in its first intermediate integral where $J_1(ka)$ is required, and replaces the boundary term $[e^{-kz}J_0(ka)]_0^\infty = -1$ by $+1$. The latter constant disappears on differentiation, but the intermediate equality is incorrect.

20.8.13 Show that

$$L\{I_0(at)\} = (s^2 - a^2)^{-1/2}, \qquad a \in \mathbb{R}, \quad s > |a|.$$

Solution.

For real a use the angular representation

$$I_0(at) = \frac{1}{\pi}\int_0^\pi e^{at\cos\theta}d\theta.$$

If a is allowed to be negative, the sufficient real convergence condition is $s > |a|$, not merely $s > a$. For a positive scale a, the original condition $s > a$ is correct. For $s > |a|$, the integrals are absolutely convergent, so

$$L\{I_0(at)\}(s) = \frac{1}{\pi}\int_0^\pi \frac{d\theta}{s - a\cos\theta}.$$

Set $u = \tan(\theta/2)$, so $\cos\theta = (1-u^2)/(1+u^2)$ and $d\theta = 2du/(1+u^2)$. Then

$$\begin{aligned} L\{I_0(at)\} &= \frac{2}{\pi}\int_0^\infty \frac{du}{(s-a)+(s+a)u^2} \\ &= \frac{2}{\pi\sqrt{(s-a)(s+a)}}\left[\tan^{-1}\left(u\sqrt{\frac{s+a}{s-a}}\right)\right]_0^\infty \\ &= \boxed{\frac{1}{\sqrt{s^2-a^2}}}. \end{aligned}$$

For complex s in $\Re s > |a|$, use the branch positive on $s > |a|$, equivalently the branch asymptotic to s at positive infinity in the denominator.

20.8.14 Verify the following Laplace transforms:

(a) $L\{j_0(at)\} = L\left\{\dfrac{\sin at}{at}\right\} = \dfrac{1}{a}\cot^{-1}\left(\dfrac{s}{a}\right)$,

(b) $L\{y_0(at)\}$ does not exist.

(c) $L\{i_0(at)\} = L\left\{\dfrac{\sinh at}{at}\right\} = \dfrac{1}{2a}\ln\dfrac{s+a}{s-a} = \dfrac{1}{a}\coth^{-1}\left(\dfrac{s}{a}\right)$,

(d) $L\{k_0(at)\}$ does not exist.

Use $a > 0$, with real $s > 0$ in (a) and $s > a$ in (c), and the real inverse-function branches there.

Solution.

Take $a > 0$; use real $s > 0$ for (a) and $s > a$ for (c). (a) Since $L\{\sin at\}(u) = a/(u^2 + a^2)$, integrate the transform parameter from s to infinity:

$$\int_s^\infty \frac{a}{u^2+a^2}du = \int_0^\infty \sin(at)\left(\int_s^\infty e^{-ut}du\right)dt = \int_0^\infty e^{-st}\frac{\sin at}{t}dt.$$

Absolute integrability after the factor e^{-st} justifies the interchange. Dividing by a and integrating yields

$$\boxed{L\{j_0(at)\} = \frac{1}{a}\left[\frac{\pi}{2} - \tan^{-1}(s/a)\right] = \frac{1}{a}\cot^{-1}(s/a)}.$$

(b) The spherical Neumann function is $y_0(at) = -\cos(at)/(at) = -1/(at) + O(t)$ near zero. Thus its Laplace integral contains a nonzero multiple of $\int_0^\epsilon dt/t$ and diverges for every s.

(c) In the convergent half-plane, $L\{\sinh at\}(u) = a/(u^2 - a^2)$. The same integration gives

$$\begin{aligned} L\{i_0(at)\} &= \int_s^\infty \frac{du}{u^2 - a^2} \\ &= \frac{1}{2a}\left[\ln(u-a) - \ln(u+a)\right]_s^\infty \\ &= \boxed{\frac{1}{2a}\ln\frac{s+a}{s-a} = \frac{1}{a}\coth^{-1}(s/a)}. \end{aligned}$$

Keep the two logarithms together when taking the infinite endpoint; the individual logarithms do not have finite limits. The manual's final logarithmic expression for (c), on p. 351, omits the factor $1/a$. Its coefficient must be $1/(2a)$, as in the textbook answer and the partial fractions shown here.

(d) In the book's spherical normalization $k_0(at) = e^{-at}/(at)$. It also has a nonintegrable $1/(at)$ singularity at zero, so its ordinary Laplace transform does not exist. The real inverse-trigonometric and inverse-hyperbolic forms presume the positive scale and the domains specified above. The $a = 0$ limits of (a) and (c) are $1/s$; for negative a it is safer to use $\tan^{-1}(a/s)/a$ in (a) and the continuous logarithmic form in (c), with $s > |a|$.

20.8.15 Develop a Laplace transform solution of Laguerre's equation,

$$tF''(t) + (1-t)F'(t) + nF(t) = 0.$$

Note that you need a derivative of a transform and a transform of derivatives. Go as far as you can with a general value of n; then (and only then) set $n = 0$.

Find the solution regular at the origin, with finite $F(0) = C$ and $F'(0)$. A logarithmically singular second solution is outside these initial-value transform rules.

Solution.

Seek the solution regular at $t = 0$, with finite $F(0) = C$ and $F'(0)$. Write $f = L\{F\}$. The needed transforms are

$$L\{F'\} = sf - C, \qquad L\{F''\} = s^2 f - sC - F'(0),$$
$$L\{tF'\} = -\frac{d}{ds}(sf - C) = -f - sf', \quad L\{tF''\} = -\frac{d}{ds}(s^2 f - sC - F'(0)) = -2sf - s^2 f' + C.$$

Adding $L\{tF''\} + L\{F'\} - L\{tF'\} + nf$ gives

$$s(1-s)f' + (n+1-s)f = 0.$$

For a nonzero solution, divide by $s(1-s)f$ and decompose:

$$\frac{f'}{f} = -\frac{n+1-s}{s(1-s)} = -\frac{n+1}{s} + \frac{n}{s-1}.$$

Integrating each logarithmic derivative gives

$$\ln f = \ln A + n\ln(s-1) - (n+1)\ln s, \qquad f(s) = A\frac{(s-1)^n}{s^{n+1}}.$$

For general n, choose the powers by continuation from real $s > 1$. The initial-value theorem gives $A = \lim_{s\to\infty} sf(s) = F(0) = C$. For $|s| > 1$ in this half-plane, expand the generalized binomial:

$$f(s) = \frac{C}{s}\left(1 - \frac{1}{s}\right)^n = C\sum_{j=0}^{\infty} \frac{(-n)_j}{j!s^{j+1}}.$$

Since $L\{t^j\} = j!/s^{j+1}$, this identifies the regular solution

$$\boxed{F(t) = C\sum_{j=0}^{\infty} \frac{(-n)_j}{(j!)^2} t^j = C\,{}_1F_1(-n; 1; t)}.$$

The transform calculation is justified, for example, in $\Re s > 1$, as in Exercise 20.7.8. If n is a nonnegative integer, the series terminates and $F = CL_n(t)$, with $f = C(s-1)^n/s^{n+1}$ already valid for $\Re s > 0$. Only now set $n = 0$: $f = C/s$, so

$$\boxed{F(t) = C, \qquad L_0(t) = 1 \text{ when } C = 1}.$$

This is the regular solution selected by the finite initial values used in transforming derivatives. The second local solution is logarithmically singular at $t = 0$ and is not included by that derivation. At $n = 0$, direct integration gives $F' = Be^t/t$; its integral is singular unless $B = 0$.

20.8.16 Show that the Laplace transform of the Laguerre polynomial $L_n(at)$ is given by

$$L\{L_n(at)\} = \frac{(s-a)^n}{s^{n+1}}, \qquad s > 0.$$

Solution.

For integer $n \geq 0$, use the finite polynomial expansion

$$L_n(x) = \sum_{j=0}^{n} (-1)^j \binom{n}{j} \frac{x^j}{j!}.$$

There is no infinite-series interchange: transform its $n+1$ terms directly in $\Re s > 0$:

$$\begin{aligned} L\{L_n(at)\}(s) &= \sum_{j=0}^{n} (-1)^j \binom{n}{j} \frac{a^j}{j!} \frac{j!}{s^{j+1}} \\ &= \frac{1}{s}\sum_{j=0}^{n} \binom{n}{j} \left(-\frac{a}{s}\right)^j = \frac{1}{s}\left(1 - \frac{a}{s}\right)^n \\ &= \boxed{\frac{(s-a)^n}{s^{n+1}}}. \end{aligned}$$

This also includes $n = 0$ and $a = 0$. Because the integrand is an exponential times a polynomial, no condition $s > a$ is needed here; any real $s > 0$ suffices.

20.8.17 Show that

$$L\{E_1(t)\} = \frac{1}{s}\ln(s+1), \qquad s > 0,$$

where

$$E_1(t) = \int_t^\infty \frac{e^{-\tau}}{\tau}\,d\tau = \int_1^\infty \frac{e^{-xt}}{x}\,dx.$$

$E_1(t)$ is the exponential integral function, first encountered in this book in Table 1.2.

Solution.

For real $s > 0$, the given representation has a nonnegative integrand, so Tonelli's theorem permits the interchange:

$$\begin{aligned} L\{E_1\}(s) &= \int_0^\infty e^{-st}\int_1^\infty \frac{e^{-xt}}{x}\,dx\,dt \\ &= \int_1^\infty \frac{dx}{x}\int_0^\infty e^{-(s+x)t}dt = \int_1^\infty \frac{dx}{x(s+x)}. \end{aligned}$$

Use $1/[x(s+x)] = s^{-1}[1/x - 1/(x+s)]$ and a common upper cutoff R:

$$\int_1^R \frac{dx}{x(s+x)} = \frac{1}{s}\big[\ln R - \ln(R+s) + \ln(1+s)\big].$$

The first two logarithms approach 0 together. Therefore

$$\boxed{L\{E_1(t)\}(s) = \frac{\ln(1+s)}{s}}.$$

The apparent singularity at $s = 0$ is removable: the limit is 1, consistent with $\int_0^\infty E_1(t)dt = 1$ obtained by reversing its defining integral.

20.8.18

(a) From Eq. (20.184) show that

$$\int_0^\infty f(x)\,dx = \int_0^\infty \frac{F(t)}{t}\,dt,$$

provided the integrals exist.

(b) From the preceding result show that

$$\int_0^\infty \frac{\sin t}{t}\,dt = \frac{\pi}{2},$$

in agreement with Eqs. (20.146) and (11.107).

In (a), justify the interchange, for example by $\int_0^\infty |F(t)|\,dt/t < \infty$; in (b), use an Abel limiting argument for the conditionally convergent integral.

Solution.

(a) Let $f(x) = \int_0^\infty e^{-xt} F(t)dt$. An interchange of improper integrals requires a convergence justification. A simple sufficient condition for the direct calculation is $\int_0^\infty |F(t)|\,dt/t < \infty$; conditionally convergent cases need a limiting argument, as in part (b). Under this absolute condition, Fubini gives

$$\begin{aligned}\int_0^\infty f(x)dx &= \int_0^\infty F(t)\left(\int_0^\infty e^{-xt}dx\right)dt \\ &= \boxed{\int_0^\infty \frac{F(t)}{t}\,dt},\end{aligned}$$

because the inner integral is $1/t$ for $t > 0$. Equivalently, the transform identity $L\{F(t)/t\}(s) = \int_s^\infty f(x)dx$ yields the same relation at the justified limit $s \downarrow 0$.

(b) For $F(t) = \sin t$, $f(x) = 1/(1+x^2)$. Insert a positive lower cutoff s in the x integration; absolute integrability then permits

$$\int_0^\infty e^{-st}\frac{\sin t}{t}dt = \int_s^\infty \frac{dx}{1+x^2} = \frac{\pi}{2} - \tan^{-1} s.$$

The original sine integral exists: it is regular at zero, and integration by parts bounds its tail from R to infinity by $2/R$. Abel's theorem thus lets $s \downarrow 0$ in the regulated integral without changing its improper value. Consequently

$$\boxed{\int_0^\infty \frac{\sin t}{t}dt = \frac{\pi}{2}}.$$

This proof uses a regulator precisely because the absolute condition of part (a) fails for $|\sin t|/t$.

20.8.19

(a) Show that

$$L\left\{\frac{\sin kt}{t}\right\} = \cot^{-1}\left(\frac{s}{k}\right).$$

(b) Using this result (with $k = 1$), prove that

$$L\{\mathrm{si}(t)\} = -\frac{1}{s}\tan^{-1} s,$$

where

$$\mathrm{si}(t) = -\int_t^\infty \frac{\sin x}{x}\,dx, \qquad \text{the sine integral.}$$

The real inverse-cotangent expression in (a) assumes $k > 0$, $s > 0$, and values in $(0, \pi/2)$. Use $s > 0$ also in (b).

Solution.

(a) For real $s > 0$, integrate the sine transform in its transform parameter:

$$L\left\{\frac{\sin kt}{t}\right\}(s) = \int_s^\infty \frac{k}{u^2+k^2}du = \boxed{\tan^{-1}(k/s)}.$$

For $k > 0$ this is $\cot^{-1}(s/k)$ with the usual value in $(0, \pi/2)$. The printed inverse-cotangent form presumes $k > 0$ and this branch. The arctangent form extends continuously to $k = 0$ and all real k.

(b) From the defining tail integral, $\mathrm{si}'(t) = \sin t/t$ and $\mathrm{si}(0+) = -\pi/2$. If $h(s) = L\{\mathrm{si}\}(s)$, the derivative transform gives

$$sh(s) - \mathrm{si}(0+) = sh(s) + \pi/2 = \tan^{-1}(1/s).$$

For $s > 0$, $\tan^{-1}(1/s) = \pi/2 - \tan^{-1} s$, so

$$\boxed{L\{\mathrm{si}(t)\}(s) = -\frac{\tan^{-1} s}{s}}.$$

The finite value at the origin and the decaying oscillatory tail of si ensure convergence in $\Re s > 0$; analytic continuation there uses the corresponding logarithmic branch of the arctangent.

20.8.20 If $F(t)$ is periodic (Fig. 20.19) with a period a so that $F(t + a) = F(t)$ for all $t \geq 0$, show that

$$L\{F(t)\} = \frac{1}{1 - e^{-as}} \int_0^a e^{-st} F(t)\, dt.$$

Note that the integration is now over only the first period of $F(t)$.

Assume $a > 0$, that F is integrable over a period, and $\Re s > 0$.

Solution.

Assume $a > 0$ and F integrable over one period. For $\Re s > 0$ split the half-line into successive periods, putting $t = na + u$ in the nth interval:

$$\begin{aligned} L\{F\}(s) &= \sum_{n=0}^{\infty} \int_{na}^{(n+1)a} e^{-st} F(t) dt \\ &= \sum_{n=0}^{\infty} e^{-nas} \int_0^a e^{-su} F(na + u) du \\ &= \left(\sum_{n=0}^{\infty} e^{-nas} \right) \int_0^a e^{-su} F(u) du. \end{aligned}$$

The geometric series converges because $|e^{-as}| < 1$. It also bounds the absolute integrals, justifying the splitting and summation. Hence

$$\boxed{L\{F(t)\}(s) = \frac{\displaystyle\int_0^a e^{-st} F(t) dt}{1 - e^{-as}}}, \qquad \Re s > 0.$$

Values assigned at isolated period boundaries do not affect the transform.

20.8.21 Find the Laplace transform of the square wave (period a) defined by

$$F(t) = \begin{cases} 1, & 0 < t < a/2, \\ 0, & a/2 < t < a. \end{cases}$$

ANS. $f(s) = \dfrac{1}{s} \dfrac{1 - e^{-as/2}}{1 - e^{-as}}$.

Take $a > 0$ and $\Re s > 0$.

Solution.

Apply Exercise 20.8.20. Only the first half of a period contributes to the numerator:

$$\int_0^a e^{-st}F(t)dt = \int_0^{a/2} e^{-st}dt = \frac{1-e^{-as/2}}{s}.$$

Therefore

$$\boxed{f(s) = \frac{1-e^{-as/2}}{s(1-e^{-as})} = \frac{1}{s(1+e^{-as/2})}}, \qquad \Re s > 0.$$

The simplification follows from $1-e^{-as} = (1-e^{-as/2})(1+e^{-as/2})$. For real $s \to \infty$, $sf(s) \to 1$, which matches $F(0+)$. For $s \downarrow 0$, $sf(s) \to 1/2$, the average value over a period; this is not a pointwise final value of the oscillating square wave.

20.8.22 Show that

(a) $L\{\cosh at \cos at\} = \dfrac{s^3}{s^4+4a^4}$,

(b) $L\{\cosh at \sin at\} = \dfrac{as^2+2a^3}{s^4+4a^4}$,

(c) $L\{\sinh at \cos at\} = \dfrac{as^2-2a^3}{s^4+4a^4}$,

(d) $L\{\sinh at \sin at\} = \dfrac{2a^2 s}{s^4+4a^4}$.

Take $a \in \mathbb{R}$ and $\Re s > |a|$.

Solution.

Take real a and $\Re s > |a|$. Define

$$D_- = (s-a)^2 + a^2 = s^2 - 2as + 2a^2, \qquad D_+ = (s+a)^2 + a^2 = s^2 + 2as + 2a^2.$$

Frequency shifting of the elementary sine and cosine transforms gives

$$L\{e^{\pm at}\cos at\} = \frac{s \mp a}{D_\mp}, \qquad L\{e^{\pm at}\sin at\} = \frac{a}{D_\mp}.$$

Also

$$D_- D_+ = (s^2+2a^2)^2 - (2as)^2 = s^4 + 4a^4.$$

Using $\cosh at = (e^{at}+e^{-at})/2$ and $\sinh at = (e^{at}-e^{-at})/2$:

(a) $L\{\cosh at \cos at\} = \dfrac{(s-a)D_+ + (s+a)D_-}{2D_-D_+} = \boxed{\dfrac{s^3}{s^4+4a^4}}$,

(b) $L\{\cosh at \sin at\} = \dfrac{a(D_+ + D_-)}{2D_-D_+} = \boxed{\dfrac{as^2+2a^3}{s^4+4a^4}}$,

(c) $L\{\sinh at \cos at\} = \dfrac{(s-a)D_+ - (s+a)D_-}{2D_-D_+} = \boxed{\dfrac{as^2-2a^3}{s^4+4a^4}}$,

(d) $L\{\sinh at \sin at\} = \dfrac{a(D_+ - D_-)}{2D_-D_+} = \boxed{\dfrac{2a^2 s}{s^4+4a^4}}$.

For the two less immediate numerators, direct expansion gives $(s-a)D_+ = s^3 + as^2 - 2a^3$ and $(s+a)D_- = s^3 - as^2 + 2a^3$; adding or subtracting proves the simplifications in (a) and (c).

20.8.23 Show that

(a) $L^{-1}\{(s^2+a^2)^{-2}\} = \dfrac{1}{2a^3}\sin at - \dfrac{t}{2a^2}\cos at,$

(b) $L^{-1}\{s(s^2+a^2)^{-2}\} = \dfrac{t}{2a}\sin at,$

(c) $L^{-1}\{s^2(s^2+a^2)^{-2}\} = \dfrac{1}{2a}\sin at + \dfrac{t}{2}\cos at,$

(d) $L^{-1}\{s^3(s^2+a^2)^{-2}\} = \cos at - \dfrac{at}{2}\sin at.$

Take real $a \neq 0$ in the displayed quotients and use their continuous limits at $a = 0$.

Solution.

Begin with $L\{\sin at/a\} = 1/(s^2+a^2)$ and $L\{\cos at\} = s/(s^2+a^2)$ for real $a \neq 0$, $\Re s > 0$. (a) Differentiate the first pair with respect to a and multiply by $-1/(2a)$:

$$-\frac{1}{2a}\frac{\partial}{\partial a}\frac{1}{s^2+a^2} = \frac{1}{(s^2+a^2)^2}, \qquad -\frac{1}{2a}\frac{\partial}{\partial a}\frac{\sin at}{a} = \boxed{\frac{\sin at}{2a^3} - \frac{t\cos at}{2a^2}}.$$

(b) Since $L\{tF\} = -dL\{F\}/ds$,

$$-\frac{1}{2}\frac{d}{ds}\frac{1}{s^2+a^2} = \frac{s}{(s^2+a^2)^2}, \qquad \boxed{L^{-1}\left\{\frac{s}{(s^2+a^2)^2}\right\} = \frac{t}{2a}\sin at}.$$

(c) Split the numerator as $s^2 = (s^2+a^2) - a^2$ and use (a):

$$\frac{s^2}{(s^2+a^2)^2} = \frac{1}{s^2+a^2} - \frac{a^2}{(s^2+a^2)^2}.$$

Its inverse is $\sin at/a - a^2[\sin at/(2a^3) - t\cos at/(2a^2)]$, or

$$\boxed{\frac{\sin at}{2a} + \frac{t}{2}\cos at}.$$

(d) Similarly $s^3/(s^2+a^2)^2 = s/(s^2+a^2) - a^2 s/(s^2+a^2)^2$. Part (b) then gives

$$\boxed{\cos at - \frac{at}{2}\sin at}.$$

At $a = 0$, interpret all quotient formulas by their continuous limits. The four answers reduce to $t^3/6$, $t^2/2$, t, and 1, the inverses of s^{-4} through s^{-1}.

20.8.24 Show that

$$L\{(t^2-k^2)^{-1/2}u(t-k)\} = K_0(ks).$$

(Here $u(t-k)$ is the unit step function.)

Hint. Try transforming an integral representation of $K_0(ks)$ into the Laplace transform integral. Take $k > 0$ and $\Re s > 0$; the step restricts the real square root to $t > k$.

Solution.

Take $k > 0$ and $\Re s > 0$. The lower endpoint $t = k$ then has an integrable square-root singularity. At $k = 0$ the kernel is $1/t$ and the ordinary transform diverges. The step function restricts the transform to

$$L\{(t^2 - k^2)^{-1/2} u(t-k)\} = \int_k^\infty \frac{e^{-st}}{\sqrt{t^2 - k^2}} dt.$$

Set $t = k\cosh v$, $v \geq 0$. Since $dt = k\sinh v\, dv$ and $\sqrt{t^2 - k^2} = k\sinh v$, the Jacobian cancels the denominator:

$$\int_0^\infty e^{-ks\cosh v} dv = \boxed{K_0(ks)}.$$

The last expression is the standard hyperbolic integral for K_0 on $\Re(ks) > 0$. The value assigned to $u(0)$ does not change this integrable transform.

20.9 Laplace Convolution Theorem

20.9.1 From the convolution theorem show that

$$\frac{1}{s} f(s) = L\left\{\int_0^t F(x)\, dx\right\},$$

where $f(s) = L\{F(t)\}$.

Solution.

In the one-sided Laplace convolution, choose $G(t) = 1$ for $t \geq 0$. Its transform is $g(s) = 1/s$. By definition,

$$(F * G)(t) = \int_0^t F(x)G(t-x)dx = \int_0^t F(x)dx.$$

The convolution theorem therefore yields

$$\boxed{L\left\{\int_0^t F(x)dx\right\}(s) = f(s)g(s) = \frac{f(s)}{s}}.$$

For locally integrable functions of exponential order this holds in a common right half-plane. Direct differentiation also checks the initial term: the primitive has value zero at $t = 0$, so its derivative transform is $s[f(s)/s] = f(s)$.

20.9.2 If $F(t) = t^a$ and $G(t) = t^b$, $a > -1$, $b > -1$,

(a) Show that the convolution $F * G$ is given by

$$F * G = t^{a+b+1} \int_0^1 y^a (1-y)^b\, dy.$$

(b) By using the convolution theorem, show that

$$\int_0^1 y^a (1-y)^b\, dy = \frac{\Gamma(a+1)\Gamma(b+1)}{\Gamma(a+b+2)} = B(a+1, b+1),$$

where B is the beta function.

Solution.

(a) For $t > 0$, substitute $x = ty$ into the convolution:

$$(F * G)(t) = \int_0^t x^a (t-x)^b dx = t^{a+b+1} \int_0^1 y^a (1-y)^b dy.$$

The endpoint conditions $a, b > -1$ make this integral finite. Call the remaining constant B. Then

$$L\{F * G\} = B \frac{\Gamma(a+b+2)}{s^{a+b+2}},$$

because $a + b + 1 > -1$.

(b) The separate gamma integrals give

$$f(s) = \frac{\Gamma(a+1)}{s^{a+1}}, \qquad g(s) = \frac{\Gamma(b+1)}{s^{b+1}}.$$

Equating $L\{F * G\} = fg$ and cancelling $s^{-(a+b+2)}$ yields

$$\boxed{\int_0^1 y^a (1-y)^b dy = \frac{\Gamma(a+1)\Gamma(b+1)}{\Gamma(a+b+2)} = B(a+1, b+1)}.$$

For arbitrary real $a, b > -1$, the factorial notation in the original question means gamma functions, $a! = \Gamma(a+1)$, not only integer factorials.

20.9.3 Using the convolution integral, calculate

$$L^{-1}\left\{\frac{s}{(s^2+a^2)(s^2+b^2)}\right\}, \qquad a^2 \neq b^2.$$

Solution.

Factor the transform into $[1/(s^2+a^2)]\,[s/(s^2+b^2)]$. The convolution theorem gives

$$H(t) = \frac{1}{a} \int_0^t \sin(ax) \cos[b(t-x)] dx.$$

Use $\sin u \cos v = [\sin(u+v) + \sin(u-v)]/2$:

$$\begin{aligned} H(t) &= \frac{1}{2a} \int_0^t \{\sin[(a-b)x + bt] + \sin[(a+b)x - bt]\} dx \\ &= \frac{1}{2a} \left[\frac{\cos bt - \cos at}{a-b} + \frac{\cos bt - \cos at}{a+b}\right] \\ &= \frac{\cos bt - \cos at}{2a} \frac{(a+b) + (a-b)}{a^2 - b^2} \\ &= \boxed{\frac{\cos at - \cos bt}{b^2 - a^2}}. \end{aligned}$$

The assumption $a^2 \neq b^2$ makes $a \pm b$ nonzero. If $a = 0$, the initial factor $\sin(ax)/a$ means x; the final formula remains valid by continuity. This agrees with the independently transformed result of Exercise 20.7.3.

20.9.4 An undamped oscillator is driven by a force $F_0 \sin \omega t$. Find the displacement $X(t)$ as a function of time, subject to initial conditions $X(0) = X'(0) = 0$. For $\omega^2 \neq \omega_0^2$, the solution is a linear combination of two simple harmonic motions, one with the frequency of the driving force and one with the frequency $\omega_0 > 0$ of the free oscillator. Find the resonant limit separately.

$$\textit{ANS. } X(t) = \frac{F_0/m}{\omega^2 - \omega_0^2}\left(\frac{\omega}{\omega_0}\sin \omega_0 t - \sin \omega t\right).$$

The displayed quotient answer applies for $\omega^2 \neq \omega_0^2$.

Solution.

Let $\omega_0^2 = k/m > 0$. The equation and initial data are

$$X'' + \omega_0^2 X = \frac{F_0}{m}\sin \omega t, \qquad X(0) = X'(0) = 0.$$

Transformation gives

$$\widehat{X}(s) = \frac{F_0}{m}\,\frac{\omega}{(s^2 + \omega^2)(s^2 + \omega_0^2)}.$$

Equivalently, the convolution response is

$$X(t) = \frac{F_0}{m\omega_0}\int_0^t \sin[\omega_0(t-u)]\sin(\omega u)du.$$

For $\omega^2 \neq \omega_0^2$, decompose

$$\widehat{X} = \frac{F_0\omega}{m(\omega_0^2 - \omega^2)}\left(\frac{1}{s^2 + \omega^2} - \frac{1}{s^2 + \omega_0^2}\right).$$

Inverting the two terms yields

$$\boxed{X(t) = \frac{F_0/m}{\omega^2 - \omega_0^2}\left(\frac{\omega}{\omega_0}\sin \omega_0 t - \sin \omega t\right)}.$$

Its value and derivative vanish at zero because the two sine slopes cancel. The printed answer requires a separate resonant limit. At $\omega = \omega_0$ it is a removable $0/0$ form and cannot be used as written. Apply l'Hôpital's rule to the ratio as $\omega \to \omega_0$. The numerator derivative is $\sin(\omega_0 t)/\omega_0 - t\cos(\omega t)$, and the denominator derivative is 2ω. Thus

$$\boxed{X_{\text{res}}(t) = \frac{F_0}{2m\omega_0^2}(\sin \omega_0 t - \omega_0 t \cos \omega_0 t)}.$$

The growing $t\cos \omega_0 t$ term is the resonant response; it is no longer a sum of two bounded oscillations at distinct frequencies. For $\omega = 0$ the driving force is identically zero and the solution is $X = 0$.

20.10 Inverse Laplace Transform

20.10.1 Derive the Bromwich integral from Cauchy's integral formula.

Hint. Apply the inverse transform L^{-1} to

$$f(s) = \frac{1}{2\pi i} \lim_{\alpha\to\infty} \int_{\beta-i\alpha}^{\beta+i\alpha} \frac{f(z)}{s-z}\, dz,$$

where $f(z)$ is analytic for $\Re e\, z \geq \beta$.

Assume decay sufficient to make the right closing arc vanish and the usual Laplace inversion hypotheses; analyticity alone does not imply the identity in the hint. Take $\Re s > \beta$.

Solution.

Choose β to the right of all singularities of the Laplace transform, and fix $\Re s > \beta$. Close the upward vertical segment $\beta - iR$ to $\beta + iR$ by a semicircle to its right. This closed contour is clockwise. The kernel $f(z)/(s-z)$ has residue $-f(s)$ at $z = s$, so Cauchy's formula with this orientation gives

$$\int_{\beta-iR}^{\beta+iR} \frac{f(z)}{s-z} dz + \int_{\text{right arc}} \frac{f(z)}{s-z} dz = -2\pi i[-f(s)] = 2\pi i f(s).$$

If, for example, $f(z) = O(|z|^{-\epsilon})$ on these arcs for some $\epsilon > 0$, the arc integral is $O(R^{-\epsilon})$. Letting $R \to \infty$ then gives the identity in the hint. For each z on the line $\Re z = \beta$,

$$\frac{1}{s-z} = \int_0^\infty e^{-st} e^{zt} dt, \qquad L_s^{-1}\{(s-z)^{-1}\}(t) = e^{zt}.$$

Apply this inverse operation to the Cauchy representation, with a convergent cutoff or regularization of the vertical integral. The result is

$$\boxed{F(t) = \frac{1}{2\pi i} \lim_{R\to\infty} \int_{\beta-iR}^{\beta+iR} e^{zt} f(z) dz}, \qquad t > 0,$$

which is the Bromwich inversion formula at a continuity point of F. Analyticity alone is not enough to discard the arc or interchange a limiting inverse transform. Decay on the closing contours and inversion hypotheses are required. A standard sufficient setting is a causal, piecewise smooth function of exponential order, with the integral interpreted by symmetric truncation (or Abel regularization). At a jump it recovers the mean of the two one-sided limits. For clarity, writing $z = \beta + i\omega$ turns this into the Fourier inversion of $e^{-\beta t} F(t)$ extended by zero to $t < 0$; Exercise 20.10.2 supplies a regularized kernel proof of the limiting step.

20.10.2 Starting with

$$\frac{1}{2\pi i} \int_{\beta-i\infty}^{\beta+i\infty} e^{st} f(s)\, ds,$$

show that by introducing

$$f(s) = \int_0^\infty e^{-sz} F(z)\, dz$$

we can convert our integral into the Fourier representation of a Dirac delta function. From this derive the inverse Laplace transform.

Use a Bromwich line to the right of the convergence abscissa, and recover $F(t)$ at positive continuity points; interpret the delta integral distributionally.

Solution.

Parameterize the line by $s = \beta + i\omega$, so $ds = i\,d\omega$. On inserting the Laplace integral, the proposed inverse becomes formally

$$\begin{aligned}\frac{1}{2\pi i}\int_{\beta-i\infty}^{\beta+i\infty} e^{st}f(s)ds &= \int_0^\infty F(z)e^{\beta(t-z)}\left[\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{i\omega(t-z)}d\omega\right]dz \\ &= \int_0^\infty F(z)e^{\beta(t-z)}\delta(t-z)dz.\end{aligned}$$

Here is a precise justification of the delta. Multiply the frequency integrand by $e^{-\epsilon\omega^2}$, with $\epsilon > 0$. Its Fourier integral is

$$K_\epsilon(u) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-\epsilon\omega^2+i\omega u}d\omega = \frac{e^{-u^2/(4\epsilon)}}{\sqrt{4\pi\epsilon}}.$$

It has mass 1, is nonnegative, and its mass outside any fixed neighborhood of zero tends to zero. If $H(z) = e^{-\beta z}F(z)$ for $z \geq 0$ and $H(z) = 0$ for $z < 0$, with $H \in L^1$, the regularized inverse is

$$e^{\beta t}\int_{-\infty}^{\infty} K_\epsilon(t-z)H(z)dz.$$

Continuity of H controls the integral near $z = t$; the vanishing Gaussian tails control the remainder. Hence its limit is $e^{\beta t}H(t) = F(t)$ at every positive continuity point. This proves

$$\boxed{F(t) = \frac{1}{2\pi i}\int_{\beta-i\infty}^{\beta+i\infty} e^{st}f(s)ds}, \qquad t > 0,$$

with the justified inversion prescription. The unsmoothed frequency integral is a distribution, not an absolutely convergent integral. At an interior jump symmetric inversion gives $[F(t-) + F(t+)]/2$; at $t = 0$ it gives $F(0+)/2$ for the causal extension. For $t < 0$ it gives zero.

20.10.3 Derive the Laplace transformation convolution theorem by use of the Bromwich integral.

Solution.

Let $f = L\{F\}$ and $g = L\{G\}$, with both functions extended by zero for negative arguments. In a common half-plane of convergence choose a Bromwich line $\Re s = \beta$. Substitute the defining integral for g into the inverse of the product:

$$\begin{aligned}L^{-1}\{fg\}(t) &= \frac{1}{2\pi i}\int_{\beta-i\infty}^{\beta+i\infty} e^{st}f(s)\left[\int_0^\infty e^{-su}G(u)du\right]ds \\ &= \int_0^\infty G(u)\left[\frac{1}{2\pi i}\int_{\beta-i\infty}^{\beta+i\infty} e^{s(t-u)}f(s)ds\right]du.\end{aligned}$$

The bracket is the causal inverse $F(t-u)$. It vanishes for $u > t$, so for $t \geq 0$ the result is

$$\boxed{L^{-1}\{f(s)g(s)\}(t) = \int_0^t F(t-u)G(u)du}.$$

Changing variables $x = t - u$ also gives $\int_0^t F(x)G(t-x)dx$, the Laplace convolution. For a direct rigorous interchange, it suffices that the vertical integral of $|f(\beta + i\omega)|$ and $\int_0^\infty e^{-\beta u}|G(u)|du$ are finite. For the usual broader class of locally integrable functions of exponential order, insert the Gaussian frequency regulator of Exercise 20.10.2, interchange the absolutely convergent integrals, and then take its approximate-identity limit at continuity points of the convolution. Values at isolated endpoints do not alter the convolution integral.

20.10.4 Find

$$L^{-1}\left\{\frac{s}{s^2-k^2}\right\}$$

(a) by a partial fraction expansion.

(b) Repeat, using the Bromwich integral.

Solution.

(a) For $k \neq 0$,

$$\frac{s}{s^2-k^2} = \frac{1}{2}\left(\frac{1}{s-k} + \frac{1}{s+k}\right).$$

Thus

$$\boxed{L^{-1}\left\{\frac{s}{s^2-k^2}\right\} = \frac{e^{kt}+e^{-kt}}{2} = \cosh kt}.$$

(b) Take $\beta > |k|$ for real k. For $t > 0$, close the Bromwich contour to the left, with positive orientation. The exponential suppresses the left arc, with the endpoint portions controlled by the usual Jordan estimate for the $O(1/s)$ rational factor. The enclosed residues are

$$\operatorname*{Res}_{s=k} \frac{se^{st}}{(s-k)(s+k)} = \frac{ke^{kt}}{2k} = \frac{e^{kt}}{2},$$

$$\operatorname*{Res}_{s=-k} \frac{se^{st}}{(s-k)(s+k)} = \frac{-ke^{-kt}}{-2k} = \frac{e^{-kt}}{2}.$$

Their sum reproduces $\cosh kt$. At $k = 0$ the transform reduces directly to $1/s$, with inverse 1, also the continuous limit of the answer. The formula gives the right-hand value at $t = 0$.

20.10.5 Find

$$L^{-1}\left\{\frac{k^2}{s(s^2+k^2)}\right\}$$

(a) by using a partial fraction expansion.

(b) Repeat using the convolution theorem.

(c) Repeat using the Bromwich integral.

ANS. $F(t) = 1 - \cos kt$.

Solution.

Assume real $k \neq 0$ during the algebra. (a) The decomposition

$$\frac{k^2}{s(s^2+k^2)} = \frac{1}{s} - \frac{s}{s^2+k^2}$$

immediately gives $F(t) = 1 - \cos kt$.

(b) Factor the transform as $[1/s]\,[k^2/(s^2+k^2)]$. The inverse factors are 1 and $k\sin kt$, so convolution gives

$$F(t) = \int_0^t k\sin(ku)du = [-\cos(ku)]_0^t = 1 - \cos kt.$$

(c) For a Bromwich line to the right of 0, close to the left for $t > 0$. The rational factor is $O(s^{-3})$, and the closing arc vanishes. At the three simple poles,

$$\begin{aligned}
\operatorname*{Res}_{s=0} \frac{k^2 e^{st}}{s(s-ik)(s+ik)} &= 1,\\
\operatorname*{Res}_{s=ik} \frac{k^2 e^{st}}{s(s-ik)(s+ik)} &= \frac{k^2 e^{ikt}}{(ik)(2ik)} = -\frac{1}{2}e^{ikt},\\
\operatorname*{Res}_{s=-ik} \frac{k^2 e^{st}}{s(s-ik)(s+ik)} &= \frac{k^2 e^{-ikt}}{(-ik)(-2ik)} = -\frac{1}{2}e^{-ikt}.
\end{aligned}$$

Summing gives

$$\boxed{F(t) = 1 - \frac{1}{2}(e^{ikt} + e^{-ikt}) = 1 - \cos kt}.$$

At $k = 0$ the transform and its inverse are both zero, which agrees with the continuous limit. All three methods give $F(0) = 0$.

20.10.6 Use the Bromwich integral to find the function whose transform is $f(s) = s^{-1/2}$. Note that $f(s)$ has a branch point at $s = 0$. The negative x-axis may be taken as a cut line. See Fig. 20.23.

Hint. A portion of the path needed to close the contour will yield nonzero contributions to the contour integral. These will need to be taken into account to get the proper value for the Bromwich integral.

ANS. $F(t) = (\pi t)^{-1/2}$.

Solution.

Choose the principal branch $s^{-1/2} = |s|^{-1/2}e^{-i\arg(s)/2}$ with $-\pi < \arg(s) < \pi$. Deform the Bromwich line to a counterclockwise contour hugging the negative real cut. The small circle around zero contributes $O(\sqrt{\rho})$ as its radius $\rho \to 0$, and for $t > 0$ the left outer arc vanishes by exponential decay and the Jordan estimate. There are no poles; the two banks supply the entire inverse. For $r > 0$ their values are

$$f(-r + i0) = -ir^{-1/2}, \qquad f(-r - i0) = +ir^{-1/2}.$$

On the counterclockwise cut contour the upper bank runs from 0 toward $-\infty$, and the lower bank returns. With $s = -r$, their sum is therefore

$$\int_0^\infty e^{-rt}\big[f(-r - i0) - f(-r + i0)\big]dr = 2i\int_0^\infty e^{-rt}r^{-1/2}dr.$$

Dividing by $2\pi i$ and substituting $u = rt$ gives

$$F(t) = \frac{1}{\pi}\int_0^\infty e^{-rt}r^{-1/2}dr = \frac{\Gamma(1/2)}{\pi\sqrt{t}} = \boxed{\frac{1}{\sqrt{\pi t}}}, \qquad t > 0.$$

Its integrable singularity at zero is compatible with a Laplace transform: direct substitution gives $\int_0^\infty e^{-st}(\pi t)^{-1/2}dt = s^{-1/2}$ for real $s > 0$, fixing the branch and normalization independently.

20.10.7 Show that

$$L^{-1}\{(s^2+1)^{-1/2}\} = J_0(t)$$

by evaluation of the Bromwich integral.

Hint. Convert your Bromwich integral into an integral representation of $J_0(t)$. Figure 20.24 shows a possible contour.

Solution.

Take a cut along the segment from $-i$ to i and choose $D(s) = \sqrt{s^2+1}$ analytic outside this segment with $D(s) \sim s$ at infinity and $D(s) > 0$ for positive real s. The endpoint singularities are inverse square roots, so small circles of radius ρ contribute $O(\sqrt{\rho})$. For $t > 0$, deformation of the Bromwich line around the cut is permitted by the large-arc estimate for $e^{st}/D(s)$. On the right bank $s = iy + 0$, $-1 < y < 1$,

$$D(iy+0) = +\sqrt{1-y^2};$$

on the left bank its sign is reversed. A counterclockwise contour travels upward along the right bank and downward along the left bank, so

$$\begin{aligned} F(t) &= \frac{1}{2\pi i}\int_{-i}^{i} e^{st}\left[\frac{1}{D(s)_{\text{right}}} - \frac{1}{D(s)_{\text{left}}}\right] ds \\ &= \frac{1}{\pi}\int_{-1}^{1} \frac{e^{iyt}}{\sqrt{1-y^2}} dy = \frac{2}{\pi}\int_{0}^{1} \frac{\cos(yt)}{\sqrt{1-y^2}} dy. \end{aligned}$$

The sine part cancels by oddness. Finally $y = \cos\theta$ converts the symmetric integral to

$$F(t) = \frac{1}{\pi}\int_0^{\pi} \cos(t\cos\theta)\, d\theta = \boxed{J_0(t)}.$$

This also has the right-hand initial value 1, as follows from $\lim_{s\to\infty} s/\sqrt{s^2+1} = 1$. The branch must be continued through the exterior of the chosen cut; taking an unrelated square-root sign on the left would spoil the jump and the factor of two.

20.10.8 Evaluate the inverse Laplace transform

$$L^{-1}\{(s^2-a^2)^{-1/2}\}$$

by each of the following methods:

(a) Expansion in a series and term-by-term inversion.

(b) Direct evaluation of the Bromwich integral.

(c) Change of variable in the Bromwich integral: $s = (a/2)(z + z^{-1})$.

Solution.

Take $a > 0$ and the branch of $\sqrt{s^2-a^2}$ asymptotic to s, with cut $[-a, a]$. The Bromwich line lies at $\Re s = \beta > a$.

(a) For $|s| > a$ on this branch,

$$(s^2-a^2)^{-1/2} = \frac{1}{s}\sum_{n=0}^{\infty} \frac{(2n)!}{4^n (n!)^2}\frac{a^{2n}}{s^{2n}}.$$

The inverse of s^{-2n-1} is $t^{2n}/(2n)!$. Thus

$$F(t) = \sum_{n=0}^{\infty} \frac{(at/2)^{2n}}{(n!)^2} = \boxed{I_0(at)}.$$

The resulting series converges for every finite t. Moreover, in $\Re s > a$ its absolute terms may be transformed and summed, reproducing the convergent binomial expansion. This verifies termwise inversion by uniqueness, without assuming uniform convergence on an arbitrary Bromwich line.

(b) For $-a < x < a$, the upper and lower bank values are

$$f(x + i0) = -\frac{i}{\sqrt{a^2 - x^2}}, \qquad f(x - i0) = +\frac{i}{\sqrt{a^2 - x^2}}.$$

The upper bank of a counterclockwise contour runs from a to $-a$, and the lower bank from $-a$ to a. Endpoint circles vanish as the square root of their radius. Hence

$$\begin{aligned} F(t) &= \frac{1}{2\pi i} \int_{-a}^{a} e^{xt} \big[f(x - i0) - f(x + i0) \big] dx \\ &= \frac{1}{\pi} \int_{-a}^{a} \frac{e^{xt}}{\sqrt{a^2 - x^2}} dx = \frac{1}{\pi} \int_0^{\pi} e^{at \cos\theta} d\theta = I_0(at). \end{aligned}$$

In the last substitution $x = a\cos\theta$, the reversed limits cancel the sign in dx.

(c) Set $s = (a/2)(z + z^{-1})$, taking the exterior inverse $z = (s + \sqrt{s^2 - a^2})/a$. Then

$$\sqrt{s^2 - a^2} = \frac{a}{2}(z - z^{-1}), \qquad \frac{ds}{\sqrt{s^2 - a^2}} = \frac{(1 - z^{-2})dz}{z - z^{-1}} = \frac{dz}{z}.$$

Deform the s contour around its cut as in (b). The upper bank maps to the upper unit semicircle from 1 to -1, and the lower bank to the lower semicircle from -1 to 1. Their union is one counterclockwise unit circle, giving

$$F(t) = \frac{1}{2\pi i} \oint_{|z|=1} \exp\left[\frac{at}{2}(z + z^{-1}) \right] \frac{dz}{z}.$$

Expand the two exponentials on this fixed circle:

$$\frac{1}{z} \sum_{n,m \geq 0} \frac{(at/2)^{n+m}}{n!m!} z^{n-m}.$$

Only $n = m$ contributes to the coefficient of z^{-1}, so the residue is $\sum_{n \geq 0} (at/2)^{2n}/(n!)^2 = I_0(at)$. For $at \neq 0$, the singularity at $z = 0$ is essential, not the simple pole called for in the manual; the residue calculation still applies. Also the image of the original vertical Bromwich line is a curve, not exactly a vertical line in z. Mapping the already deformed cut contour avoids that unwarranted replacement. The answer is even in a, so real negative a gives the same result with $|a|$. At $a = 0$ the transform is $1/s$ and its inverse is 1.

20.10.9 Show that

$$L^{-1}\left\{\frac{\ln s}{s}\right\} = -\ln t - \gamma,$$

where $\gamma = 0.5772\ldots$ is the Euler–Mascheroni constant.

Solution.

For real $q > 0$ and $s > 0$, the gamma substitution gives

$$L\left\{\frac{t^{q-1}}{\Gamma(q)}\right\}(s) = s^{-q}.$$

Differentiate with respect to q near $q = 1$. Exponential decay at infinity and local integrability of $t^{q-1}|\ln t|$ near zero justify differentiation inside the transform:

$$L\left\{\frac{t^{q-1}}{\Gamma(q)}[\ln t - \psi(q)]\right\}(s) = -(\ln s)s^{-q}, \qquad \psi(q) = \frac{\Gamma'(q)}{\Gamma(q)}.$$

Since $\Gamma(1) = 1$ and $\psi(1) = -\gamma$, setting $q = 1$ and multiplying by -1 yields

$$\boxed{L^{-1}\left\{\frac{\ln s}{s}\right\}(t) = -\ln t - \gamma}, \qquad t > 0.$$

For complex s in $\Re s > 0$, take the branch of $\ln s$ that is real on $s > 0$. The logarithmic singularity of the answer at $t = 0$ is integrable, so it causes no failure of the ordinary Laplace transform.

20.10.10 Evaluate the Bromwich integral for

$$f(s) = \frac{s}{(s^2 + a^2)^2}.$$

Solution.

For real $a \neq 0$, close the Bromwich line to the left when $t > 0$. The integrand

$$\frac{se^{st}}{(s - ia)^2(s + ia)^2}$$

has double poles at $s = \pm ia$ and a rational factor $O(s^{-3})$. The closing arc vanishes. At $s = ia$, the residue is the derivative of the remaining analytic factor:

$$\begin{aligned} R_+ &= \frac{d}{ds}\frac{se^{st}}{(s + ia)^2}\bigg|_{s=ia} \\ &= e^{iat}\left[\frac{1 + iat}{(2ia)^2} - \frac{2ia}{(2ia)^3}\right] = e^{iat}\left[-\frac{1 + iat}{4a^2} + \frac{1}{4a^2}\right] = -\frac{it}{4a}e^{iat}. \end{aligned}$$

Similarly,

$$R_- = \frac{d}{ds}\frac{se^{st}}{(s - ia)^2}\bigg|_{s=-ia} = \frac{it}{4a}e^{-iat}.$$

Summing,

$$F(t) = R_+ + R_- = \frac{it}{4a}(e^{-iat} - e^{iat}) = \boxed{\frac{t}{2a}\sin at}.$$

At $a = 0$ the two double poles merge and the reduced transform is s^{-3}; its inverse is $t^2/2$, the continuous limit of the formula above.

20.10.11 Heaviside expansion theorem. If the transform $f(s)$ may be written as a ratio

$$f(s) = \frac{g(s)}{h(s)},$$

where $g(s)$ and $h(s)$ are analytic functions, with $h(s)$ having simple, isolated zeros at $s = s_i$, and with a vanishing left closing-contour contribution and a convergent residue sum, show that

$$F(t) = L^{-1}\left\{\frac{g(s)}{h(s)}\right\} = \sum_i \frac{g(s_i)}{h'(s_i)}\, e^{s_i t}.$$

Hint. See Exercise 11.6.3.

A sufficient elementary setting is a strictly proper rational transform with finitely many simple poles, for $t > 0$. The unqualified analytic-function assertion in the book is false, as the counterexample below shows.

Solution.

At a simple zero s_i of h,

$$h(s) = h'(s_i)(s - s_i) + O((s - s_i)^2), \qquad h'(s_i) \neq 0.$$

The residue of the inverse-transform integrand is therefore

$$\operatorname*{Res}_{s=s_i} \frac{e^{st} g(s)}{h(s)} = \lim_{s \to s_i} \frac{(s - s_i) e^{st} g(s)}{h(s)} = \frac{g(s_i)}{h'(s_i)} e^{s_i t}.$$

If a left closing contour has vanishing contribution and the sum of residues converges to the contour limit, the residue theorem gives

$$\boxed{F(t) = \sum_i \frac{g(s_i)}{h'(s_i)} e^{s_i t}}, \qquad t > 0.$$

A particularly simple sufficient case is a strictly proper rational function g/h with finitely many simple poles. Zeros cancelled by g have zero residue and contribute nothing. Analytic g, h and simple zeros of h alone do not imply the stated expansion: one must also control the closing contour, and an infinite residue sum needs its own convergence prescription. For a concrete counterexample satisfying the printed analytic conditions, choose $g(s) = e^{-s}$, $h(s) = s + 1$. Its only pole is -1, so the proposed unrestricted sum would be e^{1-t}. But direct transformation gives

$$\int_1^\infty e^{-st} e^{-(t-1)} dt = \frac{e^{-s}}{s+1},$$

whose causal inverse is $e^{-(t-1)} u(t-1)$, zero for $0 < t < 1$. For these times the factor $e^{s(t-1)}$ grows on a left closing arc, explaining exactly why the omitted hypothesis fails. A polynomial part of a rational transform likewise requires impulses at zero rather than this ordinary exponential sum.

20.10.12 Using the Bromwich integral, invert $f(s) = s^{-2} e^{-ks}$, $k \geq 0$. Express $F(t) = L^{-1}\{f(s)\}$ in terms of the (shifted) unit step function $u(t - k)$.

ANS. $F(t) = (t - k)u(t - k)$.

Solution.

For the stated causal Laplace shift take $k \geq 0$. The inversion integral is

$$F(t) = \frac{1}{2\pi i} \int_{\beta - i\infty}^{\beta + i\infty} \frac{e^{s(t-k)}}{s^2} ds, \qquad \beta > 0.$$

If $t < k$, close to the right, where the exponential decays. No poles are enclosed, so $F(t) = 0$. If $t > k$, close to the left and enclose the double pole at $s = 0$. Its residue is

$$\frac{d}{ds} e^{s(t-k)} \bigg|_{s=0} = t - k.$$

Both contours have vanishing arcs because of exponential decay and the s^{-2} factor. At $t = k$ the value is zero, either by continuity or by the zero residue of s^{-2}. Thus

$$\boxed{F(t) = (t-k)u(t-k)}.$$

A forward check with $v = t - k$ gives

$$\int_k^\infty e^{-st}(t-k)dt = e^{-ks} \int_0^\infty e^{-sv} v\, dv = \frac{e^{-ks}}{s^2}.$$

If $k < 0$, integrating the proposed ramp only over $t \geq 0$ instead gives $s^{-2} - k/s$; this confirms the need for the nonnegative delay in the one-sided formula.

20.10.13 You have a Laplace transform:

$$f(s) = \frac{1}{(s+a)(s+b)}, \qquad a \neq b.$$

Invert this transform by each of three methods:

(a) Partial fractions and use of tables,

(b) Convolution theorem,

(c) Bromwich integral.

$$\textit{ANS. } F(t) = \frac{e^{-bt} - e^{-at}}{a-b}, \quad a \neq b.$$

Solution.

(a) Write

$$\frac{1}{(s+a)(s+b)} = \frac{1}{a-b}\left(\frac{1}{s+b} - \frac{1}{s+a}\right).$$

The exponential transform pair gives

$$F(t) = \frac{e^{-bt} - e^{-at}}{a-b}.$$

(b) The inverse factors are e^{-at} and e^{-bt}. Their convolution is

$$F(t) = \int_0^t e^{-au} e^{-b(t-u)} du = e^{-bt} \int_0^t e^{-(a-b)u} du$$
$$= e^{-bt} \frac{1 - e^{-(a-b)t}}{a-b} = \frac{e^{-bt} - e^{-at}}{a-b}.$$

(c) Choose $\beta > \max(-\Re a, -\Re b)$ and close the Bromwich line to the left for $t > 0$. The rational factor is $O(s^{-2})$, so the arc vanishes. The simple residues are

$$\operatorname*{Res}_{s=-a} \frac{e^{st}}{(s+a)(s+b)} = \frac{e^{-at}}{b-a}, \qquad \operatorname*{Res}_{s=-b} \frac{e^{st}}{(s+a)(s+b)} = \frac{e^{-bt}}{a-b}.$$

Their sum gives the same result:

$$\boxed{F(t) = \frac{e^{-bt} - e^{-at}}{a-b}}.$$

Its initial value is zero and its initial slope is 1. The excluded coincident case has the regular limit te^{-at}, also obtained from the double pole $1/(s+a)^2$.

21 Chapter 21. Integral Equations

Problems

21.1 Introduction

21.1.1 Starting with the ODE, integrate twice and derive the Volterra integral equation corresponding to

(a) $y''(x) - y(x) = 0; \quad y(0) = 0, \quad y'(0) = 1.$

$$\textit{ANS. } y = \int_0^x (x-t)y(t)\,dt + x.$$

(b) $y''(x) - y(x) = 0; \quad y(0) = 1, \quad y'(0) = -1.$

$$\textit{ANS. } y = \int_0^x (x-t)y(t)\,dt - x + 1.$$

Check your results with Eq. (21.11).

Solution.

Since $y'' = y$, one integration gives

$$y'(u) = y'(0) + \int_0^u y(t)\,dt.$$

A second integration gives

$$y(x) = y(0) + xy'(0) + \int_0^x du \int_0^u y(t)\,dt.$$

For $x \geq 0$ the triangular integration region is $0 \leq t \leq u \leq x$. Reverse the order:

$$\int_0^x y(t) \left(\int_t^x du \right) dt = \int_0^x (x-t)y(t)\,dt.$$

Thus the general initial-value formula is

$$y(x) = y(0) + xy'(0) + \int_0^x (x-t)y(t)\,dt.$$

Inserting the two pairs of initial data yields, respectively,

$$\boxed{y(x) = x + \int_0^x (x-t)y(t)\,dt}, \qquad \boxed{y(x) = 1 - x + \int_0^x (x-t)y(t)\,dt}.$$

To check Eq. (21.11), its ODE coefficients are $A = 0$, $B = -1$, $g = 0$, $a = 0$. Hence $K = (t-x)(B - A') - A = x - t$ and $f = y(0) + xy'(0)$, exactly as above. Oriented integrals give the same identities for $x < 0$.

21.1.2 Starting with the given answers of Exercise 21.1.1, differentiate and recover the original ODEs and the boundary conditions.

Solution.

Write either answer as

$$y(x) = a + bx + \int_0^x (x - t)y(t)\,dt,$$

where $(a, b) = (0, 1)$ or $(1, -1)$. Leibniz's rule includes an upper-endpoint term, which vanishes here:

$$y'(x) = b + (x - x)y(x) + \int_0^x \frac{\partial}{\partial x}\big[(x - t)y(t)\big]dt = b + \int_0^x y(t)\,dt.$$

Differentiate once more:

$$\boxed{y''(x) = y(x)}.$$

At $x = 0$ both integrals have zero length, so

$$y(0) = a, \qquad y'(0) = b.$$

These are precisely the two original initial conditions. Continuity of y in the integral equation gives the differentiability used here.

21.1.3 Given $\varphi(x) = x - \displaystyle\int_0^x (t - x)\varphi(t)\,dt$,

solve this integral equation by converting it to an ODE (plus boundary conditions) and solving the ODE (by inspection).

Solution.

First account for the sign:

$$\varphi(x) = x + \int_0^x (x - t)\varphi(t)\,dt.$$

Differentiating, the upper-endpoint term is zero:

$$\varphi'(x) = 1 + \int_0^x \varphi(t)\,dt, \qquad \varphi''(x) = \varphi(x).$$

The equation and its first derivative at zero give $\varphi(0) = 0$, $\varphi'(0) = 1$. The general solution is $Ae^x + Be^{-x}$, and therefore

$$A + B = 0, \qquad A - B = 1, \qquad A = \frac{1}{2}, \quad B = -\frac{1}{2}.$$

Hence $\boxed{\varphi(x) = \sinh x}$. Directly,

$$\int_0^x (x - t)\sinh t\,dt = [(x - t)\cosh t]_0^x + \int_0^x \cosh t\,dt = -x + \sinh x,$$

so substitution gives $x + (-x + \sinh x) = \sinh x$.

21.1.4 Show that the homogeneous Volterra equation of the second kind

$$\psi(x) = \lambda \int_0^x K(x,t)\psi(t)\,dt$$

has no solution (apart from the trivial solution $\psi = 0$). Assume that ψ is continuous and K is continuous on every finite triangle $0 \le t \le x \le L$.

Hint. On a fixed triangle, set $M = \max|K|$, $C = \max|\psi|$, and iterate the integral bound to obtain $|\psi(x)| \le C(|\lambda|Mx)^n/n!$.

Solution.

Use the usual continuous-kernel setting on a finite triangle $0 \le t \le x \le L$. Put

$$M = \max_{0\le t\le x\le L} |K(x,t)|, \qquad C = \max_{0\le x\le L} |\psi(x)|.$$

The equation first gives $|\psi(x)| \le C|\lambda|Mx$. Suppose $|\psi(t)| \le C(|\lambda|Mt)^n/n!$. Substituting this bound into the same equation gives

$$|\psi(x)| \le |\lambda|M \int_0^x C\frac{(|\lambda|Mt)^n}{n!}\,dt = C\frac{(|\lambda|Mx)^{n+1}}{(n+1)!}.$$

Starting with $n = 0$, induction therefore proves this bound for every integer $n \ge 0$. For fixed x, the ratio of successive bounds is $|\lambda|Mx/(n+1) \to 0$, so the bound tends to zero. Thus

$$\boxed{\psi(x) = 0 \quad (0 \le x \le L)}$$

for every finite λ. Taking arbitrary finite L covers an unbounded interval whenever the kernel is locally bounded and continuous.

For the original textbook's Maclaurin hint, $\psi(0) = 0$ follows at once. Repeated differentiation gives only integrals involving ψ and endpoint terms involving its lower derivatives; induction makes every $\psi^{(n)}(0)$ zero. A vanishing Maclaurin series proves the conclusion only if ψ is analytic, not merely indefinitely differentiable. For example, the smooth extension of e^{-1/x^2} by zero at $x = 0$ has all derivatives zero there but is not the zero function. The bounded-kernel estimate above supplies the missing uniqueness argument without assuming analyticity.

21.2 Some Special Methods

21.2.1 The kernel of a Fredholm equation of the second kind,

$$\varphi(x) = f(x) + \lambda \int_{-\infty}^{\infty} K(x,t)\varphi(t)\,dt,$$

is of the form $k(x-t)$.[6] Assuming that the required transforms exist, show that

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{F(t)e^{-ixt}}{1 - \sqrt{2\pi}\,\lambda K(t)}\,dt.$$

[6] This kernel and a range $0 \le x < \infty$ are the characteristics of integral equations of the Wiener–Hopf type. Details will be found in Chapter 8 of Morse and Feshbach (1953); see the Additional Readings.

$F(t)$ and $K(t)$ are the Fourier transforms of $f(x)$ and $k(x)$, respectively. Assume also that the denominator has no real zeros and that the quotient in the displayed formula admits Fourier inversion.

Solution.

Use the book's symmetric Fourier convention

$$\widehat{g}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} g(x) e^{i\omega x} dx, \qquad g(x) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \widehat{g}(\omega) e^{-i\omega x} d\omega.$$

Let $\Phi = \widehat{\varphi}$, $F = \widehat{f}$, $K = \widehat{k}$. To keep track of normalization, set $u = x - t$ in the transform of the convolution:

$$\begin{aligned} \widehat{k * \varphi}(\omega) &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \int_{\mathbb{R}} k(x-t)\varphi(t) e^{i\omega x}\, dt\, dx \\ &= \frac{1}{\sqrt{2\pi}} \left(\int_{\mathbb{R}} k(u) e^{i\omega u} du \right) \left(\int_{\mathbb{R}} \varphi(t) e^{i\omega t} dt \right) = \sqrt{2\pi} K(\omega)\Phi(\omega). \end{aligned}$$

Consequently

$$[1 - \sqrt{2\pi}\lambda K(\omega)]\Phi(\omega) = F(\omega).$$

Divide and invert to obtain

$$\boxed{\varphi(x) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \frac{F(\omega) e^{-i\omega x}}{1 - \sqrt{2\pi}\lambda K(\omega)}\, d\omega}.$$

Existence of the original transforms alone does not justify division: the quotient must belong to the intended inverse-transform class. Zeros of the denominator require compatibility with F and may permit homogeneous components supported on those zeros in a distributional setting. The displayed formula applies when these obstructions are absent.

21.2.2

(a) The kernel of a Volterra equation of the first kind,

$$f(x) = \int_0^x K(x,t)\varphi(t)\, dt,$$

has the form $k(x-t)$. Assuming that the required transforms exist, show that

$$\varphi(x) = \frac{1}{2\pi i} \int_{\gamma - i\infty}^{\gamma + i\infty} \frac{F(s)}{K(s)} e^{xs}\, ds,$$

where $F(s)$ and $K(s)$ are, respectively, the Laplace transforms of $f(x)$ and $k(x)$.

(b) In terms of the notation of part (a), show that the Volterra equation of the second kind,

$$\varphi(x) = f(x) + \lambda \int_0^x K(x,t)\varphi(t)\, dt,$$

has solution

$$\varphi(x) = \frac{1}{2\pi i} \int_{\gamma - i\infty}^{\gamma + i\infty} \frac{F(s)}{1 - \lambda K(s)} e^{xs}\, ds.$$

Assume each quotient is a Laplace transform for which the Bromwich inversion is valid. Choose γ in its right half-plane of analyticity, to the right of its singularities.

Solution.

For $x \geq 0$ use

$$F(s) = \int_0^\infty e^{-sx} f(x)dx, \quad K(s) = \int_0^\infty e^{-sx} k(x)dx, \quad \Phi(s) = \int_0^\infty e^{-sx}\varphi(x)dx.$$

In a right half-plane where the integrals can be interchanged, put $u = x - t$:

$$\begin{aligned}\int_0^\infty e^{-sx} \int_0^x k(x-t)\varphi(t)dt\,dx &= \int_0^\infty \int_0^\infty e^{-s(u+t)} k(u)\varphi(t)\,du\,dt \\ &= K(s)\Phi(s).\end{aligned}$$

(a) The first-kind equation gives $F = K\Phi$, hence

$$\Phi(s) = \frac{F(s)}{K(s)}, \qquad \boxed{\varphi(x) = \frac{1}{2\pi i}\int_{\gamma-i\infty}^{\gamma+i\infty} \frac{F(s)}{K(s)} e^{sx} ds}.$$

(b) The second-kind equation gives $\Phi = F + \lambda K\Phi$, so

$$\Phi(s) = \frac{F(s)}{1-\lambda K(s)}, \qquad \boxed{\varphi(x) = \frac{1}{2\pi i}\int_{\gamma-i\infty}^{\gamma+i\infty} \frac{F(s)}{1-\lambda K(s)} e^{sx} ds}.$$

In each case the quotient must be an admissible Laplace transform; its singularities determine the inversion line $\Re s = \gamma$. An uncancelled zero of a divisor cannot be ignored. When classical inversion hypotheses hold, choose γ to the right of all quotient singularities and use the ordinary Bromwich inverse.

21.2.3 Using the Laplace transform solution (Exercise 21.2.2), solve

(a) $\varphi(x) = x + \int_0^x (t-x)\varphi(t)\,dt.$

ANS. $\varphi(x) = \sin x.$

(b) $\varphi(x) = x - \int_0^x (t-x)\varphi(t)\,dt.$

ANS. $\varphi(x) = \sinh x.$

Check your results by substituting back into the original integral equations.

Solution.

Write $k(u) = -u$ in (a), and $k(u) = u$ in (b). Since $\mathcal{L}\{x\} = 1/s^2$,

	transformed equation	$\Phi(s)$
(a)	$\Phi = s^{-2} - s^{-2}\Phi$	$(s^2+1)^{-1}$
(b)	$\Phi = s^{-2} + s^{-2}\Phi$	$(s^2-1)^{-1}$

The inverses are

$$\boxed{\varphi_a(x) = \sin x, \qquad \varphi_b(x) = \sinh x}.$$

For an explicit substitution check, integration by parts gives

$$\int_0^x (x-t)\sin t\,dt = [-(x-t)\cos t]_0^x - \int_0^x \cos t\,dt = x - \sin x,$$

and

$$\int_0^x (x-t)\sinh t\,dt = [(x-t)\cosh t]_0^x + \int_0^x \cosh t\,dt = \sinh x - x.$$

Equation (a) therefore reads $x - (x - \sin x) = \sin x$; (b) reads $x + (\sinh x - x) = \sinh x$.

21.2.4 Reformulate the equations of Example 21.2.1 for integrals on the range $(0, \infty)$ using Fourier cosine transforms. Take the unknown and convolution kernel to be even functions on the full line, and retain the reflected kernel in the half-line equation.

Solution.

Take k and φ even on the full line, so their convolution f is even as well. Splitting the full-line equation of Example 21.2.1 into positive and negative t gives, for $x \geq 0$,

$$\boxed{f(x) = \int_0^\infty [k(x-t) + k(x+t)]\varphi(t)\,dt}.$$

Here the negative half-line was changed by $t \mapsto -t$. A half-line cosine formulation requires these even extensions and the reflected kernel $k(x+t)$. Simply replacing the full-line limits by $0, \infty$ while keeping only $k(x-t)$ does not yield the same convolution theorem.

Define the self-inverse cosine transform

$$G_c(\omega) = \sqrt{\frac{2}{\pi}} \int_0^\infty g(x)\cos(\omega x)dx.$$

For even g, this equals its symmetric full-line Fourier transform. Thus the equations of the example become

$$F_c(\omega) = \sqrt{2\pi} K_c(\omega)\Phi_c(\omega), \qquad \Phi_c(\omega) = \frac{F_c(\omega)}{\sqrt{2\pi}K_c(\omega)}.$$

For completeness, inversion of the first relation gives

$$f(x) = 2\int_0^\infty K_c(\omega)\Phi_c(\omega)\cos(\omega x)d\omega.$$

Inverting the second gives the desired half-line solution:

$$\boxed{\varphi(x) = \frac{1}{\pi}\int_0^\infty \frac{F_c(\omega)}{K_c(\omega)}\cos(\omega x)d\omega}.$$

The quotient and inversion need the same existence conditions as in Exercise 21.2.1. If a half-line kernel is instead defined as one half of the reflected sum, the corresponding convolution multiplier is one half as large; the normalization must be adjusted accordingly.

21.2.5 Given the Fredholm integral equation,

$$e^{-x^2} = \int_{-\infty}^{\infty} e^{-(x-t)^2} \varphi(t)\, dt,$$

apply the Fourier convolution technique of Example 21.2.1 to solve for $\varphi(t)$ as a tempered distribution.

Solution.

The symmetric Fourier transform of e^{-x^2} is

$$\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-x^2 + i\omega x} dx = \frac{1}{\sqrt{2}} e^{-\omega^2/4}.$$

The convolution theorem therefore gives

$$\frac{1}{\sqrt{2}} e^{-\omega^2/4} = \sqrt{2\pi} \left(\frac{1}{\sqrt{2}} e^{-\omega^2/4} \right) \Phi(\omega).$$

The Gaussian factor never vanishes, so $\Phi(\omega) = 1/\sqrt{2\pi}$. But this is precisely the Fourier transform of the unit point mass at zero:

$$\boxed{\varphi(t) = \delta(t)}.$$

Indeed, the original equation is checked without transforms:

$$\int_{\mathbb{R}} e^{-(x-t)^2} \delta(t)\, dt = e^{-x^2}.$$

The answer is a distribution. There is no $L^1(\mathbb{R})$ solution, since its Fourier transform would tend to zero at infinity; nor an $L^2(\mathbb{R})$ solution, since a nonzero constant transform is not square integrable. In the tempered-distribution class the nonzero Gaussian multiplier makes the solution unique.

21.2.6 Solve Abel's equation,

$$f(x) = \int_0^x \frac{\varphi(t)}{(x-t)^\alpha}\, dt, \qquad 0 < \alpha < 1,$$

by the following method:

(a) Multiply both sides by $(z-x)^{\alpha-1}$ and integrate with respect to x over the range $0 \le x \le z$.

(b) Reverse the order of integration and evaluate the integral on the right-hand side (with respect to x) by recognizing it as a beta function.

Note.

$$\int_t^z \frac{dx}{(z-x)^{1-\alpha}(x-t)^\alpha} = B(1-\alpha, \alpha) = \Gamma(\alpha)\Gamma(1-\alpha) = \frac{\pi}{\sin \pi\alpha}.$$

Solution.

For locally integrable φ, the singular kernels are integrable when $0 < \alpha < 1$. Multiply the equation by $(z-x)^{\alpha-1}$ and integrate:

$$I(z) := \int_0^z (z-x)^{\alpha-1} f(x) dx = \int_0^z dx \int_0^x (z-x)^{\alpha-1} (x-t)^{-\alpha} \varphi(t) dt.$$

Absolute integrability on this triangle follows from the beta integral below applied to $|\varphi|$. Reversing the order,

$$I(z) = \int_0^z \varphi(t) \left[\int_t^z (z-x)^{\alpha-1}(x-t)^{-\alpha} dx \right] dt.$$

Put $x = t + (z-t)u$. All powers of $z-t$ cancel:

$$\begin{aligned} \int_t^z (z-x)^{\alpha-1}(x-t)^{-\alpha} dx &= \int_0^1 (1-u)^{\alpha-1} u^{-\alpha} du \\ &= B(1-\alpha, \alpha) = \Gamma(1-\alpha)\Gamma(\alpha) = \frac{\pi}{\sin \pi\alpha}. \end{aligned}$$

Hence

$$I(z) = \frac{\pi}{\sin \pi\alpha} \int_0^z \varphi(t) dt, \qquad \boxed{\varphi(z) = \frac{\sin \pi\alpha}{\pi} \frac{d}{dz} \int_0^z (z-x)^{\alpha-1} f(x) dx}.$$

The derivative exists almost everywhere under the integral equation's local-integrability hypothesis.

If f is continuously differentiable up to zero, remove the apparent singular endpoint before differentiating. Integration by parts gives

$$I(z) = \frac{f(0) z^\alpha}{\alpha} + \frac{1}{\alpha} \int_0^z (z-x)^\alpha f'(x) dx,$$

and thus

$$\boxed{\varphi(z) = \frac{\sin \pi\alpha}{\pi} \left[f(0) z^{\alpha-1} + \int_0^z (z-x)^{\alpha-1} f'(x) dx \right]}.$$

This also proves sufficiency for such f: convolving either term with $x^{-\alpha}$ and using the same beta integral yields respectively $f(0)$ and $\int_0^z f'(x)dx = f(z) - f(0)$.

21.2.7 Given the generalized Abel equation with $f(x) = 1$,

$$1 = \int_0^x \frac{\varphi(t)}{(x-t)^\alpha} dt, \qquad 0 < \alpha < 1, \quad x > 0,$$

solve for $\varphi(t)$ and verify that $\varphi(t)$ is a solution of the given equation.

$$\textit{ANS.}\ \ \varphi(t) = \frac{\sin \pi\alpha}{\pi} t^{\alpha-1}.$$

Solution.

Set $f = 1$ in the preceding inversion formula:

$$\int_0^x (x-t)^{\alpha-1} dt = \frac{x^\alpha}{\alpha},$$

so

$$\boxed{\varphi(x) = \frac{\sin \pi\alpha}{\pi} x^{\alpha-1}}, \qquad x > 0.$$

Direct substitution, with $t = xu$, yields

$$\begin{aligned} \int_0^x \frac{\varphi(t)}{(x-t)^\alpha} dt &= \frac{\sin \pi\alpha}{\pi} x^{\alpha-1-\alpha+1} \int_0^1 u^{\alpha-1}(1-u)^{-\alpha} du \\ &= \frac{\sin \pi\alpha}{\pi} \Gamma(\alpha)\Gamma(1-\alpha) = 1. \end{aligned}$$

Both endpoint singularities in the u integral are integrable. The equation is valid for $x > 0$, with right-hand limit one as $x \downarrow 0$. It cannot also be imposed as an ordinary zero-length integral at $x = 0$, which is zero; the solution itself is singular there.

21.2.8 A Fredholm equation of the first kind has a kernel $e^{-(x-t)^2}$:

$$f(x) = \int_{-\infty}^{\infty} e^{-(x-t)^2} \varphi(t)\, dt.$$

Assume a solution exists in $L^2(\mathbb{R}, e^{-t^2} dt)$. Show that the solution is the following series, converging in that weighted norm:

$$\varphi(x) = \frac{1}{\sqrt{\pi}} \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{2^n n!} H_n(x),$$

in which $H_n(x)$ is an nth-order Hermite polynomial.

Solution.

Use the physicists' Hermite generating function:

$$e^{-(x-t)^2} = e^{-t^2} e^{2tx - x^2} = e^{-t^2} \sum_{n=0}^{\infty} H_n(t) \frac{x^n}{n!}.$$

Taking the nth derivative with respect to x at zero in the integral equation gives

$$f^{(n)}(0) = \int_{\mathbb{R}} e^{-t^2} H_n(t) \varphi(t) dt.$$

Now write $\varphi = \sum_n a_n H_n$. Hermite orthogonality is

$$\int_{\mathbb{R}} e^{-t^2} H_n(t) H_m(t) dt = \sqrt{\pi}\, 2^n n!\, \delta_{nm}.$$

Multiplying the expansion by $e^{-t^2} H_n$ and integrating therefore gives

$$a_n = \frac{1}{\sqrt{\pi}\, 2^n n!} \int_{\mathbb{R}} e^{-t^2} H_n(t) \varphi(t) dt = \frac{f^{(n)}(0)}{\sqrt{\pi}\, 2^n n!}.$$

Consequently

$$\boxed{\varphi(x) = \frac{1}{\sqrt{\pi}} \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{2^n n!} H_n(x)}.$$

The original unrestricted statement needs a function-space and convergence qualification. If $\varphi \in L^2(\mathbb{R}, e^{-t^2} dt)$, the formula is its convergent weighted-L^2 Hermite expansion; it is not automatically a pointwise series for arbitrary f. Here is the convergence justification. Write $z = a + ib$ and $\|\varphi\|_w^2 = \int e^{-t^2} |\varphi(t)|^2 dt$. Cauchy–Schwarz gives

$$\begin{aligned} \left| \int_{\mathbb{R}} e^{-(z-t)^2} \varphi(t) dt \right| &\le \|\varphi\|_w \left(\int_{\mathbb{R}} e^{-t^2 + 4at - 2\Re(z^2)} dt \right)^{1/2} \\ &= \pi^{1/4} e^{|z|^2} \|\varphi\|_w. \end{aligned}$$

Derivatives of the Gaussian add only polynomial factors in $t - z$. On a compact set of z, these are still dominated by an integrable Gaussian times a polynomial after the same Cauchy–Schwarz estimate. Differentiation under the integral is therefore valid to every order, locally uniformly, and the convolution is entire. Parseval's identity for the Hermite basis gives

$$\|\varphi\|_w^2 = \sum_{n=0}^{\infty} |a_n|^2 \sqrt{\pi}\, 2^n n!.$$

Substituting the coefficients found above gives the necessary condition

$$\sum_{n=0}^{\infty} \frac{|f^{(n)}(0)|^2}{\sqrt{\pi}\, 2^n n!} < \infty.$$

Conversely, if f is entire and this condition holds, the orthogonal series defines a weighted-L^2 function. Its Gaussian convolution has the same Taylor coefficients as f, and hence equals f.

The limitation is already visible in Exercise 21.2.5: $f(x) = e^{-x^2}$ has distributional solution δ. At $x = 0$, its proposed even terms are obtained from $f^{(2j)}(0) = (-1)^j(2j)!/j!$ and $H_{2j}(0) = (-1)^j(2j)!/j!$:

$$\frac{1}{\sqrt{\pi}} \frac{f^{(2j)}(0) H_{2j}(0)}{2^{2j}(2j)!} = \frac{1}{\sqrt{\pi}} \frac{\binom{2j}{j}}{4^j} \sim \frac{1}{\pi\sqrt{j}}.$$

The positive series diverges, so that example cannot be interpreted as an ordinary pointwise Hermite solution.

21.2.9 Solve the integral equation

$$f(x) = \int_{-1}^{1} \frac{\varphi(t)}{(1 - 2xt + x^2)^{1/2}}\, dt, \qquad -1 \le x \le 1,$$

for the unknown function $\varphi(t)$, with $s = 0, 1, 2, \ldots$, if

(a) $f(x) = x^{2s}$,

(b) $f(x) = x^{2s+1}$.

ANS. (a) $\varphi(t) = \dfrac{4s+1}{2} P_{2s}(t)$, (b) $\varphi(t) = \dfrac{4s+3}{2} P_{2s+1}(t)$.

Solution.

Let $N = 2s$ in (a) and $N = 2s + 1$ in (b), with $s = 0, 1, \ldots$. For $|x| < 1$, the generating function converges uniformly in $t \in [-1, 1]$:

$$(1 - 2xt + x^2)^{-1/2} = \sum_{n=0}^{\infty} P_n(t) x^n.$$

The equation with $f(x) = x^N$ therefore requires

$$\int_{-1}^{1} P_n(t)\varphi(t)dt = \delta_{nN}.$$

If $\varphi = \sum_m a_m P_m$ in $L^2[-1, 1]$, orthogonality gives

$$\frac{2a_n}{2n+1} = \delta_{nN}, \qquad a_n = \frac{2n+1}{2}\delta_{nN}.$$

Thus

$$\boxed{\varphi_a(t) = \frac{4s+1}{2} P_{2s}(t)}, \qquad \boxed{\varphi_b(t) = \frac{4s+3}{2} P_{2s+1}(t)}.$$

Substitution gives

$$\frac{2N+1}{2} \sum_{n=0}^{\infty} x^n \int_{-1}^{1} P_n(t) P_N(t) dt = x^N,$$

which checks both formulas. Completeness of the Legendre polynomials proves uniqueness in L^2.

The endpoint values follow by a limit of these finite-polynomial solutions. For $1/2 \le x < 1$,

$$1 - 2xt + x^2 = (1-x)^2 + 2x(1-t) \ge 1 - t.$$

The bounded numerator is consequently dominated by a constant times $(1-t)^{-1/2}$, which is integrable. Dominated convergence gives $f(1) = 1$. Reflection gives $f(-1) = (-1)^N$, so the solutions hold at both stated endpoints as improper integrals.

21.2.10 Find the eigenvalues and eigenfunctions of

$$\varphi(x) = \lambda \int_{-1}^{1} (t - x)\varphi(t)\, dt.$$

Solution.

The image of the kernel is linear in x:

$$\int_{-1}^{1} (t-x)\varphi(t)dt = \int_{-1}^{1} t\varphi(t)dt - x\int_{-1}^{1} \varphi(t)dt.$$

Every nonzero solution must therefore have the form $A + Bx$. The two required moments are

$$\int_{-1}^{1} (A + Bt)dt = 2A, \qquad \int_{-1}^{1} t(A + Bt)dt = \frac{2B}{3}.$$

Equating constant and linear coefficients gives

$$A = \frac{2\lambda}{3}B, \qquad B = -2\lambda A.$$

A nonzero solution has $A \neq 0$; eliminating B yields

$$1 = -\frac{4}{3}\lambda^2, \qquad \boxed{\lambda_{\pm} = \pm\frac{i\sqrt{3}}{2}}.$$

Choose $A = 1$:

$$\boxed{\varphi_+(x) = 1 - i\sqrt{3}\, x, \qquad \varphi_-(x) = 1 + i\sqrt{3}\, x}.$$

Their nonzero scalar multiples give the respective one-dimensional eigenspaces. No real λ permits a nontrivial solution. The imaginary characteristic values are consistent with the real antisymmetric kernel $K(x,t) = -K(t,x)$; the self-adjoint reality theorem does not apply.

21.2.11 Find the eigenvalues and eigenfunctions of

$$\varphi(x) = \lambda \int_{0}^{2\pi} \cos(x - t)\varphi(t)\, dt.$$

$$\textit{ANS. } \lambda_1 = \lambda_2 = \frac{1}{\pi}, \qquad \varphi(x) = A\cos x + B\sin x.$$

Solution.

Use

$$\cos(x - t) = \cos x \cos t + \sin x \sin t.$$

Thus every nonzero solution is $\varphi(x) = A\cos x + B\sin x$. Over $[0, 2\pi]$,

$$\int_0^{2\pi} \cos^2 t\,dt = \pi, \quad \int_0^{2\pi} \sin^2 t\,dt = \pi, \quad \int_0^{2\pi} \sin t\cos t\,dt = 0.$$

The integral is consequently $\pi A\cos x + \pi B\sin x = \pi\varphi(x)$. The equation becomes $(1-\lambda\pi)\varphi = 0$, so

$$\boxed{\lambda = \frac{1}{\pi}, \qquad \varphi(x) = A\cos x + B\sin x, \quad (A,B) \neq (0,0)}.$$

The characteristic value has a two-dimensional eigenspace. A function orthogonal to both sine and cosine is annihilated by the integral operator; it does not solve $\varphi = \lambda K\varphi$ nontrivially for any finite λ.

21.2.12 Find the eigenvalues and eigenfunctions of

$$y(x) = \lambda \int_{-1}^{1} (x-t)^2 y(t)\,dt.$$

Hint. This problem may be treated by the separable-kernel method or by a Legendre expansion.

Solution.

The kernel has degree two in x, so write $y(x) = A + Bx + Cx^2$. Its three moments are

$$\int_{-1}^{1} y(t)dt = 2A + \frac{2C}{3}, \quad \int_{-1}^{1} ty(t)dt = \frac{2B}{3}, \quad \int_{-1}^{1} t^2 y(t)dt = \frac{2A}{3} + \frac{2C}{5}.$$

Expanding $(x-t)^2 = x^2 - 2xt + t^2$ gives

$$\begin{pmatrix} A \\ B \\ C \end{pmatrix} = \lambda \begin{pmatrix} 2/3 & 0 & 2/5 \\ 0 & -4/3 & 0 \\ 2 & 0 & 2/3 \end{pmatrix} \begin{pmatrix} A \\ B \\ C \end{pmatrix}.$$

The odd equation is $B = -4\lambda B/3$, giving

$$\boxed{\lambda_0 = -\frac{3}{4}, \qquad y_0(x) = x}.$$

For the even sector, put $\mu = 1/\lambda$. Its determinant is

$$\begin{vmatrix} 2/3-\mu & 2/5 \\ 2 & 2/3-\mu \end{vmatrix} = (2/3-\mu)^2 - \frac{4}{5} = 0,$$

hence $\mu_\pm = 2/3 \pm 2/\sqrt{5}$. Taking $C = 1$, the last row says $2A + 2/3 = \mu_\pm$, so $A = \pm 1/\sqrt{5}$. Therefore

$$\boxed{\begin{aligned} \lambda_+ &= \frac{-15+9\sqrt{5}}{8}, \quad y_+(x) = x^2 + \frac{1}{\sqrt{5}}, \\ \lambda_- &= \frac{-15-9\sqrt{5}}{8}, \quad y_-(x) = x^2 - \frac{1}{\sqrt{5}}. \end{aligned}}$$

For example, rationalizing $1/(2/3 + 2/\sqrt{5})$ gives $(-15+9\sqrt{5})/8$. The matrix substitution verifies all three eigenpairs. Since its image contains only quadratic polynomials, the calculation exhausts all nonzero finite-characteristic-value solutions.

In the alternative Legendre normalization, $y_\pm$ are proportional to $P_0 + (4\lambda_\pm/3)P_2$. Indeed the latter has constant term $1 - 2\lambda_\pm/3$ and x^2 coefficient $2\lambda_\pm$, whose ratio is $1/(2\lambda_\pm) - 1/3 = \pm 1/\sqrt{5}$.

21.2.13 Use the separable-kernel technique to show that

$$\psi(x) = \lambda \int_0^\pi \cos x \sin t \, \psi(t) \, dt$$

has no solution (apart from $\psi = 0$). Explain this result in terms of separability and symmetry.

Solution.

Put

$$c = \int_0^\pi \sin t \, \psi(t) dt.$$

Separability forces $\psi(x) = \lambda c \cos x$. Substituting this form into the definition of c,

$$c = \lambda c \int_0^\pi \sin t \cos t \, dt = \frac{\lambda c}{2} \left[\sin^2 t\right]_0^\pi = 0.$$

Thus $\boxed{\psi = 0}$ for every finite λ.

The cancellation also follows from reflection about $\pi/2$: $\sin(\pi - t) = \sin t$ and $\cos(\pi - t) = -\cos t$, so the product has zero integral. In operator language, K has rank one but

$$K^2 f(x) = \cos x \left(\int_0^\pi \sin t \cos t \, dt \right) \left(\int_0^\pi \sin u \, f(u) du \right) = 0.$$

The operator itself is nonzero, since $K(\sin t) = (\pi/2) \cos x$. Its kernel is not symmetric under $x \leftrightarrow t$, so a nonzero nilpotent operator is allowed. A nonzero real symmetric rank-one kernel $c\,u(x)u(t)$ would instead have nonzero operator eigenvalue $c \int u^2$, and hence a nontrivial finite characteristic value.

21.2.14 Given $\varphi(x) = \lambda \int_0^1 (1 + xt)\varphi(t) \, dt$,

solve for the eigenvalues and the eigenfunctions by the separable-kernel technique.

Solution.

Introduce the two moments

$$c_0 = \int_0^1 \varphi(t) dt, \qquad c_1 = \int_0^1 t\varphi(t) dt.$$

The equation is $\varphi(x) = \lambda(c_0 + x c_1)$. Inserting it into those moment definitions gives

$$c_0 = \lambda \left(c_0 + \frac{c_1}{2} \right), \qquad c_1 = \lambda \left(\frac{c_0}{2} + \frac{c_1}{3} \right).$$

A nonzero moment vector requires

$$0 = \begin{vmatrix} 1 - \lambda & -\lambda/2 \\ -\lambda/2 & 1 - \lambda/3 \end{vmatrix} = (1 - \lambda)(1 - \lambda/3) - \lambda^2/4 = 1 - \frac{4}{3}\lambda + \frac{1}{12}\lambda^2.$$

Thus $\lambda^2 - 16\lambda + 12 = 0$, giving

$$\boxed{\lambda_\pm = 8 \mp 2\sqrt{13}}.$$

The constant term of a nonzero eigenfunction cannot vanish: otherwise $\varphi = Bx$ would require $0 = \lambda B/2$, forcing $B = 0$. Normalize that term to one, writing $\varphi_\pm = 1 + b_\pm x$. The constant equation $1 = \lambda(1 + b/2)$ yields $b = 2/\lambda - 2$, hence

$$\boxed{\varphi_\pm(x) = 1 + \frac{-2 \pm \sqrt{13}}{3} x}.$$

Numerically the pairs are $(0.7888974491,\ 1{+}0.5351837585x)$ and $(15.2111025509,\ 1{-}1.8685170918x)$. Both equations for the moments are satisfied, and the image of K is two-dimensional, so these are all the nontrivial eigenspaces.

21.2.15 Knowing the form of the solutions of an integral equation can be a great advantage. For

$$\varphi(x) = \lambda \int_0^1 (1 + xt)\varphi(t)\, dt,$$

assume $\varphi(x)$ to have the form $1 + bx$. Substitute into the integral equation. Integrate and solve for b and λ.

Solution.

Substitute the proposed function before making any spectral assumption:

$$\int_0^1 (1 + xt)(1 + bt)dt = 1 + \frac{b}{2} + x\left(\frac{1}{2} + \frac{b}{3}\right).$$

Equating coefficients in $1 + bx = \lambda[1 + b/2 + x(1/2 + b/3)]$ gives

$$1 = \lambda(1 + b/2), \qquad b = \lambda(1/2 + b/3).$$

The first equation rules out $b = -2$. Eliminating λ,

$$b(1 + b/2) = 1/2 + b/3, \quad \frac{b^2}{2} + \frac{2b}{3} - \frac{1}{2} = 0, \quad 3b^2 + 4b - 3 = 0.$$

The quadratic formula therefore gives

$$b_\pm = \frac{-4 \pm \sqrt{16 + 36}}{6} = \frac{-2 \pm \sqrt{13}}{3}.$$

Finally,

$$\lambda_\pm = \frac{1}{1 + b_\pm/2} = \frac{6}{4 \pm \sqrt{13}} = \boxed{8 \mp 2\sqrt{13}}.$$

Thus $\boxed{\varphi_\pm(x) = 1 + b_\pm x}$, with arbitrary nonzero overall scale. The ansatz loses no eigenfunctions: the integral's image is linear, and an eigenfunction with zero constant term must vanish, as the constant coefficient equation shows.

21.2.16 The equation

$$f(x) = \int_a^b K(x, t)\varphi(t)\, dt$$

has a degenerate kernel $K(x, t) = \sum_{i=1}^{n} M_i(x)N_i(t)$.

(a) Show that this integral equation has no solution unless $f(x)$ can be written as

$$f(x) = \sum_{i=1}^{n} f_i M_i(x),$$

where the f_i are constants.

(b) Show that to any solution $\varphi(x)$ we may add $\psi(x)$, provided that $\psi(x)$ has zero bilinear moments against every $N_i(x)$:

$$\int_a^b N_i(x)\psi(x)\,dx = 0 \qquad \text{for all } i.$$

Solution.

(a) The finite sum may be brought outside the integral:

$$\int_a^b K(x,t)\varphi(t)dt = \sum_{i=1}^{n} M_i(x) \underbrace{\int_a^b N_i(t)\varphi(t)dt}_{c_i}.$$

Each c_i is independent of x. Hence the equation requires

$$\boxed{f \in \operatorname{span}\{M_1, \ldots, M_n\}}, \qquad f(x) = \sum_i c_i M_i(x).$$

This is a necessary condition; the moment equations for the c_i must also be compatible to obtain a solution.

(b) If every specified moment of ψ is zero, then

$$K(\varphi+\psi)(x) = K\varphi(x) + \sum_i M_i(x) \int_a^b N_i(t)\psi(t)dt = f(x) + 0.$$

Thus every such $\boxed{\varphi+\psi}$ is another solution.

The manual adds that ψ must itself be a linear combination of the M_i. That extra restriction is false for this first-kind equation. Only the image f, not the unknown or its nullspace additions, is constrained to that span. For a direct counterexample, take $K(x,t) = 1$ on $[0,1]$, $f = 1$, $\varphi = 1$, and $\psi(t) = t - \frac{1}{2}$. Then

$$\int_0^1 \psi(t)dt = 0, \qquad \int_0^1 [1+\psi(t)]dt = 1,$$

although ψ is not in the span of the constant $M_1 = 1$. The displayed zero moments in the problem are the precise conditions used here; with complex functions they are bilinear moments as written.

21.2.17 A Kirchhoff diffraction theory analysis of a laser leads to the integral equation

$$v(\mathbf{r}_2) = \gamma \iint K(\mathbf{r}_1, \mathbf{r}_2) v(\mathbf{r}_1)\, dA.$$

The unknown, $v(\mathbf{r}_1)$, gives the geometric distribution of the radiation field over one mirror surface; the range of integration is over the surface of that mirror. For square confocal spherical mirrors, the integral equation becomes

$$v(x_2, y_2) = \frac{-i\gamma e^{ikb}}{\lambda b} \int_{-a}^{a}\int_{-a}^{a} e^{-(ik/b)(x_1x_2+y_1y_2)}\, v(x_1, y_1)\, dx_1\, dy_1,$$

in which b is the centerline distance between the laser mirrors. This can be put in a somewhat simpler form by the substitutions

$$\frac{kx_i^2}{b} = \xi_i^2, \qquad \frac{ky_i^2}{b} = \eta_i^2, \qquad \text{and} \qquad \frac{ka^2}{b} = \frac{2\pi a^2}{\lambda b} = \alpha^2.$$

(a) Show that the variables separate and we get two integral equations.

(b) Show that the new limits, $\pm\alpha$, may be approximated by $\pm\infty$ for fixed low-order modes when $a^2 \gg \lambda b$. Explain why $a \gg \lambda$ alone is insufficient.

(c) Solve the resulting integral equations.

Solution.

(a) Choose signed scaled coordinates

$$\xi = \sqrt{k/b}\, x, \qquad \eta = \sqrt{k/b}\, y, \qquad \alpha = a\sqrt{k/b}.$$

This choice retains the signs that were suppressed by the squared relations. The Jacobian is $dx_1 dy_1 = (b/k) d\xi_1 d\eta_1$, and $k\lambda = 2\pi$. The equation becomes

$$v(\xi_2, \eta_2) = \frac{-i\gamma e^{ikb}}{2\pi} \int_{-\alpha}^{\alpha}\!\!\int_{-\alpha}^{\alpha} e^{-i(\xi_1\xi_2 + \eta_1\eta_2)} v(\xi_1, \eta_1) d\xi_1 d\eta_1.$$

Define

$$(\mathcal{F}_\alpha h)(u) = \frac{1}{\sqrt{2\pi}} \int_{-\alpha}^{\alpha} e^{-ius} h(s) ds.$$

For a separated mode $v(\xi, \eta) = X(\xi)Y(\eta)$ the equation is

$$XY = -i\gamma e^{ikb} (\mathcal{F}_\alpha X)(\mathcal{F}_\alpha Y).$$

The quotient in ξ must be constant independently of the quotient in η. Thus the two one-dimensional equations and their consistency condition are

$$\boxed{\mathcal{F}_\alpha X = \rho X, \qquad \mathcal{F}_\alpha Y = \sigma Y, \qquad 1 = -i\gamma e^{ikb} \rho\sigma}.$$

These identities also make sense at isolated zeros of X, Y by continuity, without dividing there.

(b) The original textbook condition $a \gg \lambda$ alone does not imply that α is large: the mirror spacing b also enters. The needed large-aperture condition for fixed low-order modes is $a^2 \gg \lambda b$, equivalently $\alpha^2 = 2\pi a^2/(\lambda b) \gg 1$. For example, $a = 10^3\lambda$ and $b = 10^9\lambda$ give $\alpha^2 = 2\pi/1000$, much less than one despite $a \gg \lambda$.

The replacement by infinite limits is an approximation for modes concentrated well inside $(-\alpha, \alpha)$. Its pointwise error obeys

$$|(\mathcal{F} - \mathcal{F}_\alpha)h(u)| \le \frac{1}{\sqrt{2\pi}} \int_{|s|>\alpha} |h(s)| ds.$$

For a fixed Hermite–Gaussian mode the tail is a polynomial times $e^{-s^2/2}$ and becomes exponentially small as $\alpha \to \infty$. For the ground Gaussian $h_0(s) = \pi^{-1/4} e^{-s^2/2}$, the relative error at $u = 0$ is exactly $\operatorname{erfc}(\alpha/\sqrt{2})$. Higher orders require the aperture to exceed their characteristic width as well.

(c) Now solve the infinite-limit equations. The normalized Hermite functions are

$$h_n(u) = \frac{H_n(u)e^{-u^2/2}}{\pi^{1/4}\sqrt{2^n n!}}.$$

The Gaussian transform gives $\mathcal{F}h_0 = h_0$. To obtain all higher orders, use $a^\dagger = (u - d/du)/\sqrt{2}$ and $h_n = (a^\dagger)^n h_0/\sqrt{n!}$. Integration by parts and differentiation under the integral give

$$\mathcal{F}(sh) = i\frac{d}{du}\mathcal{F}h, \qquad \mathcal{F}(h') = iu\,\mathcal{F}h.$$

Therefore

$$\mathcal{F}(a^\dagger h) = \frac{i}{\sqrt{2}}\left(\frac{d}{du} - u\right)\mathcal{F}h = -i\,a^\dagger \mathcal{F}h,$$

and induction yields

$$\boxed{\mathcal{F}h_n = (-i)^n h_n}.$$

A separated two-dimensional mode is consequently

$$\boxed{v_{nm}(\xi,\eta) = C\,h_n(\xi)h_m(\eta), \qquad \gamma_{nm} = i^{\,n+m+1}e^{-ikb}}.$$

Indeed $-i\gamma_{nm}e^{ikb}(-i)^{n+m} = 1$. In physical coordinates the shape is proportional to

$$H_n\!\left(\sqrt{k/b}\,x\right) H_m\!\left(\sqrt{k/b}\,y\right) e^{-k(x^2+y^2)/(2b)}.$$

Hermite functions form a complete $L^2(\mathbb{R})$ basis. The full one-dimensional eigenspace for phase $(-i)^r$ consists of convergent combinations of h_n with $n \equiv r \pmod 4$. Likewise, two-dimensional combinations with fixed $n + m$ mod 4 share the same γ. These are the infinite-aperture solutions; for finite α the truncated transform must be solved instead, and the tail bound quantifies the stated approximation for a chosen mode.

21.3 Neumann Series

21.3.1 Using the Neumann series, solve

(a) $\varphi(x) = 1 - 2\displaystyle\int_0^x t\varphi(t)\,dt,$

ANS. (a) $\varphi(x) = e^{-x^2}$.

(b) $\varphi(x) = x + \displaystyle\int_0^x (t - x)\varphi(t)\,dt,$

(c) $\varphi(x) = x - \displaystyle\int_0^x (t - x)\varphi(t)\,dt.$

Solution.

(a) Define $(Tg)(x) = -2\int_0^x t\,g(t)dt$. The Neumann series is $\sum_{n\ge0} T^n 1$. We prove the individual terms: if $T^n 1 = (-1)^n x^{2n}/n!$, then

$$T^{n+1}1 = -2\int_0^x t\,\frac{(-1)^n t^{2n}}{n!}dt = \frac{(-1)^{n+1}x^{2n+2}}{(n+1)!}.$$

The initial case is $T^0 1 = 1$, so

$$\boxed{\varphi_a(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n!} = e^{-x^2}}.$$

Substitution gives $1 - 2\int_0^x te^{-t^2}dt = 1 + (e^{-x^2} - 1) = e^{-x^2}$.

(b) Put $(Jg)(x) = \int_0^x (x-t)g(t)dt$, so the equation is $\varphi = x - J\varphi$. The elementary monomial formula is

$$J(t^p)(x) = x\frac{x^{p+1}}{p+1} - \frac{x^{p+2}}{p+2} = \frac{x^{p+2}}{(p+1)(p+2)}, \qquad p > -1.$$

Starting with $J^0 x = x$, induction gives $J^n x = x^{2n+1}/(2n+1)!$. Hence

$$\boxed{\varphi_b(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = \sin x}.$$

(c) The equation is $\varphi = x + J\varphi$, so all the same terms have positive signs:

$$\boxed{\varphi_c(x) = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!} = \sinh x}.$$

The factorials give uniform convergence on every bounded x interval, justifying termwise integration. Direct substitutions for (b),(c) were evaluated explicitly in Exercise 21.2.3.

21.3.2 Solve

$$\psi(x) = x + \int_0^1 (1 + xt)\psi(t)\,dt$$

Investigate the following methods:

(a) Show that the ordinary Neumann series diverges, so it cannot supply the solution here.

(b) Solve by the separable-kernel technique.

(c) Solve by educated guessing.

Solution.

Let $(Kg)(x) = \int_0^1 (1 + xt)g(t)dt$.

(a) The ordinary Neumann series prescribed by $\psi = x + K\psi$ is

$$x + Kx + K^2 x + \cdots .$$

For $g(x) = A + Bx$,

$$Kg(x) = A + \frac{B}{2} + x\left(\frac{A}{2} + \frac{B}{3}\right),$$

so

$$Kx = \frac{1}{2} + \frac{x}{3}, \qquad K^2 x = \frac{2}{3} + \frac{13x}{36}.$$

Write $K^n x = A_n + B_n x$. Its coefficients obey

$$A_{n+1} = A_n + \frac{B_n}{2}, \qquad B_{n+1} = \frac{A_n}{2} + \frac{B_n}{3}, \quad (A_0, B_0) = (0, 1).$$

All coefficients are nonnegative, and $A_n \geq A_1 = 1/2$ for every $n \geq 1$. Thus $K^n x \geq 1/2$ on $[0,1]$ for $n \geq 1$: the terms do not even tend to zero.

The original part (a) cannot produce an ordinary convergent Neumann series for the equation as printed. The solution exists, but this particular expansion diverges. The manual's answer -2 does not address this obstruction. For clarity, introducing a parameter z gives the convergent resolvent expansion near zero:

$$\psi_z = x + zK\psi_z, \qquad \psi_z(x) = \frac{z/2 + (1-z)x}{1 - \frac{4}{3}z + \frac{1}{12}z^2}.$$

Indeed, this is the inverse of $I - z\begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{pmatrix}$ applied to $(0,1)^T$. Its Taylor series is the Neumann series and has nearest pole $z = 8 - 2\sqrt{13} < 1$. Evaluating the rational continuation at $z = 1$ gives -2, but that continuation is not the sum of the divergent series at $z = 1$.

(b) Separability forces $\psi = A + Bx$. Comparing coefficients gives

$$A = A + \frac{B}{2}, \qquad B = 1 + \frac{A}{2} + \frac{B}{3}.$$

The first equation gives $B = 0$ and the second then gives $A = -2$. Thus $\boxed{\psi(x) = -2}$.

(c) Guess a constant c. Substitution produces

$$c = x + c\int_0^1 (1 + xt)dt = c + x(1 + c/2).$$

The coefficient of x vanishes exactly when $c = -2$. Finally, $x + \int_0^1 (1 + xt)(-2)dt = x - 2 - x = -2$, an explicit check of the solution of the original equation.

21.3.3 Solve

$$\varphi(x) = 1 + \lambda^2 \int_0^x (x - t)\varphi(t)\, dt$$

by each of the following methods:

(a) Reduction to an ODE (find the boundary conditions),

(b) The Neumann series,

(c) The use of Laplace transforms.

ANS. $\varphi(x) = \cosh \lambda x$.

Solution.

(a) Differentiate the equation, including its zero endpoint term:

$$\varphi'(x) = \lambda^2 \int_0^x \varphi(t)dt, \qquad \varphi''(x) = \lambda^2 \varphi(x).$$

The initial data are $\varphi(0) = 1$, $\varphi'(0) = 0$. For $\lambda \neq 0$, write $\varphi = Ae^{\lambda x} + Be^{-\lambda x}$. Then $A + B = 1$ and $\lambda(A - B) = 0$, so $A = B = 1/2$. For $\lambda = 0$ the original equation directly gives $\varphi = 1$. In both cases

$$\boxed{\varphi(x) = \cosh(\lambda x)}.$$

(b) With J from Exercise 21.3.1, the equation is $\varphi = 1 + \lambda^2 J\varphi$. Starting with $J^0 1 = 1$,

$$J\left(\frac{t^{2n}}{(2n)!}\right) = \frac{x^{2n+2}}{(2n+2)!},$$

so its Neumann series is

$$\sum_{n=0}^{\infty} \lambda^{2n} J^n 1 = \sum_{n=0}^{\infty} \frac{(\lambda x)^{2n}}{(2n)!} = \cosh(\lambda x).$$

It converges uniformly on every finite x interval for every fixed complex λ.

(c) Since $\mathcal{L}\{1\} = 1/s$ and $\mathcal{L}\{x\} = 1/s^2$,

$$\Phi(s) = \frac{1}{s} + \frac{\lambda^2}{s^2}\Phi(s), \qquad \Phi(s) = \frac{s}{s^2 - \lambda^2}.$$

For $\lambda \neq 0$,

$$\frac{s}{s^2 - \lambda^2} = \frac{1}{2}\left(\frac{1}{s-\lambda} + \frac{1}{s+\lambda}\right),$$

whose inverse is the same hyperbolic cosine. At $\lambda = 0$ the transform is $1/s$. As a direct check, for $\lambda \neq 0$,

$$\int_0^x (x-t)\cosh(\lambda t)\,dt = \frac{\cosh(\lambda x) - 1}{\lambda^2};$$

the continuous limit is $x^2/2$, consistent with the zero-parameter case.

21.3.4

(a) In Eq. (21.59), take $V = V_0$, independent of t. Without using Eq. (21.60), show that Eq. (21.59) leads directly to

$$U(t - t_0) = \exp\left[-\frac{i}{\hbar}(t - t_0)V_0\right].$$

(b) Repeat for Eq. (21.60) without using Eq. (21.59).

For the power-series derivations, take V_0 to be a scalar, matrix, or bounded operator. State the qualification for an unbounded self-adjoint operator.

Solution.

(a) With $V(t) = V_0$, Eq. (21.59) is

$$U(t, t_0) = I - \frac{i}{\hbar}\int_{t_0}^{t} V_0 U(s, t_0)\,ds.$$

Differentiating gives

$$\partial_t U = -\frac{i}{\hbar} V_0 U, \qquad U(t_0, t_0) = I.$$

Put $\tau = t - t_0$ and $U = \sum_{n\geq 0} C_n \tau^n$. Equating powers gives

$$C_0 = I, \qquad (n+1)C_{n+1} = -\frac{i}{\hbar} V_0 C_n.$$

Induction yields $C_n = (-iV_0/\hbar)^n/n!$, so

$$\boxed{U(t, t_0) = \exp[-i(t - t_0)V_0/\hbar]}.$$

It depends only on the time difference because the constant-coefficient initial-value solution just derived has that form.

(b) Independently, Eq. (21.60) is the ordered integral series

$$U = I + \sum_{n=1}^{\infty} \left(-\frac{i}{\hbar}\right)^n \int_{t_0}^{t} ds_1 \int_{t_0}^{s_1} ds_2 \cdots \int_{t_0}^{s_{n-1}} ds_n \, V(s_1)V(s_2)\cdots V(s_n).$$

For constant V_0 the product is V_0^n. The remaining scalar simplex integral is $J_n(\tau) = \tau^n/n!$: $J_0 = 1$, and

$$J_{n+1}(\tau) = \int_0^{\tau} J_n(u)du = \int_0^{\tau} \frac{u^n}{n!}du = \frac{\tau^{n+1}}{(n+1)!}.$$

Substitution sums to precisely the same exponential, without using Eq. (21.59). Oriented integrals give the formula also for $\tau < 0$. For a scalar, matrix, or bounded operator V_0 both series converge on finite time intervals. For an unbounded self-adjoint quantum operator, the exponential is defined by its spectral calculus; an operator power series on every state cannot be assumed without domain and convergence conditions.

21.4 Hilbert–Schmidt Theory

21.4.1 In the Fredholm equation

$$\varphi(x) = \lambda \int_a^b K(x,t)\varphi(t)\, dt,$$

assume that the kernel $K(x,t)$ is self-adjoint or Hermitian:

$$K(x,t) = K^*(t,x).$$

Assume a Hilbert–Schmidt kernel, $\int_a^b\int_a^b |K(x,t)|^2\, dx\, dt < \infty$, and nonzero L^2 eigenfunctions. Extend the analysis of the present section to show that

(a) the eigenfunctions are orthogonal, in the sense that

$$\int_a^b \varphi_m^*(x)\varphi_n(x)\, dx = 0, \qquad m \neq n \ (\lambda_m \neq \lambda_n).$$

(b) the eigenvalues are real.

Solution.

Let $\langle u, v\rangle = \int_a^b u(x)^* v(x)dx$ and assume the kernel defines a self-adjoint operator on this L^2 space (a Hermitian Hilbert–Schmidt kernel is sufficient). The double-integral identity is

$$\begin{aligned}\langle \varphi_m, K\varphi_n\rangle &= \int_a^b\int_a^b \varphi_m(x)^* K(x,t)\varphi_n(t)\, dt\, dx \\ &= \int_a^b \left[\int_a^b K(t,x)\varphi_m(x)dx\right]^* \varphi_n(t)dt = \langle K\varphi_m, \varphi_n\rangle.\end{aligned}$$

A nonzero eigenfunction in the convention $\varphi_n = \lambda_n K\varphi_n$ has $\lambda_n \neq 0$. Consequently

$$\frac{1}{\lambda_n}\langle \varphi_m, \varphi_n\rangle = \frac{1}{\lambda_m^*}\langle \varphi_m, \varphi_n\rangle,$$

or

$$(\lambda_m^* - \lambda_n)\langle\varphi_m, \varphi_n\rangle = 0.$$

Take $m = n$. Since $\|\varphi_n\|^2 > 0$, this gives $\boxed{\lambda_n^* = \lambda_n}$, proving (b). Now for $\lambda_m \neq \lambda_n$ the factor is nonzero, so

$$\boxed{\langle\varphi_m, \varphi_n\rangle = 0},$$

which proves (a). Within a degenerate eigenspace orthogonality need not hold automatically, but Gram–Schmidt produces an orthonormal basis without changing its eigenvalue.

21.4.2

(a) Show that the eigenfunctions of Exercise 21.2.12 are orthogonal.

(b) Show that the eigenfunctions of Exercise 21.2.14 are orthogonal.

Solution.

(a) Use the exact eigenfunctions found in Exercise 21.2.12:

$$y_0 = x, \qquad y_+ = x^2 + \frac{1}{\sqrt{5}}, \qquad y_- = x^2 - \frac{1}{\sqrt{5}}.$$

Odd parity gives $\int_{-1}^{1} y_0 y_\pm dx = 0$. For the two even functions,

$$\int_{-1}^{1} y_+ y_- dx = \int_{-1}^{1} \left(x^4 - \frac{1}{5}\right) dx = \frac{2}{5} - \frac{2}{5} = 0.$$

All three distinct eigenfunctions are therefore mutually orthogonal.

(b) Write $\varphi_\pm = 1 + b_\pm x$, where $b_\pm = (-2 \pm \sqrt{13})/3$. Their sum and product are

$$b_+ + b_- = -\frac{4}{3}, \qquad b_+ b_- = -1.$$

It follows that

$$\int_0^1 \varphi_+ \varphi_- dx = 1 + \frac{b_+ + b_-}{2} + \frac{b_+ b_-}{3} = 1 - \frac{2}{3} - \frac{1}{3} = \boxed{0}.$$

Using the exact coefficients is necessary to see exact zero; rounded decimal eigenfunctions only approximate this cancellation.

21.4.3 Use the Hilbert–Schmidt method to solve the inhomogeneous integral equation

$$\varphi(x) = x + \frac{1}{2}\int_{-1}^{1} (t + x)\varphi(t)\, dt.$$

The corresponding homogeneous integral equation was treated in Example 21.2.3.

Note. The application of the Hilbert–Schmidt technique here is somewhat like using a shotgun to kill a mosquito, especially when the equation can be solved quickly by expanding in Legendre polynomials.

Solution.

For $K(x, t) = t + x$, choose normalized eigenfunctions

$$u_\pm(x) = \frac{1}{2}(1 \pm \sqrt{3}\, x).$$

Their norms are one, because $\int_{-1}^{1}(1+3x^2)dx/4 = 1$, and their mutual inner product is $\int_{-1}^{1}(1-3x^2)dx/4 = 0$. Evaluate the kernel action:

$$Ku_{\pm}(x) = \int_{-1}^{1}(t+x)\frac{1\pm\sqrt{3}\,t}{2}dt = x \pm \frac{1}{\sqrt{3}} = \pm\frac{2}{\sqrt{3}}u_{\pm}(x).$$

Thus the characteristic values are $\lambda_{\pm} = \pm\sqrt{3}/2$.

The inhomogeneous term $f(x) = x$ has coefficients

$$b_{\pm} = \langle u_{\pm}, x\rangle = \pm\frac{\sqrt{3}}{2}\frac{2}{3} = \pm\frac{1}{\sqrt{3}}.$$

For the requested coupling $\lambda = 1/2$, Hilbert–Schmidt projection gives

$$a_{\pm} = \frac{b_{\pm}}{1-\lambda/\lambda_{\pm}}.$$

Specifically,

$$a_{+} = \frac{1}{\sqrt{3}-1} = \frac{\sqrt{3}+1}{2}, \qquad a_{-} = -\frac{1}{\sqrt{3}+1} = \frac{1-\sqrt{3}}{2}.$$

Reconstructing,

$$\varphi = a_{+}u_{+} + a_{-}u_{-} = \frac{a_{+}+a_{-}}{2} + \frac{\sqrt{3}(a_{+}-a_{-})}{2}x = \boxed{\frac{1+3x}{2}}.$$

Directly $\int_{-1}^{1}\varphi(t)dt = 1$ and $\int_{-1}^{1}t\varphi(t)dt = 1$, so the right side of the original equation is $x+(1+x)/2 = (1+3x)/2$, as required.

21.4.4 The Fredholm integral equation

$$\varphi(x) = \lambda\int_{0}^{\infty} e^{-xt}\varphi(t)\,dt, \qquad x > 0,$$

has an infinite number of solutions, of which one is

$$\varphi(x) = x^{-1/2}, \qquad \lambda = \pi^{-1/2}.$$

Verify that this is a solution and that it is not normalizable.

Note. These are improper-integral solutions, not normalizable Hilbert–Schmidt eigenfunctions. The kernel fails the Hilbert–Schmidt integrability condition; an infinite interval alone would not establish that failure. Its Taylor series cannot be integrated term by term against the displayed power solution.

Solution.

For $x > 0$, set $u = xt$, so $dt = du/x$ and $t^{-1/2} = x^{1/2}u^{-1/2}$. Then

$$\int_{0}^{\infty} e^{-xt}t^{-1/2}dt = x^{-1/2}\int_{0}^{\infty} e^{-u}u^{-1/2}du = \sqrt{\pi}\,x^{-1/2}.$$

Multiplication by $\lambda = \pi^{-1/2}$ gives $\boxed{\varphi(x) = x^{-1/2}}$, verifying the equation.

Its truncated squared norm is

$$\int_\epsilon^R |\varphi(x)|^2 dx = \int_\epsilon^R \frac{dx}{x} = \log R - \log \epsilon.$$

It diverges as $R \to \infty$ and as $\epsilon \downarrow 0$. No nonzero constant multiple is normalizable on $(0, \infty)$.

This is a pointwise improper-integral solution for $x > 0$, not an L^2 eigenfunction to which the chapter's compact Hilbert–Schmidt conclusions automatically apply. The obstruction to that particular hypothesis can be checked directly:

$$\int_0^\infty \int_0^\infty |e^{-xt}|^2 dx\, dt = \int_0^\infty \frac{dt}{2t} = \infty.$$

Thus the kernel is not Hilbert–Schmidt; this calculation identifies the failed hypothesis more precisely than the infinite interval alone. Nor can its Taylor series be integrated term by term against $t^{-1/2}$: the coefficient of $(-x)^m/m!$ would contain $\int_0^\infty t^{m-1/2} dt$, which diverges for every integer $m \geq 0$.

The statement about infinitely many improper-integral solutions can also be verified explicitly. For $0 < a < 1$, the gamma integral gives

$$K(t^{-a})(x) = \Gamma(1-a)x^{a-1}, \qquad K(t^{a-1})(x) = \Gamma(a)x^{-a}.$$

Put $A = \sqrt{\Gamma(a)}$, $B = \sqrt{\Gamma(1-a)}$, and $\mu = AB$. Since $\Gamma(a) = A^2$ and $\Gamma(1-a) = B^2$,

$$K(At^{-a} \pm Bt^{a-1})(x) = AB^2 x^{a-1} \pm BA^2 x^{-a} = \pm\mu(Ax^{-a} \pm Bx^{a-1}).$$

Therefore

$$\varphi_{a,\pm}(x) = \sqrt{\Gamma(a)}x^{-a} \pm \sqrt{\Gamma(1-a)}x^{a-1}, \qquad \lambda_{a,\pm} = \pm\sqrt{\frac{\sin \pi a}{\pi}},$$

where the reflection formula gives $\mu^2 = \pi / \sin \pi a$. The minus combination at $a = 1/2$ is zero and must be excluded; the plus combination there is a nonzero multiple of the requested $x^{-1/2}$. These power combinations belong to the improper-integral solution class, not to the normalizable L^2 eigenfunctions.

21.4.5 Given

$$y(x) = x + \lambda \int_0^1 xt\, y(t)\, dt :$$

(a) Determine $y(x)$ as a Neumann series.

(b) Find the range of λ for which your Neumann series solution is convergent. Compare with the value obtained from

$$|\lambda| \cdot |K|_{\max} < 1.$$

(c) Find the eigenvalue and the eigenfunction of the corresponding homogeneous integral equation.

(d) By the separable-kernel method show that the solution is

$$y(x) = \frac{3x}{3-\lambda}, \qquad \lambda \neq 3.$$

(e) Find $y(x)$ by the Hilbert–Schmidt method. Also decide whether a solution exists at $\lambda = 3$.

Solution.

Let $(Kg)(x) = x\int_0^1 t\,g(t)dt$. Its action on the inhomogeneous term is $Kx = x/3$, so induction gives $K^n x = x/3^n$.

(a) The Neumann series is therefore

$$\boxed{y(x) = x\sum_{n=0}^{\infty}(\lambda/3)^n}.$$

(b) Apart from the identically zero value at $x = 0$, this geometric series converges precisely when $\boxed{|\lambda| < 3}$. For $|\lambda| = 3$ its terms do not tend to zero, and for larger modulus they grow. On $[0,1]^2$, $|K|_{\max} = 1$ and the interval length is one, so the stated sufficient bound only gives $|\lambda| < 1$. It is sufficient, not necessary.

(c) A nonzero homogeneous solution lies in the span of x. Putting $y = Cx$ gives $Cx = \lambda Cx/3$, hence

$$\boxed{\lambda_1 = 3, \qquad y_1(x) = Cx, \quad C \neq 0}.$$

(d) For the inhomogeneous equation put $c = \int_0^1 ty(t)dt$, so $y = x(1 + \lambda c)$. Taking its moment,

$$c = \frac{1+\lambda c}{3}, \qquad (3-\lambda)c = 1.$$

For $\lambda \neq 3$,

$$\boxed{y(x) = x\left(1 + \frac{\lambda}{3-\lambda}\right) = \frac{3x}{3-\lambda}}.$$

The formula excludes $\lambda = 3$. At that value the moment equation becomes $0 = 1$, so the inhomogeneous equation has no solution. The rational solution at other λ exists even when its original Neumann series diverges.

(e) Normalize the homogeneous eigenfunction: $u(x) = \sqrt{3}\,x$, since $\int_0^1 u^2 dx = 1$. The coefficient of $f = x$ is $b = \int_0^1 u(x)x\,dx = 1/\sqrt{3}$. Hilbert–Schmidt projection gives

$$y(x) = \frac{b\lambda_1}{\lambda_1 - \lambda}u(x) = \frac{(1/\sqrt{3})3}{3-\lambda}\sqrt{3}\,x = \frac{3x}{3-\lambda}.$$

The pole is consistent with the nonzero overlap b at resonance.

21.4.6 In Exercise 21.2.11 it was found that the integral equation

$$\varphi(x) = \lambda\int_0^{2\pi}\cos(x-t)\varphi(t)\,dt$$

had (unnormalized) eigenfunctions $\cos x$ and $\sin x$, both with eigenvalue $\lambda_i = 1/\pi$. Show that the kernel of this integral equation has an expansion of the form

$$K(x,t) = \sum_{n=1}^{2}\frac{\varphi_n(x)\varphi_n(t)}{\lambda_n},$$

where φ_n in this expansion are the unit-norm versions of the stated eigenfunctions.

Solution.

The norms of the stated functions are

$$\int_0^{2\pi} \cos^2 x\, dx = \int_0^{2\pi} \sin^2 x\, dx = \pi,$$

and their mutual integral is zero. Therefore use

$$u_1(x) = \frac{\cos x}{\sqrt{\pi}}, \qquad u_2(x) = \frac{\sin x}{\sqrt{\pi}}.$$

With $\lambda_1 = \lambda_2 = 1/\pi$,

$$\sum_{n=1}^{2} \frac{u_n(x)u_n(t)}{\lambda_n} = \pi\left[\frac{\cos x \cos t}{\pi} + \frac{\sin x \sin t}{\pi}\right]$$
$$= \cos x \cos t + \sin x \sin t = \boxed{\cos(x-t) = K(x,t)}.$$

The functions used in the displayed kernel expansion must be normalized. Substituting the initially stated unnormalized $\cos x, \sin x$ would instead give $\pi K(x,t)$. Equivalently, retain unnormalized functions but divide each term by its squared norm as well as by its characteristic value.

21.4.7 The integral equation $\varphi(x) = \lambda \int_0^1 (1 + xt)\varphi(t)\, dt$

has eigenvalues $\lambda_{1,2} = 8 \mp 2\sqrt{13}$ (approximately 0.7889 and 15.211). The corresponding unnormalized eigenfunctions are $\varphi_{1,2}(x) = 1 + (-2 \pm \sqrt{13})x/3$, approximately $1 + 0.5352x$ and $1 - 1.8685x$.

(a) Show that these eigenfunctions are orthogonal over the interval $[0, 1]$.

(b) Normalize the eigenfunctions to unity.

(c) Using the normalized functions from part (b), show that

$$K(x,t) = \frac{\varphi_1(x)\varphi_1(t)}{\lambda_1} + \frac{\varphi_2(x)\varphi_2(t)}{\lambda_2}.$$

ANS. (b) $\varphi_1(x) = 0.7831 + 0.4191x$,
$\varphi_2(x) = 1.8403 - 3.4386x$.

Solution.

Use the exact values behind the displayed decimals:

$$b_\pm = \frac{-2 \pm \sqrt{13}}{3}, \qquad \lambda_\pm = 8 \mp 2\sqrt{13}, \qquad v_\pm(x) = 1 + b_\pm x.$$

The supplied decimal eigenpairs are approximations, so exact orthogonality and exact kernel reconstruction use these radical values. After normalization, the eigenfunctions in part (c) are the unit-norm ones, not the original unnormalized linear functions.

(a) Since $b_+ + b_- = -4/3$ and $b_+ b_- = -1$,

$$\int_0^1 v_+(x)v_-(x)dx = 1 + \frac{b_+ + b_-}{2} + \frac{b_+ b_-}{3} = 0.$$

(b) Their squared norms are

$$N_\pm^2 = \int_0^1 (1 + b_\pm x)^2 dx = 1 + b_\pm + \frac{b_\pm^2}{3}.$$

The root equation $3b_\pm^2 + 4b_\pm - 3 = 0$ reduces this to

$$N_\pm^2 = \frac{4}{3} + \frac{5b_\pm}{9} = \frac{26 \pm 5\sqrt{13}}{27}.$$

Thus the normalized functions are

$$\boxed{u_\pm(x) = \sqrt{\frac{27}{26 \pm 5\sqrt{13}}} \left(1 + \frac{-2 \pm \sqrt{13}}{3} x\right).}$$

Their numerical forms are

$$u_+(x) = 0.7831024897 + 0.4191037337x, \qquad u_-(x) = 1.8403126068 - 3.4386555600x.$$

(c) Put $A_\pm = 1/(\lambda_\pm N_\pm^2)$. Rationalizing the radicals gives

$$A_\pm = \frac{1}{2} \pm \frac{1}{\sqrt{13}}.$$

In the sum of kernel terms, the constant, $x + t$, and xt coefficients are respectively $\sum A_\pm$, $\sum A_\pm b_\pm$, and $\sum A_\pm b_\pm^2$. Evaluate each:

$$\sum A_\pm = 1, \qquad \sum A_\pm b_\pm = \frac{1}{2}\left(-\frac{4}{3}\right) + \frac{1}{\sqrt{13}} \frac{2\sqrt{13}}{3} = 0,$$

and, using $b_\pm^2 = 1 - \frac{4}{3} b_\pm$,

$$\sum A_\pm b_\pm^2 = 1 - \frac{4}{3}(0) = 1.$$

Consequently

$$\boxed{\frac{u_+(x)u_+(t)}{\lambda_+} + \frac{u_-(x)u_-(t)}{\lambda_-} = 1 + xt.}$$

21.4.8 An alternate form of the solution to the inhomogeneous integral equation, Eq. (21.81), is

$$\varphi(x) = f_0(x) + \sum_{i=1}^{\infty} \frac{b_i \lambda_i}{\lambda_i - \lambda} \varphi_i(x).$$

Here φ_i are orthonormal eigenfunctions for the nonzero operator eigenvalues, $b_i = \langle \varphi_i, f \rangle$, f_0 is the projection of f onto $\ker K$, and $\lambda \neq \lambda_i$. The original pure sum applies when $f_0 = 0$.

(a) Derive this form without using Eq. (21.89).

(b) Show that this form and Eq. (21.89) are equivalent.

Solution.

Let K be the self-adjoint compact integral operator in the chapter, and let its normalized nonzero-eigenvalue functions satisfy $K\varphi_i = \varphi_i/\lambda_i$.

(a) Decompose both functions into their nullspace components and their components along the nonzero-eigenvalue eigenfunctions:

$$f = f_0 + \sum_i b_i\varphi_i, \qquad \varphi = \varphi_0 + \sum_i a_i\varphi_i.$$

The projection of $K\varphi$ onto $\ker K$ is zero, since K is self-adjoint: for $h \in \ker K$, $\langle h, K\varphi\rangle = \langle Kh, \varphi\rangle = 0$. Projecting the integral equation onto that nullspace gives $\varphi_0 = f_0$. Project $\varphi = f + \lambda K\varphi$ onto φ_i:

$$a_i = b_i + \frac{\lambda}{\lambda_i}a_i, \qquad (\lambda_i - \lambda)a_i = b_i\lambda_i.$$

For $\lambda \neq \lambda_i$ this gives

$$\boxed{\varphi(x) = f_0(x) + \sum_i \frac{b_i\lambda_i}{\lambda_i - \lambda}\varphi_i(x)},$$

without using Eq. (21.89).

(b) Split the scalar factor:

$$\frac{\lambda_i}{\lambda_i - \lambda} = 1 + \frac{\lambda}{\lambda_i - \lambda}.$$

Summing the first terms gives $f - f_0$; adding the explicit nullspace term restores f, so

$$\varphi(x) = f(x) + \lambda\sum_i \frac{\varphi_i(x)}{\lambda_i - \lambda}\underbrace{\int_a^b \varphi_i(t)^* f(t)dt}_{b_i}.$$

This is Eq. (21.89), with the conjugate included for a complex orthonormal basis.

The original textbook's pure eigenfunction sum assumes that f has no component in the nullspace of K. Nonzero-eigenvalue eigenfunctions need not span the whole L^2 space, for example for the finite-rank kernels in this chapter. In general an additional unchanged nullspace component is required. Thus Eq. (21.89) already retains f_0 in its explicit f term, while the original displayed pure sum required the extra condition $f_0 = 0$.

For a concrete check, take $K(x,t) = xt$ on $[0,1]$, $f = 1$, and $\lambda = 1$. Here $u = \sqrt{3}x$, $\lambda_1 = 3$, $b = \sqrt{3}/2$, and $f_0 = 1 - 3x/2$. The corrected formula gives

$$\varphi = f_0 + \frac{3b}{3-1}u = 1 - \frac{3}{2}x + \frac{9}{4}x = \boxed{1 + \frac{3}{4}x}.$$

Indeed $1 + x\int_0^1 t(1 + 3t/4)dt = 1 + 3x/4$. The sum without f_0 would give $9x/4$, which fails the equation.

For a compact self-adjoint operator the nonzero operator eigenvalues $\mu_i = 1/\lambda_i$ can accumulate only at zero. For a fixed finite λ, all but finitely many satisfy $|\lambda\mu_i| < 1/2$, so their multipliers obey $|1/(1 - \lambda\mu_i)| \leq 2$. The finitely many remaining denominators are nonzero away from resonance and therefore have a positive minimum modulus. Hence all multipliers are bounded, and Parseval's identity proves convergence of the solution expansion in L^2 whenever $f \in L^2$. At $\lambda = \lambda_j$, the projected equation instead requires $b_j = 0$ for every eigenfunction in that eigenspace; if so, its solution coefficients are free, and otherwise there is no solution.

22 Chapter 22. Calculus of Variations

Problems

22.1 Euler Equation

22.1.1 For $dy/dx \equiv y_x \neq 0$, show the equivalence of the two forms of Euler's equation:

$$\frac{\partial f}{\partial y} - \frac{d}{dx}\frac{\partial f}{\partial y_x} = 0$$

and

$$\frac{\partial f}{\partial x} - \frac{d}{dx}\left(f - y_x \frac{\partial f}{\partial y_x}\right) = 0.$$

Solution.

The two partial derivatives in the original textbook statements are interchanged. The correct equivalent forms are

$$\boxed{\frac{\partial f}{\partial y} - \frac{d}{dx}\frac{\partial f}{\partial y_x} = 0}, \qquad \boxed{\frac{\partial f}{\partial x} - \frac{d}{dx}\left(f - y_x \frac{\partial f}{\partial y_x}\right) = 0}.$$

To prove the equivalence, put $p = y_x$, so $p' = y_{xx}$. The chain rule gives

$$\frac{df}{dx} = f_x + f_y p + f_p p'.$$

Therefore

$$\begin{aligned}\frac{d}{dx}(f - pf_p) &= f_x + f_y p + f_p p' - p' f_p - p\frac{df_p}{dx} \\ &= f_x + p\left(f_y - \frac{df_p}{dx}\right),\end{aligned}$$

and consequently

$$f_x - \frac{d}{dx}(f - pf_p) = -p\left(f_y - \frac{df_p}{dx}\right).$$

For $p \neq 0$, either corrected expression vanishes exactly when the other does. At $p = 0$ the second can vanish without implying the first.

The printed versions are not generally equivalent. For $f = y$ and $y(x) = 2x$, the first printed expression is $f_x - df_p/dx = 0$, whereas the second is $f_y - d(f - pf_p)/dx = 1 - 2 = -1$. This also shows why the stated nonzero-slope condition does not repair the swapped derivatives.

22.1.2 Derive Euler's equation by expanding the integrand of

$$J(\alpha) = \int_{x_1}^{x_2} f\big(y(x,\alpha), y_x(x,\alpha), x\big)\, dx$$

in powers of α. Keep both endpoint values fixed and assume f is C^3 near the smooth trial curves, so the stated Taylor remainder is valid.

Note. The stationary condition is $\partial J(\alpha)/\partial\alpha = 0$, evaluated at $\alpha = 0$. The terms quadratic in α may be useful in establishing the nature of the stationary solution (maximum, minimum, or saddle point).

Solution.

Choose $y(x,\alpha) = y(x) + \alpha\eta(x)$ with $\eta(x_1) = \eta(x_2) = 0$, so $y_x(x,\alpha) = y'(x) + \alpha\eta'(x)$. Taylor expansion about $\alpha = 0$ gives

$$f(y+\alpha\eta, y'+\alpha\eta', x) = f + \alpha(f_y\eta + f_{y_x}\eta') + \frac{\alpha^2}{2}\left(f_{yy}\eta^2 + 2f_{yy_x}\eta\eta' + f_{y_xy_x}(\eta')^2\right) + O(\alpha^3).$$

All derivatives on the right are evaluated on the unperturbed curve. Integrating the linear term by parts,

$$J'(0) = \int_{x_1}^{x_2}(f_y\eta + f_{y_x}\eta')dx = [f_{y_x}\eta]_{x_1}^{x_2} + \int_{x_1}^{x_2}\left(f_y - \frac{d}{dx}f_{y_x}\right)\eta\,dx.$$

The boundary term is zero. Because the interior variation is arbitrary, stationarity gives

$$\boxed{f_y - \frac{d}{dx}f_{y_x} = 0}.$$

The coefficient of α^2 is one half of

$$J''(0) = \int_{x_1}^{x_2}\left(f_{yy}\eta^2 + 2f_{yy_x}\eta\eta' + f_{y_xy_x}(\eta')^2\right)dx.$$

Its sign must be tested over all admissible variations; the vanishing first variation alone does not classify the extremum.

22.1.3 Find the Euler equation corresponding to Eq. (22.14) if $f = f(y_{xx}, y_x, y, x)$, assuming that y and y_x have fixed values at the endpoints of their interval of definition.

ANS. $\dfrac{d^2}{dx^2}\left(\dfrac{\partial f}{\partial y_{xx}}\right) - \dfrac{d}{dx}\left(\dfrac{\partial f}{\partial y_x}\right) + \dfrac{\partial f}{\partial y} = 0.$

Solution.

Write $y_\alpha = y + \alpha\eta$. Fixing both y and y' at each endpoint requires $\eta = \eta' = 0$ there. The first variation is

$$\delta J = \int_{x_1}^{x_2}(f_y\eta + f_{y_x}\eta' + f_{y_{xx}}\eta'')dx.$$

The first derivative term gives

$$\int f_{y_x}\eta'dx = [f_{y_x}\eta] - \int \frac{d}{dx}f_{y_x}\,\eta\,dx.$$

For the second derivative term, integrate twice:

$$\int f_{y_{xx}}\eta''dx = [f_{y_{xx}}\eta'] - \int \frac{d}{dx}f_{y_{xx}}\eta'dx = [f_{y_{xx}}\eta' - (df_{y_{xx}}/dx)\eta] + \int \frac{d^2}{dx^2}f_{y_{xx}}\eta\,dx.$$

Every displayed boundary term vanishes. Thus

$$\delta J = \int_{x_1}^{x_2}\left[f_y - \frac{d}{dx}f_{y_x} + \frac{d^2}{dx^2}f_{y_{xx}}\right]\eta\,dx.$$

The fundamental lemma yields

$$\boxed{\frac{d^2}{dx^2}\frac{\partial f}{\partial y_{xx}} - \frac{d}{dx}\frac{\partial f}{\partial y_x} + \frac{\partial f}{\partial y} = 0}.$$

22.1.4 The integrand $f(y, y_x, x)$ of Eq. (22.2) has the form

$$f(y, y_x, x) = f_1(x, y) + f_2(x, y)y_x.$$

(a) Show that the Euler equation leads to

$$\frac{\partial f_1}{\partial y} - \frac{\partial f_2}{\partial x} = 0.$$

(b) For C^1 functions f_1, f_2, distinguish this equality along one stationary curve from an identity on a simply connected region. Under which condition is J independent of the path?

Solution.

(a) For $f = f_1(x, y) + f_2(x, y)y'$,

$$f_y = f_{1y} + f_{2y}y', \qquad f_{y'} = f_2, \qquad \frac{d}{dx}f_{y'} = f_{2x} + f_{2y}y'.$$

The y' terms cancel in Euler's equation, leaving

$$\boxed{f_{1y} - f_{2x} = 0 \quad \text{along a stationary curve}}.$$

(b) The integral can be written

$$J = \int (f_1\,dx + f_2\,dy).$$

If the equality $f_{1y} = f_{2x}$ holds identically on a simply connected region, this differential form is exact: there is a potential F with $F_x = f_1$, $F_y = f_2$. Then

$$\boxed{J = F(x_2, y_2) - F(x_1, y_1)}$$

is independent of the path in that region.

An Euler equation satisfied only on one curve does not imply path independence throughout the region. The identity on the region, together with an appropriate topology, is needed for that conclusion. For example, $f_1 = y^2$, $f_2 = 0$ has stationary curve $y = 0$, but $\int y^2 dx$ changes when the interior path changes. This example also anticipates the correction required in the next exercise.

22.1.5 Show that the condition that $J = \int f(x, y)\,dx$ has a stationary value

(a) leads to $f_y(x, y(x)) = 0$ along the stationary curve and

(b) imposes no separate condition $f_x = 0$.

Dependence on derivatives of y is not essential for a meaningful variational problem. Give a smooth minimizing solution for an integrand that depends on y.

Note. The situation will change when constraints are introduced (compare to Exercise 22.4.6).

Solution.

There is no y' dependence, so the first variation is simply

$$\delta J = \int_{x_1}^{x_2} f_y(x, y(x))\eta(x)\,dx.$$

Consequently the correct stationarity condition is the algebraic equation

$$\boxed{f_y(x, y(x)) = 0 \quad \text{for every interior } x}.$$

This is a condition along the selected curve. It does not force $f(x, y)$ to be independent of y, and absence of derivative dependence does not rule out meaningful, smooth minimizing solutions. The original textbook conclusion in part (a) and the following no-solution statement are false.

For an explicit smooth counterexample take

$$f(x, y) = (y - \sin x)^2$$

and fix the endpoint values to those of $\sin x$. Then $f_y = 2(y - \sin x)$, so the stationary function is $\boxed{y(x) = \sin x}$. Moreover,

$$J[y] = \int_{x_1}^{x_2} (y - \sin x)^2 dx \geq 0, \qquad J[\sin x] = 0,$$

which proves a global minimum despite explicit dependence on y.

If f is independent of y, every path has the same J. If f_y never vanishes, no stationary path exists. Otherwise, smooth root branches of $f_y = 0$ may furnish stationary paths, subject to the endpoint conditions. Regarding (b), there is no additional equation $f_x = 0$: x is the independent variable. But x can affect the selected curve through $f_y(x, y) = 0$, as the counterexample demonstrates.

22.1.6 A soap film stretched between two rings of unit radius centered at $\pm x_0$ will have its closest approach to the x-axis at $x = 0$, with the distance from the axis given by c_1, with x_0 and c_1 related by Eq. (22.26) or Eq. (22.29).

(a) Show that dc_1/dx_0 becomes infinite when $x_0 \sinh(x_0/c_1) = 1$, indicating that the soap film becomes unstable if x_0 is increased beyond the value satisfying this condition.

(b) Show that the condition of part (a) is equivalent to

$$\frac{x_0}{c_1} = \coth \frac{x_0}{c_1}.$$

(c) Solve the transcendental equation of part (b) to obtain the critical value of x_0/c_1 and show that the separate values of x_0 and c_1 are then approximately $x_0 \approx 0.6627$ and $c_1 \approx 0.5524$.

Solution.

Put $c = c_1 > 0$ and $u = x_0/c$. The boundary equation is $c \cosh u = 1$.

(a) Implicit differentiation with respect to x_0 gives

$$c' \cosh u + c \sinh u \left(\frac{1}{c} - \frac{x_0 c'}{c^2} \right) = 0,$$

hence

$$c'(\cosh u - u \sinh u) = -\sinh u, \qquad \boxed{\frac{dc}{dx_0} = -\frac{c \sinh u}{1 - x_0 \sinh u}}.$$

The numerator is nonzero for separated rings. Thus the derivative diverges at $x_0 \sinh u = 1$.

(b) Divide that condition by $c\cosh u = 1$:

$$u\tanh u = 1, \qquad \boxed{u = \coth u}.$$

Equivalently, parametrize the two branches by

$$c = \operatorname{sech} u, \qquad x_0 = u\operatorname{sech} u.$$

Then

$$\frac{dx_0}{du} = \operatorname{sech} u(1 - u\tanh u).$$

Because $u\tanh u$ increases strictly from zero to infinity, there is one critical point, a maximum. Beyond it no positive-neck catenoid satisfies the two boundary conditions.

(c) Newton iteration for $F(u) = u\tanh u - 1$ is

$$u_{j+1} = u_j - \frac{u_j\tanh u_j - 1}{\tanh u_j + u_j\operatorname{sech}^2 u_j}.$$

Starting at $u_0 = 1.2$ gives

$$\boxed{u_c = 1.19967864025773, \quad c_c = 0.552434124530883, \quad x_{0c} = 0.662743419349182}.$$

The manual's $c_c = 0.5523434$ transposes digits; substituting $c_c = \operatorname{sech} u_c$ gives the value above, which agrees with the textbook's rounded 0.5524.

The stability threshold can also be seen in the second variation. For $f = y\sqrt{1+p^2}$, evaluated at $y = c\cosh(x/c)$ and $p = \sinh(x/c)$, the Hessian entries are

$$f_{yy} = 0, \qquad f_{yp} = \frac{p}{\sqrt{1+p^2}} = \tanh(x/c), \qquad f_{pp} = \frac{y}{(1+p^2)^{3/2}} = c\operatorname{sech}^2(x/c).$$

Thus, for a fixed-endpoint variation,

$$\delta^2 J = \int_{-x_0}^{x_0} \left[2\tanh(x/c)\eta\eta_x + c\operatorname{sech}^2(x/c)\eta_x^2\right] dx.$$

The first term is $\tanh(x/c)(\eta^2)_x$. Integration by parts has zero endpoint term and leaves $-c^{-1}\int \operatorname{sech}^2(x/c)\eta^2 dx$. Now $x = cu$, $dx = c\,du$, and $\eta_x = \eta_u/c$ give

$$\delta^2 J = \int_{-u_0}^{u_0} \operatorname{sech}^2 u\,(\eta_u^2 - \eta^2) du.$$

The function $w = \cosh u - u\sinh u$ satisfies $(\operatorname{sech}^2 u\, w')' + \operatorname{sech}^2 u\, w = 0$. For $u_0 < u_c$, $w > 0$. Put $p(u) = \operatorname{sech}^2 u$ and $\eta = wv$. Expanding the square and using $(pw')' = -pw$ gives

$$p(\eta'^2 - \eta^2) = pw^2v'^2 + (pww'v^2)'.$$

The boundary term is zero because $\eta = 0$ and $w > 0$ there. Consequently

$$\delta^2 J = \int_{-u_0}^{u_0} \operatorname{sech}^2 u\, w^2(v')^2 du > 0$$

for a nonzero fixed-endpoint variation. At $u_0 = u_c$, w itself vanishes at the endpoints and gives a zero second variation. This identifies the loss of strict stability on the shallow branch.

22.1.7 A soap film is stretched across the space between two rings of unit radius centered at $\pm x_0$ on the x-axis and perpendicular to the x-axis. Using the solution developed in Example 22.1.3, set up the transcendental equations for the condition that x_0 is such that the area of the curved surface of rotation equals the area of the two rings (Goldschmidt discontinuous solution). Solve for x_0.

Solution.

For $y = c\cosh(x/c)$ between $x = \pm x_0$, put $u = x_0/c$. The endpoint condition is $c\cosh u = 1$. Since $y' = \sinh(x/c)$, the full surface area is

$$\begin{aligned} A &= 2\pi \int_{-x_0}^{x_0} y\sqrt{1+y'^2}\,dx = 4\pi c^2 \int_0^u \cosh^2 v\,dv \\ &= \pi c^2[\sinh(2u) + 2u]. \end{aligned}$$

The two unit disks have area 2π. Equating the areas and using $c^2 = \operatorname{sech}^2 u$ gives

$$\sinh(2u) + 2u = 2\cosh^2 u = 1 + \cosh(2u),$$

or the simpler single equation

$$\boxed{2u - 1 = e^{-2u}}.$$

Its left-minus-right derivative is $2 + 2e^{-2u} > 0$, so it has exactly one positive root. Newton iteration, or bisection on $(0.6, 0.7)$, gives

$$u = 0.639232271380537, \qquad c = \operatorname{sech} u = 0.825517453652511.$$

Thus the required half-separation is

$$\boxed{x_0 = u\operatorname{sech} u = 0.527697396962572}.$$

The full ring separation is $2x_0 = 1.05539479392514$. The root is below u_c from Exercise 22.1.6, so the equality occurs on the shallow catenoid branch.

22.1.8 In Example 22.1.1, expand $J[y(x,\alpha)] - J[y(x,0)]$ in powers of α. The term linear in α leads to the Euler equation and to the straight-line solution, Eq. (22.16). Investigate the α^2 term and show that the stationary value of J, the straight-line distance, is a minimum.

Solution.

Let $y_*(x) = y_1 + m(x - x_1)$, where $m = (y_2 - y_1)/(x_2 - x_1)$ and $x_2 > x_1$. For $y_\alpha = y_* + \alpha\eta$, with $\eta(x_1) = \eta(x_2) = 0$,

$$J[y_\alpha] = \int_{x_1}^{x_2} \sqrt{1 + (m + \alpha\eta')^2}\,dx.$$

The first two derivatives of $g(p) = \sqrt{1+p^2}$ are $g'(p) = p/\sqrt{1+p^2}$ and $g''(p) = (1+p^2)^{-3/2}$. Therefore

$$J[y_\alpha] - J[y_*] = \alpha\frac{m}{\sqrt{1+m^2}}\int_{x_1}^{x_2}\eta' dx + \frac{\alpha^2}{2(1+m^2)^{3/2}}\int_{x_1}^{x_2}(\eta')^2 dx + O(\alpha^3).$$

The linear integral is $\eta(x_2) - \eta(x_1) = 0$. The quadratic coefficient is strictly positive for any nonzero admissible η: zero would require $\eta' = 0$, and the endpoints would then require $\eta = 0$. Thus the straight line is a strict local minimum.

It is also the global minimum, since strict convexity gives

$$g(m + \eta') \geq g(m) + g'(m)\eta'.$$

After integration the last term again vanishes, so $J[y_* + \eta] \geq J[y_*]$, with equality only for $\eta = 0$.

22.1.9 Take $x_1 < x_2$, real-valued paths, and fixed endpoint values. In (b), allow endpoint jumps, and compare with the continuous class.

(a) Show that the integral

$$J = \int_{x_1}^{x_2} f(y, y_x, x)\, dx, \quad \text{with} \quad f = y(x),$$

has no extreme values.

(b) If $f(y, y_x, x) = y^2(x)$, find a discontinuous solution similar to the Goldschmidt solution for the soap-film problem.

Solution.

(a) Let η be a smooth function of compact support in (x_1, x_2) with $\int \eta\, dx \neq 0$. For $f = y$,

$$J[y + \alpha\eta] - J[y] = \alpha \int_{x_1}^{x_2} \eta(x) dx.$$

Arbitrarily small positive and negative α change J in opposite directions. Thus no admissible curve is even a local extremum. Taking $|\alpha| \to \infty$ also shows that J is unbounded above and below.

(b) For $f = y^2$, nonnegativity gives $J \geq 0$. In a class allowing endpoint jumps, set

$$y(x) = 0 \quad (x_1 < x < x_2),$$

while assigning the prescribed values at the two endpoints. Changing isolated points does not change the integral, so this discontinuous path has $\boxed{J_{\min} = 0}$.

If continuity is required and either endpoint value is nonzero, this is an infimum, not an attained minimum. If both endpoint values are zero, the ordinary smooth function $y = 0$ is already the minimum. To see the infimum explicitly, let the endpoint values be A, B. Join A to zero within a boundary layer of width ϵ, keep zero in the interior, and join zero to B in the last ϵ. Smooth profiles bounded respectively by $|A|, |B|$ give

$$0 \leq J[y_\epsilon] \leq \epsilon(A^2 + B^2) \longrightarrow 0.$$

Any continuous function with a nonzero endpoint is nonzero on an interval of positive length, so its integral of y^2 is strictly positive.

22.1.10 Fermat's principle of optics states that a light ray in a medium for which n is the (position-dependent) index of refraction will follow the path $y(x)$ for which

$$\int_{x_1, y_1}^{x_2, y_2} n(y, x)\, ds$$

is stationary. For $y_2 = y_1 = 1$, $-x_1 = x_2 = 1$, find the ray path and determine whether the minimum is attained if

(a) $n = e^y$,

(b) $n = a(y - y_0)$, $\quad y > y_0$, with $a > 0$ and $y_0 < 1$.

Solution.

Use $ds = \sqrt{1+y'^2}dx$. Since neither index depends explicitly on x, the Beltrami first integral is

$$f - y' f_{y'} = \frac{n(y)}{\sqrt{1+y'^2}} = C.$$

(a) For $n = e^y$, write $e^y = C\sec(x-b)$ on a branch where the cosine is positive. Indeed,

$$y = \log C - \log\cos(x-b), \quad y' = \tan(x-b),$$

so $e^y/\sqrt{1+y'^2} = C$. Equal endpoint values at $x = \pm 1$ give $b = 0$ on the branch covering the entire interval. The condition $y(1) = 1$ gives $C = e\cos 1$. Thus

$$\boxed{y(x) = 1 + \log\cos 1 - \log\cos x, \qquad -1 \le x \le 1}.$$

This is a global minimum, not just a stationary path. Set

$$U = e^y \cos x, \qquad V = e^y \sin x.$$

Direct differentiation gives $dU^2 + dV^2 = e^{2y}(dx^2 + dy^2)$. The optical length is therefore ordinary Euclidean length in the (U, V) plane. The endpoints are $(e\cos 1, \pm e\sin 1)$, and their straight segment has $U = e\cos 1$, precisely the curve above. Its length is $2e\sin 1$, the lower bound for any joining path.

The manual's inverse formula uses $1 + \pi^2/16$ where $\sec^2 1$ is required. At $y = 1$ it gives $x = \arctan(\pi/4)$ instead of $x = 1$, so it fails the stated endpoint condition.

(b) A positive index requires $a > 0$ and the endpoints require $H := 1 - y_0 > 0$. Put $u(x) = y(x) - y_0$. The first integral is

$$\frac{u}{\sqrt{1+u'^2}} = c > 0, \qquad u'^2 = \frac{u^2}{c^2} - 1.$$

Separate variables:

$$dx = \frac{c\,du}{\sqrt{u^2 - c^2}}, \qquad x - b = c\ \operatorname{arcosh}(u/c).$$

The two symmetric halves give

$$\boxed{y(x) = y_0 + c\cosh(x/c), \qquad c\cosh(1/c) = H}.$$

The horizontal shift b is zero because the endpoint heights are equal. With $w = 1/c$, its boundary equation is

$$\frac{1}{H} = w\operatorname{sech} w.$$

Exercise 22.1.6 shows that the maximum of the right side is 0.662743419349182. Consequently regular stationary curves exist only for

$$H \ge 1.50887956153832 \quad \text{or} \quad y_0 \le -0.50887956153832.$$

Above that threshold there are two branches; the smaller w, larger c branch is the locally minimizing one.

Part (b) has no single smooth minimizing ray for arbitrary $y_0 < 1$. Also, Fermat's variational statement is generally stationarity; an available stationary ray need not be the absolute minimum. The distinction can be quantified here. The catenary optical length is

$$J_{\mathrm{cat}} = ac^2\left[w + \frac{1}{2}\sinh(2w)\right], \qquad \frac{J_{\mathrm{cat}}}{aH^2} = w\operatorname{sech}^2 w + \tanh w.$$

To see the competing boundary cost, descend vertically from H to a small $\delta > 0$, travel the horizontal distance 2 at that height, and ascend to H. This piecewise path has

$$J_\delta = a\left(2\int_\delta^H u\,du + 2\delta\right) = a(H^2 - \delta^2 + 2\delta) \longrightarrow aH^2.$$

Steep smooth endpoint layers approximate its vertical portions, so the same limit is available within positive smooth graphs. Equality of the catenary cost with this limit occurs at the same root as Exercise 22.1.7:

$$2w - 1 = e^{-2w}, \qquad w = 0.639232271380537.$$

Thus the shallow catenary beats that boundary limit only for

$$H > 1.89502545541442.$$

For a direct global proof, let $d = \min u > 0$ on a competing path. Cauchy–Schwarz applied to the vectors $(d, \sqrt{u^2 - d^2})$ and $(1, |u'|)$ gives

$$u\sqrt{1 + u'^2} \geq d + \sqrt{u^2 - d^2}\,|u'|.$$

The path descends from H to d and returns to H, so integration over its horizontal span of length 2 yields

$$\frac{J}{a} \geq 2d + 2\int_d^H \sqrt{u^2 - d^2}\,du =: B(d) = 2d + H\sqrt{H^2 - d^2} - d^2\operatorname{arcosh}(H/d).$$

For a path with additional rises and falls, the absolute-value integral is only larger. Differentiation gives

$$B'(d) = 2 - 2d\operatorname{arcosh}(H/d), \qquad B(0^+) = H^2.$$

The product $d\operatorname{arcosh}(H/d)$ vanishes at both ends $d = 0, H$ and has a single maximum: putting $d = H\operatorname{sech} w$ reduces it to $Hw\operatorname{sech} w$. When two critical points exist, the smaller d is a local maximum of B and the larger d is its local minimum. At that minimum $d\operatorname{arcosh}(H/d) = 1$, exactly the shallow-catenary endpoint equation; equality in the preceding bound holds for that catenary. The endpoint $d = H$ cannot minimize B, since $B'(H^-) = 2$. Thus the global lower bound is the smaller of the shallow-catenary cost and H^2. The former is attained precisely for $H \geq 1.89502545541442\ldots$; at equality it ties the boundary limit. For smaller H, every $d > 0$ gives a strictly larger cost than H^2, so no smooth positive-index global minimizer exists. The infimum aH^2 is approached by the endpoint-layer paths described above. This also agrees with the catenoid problem after scaling both coordinates by H, for which the unit-ring half-separation is $1/H$.

In particular, the manual's claim of a global catenary minimum at $y_0 = -0.70$ is incorrect: $H = 1.70$ lies below the cost-crossing value 1.8950254554.

22.1.11 A particle moves, starting at rest, from point A on the surface of the Earth to point B (also on the surface) by sliding frictionlessly through a tunnel. Find the differential equation satisfied by the path if the transit time is to be a minimum. Assume the Earth to be a nonrotating sphere of uniform density. First take distinct nonantipodal endpoints, so $0 < r_0 < R$; treat the antipodal diameter as a limit.

Hint. The gain in kinetic energy of a particle of mass m a distance $r < R$ from the center of the Earth, with R the Earth's radius, is $\frac{1}{2}mg(R^2 - r^2)/R$, where g is the gravitational acceleration at the Earth's surface. It is convenient to describe the path of the particle (in the plane through A, B, and the center of the Earth) by plane polar coordinates (r, θ), with A at $(R, -\varphi)$ and B at (R, φ).

ANS. Letting r_0 be the minimum value of r (reached at $\theta = 0$), Eq. (22.21) yields $r_\theta^2 = \dfrac{r^2R^2(r^2 - r_0^2)}{r_0^2(R^2 - r^2)}$ (the constant in that equation has the value such that $r_\theta = 0$ at $\theta = 0$).

The solution for the path is a hypocycloid, generated by a circle of radius $\frac{1}{2}(R - r_0)$ rolling inside the circle of radius R. You might like to show that the transit time is

$$t = \pi \frac{(R^2 - r_0^2)^{1/2}}{(Rg)^{1/2}}.$$

For details see P. W. Cooper, *Am. J. Phys.* **34**: 68 (1966); G. Veneziano, et al., **34**: 701 (1966).

Solution.

Inside a uniform sphere the inward gravitational force is $F_r = -mgr/R$. Thus

$$U(r) - U(R) = \frac{mg}{2R}(r^2 - R^2).$$

The positive quantity in the printed hint is the *decrease* in potential energy from the surface, and hence the kinetic energy gained. It is not the potential energy itself with surface value zero. Its opposite sign would give an outward force. Energy conservation for release from rest gives

$$\frac{1}{2}mv^2 = \frac{mg}{2R}(R^2 - r^2), \qquad v = \sqrt{\frac{g}{R}}\sqrt{R^2 - r^2}.$$

Writing the path as $r(\theta)$, its travel time is

$$T = \sqrt{\frac{R}{g}} \int_{-\varphi}^{\varphi} \frac{\sqrt{r_\theta^2 + r^2}}{\sqrt{R^2 - r^2}}\, d\theta.$$

The integrand has no explicit θ dependence. With $p = r_\theta$, the Beltrami constant, omitting the constant $\sqrt{R/g}$, is

$$f - pf_p = \frac{r^2}{\sqrt{R^2 - r^2}\sqrt{r^2 + p^2}}.$$

At the minimum radius $r = r_0 > 0$, symmetry gives $p = 0$, so this constant is $r_0/\sqrt{R^2 - r_0^2}$. Squaring and rearranging,

$$r^4(R^2 - r_0^2) = r_0^2(R^2 - r^2)(r^2 + p^2),$$

$$\boxed{r_\theta^2 = \frac{r^2R^2(r^2 - r_0^2)}{r_0^2(R^2 - r^2)}}.$$

To verify the stated geometric identification, put $d = (R - r_0)/2$ and consider the hypocycloid

$$z(u) = (R - d)e^{iu} + d\, e^{-i(R-d)u/d}.$$

It is generated by a radius-d circle rolling within the radius-R circle. Set $\delta = Ru/d$. Direct products and derivatives give

$$r^2 = |z|^2 = R^2 - 4d(R - d)\sin^2(\delta/2),$$
$$\frac{dr}{du} = -\frac{R(R-d)\sin\delta}{r}, \qquad \frac{d\theta}{du} = \frac{(R-d)r_0(1-\cos\delta)}{r^2}.$$

Their squared ratio is

$$\left(\frac{dr}{d\theta}\right)^2 = \frac{R^2 r^2}{r_0^2}\cot^2(\delta/2) = \frac{R^2 r^2 (r^2 - r_0^2)}{r_0^2(R^2 - r^2)},$$

since $4d(R - d) = R^2 - r_0^2$. Rotating this arc puts its minimum radius at $\theta = 0$ and its two surface cusps at $\pm\varphi$.

The transit time can be evaluated without integrating the path explicitly. On either half,

$$\frac{d\theta}{dr} = \frac{r_0\sqrt{R^2 - r^2}}{Rr\sqrt{r^2 - r_0^2}}, \qquad \frac{ds}{dr} = \frac{r\sqrt{R^2 - r_0^2}}{R\sqrt{r^2 - r_0^2}}.$$

Consequently, with $\Delta = R^2 - r_0^2$,

$$T = 2\sqrt{\frac{R}{g}}\frac{\sqrt{\Delta}}{R}\int_{r_0}^{R} \frac{r\, dr}{\sqrt{r^2 - r_0^2}\sqrt{R^2 - r^2}}.$$

Set $r^2 = r_0^2 + \Delta\sin^2 v$; the integrand becomes dv and the limits become $0, \pi/2$. Hence

$$\boxed{T = \pi\sqrt{\frac{R^2 - r_0^2}{Rg}}}.$$

The same substitution in the angular integral gives

$$\varphi = \frac{r_0}{R}\left[R^2\int_0^{\pi/2} \frac{dv}{r_0^2\cos^2 v + R^2\sin^2 v} - \frac{\pi}{2}\right] = \frac{\pi}{2}\left(1 - \frac{r_0}{R}\right).$$

Here the integral equals $\pi/(2r_0R)$ after $\tan v$ substitution. The antipodal case is the limiting diameter $r_0 = 0$, with $T = \pi\sqrt{R/g}$; the displayed polar ODE assumes $r_0 > 0$.

22.1.12 A ray of light follows a straight-line path in a first homogeneous medium, is refracted at an interface, and then follows a new straight-line path in the second medium. See Fig. 22.7. Use Fermat's principle of optics to derive Snell's law of refraction:

$$n_1 \sin\theta_1 = n_2 \sin\theta_2.$$

Hint. Keep the points (x_1, y_1) and (x_2, y_2) fixed and vary x_0 to satisfy Fermat's principle.

Note. This is not an Euler equation problem, because the light path is not differentiable at x_0.

Solution.

Take the interface to be $y = 0$, with fixed endpoints (x_1, y_1) above it and (x_2, y_2) below it. The crossing point is $(x_0, 0)$. Up to the constant vacuum light speed, the optical travel time is

$$J(x_0) = n_1\sqrt{(x_0 - x_1)^2 + y_1^2} + n_2\sqrt{(x_2 - x_0)^2 + y_2^2}.$$

Differentiate only the crossing position:

$$J'(x_0) = n_1 \frac{x_0 - x_1}{\sqrt{(x_0 - x_1)^2 + y_1^2}} - n_2 \frac{x_2 - x_0}{\sqrt{(x_2 - x_0)^2 + y_2^2}}.$$

The two ratios are the signed sines of the angles to the interface normal. Thus $J'(x_0) = 0$ gives

$$\boxed{n_1 \sin\theta_1 = n_2 \sin\theta_2}.$$

For positive indices and endpoints off the interface,

$$J''(x_0) = \frac{n_1 y_1^2}{[(x_0 - x_1)^2 + y_1^2]^{3/2}} + \frac{n_2 y_2^2}{[(x_2 - x_0)^2 + y_2^2]^{3/2}} > 0.$$

This proves the stationary crossing is the unique minimum for these fixed endpoints. No differentiability across the corner was assumed.

22.1.13 A second soap-film configuration for the unit-radius rings at $x = \pm x_0$ consists of a circular disk, radius a, in the $x = 0$ plane and two catenoids of revolution, one joining the disk and each ring. One catenoid may be described by

$$y = c_1 \cosh\left(\frac{x}{c_1} + c_3\right).$$

(a) Impose boundary conditions at $x = 0$ and $x = x_0$.

(b) Require that the catenoids form an angle of 120° where they join the central disk. Express this third boundary condition mathematically. For films of equal surface tension and a freely moving junction, this is the equilibrium condition.

(c) Show that the total area of catenoids plus central disk is then

$$A = \pi c_1^2 \left[\sinh\left(\frac{2x_0}{c_1} + 2c_3\right) + \frac{2x_0}{c_1}\right].$$

Note. Stability depends on the branch and the allowed variations; see Exercise 22.1.14. The area is larger than that of the *shallow* simple catenoid whenever both exist.

ANS. (a) $\begin{cases} 1 = c_1 \cosh\left(\dfrac{x_0}{c_1} + c_3\right) \\ a = c_1 \cosh c_3 \end{cases}$ (b) $\dfrac{dy}{dx} = \tan 30^\circ = \sinh c_3.$

Solution.

(a) For the right-hand catenoid $y = c\cosh(x/c + c_3)$, the boundary equations are

$$\boxed{a = c\cosh c_3, \qquad 1 = c\cosh(x_0/c + c_3)}.$$

(b) At the junction, the central disk points inward in the negative radial direction. The outward right-catenoid unit tangent is $(1, y')/\sqrt{1 + y'^2}$. A 120° angle therefore requires

$$-\frac{y'}{\sqrt{1 + y'^2}} = \cos 120^\circ = -\frac{1}{2},$$

which gives $y' = 1/\sqrt{3} = \tan 30^\circ$. Since $y' = \sinh(x/c + c_3)$,

$$\boxed{\sinh c_3 = \frac{1}{\sqrt{3}}, \qquad c_3 = \operatorname{arsinh}(1/\sqrt{3}) = \tfrac{1}{2}\log 3, \qquad a = \frac{2c}{\sqrt{3}}}.$$

The same angle follows by varying the junction radius. In the reflection-symmetric class the area is $A = 4\pi \int_0^{x_0} y\sqrt{1+y'^2}\,dx + \pi y(0)^2$. After imposing the interior Euler equation, its first variation is

$$\delta A = \left[-4\pi a \frac{y'(0)}{\sqrt{1+y'(0)^2}} + 2\pi a\right] \delta a.$$

Since $a > 0$ and δa is free, this again gives $y'(0)/\sqrt{1+y'(0)^2} = 1/2$. Thus the angle is necessary for this free-junction equilibrium, although one can join mathematical surface pieces at other prescribed angles.

(c) Put $v = x_0/c + c_3$. There are two catenoids and one disk. Their total area is

$$\begin{aligned} A &= 4\pi \int_0^{x_0} y\sqrt{1+y'^2}\,dx + \pi a^2 \\ &= 4\pi c^2 \int_{c_3}^{v} \cosh^2 u\,du + \pi c^2 \cosh^2 c_3 \\ &= \pi c^2[2(v - c_3) + \sinh 2v - \sinh 2c_3 + \cosh^2 c_3]. \end{aligned}$$

The angle condition gives $\cosh c_3 = 2/\sqrt{3}$, so $\sinh 2c_3 = 4/3 = \cosh^2 c_3$. These last terms cancel:

$$\boxed{A = \pi c^2 \left[\sinh\left(\frac{2x_0}{c} + 2c_3\right) + \frac{2x_0}{c}\right]}.$$

The printed expression is missing the overall factor π. It equals A/π, not the total physical area. The surface-of-revolution factor $2\pi y\,ds$ and the disk area πa^2 fix the normalization.

22.1.14 For the soap film described in Exercise 22.1.13, find (numerically) the maximum value of x_0.

Note. This calls for a calculator with hyperbolic functions or a table of hyperbolic cotangents.

ANS. $x_{0\,\max} = 0.4078$.

Solution.

Retain the angle condition from Exercise 22.1.13: $c_3 = \frac{1}{2}\log 3$. Write

$$v = \frac{x_0}{c} + c_3, \qquad c = \operatorname{sech} v, \qquad x_0 = (v - c_3)\operatorname{sech} v, \quad v > c_3.$$

Differentiate:

$$\frac{dx_0}{dv} = \operatorname{sech} v[1 - (v - c_3)\tanh v].$$

Thus the maximum satisfies

$$\boxed{v - c_3 = \coth v}.$$

The function $v - c_3 - \coth v$ has derivative $1 + \operatorname{csch}^2 v > 0$, is negative at $v = c_3$, and tends to infinity. It has exactly one root. Moreover x_0 tends to zero at both ends of the parameter range, so this root is the global maximum.

Newton's update is

$$v_{j+1} = v_j - \frac{v_j - c_3 - \coth v_j}{1 + \operatorname{csch}^2 v_j}.$$

Starting with $v_0 = 1.6$ yields

$$v = 1.62926936306647, \qquad c = 0.377627833193942,$$

and hence

$$\boxed{x_{0\max} = (v - c_3)\operatorname{sech} v = 0.407824170219077}.$$

This rounds to the supplied 0.4078.

The two branches must not both be called stable. To check this, allow reflection-symmetric, axisymmetric variations of the two catenoids, with the junction remaining in $x = 0$ and its radius free. Set $u = x/c + c_3$ and $p(u) = \operatorname{sech}^2 u$. For the area in Exercise 22.1.13, the mixed term in the second variation gives the lower-end contribution $-\tanh(c_3)\eta(c_3)^2 = -\eta(c_3)^2/2$ after division by 4π. The disk's second variation contributes $+\eta(c_3)^2/2$, so

$$Q[\eta] = \frac{\delta^2 A}{4\pi} = \int_{c_3}^{v} p(u)(\eta'^2 - \eta^2)\, du, \qquad \eta(v) = 0, \quad \eta(c_3) \text{ free}.$$

The function

$$w(u) = \cosh u - (u - c_3)\sinh u$$

satisfies $(pw')' + pw = 0$ and $w'(c_3) = 0$. Its first zero is $v_c = 1.629269\ldots$, the root just found. For $v < v_c$, $w > 0$; putting $\eta = wz$ gives the same square identity as in Exercise 22.1.6. Both boundary terms vanish, so

$$Q[\eta] = \int_{c_3}^{v} pw^2 z'^2\, du > 0 \qquad (\eta \neq 0).$$

To pass from this quadratic test to a local minimum, note that $z(v) = 0$ implies $z(u) = -\int_u^v z'(s)ds$, so $\int z^2 \le (v - c_3)^2 \int z'^2$ by Cauchy–Schwarz. Because p, w have positive lower bounds and w, w' are bounded, $\eta = wz$ gives $Q[\eta] \ge \kappa \int (\eta^2 + \eta'^2)du$ for some $\kappa > 0$. The Taylor remainder is smaller than this bound for sufficiently small $\|\eta\|_{C^1}$, by uniform continuity of the integrand's Hessian. This proves strict local minimality on the shallow branch in the stated class. At $v = v_c$, $\eta = w$ is a zero mode; the second variation alone does not classify higher orders.

For $v > v_c$, extend w by zero from v_c to v, calling the result η_0. It is an admissible piecewise smooth variation with $Q[\eta_0] = 0$. Choose a smooth h with $h(v) = 0$ and $h(v_c) > 0$. Integration by parts on $[c_3, v_c]$ gives

$$Q[\eta_0, h] := \int_{c_3}^{v} p(\eta_0' h' - \eta_0 h)\, du = p(v_c)w'(v_c)h(v_c) < 0.$$

Therefore $Q[\eta_0 + \epsilon h] = 2\epsilon Q[\eta_0, h] + \epsilon^2 Q[h] < 0$ for small positive ϵ. Smoothing the corner preserves this strict inequality. The deep branch is unstable even in this restricted class.

The area comparison in Exercise 22.1.13 also needs its branch specified. Write the simple catenoid as $x_0 = u\operatorname{sech} u$ and $A_s = 2\pi(\tanh u + u\operatorname{sech}^2 u)$, taking its shallow root $u < u_c$ from Exercise 22.1.6. Let A_t denote the three-film area. Along either family, direct differentiation gives

$$\frac{dA_t}{dx_0} = 4\pi \operatorname{sech} v, \qquad \frac{dA_s}{dx_0} = 4\pi \operatorname{sech} u.$$

At a common separation, $u < v$: for $v \le u_c$, use the increasing function $u \operatorname{sech} u$ and $x_0 = (v - c_3)\operatorname{sech} v < v \operatorname{sech} v$; for $v > u_c$ it is immediate. Thus $A_t - A_s$ decreases with x_0 on the shallow three-film branch. At its fold,

$$A_t = 2\pi\left[\tanh v_c + \coth v_c \operatorname{sech}^2 v_c\right] = 2\pi \coth v_c > 2\pi.$$

Meanwhile $x_{0\max} < 1/2 < 0.527697\ldots$, the area crossing in Exercise 22.1.7, so $A_s < 2\pi$ throughout this range. The bound $x_0 < 1/2$ follows without rounding from

$$\tfrac{1}{2}\cosh v - v + c_3 \ge \tfrac{1}{4}(v-2)^2 + c_3 - \tfrac{1}{2} > 0.$$

The shallow-branch difference is therefore positive even at its minimum, the fold. On the deep three-film branch, $v - c_3 > \coth v$, so $A_t > 2\pi \coth v > 2\pi > A_s$ directly. This proves the stated comparison for both three-film branches. It is not a comparison with the deep simple catenoid.

22.1.15 Find the curve of quickest descent from $(0,0)$ to (x_0, y_0) for a particle that, starting from rest, slides under gravity and without friction. Assume $x_0, y_0 > 0$ with y measured downward. For the special endpoint ratio $x_0/y_0 = \pi/2$, show that the ratio of times taken by the particle along a straight line joining the two points compared to along the curve of quickest descent is $(1 + 4/\pi^2)^{1/2}$.

Hint. Take y to increase downwards. Apply Eq. (22.21) to obtain $y_x^2 = (1 - c^2 y)/c^2 y$, where c is an integration constant. It is helpful to make the substitution $c^2 y = \sin^2(\varphi/2)$ and take $(x_0, y_0) = (\pi/2c^2,\ 1/c^2)$.

Solution.

Take y positive downward. Conservation of energy gives $v = \sqrt{2gy}$, so

$$T[y] = \frac{1}{\sqrt{2g}} \int \frac{\sqrt{1 + y'^2}}{\sqrt{y}}\, dx.$$

The Beltrami first integral, with its constant written $c > 0$, is

$$\frac{1}{\sqrt{y}\sqrt{1 + y'^2}} = c, \qquad y'^2 = \frac{1 - c^2 y}{c^2 y}.$$

Put $c^2 y = \sin^2(\phi/2)$. Then

$$y = \frac{1 - \cos\phi}{2c^2}, \qquad \frac{dy}{d\phi} = \frac{\sin\phi}{2c^2}, \qquad \frac{dx}{d\phi} = \frac{1 - \cos\phi}{2c^2}.$$

Choosing $x = 0$ at $\phi = 0$ gives the cycloid

$$\boxed{x = \frac{\phi - \sin\phi}{2c^2}, \qquad y = \frac{1 - \cos\phi}{2c^2}}.$$

For arbitrary $x_0, y_0 > 0$, choose $0 < \phi_0 < 2\pi$ from

$$\frac{x_0}{y_0} = \frac{\phi_0 - \sin\phi_0}{1 - \cos\phi_0}, \qquad \frac{1}{2c^2} = \frac{y_0}{1 - \cos\phi_0}.$$

The ratio increases from zero to infinity on this interval; its derivative has numerator $2(1 - \cos\phi) - \phi\sin\phi > 0$. For $0 < \phi < \pi$, this follows from $2\tan(\phi/2) > \phi$; for $\pi \le \phi < 2\pi$ the sign is immediate.

With $A = 1/(2c^2)$,

$$ds = 2A\sin(\phi/2)d\phi, \qquad v = 2\sqrt{gA}\sin(\phi/2),$$

so $dt = \sqrt{A/g}\,d\phi$ and

$$T_{\text{cyc}} = \phi_0\sqrt{A/g}.$$

One can verify the minimizing character by a calibration. Regard $S(x,y) = \phi\sqrt{A/g}$ as a function defined by the first-arch endpoint equations. Their differentials are

$$dx = (\phi - \sin\phi)\,dA + A(1-\cos\phi)\,d\phi, \qquad dy = (1-\cos\phi)\,dA + A\sin\phi\,d\phi,$$

whereas $dS = \phi\,dA/(2\sqrt{gA}) + \sqrt{A/g}\,d\phi$. Equating the coefficients of $dA, d\phi$ in $dS = S_x dx + S_y dy$ gives

$$S_x = \frac{1}{2\sqrt{gA}}, \qquad S_y = \frac{\cot(\phi/2)}{2\sqrt{gA}}.$$

Indeed, use $(1-\cos\phi)\cot(\phi/2) = \sin\phi$ in the first coefficient and $\sin\phi\cot(\phi/2) = 1+\cos\phi$ in the second. Consequently $|\nabla S|^2 = \csc^2(\phi/2)/(4gA) = 1/(2gy)$. For any joining path in $y>0$, $dS \le |\nabla S|\,ds = dt$; integrating from the initial limiting point, where $S=0$, proves $T \ge S(x_0,y_0)$. Equality holds along the cycloid, whose tangent is parallel to ∇S. Reflection handles paths to negative x.

The fixed numerical time ratio in the question applies to the special endpoint in the hint, not to arbitrary (x_0,y_0). For that endpoint $\phi_0 = \pi$, so $x_0 = \pi/(2c^2)$, $y_0 = 1/c^2$, and $T_{\text{cyc}} = \pi/(c\sqrt{2g})$. The straight segment has length

$$L = \sqrt{x_0^2 + y_0^2} = \frac{1}{c^2}\sqrt{1+\pi^2/4}$$

and constant tangential acceleration gy_0/L. Starting from rest,

$$T_{\text{line}} = \sqrt{\frac{2L^2}{gy_0}} = \frac{1}{c}\sqrt{\frac{2}{g}}\sqrt{1+\frac{\pi^2}{4}}.$$

Therefore

$$\boxed{\frac{T_{\text{line}}}{T_{\text{cyc}}} = \frac{2}{\pi}\sqrt{1+\frac{\pi^2}{4}} = \sqrt{1+\frac{4}{\pi^2}}}.$$

22.2 More General Variations

22.2.1 Take $m>0$ and fixed positions at fixed times $t_1 < t_2$.

(a) Develop the equations of motion corresponding to $L = \frac{1}{2}m(\dot{x}^2+\dot{y}^2)$.

(b) In what sense do your solutions minimize the integral $\displaystyle\int_{t_1}^{t_2} L\,dt$?

Compare the result for your solution with $x =$ constant, $y =$ constant.

Solution.

(a) Since L is independent of x, y,

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x} = m\ddot{x} = 0, \qquad m\ddot{y} = 0.$$

Thus $x = A + Bt$, $y = C + Dt$. For fixed endpoint positions $\mathbf{r}_1, \mathbf{r}_2$ and times $t_1 < t_2$, the unique path is

$$\boxed{\mathbf{r}_*(t) = \mathbf{r}_1 + \frac{t - t_1}{t_2 - t_1}(\mathbf{r}_2 - \mathbf{r}_1)}.$$

(b) Write another admissible path as $\mathbf{r} = \mathbf{r}_* + \boldsymbol{\eta}$, with $\boldsymbol{\eta}(t_1) = \boldsymbol{\eta}(t_2) = 0$. Its action is

$$S[\mathbf{r}] = S[\mathbf{r}_*] + m\dot{\mathbf{r}}_* \cdot \int_{t_1}^{t_2} \dot{\boldsymbol{\eta}}\, dt + \frac{m}{2}\int_{t_1}^{t_2} |\dot{\boldsymbol{\eta}}|^2 dt.$$

The middle term is zero because the velocity is constant and the endpoint variation is zero. Therefore

$$\boxed{S[\mathbf{r}] - S[\mathbf{r}_*] = \frac{m}{2}\int_{t_1}^{t_2} |\dot{\boldsymbol{\eta}}|^2 dt \geq 0}, \qquad S[\mathbf{r}_*] = \frac{m|\mathbf{r}_2 - \mathbf{r}_1|^2}{2(t_2 - t_1)}.$$

Equality requires the same path. A constant path has zero action, but is an admissible competitor only if the prescribed endpoint positions coincide; in that case it is precisely the minimizing solution.

22.2.2 From the Lagrangian equations of motion, Eq. (22.38), show that a system in stable equilibrium has a minimum potential energy. Assume a natural mechanical Lagrangian with positive-definite kinetic energy and a nondegenerate equilibrium.

Solution.

For a natural mechanical system take $T = \frac{1}{2}M_{ij}(q)\dot{q}_i\dot{q}_j$ with positive-definite mass matrix, and $L = T - V(q)$. At a rest equilibrium $q = q_0$, $\dot{q} = \ddot{q} = 0$. The Lagrange equations reduce to

$$\boxed{\frac{\partial V}{\partial q_i}(q_0) = 0}.$$

Write $q = q_0 + \eta$ and $K_{ij} = \partial_i\partial_j V(q_0)$. To first order,

$$M(q_0)\ddot{\eta} + K\eta = 0.$$

Choose normal coordinates by diagonalizing the real symmetric matrix $M^{-1/2}KM^{-1/2}$. Each coordinate satisfies

$$\ddot{Q}_j + \omega_j^2 Q_j = 0.$$

A negative $\omega_j^2 = -\kappa^2$ yields $Q_j = Ae^{\kappa t} + Be^{-\kappa t}$ and hence instability under arbitrarily small perturbations. In the nondegenerate stable case, all $\omega_j^2 > 0$, so K is positive definite. Taylor expansion gives

$$V(q_0 + \eta) = V(q_0) + \frac{1}{2}\eta^T K\eta + O(|\eta|^3) > V(q_0)$$

for sufficiently small nonzero η. Thus the equilibrium has a strict local minimum of potential energy.

The usual conclusion assumes a natural mechanical system with positive kinetic energy. Linear stability alone gives only a positive-semidefinite Hessian when zero modes occur; higher-order terms then decide the result. Lagrange's equations do not turn stationarity by itself into a minimum. For example, $V(q) = q^4$ has a stable minimum despite $V''(0) = 0$, whereas $V(q) = -q^4$ has the same zero Hessian and is unstable. Conversely, a strict potential well together with positive kinetic energy confines sufficiently small-energy motions: energy conservation prevents them from reaching a nearby boundary on which V is higher. This is the energy reason for stable small oscillations about a minimum.

22.2.3 Write out the Lagrangian equations of motion of a particle in spherical coordinates for potential V equal to a constant. Identify the terms corresponding to (a) centrifugal force and (b) Coriolis force. Write the coordinate equations on $r > 0$, $0 < \theta < \pi$.

Solution.

In spherical coordinates, with $V = V_0$,

$$L = \frac{m}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\varphi}^2\right) - V_0.$$

For r the derivatives are $L_{\dot{r}} = m\dot{r}$ and $L_r = mr(\dot{\theta}^2 + \sin^2\theta\dot{\varphi}^2)$. For θ they are $L_{\dot{\theta}} = mr^2\dot{\theta}$ and $L_\theta = mr^2\sin\theta\cos\theta\dot{\varphi}^2$. For φ, $L_\varphi = 0$ and $L_{\dot{\varphi}} = mr^2\sin^2\theta\dot{\varphi}$. Differentiating the momenta and dividing by the appropriate scale factors gives

$$\boxed{\begin{aligned}\ddot{r} &= r\dot{\theta}^2 + r\sin^2\theta\,\dot{\varphi}^2,\\ r\ddot{\theta} &= r\sin\theta\cos\theta\,\dot{\varphi}^2 - 2\dot{r}\dot{\theta},\\ r\sin\theta\,\ddot{\varphi} &= -2\dot{r}\sin\theta\,\dot{\varphi} - 2r\cos\theta\,\dot{\theta}\dot{\varphi}.\end{aligned}}$$

Equivalently, $mr^2\sin^2\theta\dot{\varphi}$ is conserved.

The squared-angular-speed terms on the right in the first two equations are the centrifugal terms of the moving spherical frame. The products with the factor two are its Coriolis-type terms: $-2m\dot{r}\dot{\theta}$, $-2m\dot{r}\sin\theta\dot{\varphi}$, and $-2mr\cos\theta\dot{\theta}\dot{\varphi}$. There is no physical external force for constant V; these terms account for the changing coordinate basis. Moving them to the left recovers the three components of the zero Cartesian acceleration.

22.2.4 The spherical pendulum consists of a mass on a wire of length l, free to move in polar angle θ and azimuth angle φ (Fig. 22.8). Measure θ from the downward vertical.

(a) Set up the Lagrangian for this physical system.

(b) Develop the Lagrangian equations of motion.

Solution.

As in Fig. 22.8, measure θ from the downward vertical. The height relative to the pivot is $z = -l\cos\theta$, and the speed squared is $l^2(\dot{\theta}^2 + \sin^2\theta\dot{\varphi}^2)$. Thus, apart from an irrelevant constant,

$$\boxed{L = \frac{ml^2}{2}(\dot{\theta}^2 + \sin^2\theta\dot{\varphi}^2) + mgl\cos\theta}.$$

The polar derivatives are

$$L_{\dot{\theta}} = ml^2\dot{\theta}, \qquad L_\theta = ml^2\sin\theta\cos\theta\dot{\varphi}^2 - mgl\sin\theta.$$

Their Euler equation is

$$\boxed{\ddot{\theta} - \sin\theta\cos\theta\,\dot{\varphi}^2 + \frac{g}{l}\sin\theta = 0}.$$

The azimuth is cyclic, so

$$\boxed{\frac{d}{dt}(ml^2\sin^2\theta\,\dot{\varphi}) = 0}.$$

Where $\sin\theta \neq 0$, the latter expands to

$$\ddot{\varphi} + 2\cot\theta\,\dot{\theta}\dot{\varphi} = 0.$$

The conserved momentum is the vertical angular momentum. At a polar coordinate singularity the conserved-momentum form is the appropriate one; division by $\sin\theta$ is unnecessary.

22.2.5 Show that the Lagrangian

$$L = m_0 c^2 \left(1 - \sqrt{1 - \frac{v^2}{c^2}}\right) - V(r)$$

leads to a relativistic form of Newton's second law of motion,

$$\frac{d}{dt}\left(\frac{m_0 v_i}{\sqrt{1 - v^2/c^2}}\right) = F_i,$$

in which the force components are $F_i = -\partial V/\partial x_i$. Assume $m_0 > 0$ and $v < c$.

Solution.

Let $v^2 = \sum_j \dot{x}_j^2 < c^2$. Since $\partial v^2/\partial \dot{x}_i = 2\dot{x}_i$,

$$\frac{\partial L}{\partial \dot{x}_i} = -m_0 c^2 \frac{1}{2\sqrt{1 - v^2/c^2}}\left(-\frac{2\dot{x}_i}{c^2}\right) = \frac{m_0 v_i}{\sqrt{1 - v^2/c^2}}.$$

The constant rest-energy term has zero derivative, and the spatial derivative is $\partial L/\partial x_i = -\partial_i V$. Consequently

$$\boxed{\frac{d}{dt}\left(\frac{m_0 v_i}{\sqrt{1 - v^2/c^2}}\right) = -\partial_i V = F_i}.$$

Writing $\gamma = (1 - v^2/c^2)^{-1/2}$ gives $\dot{\gamma} = \gamma^3(\mathbf{v}\cdot\mathbf{a})/c^2$, so the same equation is

$$m_0 \gamma a_i + \frac{m_0 \gamma^3}{c^2} v_i (\mathbf{v}\cdot\mathbf{a}) = F_i.$$

In the limit $v/c \to 0$, this reduces to $m_0 a_i = F_i$.

22.2.6 The Lagrangian for a particle with charge q in an electromagnetic field described by scalar potential φ and vector potential $\mathbf{A}$ is

$$L = \frac{1}{2} m v^2 - q\varphi + q\mathbf{A}\cdot\mathbf{v}.$$

Find the equation of motion of the charged particle.

Hint. $(d/dt)A_j = \partial A_j/\partial t + \sum_i (\partial A_j/\partial x_i)\dot{x}_i$. The dependence of the force fields $\mathbf{E}$ and $\mathbf{B}$ on the potentials φ and $\mathbf{A}$ is developed in Section 3.9; see in particular Eq. (3.108).

ANS. $m\ddot{x}_i = q[\mathbf{E} + \mathbf{v}\times\mathbf{B}]_i$.

Solution.

Treat the potentials as functions of both position and time. The canonical momentum and spatial derivative are

$$\frac{\partial L}{\partial \dot{x}_i} = m\dot{x}_i + qA_i, \qquad \frac{\partial L}{\partial x_i} = -q\,\partial_i\varphi + q\sum_j \dot{x}_j \partial_i A_j.$$

Along the trajectory,

$$\frac{dA_i}{dt} = \partial_t A_i + \sum_j \dot{x}_j \partial_j A_i.$$

The Lagrange equation therefore gives

$$m\ddot{x}_i = q(-\partial_i\varphi - \partial_t A_i) + q\sum_j \dot{x}_j(\partial_i A_j - \partial_j A_i).$$

Define $\mathbf{E} = -\nabla\varphi - \partial_t\mathbf{A}$ and $\mathbf{B} = \nabla\times\mathbf{A}$. For example, the last sum at $i = x$ is

$$v_y(\partial_x A_y - \partial_y A_x) + v_z(\partial_x A_z - \partial_z A_x) = v_y B_z - v_z B_y.$$

The other two components follow by cyclic permutation. Thus

$$\boxed{m\ddot{\mathbf{r}} = q(\mathbf{E} + \mathbf{v}\times\mathbf{B})}.$$

The derivatives acting on the position-dependent vector potential are essential; omitting them would omit the magnetic force.

22.2.7 Consider a system in which the Lagrangian is given by

$$L(q_i, \dot{q}_i) = T(q_i, \dot{q}_i) - V(q_i),$$

where q_i and $\dot{q}_i$ represent sets of variables. The potential energy V is independent of velocity and neither T nor V has any explicit time dependence.

(a) Show that

$$\frac{d}{dt}\left(\sum_j \dot{q}_j\frac{\partial L}{\partial \dot{q}_j} - L\right) = 0.$$

(b) The constant quantity

$$\sum_j \dot{q}_j\frac{\partial L}{\partial \dot{q}_j} - L$$

defines the Hamiltonian H. Show that under the preceding assumed conditions, H satisfies $H = T + V$, and is therefore the total energy.

Note. The kinetic energy T is a homogeneous quadratic function of the $\dot{q}_i$.

Solution.

(a) Put $p_j = \partial L/\partial\dot{q}_j$ and $H = \sum_j \dot{q}_j p_j - L$. Its full derivative is

$$\begin{aligned}\frac{dH}{dt} &= \sum_j(\ddot{q}_j p_j + \dot{q}_j\dot{p}_j) - \sum_j\left(L_{q_j}\dot{q}_j + L_{\dot{q}_j}\ddot{q}_j\right) - L_t \\ &= \sum_j \dot{q}_j(\dot{p}_j - L_{q_j}) - L_t.\end{aligned}$$

The Lagrange equations make the sum zero, and the stated absence of explicit time dependence makes $L_t = 0$. Hence

$$\boxed{\frac{dH}{dt} = 0}.$$

(b) Use the note's homogeneous quadratic kinetic energy:

$$T = \frac{1}{2}\sum_{i,j} M_{ij}(q)\dot{q}_i\dot{q}_j, \qquad M_{ij} = M_{ji}.$$

Then

$$\sum_j \dot{q}_j \frac{\partial T}{\partial \dot{q}_j} = \sum_{i,j} M_{ij}\dot{q}_i\dot{q}_j = 2T.$$

Since V has no velocity dependence, $L_{\dot{q}_j} = T_{\dot{q}_j}$, and therefore

$$\boxed{H = 2T - (T - V) = T + V}.$$

The quadratic homogeneity is used specifically in (b); conservation of the Legendre-transform expression in (a) does not require it.

22.2.8 The Lagrangian for a vibrating string (small-amplitude vibrations) is

$$L = \int \left(\frac{1}{2}\rho u_t^2 - \frac{1}{2}\tau u_x^2\right) dx,$$

where ρ is the (constant) linear mass density and τ is the (constant) tension. The x-integration is over the length of the string. Show that application of Hamilton's principle to the Lagrangian density (the integrand), now with two independent variables, leads to the classical wave equation,

$$\frac{\partial^2 u}{\partial x^2} = \frac{\rho}{\tau}\frac{\partial^2 u}{\partial t^2}.$$

Assume $\rho, \tau > 0$ and variations vanishing on the spatial and temporal boundaries.

Solution.

Hamilton's principle applies to the action

$$S[u] = \int_{t_1}^{t_2} dt \int_{x_1}^{x_2} dx \left(\frac{1}{2}\rho u_t^2 - \frac{1}{2}\tau u_x^2\right).$$

Let $\delta u = \eta$ vanish at the fixed spatial and temporal boundaries. Then

$$\delta S = \iint (\rho u_t \eta_t - \tau u_x \eta_x)\, dx\, dt.$$

Integrating the first term by parts in t and the second in x, all boundary terms vanish and

$$\delta S = \iint (-\rho u_{tt} + \tau u_{xx})\eta\, dx\, dt.$$

Arbitrariness of the interior variation gives

$$\boxed{u_{xx} = \frac{\rho}{\tau} u_{tt}},$$

with wave speed $\sqrt{\tau/\rho}$ for positive tension and density. The constancy of ρ, τ permits moving them outside the derivatives.

22.2.9 Show that the stationary value of the total energy of the electrostatic field of Example 22.2.3 is a minimum. The region is charge-free and connected, the boundary potential is fixed, and the permittivity is a positive constant.

Hint. Investigate the α^2 terms of J.

Solution.

Let φ_* be the harmonic potential satisfying the fixed boundary values in the charge-free region Ω: $\nabla^2\varphi_* = 0$. For any admissible variation $\varphi = \varphi_* + \alpha\eta$ with $\eta = 0$ on the boundary, the physical field energy is

$$U[\varphi] = \frac{\varepsilon}{2}\int_\Omega |\nabla\varphi|^2 dV.$$

Expansion is exact because the functional is quadratic:

$$U[\varphi_* + \alpha\eta] - U[\varphi_*] = \alpha\varepsilon\int_\Omega \nabla\varphi_*\cdot\nabla\eta\, dV + \frac{\alpha^2\varepsilon}{2}\int_\Omega |\nabla\eta|^2 dV.$$

The cross integral is

$$\int_{\partial\Omega}\eta\,\partial_n\varphi_*\, dS - \int_\Omega \eta\,\nabla^2\varphi_*\, dV = 0.$$

Thus

$$\boxed{U[\varphi_* + \alpha\eta] - U[\varphi_*] = \frac{\alpha^2\varepsilon}{2}\int_\Omega |\nabla\eta|^2 dV \geq 0}.$$

For $\varepsilon > 0$ in a connected region with fixed boundary potential, equality requires $\nabla\eta = 0$ and hence $\eta = 0$. This proves a strict global minimum. The book's J omits only the positive constant $\varepsilon/2$, so the conclusion is identical.

22.2.10 The Lagrangian (per unit volume) of an electromagnetic field with a charge density ρ and current density $\mathbf{J}$ is given by

$$L = \frac{1}{2}\left(\varepsilon_0 E^2 - \frac{1}{\mu_0}B^2\right) - \rho\varphi + \mathbf{J}\cdot\mathbf{A}.$$

Show that Lagrange's equations lead to two of Maxwell's equations. (The remaining two are a consequence of the definition of $\mathbf{E}$ and $\mathbf{B}$ in terms of $\mathbf{A}$ and φ.)

Hint. Take φ and the components of $\mathbf{A}$ as dependent variables; and x, y, z, and t as independent variables. $\mathbf{E}$ and $\mathbf{B}$ are given in terms of $\mathbf{A}$ and φ by Eq. (3.108). Use prescribed sources satisfying $\partial_t\rho + \nabla\cdot\mathbf{J} = 0$ and variations vanishing on the spacetime boundary.

Solution.

Use

$$\mathbf{E} = -\nabla\varphi - \partial_t\mathbf{A}, \qquad \mathbf{B} = \nabla\times\mathbf{A},$$

with prescribed sources $\rho, \mathbf{J}$. The action is the spacetime integral of the given density. Its variation is

$$\delta S = \int dt\, d^3x\left[\varepsilon_0\mathbf{E}\cdot\delta\mathbf{E} - \frac{1}{\mu_0}\mathbf{B}\cdot\delta\mathbf{B} - \rho\,\delta\varphi + \mathbf{J}\cdot\delta\mathbf{A}\right].$$

First vary only φ. Then $\delta\mathbf{E} = -\nabla\delta\varphi$ and $\delta\mathbf{B} = 0$. Spatial integration by parts gives

$$\delta_\varphi S = \int(\varepsilon_0\nabla\cdot\mathbf{E} - \rho)\delta\varphi\, dt\, d^3x.$$

Boundary variations vanish. Hence

$$\boxed{\nabla \cdot \mathbf{E} = \rho/\varepsilon_0}.$$

In component Euler notation, this uses $L_\varphi = -\rho$ and $L_{\partial_i \varphi} = -\varepsilon_0 E_i$.

Next vary only $\mathbf{A}$. Now $\delta\mathbf{E} = -\partial_t \delta\mathbf{A}$ and $\delta\mathbf{B} = \nabla \times \delta\mathbf{A}$. The time term becomes

$$-\int \varepsilon_0 \mathbf{E} \cdot \partial_t \delta\mathbf{A} = \int \varepsilon_0 \partial_t \mathbf{E} \cdot \delta\mathbf{A}.$$

For the magnetic term, the identity

$$\nabla \cdot (\delta\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \delta\mathbf{A}) - \delta\mathbf{A} \cdot (\nabla \times \mathbf{B})$$

shows that its volume integral becomes $-\mu_0^{-1} \int (\nabla \times \mathbf{B}) \cdot \delta\mathbf{A}$. The three independent components therefore give

$$\mathbf{J} + \varepsilon_0 \partial_t \mathbf{E} - \mu_0^{-1} \nabla \times \mathbf{B} = 0,$$

or

$$\boxed{\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \partial_t \mathbf{E}}.$$

Finally the definitions themselves give $\nabla \cdot \mathbf{B} = \nabla \cdot (\nabla \times \mathbf{A}) = 0$ and $\nabla \times \mathbf{E} = -\partial_t (\nabla \times \mathbf{A}) = -\partial_t \mathbf{B}$, the remaining two Maxwell equations.

22.3 Constrained Minima/Maxima

22.3.0 The following problems are to be solved by using Lagrangian multipliers.

Solution.

This numbered entry is the source's instruction for the following problems. For an objective $F(q)$ constrained by $G(q) = C$, a permissible displacement satisfies $\nabla G \cdot \delta q = 0$. At a constrained stationary point, $\nabla F \cdot \delta q = 0$ for every such tangent displacement. If $\nabla G \neq 0$, the two normal vectors must therefore be proportional:

$$\boxed{\nabla F - \lambda \nabla G = 0, \qquad G = C}.$$

For several independent constraints, ∇F is a linear combination of their gradients. The multiplier equations find stationary candidates; one must still examine boundary behavior or use an inequality to decide whether each candidate is a maximum, minimum, or neither. The following solutions include that classification.

22.3.1 The ground-state energy of a quantum particle of mass m in a pillbox (right-circular cylinder) is given by

$$E = \frac{\hbar^2}{2m} \left(\frac{(2.4048)^2}{R^2} + \frac{\pi^2}{H^2} \right),$$

in which R is the radius and H is the height of the pillbox. Find the ratio of R to H that will minimize the energy for a fixed positive volume. Here 2.4048 is the stated rounded approximation to the first positive zero of J_0.

Solution.

Put $j = 2.4048$, retain the positive constant $\hbar^2/(2m)$ outside the minimization, and impose $\pi R^2 H = V$. For

$$\mathcal{F} = \frac{j^2}{R^2} + \frac{\pi^2}{H^2} + \lambda(\pi R^2 H - V),$$

the two multiplier equations are

$$-\frac{2j^2}{R^3} + 2\lambda\pi RH = 0, \qquad -\frac{2\pi^2}{H^3} + \lambda\pi R^2 = 0.$$

Hence

$$\lambda = \frac{j^2}{\pi R^4 H} = \frac{2\pi}{R^2 H^3}, \qquad j^2 H^2 = 2\pi^2 R^2.$$

The requested ratio is

$$\boxed{\frac{R}{H} = \frac{2.4048}{\pi\sqrt{2}} = 0.541270169274\ldots}.$$

To classify it, eliminate $H = V/(\pi R^2)$:

$$\frac{2mE}{\hbar^2} = \frac{j^2}{R^2} + \frac{\pi^4 R^4}{V^2}.$$

This diverges as $R \downarrow 0$ or $R \to \infty$, and its derivative vanishes only at $R^6 = j^2 V^2/(2\pi^4)$. Thus the stationary ratio gives the unique global minimum. The manual replaces the supplied 2.4048 by 2.048 in both its derivative and final ratio. The coefficient in the original problem is 2.4048, which is used throughout above.

22.3.2 Under the textbook's stipulated postal rule, the total length plus girth may not exceed 36 inches. Using Lagrange multipliers, find the dimensions of the rectangular parallelepiped of maximum volume subject to this constraint.

Solution.

Use the stated 36-inch mathematical constraint. Let the length be L and the two transverse widths be w, h. A maximum uses the whole allowance: otherwise increasing L raises the volume while remaining admissible. Thus

$$V = Lwh, \qquad L + 2w + 2h = 36.$$

For $\mathcal{F} = Lwh - \lambda(L + 2w + 2h - 36)$,

$$wh = \lambda, \qquad Lh = 2\lambda, \qquad Lw = 2\lambda.$$

All dimensions are positive, so division by wh gives $L = 2w = 2h$. The constraint becomes $L + L + L = 36$, hence

$$\boxed{L = 12 \text{ in}, \qquad w = h = 6 \text{ in}, \qquad V_{\max} = 432 \text{ in}^3}.$$

For a global check, the three positive numbers $L, 2w, 2h$ have fixed sum 36, and their product is $4V$. The arithmetic–geometric mean inequality gives

$$4V \le (36/3)^3 = 1728,$$

with equality exactly at $L = 2w = 2h$. The identified length is also the longest side, consistently with the length-plus-girth convention.

22.3.3 A thermal nuclear reactor is subject to the constraint

$$\varphi(a,b,c) = \left(\frac{\pi}{a}\right)^2 + \left(\frac{\pi}{b}\right)^2 + \left(\frac{\pi}{c}\right)^2 = B^2, \quad \text{a constant},$$

where the reactor is a rectangular parallelepiped of sides a, b, and c. Take $a, b, c, B > 0$. Find the ratios of a, b, and c that minimize the reactor volume. Show also that no maximum exists.

ANS. $a = b = c$, cube.

Solution.

For positive side lengths, extremizing abc is equivalent to extremizing $\log(abc)$. Apply a multiplier to

$$\mathcal{F} = \log a + \log b + \log c + \lambda\left[\pi^2(a^{-2} + b^{-2} + c^{-2}) - B^2\right].$$

The three equations are

$$\frac{1}{a} - \frac{2\lambda\pi^2}{a^3} = 0, \qquad \frac{1}{b} - \frac{2\lambda\pi^2}{b^3} = 0, \qquad \frac{1}{c} - \frac{2\lambda\pi^2}{c^3} = 0.$$

Thus $a^2 = b^2 = c^2 = 2\lambda\pi^2$ and positivity yields $a = b = c$. The constraint gives

$$a = b = c = \frac{\sqrt{3}\pi}{B}.$$

The cube is the *minimum*-volume reactor under this constraint. The printed request for a maximum is false: the volume has no finite upper bound. To prove both statements, set $u = a^{-2}$, $v = b^{-2}$, $w = c^{-2}$, so $u + v + w = B^2/\pi^2$. Then

$$uvw \le \left(\frac{B^2}{3\pi^2}\right)^3, \qquad abc = (uvw)^{-1/2} \ge \boxed{\frac{3\sqrt{3}\pi^3}{B^3}}.$$

Equality occurs only at $u = v = w$, i.e. the cube.

For an explicit unbounded family, take

$$a = b = L, \qquad c = \frac{\pi}{\sqrt{B^2 - 2\pi^2/L^2}}, \qquad L > \frac{\sqrt{2}\pi}{B}.$$

It satisfies the original constraint exactly, while $abc = \pi L^2/\sqrt{B^2 - 2\pi^2/L^2} \to \infty$. This resolves the original maximum question as well as identifying the correctly classified stationary cube.

22.3.4 For a lens of focal length f, the object distance p and the image distance q are related by $1/p + 1/q = 1/f$. Find the minimum object-image distance $(p + q)$ for fixed f. Assume real object and image (p and q both positive).

Solution.

The assumptions $p, q > 0$ imply $f > 0$. Use

$$\mathcal{F} = p + q + \lambda\left(\frac{1}{p} + \frac{1}{q} - \frac{1}{f}\right).$$

Differentiation gives $1 - \lambda/p^2 = 0$ and $1 - \lambda/q^2 = 0$. Thus $p^2 = q^2$, so $p = q$. The lens equation then gives $p = q = 2f$ and

$$\boxed{(p+q)_{\min} = 4f}.$$

For a global proof, put $S = p + q$. The constraint implies $pq = fS$. Since $(p - q)^2 \ge 0$,

$$S^2 \ge 4pq = 4fS.$$

Divide by the positive S to obtain $S \ge 4f$, with equality only for the multiplier solution.

22.3.5 You have an ellipse $(x/a)^2 + (y/b)^2 = 1$. Find the inscribed rectangle of maximum area. Show that the ratio of the area of the maximum-area rectangle to the area of the ellipse is $2/\pi = 0.6366$.

Solution.

First take the rectangle's vertices to be $(\pm x, \pm y)$ with $x, y > 0$. Its area is $A = 4xy$ and its constraint is $x^2/a^2 + y^2/b^2 = 1$. For

$$\mathcal{F} = 4xy - \lambda(x^2/a^2 + y^2/b^2 - 1),$$

the equations are

$$4y = \frac{2\lambda x}{a^2}, \qquad 4x = \frac{2\lambda y}{b^2}.$$

Multiplying respectively by x, y gives $x^2/a^2 = y^2/b^2$. Each is therefore $1/2$, so

$$\boxed{x = \frac{a}{\sqrt{2}}, \quad y = \frac{b}{\sqrt{2}}, \quad A_{\max} = 2ab}.$$

The side lengths are $\sqrt{2}a, \sqrt{2}b$ and

$$\boxed{\frac{A_{\max}}{A_{\text{ellipse}}} = \frac{2ab}{\pi ab} = \frac{2}{\pi} = 0.6366197724\ldots}.$$

This is also the maximum if the rectangle's orientation is unrestricted. Let its center be h, its orthonormal directions be e_1, e_2, and its half-lengths be u, v. For the ellipse matrix $Q = \operatorname{diag}(a^{-2}, b^{-2})$, averaging the four vertex inequalities gives

$$h^T Q h + u^2 q_1 + v^2 q_2 \le 1, \qquad q_i = e_i^T Q e_i.$$

Thus $2uv\sqrt{q_1 q_2} \le 1$. In this rotated basis, $q_1 q_2 = \det Q + (e_1^T Q e_2)^2 \ge 1/(a^2 b^2)$, and hence $4uv \le 2ab$. The aligned multiplier solution attains this bound, proving the global claim.

22.3.6 A rectangular parallelepiped is inscribed in an ellipsoid of semiaxes a, b, and c. Maximize the volume of the inscribed rectangular parallelepiped. Show that the ratio of the maximum volume to the volume of the ellipsoid is $\dfrac{2}{\pi\sqrt{3}} \approx 0.367$.

Solution.

For an aligned box let the positive half-lengths be x, y, z. Maximize $V = 8xyz$ subject to $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$. The multiplier equations give

$$8yz = \frac{2\lambda x}{a^2}, \quad 8xz = \frac{2\lambda y}{b^2}, \quad 8xy = \frac{2\lambda z}{c^2}.$$

Multiplying by x, y, z respectively makes all left sides equal. Thus $x^2/a^2 = y^2/b^2 = z^2/c^2 = 1/3$, giving

$$\boxed{x = \frac{a}{\sqrt{3}}, \quad y = \frac{b}{\sqrt{3}}, \quad z = \frac{c}{\sqrt{3}}, \qquad V_{\max} = \frac{8abc}{3\sqrt{3}}}.$$

Dividing by the ellipsoid's $4\pi abc/3$ yields

$$\boxed{\frac{V_{\max}}{V_{\text{ellipsoid}}} = \frac{2}{\pi\sqrt{3}} = 0.3675525969\ldots}.$$

The inline ratio must be read as $2/(\pi\sqrt{3})$; $(2/\pi)\sqrt{3}$ would exceed one and would not match the stated decimal approximation.

For completeness, the same bound covers arbitrary orientations and centers. With $Q = \mathrm{diag}(a^{-2}, b^{-2}, c^{-2})$, orthonormal box directions e_i and half-lengths u_i, the average of the eight vertex inequalities gives $\sum_i q_i u_i^2 \le 1$, where $q_i = e_i^T Q e_i$; the nonnegative center term can be discarded. The arithmetic–geometric mean bound then gives $\prod_i (q_i u_i^2) \le 1/27$. The Gram determinant inequality gives $\prod_i q_i \ge \det Q = 1/(a^2 b^2 c^2)$: the squared volume of the vectors $Q^{1/2} e_i$ is at most the product of their squared lengths. Therefore

$$8u_1u_2u_3 \le \frac{8}{\sqrt{27}\sqrt{q_1q_2q_3}} \le \frac{8abc}{3\sqrt{3}}.$$

The aligned box attains equality.

22.3.7 Find the maximum value of the directional derivative of $\varphi(x, y, z)$, a real differentiable function at a fixed point,

$$\frac{d\varphi}{ds} = \frac{\partial\varphi}{\partial x}\cos\alpha + \frac{\partial\varphi}{\partial y}\cos\beta + \frac{\partial\varphi}{\partial z}\cos\gamma,$$

subject to the constraint,

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1.$$

Solution.

At the point in question write $\mathbf{g} = \nabla\varphi$ and $\mathbf{n} = (\cos\alpha, \cos\beta, \cos\gamma)$. Maximize $D = \mathbf{g}\cdot\mathbf{n}$ subject to $\mathbf{n}\cdot\mathbf{n} = 1$. The multiplier equations for $D - \lambda(\mathbf{n}\cdot\mathbf{n} - 1)$ are

$$g_i - 2\lambda n_i = 0.$$

If $\mathbf{g} \neq 0$, normalization gives $4\lambda^2 = |\mathbf{g}|^2$, and hence

$$\mathbf{n}_\pm = \pm\frac{\mathbf{g}}{|\mathbf{g}|}, \qquad D_\pm = \pm|\mathbf{g}|.$$

The maximum is therefore

$$\boxed{\left(\frac{d\varphi}{ds}\right)_{\max} = |\nabla\varphi|, \qquad \mathbf{n}_{\max} = \frac{\nabla\varphi}{|\nabla\varphi|}}.$$

Cauchy–Schwarz proves the global bound $\mathbf{g}\cdot\mathbf{n} \le |\mathbf{g}||\mathbf{n}| = |\mathbf{g}|$. If $\nabla\varphi = 0$, every unit direction gives zero, so every direction is a maximizer and no preferred direction is defined.

22.4 Variation with Constraints

22.4.1 A particle of mass m is on a frictionless horizontal surface. In terms of plane polar coordinates (r, θ), it is constrained to move so that $\theta = \omega t$ (accomplished by pushing it with a rotating radial arm against which it can slide frictionlessly). With the initial conditions

$$t = 0, \qquad r = r_0, \qquad \dot{r} = 0:$$

(a) Find the radial position as a function of time.

ANS. $r(t) = r_0 \cosh\omega t.$

(b) Find the force exerted on the particle by the constraint.

ANS. $F^{(c)} = 2m\dot{r}\omega = 2mr_0\omega^2 \sinh \omega t.$

Solution.

Use polar coordinates with the positive tangential direction $\mathbf{e}_\theta$. Before imposing the constraint,

$$L = \frac{m}{2}(\dot{r}^2 + r^2\dot{\theta}^2), \qquad \phi = \theta - \omega t = 0.$$

With the convention $L + \lambda\phi$, the two Lagrange equations are

$$\frac{d}{dt}(m\dot{r}) - mr\dot{\theta}^2 = 0, \qquad \frac{d}{dt}(mr^2\dot{\theta}) = \lambda.$$

The multiplier in the second equation is a torque, since its coordinate is an angle.

(a) Substitute $\dot{\theta} = \omega$ into the first equation:

$$\ddot{r} - \omega^2 r = 0, \qquad r = A\cosh(\omega t) + B\sinh(\omega t).$$

The initial conditions give $A = r_0$, $B = 0$, so

$$\boxed{r(t) = r_0 \cosh(\omega t)}, \qquad \dot{r} = r_0\omega \sinh(\omega t).$$

The $\omega = 0$ limit is $r = r_0$.

(b) The frictionless arm pushes perpendicular to itself, along $\mathbf{e}_\theta$. Since $rF_\theta = \lambda$,

$$\lambda = \omega\frac{d}{dt}(mr^2) = 2m\omega r\dot{r}, \qquad \boxed{\mathbf{F}^{(c)} = 2m\omega\dot{r}\,\mathbf{e}_\theta = 2mr_0\omega^2 \sinh(\omega t)\,\mathbf{e}_\theta}.$$

There is no radial applied force: the radial acceleration component $\ddot{r} - r\omega^2$ vanishes. As an energy check,

$$\frac{d}{dt}\frac{m}{2}(\dot{r}^2 + r^2\omega^2) = m\dot{r}(\ddot{r} + r\omega^2) = 2m\omega^2 r\dot{r} = \mathbf{F}^{(c)} \cdot \mathbf{v}.$$

Thus the motor turning the arm supplies the increasing kinetic energy.

22.4.2 A point mass m is moving on a flat, horizontal, frictionless plane. The mass is constrained by a string to move radially inward at a constant rate. Using plane polar coordinates (r, θ), $r = r_0 - kt$, with $m, r_0, k > 0$ and $0 \le t < r_0/k$:

(a) Set up the Lagrangian.

(b) Obtain the constrained Lagrange equations.

(c) Solve the θ-dependent Lagrange equation to obtain $\omega(t)$, the angular velocity. What is the physical significance of the constant of integration that you get from your "free" integration?

(d) Using the $\omega(t)$ from part (c), solve the r-dependent (constrained) Lagrange equation to obtain $\lambda(t)$. In other words, explain what is happening to the force of constraint as $r \to 0$.

Solution.

(a) Take r and θ as independent coordinates initially:

$$L = \frac{m}{2}(\dot{r}^2 + r^2\dot{\theta}^2), \qquad \phi(r,t) = r - r_0 + kt = 0.$$

Here $k > 0$, and the prescribed motion applies for $0 \le t < r_0/k$.

(b) Using $L + \lambda\phi$ gives

$$\boxed{m\ddot{r} - mr\dot{\theta}^2 = \lambda}, \qquad \boxed{\frac{d}{dt}(mr^2\dot{\theta}) = 0}, \qquad r = r_0 - kt.$$

Because $\partial\phi/\partial r = 1$, λ is the signed radial force, positive outward. The physical string tension is $-\lambda$.

(c) Integrating the cyclic-coordinate equation once,

$$mr^2\dot{\theta} = \ell, \qquad \boxed{\omega(t) = \dot{\theta} = \frac{\ell}{m(r_0 - kt)^2} = \omega_0\frac{r_0^2}{(r_0 - kt)^2}}.$$

The constant $\ell = mr_0^2\omega_0$ is the conserved angular momentum about the hole through which the string is drawn. The manual prints the third power in the denominator of $\omega(t)$. It must be the second power: the printed expression would make $mr^2\omega = \ell/r$ vary with time. The constraint force is radial and therefore has zero torque about that point. A second integration, if desired, gives

$$\theta(t) - \theta_0 = \frac{\ell}{m}\int_0^t \frac{d\tau}{(r_0 - k\tau)^2} = \frac{\ell}{mk}\left(\frac{1}{r_0 - kt} - \frac{1}{r_0}\right).$$

(d) Since $\ddot{r} = 0$, the radial equation becomes

$$\boxed{\lambda(t) = -mr\omega^2 = -\frac{\ell^2}{m(r_0 - kt)^3}}, \qquad T_{\text{string}} = \frac{\ell^2}{mr^3}.$$

For $\ell \neq 0$, the tension tends to $+\infty$ as $r \to 0^+$. Maintaining the specified radial speed therefore requires an unbounded pulling force in that limit. The divergence requires nonzero angular momentum. If $\ell = 0$, then $\omega = 0$ and $\lambda = 0$ throughout the prescribed inward motion; the endpoint $r = 0$ itself is a polar-coordinate singularity.

22.4.3 A flexible cable is suspended from two fixed points. The length of the cable is fixed. Find the curve that will minimize the total gravitational potential energy of the cable. Assume a uniform cable in uniform gravity, supports at different horizontal positions, and a cable length greater than the straight-line distance between the supports.

ANS. Hyperbolic cosine.

Solution.

The hyperbolic-cosine answer assumes a uniform cable in a uniform gravitational field. A nonuniform mass per unit length generally produces a different curve. Let the constant mass per unit length be ρ, let y increase upward, and suppose the endpoints have $x_1 < x_2$. For a graph $y(x)$,

$$U = \rho g\int_{x_1}^{x_2} y\sqrt{1 + y'^2}\,dx, \qquad L_0 = \int_{x_1}^{x_2}\sqrt{1 + y'^2}\,dx.$$

Introduce a constant multiplier λ and vary

$$\int_{x_1}^{x_2} (\rho g y + \lambda)\sqrt{1+y'^2}\,dx.$$

There is no explicit x dependence. Its Beltrami first integral is

$$(\rho g y + \lambda)\sqrt{1+y'^2} - y'\frac{(\rho g y+\lambda)y'}{\sqrt{1+y'^2}} = \frac{\rho g y+\lambda}{\sqrt{1+y'^2}} = \rho g c.$$

For the hanging solution $c > 0$. Put $Y = y + \lambda/(\rho g)$:

$$Y = c\sqrt{1+Y'^2}, \qquad \frac{dY}{\sqrt{Y^2-c^2}} = \pm\frac{dx}{c}.$$

Since $\int dY/\sqrt{Y^2-c^2} = \operatorname{arcosh}(Y/c)$, joining the two sides of the lowest point gives

$$\boxed{y(x) = c\cosh\frac{x-b}{c} + d}, \qquad d = -\frac{\lambda}{\rho g}.$$

The three constants are determined by the two endpoint values and

$$L_0 = c\left[\sinh\frac{x_2-b}{c} - \sinh\frac{x_1-b}{c}\right].$$

The manual repeats x_2 in both terms of this length formula. The second evaluation is at x_1; otherwise the displayed length would be zero.

To verify that the hanging solution minimizes energy, rather than merely makes it stationary, parameterize it by arc length $s \in [0, L_0]$. There is a constant s_0 such that

$$s - s_0 = c\sinh\frac{x-b}{c}, \qquad \mathbf{T}(s) = \rho g\big(c,\ s-s_0\big), \qquad \mathbf{r}_*'(s) = \frac{\mathbf{T}(s)}{|\mathbf{T}(s)|}.$$

Thus $\mathbf{T}' = \rho g\,\mathbf{e}_y$. Any other cable of the same length with the same endpoints has an arc-length parameterization $\mathbf{r}(s)$ with $|\mathbf{r}'| = 1$. Integration by parts gives

$$\begin{aligned} U[\mathbf{r}] - U[\mathbf{r}_*] &= \int_0^{L_0} \mathbf{T}'\cdot(\mathbf{r}-\mathbf{r}_*)\,ds \\ &= -\int_0^{L_0} \mathbf{T}\cdot(\mathbf{r}'-\mathbf{r}_*')\,ds \\ &= \int_0^{L_0} \big(|\mathbf{T}| - \mathbf{T}\cdot\mathbf{r}'\big)\,ds \ge 0, \end{aligned}$$

where the endpoint term vanishes and the last inequality is Cauchy–Schwarz. Equality requires the same tangent everywhere, hence the same curve with the fixed endpoints. A curved cable needs L_0 greater than the straight-line distance between the endpoints; equality gives the taut straight segment, and a shorter cable cannot join the endpoints. For vertically aligned supports the smooth graph assumption fails. In the ideal zero-thickness model a slack cable has a limiting vertical fold. If its endpoint heights are y_1, y_2 and $L_0 > |y_2 - y_1|$, its lowest point is

$$y_b = \frac{y_1 + y_2 - L_0}{2},$$

since $(y_1 - y_b) + (y_2 - y_b) = L_0$. The same calibration proof applies with horizontal tension zero and $\mathbf{T} = \rho g(s-s_0)\mathbf{e}_y$: the cable goes downward for $s < s_0 = y_1 - y_b$ and upward for $s > s_0$.

22.4.4 A fixed volume of water is rotating in a cylinder with constant angular velocity ω. Find the curve of the water surface that will minimize the total potential energy of the water in the combined gravitational-centrifugal force field. Assume constant density, uniform gravity, a flat-bottomed cylinder, no overflow, and negligible surface tension. Allow a dry central region if required by nonnegative water depth.

ANS. Parabola.

Solution.

In the manual's energy integral the centrifugal term is missing the square on ω. Also, its equation $2gry - \omega^2 r^3 - 2\lambda r = 0$ gives $y = (\omega^2 r^2 + 2\lambda)/(2g)$, with a plus sign for its stated multiplier convention. The derivation below uses $U + \lambda V$. Work in the frame rotating with the water. The potential per unit mass for gravity plus the centrifugal force is

$$\Phi(r, z) = gz - \frac{1}{2}\omega^2 r^2,$$

since $-\nabla\Phi = -g\mathbf{e}_z + \omega^2 r\mathbf{e}_r$. Let the cylinder have radius R, a flat bottom $z = 0$, and water height $h(r)$. The density ρ is constant. Integrating through the depth gives

$$U[h] = 2\pi\rho \int_0^R \left(\frac{g}{2}h^2 - \frac{\omega^2 r^2}{2}h\right) r\, dr, \qquad V = 2\pi \int_0^R h(r) r\, dr.$$

For $h > 0$, stationarity of $U + \lambda V$ gives the pointwise equation

$$\rho\left(gh - \frac{\omega^2 r^2}{2}\right) + \lambda = 0.$$

Consequently the free surface is

$$\boxed{z = h(r) = h_0 + \frac{\omega^2 r^2}{2g}},$$

a paraboloid of revolution, whose section through the axis is a parabola. When the bottom is completely wet, the volume fixes

$$V = \pi R^2 h_0 + \frac{\pi\omega^2 R^4}{4g}, \qquad \boxed{h_0 = \frac{V}{\pi R^2} - \frac{\omega^2 R^2}{4g}}.$$

For a volume-preserving perturbation η, the first-order energy change vanishes because $\rho(gh - \omega^2 r^2/2)$ is constant, and

$$U[h + \eta] - U[h] = \frac{\rho g}{2} \int_{\text{bottom}} \eta^2\, dA \geq 0.$$

This proves the minimum.

The full-disk formula requires $h_0 \geq 0$ and no overflow. If it predicts a negative depth, a dry central region appears. The physical minimizer is $h(r) = \max\{0, C + \omega^2 r^2/(2g)\}$, with C chosen from the volume; the wetted free surface remains parabolic. For example, if its inner wet radius is r_d, then $h = \omega^2(r^2 - r_d^2)/(2g)$ on $r_d \leq r \leq R$, and direct integration gives $V = \pi\omega^2(R^2 - r_d^2)^2/(4g)$. The same minimum follows from convexity of the quadratic integrand, with the inequality $h \geq 0$ imposed.

22.4.5 Consider simple closed regular plane curves enclosing positive area.

(a) Show that for a fixed-length perimeter the plane figure with maximum area is a circle.

(b) Show that for a fixed planar area the boundary with minimum perimeter is a circle.

Hint. The geometric radius of curvature R is given by

$$R = \frac{(r^2 + r_\theta^2)^{3/2}}{\left| r r_{\theta\theta} - 2r_\theta^2 - r^2 \right|}.$$

Note. The problems of this section, variation subject to constraints, are often called *isoperimetric*. The term arose from problems of maximizing area subject to a fixed perimeter, as in part (a) of this problem.

Solution.

Let a regular simple closed curve be positively oriented and write $\mathbf{r}(u) = (x(u), y(u))$, $v = \sqrt{x'^2 + y'^2}$. Its area and length are

$$A = \frac{1}{2} \oint (xy' - yx')\, du, \qquad L = \oint v\, du.$$

Both parts lead to the same stationary curves: vary $A + \lambda L$ in part (a), or $L + \mu A$ in part (b), with a constant multiplier. For $A + \lambda L$, the x Euler equation is

$$\frac{y'}{2} - \frac{d}{du}\left(-\frac{y}{2} + \lambda \frac{x'}{v} \right) = 0, \quad \text{so} \quad \lambda \frac{d}{du}\frac{x'}{v} = y'.$$

The y equation similarly gives

$$-\frac{x'}{2} - \frac{d}{du}\left(\frac{x}{2} + \lambda \frac{y'}{v} \right) = 0, \quad \text{so} \quad \lambda \frac{d}{du}\frac{y'}{v} = -x'.$$

Use arc length s, so $x_s^2 + y_s^2 = 1$. Integrating these equations,

$$x + \lambda y_s = C_1, \qquad y - \lambda x_s = C_2.$$

Squaring and adding yields

$$(x - C_1)^2 + (y - C_2)^2 = \lambda^2.$$

Thus every regular extremal is a circle.

It remains to prove the requested global maximum and minimum. Translate the curve so $\int_0^L \mathbf{r}(s)\, ds = 0$. For a mean-zero L-periodic function f, its Fourier expansion gives

$$\int_0^L f'^2\, ds = \frac{L}{2} \sum_{n \geq 1} \left(\frac{2\pi n}{L} \right)^2 (a_n^2 + b_n^2) \geq \left(\frac{2\pi}{L} \right)^2 \frac{L}{2} \sum_{n \geq 1} (a_n^2 + b_n^2) = \left(\frac{2\pi}{L} \right)^2 \int_0^L f^2\, ds.$$

Apply this to both coordinates. Because $|\mathbf{r}_s| = 1$,

$$\int_0^L |\mathbf{r}|^2\, ds \leq \frac{L^2}{4\pi^2} \int_0^L |\mathbf{r}_s|^2\, ds = \frac{L^3}{4\pi^2}.$$

Cauchy–Schwarz now gives the isoperimetric inequality:

$$2A = \int_0^L (xy_s - yx_s)\,ds \le \left(\int_0^L |\mathbf{r}|^2\,ds\right)^{1/2}\left(\int_0^L |\mathbf{r}_s|^2\,ds\right)^{1/2} \le \frac{L^2}{2\pi}.$$

Equality holds for a circle, for which $\mathbf{r}_s$ is a constant multiple of the 90° rotation of $\mathbf{r}$. Therefore

$$\boxed{A_{\max} = \frac{L^2}{4\pi} \quad \text{at fixed } L}, \qquad \boxed{L_{\min} = 2\sqrt{\pi A} \quad \text{at fixed } A}.$$

The original textbook hint's expression is a signed radius of curvature with the opposite sign to the usual counterclockwise convention: for $r = \text{constant} > 0$ it gives $R = -r$. A geometric radius requires the absolute value of its denominator. Its sign does not affect the conclusion that constant curvature gives a circle.

22.4.6 Show that requiring J, given by

$$J = \int_a^b \int_a^b K(x,t)\varphi(x)\varphi(t)\,dx\,dt,$$

to have a stationary value subject to the normalizing condition

$$\int_a^b \varphi^2(x)\,dx = 1$$

leads to a Hilbert–Schmidt integral equation of the form given in Eq. (21.64), provided the corresponding integral-operator eigenvalue is nonzero. State separately what happens in the nullspace.

Note. The kernel $K(x,t)$ is symmetric. Take real $\varphi \in L^2[a,b]$ and a real $K \in L^2([a,b]^2)$, so the quadratic functional is well defined.

Solution.

Assume real functions and a real symmetric kernel, $K(x,t) = K(t,x)$, on $[a,b]$, with integrals for which the following variations are valid. Define

$$(T\phi)(x) = \int_a^b K(x,t)\phi(t)\,dt, \qquad \mathcal{A}[\phi] = J[\phi] - \mu\left(\int_a^b \phi^2\,dx - 1\right).$$

Under $\phi \mapsto \phi + \epsilon\eta$,

$$\left.\frac{dJ}{d\epsilon}\right|_0 = \int_a^b \int_a^b K(x,t)[\eta(x)\phi(t) + \phi(x)\eta(t)]\,dx\,dt$$
$$= 2\int_a^b \eta(x)(T\phi)(x)\,dx.$$

In the second term, interchange x, t and use the symmetry of K. The normalization contributes $2\int \phi\eta\,dx$. Hence

$$\delta\mathcal{A} = 2\int_a^b \eta(x)\big[(T\phi)(x) - \mu\phi(x)\big]\,dx = 0.$$

Arbitrariness of η gives

$$\boxed{\int_a^b K(x,t)\phi(t)\,dt = \mu\phi(x)}.$$

Multiplying by $\phi(x)$ and integrating shows that $J = \mu \int \phi^2 = \mu$. For $\mu \neq 0$, set $\lambda = 1/\mu$:

$$\boxed{\phi(x) = \lambda \int_a^b K(x,t)\phi(t)\,dt}.$$

The textbook's reference to Eq. (20.64) should be Eq. (21.64), the homogeneous Hilbert–Schmidt equation. Moreover, the reciprocal form assumes $\mu \neq 0$. A normalized element of the kernel of T is also stationary, with $J = \mu = 0$, and cannot be represented by a finite reciprocal λ. For instance, $K = 1$ on $[0,1]^2$ and $\phi(x) = \sqrt{12}(x - \frac{1}{2})$ give $\int \phi^2 = 1$, $\int \phi = 0$, and $T\phi = 0$. The undivided eigenvalue equation above includes this case.

22.4.7 An unknown function satisfies the differential equation

$$y'' + \left(\frac{\pi}{2}\right)^2 y = 0$$

and the boundary conditions $y'(0) = 0$, $y(1) = 0$, with $y(0) = 1$ fixing the normalization. Regard this as the ground-state problem for $-y'' = \lambda y$, whose exact eigenvalue is $(\pi/2)^2$.

(a) Calculate the approximation $\lambda = F[y_{\text{trial}}]$ for $y_{\text{trial}} = 1 - x^2$.

(b) Compare with the exact eigenvalue.

ANS. (a) $\lambda = 2.5$. (b) $\lambda/\lambda_{\text{exact}} = 1.013$.

Solution.

The nonhomogeneous values $y(0) = 1$, $y(1) = 0$ alone do not define a Sturm–Liouville eigenvalue problem with variable λ. The intended Rayleigh problem has $y'(0) = 0$, $y(1) = 0$, with $y(0) = 1$ fixing the scale of the eigenfunction. The given exact solution and both trial families in their optimum ranges obey that additional Neumann condition. Indeed, the quotient to be used is

$$F[y] = \frac{\int_0^1 y'^2\,dx}{\int_0^1 y^2\,dx}, \qquad y(1) = 0.$$

Varying its numerator minus λ times its denominator gives

$$2[y'\eta]_0^1 - 2\int_0^1 (y'' + \lambda y)\eta\,dx = 0.$$

The endpoint 1 has $\eta(1) = 0$, whereas the free endpoint 0 requires $y'(0) = 0$. The equation is $-y'' = \lambda y$.

(a) For $y_{\text{trial}} = 1 - x^2$,

$$\int_0^1 y'^2\,dx = 4\int_0^1 x^2\,dx = \frac{4}{3},$$

and

$$\int_0^1 y^2\,dx = \int_0^1 (1 - 2x^2 + x^4)\,dx = 1 - \frac{2}{3} + \frac{1}{5} = \frac{8}{15}.$$

Consequently

$$\boxed{\lambda_{\text{trial}} = \frac{4/3}{8/15} = \frac{5}{2} = 2.5}.$$

(b) The conditions $y'(0) = 0$, $y(1) = 0$ give $y = C\cos(\sqrt{\lambda}x)$ and $\cos\sqrt{\lambda} = 0$. The lowest eigenvalue and its normalized-at-zero eigenfunction are

$$\lambda_{\text{exact}} = \frac{\pi^2}{4}, \qquad y_{\text{exact}} = \cos\frac{\pi x}{2}.$$

Thus

$$\boxed{\frac{\lambda_{\text{trial}}}{\lambda_{\text{exact}}} = \frac{10}{\pi^2} = 1.013211836\ldots},$$

an excess of approximately 1.32118%.

22.4.8 In Exercise 22.4.7 use a trial function $y = 1 - x^n$, with $n > 1/2$ for finite energy.

(a) Find the value of n that will minimize $F[y_{\text{trial}}]$.

(b) Show that the optimum value of n drives the ratio $\lambda/\lambda_{\text{exact}}$ down to 1.003.

ANS. (a) $n = 1.7247$.

Solution.

Use the Rayleigh quotient and boundary interpretation established in Exercise 22.4.7. The trial $y = 1 - x^n$ has finite derivative energy precisely for $n > \frac{1}{2}$, because

$$N = \int_0^1 y'^2\,dx = n^2\int_0^1 x^{2n-2}\,dx = \frac{n^2}{2n-1}.$$

Its squared norm is

$$\begin{aligned} D &= \int_0^1 (1 - 2x^n + x^{2n})\,dx \\ &= 1 - \frac{2}{n+1} + \frac{1}{2n+1} \\ &= \frac{(n+1)(2n+1) - 2(2n+1) + (n+1)}{(n+1)(2n+1)} = \frac{2n^2}{(n+1)(2n+1)}. \end{aligned}$$

Therefore

$$F(n) = \frac{(n+1)(2n+1)}{2(2n-1)}.$$

Differentiating without dropping the denominator derivative,

$$F'(n) = \frac{(4n+3)(4n-2) - 4(2n^2+3n+1)}{(4n-2)^2} = \frac{4n^2 - 4n - 5}{2(2n-1)^2}.$$

The critical points solve $4n^2 - 4n - 5 = 0$, so the admissible one is

$$\boxed{n_{\text{opt}} = \frac{1+\sqrt{6}}{2} = 1.724744871\ldots}.$$

The other root is negative. The derivative changes from negative to positive at n_{opt}, and $F(n) \to \infty$ at both ends of $(\frac{1}{2}, \infty)$, so this is the global minimum in the trial family. Substitution yields

$$F_{\min} = \frac{5+2\sqrt{6}}{4}, \qquad \boxed{\frac{F_{\min}}{\pi^2/4} = \frac{5+2\sqrt{6}}{\pi^2} = 1.002976318\ldots}.$$

The energy error is therefore approximately 0.297632%.

The variational domain here is $n > \frac{1}{2}$, not merely $n > 0$: smaller positive n satisfy the endpoint values but have infinite derivative energy. The quadratic-form quotient $\int y'^2 / \int y^2$ remains the appropriate expression throughout this domain; replacing it by $-\int yy'' / \int y^2$ without checking the boundary term is invalid for some of these trials. At the optimum $n > 1$, $y'(0) = 0$, so that boundary term vanishes.

22.4.9 A quantum mechanical particle in a sphere (Example 14.7.1) satisfies

$$\nabla^2 \psi + k^2 \psi = 0,$$

with $k^2 = 2mE/\hbar^2$. The boundary condition is that $\psi = 0$ at $r = a$, where a is the radius of the sphere. For the ground state [where $\psi = \psi(r)$], try an approximate wave function,

$$\psi_a(r) = 1 - \left(\frac{r}{a}\right)^2,$$

and calculate an approximate eigenvalue k_a^2. Take $a > 0$ and impose regularity at the origin.

Hint. To determine $p(r)$ and $w(r)$, put your equation in self-adjoint form (in spherical polar coordinates).

ANS. $k_a^2 = \dfrac{10.5}{a^2}$, $\quad k_{\text{exact}}^2 = \dfrac{\pi^2}{a^2}$.

Solution.

For a radial function, the differential equation is

$$\frac{1}{r^2}\frac{d}{dr}(r^2\psi') + k^2\psi = 0, \quad \text{or} \quad (r^2\psi')' + k^2 r^2 \psi = 0.$$

Thus the self-adjoint coefficients are $p(r) = r^2$, $w(r) = r^2$, $q(r) = 0$. Multiplication by ψ and integration by parts gives

$$k^2 = \frac{\int_0^a r^2 \psi'^2 \, dr}{\int_0^a r^2 \psi^2 \, dr}.$$

The boundary term $[r^2 \psi \psi']_0^a$ vanishes at a because $\psi(a) = 0$, and at the origin for the regular radial solution.

For $\psi_{\text{trial}} = 1 - r^2/a^2$,

$$N = \frac{4}{a^4}\int_0^a r^4 \, dr = \frac{4a}{5},$$

whereas

$$\begin{aligned} D &= \int_0^a \left(r^2 - \frac{2r^4}{a^2} + \frac{r^6}{a^4}\right) dr \\ &= a^3\left(\frac{1}{3} - \frac{2}{5} + \frac{1}{7}\right) = \frac{8a^3}{105}. \end{aligned}$$

Therefore

$$\boxed{k_{\text{trial}}^2 = \frac{4a/5}{8a^3/105} = \frac{21}{2a^2} = \frac{10.5}{a^2}}.$$

The exact regular solution is proportional to $\sin(kr)/(kr)$. The smallest positive zero at $r = a$ is $ka = \pi$, so

$$k_{\text{exact}}^2 = \frac{\pi^2}{a^2}, \qquad \frac{k_{\text{trial}}^2}{k_{\text{exact}}^2} = \frac{21}{2\pi^2} = 1.063872428\ldots.$$

The resulting energy is approximately 6.38724% above the exact ground-state value, since $E = \hbar^2 k^2/(2m)$.

22.4.10 The wave equation for a quantum mechanical oscillator may be written as

$$\frac{d^2\psi(x)}{dx^2} + (\lambda - x^2)\psi(x) = 0,$$

with $\lambda = 1$ for the ground state; see Eq. (18.17). Take

$$\psi_{\text{trial}} = \begin{cases} 1 - \dfrac{x^2}{a^2}, & x^2 \le a^2 \\ 0, & x^2 > a^2 \end{cases}$$

for the ground-state wave function (with a^2 an adjustable parameter) and calculate the corresponding ground-state energy. How much error do you have? Take $a > 0$ and use the Rayleigh quotient with kinetic term $\int |\psi'|^2 dx$; the trial has derivative jumps at its cutoffs.

Note. Your parabola is really not a very good approximation to a Gaussian exponential. What improvements can you suggest?

Solution.

The dimensionless Hamiltonian is $H = -d^2/dx^2 + x^2$, so its Rayleigh quotient is

$$\lambda[\psi] = \frac{\int_{\mathbb{R}} (\psi'^2 + x^2\psi^2)\, dx}{\int_{\mathbb{R}} \psi^2\, dx}.$$

The stated trial is continuous, but its derivative jumps at $\pm a$. It belongs to the quadratic-form domain, and the quotient above is valid. Treating its second derivative as an ordinary function across those joins would omit delta-function terms. All three integrals reduce to $[-a, a]$:

$$\begin{aligned} D &= 2\int_0^a \left(1 - \frac{2x^2}{a^2} + \frac{x^4}{a^4}\right) dx = 2a\left(1 - \frac{2}{3} + \frac{1}{5}\right) = \frac{16a}{15}, \\ N_{\text{kin}} &= 2\int_0^a \frac{4x^2}{a^4}\, dx = \frac{8}{3a}, \\ N_{\text{pot}} &= 2\int_0^a \left(x^2 - \frac{2x^4}{a^2} + \frac{x^6}{a^4}\right) dx = 2a^3\left(\frac{1}{3} - \frac{2}{5} + \frac{1}{7}\right) = \frac{16a^3}{105}. \end{aligned}$$

Hence

$$\lambda(a) = \frac{5}{2a^2} + \frac{a^2}{7}.$$

Put $A = a^2 > 0$. Then

$$\frac{d\lambda}{dA} = -\frac{5}{2A^2} + \frac{1}{7} = 0 \quad \Longrightarrow \quad A^2 = \frac{35}{2}, \qquad \boxed{a = \left(\frac{35}{2}\right)^{1/4}}.$$

The second derivative $5/A^3$ is positive. At the minimum the two energy terms are equal:

$$\boxed{\lambda_{\min} = 2\sqrt{\frac{5}{14}} = \sqrt{\frac{10}{7}} = 1.195228609\ldots}.$$

Thus the relative error is

$$\boxed{100(\lambda_{\min}-1)\% = 19.5228609\ldots\%}.$$

In conventional oscillator units, $E_{\text{trial}} = (\hbar\omega/2)\sqrt{10/7}$. The solution manual's final value $2\sqrt{10/7}$ has an extra factor of two. Substitution into its own expression $5/(2a^2) + a^2/7$ gives $\sqrt{10/7}$.

One improvement is to make both the value and slope vanish at the cutoff:

$$\psi = (1 - x^2/a^2)^2 \quad (|x| \le a), \qquad \psi = 0 \quad (|x| > a).$$

Writing $t = x/a$, the needed polynomial integrals are

$$\int_{-1}^{1}(1-t^2)^4 dt = \frac{256}{315}, \quad \int_{-1}^{1} 16t^2(1-t^2)^2 dt = \frac{256}{105}, \quad \int_{-1}^{1} t^2(1-t^2)^4 dt = \frac{256}{3465}.$$

They give

$$\lambda(a) = \frac{3}{a^2} + \frac{a^2}{11}, \qquad a^4 = 33, \qquad \lambda_{\min} = 2\sqrt{\frac{3}{11}} = 1.044465936\ldots,$$

only 4.44659% high. A still better family is the Gaussian $\psi = e^{-\beta x^2/2}$, $\beta > 0$. Since $\langle x^2\rangle = 1/(2\beta)$ and $\psi' = -\beta x\psi$,

$$\lambda(\beta) = \frac{\beta}{2} + \frac{1}{2\beta} \ge 1,$$

with equality at $\beta = 1$, which recovers the exact ground state.

22.4.11 The Schrödinger equation for a central potential may be written as

$$\mathcal{L}u(r) + \frac{\hbar^2 l(l+1)}{2Mr^2}\,u(r) = Eu(r).$$

The $l(l+1)$ term, the angular momentum barrier, comes from splitting off the angular dependence. Compare Eq. (9.80) (divide that equation through by $-r^2$). Use the Rayleigh–Ritz technique to show that $E > E_0$ for $l > 0$ and normalizable bound states, where E_0 is the lowest energy eigenvalue of $\mathcal{L}u_0 = E_0u_0$ corresponding to $l = 0$. This means that the ground state will have $l = 0$, zero angular momentum. Assume a self-adjoint radial Hamiltonian bounded below, with the usual common quadratic-form domain.

Hint. For a discrete orthonormal eigenbasis, expand $u(r) = \sum_{i=0}^{\infty} c_i u_i(r)$, where $\mathcal{L}u_i = E_i u_i$ and $E_i \ge E_0$. Do not require $c_0 = 1$; a spectral integral handles any continuous part of the spectrum.

Solution.

The strict inequality is for $l > 0$ and a normalizable bound eigenstate, with E_0 the lowest eigenvalue of the $l = 0$ radial operator. For $l = 0$ the ground state has equality. A continuum threshold need not have a strict gap above E_0. Normalize u by $\int_0^\infty |u|^2 dr = 1$ and assume it belongs to the quadratic-form domain of the radial Hamiltonian, including the centrifugal term. For form-domain functions, $\langle u, Lu\rangle$ below denotes the associated quadratic form; it agrees with the operator inner product when u is in the operator domain. The variational lower bound for L is

$$\langle u, Lu\rangle \ge E_0\langle u, u\rangle = E_0.$$

For the discrete expansion in the hint, this follows directly from $u = \sum_{i\ge 0} c_i u_i$ in an orthonormal eigenbasis:

$$\sum_i |c_i|^2 = 1, \qquad \langle u, Lu\rangle = \sum_i |c_i|^2 E_i = E_0 + \sum_{i\ge 1} |c_i|^2 (E_i - E_0) \ge E_0.$$

There is no need to assume $c_0 \neq 0$ or to divide by it. If there is continuous spectrum, the same argument uses its spectral integral; the defining lower bound of L suffices.

Multiply the radial equation by u^* and integrate:

$$E = \langle u, Lu \rangle + \frac{\hbar^2 l(l+1)}{2M} \int_0^\infty \frac{|u(r)|^2}{r^2}\, dr.$$

For a nonzero normalizable state the last integral is strictly positive: its integrand is nonnegative, and $r^{-2} > 0$ at every finite $r > 0$. For $l > 0$ its coefficient is also positive. Thus

$$\boxed{E > E_0 \qquad (l > 0)}.$$

Every bound state with nonzero angular momentum lies above the $l = 0$ ground state. Consequently, if a bound ground state exists for the central-potential problem, it has $l = 0$.

23 Chapter 23. Probability and Statistics

Problems

Conventions for the card and dice exercises: use a standard 52-card deck without jokers and uniform draws, reshuffling after a replacement. Dice rolls and coin tosses are independent, and a fair die has six equally likely faces.

23.1 Probability: Definitions, Simple

23.1.1 A card is drawn from a shuffled deck. (a) What is the probability that it is black, (b) a red nine, (c) or a queen of spades?

Solution.

Use the standard 52-card deck, with four suits of 13 cards and no jokers. Every card has probability $1/52$. There are 26 black cards, two red nines (hearts and diamonds), and one queen of spades. Thus

$$\boxed{P(\text{black}) = \frac{26}{52} = \frac{1}{2}, \quad P(\text{red nine}) = \frac{2}{52} = \frac{1}{26}, \quad P(Q\spadesuit) = \frac{1}{52}.}$$

These are three separate questions; their events need not be disjoint.

23.1.2 Find the probability of drawing two kings from a shuffled deck of cards (a) if the first card is put back before the second is drawn, and (b) if the first card is not put back after being drawn.

Solution.

For a standard 52-card deck there are four kings. Use $P(K_1 \cap K_2) = P(K_1)P(K_2 \mid K_1)$. (a) Replacement restores all 52 cards and all four kings, so

$$\boxed{P = \frac{4}{52}\frac{4}{52} = \frac{1}{169}.}$$

(b) Without replacement a first king leaves three kings among 51 cards:

$$\boxed{P = \frac{4}{52}\frac{3}{51} = \frac{1}{221}.}$$

Equivalently, the unordered count is $\binom{4}{2}/\binom{52}{2} = 1/221$.

23.1.3 When two fair dice are thrown, what is the probability that their sum is (a) less than 4, or (b) greater than or equal to 4 but less than 6?

Solution.

Interpret the observed number as the *sum* $S = D_1 + D_2$, as in Example 23.2.1. The 36 ordered pairs (D_1, D_2) are equiprobable. (a) $S < 4$ consists of $(1,1), (1,2), (2,1)$, hence

$$\boxed{P(S < 4) = 3/36 = 1/12.}$$

(b) $4 \le S < 6$ means $S = 4$ or 5. Sum 4 has the three pairs $(1,3), (2,2), (3,1)$; sum 5 has the four pairs $(1,4), (2,3), (3,2), (4,1)$. Therefore

$$\boxed{P(4 \le S < 6) = (3+4)/36 = 7/36.}$$

23.1.4 Rolling three fair dice, what is the probability of obtaining six points?

Solution.

There are $6^3 = 216$ equiprobable ordered triples. Sum 6 has the unordered patterns $(1,1,4)$, $(1,2,3)$ and $(2,2,2)$, with respectively $3!/2! = 3$, $3! = 6$ and 1 distinct orderings. Consequently

$$\boxed{P(D_1 + D_2 + D_3 = 6) = \frac{3+6+1}{216} = \frac{5}{108}.}$$

Alternatively, subtract 1 from each die; the nonnegative triples summing to 3 number $\binom{3+2}{2} = 10$, and none violates the upper bound 5.

23.1.5 Determine the probability $P(A \cap B \cap C)$ in terms of $P(A)$, $P(B)$, $P(C)$, $P(A \cup B)$, $P(A \cup C)$, $P(B \cup C)$, and $P(A \cup B \cup C)$.

Solution.

For two events, $P(A \cap B) = P(A) + P(B) - P(A \cup B)$. Substitute this identity for each pair in inclusion–exclusion:

$$\begin{aligned} P(A \cup B \cup C) = {} & P(A) + P(B) + P(C) \\ & - [2P(A) + 2P(B) + 2P(C) \\ & \quad - P(A \cup B) - P(A \cup C) - P(B \cup C)] + P(A \cap B \cap C). \end{aligned}$$

Solving gives

$$\boxed{\begin{aligned} P(A \cap B \cap C) = {} & P(A \cup B \cup C) + P(A) + P(B) + P(C) \\ & - P(A \cup B) - P(A \cup C) - P(B \cup C). \end{aligned}}$$

The manual's conditional-probability product is valid when its conditional probabilities are defined, but does not express the answer in terms of the seven probabilities requested here.

23.1.6 Determine directly or by mathematical induction (Section 1.4) the probability of a distribution of N (Maxwell–Boltzmann) particles in k boxes with N_1 in Box 1, N_2 in Box 2, ..., N_k in the kth box for any numbers $N_j \geq 1$ with $N_1 + N_2 + \cdots + N_k = N$, $k < N$. Treat each box as one single-particle state. Use independent uniform box choices for Maxwell–Boltzmann particles and uniform allowed occupation states for Bose–Einstein particles; compare with conditioning on all boxes being nonempty. For Fermi–Dirac particles, explain why these parameters do not define a normalized ensemble.

Solution.

A probability requires a specified ensemble and its weights. For this exercise, use independent uniform box choices for classical particles, and uniformly weighted allowed occupation states for identical particles. The manual lists incorrect counts instead of normalized probabilities. With $N > k$, the specified Fermi ensemble has no allowed state at all.

For Maxwell–Boltzmann particles label the particles $1, \ldots, N$. There are k^N assignments. To obtain the stated occupation vector, choose its labels successively:

$$\binom{N}{N_1}\binom{N-N_1}{N_2}\cdots\binom{N-N_1-\cdots-N_{k-1}}{N_k} = \frac{N!}{N_1!\cdots N_k!}.$$

Every assignment has probability k^{-N}, so

$$\boxed{P_{\mathrm{MB}}(N_1,\dots,N_k) = \frac{N!}{k^N \prod_j N_j!}.}$$

This is the probability of that positive occupation vector in the *unrestricted* ensemble, which also permits empty boxes. If one conditions on all boxes being nonempty, the denominator k^N must instead be the number of onto assignments,

$$S_{N,k} = \sum_{r=0}^{k} (-1)^r \binom{k}{r} (k-r)^N, \qquad P_{\mathrm{MB}}(\boldsymbol{N} \mid \text{nonempty}) = \frac{N!}{S_{N,k} \prod_j N_j!}.$$

The sum follows by excluding assignments missing at least one box.

For Bose–Einstein particles each occupation vector is one state. Place $k-1$ separators among N indistinguishable particles: the number of nonnegative occupation vectors is $\binom{N+k-1}{k-1}$. Thus, under the stated uniform-state convention,

$$\boxed{P_{\mathrm{BE}}(\boldsymbol{N}) = \binom{N+k-1}{k-1}^{-1}.}$$

If nonempty boxes are an imposed condition, first put one particle in each box and distribute $N-k$ more; there are $\binom{N-1}{k-1}$ states, giving its reciprocal instead.

For Fermi–Dirac particles a single box can contain only 0 or 1 particle. For $0 \le N \le k$ there are $\binom{k}{N}$ states, each with probability $\binom{k}{N}^{-1}$ under uniform weighting. Here $N > k$, so at least one $N_j > 1$: its multiplicity is zero and the entire allowed sample space is empty. There is *no normalized FD probability distribution* for these parameters. If positivity is retained but $N = k$ is allowed, the sole state is $(1,\dots,1)$ with probability 1. Physical energy-dependent ensembles would require additional weights.

23.1.7 Show that $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$.

Solution.

For any outcome, put $a = 1_A$, $b = 1_B$, $c = 1_C$, each 0 or 1. The indicator of membership in at least one event is

$$1_{A \cup B \cup C} = 1 - (1-a)(1-b)(1-c) = a + b + c - ab - ac - bc + abc.$$

Take expectations and use $\langle 1_E \rangle = P(E)$ and $1_A 1_B = 1_{A \cap B}$. This gives

$$\boxed{\begin{aligned} P(A \cup B \cup C) = {} & P(A) + P(B) + P(C) - P(A \cap B) \\ & - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C). \end{aligned}}$$

No independence or disjointness assumption is needed.

23.1.8 Determine the probability that a positive integer $n \le 100$, chosen uniformly, is divisible by a fixed prime number $p \le 100$. Verify your result for $p = 3, 5, 7$.

Solution.

Choose n uniformly from $1, \dots, 100$ and fix the prime p. Its multiples in this set are $p, 2p, \dots, \lfloor 100/p \rfloor p$; therefore

$$\boxed{P(p \mid n) = \frac{\lfloor 100/p \rfloor}{100}.}$$

For $p = 3, 5, 7$ the answers are respectively $33/100$, $20/100$, and $14/100$. The approximation $1/p$ has error

$$0 \le \frac{1}{p} - P(p \mid n) < \frac{1}{100}.$$

Here the notation $p \mid n$ means divisibility, not conditional probability. If "a prime" meant *some* prime rather than a fixed p, every $n = 2, \dots, 100$ qualifies, and the different question has answer $99/100$.

23.1.9 Put two particles obeying Maxwell–Boltzmann (Fermi–Dirac, or Bose–Einstein) statistics in three boxes. How many ways of doing so are there in each case? Each box represents a single-particle state; use the counting conventions of Exercise 23.1.6, now with $N = 2$, $k = 3$.

Solution.

For Maxwell–Boltzmann statistics label the particles a, b. Each chooses any of three boxes, so the ordered box pair has $3^2 = 9$ possibilities. For Fermi–Dirac statistics choose two distinct boxes for the two indistinguishable particles: $\binom{3}{2} = 3$ states,

$$(1,1,0),\ (1,0,1),\ (0,1,1).$$

For Bose–Einstein statistics also allow the three doubly occupied states $(2,0,0), (0,2,0), (0,0,2)$. Thus

$$\boxed{W_{\mathrm{MB}} = 9, \qquad W_{\mathrm{FD}} = 3, \qquad W_{\mathrm{BE}} = 6.}$$

The BE count also equals $\binom{2+3-1}{2} = 6$.

23.2 Random Variables

23.2.1 Verify the probabilities for the outcomes of the two-card draws in Example 23.2.4, and by direct computation of the mean and variance check the results given in that example.

Solution.

Example 23.2.4 draws two *distinct* cards from the four cards numbered 1 through 4, without replacement. The six equiprobable unordered draws and their sums are

draw	$\{1,2\}$	$\{1,3\}$	$\{1,4\}$	$\{2,3\}$	$\{2,4\}$	$\{3,4\}$
X	3	4	5	5	6	7

Each draw has probability $1/6$, hence $P(3) = P(4) = P(6) = P(7) = 1/6$ and $P(5) = 2/6 = 1/3$. Direct summation gives

$$\langle X \rangle = \frac{3+4+5+5+6+7}{6} = 5, \qquad \langle X^2 \rangle = \frac{9+16+25+25+36+49}{6} = \frac{80}{3}.$$

Either subtract 25 or evaluate the centered squares directly:

$$\boxed{\operatorname{Var} X = \frac{(-2)^2 + (-1)^2 + 0^2 + 0^2 + 1^2 + 2^2}{6} = \frac{5}{3}.}$$

23.2.2 Show that adding a constant c to a random variable X changes the expectation value $\langle X \rangle$ by that same constant but not the variance. Show also that multiplying a random variable by a real constant c multiplies the mean by c and the variance by c^2. Show that the random variable $X - \langle X \rangle$ has mean value zero. Assume X is real with a finite second moment.

Solution.

Multiplication by a real constant c multiplies the mean by c, the variance by c^2, and the standard deviation by $|c|$. The original statement's claim of a factor c for the variance is false. Write $\mu = \langle X \rangle$ and assume a finite second moment. Linearity of expectation and $\langle 1 \rangle = 1$ give

$$\langle X + c \rangle = \mu + c, \qquad \operatorname{Var}(X + c) = \langle (X + c - \mu - c)^2 \rangle = \operatorname{Var} X.$$

Likewise

$$\langle cX \rangle = c\mu, \qquad \operatorname{Var}(cX) = \langle (cX - c\mu)^2 \rangle = c^2 \operatorname{Var} X.$$

These formulas also include $c = 0$ and negative c. Finally $\boxed{\langle X - \langle X \rangle \rangle = \mu - \mu = 0}$. For a quick contradiction to the printed variance rule, $c = -1$ leaves every squared deviation unchanged, rather than making a positive variance negative.

23.2.3 Using the definition given in Eq. (23.27) for the variance σ^2 of a continuous random variable, show that

$$\sigma^2 = \langle X^2 \rangle - \langle X \rangle^2.$$

Assume a finite second moment.

Solution.

Let $f(x) \geq 0$, $\int_{-\infty}^{\infty} f(x)\,dx = 1$, and suppose $\int x^2 f(x)\,dx < \infty$. Writing $\mu = \langle X \rangle$, expand before integrating:

$$\begin{aligned}
\sigma^2 &= \int_{-\infty}^{\infty} (x - \mu)^2 f(x)\,dx \\
&= \int x^2 f(x)\,dx - 2\mu \int x f(x)\,dx + \mu^2 \int f(x)\,dx \\
&= \langle X^2 \rangle - 2\mu^2 + \mu^2 = \boxed{\langle X^2 \rangle - \langle X \rangle^2}.
\end{aligned}$$

All undisplayed limits in the second line are $-\infty, \infty$.

23.2.4 A velocity $v_j = x_j/t_j$ is measured by recording the distances x_j at the corresponding times t_j. Show that $\bar{x}/\bar{t}$ is a good approximation for the average velocity v, provided that all the errors are small: $|x_j - \bar{x}| \ll |\bar{x}|$ and $|t_j - \bar{t}| \ll |\bar{t}|$. Here $\bar{v} = N^{-1} \sum_j x_j/t_j$ is the arithmetic mean of the individual velocities, with $\bar{x}, \bar{t} \neq 0$.

Solution.

Let there be N measurements, $\bar{v} = N^{-1} \sum_j x_j/t_j$, and write $x_j = \bar{x}(1 + a_j)$, $t_j = \bar{t}(1 + b_j)$. The stated relative-error hypotheses require $\bar{x} \neq 0$, $\bar{t} \neq 0$; also $\sum a_j = \sum b_j = 0$. An exact algebraic identity is

$$\frac{1 + a_j}{1 + b_j} = 1 + a_j - b_j + \frac{b_j^2 - a_j b_j}{1 + b_j}.$$

Average it. The two linear terms cancel, leaving

$$\boxed{\bar{v} = \frac{\bar{x}}{\bar{t}}\left[1 + \frac{1}{N}\sum_j \frac{b_j^2 - a_j b_j}{1 + b_j}\right].}$$

If $|a_j| \le \epsilon_x$ and $|b_j| \le \epsilon_t < 1$, then

$$\left|\frac{\bar{v}}{\bar{x}/\bar{t}} - 1\right| \le \frac{\epsilon_t^2 + \epsilon_x \epsilon_t}{1 - \epsilon_t}.$$

Thus the relative difference is second order in the small relative errors. In dimensional form its leading term is

$$\bar{v} = \frac{\bar{x}}{\bar{t}} + \frac{\bar{x}}{\bar{t}^3}\overline{(t-\bar{t})^2} - \frac{1}{\bar{t}^2}\overline{(x-\bar{x})(t-\bar{t})} + O(\epsilon^3)\frac{\bar{x}}{\bar{t}}.$$

No statistical independence of the distance and time errors was used.

23.2.5 Redefine the random variable Y in Example 23.2.5 as the number of fours through nines. Then determine the correlation of the X and Y random variables for the drawing of two cards (with replacement, as in the example).

Solution.

In one draw, the disjoint categories are honors (10,J,Q,K,A), fours through nines, and twos or threes. Their probabilities are $p = 5/13$, $q = 6/13$, $r = 2/13$. For two independent draws,

$$P(X = x, Y = y) = \frac{2!}{x!y!(2-x-y)!} p^x q^y r^{2-x-y}, \quad x, y \ge 0,\ x + y \le 2.$$

The complete joint table is

$169P(x,y)$	$y=0$	$y=1$	$y=2$
$x=0$	4	24	36
$x=1$	20	60	0
$x=2$	25	0	0

It sums to 169. Directly from its rows and columns,

$$\begin{aligned}
\langle X \rangle &= (20 + 60 + 2 \cdot 25)/169 = 10/13, \\
\langle X^2 \rangle &= (20 + 60 + 4 \cdot 25)/169 = 180/169, \\
\langle Y \rangle &= (24 + 60 + 2 \cdot 36)/169 = 12/13, \\
\langle Y^2 \rangle &= (24 + 60 + 4 \cdot 36)/169 = 228/169.
\end{aligned}$$

Only $(x, y) = (1, 1)$ contributes to XY, so $\langle XY \rangle = 60/169$. Consequently

$$\operatorname{Var} X = \frac{80}{169}, \quad \operatorname{Var} Y = \frac{84}{169}, \quad \operatorname{cov}(X, Y) = \frac{60 - 120}{169} = -\frac{60}{169},$$

$$\boxed{\rho_{XY} = \frac{-60}{\sqrt{80 \cdot 84}} = -\sqrt{\frac{15}{28}} \simeq -0.731925.}$$

The draws are independent, but the two category counts are negatively correlated because they compete for the same two positions.

23.2.6 The probability that a particle of an ideal gas travels a distance x between collisions is proportional to $e^{-x/f}dx$, where f is the constant mean free path. Verify that f is the average distance between collisions, and determine the probability of a free path of length $l \geq 3f$. Take $f > 0$ and distances $x \geq 0$.

Solution.

Distances satisfy $x \geq 0$ and $f > 0$. Normalize $p(x) = Ce^{-x/f}$:

$$1 = C\int_0^\infty e^{-x/f}\,dx = Cf, \qquad p(x) = f^{-1}e^{-x/f}.$$

With $u = x/f$, integration by parts gives

$$\langle X\rangle = f\int_0^\infty ue^{-u}\,du = f\left([-ue^{-u}]_0^\infty + \int_0^\infty e^{-u}\,du\right) = \boxed{f}.$$

The manual drops the factor f in its middle integral after rescaling. The dimensionless integral is $\int_0^\infty ue^{-u}du = 1$; the physical mean is f times that integral. The requested tail, including an endpoint of zero point probability, is

$$\boxed{P(X \geq 3f) = \int_{3f}^\infty \frac{e^{-x/f}}{f}\,dx = e^{-3} \simeq 0.0497871.}$$

23.2.7 Determine the probability density for a particle in simple harmonic motion in the interval $-A \leq x \leq A$. For $A > 0$, observe the particle at a uniformly random time in a full period. State the degenerate result when $A = 0$.

Hint. The probability that the particle is between x and $x + dx$ is proportional to the time it takes to travel across the interval.

Solution.

Assume observation at a uniformly random time in a full period, with $A > 0$ and $\omega > 0$. For $x = A\cos\omega t$,

$$\left|\frac{dx}{dt}\right| = \omega\sqrt{A^2 - x^2}, \qquad T = \frac{2\pi}{\omega}.$$

Each interior interval is crossed twice per period. Therefore its fraction of the total time is

$$p(x)\,dx = \frac{2}{T}\frac{dx}{\omega\sqrt{A^2 - x^2}}, \qquad \boxed{p(x) = \frac{1}{\pi\sqrt{A^2 - x^2}} \quad (|x| < A),}$$

and $p = 0$ outside. The endpoint divergences are integrable:

$$\int_{-A}^{A} p(x)\,dx = \frac{1}{\pi}[\arcsin(x/A)]_{-A}^{A} = 1.$$

There are no point masses at the turning points. For $A = 0$ the distribution degenerates to a unit point mass at $x = 0$.

23.3 Binomial Distribution

23.3.1 Show that the variable $X = x$, defined as the number of heads in n fair coin tosses, is a random variable and determine its probability distribution. Describe the sample space. What are its mean value, the variance, and the standard deviation? Plot the probability function $P(x) = [n!/x!(n-x)!]2^{-n}$ for $n = 10, 20$, and 30 using graphics software.

Solution.

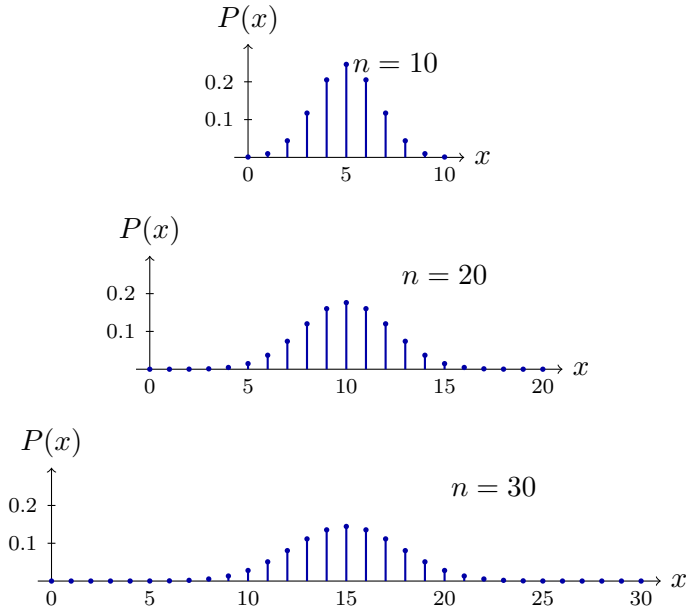
Use independent fair tosses. The elementary sample space is $\Omega = \{H, T\}^n$, with each ordered string having probability 2^{-n}. The map $X(\omega) = \sum_{j=1}^{n} 1_{\{\omega_j = H\}}$ is an integer-valued random variable on this finite space, taking values $0, \ldots, n$. Choosing the x head positions gives exactly $\binom{n}{x}$ strings, hence

$$\boxed{P(X = x) = \binom{n}{x} 2^{-n}, \quad 0 \le x \le n.}$$

The sum of these probabilities is $2^{-n}(1+1)^n = 1$. For each indicator I_j, $\langle I_j \rangle = 1/2$ and $\operatorname{Var} I_j = \langle I_j^2 \rangle - 1/4 = 1/4$. Independence makes the cross covariances zero, so

$$\boxed{\langle X \rangle = n/2, \qquad \operatorname{Var} X = n/4, \qquad \sigma_X = \sqrt{n}/2.}$$

The requested plots below show the probability *mass* at each integer, using exactly the displayed finite formula. The horizontal axis has the same scale in all three plots.



23.3.2 Plot the binomial probability function for the probabilities $p = 1/6$, $q = 5/6$, and $n = 6$ throws of a die.

Solution.

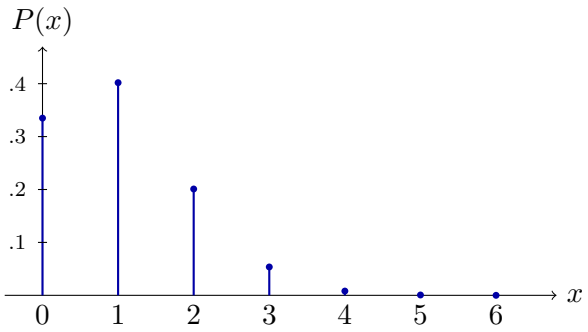
Define a success as a specified face, say a six, on each independent fair-die throw. The number X of successes in six throws has

$$P(X = x) = \binom{6}{x} (1/6)^x (5/6)^{6-x} = \frac{\binom{6}{x} 5^{6-x}}{46656}, \quad x = 0, \ldots, 6.$$

All values, and thus the plot without interpolation, are

x	0	1	2	3	4	5	6
$46656P(x)$	15625	18750	9375	2500	375	30	1

The numerator sum is 46656. The mean is $6/6 = 1$ and the largest mass is at $x = 1$.



23.3.3 A hardware company knows that the probability of mass-producing nails includes a small probability $p = 0.03$ of defective nails (usually without a sharp tip). What is the probability of finding more than two defective nails in its commercial box of 100 nails? Assume independent defects with the same probability for each nail.

Solution.

Assume nail defects are independent, with common probability 0.03. Then X, the number of defective nails, is binomial with $n = 100$. Use the complement of $X = 0, 1, 2$:

$$\begin{aligned} P(X > 2) &= 1 - \sum_{j=0}^{2} \binom{100}{j} (0.03)^j (0.97)^{100-j} \\ &= 1 - (0.97)^{100} - 100(0.03)(0.97)^{99} - 4950(0.03)^2(0.97)^{98} \\ &= \boxed{0.5802249170 \simeq 58.02\%.} \end{aligned}$$

The independence assumption matters: a defect rate alone would not determine the box probability if production defects were correlated.

23.3.4 Four cards are drawn from a shuffled bridge deck. What is the probability that they are all red? That they are all hearts? That they are honors? Compare the probabilities when each card is put back at a random place before drawing the next card, with the probabilities when the cards are not replaced in the deck.

Solution.

There are 26 red cards,13 hearts and 20 honors (10,J,Q,K,A). If a category contains K cards, replacement gives four independent successes with probability $(K/52)^4$. Without replacement the conditional success probabilities successively decrease:

$$P_K = \frac{K}{52}\frac{K-1}{51}\frac{K-2}{50}\frac{K-3}{49} = \frac{\binom{K}{4}}{\binom{52}{4}}.$$

Consequently

category	with replacement	without replacement
red	$1/16$	$\binom{26}{4}/\binom{52}{4}$
hearts	$1/256$	$\binom{13}{4}/\binom{52}{4}$
honors	$(5/13)^4$	$\binom{20}{4}/\binom{52}{4}$

For $0 < K < 52$ and $j = 1, 2, 3$, $(K-j)/(52-j) < K/52$, since $52(K-j) < K(52-j)$ is equivalent to $K < 52$. Thus all three probabilities are strictly smaller without replacement.

23.3.5 Show that for the binomial distribution of Eq. (23.42), the mean value of x is np; find the exact integer mode or modes.

Solution.

The mean is np, but the exact most probable integer need not be np. The manual differentiates the moment-generating function, which computes the mean and does not prove the printed mode claim. To establish the mean, write $X = I_1 + \cdots + I_n$, where each success indicator has $P(I_j = 1) = p$ and $P(I_j = 0) = 1 - p$. Then

$$\langle I_j \rangle = 1 \cdot p + 0 \cdot (1-p) = p, \qquad \boxed{\langle X \rangle = \sum_{j=1}^{n} \langle I_j \rangle = np}.$$

To find the mode for $0 < p < 1$, write $q = 1 - p$ and compare consecutive positive masses:

$$\frac{P(x+1)}{P(x)} = \frac{n-x}{x+1}\frac{p}{q}.$$

It exceeds 1 exactly when

$$(n-x)p > (x+1)(1-p) \quad \Longleftrightarrow \quad x + 1 < (n+1)p.$$

If $(n+1)p$ is not an integer, the sequence increases up to $\lfloor (n+1)p \rfloor$ and decreases thereafter. If $(n+1)p = k$ is an integer, $P(k) = P(k-1)$ and these are the two maxima. Hence

$$\boxed{\begin{cases} \text{unique mode } \lfloor (n+1)p \rfloor, & (n+1)p \notin \mathbb{Z}, \\ \text{modes } k-1, k, & (n+1)p = k \in \{1, \ldots, n\}. \end{cases}}$$

The endpoint cases $p = 0, 1$ have sole modes $0, n$, respectively; $n = 0$ is the point mass at 0. For example,$n = 3, p = 1/2$ has mean 1.5 but modes 1 and 2. For large n, a continuous central approximation places the peak near np; it is not the exact integer-mode statement.

23.4 Poisson Distribution

23.4.1 Radioactive decays for long-lived isotopes are governed by the Poisson distribution. In a Rutherford–Geiger experiment, the numbers of emitted α particles are counted in each of $n = 2608$ time intervals of 7.5 seconds each. In Table 23.1 n_i is the number of time intervals in which i particles were emitted. Determine the average number λ of particles emitted per time interval, and compare the n_i of Table 23.1 with np_i computed from the Poisson distribution with mean value λ.

Table 23.1 Data for Exercise 23.4.1

i	0	1	2	3	4	5	6	7	8	9	10
n_i	57	203	383	525	532	408	273	139	45	27	16

Solution.

The total number of recorded intervals and total particle count are

$$N = 57 + 203 + 383 + 525 + 532 + 408 + 273 + 139 + 45 + 27 + 16 = 2608,$$
$$S = 203 + 766 + 1575 + 2128 + 2040 + 1638 + 973 + 360 + 243 + 160 = 10086.$$

Thus the fitted mean per7.5-second interval is

$$\boxed{\widehat{\lambda} = S/N = 10086/2608 = 3.8673312883.}$$

Use $E_0 = Ne^{-\widehat{\lambda}}$, followed by the exact recurrence $E_{i+1} = \widehat{\lambda}E_i/(i+1)$, to compute $E_i = Np_i$.

i	observed n_i	Poisson expected Np_i
0	57	54.544
1	203	210.940
2	383	407.887
3	525	525.811
4	532	508.372
5	408	393.208
6	273	253.444
7	139	140.022
8	45	67.689
9	27	29.086
10	16	11.249
> 10	0	5.749

The final expected tail is $2608 - \sum_{i=0}^{10} E_i$; it is required because the Poisson model has unbounded support. Observed values fluctuate around their expectations rather than agreeing termwise. The manual's entry 55.7 at $i = 0$ is inconsistent with the stated data and fitted mean; the direct value is 54.544. Its displayed formula for np_i also omits the factor n on the right: $p_i = e^{-\lambda}\lambda^i/i!$, whereas the expected count is np_i. No renormalization to $i \le 10$ has been made.

23.4.2 Derive the standard deviation of a Poisson distribution of mean value μ.

Solution.

For $\mu > 0$, the Poisson mass is $p_n = e^{-\mu}\mu^n/n!$, $n = 0, 1, \ldots$. Shift the factorial sums:

$$\langle X \rangle = e^{-\mu} \sum_{n=1}^{\infty} \frac{n\mu^n}{n!} = \mu e^{-\mu} \sum_{m=0}^{\infty} \frac{\mu^m}{m!} = \mu,$$
$$\langle X(X-1) \rangle = e^{-\mu} \sum_{n=2}^{\infty} \frac{n(n-1)\mu^n}{n!} = \mu^2 e^{-\mu} \sum_{m=0}^{\infty} \frac{\mu^m}{m!} = \mu^2.$$

Since $X^2 = X(X-1) + X$, $\langle X^2 \rangle = \mu^2 + \mu$ and

$$\boxed{\operatorname{Var} X = \mu, \qquad \sigma_X = \sqrt{\mu}.}$$

At $\mu = 0$ the distribution is concentrated at 0 and the same result holds.

23.4.3 The number of α-particles emitted by the decay of a radium sample is counted per minute for 40 hours. The total number is 5000. How many 1-minute intervals are expected with (a) 2, and (b) 5 α-particles? Use a constant-rate Poisson model with its rate estimated from the observed total.

Solution.

There are $40 \cdot 60 = 2400$ one-minute intervals. Their fitted mean is $\mu = 5000/2400 = 25/12$ decays per minute. The expected number of intervals containing exactly k decays is

$$E_k = 2400 P(X = k) = 2400 e^{-25/12} \frac{(25/12)^k}{k!}.$$

In particular,

$$\boxed{E_2 = 648.5128721 \simeq 649, \qquad E_5 = 97.73354192 \simeq 98 \text{ intervals.}}$$

These are expectations under the fitted Poisson model, not guaranteed integer observations. The manual calls the 2400 intervals seconds rather than minutes and stops at single-interval probabilities. The question asks for the corresponding expected numbers of intervals.

23.4.4 For a radioactive sample, 10 decays are counted on average in 100 seconds. Use the Poisson distribution to estimate the probability of counting 3 decays in 10 seconds.

Solution.

The rate is $10/100 = 0.1$ decay per second, so a 10-second interval has Poisson mean $\mu = 1$. Therefore

$$\boxed{P(X = 3) = e^{-1} \frac{1^3}{3!} = \frac{1}{6e} = 0.0613132402 \simeq 6.13\%.}$$

The time rescaling changes the mean, not the factorial in the mass formula.

23.4.5 ^{238}U has a half-life of 4.51×10^9 years. Its decay series ends with the stable lead isotope ^{206}Pb. The ratio of the number of ^{206}Pb to ^{238}U atoms in a rock sample is measured as 0.0058. Estimate the age of the rock assuming that all the lead in the rock is from the initial decay of the ^{238}U, which determines the rate of the entire decay process, because the subsequent steps take place

far more rapidly. Assume a closed sample, with no uranium or lead gained or lost since formation and no initial lead.

Hint. This is not a Poisson distribution problem, but is an application of the decay law $N(t) = Ne^{-\lambda t}$, where λ, the decay constant, is related to the half-life T by $T = \ln 2/\lambda$.

ANS. 3.8×10^7 years.

Solution.

Let U_0 be the initial uranium count and $U(t) = U_0 e^{-\lambda t}$ the remaining count. Under the stated closed-sample, zero-initial-lead and rapid-daughter-decay assumptions, $L(t) = U_0 - U(t)$. Thus the measured ratio is

$$R = \frac{L(t)}{U(t)} = \frac{U_0}{U(t)} - 1 = e^{\lambda t} - 1.$$

Solving, and using the *given* half-life T,

$$\boxed{t = \frac{\ln(1+R)}{\lambda} = \frac{4.51 \times 10^9}{\ln 2} \ln(1.0058) = 3.7628998 \times 10^7 \text{ years} \simeq 3.8 \times 10^7 \text{ years.}}$$

One uranium decay ultimately yields one lead atom in this accounting; the intermediate decay steps are not additional initial uranium atoms.

23.4.6 The probability of hitting a target in one shot is known to be 20%. If five shots are fired independently, what is the probability of striking the target at least once?

Solution.

The probability that one shot misses is $1 - 0.2 = 0.8$. By independence, all five miss with probability $0.8^5 = 0.32768$. Its complement is

$$\boxed{P(\text{at least one hit}) = 1 - 0.8^5 = 0.67232 = 67.232\%.}$$

This is an exact binomial complement; a Poisson approximation is unnecessary.

23.5 Gauss' Normal Distribution

23.5.1 Show that the variance of the normal distribution given by Eq. (23.56) is σ^2, the symbol in that equation. Take $\sigma > 0$.

Solution.

For $\sigma > 0$, write the density explicitly:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}.$$

With $z = (x-\mu)/\sigma$, normalization is $(2\pi)^{-1/2} \int_{-\infty}^{\infty} e^{-z^2/2}\, dz = 1$. Oddness gives $\langle X - \mu \rangle = 0$, so the mean really is μ. For the variance, integrate by parts using $d(e^{-z^2/2}) = -ze^{-z^2/2}\, dz$:

$$\begin{aligned} \operatorname{Var} X &= \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^2 e^{-z^2/2}\, dz \\ &= \frac{\sigma^2}{\sqrt{2\pi}} \left([-ze^{-z^2/2}]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} e^{-z^2/2}\, dz \right) = \boxed{\sigma^2}. \end{aligned}$$

The boundary term is zero because the Gaussian dominates the linear factor.

23.5.2 Show that Eq. (23.62) can be obtained by manipulation of the formula Eq. (23.61) for $W(v)$. Use $0 < p < 1$, $q = 1 - p$, $n > 0$, and $-pn < v < qn$; both formulas represent the same Stirling approximation.

Solution.

Let $0 < p < 1$, $q = 1 - p$, $s = pn + v$, and $n - s = qn - v$. The Stirling approximation in Eq.(23.61) is

$$W(v) = \frac{p^s q^{n-s} n^{n+1/2} e^{-n+s+(n-s)}}{\sqrt{2\pi}(pn+v)^{s+1/2}(qn-v)^{n-s+1/2}}.$$

The exponential is $e^0 = 1$. Factor the two denominator powers:

$$(pn+v)^{s+1/2} = (pn)^{s+1/2}(1+v/(pn))^{s+1/2},$$

$$(qn-v)^{n-s+1/2} = (qn)^{n-s+1/2}(1-v/(qn))^{n-s+1/2}.$$

The remaining constant quotient is

$$\frac{p^s q^{n-s} n^{n+1/2}}{p^{s+1/2} q^{n-s+1/2} n^{(s+1/2)+(n-s+1/2)}} = p^{-1/2} q^{-1/2} n^{-1/2}.$$

Therefore

$$\boxed{W(v) = \frac{(1+v/(pn))^{-(pn+v+1/2)}(1-v/(qn))^{-(qn-v+1/2)}}{\sqrt{2\pi pqn}}.}$$

The rearrangement is exact for the Stirling expression; its identification with the original binomial mass is asymptotic, requiring both s and $n-s$ large. For real logarithms also require $-pn < v < qn$.

23.5.3 With $W(v)$ the expression in Eq. (23.62), show that the expansion of $\ln W(v)$ in powers of v leads to Eq. (23.63). Take $p, q > 0$ fixed, $p + q = 1$, and $|v| < \min(pn, qn)$. For the Gaussian limit use the central scaling $v = O(\sqrt{n})$.

Solution.

Take the logarithm of the preceding result:

$$\begin{aligned}\ln W(v) = &-\tfrac{1}{2}\ln(2\pi pqn) - (pn+v+\tfrac{1}{2})\ln(1+v/(pn)) \\ &-(qn-v+\tfrac{1}{2})\ln(1-v/(qn)).\end{aligned}$$

For $|v| < \min(pn, qn)$ use $\ln(1+u) = u - u^2/2 + O(u^3)$. Keeping all terms through v^2, including the half-power factors,

$$-(pn+v+\tfrac{1}{2})\ln(1+v/(pn)) = -v - \frac{v}{2pn} - \frac{v^2}{2pn} + \frac{v^2}{4p^2n^2} + O(v^3),$$

$$-(qn-v+\tfrac{1}{2})\ln(1-v/(qn)) = v + \frac{v}{2qn} - \frac{v^2}{2qn} + \frac{v^2}{4q^2n^2} + O(v^3).$$

The $-v$ and v cancel. Collecting the rest yields Eq.(23.63):

$$\boxed{\begin{aligned}\ln W(v) = &-\tfrac{1}{2}\ln(2\pi pqn) - \frac{v}{2n}\left(\frac{1}{p} - \frac{1}{q}\right) \\ &- \frac{v^2}{2n}\left(\frac{1}{p} + \frac{1}{q}\right) + \frac{v^2}{4n^2}\left(\frac{1}{p^2} + \frac{1}{q^2}\right) + O(v^3).\end{aligned}}$$

Here $O(v^3)$ first denotes a Taylor remainder for fixed n, p, q. For the limiting argument hold $p, q > 0$ fixed and take $v = O(\sqrt{n})$. The remainder is then $O(|v|^3/n^2 + |v|^3/n^3) = O(n^{-1/2})$, while the displayed linear and last quadratic corrections also vanish. Since $1/p + 1/q = 1/(pq)$, the surviving form is

$$W(v) \sim \frac{1}{\sqrt{2\pi npq}} e^{-v^2/(2npq)}.$$

This central scaling does not assert a uniformly accurate Gaussian approximation all the way to the extreme binomial tails.

23.5.4 What is the probability for a normally distributed random variable to differ by more than 4σ from its mean value? Compare your result with the corresponding one from Chebyshev's inequality. Explain the difference in your own words. Assume $\sigma > 0$.

Solution.

Standardize $Z = (X - \mu)/\sigma$, for $\sigma > 0$. Symmetry and $u = z/\sqrt{2}$ give

$$P(|X - \mu| > 4\sigma) = \frac{2}{\sqrt{2\pi}} \int_4^\infty e^{-z^2/2}\, dz = \operatorname{erfc}(2\sqrt{2}) = \boxed{6.33424837 \times 10^{-5}}.$$

This is about 0.00633425%. Chebyshev's inequality gives only

$$P(|X - \mu| > 4\sigma) \le \frac{1}{4^2} = 0.0625 = 6.25\%.$$

The bound is about 987 times the exact Gaussian tail. Chebyshev uses only the variance and applies to every distribution with that variance; the Gaussian computation uses its entire density, including its rapidly decaying tails. A general upper bound need not be attained by this particular density. The manual's trailing approximation $6.3 \times 10^{-5} \approx 10^{-3}$ is numerically incorrect.

23.5.5 An instructor grades a final exam of a large undergraduate class, obtaining the mean value of points m and the variance σ^2. Assuming a normal distribution for the random variable M of points, he defines a grade F when $M < m - 3\sigma/2$, D when $m - 3\sigma/2 < M < m - \sigma/2$, C when $m - \sigma/2 < M < m + \sigma/2$, B when $m + \sigma/2 < M < m + 3\sigma/2$, and A when $M > m + 3\sigma/2$. What is the percentage of As, Fs; Bs, Ds; and Cs? Redesign the cutoffs so that there are equal percentages of As and Fs (5%), 25% Bs and Ds, and 40% Cs. Assume $\sigma > 0$.

Solution.

Use the corrected notation: m is the mean, M the random score, and $Z = (M - m)/\sigma$ has standard normal CDF

$$\Phi(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-u^2/2}\, du = \tfrac{1}{2}[1 + \operatorname{erf}(z/\sqrt{2})].$$

The original cutoffs give

$$\begin{aligned} P(A) = P(F) &= 1 - \Phi(1.5) = 0.0668072013, \\ P(B) = P(D) &= \Phi(1.5) - \Phi(0.5) = 0.2417303375, \\ P(C) &= \Phi(0.5) - \Phi(-0.5) = 0.3829249225. \end{aligned}$$

Thus the percentages are $\boxed{6.68072\%,\ 24.17303\%,\ 38.29249\%}$ for each extreme grade, each adjacent grade, and C, respectively. They sum as $2(6.68072) + 2(24.17303) + 38.29249 = 100\%$, up to rounding.

For the redesign let the positive thresholds be $a > b > 0$. The outer 5% requires $\Phi(a) = 0.95$, and the next 25% requires $\Phi(a) - \Phi(b) = 0.25$, hence $\Phi(b) = 0.70$. Therefore

$$a = \Phi^{-1}(0.95) = 1.644853627, \qquad b = \Phi^{-1}(0.70) = 0.524400513.$$

The new four score boundaries, in order, are

$$\boxed{m - 1.644853627\sigma, \quad m - 0.524400513\sigma, \quad m + 0.524400513\sigma, \quad m + 1.644853627\sigma.}$$

They yield 5%,25%,40%,25%,5%. Any assignment of an exact boundary to one adjacent grade gives the same probabilities because a continuous normal variable has no point masses.

23.6 Transformations of Random Variables

23.6.1 Let $X_1, X_2, \ldots, X_n$ be independent normal random variables with the same mean $\bar{x}$ and variance σ^2. Show that

$$\frac{\sum_i X_i/n - \bar{x}}{\sigma/\sqrt{n}}$$

is normal with mean zero and variance 1. Take integer $n \geq 1$ and $\sigma > 0$.

Solution.

Take $n \geq 1$ and $\sigma > 0$. To track the distribution, use the characteristic function, not just its first two moments. For a centered normal variable of variance σ^2, write $\phi(t) = \langle e^{itX} \rangle$. Since $xf(x) = -\sigma^2 f'(x)$, differentiation followed by integration by parts gives

$$\phi'(t) = i \int x e^{itx} f(x)\, dx = -i\sigma^2 \int e^{itx} f'(x)\, dx = -\sigma^2 t \phi(t).$$

The boundary term vanishes. With $\phi(0) = 1$ this yields $\phi(t) = e^{-\sigma^2 t^2/2}$; translation by $\bar{x}$ adds $e^{it\bar{x}}$. Independence therefore gives

$$\phi_{\overline{X}}(t) = \prod_{j=1}^{n} \phi_{X_j}(t/n) = \left[e^{it\bar{x}/n - \sigma^2 t^2/(2n^2)} \right]^n = e^{it\bar{x} - \sigma^2 t^2/(2n)}.$$

Thus $\overline{X} = n^{-1} \sum X_j$ is normal with mean $\bar{x}$ and variance σ^2/n. For the requested variable,

$$Z = \frac{\overline{X} - \bar{x}}{\sigma/\sqrt{n}}, \qquad \phi_Z(t) = e^{-t^2/2}, \qquad \boxed{Z \sim N(0,1).}$$

The corrected denominator is $\sigma/\sqrt{n}$. The printed $\sigma\sqrt{n}$ would give variance $1/n^2$ instead. For $\sigma = 0$ all observations are constant and this standardization is undefined, so positivity is necessary.

23.6.2 If the random variable X is normal with mean value 29 and standard deviation 3, what can you say about the distributions of $2X - 1$ and $3X + 2$?

Solution.

For $Y = aX + b$ with $a \neq 0$, change variables $x = (y - b)/a$:

$$f_Y(y) = \frac{1}{|a|} \frac{1}{3\sqrt{2\pi}} \exp\left[-\frac{((y-b)/a - 29)^2}{18} \right] = \frac{1}{3|a|\sqrt{2\pi}} \exp\left[-\frac{(y - (29a + b))^2}{2(3a)^2} \right].$$

This identifies both full distributions:

$$\boxed{2X-1\sim N(57,36),\quad \sigma=6;\qquad 3X+2\sim N(89,81),\quad \sigma=9.}$$

The second argument of $N(\mu,\tau^2)$ here denotes variance. The manual multiplies variance by a and reports $3\sqrt{2}$ and $3\sqrt{3}$ as standard deviations. The correct variance factor is a^2, so the standard-deviation factor is $|a|$.

23.6.3 For a normal distribution of mean value m and variance σ^2, find the distance r such that half the area under the bell shape is between $m-r$ and $m+r$. Assume $\sigma>0$ and seek $r\geq 0$.

Solution.

For $\sigma>0$, standardize $z=(x-m)/\sigma$ and use symmetry:

$$\frac{1}{2}=P(m-r\leq X\leq m+r)=2\Phi(r/\sigma)-1=\operatorname{erf}\left(\frac{r}{\sigma\sqrt{2}}\right).$$

Hence $\Phi(r/\sigma)=3/4$, and strict monotonicity gives the unique nonnegative answer

$$\boxed{r=\sigma\Phi^{-1}(3/4)=\sigma\sqrt{2}\,\operatorname{erf}^{-1}(1/2)=0.6744897502\sigma.}$$

This interval contains the central half of the probability mass.

23.6.4 If $\langle X\rangle$, $\langle Y\rangle$ are the average values of two independent random variables X, Y, what is the expectation value of the product XY? Assume finite absolute first moments.

Solution.

Assume finite absolute first moments. Independence gives the product joint measure, and also $\langle|XY|\rangle=\langle|X|\rangle\langle|Y|\rangle<\infty$, so the integrals may be separated. When densities exist, this reads

$$\boxed{\begin{aligned}\langle XY\rangle&=\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}xy\,p_X(x)p_Y(y)\,dx\,dy\\&=\left(\int xp_X(x)\,dx\right)\left(\int yp_Y(y)\,dy\right)=\langle X\rangle\langle Y\rangle.\end{aligned}}$$

For general laws the same calculation uses their product probability measure; discrete laws give sums. The manual omits the factors xy, x, and y in its three integrals. Without them those integrals are normalization integrals equal to 1, not the stated expectations. It follows in particular that $\operatorname{cov}(X,Y)=0$ when second moments exist.

23.6.5 If X and Y are two independent random variables with different probability densities and the function $f(x,y)$ has derivatives of any order, explain why the two means alone do not determine $\langle f(X,Y)\rangle$. Derive a finite Taylor expansion in central moments, including its remainder, and state conditions for an infinite moment expansion. Give the covariance and correlation of X,Y; also derive the corresponding formulas for transformed variables $G=g(X,Y)$ and $H=h(X,Y)$ when the required moments exist.

Solution.

The two means alone do not determine $\langle f(X,Y)\rangle$. Higher moments are needed. Existence of derivatives of every order also does not justify an infinite Taylor expansion or its termwise

expectation. The manual's weighted integral is not the ordinary covariance of the independent variables.

Let $\mu = \langle X \rangle$, $\nu = \langle Y \rangle$, $U = X - \mu$, $V = Y - \nu$, and define central moments $M_r = \langle U^r \rangle$, $N_s = \langle V^s \rangle$ when finite. Then $M_0 = N_0 = 1$, $M_1 = N_1 = 0$, and independence implies $\langle U^r V^s \rangle = M_r N_s$. The always valid starting point, when $\langle |f(X,Y)| \rangle < \infty$, is

$$\langle f(X,Y) \rangle = \iint f(x,y) p_X(x) p_Y(y)\, dx\, dy.$$

For a finite Taylor order k, apply one-variable Taylor's theorem to $g(t) = f(\mu + tU, \nu + tV)$ at $t = 0$, evaluated at $t = 1$. The chain rule and binomial theorem give

$$g^{(j)}(t) = \sum_{r+s=j} \frac{j!}{r!s!} U^r V^s f_{rs}(\mu + tU, \nu + tV).$$

Dividing $g^{(j)}(0)$ by $j!$ gives the following coefficients:

$$f(X,Y) = \sum_{r+s \le k} \frac{f_{rs}(\mu,\nu)}{r!s!} U^r V^s + R_k,$$

$$R_k = \frac{1}{k!} \int_0^1 (1-t)^k (U\partial_x + V\partial_y)^{k+1} f(\mu + tU, \nu + tV)\, dt, \quad f_{rs} = \partial_x^r \partial_y^s f.$$

Assuming integrability of these terms, taking expectations gives the rigorous finite form

$$\boxed{\langle f(X,Y) \rangle = \sum_{r+s \le k} \frac{f_{rs}(\mu,\nu)}{r!s!} M_r N_s + \langle R_k \rangle.}$$

For example, through second order,

$$\langle f(X,Y) \rangle = f(\mu,\nu) + \tfrac{1}{2} f_{20}(\mu,\nu)\sigma_X^2 + \tfrac{1}{2} f_{02}(\mu,\nu)\sigma_Y^2 + \langle R_2 \rangle.$$

The mixed term is zero because $M_1 N_1 = 0$. If the Taylor series equals f on the support and is integrable term by term (for example, $\sum_{r,s} |f_{rs}(\mu,\nu)| \langle |U|^r \rangle \langle |V|^s \rangle / (r!s!) < \infty$), the limit is

$$\langle f(X,Y) \rangle = \sum_{r,s \ge 0} \frac{f_{rs}(\mu,\nu)}{r!s!} M_r N_s.$$

Both stated conditions matter. For a counterexample to determination by the means, take $f(x,y) = x^2$, independent Y uniform on $[-3,3]$, and compare X uniform on $[-1,1]$ with X uniform on $[-2,2]$. All means are zero, but the expectations are $1/3$ and $4/3$. For a counterexample to smoothness sufficing, the smooth function $f(x,y) = e^{-1/x^2}$ for $x \neq 0$, extended by 0 at $x = 0$, has every derivative at $(0,0)$ zero: each derivative is an exponential times a polynomial in $1/x$, whose limit is 0. For centered independent normal X, Y, its expectation is nevertheless strictly positive.

For the variables actually named in the question,

$$\boxed{\operatorname{cov}(X,Y) = M_1 N_1 = 0, \qquad \rho_{XY} = 0}$$

provided both variances are finite and nonzero; correlation is undefined if either variance is zero. For the transformed variables $G = g(X,Y)$ and $H = h(X,Y)$, write their Taylor coefficients as

$g_{rs}/(r!s!)$ and $h_{uv}/(u!v!)$, all evaluated at (μ,ν). Under convergence and interchange conditions for the product as well,

$$\boxed{\operatorname{cov}(G,H)=\sum_{r,s,u,v\ge 0}\frac{g_{rs}h_{uv}}{r!s!u!v!}\big[M_{r+u}N_{s+v}-M_rN_sM_uN_v\big].}$$

This follows by expanding $\langle GH\rangle-\langle G\rangle\langle H\rangle$; setting $h=g$ gives the variance. The correlation is this covariance divided by $\sqrt{\operatorname{Var}G\operatorname{Var}H}$, when positive. To lowest order in small fluctuations it becomes

$$\operatorname{cov}(G,H)\simeq g_xh_x\sigma_X^2+g_yh_y\sigma_Y^2.$$

Finally the manual's object $\langle UVf(X,Y)\rangle$ would instead expand as $\sum f_{rs}M_{r+1}N_{s+1}/(r!s!)$, with leading term $f_{11}\sigma_X^2\sigma_Y^2$. It is a weighted mixed moment, not a covariance or a normalized correlation of X,Y.

23.6.6 Let $f(x,y)$ be the joint probability density of two random variables X, Y. Find the variance $\sigma^2(aX+bY)$, where a, b are constants. What happens when X, Y are independent? Take real a,b,X,Y and finite second moments.

Solution.

Set $\mu=\langle X\rangle$, $\nu=\langle Y\rangle$ and assume finite second moments. Since $\langle aX+bY\rangle=a\mu+b\nu$,

$$\begin{aligned}\operatorname{Var}(aX+bY)&=\iint[a(x-\mu)+b(y-\nu)]^2f(x,y)\,dx\,dy\\&=a^2\iint(x-\mu)^2f\,dx\,dy+b^2\iint(y-\nu)^2f\,dx\,dy\\&\quad+2ab\iint(x-\mu)(y-\nu)f\,dx\,dy\\&=\boxed{a^2\sigma_X^2+b^2\sigma_Y^2+2ab\operatorname{cov}(X,Y)}.\end{aligned}$$

All integrations are over $\mathbb{R}^2$. For independent X,Y, $f(x,y)=p_X(x)p_Y(y)$ makes the final integral the product of two zero centered means. Thus $\boxed{\operatorname{Var}(aX+bY)=a^2\sigma_X^2+b^2\sigma_Y^2}$ in that case.

23.6.7 Obtain an addition theorem for the distribution of a random variable $Y=X_1+X_2$ where X_1 and X_2 are independent Gauss normal random variables with different mean values μ_j and variances σ_j^2.

ANS. Y is normal with mean $\mu_1+\mu_2$ and variance $\sigma_1^2+\sigma_2^2$.

Solution.

The stated answer requires independence, as in the addition theorem being used. Normal marginal distributions alone are insufficient. The manual's final exponential also omits a factor 1/2. Let $X_j\sim N(\mu_j,\sigma_j^2)$ independently. Their characteristic functions, derived in Exercise 23.6.1, multiply:

$$\begin{aligned}\phi_Y(t)&=\langle e^{it(X_1+X_2)}\rangle=\langle e^{itX_1}\rangle\langle e^{itX_2}\rangle\\&=e^{it\mu_1-\sigma_1^2t^2/2}e^{it\mu_2-\sigma_2^2t^2/2}=e^{it(\mu_1+\mu_2)-(\sigma_1^2+\sigma_2^2)t^2/2}.\end{aligned}$$

The characteristic function identifies the complete distribution:

$$\boxed{Y\sim N(\mu_1+\mu_2,\sigma_1^2+\sigma_2^2).}$$

One can also check it directly by convolution. With $S = \sigma_1^2 + \sigma_2^2$ and $h = y - \mu_1 - \mu_2$, set $u = x - \mu_1$. Completing the square gives

$$\frac{u^2}{\sigma_1^2} + \frac{(h-u)^2}{\sigma_2^2} = \frac{S}{\sigma_1^2\sigma_2^2}\left(u - \frac{\sigma_1^2 h}{S}\right)^2 + \frac{h^2}{S}.$$

The Gaussian convolution integral is consequently

$$f_Y(y) = \frac{e^{-h^2/(2S)}}{2\pi\sigma_1\sigma_2}\sqrt{\frac{2\pi\sigma_1^2\sigma_2^2}{S}} = \frac{e^{-h^2/(2S)}}{\sqrt{2\pi S}}.$$

This calculation assumes positive variances; a zero variance just adds a constant, and both zero give a point mass. Without independence, take $X_1 = Z$, $X_2 = 1 + 2Z$ for $Z \sim N(0,1)$. Then the sum has variance 9, whereas the printed sum of marginal variances is $1 + 4 = 5$. For jointly normal dependent variables the correct variance adds $2\,\mathrm{cov}(X_1, X_2)$.

23.6.8 Show that the Fourier transform of the gamma-distribution probability density, Eq. (23.83), has the functional form given in Eq. (23.84). Take $p, \sigma > 0$ and use the book's positive-exponent, $1/\sqrt{2\pi}$ Fourier convention.

Solution.

For $p > 0$, $\sigma > 0$, Eq.(23.83) is

$$g(y) = \frac{y^{p-1}e^{-y/(2\sigma^2)}}{(2\sigma^2)^p\Gamma(p)}, \quad y > 0,$$

and zero elsewhere. In the book's positive-exponent Fourier convention,

$$\widehat{g}(t) = \frac{1}{\sqrt{2\pi}(2\sigma^2)^p\Gamma(p)}\int_0^\infty y^{p-1}e^{-ay}\,dy, \qquad a = \frac{1}{2\sigma^2} - it.$$

For positive real a, $u = ay$ gives the integral $\Gamma(p)a^{-p}$. Both the integral and a^{-p} are holomorphic in $\Re a > 0$, using the logarithm continued from positive real a; exponential domination justifies differentiation on its compact subsets. The identity theorem extends the formula throughout that half-plane, which contains every a above for real t. Substitution gives

$$\boxed{\widehat{g}(t) = \frac{1}{\sqrt{2\pi}}(1 - 2i\sigma^2 t)^{-p}.}$$

At $t = 0$ it is $1/\sqrt{2\pi}$, confirming normalization. The characteristic function is $\sqrt{2\pi}\widehat{g}(t)$; it has value 1 at 0 and gives mean $2p\sigma^2$ on differentiation.

23.7 Statistics

23.7.1 Let ΔA be the error of a measurement of A, etc. Use first-order propagation of uncorrelated errors at nonzero nominal values to show that

$$\left(\frac{\sigma(C)}{C}\right)^2 \simeq \left(\frac{\sigma(A)}{A}\right)^2 + \left(\frac{\sigma(B)}{B}\right)^2$$

holds for the product $C = AB$ and the ratio $C = A/B$.

Solution.

The stated rule is first-order error propagation for uncorrelated errors, evaluated at nonzero nominal A, B. Correlations add a cross term, and nonlinear propagation need not satisfy this rule exactly. Write the errors as δA, δB, with zero means. For the product, first-order Taylor expansion gives

$$\delta C = B\delta A + A\delta B, \qquad \frac{\delta C}{C} = \frac{\delta A}{A} + \frac{\delta B}{B}.$$

For the ratio, the two derivatives are $\partial_A C = 1/B$ and $\partial_B C = -A/B^2$, so

$$\delta C = \frac{\delta A}{B} - \frac{A\delta B}{B^2}, \qquad \frac{\delta C}{C} = \frac{\delta A}{A} - \frac{\delta B}{B}.$$

Square each relation and average:

$$\frac{\sigma_C^2}{C^2} \simeq \frac{\sigma_A^2}{A^2} + \frac{\sigma_B^2}{B^2} \pm \frac{2\,\mathrm{cov}(A,B)}{AB},$$

where $+$ is for the product and $-$ for the ratio. With zero covariance,

$$\boxed{\left(\frac{\sigma_C}{C}\right)^2 \simeq \left(\frac{\sigma_A}{A}\right)^2 + \left(\frac{\sigma_B}{B}\right)^2.}$$

To see the scope of the approximation, for independent random A, B with nonzero means μ_A, μ_B the exact product formula is

$$\begin{aligned}\mathrm{Var}(AB) &= \langle A^2\rangle\langle B^2\rangle - \mu_A^2\mu_B^2 \\ &= (\mu_A^2 + \sigma_A^2)(\mu_B^2 + \sigma_B^2) - \mu_A^2\mu_B^2 \\ &= \mu_B^2\sigma_A^2 + \mu_A^2\sigma_B^2 + \sigma_A^2\sigma_B^2.\end{aligned}$$

First-order propagation omits the last, higher-order term. For a ratio it additionally requires fluctuations small enough that the denominator stays away from zero in the regime being approximated.

23.7.2 Find the mean value and standard deviation of the sample of measurements $x_1 = 6.0$, $x_2 = 6.5$, $x_3 = 5.9$, $x_4 = 6.1$, $x_5 = 6.2$. If the point $x_6 = 6.1$ is added to the sample, how does the change affect the mean value and standard deviation? Use the sample standard deviation with divisor $N - 1$; state separately the empirical-distribution convention with divisor N.

Solution.

Use the sample standard deviation of Eq.(23.95), with divisor $N - 1$. For the five given measurements,

$$\bar{x}_5 = \frac{6.0 + 6.5 + 5.9 + 6.1 + 6.2}{5} = \frac{30.7}{5} = 6.14.$$

The squared-deviation sum is

$$S_5 = (-.14)^2 + (.36)^2 + (-.24)^2 + (-.04)^2 + (.06)^2 = .0196 + .1296 + .0576 + .0016 + .0036 = .212.$$

Thus

$$\boxed{\bar{x}_5 = 6.14, \quad s_5^2 = .212/4 = .053, \quad s_5 = .230217289.}$$

Adding $x_6 = 6.1$ gives $\bar{x}_6 = 36.8/6 = 92/15 = 6.133333333$. To compute its spread exactly, write the six deviations with denominator 30:

$$\left(-\frac{4}{30}, \frac{11}{30}, -\frac{7}{30}, -\frac{1}{30}, \frac{2}{30}, -\frac{1}{30}\right).$$

They sum to zero and their squared sum is $S_6 = (16 + 121 + 49 + 1 + 4 + 1)/900 = 192/900 = 16/75$. Therefore

$$\boxed{s_6^2 = \frac{16/75}{5} = \frac{16}{375}, \qquad s_6 = .206559112.}$$

The mean decreases by $1/150 = 0.006666667$ and the sample standard deviation decreases by about 0.0236582. If one instead describes the empirical distribution with divisor N as in Eq.(23.23), its standard deviations are $\sqrt{.212/5} = .205912603$ and $\sqrt{(16/75)/6} = .188561808$; these are different conventions. The manual omits the original $x_4 = 6.1$, computing four then five observations. The exercise supplies five then six.

23.7.3 Carry out a χ^2 analysis of the fit corresponding to Fig. 23.10 using the same points as in Example 23.7.2, but with the errors now associated with the t_i rather than the y_i. Keep the fit through the origin. Treat the ordinates as exact and the quoted uncertainties as known standard deviations of independent normal time errors.

Solution.

The actual textbook Example 23.7.2 has $y_3 = 2.7$. The manual substitutes 3.0 and consequently obtains a different fit. Use the textbook data throughout. With the errors now assigned to time, the exact ordinates are $y = (0.8, 1.5, 2.7)$ and the measured times are $t = (1, 2, 3)$, with independent standard deviations $\sigma = (0.1, 0.05, 0.2)$. The physical constraint remains that the line passes through the origin. Fit $t = by$ by minimizing

$$\chi^2(b) = \frac{(1 - .8b)^2}{.1^2} + \frac{(2 - 1.5b)^2}{.05^2} + \frac{(3 - 2.7b)^2}{.2^2}.$$

Differentiate and set the result to zero:

$$0 = -2\sum_j \frac{y_j(t_j - by_j)}{\sigma_j^2}, \qquad \widehat{b} = \frac{N}{D}, \quad N = \sum_j \frac{y_j t_j}{\sigma_j^2}, \quad D = \sum_j \frac{y_j^2}{\sigma_j^2}.$$

The individual terms give

$$N = 80 + 1200 + 202.5 = 1482.5, \qquad D = 64 + 900 + 182.25 = 1146.25.$$

Since $\chi^{2\prime\prime}(b) = 2D > 0$, this is the unique minimum:

$$\boxed{\widehat{b} = \frac{1186}{917} = 1.2933478735, \qquad y = \frac{917}{1186}t = 0.7731871838t.}$$

For its uncertainty, differentiate the estimator with respect to each time: $\partial\widehat{b}/\partial t_j = y_j/(D\sigma_j^2)$. Because the estimator is linear, independent-error propagation is exact:

$$\operatorname{Var}\widehat{b} = \sum_j \frac{y_j^2\sigma_j^2}{D^2\sigma_j^4} = \frac{1}{D} = \frac{4}{4585}, \qquad \boxed{\sigma_b = 0.0295365880.}$$

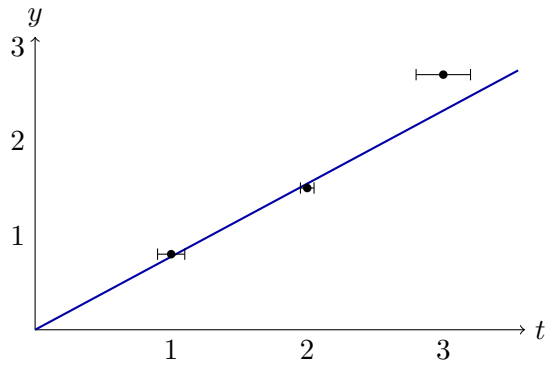
Completing the quadratic gives

$$\chi^2(b) = \chi^2_{\min} + D(b-\widehat{b})^2, \quad \chi^2_{\min} = \sum_j t_j^2/\sigma_j^2 - N^2/D = 1925 - \frac{1482.5^2}{1146.25} = \frac{6980}{917} = 7.6117775354.$$

For independent normal measurement errors with the specified variances, write $z_j = t_j/\sigma_j$, $v_j = y_j/\sigma_j$. Minimizing $\|z - bv\|^2$ projects z onto the one-dimensional span of v. An orthogonal rotation taking v to the first coordinate leaves the standardized normal errors independent with unit variance. The residual squared norm uses the other two coordinates, so it has $3-1=2$ degrees of freedom. Its upper-tail probability is

$$\boxed{P(\chi^2_2 \ge \chi^2_{\min}) = e^{-\chi^2_{\min}/2} = 0.0222394227.}$$

Thus a residual at least this large occurs about 2.22% of the time under this model and the stated errors. The fit has reduced $\chi^2 = 3.805888768$; this alone does not identify which model assumption is responsible for the discrepancy.



If a 95% confidence interval for b is also desired, known normal measurement errors give $\widehat{b} \pm 1.959963985\sigma_b$:

$$b \in [1.235457225, 1.351238522].$$

Taking reciprocals reverses the positive endpoints, giving $dy/dt \in [0.740061790, 0.809416935]$. Here σ_b already is the estimator's standard deviation; it must not be divided by $\sqrt{3}$. With known individual variances there is no additional Student-t factor either.

23.7.4 Using the data from Exercise 23.7.2 (including the point x_6), find the 90% and 95% confidence intervals for the mean of the x_i. Assume independent observations from one normal population with unknown mean and variance.

Solution.

Assume the six measurements are an independent sample from a normal population with unknown mean μ and unknown variance σ^2. Exercise 23.7.2 gives

$$N = 6, \quad \bar{x} = \frac{92}{15}, \quad s^2 = \frac{1}{N-1}\sum_{j=1}^{N}(x_j - \bar{x})^2 = \frac{16}{375}.$$

The standard error of the mean is therefore

$$\frac{s}{\sqrt{N}} = \sqrt{\frac{16}{375 \cdot 6}} = \sqrt{\frac{8}{1125}} = 0.0843274043.$$

To distinguish sample size from degrees of freedom, standardize the six observations by their true μ, σ and rotate their Gaussian coordinate vector orthogonally so that its first coordinate is the normalized sum. Its density is proportional to $\exp(-\sum z_j^2/2)$, which is unchanged by this rotation and factors over the new coordinates. Thus

$$Z = \frac{\sqrt{N}(\overline{X} - \mu)}{\sigma} \sim N(0,1), \qquad Q = \frac{(N-1)S^2}{\sigma^2} \sim \chi^2_{N-1}$$

are independent: the residual squared norm uses the other five coordinates. Their ratio is

$$T = \frac{Z}{\sqrt{Q/(N-1)}} = \frac{\overline{X} - \mu}{S/\sqrt{N}} \sim t_5.$$

Let c_p have $P(T \le c_p) = p$. By symmetry, $P(-c_p \le T \le c_p) = 2p - 1$. Solving these inequalities for μ gives the confidence interval $\bar{x} \pm c_p s/\sqrt{N}$. For 90% and 95% use respectively $p = .95, .975$:

$$c_{.95} = 2.0150483733, \qquad c_{.975} = 2.5705818356.$$

They can be obtained by integrating the normalized density

$$f_5(t) = \frac{\Gamma(3)}{\sqrt{5\pi}\,\Gamma(5/2)}(1 + t^2/5)^{-3}$$

and solving $\int_{-\infty}^{c_p} f_5(t)\,dt = p$. Multiplication by the standard error yields

$$\boxed{\begin{array}{ll} 90\%: & 6.133333333 \pm 0.169923799 = [5.963409535, 6.303257132], \\ 95\%: & 6.133333333 \pm 0.216770494 = [5.916562840, 6.350103827]. \end{array}}$$

The sample size is 6; the Student distribution has 5 degrees of freedom. The manual's use of $\sqrt{5}$ in the standard error confuses these quantities and makes both intervals too wide. The stated coverage is for repeated sampling under the normal-sample assumptions, rather than a probability distribution assigned to the fixed unknown population mean.